HANDBOOK OF CONVEX GEOMETRY

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Preface

One aim of this Handbook is to survey convex geometry, its many ramifications and its relations with other areas of mathematics. We believe that it will be a useful tool for the expert. A second aim is to give a high level introduction to most branches of convexity and its applications, showing the major ideas, methods and results. We hope that because of this feature the Handbook will act as an appetizer for future researchers in convex geometry. For them the many explicitly or implicitly stated problems should turn out to be a valuable source of inspiration. Third, the Handbook should be useful for mathematicians working in other areas as well as for econometrists, computer scientists, crystallographers, phycisists and engineers who are looking for geometric tools for their own work. In particular, mathematicians specializing in optimization, functional analysis, number theory, probability theory, the calculus of variations and all branches of geometry should profit from the Handbook.

The famous treatise "Theorie der konvexen Körper" by Bonnesen and Fenchel presented in 164 pages an almost complete picture of convexity as it appeared around 1930. While a similarly comprehensive report today seems to be out of reach, the Handbook deals with most of the more important topics of convexity and its applications. By comparing the Handbook with the survey of Bonnesen and Fenchel and with more recent collections of surveys of particular aspects of geometric convexity such as the AMS volume edited by Klee (1963), the Copenhagen Colloquium volume edited by Fenchel (1967), the two green Birkhäuser volumes edited by Tölke and Wills (1979) and Gruber and Wills (1983), respectively, and the New York Academy volume edited by Goodman, Lutwak, Malkevitch and Pollack (1985), the reader may see where progress was made in recent years.

During the planning stage of the Handbook, which started in 1986, we got generous help from many prominent convex geometers, in particular Peter McMullen, Rolf Schneider and Geoffrey Shephard. The discussion of the list of contents and of prospective authors turned out to be difficult. Both of us are obliged to the authors who agreed to contribute to the Handbook. In the cooperation with them we got much encouragement and the professional contacts furthered our good personal relations with many of them. The manuscripts which we finally received turned out to be much more diverse than we had anticipated. They clearly exhibit the most different characters and scientific styles of the authors and this should make the volume even more attractive.

There are several researchers in geometric convexity whom we invited to contribute to the Handbook but who for personal, professional or other reasons – regretfully – were not able to participate. The reader will also note that one area or another is missing in the list of contents. Examples are elementary geometry of normed planes, axiomatic convexity, and Choquet theory, but this should not diminish the usefulness of the Handbook.

Some fields such as computational and algorithmic aspects or lattice point results are dealt with in several chapters. The subjects covered are organized in five parts which in some sense reflect the fact that convexity is situated between analysis, geometry and discrete mathematics. This organization clearly has some disadvantages but we think that it will be helpful for the reader who wants to orient himself.

In the editing of the Handbook we received much help, in particular from Dr. M. Henk, Dr. J. Müller, Professor S. Hildebrandt, Professor L. Payne and Professor F. Schnitzer. We are most grateful to Ms. S. Clarius and Ms. E. Rosta who typed about 1000 letters to the authors, to colleagues whom we asked for advice and to the Publishers. We most gratefully acknowledge the friendly coopcrations and expert support of Dr. A. Sevenster and Mr. W. Maas from Elsevier Science Publishers.

Finally we should like to express our sincere hope that the readers will appreciate the great effort of so many prominent authors and will find the Handbook useful for their scientific work.

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CHAPTER 0

History of Convexity

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Contents

1. Introduction 3
2. Antiquity 3
3. Modern times up to the 18th century 4
4. The 19th century with glimpses into the 20th century 5
5. From the 19th to the 20th century and the 20th century 7

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1. Introduction

The history of convexity is a topic with many facets. The majority of these are easily recognizable, but there are facets which are almost hidden, for example, in the contexts of functional analysis, calculus of variations, complex function theory of one or more variables, graph theory or crystallography.

The first contributions to convexity date back to antiquity but the systematic study started only at the turn of the 19th to the 20th century.

Due to its position between geometry, analysis and discrete mathematics, convexity has many relations with other areas. This is even more apparent in recent years. It is surprising to observe that many eminent mathematicians contributed one or another result to convexity while working mainly in other areas. Examples not mentioned below are Lebesgue, A. Kneser, Lindelöf, Sierpiński, Perron and van der Waerden.

A large body of results about convexity is rather special, but there also exist some areas of a systematic nature, for example, the Brunn–Minkowski theory on mixed volumes, the combinatorial theory of convex polytopes, dissection results and the intrinsic geometry of convex surfaces. It is in this sense that convexity resembles elementary number theory. Since any survey or historical outline necessarily stresses the more systematic aspects, it is not possible to give a complete and objective picture of the whole area. Further, the reader will note a basic difficulty encountered by historians of mathematics: how should a highly complex structure such as convexity be presented realistically in terms of a – necessarily – linear language? Many more recent developments are touched only superficially, or not at all.

Earlier historical accounts of convexity are due to Fenchel ("Convexity through the ages", in: "Convexity and its applications", Basel 1983) and Gruber ("Zur Geschichte der Konvexgeometrie und der Geometrie der Zahlen", in: "DMV-Festschrift", Wiesbaden 1990). The latter addresses a general public and contains some original citations, biographical notes, photographs and many references. Historical surveys dealing with the isoperimetric problem and polytopes were given by Porter ("A history of the classical isoperimetric problem", in: "Contributions to the calculus of variations", Chicago 1933) and Malkevitch ("Milestones in the history of polyhedra", in: "Shaping space", Basel 1988). A thorough discussion of Descartes results on polytopes is due to Federico ("Descartes on polyhedra", New York 1982).

Due to the restricted space only a few results about convexity will be stated explicitly. Since this Handbook contains most of the more important contributions to convex geometry, the interested reader is referred to the pertinent sections.

2. Antiquity

Surprisingly, the known contributions of antiquity to convexity all led to impor-

tant modern developments, even when originally they appeared to be of a rather particular type.

The "Elements" of Euclid (about 300 B.C.) contain several contributions to convexity. Among these are the axioms for equality ("Elements", Book 1), the implicitly defined and frequently used concepts of equidissectable and equicomplementable polygons and polytopes (Books 1,12), theorems on polytopes (Books 11–13) and the derivation of regular polytopes (Book 13). The latter were known already to the Pythagoreans and to Theaetet (410?–369 B.C.), and the Etruscans seem to have known the dodecahedron even earlier.

Archimedes (287?–212 B.C.) was the first to give a precise definition of a convex curve or surface (in his book "On the sphere and cylinder"). Other contributions of Archimedes to what now is considered convex geometry include his statements – in part with proofs or mechanical explanations – on convex plane figures and solids (see "Measurement of a circle", "On the sphere and cylinder" and "The method of mechanical theorems"). The exhaustion method which Archimedes applied masterly was used earlier already by Eudoxus (about 400–about 350 B.C.) and perhaps even by Democrit (about 460–about 370 B.C.). For another application see Euclid (Book 12). Further, the postulates and results of Archimedes concerning the center of gravity ("On the equilibrium of planes or centers of gravity of planes") and his description of the 13 semiregular polytopes [lost, but referred to by Pappus (about 320 A.D.)] deserve mentioning. (A convex polytope is semiregular if its facets are regular polygons of at least two types and the group of isometries is transitive on the vertices.)

Of the contributions to the isoperimetric problem those of Zenodorus (between 200 B.C. and 90 A.D.) seem to be the most important ones. He shows that among all convex n-gons of given perimeter the regular n-gon has the largest area (assuming that a maximizing n-gon exists).

3. Modern times up to the 18th century

Kepler (1571–1630) rediscovered the Archimedean solids or semiregular polytopes (in "Harmonices mundi" 1619) and showed that there are only 13 of them. He also specified two of the four regular star polytopes. The remaining two were discovered by Poinsot (1777–1859). One of Kepler's star polytopes was found already earlier, probably by Uccello (1397?–1475) and is contained in a mosaic in San Marco in Venice and one of Poinsot's polytopes was anticipated in a slightly imprecise form by Jamnitzer (1508–1585). Kepler also described two of the duals of the Archimedian solids, usually called isohedra. Catalan (1814–1894) gave a systematic account of all 13 isohedra. More recently, Coxeter (born 1907) substantially enlarged and summarized the knowledge on regular polytopes of various kinds ("Regular polytopes", London 1948, "Regular complex polytopes", Cambridge 1974). Kepler considered planar tilings and gave a picture of the densest lattice packing of balls of equal radii in $\mathbb{E}^3$. The first precise determination of the densest lattice packing of circular discs in $\mathbb{E}^2$ is due to Lagrange (1736–1813) ("Recherches d'arithmétique" 1773).

Euler (1707–1783) stated the following relations between the numbers f_0 of vertices, f_1 of edges and f_2 of facets of a convex polytope in $\mathbb{E}^3$:

$$f_0 - f_1 + f_2 = 2, \qquad f_0 \leqslant 2f_2 - 4, \qquad f_2 \leqslant 2f_0 - 4 \tag{1}$$

("Elementa doctrina solidorum" 1752/53). Euler's proof is topological but seems to lack some tool of the type of Jordan's curve theorem for polygons (or the "shelling" procedure of convex polytopes due to Bruggesser and Mani). The first completely rigorous proof is due to Legendre (1752–1833) in 1794. It makes use of spherical trigonometry. Descartes (1596–1650) found results (dealing with angles of a polytope) of which (1) is an immediate consequence.

The geometric results on the isoperimetric property of circles and balls up to the 18th century in essence do not surpass the results achieved already in antiquity. Besides there are numerous analytic proofs. In all of them the existence of solutions is tacitly assumed.

The contributions to regular polytopes and Euler's polytope formula are still of interest, whereas the analytic proofs of the isoperimetric inequality seem to have no influence on modern convexity theory.

4. The 19th century with glimpses into the 20th century

In the early 19th century, indications of systematic later developments can already be recognized. For this reason topics such as optimization, and isoperimetric problems are discussed in this section.

Fourier (1768–1830) first specified (in "Solution d'une question particulière du calcul des inégalités" 1826) conditions for the solvability of linear inequalities (theorem of Fourier–Kuhn) and developed an algorithm for the determination of all solutions (Fourier elimination). Later contributions of this kind are due to Gordan 1873, Farkas 1902, Stiemke 1915, and Motzkin 1936. Linear programming was developed mainly by Kantorovič (1912–1986) and Dantzig (born 1914); other workers in this area are Gomory, Fan, Gale, Kuhn, Tucker, Minty, Klee, Shor, Khachyan and Karmarkar.

Using the concept of (point) lattice, Gauss (1777–1855) determined in a review of a book ("Untersuchungen über die Eigenschaften der positiven ternären quadratischen Formen von Ludwig August Seeber", Freiburg 1831) the densest lattice packing of balls in $\mathbb{E}^3$ [Seeber (1793–1855)]. Gauss' proof is an example of an extremely elegant argument which – a prima vista – gives no insight why it works. The corresponding results for dimensions 4, 5 are due to Korkin (1837–1908) and Zolotarev (1847–1878) 1872, 1877 and for dimensions 6, 7, 8 to Blichfeldt 1925–1934.

Cauchy's (1798–1857) fundamental contributions to convexity are the following: in the "Elements", Book 11, Euclid essentially states in "Definitions" 9, 10 that two convex polytopal surfaces in $\mathbb{E}^3$ coincide up to proper or improper rigid motions if there is a homeomorphism between these surfaces the restriction of which to any facet is a rigid motion. Legendre pointed out to Cauchy that this statement requires a proof and the latter provided a most elegant proof (in "Sur les polygons

et polyèdres" 1813). Minor gaps in Cauchy's proof were corrected by Lebesgue in 1909 and Steinitz in 1934 (posthumous).

Cauchy's surface area formula ("Memoire sur la rectification des courbes et la quadrature des surfaces courbes" 1850) says that the surface area of a convex body in $\mathbb{E}^3$ is four times the mean value of the areas of its projections onto the 2-dimensional subspaces of $\mathbb{E}^3$. This result is a particular case of Kubota's 1925 formulae for the calculation of quermassintegrals.

In the context of his work on quadratic forms Dirichlet (1805–1859) introduced in 1850 the concept of Dirichlet-cell, now also known as Voronoi-cell.

Von Staudt (1798–1867) contributed to convexity the concept of polar convex body (in his famous treatise "Geometrie der Lage", Nürnberg 1847, on projective geometry). His results were later rediscovered by Minkowski (posthumous) and Steinitz in the years from 1913 to 1916.

The impact of the work of Steiner (1796–1863) is more easily recognizable. He gave a series of proofs of the isoperimetric property of circles and balls using, amongst others, Steiner symmetrization and the four hinge method (see, e.g., "Einfache Beweise der isoperimetrischen Hauptsätze" 1836, "Über Maximum und Minimum bei den Figuren in der Ebene, auf der Kugel und im Raum überhaupt" 1841). (The four hinge method seems to have been anticipated by Cramer in 1752 and the idea of symmetrization first appeared in work of L'Huillier in the 1780s; see Danilova, "Development of elementary methods in the theory of isoperimeters in the 18th and the first third of the 19th centuries" 1985.) Steiner and his predecessors all ignored the existence problem as was pointed out by Dirichlet. The solution of the isoperimetric problem was finally achieved by Edler ("Vervollständigung der Steinerschen elementargeometrischen Beweise für den Satz, daß der Kreis größeren Flächeninhalt besitzt als jede andere ebene Figur gleich großen Umfangs" 1882) and by Schwarz (1843–1921) ("Beweis des Satzes, daß die Kugel kleinere Oberfläche besitzt als jeder andere Körper gleichen Volumens" 1884) and Weierstrass (1815–1897) (in his posthumously published lectures). Among the various early alternative solutions of the isoperimetric problem we mention those of Minkowski ("Über die Begriffe Länge, Oberfläche und Volumen" 1901) and Blaschke (in: "Kreis und Kugel", Leipzig 1916). Minkowski used his new concept of surface area and the Brunn–Minkowski theorem; Blaschke used Steiner symmetrization and his selection theorem. More modern contributions are due to Gross (1886–1918), Bonnesen (1873–1935) and, in particular, E. Schmidt (1876–1959) ("Die Brunn–Minkowskische Ungleichung und ihr Spiegelbild sowie die isoperimetrische Eigenschaft der Kugel in der euklidischen und nichteuklidischen Geometrie I,II" 1948, 1949) and to Dinghas (1908–1974). In recent years, geometric isoperimetric problems on manifolds and minimal surfaces were investigated. Examples are the Federer–Fleming isoperimetric inequality of 1960 and its precise form due to Almgren ("Optimal isoperimetric inequalities" 1986). For detailed expositions see Burago and Zalgaller ("Geometric inequalities", New York 1988) or Osserman 1978, 1980.

A second contribution of Steiner to convexity is his formula for the volume of parallel bodies of a convex body ("Über parallele Flächen" 1840). $V(C+\lambda B^d)$ $(\lambda \geqslant$

0) is a polynomial in λ of degree d, the coefficients of which are (up to constants) the quermassintegrals or intrinsic volumes. Their importance is reflected in the "functional theorems" of Blaschke ("Vorlesungen über Integralgeometrie", Berlin 1937) and Hadwiger in 1950. A generalization of the functional theorem was given by Schneider in 1971. Steiner ("Von dem Krümmungs-Schwerpuncte ebener Curven" 1838) introduced the curvature centroid or Steiner point $\operatorname{st} C$ of a convex body C. It satisfies $\operatorname{st}(C+D) = \operatorname{st} C + \operatorname{st} D$ for any convex bodies C, D.

Schläfli's (1814–1895) results on higher-dimensional polytopes ("Theorie der vielfachen Kontinuität" 1852) were published only in Bern 1901 and remained largely unnoticed. He extended Euler's polytope formula to higher dimensions as did independently of him Poincaré (1854–1912). Schläfli's arguments were completed by the shelling method of Bruggesser and Mani in 1972 and Poincaré's result required justifications from homology theory which were obtained in the 1930s. The first elementary proofs in higher dimensions are due to Hadwiger in 1955 and Grünbaum in 1967.

Extremum problems for physical quantities such as torsional rigidity, electrostatic capacity or principal frequencies of a membrane which may be treated by geometric methods such as Steiner and Schwarz symmetrization were first considered by de Saint-Venant (1799–1886) in 1856 and later on by Rayleigh (1842–1919), Hadamard (1865–1963), Poincaré, Faber (1877–1966), Pòlya (1887–1985) and Szegö (1895–1985). More recent work is due to Payne, Osserman, Protter, Talenti, Hersch and others. These results form a bridge from convexity to partial differential equations. See Bandle ("Isoperimetric inequalities and applications", Boston 1980). Another such bridge is described in Pogorelov ("Extrinsic geometry of convex surfaces", Moscow 1969). The existence of a convex surface with prescribed Gaussian curvature is connected with the existence of solutions of Monge–Ampère equations.

5. From the 19th to the 20th century and the 20th century

Finally, at the turn of the 19th century convexity became an independent branch of mathematics with its own problems, methods and theories and only a few easily recognizable connections with other areas (for example, with differential geometry and number theory). Up to the 1930s, systematic results seem to have been predominant. Since then there appears to be a balance between more theoretical results and a great body of interesting but special contributions to convexity. Since the 1950s, a multitude of additional relations to other areas became apparent, e.g., to optimization, convex analysis, control theory, crystallography, coding-theory, functional analysis, algebraic geometry and probability theory.

The major contribution of Brunn (1862–1939) to convexity is the Brunn–Minkowski inequality:

$$V(C+D)^{1/d} \geqslant V(C)^{1/d} + V(D)^{1/d} \tag{2}$$

for proper convex bodies C, D, where equality holds if and only if C, D are positive

homothetic ("Über Ovale und Eiflächen", Munich 1887, "Über Kurven ohne Wendepunkte", Munich 1889). (2) is valid for quermassintegrals when $1/d$ is replaced by $1/(d-i)$ for the ith quermassintegral, see Aleksandrov 1937 and Fenchel 1936. Generalizations of (2) to more general sets, even to arbitrary measurable sets are due to Lyusternik 1935, Schmidt 1948, Hadwiger and Ohmann 1958, Henstock and Macbeath 1953 and Das Gupta 1980. In recent years, stability problems related to (2) but also to packings, coverings and the Aleksandrov–Fenchel inequality have been considered by Diskant, Fuglede, Groemer and Schneider.

Minkowski (1864–1909) systematically developed convexity theory (in particular in "Volumen und Oberfläche" 1903 and "Theorie der konvexen Körper, insbesondere Begründung ihres Oberflächenbegriffs", posthumous). His theorem on mixed volumes says that for convex bodies $C_1, \ldots, C_n$ in $\mathbb{E}^d$ there are coefficients $V(C_{i_1}, \ldots, C_{i_d})$, symmetric in the indices, the mixed volumes, such that for $\lambda_1, \ldots, \lambda_n \geqslant 0$

$$V(\lambda_1 C_1 + \cdots + \lambda_n C_n) = \sum_{i,j} V(C_{i_j}, \ldots, C_{i_d}) \lambda_{i_1} \cdots \lambda_{i_d}.$$

His inequalities for $d = 3$ between the mixed volumes were extended by Fenchel (1905–1988) 1936 and Aleksandrov 1937, 1938 to general d:

$$V(C_1, C_2, C_3, \ldots, C_d)^2 \geqslant V(C_1, C_1, C_3, \ldots, C_d) V(C_2, C_2, C_3, \ldots, C_d).$$

They have numerous applications and their connections to algebraic geometry (Hodge index theorem) were discovered by Teissier 1979 and Hovanski 1979. The equality case is still open.

Minkowski ("Allgemeine Lehrsätze über die konvexen Polyeder" 1897) considered the determination of a convex polytope by the areas and exterior normal vectors of its facets. This is related to the determination of a smooth convex body by its Gauss curvature (considered as a function of the exterior normal vector) ("Über die geschlossenen konvexen Flächen" 1901). Generalizations and related results of Minkowski's theorem in the context of differential geometry were given amongst others by Nirenberg in 1953, Chern 1959, Cheng–Yau 1976 and Pogorelov. The latter's results are presented in his book "The Minkowski multidimensional problem", Washington 1978. Basic results in this area are also due to Busemann, see his treatise "Convex surfaces", New York 1958. On the other hand, Aleksandrov 1937, 1938 and Fenchel and Jessen 1938 proved existence theorems for convex surfaces where instead of the Gauss curvature a corresponding measure is used. For a survey see Schneider ("Boundary structure and curvature of convex bodies" 1979). Related to Minkowski's problem is Christoffel's problem (1865): determine a smooth convex body by the sum of its principal radii of curvature (or, more generally, by a symmetric function of these). Contributions to these problems in the context of differential geometry and convexity are due to Aleksandrov 1937, 1962, Fenchel and Jessen 1938, and Oliker 1979. Related results were given by Voss, Leichtweiss, and Firey. Minkowski's and Christoffel's problem are connected with Weyl's problem, see below.

Refining estimates of Hermite for the arithmetic minimum of a positive quadratic form, Minkowski found in 1893 the following fundamental theorem (see also his books "Geometrie der Zahlen", Leipzig 1896, "Diophantische Approximationen", Leipzig 1907): let C be a convex body in $\mathbb{E}^d$, symmetric with respect to o and let L be a lattice of determinant $d(L)$ such that $V(C) \geqslant 2^d d(L)$. Then C contains a pair $\pm l \neq o$ of points of L. This result has far-reaching number-theoretic implications. Generalizations are Minkowski's theorem on successive minima, Blichfeldt's (1873–1945) theorem ("A new principle in the geometry of numbers" 1914) and Siegel's (1896–1981) result of 1935 employing Fourier series. In recent years a series of conditions have been specified that a convex body contains a point of a given lattice; for example see Chalk–Yates 1964–1970 and Bokowski–Hadwiger–Wills 1972.

Two problems of Minkowski aroused considerable attention: his conjecture on lattice tiling with cubes, solved by Hajós (1912–1972) in 1941 (and slightly later by Hlawka who decided not to publish it when he saw Hajós' paper) and his conjecture on the existence of admissible lattices of determinant 1 of star bodies of volume $\zeta(d)$, solved by Hlawka (born 1916) in 1943. Deep contributions to the latter result were given by Siegel in 1945, Weil 1946, Rogers 1954, Malyšev 1952, Macbeath and Rogers 1958 and Schmidt between 1956 and 1963.

Important impulses to the geometry of numbers are due also to Mordell (1888–1972), Davenport (1907–1969), Mahler (1903–1988), Bambah, Cassels, Rankin, M. Kneser, Woods, Skubenko, Zassenhaus and Hans-Gill.

In the last two decades, geometry of numbers got again much attention. One reason for this might be the emerging of relations with other areas of mathematics such as coding, spherical harmonics, modular functions, numerical integration, algebraic geometry and of new connections with old relatives. Areas of particular interest are zeta-functions, lattice packing of balls, lattice polytopes, reduction theory, and geometric algorithms. At present, the geometry of numbers appears to be much more geometric than it was between 1920 and 1960.

The zeta-function for lattices was originally introduced by Epstein. It was rediscovered by Sobolev who noted its connection with numerical integration ("Introduction to the theory of cubature formulae", Moscow 1974). The zeta-function was investigated intensively by the English and the Soviet schools of the geometry of numbers.

Essentially the best known asymptotic lower bounds for the maximum density of lattice packing of balls are obtained from the Minkowski–Hlawka theorem and its refinements. Upper bounds are due to Blichfeldt 1929, Sidelnikov 1973, 1975, and Kabatjanski and Levenštein 1978. Explicit descriptions of dense lattice and more general periodic packings of balls based on error correcting binary codes were specified by Leech and Sloane in 1971, Sloane 1977, Bos 1982, Litsyn and Tsfasman 1985, Conway and Sloane 1987 and Rush in 1989, the latter finally reaching the Minkowski–Hlawka bound $2^{-d+o(d)}$ $(d \to \infty)$. Many results deal with the Leech lattice. For references see Conway and Sloane ("Sphere packings, lattices and groups", New York 1987).

A lattice polytope is a polytope the vertices of which have integer coordinates.

Questions dealing with the determination of volumes, analogs of mixed volumes, inversion laws and analogs of the functional theorems of Blaschke and Hadwiger have been considered by Reeve 1957, Macdonald 1963, Ehrhart 1967, McMullen 1975, 1977, Wills 1980, and Betke and Kneser 1985. Relations to algebraic geometry are provided by particular lattice polyhedra, called after Newton.

In reduction theory, given a lattice one selects one or more particular bases of it. There exist several sets of rules for selections of bases due to Hermite, Korkin and Zolotarev, Selling, Minkowski, Venkov and, more recently, Lenstra–Lenstra–Lovász ("Factoring polynomials with rational coefficients" 1982). Important modern results dealing with Minkowski and Venkov reduction were given by Tammela in 1973 and Ryškov also in 1973 and the latter's students. The reduction algorithm of Lenstra–Lenstra–Lovász has applications in computational geometry, Diophantine approximation, cryptography, factoring integer polynomials, and was used by Odlyzko and te Riele (in "Disproof of the Mertens conjecture" 1985). The Korkin–Zolotarev reduction has implications for packing and covering as shown recently by Lagarias, Lenstra and Schnorr.

Algorithmic aspects of convexity and geometry of numbers and, in particular, complexity problems were studied amongst others by the Lenstras, Lovász, Grötschel, Schrijver, Kannan, Lagarias, and Schnorr. A major problem is that of calculating the volume of a convex body which is defined by means of an oracle, see Lovász ("An algorithmic theory of numbers, graphs and convexity", Philadelphia 1986), Edelsbrunner ("Algorithms in combinatorial geometry", Berlin 1987) and Grötschel et al. ("Geometric algorithms and combinatorial optimization", New York 1988). Algorithmic and constructive problems and results have influenced the thinking even of some of those workers in convexity and geometry of numbers who never will contribute to algorithmic problems.

Voronoi (1868–1908) created in 1908 and 1909 (posthumous) a geometric theory of positive quadratic form, using earlier ideas and results of Dirichlet, Korkin–Zolotarev and Fedorov (1853–1919). Given a lattice L, he classifies L according to the "type" of its Dirichlet–Voronoi cell $\{x: |x| \leqslant |x - l| \text{ for all } l \in L\}$. The concept of Dirichlet–Voronoi cell turned out to be of importance for packing and covering with balls, quantization of data, numerical integration, computational geometry, crystallography, metallurgy, geography and botany. It is an open problem of Voronoi whether any polytope which admits a lattice tiling essentially is a Dirichlet–Voronoi cell. The answer is yes for $d \leqslant 4$ as shown by Delone in 1929. Voronoi and later on Žitomirskii in 1929 made progress in the general case. Venkov 1954 and McMullen 1980 showed that any polytope that admits a tiling by translates also admits a lattice tiling. Generalizations of this are due to Aleksandrov 1954 and Groemer 1962. Voronoi's fundamental criterion for lattice packings of balls which locally have maximum density seems to be accessible to computer calculations.

Dehn (1878–1952) specified (in "Über den Rauminhalt" 1901) necessary conditions for two polytopes in $\mathbb{E}^3$ to be equidissectable with respect to the group of isometries. These yield that a cube and a regular tetrahedron of the same volume are not equidissectable, thus solving Hilbert's third problem. (For $d = 2$ equality of

areas implies equidissectability.) Predecessors of Dehn were Rausenberger 1893, Bricard 1896 and Sforza 1897. Dehn's conditions were extended to general d by Hadwiger 1954, and Sydler 1965 and Jessen 1968, 1972 showed their sufficiency for $d \leqslant 4$. In case of the group of translations a corresponding set of necessary conditions was specified by Hadwiger in 1968, and Jessen–Thorup 1978 and Sah 1979 proved that they are sufficient. Other groups which have been investigated are the affine group, the equiaffine group (volume preserving), the group of all equiaffinities mapping the integer lattice $\mathbb{Z}^d$ onto itself and the group of all translations with vectors from $\mathbb{Z}^d$.

A valuation φ on a class of convex bodies is a real function satisfying

$$\varphi(C \cup D) + \varphi(C \cap D) = \varphi(C) + \varphi(D)$$

whenever $C, D, C \cup D, C \cap D$ belong to the class. Examples are quermassintegrals, lattice points enumerators and functions appearing in the conditions for equidissectability. A general theory was developed in the last two decades; contributors are McMullen ("Valuations and Euler-type relations on certain classes of convex polytopes" 1977) and Schneider.

The convexity results of Blaschke (1885–1962) are of two types. On the one hand there is a wealth of special results, e.g., characterizations of balls and ellipsoids. In "Räumliche Variationsprobleme mit symmetrischer Transveralitätsbedingung" 1916 (see also "Kreis und Kugel", Leipzig 1916) Blaschke characterized ellipsoids among all convex bodies by the property that the shadow boundaries with respect to parallel illuminations are planar. The incentive for this came from Carathéodory who asked whether (in modern terms) among all normed spaces the orthogonality relation for normed spaces is symmetric precisely for Euclidean spaces. The answer is yes for $d \geqslant 3$; for $d = 2$ the corresponding norms are the Radon norms ("Über eine besondere Art ebener konvexer Kurven", Radon 1916). Blaschke's characterization of ellipsoids has many further applications; a classical one is due to Kakutani from 1939. A concept with several applications is the Blaschke symmetrization. The precise upper estimate for the product $V(C)V(C^*)$, where C^* is the polar body of the convex body C with center of gravity o, due to Blaschke 1917 ($d \leqslant 3$) and Santaló 1949 (general d), has attracted much interest. An important step to verify the corresponding lower bound for o-symmetric C (conjectured by Mahler) is due to Bourgain–Milman 1987.

On the other hand, Blaschke proved also some convexity results of a systematic character. The basic topological result on the space of convex bodies is the selection theorem of Blaschke (published in "Kreis und Kugel" 1914, and 1916 in the booklet having the same title): let the space of all compact convex subsets in $\mathbb{E}^d$ be endowed with the (Pompeiu–)Hausdorff(–Blaschke) metric. Then each bounded sequence contains a convergent subsequence. The original proof was based on a first version of the Arzelà–Ascoli theorem due to Hilbert, the second one followed a suggestion of Carathéodory to use, in essence, ε-nets. Among the numerous generalizations we mention those of Urysohn in 1921/22, Vietoris 1922 and Michael 1951. Bol and Heil proved in 1965, resp. 1967, that each of the theorems of Blaschke and

Arzelà–Ascoli is a consequence of the other one. Classical applications guarantee the existence of solutions of extremum problems. The selection theorem is the basis for Baire category results in convexity as developed by Klee, Zamfirescu and Gruber. A related compactness result is due to Macbeath ("A compactness theorem for affine equivalence classes of convex sets" 1951).

Convexity has many relations with (global) differential geometry. Links are curvature (measures), geodesics, corresponding characterizations of quadrics, ellipsoids and balls, see Blaschke "Vorlesungen über Differentialgeometrie I", Berlin 1921 and the books of Aleksandrov of 1948, Busemann 1958 and Pogorelov 1969. Surprisingly, in many cases corresponding problems in differential geometry and convexity (for example, the characterization of balls by the property that the geodesics are all planar) require completely different proofs.

In the second decade of this century, Blaschke and his friends Pick, Reidemeister, Winternitz, Berwald and Radon started the investigation of differential geometric properties of surfaces which are invariant with respect to (area preserving or even all) affinities. They introduced the concept of affine surface area and proved the affine isoperimetric inequality (Blaschke 1916 ($d \leqslant 3$); for general d compare Santaló 1949, Deicke 1953). See Blaschke, "Vorlesungen über Differentialgeometrie II", Berlin 1923. In recent years, these investigations were continued by Lutwak, Petty, Santaló and others; see the articles "On the Blaschke–Santaló inequality", "Affine isoperimetric problems" and "On some invariants under similitudes for convex bodies" which appeared in the volume "Discrete geometry and convexity", New York 1985. Extensions of the concept of affine surface area to arbitrary convex surfaces are due to Leichtweiss, Schütt, and Lutwak. Tools from affine differential geometry are of increasing importance for convexity, for example, for asymptotic approximation of convex bodies by polytopes (Schneider, Gruber).

Together with a group of collaborators Blaschke systematically developed integral geometry, see his booklet "Vorlesungen über Integralgeometrie", Berlin 1935, 1955. Earlier sporadic results are due to de Buffon (1707–1788) in 1733, Crofton (1826–1915), Sylvester (1814–1897) and Poincaré. Blaschke's fundamental kinematik formula of 1936 ($d \leqslant 3$) was generalized by Chern (born 1911) in 1956 to arbitrary dimensions. More recent contributions to this area and, more generally, to geometric probability and stereology were given, amongst others by Chern, Fenchel, Hadwiger and Kendall, Miles, Moran, Santaló, Stoka, Schneider, Ambartzumian and Weil. In recent years, problems connected with random polytopes attracted interest. Santaló collected much material on integral geometry in his treatise "Integral geometry and geometric probability" Reading, MA 1976.

Carathéodory (1873–1950) seems to have known a lot on convexity and contributed to it through personal conversation (selection theorem, geodesics on convex surfaces), but published only a few results. Among these is his theorem that each point of the convex hull of a set in $\mathbb{E}^d$ belongs to a simplex with vertices in this set. ("Über den Variabilitätsbereich der Koeffizienten von Potenzreihen, die gegebene Werte nicht annehmen" 1907). His fundamental contributions to measure theory (later developed by Hausdorff and Besicovitch) and calculus of variations are of importance for many more refined modern convexity results.

Carathéodory's theorem just cited together with theorems of Helly (1884–1943) ("Über Mengen konvexer Körper mit gemeinschaftlichen Punkten" 1923) and Radon (1887–1956) ("Mengen convexer Körper, die einen gemeinsamen Punkt enthalten" 1921) form the cornerstones of the vast area of combinatorial convexity. Helly's result says that any finite system of convex sets in $\mathbb{E}^d$ with the property that each subsystem of at most $d+1$ sets has non-empty intersection has itself non-empty intersection. A singular result of combinatorial convexity was found already in 1903 by Kirchberger. Important later contributions are due to Krasnoselskii 1954, Tverberg 1966, Eckhoff 1968 and many others. For references see Danzer–Grünbaum–Klee ("Helly's theorem and its relatives" 1963) and Eckhoff ("Radon's theorem revisited" 1979).

Helly also anticipated a special form of the Hahn–Banach (separation) theorem.

Geometrically oriented results of crystallography from about 1910 due to Bieberbach (1886–1982) and Frobenius (1847–1917) confirmed Hilbert's conjecture that in any dimension there are only finitely many space group types. An interesting explanation based on a minimum principle why crystals have the form of polytopes was stated by Wulff (1862–1925) and confirmed by Dinghas in 1944.

Steinitz (1871–1928) developed systematically the combinatorial theory of polytopes in $\mathbb{E}^3$. As a converse to Euler's relations (1) he proved in his posthumous book with Rademacher "Vorlesungen über die Theorie der Polyeder", Berlin 1934, that for any triple f_0, f_1, f_2 of non-negative integers satisfying (1) there is a convex polytope in $\mathbb{E}^3$ with f_0 vertices, f_1 edges and f_2 facets. The corresponding characterization for f-vectors of general convex polytopes in $\mathbb{E}^d$ $(d \geqslant 4)$ is still missing. For simplicial polytopes a characterization of the f-vectors was conjectured by McMullen in 1971 and proved jointly by Billera and Lee 1980 and Stanley 1981. The conditions of McMullen include the Brückner 1893 – Dehn 1905 – Sommerville 1927 linear equations for f-vectors. For general convex polytopes the "upper bound conjecture" of Motzkin 1957 on the maximum values of $f_1, \ldots, f_{d-1}$, given f_0, was confirmed by McMullen in 1971, and Barnette 1971–73 confirmed the corresponding lower bound conjecture for simplicial polytopes.

A second contribution of Steinitz says that any finite 3-connected planar graph without loops is isomorphic to the edge graph of a convex polytope in $\mathbb{E}^3$. No extension of this result to higher dimensions has been achieved so far. As a consequence of Steinitz' theorem each convex polytope in $\mathbb{E}^3$ is combinatorially isomorphic to a convex polytope with vertices having rational coordinates only, thus solving a problem of Klee for $d = 3$. For $d \geqslant 6$ this is not true as shown by Sturmfels in 1987.

Delone (1890–1980) continued earlier work of Fedorov and Voronoi on tiling problems related to crystallography and on lattice tilings. He gave an upper estimate for the number of facets of a convex polytope admitting a tiling of $\mathbb{E}^d$ by means of a crystallographic group. In $\mathbb{E}^3$ his bound is 390 which is almost certainly too large. Engel specified in 1980 a crystallographic tiling polytope in $\mathbb{E}^3$ having 38 facets. It seems to be unknown whether for convex polytopes that admit a tiling by congruent images there is a finite upper bound for the number of facets $(d \geqslant 3)$; for $d = 2$ no convex n-gon with $n \geqslant 7$ admits a tiling. Reinhardt (1895–1941) gave

an example of a non-convex polytope in $\mathbb{E}^3$ which admits a tiling of $\mathbb{E}^3$ with congruent copies but no tiling by means of a group of rigid motions. Planar examples were given by Heesch in 1935 and Kershner in 1968. Refining earlier work of Stein, Szabó 1987 specified a star polytope in $\mathbb{E}^3$ admitting a tiling with translates, but no tiling by means of a crystallographic group. A surprising result of Schulte 1984 says that every convex (simplicial) polytope in $\mathbb{E}^3$ gives rise to a locally finite (face to face) tiling of $\mathbb{E}^3$ with combinatorially isomorphic images. Penrose's 1974 tilings have attracted a good deal of interest, see also the comprehensive book of Grünbaum and Shephard ("Tilings and patterns", San Francisco 1986) on planar tilings.

Further results of Delone – in part with Ryškov and their students – include work on the lattice zeta-function, coverings with balls and the algorithm of divided cells for lattice parallelograms (related to geometric interpretations of continued fractions due to Klein and Minkowski).

In his treatise "Lagerungen in der Ebene, auf der Kugel und im Raum", Berlin 1953, L.Fejes Tóth (born 1918) presented results on packing and covering of $\mathbb{E}^2, \mathbb{E}^3$ with convex bodies, due to Thue (1863–1923) 1892, Kershner 1939, himself, Bambah, Rogers and others, and then systematically develops the theory of packing and covering in $\mathbb{E}^2, \mathbb{E}^3$. This booklet strongly influenced the later development of "discrete geometry" (as did Blaschke's "Kreis und Kugel" and Bonnesen and Fenchel's "Theorie der konvexen Körper", Berlin 1934, in the corresponding contexts). The basic problems, e.g., densest packings and thinnest coverings with translates of a convex body, are solved only for $d = 2$ by Fejes Tóth 1950, Rogers 1951, and Bambah and Rogers 1952. In higher dimensions surprisingly small asymptotic upper bounds for coverings are due to Rogers 1957, 1959. Except for the case of balls (where coding results apply), fast algorithms providing dense (lattice) packings and (lattice) coverings of a given convex body ($d > 2$) are still missing. Important contributors to discrete geometry are Florian, G. Fejes Tóth, Böröczky and other members of the Hungarian school.

At present, one has the impression that the three main branches of convexity, namely combinatorial convexity, discrete geometry and the analytically oriented part of convexity are drifting rapidly apart. It will be interesting to observe whether this will continue in the future.

Of the convexity results of A.D. Aleksandrov (born 1912) we mention the Aleksandrov–Fenchel inequalities, the corresponding results on mixed discriminants 1938, the almost-everywhere-twice differentiability of a convex function 1939 and the results on surface area measures. The intrinsic geometry of, possibly, non-differentiable convex surfaces in $\mathbb{E}^3$ is presented in Aleksandrov's monographs "Intrinsic geometry of convex surfaces", Moscow 1948, "Convex polyhedra", Moscow 1950. Many results of Riemannian geometry (geodesics, curvature, Gauss–Bonnet theorem, determination of a surface by its curvature, etc.) are transferred to this more general situation; see also Pogorelov "Extrinsic geometry of convex surfaces", Moscow 1969. A central result of Aleksandrov 1942 and Pogorelov 1952 is the following: let M be a metric manifold homeomorphic to S^2 which has (a suitably defined) non-negative curvature. Then M is isometric to a closed convex

surface S in $\mathbb{E}^3$ if both are endowed with their intrinsic metric. S is unique up to rigid motions. A particular case is Cauchy's rigidity theorem. The corresponding differential geometric problem of Weyl 1916 was solved by Lewy in 1935 (in the C^ω-case); see also Blaschke and Herglotz 1937, and Nirenberg 1953.

Some of Hadwiger's (1908–1981) results have been mentioned already. His extension of the Euler characteristic to the Konvexring (i.e., the system of finite unions of compact convex sets) in 1955, later continued by Groemer, his result on lattice coverings and number of neighbors 1969 and the great body of results on valuations and on integral geometry are examples of further contributions. A book of lasting influence is Hadwiger's "Vorlesungen über Oberfläche, Inhalt und Isoperimetrie", Berlin 1957.

On the borderline between convexity and functional analysis lies the theorem of Dvoretzky in 1961 which says that any convex body of sufficiently large dimension has lower-dimensional planar sections which are almost spherical. This result triggered a series of theorems on finite-dimensional normed spaces in which – so to say – asymptotic properties are investigated as $d \to \infty$. Of particular interest is the Banach–Mazur distance between different (subspaces or quotients of) normed spaces. A central role is played by the Löwner–Behrend–John minimal ellipsoid of a convex body. Further objects of interest are the Blaschke–Santaló inequality and other volume estimates. Surveys of the work of Lindenstrauss, Milman, Figiel, Pelczynski, König, Tzafriri, Gluskin, Pajor, Bourgain, Ball and others are contained in the monographs of Tomczak-Jaegermann ("Banach–Mazur distances and finite-dimensional operator ideals", Harlow 1989) and Pisier ("The volume of convex bodies and Banach space geometry", Cambridge 1989).

Other links between convexity and functional analysis are fixed points (Brouwer's theorem and its infinite-dimensional extension), extreme points (Minkowski, Krein and Milman, Bauer, Klee), Choquet and Bauer simplices, results of Phelps, stable convex sets (Papadopoulou), Lyapunov's convexity theorem on vector valued measures (related to zonoids), results of Pták on finite-dimensional operators and Schaeffer's geometry of the unit sphere of a normed space.

Among all areas of mathematics related to convexity, in the last three decades functional analysis, optimization and algorithmic theory seem to have had the strongest impact on convexity. But, of course, there are also many other influences. Part of the fascination of convexity is due to these interrelations. Another reason for this might be the fact that many problems can be stated easily and the results are frequently appealing from the aesthetic viewpoint.

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CHAPTER 1.1

Characterizations of Convex Sets

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Contents

1. The classical characterizations 21
2. Characteristic properties of convex sets in analysis and differential geometry 22
3. Topology. Combinatorics 25
4. Extensions of the notion of a convex set. Realizations 29
5. Measures of convexity 32
6. Looking out 33
References 35

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1. The classical characterizations

We begin with three characteristic properties of closed convex sets in Euclidean spaces: they have a supporting hyperplane at each boundary point, they are locally convex, and they possess a nearest point map. We also discuss some extensions to arbitrary real vectorspaces.

A subset X of a real vectorspace V is called convex if it contains, with every pair p, q of its points, the entire segment $[p,q] = \{\alpha p + (1-\alpha)q\colon 0 \leqslant \alpha \leqslant 1\}$. Examples are the closed hyperplanes in a real topological vectorspace V, sets of the form $H = \{x \in V\colon f(x) = \alpha\}$, where $f\colon V \to \mathbb{R}$ is a continuous nonzero linear functional, and α a real number. The convex sets $H^+ = \{x \in V\colon f(x) \geqslant \alpha\}$ and $H^- = \{x \in V\colon f(x) \leqslant \alpha\}$ are the closed halfspaces bounded by H in V. H is called a supporting hyperplane of $X \subset V$ at the point $p \in X$ if $p \in H$, and if, furthermore, X is entirely contained in one of the closed halfspaces bounded by H. Let $\mathbb{R}^d$ be the d-dimensional Euclidean space.

Theorem 1.1. *A closed set $X \subset \mathbb{R}^d$ is convex if and only if it possesses a supporting hyperplane at each of its boundary points.*

A proof of this result, explicitly worked out for the case $d = 3$, can be found in Minkowski (1911, Zweiter Band, pp. 137–140). It uses polyhedral approximation. A second proof, for arbitrary d, is contained in Brunn (1928).

We now present an extension of these considerations to infinite dimensional vectorspaces, following Klee (1951), and Bishop and Phelps (1963).

Theorem 1.2. *Let C be a closed subset with nonempty interior, in some real Hausdorff topological vectorspace V. C is convex if and only if it possesses a supporting hyperplane at each of its boundary points.*

As described in Klee (1951), the above characterization is related to the Hahn–Banach theorem. If the assumption about the nonempty interior of C is dropped, then it can happen that no supporting hyperplane exists at all (Klee 1958), even if V is complete, metrizable and locally convex (Peck 1971, Phelps 1974). However, according to Bishop and Phelps (1963), a closed convex subset C of a Banach space V has a supporting hyperplane at every point in some dense subset of its boundary. Further developments of these topics are described in Valentine (1960) and Phelps (1974).

Convexity is a local property, at least in Euclidean spaces. We say that the subset X of the real topological vectorspace V is locally convex at $p \in V$ if there exist a neighbourhood U of p in V and a convex set $C \subset V$ such that $X \cap U = C \cap U$.

Theorem 1.3. *A closed and connected subset X of $\mathbb{R}^d$ is convex if and only if X is locally convex, at every point $p \in X$.*

This result is due to Tietze (1928). Klee (1951) has extended the above characterization to infinite dimensions.

Theorem 1.4. *Let C be a closed connected subset of some real topological vectorspace. C is convex if and only if C is locally convex at every point $p \in C$.*

A further strengthening of this result has been found by Sacksteder, Straus and Valentine (1961).

Let $V = (V, \| \|)$ be a normed vectorspace over the reals. We set, for $M \subset V$ and $x \in V$, $d(x, M) = \inf\{\|x - y\|: y \in M\}$, and say that M is a Chebyshev set in V if there exists, for every $x \in V$, a unique point $n(x)$ in M such that $\|x - n(x)\| = d(x, M)$. The resulting map $n: V \to M$ is the nearest point map of M. The fact that closed convex subsets of Euclidean spaces enjoy the Chebyshev property has been familiar to geometers early on. First steps in the opposite direction were undertaken by Motzkin (1935a,b).

Theorem 1.5. *A compact subset of the Euclidean plane $\mathbb{R}^2$ is a Chebyshev set if and only if it is convex.*

Contrary to the first two characterizations of convex sets, the Chebyshev property does not seem to allow easy generalizations, not even to finite dimensional Banach spaces. Much of the work devoted to this problem up to this time is described in Cudia (1963). For example, if V is a finite dimensional smooth and strictly convex Banach space, then the class of all Chebyshev sets in V coincides with the class of all closed convex subsets of V. However, many interesting questions remain open. We mention two of them.

Question 1.1. Let C be a Chebyshev set in some Hilbert space. Must C be convex?

Question 1.2. Let V be a finite-dimensional Banach space, with unit ball B. Assume that, given any exposed point p of B, there exists a unique supporting hyperplane of B at p. Is every Chebyshev set in V necessarily convex?

A description of Question 1 in a wider context, and an account of many problems related to it, can be found in Vlasov (1973). Question 2 has been answered in the affirmative, if the dimension d of V is equal to 3, by Brøndsted (1966), and, for $d = 4$, by Brown (1980). Klee (1981) constructs strange nonconvex Chebyshev sets in perfectly harmless looking Banach spaces, under appropriate assumptions about the underlying set theory. An important class of sets has emerged in an attempt to expand the horizon of classical convexity: the sets of positive reach, which enjoy a local Chebyshev property, see Federer (1959), Kleinjohann (1981), and Bangert (1982).

2. Characteristic properties of convex sets in analysis and differential geometry

Most of the results presented here can also be found, in a wider context, in one of the chapters 1.8, 4.1 and 4.2.

We therefore restrict ourselves to a brief description of those aspects which reveal characteristic properties of convex sets, and also to real, rather than complex analysis. We consider five topics: convex functions, the Weyl problem, the Minkowski problem, convexity on the Grassmann cones, and quasiconvex functions.

Let U be a convex subset of some real vectorspace V. A function $f: U \to \mathbb{R}$ is called convex, if for every pair x, y in U and every number $\alpha \in [0,1]$, we have

$$f(\alpha x + (1-\alpha)y) \leqslant \alpha f(x) + (1-\alpha) f(y).$$

Notice that the epigraph $E_+(f) = \{(x,y) \in U \times \mathbb{R}: y \geqslant f(x)\}$ is a convex subset of $V \times \mathbb{R}$. This illustrates that characterizations of convex sets and of convex functions are closely related. The following is one of the classical characterizations of C^2-functions, that is, functions which are twice continuously differentiable, compare Roberts and Varberg (1973, p. 103).

Theorem 2.1. *Let $U \subset \mathbb{R}^d$ be open and convex, and $f: U \to \mathbb{R}$ a function of class C^2. f is convex if and only if the Hessian matrix $Hf(x)$ is positive semidefinite, for every $x \in U$.*

Notice that the graph $\Gamma_f = \{(x, f(x)): x \in U\}$ is a submanifold of $\mathbb{R}^{d+1}$, and f is convex if and only if all sectional curvatures of Γ_f, with respect to the standard Riemannian metric, are nonnegative (Spivak 1975).

The Weyl problem asks for a statement in the opposite direction, see Yau (1982, p. 681).

Question 2.1. Let M be a smooth $(d-1)$-dimensional manifold, with positive sectional curvature. Does there exist a locally isometric immersion of M into $\mathbb{R}^{\binom{d}{2}}$?

In the case $d = 2$, and if M is a sphere, a much stronger statement has been established by Aleksandrov (1955). We give a brief description of his work in section 4. Although the Weyl problem still seems far from a solution, there are a number of interesting results related to it.

Theorem 2.2. *Let $f: M \to \mathbb{R}^{d+1}$ be a smooth immersion of a compact d-manifold. f is a diffeomorphism onto a convex surface if and only if f is locally convex.*

By a convex surface we understand the boundary of a convex compact set with nonempty interior, and the local convexity of f can be expressed by means of the Hessian, or by using sectional curvatures. Theorem 2.2, a close relative to Theorem 1.1, has a long history. In the above form it is due to Sacksteder (1960), who also gives an account of earlier work. Even a sketchy description of the vast literature which has been devoted since then to questions of a similar nature, is not possible here. We just mention, as one recent beautiful result, a sphere theorem by Gromov, in manifolds of nonnegative sectional curvature, see Eschenburg (1986). Related studies have been undertaken by Labourie (1987). If we allow submanifolds of

higher codimension, then the conditions on positive curvature or local convexity have much weaker impact on the geometric shape, compare Polikanova (1988), Micallef and Moore (1988), and Otsu, Shiohama and Yamaguchi (1989). In this context, the following problem, known as the Hopf Conjecture, still seems open (Hsiang and Kleiner 1989).

Question 2.2. Does there exist a Riemannian metric with strictly positive sectional curvature on $S^2 \times S^2$?

Given a smooth and strictly convex hypersurface F in $\mathbb{R}^d$, the map $n: F \to S^{d-1} = \{x \in \mathbb{R}^d : \|x\| = 1\}$ which associates to every point $p \in F$ the outer unit normal to F at p, is a diffeomorphism. Thus, if we associate to $x \in S^{d-1}$ the Gaussian curvature $K(x)$ of F at $n^{-1}(x)$, we obtain a nonnegative function $K: S^{d-1} \to \mathbb{R}$. Let us call K the Gauss–Kronecker curvature of F. The Minkowski problem asks for a characterization of all positive functions on S^{d-1} which are Gauss–Kronecker curvatures of convex hypersurfaces. A celebrated answer has been given by Pogorelov.

Theorem 2.3. *Let $K: S^{d-1} \to \mathbb{R}$ be a positive function, of class C^r, $r \geqslant 3$. K is the Gauss–Kronecker curvature of a convex hypersurface F if and only if*

$$\int (K(x))^{-1} \cdot x \, d\omega(x) = 0.$$

F is unique, up to a translation, and of Hölder class $C^{(r+1),\alpha}$, for some $\alpha > 0$.

The integration is to be taken over S^{d-1}, with respect to the standard surface element $d\omega$. Pogorelov (1978) presents a proof of this, and of related results, together with a historic account.

An extension of this theory to curvature measures, instead of curvature functions, on the one hand, and to arbitrary convex surfaces on the other hand, is discussed in Gluck (1975) and in Schneider (1979a). Schneider reminds us that the Gauss–Kronecker curvature corresponds to one of $d-1$ possible curvature measures on S^{d-1}, and that a similar characterization had only be found, so far, in one other case. Schneider (1979a) describes a related, not yet fully solved, characterization problem for the case where, instead of the normal map, the radial projection between a convex surface and the unit sphere is studied. Compare Oliker (1986) for a recent contribution.

Some time ago a theory of convexity on Grassmann cones was initiated by Busemann, Ewald, Shephard, Phadke, and other authors. Let V_r^d be the vectorspace of all r-vectors in $\mathbb{R}^d$, and G_r^d the collection of all simple r-vectors of unit length in V_r^d. A function $f: G_r^d \to \mathbb{R}$ is called convex if it can be extended to a convex function on the convex hull of G_r^d in the vectorspace V_r^d. Remember that there corresponds, to every $x \in G_r^d$, a unique oriented r-dimensional linear subspace $\bar{x}$ of $\mathbb{R}^d$, and that the correspondence $x \mapsto \bar{x}$ is bijective. Given a compact and convex subset C of $\mathbb{R}^d$ and a positive integer r, we consider the projection function

$p_r(C): G_r^d \to \mathbb{R}$, which associates to every $x \in G_r^d$ the r-dimensional volume of the orthogonal projection of C onto $\overline{x}$. Among the many open questions which have been studied in this context, let us mention one characterization problem.

Question 2.3. For every pair of integers d and $r \leqslant d$, describe the class of those convex and compact subsets C of $\mathbb{R}^d$ whose projection function $p_r(C)$ is convex.

Busemann, Ewald and Shephard (1963) show that $p_r(C)$ is convex, for every r, whenever C is a zonoid in $\mathbb{R}^d$. References to further contributions in this still fascinating area can be found, up to then, in Busemann and Shephard (1967). See also Phadke (1972) and Silverman (1971).

We conclude this section with an entirely different characterization of convex functions, which was initiated by Jensen (1906). Let U be a convex subset of some normed linear space V. A function $f: U \to \mathbb{R}$ is called midconvex in U if, for every pair x, y in U,

$$f(\tfrac{1}{2}(x+y)) \leqslant \tfrac{1}{2}(f(x)+f(y)).$$

Theorem 2.4. *Let U be an open and convex subset of a normed real vectorspace V. A midconvex function $f: U \to \mathbb{R}$ is convex if and only if there exists a point $x \in U$ such that f is bounded in some neighbourhood of x.*

You may consult Roberts and Varberg (1973) for a proof, and for a list of references to a large number of further characterizations of convex functions and convex sets in this context.

3. Topology. Combinatorics

Here we try to give an account of the numerous characterizations of convex sets and convex functions in Euclidean spaces which have appeared during the past few decades. We consider four aspects: contact with affine subspaces, contact with some other classes of sets, metric properties, other criteria. First we consider the contacts with affine subspaces.

Theorem 3.1. *Let $X \subset \mathbb{R}^d$ be a nonempty compact set. X is convex if and only if X, as well as every intersection of X with a supporting hyperplane, is a homology cell.*

Theorem 3.1 was first established by Liberman (1943), then, in a slightly weaker form by Kuhn (1954) in the context of game theory. Bourgin (1965) gave a short proof and asked whether a similar result holds in arbitrary Banach spaces.

An entirely different description of compact convex sets by means of their support elements is due to Leichtweiss (1965). A full characterization of convexity in this context still seems to be an open problem, if we go beyond the planar case.

Denote by $G_k(V)$, $k \in \mathbb{N}$, the collection of all k-dimensional affine subspaces of the vectorspace V. If V is real and finite-dimensional, then $G_k(V)$ carries a natural topology.

Theorem 3.2. *Let X be a compact subset of $\mathbb{R}^d$. X is convex if and only if there exists an integer $k \in [1, d-1]$ such that $X \cap E$ is either empty or a homology cell, for every $E \in G_k(\mathbb{R}^d)$.*

Theorem 3.2 has an immediate extension to infinite dimensions. A first proof is due to Aumann (1936), another proof and, at the same time, a characterization of star shaped sets, was found by Kosinski (1962). Kosinski (1963) asks whether one still obtains simple geometric characterizations by requiring that the reduced homology groups $H_i(X \cap E)$ vanish, for all $i \in [0, l]$, where $l < k$ is another positive integer.

Theorem 3.3. *Let $X \subset \mathbb{R}^d$ be a Euclidean polyhedron or, more generally, a union of some finite collection of compact and convex sets. X is convex if and only if there exist integers $k \in [1, d-1]$ and $c \in \mathbb{Z}$ such that the Euler characteristic $\chi(E \cap X)$ is equal to c, for every $E \in G_k(\mathbb{R}^d)$ with $E \cap X \neq \emptyset$.*

This characterization of convexity was discovered by Hadwiger (1968). Before we can extend this result to wider classes of sets, we would have to find an appropriate concept of the Euler characteristic.

Theorem 3.4. *Let $A \subset \mathbb{R}^d$ be an open set. Denote, for some $k \in [1, d-1]$, by $G_k(A)$ the set of all $E \in G_k(\mathbb{R}^d)$ for which $A \cap E \neq \emptyset$. A is convex if and only if there exists a continuous map $\kappa : G_k(A) \to A$ such that $\kappa(X) \in X \cap A$ for every X.*

Stein (1958), who stated Theorem 3.4, also raised the following problem: give necessary and sufficient conditions for A such that $\kappa(X)$ can be chosen to be the barycenter of $X \cap A$, with respect to some mass distribution on A.

A characterization which involves, together with the convex surfaces in $\mathbb{R}^3$, two types of nonconvex quadratic surfaces, has been obtained by Verner (1963). It is based on the observation that their planar sections are unions of finitely many, essentially disjoint, convex discs and convex curves. Falconer (1976) shows that a compact set $X \subset \mathbb{R}^d$ with positive volume is convex if and only if all its Steiner symmetrizations with respect to varying hyperplanes are convex. Similar statements may be true for other types of symmetrizations.

Let us also mention here the construction by Gizaslewic (1981) of a compact convex set $C \subset \mathbb{R}^{d+2}$ with the property that every compact convex subset of the unit ball in $\mathbb{R}^d$ is congruent to a section of C. A related question of Mazur, involving central symmetry, has been answered in the negative by Grünbaum (1958). We will not begin to describe here the rich literature on universal convex figures in infinite dimensions.

Consider a one-parameter family ϕ of convex surfaces in $\mathbb{R}^d$. Is there a convex surface C in $\mathbb{R}^{d+1}$ such that the members of ϕ are parallel, or congruent, or affinely equivalent, to the sections of C by hyperplanes parallel to $\mathbb{R}^d$? A related question was raised by de Finetti (1949).

Question 3.1. Let $(F_\lambda)_{0 \leqslant \lambda < \infty}$ be a continuous family of compact and convex sets in $\mathbb{R}^d$, such that F_λ belongs to the interior of F_μ, whenever $\lambda < \mu$. Does there exist a convex function φ such that $F_\lambda = \{x \in \mathbb{R}^d\colon\ \varphi(x) \leqslant \lambda\}$, for every λ?

After some first contributions by de Finetti and by Massera and Schäffer (1953), Fenchel (1956) gave a complete characterization of such families, under appropriate smoothness conditions.

We conclude this section with a theorem by Zelinskii (1980): a compact set $X \subset \mathbb{R}^d$ is convex if and only if there exists an integer $i \in [1, d-1]$ such that every $p \in \mathbb{R}^d \setminus X$ belongs to some flat $F \in G_i(\mathbb{R}^d)$ with $F \cap X = \emptyset$, and every orthogonal projection of X onto a flat $G \in G_{i+1}(\mathbb{R}^d)$ is convex.

We now consider characterizations of convex sets, based on appropriate statements about self-intersections. A beautiful result of this kind is Dvoretzky's (1955) converse of Helly's Theorem.

Let us say that a family $\mathcal{K}$ of subsets of $\mathbb{R}^d$ has the Helly property, if either $\bigcap \mathcal{K} \neq \emptyset$, or else $\bigcap \mathcal{L} = \emptyset$ for some subfamily $\mathcal{L}$ of $\mathcal{K}$ with $d+1$ members.

Theorem 3.5. *Let $\mathcal{K}$ be a family of no less than $d+2$ compact sets in $\mathbb{R}^d$, none of them contained in a hyperplane. All members of $\mathcal{K}$ are convex if and only if every family $\mathcal{L}$ in $\mathbb{R}^d$, consisting of regular affine images of the members of $\mathcal{K}$, has the Helly property.*

Related to Theorem 3.5 are two characterizations of convex sets, the first one by Guggenheimer (1971), the second one by Rublev (1967). Let $\mathcal{K}$ be a family of compacta, as in Theorem 3.5. All members of $\mathcal{K}$ are convex if and only if $\mathcal{K} \cup Y$ has the Helly property, where Y is the family of all d-dimensional simplexes in $\mathbb{R}^d$. Given a nonconvex compact subset X of $\mathbb{R}^d$, denote by $r(X)$ the smallest positive integer n with the property that n properly chosen congruent copies of X have a nonconnected intersection. Then it is stated that $r(X) \leqslant 4^{d-1}$ for every d and X. Similar estimates concerning the higher homotopy groups of the intersections could be imagined.

An interesting question, which has not been answered yet in its full generality, has been posed by Leichtweiss and Schöpf (1982).

Question 3.2. Let $A \subset \mathbb{R}^d$ be a closed set, perhaps with nonempty interior. Is it true that either A or $\mathbb{R}^d \setminus A$ is convex if and only if $A \setminus (A + t)$ has the homotopy type of a point, for every $t \in \mathbb{R}^d$?

Leichtweiss and Schöpf show that closed convex sets A have nullhomotopic differences in the above sense, but that the closedness of A is indispensable. In the case $d = 2$ the above question, and even a stronger version thereof, has been established by Nishiura and Schnitzer (1971), and by Imrich and Schnitzer (1975).

There are a number of characterizations of convexity, involving the contact with other figures. Juul (1975) shows that a closed and connected subset S of $\mathbb{R}^2$ is a

convex curve if and only if S does not meet the interior of the convex hull of any triplet contained in S. A similar characterization, for arbitrary compact $S \subset \mathbb{R}^2$, has been found by Kramer (1978). I have not encountered an extension of these results to higher dimensions.

Let us say that a convex set $S \subset \mathbb{R}^d$ is a spindle if its boundary arises from some convex circular arc by rotating it around the axis determined by its endpoints, which we shall call the vertices of the spindle S. Zamfirescu (1971/72) shows that a closed set $M \subset \mathbb{R}^d$ is convex if and only if the interior of every spindle S, whose vertices belong to M, has a nonempty intersection with M. Borwein and O'Brien (1978) shows that convexity can be characterized by a cancellation property, with respect to Minkowski addition.

Theorem 3.6. *A compact subset B of $\mathbb{R}^d$ is convex if and only if, given any pair A, C of compact convex subsets of $\mathbb{R}^d$, the equation $A + B = A + C$ implies that $B = C$.*

Straus and Valentine (1952), and Valentine (1953, 1960) characterize convexity of a set X by various geometric conditions. We describe one result from Valentine (1953) as an example. Let S be a subset of $\mathbb{R}^d$. $V \subset S$ is called a set of visibility for S if, given any $p \in S$, there exists a point $q \in V$ such that the entire line segment $[p, q]$ is contained in S. A compact set $S \subset \mathbb{R}^2$ is convex, or else star shaped with a single point as its kernel if and only if every point $x \in S$ is contained in a unique minimal closed set $V(x)$ of visibility for S.

We now present a few more metric characterizations. Let $A \subset \mathbb{R}^d$ be a nonempty and closed set. We set, for $x \in \mathbb{R}^d \setminus A$, $r_A(x) = \inf\{||x - y||\colon y \in A\}$ and denote by $K_A(x)$ the closed ball with center x and radius $r_A(x)$. Every point of $A \cap K_A(x)$ is called a foot of x on A. We say that x is skeletal for A, if there is no point $y \neq x$ in $\mathbb{R}^d \setminus A$ such that $K_A(x) \subset K_A(y)$. The set of all skeletal points for A is denoted by $S(A)$; it contains all points with more than one foot on A. We denote by $\operatorname{conv}(X)$ the convex hull of $X \subset \mathbb{R}^d$. $f|_X$ is the restriction of the map $f: Y \to Z$ to the subset X of Y.

Theorem 3.7 (Zamfirescu 1971/72). *A closed set $A \subset \mathbb{R}^d$ is convex if and only if there exists a number $\gamma > 0$ such that $K(A, \gamma) = \{x \in \mathbb{R}^d\colon r_A(x) \leqslant \gamma\}$ is connected and, furthermore, for every pair x, y in A with $||x - y|| \leqslant 2\gamma$ the line segment $[x, y]$ belongs to A.*

If we assume A to be connected, then the condition on $K(A, \gamma)$ is redundant, and we obtain a classical characterization, see Valentine (1964).

Theorem 3.8 (Klee 1949). *A closed nonvoid subset A of $\mathbb{R}^d$ is convex if and only if any foot on S of some point $p \notin S$ is also a foot on S of some unbounded set of points.*

Theorem 3.9 (Calabi and Hartnett 1968a). *Let A and B be closed nonempty subsets of $\mathbb{R}^d$. Then $\operatorname{conv}(A) \cap (\mathbb{R}^d \setminus A) = \operatorname{conv}(B) \cap (\mathbb{R}^d \setminus B)$ if and only if $(S(A), r_A|_{S(A)}) = (S(B), r_B|_{S(B)})$.*

Notice that Theorem 3.9 contains Motzkins characterization of convex sets, Theorem 1.5, as a special case. The authors encourage the further study of the possible skeletal structures $(S(A), r_A|_{S(A)})$, and an extension to other situations. A charming illustration of their concept is given in Calabi and Hartnett (1968b).

The assignment, to give a topological characterization of convex sets, could be understood in quite a different way: give a list of models for those topological spaces that are homeomorphic to some closed convex set (Corson and Klee 1963) or, if this appears too easy, to the set of all extreme points of some closed convex set. There exists a rich literature on this topic, but a careful survey would go beyond my abilities. The interested reader may perhaps use the following references as a starting point: Klee (1955, 1958), Björck (1958), Collier (1975), Roberts (1977), who constructed a complete metrizable real topological vectorspace and, in it, a nonempty compact convex set without extreme points, Bronshtein (1981, 1985). For a similar question in the context of Riemannian geometry, see Bangert (1978).

We conclude this section with four characterizations of convexity, which have escaped our attempt toward a systematic presentation.

Golab (1951) characterizes the convex curves among all Jordan curves in the plane by conditions on their tangent rays. Brooks (1971) associates to every bounded open set $O \subset \mathbb{R}^2$ and to every point p on the unit circle S^1 the average length $l(O,p)$ of intersections of lines in the direction of p with O. $p \mapsto l(O,p)$ defines the distance function of some centrally symmetric convex set, and each centrally symmetric convex set is obtained in this way. This procedure may be generalized. Peterson (1967) shows that every simplicial complex C in $\mathbb{R}^2$, whose underlying polyhedron is convex, admits a subdivision with the property that the underlying polyhedron of the star at every vertex of C is also convex. The corresponding question, which could lead, in a roundabout way, to a combinatorial characterization of convexity, still seems open in higher dimensions. Shultz (1974) relates convexity to quantum physics in showing that C is a compact and convex subset of some locally convex topological vectorspace if and only if C is affinely homeomorphic to the set of all states on a σ-orthomodular lattice.

4. Extensions of the notion of a convex set. Realizations

Every characterization of convex sets can be meaningful in the context of generalized convexity: either as a germ for a new system of axioms, or else as a test for the discrepancy between generalized convex sets and familiar ones. We pursue this observation under the following aspects: metric convexity, hyperconvexity, barycentric structures, and other topics. This account will give, at best, a fragmentary image of generalized convexity, since it is concentrated on those questions that are related to characterizations. The interested reader finds more information on generalized convexity in the classical survey Danzer, Grünbaum and Klee (1963).

According to Menger (1928), a subset A of some metric space $M = (M,d)$ is called metrically convex if there exists for every pair x and $y \neq x$ of points in A a point z in $A \setminus \{x,y\}$ such that $d(x,z)+d(z,y) = d(x,y)$. A geodesic segment in M,

with endpoints x and y, is a map γ from some closed interval $[a,b] \subset \mathbb{R}$ into M such that $\gamma(a) = x$, $\gamma(b) = y$, and $\gamma|_U$ is an isometry, for each sufficiently small subinterval U of $[a,b]$.

Busemann (1942, 1955) continues, and expands, Mengers theory of convexity in metric spaces. He introduces the concept of a geodesic space, that is, a locally compact, convex metric space which satisfies a condition on unique local prolongation of segments. A closed subset A in a geodesic space M is called geodesically convex if there exists, for every pair of points x,y in A, a unique geodesic segment $\gamma : [a,b] \to M$ with endpoints x and y and if, furthermore, $\gamma(t) \in A$, for every $t \in [a,b]$. Among other characterizations of convex sets, Busemann (1955) establishes the following relative of Theorem 1.5. Notice that there is a natural definition of Chebyshev sets in any metric space.

Theorem 4.1. *Let M be a geodesic space, whose topological dimension is at least 3. Assume that all geodesic segments in M can be extended indefinitely. If the Chebyshev sets in M coincide with the closed convex sets, then M is a finite dimensional Banach space.*

Further characterizations of metrically convex sets have been given by Bantegnie (1968), German and Soltan (1974), Lassak (1977), and by Boltjanskii and Soltan (1978). Busemann and Phadke (1979), and Freese and Andalefte (1985) study geodesic spaces with convexity conditions for the metric balls. See also Busemann and Phadke (1987). An interesting companion of Theorem 1.4 has been discovered by Szenthe (1959). Let us say that a metric space M is straight if every local isometry from a closed interval $I \subset \mathbb{R}$ into M is a global isometry.

Theorem 4.2. *Every closed connected metrically locally convex subset of a locally compact metric space M is metrically convex in M if and only if M is straight.*

Another important contribution to metric convexity and to the study of geodesic metric spaces has been, and is being, made by Russian mathematicians. Aleksandrov (1955) calls a metric space (M,d) intrinsic if, for every pair of points x,y in M, $d(x,y)$ is the infimum of all arclengths, with respect to d, of curves connecting x to y in M. If the intrinsic metric space M is a two-dimensional manifold, then, according to Aleksandrov, we can define an intrinsic curvature at every point $p \in M$. Examples of positively curved surfaces are the convex surfaces, the boundaries of compact convex subsets of $\mathbb{R}^3$ with nonempty interior, provided with the natural intrinsic metric. The following converse by Aleksandrov and Pogorelov is one of the most ingenious characterizations of convex sets.

Theorem 4.3 (Aleksandrov 1955). *An intrinsic metric space, homeomorphic to the standard sphere S^2, is isometric to a convex surface S if and only if M has a nonnegative curvature function. Furthermore, S is unique, up to an isometry of $\mathbb{R}^3$.*

Among the many refinements and extensions of this result, also in spaces of constant positive or negative curvature, we only mention Zilberberg (1963), who

established a realization theorem for higher-dimensional polyhedral manifolds with positive curvature.

Recently, the theory of geodesic metric spaces has found many exciting and unexpected extensions, even into the seemingly most remote areas of group theory. We recommend the monographs by Gromov (1981, 1987) and Shapiro (1989) for further reading.

Convexity of a set $X \subset \mathbb{R}^n$ can be defined by requiring that the intersection of X with every straight line in $\mathbb{R}^n$ should be connected. If we replace the straight lines by some other family of curves, of sets, then we are led to a new concept of generalized convexity. The word hyperconvexity has been used by Danzer, Grünbaum and Klee (1963) in that context.

Let us say that $L \subset \mathbb{R}^2$ is a pseudoline if there exists a homeomorphism $h : \mathbb{R} \to L$ such that $||h(x)|| \to \infty$, whenever $|x| \to \infty$. Drandell (1952) considers a family $\mathcal{C}$ of pseudolines in $\mathbb{R}^2$ with the property that, given any pair p and $q \neq p$ of points in $\mathbb{R}^2$, there exists a unique element $L(p,q)$ in $\mathcal{C}$ such that $\{p,q\} \subset L$. He calls a set $X \subset \mathbb{R}^2$ $\mathcal{C}$-convex if, for every pair p and $q \neq p$ in X, the bounded component of $L(p,q) \setminus \{p,q\}$ is contained in X. The notion of a $\mathcal{C}$-supporting curve for X can be defined in a straightforward way. With this terminology, Drandell gives the following extension of Theorem 1.1.

Theorem 4.4. *Let $X \subset \mathbb{R}^2$ be compact, with nonempty interior. X is $\mathcal{C}$-convex if and only if every boundary point of X is contained in a $\mathcal{C}$-supporting curve for X.*

Related studies have been pursued by Zamfirescu (1971).

Dol'nikov (1988) presents an exciting extension of these concepts to higher dimensions. He uses methods from algebraic topology to obtain a common generalization of Helly's Theorem and of the antipodal point theorem by Borsuk and Ulam.

The concept of hyperconvexity, as described in Danzer, Grünbaum and Klee (1963) is studied in Blanc (1943). To every real number $\rho > 0$ and every pair of points p,q in $\mathbb{R}^2$ with $0 < ||p-q|| < 2\rho$ we associate the lens $L(p,q,\rho)$. $L(p,q,\rho)$ is the intersection of the two circular discs with radius ρ that contain p and q on their boundary.

$X \subset \mathbb{R}^2$ is called hyperconvex if it contains, together with every pair of points p,q, all lenses $L(p,q,\rho)$. Blanc gives a characterization of hyperconvex sets. His studies are continued by Santaló (1948) and, in a wider context, by Blind, Valette and Zamfirescu (1980), who give a characterization of unbounded rectangular convex sets in the plane $\mathbb{R}^2$.

Santaló (1969) introduces a related concept for the hyperbolic plane H^2, where the circular discs are replaced by horocycles. Pinkall (1984) establishes an isoperimetric inequality for horocyclically convex curves in H^2, by associating to each such curve C a centrally symmetric convex curve $\overline{C}$ in the affine plane $\mathbb{R}^2$. We have not been able to see whether it would be an interesting task to characterize the class of all such curves $\overline{C}$.

In many cases the underlying scalar domain for the barycentric calculus is an

ordered division ring R, and the question, whether a given barycentric structure is isomorphic to the standard structure in a convex subset of some R-module, can be quite interesting (Valette 1967). We shall call this the question of realization.

Similar questions arise in the theory of abstract convexity. Bryant and Webster (1973) give a criterion for a convexity structure to ensure enough supporting hyperplanes in the sense of Theorem 1.1, but, beyond that, we have not yet found many analogues of the classical characterizations for convex sets. The interested reader may consult one of the monographs Kay and Womble (1971), Edelman and Jamison (1985), Soltan (1984), Edelman (1986) and Duchet (1987) as an introduction.

We are closing this section with a few characterizations which we found interesting, but could not place under any of the above headings.

Given $X \subset \mathbb{R}^d$ and $\varepsilon > 0$, define the ε-convex hull of X, $C_\varepsilon(X)$, by requiring that its complement should be the union of all open Euclidean balls of diameter ε that are disjoint to X. Notice that the convex hull of X is the closure of $\bigcup\{C_\varepsilon(X)\colon \varepsilon > 0\}$, and the closed hull of X coincides with $\bigcap\{C_\varepsilon(X)\colon \varepsilon > 0\}$. X is called ε-convex if it coincides with $C_\varepsilon(X)$. One has a feeling that ε-convex sets cannot be too irregular, but a full characterization is still missing.

Question 4.1 (Perkal 1956)**.** Is an ε-convex set locally contractible? Can it be triangulated?

Zamfirescu (1972/73) and Bielawsky (1987) both investigate a structure which they call simplicial convexity. While Zamfirescu's construction is of a more geometric nature, Bielawsky deals with topological, singular simplexes. As a result, their characterizations have quite different viewpoints, simplicially convex sets on the one hand, spaces with a simplicial structure on the other hand.

McNulty and Taylor (1975) introduce a notion of k-convexity for subsets of the extended integers. They prove a k-convexity criterion, with applications in algebra, logic and combinatorics.

Finally we would like to mention a rich literature on fuzzy convex sets, including characterization theorems, compare Lowen (1980) and Pei (1984) as sources of further reading.

5. Measures of convexity

Imagine a collection C of sets, a function $m\colon C \to \mathbb{R}$ and a real number α. It may happen that $X \in C$ is convex if and only if $m(X) = \alpha$. In this case m can serve as a means to characterize convex sets, a measure of convexity. Schneider (1979b) asks for all affine invariant measures of convexity on the collection $C = C^d$ of all compact subsets of $\mathbb{R}^d$, in particular, whether there exists such a measure which is continuous, with respect to the Hausdorff metric. One affine invariant measure m of convexity on C^d, given by

$$m(X) = \inf\{\lambda \geqslant 0\colon X + \lambda \mathrm{conv}(X) \text{ is convex }\},$$

has been studied in Schneider (1975). Beer (1973, 1974) considers the class C of all subsets $X \subset \mathbb{R}^d$ which are measurable with respect to the Lebesque measure $\mathcal{L}^d$, and the measure $m : C \to [0,1]$, given by

$$m(X) = (\mathcal{L}^d(X))^{-2} \cdot \int_X \mathcal{L}^d(S(p))\, \mathrm{d}\mathcal{L}^d(p),$$

where the set of visibility $S(p)$ is given, for $p \in X$, by $S(p) = \{q \in X : [p,q] \subset X\}$.

Another class of motion invariant convexity measures has been studied by Cassels (1975), Khan (1977), and Wegmann (1980). We describe one of them. Given $X \subset \mathbb{R}^d$, we denote by $\mathfrak{X}(X)$ the collection of all random vectors that take a finite number of values a_i, $1 \leqslant i \leqslant n$ in X, each with a certain probability p_i. $a^* = p_1 a_1 + \cdots + p_n a_n$ is the expected value, and $V(a) = p_1 \|a_1 - a^*\|^2 + \cdots + p_n \|a_n - a^*\|^2$ the variance, of a. By setting

$$m(X) = \sup\{\inf\{V(a) : a \in \mathfrak{X}(X),\ a^* = p\} : p \in \operatorname{conv}(X)\}$$

we obtain a convexity measure on the space C^d of all compact sets in $\mathbb{R}^d$.

Let C be the collection of all smooth and closed curves in $\mathbb{R}^d$. For $c \in C$ we denote by $m(c)$ the integral of the absolute value of its curvature.

Theorem 5.1 (Fenchel 1929). *With the above notation, $m(c) \geqslant 2\pi$, for every $c \in C$, and equality holds if and only if c is a planar convex curve.*

Thus m is a motion invariant measure of convexity. An extension of Fenchels characterization to higher-dimensional immersed submanifolds has been found by Chern and Lashof. The monograph Burago and Zalgaller (1980) puts these characterizations in a wider context. If we restrict the curvature integral to the class of all immersions in a fixed homotopy class, then a number of more delicate questions arise, in the context of global differential geometry and algebraic topology, remember the analogues of Fenchels inequality for knotted curves by Fary and Milnor, and a recent study concerning higher-dimensional immersions by Sharpe (1988).

6. Looking out

Some of the most interesting questions of characterization arise when we consider a mapping f from a collection C of convex sets into another set X, and try to describe $f(C)$ in X. One could perhaps say that character appears most clearly while transcending. We only give three examples here, to illustrate this.

Let us say that a graph is 3-polytopal if it is isomorphic to the collection of all vertices and edges of some three-dimensional convex polytope.

Theorem 6.1 (Steinitz and Rademacher 1934). *A finite graph is 3-polytopal if and only if it is planar and triply connected.*

In a similar way one can try to characterize those simplicial or, more generally, polyhedral cell complexes which are isomorphic to the boundary complex, that is, the collection of all proper faces, of some d-dimensional convex polytope. Let us call such a complex d-polytopal.

Surely a d-polytopal complex must be a combinatorial $(d-1)$-sphere: its underlying polyhedron is homeomorphic to $S^{d-1} = \{x \in \mathbb{R}^d : ||x|| = 1\}$.

Question 6.1. Characterize, among all combinatorial $(d-1)$-spheres, the d-polytopal ones.

As mentioned in Grünbaum (1967) there exists, according to Tarski, an algorithm to decide this question, known as the Steinitz problem. A presentation of this problem, and of questions related to it, can be found in Ewald et al. (1979) and in Bokowski and Sturmfels (1989). They show that the Steinitz problem is algorithmically as hard, and as easy, as the problem to determine whether a finite set of polynomial equations in d variables, with integer coefficients, has a common solution in $\mathbb{R}^d$. Nevertheless we can go one step further, and try to describe the configuration spaces of polytopes. If $P \subset \mathbb{R}^d$ is an ordered d-polytope, that is, a convex d-dimensional polytope, together with a total ordering of its vertex set, we denote by $[P]$ the space of all ordered polytopes $Q \subset \mathbb{R}^d$ that are isomorphic to P, together with the topology induced by the Hausdorff metric. The basic configuration space $[P]_0$ is the quotient of $[P]$ with respect to the natural action of the affine group upon $[P]$. An easy consideration shows that $[P]_0$ is, in every case, a simple semialgebraic variety, the set of all solutions of a collection of polynomial equations and inequalities, with integer coefficients, in some Euclidean space.

Indeed many people believed, perhaps remembering Steinitz's positive result for the case $d = 3$, that $[P]_0$ might always be contractible, until we became acquainted with the following striking result by Mnëv (1989).

Theorem 6.2. *Given any simple semialgebraic variety V, there exists a convex polytope P whose basic configuration space $[P]_0$ has the same homotopy type as V.*

If we project the boundary complex of a polytope P radially from some inner point x onto a sphere S with midpoint x, we obtain a spherically convex polytope on S. Shephard (1971) has characterized those spherically convex polytopes which can be obtained in this way, by radial projection, from an Euclidean convex polytope. Many of the above questions can be asked again for spherically convex polytopes. This might be interesting, since the basic configuration spaces of simplicial, spherically convex polytopes are related to the problem of smoothing piecewise linear manifolds, see Cairns (1940).

Our next example concerns f-vectors. Denote by $\mathcal{P}$ the set of all d-dimensional convex polytopes, and define a mapping $f: \mathcal{P} \to \mathbb{Z}^d$ by $f(P) = (f_0(P), \ldots, f_{d-1}(P))$, where $f_i(P)$ denotes the number of all i-dimensional faces of P. Also, let $\mathcal{P}_S \subset \mathcal{P}$ be the set of all simplicial polytopes. One of the highlights in combinatorial geometry is the characterization of $f(\mathcal{P}_S)$ which had been conjectured by McMullen (1971), and was achieved by Stanley (1980), and Billera and Lee (1979).

A fascinating account of the history, and of the ideas involved, can be found in Stanley (1985) and in Björner (1986). Chapter 2.3 in this Handbook is warmly recommended for further reading about this theme, also on the attempts to characterize the f-vectors, or other combinatorial invariants, of nonsimplicial convex polytopes.

Instead of the f-vectors of convex polytopes we might try to characterize those vectors whose ith coordinate is the ith intrinsic volume of some convex compact set in $\mathbb{R}^d$. This question, however, still seems rather hopeless, even in the case $d = 3$. Compare Hadwiger (1957, p. 284) and, for one positive result, Ohmann (1953).

Koszul (1965) characterizes all compact, connected, locally flat manifolds with an affine connection, whose universal covering space is isomorphic to an open convex, linefree subset of some Euclidean space, provided with its standard affine connection. Goldmann (1990) studies the space of all these convex characters which are homeomorphic to a given compact surface, in the spirit of Teichmüller theory.

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CHAPTER 1.2

Mixed Volumes

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Contents

1. Elementary convexity 45
2. Mixed volumes 46
3. Quermassintegrals and intrinsic volumes 49
4. Mixed surface area measures 52
5. Symmetrization 54
6. Fundamental inequalities 57
7. Equality in the Aleksandrov–Fenchel inequality 60
8. Additional inequalities 64
References 66

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HANDBOOK OF CONVEX GEOMETRY
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On December 10, 1900, Hermann Minkowski wrote to David Hilbert that his study of surface area and volume in three-dimensional space was complete, and his most important improvement was the introduction of an idea associated with three convex bodies which he was provisionally calling their *mixed volume*. The letter is reproduced in Minkowski (1973, pp. 133–136).

In this article we begin with an investigation of the properties of mixed volumes emphasizing their geometric significance. Many readers will be familiar with a special class of mixed volumes – those involving an arbitrary convex body and a ball. Their classical name is the *quermassintegrals*, and they include volume, surface area, total mean curvature and mean width.

The study of mixed volumes led immediately to the discovery of inequalities among them. By 1937, Aleksandrov and Fenchel had independently shown that a large number of these, including the classical isoperimetric inequality, could be expressed as consequences of a single inequality for mixed volumes – the inequality which bears their name. To study this result we include sections on mixed surface area measures and symmetrization. This article concludes with an analysis of a central unsolved problem in convexity, the general equality conditions for the Aleksandrov–Fenchel inequality, and with a brief presentation of further inequalities.

1. Elementary convexity

Our setting is d-dimensional Euclidean space $\mathbb{E}^d$, and $\mathbb{R}$ represents the real numbers. The inner product of $x, y \in \mathbb{E}^d$ is denoted by $\langle x, y\rangle$ and $\|\cdot\|$ is its corresponding norm. The unit ball centered at the origin o is $B = \{x: \|x\| \leqslant 1\}$, and $S^{d-1} = \{x: \|x\| = 1\}$ is the unit sphere. The Borel sets of $\mathbb{E}^d$ and S^{d-1} are denoted by $\mathcal{B}(\mathbb{E}^d)$ and $\mathcal{B}(S^{d-1})$, and the usual spherical Lebesgue measure on $\mathcal{B}(S^{d-1})$ is denoted by $s(\cdot)$.

Further discussion of the elementary notions of convexity to be introduced in this section may be found in Bonnesen and Fenchel (1934, 1987), Eggleston (1969), Hadwiger (1955, 1957), Leichtweiss (1980), and Valentine (1964).

A subset K of $\mathbb{E}^d$ is *convex* if $(1-\lambda)x + \lambda y \in K$ for all $x, y \in K$ and $0 < \lambda < 1$. The family of compact, convex subsets of $\mathbb{E}^d$ is denoted by $\mathcal{K}^d$ (we include the empty set), and those with nonempty interior are members of $\mathcal{K}_o^d$ and called *convex bodies*. The family $\mathcal{K}^d$ is metrized by the Hausdorff metric.

The *Minkowski sum* and *scalar multiple* of convex sets K, $L \in \mathcal{K}^d$ and $\lambda \geqslant 0$ are defined by $K + L = \{k + l: k \in K, l \in L\}$ and $\lambda K = \{\lambda k: k \in K\}$, respectively. A *homothet* of a convex set is of the form $x + \lambda K$ for $x \in \mathbb{E}^d, \lambda > 0$, and the orthogonal projection of K into a $(d-1)$-dimensional subspace with normal $u \in S^{d-1}$ is denoted by K^u.

The *support function* of a convex set K is a real valued function on $\mathbb{E}^d$ defined by $h(K, x) = \max\{\langle x, y\rangle: y \in K\}$. When restricted to S^{d-1} the support function

is the signed distance from the origin to the *support hyperplane* of K with the corresponding unit outer normal. For all x, $y \in \mathbb{E}^d$ and $\lambda \geqslant 0$, $h(K, \lambda x) = \lambda h(K, x)$, and $h(K, x+y) \leqslant h(K, x)+h(K, y)$. Convex bodies are characterized by their support functions, that is, any real valued function on $\mathbb{E}^d$ with these two properties is the support function of some convex set. A convex body is said to be *smooth* if there is only one support hyperplane to the body at each boundary point, and $K \in \mathcal{K}_d^d$ is said to be *strictly convex* if its boundary contains no line segments.

Many of our results will make use of the *Blaschke selection theorem*: Every infinite collection of convex bodies contained within a bounded region of $\mathbb{E}^d$ has a subsequence which converges to a nonempty convex set. For further discussion of the theorem and convergence see Gruber in this volume.

A family of convex sets $K(\lambda)$, $\alpha \leqslant \lambda \leqslant \beta$, is called *linear* or *concave* if, for all $\alpha \leqslant \lambda_1 \leqslant \lambda_2 \leqslant \beta$ and $0 \leqslant \theta \leqslant 1$, $(1-\theta)K(\lambda_1) + \theta K(\lambda_2)$ equals or is contained in $K((1-\theta)\lambda_1 + \theta\lambda_2)$, respectively. The intersections of a convex set with a family of parallel hyperplanes, and the inner parallel sets of a convex body are two examples of concave families. In sections 5 and 6 we shall use a concave family to construct a convex set of higher dimension as follows. Let $K(\lambda) \in \mathcal{K}^d$, $\alpha \leqslant \lambda \leqslant \beta$, be a concave family of $(d-1)$-dimensional sets which lie in the subspace with coordinate $x_d = 0$. Translate $K(\lambda)$ by changing the x_d coordinate to λ. The union of these translates is a convex set in $\mathcal{K}^d$.

We conclude with the definition of several quantities associated with convex sets. The *inradius* r, and *circumradius* R of K with respect to L are defined to be the largest scalar for which a homothet of L is contained in K, and the smallest scalar for which a homothet of L contains K, respectively. Usually we take $L = B$ in the definitions of r and R. The *diameter* D is the maximum distance between pairs of points of a convex body, and the *minimal width* w is the minimum distance between parallel support hyperplanes of the body.

2. Mixed volumes

Historically different approaches have been taken in defining the volume of a convex body. One early approach used approximation by polytopes. This was carried out for planar convex sets by Hilbert (1899), and for $d = 3$ by Minkowski (1911, section 7), also see Blaschke (1916a, section 17). These approaches and their inherent difficulties are discussed by Hadwiger (1957), and Boltianskii (1978). Since the assumption of convexity guarantees Lebesgue measurability, we define the *volume* of a convex set $K \in \mathcal{K}^d$ to be its d-dimensional Lebesgue measure and it is denoted by $V(K)$. In fact K has volume in the sense of Peano and Jordan.

For $d = 2$ volume is traditionally called *area*, and $V(B)$ is denoted by κ_d for all d. If a convex set $K \in \mathcal{K}^d$ lies in a hyperplane, we shall let $V_{d-1}(K)$ represent its $(d-1)$-dimensional volume. In the next section it will be shown that this notation is consistent with the definition of the $(d-1)$st intrinsic volume which shares the notation.

The following theorem is due to Minkowski (1911, sections 20–22), for $d = 3$. It is a generalization of a formula due to Steiner (1840), see section 3 of this article.

Theorem 2.1 (Minkowski). *The volume of the linear family,* $K = \lambda_1 K_1 + \cdots + \lambda_r K_r$, $K_i \in \mathcal{K}^d$, *r a positive integer, is a dth degree homogeneous polynomial in* $\lambda_1, \ldots, \lambda_r$. *That is,*

$$V(K) = \sum_{i_1=1}^{r} \cdots \sum_{i_d=1}^{r} \lambda_{i_1} \cdots \lambda_{i_d} V(K_{i_1}, \ldots, K_{i_d}),$$

where the coefficients $V(K_{i_1}, \ldots, K_{i_d})$ *are chosen to be invariant under permutations of their arguments. The coefficient* $V(K_1, \ldots, K_d)$ *is called the* mixed volume *of the convex sets* $K_1, \ldots, K_d$.

Theorem 2.1 is the first of several results which may be proved by approximation with polytopes. It is quite easily established for this class of convex sets because the projection of the face of a Minkowski sum of polytopes is the sum of the projections. The proof may be found in Bonnesen and Fenchel (1934, p. 31), (1987, p. 35), and Leichtweiss (1980, pp. 164–165).

The following properties of mixed volumes are presented in order of implication. The proofs may be found in Bonnesen and Fenchel (1934, pp. 40–44), (1987, pp. 44–49), and Leichtweiss (1980, pp. 166–177).

Property 2.2. $V(K, \ldots, K) = V(K)$.

Property 2.3. *If A is an affine transformation whose determinant is denoted by* det *A, then*

$$V(A(K_1), \ldots, A(K_d)) = |\det A| V(K_1, \ldots, K_d).$$

As in the next property, we shall often use the notation $\mathcal{C}$ to represent an ordered collection of convex sets. It follows from Property 2.4 that the mixed volumes are linear in each argument.

Property 2.4. *If* $K = \lambda_1 K_1 + \cdots + \lambda_r K_r$, *r a positive integer, and* $\mathcal{C} = L_{j+1}, \ldots, L_d$, $1 \leqslant j \leqslant d-1$, *then*

$$V(\underbrace{K, \ldots, K}_{j}, \mathcal{C}) = \sum_{i_1=1}^{r} \cdots \sum_{i_j=1}^{r} \lambda_{i_1} \cdots \lambda_{i_j} V(K_{i_1}, \ldots, K_{i_j}, \mathcal{C}).$$

Property 2.5. $V(K_1, \ldots, K_d)$ *varies continuously with* $K_1, \ldots, K_d$.

Property 2.6.

$$V(K, \ldots, K, L) = \frac{1}{d} \lim_{\lambda \to 0^+} \frac{V(K + \lambda L) - V(K)}{\lambda}.$$

Property 2.7. *If P is a polytope with facets* F_i, *and corresponding unit outer normals* u_i, $i = 1, \ldots, r$, *then*

$$V(K, P, \ldots, P) = \frac{1}{d} \sum_{i=1}^{r} h(K, u_i) V_{d-1}(F_i).$$

Property 2.8. *The mixed volumes are monotone and nonnegative. That is, if* $K \subset L$ *and* $\mathscr{C} = K_2, \ldots, K_d$, *then* $0 \leqslant V(K, \mathscr{C}) \leqslant V(L, \mathscr{C})$.

A proof of monotonicity is similar to the proof of Theorem 2.1 and uses Property 2.4; nonnegativity follows.

Property 2.9. *The mixed volumes are* valuations. *That is, if* K, L, $K \cup L$, *and* $\mathscr{C} = K_2, \ldots, K_d \in \mathscr{K}^d$, *then*

$$V(K, \mathscr{C}) + V(L, \mathscr{C}) = V(K \cap L, \mathscr{C}) + V(K \cup L, \mathscr{C}).$$

The previous property is proved using the valuation property of volume, and the following result. For all $M \in \mathscr{K}^d$,

$$(K \cap L) + M = (K \cap M) + (L \cap M),$$
$$(K \cup L) + M = (K \cup M) + (L \cup M).$$

The convexity of the union is required for one inclusion in the second expression above. The valuation property was first observed by Blaschke (1949, section 37), for the quermassintegrals, defined in section 3 below, see also Hadwiger (1957, pp. 210–211), but as is noted by McMullen and Schneider (1983, p. 178), it has been neglected for mixed volumes. For an extensive discussion of valuations see McMullen and Schneider (1983); this article is updated by McMullen in this Handbook. McMullen and Schneider (1983, p. 179), include a nice derivation of the following property due to Groemer (1977, p. 160).

Property 2.10. *If* K, L, $K \cup L$, *and* $\mathscr{C} = K_3, \ldots, K_d \in \mathscr{K}^d$, *then*

$$V(K, L, \mathscr{C}) = V(K \cap L, K \cup L, \mathscr{C}).$$

For any $u \in S^{d-1}$ let $U \in \mathscr{K}^d$ be the line segment from o to u.

Property 2.11. *If* $u \in S^{d-1}$ *and* $\mathscr{C} = K_2, \ldots, K_d \in \mathscr{K}^d$, *then* $h(u) = V(U, \mathscr{C})$ *determines the support function of a convex set.*

This property follows from the characterization result for support functions, see section 1, and the properties of mixed volumes. If $K = K_2 = \cdots = K_d$, then it is not difficult to show that $V(U, K, \ldots, K) = V_{d-1}(K^u)$, and the *projection body* of K is the body whose support function restricted to S^{d-1} is $V_{d-1}(K^u)$.

We conclude with a property whose derivation may be found in Weil (1974).

Property 2.12. *If* K, $L \in \mathscr{K}^d$, *and* $K + L = M$, *then*

$$V(K) = \sum_{i=0}^{d} \binom{d}{i} (-1)^i V(\underbrace{M, \ldots, M}_{d-i}, \underbrace{L, \ldots, L}_{i}).$$

3. Quermassintegrals and intrinsic volumes

Of significant geometric interest are the mixed volumes of a fixed convex set $K \in \mathcal{K}^d$ and the unit ball B. In the literature they have two representations, the *quermassintegrals* $W_i(K)$, and the *intrinsic volumes* $V_i(K)$, and we shall use both throughout. They are defined by

$$V(\underbrace{K,\dots,K}_{d-i},\underbrace{B,\dots,B}_{i}) = W_i(K) = \frac{\kappa_i}{\binom{d}{i}} V_{d-i}(K), \quad i = 0,\dots,d.$$

The latter term was introduced by McMullen (1975). There are circumstances under which each is preferable. For example, the quermassintegrals provide an elegant statement of the Aleksandrov–Fenchel inequality for a convex body and B, see (6.5), while the intrinsic volumes have a geometric interpretation dependent on the dimension of the set and not the space in which it lies, see Property 3.1.

The properties of mixed volumes will extend to the quermassintegrals and intrinsic volumes. In particular, they are monotone, and W_i and V_i are homogeneous of degrees $d-i$ and i, respectively. It also follows that $W_0(K) = V_d(K)$ is the d-dimensional volume of K, $V_0(K)$ is the Euler characteristic function, and $W_d(K) = \kappa_d$ for all K.

For $\lambda \geqslant 0$ the body $K + \lambda B$ is called the *outer parallel body* of K at the distance λ. Theorem 2.1 for outer parallel bodies is the *Steiner formula* (Steiner 1840):

$$V(K+\lambda B) = \sum_{i=0}^{d} \binom{d}{i} \lambda^i W_i(K) = \sum_{i=0}^{d} \kappa_{d-i} \lambda^{d-i} V_i(K).$$

The reader unfamiliar with quermassintegrals may find it instructive to investigate the Steiner formula for polytopes. The outer parallel set of a polytope P at a distance λ is the union of P and the "brush set" at distance λ. The bristles of the brush set are the outer normals of length λ to each boundary point of P; see section 4 for the definition. If a face is i-dimensional, the volume of the union of the bristles normal to that face is homogeneous of degree $d-i$ in λ. It follows that

$$V_i(P) = \frac{\binom{d}{i}}{\kappa_{d-1}} W_{d-i}(P) = \sum_F \gamma(F,P) V_i(F), \quad i = 0,\dots,d,$$

where the sum is over the i-dimensional faces F of P, and γ is the exterior angle at each face; for further discussion of this formula see McMullen (1975).

In the formula above for polytopes we make use of the fact that the intrinsic volumes are normalized so that the intrinsic volume of a j-dimensional set is its j-dimensional volume. In fact one can establish the following:

Property 3.1. *If K is a j-dimensional set in $\mathcal{K}^d$, $0 \leqslant j < d$, then*

$$W_i(K) = V_{d-i}(K) = 0, \quad i = 0,\dots,d-j-1,$$

and

$$\frac{\binom{i+j}{j}}{\binom{i+j}{d}} \frac{\kappa_{i+j-d}}{\kappa_i} W_i(K), \quad i = d-j, \ldots, d,$$

is the $(i+j-d)$*th* j*-dimensional quermassintegral of* K*. In particular,*

$$\left(\binom{d}{j}/\kappa_{d-j}\right) W_{d-j}(K) = V_j(K) \text{ is the } j\text{-dimensional volume of } K.$$

Note that for $j = d-1$, $V_{d-1}(K)$ is consistent with our earlier definition, and the geometric significance of this intrinsic volume is established. If $K \in \mathcal{K}^d$, then $dW_1(K) = 2V_{d-1}(K)$ is the $(d-1)$-dimensional *surface area* of K, and it is also denoted by $S(K)$. For $d = 2$ it is commonly called *perimeter*. This definition is due to Minkowski (1911, section 24); notice that he has defined surface area to be a constant multiple of the quermassintegral which is homogeneous of degree $(d-1)$. The simplicity of the definition should be evident to the reader, its elegance is apparent given the following, much later, result.

Theorem 3.2 (Hadwiger). *If* $\phi : \mathcal{K}^d \to \mathbb{R}$ *is a valuation, continuous and invariant under rigid motion, then there exist* $\alpha_i \in \mathbb{R}$, $i = 0, \ldots, d$, *such that*

$$\phi = \sum_{i=0}^{d} \alpha_i W_i.$$

The earliest characterization theorem for quermassintegrals is due to Blaschke (1949, section 43). These results are discussed by McMullen in this volume. Also see Hadwiger (1957, sections 6.1.7 and 6.1.10), Leichtweiss (1980, section 17), and McMullen and Schneider (1983, sections 13–17).

As with volume there have been many approaches to defining the surface area of a convex set. It follows from (2.7) for $B = K$ that our definition is equivalent to the usual definition of surface area for polytopes. Steiner (1840) used the outer parallel sets of K to define surface area as the right-hand side of the formula in Property 2.6 for $B = L$. For other approaches see Bonnesen and Fenchel (1934, 1987, section 31), and Hadwiger (1957, p. 207).

The Steiner formula yields one of several distinctly different representations of the quermassintegrals. Analogous to Property 2.6 is

$$d(d-1)\cdots(d-i+1)W_i(K) = \frac{d^{d-i}}{d\lambda^{d-i}} V(K + \lambda B)|_{\lambda=0}, \quad i = 1, \ldots, d. \tag{3.3}$$

In the next section we shall derive a formula for the ith quermassintegral involving the area functions of a convex body. The ith quermassintegral also measures the i-dimensional flats which intersect a convex set, and for sufficiently smooth bodies

it is proportional to the mean value of the $(d-i)$th elementary symmetric function of the principal radii of curvature. Discussion of these results is beyond the scope of this article and we refer the reader to articles in this volume by Schneider, Schneider and Wieacker, and Weil and Wieacker; here we present a hint of the results. Cauchy (1841, $d = 2$, 1850, $d = 3$), showed that the surface area of a convex set was proportional to the mean value of the $(d-1)$-dimensional volume of its projections. The result was later proved by Minkowski (1911, section 26), and generalized to the other intrinsic volumes by Kubota (1925).

Theorem 3.4 (Cauchy's surface area formula). *If* $K \in \mathcal{K}^d$, *then*

$$S(K) = \frac{1}{\kappa_{d-1}} \int_{S^{d-1}} V_{d-1}(K^u)\ \mathrm{d}s(u).$$

It suffices to establish the formula for a polytope P with facets F_i, and corresponding unit outer normals, u_i, $i = 1, \ldots, r$. For almost all u in S^{d-1}

$$2V_{d-1}(P^u) = \sum_{i=1}^{r} V_{d-1}(F_i^u).$$

If we apply the cosine principle to the right-hand side along with

$$\int_{S^{d-1}} |u \cdot u_i|\ \mathrm{d}s(u) = 2\kappa_{d-1},$$

the proof is complete.

We conclude our discussion of the geometric significance of the quermassintegrals with the following identities:

$$dW_2(K) = (\pi d / \tbinom{d}{2}) V_{d-2}(K) \text{ is the total mean curvature of } K,$$

and

$$(2/\kappa_d) W_{d-1}(K) = (2\kappa_{d-1}/d\kappa_d) V_1(K) \text{ is the mean width of } K.$$

In addition to the formulae which appear in the references on integral geometry cited above, the quermassintegrals have integral representations involving the support function. A simple example for $d = 3$ is

$$S(K) = \int_{S^2} [h^2(K,u) - \tfrac{1}{2}\nabla h(K,u)]\ \mathrm{d}s(u),$$

where ∇ is the first Beltrami differential operator on S^2; see Bonnesen and Fenchel (1934, 1987, section 39). Generalizations of this formula may be found in Bokowski and Heil (1986, p. 82), and Heil (1987, pp. 43–44).

There are many extensions of the mixed and intrinsic volumes; they are also beyond the scope of this article, and are considered in Burago and Zalgaller (1988,

section 26), McMullen and Schneider (1983, section 5), and McMullen in this volume. Results which are applicable to a particular class of convex sets, for example polytopes or sets of constant width, may also be found in the appropriate article in this volume.

The evaluation of mixed or intrinsic volumes is often difficult. The intrinsic volumes of elementary bodies are computed in Hadwiger (1949, p. 120), (1955, pp. 35–37), (1957, section 6.1.9), and Santaló (1976, pp. 224–227, pp. 229–232). These formulae may be used in conjunction with the valuation property (2.9) to evaluate the intrinsic volumes of bodies which are either truncations or unions of simple bodies.

4. Mixed surface area measures

Here we introduce the mixed surface area measures which may be viewed as a "local" generalization of the mixed volumes. We shall follow Aleksandrov's (1937a) original work to demonstrate how it parallels Minkowski's definition of the mixed volumes. The section concludes with the classification of support hyperplanes.

We began the study of mixed volumes by defining the volume of convex set. Aleksandrov began with the definition of the surface area measure: For $K \in \mathcal{K}^d$, and $\beta \in \mathcal{B}(\mathbb{E}^d)$, $\sigma(K,\beta)$ is the set of all $u \in S^{d-1}$ such that u is an outer normal to K at a boundary point of $K \cap \beta$. It follows that $\sigma^{-1}(K,\omega)$, $\omega \in \mathcal{B}(S^{d-1})$, is the set of boundary points of K at which there exists an exterior normal in ω. It is not difficult to show that there exists a measure denoted by $S_{d-1}(K,\omega)$ on $\mathcal{B}(S^{d-1})$ which is the $(d-1)$-dimensional Hausdorff measure of $\sigma^{-1}(K,\omega)$. It follows from Property 2.7 that

$$V(L,K,\ldots,K) = \frac{1}{d}\int_{S^{d-1}} h(L,u)\,\mathrm{d}S_{d-1}(K,u).$$

The Riesz representation theorem may be used to guarantee the existence of the *mixed surface area measures* $S(\mathscr{C},\cdot)$ on $\mathcal{B}(S^{d-1})$ for $\mathscr{C} = K_1,\ldots,K_{d-1} \in \mathcal{K}^d$ which are defined by

$$V(L,\mathscr{C}) = \frac{1}{d}\int_{S^{d-1}} h(L,u)\,\mathrm{d}S(\mathscr{C},u) \tag{4.1}$$

for all $L \in \mathcal{K}^d$. The analogue to Theorem 2.1 is

$$S_{d-1}\Big(\sum_{i=1}^{r}\lambda_i K_i,\omega\Big) = \sum_{i_1=1}^{r}\cdots\sum_{i_{d-1}=1}^{r}\lambda_{i_1}\cdots\lambda_{i_{d-1}}\,S(K_{i_1},\ldots,K_{i_{d-1}},\omega),$$

and the analogues to the intrinsic volumes are

$$S_i(K,\cdot) = S(\underbrace{K,\ldots,K}_{i},\underbrace{B,\ldots,B}_{d-i-1},\cdot), \quad i = 0,\ldots,d-1,$$

which are called *area functions* or *curvature measures of the second kind.* Note that $S_0(K,\cdot) = S_{d-1}(B,\cdot) = s(\cdot)$. Representations for the quermassintegrals are

$$W_i(K) = \frac{1}{d}\int_{S^{d-1}} h(K,u)\, \mathrm{d}S_{d-i-1}(K,u), \quad i = 0,\ldots,d-1,$$

$$= \frac{1}{d}\int_{S^{d-1}} \mathrm{d}S_{d-i}(K,u), \quad i = 1,\ldots,d. \tag{4.2}$$

The mixed surface area measures were introduced by Aleksandrov (1937a, section 4), also see Aleksandrov (1939), and by Fenchel and Jessen (1938). An excellent and more thorough reference on this topic is Schneider (1979); this reference is updated by Schneider in this Handbook. We have followed Aleksandrov's earlier approach to define the mixed surface area measures (which varies little from Fenchel and Jessen), and shall use the method of the second authors to obtain a "local version" of the Steiner formula. We first define the *local outer parallel set* or *brush set* of $K \in \mathcal{K}^d$ and $\omega \in \mathcal{B}(S^{d-1})$ at a distance $\lambda > 0$; it is denoted by $B_\lambda(K,\omega)$. For $x \in \mathbb{E}^d$ if $p(K,x)$ is the unique closest point of K to x, then

$$B_\lambda(K,\omega) = \{x\colon\ 0 < \|x - p(K,x)\| \leqslant \lambda \ \text{ and } \ \frac{x - p(K,x)}{\|x - p(K,x)\|} \in \omega\}.$$

The local version of Steiner's formula is

$$V(B_\lambda(K,\omega)) = \sum_{i=0}^{d-1} \frac{1}{d}\binom{d}{i}\lambda^{d-i} S_i(K,\omega).$$

Choose $K \in \mathcal{K}^d$ and $u \in S^{d-1}$. At every boundary point $k \in K$ there is a cone (with apex o) of outer normal vectors to the boundary of K at k, and those cones which contain u may not all be identical. It is the case that u is contained in a unique relatively open face of these cones, we call this face the *touching cone* of K at u, and it is denoted by $T(K,u)$, see Rockafellar (1970, Theorem 18.2). Choose $\mathcal{C} = K_1,\ldots,K_{d-1} \in \mathcal{K}^d$, then $u \in S^{d-1}$ is said to be $\mathcal{C}$-*extreme* if there exist $(d-1)$-dimensional linear subspaces, $H_1,\ldots,H_{d-1}$ in $\mathbb{E}^d$, such that $T(K_i,u) \subset H_i$, $i = 1,\ldots,d-1$, and the dimension of $H_1 \cap \cdots \cap H_{d-1}$ is 1.

To illustrate this definition in $\mathbb{E}^3$ choose polytopes P and Q. Any common outer normals to facets of P and Q are (P,Q)-extreme, and if the Minkowski sum of two edges of P and Q creates a new facet in $P + Q$, then the outer normal to that facet is also (P,Q)-extreme. Also note that if each K_i in $\mathcal{C}$ is smooth, then the set of $\mathcal{C}$-extreme directions is S^{d-1}; in particular, the set of $(B,\ldots,B)$-extreme directions is S^{d-1}.

For any $K \in \mathcal{K}^d u \in S^{d-1}$ is said to be an *i-extreme direction*, $i = 0,\ldots,d-1$, of K if the dimension of $T(K,u) \leqslant i+1$. Similarly the support hyperplane to K with outer normal u is called i-extreme. It follows that if u is i-extreme, it is also j-extreme for $j = i,\ldots,d-1$, and it is

$$(\underbrace{K,\ldots,K}_{d-i-1},\underbrace{L,\ldots,L}_{i})\text{-extreme}$$

for any smooth convex body L. An outer normal u to a polytope is i-extreme if it is an outer normal to a face of maximum dimension $d-i-1$. $K \in \mathcal{K}_d^d$ is said to be an *i-tangential body*, $i = 0, \ldots, d-1$, of $L \in \mathcal{K}_d^d$ if $L \subset K$ and every $(d-i-1)$-extreme support hyperplane to K also supports L.

The properties of tangential bodies have been investigated by Favard (1933), Bol (1943a,b), and Schneider (1978). Favard defines the tangential bodies by classifying the boundary points, and this is discussed by Schneider in this volume. The definitions and enumeration above are due to Schneider who shows that they are equivalent to those of Favard. The work of Bol is discussed in sections 6 and 7.

The tangential bodies are extremal bodies for certain inequalities, see Theorem 6.3 and Conjecture 7.7. For illustrations let L be a ball. The $(d-1)$-tangential polytopes of a ball are polytopes whose facets are tangential to the ball. The *cap bodies ("Kappenkörper")* are the convex hull of a ball and countably many points exterior to it such that the line segment joining any two of these points intersects the ball; these are the 1-tangential bodies. The 0-tangential body is the ball itself. For another example, K is a $(d-2)$-tangential body of any convex set L, if all $(K, \ldots, K, B)$-extreme support hyperplanes of K also support L.

5. Symmetrization

Many geometric inequalities have been established by modifying the body in one or a series of ways, and in most cases this results in a body with increased symmetry. Here we define various techniques, establish the properties of each, and cite some applications.

Steiner (1838, 1842) introduced the symmetrization defined below in order to establish the isoperimetric inequality, see Talenti in this volume.

Definition 5.1. Let $K \in \mathcal{K}^d$, and H a hyperplane in $\mathbb{E}^d$. For each line ℓ, ℓ orthogonal to H, and $\ell \cap K \neq \emptyset$, construct the line segment of ℓ symmetric about H with length $V_1(\ell \cap K)$. The union of these line segments is the *Steiner symmetral* of K with respect to H, denoted by $S(K, H)$.

Properties 5.2.

(a) $S(K, H) \in \mathcal{K}^d$.

(b) *Volume is unchanged.*

(c) *Diameter is not increased.*

(d) *If $\{K_\lambda\}$ is a concave family, $\{S(K_\lambda, H)\}$ is a concave family.*

(e) V_i, $1 \leqslant i \leqslant d-1$ *are not increased.*

To see that $S(K, H)$ is convex, first notice that the Steiner symmetral of any subset of K is a subset of the Steiner symmetral of K. Choose s' and t' in $S(K, H)$, and denote the end points of the orthogonal line segments to which they belong by s_1', s_2', and t_1', t_2'. The vertices of the isosceles trapezoid s_1', s_2', t_1', t_2' are the image of the vertices of a trapezoid in K, call them s_1, s_2, t_1, t_2. Since the convex hull of s_1, s_2, t_1, t_2 is in K, the convex hull of the isosceles trapezoid is in $S(K, H)$. Property

(c) is established by observing that the diagonal of trapezoid in K is not shorter than the diagonal of the isosceles trapezoid of $S(K,H)$. Recall from section 1 that a concave family $K(\lambda)$ in $\mathcal{K}^d$ may be arranged as the parallel d-dimensional slices of a $(d+1)$-dimensional body. Steiner symmetrization in $\mathbb{E}^{d+1}$ results in a convex body and thus the last two properties are established.

More detailed treatments, including further properties and proofs, may be found in Blaschke (1916a, sections 15, 19, 22, 25; 1930, section 114; 1923, sections 22, 24, 26, 72, 73), Burago and Zalgaller (1988, pp. 77–83), Eggleston (1969, pp. 90–96), and Leichtweiss (1980, section 18). The applications of Steiner symmetrizations are too numerous to summarize; they usually involve multiple symmetrizations and the following result. Choose $K \in \mathcal{K}_d^d$. If $\mathcal{S}$ is the set of convex bodies which result from a finite number of applications of Steiner symmetrization to K, then there exists a sequence in $\mathcal{S}$ which converges to a ball with volume $V(K)$. This result is valid for nonconvex sets, and is due to Gross (1917) and Lusternik (1935). Mani-Levitska (1986) has proven a probabilistic version of it. Bourgain, Lindenstrauss and Milman (1989) establish a stability result by obtaining estimates on the distance from the kth sequential Steiner symmetral of an arbitrary body to a ball. A forerunner of this result is due to Hadwiger (1952).

Verification of the properties of the next two techniques are left to the reader.

Definition 5.3. The *central symmetrization* $C(K)$ of a convex set $K \in \mathcal{K}^d$ is defined by

$$C(K) = \tfrac{1}{2}(K + (-K)).$$

Definition 5.4. The *Blaschke symmetrization* $B(K,H)$ of a convex set $K \in \mathcal{K}^d$, with respect to a hyperplane H in $\mathbb{E}^d$, is defined by

$$B(K,H) = \tfrac{1}{2}(K + K^H),$$

where K^H is the reflection of K in H.

Properties 5.5.

(a) $C(K)$, $B(K,H) \in \mathcal{K}^d$.
(b) V_1 *is unchanged.*
(c) *Diameter is constant for* $C(K)$, *and not increased for* $B(K,H)$.
(d) *If* $\{K_\lambda\}$ *is a concave family,* $\{C(K_\lambda)\}$, $\{B(K_\lambda,H)\}$ *are concave families.*
(e) V_i, $2 \leqslant i \leqslant d$ *are not decreased.*

Central symmetrization was used by Kubota (1923, 1925) to establish inequalities involving the quermassintegrals, diameter and minimal width, see section 8. Further properties of central symmetrization may be found in Bonnesen and Fenchel (1934, 1987, section 42), and Leichtweiss (1980, section 20). The second symmetrization was introduced by Blaschke (1916a, pp. 103–104), and its properties may also be found in Hadwiger (1957, pp. 259–262). Stability results similar to those mentioned above for Steiner symmetrization have been found for Blaschke symmetrization by Bourgain, Lindenstrauss and Milman (1988).

Steiner symmetrization can be generalized by constructing i-dimensional, $i = 1, \ldots, d-1$, balls orthogonal to a fixed $(d-i)$-dimensional flat. These symmetrizations are discussed by Dinghas (1961) and Wills (1969). For $i = 1$ the procedure is Steiner symmetrization. For $i = d-1$ a body of rotation is created, and it is considered next. Properties (a) to (e) of Steiner symmetrization are valid.

Definition 5.6. Let $K \in \mathcal{K}^d$, and ℓ a line in $\mathbb{E}^d$. For each hyperplane H, H orthogonal to ℓ, and $H \cap K \neq \emptyset$ construct the $(d-1)$-dimensional ball in H with center on ℓ whose $(d-1)$-dimensional volume is equal to $V_{d-1}(H \cap K)$. The union of these balls is the *Schwarz rounding* of K with respect to ℓ, denoted by $R(K, \ell)$.

Theorem 5.7 (Blaschke). *The Schwarz rounding of a convex body $K \in \mathcal{K}^d$ with respect to any line is a convex set.*

To verify Theorem 5.7 assume that the origin is on the intersection of the line and the interior of K. Choose hyperplanes $H_1, \ldots, H_{d-1}$ containing the line and inclined at angles which are irrational multiples of π. Let $K_0 = K$, and create the Steiner symmetrals, $\{K_i\}$, $i \geqslant 1$, where $K_i = S(K_{i-1}, H_{i \bmod(d-1)})$. From the properties of Steiner symmetrization and the Blaschke selection theorem it follows that the $\{K_i\}$ converge to a convex body. This body is a body of rotation about the line, and we can conclude that it is the Schwarz rounding.

Theorem 5.7 was first established by Blaschke (1916a, sections 21–22), and we shall use it to prove the Brunn–Minkowski Theorem 6.2 below. Rounding was introduced by Schwarz (1884), and is not restricted to convex sets. Further details may be found in Bandle (1980) (for the general case), Blaschke (1916a), Bonnesen and Fenchel (1934, 1987, section 41), and Leichtweiss (1980, section 19).

The next technique is included because of the important role it plays in establishing the characterization theorems, see Theorem 3.2.

Definition 5.8. Let $K \in \mathcal{K}^d$, and r a positive integer. For any collection of r rotations δ_i about a fixed point of $\mathbb{E}^d$, and any set of r positive integers λ_i, $\sum \lambda_i = 1$, $i = 1, \ldots, r$, let

$$A(K) = \sum_{i=1}^{r} \lambda_i \delta_i(K).$$

The body $A(K)$ is said to be the result of *rotation averaging* of K.

The properties of rotation averaging may be found in Hadwiger (1957, section 4.5.2–3), and Leichtweiss (1980, pp. 155–158). A theorem of Hadwiger (1955, p. 27), guarantees the existence of a sequence of rotation averagings of K which converge to a ball; the volume of the ball is greater than or equal to $V(K)$.

There are many other symmetrization techniques, and they are discussed by Bonnesen and Fenchel (1934, 1987, chapter 9), Burago and Zalgaller (1988, section 9.2), and Leichtweiss (1980, chapter 4). We note that Heil (1987) has recently used *stiffening*, and Gage (1990) has applied *annular symmetrization*.

6. Fundamental inequalities

Here we consider the inequalities due to Brunn, Minkowski, Aleksandrov and Fenchel. Every inequality in this section is a consequence of the Aleksandrov–Fenchel inequality 6.1, therefore we shall begin with it and then proceed chronologically. For each subsequent inequality we give an indication of an early proof and demonstrate the derivation from Theorem 6.1. The general equality condition in the Aleksandrov–Fenchel inequality, which remains a central unsolved problem in convexity, is discussed in the next section. The isoperimetric inequality is considered separately in this volume by Talenti.

Theorem 6.1 (Aleksandrov and Fenchel). *For* $K, L, K_3, \ldots, K_d \in \mathcal{K}^d$,

$$V(K, L, K_3, \ldots, K_d)^2 \geqslant V(K, K, K_3, \ldots, K_d)V(L, L, K_3, \ldots, K_d).$$

The proof is due to Aleksandrov (1937b, 1938); Fenchel (1936a) sketched an alternate proof, see the comments of Busemann (1958, p. 51), and Hadwiger (1957, p. 290). In Aleksandrov's first proof the inequality was established for polytopes; it is reproduced by Leichtweiss (1980). The second proof, reproduced by Busemann (1958) and by Leichtweiss in this volume, established the result for bodies with sufficiently smooth boundaries. In both cases the inequality is carried through for general convex sets by approximation. For this reason the equality condition is lost.

There has been a recent revival of interest in the inequality. Connections between convexity and algebraic geometry have led to proofs of the inequality by Khovanskij (1978) for $d = 2$, and Teissier (1979, 1980/81) for all d. (The proof due to Fedotov (1979) is in error.) This work is discussed by Ewald in this volume, Burago and Zalgaller (1988, section 27), and Oda (1988). Egorychev (1981) used a special case of the closely related Aleksandrov (1938) inequality for mixed discriminants of quadratic forms to prove the van der Waerden conjecture. Falikman (1981) independently derived a more special case of the inequality sufficient for his proof of the same conjecture. For a discussion see Lagarias (1982). Two combinatorial applications of these inequalities are due to Stanley (1981).

We now consider its predecessors.

Theorem 6.2 (Brunn and Minkowski). *If* $K(\lambda) = (1 - \lambda)K + \lambda L$ *for* $K, L \in \mathcal{K}^d$, *and* $0 \leqslant \lambda \leqslant 1$, *then*

$$V(K(\lambda))^{1/d} \geqslant (1 - \lambda)V(K)^{1/d} + \lambda V(L)^{1/d}.$$

Equality holds if and only if K and L lie in parallel hyperplanes or are homothetic.

The following proof is due to Blaschke (1916a, sections 21–22), and makes use of Schwarz rounding, see Definition 5.6. From the linear family $K(\lambda) \in \mathcal{K}^d$, $0 \leqslant \lambda \leqslant 1$, form the convex body in $\mathcal{K}^{d+1}$, as described in section 1, and call it K'. Choose a line ℓ in $\mathbb{E}^{d+1}$ which is orthogonal to the parallel hyperplanes in which the translated $K(\lambda)$ lie. The Schwarz rounding $R(K', \ell)$ is a convex body in $\mathbb{E}^{d+1}$. The nonempty

intersections of $R(K',\ell)$ with hyperplanes perpendicular to ℓ are d-dimensional balls with radius $V(K(\lambda))^{1/d}$, $0 \leqslant \lambda \leqslant 1$, and the inequality follows. The first equality condition guarantees that the volume of the $K(\lambda)$, $0 \leqslant \lambda \leqslant 1$, are zero; verification of the second equality condition is left to the reader.

A simple statement of the theorem is that the dth root of the volume of a linear family is a concave function. Bonnesen (1932) first employed this property of volume to show that the Minkowski inequalities, see Theorem 6.3 below, are a consequence of the Brunn–Minkowski theorem. Aleksandrov (1937b, section 4), reversed the argument to derive the Brunn–Minkowski theorem. Let

$$f(\lambda) = V(K(\lambda))^{1/d} - (1-\lambda)V(K)^{1/d} - \lambda V(L)^{1/d},$$

where $K(\lambda) = (1-\lambda)K + \lambda L$, $0 \leqslant \lambda \leqslant 1$. We wish to show that the second derivative f'' is nonpositive, and it suffices to show this at $\lambda = 0$. To verify the claim observe that $K(\lambda)$ is the zeroth body in the linear family from $K(\lambda)$ to L. Using Property 2.4 we have

$$f''(0) = (d-1)V(K)^{(2d-1)/d}(V(L)V(K,K,L,\ldots,L) - V(K,L,\ldots,L)^2).$$

Brunn (1887, chapter III; 1889, chapter III), gave the first proof of the inequality in Theorem 6.2, and Minkowski (1910, sections 56–57), followed with an analytical proof. The original proof of the equality condition is due to Minkowski, see the later publication Minkowski (1910), and it was followed by Brunn (1894). For other proofs see Bonnesen and Fenchel (1934, 1987, section 48), and Leichtweiss (1980, section 21). The Brunn–Minkowski theorem is valid for compact Lebesgue measurable sets, not necessarily convex. This was first proved by Lusternik (1935) using symmetrization, and the proof for the equality cases were corrected by Henstock and Macbeath (1953). Alternate proofs and more general results are due to Ohmann (1955), Hadwiger and Ohmann (1956), and Knothe (1957); also see Burago and Zalgaller (1988, chapter 2), Dinghas (1961), and Hadwiger (1957, section 5.2).

Generalizations of the theorem are numerous, for example the proof of Blaschke sketched above is valid for concave families, and the proof of Aleksandrov is actually used to show that the ith root of the mixed volume

$$V(K(\lambda),\ldots,K(\lambda),K_{i+1},\ldots,K_d)$$

for a linear family $K(\lambda)$, $K_{i+1},\ldots,K_d \in \mathcal{K}^d$ and $i = 1,\ldots,d$ is a concave function. This was established independently by Fenchel (1936b). It is a consequence that the ith root of the ith intrinsic volume is concave. Generalizations are presented in Bonnesen and Fenchel (1934, 1987, section 50), Burago and Zalgaller (1988, section 20.4), and Leichtweiss (1980, section 21). The stability of this inequality is discussed by Groemer in this volume. There are also Brunn–Minkowski type theorems for many of the extensions of the quermassintegrals discussed by McMullen in this volume.

The earliest forms of (6.1) involved only two bodies.

Theorem 6.3 (Minkowski). *If* $K, L \in \mathcal{K}_d^d$, *then*

$$V(K,\ldots,K,L)^2 \geqslant V(K,\ldots,K,L,L)V(K),$$

where equality holds if and only if K *is a* $(d-2)$*-tangential body of a homothet of* L, *and*

$$V(K,\ldots,K,L)^d \geqslant V(L)V(K)^{d-1},$$

where equality holds if and only if K *and* L *are homothetic.*

The restriction to $\mathcal{K}_d^d$ is not required by the inequalities, but rather to simplify the statement of the equality conditions. As with the Brunn–Minkowski theorem, the equality conditions in $\mathcal{K}_d^d$ are unrelated to the equality conditions for lower dimensional bodies, and the description of such situations will be increasingly difficult as the inequalities become more general.

Each Minkowski inequality is valid for all K and L, therefore the bodies may be interchanged to obtain a "second" inequality. It is traditional to state both versions of each. As was mentioned earlier, Bonnesen (1932) showed that both are consequences of the Brunn–Minkowski theorem; the second Minkowski inequality is equivalent to the first derivative of f being nonnegative at $\lambda = 0$. The first Minkowski inequality is a special case of the Aleksandrov–Fenchel inequality Theorem 6.1, and the second follows from Theorem 6.1 by successive application for varying $(d-2)$-tuples consisting only of K and L. These inequalities and the equality condition for the second were established for $d = 3$ by Minkowski (1903, sections 6–7), and the second was generalized for all d by Süss (1931). The equality condition for the first was conjectured by Minkowski, and verified by Bol (1943a) using the method of inner parallel bodies.

Before proceeding we shall investigate these inequalities when either K or L is a ball, and $d = 2$ and 3. In the plane there is exactly one inequality: The perimeter squared is greater than or equal to 4π times area, and in $\mathcal{K}_2^2$ equality holds for balls only. This is the isoperimetric inequality.

For $d = 3$ and $K \in \mathcal{K}_3^3$ let V be volume, S surface area and M the total mean curvature (or for bodies not sufficiently smooth, M is proportional to the mean width with factor 2π). There are four inequalities:

(a) $S^2 \geqslant 3MV$. Equality holds for cap bodies and this inequality is not valid in general for nonconvex sets. For example, it is not valid for smooth sets lying between the cap body which is the convex hull of B and one point exterior to it, and a "spiked sphere" which is the nonconvex union of B and the line segment from o to the same point.

(b) $M^2 \geqslant 4\pi S$. Equality holds only for balls, and this inequality is also not valid in general for nonconvex sets. For example, given a family of tori with fixed outer radius, the inequality becomes invalid as the radius of the revolved disc increases.

(c) $S^3 \geqslant 36\pi V$. This is the 3-dimensional isoperimetric or isepiphanic inequality, see Bonnesen (1929). It is valid in general, and equality holds for the multi-spiked spheres. For convex sets it is a consequence of (a) followed by (b).

(d) $M^3 \geqslant 48\pi^2 V$. This inequality is valid in general, equality holds only for balls. For convex sets it is a consequence of (b) followed by (a).

Blaschke (1916b) observed that there was a missing inequality for $d = 3$: minimize V when S and M are given. The so-called missing boundary of the *Blaschke diagram* and its higher dimensional analogues remain undiscovered. For a discussion of this problem see Hadwiger (1955, sections 28–29), Bieri (1948/49), and Sangwine-Yager (1989). Partial results are due to Groemer (1965) for $d = 3$, and extended for all d by Firey (1964).

To continue the consideration of two body inequalities we have

$$r \leqslant_1 \frac{V(K)}{V(K,\ldots,K,L)} \leqslant_2 \frac{V(K,\ldots,K,L)}{V(K,\ldots,K,L,L)} \leqslant_3 \cdots \leqslant_d \frac{V(K,L,\ldots,L)}{V(L)}, \tag{6.4}$$

where r is the inradius of K with respect to L, see section 1. The first inequality follows from (2.8), the second and the last are the first inequality of Minkowski, and the remainder are special cases of Theorem 6.1. The chain of inequalities was conjectured by Favard (1933, p. 275). For the quermassintegrals we have

$$W_i^2 \geqslant W_{i-1} W_{i+1}, \quad i = 1,\ldots,d-1. \tag{6.5}$$

See the next section, in particular Conjectures 7.7 and 7.8, for a discussion of equality.

Minkowski (1903, section 7), obtained Theorem 6.1 for $d = 3$ and three distinct convex sets. The first higher dimensional special case of Theorem 6.1 is due to Kubota (1925) where $K_3 = \cdots = K_d = B$. Kubota was able to show that for $K, L \in \mathcal{K}_d^d$ equality holds if and only if K and L are homothetic.

The last special case of the Aleksandrov–Fenchel inequality (Theorem 6.1) which we shall present is a generalization of the second Minkowski inequality (6.3). If $i = 2,\ldots,d$ and $\mathcal{C} = K_{i+1},\ldots,K_d \in \mathcal{K}^d$, then

$$V(K_1,\ldots,K_i,\mathcal{C})^i \geqslant \prod_{j=1}^{i} V(K_j,\ldots,K_j,\mathcal{C}).$$

The inequality follows from multiple applications of Theorem 6.1. The higher power inequalities involving only the quermassintegrals have been investigated by Gritzmann (1988).

7. Equality in the Aleksandrov–Fenchel inequality

In preparation for considering the equality conditions in the Aleksandrov–Fenchel inequality we shall first investigate further properties of the mixed surface area measures. Consideration of the equality conditions for a body and a ball will precede a general conjecture for $\mathcal{K}_d^d$, and we conclude with a discussion of lower dimensional sets. The following equivalence lemma is useful.

Lemma 7.1. *Let K, L and $\mathscr{C} = K_3, \ldots, K_d \in \mathscr{K}_d^d$. The following are equivalent:*
(a) *equality in the Aleksandrov–Fenchel inequality (Theorem 6.1),*
(b) *equality in*

$$\frac{V(K,K,\mathscr{C})}{V(M,K,\mathscr{C})^2} - 2\frac{V(K,L,\mathscr{C})}{V(M,K,\mathscr{C})V(M,L,\mathscr{C})} + \frac{V(L,L,\mathscr{C})}{V(M,L,\mathscr{C})^2} \leqslant 0,$$

for one $M \in \mathscr{K}_d^d$,
(c) *$V(M,K,\mathscr{C})$ and $V(M,L,\mathscr{C})$ are proportional for all $M \in \mathscr{K}^d$,*
(d) *$S(K,\mathscr{C},\cdot)$ and $S(L,\mathscr{C},\cdot)$ are proportional.*

This lemma dates to the inequality itself. Parts are due to Favard (1933), Fenchel (1936) and Aleksandrov (1937); also see Fenchel and Jessen (1938) and Leichtweiss (1980, section 24). (The statement of (c) is corrected from Schneider 1985.) Parts (b) and (c) follow by applying Theorem 6.1 to the sums of convex bodies. In the more recent results which follow, the support of a measure μ is denoted by $\operatorname{supp}\mu$.

Lemma 7.2. *If $L \in \mathscr{K}_d^d$ is smooth and strictly convex, and $\mathscr{C} = K_1, \ldots, K_{d-2} \in \mathscr{K}_d^d$, then*

$$\operatorname{supp} S(K,\mathscr{C},\cdot) \subset \operatorname{supp} S(L,\mathscr{C},\cdot)$$

for all $K \in \mathscr{K}_d^d$.

Lemma 7.3. *Let $K, L, \mathscr{C} = K_1, \ldots, K_{d-2} \in \mathscr{K}_d^d$. If there exists a homothet $\lambda L + x$ of L such that*

$$h(K,u) = h(\lambda L + x, u) \quad \text{for all } u \in \operatorname{supp} S(B,\mathscr{C},\cdot),$$

then $S(K,\mathscr{C},\cdot) = \lambda S(L,\mathscr{C},\cdot)$.

Lemma 7.2 is proved by contradiction. For the second lemma, choose $M \in \mathscr{K}^d$, then $\operatorname{supp} S(M,\mathscr{C},\cdot) \subset \operatorname{supp} S(B,\mathscr{C},\cdot)$, and

$$\begin{aligned}
&\int_{S^{d-1}} h(M,u)\, \mathrm{d}[S(K,\mathscr{C},u) - \lambda S(L,\mathscr{C},u)] \\
&= dV(M,K,\mathscr{C}) - d\lambda V(M,L,\mathscr{C}) \\
&= dV(M,K,\mathscr{C}) - dV(M,\lambda L + x,\mathscr{C}) \\
&= \int_{S^{d-1}} [h(K,u) - h(\lambda L + x,u)]\, \mathrm{d}S(M,\mathscr{C},u).
\end{aligned}$$

These lemmas and the proof sketched above are due to Schneider (1985).

Conjecture 7.4. The converse of Lemma 7.3 is true.

The conjecture is due to Loritz (1979) for $L = B$. Schneider (1990) has verified it when the bodies of $\mathscr{C}$ are smooth. This is a significant improvement of the earlier result of Aleksandrov (1938) which required that the convex bodies of $\mathscr{C}$ have analytic support functions.

Conjecture 7.5. If $\mathscr{C} = K_1, \ldots, K_{d-1} \in \mathscr{K}_d^d$, then $\operatorname{supp} S(\mathscr{C}, \cdot)$ is the closure of the set of $\mathscr{C}$-extreme directions.

This conjecture is due to Schneider (1985), where it is shown to be valid when the bodies of $\mathscr{C}$ are polytopes, and when the first i, $i = 1, \ldots, d-1$, bodies of $\mathscr{C}$ are identical and the last $d - i - 1$ are smooth and strictly convex. Therefore it is known for the area functions, $S_i(\cdot, \cdot)$. Partial results are obtained for zonoids by Schneider (1988).

We now consider equality in (6.5). The connection between the area functions and tangential bodies is described in the following lemma.

Lemma 7.6. *Let $K \in \mathscr{K}_d^d$, and let r be the inradius of K with respect to B. For any i, $i = 1, \ldots, d$, the following are equivalent:*

(a) $W_{i-1}(K) = rW_i(K)$,

(b) $V(K) = rW_1(K) = \cdots = r^i W_i(K)$,

(c) $i > 1$, $S_{d-1}(K, \cdot) = r^{i-1} S_{d-i}(K, \cdot)$,

(d) $i > 1$, $S_{d-1}(K, \cdot) = rS_{d-2}(K, \cdot) = \cdots = r^{i-1} S_{d-i}(K, \cdot)$,

(e) *K is a $(d-i)$-tangential body of a ball of radius r.*

Favard (1933) established the equivalence of (a), (b) and (e), and the remainder of the lemma is due to Schneider (1978). It is a consequence of the lemma that the converse of the following conjecture is true; the conjecture is also due to Favard (1933, p. 275). (The case $d = 3$ and $i = 1$ was first posed by Minkowski.)

Conjecture 7.7. Let $K \in \mathscr{K}_d^d$. If equality holds in (6.5), $i = 1, \ldots, d-1$, then K is a $(d - i - 1)$-tangential body of a ball.

The conjecture is known for:

(a) $i = 1$, $d - 1$, Bol (1943a). The method of *inner parallel bodies* employed by Bol is limited to these values of i because the derivative of the volume of the inner parallel sets satisfies a property analogous to Property 2.6, but its generalization, analogous to (3.3), is not valid (only an inequality holds). Also see Knothe (1949). (It is often said in the literature that equality holds when $i = 1$ for cap bodies, but this is true only for $d = 3$.)

(b) All i, K centrally symmetric, Schneider (1985). The result is established by extending a result of Favard which relates the equality of certain mixed surfaced area measures and equality in Theorem 6.1 for projections.

Now consider (6.4), where neither body is a ball. It follows from a result of Favard (1933), analogous to Lemma 7.6 (b) and (e), that equality holds in the ith inequality sign in (6.4), $i = 1, \ldots, d$, if either K is a $(d-i)$-tangential body of a homothet of L or L is an $(i-2)$-tangential body, $i \neq 1$, of a homothet of K, and

he conjectured that the converse was also true. Recently Bol's (1943a) counter-conjecture that for $d > 3$ and $i = 3, \ldots, d-1$ equality might hold for a larger class of bodies has been confirmed by Schneider, private communication. Let K be a 4-dimensional cap body formed by the convex hull of B and a single point exterior to it, and let L be the reflection of K through o. Clearly neither K nor L is a 2-tangential body of any homothet of the other, and the mixed volumes $V(K,K,K,L)$ and $V(K,L,L,L)$ are equal. With some work one can also show that $V(K,K,L,L) = V(K,K,K,L)$. The symmetry simplifies the computation; the example does not depend on it.

We now have the general conjecture for $\mathcal{K}_d^d$. It is due to Schneider (1985), and it is consistent with all of the preceding conjectures and the 4-dimensional example above.

Conjecture 7.8. If K, L, $\mathscr{C} = K_3, \ldots, K_d \in \mathcal{K}_d^d$, then equality holds in the Aleksandrov–Fenchel inequality 6.1 if and only if suitable homothets of K and L have the same $(B,\mathscr{C})$-extreme support hyperplanes.

The cases where equality is known in general are:

(a) If the convex bodies of $\mathscr{C}$ are strongly combinatorially isomorphic simple polytopes, and the facets of K and L are parallel to the facets of the polytopes of $\mathscr{C}$, then equality holds if and only if K and L are homothetic. Polytopes which are strongly combinatorially isomorphic share the same sets of i-extreme directions for all i, and the $(B,\mathscr{C})$-extreme directions are those orthogonal to the facets. This result is due to Aleksandrov (1937).

(b) If the bodies of $\mathscr{C}$ are all K we have the first Minkowski inequality, Theorem 6.3. Conjecture 7.8 is equivalent to the statement that K is a $(d-2)$-tangential body of a homothet of L. The proof is due to Bol (1943a).

(c) Assume equality holds for K, L and $\mathscr{C}$ fixed. Schneider (1990) shows that equality must also hold for K, L and $\mathscr{C}'$, where each K_i' of $\mathscr{C}'$ is adapted to K_i of $\mathscr{C}$, $i = 3, \ldots, d$. A convex body K' is *adapted* to K if for all boundary points $k \in K$ there is a boundary point $k' \in K'$ such that the cone of normal vectors to K at k, $N(K,k)$, is contained in $N(K',k')$. The proof relies on a representation for the derivative of the support function of the *inner* parallel bodies analogous to Property 2.6. The equality condition generalizes an earlier result, Schneider (1985), where summand replaced adapted.

(d) If the convex bodies of $\mathscr{C}$ are smooth, then equality holds if and only if K and L are homothetic (no restrictions on K and L). In this case all directions are $(B,\mathscr{C})$-extreme. This result is due to Schneider (1990). It follows from (c) and the earlier result of Minkowski (1903), $d = 3$, and Kubota (1925), all d, where the bodies of $\mathscr{C}$ are all balls.

(e) If the convex bodies of $\mathscr{C}$ are zonoids, Schneider (1988) shows that Conjecture 7.8 is true for K and L centrally symmetric.

Now we consider the equality conditions in Theorem 6.1 for lower dimensional bodies. For $d = 2$ equality holds in three cases: K and L have nonempty interior and are homothetic, either K or L consists of a single point, and K and L are

parallel line segments. Thus for lower dimensional bodies equality holds if and only if both sides of the inequality in Theorem 6.1 are zero. Ewald (1988) shows, with the following example, that the situation in higher dimensions is more complicated. In $\mathbb{E}^3$ let K, L and M be unit squares in each of the coordinate planes. The Minkowski sum of any two distinct squares is a rectangular parallelepiped, and the mixed volumes are easily computed. Equality holds in Theorem 6.1, $K_3 = M$, but all of the mixed volumes are nonzero, and Conjecture 7.8 is not satisfied. Therefore the restriction to $\mathcal{K}_d^d$ in Conjecture 7.8 is necessary, and we can assume that lower dimensional equality conditions will be complicated.

8. Additional inequalities

It is not possible within this article to present a thorough review of all inequalities involving mixed volumes. Many are considered in succeeding articles, and some discussions begun here are concluded elsewhere in this volume. There is an extensive consideration of inequalities in Burago and Zalgaller (1988, chapter 4).

We begin with *Bonnesen's inequality* in the plane because it sharpens the Aleksandrov–Fenchel inequality 6.1 or isoperimetric inequality. If K, $L \in \mathcal{K}^2$, and r and R are the relative inradius and circumradius of K with respect to L, see section 1, then

$$0 \geqslant V(K) - 2\lambda V(K, L) + \lambda^2 V(L), \quad r \leqslant \lambda \leqslant R.$$

Equality holds at $\lambda = r$ if and only if rL is a summand of K, and at $\lambda = R$ if and only if K and L are homothetic. Bonnesen (1929) proved the result for $L = B$, and the general result is due to Blaschke (1949), also see Flanders (1968). It is an immediate consequence of the inequality that

$$V(K, L)^2 - V(K)V(L) \geqslant (\tfrac{1}{2}V(L)(R - r))^2.$$

There is no known generalization of Bonnesen's inequality to higher dimensions which preserves the dual properties of sharpening the corresponding quadratic inequality and being valid on the interval $r \leqslant \lambda \leqslant R$. In fact, many extensions to higher dimensions only sharpen weaker inequalities, for example

$$\left(\frac{S}{\kappa_d}\right)^{d/(d-1)} - \frac{V}{\kappa_d} \geqslant \left(\left(\frac{S}{\kappa_d}\right)^{1/(d-1)} - r\right)^d,$$

where V is volume, S is surface area, and r is the inradius with respect to B. This inequality is due to Hadwiger (1948) and Dinghas (1948). A Bonnesen-style inequality which does sharpen (6.5) for $i = d - 1$ is

$$0 \geqslant W_{d-2}(K) - 2\lambda(u)W_{d-1}(K) + \lambda(u)^2\kappa_d,$$

where $\lambda(u)$ is one-half of the mean width of K^u, $u \in S^{d-1}$. It is due to Benson (1970) for $d = 2$, and in general to Chakerian (1971). For further discussion of the

role Bonnesen-style inequalities play in the isoperimetric inequality and stability theory see Talenti and Groemer, respectively, in this Handbook.

A family of Bonnesen-style inequalities which provide sharp bounds for the inradius and circumradius with respect to the unit ball include

$$0 \geqslant V - rS + (d-1)r^d \kappa_d,$$

and

$$0 \leqslant (d-1)V - 2RS + (d+1)R^d \kappa_d.$$

The inradius inequality was conjectured by Wills (1970) and proved by Bokowski (1973) and Diskant (1973); sharper versions are due to Osserman (1979) and Sangwine-Yager (1988). The circumradius inequality for $d = 2$ is due to Favard (1929), and the inequality above is part of a family of inequalities due to Bokowski and Heil (1986). Equality holds in both only for balls.

We state the following conjecture.

Conjecture 8.1. If $a_1 \leqslant \cdots \leqslant a_d$ are the real parts of the roots of

$$\sum_{i=0}^{d} \binom{d}{i} (-\lambda)^i W_i(K),$$

$K \in \mathcal{K}^d$ and $\lambda \in \mathbb{R}$, then $0 < a_1 \leqslant r \leqslant R \leqslant a_d$, where r and R are the radii of K relative to B.

The algebraic geometry equivalent of this conjecture is due to Teissier (1982), also see Oda (1988). The reader may confirm that the "alternating" Steiner polynomials of cap bodies, $d = 3$, do have complex roots which satisfy the conjecture. While no progress has been made on it to date some results have been obtained by Wills (1990) for a closely related polynomial.

The Aleksandrov–Fenchel inequality 6.1 may be used to derive inequalities involving the quermassintegrals and D and w, see section 1 for definitions. Here we consider only the diameter and refer the reader to other articles in this volume, and Bonnesen and Fenchel (1934, 1987, chapters 10 and 12). The *Bieberbach inequality* (Bieberbach 1915) states that of all convex bodies of fixed volume, the ball has the smallest diameter:

$$\left(\frac{D}{2}\right)^d \geqslant \frac{V}{\kappa_d}.$$

The inequality follows from the *Urysohn inequality* (Urysohn 1924), which is the second Minkowski inequality 6.3 with $K = B$. The *Kubota formula* (Kubota 1925) is

$$\left(\frac{D}{2}\right)^{d-1} \geqslant \frac{S}{d\kappa_d}.$$

This inequality similarly follows from repeated applications of the Aleksandrov–Fenchel inequality. Equality holds in both if and only if K is a ball, except for the Kubota formula, $d = 2$, when equality holds for sets of constant width. An upper bound on the diameter is

$$\frac{S^{d-1}}{(dV)^{d-2}} \geqslant \kappa_{d-1} D.$$

This inequality was conjectured by Rieger (1985) and proved by Gritzmann, Wills and Wrase (1987). It has been generalized to other quermassintegrals by Gritzmann (1988).

There are numerous inequalities involving the sections or projections of a convex body and they are discussed elsewhere in this volume. One such result is the Pythagorean inequality for mixed volumes of Firey (1960) which generalizes a result of Blaschke (1916a, p. 148). Choose $K_1, \ldots, K_{d-1} \in \mathcal{K}^d$, $u \in S^{d-1}$, and let $u_1, \ldots, u_d \in S^{d-1}$ be any d mutually perpendicular directions. The line segment from o to $u \in S^{d-1}$ is denoted by U, and

$$V(K_1, \ldots, K_{d-1}, U)^2 \leqslant \sum_{i=1}^{d} V(K_1, \ldots, K_{d-1}, U_i)^2.$$

The following result of Weil (1974) "reverses" monotonicity, Property 2.8. For $K, L \in \mathcal{K}^d$, L is contained in a translate of K if and only if $V(L, \mathscr{C}) \leqslant V(K, \mathscr{C})$ for all $(d-1)$-tuples $\mathscr{C}$ of $\mathcal{K}^d$.

We conclude with an open problem due to Betke and Weil (1991). Let $k_1, \ldots, k_r$ be positive integers such that $k_1 + \cdots + k_r = d$, and let $K_1, \ldots, K_r \in \mathcal{K}^d$. Find the positive constant $c(k_1, \ldots, k_r)$ such that

$$V(\underbrace{K_1, \ldots, K_1}_{k_1}, \ldots, \underbrace{K_r, \ldots, K_r}_{k_r}) \leqslant c(k_1, \ldots, k_r) V_{k_1}(K_1) \cdots V_{k_r}(K_r).$$

For $d = 2$, Betke and Weil show that $c(1,1) = \frac{1}{8}$, and that equality holds if and only if the two convex sets are orthogonal line segments. They conjecture that in the general case equality will hold when the K_i are k_i-dimensional and orthogonal.

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CHAPTER 1.3

The Standard Isoperimetric Theorem

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Contents

1. Introduction 75
   1.1. General remarks 75
   1.2. Archetypes 76
   1.3. Remarks about Dido's problem 77
2. The isoperimetric theorem for the Euclidean plane 79
   2.1. Statement 79
   2.2. Proof 79
   2.3. Remarks 84
3. The isoperimetric theorem for Euclidean n-dimensional space 84
   3.1. Statement 84
   3.2. Organization of the proof 86
   3.3. Beginning the proof 86
   3.4. Minimizing sequences 87
   3.5. Caccioppoli sets and their perimeters 90
   3.6. The isoperimetric theorem in one dimension 102
   3.7. Existence of a minimizer 105
   3.8. Steiner symmetrization 106
   3.9. Identification of minimizers 118
   3.10. Conclusions 119
References 120

HANDBOOK OF CONVEX GEOMETRY
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A highly informative prototypal theorem says that among all subsets of Euclidean n-dimensional space $\mathbb{R}^n$ having a prescribed measure the balls take the smallest perimeter. The present chapter is devoted to a thorough treatment of this theorem. The proof offered here stems from direct methods of the calculus of variations, and consists of two main steps. (i) A set which renders the perimeter a minimum does actually exist in the class of all subsets of $\mathbb{R}^n$ that have a prescribed measure and lie within a given large ball. (ii) Any relevant minimizer is virtually a ball. Step (i) – existence of minimizers – is a matter of building up an appropriate perimeter, i.e., a set function which continues the elementary perimeter – the one fitting polyhedra and nice subsets of $\mathbb{R}^n$ – moreover makes semicontinuity and compactness arguments work. Step (ii) – identification of minimizers – involves Steiner symmetrizations. The theory of $BV(\mathbb{R}^n)$, a space of functions whose derivatives are measures, and the parallel theory of Caccioppoli sets, the subsets of $\mathbb{R}^n$ whose characteristic functions are in $BV(\mathbb{R}^n)$, provide the appropriate framework.

For the sake of completeness, an alternative proof, due to Hurwitz and Lebesgue, is given in case the dimension n is 2.

1. Introduction

1.1. General remarks

According to usage, *isoperimetric theorem* is a nickname for propositions about extreme values – typically, minima or maxima subject to side conditions – of set functions, involving such quantities as area or volume, perimeter, diameter, eigenvalues, etc., that have a special geometric or physical significance. The following are examples – scattered examples – of isoperimetric theorems.

(1) Among all polygons, that have a prescribed number of sides and are inscribed in a circle, the regular polygons are the ones that maximize area and perimeter.

(2) Among all polygons, having a prescribed number of sides and a prescribed area, the regular polygons are precisely the ones which render the perimeter a minimum.

(3) Let K be a plane compact convex set – containing the origin – and let

$$K' = \{ y \in \mathbb{R}^2 : \langle x, y \rangle \leq 1 \text{ for all } x \text{ from } K \} ,$$

a polar set of K – $\langle \, , \rangle$ denotes the inner product between vectors throughout. Then

$$\tfrac{27}{4} \leq (\text{area } K) \cdot (\text{area } K') \leq \pi^2 .$$

The extreme values are achieved if K is a triangle and an ellipse, respectively.

(4) Let G be any bounded open subset of Euclidean n-dimensional space $\mathbb{R}^n$ and let $m(G)$, the Lebesgue measure of G, be fixed. Then the bass note of G (i.e., the least eigenvalue of the following boundary value problem: $\Delta u + \lambda u = 0$ in G, $u = 0$ on the boundary of G) is a minimum if G is a ball.

Theorem (1) is an elementary exercise – a wealth of related questions is in Kazarinoff (1961) and Mitrinović, Pecarić and Volenec (1989). A proof of Theorem (2) – perhaps the predominant isoperimetric theorem for polygons – is in Chisini (1927). Note that Theorem (2) amounts to an *isoperimetric inequality*, i.e., area A and perimeter P of a polygon obey

$$P^2 \geq 4n \cdot \tan(\pi/n) \cdot A ,$$

and equality holds if and only if the polygon is regular – here n = number of sides. Theorem (3) follows from results by Mahler (1939a,b) and Santaló (1949) – see also Lekkerkerker (1969), Santaló (1976). Interestingly enough, Theorem (3) is related to estimates of solutions to elliptic partial differential equations, see Aleksandrov (1965). Theorem (4) is a generalization of a famous result by Faber (1923) and Krahn (1925), a typical example of the so-called isoperimetric theorems of mathematical physics. We refer to Pòlya and Szegö (1951), Payne (1967), Bandle (1980), Mossino (1984), Hersch (1988), and Payne (1991) for more information on such a category of isoperimetric theorems. Accounts of isoperimetric theorems are in Hadwiger (1957), Berger, Gauduchon and Mazet (1971), Berger (1973), Osserman (1978), Osserman (1979), Osserman and Weinstein (1980), Berard and Meyer (1982), Burago and Zalgaller (1988), Gallot (1986) and in other chapters of this Handbook.

1.2. Archetypes

In this chapter we focus our attention on two archetypes: the isoperimetric theorem for the Euclidean plane and the isoperimetric theorem for Euclidean n-dimensional space $\mathbb{R}^n$. The isoperimetric theorem for the Euclidean plane says that a plane curve, which encloses a prescribed area and has the shortest length, is a circle; alternatively, a plane curve, which has a prescribed length and encloses the largest area, is a circle. Intriguing from time immemorial – Archimedes is reported to have been familiar with it, as Zenodorus was and Kepler, Galilei, James Bernoulli were later – the isoperimetric theorem for the Euclidean plane was considered by Euler and intensively worked out during the last century by such people as Sturm, Schwarz, Steiner, Weierstrass. A proof is ascribed to Edler, who in 1882 was able to improve Steiner's arguments. Exhaustive proofs appeared during the early years of this century, contributors include Hurwitz (1901), Minkowski (1903), Lebesgue (1906), Carathéodory and Study (1910), Blaschke (1916), Bonnesen (1929), and Blaschke (1930). See Porter (1933) and Gericke (1982) for more detailed historical comments. Comprehensive treatments of the isoperimetric theorem for Euclidean spaces of dimension n greater than 2 came out subsequently, in parallel with developments in geometric measure theory and

calculus of variations, Tonelli (1915), Schmidt (1939), Radò (1947), Schmidt (1948), Schmidt (1949), and De Giorgi (1958).

Proofs of these theorems are the subject of the present chapter.

1.3. Remarks about Dido's problem

Let us prepare the way by making a comment on *Dido's problem*. The aim of this comment is to make the role of the calculus of variations immediately clear. Dido's problem consists in finding a curve which lies in a given half-plane, lays its terminals on the edge, has a prescribed length, and is such that the region, enclosed by curve and edge, has the largest area. The solution is a semicircle – Queen Dido is reported to have been aware of it and to have established the early boundaries of Carthage accordingly. Though pertaining the special area of *relative* isoperimetric theorems, cf. Federer and Fleming (1960), Maz'ja (1985), Burago and Zalgaller (1988), Dido's problem parallels the isoperimetric theorem, as a reflection of competing curves in the edge shows. However, in the case where the competing curves do not zigzag too much, i.e., are restricted to be *graphs*, Dido's problem can be recast as a straightforward problem of the calcalus of variations, then solved in an independent and easy way. An apropos theorem may sound as follows.

Theorem. *Let u be a real-valued function defined in an interval $[-a, a]$; assume some smoothness, e.g.,*

$$u \text{ is absolutely continuous in } [-a, a]\,, \tag{1}$$

and assume u vanishes at both end points, i.e.,

$$u(-a) = u(a) = 0\,. \tag{2}$$

Let

$$L = \int_{-a}^{a} \sqrt{1 + (u')^2}\, \mathrm{d}x\,,$$

the length of the graph of u; and

$$A = \int_{-a}^{a} u\, \mathrm{d}x\,.$$

Then

$$L^2 \geqslant 2\pi A\,; \tag{3}$$

equality holds in (3) *if and only if the graph of u is a semicircle, i.e.,*

$$u(x) = \sqrt{a^2 - x^2}\,. \tag{4}$$

Proof. We claim that under conditions (1) and (2)

$$\frac{1}{a}\int_{-a}^{a}\left(\sqrt{1+(u')^2}+\frac{x}{a}u'\right)\mathrm{d}x \geqslant \tfrac{1}{2}\pi\,, \tag{5}$$

and equality takes place if and only if u obeys (4). In other words, the functional, whose domain is the set of real-valued functions satisfying conditions (1) and (2) and whose value at a member u from such a set is the left-hand side of (5), achieves a minimum; $\frac{1}{2}\pi$ is such a minimum and (4) defines the minimizer – no other minimizer exists. We are faced by the problem of the calculus of variations we alluded to.

Note that an integration by parts and hypothesis (2) give

$$\int_{-a}^{a} x(-u')\,\mathrm{d}x = A\,.$$

Hence inequality (5) reads

$$-Aa^{-2}+La^{-1} \geqslant \tfrac{1}{2}\pi\,,$$

or

$$L^2-2\pi A \geqslant (L-\pi a)^2\,.$$

Inequality (3) follows. On the other hand, equality in (3) yields $L^2-2\pi A = (L-\pi a)^2$ necessarily, hence forces (5) to be an equality; the assertion about eq. (4) follows.

The problem in hand is an easy exercise of the calculus of variations. Indeed, the relevant integrand is given by

$$f(x,u') = \sqrt{1+(u')^2}+\frac{x}{a}u'\,,$$

therefore

$$\frac{\partial^2 f}{\partial (u')^2}(x,u') > 0\,,$$

thus $u' \to f(x,u')$ is *strictly convex*. Convexity is a decisive ingredient in most minimization problems; in the present situation we have

$$f(x,w') > f(x,u') + (w'-u')\frac{\partial f}{\partial u'}(x,u')$$

unless $w' = u'$. Select function u in such a way that

$$\frac{\partial f}{\partial u'}(x,u') = \text{constant}\,, \quad u(-a) = u(a) = 0\,. \tag{6}$$

Then necessarily

$$\int_{-a}^{a} f(x, w')\,\mathrm{d}x > \int_{-a}^{a} f(x, u')\,\mathrm{d}x$$

unless $w = u$. Here w denotes any real-valued function, absolutely continuous in $[-a, a]$ and vanishing at the end points. Thus u is the sought minimizer. Integration shows that (6) – a trivial boundary value problem – implies eq. (4); eq. (4) implies in turn

$$\int_{-a}^{a} f(x, u')\,\mathrm{d}x = \tfrac{1}{2}\pi\,.$$

This completes the proof. □

2. The isoperimetric theorem for the Euclidean plane

2.1. Statement

The theorem in question may be conveniently presented in the following form.

Theorem. *Let C be a Jordan (i.e., simple closed) curve in Euclidean plane* $\mathbb{R}^2$. *Assume C is smooth enough, e.g., absolutely continuous. Denote by L the* length *of C and denote by A the* area *of the region interior to C (i.e., the connected component of* $\mathbb{R}^2 \backslash C$, *which is bounded). Then*

(i) *the inequality*

$$L^2 \geqslant 4\pi A$$

holds;

(ii) *equality takes place in such inequality if and only if C is a circle.*

2.2. Proof

We present here a proof by Hurwitz (1901), Hurwitz (1902), and Lebesgue (1906). Despite the geometric character of the theme, this proof is completely analytic; it basically rests on the following two ingredients: (i) Gauss–Green formulas, allowing one to express area A of the region, interior to C, in terms of the geometry of C only. (ii) A variational characterization of eigenvalues and eigenfunctions of Laplace–Beltrami operator on the one-dimensional torus, i.e., the second-order linear differential operator $-\mathrm{d}^2/\mathrm{d}t^2$, acting on smooth 2π-periodic functions.

Step 1: By Gauss–Green formulas,

$$A = \tfrac{1}{2} \oint_C (-y + y_0)\,\mathrm{d}x + (x - x_0)\,\mathrm{d}y\,, \tag{1}$$

provided C is oriented counterclockwise. Here x_0 and y_0 are arbitrary constants; for a reason, that will be clear later, we let x_0 and y_0 be the coordinates of the *center of mass* of C, i.e.,

$$x_0 = \frac{1}{L}\int_C x\sqrt{dx^2 + dy^2}\,, \qquad y_0 = \frac{1}{L}\int_C y\sqrt{dx^2 + dy^2}\,. \tag{2}$$

Consider the parametric representation of C:

$$x = x(s)\,, \qquad y = y(s)\,, \tag{3a}$$

where parameter s is the *arclength*. Thus

$$0 \leqslant s \leqslant L \tag{3b}$$

and

$$(x'(s))^2 + (y'(s))^2 = 1\,. \tag{3c}$$

We have

$$x_0 = \frac{1}{L}\int_0^L x(s)\,ds\,, \qquad y_0 = \frac{1}{L}\int_0^L y(s)\,ds\,, \tag{4}$$

and

$$A = \tfrac{1}{2}\int_0^L [(-y(s) + y_0)x'(s) + (x(s) - x_0)y'(s)]\,ds\,. \tag{5}$$

Formula (5) and Cauchy–Schwarz inequality yield

$$A \leqslant \tfrac{1}{2}\left\{\int_0^L [(-y(s) + y_0)^2 + (x(s) - x_0)^2]\,ds\right\}^{1/2} \times \left\{\int_0^L [(x'(s))^2 + (y'(s))^2]\,ds\right\}^{1/2}. \tag{6}$$

By (5) and Cauchy–Schwarz inequality, equality takes place in (6) if and only if the vector, whose components are $-y(s) + y_0$ and $x(s) - x_0$, is parallel to the tangent vector $(x'(s), y'(s))$, i.e.,

$$(x(s) - x_0)x'(s) + (y(s) - y_0)y'(s) = 0\,; \tag{7a}$$

equivalently

$$(x(s) - x_0)^2 + (y(s) - y_0)^2 = \text{constant}\,, \tag{7b}$$

i.e., C is a circle.

Formulas (3c) and (6) imply

$$A \leq \frac{\sqrt{L}}{2} \left\{ \int_0^L (x(s) - x_0)^2 \, ds + \int_0^L (y(s) - y_0)^2 \, ds \right\}^{1/2}, \tag{8}$$

and equality holds in (8) if and only if C is a circle.

Step 2: Consider the following eigenvalue problem:

$$u''(t) + \lambda u(t) = 0 \quad \text{for } 0 < t < 2\pi, \tag{9a}$$

$$u(0) = u(2\pi) \quad \text{and} \quad u'(0) = u'(2\pi). \tag{9b}$$

Integrations show that the eigenvalues $\lambda_0, \lambda_1, \lambda_2, \ldots$ of (9) – all the eigenvalues – are given by

$$\lambda_k = k^2, \quad k = 0, 1, 2, \ldots. \tag{10a}$$

The first eigenvalue, 0, is simple and any associated eigenfunction is constant. Eigenvalues $\lambda_1, \lambda_2, \ldots$ have multiplicity two; the eigenspace corresponding to λ_k consists exactly of the trigonometric homogeneous polynomials of degree k, i.e., is the linear span of $\cos(kt)$ and $\sin(kt)$. Thus the set of eigenfunctions of (9) is

$$\cos(kt), \sin(kt), \quad k = 0, 1, 2, \ldots, \tag{10b}$$

precisely the trigonometric system.

On the other hand, the variational theory of Sturm–Liouville problems, see Courant and Hilbert (1953) for instance, says that the eigenvalues and the eigenfunctions of (9) are – nay, can be characterized as – minima and minimizers of

$$\int_0^{2\pi} (u')^2 \, dt \Big/ \int_0^{2\pi} u^2 \, dt, \tag{11}$$

a *Rayleigh quotient*. Specifically: (i) λ_k = minimum of Rayleigh quotient (11) under the following conditions: u is absolutely continuous and its derivative is square integrable, $u(0) = u(2\pi)$, u is orthogonal to the eigenspaces corresponding to $\lambda_0, \lambda_1, \ldots, \lambda_{k-1}$. (ii) Any relevant minimizer is in the eigenspace corresponding to λ_k. Here $k = 1, 2, 3, \ldots$.

In case $k = 1$ the above statements result in the following proposition, known as *Wirtinger inequality*.

(i) *The following inequality*

$$\int_0^{2\pi} u^2 \, dt \leq \int_0^{2\pi} (u')^2 \, dt \tag{12a}$$

holds provided u is absolutely continuous, u' is square integrable,

$$u(0) = u(2\pi)\,, \qquad \int_0^{2\pi} u\,\mathrm{d}t = 0\,; \tag{12b}$$

(ii) *Under such hypotheses, equality takes place in* (12a) *if and only if*

$$u(t) = a\cos(t) + b\sin(t) \tag{12c}$$

for some constants a and b.

An exhaustive proof of Wirtinger inequality consists in showing, via direct methods of the calculus of variations, that the Rayleigh quotient (11) achieves a minimum in an appropriate function space, and the minimum value and any relevant minimizer are an eigenvalue and an eigenfunction of problem (9). A shorter, though less deep, proof rests on the Parseval's theorem for Fourier series, and is sketched below for the sake of completeness.

Suppose u is absolutely continuous in $[0, 2\pi]$ and u' is square integrable in the same interval. By Parseval's theorem, i.e., the completeness of trigonometric system (10b),

$$\int_0^{2\pi} u^2\,\mathrm{d}t = 2\pi\sum_{-\infty}^{+\infty}|a_k|^2\,, \qquad \int_0^{2\pi}(u')^2\,\mathrm{d}t = 2\pi\sum_{-\infty}^{+\infty}|b_k|^2\,,$$

where

$$a_k = \frac{1}{2\pi}\int_0^{2\pi} u(t)\,\mathrm{e}^{-\mathrm{i}kt}\,\mathrm{d}t\,,$$

a Fourier coefficient of u, and

$$b_k = \frac{1}{2\pi}\int_0^{2\pi} u'(t)\,\mathrm{e}^{-\mathrm{i}kt}\,\mathrm{d}t\,,$$

a Fourier coefficient of the derivative (although functions involved here are real-valued, the complex form of Fourier coefficients is notationally convenient). An integration by parts gives

$$b_k = \mathrm{i}ka_k\,, \quad k = 0, \pm1, \pm2, \ldots,$$

thanks to the first hypothesis (12b). The second hypothesis (12b) yields $a_0 = 0$. Hence Rayleigh quotient (11) equals

$$\frac{(\Sigma_{-\infty}^{-1} + \Sigma_{+1}^{+\infty})k^2|a_k|^2}{(\Sigma_{-\infty}^{-1} + \Sigma_{+1}^{+\infty})|a_k|^2}\,.$$

In conclusion, the Rayleigh quotient $\geqslant 1$ and equals 1 if and only if $a_k = 0$ ($k = \pm 2, \pm 3, \pm 4, \ldots$). Wirtinger inequality is proved. □

Step 3: Wirtinger inequality is now applied for estimating the right-hand side of (8). Note that

$$x(0) = x(L)\,, \qquad y(0) = y(L)$$

since C is a closed curve; moreover,

$$\int_0^L (x(s) - x_0)\,\mathrm{d}s = 0\,, \qquad \int_0^L (y(s) - y_0)\,\mathrm{d}s = 0\,,$$

because of eqs. (4). Thus Wirtinger inequality and a trivial change of scale give

$$\int_0^L (x(s) - x_0)^2\,\mathrm{d}s \leqslant \left(\frac{L}{2\pi}\right)^2 \int_0^L (x'(s))^2\,\mathrm{d}s\,,$$

$$\int_0^L (y(s) - y_0)^2\,\mathrm{d}s \leqslant \left(\frac{L}{2\pi}\right)^2 \int_0^L (y'(s))^2\,\mathrm{d}s\,,$$

consequently

$$\int_0^L (x(s) - x_0)^2\,\mathrm{d}s + \int_0^L (y(s) - y_0)^2\,\mathrm{d}s \leqslant \frac{L^3}{(2\pi)^2}\,, \tag{13}$$

as (3c) is in force.

Inequalities (8) and (13) yield

$$A \leqslant L^2/4\pi\,, \tag{14}$$

the required isoperimetric inequality.

When does equality hold in (14)? Inequality (14) has been derived from two predecessors only: inequalities (8) and (13). Statements (12) about Wirtinger inequality tell us that equality takes place in (13) if and only if

$$\begin{aligned} x(s) &= x_0 + \frac{L}{2\pi}\cos\left(\frac{2\pi}{L}s + \varphi\right), \\ y(s) &= y_0 + \frac{L}{2\pi}\sin\left(\frac{2\pi}{L}s + \varphi\right) \end{aligned} \tag{15}$$

for some constant φ, i.e., C is a circle – in the derivation of (15) condition (3c) must be taken into account. On the other hand, equality takes place in (8) under the same conditions, as already shown. In conclusion, equality holds in isoperimetric inequality (14) if and only if C is a circle. □

2.3. Remarks

(i) The proof of Hurwitz and Lebesgue can easily be carried out in any dimension n and results in the following theorem, compare with Reilly (1977) and Chavel (1978).

Let G be a bounded open subset of Euclidean n-dimensional space $\mathbb{R}^n$ and let the boundary of G be a smooth $(n-1)$-dimensional manifold M. Then

$$\frac{L}{A} \geqslant \frac{n}{\sqrt{n-1}} \sqrt{\lambda(M)},$$

and equality holds if and only if G is a ball. Here A is the measure of G, L is the $(n-1)$-dimensional measure of M and $\lambda(M)$ is the first nonzero eigenvalue of the Laplace–Beltrami operator on M.

(ii) Gauss–Green formulas are a key to the proof by Hurwitz and Lebesgue. A proof of the isoperimetric theorem in $\mathbb{R}^n$, based on Gauss–Green formulas, is outlined in Berger and Gostiaux (1988).

3. The isoperimetric theorem for the Euclidean n-dimensional space

3.1. Statement

The isoperimetric theorem for Euclidean n-dimensional space $\mathbb{R}^n$ – the *standard isoperimetric theorem* – basically asserts that a subset G of $\mathbb{R}^n$ such that $m(G)$ is a prescribed constant and $p(G)$ is a minimum, is a ball. Here $n = 2, 3, \ldots$; m and p denote n-dimensional *measure* and *perimeter*, respectively. m is the customary measure, informing about content. p is a surrogate of the elementary perimeter, i.e., a set function whose value on any smooth subset G of $\mathbb{R}^n$ agrees with the standard $(n-1)$-dimensional measure of ∂G (henceforth a subset G of $\mathbb{R}^n$ is qualified *smooth* if G is open, ∂G is a smooth manifold of codimension one, and G lies on one side of ∂G only).

The above statement amounts to an inequality between measure m and perimeter p, the *isoperimetric inequality* for Euclidean space $\mathbb{R}^n$, a prototypal isoperimetric inequality. The latter basically says that every subset G of $\mathbb{R}^n$ of finite measure satisfies

$$p(G) \geqslant n\kappa_n^{1/n}[m(G)]^{1-1/n}, \tag{1}$$

and the balls are the only subset of $\mathbb{R}^n$ which render (1) an equality. As usual,

$$\kappa_n = \frac{\pi^{n/2}}{\Gamma(n/2+1)}, \tag{2}$$

the *measure of the unit n-dimensional ball*, so that the perimeter of any n-dimensional ball B equals $n\kappa_n^{1/n}[m(B)]^{1-1/n}$.

There are three basic ingredients of the isoperimetric theorem for Euclidean space $\mathbb{R}^n$: measure m, perimeter p, and the collection of competing sets. Various versions of the theorem are available, which differ in the specification of such ingredients: Minkowski measures might be involved, as in Osserman (1978) or Federer (1969); on the other hand, the competition might be restricted to convex sets, as in Eggleston (1958). In this paper we let

$$m = \text{the standard Lebesgue measure}, \tag{3}$$

and

$$p = \text{perimeter à la De Giorgi}. \tag{4}$$

Accordingly, we let the competing sets be all the Lebesgue measurable – measurable, in short – subset of $\mathbb{R}^n$.

Recall the following properties, see section 3.4 for details.

(i) p is defined by:

$$p(G) = \sup\left\{\left|\int_G (\operatorname{div} f)\,dx\right| : f \text{ is a smooth compactly supported vector field, } \max|f| \leq 1\right\} \tag{5}$$

for every measurable subset G of $\mathbb{R}^n$. Here $|\cdot|$ denotes both absolute value of numbers and length of vectors. This notation will be used consistently throughout.

(ii) Suppose G has a finite measure. Then $p(G)$ is finite if and only if G is a *Caccioppoli set*, i.e.,

$$m((G+h)\triangle G) = O(|h|) \tag{6}$$

for any vector h in $\mathbb{R}^n$. Here $G + h = \{x \in \mathbb{R}^n : x - h \in G\}$, a translation of G, and $\triangle$ stands for *symmetric difference*. If G is a Caccioppoli set,

$$p(G) = \inf\{\liminf p(E_k) : \{E_k\} \text{ is a sequence of smooth subsets of } \mathbb{R}^n \text{ such that } m(G\triangle E_k) \to 0\}.$$

(iii) The following inequality

$$p(G) \leq H_{n-1}(\partial G) \tag{7}$$

holds for any measurable subset G of $\mathbb{R}^n$. Throughout this paper H stands for *Hausdorff measures* in $\mathbb{R}^n$.

Theorem. *Let G be any measurable subset of $\mathbb{R}^n$ such that $m(G)$ is finite. Then*
(i) *inequality* (1) *holds;*
(ii) *equality takes place in* (1) *if and only if G either is a ball or is equivalent to a ball, i.e., a ball B exists such that $m(G \triangle B) = 0$.*

The above form of the isoperimetric theorem appears in De Giorgi (1958).

Corollary. *Let G be a measurable subset of $\mathbb{R}^n$ such that $H_n(G)$ is finite. Then*

$$H_{n-1}(\partial G) \geq n\kappa_n^{1/n}[H_n(G)]^{1-1/n} . \tag{8}$$

Proof. Notice inequality (7) and that H_n, the n-dimensional Hausdorff measure in $\mathbb{R}^n$, coincides with the outer Lebesgue measure. □

We refer to Federer (1969) and Rogers (1970) for information about Hausdorff measures. For the sake of completeness let us recall that the *m-dimensional Hausdorff measure* of a subset E of $\mathbb{R}^n$, $H_m(E)$, is given by

$$H_m(E) = \frac{\pi^{m/2}}{2^m \Gamma(m/2+1)} \times \liminf_{r \downarrow 0} \left\{ \sum_{k=1}^{\infty} (\operatorname{diam} E_k)^m : \bigcup_{k=1}^{\infty} E_k \supseteq E, \operatorname{diam} E_k \leq r \text{ for any } k \right\} . \tag{9}$$

3.2. *Organization of the proof*

A proof of the isoperimetric theorem, as stated in section 3.1, is recorded in sections 3.3 to 3.10. This proof is largely modeled on the quoted paper by De Giorgi. For convenience, the proof is organized in several steps. It also includes a treatment of related material: section 3.5, on the so-called Caccioppoli sets and perimeter à la De Giorgi.

3.3. *Beginning the proof*

In this section we make some preparatory comments.

Let us cope with the problem of rendering $p(G)$ a minimum within a collection of subsets G of $\mathbb{R}^n$ such that $m(G)$ equals a given constant – a problem of the calculus of variations, where perimeter p is the functional involved and subsets of $\mathbb{R}^n$ enter the lists. A *direct method* is in order. As a rule, direct methods of the calculus of variations rest on *Weierstrass' principle*: a minimum value is achieved if some topology exists which makes the appropriate level sets compact and simultaneously the function in hand lower semicontinuous. Thus a crucial point is to match a *class of competing sets* and a *notion of perimeter* in such a way that

some topology causes any minimizing sequence to include converging subsequences and causes the perimeter to be lower semicontinuous.

For technical reasons that will appear presently, we restrict *the competing sets to be bounded*. Since the isoperimetric inequality is invariant under translations and dilations, we can equivalently assume that *the competing sets are all confined in some fixed ball*. Further effective hypotheses will be detailed later. We choose to commence by playing with smooth sets only – thus temporarily setting troubles aside – and let the ultimate framework be dictated by developments. In other words, we warm up by considering

$$\inf p(G) \tag{1}$$

under the following constraints:

$$m(G) = M\,, \tag{2}$$

a fixed positive constant,

$$G \subseteq \text{Ball}\,, \tag{3}$$

a large fixed ball, and

$$G \text{ is smooth}\,. \tag{4}$$

Of course, we understand that Ball is big enough for condition (3) to be consistent with (2).

3.4. Minimizing sequences

In this section we address ourselves to the following question: does a minimizing sequence have some convergence property?

As perimeters of smooth sets are positive, the infimum defined by (1)–(4), section 3.3, is finite, we call it P. Consider *a minimizing sequence* $\{G_k\}_{k=1,2,3,\ldots}$, i.e., a sequence of subsets G_k of $\mathbb{R}^n$ such that

$$m(G_k) = M\,, \tag{1a}$$

$$G_k \subseteq \text{Ball}\,, \tag{1b}$$

$$p(G_k) \to P \quad \text{as } k \to +\infty\,, \tag{1c}$$

$$G_k \text{ is smooth}\,. \tag{1d}$$

Let

$$u_k = \chi_{G_k}\,, \tag{2a}$$

the characteristic function of G_k. In other words,

$$u_k(x) = \begin{cases} 1 & \text{if } x \in G_k \,, \\ 0 & \text{if } x \in \mathbb{R}^n \setminus G_k \,. \end{cases} \tag{2b}$$

We claim that

$$\int_{\mathbb{R}^n} |u_k| \, dx = M \,, \tag{3a}$$

$$\text{sprt } u_k \subseteq \text{Ball} \,, \tag{3b}$$

$$\int_{\mathbb{R}^n} |u_k(x+h) - u_k(x)| \, dx \leqslant |h| \times (\text{constant independent of } h \text{ and } k) \tag{3c}$$

for any vector h in $\mathbb{R}^n$. Notice that support is abbreviated sprt throughout. Properties (3a) and (3b) are mere copies of (1a) and (1b), respectively. Property (3c) follows from (1c) via the lemma below.

Lemma. *If u_k is the characteristic function of G_k, i.e. is given by* (2a) *or* (2b), *then*

$$\int_{\mathbb{R}^n} |u_k(x+h) - u_k(x)| \, dx \leqslant |h| \times p(G_k)$$

for any vector h in $\mathbb{R}^n$.

Proof. Let ϕ be any *test* (i.e., smooth and compactly supported) function. We have

$$\int_{\mathbb{R}^n} [u_k(x+h) - u_k(x)]\phi(x) \, dx = \int_{G_k} [\phi(x-h) - \phi(x)] \, dx \,,$$

and

$$\phi(x-h) - \phi(x) = -\int_0^1 \sum_{i=1}^n \frac{\partial \phi}{\partial x_i} (x - th) h_i \, dt \,,$$

where $h_1, \ldots, h_n$ are the components of h. Hence

$$\int_{\mathbb{R}^n} [u_k(x+h) - u_k(x)]\phi(x) \, dx$$

$$= -\int_{\partial G_k} \left\{ \int_0^1 \phi(x - th) \, dt \right\} \langle h, \text{nrml} \rangle H_{n-1}(dx)$$

by the Gauss–Green theorem – here nrml is a shorthand for unit outer normal.

Recall that if m is an integer between 0 and n and E is a sufficiently smooth m-dimensional submanifold of $\mathbb{R}^n$, e.g., the range of a Lipschitz continuous map from a closed m-dimensional ball into $\mathbb{R}^n$, then $H_m(E)$, the m-dimensional Hausdorff measure of E, agrees with the elementary measure of E. Therefore,

$$\left| \int_{\mathbb{R}^n} [u_k(x+h) - u_k(x)]\phi(x)\,dx \right| \leq |h| \cdot p(G_k) \cdot \max |\phi| .$$

The conclusion follows. □

Properties (3a–c) enable to apply a *compactness criterion* for Lebesgue space $L^1(\mathbb{R}^n)$, see, e.g., Dunford and Schwartz (1958, Theorem IV.8.21). Our hypothesis about the boundedness of competing sets plays here its major role. Thus a subsequence of $\{u_k\}$, still denoted $\{u_k\}$, and an integrable function u exist such that

$$\int_{\mathbb{R}^n} |u_k - u|\,dx \to 0 \quad \text{as } k \to +\infty . \tag{4}$$

Clearly, (3a–c) and (4) yield

$$\int_{\mathbb{R}^n} |u|\,dx = M , \tag{5a}$$

$$m(\{x \in \mathbb{R}^n : u(x) \neq 0 \text{ and } x \not\in \text{Ball}\}) = 0 , \tag{5b}$$

$$\int_{\mathbb{R}^n} |u(x+h) - u(x)|\,dx = O(|h|) . \tag{5c}$$

Observe that u equals a characteristic function almost everywhere. More precisely, if G is defined by:

$$G = \{x \in \mathbb{R}^n : u(x) = 1\} , \tag{6}$$

then

$$u = \chi_G , \tag{7}$$

the characteristic function of G, almost everywhere. In fact, (4) implies

$$m(\{x \in \mathbb{R}^n : |u_k(x) - u(x)| \geq \varepsilon\}) \to 0 \quad \text{as } k \to +\infty$$

for every positive ε, a convergence in measure. As is easy to see, the convergence in measure yields

$$m(\{x \in \mathbb{R}^n : a < u(x) < b\}) \leq \liminf_{k \to +\infty} m(\{x \in \mathbb{R}^n : a < u_k(x) < b\}) ,$$

no matter how a and b are. Hence, (2), (4) and (6) yield (7).

In conclusion, we have shown the following. *A sequence of subsets of* $\mathbb{R}^n$, $\{G_k\}$, *and a measurable subset of* $\mathbb{R}^n$, G, *exist such that* $\{G_k\}$ *satisfies* (1) *and*

$$m(G_k \triangle G) \to 0 \quad \text{as } k \to +\infty . \tag{8}$$

Set G *enjoys the following properties*:

$$m(G) = M , \tag{9a}$$

$$m(G \backslash \text{Ball}) = 0 , \tag{9b}$$

$$m((G + h) \triangle G) = \mathrm{O}(|h|) . \tag{9c}$$

Here h is any vector from $\mathbb{R}^n$ and

$$G + h = \{x \in \mathbb{R}^n : x - h \in G\} ,$$

a translation of G.

Recall that the symmetric difference of two sets A and B,

$$A \triangle B ,$$

is either side of the equality $(A \backslash B) \cup (B \backslash A) = (A \cup B) \backslash (A \cap B)$. Thus

$$\chi_{A \triangle B} = |\chi_A - \chi_B| ,$$

$$m(A \triangle B) = \int_{\mathbb{R}^n} |\chi_A(x) - \chi_B(x)| \, \mathrm{d}x ,$$

$$m((G + h) \triangle G) = \int_{\mathbb{R}^n} |\chi_G(x - h) - \chi_G(x)| \, \mathrm{d}x .$$

Equation (9c) shows that small translations do not modify G very much: importantly, the *first power* of $|h|$ appears in the right-hand side of (9c). We will paraphrase equation (9c) by saying that G is a *Caccioppoli set*. Equation (9c) is crucial to our arguments to such a point that we raise it to the rank of a defining property. The next section will be devoted to investigate Caccioppoli sets.

3.5. Caccioppoli sets and their perimeters

Roughly speaking, Caccioppoli sets are those subsets of Euclidean space $\mathbb{R}^n$ for which Gauss–Green formulas hold. Caccioppoli sets play a role in branches of real and functional analysis, e.g., geometric measure theory and calculus of variations. The relevant theory – which overlaps with the theory of $\mathrm{BV}(\mathbb{R}^n)$, the space of functions of bounded variation in $\mathbb{R}^n$ – has its roots in a 1952 paper by Caccioppoli, and was developed by De Giorgi (1954), De Giorgi (1955), Fleming (1957), Krickeberg (1957), Pauc (1957), Fleming (1960), Miranda (1964a),

Miranda (1964b) and others. Comprehensive treatments appear in Federer (1969), Giusti (1984), Maz'ja (1985) and Ziemer (1989).

In this section, we itemize some basic definitions and propositions about Caccioppoli sets. Note that, though different looking, our definition of these sets, item (i) below, proves equivalent to the more standard ones.

(i) Any Lebesgue measurable subset G of $\mathbb{R}^n$ such that $m(G)$ is finite and

$$m((G + h)\triangle G) = \mathrm{O}(|h|) \tag{1}$$

is called a *Caccioppoli set.*

(ii) *Suppose G is a Caccioppoli subset of $\mathbb{R}^n$. Then*

$$\text{grad } \chi_G\,, \tag{2}$$

the gradient of the characteristic function of G is a vector-valued regular Borel measure of bounded variation. More precisely, a regular countably additive measure μ exists such that

$$\mu \textit{ is defined in the algebra of Borel subsets of } \mathbb{R}^n, \textit{ and takes it values in } \mathbb{R}^n\,; \tag{3a}$$

$$\textit{the total variation of } \mu \textit{ is bounded}\,; \tag{3b}$$

$$-\int_G \operatorname{div} f(x)\,\mathrm{d}x = \int_{\mathbb{R}^n} \langle f(x),\, \mu(\mathrm{d}x)\rangle \textit{ for every test (i.e., smooth and compactly supported) vector field } f\,. \tag{3c}$$

Proof. Suppose G is a measurable subset of $\mathbb{R}^n$, $m(G)$ is finite and (1) holds. Consider the vector-valued kernel $\boldsymbol{J}$ defined by:

$$\boldsymbol{J}(x) = \begin{cases} -(n+1)\kappa_n^{-1}\,\dfrac{x}{|x|} & \text{if } |x| < 1\,, \\ 0 & \text{if } |x| \geqslant 1\,, \end{cases} \tag{4a}$$

and let

$$\boldsymbol{J}_\varepsilon(x) = \varepsilon^{-n-1}\boldsymbol{J}(x/\varepsilon) \tag{4b}$$

for every positive ε.

We claim that

$$\int_{\mathbb{R}^n} |\boldsymbol{J}_\varepsilon * \chi_G|\,\mathrm{d}x \leqslant \text{constant independent of } \varepsilon\,, \tag{5}$$

where star denotes convolution, i.e.,

$$J_\varepsilon * \phi(x) = \int_{\mathbb{R}^n} J_\varepsilon(x-y)\phi(y)\,dy\,.$$

In fact,

$$\int_{\mathbb{R}^n} J_\varepsilon(x)\,dx = 0$$

since J is odd; hence

$$(J_\varepsilon * \chi_G)(x) = \int_{\mathbb{R}^n} J_\varepsilon(y)[\chi_G(x-y) - \chi_G(x)]\,dy\,.$$

Consequently,

$$\begin{aligned}\int_{\mathbb{R}^n} |J_\varepsilon * \chi_G(x)|\,dx &\leq \int_{\mathbb{R}^n} |J_\varepsilon(y)|\,dy \int_{\mathbb{R}^n} |\chi_G(x-y) - \chi_G(x)|\,dx \\ &= \int_{\mathbb{R}^n} |J_\varepsilon(y)|\, m((G+y)\triangle G)\,dy \\ &\leq \text{constant} \times \int_{\mathbb{R}^n} |J_\varepsilon(y)|\,|y|\,dy\,,\end{aligned}$$

by (1). As

$$\int_{\mathbb{R}^n} |J_\varepsilon(y)|\,|y|\,dy = n\,,$$

the claim is demonstrated.

Inequality (5) has a decisive consequence: a sequence $\{\varepsilon_k\}$ of positive numbers and a *countably additive regular Borel measure μ of bounded total variation* exist such that $\varepsilon_k \downarrow 0$, μ satisfies (3a) and (3b), and

$$\int_{\mathbb{R}^n} \langle f(x), J_\varepsilon * \chi_G(x)\rangle\,dx \to \int_{\mathbb{R}^n} \langle f(x), \mu(dx)\rangle \tag{6}$$

for any test vector field f as ε approaches zero along that sequence.

In fact, let $M(\mathbb{R}^n)$ consist of all real-valued countably additive regular measures defined in the σ-field of all Borel subsets of $\mathbb{R}^n$ – a Banach space if the relevant norm is assumed to be the supremum of the total variation. Principles of functional analysis guarantee that *any bounded subset of $M(\mathbb{R}^n)$ is relatively compact under a suitable weak topology* – a sequence $\{\mu_k\}$ from $M(\mathbb{R}^n)$ converges to zero in such a topology if and only if $\int \phi\mu_k(dx) \to 0$ for any continuous real-valued function ϕ vanishing at infinity. These principles are:

(a) Any closed ball from X^*, the dual space of a Banach space X, is closed under the weak-star topology of X^* – Alaoglu theorem, see Dunford and Schwartz (1958, Theorem V.4.2).

(b) $M(\mathbb{R}^n)$ is exactly the dual space of the space of all continuous real-valued functions defined in $\mathbb{R}^n$ and vanishing at infinity – Riesz representation theorem, see Rudin (1966, Theorem 6.19).

Let us show that such a measure μ is exactly the gradient of the characteristic function of G. Actually, one may easily check that, if ϕ is a scalar-valued test function, then

$$\boldsymbol{J}_\varepsilon * \phi = \operatorname{grad} \phi + \frac{n+1}{2(n+2)(n+3)} \varepsilon^2 \operatorname{grad}(\Delta\phi) + \cdots,$$

so that $\boldsymbol{J}_\varepsilon * \phi$ approaches grad ϕ as ε approaches zero. On the other hand, let $\boldsymbol{f}$ be any test vector field. We have

$$\int_{\mathbb{R}^n} \langle \boldsymbol{f}(y), \boldsymbol{J}_\varepsilon(y-x)\rangle \, \mathrm{d}y \to -\operatorname{div} \boldsymbol{f}(x) \quad \text{as } \varepsilon \downarrow 0,$$

and

$$\left| \int_{\mathbb{R}^n} \langle \boldsymbol{f}(y), \boldsymbol{J}_\varepsilon(y-x)\rangle \, \mathrm{d}y \right| \leqslant \max |\operatorname{div} \boldsymbol{f}|$$

inasmuch as

$$\begin{aligned}
&\int_{\mathbb{R}^n} \langle \boldsymbol{f}(y), \boldsymbol{J}_\varepsilon(y-x)\rangle \, \mathrm{d}y \\
&\quad = -\varepsilon^{-n-1} \int_0^\varepsilon \mathrm{d}(t^{n+1}) \int_{\{y\in\mathbb{R}^n:\, |y-x|=t\}} \left\langle \boldsymbol{f}(y), \frac{y-x}{|y-x|} \right\rangle \frac{H_{n-1}(\mathrm{d}y)}{\kappa_n t^n} \\
&\quad = -\varepsilon^{-n-1} \int_0^\varepsilon \mathrm{d}(t^{n+1}) \int_{\{y\in\mathbb{R}^n:\, |y-x|<t\}} \operatorname{div} \boldsymbol{f}(x) \frac{\mathrm{d}x}{\kappa_n t^n},
\end{aligned}$$

by Gauss–Green formulas. Thus

$$\int_G \mathrm{d}x \int_{\mathbb{R}^n} \langle \boldsymbol{f}(y), \boldsymbol{J}_\varepsilon(y-x)\rangle \, \mathrm{d}y \to -\int_G \operatorname{div} \boldsymbol{f}(x) \, \mathrm{d}x \quad \text{as } \varepsilon \downarrow 0, \tag{7}$$

by the dominated convergence theorem. Clearly,

$$\int_G \mathrm{d}x \int_{\mathbb{R}^n} \langle \boldsymbol{f}(y), \boldsymbol{J}_\varepsilon(y-x)\rangle \, \mathrm{d}y = \int_{\mathbb{R}^n} \langle \boldsymbol{f}(x), \boldsymbol{J}_\varepsilon * \chi_G(x)\rangle \, \mathrm{d}x,$$

therefore (7) yields

$$\int_{\mathbb{R}^n} \langle \boldsymbol{f}(x), \boldsymbol{J}_\varepsilon * \chi_G(x)\rangle \, \mathrm{d}x \to -\int_G \operatorname{div} \boldsymbol{f}(x) \, \mathrm{d}x \quad \text{as } \varepsilon \downarrow 0. \tag{8}$$

Relations (6) and (8) yield (3c). This completes the proof. □

(iii) *Suppose G is a measurable subset of $\mathbb{R}^n$, $m(G)$ is finite, and the gradient of the characteristic function of G is a vector-valued Borel measure of bounded total variation, i.e., a countably additive measure μ exists such that* (3a–c) *hold. Then G is a Caccioppoli set.*

Proof. Let ϕ be any real-valued test function. We have

$$\begin{aligned}&\int_{\mathbb{R}^n} [\chi_G(x-h) - \chi_G(x)]\phi(x)\,dx \\ &= \int_G [\phi(x+h) - \phi(x)]\,dx \\ &= \int_G \operatorname{div}\left(h \int_0^1 \phi(x+th)\,dt\right) dx\end{aligned}$$

trivially, hence

$$\int_{\mathbb{R}^n} [\chi_G(x-h) - \chi_G(x)]\phi(x)\,dx = \int_{\mathbb{R}^n} \left(\int_0^1 \phi(x+th)\,dt\right) \langle h, \mu(dx)\rangle$$

by (3c). Consequently,

$$\left|\int_{\mathbb{R}^n} [\chi_G(x-h) - \chi_G(x)]\phi(x)\,dx\right| \leq \max|\phi| \cdot |h| \cdot (\text{tot. var. of } \mu \text{ on } \mathbb{R}^n)\,.$$

Therefore,

$$\int_{\mathbb{R}^n} |\chi_G(x-h) - \chi_G(x)|\,dx \leq |h| \cdot (\text{tot. var. of } \mu \text{ on } \mathbb{R}^n)\,,$$

by the arbitrariness of ϕ. The conclusion follows. □

Remark. The above proposition guarantees that any smooth subset of $\mathbb{R}^n$ is a Caccioppoli set. In fact, if G is a smooth subset of $\mathbb{R}^n$ the gradient of the characteristic function of G is the measure given by

$$\operatorname{grad} \chi_G(E) = -\int_{E \cap \partial G} (\text{outer unit normal to } \partial G)\, H_{n-1}(dx)$$

for every Borel subset E of $\mathbb{R}^n$.

(iv) *Let G be a Caccioppoli subset of $\mathbb{R}^n$. By definition, the perimeter of G, $p(G)$, is the total variation of* grad χ_G *on $\mathbb{R}^n$. We have*

$$p(G) = \sup\left\{\left|\int_G \operatorname{div} f\,dx\right| : f \textit{ is a test vector field}, \max|f| \leq 1\right\}. \tag{10}$$

This formula is a consequence of (3a–c); in fact, the total variation of a vector-valued Borel measure $\boldsymbol{\mu}$ obeys:

$$\text{tot. var. of } \boldsymbol{\mu} \text{ on } E$$
$$= \sup\left\{\left|\int_{\mathbb{R}^n} \langle \boldsymbol{f}(x), \boldsymbol{\mu}(\mathrm{d}x)\rangle\right| : \boldsymbol{f} \text{ is a test vector field, } \operatorname{sprt} \boldsymbol{f} \subseteq E, \max|\boldsymbol{f}| \leqslant 1\right\}$$

whenever E is open.

Remark. Formula (10) guarantees that $p(G)$ agrees with the customary $(n-1)$-dimensional measure of ∂G whenever G is smooth, for

$$\int_G \operatorname{div} \boldsymbol{f}\, \mathrm{d}x = \int_{\partial G} \langle \boldsymbol{f}, \text{outer unit normal to } \partial G\rangle\, H_{n-1}(\mathrm{d}x)$$

if G is smooth and $\boldsymbol{f}$ is a test vector field.

(v) *Let G be a measurable subset of $\mathbb{R}^n$. Assume $m(G)$ is finite and*

$$\sup\left\{\left|\int_G \operatorname{div} \boldsymbol{f}\, \mathrm{d}x\right| : \boldsymbol{f} \text{ is a test vector field, } \max|\boldsymbol{f}| \leqslant 1\right\} \tag{11}$$

is finite too. Then G is a Caccioppoli set.

A proof of this proposition mimics the arguments of item (iii).

Remark. Let G be any measurable subset of $\mathbb{R}^n$; by definition, (11) is the perimeter, $p(G)$ of G. This definition is consistent with the previous one – see formula (10). Items (iv) and (v) guarantee that Caccioppoli sets are exactly those measurable subsets of $\mathbb{R}^n$ whose measure and perimeter are both finite.

(vi) *If G is a Caccioppoli subset of $\mathbb{R}^n$, then*

$$m((G+h)\triangle G) \leqslant |h| \cdot p(G)\,. \tag{12}$$

This inequality is a byproduct of Proposition (iii) and Definition (iv). Incidentally, inequality (12) generalizes a lemma from section 3.4.

Remark. The collection of measurable subsets of $\mathbb{R}^n$ is a commutative ring if symmetric difference and intersection are taken as the addition and multiplication operations of the ring. Consider the quotient of such a ring modulo the ideal of

sets of measure zero – a measure ring. The collection of sets having a finite measure from the measure ring in question is a metric space if the distance between A and B is defined as the measure of the symmetric difference of A and B. Thus inequality (12) has the following corollary. *Any collection of Caccioppoli sets, which are contained in some given box and whose perimeters are all bounded by some given number, is relatively compact.* A proof parallels verbatim arguments from section 3.3.

(vii) *Let* $\{G_k\}$ *be a sequence of Caccioppoli sets, let* G *be measurable. Assume*

$$m(G_k \triangle G) \to 0 \quad \text{as } k \to +\infty\,, \tag{13a}$$

and

$$p(G_k) \leqslant \text{constant}\,. \tag{13b}$$

Then G *is a Caccioppoli set, and*

$$p(G) \leqslant \liminf_{k \to +\infty} p(G_k)\,. \tag{13c}$$

Remark. The statement here above amounts to saying that *perimeter* p *is a lower semicontinuous function.* The underlying topology is defined in item (vi).

Proof. Inequality (12) and hypothesis (13b) yield

$$m((G_k + h)\triangle G) \leqslant |h| \times (\text{constant independent of } h \text{ and } k)\,.$$

On the other hand,

$$m((G_k + h)\triangle G) \to m((G + h)\triangle G) \quad \text{as } k \to +\infty$$

by hypothesis (13a). Recall that $|m(A) - m(B)| \leqslant m(A \triangle B)$. Thus

$$m((G + h)\triangle G) \leqslant |h| \times \text{constant}\,,$$

proving that G is Caccioppoli.

Let $\boldsymbol{f}$ be any test vector field. We have

$$\int_{G_k} \operatorname{div} \boldsymbol{f}\, dx \to \int_G \operatorname{div} \boldsymbol{f}\, dx \quad \text{as } k \to +\infty\,,$$

since hypothesis (13a) tells us that χ_{G_k} converges in $L^1(\mathbb{R}^n)$ to χ_G. Therefore,

$$\left| \int_G \operatorname{div} \boldsymbol{f}\, dx \right| \leqslant \max |\boldsymbol{f}| \cdot \liminf_{k \to +\infty} p(G_k)\,,$$

by the very definition of perimeter. The assertion follows. □

(viii) *Suppose G is a Caccioppoli subset of $\mathbb{R}^n$. Then*

$$p(G) \geqslant \int_{\mathbb{R}^n} |\operatorname{grad} J_\varepsilon * \chi_G| \, dx \tag{14a}$$

for every positive ε, and

$$p(G) = \lim_{\varepsilon \downarrow 0} \int_{\mathbb{R}^n} |\operatorname{grad} J_\varepsilon * \chi_G| \, dx \, . \tag{14b}$$

Here J_ε is any mollifier, i.e.,

$$J_\varepsilon(x) = \varepsilon^{-n} J(x/\varepsilon) \, , \tag{15a}$$

$$J(x) \geqslant 0 \, , \tag{15b}$$

$$\int_{\mathbb{R}^n} J(x) \, dx = 1 \, , \tag{15c}$$

$$J \text{ is infinitely differentiable} \, , \tag{15d}$$

$$\operatorname{sprt} J \subseteq \{x \in \mathbb{R}^n : |x| \leqslant 1\} \, . \tag{15e}$$

Proof. Denote $J_\varepsilon * \chi_G$ by u_ε and observe that u_ε is infinitely differentiable. Thus integrations by parts give

$$-\int_{\mathbb{R}^n} u_\varepsilon \cdot \operatorname{div} f \, dx = \int_{\mathbb{R}^n} \langle f, \operatorname{grad} u_\varepsilon \rangle \, dx \, , \tag{16}$$

if f is a test vector field.

Clearly,

$$\int_{\mathbb{R}^n} u_\varepsilon \cdot \operatorname{div} f \, dx = \int_{\mathbb{R}^n} J_\varepsilon(y) \, dy \int_G \operatorname{div} f(x + y) \, dx \, ;$$

therefore

$$\left| \int_{\mathbb{R}^n} u_\varepsilon \cdot \operatorname{div} f \, dx \right| \leqslant \max |f| \cdot p(G) \, ,$$

by formula (10) and properties (15a–c) of mollifier J_ε. This inequality and eq. (16) give

$$\left| \int_{\mathbb{R}^n} \langle f, \operatorname{grad} u_\varepsilon \rangle \, dx \right| \leqslant \max |f| \cdot p(G)$$

for every test vector field f, proving (14a).

We have

$$u_\varepsilon(x) - \chi_G(x) = \int_{\mathbb{R}^n} J_\varepsilon(y)[\chi_G(x - y) - \chi_G(x)] \, dx \, ,$$

hence

$$\int_{\mathbb{R}^n} |u_\varepsilon - \chi_G| \, dx \leq \varepsilon p(G) \int_{\mathbb{R}^n} J(x)|x| \, dx \tag{17}$$

by inequality (12) and properties of mollifier J_ε. Thus u_ε converges to χ_G fast enough in $L^1(\mathbb{R}^n)$ as $\varepsilon \downarrow 0$.

Property (17) yields

$$\int_G \operatorname{div} \boldsymbol{f} \, dx = \lim_{\varepsilon \downarrow 0} \int_{\mathbb{R}^n} u_\varepsilon \cdot \operatorname{div} \boldsymbol{f} \, dx$$

if $\boldsymbol{f}$ is a test vector field. Equation (16) yields

$$\left| \int_{\mathbb{R}^n} u_\varepsilon \cdot \operatorname{div} \boldsymbol{f} \, dx \right| \leq \max |\boldsymbol{f}| \cdot \int_{\mathbb{R}^n} |\operatorname{grad} u_\varepsilon| \, dx \, .$$

Then we must have

$$\left| \int_G \operatorname{div} \boldsymbol{f} \, dx \right| \leq \max |\boldsymbol{f}| \cdot \liminf_{\varepsilon \downarrow 0} \int_{\mathbb{R}^n} |\operatorname{grad} u_\varepsilon| \, dx \, ,$$

showing that

$$p(G) \leq \liminf_{\varepsilon \downarrow 0} \int_{\mathbb{R}^n} |\operatorname{grad} u_\varepsilon| \, dx$$

because of formula (10).

Since (14a) is true, (14b) follows. The proof is done. □

(ix) *Suppose G is a Caccioppoli subset of $\mathbb{R}^n$. Then a sequence of subsets of $\mathbb{R}^n$, $\{E_k\}$, exists such that*

$$m(G \triangle E_k) \to 0 \quad \text{as } k \to +\infty \, , \tag{20a}$$

$$p(E_k) \to p(G) \quad \text{as } k \to +\infty \, , \tag{20b}$$

$$E_k \text{ is smooth} \, . \tag{20c}$$

Proof. Let J_ε be a mollifier as in item (viii), and let

$$u_\varepsilon = J_\varepsilon * \chi_G \, ,$$

a smoothing of the characteristic function of G. Our proof consists in showing that a suitable selection of

$$\{x \in \mathbb{R}^n : u_\varepsilon(x) > t\} \, , \tag{21}$$

the level sets of u_ε, provides us with the required sequence.

Properties (15b,c) of mollifier J_ε yield $0 \leqslant u_\varepsilon(x) \leqslant 1$ for every x from $\mathbb{R}^n$. Deduce

$$\{x \in \mathbb{R}^n : u_\varepsilon(x) > t\} \triangle G \subseteq \{x \in \mathbb{R}^n : |u_\varepsilon(x) - \chi_G(x)| \geqslant \min(t, 1-t)\},$$

and recall from item (viii) that

$$\int_{\mathbb{R}^n} |u_\varepsilon - \chi_G| \, dx \leqslant \varepsilon \cdot p(G) \cdot \int_{\mathbb{R}^n} J(x)|x| \, dx .$$

Hence

$$m(\{x \in \mathbb{R}^n : u_\varepsilon(x) > t\} \triangle G) \leqslant \varepsilon \frac{p(G)}{\min(t, 1-t)} \int_{\mathbb{R}^n} J(x)|x| \, dx$$

provided $0 < t < 1$, in particular

$$m(\{x \in \mathbb{R}^n : u_\varepsilon(x) > t\} \triangle G) \to 0 \quad \text{as } \varepsilon \downarrow 0 , \tag{22}$$

provided $0 < t < 1$.

Equation (22) and Proposition (vii) imply

$$p(G) \leqslant \liminf_{\varepsilon \downarrow 0} p(\{x \in \mathbb{R}^n : u_\varepsilon(x) > t\}) \tag{23a}$$

if t belongs to $]0, 1[$.

On the other hand, Fleming and Rishel coarea formula (see the end of this subsection) tells us that

$$\int_{\mathbb{R}^n} |\operatorname{grad} u_\varepsilon| \, dx = \int_0^1 p(\{x \in \mathbb{R}^n : u_\varepsilon(x) > t\}) \, dt .$$

Hence Proposition (viii) gives

$$p(G) \geqslant \int_0^1 \liminf_{\varepsilon \downarrow 0} p(\{x \in \mathbb{R}^n : u_\varepsilon(x) > t\}) \, dt , \tag{23b}$$

by Fatou's theorem.

Inequalities (23a,b) imply

$$p(G) = \liminf_{\varepsilon \downarrow 0} p(\{x \in \mathbb{R}^n : u_\varepsilon(x) > t\}) \tag{24}$$

provided t is in $]0, 1[$.

Clearly, eqs. (22) and (24) continue to hold if ε goes to zero along a sequence, the sequence $1, \frac{1}{2}, \frac{1}{3}, \ldots, 1/k, \ldots$, for instance. As u_ε is infinitely differentiable, Sard's theorem – the critical values of a smooth map make up a set of measure zero – and the customary implicit function theorem guarantee that the set of those levels t, such that $0 < t < 1$ and (21) fails to be smooth, is negligible. Thus a level t

exists such that $0<t<1$ and (21) is a smooth set for $\varepsilon = 1, \frac{1}{2}, \frac{1}{3}, \ldots, 1/k, \ldots$. The conclusion follows. □

(x) *If G is a Caccioppoli subset of $\mathbb{R}^n$, then $p(G) = \inf\{\liminf p(E_k)\colon \{E_k\}$ is a sequence of smooth subsets of $\mathbb{R}^n$ such that $m(G \triangle E_k) \to 0\}$.*

This formula, somewhat reminiscent of the area à la Lebesgue, follows immediately from Propositions (vii) and (ix).

(xi) The following propositions are among the most significant ones in the theory of Caccioppoli sets. We quote them without proof; proofs can be found in De Giorgi (1955), Federer (1969), Maz'ja (1985) and Ziemer (1989).

Let G be a subset of $\mathbb{R}^n$ such that $m(G)$ is finite. Then G is a Caccioppoli set if and only if $\partial^* G$, the measure-theoretic *or* reduced boundary *of G, has a finite $(n-1)$-dimensional Hausdorff measure.*

By definition, $\partial^* G = \{x \in \mathbb{R}^n\colon 0 < (\text{density of } G \text{ at } x) < 1\}$. As the density of a set equals 1 at any interior point of the set and vanishes at any exterior point, the measure-theoretic boundary is included in the topological boundary. Recall that the density of G at x is the limit of

$$(\kappa_n r^n)^{-1} m(G \cap \{y \in \mathbb{R}^n\colon |y - x| < r\})$$

as r approaches zero.

Suppose G is a Caccioppoli subset of $\mathbb{R}^n$. Then grad χ_G *is absolutely continuous with respect to the $(n-1)$-dimensional Hausdorff measure H_{n-1}. The relevant Radon–Nikodym derivative* – a vector field $\boldsymbol{n}$ – *enjoys the following properties*:

(a) *Support of $\boldsymbol{n} \subseteq \partial^* G$, the measure-theoretic boundary of G.*

(b) $H_{n-1}(\partial^* G \setminus \{x \in \mathbb{R}^n\colon \boldsymbol{n}(x)$ *is a unit vector*$\}) = 0$.

(c) *For any x from $\partial^* G$ such that $\boldsymbol{n}(x)$ is a unit vector, $\boldsymbol{n}(x)$ is the outer normal to G at x.*

By definition, a unit vector h from $\mathbb{R}^n$ is the outer normal to set G at point x if

$$\{y \in \mathbb{R}^n\colon \langle y - x, h\rangle > 0\} \cap G \quad \text{and} \quad \{y \in \mathbb{R}^n\colon \langle y - x, h\rangle < 0\} \setminus G$$

both have density zero at x.

The above propositions plainly imply the following corollary.

Suppose G is a Caccioppoli subset of $\mathbb{R}^n$. Then the perimeter of G obeys

$$p(G) = H_{n-1}(\partial^* G)\,.$$

Moreover, a vector field $\boldsymbol{n}$ *exists which enjoys properties* (a)–(c) *above and is such that the following equation,*

$$\int_G \operatorname{div} \boldsymbol{f}(x)\, dx = \int_{\partial^* G} \langle \boldsymbol{f}(x), \boldsymbol{n}(x) \rangle H_{n-1}(dx)\,,$$

a Gauss–Green formula, holds for any test vector field $\boldsymbol{f}$.

(xii) Here we parenthetically summarize considerations about a special function space closely related to Caccioppoli sets: $\mathrm{BV}(\mathbb{R}^n)$.

The collection of all members u of $L^1(\mathbb{R}^n)$ such that

$$\int_{\mathbb{R}^n} |u(x+h) - u(x)|\, dx = \mathrm{O}(|h|)$$

is called $\mathrm{BV}(\mathbb{R}^n)$. A member u of $L^1(\mathbb{R}^n)$ is in $\mathrm{BV}(\mathbb{R}^n)$ if and only if grad u, the gradient of u in the sense of distributions, is a vector-valued regular Borel measure of bounded total variation. Let u be in $\mathrm{BV}(\mathbb{R}^n)$; by definition, the total variation of u is the supremum of the total variation of grad u. The following formula

$$\text{tot. var. } u = \sup\left\{ \left| \int_{\mathbb{R}^n} u \cdot \operatorname{div} \boldsymbol{f}\, dx \right| : \boldsymbol{f} \text{ is a test vector field, } \max |\boldsymbol{f}| \leqslant 1 \right\}$$

holds for every u in $\mathrm{BV}(\mathbb{R}^n)$. A member u of $L^1(\mathbb{R}^n)$ is in $\mathrm{BV}(\mathbb{R}^n)$ if and only if the right-hand side of the last formula is finite.

Thus a subset of G of $\mathbb{R}^n$ is a Caccioppoli set if and only if χ_G, the characteristic function of G, is in $\mathrm{BV}(\mathbb{R}^n)$; the total variation of χ_G is nothing but the perimeter of G. Basic properties of $\mathrm{BV}(\mathbb{R}^n)$ are exact counterparts of properties of Caccioppoli sets. For instance, if u is in $\mathrm{BV}(\mathbb{R}^n)$ we have

$$\int_{\mathbb{R}^n} |u(x+h) - u(x)|\, dx \leqslant |h| \times \text{tot. var. } u$$

for any h from $\mathbb{R}^n$; furthermore,

$$\text{tot. var. } u \geqslant \int_{\mathbb{R}^n} |\operatorname{grad} J_\varepsilon * u|\, dx\,,$$

where ε is any positive number and J_ε is a mollifier, and

$$\text{tot. var. } u = \lim_{\varepsilon \downarrow 0} \int_{\mathbb{R}^n} |\operatorname{grad} J_\varepsilon * u|\, dx\,.$$

Recall the coarea formula of Fleming and Rishel. Let u be in $\mathrm{BV}(\mathbb{R}^n)$; then

$$\text{tot. var. } u = \int_0^{+\infty} p(\{x \in \mathbb{R}^n : |u(x)| > t\})\, dt\,.$$

If u is Lipschitz continuous, this formula may be replaced by

$$\int_{\mathbb{R}^n} |\text{grad } u(x)| \, dx = \int_0^{+\infty} H_{n-1}(\{x \in \mathbb{R}^n : |u(x)| = t\}) \, dt ,$$

the Federer coarea formula. Coarea formulas are a counterpart of Cavalieri's principle, which may be stated as

$$\int_{\mathbb{R}^n} |u(x)| \, dx = \int_0^{+\infty} m(\{x \in \mathbb{R}^n : |u(x)| > t\}) \, dt ;$$

in a sense, coarea formulas amount to saying that distance between level surfaces and gradient length are inversely proportional. Proofs of coarea formulas appear in Fleming and Rishel (1960) and Federer (1969).

3.6. The isoperimetric theorem in one dimension

The following proposition will come in useful and may have some interest per se.

Theorem. *Let G be a Caccioppoli subset of the real line $\mathbb{R}$. Assume $m(G)$, the one-dimensional Lebesgue measure of G, is positive. Then*

$$p(G) \geq 2 .$$

Equality holds if and only if G is equivalent to a bounded interval.

Proof. By Proposition (vii), section 3.5, the perimeter of G, $p(G)$, obeys

$$p(G) = \lim_{\varepsilon \downarrow 0} \int_{-\infty}^{+\infty} |u_\varepsilon'| \, dx ,$$

where

$$u_\varepsilon = J_\varepsilon * \chi_G ,$$

a mollified version of the characteristic function of G. As u_ε is smooth,

$$\int_{-\infty}^{+\infty} |u_\varepsilon'| \, dx$$
$$= \sup \left\{ \sum_{k=1}^{m} |u_\varepsilon(x_k) - u_\varepsilon(x_{k-1})| : -\infty < x_0 < x_1 < \cdots < x_m < \infty \right\} ,$$

the total variation of u_ε. A property of mollifiers yields

$$u_\varepsilon(x) \to \chi_G(x) \quad \text{as } \varepsilon \downarrow 0$$

provided x is a Lebesgue point of χ_G, i.e., either x is in G and

$$\lim_{\varepsilon \downarrow 0} \frac{1}{2\varepsilon} m(G \cap [x - \varepsilon, x + \varepsilon]) ,$$

the *density* of G at x, is 1 or x is out of G and the density of G at x is 0. Thus

$$p(G) \geqslant \sum_{k=1}^{m} |\chi_G(x_k) - \chi_G(x_{k-1})| , \tag{1a}$$

where $x_0, x_1, \ldots, x_m$ is any sequence satisfying

$$x_0 < x_1 < \cdots < x_m \tag{1b}$$

and

$$x_0, x_1, \ldots, x_m \text{ belong to the Lebesgue set of } \chi_G . \tag{1c}$$

Incidentally, one might show that

$$p(G) = \sup \left\{ \sum_{k=1}^{m} |\chi_G(x_k) - \chi_G(x_{k-1})| : \{x_k\}_{k=0,\ldots,m} \text{ increases and is included in the Lebesgue set of } \chi_G \right\} , \tag{2}$$

i.e., the perimeter of a Caccioppoli subset G of real line $\mathbb{R}$ is nothing but the *essential variation* of χ_G.

Let

$$L = \text{Lebesgue set of } \chi_G , \tag{3}$$

and

$$H = L \cap G , \qquad K = L \backslash G . \tag{4}$$

A standard theorem on differentiation says that

$$m(\mathbb{R} \backslash L) = 0 . \tag{5}$$

As G is the disjoint union of H and $G \backslash L$, and $G \backslash L$ is a subset of $\mathbb{R} \backslash L$, (5) implies

$$m(H) = m(G) , \tag{6}$$

therefore assumptions yield

$$m(H) < \infty \tag{7}$$

and

$$m(H) > 0 . \tag{8}$$

Clearly, both $]-\infty, \inf K[$ and $]\sup K, +\infty[$ are contained in the complement of K; the latter equals $H \cup (\mathbb{R} \setminus L)$, since $K = L \setminus H$. Hence (5) and (7) imply

$$\inf K = -\infty , \qquad \sup K = +\infty . \tag{9}$$

Thanks to properties (8) and (9), we can select a point x_1 from H, then two points x_0 and x_2 from K such that $x_0 < x_1 < x_2$. Evaluated at x_0, x_1, x_2 – the triple now assembled, an increasing sequence included in Lebesgue set L – the right-hand side of (1a) is 2. Thus (1) gives

$$p(G) \geqslant 2 , \tag{10}$$

the isoperimetric inequality in question.

The very definitions tell us that the perimeter of an interval, having a positive and finite length, is exactly 2 – the number of boundary points. On the other hand, two Caccioppoli sets, whose symmetric difference has measure zero, have the same perimeter. Thus equality holds in (10) if G is, or is equivalent to, a bounded interval not reduced to a point.

Now assume G renders (10) an equality, i.e.,

$$p(G) = 2 . \tag{11}$$

We claim:

$$]\inf H, \sup H[\cap K \text{ is empty} . \tag{12}$$

In fact, if (12) is violated, there exist a point in K, call it x_2, and two points in H, call them x_1 and x_3, such that $x_1 < x_2 < x_3$. Let x_0 and x_4 be points from K such that $x_0 < x_1$ and $x_3 < x_4$. These points exist because of (9). Evaluated at x_0, x_1, x_2, x_3, x_4 – the sequence thus assembled, an admissible one – the right-hand side of (1a) is 4. Hence (1a–c) give $p(G) \geqslant 4$, a contradiction.

Property (12) yields

$$]\inf H, \sup H[\subseteq H \cup (\mathbb{R} \setminus L) ,$$

the complement of K. On the other hand,

$$H \subseteq [\inf H, \sup H] .$$

Consequently

$$m(H \triangle]\inf H, \sup H[) = 0 , \tag{13}$$

since the measure of $\mathbb{R}\backslash L$ is zero, see (5). In other words, H is equivalent to

$$]\inf H, \sup H[\ . \tag{14}$$

We deduce also that $\sup H - \inf H = m(H)$, i.e., (14) is a genuine interval, compare with (7) and (8).

As H is a subset of G and (6) holds

$$m(G \triangle H) = 0\ . \tag{15}$$

Equations (13) and (15) give

$$m(G \triangle]\inf H, \sup H[) = 0\ .$$

In conclusion, (11) forces G to be equivalent to interval (14).

The proof is complete. □

3.7. Existence of a minimizer

In this section, the machinery of Caccioppoli sets is applied for solving the variational problem raised in section 3.2.

Let us come back to the situation we faced at the end of the section 3.4. A minimizing sequence $\{G_k\}$, i.e., a sequence of subsets of $\mathbb{R}^n$ satisfying properties (1a–1d), and a subset G of $\mathbb{R}^n$ are in hand such that G is a limit of $\{G_k\}$, i.e., relation (8) holds. G satisfies properties (9a–b), moreover G is a Caccioppoli set, i.e., satisfies (9c).

Is such a set G a minimizer? Now we are in a position to state the question in such a form that the answer is affirmative. The key device is to give up smoothness and cast the class of competing sets properly. Let us focus our problem as follows:

$$p(G) \text{ is minimum}\ , \tag{1a}$$

under the conditions:

$$m(G) = M\ , \tag{1b}$$

$$m(G\backslash \text{Ball}) = 0\ , \tag{1c}$$

$$G \text{ is a Caccioppoli set}\ . \tag{1d}$$

We claim that the *set G, obtained by the process of section* 3.4 *recalled above, is a solution to problem* (1a–d). In fact, properties (1b–d) have been checked already. Property (1c), section 3.4, of minimizing sequences and Proposition (vii), section 3.5 – semicontinuity of perimeter p – give

$$p(G) \leqslant P\ . \tag{2}$$

Recall that P was defined by (1)–(4), of section 3.3. Importantly, the infimum of perimeter p stays unchanged if the class of competing sets is enriched by Caccioppoli sets, i.e.,

$$P = \inf \{ p(G) \colon G \text{ is a subset of } \mathbb{R}^n \text{ satisfying (1b–d)} \} . \tag{3}$$

Equation (3) is an easy consequence of the definition of P and Proposition (ix), section 3.5. Relations (2) and (3) give

$$p(G) = P , \tag{4}$$

proving the claim.

Thus the existence of a minimizer is demonstrated.

By the way, taking for granted the apparatus we set up in section 3.5 may allow one to bypass our approach and get a solution to problem (1) straightforwardly, just by invoking Weierstrass' principle together with Propositions (vi) and (vii) of section 3.5.

3.8. Steiner symmetrization

Jacob Steiner, one of the major early investigators of the isoperimetric theorem, introduced some 150 years ago an ingenious and elegant technique – see pages 75–91, Vol. 2 of Steiner's Gesammelte Werke. This technique, which nowadays is known as Steiner symmetrization and has a wide range of remarkable applications, allows one to accomplish a decisive step in the proof of the isoperimetric theorem: the identification of minimizers.

Suppose V is a hyperplane in $\mathbb{R}^n$. The *Steiner symmetrization with respect to* V is the operation which converts any measurable subset X of $\mathbb{R}^n$ into the subset Y of $\mathbb{R}^n$ defined by the following rule: for every straight line L orthogonal to V, let $L \cap Y$ be empty if L has an empty intersection with X; for every straight line L, that is orthogonal to V and does meet X, let $L \cap Y$ be

the point where L meets V ,

the segment of line L whose midpoint is in V and whose length equals the one-dimensional measure of $L \cap Y$,

the whole line L ,

according to whether the one-dimensional measure of $L \cap X$ is zero, positive, or infinite. Figure 1 – taken from a paper by Steiner – and fig. 2 may help to grasp the situation.

Salient properties of Steiner symmetrization are summarized in the following theorem.

Theorem. *Let X be a measurable subset of $\mathbb{R}^n$ and let Y be the Steiner symmetrized of X with respect to a hyperplane V. The following properties hold*:

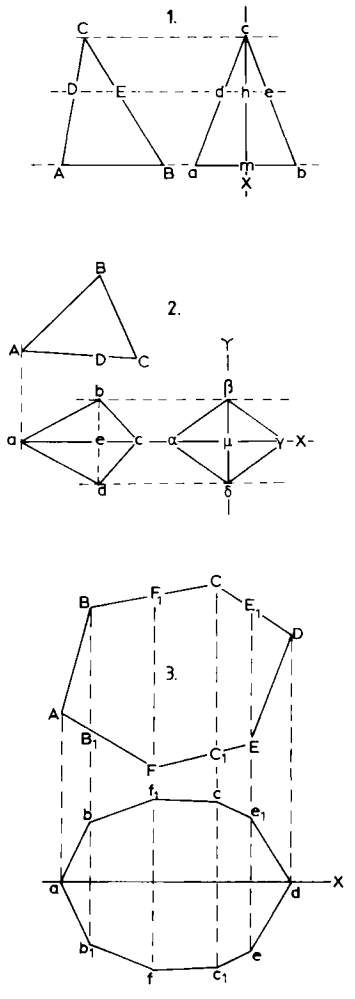

Figure 1. J. Steiner, Einfache Beweise der isoperimetrischen Hauptsätze, figs. 1–3, Gesammelte Werke II Band, Reimer 1882.

(i) *Y is symmetric about V, i.e., Steiner symmetrization brings indeed a symmetry.*

(ii) $m(X) = m(Y)$, *i.e., Steiner symmetrization preserves measure.*

(iii) *Assume X is a Caccioppoli set. Then Y is a Caccioppoli set too and* $p(X) \geq p(Y)$, *i.e., Steiner symmetrization diminishes perimeter.*

(iv) *Assume X is a Caccioppoli set and the measure of X is positive. Then either* $p(X) > p(Y)$ *or X is equivalent to a Caccioppoli set W such that the intersection of W and any straight line orthogonal to V is either empty or a segment.*

(v) *Assume X is convex and the measure of X is positive and finite. Then either* $p(X) > p(Y)$ *or X is a translate of Y.*

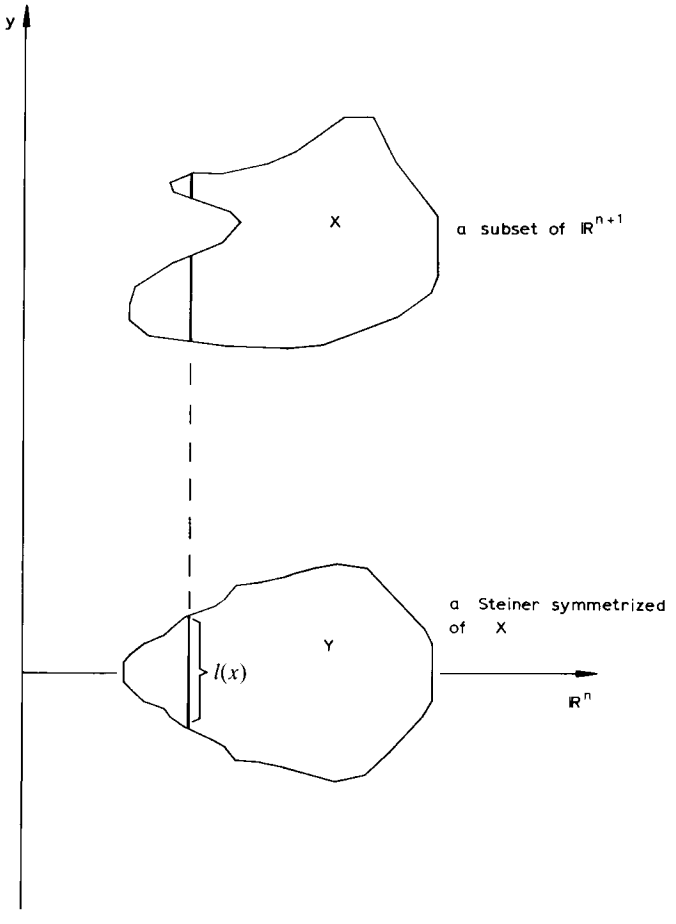

Figure 2.

Proof. Property (i) follows from the very definition of Steiner symmetrization. Property (ii) follows from Fubini's theorem. Incidentally, Property (ii) might be considered an analog of Cavalieri's principle.

The proofs of (iii) and (iv) require some lemmas. For convenience, throughout the remainder of this section we switch notations slightly: we replace $\mathbb{R}^n$ by $\mathbb{R}^{n+1}$ and label (x, y) the points in $\mathbb{R}^{n+1}$. Thus y is the last coordinate and x stands for $(x_1, \ldots, x_n)$, the array of coordinates no. 1 to n. Assume

$$X \text{ is a Caccioppoli subset of } \mathbb{R}^{n+1}, \tag{1}$$

and

$$Y \text{ is the Steiner symmetrized of } X \text{ with respect to } \mathbb{R}^n. \tag{2}$$

Here we think of $\mathbb{R}^n$ as the hyperplane in $\mathbb{R}^{n+1}$ whose equation reads $y = 0$. Define:

$$\text{sctn}_x(X) = \{ y \in \mathbb{R} : (x, y) \in X \} , \tag{3}$$

the intersection between X and the straight line orthogonal to $\mathbb{R}^n$ having foot at x, and

$$l(x) = \text{the one-dimensional measure of } \text{sctn}_x(X) , \tag{4}$$

where x is any point in $\mathbb{R}^n$. Clearly,

$$Y = \{(x, y) \in \mathbb{R}^{n+1} : x \in \text{projection of } X \text{ on } \mathbb{R}^n, |y| \leqslant \tfrac{1}{2} l(x)\} , \tag{5}$$

see fig. 2.

Lemma 1. *For every Borel subset B of $\mathbb{R}^n$,*

$$\textit{tot. var. of } (\partial/\partial y)\chi_X \textit{ on } B \times \mathbb{R} = \int_B \mathrm{d}x \textit{ perimeter of } \text{sctn}_x(X) , \tag{6}$$

where $(\partial/\partial y)\chi_X$ stands for the last component of grad χ_X *and perimeter stands for one-dimensional perimeter. Thus, for almost every x from $\mathbb{R}^n$, $\text{sctn}_x(X)$ is a Caccioppoli subset of $\mathbb{R}$.*

Proof. Since the measures involved in eq. (6) are *regular*, there is no handicap in assuming that B is *open*. Recall that the total variation of a regular scalar-valued Borel measure is itself a regular measure, and a nonnegative Borel measure is regular if and only if the *measure* of B equals $\inf\{\text{measure of } C : C \text{ open}, C \supseteq B\}$ for every Borel set B. The following is a key formula:

$$\begin{aligned} &\text{tot. var. of } (\partial/\partial y)\chi_X \text{ on } B \times \mathbb{R} \\ &\quad = \sup\left\{ \left| \int_{\mathbb{R}^{n+1}} \chi_X(\partial\phi/\partial y)\, \mathrm{d}x\, \mathrm{d}y \right| : \phi \text{ is a scalar-valued test function}, \right. \\ &\qquad\qquad \left. \text{sprt } \phi \subseteq B \times \mathbb{R}, \max |\phi| = 1 \right\} . \end{aligned}$$

We have

$$\left| \int_{-\infty}^{+\infty} \chi_X(x, y) \frac{\partial \phi}{\partial y}(x, y)\, \mathrm{d}y \right| \leqslant \max |\phi| \cdot \text{perimeter of } \text{sctn}_x(X)$$

by the very definition of perimeter, if x is any point from $\mathbb{R}^n$ and ϕ is any test

function. Consequently,

$$\text{tot. var. of } (\partial/\partial y)\chi_X \text{ on } B \times \mathbb{R} \leqslant \int_B \mathrm{d}x \text{ perimeter of } \mathrm{sctn}_x(X)\,. \tag{6a}$$

On the other hand, quite the same arguments used in the proof of item (viii), section 3.5, demonstrate the following two facts. First, total variations of partial derivatives decrease under mollifications. Thus

$$\text{tot. var. of } (\partial/\partial y)\chi_X \text{ on } B \times \mathbb{R} \geqslant \int_B \mathrm{d}x \int_{-\infty}^{+\infty} \left| \frac{\partial}{\partial y} u_\varepsilon \right| \mathrm{d}y\,,$$

where u_ε is defined by:

$$u_\varepsilon(x, y) = \int_{-\infty}^{+\infty} J_\varepsilon(y - t)\chi_X(x, t)\,\mathrm{d}t\,,$$

and J_ε is a mollifier, just a one-dimensional mollifier. Second,

$$\lim_{\varepsilon \downarrow 0} \int_{-\infty}^{+\infty} \left| \frac{\partial}{\partial y} u_\varepsilon(x, y) \right| \mathrm{d}y = \text{perimeter of } \mathrm{sctn}_x(X)$$

for every x such that the length of $\mathrm{sctn}_x(X)$ is finite, irrespective of whether $\mathrm{sctn}_x(X)$ is a Caccioppoli set or not. Therefore,

$$\text{tot. var. of } (\partial/\partial y)\chi_X \text{ on } B \times \mathbb{R} \geqslant \int_B \mathrm{d}x \text{ perimeter of } \mathrm{sctn}_x(X) \tag{6b}$$

by Fatou's theorem.

Inequalities (6a) and (6b) imply (6). □

Lemma 2. *Function l belongs to* $\mathrm{BV}(\mathbb{R}^n)$ *and*

$$\text{tot. var. of } (\partial/\partial x_i)\chi_X \text{ on } B \times \mathbb{R} \geqslant \text{tot. var. of } \partial l/\partial x_i \text{ on } B\,, \tag{7}$$

where $i = 1, 2, \ldots, n$, *and* $(\partial/\partial x_i)\chi_X$ *and* $\partial l/\partial x_i$ *denote the ith components of* grad χ_X *and* grad l, *respectively. Furthermore*

$$\text{tot. var. of } (\partial/\partial y)\chi_X \text{ on } B \times \mathbb{R} \geqslant 2m(B \cap \{x \in R^n : 0 < l(x) < \infty\})\,. \tag{8}$$

Here B stands for any Borel subset of $\mathbb{R}^n$.

Proof. The following formula

$$l(x) = \int_{-\infty}^{+\infty} \chi_X(x, y)\,\mathrm{d}y$$

and the very definitions of Caccioppoli sets and $\mathrm{BV}(\mathbb{R}^n)$ tell us immediately that l belongs indeed to $\mathrm{BV}(\mathbb{R}^n)$.

Let us prove inequality (7). Since the measures involved, $(\partial/\partial x_i)\chi_X$ and $\partial l/\partial x_i$, are *regular*, there is no handicap in assuming that B is *open*. We have clearly

$$\text{tot. var. of } (\partial/\partial x_i)\chi_X \text{ on } B \times \mathbb{R} \geqslant \left| \int_{\mathbb{R}^{n+1}} \chi_X(x, y) \frac{\partial \phi}{\partial x_i}(x)\, \mathrm{d}x\, \mathrm{d}y \right| .$$

Here ϕ is any scalar-valued smooth function such that sprt ϕ is compact and contained in B, $\max|\phi| = 1$ – a function of x only, independent of y. The right-hand side of the last inequality equals

$$\left| \int_{\mathbb{R}^n} l(x) \frac{\partial \phi}{\partial x_i}(x)\, \mathrm{d}x \right| ,$$

thus

$$\text{tot. var. of } (\partial/\partial x_i)\chi_X \text{ on } B \times \mathbb{R} \geqslant \left| \int_{\mathbb{R}^n} l(x) \frac{\partial \phi}{\partial x_i}(x)\, \mathrm{d}x \right| .$$

Inequality (7) follows, thanks to the arbitrariness of ϕ.

Inequality (8) is an immediate corollary of Lemma 1 and the isoperimetric inequality in one dimension, section 3.6. In fact, the latter gives

$$\text{perimeter of } \mathrm{sctn}_x(X) \geqslant 2$$

for every x such that $l(x)$, the length of $\mathrm{sctn}_x(X)$, is positive and finite. □

Lemma 3. *Denote by* $\boldsymbol{\alpha}$ *the vector-valued measure whose components* $\alpha_1, \ldots, \alpha_n$, α_{n+1} *are defined in the following way*:

$$\alpha_i(B) = \text{tot. var. of } (\partial/\partial x_i)\chi_X \text{ on } B \times \mathbb{R}, \quad i = 1, 2, \ldots, n, \tag{9a}$$

$$\alpha_{n+1}(B) = \text{tot. var. of } (\partial/\partial y)\chi_X \text{ on } B \times \mathbb{R} \tag{9b}$$

for any Borel subset B *of* $\mathbb{R}^n$. *Then*

$$\textit{perimeter of } X \geqslant \textit{tot. var. of } \boldsymbol{\alpha} \textit{ on } \mathbb{R}^n . \tag{10}$$

Proof. Consider the polar decomposition of grad χ_X. In other words, let μ be the *total variation of* grad χ_X and let $\boldsymbol{f}$ be the *unit vector field* such that

$$\text{grad } \chi_X \text{ on } E = \int_E \boldsymbol{f}\, \mu(\mathrm{d}x\, \mathrm{d}y)$$

for every Borel subset E of $\mathbb{R}^{n+1}$. An explicit representation of μ, as well as a

characterization of f, are available from item (xi), section 3.5, but irrelevant here. We merely need the existence of a polar decomposition, as follows from the Radon–Nikodym theorem. We have

$$\alpha_i(B) = \int_{B\times\mathbb{R}} |f_i|\ \mu(\mathrm{d}x\,\mathrm{d}y)\,,$$

if $i = 1, \ldots, n$, $n+1$ and B is a Borel subset of $\mathbb{R}^n$. Hence

$$|\boldsymbol{\alpha}(B)| \leqslant \int_{B\times\mathbb{R}} |\boldsymbol{f}|\ \mu(\mathrm{d}x\,\mathrm{d}y)\,,$$

via Minkowski inequality. As $\boldsymbol{f}$ is unitary,

$$|\boldsymbol{\alpha}(B)| \leqslant \mu(B\times\mathbb{R})$$

for every Borel subset B of $\mathbb{R}^n$. Inequality (10) follows. □

Lemma 4. *The Steiner symmetrized of X, Y, is a Caccioppoli set too. If B is any Borel subset of $\mathbb{R}^n$, then*

$$\textit{tot. var. of } (\partial/\partial x_i)\chi_Y \textit{ on } B\times\mathbb{R} = \textit{tot. var. of } \partial l/\partial x_i \textit{ on } B \tag{11}$$

for $i = 1, 2, \ldots, n$ and

$$\textit{tot. var. of } (\partial/\partial y)\chi_Y \textit{ on } B\times\mathbb{R} = 2m(B\cap\{x\in\mathbb{R}^n : 0 < l(x) < \infty\})\,. \tag{12}$$

In other words, equality holds in (7) *and* (8) *if X is replaced by Y.*

Proof. We showed in Lemma 2 that the function l is in $\mathrm{BV}(\mathbb{R}^n)$; then formula (5) and the very definition of Caccioppoli sets tell us that Y is indeed a Caccioppoli set.

A proof of (11) runs as follows. Since the measures involved, $(\partial/\partial x_i)\chi_Y$ and $\partial l/\partial x_i$, are *regular*, there is no handicap in assuming that B is *open*. Let ϕ be any scalar-valued smooth function such that sprt ϕ is compact and contained in $B\times\mathbb{R}$, $\max|\phi| = 1$. We have

$$\left|\int_{\mathbb{R}^n} \mathrm{d}x \int_{-l(x)/2}^{+l(x)/2} \frac{\partial\phi}{\partial x_i}\,\mathrm{d}y\right| \leqslant \text{tot. var. of } \partial l/\partial x_i \text{ on } B\,.$$

In fact,

$$\text{tot. var. of } \partial l/\partial x_i \text{ on } B = \int_B \left|\frac{\partial l}{\partial x_i}\right| \mathrm{d}x\,,$$

and

$$\int_{-l(x)/2}^{+l(x)/2} \frac{\partial \phi}{\partial x_i}(x, y)\,dy = \frac{\partial}{\partial x_i}\int_{-l(x)/2}^{+l(x)/2} \phi(x, y)\,dy - \tfrac{1}{2}\left[\phi(x, l(x)/2) + \phi(x, -l(x)/2)\right]\frac{\partial l}{\partial x_i}(x),$$

hence

$$\int_{\mathbb{R}^n} dx \int_{-l(x)/2}^{+l(x)/2} \frac{\partial \phi}{\partial x_i}\,dy = -\tfrac{1}{2}\int_{\mathbb{R}^n} \left[\phi(x, l(x)/2) + \phi(x, -l(x)/2)\right]\frac{\partial l}{\partial x_i}(x)\,dx,$$

consequently

$$\left|\int_{\mathbb{R}^n} dx \int_{-l(x)/2}^{+l(x)/2} \frac{\partial \phi}{\partial x_i}\,dy\right| \leqslant \int_{\text{projection of sprt } \phi \text{ on } \mathbb{R}^n} \left|\frac{\partial l}{\partial x_i}\right| dx$$

if l is continuously differentiable. If l is not continuously differentiable, but merely in $\mathrm{BV}(\mathbb{R}^n)$, replace l by a mollified version of l, use the arguments above, then pass to the limit. Observe that if J_ε is a mollifier and $l_\varepsilon = J_\varepsilon * l$, then

$$\int_{\mathbb{R}^n} dx \int_{-l_\varepsilon(x)/2}^{+l_\varepsilon(x)/2} \frac{\partial \phi}{\partial x_i}\,dy \to \int_{\mathbb{R}^n} dx \int_{-l(x)/2}^{+l(x)/2} \frac{\partial \phi}{\partial x_i}\,dy$$

as $\varepsilon \downarrow 0$ and

$$\int_{\text{projection of sprt } \phi \text{ on } \mathbb{R}^n} \left|\frac{\partial l_\varepsilon}{\partial x_i}\right| dx \leqslant \text{tot. var. of } \partial l/\partial x_i \text{ on } B$$

as ε is small enough.

Formula (5) gives

$$\int_{\mathbb{R}^{n+1}} \chi_Y \frac{\partial \phi}{\partial x_i}\,dx\,dy = \int_{\mathbb{R}^n} dx \int_{-l(x)/2}^{+l(x)/2} \frac{\partial \phi}{\partial x_i}\,dy,$$

thus the preceding matter yields

$$\left|\int_{\mathbb{R}^{n+1}} \chi_Y \frac{\partial \phi}{\partial x_i}\,dx\,dy\right| \leqslant \text{tot. var. of } \partial l/\partial x_i \text{ on } B.$$

We deduce

$$\text{tot. var. of } (\partial/\partial x_i)\chi_Y \text{ on } B \times \mathbb{R} \leqslant \text{tot. var. of } \partial l/\partial x_i \text{ on } B \tag{11a}$$

by the arbitrariness of ϕ. The reverse inequality

$$\text{tot. var. of } (\partial/\partial x_i)\chi_Y \text{ on } B \times \mathbb{R} \geqslant \text{tot. var. of } \partial l/\partial x_i \text{ on } B \tag{11b}$$

follows from Lemma 2. Inequalities (11a) and (11b) imply eq. (11).

In the derivation of (11b) the formula

$$\operatorname{sctn}_x(Y) = \begin{cases} [-l(x)/2, +l(x)/2] & \text{if } x \text{ belongs to the projection of } X \text{ on } \mathbb{R}^n, \\ \emptyset & \text{otherwise,} \end{cases}$$

a consequence of (5), has been taken into account. The same formula implies eq. (12), via Lemma 1.

Lemma 5. *Denote by* $\boldsymbol{\beta}$ *the vector-valued measure whose components* $\beta_1, \ldots, \beta_n$, β_{n+1} *are defined in the following way*:

$$\beta_i(B) = \textit{tot. var. of } (\partial/\partial x_i)\chi_Y \textit{ on } B \times \mathbb{R}, \quad i = 1, 2, \ldots, n, \tag{13a}$$

$$\beta_{n+1}(B) = \textit{tot. var. of } (\partial/\partial y)\chi_Y \textit{ on } B \times \mathbb{R} \tag{13b}$$

for every Borel subset B *of* $\mathbb{R}^n$. *Then*

$$\textit{perimeter of } Y = \textit{tot. var. of } \boldsymbol{\beta} \textit{ on } \mathbb{R}^n. \tag{14}$$

In other words, equality holds in (10) *if* X *is replaced by* Y.

Proof. Suppose for a moment that l is continuously differentiable. Then Lemma 4 gives

$$\text{tot. var. of } \boldsymbol{\beta} \text{ on } \mathbb{R}^n = \int_{\mathbb{R}^n} \sqrt{4\chi_{\{x:\, 0<l(x)<\infty\}} + |\operatorname{grad} l|^2}\, dx.$$

On the other hand, usual calculus rules give

$$\begin{aligned} &\int_{\mathbb{R}^n} dx \int_{-l(x)/2}^{+l(x)/2} \operatorname{div} \boldsymbol{f}\, dy \\ &\quad = \int_{\mathbb{R}^n} \left\{ \sum_{i=1}^{n} [f_i(x, l(x)/2) + f_i(x, -l(x)/2)]\, \tfrac{1}{2}\, \frac{\partial l}{\partial x_i} \right. \\ &\qquad \left. + f_{n+1}(x, l(x)/2) - f_{n+1}(x, -l(x)/2) \right\} dx \end{aligned}$$

if $\boldsymbol{f}$ is a test vector field. Consequently, Cauchy–Schwarz inequality yields

$$\left| \int_Y \operatorname{div} \boldsymbol{f}\, dx\, dy \right| \leqslant \max |\boldsymbol{f}| \times \text{tot. var. of } \boldsymbol{\beta} \text{ on } \mathbb{R}^n.$$

The last inequality continues to hold even if l is not continuously differentiable, but merely belongs to $\mathrm{BV}(\mathbb{R}^n)$. A relevant proof might consist in a suitable refinement of an argument used in Lemma 4.

Thus we have shown

$$\text{perimeter of } Y \leqslant \text{tot. var. of } \boldsymbol{\beta} \text{ on } \mathbb{R}^n .$$

The reverse inequality follows from Lemma 3. Equation (14) is proved. □

Proof of the main proposition (*Continued*). Proposition (iii) follows immediately from previous lemmas. In fact, definitions (9) and (13), Lemma 2 and Lemma 4 give

$$\alpha_i(B) \geqslant \beta_i(B) \geqslant 0 \tag{15}$$

for $i = 1, \ldots, n, n+1$ and any Borel subset B of $\mathbb{R}^n$. Then we must have

$$v(\boldsymbol{\alpha}, B) \geqslant v(\boldsymbol{\beta}, B) \tag{16}$$

for every Borel set B. Here we use the abbreviation: v(measure, set) = tot. var. of the measure on the set. In particular

$$v(\boldsymbol{\alpha}, \mathbb{R}^n) \geqslant v(\boldsymbol{\beta}, \mathbb{R}^n) ,$$

which implies

$$\text{perimeter of } X \geqslant \text{perimeter of } Y$$

via Lemma 3 and 5.

In order to prove Proposition (iv), we suppose

$$\text{perimeter of } X = \text{perimeter of } Y \tag{17}$$

and infer successively a number of consequences.

First,

$$v(\boldsymbol{\alpha}, B) = v(\boldsymbol{\beta}, B) \tag{18}$$

for every Borel set B. In fact, inequality (16) ensures that $v(\boldsymbol{\alpha}, \mathbb{R}^n) - v(\boldsymbol{\beta}, \mathbb{R}^n) \geqslant v(\boldsymbol{\alpha}, B) - v(\boldsymbol{\beta}, B) \geqslant 0$. Equation (17), Lemma 3 and Lemma 5 force $v(\boldsymbol{\alpha}, \mathbb{R}^n)$ to equal $v(\boldsymbol{\beta}, \mathbb{R}^n)$. Equation (18) follows.

Second,

$$\boldsymbol{\alpha} = \boldsymbol{\beta} . \tag{19}$$

In fact, let μ be an alias for the *total variation* of $\boldsymbol{\alpha}$. Let $\boldsymbol{a}$ and $\boldsymbol{b}$ the Radon–Nikodym derivatives of $\boldsymbol{\alpha} - \boldsymbol{\beta}$ and $\boldsymbol{\beta}$ with respect to μ, respectively, i.e.,

$$\boldsymbol{\alpha}(B) - \boldsymbol{\beta}(B) = \int_B \boldsymbol{a}(x)\mu(\mathrm{d}x)\,, \qquad \boldsymbol{\beta}(B) = \int_B \boldsymbol{b}(x)\mu(\mathrm{d}x)$$

for every Borel set B. The following equation:

$$|\boldsymbol{\alpha}(B) - \boldsymbol{\beta}(B)|^2 + |\boldsymbol{\beta}(B)|^2 = |\boldsymbol{\alpha}(B)|^2 - 2\langle \boldsymbol{\beta}(B), \boldsymbol{\alpha}(B) - \boldsymbol{\beta}(B)\rangle$$

and inequalities (15) give

$$|\boldsymbol{a}(B) - \boldsymbol{\beta}(B)|^2 + |\boldsymbol{\beta}(B)|^2 \leqslant |\boldsymbol{\alpha}(B)|^2\,.$$

Obviously $|\boldsymbol{\alpha}(B)| \leqslant \mu(B)$. Hence

$$\frac{1}{\mu(B)}\left[\left|\int_B \boldsymbol{a}(x)\mu(\mathrm{d}x)\right|^2 + \left|\int_B \boldsymbol{b}(x)\mu(\mathrm{d}x)\right|^2\right]^{1/2} \leqslant 1$$

for every Borel set B such that $\mu(B)$ is positive. It is an easy matter to deduce that

$$|\boldsymbol{a}(x)|^2 + |\boldsymbol{b}(x)|^2 \leqslant 1$$

for μ-almost every x, i.e., the set of points x for which the last inequality fails is annihilated by μ. See Rudin (1966, Theorem 1.40) for details. Equation (18) tells us that μ is the *total variation* of $\boldsymbol{\beta}$ too, consequently $|\mathbf{b}(x)| = 1$ for μ-almost every x. In conclusion, $\boldsymbol{a}(x) = 0$ for μ-almost every x, an equivalent form of (19).

Third,

$$\text{perimeter of } \mathrm{sctn}_x(X) = 2 \tag{20}$$

for almost every point x such that $0 < l(x) < \infty$. In fact, eq. (19) yields in particular $\alpha_{n+1} = \beta_{n+1}$. Definitions (9b) and (13b), Lemmas 1 and 4 then give

$$\int_B \mathrm{d}x \text{ perimeter of } \mathrm{sctn}_x(X) = 2m(B \cap \{x \in \mathbb{R}^n : 0 < l(x) < \infty\})$$

for every Borel set B. Observe that the perimeter of $\mathrm{sctn}_x(X)$ is zero if x is such that $l(x) = 0$; on the other hand, the set of points x such that $l(x) = \infty$ has measure zero. Equation (20) follows.

Fourth

$$\mathrm{sctn}_x(X) \text{ is equivalent to a segment} \tag{21}$$

for almost every point x such that $0 < l(x) < \infty$. This follows from (20) and the proposition of section 3.6.

In conclusion, adding to and deleting from X appropriate sets of measure zero produce a set W such that for every x either $\mathrm{sctn}_x(W)$ is empty or $\mathrm{sctn}_x(W)$ is a segment. Property (iv) is demonstrated.

A proof of (v) runs as follows. As the boundary of a convex subset of $\mathbb{R}^{n+1}$ has $(n+1)$-dimensional measure zero, set X in question and the closure of X have the same perimeter, thus there is no handicap in additionally assuming, e.g., that X is closed. We have

$$X = \{(x, y) \in \mathbb{R}^{n+1} : x \in B, \mathrm{fl}(x) \leq y \leq \mathrm{cl}(x)\} , \tag{22}$$

where B, the basement, is a convex closed subset of $\mathbb{R}^n$; fl, the floor, is convex and cl, the ceiling, is concave – both are real-valued locally Lipschitz continuous functions. Incidentally, a convex set, whose measure is positive and finite, is bounded, thus B, fl and cl are bounded. Formula (22) gives

$$\mathrm{sctn}_x(X) = \begin{cases} [\mathrm{fl}(x), \mathrm{cl}(x)] & \text{if } x \in B , \\ \emptyset & \text{otherwise} , \end{cases}$$

consequently

$$l(x) = \begin{cases} \mathrm{cl}(x) - \mathrm{fl}(x) & \text{if } x \in B , \\ 0 & \text{otherwise} . \end{cases}$$

Consider Y, the Steiner symmetrized of X with respect to $\mathbb{R}^n$. Clearly,

$$Y = \{(x, y) \in \mathbb{R}^{n+1} : x \in B, |y| \leq l(x)/2\} , \tag{23}$$

a convex closed bounded set. The perimeter of Y may be evaluated via Lemma 5. As a matter of fact, usual calculus rules suffice: Y is nice enough for elementary forms of Gauss–Green formulas to work. In one way or another one deduces from (23) that

$$\text{perimeter of } Y = 2J(l/2) + \text{a remainder} , \tag{24}$$

where J is defined by:

$$J(\phi) = \int_B j(\mathrm{grad}\ \phi)\, \mathrm{d}x , \tag{25a}$$

$$j(\xi_1, \ldots, \xi_n) = \sqrt{1 + \xi_1^2 + \cdots + \xi_n^2} , \tag{25b}$$

and the remainder term comes out due to possible jump discontinuities of l across the boundary of B.

By similar arguments one deduces from (22) that

$$\text{perimeter of } X = J(\mathrm{fl}) + J(\mathrm{cl}) + \text{a remainder} . \tag{26}$$

Importantly, the remainder term in (26) is quite the same as the remainder term in (24).

Thus

$$\text{perimeter of } X - \text{perimeter of } Y = J(\text{fl}) + J(\text{cl}) - 2J(l/2)\,.$$

Observe that grad$(l/2)$ is the midpoint of the segment joining grad$(-\text{fl})$ and grad(cl). Formulas (25a) and (25b) show that functional J is *even* and function j is *strictly convex*. Hence

$$J(\text{fl}) + J(\text{cl}) \geqslant 2J(l/2)\,,$$

and equality holds if and only if

$$\text{grad}(-\text{fl}) = \text{grad}(\text{cl})\,.$$

In conclusion,

$$\text{perimeter of } X \geqslant \text{perimeter of } Y$$

and equality holds if and only if

$$\text{fl} + \text{cl} = \text{constant}\,.$$

Assertion (v) is demonstrated. The proof is complete. □

3.9. Identification of minimizers

Let G be a minimizing set, i.e., a solution to variational problem (1a–d) of section 3.7. In this section we show that G *is equivalent to a ball.*

Consider any hyperplane V passing through the center of Ball – the big ball involved in condition (1c), section 3.7. Consider the Steiner symmetrized of G with respect to V, L say. By Proposition (ii), section 3.8, $m(L) = m(G)$; clearly, $L \subseteq$ Ball, thus L is a competing set. By Proposition (iii), section 3.8, the perimeter of L is less than or equal to the perimeter of G. Since G is a minimizer, we must have $p(L) = p(G)$. By Proposition (iv), section 3.8, G is equivalent to a Caccioppoli set whose intersection with any straight line orthogonal to V is either empty or a segment.

But V was arbitrary. Thus G is equivalent to a set whose intersection with any straight line is either empty or a segment: a *convex* set.

Having convexity in hand, use again the argument above, but now invoke Propositions (i) and (v), section 3.8. Deduce that G is equivalent to a *convex* set which is *symmetric* about *every* hyperplane passing through a certain point: a *ball*.

The variational problem (1a–d) of section 3.7, is completely solved.

3.10. Conclusions

The proof of the isoperimetric theorem can be accomplished in the following way.

Consider a solution of the variational problem (1a–d) of section 3.7, and name it B. In section 3.7 we proved that B exists, and in section 3.9 we learned that B is equivalent to a ball. If G is any competing set we must have $p(G) \geqslant p(B)$. On the other hand, $m(G) = m(B)$, and

$$p(B) = n\kappa_n^{1/n}[m(B)]^{1-1/n}$$

since B is equivalent to a ball. Recall indeed that equivalent Caccioppoli sets have the same perimeter, and κ_n is the measure of the n-dimensional unit ball, see formula (2) of section 3.1. Thus any competing set G satisfies

$$p(G) \geqslant n\kappa_n^{1/n}[m(G)]^{1-1/n}\,, \tag{1}$$

the isoperimetric inequality. Observe that inequality (1) does not depend on Ball, the big ball constraining the competing sets, see condition (1c) of section 3.7. Thus any Caccioppoli set G satisfies inequality (1), provided G is bounded.

We want to show that *any* measurable subset G of R^n satisfies inequality (1), provided $m(G)$ is *finite*, irrespective of whether G is bounded or not. We can restrict our attention to Caccioppoli sets, since inequality (1) holds trivially if G is not Caccioppoli. Since any Caccioppoli set G is associated with a sequence of smooth sets E_k such that $p(E_k) \to p(G)$ and $m(E_k) \to m(G)$, see Proposition (ix), section 3.5, there is no handicap in carrying a proof for *smooth* sets.

Let G be a nice subset of $\mathbb{R}^n$ and let the measure of G be finite. Denote by B_r any ball of radius r. The results we have in hand tell us that

$$p(G \cap B_r) \geqslant n\kappa_n^{1/n}[m(G \cap B_r)]^{1-1/n}\,, \tag{2}$$

for every r, since the intersection of G and B_r is bounded. Clearly

$$m(G \cap B_r) \to m(G) \quad \text{as } r \to +\infty\,. \tag{3}$$

On the other hand, $p(G \cap B_r)$ agrees with the customary $(n-1)$-dimensional measure of $\partial(G \cap B_r)$ since we are dealing with innocent sets. As the boundary of $G \cap B_r$ is included in the union of ∂G and $G \cap \partial B_r$,

$$p(G \cap B_r) \leqslant p(G) + H_{n-1}(G \cap \partial B_r)\,. \tag{4}$$

Formulas for evaluating integrals in polar coordinates – Federer coarea formula too – give

$$m(G) = \int_0^{+\infty} H_{n-1}(G \cap \partial B_r)\,\mathrm{d}r\,.$$

In particular, the right-hand side is finite – the assumption about $m(G)$ comes in here. Hence a sequence of radii r_k exists such that $r_k \to +\infty$ and

$$H_{n-1}(G \cap \partial B_r) \to 0 \tag{5}$$

as r goes to $+\infty$ along that sequence. Putting together (2)–(5) yields the target inequality (1).

It remains to show that the subsets of $\mathbb{R}^n$ that have finite measure and render (1) an equality are equivalent to balls. This can be done by repeating verbatim the arguments of section 3.9. Note that boundedness was not involved in sections 3.8 and 3.9. This completes the proof. □

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CHAPTER 1.4

Stability of Geometric Inequalities

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Contents

1. Introduction 127
   1.1. Description of the problem 127
   1.2. Notations 129
   1.3. Metrics on $\mathbb{C}^d$ 130
2. Bonnesen's inequality and its consequences 130
   2.1. The isoperimetric inequality 130
   2.2. Mixed area inequalities 131
3. The Brunn–Minkowski inequality 133
   3.1. The problem 133
   3.2. Stability estimates for the Brunn–Minkowski inequality 134
   3.3. Central symmetrization 135
4. Inequalities for mixed volumes 137
   4.1. Minkowski's inequality for the mixed volume $V(K, \ldots, K, L)$ 137
   4.2. Minkowski's inequality for the mixed volume $V(K, L, B^d, \ldots, B^d)$ 138
   4.3. The Aleksandrov–Fenchel inequality 139
5. Inequalities for the mean projection measures and the isoperimetric inequality 140
   5.1. Inequalities between two mean projection measures 140
   5.2. The isoperimetric inequality 141
   5.3. The inequalities of Urysohn and Bieberbach 143
6. Cap bodies, bodies of constant width, equichordal sets, packings and coverings 144
   6.1. Stability problems related to cap bodies 144
   6.2. Bodies of constant width 144
   6.3. Equichordal sets 146
   6.4. Packings and coverings 146
7. Projections of convex bodies 146
   7.1. The dependence of convex bodies on their projections 147
   7.2. Projection bodies 147
References 148

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HANDBOOK OF CONVEX GEOMETRY
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Most geometric inequalities concerning convex bodies have the property that the occurrence of the equality sign characterizes geometrically significant objects, like balls, ellipsoids or simplexes. This fact suggests the following stability problem: If for some convex body the given inequality is satisfied so that it is not very different from an equality what can be said about the deviation of the body from these objects? Problems of this kind appear already in the work of Minkowski and Bonnesen but have been investigated more systematically only in recent times. The present article is a survey of stability results for geometric inequalities and provides pertinent references. Because of space limitations it was not possible to include complete proofs but occasionally some indications regarding the methods of proof will be provided.

1. Introduction

1.1. Description of the problem

Throughout this survey, $\mathbf{E}^d$ denotes the Euclidean d-dimensional space and $\mathbb{C}^d$ the class of all *convex bodies* in $\mathbf{E}^d$, i.e., of all non-empty compact convex subsets of $\mathbf{E}^d$. It is always assumed that $d \geq 2$ (although some of the results that will be discussed are trivially true for $d = 1$).

Let Φ be a real-valued function on some subclass $\boldsymbol{S}$ of $\mathbb{C}^d$ and let us assume that all convex bodies $K \in \boldsymbol{S}$ satisfy the geometric inequality

$$\Phi(K) \geq 0 . \tag{1}$$

For example, if $d = 2$, $\boldsymbol{S} = \mathbb{C}^2$, and $\Phi(K) = P(K)^2 - 4\pi A(K)$, where $A(K)$ denotes the area and $P(K)$ the perimeter of K, then (1) is the isoperimetric inequality

$$P(K)^2 - 4\pi A(K) \geq 0 . \tag{2}$$

Or, assuming again that $d = 2$ and setting $\alpha_0 = (\pi - \sqrt{3})/2$ we may define $\Phi(K)$ by $\Phi(K) = A(K) - \alpha_0$ and let $\boldsymbol{S}$ denote the class of all plane convex bodies of constant width 1. Then (1) becomes the Lebesgue–Blaschke inequality

$$A(K) - \alpha_0 \geq 0 . \tag{3}$$

Let now $\boldsymbol{S}^=$ denote the class of all those elements L of $\boldsymbol{S}$ with the property that $\Phi(L) = 0$. Thus in the example concerning (2) the class $\boldsymbol{S}^=$ consists of all circular discs and in the second example $\boldsymbol{S}^=$ consists of the Reuleaux triangles of width 1. Informally stated, the stability problem associated with the geometric inequality (1) is the problem to estimate (in terms of some metric or similar deviation measure on $\boldsymbol{S}$) the deviation of a $K \in \boldsymbol{S}$ from the members of $\boldsymbol{S}^=$ if (1) is close to

being an equality in the sense that $\Phi(K) \leqslant \varepsilon$ (for some $\varepsilon \geqslant 0$). Hence, in the case of inequality (2) the problem is to determine how much a $K \in \mathbb{C}^2$ deviates from a circular disc if the isoperimetric deficit $P(K)^2 - 4\pi A(K)$ is at most ε. In the case of inequality (3) one meets the problem to estimate the deviation of a convex body (in $\mathbf{E}^2$) from a Reuleaux triangle if its area is known to be at most $\alpha_0 + \varepsilon$.

To state the problem more formally we call a real-valued function $\theta(x)$ that is defined on $[0, \infty)$ a *stability function* if $\theta(x) > 0$ for $x \neq 0$ and

$$\lim_{x \to 0+} \theta(x) = \theta(0) = 0 . \tag{4}$$

Let μ be a metric on $\boldsymbol{S}$. The *stability problem associated with a geometric inequality* [of the type (1)] consists of finding a stability function θ such that for any $\varepsilon \geqslant 0$ the condition

$$\Phi(K) \leqslant \varepsilon \tag{5}$$

implies the existence of an $L \in \boldsymbol{S}^=$ such that

$$\mu(K, L) \leqslant c(K)\theta(\varepsilon) . \tag{6}$$

The factor $c(K)$ may depend on K and d, but for fixed d and any given two numbers r_0, R_0 with $0 < r_0 < R_0$ it should be a bounded function on the class of all K whose inradius is at least r_0 and whose circumradius is at most R_0. With very few exceptions the actually occurring stability function will be of the form x^λ with some $\lambda > 0$.

We note that the stability statement that (5) implies (6) is actually a strengthened form of the original inequality (1). In other words, (1) is a consequence of the implication (5) $\Rightarrow$ (6). To see this let us assume that (5) $\Rightarrow$ (6) and that (1) were not true. Then, (5) is satisfied with $\varepsilon = 0$ and from (4) and (6) it would follow that $K = L$; but this would yield $\Phi(K) = \Phi(L) = 0$ in contradiction to the assumption that (1) is not satisfied. The situation becomes even more transparent if we assume that $c(K) \neq 0$ and that θ is a strictly increasing function. Choosing $\varepsilon = \Phi(K)$ we see that (6) can be written in the form

$$\Phi(K) \geqslant \theta^{-1}(\mu(K, L)/c(K)) . \tag{7}$$

This strengthened form of (1) is to be interpreted in the sense that for any $K \in \boldsymbol{S}$ there is an $L \in \boldsymbol{S}^=$ such that (7) holds. Conversely, any inequality of the type (7) leads immediately to a corresponding stability statement. An inequality of the form (7) will occasionally be referred to as a *stability inequality* corresponding to (1) and we will frequently take advantage of the possibility to formulate stability results either as an implication of the form (5) $\Rightarrow$ (6) or by stating a corresponding stability inequality.

Already Minkowski (1903) has noticed that if one wishes to prove a stability inequality of the form (7) (with a continuous θ) one only has to consider convex

bodies having strong regularity properties. The general case, including the conditions for equality in the original inequality (1), are then obtained by standard approximation techniques. Similarly, it would suffice to prove (7) for polytopes.

There are many variations and generalizations of the stability problem just formulated. One of the most frequently occurring generalization concerns inequalities of the form

$$\Phi(K_1, K_2) \geqq 0,$$

where Φ is defined for $K_1 \in \boldsymbol{S}_1$, $K_2 \in \boldsymbol{S}_2$ with $\boldsymbol{S}_1 \subset \mathbb{C}^d$, $\boldsymbol{S}_2 \subset \mathbb{C}^d$. In most situations the case of equality is characterized either by a statement describing the classes to which K_1 and K_2 must belong to, or, more frequently, by a relationship that must exist between K_1 and K_2. For example, an often appearing condition is that K_1 and K_2 should be homothetic. In order to formulate a corresponding stability problem one must of course somehow quantify the strength of this relationship. Instead of describing here a formal theory to deal with such situations we rather will explain the pertinent features in the actually occurring concrete cases.

There exists an extensive literature on improvements of classical inequalities of the form (1) where 0 is replaced by some expression that is, more or less obviously, at least 0. For example, many of the so-called "Bonnesen-style inequalities" in higher dimensions are of this type (see Osserman 1979 and Diskant 1989). For the purpose of the present article we will not consider such improvements as stability results unless there appears an expression on the right-hand side of (1) that indicates the deviation from the elements of $\boldsymbol{S}^=$ in terms of some metric or similar deviation measure.

1.2. Notations

The following notations will be used consistently. For any $K \in \mathbb{C}^d$ we let $V(K)$ denote the volume and $S(K)$ the surface area of K. As usual, $W_i(K)$ signifies the ith mean projection measure (Quermaßintegral) of K, and $V(K_1, \ldots, K_d)$ denotes the mixed volume of d convex bodies $K_1, \ldots, K_d$ from $\mathbb{C}^d$. The circumradius and inradius of K will be denoted by $R(K)$ and $r(K)$, respectively. $D(X)$ denotes the diameter of a subset X of $\mathbf{E}^d$. The unit sphere in $\mathbf{E}^d$ centered at the origin o of $\mathbf{E}^d$ will be denoted by S^{d-1} and the corresponding unit ball by B^d. We let σ_d denote the surface area and κ_d the volume of B^d. As a linguistic simplification we refer to plane convex bodies as *convex domains* and occasionally we say "circle" when we actually mean a circular disc. If K is a convex domain we often write $A(K)$ [instead of $V(K)$] for the area of K and $P(K)$ [instead of $S(K)$] for the perimeter of K.

We frequently use the standard definitions and theorems of the theory of convex sets without citing in each case particular references. In such cases the pertinent information can be obtained from well-known monographs dealing with this subject, like the books of Bonnesen and Fenchel (1934) or Leichtweiß

(1980). Important references for the general area of geometric inequalities are the books of Burago and Zalgaller (1988), and Mitrinović, Pečarić and Volenec (1989).

1.3. Metrics on $\mathbb{C}^d$

As suitable metrics on $\mathbb{C}^d$ only two possibilities will be of interest for our purpose, namely the Hausdorff metric, which will be denoted by δ, and the case $p=2$ of the L_p-metric. If $K, L \in \mathbb{C}^d$ and $p>0$ the *L_p-distance* between two convex bodies is defined by:

$$\delta_p(K, L) = \left(\int_{S^{d-1}} |h_K(u) - h_L(u)|^p \, d\sigma(u) \right)^{1/p},$$

where h_K and h_L denote the respective support functions of K and L. The symbol $d\sigma(u)$ indicates integration with respect to the (Lebesgue) surface area measure on S^{d-1}. Vitale (1985) has shown that δ and δ_p are related by the inequalities

$$c_{d,p} D(K \cup L)^{1-d} \delta(K, L)^{p+d-1} \leq \delta_p(K, L)^p \leq \sigma_d \delta(K, L)^p, \tag{8}$$

where $c_{d,p} = \sigma_{d-1}\Gamma(p+1)\Gamma(d-1)/\Gamma(d+p)$.

For the important special case when L is a ball and $p=2$, Groemer and Schneider (1991) have shown the following improvement of (8):

If L is a ball of radius ρ whose center is contained in K, and if (for some $c>0$) $\delta(K, L) \leq c\rho$, then

$$k_d(c) \rho^{-(d-1)/2} \delta(K, L)^{(d+3)/2} \leq \delta_2(K, L)^2, \tag{9}$$

where

$$k_d(c) = \frac{\sigma_{d-1}}{(d^2-1)(d+3)} \min\left\{ \frac{3}{\pi^2 d(d+2)2^d}, \frac{16(c+2)^{(d-3)/2}}{(c+1)^{d-2}} \right\}. \tag{10}$$

For the formulation of some stability results it is often convenient to introduce the *translative Hausdorff distance*. If $K, L \in \mathbb{C}^p$ it is defined by

$$\tilde{\delta}(K, L) = \inf\{\delta(K, L + x): x \in \mathbf{E}^d\}.$$

Concerning other types of metrics that have recently emerged in connection with stability problems see Arnold and Wellerding (1992) and Schneider (1992).

2. Bonnesen's inequality and its consequences

2.1. The isoperimetric inequality

In a certain sense one can interpret a well-known inequality of Bonnesen as a

stability result for the isoperimetric inequality (2). Bonnesen's inequality can be written as

$$P(K)^2 - 4\pi A(K) \geqslant 4\pi(\rho^0(K) - \rho_0(K))^2 , \tag{11}$$

where $\rho^0(K)$ and $\rho_0(K)$ are, respectively, the radii of the outer and the inner circle of an annulus of minimal width that contains the boundary of K. Clearly, $\rho^0(K) - \rho_0(K)$ can be considered as a kind of deviation measure of K from a circle. Since we have obviously $\rho^0(K) - \rho_0(K) \geqslant 2\delta(K, C)$, where C is a circle halfway between the inner and the outer circle of the annulus, it follows that (11) implies

$$P(K)^2 - 4\pi A(K) \geqslant 16\pi\delta(K, C)^2 . \tag{12}$$

Equation (12) is already a very sharp stability estimate of the type (7). The inequality (11) has been proved by Bonnesen (1929). He also proved a similar inequality with $\rho^0(K) - \rho_0(K)$ replaced by $R(K) - r(K)$ and 4π replaced by π^2. An earlier (and weaker) form of an inequality of this kind is due to Bernstein (1905). For further investigations regarding this kind of inequalities see Bonnesen and Fenchel (1934), Boll (1939), Knothe (1957), Osserman (1979), Diskant (1989), Gage (1990), Fuglede (1991), and Wallen (1991). Using the Fourier series development of the support function of K one can deduce the following stability inequality that is of the same type as (12) but involves the L_2-distance between K and a disc C that has diameter $P(K)/\pi$ and center at the Steiner point of K:

$$P(K)^2 - 4\pi A(K) \geqslant 6\pi\delta_2(K, C)^2 .$$

This inequality is best possible in the sense that equality may hold without both sides being zero. In fact there are domains of this kind with C being the unit circle and $\delta_2(K, C)$ being positive but arbitrarily small. This shows that the exponent 2 in any inequality of the form (12) is best possible even if one permits instead of 16π a factor that may depend on K (but is bounded away from 0 for all K with $r(K)$ above a given positive bound). For more details and various related results of this type, see Groemer (1990b) and chapter 4.8, sections 2.1 and 2.2. If one employs instead of Fourier series spherical harmonics one can prove analogues of such inequalities for the d-dimensional case; see sections 5.1, 5.2 and chapter 4.8, section 3.7. There exist various generalizations of Bonnesen's inequality for convex bodies in $\mathbf{E}^3$ or even $\mathbf{E}^d$. Pertinent references concerning this subject area are Osserman (1979), Diskant (1989), Boll (1939), Knothe (1957), and Wallen (1991). But most of these inequalities cannot be interpreted as stability statements in the sense as we use this term here (although they may serve as a basis for proving such theorems). See also section 5.2.

2.2. Mixed area inequalities

Let $A(K, L)$ denote the *mixed area* (i.e., the mixed volume $V(K, L)$ in the case

$d=2$) of K and L. As a generalization of the isoperimetric inequality one can show that

$$A(K, L)^2 - A(K)A(L) \geqslant 0 ,$$

where equality holds if and only if K and L are homothetic. Minkowski, not only proved this inequality but he also established a stability version of it of the following kind (Minkowski 1903, p. 249).

Let K' and L' be homothetic copies of K and L (respectively) such that $A(K') = A(L') = 1$ and both K' and L' have o as centroid. Let h_1, h_2 be the respective support functions of K' and L'. As a measure for strength of the homothetic relationship between K and L Minkowski introduces the expression $H_0 = \max\{h_1(u)/h_2(u): u \in S^1\}$ and he essentially proves that

$$A(K, L)^2 - A(K)A(L) \geqslant A(K)A(L) \frac{(H_0 - 1)^4}{972\, H_0^3} . \tag{13}$$

If r is the radius of a circle that is contained in both K' and L', and if R is the radius of a circle that contains both K' and L', then it is clear that $H_0 \leqslant R/r$ and

$$\begin{aligned} H_0 - 1 &= \max\{(h_1(u) - h_2(u))/h_2(u): u \in S^1\} \\ &\geqslant \max\{(h_1(u) - h_2(u))/R: u \in S^1\} . \end{aligned}$$

Interchanging, if necessary, the roles of K' and L' one can draw the conclusion that $H_0 - 1 \geqslant \delta(K', L')/R$. Hence (13) implies the following more conventional form of a stability result that uses $\delta(K', L')$ as a deviation measure:

$$A(K, L)^2 - A(K)A(L) \geqslant A(K)A(L) \frac{r^3}{972\, R^7} \delta(K', L')^4 . \tag{14}$$

However, this stability inequality is mainly of historic interest since it is apparently the first stability result for geometric inequalities that has been published. Substantially better results are now available. These can be derived from the following natural generalization of Bonnesen's inequality (11).

If one defines

$$\begin{aligned} \Lambda &= \min\{t: L \subset tK + p, t > 0, p \in \mathbf{E}^2\} , \\ \lambda &= \max\{s: sK + q \subset L, s > 0, q \in \mathbf{E}^2\} , \end{aligned}$$

then

$$A(K, L)^2 - A(K)A(L) \geqslant \tfrac{1}{4} A(K)^2 (\Lambda - \lambda)^2 .$$

The first published proof of this inequality appears to be due to Blaschke

(1921) (who mentions a prior unpublished proof of Bonnesen). Further proofs can be found in the book of Blaschke (1949) (where integral geometric ideas are used) and in the articles of Boll (1939), Flanders (1968), and Wallen (1987) (who shows how Wirtinger's inequality can be employed for this purpose). From this mixed area inequality it is easy to deduce (cf. Groemer 1990b) the following inequality which is a stability inequality of the usual type and enables one to better compare the generalized inequality of Bonnesen with Minkowski's result (14).

$$A(K, L)^2 - A(K)A(L) \geqslant \frac{1}{4} \frac{A(K)^2 A(L)}{D(K)^2} \delta(K', L')^2 . \tag{15}$$

K' and L' denote again suitable homothetic copies of K and L (respectively) of unit area.

As before, there are similar inequalities for δ_2 that can be obtained by using Fourier series and these methods generalize to the d-dimensional case. In section 4.2 generalizations of this type are discussed and section 4.1 deals with other extensions of (15) to $\mathbf{E}^d$. See also chapter 4.8, sections 3.7 and 3.8.

3. The Brunn–Minkowski inequality

3.1. The problem

If K, L is any pair of convex bodies in $\mathbf{E}^d$ and $x \in [0, 1]$ the Brunn–Minkowski inequality can be written in the form

$$V(xK + (1 - x)L)^{1/d} \geqslant xV(K)^{1/d} + (1 - x)V(L)^{1/d} . \tag{16}$$

Under the assumption that $V(K) > 0$, $V(L) > 0$ and $0 < x < 1$ equality holds for a particular value of x if and only if K and L are homothetic. On the other hand, if K and L are homothetic, then (16) is valid with equality for all $x \in [0, 1]$. The stability questions regarding (16) are easier to describe and there is essentially no loss of generality if one assumes that $V(K) = V(L) = 1$. Under this assumption (16) simplifies to $V(xK + (1 - x)L) \geqslant 1$ and the problem is to estimate the strength of the homothetic relationship (actually a translative relationship because of the volume restriction) between K and L if

$$V(xK + (1 - x)L) \leqslant 1 + \varepsilon \tag{17}$$

(or $V(xK + (1 - x)L)^{1/d} \leqslant 1 + \varepsilon$).

This problem can be interpreted in two different ways. One may assume that (17) holds for all $x \in [0, 1]$, or one may impose the weaker assumption that (17) holds for only one particular value of $x \in (0, 1)$.

3.2. *Stability estimates for the Brunn–Minkowski inequality*

Corresponding to the two types of problems just mentioned two stability estimates have been proved. The first one is due to Diskant (1973) (also presented in Diskant 1989), the other has been established by Groemer (1988b). Groemer's result has the additional feature that the pertinent constants are explicitly evaluated and the necessary translations are described.

Diskant's result can be formulated as follows: Under the assumption that $V(K) = V(L) = 1$ there exist an ε_0 and a c, both depending only on d and the circumradius and inradius of K, with the property that if (17) holds for all $x \in [0, 1]$ and some ε with $0 \leqslant \varepsilon < \varepsilon_0$, then

$$\tilde{\delta}(K, L) \leqslant c\varepsilon^{1/d}. \tag{18}$$

($\tilde{\delta}$ is the translative Hausdorff distance as defined in section 1.2.)

To prove this inequality Diskant proceeds by first establishing a stability estimate for an inequality for mixed volumes of the type discussed in section 4.1 and by deriving the desired result from this estimate. In further papers (Diskant 1975, 1982, and 1989) he proved analogues of his result for the surface area (if $d \geqslant 5$) and for all the mean projection measures W_k if one of the bodies is a ball, or if $d \geqslant 5$, $k \geqslant 4$ and an additional condition regarding the projections of the bodies onto a suitable hyperplane is satisfied. A strong stability version of the Brunn–Minkowski inequality for W_{d-2} has recently been proved by Groemer (1992a), and subsequently Schneider (1992) has established stability results for all W_k with $1 \leqslant k \leqslant d-2$, and even more general mixed volumes, without any restrictions on the convex bodies.

The principal stability result regarding (17) of Groemer is the following statement: Suppose that $V(K) = V(L) = 1$ and that the respective diameters of K and L are at most D_0. Let L' denote the translate of L whose centroid coincides with the centroid of K. Furthermore, let an $\varepsilon \geqslant 0$ be given and assume that (17) holds for a particular value $x = x_0$ where $0 < x_0 < 1$. Then,

$$\delta(K, L') \leqslant \lambda_d \left(\frac{1}{\sqrt{x_0(1-x_0)}} + 2 \right) D_0 \varepsilon^{1/(d+1)} \tag{19}$$

with $\lambda_d = 3^{(d-1)/d} 2^{(d+2)/(d+1)} d^{d/d+1} < 6d$.

The proof of this theorem is obtained by applying strengthened versions of classical inequalities in the context of a previously known proof of the Brunn–Minkowski theorem. We finally cite from Groemer (1988b) a stability inequality corresponding to the Brunn–Minkowski inequality (16) that is not restricted to normalized bodies.

Let K, L be two convex bodies in $\mathbf{E}^d$ with nonempty interiors, and let D_0 be such that $D(K) \leqslant V(K)^{1/d} D_0$ and $D(L) \leqslant V(L)^{1/d} D_0$. Let K', L' be homothetic copies of K, L (respectively) so that K' and L' have coinciding centroids and $V(K') = V(L') = 1$. Furthermore, set

$$M = \max\{V(K)^{1/d}, V(L)^{1/d}\}$$

and

$$m = \min\{V(K)^{1/d}, V(L)^{1/d}\} .$$

Then, for all $x \in [0, 1]$,

$$V(xK + (1-x)L)^{1/d} \geqslant xV(K)^{1/d} + (1-x)V(L)^{1/d} + \omega\delta(K', L')^{d+1} ,$$

where

$$\omega = m\left(\gamma_d\left(\frac{M}{m}\,\frac{1}{\sqrt{x(1-x)}} + 2\right)D_0\right)^{-(d+1)}$$

and

$$\gamma_d = (3 + 2^{-13})3^{-1/d}2^{(d+2)/(d+1)}d < 6.00025\, d .$$

If $d = 2$, stronger results can be derived from (15). For example, Groemer (1990b) has shown that if $K, L \in \mathbb{C}^2$, $A(K) = A(L) = 1$, and if one sets $D_0 = \max\{D(K), D(L)\}$, then the following inequality holds for all $x \in [0, 1]$:

$$A(xK + (1-x)L) \geqslant 1 + \frac{4}{17}\,\frac{x(1-x)}{D_0^2}\,\tilde{\delta}(K, L)^2 . \tag{20}$$

We finally add another remark regarding Diskant's result (18). If for some particular value $x_0 \in (0, 1)$ we have $V(x_0K + (1-x_0))^{1/d} \leqslant 1 + \varepsilon$ it easily follows from the concavity of the left-hand side of this inequality that

$$V(xK + (1-x)L)^{1/d} \leqslant 1 + \max\{1/x_0, 1/(1-x_0)\}\varepsilon \quad (\text{for all } x \in [0, 1]) .$$

Hence one can derive an estimate of the form (18) with ε replaced by $\max\{1/x_0, 1/(1-x_0)\}\varepsilon$. However, one also must pay attention to the fact that ε is now restricted by the more involved inequality $\varepsilon \leqslant \min\{x_0, 1-x_0\}\varepsilon_0$.

3.3. Central symmetrization

If some $K \in \mathbb{C}^d$ is given one can use stability versions of the Brunn–Minkowski inequality to obtain stability estimates for the volume of the body K^* that is obtained from K by central symmetrization, i.e., the convex body $K^* = \frac{1}{2}(K + (-K))$. The inequality under consideration, namely

$$V(K^*) \geqslant V(K) ,$$

is an obvious consequences of the Brunn–Minkowski theorem which shows also that under the assumption $V(K) > 0$ equality holds if and only if K is centrally

symmetric. Corresponding to this inequality one can deduce from (19) the following stability theorem (see Groemer 1988b): For any $K \in \mathbb{C}^d$ with $V(K) = 1$ there exists a centrally symmetric convex body Z (which actually is a translate of K^*) such that

$$V(K^*) \geqq 1 + (2\lambda_d D(K))^{-(d+1)} \delta(K, Z)^{d+1},$$

with λ_d as in (19). Using Diskant's stability version of the Brunn–Minkowski inequality one can obtain an inequality of this type with $d+1$ replaced by d under an additional condition on the size of $V(K^*) - 1$ (and with undetermined constants).

Again, if $d = 2$ a stronger result can be proved (cf. Groemer 1990b). In fact from (20) it follows easily that if $K \in \mathbb{C}^d$ and $A(K) = 1$, there exists a centrally symmetric convex domain Z (namely a translate of K^*) such that

$$A(K^*) \geqq 1 + \frac{4}{17 D(K)^2} \delta(K, Z)^2 .$$

It is known that among all convex bodies K of $\mathbf{E}^d$ with $V(K) = 1$ the minimum of $V(K^*)$ is attained for simplices but no stability statements for the corresponding inequality have been proved.

Analogously to the inequality $V(K^*) \geqq V(K)$ one can, as first shown by Kubota (1925a), prove similar inequalities for the mean projection measures. More precisely, if $K \in \mathbb{C}^d$ and $j = 0, 1, \ldots, d$, then

$$W_j(K^*) \geqq W_j(K) .$$

If $j = d$ or $j = d-1$ this is trivially true with equality, but if $j < d-1$ and $W_j(K) > 0$ equality holds if and only if K is centrally symmetric. Hence in the latter case one can again consider the associated stability problem. Using Kubota's induction formulas for the mean projection measures and the above stability result for the volume, Groemer (1991a) has shown the following stability statement for the mean projection measures: Let K be a convex body in $\mathbf{E}^d$ and let j be an integer with $0 \leqq j \leqq d-2$. Then there exists a positive $\gamma_{d,j}$, depending only on d and j, such that

$$W_j(K^*) \geqq W_j(K) + \gamma_{d,j} r(K)^{d-j} \left(\frac{\delta(K, Z)}{D(K)} \right)^{(j+1)(d+1) - j(j+5)/2},$$

where Z is again a suitable centrally symmetric convex body. In the case $j = d-2$ a substantially better inequality can be proved (see Groemer, 1992a). See also the recent work of Schneider (1992) on the general Brunn–Minkowski inequality that enables one to deduce strong stability results of this type for all W_k with $1 \leqq k \leqq d-2$.

Several other applications of the stability version of the Brunn–Minkowski inequality will be discussed in the following sections.

4. Inequalities for mixed volumes

In this section we survey stability results regarding some of the classical inequalities for mixed volumes. More special inequalities, like the isoperimetric inequality, will be considered separately in later sections since in most cases better estimates are available for these more specific inequalities.

4.1. Minkowski's inequality for the mixed volume $V(K, \ldots, K, L)$

If K and L are convex bodies in $\mathbf{E}^d$ we let $V_1(K, L)$ denote the mixed volume $V(K, \ldots, K, L)$, where K is repeated $d-1$ times. We will discuss here the stability of Minkowski's inequality

$$V_1(K, L)^d - V(K)^{d-1}V(L) \geq 0\,, \tag{21}$$

where the equality sign holds if and only if K and L are homothetic. This inequality contains many important and well-known inequalities as special cases. For example, if we let L or K be the unit ball in $\mathbf{E}^d$ we obtain respectively the isoperimetric inequality and an inequality for the volume and the mean width that is known as Urysohn's inequality. (Stability properties of these two inequalities are the subject of sections 5.2 and 5.3.) We note that if $d = 2$, then (21) becomes the mixed area inequality already discussed in section 2.2. Similarly as in the two-dimensional case, already Minkowski has considered the stability problem associated with this inequality in the case $d = 3$. His result is of the same nature as (13) with the exponent 4 replaced by 6. However, since much better inequalities are known we will not explicitly spell out Minkowski's estimate although, historically, it is of interest. The following statement has been proved by Groemer (1990a):

Let K and L be two d-dimensional convex bodies of volume 1 with coinciding centroids. Let D_0 be defined by $D_0 = \max\{D(K), D(L)\}$. Then

$$V_1(K, L) \geq 1 + \frac{\gamma_d}{dD_0^{d+1}}\,\delta(K, L)^{d+1} \tag{22}$$

with

$$\gamma_d = \frac{2d}{(2\lambda_d)^{d+1}}$$

and λ_d as in (19).

The proof of this inequality is obtained from the stability version (19) of the Brunn–Minkowski inequality and a simple property of concave functions. A similar result with $d+1$ replaced by d, but with an additional condition on the size of $V_1(K, L) - 1$ and not explicitly given constants, has been proved by Diskant (1989). See also Diskant (1972, 1973). A slightly weaker form of (22)

(with the exponent $d+1$ replaced by $d+2$ and without an explicit evaluation of the constants) is contained in the much earlier investigations of Volkov (1963) that concern primarily stability problems for area measures. Another stability statement of Diskant (1972) uses instead of the Hausdorff metric the expression

$$q(K, L) = \max\{\lambda: \lambda K + x \subset L, \lambda \geqslant 0, x \in \mathbf{E}^d\}$$

as a measure for the homothetic deviation. It is shown that if $K, L \in \mathbb{C}^d$, then there exists an $\varepsilon_0 > 0$ and a $c > 0$, depending only on d and the inradii and circumradii of K and L, such that for any $\varepsilon \in [0, \varepsilon_0]$ the inequality

$$V_1(K, L)^d - V(K)^{d-1} V(L) \leqslant \varepsilon$$

implies

$$q(K, sL) \geqslant 1 + c\varepsilon^d \quad \text{with } s = (V(K)/V(L))^{1/d}.$$

A different proof of a weaker form of this statement has been given by Bourgain and Lindenstrauss (1988b). See also the survey article Diskant (1989).

4.2. Minkowski's inequality for the mixed volume $V(K, L, B^d, \ldots, B^d)$

To abbreviate the notation let us write $V(K, L, B^d, \ldots, B^d) = V_{11}(K, L)$. We note that $V_{11}(K, K) = W_{d-2}(K)$ and that

$$V_{11}(K, B^d) = V_{11}(B^d, K) = W_{d-1}(K) = \tfrac{1}{2}\kappa_d \bar{w}(K),$$

where $\bar{w}(K)$ is the mean width of K. The inequality that concerns us here is

$$V_{11}(K, L)^2 - V_{11}(K, K) V_{11}(L, L) \geqslant 0. \tag{23}$$

The stability problem associated with (23) has been investigated by Schneider (1989), and Goodey and Groemer (1990). In both articles the essential tool for proving such stability statements is the development of the pertinent support functions into series of spherical harmonics. However, Schneider uses spherical harmonics to establish a generalization of Wirtinger's inequality and then uses this inequality to deduce stability statements. The approach of Goodey and Groemer is more direct and is inspired by a proof of (23) due to Kubota (1925b) (see also chapter 4.8, sections 3.7 and 3.8). As in most cases, spherical harmonics yield primarily estimates in terms of the L_2-distance δ_2. Using (8) (for $p = 2$) or, if possible, (9) one can of course convert these results to stability statements for the Hausdorff metric. The following theorem is the principal result:

Assume that $K, L \in \mathbb{C}^p$ and that K', L' are homothetic copies of K and L (respectively) so that $\bar{w}(K') = \bar{w}(L')$ and that K' and L' have coinciding Steiner

points. Then

$$V_{11}(K, L)^2 - V_{11}(K, K)V_{11}(L, L) \geqslant \frac{d+1}{d(d+1)}\, \bar{w}(K)^2 W_{d-2}(L)\delta_2(K', L')^2 . \tag{24}$$

The inequality is best possible in the sense that equality may hold without both sides being zero.

4.3. The Aleksandrov–Fenchel inequality

Inequality (23) permits the following generalization which holds for all convex bodies K, L, $C_1, \ldots, C_{d-2}$ in $\mathbf{E}^d$ and is known as the *Aleksandrov–Fenchel inequality*.

$$V(K, L, C_1, \ldots, C_{d-2})^2 - V(K, K, C_1, \ldots, C_{d-2})V(L, L, C_1, \ldots, C_{d-2}) \geqslant 0 . \tag{25}$$

Since the case $d = 2$ has already been discussed in section 2.2 we assume now that $d > 2$. The exact conditions for equality in (25) are not yet known. However, it is clear that a sufficient condition for equality is that K and L are homothetic, and it has been shown by Schneider that if each C_i has a unique support plane at every boundary point, then this condition is also necessary. It is also known that without any regularity conditions on the convex bodies there are other possibilities for equality to occur. [For example, in (34), which is a special case of (25), equality holds not only if K is a ball.] Thus any stability statement regarding (25) that claims a homothetic relationship between K and L must impose some additional conditions on the convex bodies $C_1, C_2, \ldots, C_{d-2}$. Schneider (1990) has proved the following stability statement concerning (25):

Let ρ, r_0, R_0 be positive numbers and assume that $K, L, C_1, \ldots, C_{d-2}$ are convex bodies in $\mathbf{E}^d$ with inradii at least r_0 and circumradii at most R_0. Furthermore, suppose that a ball of radius ρ can roll freely in each C_i. Then the left-hand side of (25) is at least

$$c\rho^{4(2^{d-2}-1)}\Big(V_{11}(K, L)^2 - V_{11}(K, K)V_{11}(L, L)\Big)^{2^{d-2}},$$

where c depends only on d, r_0, and R_0.

For this expression one can of course determine a lower bound in terms of $\delta_2(K', L')$ or $\delta(K', L')$ by the use of (24). Schneider (l.c.) shows also that if k of the bodies $C_1, \ldots, C_{d-2}$ are balls it is possible to obtain better estimates [with exponent 2^{d-2} replaced by 2^{d-2-k}]. As an application of the above result he obtains an estimate for the deviation of a convex body from a unit sphere if the boundary of the body is of class C^2 and its mean curvature is close to 1. See also chapter 1.8, section 6 and Arnold (1992).

5. Inequalities for the mean projection measures and the isoperimetric inequality

5.1. *Inequalities between two mean projection measures*

We consider here inequalities of the form

$$\kappa_d^{i-j} W_j(K)^{d-i} - W_i(K)^{d-j} \geqslant 0\,, \tag{26}$$

which are known to be valid for $0 \leqslant i \leqslant j \leqslant d$. If $0 \leqslant i < j < d$, equality holds if and only if K is a ball. If $d = 2$, $i = 0$, $j = 1$ this is the isoperimetric inequality, and if $d = 3$ one obtains the isoperimetric inequality and two additional inequalities that have been discovered by Minkowski. For arbitrary dimension d and $i = d-2$, $j = d-1$ it is an immediate consequence of (24) that

$$\frac{1}{\kappa_d} W_{d-1}(K)^2 - W_{d-2}(K) \geqslant \frac{d+1}{d(d-1)}\, \delta_2(K, B))^2\,, \tag{27}$$

where B denotes the *Steiner ball* of K, i.e., the d-dimensional ball whose center is at the Steiner point of K and whose diameter is $\bar{w}(K)$. Regarding the proof of (27) see also chapter 4.8, section 3.7. Goodey and Groemer (1990) have pointed out that (due to a result of Schneider) for $d > 3$ equality holds in (27) if and only if K can rotate arbitrarily in a regular simplex without losing contact with the facets of the simplex. A very remarkable feature of the stability inequality (27) is the fact that it enables one to deduce stability results for all the inequalities (26) for all relevant values of i and j. For example, if we write $W_k(K) = W_k$ and use the well-known inequalities $W_k^2 \geqslant W_{k-1} W_{k+1}$ we see that

$$\begin{aligned}\frac{W_{i+1}^{d-i}}{W_i^{d-i-1} W_d} &= \left(\frac{W_{i+1}^2}{W_i W_{i+2}}\right)^{d-i-1} \left(\frac{W_{i+2}^2}{W_{i+1} W_{i+3}}\right)^{d-i-2} \cdots \left(\frac{W_{d-1}^2}{W_{d-2} W_d}\right) \\ &\geqslant \frac{W_{d-1}^2}{W_{d-2} W_d}\,.\end{aligned}$$

Hence, any stability estimate for $W_{d-1}^2 - W_{d-2} W_d$, and in particular the stability inequality (27), will yield a stability estimate for $W_{i+1}^{d-i} - W_i^{d-(i+1)} W_d$. Using such "telescoping products" Groemer and Schneider (1991) proved the following theorem:

Let K be a convex body in $\mathbf{E}^d$ with Steiner ball B, and let $s(x, y)$ be defined by

$$s_k(x, y) = \sum_{\nu=0}^{k-1} x^\nu y^{k-\nu-1}\,.$$

If i and j are two integers such that $0 \leqslant i < j < d$, then

$$\begin{aligned}\kappa_d^{i-j} W_j(K)^{d-i} - W_i(K)^{d-j} &\geqslant \gamma_{d,i,j}(K) \delta_2(K, B)^2 \\ &\geqslant \lambda_{d,i,j}(K) \delta(K, B)^{(d+3)/2}\,,\end{aligned} \tag{28}$$

where

$$\gamma_{d,i,j}(K) = \frac{d+1}{d(d-1)}\, s_{j-1}(W_{d-1}(K)^2/\kappa_d, W_{d-2}(K))\, \frac{W_i(K)^{d-j}}{W_{d-2}(K)^{j-1}},$$

$$\lambda_{d,i,j}(K) = k_d(\sigma_d/2\kappa_{d-1})\gamma_{d,i,j}(K)\left(\frac{\kappa_d}{W_{d-1}(K)}\right)^{(d-1)/2},$$

and $k_d(\cdot)$ is given by (10).

The second inequality in (28) is of course obtained from the first part by an application of the conversion formula (9). A result of the same type as (28), but of a less explicit nature and with a much larger exponent $[2^{d-2}(d!)$ instead of $(d+3)/2]$ has been proved by Diskant (1975, 1989).

There exist various inequalities for the average value of certain mean projection measures of particular types of convex bodies that contain or are contained in a given $K \in \mathbb{C}^d$. Recently stability statements have been proved for two such inequalities. One of these (Groemer 1992b) concerns the volume of simplices contained in K, the other (Groemer 1992c) the mean value of W_{d-2} of circumscribed cylinders of K.

5.2. The isoperimetric inequality

The isoperimetric inequality (i.e., its generalized version for $\mathbf{E}^d$) is undoubtedly one of the best known geometric inequalities. If $K \in \mathbb{C}^d$, it can be written in the form

$$\left(\frac{S(K)}{\sigma_d}\right)^d - \left(\frac{V(K)}{\kappa_d}\right)^{d-1} \geqslant 0.$$

Since the case $d = 2$ has already been discussed in section 2.1 we assume now that $d > 2$ and we consider primarily stability inequalities of the form

$$\left(\frac{S(K)}{\sigma_d}\right)^d - \left(\frac{V(K)}{\kappa_d}\right)^{d-1} \geqslant \beta\delta(K, B)^\alpha, \tag{29}$$

where B denotes a suitable ball and α depends only on d whereas β may also depend on $R(K)$, $r(K)$, and $D(K)$. Most of the earlier work on the stability of the isoperimetric inequality has been performed in the more general context of mixed volumes (see section 4.1). Thus, in the notation of (29) the result of Volkov (1963) yields the exponent $\alpha = d + 2$, and the work of Diskant (1973, 1989) shows that one may take $\alpha = d$. Osserman (1987) gave a proof of a stability version of the isoperimetric inequality that depends on a d-dimensional form of a Bonnesen-type inequality. It yields an estimate of the form (29) with $\alpha = d(d+1)/2$. Substantially better inequalities, essentially (29) with $\alpha = (d+3)/2$, can be obtained from (28) by setting $i = 0$, $j = 1$. If the resulting inequality is expressed in terms of V and S it can be written as

$$\left(\frac{S(K)}{\sigma_d}\right)^d - \left(\frac{V(K)}{\kappa_d}\right)^{d-1} \geqslant \eta_d \frac{V(K)^{d-1}}{W_{d-2}(K)} \delta_2(K, B)^2$$
$$\geqslant \tau_d \frac{V(K)^{d-1}}{W_{d-2}(K) W_{d-1}(K)^{(d-1)/2}} \delta(K, B)^{(d+3)/2}, \tag{30}$$

where

$$\eta_d = \frac{d+1}{d(d-1)} \frac{1}{\kappa_d^{d-1}}, \qquad \tau_d = \frac{d+1}{d(d-1)} \frac{k_d(\sigma_d/2\kappa_{d-1})}{\kappa_d^{(d-1)/2}},$$

with $k_d(\cdot)$ as defined by (10). Setting $\mu_d = \tau_d/2^{(d+3)/2}$ and using well-known estimates from the theory of convex sets and the fact that $R(K) - r(K) \leqslant 2\delta(K, B)$ one easily deduces from (30) the following Bonnesen-type inequalities:

$$\left(\frac{S(K)}{\sigma_d}\right)^d - \left(\frac{V(K)}{\kappa_d}\right)^{d-1}$$
$$\geqslant \mu_d \kappa_d^{(d-3)/2} r(K)^{(2d^2-3d-3)/2} (R(K) - r(K))^{(d+3)/2},$$
$$\left(\frac{S(K)}{\sigma_d}\right)^d - \left(\frac{V(K)}{\kappa_d}\right)^{d-1}$$
$$\geqslant \mu_d V(K)^{(d+3)(d-2)/2} S(K)^{-(d^3-3)/2} (R(K) - r(K))^{(d+3)/2}.$$

Thus it is possible to estimate the isoperimetric deficit solely in terms of $R(K)$ and $r(K)$, and to estimate $R(K) - r(K)$ in terms of $V(K)$ and $S(K)$ only.

Recently, Fuglede (1989) has proved a stability estimate for the isoperimetric inequality with even better exponents. If K is a convex body in $\mathbf{E}^d$ of volume κ_d let us set $\Delta(K) = (S(K)/\sigma_d) - 1$. Fuglede's principal result can be formulated as follows:

Assume that $K \in \mathbb{C}^d$, $V(K) = \kappa_d$, and that K has o as centroid. Then there exists a $b_d > 0$ and $r_d > 0$ (depending only on d) such that

$$\delta(K, B^d) \leqslant b_3 \left(\Delta(K) \log \frac{1}{\Delta(K)}\right)^{1/2} \quad \text{if } d = 3$$

and

$$\delta(K, B^d) \leqslant b_d \Delta(K)^{2/(d+1)} \quad \text{if } d > 3,$$

provided that $\Delta(K) \leqslant r_d$.

Moreover, possible values for r_d and b_d are explicitly exhibited. To compare this result with inequality (30) we note that under the assumption that $\Delta(K)$ is small enough it follows from (30) that

$$\delta(K, B) \leqslant c_d \Delta(K)^{2/(d+3)},$$

where B denotes the Steiner ball of K and c_d depends only on d Fuglede (l.c.) gives examples that show that the bounds on the right-hand side of his inequalities are best possible with respect to their order of magnitude. For $d>3$ a similar example has been given by Groemer and Schneider (1991). The proof of Fuglede's stability statements is of an analytic nature and uses the expansion of the radial function of K into a series of spherical harmonics. It is possible to combine Fuglede's result with (30) to derive inequalities of the same strength as those of Fuglede but without the restriction $\Delta(K) \leq r_d$. Fuglede (1986, 1989) has also studied the stability problem of the isoperimetric inequality without assuming convexity. In this case additional conditions have to be imposed, e.g., the assumption that the body is in a certain sense already nearly spherical. Finally, we mention that Fuglede's inequalities have recently been used to establish stability results for inequalities regarding the potential theoretic capacity of convex bodies. See Fuglede (1992) and the references listed there.

5.3. *The inequalities of Urysohn and Bieberbach*

Urysohn's inequality can be written as

$$(\tfrac{1}{2}\bar{w}(K))^d - \frac{V(K)}{\kappa_d} \geq 0 .$$

Equality holds exactly for balls. From (28) one can immediately deduce the following associated stability inequality.

$$\begin{aligned}(\tfrac{1}{2}\bar{w}(K))^d - \frac{V(K)}{\kappa_d} &\geq \frac{1}{\kappa_d}\,\gamma_{d,0,d-1}(K)\delta_2(K, B)^2 \\ &\geq \frac{1}{\kappa_d}\,\lambda_{d,0,d-1}(K)\delta(K, B)^{(d+3)/2} . \end{aligned} \tag{31}$$

Weaker inequalities of this kind (but with less involved factors) have been proved by Groemer (1988b, 1990b). Any inequality for mixed volumes of the type discussed in section 4.1 yields of course a stability version of Urysohn's inequality.

Clearly, $\bar{w}$ in Urysohn's inequality can be replaced by $D(K)$. If this is done the result is Bieberbach's inequality relating the volume and diameter of a convex body. Since (31) holds also if $\bar{w}(K)$ is replaced by $D(K)$ one obtains immediately a stability statement for Bieberbach's inequality.

If $d=2$ we have $\bar{w}(K) = P(K)/\pi$ and it is clear that Urysohn's inequality is essentially the same as the isoperimetric inequality. Hence, the stability statements (11) or (12) can also be interpreted as stability statements for Urysohn's inequality and [after replacing $\bar{w}(K)$ by $D(K)$] for Bieberbach's inequality. See also Groemer (1988c). For example, (12) shows that there is a circle, say C, such that

$$D(K)^2 - \frac{4}{\pi}A(K) \geq \frac{8}{\pi}\,\delta(K, C)^2 . \tag{32}$$

6. Cap bodies, bodies of constant width, equichordal sets, packings and coverings

6.1. Stability problems related to cap bodies

A *cap body* in $\mathbf{E}^d$ is defined as the convex hull of a d-dimensional ball and a countable set of points outside the ball, provided that all line segments joining any two of these points meet the ball. The importance of cap bodies is that they appear naturally as the extremal bodies that satisfy certain geometric inequalities. Two typical examples in this respect are Favard's inequality

$$W_{d-2}(K) - r(K)W_{d-1}(K) \geqslant 0\,, \tag{33}$$

and the inequality

$$W_1(K)^2 - W_0(K)W_2(K) \geqslant 0 \tag{34}$$

which is due to Minkowski (for $d = 3$). In both cases equality occurs exactly if K is a cap body. For more details regarding these and similar inequalities see Bonnesen and Fenchel (1934) and Sangwine–Yager (1991), where also further references are given.

Using area measures Sangwine–Yager (l.c.) has been able to prove the following stability theorem associated with (33):

If for some $\varepsilon \geqslant 0$

$$W_{d-2}(K) - r(K)W_{d-1}(K) \leqslant \varepsilon\,,$$

then there exists a cap body C such that

$$\delta(K, C) \leqslant \mu_d R(K)^{(d-2)/d}\varepsilon^{1/d}\,,$$

where μ_d depends only on d.

In addition, in the case $d = 3$, Sangwine–Yager proves a kind of a stability inequality corresponding to (34). However, in this result there appears the volume of an inner parallel body of K which may degenerate and thereby cause the stability inequality to degenerate to the original inequality (34).

6.2. Bodies of constant width

We first discuss results that concern only the two-dimensional case and start with the Lebesgue–Blaschke inequality (3) which has already been mentioned in section 1.1 to describe an example of a stability problem. Writing again $\alpha_0 = (\pi - \sqrt{3})/2$, Groemer (1988c) has shown that if M is a domain of constant width 1 and if an $\varepsilon \geqslant 0$ is given, then the condition

$$A(M) \leqslant \alpha_0 + \varepsilon$$

implies that there exists a Reuleaux triangle T such that

$$\delta(M, T) \leq (10\varepsilon)^{1/2} .$$

Combining this result with the stability inequality (32) one can deduce that any convex domain M of constant width 1 has the property that

$$\alpha_0 + \tfrac{1}{10}\,\delta(M, T)^2 \leq A(M) \leq \tfrac{1}{4}\pi - 4\delta(M, C)^2 ,$$

where T is a suitable Reuleaux triangle and C an appropriate circle.

Another classical theorem about convex domains of constant width is the theorem of Barbier which states that all convex domains of constant width w_0 have perimeter πw_0. Since any convex domain, say K, of diameter 1 is contained in a domain of constant width 1 Barbier's theorem implies that

$$P(K) \leq \pi , \tag{35}$$

with equality if and only if K is of constant width 1. Groemer (1988c) has proved the following result regarding the stability problem associated with (35):

If K is a convex domain of diameter 1 and if $P(K) \geq \pi - \varepsilon$ (for some $\varepsilon \geq 0$) there exists a convex domain M of constant width 1 such that

$$\delta(K, M) \leq \tfrac{24}{25}\varepsilon^{2/3} .$$

Moreover, the exponent $\frac{2}{3}$ cannot be replaced by any larger constant.

This result shows that even trivial inequalities, like (35) can lead to interesting stability problems. For d-dimensional convex bodies the corresponding problem is to estimate the deviation of a convex body K of diameter 1 from a convex body of constant width if for some $\varepsilon \geq 0$

$$\bar{w}(K) \geq 1 - \varepsilon .$$

Following an idea of Goodey (unpublished) one can obtain a stability estimate by the following argument: $1 - \bar{w}(K) \leq \varepsilon$ can also be interpreted as

$$\|1 - w_K(u)\|_1 \leq \varepsilon\sigma_d ,$$

where $w_K(u)$ denotes the width of K in the direction u and $\|\cdot\|_1$ denotes the 1-norm. Since both 1 and $w_K(u)$ are support functions of convex bodies one can use Vitale's inequality (8) (in the case when $p = 1$) and infer that $|1 - w_K(u)| \leq \eta_d \varepsilon^{1/d}$ with η_d depending on d only. From this inequality it is not difficult to deduce (see Goodey and Groemer 1990, Proposition 1) that there exists a convex body, say M, of constant width such that $\delta(K, M) \leq 3\eta_d \varepsilon^{1/d}$.

6.3. *Equichordal sets*

A convex body K in $\mathbf{E}^d$ is called an *equichordal set* if there is a point p in K such that all chords of K that contain p have equal length. It can be shown that the volume of an equichordal set K of chord length 1 has the property that $V(K) \geqslant \kappa_d/2^d$ (with equality if and only if K is a ball) and that $V(K) < \frac{1}{2}\kappa_d$, where $\frac{1}{2}\kappa_d - V(K)$ can be made arbitrarily small if K is sufficiently close to a *semiball*, i.e., the intersection of a ball with a half space whose boundary contains the center of the ball. The following stability estimates associated with these two inequalities (together with some generalizations and better but more involved constants) have been proved by Groemer (1988a): There exist a ball B and a semiball Q such that

$$\frac{\kappa_d}{2^d} + \left(\frac{1}{4\sqrt{d-1}}\right)^{d+1} \delta(K, B)^{d+1} \leqslant V(K) \leqslant \frac{\kappa_d}{2} - \left(\frac{1}{2\sqrt{d}}\right)^d \delta(K, Q)^d .$$

6.4. *Packings and coverings*

Let K be a convex body in $\mathbf{E}^d$ and let $\tau(K)$ denote the translative packing constant of K (i.e., the maximum density of packings by translates of K). Obviously $\tau(K) \leqslant 1$ and it is known that equality implies that K is centrally symmetric. The following stability statement regarding this inequality can easily be deduced from the results mentioned in section 3.3 (see Groemer 1990c):

If $V(K) = 1$ there exists a centrally symmetric convex body C such that

$$\delta(K, C) \leqslant 12dD(K)\left(\frac{1-\tau(K)}{\tau(K)}\right)^{1/d+1} . \tag{36}$$

A stronger form of this theorem can be proved for lattice packings of translates of K. In this case C can be determined so that it is a centrally symmetric polytope with at most $2(2^d - 1)$ facets and that an inequality of the form (36) holds (with $12d$ replaced by a different factor and $\tau(K)$ denoting the lattice packing constant, i.e., the maximum density of lattice packings by translates of K). Further improvements are possible if $d = 2$, and under this assumption one can also prove analogous theorems for coverings.

7. Projections of convex bodies

In recent times, various theorems have been proved that deal with the stability of the relationship between convex bodies and their orthogonal projections. Although these stability theorems cannot be formulated in a natural way as geometric inequalities we informally describe here some of these results since they are in spirit close to stability statements for inequalities. A similar situation occurs with respect to stability results for area measures. Since this latter subject is

covered in chapter 1.8, section 6 (see also Diskant 1989) we omit here a discussion of this topic. Most of the work for projections involves the use of spherical harmonics and if this is the case references to this subject may also be found in chapter 4.8, section 3.4.

7.1. The dependence of convex bodies on their projections

If $K \in \mathbb{C}^p$ we let K_u denote the orthogonal projection of K onto the $(d-1)$-dimensional linear subspace of $\mathbf{E}^d$ that is orthogonal to the unit vector u. A basic stability problem regarding projections is to estimate the translative Hausdorff distance between two convex bodies K, L, if for all $u \in S^{d-1}$ the translative distance between K_u and L_u is below a given bound. This problem has first been formulated by Golubyatnikov (1982). Groemer (1987) has shown that if for a sufficiently large set of unit vectors u the condition $\tilde{\delta}(K_u, L_u) \leq \varepsilon$ is satisfied, then $\tilde{\delta}(K, L) \leq (1+2\sqrt{2})\varepsilon$. As already noted by Golubyatnikov (l.c.) the factor $1+2\sqrt{2}$ can be removed if K and L are centrally symmetric. Groemer's article contains also results concerning the more general situation when it is only known that the projections are nearly homothetic. Golubyatnikov (1992) has recently investigated the analogous stability problem that arises when the projections are only given up to rigid motions and suitable restrictive conditions on the bodies are satisfied.

7.2. Projection bodies

We let W_j' denote the jth mean projection measure in $\mathbf{E}^{d-1}$. If some $K \in \mathbb{C}^d$ is given and K_u is defined as before it is a well-known fact that $W_j'(K_u)$, considered as a function of u, is the support function of a convex body. This convex body is called the *projection body of K of order $d-j-1$*. The projection body does of course not uniquely determine the original body, but according to a well-known theorem of Aleksandrov the relationship between projection bodies (of a given order $j>0$) and the original bodies is one-to-one if only d-dimensional convex bodies that are centrally symmetric with respect to o are considered. This leads immediately to the problem of estimating the deviation of two convex bodies in terms of the distance of the corresponding projection bodies. It can be shown (see Goodey 1986) that for $j>1$ and without any condition prohibiting degeneracy this relationship between (centrally symmetric) convex bodies and their projection bodies of order j is in fact unstable. But under appropriate restrictions on the size of the given body [in terms of $r(K)$ and $R(K)$ or $D(K)$] stability theorems have been proved for the cases $j=0$ and $j=d-2$.

Stability results for first-order projection bodies have been obtained for $d=3$ by Campi (1986) (who published related work already in the paper Campi 1981) and in the general case by Goodey and Groemer (1990). This problem is also of interest with respect to a theorem of Minkowski (1904–1906) stating that a three-dimensional convex body is of constant width if and only if all of its orthogonal projections onto planes have equal perimeter. The stability result just

cited implies that a d-dimensional convex body must be close to a convex body of constant width if all orthogonal projections onto hyperplanes have approximately the same mean width. For more details see Goodey and Groemer (l.c.) and chapter 4.8, section 3.2.

In the case of projection bodies of order $d-1$ one is concerned with the question whether the translative distance between two d-dimensional convex bodies that are centrally symmetric with respect to o must be small if the $(d-1)$-dimensional volumes of all orthogonal projections unto hyperplanes differ only slightly. For convex bodies in $\mathbf{E}^3$ that have an axis of rotational symmetry, Campi (1986) has found a stability estimate. Without this condition corresponding stability results have been proved for $d=3$ by Campi (1988) and for all $d \geqslant 3$ by Bourgain and Lindenstrauss (1988a, b) (see also the outline by Bourgain 1988 and the exposition of Burger 1990). Additional stability results loosely connected with projections can be found in an article of Groemer (1991b) on the stability of Cauchy's surface area formula, and in a paper of Anikov and Stepanov (1981) dealing with a combination of an illumination and projection problem. Again, these results do actually not concern inequalities but the general ideas and methods of proof show some similarity to stability investigations for inequalities.

Regarding the role of spherical harmonics in the proofs of several results of this section, see chapter 4.8. In particular we mention that in chapter 4.8, sections 2.2, 3.3, 3.4, and 3.5 various stability results are discussed.

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CHAPTER 1.5

Selected Affine Isoperimetric Inequalities

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Contents

Introduction 153
1. Inequalities for random simplices in a convex body 153
2. The Busemann–Petty centroid inequality 155
3. The Petty projection inequality 156
4. Petty's affine projection inequality 157
5. Relationship between the Busemann–Petty centroid and Petty projection inequalities 158
6. Relationship between the Busemann–Petty centroid inequality and Petty's affine projection 999
6. inequality 159
7. The Busemann intersection inequality and its relatives 160
8. Petty's geominimal surface area inequality and its relatives 163
9. The curvature image inequality 164
10. The affine isoperimetric inequality 164
11. The Blaschke–Santaló inequality 165
12. Open problems 167
    12.1. The complementary Blaschke–Santaló inequality 167
    12.2. The conjectured projection inequalities of Petty and Schneider 168
    12.3. Inequalities for the affine Quermassintegrals 168
    12.4. The curvature image conjecture 169
    12.5. The polar projection problem 169
    12.6. The polar intersection problem 170
    12.7. The second projection question 170
    12.8. The second intersection conjecture 170
    12.9. The projection centroid conjectures 170
    12.10. The Petty body conjectures 170
References 171

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Introduction

An affine isoperimetric inequality in this article will invariably mean a *sharp* affine isoperimetric inequality. Affine isoperimetric inequalities have always – or should always have – occupied a central position in the field of geometric convexity. There have been numerous applications of these inequalities to fields such as Ordinary and Partial Differential Equations [see, for example, Aleksandrov (1963), the book of Guggenheimer (1977), Petty and Barry (1962), Guggenheimer (1969, 1971), Firey (1974), Heil (1975)], Functional Analysis [see, for example, the tract of Pisier (1989), and the volumes of Lindenstrauss and Milman (1987, 1988, 1989)], the Geometry of Numbers [see, for example, the books of Gruber and Lekkerkerker (1987) and Erdős, Gruber and Hammer (1989), and also Kannan and Lovász (1988), Henk and Wills (1991), and Wills (1992)], Discrete Geometry and Polytopal Approximations [see, for example, the book of Fejes Tóth (1953), the survey of Gruber (1983), Gruber (1988, 1991, 1992a,b), and Schneider (1986)], Stereology and Stochastic Geometry [see, for example, the book of Santaló (1976), the survey of Schneider (1988), Santaló (1988), and Vitale (1991)], and to the field of Minkowskian Geometry [see, for example, the book of Busemann and Phadke (1987), Petty (1952), Busemann and Petty (1956), Guggenheimer (1966), Holmes and Thompson (1979) and Thompson (1992)].

While many of the inequalities examined in this survey have older roots, an attempt has been made to focus on inequalities established in the second half of this century. Since attention is restricted to sharp affine isoperimetric inequalities, asymptotic affine inequalities such as treated in the book of Pisier (1989) are (with one small exception) ignored. The selection of those few inequalities which are to be considered is merely a reflection of the author's personal tastes. As a result many very important sharp affine isoperimetric inequalities are not even mentioned. For example, it would be difficult to name an affine inequality more beautiful than the Rogers–Shephard (1957, 1958) difference body inequality [and its extensions by Chakerian (1967), Schneider (1970), and others]. Yet it, as well as many others, will not be examined here. The reader interested in a broad view of the subject is referred to the books of Burago and Zalgaller (1988), Santaló (1976), Pisier (1989), the volumes of Lindenstrauss and Milman (1987, 1988, 1989, 1991), the surveys (and references therein) of Grünbaum (1963), Gruber and Höbinger (1976), Gruber (1983), Leichtweiß (1985), Petty (1983), Schneider (1988), and Simon (1985).

1. Inequalities for random simplices in a convex body

A convex body in Euclidean n-space, $\mathbb{R}^n$, is a compact, convex subset of $\mathbb{R}^n$ that has non-empty interior. For the set of convex bodies in $\mathbb{R}^n$, write $\mathcal{K}^n$, and for the set of centered (symmetric about the origin) convex bodies, $\mathcal{K}^n_c$.

Write $[x_0, x_1, \ldots, x_n]$ for the volume of the simplex in $\mathbb{R}^n$ whose vertices are the

points $x_0, x_1, \ldots, x_n \in \mathbb{R}^n$, and ω_i for the i-dimensional volume of the unit ball in $\mathbb{R}^i$.

In solving Sylvester's Four Point Problem [see the discussion and references in Santaló (1976, 1985), Schneider (1988), and Affentranger (1988)], Blaschke (1917a) [see also Blaschke (1923)] proved that for a body $K \in \mathcal{K}^2$, with area $A(K)$,

$$\frac{1}{A(K)^3} \int_K \int_K \int_K [x_0, x_1, x_2] \, dx_0 \, dx_1 \, dx_2 \geqslant (35/48\pi^2) A(K),$$

with equality if and only if K is an ellipse. The integrals above are over the interiors of K, and dx_i denotes the area element at the point $x_i \in K$. Blaschke also showed that for bodies of unit area this integral assumes its maximum value only for triangles.

The Blaschke–Groemer inequality states that for a convex body $K \in \mathcal{K}^n$, with volume $V(K)$,

$$\frac{1}{V(K)^{n+1}} \int_K \cdots \int_K [x_0, x_1, \ldots, x_n] \, dx_0 \, dx_1 \cdots dx_n \geqslant \kappa_n V(K), \tag{BG}$$

with equality if and only if K is an ellipsoid. The $n+1$ integrals above are over the interiors of K, and dx_i denotes the volume element at $x_i \in K$. The inequality (for $n > 2$) is due to Groemer (1973). The computation for the expected value of the volume of a random simplex in the unit ball,

$$\kappa_n = 2^{-n} \binom{n+1}{\frac{1}{2}(n+1)}^{n+1} \binom{(n+1)^2}{\frac{1}{2}(n+1)^2}^{-1},$$

is due to Kingman (1969). In Groemer (1974, 1982) many extensive generalizations of the Blaschke–Groemer inequality are established. The survey of Schneider (1988) contains a number of related inequalities. The complementary inequality to the Blaschke–Groemer inequality is as yet unknown (for $n > 2$): what bodies of unit volume maximize the integral in (BG)? See Klee (1969) and Schneider (1988) for a discussion.

Closely related to the Blaschke–Groemer inequality is the Busemann random simplex inequality, which states that for $K \in \mathcal{K}^n$, and $x_0 \in K$,

$$\frac{1}{V(K)^n} \int_K \cdots \int_K [x_0, x_1, \ldots, x_n] \, dx_1 \cdots dx_n \geqslant \frac{2\omega_{n+1}^{n-1}}{(n+1)! \, \omega_n^{n+1}} V(K), \tag{BRS}$$

with equality if and only if K is an ellipsoid centered at x_0.

An extension of Busemann's random simplex inequality was recently obtained by Bourgain et al. (1988).

The proofs of all of these inequalities follow the method Blaschke used to establish the case $n = 2$ of the Blaschke–Groemer inequality: the integrals in question are not increased by Steiner symmetrization.

2. The Busemann–Petty centroid inequality

Write $h(Q, \cdot): S^{n-1} \to \mathbb{R}$ for the support function of $Q \in \mathcal{K}^n$; i.e., for $u \in S^{n-1}$, $h(Q,u) = \max\{u \cdot x: x \in K\}$, where $u \cdot x$ denotes the usual inner product of u and x in $\mathbb{R}^n$.

The radial function, $\rho(K, \cdot): S^{n-1} \to [0,\infty)$, of a compact, star-shaped (about the origin) $K \subset \mathbb{R}^n$, is defined, for $u \in S^{n-1}$, by $\rho(K,u) = \max\{\lambda \geqslant 0: \lambda u \in K\}$. If $\rho(K, \cdot)$ is positive and continuous, call K a star body (about the origin), and write $\mathcal{S}_o^n$ for the set of star bodies (about the origin) of $\mathbb{R}^n$. The set of convex star bodies (i.e., convex bodies containing the origin in their interiors) will be denoted by $\mathcal{K}_o^n$.

The centroid body $\Gamma K \in \mathcal{K}_o^n$, of the star body $K \in \mathcal{S}_o^n$, is the convex body whose support function, for $u \in S^{n-1}$, is given by

$$h(\Gamma K, u) = \frac{1}{V(K)} \int_K |u \cdot x| \, dx .$$

Centroid bodies were defined and studied by Petty (1952, 1955, 1961a). They are also investigated in Milman and Pajor (1989) and Lutwak (1990). They have proven to be remarkably powerful tools in establishing a number of affine isoperimetric inequalities. Centroid bodies derive their name from the fact that for $K \in \mathcal{K}_e^n$, the boundary of the centroid body of K is the locus of the centroids of the halves of K which are obtained by cutting K by hyperplanes through the origin.

Petty (1961a) obtained numerous interesting results regarding centroid bodies. One of these is an integral representation of the mixed volume of centroid bodies. A special case of this integral representation gives the volume of the centroid body of $K \in \mathcal{S}_o^n$ as

$$V(\Gamma K) = \frac{2^n}{V(K)^n} \int_K \cdots \int_K [0, x_1, \ldots, x_n] \, dx_1 \cdots dx_n .$$

A proof of this can also be found in Milman and Pajor (1989). Extensions (to generalized zonoids) are given in Schneider and Weil (1983). This integral representation allows Petty to reformulate Busemann's random simplex inequality as the Busemann–Petty centroid inequality: if $K \in \mathcal{S}_o^n$, then

$$V(\Gamma K) \geqslant (2\omega_{n-1}/(n+1)\omega_n)^n V(K), \tag{BPC}$$

with equality if and only if K is an ellipsoid centered at the origin.

Given Petty's integral representation, for the volume of a centroid body, it is easily seen how the Busemann–Petty centroid inequality, for $\mathcal{K}^n$, follows from (in fact, is) the Busemann random simplex inequality. It will be shown that once the Busemann–Petty centroid inequality has been established for $\mathcal{K}_e^n$ (in fact, the class of polars of zonoids will more than suffice), then the Busemann–Petty centroid inequality (and hence also Busemann's random simplex inequality), with the equality conditions, is easily obtained for all of $\mathcal{S}_o^n$.

3. The Petty projection inequality

For $K, L \in \mathcal{K}^n$, and $\varepsilon \geqslant 0$, the Minkowski linear combination $K + \varepsilon L \in \mathcal{K}^n$ is defined by $h(K + \varepsilon L, \cdot) = h(K, \cdot) + \varepsilon h(L, \cdot)$. The mixed volume $V_1(K, L) = V(K, \ldots, K, L)$ of $K, L \in \mathcal{K}^n$ can be defined (see Bonnesen and Fenchel 1934, Busemann 1958, or Leichtweiß 1980) by

$$nV_1(K, L) = \lim_{\varepsilon \to 0} \frac{V(K + \varepsilon L) - V(K)}{\varepsilon} = \int_{\partial K} h(L, v)\, \mathrm{d}v,$$

where the integration is over the boundary, ∂K, of K, and $\mathrm{d}v$ denotes the $(n-1)$-dimensional volume element at the point of ∂K whose outer unit normal is $v \in S^{n-1}$.

The projection body, $\Pi K \in \mathcal{K}_e^n$, of the convex body $K \in \mathcal{K}^n$, is the convex body whose support function, for $u \in S^{n-1}$, is given by

$$h(\Pi K, u) = V_{n-1}(K|\xi_u) = \tfrac{1}{2} \int_{\partial K} |u \cdot v|\, \mathrm{d}v,$$

where V_{n-1} denotes $(n-1)$-dimensional volume, and $K|\xi_u$ is the (image of the) orthogonal projection of K onto ξ_u, the hyperplane orthogonal to u.

From the integral representations given above of the projection operator and the mixed volume V_1, it follows that, for $K, L \in \mathcal{K}^n$,

$$V_1(K, \Pi L) = V_1(L, \Pi K). \tag{3.1}$$

Petty (1967) proved that for $\phi \in \mathrm{SL}(n)$,

$$\Pi \phi K = \phi^{-t} \Pi K,$$

where ϕ^{-t} denotes the inverse of the transpose of ϕ. Alternate proofs can be found in Bourgain and Lindenstrauss (1988), and Lutwak (1990). The range of $\Pi : \mathcal{K}^n \to \mathcal{K}^n$ is denoted by $\mathcal{Z}^n$, and its members are known as zonoids (or projection bodies). The range of $\Gamma : \mathcal{S}_o^n \to \mathcal{K}^n$ is a proper subset of $\mathcal{Z}^n$.

The polar body, K^*, of $K \in \mathcal{K}_o^n$, is the convex body defined by

$$K^* = \{ x \in \mathbb{R}^n : x \cdot y \leqslant 1 \text{ for all } y \in K \}.$$

For $\phi \in \mathrm{SL}(n)$, obviously $(\phi K)^* = \phi^{-t} K^*$. Write $\mathcal{Z}_*^n$ for the set of polars of zonoids, and $\Pi^* : \mathcal{K}^n \to \mathcal{Z}_*^n$ for the function $Q \mapsto (\Pi Q)^* = \Pi^* Q$, and $\Gamma^* : \mathcal{S}_o^n \to \mathcal{Z}_*^n$ for the function $Q \mapsto (\Gamma Q)^* = \Gamma^* Q$.

The Petty projection inequality states that for $K \in \mathcal{K}^n$,

$$V(K)^{n-1} V(\Pi^* K) \leqslant (\omega_n / \omega_{n-1})^n, \tag{PP}$$

with equality if and only if K is an ellipsoid. This inequality is due to Petty (1972). An alternate proof and extensions are given in Lutwak (1985, 1986a,b). Petty

obtains his inequality by using (among other inequalities) the Busemann–Petty centroid inequality. A proof will be given below where it will also be shown that to obtain this inequality, with its equality conditions, for all bodies in $\mathcal{K}^n$, it is only necessary to establish it for the bodies in $\mathcal{L}^n$.

The Petty projection inequality is a strengthened form of the classical isoperimetric inequality. To see this consider the following inequality for real p:

$$(V(K)/\omega_n)^{(n-1)/n} \leqslant \omega_{n-1}^{-1} \left\{ \frac{1}{n\omega_n} \int_{S^{n-1}} V_{n-1}(K|\xi_u)^p \, \mathrm{d}u \right\}^{1/p},$$

with equality if and only if K is a ball. For $p = 1$, by Cauchy's surface area formula, the right-hand side of this inequality is proportional to the surface area of K, and the inequality is just the classical isoperimetric inequality between the volume and surface area of a convex body. The Hölder inequality shows that smaller values of p provide stronger inequalities. How small can p get (with the inequality still holding for all K)? The Petty projection inequality provides the answer: the inequality (with the same equality conditions) is valid for all $p > -n$. For $p = -n$, the inequality is the Petty projection inequality. Hence, for $p = -n$, the inequality still holds, but the equality conditions change to those of the Petty projection inequality. For $p < -n$, the inequality is not valid (for all K).

Recently, Zhang (1991) has obtained the complementary inequality to the Petty projection inequality: for $K \in \mathcal{K}^n$,

$$\binom{2n}{n} n^{-n} \leqslant V(K)^{n-1} V(\Pi^* K), \tag{ZPI}$$

with equality if and only if K is a simplex. This inequality was conjectured by Ball (1991).

4. Petty's affine projection inequality

A body $K \in \mathcal{K}^n$ is said to have a continuous curvature function $f(K, \cdot): S^{n-1} \to (0, \infty)$, if the integral representation

$$V_1(K, Q) = \frac{1}{n} \int_{S^{n-1}} h(Q, u) f(K, u) \, \mathrm{d}u,$$

holds for all $Q \in \mathcal{K}^n$. Let $\mathcal{F}^n$ denote the class of all convex bodies with a positive continuous curvature function. It is known that if ∂K is C^2 and has everywhere positive principal curvatures, then $f(K, u)$ is the reciprocal Gauss curvature of ∂K at the point of ∂K whose outer unit normal is u.

The affine surface area, $\Omega(K)$, of the body $K \in \mathcal{F}^n$, is defined by Petty (1967) as

$$\Omega(K) = \int_{S^{n-1}} f(K, u)^{n/(n+1)} \, \mathrm{d}u.$$

It is known (see, for example, Lutwak 1991a) that Ω is an affine functional; i.e., $\Omega(\phi K) = \Omega(K)$, for $K \in \mathcal{F}^n$, and $\phi \in \mathrm{SL}(n)$. Recently, the domain of $\Omega : \mathcal{F}^n \to [0, \infty)$ has been extended to all of $\mathcal{K}^n$ by Leichtweiß (1986, 1988, 1989). Later extensions were given by Schütt and Werner (1990), Schütt (1992), and Lutwak (1991a) [see Leichtweiß (1991) for a discussion]. All of the inequalities treated in this article which involve affine surface area hold for extended affine surface area – with the same equality conditions. However, these extended inequalities will not be examined. Since in all reasonable extensions of affine surface area the affine surface area of polytopes is 0, it is quite surprising how often the concept of affine surface area arises in problems involving polytopal approximations [see, for example, Bárány and Larman (1988), Gruber (1983, 1988, 1991, 1992b), Schneider (1986), Schütt (1991)].

Petty's affine projection inequality states that for $K \in \mathcal{F}^n$,

$$V(\Pi K) \geqslant \frac{\omega_{n-1}^n}{n^{n+1}\omega_n^n} \Omega(K)^{n+1}, \tag{PAP}$$

with equality if and only if K is an ellipsoid.

This inequality was obtained by Petty (1967) by using the Busemann–Petty centroid inequality. It will be shown that, when viewed from the proper perspective, it is the Busemann–Petty centroid inequality!

5. Relationship between the Busemann–Petty centroid and Petty projection inequalities

For $K, L \in \mathcal{S}_o^n$, and $\varepsilon > 0$, the harmonic linear combination $K \hat{+} \varepsilon \cdot L \in \mathcal{S}_o^n$ is defined by $\rho(K \hat{+} \varepsilon \cdot L, \cdot)^{-1} = \rho(K, \cdot)^{-1} + \varepsilon \rho(L, \cdot)^{-1}$. The dual mixed volume $\tilde{V}_{-1}(K, L)$ of $K, L \in \mathcal{S}_o^n$, can be defined by

$$\begin{aligned} -n\tilde{V}_{-1}(K, L) &= \lim_{\varepsilon \to 0} \frac{V(K \hat{+} \varepsilon \cdot L) - V(K)}{\varepsilon} \\ &= \frac{1}{n} \int_{S^{n-1}} \rho(K, v)^{n+1} \rho(L, v)^{-1} \, dv. \end{aligned}$$

The Minkowski mixed volume inequality, for V_1, is that for $K, L \in \mathcal{K}^n$,

$$V_1(K, L)^n \geqslant V(K)^{n-1} V(L), \tag{5.1}$$

with equality if and only if K and L are homothetic.

The dual mixed volume inequality, for $\tilde{V}_{-1}$, is that for $K, L \in \mathcal{S}_o^n$,

$$\tilde{V}_{-1}(K, L)^n \geqslant V(K)^{n+1} V(L)^{-1}, \tag{5.2}$$

with equality if and only if K and L are dilates.

It is easily shown Lutwak (1985, 1990) that for $K \in \mathcal{K}^n$, and $L \in \mathcal{S}_o^n$,

$$V_1(K, \Gamma L) = \frac{2}{(n+1)V(L)} \tilde{V}_{-1}(L, \Pi^* K). \tag{5.3}$$

Since obviously $V_1(Q, Q) = V(Q)$, and $\tilde{V}_{-1}(Q, Q) = V(Q)$, it follows from (5.3) that

$$V(\Gamma L) = (2/(n+1))\tilde{V}_{-1}(L, \Pi^* \Gamma L)/V(L) \tag{5.4}$$

and

$$V_1(K, \Gamma \Pi^* K) = 2/(n+1). \tag{5.5}$$

For $K \in \mathcal{K}^n$, define

$$\mathrm{pp}(K) = \frac{(\omega_n/\omega_{n-1})^n}{V(K)^{n-1} V(\Pi^* K)},$$

and for $K \in \mathcal{S}_o^n$, define

$$\mathrm{bpc}(K) = ((n+1)\omega_n/2\omega_{n-1})^n V(\Gamma K)/V(K).$$

Thus, the Busemann–Petty centroid inequality states that for $K \in \mathcal{S}_o^n$, one has $\mathrm{bpc}(K) \geqslant 1$, with equality if and only if K is an ellipsoid centered at the origin, while the Petty projection inequality states that for $K \in \mathcal{K}^n$, one has $\mathrm{pp}(K) \geqslant 1$, with equality if and only if K is an ellipsoid. For $L \in \mathcal{S}_o^n$, (5.4) and the dual mixed volume inequality (5.2) yield

$$\mathrm{bpc}(L) \geqslant \mathrm{pp}(\Gamma L), \tag{5.6}$$

with equality if and only if L and $\Pi^* \Gamma L$ are dilates. For $K \in \mathcal{K}^n$, (5.5) and the Minkowski inequality (5.1) yield

$$\mathrm{pp}(K) \geqslant \mathrm{bpc}(\Pi^* K), \tag{5.7}$$

with equality if and only if K and $\Gamma \Pi^* K$ are dilates. Thus, the Petty projection inequality for $\mathcal{Z}^n$ will quickly yield the Busemann–Petty centroid inequality, with the equality conditions, for all of $\mathcal{S}_o^n$. On the other hand, the Busemann–Petty centroid inequality for $\mathcal{Z}_*^n$ will quickly yield the Petty projection inequality, with equality conditions, for all of $\mathcal{K}^n$.

6. Relationship between the Busemann–Petty centroid inequality and Petty's affine projection inequality

Let $\mathcal{S}_c^n$ denote the set of star bodies whose centroids are at the origin. The (weak) solution of the Minkowski problem shows that for each $K \in \mathcal{S}_c^n$, there exists a $\Lambda K \in \mathcal{F}^n$, unique up to translation, such that

$$f(\Lambda K, \cdot) = \omega_n \rho(K, \cdot)^{n+1}/V(K).$$

Call ΛK the curvature image of K. It is easily seen that $\Lambda: \mathcal{S}_c^n \to \mathcal{F}^n/\mathbb{R}^n$, is bijective and that for $K \in \mathcal{S}_c^n$, and $L \in \mathcal{K}_o^n$,

$$V_1(\Lambda K, L) = \omega_n \tilde{V}_{-1}(K, L^*)/V(K). \tag{6.1}$$

From this identity, and the invariance of V_1 and $\tilde{V}_{-1}$ under simultaneous unimodular centro-affine transformations, it follows (see Lutwak 1991a) that for $\phi \in \mathrm{SL}(n)$,

$$\Lambda \phi K = \phi^{-t} \Lambda K.$$

Note that ΛK is an ellipsoid if and only if K is an ellipsoid. From the definition of the curvature image and the definition of affine surface area:

$$\Omega(\Lambda K)^{n+1} = n^{n+1} \omega_n^n V(K). \tag{6.2}$$

By rewriting the integrals in the definitions of the centroid and projection operators (as integrals over S^{n-1}), one sees that for $K \in \mathcal{S}_o^n$,

$$\Pi \Lambda K = \tfrac{1}{2}(n+1)\omega_n \Gamma K. \tag{6.3}$$

For convex K this result is due to Petty (1974, Theorem 3.11).

From the identities (6.2) and (6.3) one sees immediately that the Busemann–Petty centroid inequality for $K \in \mathcal{S}_c^n$ is just Petty's affine projection inequality for $\Lambda K \in \mathcal{F}^n$.

7. The Busemann intersection inequality and its relatives

Just as the projection body ΠK, of $K \in \mathcal{K}^n$, can be defined by $h(\Pi K, u) = V_{n-1}(K|\xi_u)$, the intersection body $\mathrm{I}K \in \mathcal{S}_o^n$, of $K \in \mathcal{S}_o^n$, is defined as the body whose radial function, for $u \in S^{n-1}$, is given by

$$\rho(\mathrm{I}K, u) = V_{n-1}(K \cap \xi_u).$$

Abbreviate $\Pi(\Pi K)$, the second projection body of K, by $\Pi^2 K$, and $\mathrm{I}(\mathrm{I}K)$, the second intersection body of K, by $\mathrm{I}^2 K$. It is easily shown (see Lutwak 1991b) that for $K \in \mathcal{S}_o^n$ and $\phi \in \mathrm{SL}(n)$,

$$\mathrm{I}\phi K = \phi^{-t} \mathrm{I}K.$$

A fundamental result of Busemann (1949a), is that $\mathrm{I}K \in \mathcal{K}_e^n$, when $K \in \mathcal{K}_e^n$. Intersection bodies play a central role in Busemann's theory of Minkowskian surface area (see Busemann 1947a,b, 1949b, 1950a,b). A dual (to Busemann's) theory of Minkowskian surface area has been developed by Holmes and Thompson (1979) (see also Johnson and Thompson 1987) in which projection bodies (rather than intersection bodies) play a central role.

The Busemann intersection inequality states that for $K \in \mathcal{S}_o^n$,

$$V(IK) \leqslant (\omega_{n-1}^n/\omega_n^{n-2})V(K)^{n-1}, \tag{BI}$$

with equality if and only if K is an ellipsoid centered at the origin. For convex bodies, this inequality is due to Busemann (1953), who derived it by using the Busemann random simplex inequality. In fact, it was in order to obtain this inequality that Busemann established his random simplex inequality.

The integral representation of the dual mixed volume $\tilde{V}(K_1, \ldots, K_n)$ of $K_1, \ldots, K_n \in \mathcal{S}_o^n$, is

$$\tilde{V}(K_1, \ldots, K_n) = \frac{1}{n} \int_{S^{n-1}} \rho(K_1, u) \cdots \rho(K_n, u)\, du.$$

For $K, L \in \mathcal{S}_o^n$, let $\tilde{V}_1(K, L) = \tilde{V}(K, \ldots, K, L)$. As an immediate analog of (3.1) one has (see Lutwak 1988b) that for $K, L \in \mathcal{S}_o^n$,

$$\tilde{V}_1(K, IL) = \tilde{V}_1(L, IK),$$

and hence, $V(IK) = \tilde{V}_1(K, I^2K)$. A direct consequence of the Hölder inequality is the dual mixed volume inequality,

$$\tilde{V}_1(K, L)^n \leqslant V(K)^{n-1} V(L), \tag{7.1}$$

with equality if and only if K and L are dilates. The dual mixed volume inequality for $\tilde{V}_1$, now shows that in order to establish the Busemann intersection inequality, with the equality conditions, for all bodies in $\mathcal{S}_o^n$, one need only establish it for those bodies that are dilates of their second intersection bodies.

The Petty body, $PK \in \mathcal{S}_o^n$, of $K \in \mathcal{S}_o^n$, is the star body whose radial function, for $u \in S^{n-1}$, is given by $\rho(PK, u) = V_{n-1}(\Gamma_u(K \cap \xi_u))$, where $\Gamma_u(K \cap \xi_u)$ is the centroid body of $K \cap \xi_u$ in the $(n-1)$-dimensional space ξ_u.

Petty (1961a) observed that the Busemann intersection inequality holds for star bodies and presented an alternate proof of it. Petty's proof is based on his representation for the volume of $K \in \mathcal{S}_o^n$,

$$V(K)^{n-1} = \frac{n!}{2^n} \tilde{V}_1(IK, PK). \tag{7.2}$$

This representation for the volume of a body is Petty's reinterpretation and extension of an integral representation given by Busemann (1953) for the volume of a convex body. A proof of this representation can also be found in Milman and Pajor (1989). Let $\iota_n = (n\omega_{n-1}/2\omega_{n-2})^{n-1}$. The Busemann–Petty centroid inequality shows that, for $K \in \mathcal{S}_o^n$,

$$IK \subset \iota_n PK, \tag{7.3}$$

with equality if and only if K is an ellipsoid centered at the origin. Since $\tilde{V}_1(Q,Q) = V(Q)$, the inclusion (7.3), the representation (7.2), and the monotonicity of dual mixed volumes yield the Busemann intersection inequality for $\mathcal{S}_o^n$.

From (7.3) it follows that $V(IK) \leqslant \iota_n^n V(PK)$. But more can be said: from the inclusion (7.3), the representation (7.2), and the monotonicity of dual mixed volumes, it follows that, for $K \in \mathcal{S}_o^n$,

$$V(IK) \leqslant (\omega_{n-1}^n/\omega_n^{n-2})V(K)^{n-1} \leqslant \iota_n^n V(PK),$$

with equality, in either inequality, if and only if K is an ellipsoid centered at the origin.

It would be of interest to find a simple path from the Busemann intersection inequality to the Busemann–Petty centroid inequality.

There is an extension of the Busemann intersection inequality to slices of arbitrary dimensions: Grinberg (1986, 1991) shows that for $K \in \mathcal{S}_o^n$, and $1 < i < n$,

$$\int_{G(n,i)} V_i(K \cap \xi)^n \, d\xi \leqslant (\omega_i^n/\omega_n^i)V(K)^i,$$

with equality if and only if K is an ellipsoid centered at the origin. Here, V_i denotes i-dimensional volume, and the integration is over the Grassmannian $G(n,i)$, of i-dimensional subspaces of $\mathbb{R}^n$, with respect to the usual invariant probability measure on $G(n,i)$. That for ellipsoids there is equality in this inequality is known as the Furstenberg–Tzkoni (1971) formula. There is also a similar affine isoperimetric inequality due to Schneider (1985, (second part of) Theorem 1).

One consequence of the Busemann intersection inequality appears to have been overlooked: for $f \in C(S^{n-1})$, let $Rf \in C(S^{n-1})$ denote the spherical Radon transform of f; i.e., for $u \in S^{n-1}$,

$$(Rf)(u) = \frac{1}{(n-1)\omega_{n-1}} \int_{S^{n-1} \cap \xi_u} f(v) \, dv,$$

where the integration is over the great circle $S^{n-1} \cap \xi_u$, and dv denotes the $(n-2)$-dimensional volume element at $v \in S^{n-1} \cap \xi_u$. For positive $f \in C(S^{n-1})$, and $p > 0$, let

$$\|f\|_p = \left\{ \frac{1}{n\omega_n} \int_{S^{n-1}} f(u)^p \, du \right\}^{1/p}.$$

The following inequality is equivalent to the Busemann intersection inequality (compare Oberlin and Stein 1982, p. 642, (3)): for positive $f \in C(S^{n-1})$,

$$\|Rf\|_n \leqslant \|f\|_{n/(n-1)}.$$

To see this just define $K \in \mathcal{S}_o^n$ by $\rho(K, \cdot)^{n-1} = f$, and note that $\rho(IK, \cdot) = \omega_{n-1} Rf$.

8. Petty's geominimal surface area inequality and its relatives

The important concept of geominimal surface area was introduced by Petty (1974). The notion of geominimal surface area serves as a bridge connecting a number of areas of geometry: affine differential geometry, relative differential geometry, and Minkowskian geometry (see Petty 1961b, 1972, 1974, Clack 1990, Leichtweiß 1988, 1989, 1990, 1991, and Lutwak 1991a, 1992a,b).

The geominimal surface area, $G(K)$, of $K \in \mathcal{K}^n$, can be defined by

$$\omega_n^{1/n} G(K) = \inf\{ nV_1(K, Q^*)V(Q)^{1/n} : Q \in \mathcal{K}_c^n \}.$$

The infimum is taken over $\mathcal{K}_c^n$, the set of convex bodies whose centroids lie at the origin. This is a slight modification of (but equivalent to) Petty's original definition. Petty (1974) shows that, unlike ordinary surface area, the geominimal surface area of a body is invariant under unimodular affine transformations of the body.

Petty's geominimal surface area inequality states that for $K \in \mathcal{K}^n$,

$$G(K)^n \leqslant n^n \omega_n V(K)^{n-1}, \tag{PGS}$$

with equality if and only if K is an ellipsoid. The inequality is due to Petty (1974). A proof will be given in section 11.

As Petty (1974) shows, many interesting inequalities are obtained by taking the infimum, in the definition of geominimal surface area, over proper affine invariant subclasses of $\mathcal{K}_c^n$. Behrend (1937), Green (1953), Petty (1961b), and Ball (1991) consider a geominimal surface area, $G(\cdot; \mathcal{E}_c^n)$, where the infimum is taken over the smallest affine invariant subset (which contains the unit ball) of $\mathcal{K}_c^n$, the set of centered ellipsoids, $\mathcal{E}_c^n$:

$$\omega_n^{1/n} G(K; \mathcal{E}_c^n) = \inf\{ nV_1(K, E^*)V(E)^{1/n} : E \in \mathcal{E}_c^n \}.$$

The Minkowski inequality (5.1) immediately yields a sharpened form of the classical isoperimetric inequality: for $K \in \mathcal{K}^n$,

$$n^n \omega_n V(K)^{n-1} \leqslant G(K; \mathcal{E}_c^n)^n,$$

with equality if and only if K is an ellipsoid. The complementary inequality is the Ball–Behrend inequality: for $K \in \mathcal{K}^n$,

$$G(K; \mathcal{E}_c^n)^n \leqslant \frac{n^{3n/2}(n+1)^{(n+1)/2}}{n!} V(K)^{n-1}. \tag{BG}$$

For $n = 2$, this inequality is due to Behrend (1937), who also showed that equality occurs only for triangles. A later proof of this was given by Gustin (1953). For $n > 2$, the inequality is due to Ball (1991). In addition to showing that simplices maximize the ratio $G(K; \mathcal{E}_c^n)^n / V(K)^{n-1}$ in $\mathcal{K}^n$, Ball also shows that in $\mathcal{K}_c^n$ this ratio is maximized by parallelotopes. It appears to be an open problem whether, for

$n > 2$, these are the only bodies that maximize this ratio (for their respective classes)?

A closely related (see Ball 1991) affine isoperimetric inequality concerns the volume ratio convex bodies. For $K \in \mathcal{K}^n$, this ratio is defined by $\mathrm{vr}(K)^n = V(K)/V(E)$, where E is the ellipsoid of maximal volume contained in K. Obviously, $1 \leqslant \mathrm{vr}(K)$, with equality if and only if K is an ellipsoid. Behrend (1938), for $n = 2$, and Ball (1991), for all $n \geqslant 2$, obtain the complementary inequality which shows that simplices have maximal volume ratio; i.e., for $K \in \mathcal{K}^n$,

$$\mathrm{vr}(K)^n \leqslant \frac{n^{n/2}(n+1)^{(n+1)/2}}{\omega_n n!}.$$

Finding the exact equality conditions in this inequality appears to be an open problem, for $n > 2$.

9. The curvature image inequality

The curvature image inequality states that for $K \in \mathcal{S}_c^n$,

$$V(\Lambda K)^{n-1} \geqslant \omega_n^{n-2} V(K), \tag{CI}$$

with equality if and only if K is an ellipsoid centered at the origin.

This inequality is a consequence of Petty's geominimal surface area inequality. Note that by (6.1), and the dual mixed volume inequality (5.2),

$$\begin{aligned}\omega_n^{1/n} G(\Lambda K) &= \inf\{ nV_1(\Lambda K, Q^*) V(Q)^{1/n} : Q \in \mathcal{K}_c^n \} \\ &= \inf\{ n\omega_n \tilde{V}_{-1}(K, Q) V(Q)^{1/n} / V(K) : Q \in \mathcal{K}_c^n \} \\ &\geqslant n\omega_n V(K)^{1/n}.\end{aligned}$$

When combined with Petty's geominimal surface area inequality, this simple observation immediately yields the curvature image inequality, with its conditions for equality.

10. The affine isoperimetric inequality

The affine isoperimetric inequality states that for $K \in \mathcal{F}^n$,

$$\Omega(K)^{n+1} \leqslant \omega_n^2 n^{n+1} V(K)^{n-1}, \tag{AI}$$

with equality if and only if K is an ellipsoid. For a somewhat more restricted class of bodies, and $n \leqslant 3$, this inequality is due to Blaschke (1916) (see also Blaschke 1923). Santaló (1949) and Deicke (1953) show how to extend it for all $n \geqslant 2$. In the form stated above the inequality is due to Petty (1985).

The affine isoperimetric inequality can be obtained from the curvature image inequality. In fact, if viewed from the proper perspective, the affine isoperimetric inequality is the curvature image inequality! To see this, note that given $K \in \mathcal{F}^n$, there exists an $L \in \mathcal{S}_c^n$, such that $K = \Lambda L$. Now by (6.2),

$$\begin{aligned}\omega_n^2 n^{n+1} V(K)^{n-1} &= \omega_n^2 n^{n+1} V(\Lambda L)^{n-1} \\ &\geqslant \omega_n^n n^{n+1} V(L) = \Omega(\Lambda L)^{n+1} = \Omega(K)^{n+1}.\end{aligned}$$

Affine inequalities involving centro-affine surface area can be found in Heil (1967) and Schneider (1968).

11. The Blaschke–Santaló inequality

For $K \in \mathcal{K}^n$, and $x \in \operatorname{int} K$, define $K^x = x + (-x + K)^*$. Thus, K^x is the polar body of K, with respect to the unit sphere centered at x. Let $c = \operatorname{Cen}(K)$ denote the centroid of $K \in \mathcal{K}^n$. The Santaló point $s = \operatorname{San}(K)$, of $K \in \mathcal{K}^n$, can be defined as the unique point $s \in \operatorname{int} K$ that satisfies the equation $\operatorname{Cen}(K^s) = s$. The Santaló point of a convex body can also be defined as the unique point $s \in \operatorname{int} K$, for which

$$V(K^s) = \inf\{ V(K^x) : x \in \operatorname{int} K \}.$$

A third characterization: the Santaló point is the unique point $s \in \operatorname{int} K$, such that

$$\int_{S^{n-1}} u(-s \cdot u + h(K,u))^{-n-1}\, du = 0.$$

There is a duality between the notions of centroid and Santaló point. If one first defines the Santaló point, one can define the centroid of $K \in \mathcal{K}^n$ as the unique point $c \in \operatorname{int} K$ that satisfies the equation $\operatorname{San}(K^c) = c$. If K is centrally symmetric, then its Santaló point and centroid coincide with its center of symmetry.

The Blaschke–Santaló inequality is usually stated as: if $K \in \mathcal{K}^n$, then

$$V(K)V(K^s) \leqslant \omega_n^2, \tag{BS1}$$

with equality if and only if K is an ellipsoid. The duality between the notions of centroid and Santaló point, and the fact that $Q^{**} = Q$, allows this inequality to be rewritten as: if $K \in \mathcal{K}^n$, then

$$V(K)V(K^c) \leqslant \omega_n^2, \tag{BS2}$$

with equality if and only if K is an ellipsoid. Equivalently, if $K \in \mathcal{K}_c^n$, then

$$V(K)V(K^*) \leqslant \omega_n^2, \tag{BS3}$$

with equality if and only if K is an ellipsoid.

The Blaschke–Santaló inequality is due to Blaschke (1917b) [see also Blaschke (1917c)] for $n \leqslant 3$, and Santaló (1949) [see also Santaló (1950)] for all $n \geqslant 2$. As noted by Leichtweiß (1983) (see also Schneider 1982a), the equality conditions obtained from the (slightly modified) proofs of Blaschke and Santaló are for bodies with a required degree of smoothness. The equality conditions in the Blaschke–Santaló inequality for bodies in $\mathcal{K}_c^n$ are due to Saint Raymond (1981), who also gave a new proof of the inequality. The equality conditions for arbitrary convex bodies are due to Petty (1985). There is also a proof of the inequality by Heil (1967), for $n = 2$, and an unpublished proof of the inequality by Calabi (1985). Recently, a new proof of the inequality was found by Meyer and Pajor (1989, 1990). Their proof uses the Brunn–Minkowski inequality and Steiner symmetrization. Meyer and Pajor (1990) also present an interesting strengthened version of the Blaschke–Santaló inequality. An extension of the Blaschke–Santaló inequality, to star-shaped sets, with the same equality conditions, can be found in Lutwak (1991a). A stronger inequality than the Blaschke–Santaló inequality was found by Ball (1988) for bodies which are affine images of bodies symmetric with respect to the coordinate hyperplanes. Ball conjectured that his inequality should hold for all bodies in $\mathcal{K}_e^n$.

Blaschke and Santaló obtain the Blaschke–Santaló inequality by using the affine isoperimetric inequality. To see how the affine isoperimetric inequality will give the Blaschke–Santaló inequality, note that from (6.1) and the Minkowski inequality (5.1) it follows that for $K \in \mathcal{K}_c^n$,

$$\omega_n^n = \omega_n^n \tilde{V}_{-1}(K, K)^n / V(K)^n = V_1(\Lambda K, K^*)^n \geqslant V(\Lambda K)^{n-1} V(K^*),$$

with equality if and only if ΛK and K^* are homothetic. But by the affine isoperimetric inequality and identity (6.2),

$$\omega_n^2 n^{n+1} V(\Lambda K)^{n-1} \geqslant \Omega(\Lambda K)^{n+1} = \omega_n^n n^{n+1} V(K),$$

with equality if and only if ΛK, and hence K, is an ellipsoid.

Thus far, the following path has been followed:

$$\text{PGS} \Longrightarrow \text{CI} \Longrightarrow \text{AI} \Longrightarrow \text{BS}.$$

That BS $\Longrightarrow$ PGS can be seen as follows: suppose $K \in \mathcal{K}^n$. Since both sides of the Petty's geominimal surface area inequality are translation invariant, it may be assumed that K is positioned so that its Santaló point is at the origin. Hence, $K^* \in \mathcal{K}_c^n$, and thus,

$$\begin{aligned} \omega_n G(K)^n &= \inf\{ n^n V_1(K, Q^*)^n V(Q) : Q \in \mathcal{K}_c^n \} \\ &\leqslant n^n V_1(K, K^{**})^n V(K^*) \\ &= n^n V(K)^{n-1} [V(K) V(K^*)], \end{aligned}$$

which completes the argument.

12. Open problems

Thirty-five years ago, Busemann and Petty (1956) published a list of ten open problems from the field of Minkowskian geometry. They reformulated all of these problems as questions regarding convex bodies in ordinary Euclidean space. The importance of the problems they posed was immediately evident. As Busemann has warned (see, for example, Busemann 1950b), progress in the geometry of Finsler spaces must be preceded by a deeper understanding of the local structure of Finsler spaces – which is Minkowskian. Unfortunately, many of the most basic questions regarding the geometry of Minkowski spaces remain unanswered.

With the exception of work such as Larman and Rogers (1975), Ball (1986), (1988), Giannopoulos (1990), Bourgain (1991), Papadimitrakis (1992) and Gardner (1992), there has been little progress to report on the Busemann–Petty problems. This is certainly not due to any lack of effort. Most of the problems are still on the top-ten-list of those working on affine extremal problems. Some of the affine problems below might prove to be less of a challenge. As indicated there is a bit of overlap with the Busemann–Petty list.

12.1. The complementary Blaschke–Santaló inequality

If $\mathcal{T}_c^n$ is a set of convex bodies whose centroids lie at the origin (i.e., $\mathcal{T}_c^n \subset \mathcal{K}_c^n$), let

$$c_n(\mathcal{T}_c^n) = \inf\{V(K)V(K^*): K \in \mathcal{T}_c^n\},$$

and let $c_n = c_n(\mathcal{K}_c^n)$. The problem here is to determine c_n and the equality conditions in the complementary Blaschke–Santaló inequality: if $K \in \mathcal{K}_c^n$, then

$$c_n \leqslant V(K)V(K^*).$$

It has been conjectured (see, for example, Aleksandrov 1967, Guggenheimer 1966, 1973, p. 315, Petty 1974, p. 86) that $c_n = (n+1)^{(n+1)}/(n!)^2$; i.e., the minimal product of the volumes of polar reciprocal bodies is attained by simplices. Mahler (1939b) proves this for $n = 2$.

It has been conjectured (see, for example, Mahler 1939a, Guggenheimer 1966, Petty 1974, Santaló 1976, and Leichtweiß 1983, 1985) that for $\mathcal{K}_e^n$, the centered (symmetric about the origin) convex bodies, $c_n(\mathcal{K}_e^n) = 4^n/n!$, with the parallelotopes and cross-polytopes being extremal. Mahler (1939b) and Reisner (1985) show that this is the case for $n = 2$. Reisner (1986) shows that for the case $n = 2$, parallelograms (the 2-dimensional parallelotopes and cross-polytopes) are the only extremal bodies. Saint Raymond (1981) showed that, for $n > 3$, the parallelotopes and cross-polytopes could not be the only extremal bodies for this problem.

For the case $\mathcal{T}_c^n = \mathcal{Z}^n$ (and thus, $\mathcal{T}_c^n = \mathcal{Z}_*^n$) there is a complete solution to the problem: Reisner (1985) has shown that $c_n(\mathcal{Z}^n) = 4^n/n!$, and then Reisner (1986) went on to show, that for the class $\mathcal{Z}^n$, equality is possible in the complementary Blaschke–Santaló inequality only for parallelotopes and cross-polytopes.

Later Gordon, Meyer and Reisner (1988) gave a very clever simple proof of the results of Reisner. Their proof is inductive and is based on the fact that for $K \in \mathcal{K}^n$,

$$\tfrac{1}{2}(n+1)V_1(K, \Gamma\Pi^* K) = V_1(K, \Pi^2 K)/V(\Pi K).$$

Note that by (5.5) and (3.1) both sides are equal to 1.

Bourgain and Milman (1985, 1987) (see also Pisier 1989) showed the existence of a constant $\lambda > 0$ (which is independent of n), such that

$$\lambda^n \omega_n^2 \leqslant c_n(\mathcal{K}_e^n).$$

In connection with this problem see also Meyer (1986) and Reisner (1987).

For the minimal product of the Quermassintegrals of polar reciprocal bodies see the discussion, results and references of Heil (1976).

There are a number of equivalent formulations of the problem of finding the complementary Blaschke–Santaló inequality. For example, Petty (1974) shows that this problem is equivalent to finding the complementary geominimal surface area inequality (PGS): the extremal bodies for both complementary inequalities must be the same!

12.2. The conjectured projection inequalities of Petty and Schneider

Petty's conjectured projection inequality (see Petty 1972) is that for $K \in \mathcal{K}^n$,

$$V(\Pi K) \geqslant (\omega_{n-1}^n / \omega_n^{n-2}) V(K)^{n-1},$$

with equality if and only if K is an ellipsoid. As can be seen from the Blaschke–Santaló inequality, this inequality is stronger than the Petty projection inequality. By using the Minkowski inequality (5.1), Petty's conjectured inequality can be reformulated as follows: for $K, L \in \mathcal{K}^n$,

$$\int_{\partial K} \int_{\partial L} |u \cdot v| \, du \, dv \geqslant \frac{2n\omega_{n-1}}{\omega_n^{(n-2)/n}} [V(K)V(L)]^{(n-1)/n},$$

with equality if and only if K and L are homothetic to polar reciprocal ellipsoids.

Schneider (1982b) conjectured that for $K \in \mathcal{K}_e^n$,

$$V(\Pi K) \leqslant 2^n V(K)^{n-1},$$

with equality if and only if K is a direct sum of one- and two-dimensional convex bodies.

12.3. Inequalities for the affine Quermassintegrals

The Quermassintegrals, $W_0(K), \ldots, W_n(K)$, of a convex body $K \in \mathcal{K}^n$, can be defined by letting $W_0(K) = V(K)$, $W_n(K) = \omega_n$, and for $0 < i < n$, by

$$W_{n-i}(K) = \frac{\omega_n}{\omega_i} \int_{G(n,i)} V_i(K|\xi) \, d\xi,$$

where $K|\xi$ denotes the (image of) the orthogonal projection of K onto $\xi \in G(n,i)$, and V_i denotes i-dimensional volume, and the integration is over $G(n,i)$, the Grassmannian of i-dimensional subspaces of $\mathbb{R}^n$, with respect to its usual invariant probability measure.

The affine Quermassintegrals, $\Phi_0(K), \ldots, \Phi_n(K)$, of a convex body $K \in \mathcal{K}^n$, can be defined by letting $\Phi_0(K) = V(K)$, $\Phi_n(K) = \omega_n$, and for $0 < i < n$, by

$$\Phi_{n-i}(K) = \frac{\omega_n}{\omega_i} \left\{ \int_{G(n,i)} V_i(K|\xi)^{-n} \, \mathrm{d}\xi \right\}^{-1/n}.$$

Obviously, $\Phi_i(K) \leqslant W_i(K)$. Grinberg (1991) shows that these new Quermassintegrals are indeed affine functionals: $\Phi_i(\phi K) = \Phi_i(K)$, for $\phi \in \mathrm{SL}(n)$, and all i.

The classical isoperimetric inequality states that for $K \in \mathcal{K}^n$, and $0 < i < n$,

$$\omega_n^{i/n} V(K)^{(n-i)/n} \leqslant W_i(K),$$

with equality if and only if K is a ball. It has been conjectured that a stronger inequality holds: for $K \in \mathcal{K}^n$, and $0 < i < n$,

$$\omega_n^{i/n} V(K)^{(n-i)/n} \leqslant \Phi_i(K),$$

with equality if and only if K is an ellipsoid.

The case $i = 1$ of this conjectured inequality is just the Petty projection inequality. The case $i = n - 1$ is a simple consequence of the Blaschke–Santaló inequality. In fact for bodies in $\mathcal{K}_e^n$ it is the Blaschke–Santaló inequality! All of the cases $1 < i < n - 1$ appear to be open. Other related conjectures can be found in Lutwak (1988a).

12.4. The curvature image conjecture

The curvature image conjecture is that for $K \in \mathcal{K}_c^n$, the bodies ΛK and K^* are homothetic only when K is an ellipsoid.

This conjecture has been verified by Petty (1985) for plane convex bodies ($n = 2$). For $n \geqslant 3$, Petty (1985) also verified the conjecture for the cases when K is a body of revolution. Petty (1985) showed that if ∂K is C^2, then the conjecture holds.

A verification would have many interesting applications, especially to Minkowskian geometry. It would, among other things, also provide a new proof of the affine isoperimetric inequality and the Blaschke–Santaló inequality.

12.5. The polar projection problem

Let $\mathrm{D}K \in \mathcal{K}_e^n$ denote the difference body of $K \in \mathcal{K}^n$, defined by $h(\mathrm{D}K, u) = h(K,u) + h(K,-u)$, for $u \in S^{n-1}$. For $n > 2$, which bodies $K \in \mathcal{K}^n$ have the property that $\Pi^* K$ and $\mathrm{D}K$ are homothetic?

Ellipsoids and simplices have this property! Martini (1991) shows that a polytope with this property must be a simplex. Martini (1989a) [see also Martini (1990)] has also shown that if $DK = nV(K)\Pi^* K$, then K must be a simplex.

Even the following special case is of interest: if $K \in \mathcal{L}_*^n$, and ΠK and K^* are homothetic, must K be an ellipsoid? Bodies with this property were investigated by Petty (1972), and conjectured to be ellipsoids (see Gruber 1987).

12.6. The polar intersection problem

Are ellipsoids the only convex $K \in \mathcal{K}_e^n$ with the property that IK and K^* are dilates? This is a restatement of Problem 5 from the list of Busemann and Petty.

12.7. The second projection question

For which $K \in \mathcal{L}^n$ are $\Pi^2 K$ and K homothetic? If the class of bodies turns out to be manageable, an answer here could, among other things, provide a solution to Petty's conjectured projection inequality: as Schneider (1987) observes, since $V(\Pi K) = V_1(K, \Pi^2 K)$, it follows from the Minkowski inequality (5.1) that

$$V(\Pi K)/V(K)^{n-1} \geqslant V(\Pi^2 K)/V(\Pi K)^{n-1},$$

with equality if and only if $\Pi^2 K$ and K are homothetic. Hence, the extremal bodies for Petty's conjectured projection inequality (Problem 12.2) must come from the class of those bodies that are homothetic to their second projection bodies. For the class of polytopes, Problem 12.7 has been completely solved by Weil (1971).

12.8. The second intersection conjecture

Are centered ellipsoids the only $K \in \mathcal{S}_o^n$ for which $I^2 K$ and K are dilates?

The situation here is the dual of that in Problem 12.7: replace the mixed volume V_1 by the dual mixed volume $\tilde{V}_1$, replace the projection operator, Π, by the intersection operator, I, and reverse the inequalities.

12.9. The projection centroid conjectures

These are two related conjectures: if $K \in \mathcal{L}^n$ is such that $\Gamma\Pi^* K$ and K are dilates must K be an ellipsoid? If $K \in \mathcal{L}_*^n$ such that $\Pi^* \Gamma K$ and K are dilates does it follow that K is an ellipsoid?

A hint of some of the implications of an affirmative answer here can be seen from the discussion in section 5.

12.10. The Petty body conjectures

The Petty body $PK \in \mathcal{S}_o^n$, of $K \in \mathcal{S}_o^n$, is the star body whose radial function, for $u \in S^{n-1}$, is given by $\rho(PK, u) = V_{n-1}(\Gamma_u(K \cap \xi_u))$, where $\Gamma_u(K \cap \xi_u)$ is the

centroid body of $K \cap \xi_u$, in ξ_u, the $(n-1)$-dimensional space orthogonal to u. If K is a centered convex body, will its Petty body, PK, be convex? Another question: Martini (1989b) and Petty (1952, p. 60) have shown that, for $K \in \mathcal{K}_o^n$, $IK \subset \Pi K$, with equality if and only if K is an ellipsoid centered at the origin. There is a similar relation between the intersection and Petty bodies (7.3). Is there a similar relation between the Petty body and the projection body?

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CHAPTER 1.6

Extremum Problems for Convex Discs and Polyhedra

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Contents

1. Introduction and notation 179
2. Some extremum problems for convex discs 181
3. Bounds for the volume of a convex polyhedron 197
4. Surface area and edge-curvature 202
5. Total length of the edges of a polyhedron 205
6. The isoperimetric problem for convex polyhedra 208
References 214

HANDBOOK OF CONVEX GEOMETRY
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Geometric extremum problems were already investigated in ancient times, and the isoperimetric problem in particular attracted the attention of mathematicians. In the 19th century considerable progress was made by the work of Steiner, Lindelöf, Schwarz and Brunn, among others. Minkowski's research on convex bodies and the application of geometric methods to number-theoretic questions gave an enormous impetus to the field of geometric extremum problems. In the present century a great number of very important results were obtained by Blaschke, Steinitz, Hadwiger and Dinghas, among many others. Beginning in the forties, the fruitful ideas and the contributions of L. Fejes Tóth formed and advanced this area of geometry in a singular way. His books (L. Fejes Tóth 1972, 1964), to which we shall refer continually, turned out to be an abundant source of inspiration to workers in this field.

In this chapter we shall be dealing with extremum problems for convex sets mainly in 2- and 3-dimensional Euclidean space. Limitations of space force us to make a selection of the problems to be considered. We shall give a survey of the results, including the pertinent references, but only occasionally indicate the methods of proof.

1. Introduction and notation

Let E^d be the d-dimensional Euclidean space. It is always assumed that $d \geq 2$. We denote the class of all compact convex and nonempty subsets of E^d by $\bar{\mathbb{C}}^d$. An element of $\bar{\mathbb{C}}^d$ with nonempty interior will be called a *convex body* (if $d = 2$, a *convex disc*) and the class of all convex bodies in E^d will be denoted by $\mathbb{C}^d$.

Let Φ be a real-valued function on some subclass $\mathcal{S}$ of $\mathbb{C}^d$ and let us assume that there is a constant α such that all convex bodies $C \in \mathcal{S}$ satisfy the geometric inequality

$$\Phi(C) \leq \alpha \,. \tag{i}$$

Let $\mathcal{S}^*$ denote the class of all elements $C \in \mathcal{S}$ such that $\Phi(C) = \alpha$. In many cases the class $\mathcal{S}^*$ may contain certain special convex bodies, such as balls, ellipsoids or regular polyhedra. Then, we say that these objects have a maximum property with respect to the function Φ.

For example, let K be a fixed circular disc of radius ρ in E^2 and let $\mathcal{S}_n$ be the class of all convex n-gons C of given area a. Put $\Phi(C) = a(K \cap C)$ and $\alpha = a(K \cap \bar{C})$, where $\bar{C}$ is a regular n-gon of area a that is concentric with K, and $a(S)$ stands for the area of the set S. Then (i) becomes the inequality

$$a(K \cap C) \leq a(K \cap \bar{C}) \,, \tag{ii}$$

which is due to L. Fejes Tóth. If ρ is not smaller than the inradius and not greater than the circumradius of $\bar{C}$, equality holds in (ii) if and only if $C = \bar{C}$.

To give a further example, we consider the class $\mathcal{S}_f$ of all 3-dimensional convex

polyhedra P with f faces and inradius r. Let $\Phi(P) = -V(P)$, where $V(P)$ denotes the volume of P. According to L. Fejes Tóth we have

$$V(P) \geqslant 2(f-2)\tan\omega_f\,(4\sin^2\omega_f - 1)r^3\,, \tag{iii}$$

where $\omega_f = \pi f/6(f-2)$. Equality holds only if $f=4$ and P is a regular tetrahedron, or $f=6$ and P is a cube, or $f=12$ and P is a regular dodecahedron. We naturally cannot expect to find a lower bound for $V(P)$ which is precise for every value of f.

In this article we shall primarily be dealing with geometric inequalities of type (i) and extremum problems involving the fundamental measures of a convex body in two and three dimensions, i.e., area, perimeter, volume, surface area and integral of mean curvature. Some of the results concern the approximation of convex discs or bodies by convex polygons or polyhedra. The approximation of a convex disc by polygons is not only of interest in itself, but has important applications to problems of packing and covering.

For the isoperimetric problem we refer to chapter 1.3 in this Handbook by Talenti. We shall restrict ourselves to a few remarks in the 2-dimensional case (section 2) and the discussion of the isoperimetric problem for convex polyhedra (section 6). The books and the review paper by L. Fejes Tóth (1964, 1972, 1984), and chapter 3.3 in this Handbook by G. Fejes Tóth and Kuperberg provide comprehensive information on extremum problems in the area of packing and covering. Geometric inequalities involving mixed volumes and quermassintegrals are covered in the books of Hadwiger (1957) and Leichtweiss (1980) and chapter 1.2 in this Handbook by Sangwine-Yager. Groemer's chapter 1.4 concerns strengthened forms of inequalities of type (i).

The book of Mitrinović, Pečarić and Volenec (1989) represents an encyclopaedic work on geometric inequalities of various levels. It contains much material that is not easily accessible elsewhere.

An important reference work, not only for the geometric inequalities themselves, but also for typical methods of their proof, is the book by Burago and Zalgaller (1988).

The following notation will be used consistently throughout this chapter. For any $K \in \bar{\mathbb{C}}^d$ let $V(K)$ denote the volume, $F(K)$ the surface area, $M(K)$ the integral of mean curvature, $W(K)$ the mean width and $W_j(K)$ $(j=0,\ldots,d)$ the jth quermassintegral of K. The circumradius and inradius of K will be denoted by $R(K)$ and $r(K)$, respectively. Where no confusion is possible, we shall simply write $V, F, \ldots$ instead of $V(K), F(K), \ldots$. The unit ball in E^d centred at the origin will be denoted by B^d, the boundary of B^d by S^{d-1} and the volume of B^d by κ_d. Note that in E^3 we have $M = 2\pi W$.

A compact subset of S^2 is said to be *convex* if it contains, with any two of its points, the smaller arc or at least one semicircular arc of a great circle connecting them. A compact convex proper subset of S^2 with nonempty interior will be called a *convex cap*. Occasionally we shall say "circle" or "sphere" if we mean a circular disc or a solid sphere, respectively. We let H^d denote the d-dimensional hy-

perbolic space. If C is a convex disc in E^2 or H^2 (a convex cap in S^2), we write $a(C)$ or a for the area of C and $p(C)$ or p for the perimeter of C. By a *convex n-gon* we mean a convex polygon with exactly n sides. Further definitions will be given where they are actually used.

In d-dimensional space, a polytope P is *circumscribed* about a convex body K if all the $(d-1)$-dimensional faces of P are tangential to K. Analogously, we say that P is *inscribed* in K if the boundary of K contains all the vertices of P.

For the standard definitions and theorems of the theory of convex sets we refer to the books of Bonnesen and Fenchel (1934), Hadwiger (1957), Eggleston (1958), or Leichtweiss (1980). The book of Grünbaum (1967) gives a detailed exposition of the theory of polytopes.

2. Some extremum problems for convex discs

Let us begin with four simple extremum properties of the regular n-gons in the Euclidean plane E^2, in the hyperbolic plane H^2 or on the 2-sphere S^2. The number n is an arbitrary (but fixed) integer with $n \geqslant 3$.

Theorem 1. *Among all convex n-gons of given inradius, the regular n-gon has minimal area and minimal perimeter.*

Theorem 2. *Among all convex n-gons of given circumradius, the regular n-gon has maximal area and maximal perimeter.*

These facts are summarized in the following inequalities. Let a and p denote the area and the perimeter of a convex n-gon P with inradius r and circumradius R. The parameter $\kappa = +1$, 0, or -1 refers to the spherical, Euclidean, or hyperbolic case, respectively.

$$[2\pi - 2n \arcsin(\cos(\sqrt{\kappa}r)\sin(\pi/n))]/\kappa \leqslant a \leqslant [2\pi - 2n \arctan(\cos(\sqrt{\kappa}R)\tan(\pi/n))]/\kappa\,, \tag{1,2}$$

$$2n \arctan(\sin(\sqrt{\kappa}r)\tan(\pi/n))/\sqrt{\kappa} \leqslant p \leqslant 2n \arcsin(\sin(\sqrt{\kappa}R)\sin(\pi/n))/\sqrt{\kappa}\,. \tag{3,4}$$

In the Euclidean case the corresponding formulae are obtained by letting $\kappa \to 0$ in the above inequalities. For instance, we have

$$n\tan(\pi/n)\cdot r^2 \leqslant a \leqslant n\sin(\pi/n)\cos(\pi/n)\cdot R^2\,, \quad P \in \mathrm{E}^2\,.$$

Equality holds in each of the formulae (1)–(4) only for regular n-gons. For proofs of both theorems see Kürschák (1887) and L. Fejes Tóth (1964, p. 153, 1972, p. 6). In the inequality (1), the area of P can be replaced by the area of the

intersection of P with the circumcircle of the regular n-gon circumscribed about the incircle of P. Similarly, in the inequality (4), the perimeter of P can be replaced by the perimeter of the convex hull of P and the incircle of the regular n-gon inscribed in the circumcircle of P (L. Fejes Tóth 1947, 1959, 1972, pp. 7, 189). The above inequalities are not independent, e.g., spherical polarity shows that (1) and (4) as well as (2) and (3) are equivalent for $\kappa = +1$.

Combining the two inequalities (1) and (2), or (3) and (4), we obtain the nice corollaries

$$\frac{R}{r} \geqslant \frac{1}{\cos(\pi/n)}, \quad \kappa = 0, \qquad \frac{\tan R}{\tan r} \geqslant \frac{1}{\cos(\pi/n)}, \quad \kappa = +1,$$

with equality only for regular n-gons.

The circle is uniquely characterized by the property that, among all plane convex discs of given perimeter, the circle has maximal area. This property is expressed by the isoperimetric inequality

$$p^2 \geqslant 4\pi a, \tag{5}$$

where a and p are the area and the perimeter of a convex disc C, and where equality holds if and only if C is a circle. Many stronger forms, generalizations, extensions to other spaces and applications gave rise to a very extensive literature. We refer to the books of Blaschke (1916), Bonnesen and Fenchel (1934), Hadwiger (1955, 1957), Leichtweiss (1980), Pólya and Szegö (1951) (for relations to physics), Santaló (1976) (for relations to integral geometry), the review paper by Osserman (1978), and the references given there. For the history of the isoperimetric problem in ancient and more recent times, see Gericke (1982).

Some integral inequalities involve the lengths of chords which bisect the area, or perimeter, of a planar convex set. If C is a convex disc and $\theta \in [0, 2\pi]$, we denote by $l_p(C; \theta)$ (and $l_a(C; \theta)$) the lengths of the perimeter (and area) bisectors of C which make an angle θ with a fixed line. Then

$$8a(C) \leqslant \int_0^{2\pi} l_p^2(C; \theta)\, \mathrm{d}\theta \leqslant 2p^2(C)/\pi$$

and

$$8a(C) \leqslant \int_0^{2\pi} l_a^2(C; \theta)\, \mathrm{d}\theta \leqslant 2p^2(C)/\pi .$$

The left-hand inequalities were established by Goodey (1982); equality occurs if and only if C is centrally symmetric. The right-hand inequalities are due to Lutwak (1980); equality occurs if and only if C is a circular disc. Note that either pair of inequalities implies (5). See also Chakerian and Goodey (1984).

The isoperimetric property of the circle in the plane was extended to the

2-sphere by Bernstein (1905) and Bonnesen (1921), and to the hyperbolic plane (in a much more general context) by Schmidt (1940); see also Bezdek (1985). Summing up, we can state the following theorem.

Theorem 3. *A circle in* S^2, E^2, *or* H^2 *has a greater area than any other compact convex set with boundary of the same length.*

The isoperimetric inequality now takes the form

$$p^2 \geqslant (4\pi - \kappa a)a\,, \tag{6}$$

with equality if and only if the set of area a and perimeter p is a circle.

If we restrict ourselves to convex n-gons, Theorem 3 can be strengthened.

Theorem 4. *A regular n-gon* ($n \geqslant 3$) *in* S^2, E^2, *or* H^2 *has a greater area than any other convex n-gon with boundary of the same length.*

Theorem 4 is equivalent to the inequality

$$a \leqslant \left(2\pi - 2n \arccos \frac{\cos(\pi/n)}{\cos(\sqrt{\kappa}p/2n)}\right)/\kappa \tag{7}$$

which, for plane convex n-gons, takes the form

$$p^2 \geqslant 4an \tan(\pi/n)\,. \tag{8}$$

Equality holds in (7) and (8) only for regular n-gons. Indirect proofs of Theorem 4 are contained in L. Fejes Tóth (1964, p. 213, 1972, p. 9) and Bezdek (1985) for $\kappa = +1$, 0, and -1, respectively.

A direct and completely elementary proof of (8) is given in L. Fejes Tóth (1964, p. 157). This proof makes use of the concept of inner parallel domains of a plane convex polygon and yields an inequality which is stronger than (8); see Bol (1943a). This method provides also an improvement of (5), namely

$$p^2 - 4\pi a \geqslant (p - 2\pi r)^2 \tag{9}$$

(r = inradius), which applies to all convex discs. Equality holds for a parallel domain of a line segment. Inequality (9) together with the analogous inequality

$$p^2 - 4\pi a \geqslant (2\pi R - p)^2 \tag{9'}$$

had been previously obtained by Bonnesen (1921), who actually used similar arguments as Bol in his proof. For alternative approaches to (9) and (9') see L. Fejes Tóth (1950a) and Santaló (1976, p. 119).

Besicovitch (1949) gave a proof of (9) for domains of area a which are bounded

by a Jordan curve of length p and have inradius r. Furthermore, he gave the first characterization of those curves for which equality holds in (9). In the class of all convex sets with given perimeter and circumradius, only the symmetrical lens bounded by two circular arcs has maximal area. This was shown by Favard (see Bonnesen 1929) and, in a different way, by Besicovitch (1952). Inequality (9′) can be deduced from Favard's theorem, and it can be proved that (9′) holds for closed Jordan curves and that equality occurs only for circles. For a comprehensive survey of Bonnesen-style inequalities see Osserman (1979).

The following isoperimetric problem with constraints, considered by Besicovitch (1952) has applications to problems of packing (L. Fejes Tóth 1957): find a convex disc C of perimeter p that is contained in a given compact convex set S and has maximal area. In a closely related paper, Singmaster and Souppouris (1978) solved the problem of maximizing the ratio $a(C)/p(C)$ for $C \subset S$. Montesinos and Segura (1990) and Longinetti (1986) discussed further variants of the isoperimetric problem.

Observe that the combination of (1) and (7) gives (3), and the combination of (4) and (7) gives (2).

The inequalities (1) and (2) can be generalized to a finite number of convex polygons. Let $P_1, \ldots, P_k$ be convex polygons (on S^2 possibly digons, i.e., convex polygons each bounded by two semicircular arcs), and let N and A be their total number of sides and their total area. Decompose every polygon P_i into triangles spanned by an inner point of P_i, an inner point of a side of P_i and an endpoint of this side. The total number of such triangles is $2N$ and the total area is A. The sums of the angles of these triangles at the vertices lying within the polygons and on the sides of the polygons are $2k\pi$ and $N\pi$, respectively. Thus the triangle $T = OBC$ defined by the data

$$\measuredangle\, O = k\pi/N\,, \qquad \measuredangle\, B = \pi/2\,, \qquad a(T) = A/2N$$

can be considered as representing the average of all triangles. The triangle T is called the *characteristic triangle* of the set $\{P_1, \ldots, P_k\}$. The side-lengths $\overline{OB} = \bar{r}$ and $\overline{OC} = \bar{R}$ are given by

$$\begin{aligned} \cos(\sqrt{\kappa}\bar{r}) &= \sin\frac{2k\pi - \kappa A}{2N}\operatorname{cosec}\frac{k\pi}{N}\,, \\ \cos(\sqrt{\kappa}\bar{R}) &= \tan\frac{2k\pi - \kappa A}{2N}\cot\frac{k\pi}{N}\,, \end{aligned} \tag{10}$$

and satisfy the inequalities

$$\begin{aligned} &0 < \bar{r} < \bar{R} < \infty \qquad \text{for } \kappa = 0, -1\,, \\ &0 < \bar{r} \leqslant \bar{R} \leqslant \pi/2 \qquad \text{for } \kappa = +1\,. \end{aligned}$$

Note that $\bar{r} = \bar{R}$ if and only if $P_1, \ldots, P_k$ are hemispheres, and that $\bar{R} = \pi/2$ if and only if $P_1, \ldots, P_k$ are digons. If the polygons are regular and congruent, then $\bar{r}$ is their inradius and $\bar{R}$ their circumradius.

The inequalities (1) and (2) are equivalent to the particular case $k = 1$ of the following theorem.

Theorem 5. *Let $K(\rho)$ be a (closed) circle of radius ρ ($<\pi/2$ if $K(\rho) \subset S^2$), and let $P_1, \ldots, P_k$ be convex polygons with total number of sides N and total area A. If $K(\rho) \subset P_i$ for $i = 1, \ldots, k$, then*

$$\rho \leqq \bar{r}\,, \tag{11}$$

and if $K(\rho) \supset P_i$ for $i = 1, \ldots, k$, then

$$\rho \geqq \bar{R}\,. \tag{12}$$

Equality occurs if and only if $P_1, \ldots, P_k$ are regular and congruent and circumscribed about $K(\rho)$ or inscribed in $K(\rho)$, respectively.

Theorem 5 can be applied to problems of packing and covering with equal circles on the sphere and in the plane; see L. Fejes Tóth (1943a,b, 1972, pp. 80, 113).

If $\bar{r} < \rho < \bar{R}$, then the circle $K(\rho)$ is neither contained in all P_i nor contains all P_i. Thus we can ask for an upper bound of the sum $\Sigma_{i=1}^{k} a(P_i \cap K(\rho))$ in terms of k, A, N and ρ. Such an upper bound is given by Theorem 6. In the following we shall repeatedly have to consider the particular case when $P_1, \ldots, P_k$ are regular and congruent polygons concentric with $K(\rho)$. We shall refer to this situation as the *regular case*.

Theorem 6. *Let $K(\rho)$ denote the circle with centre O and radius ρ ($<\pi/2$ if $K(\rho) \subset S^2$). Let $P_1, \ldots, P_k$ be convex polygons as in Theorem 5, and let $T = OBC$ be the characteristic triangle of the set $\{P_1, \ldots, P_k\}$. Then*

$$\sum_{i=1}^{k} a(P_i \cap K(\rho)) \leqq 2Na(T \cap K(\rho)) \tag{13}$$

with equality in the regular case. If moreover $\bar{r} \leqq \rho \leqq \bar{R}$, then equality holds only in the regular case.

Obviously, Theorem 6 implies Theorem 5. Theorem 6 originates in two papers by L. Fejes Tóth (1948a, 1950b) and will be generalized in Theorem 7. Two different proofs of Theorem 6 are due to G. Fejes Tóth (1973) and A. Florian (1975).

In the particular case $k = 1$, $2N$ copies of the triangle T fit together to form a regular N-gon of area A. Thus we have the following corollary.

Corollary 1. *Let $K(\rho)$ be a circle of radius ρ ($<\pi/2$ if $K(\rho) \subset S^2$), and let P be a convex n-gon. If $\bar{P}$ is a regular n-gon concentric with $K(\rho)$ and such that $a(\bar{P}) = a(P)$, then*

$$a(P \cap K(\rho)) \leqslant a(\bar{P} \cap K(\rho)) . \tag{14}$$

If $\bar{r} \leqslant \rho \leqslant \bar{R}$, where $\bar{r}$ and $\bar{R}$ are the inradius and the circumradius of $\bar{P}$, then equality holds only for $P = \bar{P}$.

An elementary proof of (14) which includes the discussion of asymptotic polygons was given by Hajós (1968).

Let D be a domain and O a fixed point. Let g be a (possibly not strictly) decreasing function defined for $x \geqslant 0$. The integral

$$M_g(D, O) = \int_D g(\overline{OX})\, \mathrm{d}a$$

is called the *moment of D with respect to O*, where $\mathrm{d}a$ denotes the area element at the variable point X.

Theorem 7. *In S^2, E^2 or H^2 let $P_1, \ldots, P_k$ be convex polygons and let N denote the total number of sides and A the total area. Let $T = OBC$ be the characteristic triangle of the set $\{P_1, \ldots, P_k\}$. If g is a decreasing function defined for $x \geqslant 0$, then*

$$\sum_{i=1}^{k} M_g(P_i, O) \leqslant 2N M_g(T, O) \tag{15}$$

with equality in the regular case. If g satisfies the additional conditions

(i) *$x_1 < \bar{r}$ implies $g(x_1) > g(\bar{R} - 0)$ (if $P_i \in S^2$, $\bar{r} < \bar{R} = \pi/2$);*

(ii) *$x_1 < \bar{r} \leqslant \bar{R} < x_2$ implies $g(x_1) > g(x_2)$ (in any other case),*

then equality holds only in the regular case.

The number $g(\bar{R} - 0)$ is defined by $\lim g(x)$ as x tends to $\bar{R}$ from the left. For an increasing function g the sign $\leqslant$ in (15) has to be replaced by $\geqslant$.

Let us point out the special case $k = 1$.

Corollary 2 (Momentum lemma). *Let P be a convex n-gon, and let $\bar{P}$ be a regular n-gon of the same area as P and with centre O. Then we have for any decreasing function g*

$$M_g(P, O) \leqslant M_g(\bar{P}, O) . \tag{16}$$

Whenever g satisfies the conditions (i), (ii) *in Theorem* 7, *equality holds only for $P = \bar{P}$.*

Theorem 7 reduces to Theorem 6 by taking for g the characteristic function of the interval $[0, \rho]$. On the other hand, inequality (15) can be deduced from (13) by first proving (15) for a decreasing step-function g and then using an appropriate approximation argument.

Proofs of Theorem 7 may be found in L. Fejes Tóth (1950b, 1972, p. 137, 1964,

p. 219), Imre (1964), G. Fejes Tóth (1973), A. Florian (1975). This theorem turned out to be very useful in discrete geometry and has various applications to problems of packing and covering (L. Fejes Tóth 1972, pp. 80, 140; Imre 1964), in the theory of convex polyhedra (Coxeter and L. Fejes Tóth 1963, L. Fejes Tóth 1948a, 1950b, 1956a, 1963, 1964, p. 308, A. Florian 1975, Linhart 1979, Tomor 1965) and in location theory (Matérn and Persson 1965). It has also points of contact with information theory (Landau and Slepian 1966) and biology (L. Fejes Tóth 1964, p. 233).

In the rest of this section we shall deal with problems in the Euclidean plane, unless otherwise stated.

Let C be a convex disc of prescribed area a, and let $a_n(C)$ denote the maximum of the areas of all convex n-gons contained in C. If K is a circle of area a, then by Theorem 2

$$a_n(K) = a \frac{n}{2\pi} \sin \frac{2\pi}{n} .$$

The following theorem characterizes the ellipses as extremal domains with respect to $a_n(C)$.

Theorem 8. *If C is a convex disc of area a, then*

$$a_n(C) \geqslant a \frac{n}{2\pi} \sin \frac{2\pi}{n} \tag{17}$$

with equality only for ellipses.

Observe that a_n/a is invariant under nonsingular affine transformations. Sas (1939, 1941) (see also L. Fejes Tóth 1972, p. 36) proved inequality (17) in an elegant analytic way. Theorem 8 confirms a conjecture by L. Fejes Tóth, who supplied the proof of uniqueness, making use of Fourier series. Long before, Blaschke (1917a, 1923, p. 49) had established Theorem 8 for $n = 3$ and proved uniqueness under the assumption that C has a smooth boundary. This restriction was removed in a subsequent paper by Gross (1918). Pucci (1982) gave a further complete proof of Theorem 8 when $n = 3$. The arguments used by Blaschke, Gross and Pucci are of geometrical nature and rest on the Steiner symmetrization. As in similar cases, the proof of uniqueness causes the main difficulty.

Let C be a convex body in Euclidean 3-space and let B be a ball. Define $\phi(C)$ to be the maximum of the volume ratio $V(P)/V(C)$, as P ranges over all tetrahedra in C. Then

$$\phi(C) \geqslant \phi(B) \tag{18}$$

with equality only for ellipsoids (Blaschke 1917b, 1923, p. 191, Gross 1918). Bianchi (1986) obtained a similar result for inscribed polyhedra with five vertices. Macbeath (1951a) proved (18) for all dimensions $d \geqslant 2$ and all convex polyhedra $P \subset C$ with a given number of vertices. But here the problem of finding all convex

bodies C that minimize $\phi(C)$ remains open. A variant of Blaschke's problem is completely solved for all dimensions in a paper of McKinney (1974).

Extremum properties of ellipsoids involving multiple integrals can be found in a review paper by Petty (1983) and in the references given there.

Let $A_n(C)$ be the minimum of the areas of all convex n-gons containing a convex disc C of given area a. It can be shown that $A_3(C) \leqslant 2a$ with equality only if C is a parallelogram (Gross 1918, Eggleston 1953, 1957b, p. 156). Reinhardt (1934) and Mahler (1947) independently discovered centrally symmetric convex discs with $A_6/a > A_6(E)/a = \sqrt{12}/\pi = 1.10265\ldots$, where E is an ellipse of area a. Mahler showed that a regular octagon has the required property by proving that in this case $A_6/a = (3+\sqrt{2})/4 = 1.10355\ldots$. For $n > 3$, the convex discs C maximizing $A_n(C)/a$ seem not to be known. Some partial results are contained in Chakerian and Lange (1971), Chakerian (1973a) and Kuperberg (1983). If C is assumed to be centrally symmetric, then $A_4(C) \leqslant 4a/3$ with equality only if C is an affine image of a regular hexagon (Petty 1955). For alternative proofs see Babenko (1988) and Pelczynski and Szarek (1991). In these papers (see also Dvoretzky and Rogers 1950) upper bounds for the ratio $V(P)/V(C)$ are established, where P is a parallelotope of minimal volume circumscribed about the central convex body C.

Theorem 8 and the results of Reinhardt and Mahler have important consequences for the problems of thinnest lattice covering and densest lattice packing with a central convex disc (L. Fejes Tóth 1972, p. 104, Gruber and Lekkerkerker 1987, p. 245).

In view of the above remarks on $A_n(C)$ the following theorem of Lázár (1947) is of interest (see also L. Fejes Tóth 1972, p. 40).

Theorem 9. *Let $A_n(C)$ be the minimal area of all convex n-gons containing the convex disc C, and let $a_n(C)$ be the maximal area of all convex n-gons contained in C. Then*

$$\frac{A_n(C)}{a_n(C)} \leqslant \frac{1}{\cos^2(\pi/n)} \tag{19}$$

with equality when C is an ellipse.

For related inequalities concerning homothetic triangles, see Lassak (1992). Based on a result of Macbeath (1951b), Butler (1971) showed that the class of d-dimensional simplices is extreme for the problem of maximizing the volume of the minimal circumscribing cylinder to a d-dimensional convex body of unit volume. According to Schneider (1973) the simplices form the only class of extremal bodies. Martini (1990) gave a different and elementary proof.

Let C be a given convex body in E^d, and let C_1 and C_2 be central convex bodies such that $C_1 \subset C \subset C_2$. Fáry and Rédei (1950) considered the problems of maximizing $V(C_1)/V(C)$ and $V(C)/V(C_2)$ and solved them when C is a simplex.

We now proceed to discuss two inequalities that are similar to (17) but involve the perimeter instead of the area.

Theorem 10. *Let $P_n(C)$ be the minimum perimeter of all convex n-gons containing the convex disc C with perimeter p, and let $p_n(C)$ be the maximum perimeter of all convex n-gons contained in C. Then*

$$P_n(C) \leq p \frac{n}{\pi} \tan \frac{\pi}{n}, \tag{20}$$

$$p_n(C) \geq p \frac{n}{\pi} \sin \frac{\pi}{n}. \tag{21}$$

Equality holds if and only if C is a circle.

Both inequalities were proved by Schneider (1971) who also showed that equality in (20) implies that C is a circle. The idea of the proof is similar to that of Theorem 8 given by Sas and L. Fejes Tóth. To prove that in (21), too, equality applies only to circles, one has to show that the equation

$$\tan(k\pi/n) = k \tan(\pi/n) \tag{22}$$

has no solution in integers $k \geq 2$, $n \geq 3$; for a proof of this result see A. Florian and Prachar (1986). Kanemitsu and Shirasaka (1988) investigated a more general Diophantine equation.

Inequality (20) can be extended to d-dimensional convex bodies of given mean width and circumscribing convex polytopes with n $(d-1)$-dimensional faces (Schneider 1967). A similar extension is possible also for inequality (21). The sphere proves to be an extremal body in either case, but the question of uniqueness remains open for $d \geq 3$.

We know only few extremum properties of non-Euclidean domains in connection with inscribed or circumscribed polygons. In the following we call attention to some unsolved problems.

Let C be a convex cap on the 2-sphere. Let $p_n(C)$ be the maximal perimeter of all convex n-gons contained in C. Let $P_n(C)$ be the minimal perimeter of all convex n-gons containing C. We denote by $a_n(C)$ and $A_n(C)$ the corresponding numbers concerning the area instead of the perimeter. If K is a circular cap of radius r, we write $P_n(K) - p_n(K) = \pi(r)$ and $A_n(K) - a_n(K) = \alpha(r)$ (the values of $\pi(r)$ and $\alpha(r)$ can be obtained from Theorems 1 and 2). The functions $\pi(r)$ and $\alpha(r)$ attain their common maximum for

$$r_{np} = \arcsin \frac{(-3 + (9 + 4\tan^2 \pi/n)^{1/2})^{1/2}}{\sqrt{2} \tan \pi/n}$$

and

$$r_{na} = \pi/2 - r_{np},$$

respectively. L. Fejes Tóth (1988) raised the questions whether

$$P_n(C) - p_n(C) \leq \pi(r_{np}) \tag{23}$$

and

$$A_n(C) - a_n(C) \leqslant \alpha(r_{na}) \tag{24}$$

hold for any convex spherical cap C. The inequalities (23) and (24) are equivalent, as can be seen by spherical polarity.

Much earlier, the same author (1956b) had proved (23) for $n = 3$, where equality holds only if C is a circle of radius r_{3p}. For $n > 3$ the problem is still open. Two further unsolved problems are the following: find (i) the maximum of $P_n(C) - p$ and $a - a_n(C)$, and (ii) the maximum of $A_n(C) - a$ and $p - p_n(C)$, extended over all convex caps $C \in S^2$ with given perimeter p and given area a, respectively.

Let us return to Euclidean space. Several minimum properties of the sphere concern the mean values of certain functionals associated with a convex body; see Chakerian (1972, 1973b), Schneider (1972), Lutwak (1977), Sangwine-Yager (1983) and the bibliography cited there. This type of problem will be illustrated by the following result of Sangwine-Yager (1980).

Let P be a polygon circumscribed about the unit circle. For any θ with $0 \leqslant \theta < 2\pi$ let $P(\theta)$ be the polygon obtained from P by rotation about the origin through the angle θ in the clockwise direction. If C is a convex disc, denote by $P(C, \theta)$ the polygon circumscribed about C with sides parallel to those of $P(\theta)$.

Theorem 11. *For any convex disc C we have*

$$\int_0^{2\pi} a(P(C, \theta))\, \mathrm{d}\theta \geqslant 2a(P)a(C)\,,$$

with equality if and only if C is a circle.

The technique developed for the proof makes use of inner parallel domains and is of interest in itself.

A number of problems discussed in this section may be viewed from the aspect of approximating a given convex disc by inscribed or circumscribed polygons. To make this viewpoint precise, we shall employ two methods of measuring the deviation of two convex sets. One of the most usual distance functions is:

The symmetric difference metric ρ^S. If K_1 and K_2 are plane convex discs, the distance $\rho^S(K_1, K_2)$ is defined by

$$\begin{aligned} \rho^S(K_1, K_2) &= a(K_1 \cup K_2) - a(K_1 \cap K_2) \\ &= a(K_1 \triangle K_2)\,, \end{aligned} \tag{25}$$

where $K_1 \triangle K_2 = (K_1 \cup K_2) \backslash (K_1 \cap K_2)$ is the symmetric difference of the two sets. The distance $a(K_1 \triangle K_2)$ is also called the *area deviation* of K_1 and K_2. If $K_1 \subset K_2$, (25) reduces to

$$\rho^{S}(K_1, K_2) = a(K_2) - a(K_1), \quad K_1 \subset K_2.$$

Now, ρ^{S} is a metric on $\mathbb{C}^2$, but not on $\bar{\mathbb{C}}^2$. Definition (25) immediately generalizes to d dimensions with the d-dimensional volume V in place in place of the area a.

We shall also make use of another concept of deviation which corresponds closely to the area deviation.

The perimeter deviation ρ^{P}. If K_1 and K_2 are compact, convex and nonempty subsets of the plane, the distance $\rho^{P}(K_1, K_2)$ is defined by

$$\rho^{P}(K_1, K_2) = 2p([K_1, K_2]) - p(K_1) - p(K_2), \tag{26}$$

where $[K_1, K_2] = \operatorname{conv}(K_1 \cup K_2)$. In the case $K_1 \subset K_2$, definition (26) reduces to

$$\rho^{P}(K_1, K_2) = p(K_2) - p(K_1), \quad K_1 \subset K_2.$$

Observe that ρ^{P} is a metric on $\bar{\mathbb{C}}^2$, but ρ^{P} should be distinguished from the distance function $\bar{\rho}^{P}$ defined by

$$\bar{\rho}^{P}(K_1, K_2) = p(K_1 \cup K_2) - p(K_1 \cap K_2),$$

which is not a metric. In fact, $\bar{\rho}^{P}$ was used by Eggleston (1957a, 1957b, p. 93) in connection with the approximation of convex discs by polygons.

Cauchy's perimeter formula states that $p(K) = \pi W(K)$ for any plane compact convex set K, where W denotes the mean width. This suggests defining the distance function

$$\rho^{W}(K_1, K_2) = 2W([K_1, K_2]) - W(K_1) - W(K_2) \tag{27}$$

for any two compact convex nonempty subsets of E^d ($d \geq 2$). Then ρ^{W} can be written in the form

$$\rho^{W}(K_1, K_2) = \frac{2}{d\kappa_d} \int_{S^{d-1}} |h(K_1, u) - h(K_2, u)| \, d\sigma(u), \tag{28}$$

where $h(K_i, u)$ is the support function of K_i in direction u, and $d\sigma(u)$ denotes the surface area element at the point $u \in S^{d-1}$. From (28) it is clear that ρ^{W} is the L_1-metric on $\mathbb{C}^d$. In the form given by (28) ρ^{W} was introduced in a paper by McClure and Vitale (1975). Several properties of ρ^{W} were discussed by Vitale (1985) (in a more general context) and, independently and in a different way, by A. Florian (1989a). For a review of various concepts of deviation between convex bodies in E^d, including a comprehensive bibliography, we refer to Gruber (1983) and chapter 1.10 in this Handbook by Gruber.

By using the metrics ρ^{S} and ρ^{P} we can restate Theorems 8 and 10 in the following form.

Let $\mathscr{C}(a)$ be the class of all convex discs of area a, and let $n \geq 3$ be given. The least area deviation between $C \in \mathscr{C}(a)$ and any convex n-gon contained in C attains its maximum if and only if C is an ellipse.

Let $\mathscr{C}(p)$ be the class of all convex discs of perimeter p, and let $n \geq 3$ be given. The least perimeter deviation between $C \in \mathscr{C}(p)$ and any convex n-gon contained in C (containing C) attains its maximum if and only if C is a circle.

Let us now turn to another group of problems. We consider the class of all convex discs with areas and perimeters bounded by given constants. Which disc of this class has *minimal* area deviation or *minimal* perimeter deviation from an n-gon? Let C be a disc from this class, and P an n-gon. First we shall restrict ourselves to the case $P \subset C$, then we shall assume that $P \supset C$, and finally we shall consider the general case, which means that the mutual position of C and P is not restricted in any way.

For positive numbers a and p satisfying the isoperimetric inequality $p^2/a \geq 4\pi$, let $\mathscr{C}(a, p)$ be the class of all convex discs with area not less than a and perimeter not greater than p. Furthermore, let $\mathscr{P}_n$ be the class of all convex m-gons with $m \leq n$. Write

$$\begin{aligned} d_1^{\mathrm{S}}(a, p, n) &= \min \rho^{\mathrm{S}}(C, P), & d_1^{\mathrm{P}}(a, p, n) &= \min \rho^{\mathrm{P}}(C, P), \\ f_1(a, p, n) &= \max a(P), & g_1(a, p, n) &= \max p(P), \end{aligned} \tag{29}$$

where the minimum and the maximum is to be taken over all discs C from $\mathscr{C}(a, p)$ and all polygons P from $\mathscr{P}_n$ with $P \subset C$. The existence of these minima and maxima follows from the Blaschke selection theorem. The functions d_1^{S} and d_1^{P} are interesting only in the case when

$$p^2/a < 4\pi t,$$

where

$$t = \frac{n}{\pi} \tan \frac{\pi}{n}.$$

Otherwise we have $d_1^{\mathrm{S}}(a, p, n) = d_1^{\mathrm{P}}(a, p, n) = 0$.

A *regular arc-sided n-gon* is defined to be a convex disc C obtained from a regular n-gon P by joining each two adjacent vertices of P by congruent circular arcs. P is called the *kernel* of C. For the following theorem see G. Fejes Tóth and A. Florian (1984).

Theorem 12. *Let $n \geq 3$ be an integer and let a and p be positive numbers such that*

$$4\pi \leq p^2/a < 4\pi t. \tag{30}$$

Let us assume that a disc $C \in \mathscr{C}(a, p)$ and a polygon $P \in \mathscr{P}_n$ with $P \subset C$ satisfy one of the conditions

(i) $\rho^{\mathrm{P}}(C, P) = d_1^{\mathrm{P}}(a, p, n)$,
(ii) $p(P) = g_1(a, p, n)$,
(iii) $a(P) = f_1(a, p, n)$,
(iv) $\rho^{\mathrm{S}}(C, P) = d_1^{\mathrm{S}}(a, p, n)$.
Then C is a regular arc-sided n-gon of area a and perimeter p, and P is the kernel of C.

It can be shown that with condition (30) there is exactly one regular arc-sided n-gon of area a and perimeter p. Observe that the cases (ii), (iii) and (iv) are consequences of case (i).

Similarly as above we define four functions

$$\begin{aligned} d_2^{\mathrm{S}}(a, p, n) &= \min \rho^{\mathrm{S}}(C, P)\,, & d_2^{\mathrm{P}}(a, p, n) &= \min \rho^{\mathrm{P}}(C, P)\,, \\ f_2(a, p, n) &= \min a(P)\,, & g_2(a, p, n) &= \min p(P)\,, \end{aligned} \tag{31}$$

where the minimum is taken over all discs $C \in \mathscr{C}(a, p)$ and all polygons $P \in \mathscr{P}_n$ with $C \subset P$.

A *smooth regular n-gon* is defined to be a smooth convex disc C obtained from a regular n-gon P by rounding off its corners with congruent circular arcs. P is called the *case* of C. Note that $C = (P_{-\sigma})_{\sigma}$, which means that C is the outer parallel domain of the inner parallel domain of P at some distance $\sigma \leqslant r(P)$.

Theorem 13. *Let a, p and n satisfy inequality* (30). *Let us assume that a disc $C \in \mathscr{C}(a, p)$ and a polygon $P \in \mathscr{P}_n$ with $P \supset C$ satisfy one of the conditions*:
(i) $\rho^{\mathrm{S}}(C, P) = d_2^{\mathrm{S}}(a, p, n)$,
(ii) $a(P) = f_2(a, p, n)$,
(iii) $p(P) = g_2(a, p, n)$,
(iv) $\rho^{\mathrm{P}}(C, P) = d_2^{\mathrm{P}}(a, p, n)$.
Then C is a smooth regular n-gon of area a and perimeter p, and P is the case of C.

Again, with condition (30) there exists exactly one smooth regular n-gon of area a and perimeter p. Theorem 13 can be deduced from a result of L. Fejes Tóth (1957), who applied a theorem by Besicovitch (1952); Hadwiger gave an alternative proof of this result; see L. Fejes Tóth and Heppes (1960). Theorems 12 and 13 have applications in the theory of packing and covering; see L. Fejes Tóth (1957, 1964, p. 175), G. Fejes Tóth and A. Florian (1984).

Denoting by $\mathscr{P}_n(a_0)$ the class of all convex m-gons with $m \leqslant n$ and given area a_0, we direct our attention to the following problem.

Problem. Find a member of $\mathscr{C}(a, p)$ and a member of $\mathscr{P}_n(a_0)$ such that their intersection has maximal area.

Accordingly, we introduce the function

$$M(a, p; n, a_0) = \max a(C \cap P)\,,$$

where the maximum extends over all discs $C \in \mathscr{C}(a, p)$ and all polygons $P \in \mathscr{P}_n(a_0)$. The function M is interesting only when $f_1(a, p, n) \leq a_0 \leq p^2 t/4\pi$ (=area of a regular n-gon circumscribing a circle of perimeter p). Otherwise we have $M = a_0$ or $M = p^2/4\pi$. Let C be the outer parallel domain of a regular arc-sided n-gon at some distance r_1. If C is not a circle, then C is bounded by n equal circular arcs of radius r_1 and n equal circular arcs of some radius $r_2 > r_1$. The lines joining the endpoints of every arc of radius r_2 enclose a regular n-gon P which we call the *central* n-gon of C. By a central n-gon of a circle C we mean any regular n-gon concentric with C.

The following theorem by A. Florian (1986a) provides the solution to the above problem.

Theorem 14. *Suppose that condition* (30) *is satisfied. Let* $C \in \mathscr{C}(a, p)$ *and* $P \in \mathscr{P}_n(a_0)$ *such that*

$$a(C \cap P) = M(a, p; n, a_0) .$$

(i) *If* $a_0 = f_1(a, p, n)$, *then* C *is a regular arc-sided* n*-gon of area* a *and perimeter* p, *and* P *is the kernel of* C.

(ii) *If* $f_1(a, p, n) < a_0 < f_2(a, p, n)$, *then* C *is an outer parallel domain of a regular arc-sided* n*-gon, and* P *is the central* n*-gon of* C. *Furthermore,* $a(C) = a$ *and* $p(C) = p$.

(iii) *If* $f_2(a, p, n) \leq a_0 \leq p^2 t/4\pi$, *then* C *is a smooth regular* n*-gon of perimeter* p, *and* P *is the case of* C.

Note that either extremal configuration corresponding to (i) and (iii) may be regarded as a degenerate parallel domain of a regular arc-sided n-gon and its central n-gon. The set of parameters $\{a, \ldots, a_0\}$ determines the respective extremal configuration up to isometry. Theorem 14 implies Theorem 13 and Corollary 1 of Theorem 6.

Any side of the central n-gon P, say A_1A_2, intersects the (possibly degenerate) parallel domain C of a regular arc-sided n-gon in a segment B_1B_2 so that $\overline{A_1B_1} = \overline{B_2A_2}$. We put

$$\overline{B_1B_2}/\overline{A_1A_2} = v(C) .$$

Let C_1 be a smooth regular n-gon with $v(C_1) = \frac{1}{2}$, and write $p(C_1) = p_1$, $a(C_1) = a_1$. Then p_1^2/a_1 clearly depends only on n. Using this notation one can deduce from Theorem 14 the following proposition.

Theorem 15. *Suppose that condition* (30) *is satisfied. There is exactly one disc* $C \in \mathscr{C}(a, p)$ *and one polygon* $P \in \mathscr{P}_n$ *such that*

$$\rho^S(C, P) = \min \rho^S(C', P') ,$$

where $C' \in \mathscr{C}(a, p)$, $P' \in \mathscr{P}_n$. *C and P are characterized by the following properties*:

(i) $a(C) = a$, $p(C) = p$;

(ii) *if* $p^2/a < p_1^2/a_1$, *then C is a parallel domain of a regular arc-sided n-gon, and P is the central n-gon of C. Furthermore*, $v(C) = \frac{1}{2}$;

(iii) *if* $p^2/a \geqslant p_1^2/a_1$, *then C is a smooth regular n-gon, and P is the case of C.*

Results analogous to Theorems 14 and 15 involving the perimeter deviation in place of the area deviation are contained in a recent paper of A. Florian (1992). Let C be an outer parallel domain of a regular arc-sided n-gon at distance r_1. We recall that C, if not a circle, is bounded by n equal circular arcs of radius r_1 and n equal circular arcs of some radius $r_2 > r_1$. The tangents at the endpoints of the ith arc of radius r_1 intersect at a point, say A_i, forming there an angle 2α $(0 < \alpha < \frac{1}{2}\pi)$. The points $A_1, \ldots, A_n$ are the vertices of a regular n-gon P which we call the *n-gon associated with C*. By an n-gon associated with a circle C we mean any regular n-gon concentric with C. Write 2β for the interior angle of P and put

$$\frac{\cos \alpha}{\cos \beta} = w(C) .$$

Let C_2 be a regular arc-sided n-gon with $w(C_2) = \frac{1}{2}$. Let $p(C_2) = p_2$, $a(C_2) = a_2$, then p_2^2/a_2 depends only on n.

We confine ourselves to formulating the counterpart of Theorem 15.

Theorem 16. *Suppose that condition* (30) *is satisfied. There exists exactly one disc* $C \in \mathscr{C}(a, p)$ *and one polygon* $P \in \mathscr{P}_n$ *such that*

$$\rho^{\mathrm{P}}(C, P) = \min \rho^{\mathrm{P}}(C', P') ,$$

where $C' \in \mathscr{C}(a, p)$, $P' \in \mathscr{P}_n$. *C and P are characterized by the following properties*:

(i) $a(C) = a$, $p(C) = p$;

(ii) *if* $p^2/a < p_2^2/a_2$, *then C is a parallel domain of a regular arc-sided n-gon, and P is associated with C. Furthermore*

$$w(C) = \tfrac{1}{2} ;$$

(iii) *if* $p^2/a \geqslant p_2^2/a_2$, *then C is a regular arc-sided n-gon, and P is the kernel of C.*

Let C be a convex disc and let

$$\rho_n^{\mathrm{P}}(C) = \min \rho^{\mathrm{P}}(C, P) , \quad P \in \mathscr{P}_n .$$

For which disc C of given perimeter p does $\rho_n^{\mathrm{P}}(C)$ attain its maximum? If P is restricted to the classes of n-gons that are either inscribed in C or circumscribed

about C, the unique solution to both problems is the circle (Theorem 10). It is very likely that the solution is also the circle if we do not restrict the mutual position of C and P.

The corresponding question, where ρ^P is replaced by ρ^S, appears entirely open.

We now turn to the problem of approximating a convex disc C, involving the affine perimeter of C instead of the ordinary perimeter. The affine length of a curve is an additive measure of arc which is invariant under area-preserving affine transformations. It was introduced by Pick and Blaschke in two different ways (Blaschke 1923, § 4). Following one of these ways, we introduce the affine length of an arbitrary convex arc. Let $AB = b$ be an arc on the boundary of a convex disc C. On b we choose a finite sequence of points $A = P_0, P_1, \ldots, P_n = B$ dissecting b into n subarcs $P_{j-1}P_j$ $(j = 1, \ldots, n)$. Let s_j be a support line of C at P_j. We choose $P_1, \ldots, P_{n-1}$ and $s_0, \ldots, s_n$ so that, for $j = 1, \ldots, n$, we have

(i) s_{j-1} and s_j intersect in a point T_j, and

(ii) the arc $P_{j-1}P_j$ is contained in the triangle

$$\triangle_j = P_{j-1}P_jT_j .$$

In this case we call $\{\triangle_1, \ldots, \triangle_n\}$ a *chain of triangles covering the arc* b. We define the *affine length* of the arc b by:

$$\lambda(b) = \inf \sum_{j=1}^{n} 2(a(\triangle_j))^{1/3} ,$$

where the infimum extends over all chains of triangles covering b; for various properties of the affine length see Blaschke (1923), L. Fejes Tóth (1951, 1972, p. 40) and A. Florian (1986b).

The affine length of the whole boundary of a convex disc C is called the *affine perimeter* of C. The affine perimeter L and the area a of a convex disc satisfy the isoperimetric inequality discovered by Blaschke (1923, § 26),

$$L^3 \leqslant 8\pi^2 a , \tag{32}$$

where equality holds only for ellipses; a different proof can be found in L. Fejes Tóth (1951, 1972, p. 51).

To state the main result we need some additional notions. Let P be an affinely regular n-gon. Let Q be the n-gon, the vertices of which are the midpoints of the sides of P. We define a *regular conic-n-gon* to be a convex disc bounded by n affinely equivalent arcs of a conic joining adjacent vertices of Q and touching the respective sides of P. We call P the *case* and Q the *kernel* of the conic-n-gon. The conic-n-gons with case P and kernel Q form a pencil joining P and Q. The pencil contains an ellipse E. The members of the pencil between Q and E are said to be *elliptic*, and those between E and P *hyperbolic*. The ellipse E is considered to be both elliptic and hyperbolic. If C is a hyperbolic member of the pencil, then the case P has minimal area among all the convex n-gons containing C. On the other hand, if C is an elliptic member of the pencil, then the kernel Q has maximal area among all the convex n-gons contained in C.

Let C be a convex disc. The following two theorems (L. Fejes Tóth 1980) provide a lower bound of the areas of all n-gons containing C, and an upper bound of the areas of all n-gons contained in C. The bounds depend on the area and the affine perimeter of C. This choice is motivated by the invariance of both parameters under area-preserving affine transformations.

Let a and L be positive numbers satisfying (32).

Theorem 17. *Let $\{C\}$ be the class of convex discs of area a and affine perimeter not less than L. The area of the smallest convex n-gon containing a disc from $\{C\}$ attains its minimum if C is a regular hyperbolic conic-n-gon with affine perimeter L.*

Theorem 18. *The area of the greatest convex n-gon contained in a disc from $\{C\}$ attains its maximum if C is a regular elliptic conic-n-gon with affine perimeter L.*

Let $\mathscr{C}'(a, L)$ denote the class of convex discs with area not greater than a and affine perimeter not less than L. Then we can state the following corollary.

Corollary 3. *Let the disc C and the polygon P range over $\mathscr{C}'(a, L)$ and $\mathscr{P}_n$ such that $C \subset P$. The area deviation $\rho^S(C, P)$ attains its minimum when C is a regular hyperbolic conic-n-gon of area a and affine perimeter L, and P is the case of C.*

A similar corollary applies when $P \subset C$ and $P \in \mathscr{P}_n$, $C \in \mathscr{C}'(a, L)$. But we do not know the minimum of $\rho^S(C, P)$ when $C \in \mathscr{C}'(a, L)$, $P \in \mathscr{P}_n$ and the mutual position of C and P is not restricted.

Theorems 17 and 18 admit applications in the theory of packing and covering with convex discs (L. Fejes Tóth and A. Florian 1982, A. Florian 1986b).

We conclude this section with some remarks on the approximation of convex spherical caps by polygons. Let P be a regular n-gon on S^2. The notions of a smooth regular n-gon with case P and a regular arc-sided n-gon with kernel P can be extended to S^2, and theorems analogous to 12 and 13 can be established (A. Florian 1989b). Since these two theorems are equivalent by spherical polarity, the situation in S^2 appears in a way more satisfactory than that in the plane. We expect that the result can be applied to problems of packing and covering with convex spherical caps. It would also be desirable to extend Theorems 15 and 16 to S^2.

Various extremum properties of regular figures, including historical remarks and relations to natural sciences, are discussed in two review papers by L. Fejes Tóth (1981, 1986).

3. Bounds for the volume of a convex polyhedron

In the theory of convex polyhedra in Euclidean 3-space three fundamental measures play a central rôle: the volume V, the surface area F, and the edge-curvature M (to be defined in section 4). Let P be a convex polyhedron with

f faces, v vertices, inradius r and circumradius R. We shall state two inequalities for the volume $V(P)$ which are analogous to the inequalities (1, 2) for the area of a convex n-gon. The corresponding inequalities for the surface area and the edge-curvature will be discussed in the next section.

Projecting the faces of the polyhedron from an inner point O^* onto the unit sphere with centre O^* we obtain a spherical tiling. Let $S_1, \ldots, S_f$ be the faces of the tiling. According to the definition given in section 2, the characteristic triangle $T = OBC$ of the set $\{S_1, \ldots, S_f\}$ is determined by

$$\sphericalangle O = \pi f/2e\,, \qquad \sphericalangle B = \pi/2\,, \qquad a(T) = \pi/e\,, \tag{33}$$

where $e = f + v - 2$ is the number of edges. Note that $\sphericalangle C = \pi v/2e$. With the polyhedron P we associate two homothetic tetrahedra $T_j = O^*O_jB_jC_j$ ($j = 1, 2$) defined by the following conditions:

(i) T_j is an *orthoscheme*, which means that O^*O_j is perpendicular to the plane $O_jB_jC_j$, and B_jC_j is perpendicular to the plane $O^*O_jB_j$;

(ii) the radial projection of the triangle $O_jB_jC_j$ onto the unit sphere with centre O^* is the characteristic triangle T;

(iii) $\overline{O^*O_1} = r$, $\overline{O^*C_2} = R$.

Two extremum properties of the five Platonic solids are contained in the following theorem.

Theorem 19. *If $V(P)$ denotes the volume, r the inradius and R the circumradius of a convex polyhedron P with f faces, e edges and v vertices, then*

$$4eV(T_1) \leqq V(P) \leqq 4eV(T_2)\,. \tag{34,35}$$

In either case, equality holds only for the five regular polyhedra.

If we write $2e/f = p$ for the average number of edges around a face and $2e/v = q$ for the average number of edges radiating from a vertex, then (34) and (35) take the form

$$\begin{aligned} &\frac{e}{3}\sin\frac{2\pi}{p}\left(\tan^2\frac{\pi}{p}\tan^2\frac{\pi}{q} - 1\right) r^3 \leqq V(P) \\ &\quad \leqq \frac{2e}{3}\cos^2\frac{\pi}{p}\cot\frac{\pi}{q}\left(1 - \cot^2\frac{\pi}{p}\cot^2\frac{\pi}{q}\right) R^3\,. \end{aligned} \tag{36, 37}$$

Combining these inequalities we obtain the following corollary.

Corollary 4. *Let r and R be the inradius and the circumradius of a convex polyhedron with f faces, e edges and v vertices, and let $2e/f = p$ and $2e/v = q$. Then*

$$\frac{R}{r} \geqslant \tan\frac{\pi}{p}\tan\frac{\pi}{q}\,, \tag{38}$$

with equality only for the five regular polyhedra.

Proofs of Theorem 19 can be found in L. Fejes Tóth (1964, p. 263, 1972, pp. 123, 137). Inequality (36) was established in this form by L. Fejes Tóth (1950b) who gave two different proofs. One is a nice application of Theorem 7. Dissect the polyhedron P into f pyramids based on the faces of P and having their common apex at the centre O^* of the insphere of P. Projecting the faces of P from O^* onto the unit sphere S with centre O^* we obtain the above-mentioned tiling with faces $S_1, \ldots, S_f$. Let $O_1', \ldots, O_f'$ be points of S such that $O^*O_1', \ldots, O^*O_f'$ are perpendicular to the faces of P. Denoting a variable point of S by X and the function $r^3/(3\cos^3 x)$ by $g(x)$, we have

$$V(P) \geqq \sum_{i=1}^{f} \int_{S_i} g(O_i'X)\,\mathrm{d}a\,.$$

Since g is a strictly increasing function, the required result follows from Theorem 7 and (33). The arguments used in two independent papers by Goldberg (1934) and L. Fejes Tóth (1948b) can be completed so as to provide a third proof of inequality (36) (A. Florian 1956a).

In a similar way the more general inequality

$$4eV(T_1 \cap K(\rho)) \leqq V(P \cap K(\rho)) \tag{34'}$$

can be obtained, where $K(\rho)$ is a sphere of radius $\rho > r$ concentric with the insphere of P. Equality holds for the five regular polyhedra and, when $\rho \geqq \overline{O^*B_1}$ $[= r\sin(\pi/p)\sec(\pi/q)]$, in no other case. For $\rho = \overline{O^*C_1}$ $[= r\tan(\pi/p)\tan(\pi/q)]$, the left-hand side of (34') reduces to $4eV(T_1)$. This improvement of (34) was observed by L. Fejes Tóth.

Inequality (37) was conjectured by L. Fejes Tóth (1950b) and proved, on the assumption that $\tan(\pi f/2e)\tan(\pi v/2e) \leqq \sqrt{3}$, by A. Florian (1956a,b). (I would like to point out that there is a mistake on p. 135, l.9 of the second paper, where the term $6\tan u \tan v$ should be replaced by $6\tan^2 u \tan^2 v$. But this does not affect the required inequality $g_{uu}g_{vv} - g_{uv}^2 > 0$, for $0 < (\pi/2 - u)/2 \leqq v < \pi/2 - u \leqq \pi/3$ except when $u = v = \pi/6$). There are only seven cases where $\tan(\pi f/2e)\tan(\pi v/2e) > \sqrt{3}$. These cases can be settled by strengthening inequality (37) (L. Fejes Tóth, 1960, H. Florian 1961).

The faces of a convex polyhedron with a given number v of vertices can be decomposed by nonintersecting diagonals into a total number of $2v - 4$ triangles. Hence, by (36) and (37),

$$\begin{aligned} &\tfrac{1}{2}\sqrt{3}(v-2)(3\tan^2\omega_v - 1)r^3 \leqq V(P) \\ &\quad \leqq \tfrac{1}{6}(v-2)\cot\omega_v(3 - \cot^2\omega_v)R^3\,, \end{aligned} \tag{39,40}$$

where

$$\omega_v = \frac{\pi v}{6(v-2)}\,.$$

Equality holds only for the regular tetrahedron, octahedron and icosahedron ($v = 4, 6, 12$); see L. Fejes Tóth (1948c).

How do we have to place v points on the unit sphere in E^3 so as to maximize the volume of their convex hull? For $v = 4$, 6 and 12, the solution is given by inequality (40). Apart from the simple case $v = 5$ (the vertices of a triangular dipyramid) the extremal configurations are known also for $v = 7$ and 8 (see Berman and Hanes 1970). The polyhedron corresponding to $v = 8$ is essentially that discovered with the aid of a computer by Grace (1963). Let P have maximal volume among all convex polyhedra with v vertices inscribed in a given sphere. L. Fejes Tóth (1972, p. 127) observed that P must be bounded solely by triangular faces.

Theorem 7 implies inequality (34) also for convex polyhedra in non-Euclidean spaces. In the following theorem by L. Fejes Tóth (1956a, 1964, p. 308), the parameter $\kappa = +1, 0, -1$ refers to the spherical, Euclidean and hyperbolic 3-space.

Theorem 20. *In S^3, E^3 or H^3 let P be a convex polyhedron of f faces, e edges and v vertices. If r denotes the inradius and $V(P)$ the volume of P, then*

$$V(P) \geqslant \frac{2e}{(\sqrt{\kappa})^3} \int_0^{\pi f/2e} \left\{ \sqrt{\kappa} r - \frac{\cos\varphi}{(k^2 - \sin^2\varphi)^{1/2}} \times \arctan\left(\tan(\sqrt{\kappa} r) \frac{(k^2 - \sin^2\varphi)^{1/2}}{\cos\varphi} \right) \right\} d\varphi , \qquad (41)$$

where

$$k = \sin\frac{\pi f}{2e} \sec\frac{\pi v}{2e} .$$

Equality holds only if P is regular.

In the case $\kappa = 0$ the right-hand side of (41) is defined by its limit as $\kappa \to 0$ which leads to inequality (36) for Euclidean polyhedra.

An extension of inequality (35) to non-Euclidean polyhedra is not known.

If the number of faces, the number of vertices and the volume of a star-polyhedron P are conveniently interpreted, then, under certain conditions on P, the inequalities (36) and (37) can be shown to hold also for star-polyhedra. Equality applies to the five Platonic solids and the four Kepler–Poinsot star-polyhedra (L. Fejes Tóth 1956c, 1964, p. 219; A. Florian 1957). The proof rests on a general inequality for spherical star-tilings which, in the special case of convex tilings, is contained in Theorem 7. Further research may possibly reduce the conditions on P.

Let P be a convex polyhedron with v vertices and volume $V(P)$. To state a further result we consider an orthoscheme $T_3 = O^* O_3 B_3 C_3$ with the properties:

(i) the radial projection of $O_3 B_3 C_3$ onto the unit sphere with centre O^* is the triangle $T' = O'B'C'$ given by

$$\sphericalangle O' = \pi/3\,, \qquad \sphericalangle B' = \pi/2\,, \qquad a(T') = 4\pi/12(v-2)\,,$$

(ii) $V(T_3) = V(P)/12(v-2)$.

The following theorem of A. Florian (1971) implies the inequalities (39) and (40).

Theorem 21. *Let $K(\rho)$ be a ball with centre O^* and radius ρ. Let P be a convex polyhedron with v vertices and volume $V(P)$, and let the tetrahedron T_3 be defined as above. Then*

$$V(P \cap K(\rho)) \leqslant 12(v-2)V(T_3 \cap K(\rho)) \tag{42}$$

with equality if $v = 4$, 6 or 12 and P is a regular tetrahedron, octahedron or icosahedron with centre O^*. When $\overline{O^*O_3} \leqslant \rho \leqslant \overline{O^*C_3}$, these are the only cases of equality.

Theorem 21 can easily be generalized to a "momentum lemma" for polyhedra with a given number of vertices similar to that for n-gons. Even so, a proper 3-dimensional analogue of Corollary 2 should apply to convex polyhedra, where both the number of faces and the number of vertices are prescribed. Let P be a convex polyhedron with f faces, e edges, v vertices and volume $V(P)$. We associate with P an orthoscheme $T_4 = O^*O_4B_4C_4$ such that

(i) the radial projection of $O_4B_4C_4$ onto the unit sphere with centre O^* is the characteristic triangle $T = OBC$ determined by (33),

(ii) $V(T_4) = V(P)/4e$.

Let $K(\rho)$ be a ball with centre O^* and radius ρ. The conjecture

$$V(P \cap K(\rho)) \leqslant 4eV(T_4 \cap K(\rho)) \tag{43}$$

is supported by the fact that (43) would imply the inequalities (34), (35) and (42) already proved. Equality holds for the five regular polyhedra. If moreover $\overline{O^*O_4} \leqslant \rho \leqslant \overline{O^*C_4}$, then these are probably the only cases of equality.

By Theorem 19 the orthoschemes T_1, T_2, T_4 which are associated with a convex polyhedron P satisfy the relation

$$T_1 \subset T_4 \subset T_2\,.$$

In either case, equality holds if and only if P is a regular polyhedron.

Several extremum properties of the regular triangle and the regular tetrahedron may be generalized to regular simplices in all dimensions. For example, it can easily be shown that among all Euclidean d-simplices of inradius r (circumradius R) only the regular simplex has minimal (maximal) volume. As a consequence we note that

$$\frac{R}{r} \geqslant d$$

(L. Fejes Tóth 1964, p. 312, Klamkin and Tsintsifas 1979). In E^d let $K(\rho)$ be a ball of radius ρ and let $\bar{P}$ be a regular simplex concentric with $K(\rho)$. For any simplex P with $V(P) = V(\bar{P})$ it can be proved that

$$V(P \cap K(\rho)) \leqslant V(\bar{P} \cap K(\rho)) .$$

If $r(\bar{P}) \leqslant \rho \leqslant R(\bar{P})$, equality occurs only for $P = \bar{P}$.

Little is known in this respect about the general cross polytope, the hypercube and the nontrivial regular convex polytopes $\{3, 4, 3\}$, $\{3, 3, 5\}$ and $\{5, 3, 3\}$ in 4-space. Some extremum properties of these polytopes were established by comparing any one of them with the topologically isomorphic convex polytopes and using Steiner symmetrization (L. Fejes Tóth 1955, 1956c, 1964, p. 312). But no methods are available for proving inequalities analogous to (39) or (40), for instance.

In Euclidean 4-space let P be a convex polytope bounded by n tetrahedra. Let S^3 be the boundary of the unit ball centred at the origin O. Let T be a regular tetrahedron contained in a hyperplane which touches S^3 at the centre of the tetrahedron. Denote by τ the volume of the spherical tetrahedron obtained from T by radial projection. We write $V(\tau)$ for the volume of the Euclidean convex hull of O and T. The arguments used by L. Fejes Tóth show that

$$V(P) \geqslant nV(2\pi^2/n)r^4 ,$$

where $r = r(P)$. Equality holds only for $\{3, 3, 3\}$ $(n = 5)$, $\{3, 3, 4\}$ $(n = 16)$ and $\{3, 3, 5\}$ $(n = 600)$. Linhart (1985) proved an analogous but more difficult inequality involving $R(P)$.

Böröczky (1987) stated that among all d-dimensional spherical simplices contained in a given sphere with radius less than $\pi/2$ the regular inscribed simplex has maximal volume. More generally, we may ask for the maximum of $V(P \cap K)$, where K is a given ball in spherical or hyperbolic d-space, and P ranges over all simplices of prescribed volume.

Recently, Linhart (1987), Filliman (1988) and Gordon, Meyer and Reisner (1988) discussed various extremum properties of zonotopes.

The papers of Kind and Kleinschmidt (1976), McMullen (1982) and Martini (1989) contain further relevant material.

4. Surface area and edge-curvature

The arguments used in the proof of inequality (34) and based on Theorem 7 yield the following theorem by L. Fejes Tóth (1950b, 1964, p. 279).

Theorem 22. *If $F(P)$ denotes the surface area and r the inradius of a convex polyhedron P with f faces, e edges and v vertices, then*

$$F(P) \geqslant 4ea(O_1B_1C_1) , \tag{44}$$

where $O_1B_1C_1$ is the base triangle of the orthoscheme T_1 defined in section 3. *Equality holds only for the Platonic solids.*

On the other hand, there are difficulties in proving the analogous inequality for F and the circumradius R, but these can be overcome under certain additional conditions on the polyhedron P. Let us suppose that every face (edge) of P contains the foot of the perpendicular drawn from the centre of the circumsphere (or insphere, respectively) to the plane of the face (line of the edge). We shall refer to that as the foot condition for the faces (edges).

L. Fejes Tóth (l.c.) proved the following inequality (45) for convex polyhedra satisfying the foot condition both for the faces and edges. By bringing the polar polyhedron of P into play, Linhart (1982) was able to reduce the conditions on P.

Theorem 23. *Let $F(P)$ denote the surface area and R the circumradius of a convex polyhedron P with f faces, e edges and v vertices. If P satisfies the foot condition for the edges, then*

$$F(P) \leqslant 4ea(O_2B_2C_2)\,, \tag{45}$$

where $O_2B_2C_2$ is the base triangle of the orthoscheme T_2 defined in section 3. *Equality holds only for the Platonic solids.*

Thus, of the two inequalities

$$e \sin \frac{2\pi}{p}\left(\tan^2 \frac{\pi}{p} \tan^2 \frac{\pi}{q} - 1\right) r^2 \leqslant F(P)$$

$$\leqslant e \sin \frac{2\pi}{p}\left(1 - \cot^2 \frac{\pi}{p} \cot^2 \frac{\pi}{q}\right) R^2 \tag{46,47}$$

corresponding to (36) and (37) only the first could be proved without restrictions. A remarkable consequence of Theorem 23 is the following proposition.

Corollary 5. *For any convex polyhedron with v vertices we have (without condition)*

$$F(P) \leqslant \tfrac{3}{2}\sqrt{3}(v-2)(1 - \tfrac{1}{3}\cot^2\omega_v)R^2 \tag{48}$$

with equality only if $v = 4$, 6 or 12 and P is a regular tetrahedron, octahedron or icosahedron.

The case when $v = 4$ was settled long ago by Krammer (1961) and Heppes (1961).

Theorem 7 can be used to extend inequality (46) to non-Euclidean spaces. The following result of Tomor (1965) is an analogue of Theorem 20.

Theorem 24. *In S^3, E^3 or H^3 let P be a convex polyhedron with f faces, e edges and v vertices. If r denotes the inradius and $F(P)$ the surface area of P, then*

$$F(P) \geqslant \frac{4e}{\kappa}\left[\frac{\pi f}{2e} - \arcsin \frac{\sin(\pi f/2e)}{(1+(k^2-1)\sin^2(\sqrt{\kappa}r))^{1/2}}\right],$$

where

$$k = \sin\frac{\pi f}{2e}\sec\frac{\pi v}{2e}.$$

Equality holds only if P is regular.

Again, under convenient foot conditions inequality (46) may be generalized to star-polyhedra. Equality holds only for the nine regular polyhedra (L. Fejes Tóth 1956c, 1964, p. 282).

The integral of mean curvature of a convex polyhedron P in E^3 can be written in the form

$$M(P) = \tfrac{1}{2}\sum s\alpha,$$

where s denotes the length of an edge, α the angle formed by the outer normals to the faces meeting in this edge and the summation extends over all edges of P. The functional $M(P)$ is called the *edge-curvature* of P (Steiner 1840, Hadwiger 1955, p. 32).

Let φ be the inner dihedral angle of the orthoscheme $T_1 = O^*O_1B_1C_1$ at the edge B_1C_1 (which is clearly equal to the inner dihedral angle of $T_2 = O^*O_2B_2C_2$ at B_2C_2). The following Theorems 25 and 26 were first established and proved for the restricted class of polyhedra satisfying the foot conditions both for faces and edges (A. Florian 1956b, L. Fejes Tóth 1964, p. 279). A. Florian (1966) was able to establish the first of these theorems in the following form (i.e., without the foot condition for edges):

Theorem 25. *Let $M(P)$ denote the edge-curvature and r the inradius of a convex polyhedron P with f faces, e edges and v vertices. If P satisfies the foot condition for the faces, then*

$$M(P) \geqslant 2e(\tfrac{1}{2}\pi - \varphi)\overline{B_1C_1} \tag{49}$$

with equality only for the Platonic solids.

For the corresponding theorem involving the circumradius of P, Linhart (1979) gave a simple and elegant proof without any condition. Two crucial points of the proof are the inclusion of the polar polyhedron and the application of Theorem 7.

Theorem 26. *If $M(P)$ denotes the edge-curvature and R the circumradius of a convex polyhedron P with f faces, e edges and v vertices, then*

$$M(P) \leqslant 2e(\pi/2 - \varphi)\overline{B_2C_2} \tag{50}$$

with equality only for the Platonic solids.

Thus, of the two inequalities

$$2e \sin \frac{\pi}{p} \left(\tan^2 \frac{\pi}{p} \tan^2 \frac{\pi}{q} - 1 \right)^{1/2} \arccos \left(\cos \frac{\pi}{q} \operatorname{cosec} \frac{\pi}{p} \right) r \leqslant M(P)$$

$$\leqslant 2e \sin \frac{\pi}{p} \left(1 - \cot^2 \frac{\pi}{p} \cot^2 \frac{\pi}{q} \right)^{1/2} \arccos \left(\cos \frac{\pi}{q} \operatorname{cosec} \frac{\pi}{p} \right) R , \qquad (51)$$

only the second could be proved unconditionally.

A consequence of Theorem 25 is the following corollary.

Corollary 6. *For any convex polyhedron of f faces and inradius r we have (without condition)*

$$M(P) \geqslant 6(f-2) \sin \omega_f (3 \tan^2 \omega_f - 1)^{1/2} \arccos(\tfrac{1}{2} \operatorname{cosec} \omega_f) r \qquad (52)$$

with equality only for $f = 4$, 6 or 12 and P is a regular tetrahedron, a cube or a regular dodecahedron.

Extremum problems for the surface area, the inradius and the mean width of 3-zonotopes are discussed by Linhart (1986, 1988). For instance, it is shown that the rhombic dodecahedron has maximal surface area and maximal inradius among all 3-zonotopes which are generated by four line segments of unit length.

5. Total length of the edges of a polyhedron

A further range of extremum problems is related to the sum of the lengths of all edges of a polyhedron.

Theorem 27. *The total length L of the edges of any convex polyhedron with inradius r satisfies*

$$L \geqslant 24r . \qquad (53)$$

Equality holds only for the cube.

The interesting feature of this remarkable result by Besicovitch and Eggleston (1957) is that, in contrast to the theorems of sections 3 and 4, neither the number of faces nor the number of vertices is prescribed. Thus the cube realizes the absolute minimum of the ratio L/r within the whole class of convex polyhedra. Theorem 27 confirms a conjecture of L. Fejes Tóth (1948d, 1950c, 1972, p. 142), who proved, inter alia, that $L > 20r$ and that (53) is true at least for convex polyhedra bounded by faces of equal area; see also Hammersley (1951).

Note that in Euclidean 4-space, among the six convex regular polyhedra of given inradius, the simplex (and not the hypercube) has least total edge-length.

Linhart (1977a) succeeded in proving another conjecture by L. Fejes Tóth (1948d, 1972, p. 144):

Theorem 28. *The total length of the edges of a convex polyhedron with triangular faces and inradius r satisfies*

$$L \geqslant 12\sqrt{6}r = 29.39\ldots r\,. \tag{54}$$

Equality holds only for the regular tetrahedron and the regular octahedron.

By means of a simple argument, L. Fejes Tóth (l.c.) had obtained the good estimate $L > 28r$.

We define the total edge-length L of a convex polytope P to be the sum of the lengths of the one-dimensional faces of P. The following theorem was conjectured and proved for simplicial polytopes by Lillington (1974). The proof of the general theorem is due to Linhart (1977b).

Theorem 29. *For any $d \geqslant 2$ the total edge-length of a d-dimensional convex polytope of circumradius R satisfies*

$$L > 2dR\,. \tag{55}$$

Note that there are long thin d-simplices with total edge-lengths arbitrarily close to $2dR$. Lillington (1977) pointed out, however, that the bound $2dR$ is not always the best possible if the magnitudes of the faces of the polytopes are restricted by a constant.

Linhart's proof of Theorem 29 is based on two inequalities of independent interest. If S_p denotes the total p-dimensional volume of the p-dimensional faces of a convex polytope P, then

$$S_p \geqslant \binom{d}{p} \frac{d-p+1}{\kappa_{d-p}} W_{d-p}(P) \tag{56}$$

(Firey and Schneider 1979). Further, if K is a d-dimensional convex body of circumradius R, then

$$W_{d-1}(K) > 2\,\frac{\kappa_{d-1}}{d}\,R\,, \tag{57}$$

and in particular for $d = 3$,

$$M(K) = 3W_2(K) > 2\pi R\,.$$

In view of a result by Schneider (1978), inequality (55) extends to any d-dimensional convex body K of circumradius R, if L is interpreted as the 1-dimensional Hausdorff measure of the 1-skeleton of K.

Let K be a convex body in Euclidean d-space. Let T_* and T^* represent the simplex of maximal total edge-length contained in K, and the simplex of minimal edge-length containing K, respectively. Lillington (1975) established lower and upper bounds, depending only on d, for each of the four ratios $L(T_*)/r(K), \ldots, L(T^*)/R(K)$. Each of these inequalities is the best possible, the extremal body in most cases being a sphere.

In Euclidean 4-space let P be a convex polytope with inradius 1. L. Fejes Tóth (1965) gave lower bounds for the total area F_2 of the two-dimensional faces of P. If P is restricted to the class of simplicial polytopes, the relevant bound is very close to the value of F_2 for the cross polytope $\{3, 3, 4\}$. But no polytope is known for which one of the bounds is attained.

Coxeter (1962) discussed the problem of minimal total edge-length for three-dimensional non-Euclidean polyhedra of given inradius r and pointed out some results to be expected. Considerations of continuity suggest that the extremum property of the hexahedron stated in Theorem 27 still holds for sufficiently small values of r. However, in spherical space, the dihedron $\{p, 2\}$ has a shorter total edge-length than the hexahedron $\{4, 3\}$ whenever $r > 0.281\ldots$. Similarly, in hyperbolic space, the dodecahedron $\{5, 3\}$ has a shorter total edge-length than the hexahedron whenever $r > 0.392\ldots$.

Coxeter and L. Fejes Tóth (1963) considered the corresponding problem for the class of triangle-faced polyhedra in spherical and hyperbolic space. In spherical space, for the special inradius $r = \arcsin \frac{1}{4} = 0.252\ldots$, the regular tetrahedron $\{3, 3\}$ and the trigonal dihedron $\{3, 2\}$ have the same edge-length 2π. It is proved that every other triangle-faced convex polyhedron with this inradius has a longer edge-length. Similarly, in hyperbolic space this extremum property holds for the octahedron $\{3, 4\}$ when $r = 0.364\ldots$, and for the icosahedron $\{3, 5\}$ when $r = 0.828\ldots$. The proof makes use of the momentum lemma (Corollary 2).

Returning to Euclidean space, we mention two inequalities providing upper bounds for the surface area and the volume of a convex polyhedron with fixed total edge-length (Aberth 1963, 1973).

Theorem 30. *For any 3-dimensional convex polyhedron with surface area F, volume V and total edge-length L we have*

$$L^2 > 6\pi F \tag{58}$$

and

$$L^3 > 432\pi V\,. \tag{59}$$

By (58), we have $\inf L^2/F \geqslant 6\pi = 18.849\ldots$, where the infimum extends over all three-dimensional convex polyhedra. Kömhoff (1970) constructed a sequence of polyhedra showing that $\inf L^2/F \leqslant 2(\sqrt{3} + 8\pi/3) = 20.219\ldots$. Similarly, by (59) we have $\inf L^3/V \geqslant 432\pi = 1357.1\ldots$, while Okuneva and Fedotov (1984) observed that $\inf L^3/V \leqslant 972\sqrt{3} = 1683.5\ldots$. This constant is attained by a right

prism whose base is an equilateral triangle with side-length equal to the height of the prism.

Some special results concerning the volume and the total edge-length are contained in Sansone (1940).

The restriction to polyhedra with triangular faces makes it possible to prove an inequality stronger than (58) (Kömhoff 1968).

Theorem 31. *Let F be the surface area and L the total edge-length of a 3-dimensional convex polyhedron P with triangular faces. Then*

$$L^2 \geqslant 12\sqrt{3}F \tag{60}$$

with equality if and only if P is a regular tetrahedron.

An analogous inequality involving the total edge-length and the volume is not known. Sansone (1928) proved that among all tetrahedra of given volume only the regular tetrahedron has minimal total edge-length; for a more recent proof see Melzak (1966). A more general minimum property of the regular d-simplex of given volume is established by Weiss (1987). Further extremum properties of the regular simplex concerning the total edge-length or the maximal edge-length were stated by Tanner (1974), Linderholm (1986), and Wode (1987).

Coxeter (1959) raised the following nice problem: find a cage of minimal total edge-length to hold a unit sphere without permitting it to slide out. By a cage we mean a frame formed by the edges of a convex polyhedron. Given any $\varepsilon > 0$, Besicovitch (1963) constructed such a cage of total edge-length less than $\gamma + \varepsilon$, where $\gamma = 8\pi/3 + 2\sqrt{3} = 11.84\ldots$, and Aberth (1963) proved that there is no cage of total edge-length $\leqslant \gamma$.

Theorem 32. *Any cage which contains a unit sphere and does not allow it to slide out has a total edge-length $L > 8\pi/3 + 2\sqrt{3}$, and this bound is the best possible.*

Let a unit sphere be enclosed in a net of an inextensible string in such a way that the sphere cannot slip out. Besicovitch (1957) proved that the length of the string is strictly greater than 3π, and this is false with any greater constant replacing 3π. Stein (1963) and Croft (1964) gave alternative proofs of this theorem.

6. The isoperimetric problem for convex polyhedra

If we restrict ourselves to convex bodies in Euclidean 3-space, the isoperimetric problem can be formulated as follows: find a convex body of given surface area which contains the largest volume. The solution is given by the isoperimetric inequality

$$F^3 - 36\pi V^2 \geqslant 0\,, \tag{61}$$

where F is the surface area and V the volume of an arbitrary convex body. Equality holds only for a sphere.

Schwarz (1884) presented the first complete solution of the isoperimetric problem for not necessarily convex bodies which are bounded by sufficiently regular surfaces. His proof makes use of an analytical method developed by Weierstrass.

The Brunn–Minkowski theorem gives a different approach to the subject. Minkowski (1903) proved several inequalities involving the mixed volumes of two convex bodies. One of them implies inequality (61) for the whole class of convex bodies, including the discussion of equality. Relation (61) is also a consequence of two further inequalities proved by Minkowski,

$$M^2 - 4\pi F \geqslant 0\,, \tag{62}$$

$$F^2 - 3MV \geqslant 0\,, \tag{63}$$

where M is the integral of mean curvature. Minkowski conjectured that in (62) equality holds only for a sphere and in (63) only for a sphere and a cap body of a sphere. Let B be a ball and let S be a finite or countably infinite set of points outside B so that the line segment joining any two points of S intersects B. The convex hull of $B \cup S$ is called a *cap body* of B. Minkowski's conjecture was confirmed by Bol (1943b) in a more general context. His proof is based on the concept of inner parallel bodies. Knothe (1949) supplied an alternative proof of this theorem.

There is a wide range of literature containing generalized or strengthened forms of the inequalities (61)–(63). These results are the work of Minkowski, Blaschke, Bonnesen, Favard, Bol, Hadwiger, Dinghas and others. We refer to the books of Bonnesen and Fenchel (1934), Hadwiger (1955, 1957), Leichtweiss (1980), the review paper of Osserman (1978) and the references given there. See also the relevant chapter of the present book.

New problems arise if we want to find the best convex body, that means the body which represents the minimum of the ratio F^3/V^2, among the convex polyhedra. In an extensive paper Steiner (1842) – and before him Lhuilier – considered n-sided prisms, pyramids and dipyramids and proved, e.g., that the best n-sided prism is regular, rectangular and circumscribed about a sphere. He pointed out that in all these cases the best polyhedron is circumscribed about a sphere which touches each face in its centroid, and raised the question whether the best polyhedron of any given combinatorial type has the same property. Furthermore, Steiner conjectured that the regular n-sided prism circumscribed about a sphere is the best polyhedron not only among the n-sided prisms but even within the whole class of isomorphic polyhedra. However, he could not prove or disprove the conjecture for any $n \geqslant 3$.

According to another conjecture of Steiner, each of the five Platonic solids is the best of all isomorphic polyhedra. Lhuilier's and Steiner's results on pyramids show that this is true for the regular tetrahedron. By using his symmetrization process Steiner was able to prove that the best n-sided dipyramid is the best

polyhedron among all isomorphic polyhedra, which confirms his conjecture in the case of the regular octahedron.

The following theorem is due to Lindelöf (1869).

Theorem 33. *Of all convex polyhedra with the same number of faces, the best polyhedron is circumscribed about a sphere which touches each face in its centroid.*

It should be noted that Lindelöf's theorem does not answer Steiner's question mentioned above, since the theorem applies to the set of all polyhedra with the same number of faces and not to a class of isomorphic polyhedra. Nevertheless, the arguments used in the proof show that a simple polyhedron which happens to be the best of its type, has to satisfy the Steiner–Lindelöf condition. (A polyhedron in which three edges meet at each vertex is called a *simple* polyhedron.) Lindelöf did not consider the existence of a best polyhedron with a prescribed number of faces.

Minkowski (1897) obtained Theorem 33 in a quite different way, at the same time showing the existence of a best polyhedron with f faces. His result is the special case $d = 3$ of the next theorem. Let P be a convex polytope in d-dimensional Euclidean space. Let P_0 be that convex polytope circumscribed about a fixed unit sphere which has the same number of $(d-1)$-dimensional faces and the same set of outer normals to these faces. P_0 is called the *form polytope* of P. Write $V(P) = V$, $F(P) = F$, $V(P_0) = V_0$ and $F(P_0) = F_0$, so that $F_0 = V_0 d$.

Theorem 34. *For any d-dimensional convex polytope P we have*

$$\frac{F^d}{V^{d-1}} \geqq \frac{F_0^d}{V_0^{d-1}} = d^d V_0 . \tag{64}$$

Equality holds only if P is circumscribed about a sphere.

Since $V_0 > \kappa_d$, inequality (64) is stronger than the *isoperimetric inequality* for convex bodies

$$F^d - d^d \kappa_d V^{d-1} \geqq 0 , \tag{65}$$

which generalizes (61) to d dimensions. Theorem 34 is contained in a theorem (Hadwiger 1957, p. 274) which applies to an arbitrary convex body K and strengthens inequality (65) whenever the boundary of K has singular points. For the case $d = 3$, see also Bol and Knothe (1948/49) and Hadwiger (1955, p. 69).

Two inequalities for three-dimensional convex polyhedra contain the measure

$$M^* = \sum s \tan \tfrac{1}{2}\alpha , \tag{66}$$

where s denotes the length of an edge, α the angle formed by the outer normals of the faces meeting in this edge, and the sum is taken over all edges of the

polyhedron. The functional M^* is similar in appearance to the edge-curvature M, but is of course not of the same importance in the theory of convex polyhedra.

Theorem 35. *For any three-dimensional convex polyhedron P we have*

$$F^2 - 3M^*V \geqslant 0\,, \tag{67}$$

$$M^{*2} - F_0F \geqslant 0\,, \tag{68}$$

where F_0 denotes the surface area of the form polyhedron of P. In either case, equality holds only if P is circumscribed about a sphere.

Inequality (67) was proved and generalized to convex bodies by Bol (1943a, b). Since $M^* > M$, (67) is stronger than (63). Note that (68) is a result of Dinghas (1947/49), see also Bol and Knothe (1948/49). The combination of both inequalities implies Theorem 34 for $d = 3$. The example of a cube circumscribed about a unit sphere shows that in relation (68) M^* generally cannot be replaced by M.

In an extensive paper, Steinitz (1927/28) dealt with special isoperimetric problems in three dimensions. A convex polyhedron P is of *circumscribable type* if there exists a polyhedron which is isomorphic to P and can be circumscribed about a sphere. Analogously, we say that P is of *inscribable type* if there exists a polyhedron which is isomorphic to P and can be inscribed in a sphere. Steinitz proved the existence of a convex simple polyhedron P which is not of circumscribable type. (For example, such a P is obtained by cutting off small pyramids from the corners of a regular dodecahedron.) Thus, there is no isoperimetrically best polyhedron within the combinatorial type of P. A characterization of those types of polyhedra which have an isoperimetrically best representative appears not to be known. Furthermore, Steinitz gave an example of a polyhedron which is the best of its own type, yet still fails to satisfy the Steiner–Lindelöf condition. This answers the relevant question of Steiner in the negative. On the other hand, Steinitz showed that in the class of convex polyhedra with v vertices (f faces) there is always a best polyhedron.

Grünbaum and Shephard (1987) discussed a number of open problems concerning polyhedra of inscribable type and related matters. Schulte (1987) proved, among others, the existence of noninscribable types of d-dimensional polytopes, for any $d \geqslant 4$. For a recent important result, see Schramm (1992).

Steinitz also examined Steiner's conjecture concerning the n-sided prisms and confirmed it for $n = 3$ (his proof seems not to have been published), yet disproved it for $n \geqslant 8$. For $n = 3$, Sucksdorff (1857) had given a very troublesome proof based on differential calculus. The case $n = 4$, the case of the cube, was later solved in accordance with the conjecture and under much more general conditions (see Theorem 37). The cases $n = 5$, 6 and 7 are still open.

The following theorem can easily be proved using Steiner symmetrization (Hadwiger 1957, p. 273).

Theorem 36. *The isoperimetric ratio of a d-parallelotope satisfies*

$$\frac{F^d}{V^{d-1}} \geqslant (2d)^d , \tag{69}$$

and that of a d-simplex satisfies

$$\frac{F^d}{V^{d-1}} \geqslant \frac{1}{d!}\, d^{3d/2}(d+1)^{(d+1)/2} . \tag{70}$$

In (69) *equality holds only for a hypercube, and in* (70) *only for a regular simplex.*

Reznikov (1991) achieved a stronger inequality than (70).

In the opposite direction, Weissbach (1987) established the inequality for d-simplices

$$F/V \leqslant c_d \Delta^{-1} ,$$

where

$$c_d = \begin{cases} 2d(d+1)(d+2)^{-1/2} , & d \text{ even} , \\ 2d^{3/2} , & d \text{ odd} , \end{cases}$$

and Δ denotes the minimal width of the simplex. Equality holds for regular simplices.

Proceeding from Theorem 34, Hadwiger (1953) established a lower bound for the isoperimetric ratio F^d/V^{d-1} of a convex d-polytope with f $(d-1)$-dimensional faces. This bound is attained only in the case $d = 2$, where Hadwiger's inequality is identical with the classic inequality (8). In the case $d = 3$ one obtains Goldberg's inequality (1934):

$$\frac{F^3}{V^2} \geqslant 36\pi \frac{f(f-1)}{(f-2)^2} .$$

L. Fejes Tóth (1948a) proved a lower bound for the volume of a convex polyhedron with f faces which is circumscribed about a unit sphere [this bound follows from inequality (36)]. Combining his result with Theorem 34 he gave the first complete proof of the following important theorem (see also L. Fejes Tóth 1964, p. 282, 1972, p. 135).

Theorem 37. *If F denotes the surface area and V the volume of a three-dimensional convex polyhedron with f faces, then*

$$\frac{F^3}{V^2} \geqslant 54(f-2)\tan\omega_f(4\sin^2\omega_f - 1) , \tag{71}$$

where $\omega_f = \pi f/6(f-2)$. *Equality holds only for the regular tetrahedron, the cube and the regular dodecahedron.*

Incomplete proofs are contained in Goldberg (1934) and L. Fejes Tóth (1948b). Theorem 37 confirms Steiner's conjectures concerning the regular 4-sided prism, the cube and the regular dodecahedron within a much wider scope. Goldberg observed that the regular octahedron and the regular icosahedron are not "best" among all polyhedra with 8 and 20 faces, respectively. He conjectured that a best polyhedron of f faces is simple.

In spite of various efforts (L. Fejes Tóth 1952), it has not yet been decided whether Steiner's conjecture that the regular icosahedron is the isoperimetrically best of its own type is true. It can be expected for any convex polyhedron with v vertices that

$$\frac{F^3}{V^2} \geqslant \frac{27\sqrt{3}}{2}(v-2)(3\tan^2\omega_v - 1)\,, \tag{72}$$

with equality only for the regular tetrahedron, octahedron and icosahedron. Inequality (72), which represents a counterpart of (71), would imply the truth of the above conjecture. We note that the cube and the regular dodecahedron are not the best polyhedra with eight and twenty vertices, respectively. This follows from the remark of L. Fejes Tóth (1948b, 1972, p. 136) that a best polyhedron of v vertices is bounded by triangles only (and satisfies an additional condition). Theorem 33 raises the question whether a best polyhedron of v vertices is always inscribed in a sphere.

It may be true that each convex polyhedron with f faces, v vertices and e edges even satisfies the inequality

$$\frac{F^3}{V^2} \geqslant 9e\sin\frac{2\pi}{p}\left(\tan^2\frac{\pi}{p}\tan^2\frac{\pi}{q} - 1\right), \tag{73}$$

where $p = 2e/f$ and $q = 2e/v$, and that equality holds only for the five Platonic solids. Inequality (73) includes (71) and (72); its proof would settle the isoperimetric problem for convex polyhedra (L. Fejes Tóth 1950b, 1964, p. 305). It can be shown that (73) is equivalent to the inequality

$$F \geqslant 4ea(O_4B_4C_4)\,, \tag{74}$$

where $O_4B_4C_4$ is the base triangle of the orthoscheme $T_4 = O^*O_4B_4C_4$ introduced in section 3. If the polyhedron is circumscribed about a sphere of radius r, then (73) is identical with (36).

Very little is known about the isoperimetric problem for non-Euclidean polytopes. One reason may be that the measurement of volume in non-Euclidean spaces is rather complicated. Moreover, there is no analogue to the Minkowski–Lindelöf Theorem 34. Making use of the momentum lemma (Corollary 2), L. Fejes Tóth (1963) proved the following theorem.

Theorem 38. *In hyperbolic 3-space, among all tetrahedra of given surface area the regular tetrahedron has maximal volume.*

Unfortunately, the proof does not work in spherical space. It can be shown that in the class of isoperimetric tetrahedra in spherical 3-space the volume attains a *relative* maximum for the corresponding regular tetrahedron (Böhm 1986).

A slight modification of the arguments used by L. Fejes Tóth shows that, in hyperbolic 3-space, inequality (74) holds for any convex polyhedron circumscribed about a sphere. Equality occurs only for the five regular polyhedra (A. Florian 1975).

There is hardly any doubt that in the hyperbolic and in the spherical space of any dimension the regular simplex has maximal volume among all the simplices of equal surface area. In the hyperbolic case, a possible proof may be based on an extension of the momentum lemma to d-dimensional simplices. A momentum theorem for simplices in hyperbolic and, of course, also in spherical space would be of independent interest.

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CHAPTER 1.7

Rigidity

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Contents

1. Introduction 225
2. Early results 226
   2.1. Cauchy's proof of his theorem 226
   2.2. Smooth analogues of Cauchy's Theorem 228
   2.3. Some counter-examples and flexes 229
   2.4. Aleksandrov–Pogorelov theory 231
3. Basic definitions and basic results 232
   3.1. Frameworks and tensegrity frameworks 232
   3.2. Rigidity and global rigidity 233
   3.3. Infinitesimal rigidity 235
   3.4. Static rigidity 235
   3.5. Projective invariance 237
   3.6. The rigidity map 238
4. Infinitesimal and static rigidity related to surfaces 240
   4.1. Grünbaum and Shephard's conjectures 240
   4.2. The infinitesimal version of Cauchy's Theorem 243
   4.3. Aleksandrov's theory 244
   4.4. Consequences of Aleksandrov's theory 246
   4.5. Kann's proof and extension of Dehn's Theorem 247
   4.6. More proofs and extensions of Dehn's Theorem 249
   4.7. Maxwell–Cremona theory and spider webs 251
   4.8. Rigidity, convexity, and combinatorics in higher dimensions 253
5. Second-order rigidity and pre-stress stability 256
   5.1. The definition of second-order rigidity 256
   5.2. Second-order rigidity and convex surfaces 257

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5.3. Pre-stress stability 258
5.4. Global rigidity and the stress matrix 261
5.5. Second-order duality 263
5.6. Polyhedral surfaces revisited 264
References 266

1. Introduction

> *Equal and similar solid figures are those contained by similar planes equal in multitude and magnitude.*

This is Definition 10 in Book XI of Euclid's Elements. (See Heath 1956 and Legendre 1794 for a discussion.) Many authors have pointed out that this is not properly a definition but a statement that should be proved. Indeed, one should be more careful about just what the hypothesis should be and what the terms mean. The two "solid figures" in fig. 1.1 might be interpreted as a counter-example to the above statement. The polyhedron on the left is cut into two pieces and the top portion, a tetrahedron, is put back, inside the other half to get the polyhedron on the right.

One way to avoid the difficulty of this example is to assume that all the objects being considered are convex. Indeed, the first substantial mathematical result concerning rigidity is the following.

Cauchy's Theorem 1.1 (Cauchy 1813). *Two convex polyhedra comprised of the same number of equal similarly placed faces are superposable or symmetric.*

Cauchy's Theorem was the subject of a great deal of scrutiny as well as inspiration for many generalizations, even though both were somewhat belated. The proof was in two parts, one geometric and one topological, both with certain flaws. The mistake in the geometric part was the more serious and was finally corrected by Steinitz in Steinitz and Rademacher (1934). The relatively minor mistake in the topological part of the proof was found by Hadamard (1907) and corrected by Lebesgue (1909). The geometric argument is especially interesting, and we will discuss it further later.

There are two main categories for generalizations coming from Cauchy's Theorem. One is in the category of polyhedra and similar discrete objects such as frameworks and hinged plates. The other generalization is in the category of appropriately smooth surfaces. There is also the question of exactly what "kind" of rigidity one is discussing, and this applies to both categories. In Cauchy's Theorem one thinks of each of the faces of the polytope as a "rigid" plate, and one stays in the configuration space of convex objects. Then the rigidity result is a statement about uniqueness in this space. On the other hand, one can also form a "linearized" definition of rigidity, called infinitesimal rigidity or first-order rigidity, in both the discrete and smooth categories. However, only in the discrete

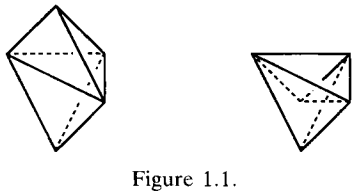


Figure 1.1.

category has it been proved generally that infinitesimal rigidity implies rigidity in the sense of there being no non-trivial continuous motion of the object preserving its appropriate metric character. Despite this, there are many similarities between the ideas in the discrete category and the smooth category. Yet other "kinds" of rigidity are second-order rigidity and pre-stress stability. These are natural extensions of first-order rigidity and they are discussed in section 5.

The study of rigid structures is large and there are many different points of view. In Connelly (1988c, 1990), Crippen and Havel (1988), Sugihara (1986), and Recski (1989) there are some interesting applications and relations to other areas that we will not explore here. Since this is a handbook on convexity we shall try to restrict this discussion to the intersection of convexity theory and the theory of rigid structures. Nevertheless, this still leaves a great deal to be covered. Because of personal interests and lack of space we shall deal less comprehensively with the smooth category. For some older results in the category of polyhedral surfaces see Stoker (1968). In Crapo and Whiteley (1995) there will be an extensive discussion of the geometry of rigid structures in the discrete category in much greater detail. In particular the chapter by Roth (1987), "The rigidity of frameworks given by convex surfaces", will provide many of the details left out here. See Roth (1981) as well. For a discussion of some of the classical results in the smooth category see Efimov (1962), Nirenberg (1963), Hicks (1964), Stoker (1969), and Spivak (1979, Vol. 5).

2. Early results

We shall start with a discussion of some of the early results of Cauchy and others that are in the spirit of elementary geometry and basic differential geometry.

2.1. Cauchy's proof of his theorem

Cauchy's proof was in two parts, a geometric part, where the convexity assumption was used in an essential way, and a topological or combinatorial part where an Euler characteristic argument is used, as well as the underlying manifold structure. I like to call the basic Geometric Lemma the "Arm Lemma" since the two polygons involved resemble an opening arm.

Lemma 2.1 (Cauchy 1813). *Let $A, B, \ldots, G$ represent a convex planar or spherical polygon, and let $A', B', \ldots, G'$ represent another such convex polygon, where for the corresponding lengths*

$$AB = A'B', BC = B'C', \ldots, FG = F'G'$$

and for the corresponding angles

$$\angle ABC \leqslant \angle A'B'C', \angle BCD \leqslant \angle B'C'D', \ldots, \angle EFG \leqslant \angle E'F'G' .$$

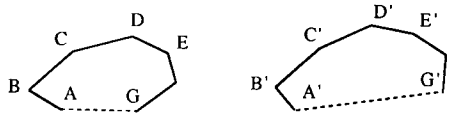


Figure 2.1.

Then for the last unmentioned length, $AG \leq A'G'$, where equality holds if and only if the two polygons are congruent. (See fig. 2.1.)

Cauchy attempted to prove this lemma by induction on the number of vertices "opening" the "arm" one joint (or vertex) at a time. The trouble is that the intermediate configurations may not be convex, ruining the induction hypothesis. This is the more serious mistake mentioned in the introduction. Figure 2.2 shows the problem.

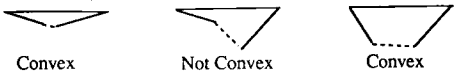


Figure 2.2.

Nevertheless, the Arm Lemma is still true, and there have been various alternate (correct) proofs that have been proposed. For example, see Danzer (1986), Steinitz and Rademacher (1934), Volkov (1956), Zalgaller (1956), and Lyusternik (1966, 1963). In Schoenberg and Zaremba (1967) there is an especially clever and elementary proof. Indeed the discussion in Chern (1967) of the Lemma of Schur (1921) is a pleasant generalization that seems to work in the case the arm is piecewise smooth, thus including both the discrete and smooth cases. See also Epstein (1985) for an analogue in the hyperbolic plane. The discussion in section 4.1 of this chapter is of a generalization replacing some of the equality constraints with inequalities, as well as with different combinatorial arrangements. This follows Connelly (1982) and provides yet another proof of the Arm Lemma, but this time only in the discrete category. There is a short discussion of other related references in Connelly (1982). An easy corollary of this Arm Lemma is the following.

Corollary 2.2. *Let P and P' be two convex planar or spherical polygons with the same number of corresponding sides of equal length. Assign a + sign or a − sign for those vertices of P where the internal angle increases or decreases, respectively, in going from P to P'. (No sign is given to those angles where there is no change.) Then either there are at least four changes in sign as one travels around P, or there are no sign assignments at all, and P and P' are congruent. (See fig. 2.3.)*

The idea of the proof is as follows. Suppose there are only two sign changes. Choose a line segment with its end points in the interior of two edges of P separating the two signs. The Arm Lemma applied to one side of P implies that

Figure 2.3.

this line segment must increase in length in going to P'. Similarly, the Arm Lemma applied to the other side in going from P' to P also implies that the line segment must have the opposite change in length, a contradiction. Thus there must be at least four sign changes, if there are any at all.

The following is Cauchy's Combinatorial Lemma.

Lemma 2.3. *Suppose that + and − signs are assigned to some of the edges of a triangulated (two-dimensional) sphere so that at each vertex with some labeled edge there are at least four changes of sign as one proceeds around the vertex. Then none of the edges are labeled.*

The proof of this lemma is an argument using the Euler characteristic of the sphere and counting the number of sign changes around each vertex as well as the number of sign changes in each region defined by the edges that are labeled with a + or a −. See Lyusternik (1963), for example, for a proof. See also Pogorelov (1956) or Sen'kin (1956) for a proof of Cauchy's Theorem from the Arm Lemma.

The proof of Cauchy's Theorem (see introduction) now follows easily. If P and P' are the two corresponding convex polyhedra, then label an edge of P a + or a − depending on whether the dihedral angle at the edge increases or decreases, respectively. By taking a small sphere centered at each vertex of P, the Corollary to the Arm Lemma implies that there must be at least four changes in sign around each vertex with at least one labeled edge coming into it. The combinatorial lemma then implies that there are no sign changes at all. Then it is easy to see that P and P' are congruent. See also Aleksandrov and Sen'kin (1956) for a proof of a generalization of Cauchy's Theorem to include some non-convex polyhedra.

These basic results of Cauchy come up again and they are applied in many other contexts.

2.2. Smooth analogues of Cauchy's Theorem

See Stoker (1969, 1978), and Liebmann (1900) as well as Busemann (1958), Sabitov (1989), and Ivanova-Karatopraklieva and Sabitov (1991) for discussions and surveys of this and related problems to follow in this section. [See also Vekua (1962), and a later edition (1988), for a basis for many of the results.]

If M is a Riemannian manifold, part of the structure that defines M is the degree of differentiability of the coordinate maps as well as the implied Riemannian metric obtained by integrating curves on the surface. The first rigidity theorems had rather strong assumptions on the differentiability of M, but after some effort these assumptions were weakened greatly. The first smooth analogue to Cauchy's Theorem was the following.

Theorem 2.4 (Cohn-Vossen 1936). *Suppose that M and M' are two isometric, compact, closed, convex, analytic surfaces in $\mathbb{R}^3$. Then M and M' are congruent (including possibly being reflected copies of each other).*

It was later shown how to remove the condition of being analytic. Notice that this result is again a statement about the uniqueness of an object in some space of convex surfaces. Later in Herglotz (1943) an extremely succinct proof of Cohn-Vossen's Theorem is presented, where the surface need only be of the class C^2 (i.e., continuous second derivatives) instead of analytic. See Hicks (1964), for example, for "modern" treatments of this proof. See Aleksandrov and Sen'kin (1955) for a short proof based on differential equations. With some imagination, one can see certain similarities between Herglotz' proof and Cauchy's proof. For example, the following Lemma comes from Herglotz.

Lemma 2.5. *Let $Q(x, y)$ and $Q'(x, y)$ be two positive definite quadratic forms in two variables with the same determinant when expressed as symmetric two-by-two matrices. Then the quadratic form given by the difference $Q - Q'$ is either indefinite or identically zero.*

The idea is that the equations $Q(x, y) = 1$ and $Q'(x, y) = 1$ represent two ellipses (centered at the origin) with the same area. If they are not identical, then there must be four changes in sign as one proceeds around the unit circle of directions in the plane. See fig. 2.4. Loosely speaking, quadratic forms are used to represent the infinitesimal "bending" near a point in the smooth surface. The above Lemma indicates that as one proceeds around the circle of directions at a point on the surface, either there are four times when there is a change as to which surface bends the most, or the surfaces are infinitesimally isometric at that point. It seems to me that this Lemma is a smooth analogue to Cauchy's Corollary 2.2 to his Arm Lemma. But in the smooth category this Lemma is quite simple. Smooth surfaces have an infinitesimal symmetry that is lacking in polyhedral surfaces. Perhaps this is why Herglotz' proof is so much easier than Cauchy's, after one gets past the fundamentals in each category. The other half of Herglotz' argument involves an integral formula, and this seems to play the role of the Euler characteristic argument in Cauchy's combinatorial lemma.

2.3. Some counter-examples and flexes

In the results mentioned above, both surfaces were assumed to be convex. The consequence was that there was only one such object in the given category. In particular, this means that there is no continuous deformation, preserving the

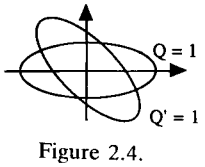


Figure 2.4.

appropriate metric structure and the convexity. A continuous deformation of the surface, preserving the metric, is what I call a *flex*. It is *non-trivial* if it is not the restriction of a rigid motion of the ambient Euclidean space.

On the other hand, if only one of the surfaces is assumed to be convex, then there can be difficulties. For instance, consider the polyhedral category, where the objects are obtained by gluing together planar polygons along their boundary. Here the metric structure is the metric function that defines the distance between two points on the surface as the length of the shortest connecting arc on the surface, similar to the smooth category. Then an example similar to fig. 1.1 exists even with a non-trivial continuous flex. Each member of the family has the same metric. Furthermore, this flex can start with a convex surface as in fig. 2.5.

The top of the tetrahedron is sliced off by a continuously moving plane and then "popped in" (i.e., reflected inside through the plane). This is something like the way one might turn a paper bag inside out. There is a "crease" that moves continuously along some of the faces of the polyhedron.

Of course in Cauchy's Theorem each face of the polyhedron is assumed to be a rigid "plate", and no such continuous creasing is allowed. So for Cauchy's hypothesis in his Theorem there is no such flex. When there is no such non-trivial continuous flex, we say that the object is *rigid*.

A natural question then comes up: is convexity really needed for the rigidity of a polyhedral surface with rigid faces? Apparently even Euler conjectured that there was no such continuous flex. This question and related results are discussed explicitly in Gluck (1975). Nevertheless a counter-example was found in Connelly (1978). This is an embedded polyhedral surface in three-space with rigid faces that flexes. See Connelly (1979) for the best discussion of this example as well as a very simple example by K. Steffan roughly based on the original example in Connelly (1978). See also the description in Kuiper (1978).

What about the same question in the smooth category? Is a smooth surface convex in three-space rigid, if it must stay in the category of smooth surfaces? Here one must be careful what the definition of smooth means. The answer in the C^1 category (continuous first derivatives) is surprising. According to Kuiper, following the ideas of Kuiper (1955a,b) (see also Nash 1954), if one takes any C^1 smooth surface, there is a non-trivial flex in the category of C^1 surfaces. In particular, the standard unit sphere in three-space must have a non-trivial C^1 flex. It turns out (see section 2.4) only the initial configuration can be convex. The surface must immediately move into some sort of "prune-like" shape. I know of no explicit description of such a flex or even of an explicit C^1 embedding other than the original sphere.

In the C^2 category of course no such examples exist. In this category one can

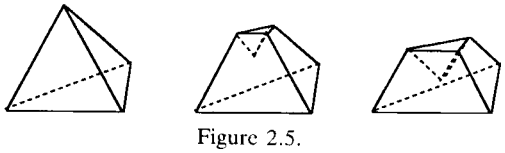


Figure 2.5.

define the Gaussian curvature of a surface. The space of C^2 closed surfaces with strictly positive Gaussian curvature is open in the space of C^2 surfaces. If the Gaussian curvature of a closed surface is positive at each point, then the surface must be convex. Hence Hergoltz' uniqueness Theorem implies that such a surface is rigid at least in the class of C^2 surfaces.

If a smooth surface is C^2 then I know of no case when a (necessarily non-convex) surface flexes staying in the C^2 category, even when the surface is immersed. (In the smooth category, a manifold is *immersed* when the differential of each coordinate map is one-to-one.)

2.4. Aleksandrov–Pogorelov theory

Although it may seem as though the smooth and piecewise-linear categories are distinct with only vague similarities, it turns out that there is a common generalization. For Cauchy-type theorems this appears to be the ultimate as far as results about the uniqueness in the class of convex surfaces. This follows the work of Aleksandrov (1958) and Pogorelov (1973), which is a recounting of earlier work.

Given any two points p and q on a convex surface in n-space, consider any continuous arc that lies in the surface that connects p and q. The length of the shortest possible such arc is called the distance between p and q. This length is always a well-defined finite number, when, say, the convex surface is the boundary of a compact convex set. This distance function is easily seen to be a metric function for the surface. See Aleksandrov and Zalgaller (1967) for a careful discussion.

If the surface is the boundary of a compact convex polytope in three-space, then intrinsically in the neighborhood of any point that is not a vertex of the polytope, the metric function is the same as the metric defined in the plane. At an interior point of an edge, intrinsically the surface is just a flat surface folded, and so its metric is the same as the metric in the plane. If any compact topological metric space has the property that all but a finite number of points have neighborhoods that are isometric to open subsets of the Euclidean plane, and the other points have neighborhoods isometric to a cone over a circle with the metric explained above, then Aleksandrov defines this as a *convex polyhedral metric.* The following existence theorem of Aleksandrov is related closely to a rigidity result that will be explained later.

Theorem 2.6 (Aleksandrov 1958). *Any convex polyhedral metric given on a manifold homeomorphic to a sphere is realizable as a compact convex polytope, possibly degenerating into a doubly covered plane polygon.*

This result is also an important steppingstone on the way to proving the following result of Pogorelov, which he called a "monotypy" theorem.

Theorem 2.7 (Pogorelov 1973). *Isometric closed convex surfaces in three-space are congruent.*

This very powerful result applies equally well to the smooth category as to the piecewise-linear category. Note how these results concentrate on the intrinsic metric rather than on any other further underlying structure. In Cauchy's Theorem, for instance, there was the further structure of the faces of the polytope, which act as rigid "plates". There are many more such important results in Pogorelov (1973).

There is also a proof of Pogorelov's result in Sen'kin (1956). An important tool for approximating convex surfaces is outlined in Volkov (1968).

A special case when one of the surfaces is assumed to be the boundary of a convex polytope was proved first in Olovianishnikoff (1946).

3. Basic definitions and basic results

In order to explain the results that follow and to provide a more unified and precise point of view in the discrete category, we discuss the concept of a (tensegrity) framework and the various kinds of associated rigidity. This also serves to set the notation. The discussion in sections 3.1 to 3.3 has evolved from the point of view of Aleksandrov (1958), Gluck (1975), Asimow and Roth (1978, 1979), Crapo (1979), Roth and Whiteley (1981), and Crapo and Whiteley (1982), which was again reformulated in Connelly (1988a,b).

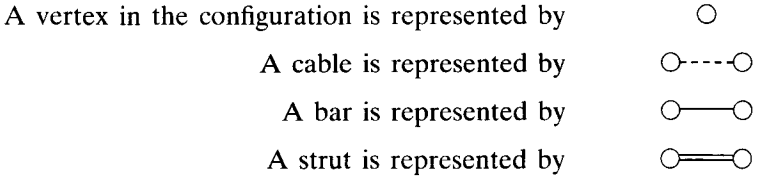
3.1. Frameworks and tensegrity frameworks

A *configuration* is a finite collection $\boldsymbol{p}$ of n labeled points $(p_1, p_2, \ldots, p_n)$, where each point p_i (also called a *vertex*) is in a fixed Euclidean space $\mathbb{R}^d$. A configuration $\boldsymbol{p} = (p_1, p_2, \ldots, p_n)$ and a configuration $\boldsymbol{q} = (q_1, q_2, \ldots, q_n)$ are *congruent* if there is a congruence of $\mathbb{R}^d$ that takes p_i to q_i for all $i = 1, 2, \ldots, n$.

A *tensegrity graph* G is an abstract graph on the vertices $1, 2, \ldots, n$, where each edge is labeled as either a cable, a strut, or a bar. The idea is that cables cannot increase in length, struts cannot decrease in length, and bars cannot change in length at all. The pair G and $\boldsymbol{p}$ is called a *tensegrity framework* and it is denoted as $G(\boldsymbol{p})$.

In case all edges (or *members*) of the graph G are only bars, $G(\boldsymbol{p})$ will be called a *bar* framework. It often turns out that many of the geometric situations that come up are conveniently and reasonably expressed as statements about tensegrity frameworks even if the primary object of interest is a bar framework.

It is also pleasant to show a framework visually. We use the following notation.



In fig. 3.1 we show some examples, a tensegrity framework and two bar frameworks in the plane. Note that there is no requirement that a vertex of the configuration be placed at the intersection where two members (thought of as line segments) cross.

3.2. Rigidity and global rigidity

We say that a tensegrity framework $G(\boldsymbol{p})$ *dominates* another tensegrity framework $G(\boldsymbol{q})$ (with the same graph) if the length of each cable of $G(\boldsymbol{q})$ is no longer than the length of the corresponding cable of $G(\boldsymbol{p})$, the length of each strut of $G(\boldsymbol{q})$ is no shorter than the length of the corresponding strut of $G(\boldsymbol{p})$, and the length of each bar of $G(\boldsymbol{q})$ is the same length as the corresponding bar of $G(\boldsymbol{p})$. Here, if $\{i, j\}$ is a member of G, then $|p_i - p_j|$ is its length, where we use vector notation throughout.

For a fixed graph G it is clear that this relation of dominance is only a partial ordering. On the other hand, when all the members of G are bars, this relation is an equivalence relation. So for bar frameworks, we say that $G(\boldsymbol{p})$ is *equivalent* to another bar framework $G(\boldsymbol{q})$ when one of them dominates the other by the above definition. In other words, $G(\boldsymbol{p})$ is *equivalent* to $G(\boldsymbol{q})$ when the corresponding bars have the same length. Note that when two configurations $\boldsymbol{p}$ and $\boldsymbol{q}$ are congruent, then $G(\boldsymbol{p})$ always dominates $G(\boldsymbol{q})$, and indeed $G(\boldsymbol{p})$ is equivalent to $G(\boldsymbol{q})$ for bar frameworks.

Fix a particular tensegrity framework $G(\boldsymbol{p})$ in $\mathbb{R}^d$. We say that $G(\boldsymbol{p})$ is *globally rigid* in $\mathbb{R}^d$ if, when $G(\boldsymbol{p})$ dominates another tensegrity framework $G(\boldsymbol{q})$ in $\mathbb{R}^d$, then $\boldsymbol{p}$ and $\boldsymbol{q}$ are congruent. For example, in fig. 3.1, the tensegrity framework on the left is globally rigid in the plane, and the other two bar frameworks are not. The framework on the right even has a continuous family of mutually non-congruent equivalent configurations. The framework in the middle has equivalent non-congruent configurations that involve the change of only the rightmost vertex, for example.

Global rigidity is a very simple and natural concept, but it is also a very strong property to impose on a framework. We will see later how it comes up in the discussion of the proof of Cauchy's Arm Lemma as well as in the resolution of some conjectures in Grünbaum and Shephard (1975).

Meanwhile, a very basic property of frameworks is simply their rigidity. We say that the family of configurations $\boldsymbol{p}(t)$ in $\mathbb{R}^d$, for $0 \leq t \leq 1$, is a (*continuous*) *flex* of the (tensegrity) framework $G(\boldsymbol{p})$ if each coordinate of each vertex is a continuous function of t, $\boldsymbol{p}(0) = \boldsymbol{p}$, and $G(\boldsymbol{p})$ dominates each $G(\boldsymbol{p}(t))$ for all $0 \leq t \leq 1$. We say that the flex $\boldsymbol{p}(t)$ is *trivial* if each $\boldsymbol{p}(t)$ is congruent to $\boldsymbol{p}$. If $G(\boldsymbol{p})$ admits only trivial

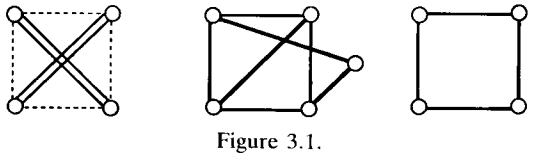


Figure 3.1.

flexes, then we say that $G(\boldsymbol{p})$ is *rigid in* $\mathbb{R}^d$. For example, the left framework and the middle framework in fig. 3.1 are rigid in the plane, whereas the rightmost framework is not. Clearly, if a (tensegrity) framework is globally rigid in $\mathbb{R}^d$, then it is rigid in $\mathbb{R}^d$. The middle framework in fig. 3.1 is a counter-example to the converse statement. See the discussion in Connelly (1988a) for some equivalent definitions.

The statement of Cauchy's Theorem can be reformulated in terms of frameworks and configurations. Each vertex of the polytope is a vertex of the configuration, and each edge of the polytope corresponds to a bar of the framework. But since each face is to be regarded as a rigid plate, it is convenient to specialize to the case when each face of the polytope is a triangle. The conclusion of Cauchy's Theorem is a statement similar to the property of global rigidity, except that the configurations are restricted to those coming from convex polytopes with triangular faces instead of all configurations in three-space. In any case, at least in the triangular face case, it turns out that Cauchy's Theorem does imply that the corresponding framework is rigid in three-space. We shall have more to say about this in section 4.

3.3. *Infinitesimal rigidity*

Even if one's main interest is only the question of whether a certain class of frameworks is rigid, one needs some methods. A very natural technique is to "linearize" the problem. That is, one tries to find some system of linear equations (or linear inequalities) that will detect rigidity. One approach is to "differentiate" the member constraints.

An *infinitesimal flex* of a framework $G(\boldsymbol{p})$ in $\mathbb{R}^d$ is an assignment of a vector p_i' in $\mathbb{R}^d$ to each vertex i of G such that for each member $\{i, j\}$ of G,

$$(p_i - p_j)\cdot(p_i' - p_j')\begin{cases} \leqslant 0 & \text{if } \{i, j\} \text{ is a cable,} \\ = 0 & \text{if } \{i, j\} \text{ is a bar,} \\ \geqslant 0 & \text{if } \{i, j\} \text{ is a strut.} \end{cases} \tag{3.1}$$

The product that is used for the vectors above is the ordinary dot product (or inner product) in $\mathbb{R}^d$. We denote the infinitesimal flex by $\boldsymbol{p}' = (p_1', p_2', \ldots, p_n')$, another configuration. We imagine p_i' as a vector based at p_i. We say that $\boldsymbol{p}'$ is *trivial* if $\boldsymbol{p}'$ is the derivative of a rigid congruence of all of $\mathbb{R}^d$ restricted to the vertices of $\boldsymbol{p}$ at $t = 0$. It turns out in dimension three that $\boldsymbol{p}'$ is trivial if, for each i, $p_i' = r \times p_i + u$, where r and u are fixed vectors in three-space and $\times$ represents the cross product. See Connelly (1988a) for a further description of such trivial infinitesimal flexes. We say that a (tensegrity) framework $G(\boldsymbol{p})$ in $\mathbb{R}^d$ is *infinitesimally rigid* if it has only trivial infinitesimal flexes.

So $\boldsymbol{p}'$ is analogous to a vector field on a smooth manifold (in the smooth category). If we have a flex $\boldsymbol{p}(t)$ of a framework $G(\boldsymbol{p})$, and each of the coordinates of $\boldsymbol{p}(t)$ are differentiable at $t = 0$, then (3.1) holds where $\boldsymbol{p}'$ is the derivative of $\boldsymbol{p}(t)$ at $t = 0$. The square of the member length $\{i, j\}$ equals

$(p_i(t) - p_j(t)) \cdot (p_i(t) - p_j(t))$. The derivative evaluated at $t = 0$ gives (3.1). One must be careful though. Even if the flex $\boldsymbol{p}(t)$ is non-trivial, it may turn out that the infinitesimal flex $\boldsymbol{p}'$ *is* trivial. Simply replace the parameter t by t^2, for example.

In a visual representation of a framework, we will use a small arrow at p_i to represent an infinitesimal flex p_i'. If there is no arrow it will be understood that $p_i' = 0$. Figure 3.2 shows two examples of bar frameworks in the plane that have non-trivial infinitesimal flexes. The infinitesimal flex for the framework on the right comes from a smooth flex of the framework. For the framework on the left, there is no such (smooth) flex. In other words, both frameworks are not infinitesimally rigid in the plane, but the one on the left is rigid nevertheless.

The following basic result helps to justify the study of infinitesimal rigidity. See Connelly (1988a), as well as Roth and Whiteley (1981) and Connelly (1980), for an explicit proof.

Theorem 3.1. *If a tensegrity framework $G(\boldsymbol{p})$ is infinitesimally rigid in $\mathbb{R}^d$, then it is rigid in $\mathbb{R}^d$.*

As far as I know there is no analogous result in the smooth category, although both the notions of rigidity and infinitesimal rigidity have appropriate analogues.

3.4. Static rigidity

The concept of infinitesimal rigidity seems to be motivated by the "kinematics" of a framework, i.e., by its motions (even if they are "infinitesimal"). For engineering structures though, often another point of view is taken. The idea is that the "stability" of a structure should have more to do with forces or "loads" acting on it and the capability of the structure to "resolve" these forces.

We say that a *load* or *force* $\boldsymbol{F}$ acting on a framework $G(\boldsymbol{p})$ in $\mathbb{R}^d$ is an assignment of a vector F_i in $\mathbb{R}^d$ to each vertex i of G. Formally $\boldsymbol{F}$ is simply another configuration in $\mathbb{R}^d$, but the words are meant to suggest a physical force. Note that the force $\boldsymbol{F}$ is really defined for the configuration $\boldsymbol{p}$, independently of the graph G.

We say that a *stress* ω for a framework $G(\boldsymbol{p})$ in $\mathbb{R}^d$ is an assignment of a real number $\omega_{ij} = \omega_{ji}$ for each member $\{i, j\}$ of G. We regard ω_{ij} as the tension or compression in the member $\{i, j\}$. We say that ω *resolves* a load $\boldsymbol{F}$ if the following vector equation holds for each vertex i of G:

$$F_i + \sum_j \omega_{ij}(p_j - p_i) = 0\,, \tag{3.2}$$

where the summation is taken over all the vertices j adjacent to i in G.

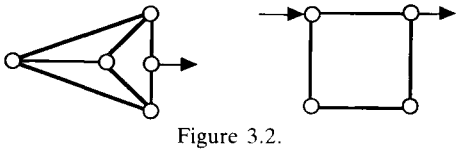


Figure 3.2.

(Alternatively we can define $\omega_{ij} = 0$ for each non-member of $\smile$, and sum over all j.) Equation (3.2) represents an equilibrium at each vertex i.

It turns out that "most" forces cannot be resolved by any framework. So we say that a force $\boldsymbol{F}$ is an *equilibrium force* (or an *equilibrium load*) at the configuration $\boldsymbol{p}$ in $\mathbb{R}^d$ if for every trivial infinitesimal flex $\boldsymbol{p}'$ at the configuration $\boldsymbol{p}$, the following holds:

$$\sum_i F_i \cdot p_i' = 0 ,$$

where the summation is taken over all vertices i. Physically this means that when $\boldsymbol{F}$ is regarded as giving velocities at the configuration $\boldsymbol{p}$, it has zero linear and angular momentum. See Connelly (1988b), Roth and Whiteley (1981), or Crapo and Whiteley (1982) for a discussion of other equivalent conditions and reformulations for what it means to be an equilibrium force.

Now it turns out that it is possible for some frameworks to resolve an equilibrium load. On the other hand, a cable cannot support compression and a strut cannot support tension, so we define one further restriction on stresses. We say that a stress ω for a framework $G(\boldsymbol{p})$ is *proper* if $\omega_{ij} \geqslant 0$ when $\{i, j\}$ is a cable, and $\omega_{ij} \leqslant 0$ when $\{i, j\}$ is a strut. (There is no condition when $\{i, j\}$ is a bar.)

We say that a tensegrity framework $G(\boldsymbol{p})$ in $\mathbb{R}^d$ is *statically rigid* if every equilibrium load $\boldsymbol{F}$ can be resolved by a proper stress ω. The frameworks in fig. 3.3 have equilibrium loads indicated. The framework on the left is not statically rigid, and the given load cannot be resolved. The framework on the right is statically rigid, and the resolving proper stress is indicated.

A very basic result now connects this definition to infinitesimal rigidity. See Connelly (1988b) or Roth and Whiteley (1981) for a proof. Section 3.6 has some information concerning a proof.

Theorem 3.2. *A tensegrity framework $G(\boldsymbol{p})$ in $\mathbb{R}^d$ is infinitesimally rigid if and only if it is statically rigid.*

We also mention one more result that is very useful for relating the static rigidity (or the infinitesimal rigidity) of a tensegrity framework to the static rigidity of a bar framework. A *self stress* for a tensegrity framework $G(\boldsymbol{p})$ is a stress that resolves the zero load. So (3.2) holds with $\boldsymbol{F} = 0$. The following is the main result of Roth and Whiteley (1981).

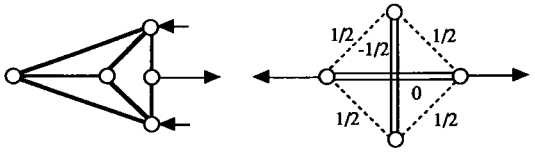


Figure 3.3.

Theorem 3.3. *Let $G(\boldsymbol{p})$ be a tensegrity framework in $\mathbb{R}^d$. Then $G(\boldsymbol{p})$ is statically rigid in $\mathbb{R}^d$ if and only if there is a proper self stress that is non-zero on each cable and strut, and $G'(\boldsymbol{p})$ is statically rigid, where G' is obtained from G by replacing each member with a bar.*

3.5. Projective invariance

It is clear that the definitions of infinitesimal rigidity and static rigidity for a framework $G(\boldsymbol{p})$ do not depend on the way in which the configuration $\boldsymbol{p}$ is realized in $\mathbb{R}^d$. In other words, if each of the points p_i of $\boldsymbol{p}$ is transformed by the same congruence $g:\mathbb{R}^d \to \mathbb{R}^d$, then the framework $G(\boldsymbol{p})$ is statically and infinitesimally rigid if and only if $G(g(\boldsymbol{p}))$ is statically and infinitesimally rigid, where $g(\boldsymbol{p}) = (g(p_1), \ldots, g(p_n))$. With a bit more thought it is easy to see directly that at least static rigidity is preserved when g is only affine linear. (Of course then infinitesimal rigidity is preserved as well.) However, even more is true.

Suppose that the points of one configuration $\boldsymbol{p}$ lie in an affine $(d\text{-}1)$-dimensional subspace S of some larger dimensional space. Let c be a point that does not lie in S, and let S' be another $(d\text{-}1)$-dimensional subspace that does not contain c. Then we can project the points of $\boldsymbol{p}$ from c to the points of another configuration $\boldsymbol{q}$ in S'. See fig. 3.4. Notice that a cable may be transformed into either a cable or a strut. To see which, suppose $g(p_i) = q_i$ and $g(p_j) = q_j$, where $\{i, j\}$ is a member of G. If c lies between p_i and q_i but not between p_j and q_j, then $\{i, j\}$ is changed from a cable to a strut or from a strut to a cable. Otherwise, when c is not between p_i and q_i and not between p_j and q_j (or c is between p_i and q_i and is between p_j and q_j) $\{i, j\}$ is left unchanged. This defines a rule for changing the tensegrity graph G to another tensegrity graph G'. Then the framework $G(\boldsymbol{p})$ is statically and infinitesimally rigid in S if and only if $G'(g(\boldsymbol{p}))$ is statically and infinitesimally rigid in S'. See Roth and Whiteley (1981) for a proof in this general situation of tensegrity frameworks. For the case of a bar framework there have been several proofs from several different points of view. For example, see Wunderlich (1980, 1982), Crapo and Whiteley (1982), Whiteley (1987a), and Wegner (1984).

In fact, with the rule described above, it is not hard to see that one does not even have to have the vertices of the configurations lie on a hyperplane. Let $c * G(\boldsymbol{p})$ denote the framework obtained by joining the vertex c with a bar to all

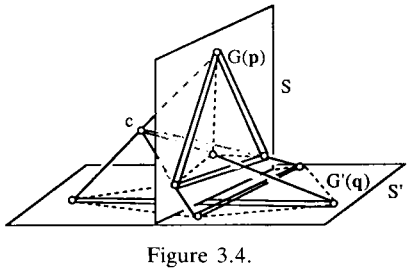


Figure 3.4.

the vertices of some framework $G(\boldsymbol{p})$ in $\mathbb{R}^d$. Suppose that $G(\boldsymbol{p})$ lies in a hyperplane S as before, but the points of $\boldsymbol{q}$ and c are only required, for each i, to be such that p_i, q_i and c are collinear, and c is distinct from both p_i and q_i. Then we have the following theorem.

Theorem 3.4. *There are natural linear isomorphisms from the infinitesimal flexes of $G(\boldsymbol{p})$ in S to the infinitesimal flexes of $c * G(\boldsymbol{p})$ in $\mathbb{R}^d$, fixing c, to the infinitesimal flexes of $c * G'(\boldsymbol{q})$ in $\mathbb{R}^d$, fixing c.*

In particular, we can see that $G(\boldsymbol{p})$ is infinitesimally rigid in the hyperplane S if and only if a projection of $\boldsymbol{p}$ into the unit sphere in $\mathbb{R}^d$ is infinitesimally rigid in the sphere. (That is, it is infinitesimally rigid in the sense that all the infinitesimal flexes tangent to the sphere are the time 0 derivatives of a rigid congruence of the sphere.)

We can see that it is reasonable to look at the problem of static and infinitesimal rigidity, at least for bar frameworks, from the point of view of projective geometry. This is essentially what is done in Crapo and Whiteley (1982). The property of static rigidity is reformulated entirely in terms of homogeneous coordinates of projective space. The question of the projective invariance is incorporated in the definitions and terms of a particular algebra, the Cayley algebra, that is especially convenient for projective geometry. One can then use the Cayley algebra to get some general algebraic information about these special configurations of bar frameworks whose space of self stress is of a larger dimension than the minimum possible for the given graph. This is done in White and Whiteley (1983). A similar approach is taken in White and Whiteley (1983), but for infinitesimal motions.

But one other case arises. In projective geometry, if we regard projective space as extended Euclidean space, there are points "at infinity". Suppose some points of our configuration are at infinity and they are to be held fixed for infinitesimal flexes. Then in the linear equations (3.1) that define infinitesimal rigidity we can replace the vector $p_i - p_j$ by a unit vector if p_j is at infinity and p_i is not. Note that $p_j' = 0$. In some cases it is convenient to transform the configuration in such a way that some of the points are at infinity, and, modulo trivial infinitesimal flexes, we can often assume that these points at infinity are fixed infinitesimally. This will be useful in section 4.5. This is also done in the smooth category where the projective transformation is called a Darboux Transformation. See Darboux (1896).

3.6. The rigidity map

We can shed a little light on what is going on in Theorem 3.2. Regard a configuration $\boldsymbol{p}$ of n points in $\mathbb{R}^d$ as a single point in $\mathbb{R}^{nd}$, the concatenation of the n column vectors of $\mathbb{R}^d$ into one single column vector. Suppose that a tensegrity graph G has e members. To each configuration $\boldsymbol{p}$ and each member $\{i, j\}$ of G we associate the square of its length $|p_j - p_i|^2$. This defines a smooth map, which we

call the *rigidity map*,

$$f: \mathbb{R}^{nd} \to \mathbb{R}^e ,$$

given by

$$f(\boldsymbol{p}) = (\ldots, |p_j - p_i|^2, \ldots) .$$

The differential of this map is a linear map $\mathrm{d}f_{\boldsymbol{p}}: \mathbb{R}^{nd} \to \mathbb{R}^e$. We define the matrix (divided by 2) of this map (with respect to the usual bases) to be the *rigidity matrix* $R(\boldsymbol{p})$. Each row of $R(\boldsymbol{p})$ corresponds to a member $\{i, j\}$ of G. The columns of $R(\boldsymbol{p})$ are organized into n sets of d columns. Furthermore, row $\{i, j\}$ of $R(\boldsymbol{p})$ consists of all zeros except for the two sets of d entries corresponding to i and j. The entries corresponding to i are the coordinates of $p_i - p_j$, and the entries corresponding to j are the coordinates of $p_j - p_i$. It is easy to see that if $\boldsymbol{q}$ is another configuration, again regarded as a column vector in $\mathbb{R}^{nd}$, then the $\{i, j\}$ coordinate of $R(\boldsymbol{p})\boldsymbol{q}$ is simply $(p_i - p_j) \cdot (q_i - q_j)$. Thus $\boldsymbol{p}'$ is an infinitesimal flex of $G(\boldsymbol{p})$ if and only if the $\{i, j\}$ coordinate of $R(\boldsymbol{p})\boldsymbol{p}'$ is non-positive for a cable, non-negative for a strut, and zero for a bar. For a bar framework this is simply the condition $R(\boldsymbol{p})\boldsymbol{p}' = 0$.

As mentioned in section 3.4, any tensegrity framework $G(\boldsymbol{p})$ has the trivial infinitesimal flexes. These will always be in the kernel of $R(\boldsymbol{p})$. In $\mathbb{R}^d$ the dimension of the space of trivial infinitesimal flexes is $\frac{1}{2}d(d+1)$. When $d = 2$, the dimension of trivial infinitesimal flexes is 3, and when $d = 3$, the dimension of trivial infinitesimal flexes is 6. We have the following interpretation of infinitesimal and static rigidity. [See Asimow and Roth (1978, 1979), and Connelly (1988b) for a discussion of this result.]

Theorem 3.5. *Let $G(\boldsymbol{p})$ be a bar framework in $\mathbb{R}^d$, where G has n vertices and they do not all lie in a hyperplane. Then $G(\boldsymbol{p})$ is infinitesimally rigid if and only if the rank of $R(\boldsymbol{p})$ is $nd - \frac{1}{2}d(d+1)$.*

So in the plane, for infinitesimal rigidity, this rank must be $2n - 3$, and in three-space it must be $3n - 6$.

The e rows of the rigidity matrix $R(\boldsymbol{p})$ correspond to the members of G and if the rank of $R(\boldsymbol{p})$ is to be of maximal [and $G(\boldsymbol{p})$ is to be infinitesimally rigid], then we must have

$$e \geqslant nd - \tfrac{1}{2}d(d+1) . \tag{3.3}$$

When G has all bars, if $e = nd - \frac{1}{2}d(d+1)$, then the rows of $R(\boldsymbol{p})$ are independent if and only if $G(\boldsymbol{p})$ is infinitesimally rigid. The row relations are related to stresses.

It is easy to check that a force $\boldsymbol{F}$, regarded as an e-dimensional row vector, is resolved by the stress ω, also regarded as an e-dimensional row vector, if and only

if $\boldsymbol{F} + \omega R(\boldsymbol{p}) = 0$. This is one approach to the duality statement of Theorem 3.2. Recall that a stress ω for a framework $G(\boldsymbol{p})$ is a self stress if the equilibrium equation as in (3.2) holds at each vertex i with $F_i = 0$. Then $\omega R(\boldsymbol{p}) = 0$ is equivalent to (3.2) for each i. Putting this information together, we get the following theorem.

Theorem 3.6. *Let $G(\boldsymbol{p})$ be a bar framework in $\mathbb{R}^d$ with n vertices and e edges, where $e = nd - \frac{1}{2}d(d+1)$. Then $G(\boldsymbol{p})$ is infinitesimally rigid if and only if $G(\boldsymbol{p})$ has only the zero self stress.*

This is one approach to an infinitesimal version of Cauchy's Theorem, which we will discuss in section 4.2.

4. Infinitesimal and static rigidity related to surfaces

Now that we have the language of frameworks and some of the basic tools, we can apply them to some problems involving convexity.

4.1. Grünbaum and Shephard's conjectures

In Grünbaum and Shephard (1975, 1978) there is a very interesting discussion of the rigidity of tensegrity frameworks in the plane. In particular, their discussion relates to some of the problems concerning Cauchy's Theorem and his Arm Lemma. Although Grünbaum and Shephard did not put their questions exactly into the language of frameworks, it is natural to do so.

Suppose that $\boldsymbol{p}$ is a configuration of points in the plane such that the vertices $(p_1, p_2, \ldots, p_n) = \boldsymbol{p}$ form the vertices, in order, of a convex polygon. (This also means of course that no three vertices are collinear.) Suppose that the edges of the polygon $\{1,2\}, \{2,3\}, \ldots, \{n-1, n\}, \{n, 1\}$ are bars. Grünbaum and Shephard conjectured that for various ways of adding extra cables (which are necessarily on the inside of the polygon) the resulting framework was rigid in the plane. The two frameworks on the left in fig. 4.1 are some examples. The framework on the right is the same as the framework in the middle, except that the bars and cables have been interchanged. It turns out that all three of the frameworks are rigid in the plane.

Grünbaum and Shephard conjectured the following:

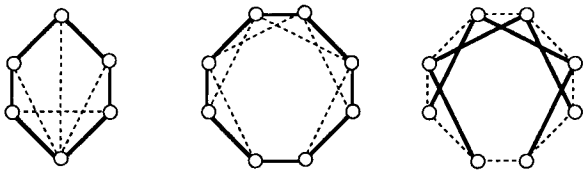


Figure 4.1.

Statement 4.1. *Let $G(\boldsymbol{p})$ be a rigid tensegrity framework in the plane, where the vertices form a convex polygon, all the external edges are bars, and cables are the only other members. Then the framework obtained by replacing all the bars by cables and the cables by bars, is also rigid in the plane.*

For example, this statement says that since the framework in the middle of fig. 4.1 is rigid, then the framework on the right is rigid. On the other hand the converse statement is not true in general. In fig. 4.2 consider the following frameworks in the plane based on a regular hexagon. The framework on the right with cables on the outside is (globally) rigid in the plane, whereas the framework on the left, obtained by interchanging the cables and bars, is not even rigid in the plane. In rigidity there is an important asymmetry between the roles that cables and bars play (or the roles that cables and struts play).

In Connelly (1982), Statement 4.1 was proved. It is a corollary of the following theorem. See section 5.4 for a further discussion.

Theorem 4.2. *Let $G(\boldsymbol{p})$ be a tensegrity framework in the plane, where the vertices form a convex polygon, all the external edges are cables, struts are the only other members, and $G(\boldsymbol{p})$ has a proper non-zero self stress. Then $G(\boldsymbol{p})$ is globally rigid in $\mathbb{R}^d$, for all $d \geq 2$.*

The framework on the left in fig. 4.3 is an example.

The framework on the right is also globally rigid in any dimension, since it is obtained from the framework on the left by simply replacing all but one of the cables by a bar.

It is also shown in Connelly (1982) that if a tensegrity framework $G(\boldsymbol{p})$ is rigid in $\mathbb{R}^d$ and G has at least one cable or strut, then $G(\boldsymbol{p})$ has a proper non-zero self stress (although it may be zero on the given cable or strut). Thus for the proof of Conjecture 4.1 we obtain a non-zero self stress ω from the rigidity of $G(\boldsymbol{p})$. Then $-\omega$ serves as the proper self stress for the framework in Theorem 4.2, where the cables and bars are interchanged.

For many frameworks coming from convex polygons it is possible to tell directly that the framework has a proper self stress and thus is globally rigid in the case of Theorem 4.2. Let $(p_1, p_2, \ldots, p_n) = \boldsymbol{p}$ be a configuration in the plane consisting of the vertices of a convex polygon as before. Let all of the external edges $\{1,2\}, \{2,3\}, \ldots, \{n-1, n\}, \{n, 1\}$ be the cables, and let $\{1,3\}$, $\{2,4\}, \ldots, \{n-2, n\}$ be the struts. (Notice that two consecutive struts in the cyclic sequence are missing.) I called such a tensegrity framework $G(\boldsymbol{p})$ a *Cauchy*

Figure 4.2.

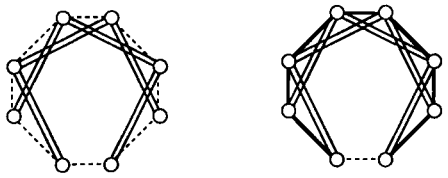


Figure 4.3.

polygon. The framework on the left in fig. 4.3 is an example of such a Cauchy polygon. It turns out (see Connelly 1982, Roth and Whiteley 1981) that any Cauchy polygon has a non-zero proper self stress. Thus it is always globally rigid.

Consider a polygonal framework such as the one on the right in fig. 4.3. When a pair of bars, adjacent to a vertex p_i, have a strut connecting the other two vertices p_{i-1} and p_{i+1}, it is equivalent to the condition that the internal angle at p_i is not permitted to decrease. This is precisely the condition for Cauchy's Arm Lemma. So if there were another configuration (even in three-space), where each angle at $p_2, p_3, \ldots, p_{n-1}$ increased or stayed the same, the distance between p_i and p_{i+1} would be fixed for $i = 1, 2, \ldots, n-1$, and the distance between p_1 and p_n would decrease, then this would contradict the global rigidity of Cauchy polygons. Thus we have yet another proof and generalization of Cauchy's Arm Lemma as mentioned in section 2.1.

Many of the frameworks mentioned in Grünbaum and Shephard (1975) were rigid but not globally rigid. For example, the middle and the left frameworks of fig. 4.1 are infinitesimally rigid in the plane by applying Theorem 3.3. Thus they are rigid, but it can be easily seen that they are not globally rigid. Also any Cauchy polygon is infinitesimally rigid in the plane. Then, if one changes every cable to a strut and every strut to a cable, it is easy to see that infinitesimal rigidity is preserved. (This is true for the infinitesimal rigidity of any framework.) The frameworks in fig. 4.2 naturally are not infinitesimally rigid.

Roth in (Roth and Whiteley 1981) suggested another approach to Statement 4.1. He conjectured the following:

Statement 4.3. *Let $G(\boldsymbol{p})$ be a rigid tensegrity framework in the plane, where the vertices form a convex polygon, all the external edges are bars, and cables are the only other members. Then $G(\boldsymbol{p})$ is infinitesimally rigid in the plane.*

As a corollary we find that the framework $G'(\boldsymbol{p})$, obtained by reversing the cables and bars, is infinitesimally rigid in the plane. (The bars can be replaced by struts in this case for the infinitesimal theory.) Thus $G'(\boldsymbol{p})$ is rigid in the plane. So this conjecture by itself implies Grünbaum and Shephard's Statement 4.1.

In Connelly and Whiteley (1990), Statement 4.3 is proved. It is also related to Theorem 4.2, but depends on a higher order analysis. An application of this theorem, for example, is that the hexagon on the left in fig. 4.2 is not rigid, since it has too few members to be infinitesimally rigid. We will say more about this in section 5.

4.2. The infinitesimal version of Cauchy's Theorem

In view of our discussion of infinitesimal rigidity, it is natural to ask whether convex polyhedra are infinitesimally rigid in three-space. Given a compact convex polytope P in three-space, one may associate to P a bar framework $G(\boldsymbol{p})$. The vertices of $G(\boldsymbol{p})$ are the vertices of P, and the edges of $G(\boldsymbol{p})$ are the edges of P. One must be careful though. With this description, it is not automatic that the faces of P are held rigid by the bars of the framework $G(\boldsymbol{p})$. Indeed, if any face of P is not a triangle, the framework $G(\boldsymbol{p})$ is not rigid (see section 4.4). For example, the bar framework associated to a cube is very flexible. Nevertheless, the following result of Dehn does show that when the faces are triangular there is no problem.

Theorem 4.4. *Let P be a compact convex polytope in three-space with all faces triangles. Then the associated bar framework $G(\boldsymbol{p})$ is infinitesimally rigid in three-space.*

See Dehn (1916) for his original proof. See also Weyl (1917). We will also describe a very simple proof that is basically in Aleksandrov (1958), repeated and clarified in Gluck (1975) and Asimow and Roth (1979).

We sketch Dehn's original proof of Theorem 4.4. His idea was to calculate the rank of the rigidity matrix $R(\boldsymbol{p})$ (as defined in section 3.5) by means of a Lagrange expansion of the determinant of the submatrix obtained by removing three rows corresponding to the edges of a triangle of a face of P and removing the nine rows corresponding to the three vertices of the same triangle. (This amounts to fixing those three points.) Each term in this expansion corresponds to an oriented graph on the remaining vertices of G (and conversely) such that there are exactly three edges coming into each vertex. Then it is possible to perform an operation on this oriented graph that creates another oriented graph with the same sign on the corresponding term in the Lagrange expansion. The proof is completed when it is shown that any such oriented graph can be obtained from any other by a sequence of such operations. All the terms must have the same sign; the determinant is non-zero; and the rank of $R(\boldsymbol{p})$ is $3n-6$ (where n is the number of vertices of G), which is what is needed to show that $G(\boldsymbol{p})$ is infinitesimally rigid in three-space.

Aleksandrov's idea, as discussed in Gluck (1975), was to calculate the row rank of $R(\boldsymbol{p})$. Since each face of P is a triangle, if there are e edges in P (and G), then $2e = 3f$, where f is the total number of triangular faces in P. Then Euler's formula for the surface of P states:

$$n - e + \tfrac{2}{3}e = 2\,,$$

where n is the number of vertices of P. This implies that

$$e = 3n - 6\,.$$

In dimension three this is precisely the minimum number of edges needed for a graph with n vertices to be infinitesimally rigid. By Theorem 3.6, $G(\boldsymbol{p})$ is infinitesimally rigid in $\mathbb{R}^3$ if and only if it has only the zero self stress.

There is a very simple argument to show that a non-zero self stress does not exist. At each vertex of $G(\boldsymbol{p})$ the equilibrium condition (3.2) (with $F_i = 0$) implies that either all the stresses at that vertex are zero, or there are at least four changes in the sign of the stresses ω_{ij} as one proceeds around the point p_i. This is because if there were exactly two changes in sign, then a plane through p_i could be found that separates those edges with a positive stress from those edges with a negative stress. See fig. 4.4. But that contradicts the equilibrium condition for the vertex i, since in the equilibrium equation (3.2), the vector components perpendicular to the plane all have the same sign. Hence the conditions for Cauchy's Combinatorial Lemma 2.3 are satisfied. None of the stresses on the edges of G are actually strictly positive or strictly negative. Thus $\omega = 0$, and $G(\boldsymbol{p})$ is statically rigid and infinitesimally rigid.

Notice how the above argument takes the place of Cauchy's somewhat more difficult Arm Lemma. Also, in both cases this is precisely the point where convexity is used in an essential way.

4.3. Aleksandrov's theory

The restriction in section 4.2 to polyhedra with only faces that are triangles is a bit unsatisfying. Cardboard models with non-triangular faces seem to hold their shape quite well, and there should be some reasonable theory to explain this. On the other hand, we must be careful in the way that we extend Dehn's Theorem 4.4. We wish to have the framework $G(\boldsymbol{p})$ based on the polytope P in that the members of P should at least be on the surface of P. A natural idea is to find a triangulation K of the surface of P, and the vertices and edges of K will give the framework $G(\boldsymbol{p})$. By a triangulation of P we mean a simplicial complex K (in the sense of piecewise linear topology), whose underlying space is exactly the set P. (In this circumstance, a *simplicial complex* K is simply a finite collection of triangles together with their edges and vertices, where any two of these triangles intersect at a common vertex, edge, or empty set. The *underlying space of* K, denoted by $|K|$, is the union of the closed triangles.) Unfortunately, though, if any of the vertices of $G(\boldsymbol{p})$ lie in the interior of a face of P, then $G(\boldsymbol{p})$ will not be infinitesimally rigid. For example, choose an infinitesimal flex $\boldsymbol{p}'$ where the only non-zero vector of $\boldsymbol{p}'$ corresponds to a vertex in the interior of a face of P, and is

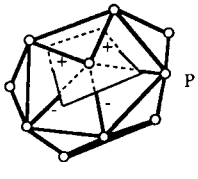


Figure 4.4.

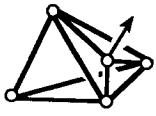


Figure 4.5.

perpendicular to the plane of that face. See fig. 4.5. Notice that in Dehn's Theorem 4.4, since each face is a triangle, there is a well-defined triangulation associated to the polytope P, assuming that there are no vertices in K and $G(\boldsymbol{p})$ beyond those in P.

The following theorem of Aleksandrov (1958) gives a good description of when convex polytopes can be used for infinitesimal rigidity.

Theorem 4.5. *Let K be a triangulation of a convex polytope P in three-space such that no vertex of K lies in the interior of a face of P. Then the bar framework $G(\boldsymbol{p})$, obtained from the vertices and edges of K, is infinitesimally rigid in three-space.*

For example, fig. 4.6 shows two triangulations of a cube that are infinitesimally rigid by Aleksandrov's Theorem. The framework on the left has only the original vertices of P as vertices of $G(\boldsymbol{p})$. The framework on the right has some extra vertices placed on the edges of P.

The proof of Aleksandrov's Theorem 4.5 is more delicate than one might first imagine. In this situation the argument that is outlined in section 4.2 has some difficulties in that the separating plane is not evident. There might possibly be exactly two changes in sign for the self stress, where all the non-zero stresses occur in one face of P. Nevertheless, it is still possible to prove infinitesimal rigidity in this situation using roughly similar ideas. See Asimow and Roth (1979) as well as Whiteley (1984) for careful proofs. There are also some extensions of Aleksandrov's Theorem to higher dimensions in Whiteley (1984). (See also section 4.8).

The techniques used in Gluck (1975) are not adequate to show this version of Aleksandrov's Theorem. Gluck defines a simplicial complex K, triangulating the convex polytope P, as *strictly convex* if for each vertex p_i of K there is a plane meeting P only at p_i. (So the triangulation on the left of fig. 4.6 is strictly convex, but the one on the right is not strictly convex.) Unfortunately in Gluck's proof of his Separating Lemma 5.3, his definition of strict convexity is not enough to insure that the edges with positive and negative stresses can be separated by a plane.

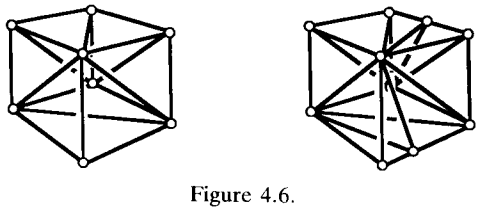


Figure 4.6.

(Grünbaum was the first to point out this problem with Gluck 1975.) However, when each face of P is a triangle, Gluck's description is more than adequate. This is also enough for Gluck's main result that $G(\boldsymbol{p})$ is rigid for an open dense set of configurations $\boldsymbol{p}$.

4.4. *Consequences of Aleksandrov's theory*

There are several corollaries/extensions of Aleksandrov's Theorem 4.5. Most of these results follow from the statement of Theorem 4.5 without too much extra work, or one can extend one's favorite proof of Theorem 4.5. The following are some examples. For more details see Connelly (1980), Whiteley (1984), Asimow and Roth (1979), and Roth (1987).

Corollary 4.6. *Let $G(\boldsymbol{p})$ be a (tensegrity) framework where its vertices include the vertices and lie on the edges of a convex polytope P in three-space; all members of $G(\boldsymbol{p})$ lie on the surface of P; and for each face F of P the framework, determined by the vertices and members of $G(\boldsymbol{p})$ that lie in F, is infinitesimally rigid in the plane of F. Then $G(\boldsymbol{p})$ is infinitesimally rigid in three-space.*

For example, fig. 4.7 shows some tensegrity frameworks in three-space that are infinitesimally rigid by this result. The planar tensegrity frameworks are infinitesimally rigid by the results in section 4.1. Only the near faces of the dodecahedron on the right are shown.

In these examples the bars can be replaced by struts and the frameworks will still be infinitesimally rigid. These examples were taken from Grünbaum and Shephard (1975).

The principle used for Corollary 4.6 is that the infinitesimal rigidity of whole structure does not depend on the particular framework that "holds" the vertices of each face infinitesimally rigid in its plane. This "subframework" provides the appropriate implied bars needed. (A bar is *implied* between two vertices i and j if adding the corresponding condition for $\boldsymbol{p}'$ for the bar $\{i, j\}$ does not change the space of infinitesimal flexes.) Aleksandrov's Theorem 4.5 provides one such framework that is infinitesimally rigid in three-space, and so we get Corollary 4.6. Other cell complexes with planar faces would work as well, as long as the analogue of Theorem 4.5 holds.

Another interesting application of Theorem 4.5 is to show that certain frameworks are *not* rigid. Suppose that $G(\boldsymbol{p})$ is a bar framework in $\mathbb{R}^d$ (with

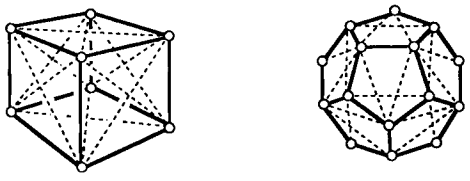


Figure 4.7.

$n \geq d+1$) which is infinitesimally rigid, but with just the minimum number of edges needed, namely $nd - \frac{1}{2}d(d+1)$. When $d=3$ this number is $3n-6$, of course. What happens when one of the bars is removed? Certainly the framework does not remain infinitesimally rigid, but we can say more. From the discussion in section 3.6, the rigidity map f for G into the space of edge lengths is locally onto. This is because the differential of f, the rigidity matrix $R(\boldsymbol{p})$ is onto since its rank $nd - \frac{1}{2}d(d+1)$ is equal to the number of edges of G. This means that any perturbation of the edge lengths that is small enough will correspond to a configuration with those given lengths. So if one edge length is shortened or lengthened continuously, there will be a configuration realizing those given lengths. It is easy to show (using the inverse function theorem) that the configuration can be chosen to vary continuously as well. Thus when one bar is removed from $G(\boldsymbol{p})$, the resulting framework is flexible. For example, when one bar is removed from either of the frameworks of fig. 4.6, it becomes flexible. Also ideas similar to those above show that if any one of the cables of the cube on the left in fig. 4.7 is removed, then the framework becomes flexible. These ideas essentially come from Aleksandrov (1958) and are used extensively in Asimow and Roth (1978). See also Whiteley (1988).

4.5. Kann's proof and extension of Dehn's Theorem

In the last several years there has been some interest in proving the infinitesimal version of Cauchy's Theorem, Dehn's Theorem. This has not been because of any lack of confidence in the result, but because there was interest in finding a generalization in some particular direction or because the result followed from some other techniques.

In Kann (1970, 1982) there is a proof of the infinitesimal rigidity of C^2 smooth convex surfaces with positive Gaussian curvature. This follows the results of Minagawa and Rado (1952), and Blaschke (1967).

In Kann (1976, 1990) this idea is applied to "polyhedral" surfaces in three-space. This is also reminiscent of the approach in Aleksandrov and Sen'kin (1956). We define a *polygonal region* as a compact polygonal disk in a plane. We say that K is a *polyhedral network* if it is a finite collection of polyhedral regions such that they have disjoint interiors, and any two such intersect either at a common vertex, a common edge, or in the empty set. Each polygonal region in K has its own affine linear structure, and any two of these structures agree on the overlap on their boundaries. We may think of K as something like a generalized simplical complex. The *underlying space* of K, denoted by $|K|$, is what you get by gluing together the polyhedral regions along their boundaries. Figure 4.8 shows such a polyhedral network when the underlying space is naturally identified with a disk in the plane.

Suppose for simplicity that the underlying space $|K|$ is a connected orientable two-dimensional surface with boundary, i.e., a sphere with handles and holes. Let g be a continuous function from the underlying space of K to three-space, where g is affine linear on each polygonal region. We will say that g represents a

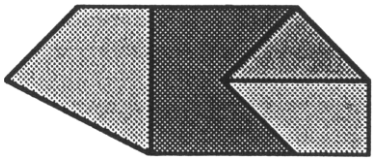


Figure 4.8.

polyhedral cap if:

(i) The affine map g is one-to-one when restricted to each polygonal region.

(ii) The directed normals to the singular surface given by g lie in a fixed hemisphere. In other words, for each polygonal region the outward pointing normal N_i has $N_i \cdot e > 0$, where e is a fixed vector in three-space.

(iii) The images under g of adjacent polygonal regions are *edge-convex*. That is, the interior one of the regions is contained in the open positive half-space determined by the normal to the plane of the other region.

Notice that condition (iii) is milder than the usual global form of convexity. We regard the image under g as a singular surface in that there can be a variety of branch points and self-intersections. Figure 4.9 shows some examples. The surface on the left is an example of singularity. The surface on the right is a spiral trough.

We can now state one form of Kann's Theorem that is a generalization of Dehn's Theorem about infinitesimal rigidity.

Theorem 4.7. *Let C be a polyhedral cap satisfying* (i)–(iii), *and let $\mathbf{p}'$ be an infinitesimal flex such that $\mathbf{p}'$ is trivial on each polygonal region and $p_i' \cdot e =$ constant for every vertex p_i on each boundary component of C separately. Then $\mathbf{p}'$ is a trivial infinitesimal flex.*

For example, if the boundary of C has just one component, then by adding a trivial flex to $\mathbf{p}'$ we may assume that $p_i' \cdot e = 0$ for each boundary component. This is called the *glidebending condition*.

Despite the special role that the boundary plays, Theorem 4.7 still does specialize to Dehn's result about the infinitesimal rigidity of triangulated polyhedra (Theorem 4.4). [Indeed, it is even possible to prove the more refined Aleksandrov version (Theorem 4.5) using these techniques.] To see this start with

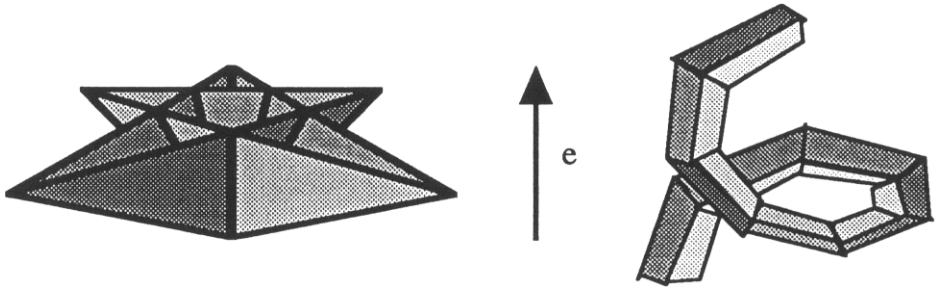


Figure 4.9.

one vertex of a triangulated convex polytope. As we mentioned in section 3.5, the infinitesimal rigidity of a bar framework is invariant under projective transformations, even when a point is sent to the "plane at infinity" under the transformation. It is always possible to find a projective transformation that sends exactly one vertex (and no other vertices or edges) of the polytope to the plane at infinity. Fix that point at infinity, so the infinitesimal flex is zero at that point. Remove the vertex now at infinity. The object that one obtains is now a polyhedral cap that satisfies the glidebending condition as well as the local convexity. Thus it is rigid by Theorem 4.7.

We sketch a proof of Theorem 4.7. Recall from section 3.3 that a trivial infinitesimal flex $\boldsymbol{p}'$ has the form, for each i, $p_i' = r \times p_i + u$, where r and u are fixed vectors in three-space. The vector r corresponds to an infinitesimal rotation and the vector u corresponds to an infinitesimal translation. Since a triangle composed of three bars is infinitesimally rigid, the infinitesimal flexes at its vertices are the restriction of a trivial infinitesimal flex. So in a triangulation of a surface as in Theorem 4.7, each triangle τ in any infinitesimal flex will have its own infinitesimal rotation vector r_τ. If $r_\tau \cdot e > 0$, then we say that the *spin* of the triangle is counterclockwise. If $r_\tau \cdot e < 0$, then we say that the *spin* of the triangle is clockwise. Add a trivial infinitesimal flex to the given infinitesimal flex so that $r_\tau = 0$ for some face τ. We do this in such a way that there remain faces where there are both kinds of spin clockwise and counterclockwise.

Divide the triangles (with a non-zero spin) of the polytope into two sets, those with counterclockwise spin and those with clockwise spin. If two adjacent triangles have opposite spins (or one spins one way and the other has no spin), then when their common edge is oriented in the direction compatible with both spins, the convexity condition implies that the infinitesimal change in the oriented edge (i, j) points "uphill", i.e., $(p_i' - p_j') \cdot e > 0$. The boundary of one of the sets of triangles has all of its edges oriented in a cycle, where the edges point uphill. This is not possible so there can be non-zero spin and the infinitesimal flex must be trivial.

4.6. More proofs and extensions of Dehn's Theorem

We mention briefly some other approaches to Dehn's Theorem.

The following is a sketch of an idea of Schramm that is basically in Schramm (1990). This also seems to be very close to the idea in Truskina (1987). Start with an infinitesimal flex $\boldsymbol{p}'$ of a convex polytope P with triangular faces. Orient some of the edges of P by the following rule. If p_i and p_j are the vertices of an edge, orient the edge from p_i to p_j if $p_i' \cdot (p_j - p_i) > 0$. This means that p_i is moving infinitesimally in the direction of p_j. By the condition of being an infinitesimal flex (3.1), $(p_i' - p_j') \cdot (p_i - p_j) = 0$, we get $p_i' \cdot (p_j - p_i) = -p_j' \cdot (p_i - p_j)$, so each edge has only one possible orientation. Since P is convex, the plane defined by $\{p \mid p_i' \cdot p = 0\}$ can separate the edges incident to p_i into at most two sets and can contain at most two edges unless $p_i' = 0$. Thus the orientation of the edges of P has the following properties:

(i) The edges oriented out of a vertex are connected as one proceeds cyclically around a vertex. Similarly the edges oriented into a vertex are connected.

(ii) Either 0, 1, 2, or all of the edges incident to a vertex are unoriented. (If all the edges incident to p_i are unoriented, then $p_i' = 0$.)

(iii) If two edges are unoriented, then either they separate both ends of the interval of edges that are oriented out of a vertex from the interval of edges that are oriented into a vertex, or they are next to each other and all the other edges are oriented out of or into the vertex.

Figure 4.10 shows an example of such an orientation of the edges in a neighborhood of a vertex. Fix a triangular face of P so that the infinitesimal flex is zero on all its vertices. Thus all the edges incident to those three vertices are unoriented.

Then a combinatorial lemma implies that there is no way of orienting the rest of the vertices consistent with the three properties above, other than leaving them all unoriented. This implies that the infinitesimal flex is zero on all the vertices, and thus the bar framework of the edges is infinitesimally rigid.

We mention some other approaches to Dehn's Theorem. Let $\{n_1, n_2, \ldots, n_k\}$ be a finite set of unit vectors in three-space. Let $a_1, a_2, \ldots, a_k$ be a corresponding set of positive real numbers such that

$$\sum_{1}^{k} a_i n_i = 0.$$

A theorem of Minkowskii states that there is a unique polytope (up to translation) for which n_i is the unit normal and a_i is the area of the ith face. See Stoker (1969, chapter X) for a discussion in the smooth category as well as Lyusternik (1963, chapter 5) for a discussion in the category of polytopes. The uniqueness parts of the proofs of these results are very similar to the proofs of Cauchy's original theorem.

Another recent proof of Dehn's Theorem is due to Filliman (1990). The idea is to translate the problem of the infinitesimal rigidity of the simplical polytope P to a corresponding statement about its polar dual P^*. Then Minkowskii's Theorem and the Brunn–Minkowskii inequality is applied to obtain the rigidity result. See Aleksandrov (1937) for a discussion of mixed volumes and the Brunn–Minkowskii inequality.

So far, most of the results have been concerned with objects that have been topologically spheres. This is natural since the idea is to use convexity to show

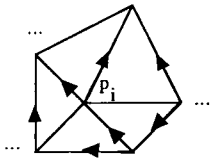


Figure 4.10.

rigidity. However, it is possible to have convexity available even for a surface other than a sphere. The only published results that I know along this line are in the smooth category. Let S and S' be two compact oriented smooth (C^3) immersed surfaces in a Riemannian three-manifold of constant sectional curvature. Assume that S and S' are both "locally convex" in the sense that the second fundamental forms of both surfaces are positive definite. Let e_3 and e_3' be unit normal vectors in S and S', respectively, and let e denote a continuous conformal vector field on an open set containing both S and S'. We assume that scalar products $e \cdot e_3 > 0$ and $e \cdot e_3' > 0$. This is analogous to the similar condition (ii) in section 4.5. With this set-up the following result of Hsuing and Liu (1977) is a generalization of Cohn-Vossen's Theorem that applies naturally to the surfaces S and S' of arbitrarily large genus.

Theorem 4.8. *Let $f: S \to S'$ be an isometry. Then the second fundamental forms of S and S' are equal under f.*

The second fundamental form is analogous to the dihedral angle in the polyhedral category.

A suggestion was made by Gromov that there should be a result similar to Theorem 4.8, but for infinitesimal rigidity in the polyhedral category. For such triangulated two-manifolds, in Gromov (1986) there is a proof of certain combinatorial inequalities that would follow from the infinitesimal rigidity of such triangulated two-manifolds. More will be said about this in section 4.7.

In Fogelsanger (1988) there is a proof that any compact connected triangulated two-manifold is infinitesimally rigid in three-space if the configuration of vertices is generic; that is, there is no polynomial relation with integer coefficients among the coordinates of the vertices. This result, however, makes no use of convexity.

See Tay and Whiteley (1985) for an introduction to generic rigidity in general.

4.7. Maxwell–Cremona theory and spider webs

So far most of the results that have been discussed have been related to Cauchy's Theorem. There is a very simple rigidity result which on the face of it does not seem to be related to convexity or Cauchy's result, but nevertheless it does have a connection to convex polytopes in three-space.

Suppose that one has a tensegrity framework that is composed entirely of cables attached to some points fixed in the plane. In Connelly (1982), such a tensegrity framework is called a *spider web* in honor of the creature that so frequently makes them. When is such a spider web rigid? For example, fig. 4.11 shows two spider webs where the one on the left is rigid, but the one on the right is not rigid. The black vertices are the fixed vertices. Spider webs with the graph as in fig. 4.11 are rigid only if the lines through the three outside cables intersect in one point. In general, a spider web is rigid if there is a proper self stress that is positive on all the cables. [The equilibrium condition (3.2) with $F_i = 0$ holds only for the vertices that are not fixed.] In fact, when the self stress condition does hold, then the

Figure 4.11.

spider web is globally rigid. This result is in Connelly (1982) and is quite easy.

The question remains: when is there a positive self stress? One answer, of course, is just when there is an appropriate solution to the equilibrium equations (3.2), with the constraint that each stress is positive.

However, there is a more geometric answer. Suppose that the fixed vertices form a convex polygon F_0 in the plane, and the cables do not cross, as in fig. 4.11. Suppose further that there is a convex polytope P in three-space that has F_0 as a face, and orthogonal projection into the plane of F_0 takes the vertices of P onto the vertices of the spider web in a one-to-one manner. Then a theorem of Maxwell (1864) and Cremona (1872) states that there is positive self stress in the spider web. Figure 4.12 shows how this is done for the spider web on the left in fig. 4.11. The polytope P is a truncated tetrahedron. This result has been reformulated, expanded and extended in Whiteley (1982), and Crapo and Whiteley (1988). In fact, the convexity of P is not relevant to the correspondence itself, which can be achieved by a purely projective construction. It is true though that since P is convex, then in this set-up the corresponding stresses will be positive.

Indeed, perhaps the proper way to think of this sort of construction is to instead consider an infinitesimal flex $\boldsymbol{p}'$ of a closed orientable polyhedral surface P_0. Each face of the surface is say a planar polygon, but the surface itself is allowed to be quite singular. Then there is a canonical self stress ω defined on the edges of P_0. Let θ'_{ij} be the infinitesimal rate of change of the dihedral angle corresponding to the edge between the vertices p_i and p_j of P_0. Then

$$\omega_{ij} = \frac{\theta'_{ij}}{|p_i - p_j|}$$

serves as the self stress defined by $\boldsymbol{p}'$. One must be careful to be sure that the infinitesimal flex $\boldsymbol{p}'$ preserves the faces of P_0 (as if they were rigid plates). This point of view is presented in Gluck (1975). A purely algebraic/projective-geometric description of ω is described in Whiteley (1982), Crapo and Whiteley (1988) and Maxwell (1869–72).

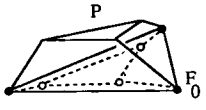


Figure 4.12.

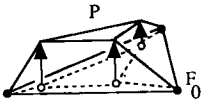


Figure 4.13.

In our setting for spider webs, the polyhedral surface P_0 is the orthogonal projection of P, contained entirely in F_0. The infinitesimal flex of a vertex p_i in P_0 is simply the vector from p_i to the corresponding vertex of P. See fig. 4.13.

One can also reverse the correspondence $\boldsymbol{p}' \to \omega$ above if the surface P_0 is simply connected. In other words, if P_0 is topologically a sphere and ω is a self stress of the edge framework of P_0, then there is an infinitesimal flex $\boldsymbol{p}'$ of P_0 which is non-trivial if ω is non-zero. This is described in Whiteley (1982), and Crapo and Whiteley (1988).

See also Hopcroft and Kahn (1991) for an explicit description of the self stress described above as well as an application of the rigidity of spider webs to convex polytopes. In computational geometry it often occurs that a polytope P is defined, where some small unwanted edges appear. One wants to change P to another "nearby" polytope that does not have those small edges. This can be accomplished by using the correspondences above and deforming the spider web. Simply increase the stress on the corresponding unwanted edges until their end points are close enough to be considered the same. Given the fixed vertices, there will be a unique spider web with any given positive stress, and in this situation the deformed spider web will be close to the original. It turns out that such a deformation of P quite often must necessarily deform almost all of the vertices. This is why some sort of global procedure as above is necessary.

In Ash et al. (1988) there is a survey of various aspects of the Maxwell–Cremona correspondence and its relation to things such as Dirichlet tessellations. (A *Dirichlet tessellation* is the collection of sets corresponding to a collection of points in the plane, where each set is the set of nearest points to one of the given points.) A finite part of the vertices and edges of a Dirichlet tessellation is a spider web. Section 5.3 discusses some of the ideas inherent to this technique.

See Sugihara (1986) for an application of the Maxwell–Cremona correspondence to scene analysis and combinatorics.

4.8. Rigidity, convexity, and combinatorics in higher dimensions

Suppose that P is a convex polytope in $\mathbb{R}^d$, where each face is a simplex. Let f_i, $i = 1, 2, 3, \ldots$, be the number of simplicial faces of P of dimension i. The Lower Bound Theorem of Barnette (1973) (see also Walkup 1970) says, among other things, that

$$f_k \geqslant \begin{cases} \binom{d}{k} f_0 - \binom{d+1}{k+1} k & \text{for } 1 \leqslant k \leqslant d-2\,, \\ (d-1)f_0 - (d+1)(d+2) & \text{for } k = d-1\,. \end{cases}$$

When $d \geq 3$, a result due to McMullen, Perles, and Walkup (see Kalai 1987, McMullen and Walkup 1971), reduces this inequality to the special case:

$$f_1 \geq f_0 d - \tfrac{1}{2} d(d+1) .$$

The number of vertices of P is f_0, and the number of edges of P is f_1. But this is the same as the inequality (3.3) that is necessary for the infinitesimal rigidity of the 1-skeleton (the vertices and edges) of P in $\mathbb{R}_d$. This observation was exploited and extended in Kalai (1983, 1987) to many other similar situations.

However, Dehn's Theorem 4.4 on the infinitesimal rigidity of convex polytopes with triangular faces is stated only when the dimension $d = 3$. Fortunately, there is a general principle that allows one to "bootstrap" rigidity results in lower dimensions up to results in higher dimensions. For example, suppose that P is a convex polytope in $\mathbb{R}^4$, where all its faces are simplices. Let p_i be any vertex of P. Then those simplices that are adjacent to p_i form a simplical complex (called the *star* of p_i in P) which may be regarded as the cone over another complex, called the *link* of p_i in P. Let S_0 be a three-dimensional support hyperplane that intersects P at p_i only. Let S be another hyperplane that is parallel to S_0 that does not contain p_i. Project the link of p_i into S from p_i. It is easy to see that this projection is a convex simplicial three-polytope in S. Thus by Dehn's Theorem 4.4 this projection is infinitesimally rigid in S. See fig. 4.14. (So that we may see things more clearly, the dimension of all the sets in the figure are decreased by one. In fig. 4.14 the link is a polygonal circle in the plane, and this is not infinitesimally rigid.)

By our discussion in section 3.5 about the projective invariance of infinitesimal rigidity and from Theorem 3.4, we see that the cone on the projection and the cone on the link (i.e., the star of p_i in P) are both infinitesimally rigid in $\mathbb{R}^4$. But the stars of all the vertices of P overlap in such a way that the whole framework can be built by gluing infinitesimally rigid pieces together along intersections such that the union is infinitesimally rigid.

Clearly this argument extends to higher dimensions, and, by induction on the dimension d, we can show that all simplical polytopes for $d \geq 3$ are infinitesimally rigid in $\mathbb{R}^d$. Thus we have a proof of the lower bound inequalities of Barnette.

A slightly more ambitious idea is to extend an appropriate version of Aleksandrov's Theorem 4.5 to higher dimensions. The following is a result in Whiteley (1984).

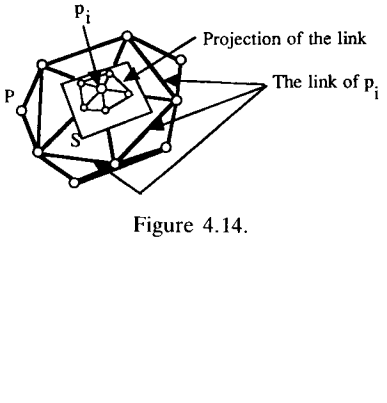


Figure 4.14.

Theorem 4.9. *Suppose that P is a convex polytope in $\mathbb{R}^d$, $d \geq 3$, and $G(\boldsymbol{p})$ is a bar framework consisting of the vertices and edges of P, together with additional bars that triangulate all the two-dimensional faces of P. Then $G(\boldsymbol{p})$ is infinitesimally rigid in $\mathbb{R}^d$.*

This result follows essentially from the bootstrap idea and the property that one can retriangulate the two-dimensional faces of P without altering its infinitesimal rigidity. As a consequence we get the following inequality:

$$f_1 + \sum_{k \geq 3} (k-3) f_2^k \geq f_0 d - \tfrac{1}{2} d(d+1) ,$$

where f_2^k is the number of two-dimensional faces of P that have k sides. This is again from Kalai (1987). Previously this inequality had been only known for polytopes that had a realization with all the coordinates as rational numbers.

In Filliman (1991) there is an attempt of a proof of the conditions of McMullen (1973) (generalizing the Lower Bound Theorem of Barnette) on the face numbers f_k of a simplicial polytope. The idea is to replace a previous difficult proof with the calculation of the rank of a matrix that generalizes the rigidity matrix of section 3.6. See Billera and Lee (1981) for a discussion. This follows a suggestion of Lee, which has generated some interest in the various proofs of Dehn's Theorem.

Even though the proof of Dehn's Theorem 4.4 is simple enough, it seems that there should be a way of proving the combinatorial inequalities of Barnette following the general outline as suggested by Kalai, but without appealing to the geometric realization that is needed for infinitesimal rigidity. In fact, this is the case, and this is done in Gromov (1986, chapter 2.4.10). (Unfortunately, some of the proofs are slightly garbled.)

Another way of thinking of the proof of the higher dimensional Dehn's Theorem is to imagine a small $(d-1)$-dimensional sphere S^{d-1} centered at p_i. The intersection of S^{d-1} with the star of p_i in P will be a triangulated spherical polytope of dimension one less than P. If we have an appropriate theorem about the infinitesimal rigidity of such objects, then we will be able to conclude that the star of p_i in P is infinitesimally rigid in $\mathbb{R}^d$. Theorem 3.4 can be interpreted as a way of transforming a rigidity result about polytopes in $\mathbb{R}^{d-1}$ to the corresponding result about polytopes in S^{d-1}.

Indeed, this correspondence has been exploited in Pogorelov (1964) to show that there is in fact a way of transforming rigidity results about pairs of objects in $\mathbb{R}^d$ to rigidity results about pairs of objects in S^d. One can then bootstrap lower dimensional rigidity results about pairs to higher dimensional results about pairs. For example, Cauchy's original result about convex polytopes (mentioned in the introduction) is about pairs of convex polytopes, and this bootstrap method allows one to generalize it to any dimension greater than three:

Theorem 4.10. *Any pair of convex polytopes in $\mathbb{R}^d$ or S^d for $d \geq 3$ that have a correspondence which is a congruence on each facet of one lower dimension, extends to congruence of the whole space.*

(Of course, a set in S^d is convex if, for every pair of points in the set, the shortest geodesic arc joining them is in the set.)

One can view our statements about infinitesimal rigidity as a statement about pairs of configurations, where both pairs are "almost the same" but differ by an "infinitesimal". With that interpretation Pogorelov's correspondence can be regarded as specializing to give the statements about infinitesimal rigidity. [In Connelly (1988a) it is shown how to deduce the statement about some pairs from the statement about infinitesimal flexes (Theorem 3.4) as well.]

5. Second-order rigidity and pre-stress stability

So far we have been only concerned with the linear or first-order theory of rigid mostly convex objects. There are many other convex objects that are still rigid, but are not first-order rigid.

5.1. The definition of second-order rigidity

Most of the definitions will be taken from Connelly (1980), and Connelly and Whiteley (1990). There is also a discussion of second-order rigidity in Rosenberg (1980). Suppose that we have a (tensegrity) framework $G(\boldsymbol{p})$ in $\mathbb{R}^d$. Let $\boldsymbol{p}'$ be an infinitesimal flex of $G(\boldsymbol{p})$, which we will also call a *first-order flex* of $G(\boldsymbol{p})$ satisfying (3.1). Let $\boldsymbol{p}'' = (p_1'', p_2'', \ldots, p_n'')$ be another configuration of vectors in $\mathbb{R}^d$ such that for each member $\{i, j\}$ of G where $(p_i - p_j)\cdot(p_i' - p_j') = 0$,

$$|p_i' - p_j'|^2 + (p_i - p_j)\cdot(p_i'' - p_j'') \begin{cases} \leqslant 0 & \text{if } \{i, j\} \text{ is a cable,} \\ = 0 & \text{if } \{i, j\} \text{ is a bar,} \\ \geqslant 0 & \text{if } \{i, j\} \text{ is a strut.} \end{cases} \tag{5.1}$$

There is no condition for the member $\{i, j\}$ when $(p_i - p_j)\cdot(p_i' - p_j') \neq 0$. [Of course, for a bar framework both (5.1) and (3.1) must hold.] We then say that the pair $(\boldsymbol{p}', \boldsymbol{p}'')$ is a *second-order flex* of $\boldsymbol{p}'$ and extends the first-order flex $\boldsymbol{p}'$. We can think of $\boldsymbol{p}'$ as the velocities of the points of the configuration, and $\boldsymbol{p}''$ as the accelerations of the points of the configuration, since (3.1) and (5.1) are obtained by differentiating the distance constraints.

We say that the tensegrity framework $G(\boldsymbol{p})$ is *second-order rigid* in $\mathbb{R}^d$ if for every non-trivial first-order flex $\boldsymbol{p}'$ of $G(\boldsymbol{p})$, there is no extension to a second-order flex $(\boldsymbol{p}', \boldsymbol{p}'')$. It is important for this definition that the first-order flex be non-trivial, since any trivial first-order flex extends to some second-order flex. For example, if $\boldsymbol{p}' = 0$, then the equations (5.1) become identical to the first-order equations (3.1) except for $\boldsymbol{p}''$ replacing $\boldsymbol{p}'$.

The basic result for second-order rigidity is the following theorem.

Theorem 5.1. *If a tensegrity framework $G(\boldsymbol{p})$ is second-order rigid in $\mathbb{R}^d$, then it is rigid in $\mathbb{R}^d$.*

This was proved for bar frameworks in Connelly (1980) and extended to tensegrity frameworks in Connelly and Whiteley (1990). The proof involves an analysis of the Taylor series expansion of any non-trivial analytic flex $\boldsymbol{p}(t)$ of $G(\boldsymbol{p})$.

Consider the left framework of fig. 3.2, which is first-order flexible and has a first-order flex indicated. This first-order flex together with the trivial first-order flexes generate all of the first-order flexes of $G(\boldsymbol{p})$. But none of the non-trivial first-order flexes extend to a second-order flex. For example, for the flex indicated in fig. 3.2, the middle vertex must have a second-order flex, an acceleration vector, that has a non-zero component in both the positive and negative y directions. This is not possible, so the second-order flex does not exist, and the framework is rigid (which is easy to see anyway for this example).

5.2. Second-order rigidity and convex surfaces

Let P be a compact convex polytope in three-space. For Aleksandrov's Theorem 4.5 it is important that no vertex be in the interior of any two-dimensional face of P. But if some of the vertices *are* chosen in the relative interior of some of the faces of P, is the resulting framework still rigid, even though it is not infinitesimally rigid? Figure 5.1 shows some examples of triangulated polytopes with vertices in the interior of some of the faces. The lighter weight edges are bars in the interior of the faces of the polytope.

It is easy to see from Aleksandrov's Theorem that any such first-order flex on a framework as in fig. 5.1 is trivial when restricted to the vertices and edges of the polytope P. So if $\boldsymbol{p}'$ is any first-order flex of the whole framework $G(\boldsymbol{p})$, by adding a trivial infinitesimal flex to $\boldsymbol{p}'$, we may assume that each p_i', in addition to being zero for all the vertices that do not lie in the interior of a face of P, is perpendicular to each face on which it lies.

If $(\boldsymbol{p}', \boldsymbol{p}'')$ is a second-order flex that is an extension of $\boldsymbol{p}'$, then it turns out that $\boldsymbol{p}''$, when restricted to the vertices of P, acts as if it were a first-order flex of a framework as in Aleksandrov's Theorem. This is a tensegrity framework that has just the vertices of P, cables between all of the vertices on the same face and bars along the edges of P. (The frameworks of fig. 4.7 are examples.) So $(\boldsymbol{p}', \boldsymbol{p}'')$ acts as if the vertices of P were fixed. Then it is easy to see that the rest of the vertices in the triangulation must have $p_i' = 0$ for the vertices in the interior of faces of P, for much the same reason as for the middle vertex of fig. 3.2 as was discussed in

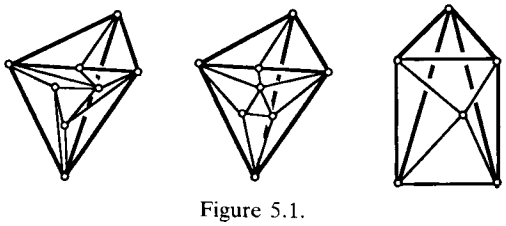


Figure 5.1.

section 5.1. This shows that $\boldsymbol{p}'$ is trivial, and $G(\boldsymbol{p})$ is second-order rigid. Thus we have the following theorem, which is in Connelly (1980).

Theorem 5.2. *Any triangulation of a convex polytope in three-space is second-order rigid.*

In fact, the same idea can be extended to show that if a convex hole is removed from the polyhedral surface, and the resulting surface with boundary is triangulated, then it is still second-order rigid. It might look something like fig. 5.2.

In the smooth category similar objects have been studied, but the analogue of the fundamental result (Theorem 5.1) is not known (except for analytic surfaces). There is a discussion of second-order rigidity in the smooth category in Efimov (1962), which is translated from the original paper Efimov (1948). See also Efimov (1957, 1962) for a general discussion of the rigidity of smooth surfaces.

In Connelly (1980, fig. 5) is an example of a bar framework in the plane that is rigid but not second-order rigid.

5.3. Pre-stress stability

Although second-order rigidity is natural mathematically, physically it may not be the concept that describes the stability of a structure most accurately. From a physical point of view, it is not enough just to declare that the distance constraints force the structure to have only one configuration locally up to rigid motions. The members of a physical framework do not behave as ideal bars or cables. There must be some way of describing the behavior of the system as it is perturbed.

A common method used to describe such behavior is to introduce energy functions. For example, if a spring at rest is displaced, then Hooke's law describes how much work it takes to displace the spring any given amount. This can be thought of as changing the potential energy that is stored in the spring. Idealized springs will be our way of modeling a bar in a framework. Mathematically the idea is that when the end points of a bar are displaced, there is a well-defined (energy) function of the lengths of the bars (or the squares of the lengths) such that each function has a strict minimum at the rest length of the bar. For a cable, its energy function is strictly monotone increasing, and for a strut its energy function is strictly monotone decreasing.

In structural mechanics a basic principle used for the stability of structures is that the configuration must be at a local minimum for the total energy functional.

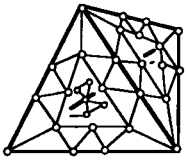


Figure 5.2.

This is often associated with "Castigliano's Principle". See Przemieniecki (1968). See especially Kötter (1912) for a method similar to the one described next. See also Kuznetsov (1991) as well as Callidine and Pellegrino (1991) for a treatment from an engineering perspective. Mathematically this principle is sound, since if the total energy function is at a unique local minimum, modulo rigid motions, then it must be at least rigid. This is because if there were another nearby non-congruent configuration that satisfied the member constraints, then each of the energy functions of the members would be no larger, and the original configuration would not be at a unique minimum. Physically, the idea is that if the structure is perturbed to a nearby configuration, then the gradient of the energy function provides the force necessary to push the configuration back to the unique minimum point, or at least keep the structure from wandering far from the desired minimum point.

The following is taken mostly from Connelly and Whiteley (1990, 1991). Suppose that $G(\boldsymbol{p})$ is a tensegrity framework in $\mathbb{R}^d$, and let H_{ij} be an energy function for the member $\{i, j\}$ as described above. The total energy is

$$H(\boldsymbol{q}) = \sum_{i,j} H_{ij}(|q_i - q_j|^2) ,$$

where $\boldsymbol{q}$ is any configuration in $\mathbb{R}^d$. We have chosen to write H as a function of the square of the lengths to make some expressions simpler that appear later.

The conditions for a configuration $\boldsymbol{p}$ to be a critical point for H turn out to be that $G(\boldsymbol{p})$ has a proper self stress ω, where $\omega_{ij} = H'_{ij}(|p_i - p_j|^2)$. Conditions (3.2) must hold with $F_i = 0$. The functional H'_{ij} is the first derivative of H_{ij}.

We also need to find conditions when H has a minimum, or at least a local minimum at a given configuration $\boldsymbol{p}$. We simply apply the second derivative test from multivariable calculus. The Hessian of H turns out to be the following nd-by-nd symmetric matrix (up to a scalar multiple):

$$\Omega \otimes I^d + R(\boldsymbol{p})^{\mathrm{T}} D R(\boldsymbol{p}) , \tag{5.2}$$

where $R(\boldsymbol{p})$ is the e-by-dn rigidity matrix as defined in section 3.6, D is the e-by-e diagonal matrix such that the second derivative $H''_{ij}(|p_i - p_j|^2)$ is the ijth diagonal entry of D, and Ω is the n-by-n *stress matrix* obtained by defining its ijth entry as

$$\Omega_{ij} = \begin{cases} -\omega_{ij} & \text{if } i \neq j , \\ \sum_k \omega_{ik} & \text{if } i = j . \end{cases} \tag{5.3}$$

Recall that $\omega_{ij} = 0$, for $\{i, j\}$ non-members of the graph G. The tensor product of Ω with the d-by-d identity matrix is denoted by $\Omega \otimes I^d$, which is obtained by replacing each 1 entry of I^d with an n-by-n block Ω. The transpose of a matrix is indicated by $(\cdot)^{\mathrm{T}}$, n is the number of vertices of G, and e is the number of members of G.

The matrix $R(\boldsymbol{p})^{\mathrm{T}} D R(\boldsymbol{p})$ is called the *stiffness matrix*, and it is always positive

semi-definite, since we insist that the second derivative $H''_{ij} > 0$ for all members $\{i, j\}$.

The second derivative test says that H has a local minimum at the configuration $\boldsymbol{p}$, if the matrix (5.2) is positive semi-definite (regarding it as quadratic form). If, in addition, the kernel of (5.2) contains only the trivial infinitesimal flexes, we say that $G(\boldsymbol{p})$ is *pre-stress stable* (*with ω as a rigidifying pre stress*). Clearly if $G(\boldsymbol{p})$ is pre-stress stable, then it is rigid.

A basic result of Connelly and Whiteley (1990) is the following theorem.

Theorem 5.3. *If a tensegrity framework $G(\boldsymbol{p})$ is infinitesimally rigid, then it is pre-stress stable. If $G(\boldsymbol{p})$ is pre-stress stable, then it is second-order rigid. Neither of these implications is reversible.*

For example, many spider webs are not first-order rigid but when they have a proper self stress that is positive on all the cables, they are pre-stress stable, essentially because the stress matrix is positive semi-definite of maximal rank. (This imagines some appropriate globally rigid framework, behind the scenes, holding the pinned vertices fixed.) Also all three of the frameworks in fig. 5.1 are not first-order rigid, but they are pre-stress stable, which is a bit harder to show. Many of the tensegrity frameworks championed by R. Buckminster Fuller are pre-stress stable. Figure 5.3 shows some of these tensegrities. See Fuller (1975). The framework on the right was one of the first discovered by Kenneth Snelson (see Snelson 1973). The bars in these tensegrity frameworks can be replaced by struts and they will still be pre-stress stable. All of the tensegrity frameworks that are rigid in Grünbaum and Shephard (1975) are surely rigid because they are pre-stress stable. Similarly most of the large collection of tensegrity frameworks in Pugh (1976) are not infinitesimally rigid and must be pre-stress stable. In Connelly and Terrell (1991) an explicit computation with the stress matrix is performed for a certain collection of tensegrity frameworks with dihedral symmetry. Those with a positive definite stress matrix are identified, and it is easy to check that they are at least pre-stress stable.

When a framework is pre-stress stable with ω as the rigidifying pre-stress, then those bars $\{i, j\}$ that have $\omega_{ij} \neq 0$ can be replaced by a cable or strut. For example, the framework on the left in fig. 5.4 is a triangulation of a tetrahedron and is pre-stress stable. The framework on the right replaces some of the bars by cables and struts. Both are pre-stress stable. If any member is removed or any bar is replaced by a cable or strut in the framework on the right, then it will not be rigid.

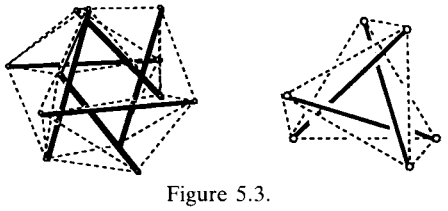


Figure 5.3.

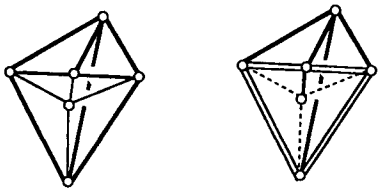


Figure 5.4.

5.4. Global rigidity and the stress matrix

It turns out that there is another geometric property associated to the stress matrix Ω that was defined in section 5.3. It is very useful for guaranteeing that a tensegrity framework is globally rigid. This analysis however does not deal directly with the rigidity matrix or the stiffness matrix.

It is helpful instead to think of the stress ω, and its associated stress matrix Ω, first, and then look for a configuration $\boldsymbol{p}$ with ω as its self stress. (The graph G is superfluous.) It is easy to check that if the configuration $\boldsymbol{q}$ is an affine image of the configuration $\boldsymbol{p}$, then any self stress for $\boldsymbol{p}$ will be a self stress for $\boldsymbol{q}$. Extending the ideas discussed in section 5.2, the following result is shown in Connelly (1982). (Recall from section 3.2, that if one tensegrity framework dominates another, it means the cables become no longer, struts become no shorter, and bars are the same length.)

Theorem 5.4. *Suppose $G(\boldsymbol{p})$ is a tensegrity framework in $\mathbb{R}^d$ with n vertices not all in a $(d\text{-}1)$-dimensional affine subspace, a self stress ω, and associated stress matrix Ω. If Ω is positive semi-definite of rank $n\text{-}d\text{-}1$, and if $G(\boldsymbol{q})$ is dominated by $G(\boldsymbol{p})$, then $\boldsymbol{q}$ is an affine image of $\boldsymbol{p}$.*

In other words, if Ω is positive semi-definite of maximal rank [it turns out that if the affine span is all of $\mathbb{R}^d$, then the kernel must be at least $(d+1)$-dimensional], then the framework is globally rigid up to affine motions.

When is there an affine image $\boldsymbol{q}$ of a configuration $\boldsymbol{p}$ such that $G(\boldsymbol{p})$ dominates $G(\boldsymbol{q})$? Since translations are rigid motions we need only consider when the affine motion is given by some linear map given by a d-by-d matrix A in some coordinate system. If $\{i, j\}$ is a member of the graph G, then

$$
\begin{aligned}
|q_i - q_j|^2 = |Ap_i - Ap_j|^2 &= (Ap_i - Ap_j)^{\mathrm{T}}(Ap_i - Ap_j) \\
&= (p_i - p_j)^{\mathrm{T}}A^{\mathrm{T}}A(p_i - p_j) \\
&\begin{cases}
\leqslant (p_i - p_j)^{\mathrm{T}}(p_i - p_j) & \text{if } \{i, j\} \text{ is a cable,} \\
= (p_i - p_j)^{\mathrm{T}}(p_i - p_j) & \text{if } \{i, j\} \text{ is a bar,} \\
\geqslant (p_i - p_j)^{\mathrm{T}}(p_i - p_j) & \text{if } \{i, j\} \text{ is a strut.}
\end{cases}
\end{aligned}
$$

This is the same as the condition

$$(p_i - p_j)^T[A^TA - I^d](p_i - p_j)\begin{cases} \leq 0 & \text{if } \{i, j\} \text{ is a cable,} \\ = 0 & \text{if } \{i, j\} \text{ is a bar,} \\ \geq 0 & \text{if } \{i, j\} \text{ is a strut.} \end{cases} \tag{5.4}$$

We can regard the symmetric matrix $A^TA - I^d$ as a quadratic form, and condition (5.4) says that all the *member directions*, defined as all scalar multiples of $(p_i - p_j)$ with $\{i, j\}$ in G, lie in the appropriate region defined by the *quadric at infinity* [regarding the coordinates in $\mathbb{R}^d$ as homogeneous coordinates for the $(d-1)$-dimensional real projective space "at infinity"]. For example, the bar directions must lie on the quadric at infinity, where the quadric is given by the symmetric matrix $A^TA - I^d$. When the dimension $d = 2$, then the quadric at infinity is simply two directions. When $d = 3$, then the projective space at infinity is the usual real projective plane, and the quadric at infinity consists of the directions that lie on any given conic, possibly degenerating into two lines.

We could leave things at this state, but there is a further simplification that is very useful. Suppose that $G(\boldsymbol{p})$ is a tensegrity framework in $\mathbb{R}^d$ with a proper self stress ω. We say that a member direction $p_i - p_j$ with $\omega_{ij} \neq 0$ is a *stressed direction*.

Theorem 5.5. *There is another non-congruent configuration $\boldsymbol{q}$, an affine image of $\boldsymbol{p}$, such that $G(\boldsymbol{q})$ is dominated by $G(\boldsymbol{p})$, if and only if for $G(\boldsymbol{p})$ the stressed directions lie on a non-zero quadric Q at infinity, as well at the condition* (5.4), *with Q replacing $A^TA - I^d$ for the non-stressed directions.*

By replacing the symmetric matrix Q by εQ for $\varepsilon > 0$ small enough, it is possible to solve $Q = A^TA - I^d$ for the matrix A. See Whiteley (1987b,c).

Suppose that ω is a proper self stress of a tensegrity framework $G(\boldsymbol{p})$ in $\mathbb{R}^d$ such that the associated stress matrix Ω is positive semi-definite of maximal rank $n - d - 1$, and no non-congruent affine image $\boldsymbol{q}$ of $\boldsymbol{p}$ is such that $G(\boldsymbol{q})$ is dominated by $G(\boldsymbol{p})$. Then we say that $G(\boldsymbol{p})$ is *ω globally rigid*. For example, all spider webs are ω globally rigid, where each scalar component of ω is positive. (But some vertices are fixed by some appropriate globally rigid tensegrity framework in the background with its own self stress and corresponding positive semi-definite stress matrix of maximal rank. The two stresses then can be combined to give a positive semi-definite stress matrix of maximal rank on all of the vertices.)

We apply the stress matrix–stiffness matrix decomposition of (5.2) to convex polygons. This follows the proof in Connelly (1982) of Theorem 4.2 about the global rigidity of convex polygons when the internal members are struts. Theorem 5.6 is the key fact needed.

Theorem 5.6. *Let $G(\boldsymbol{p})$ be a tensegrity framework in the plane, where the vertices form a convex polygon, all the external edges are cables, struts are the only other members, and $G(\boldsymbol{p})$ has a proper non-zero self stress ω. Then the stress matrix Ω is positive semi-definite of rank $n - 3$. Thus $G(\boldsymbol{p})$ is ω globally rigid.*

It is clear that there are more than two stressed directions for such polygons, and thus Theorem 5.6 together with (5.4) imply that such polygons are globally rigid in any dimension greater than one. From Connelly (1982) it is shown that if such a framework (other than a triangle) is rigid at all, then it must have a non-zero proper self stress. Thus any such rigid polygonal framework with cables on the outside edges and struts on the inside must be globally rigid even in higher dimensional Euclidean spaces.

Many, but not all, of the tensegrity frameworks in Pugh (1976) are also ω globally rigid. Thus they are also globally rigid in higher dimensional Euclidean spaces.

5.5. Second-order duality

The following equivalence is useful for understanding the relation among global rigidity, second-order rigidity, and pre-stress stability. This is the second-order stress test in Connelly and Whiteley (1990). It follows from an application of the "Farkas Alternative" as used in linear programming duality.

Theorem 5.7 (Second-order stress test). *A tensegrity framework $G(\boldsymbol{p})$ in $\mathbb{R}^d$ is second-order rigid if and only if for every non-trivial first-order flex $\boldsymbol{p}'$ there exists a proper self stress ω of $G(\boldsymbol{p})$ such that*

$$(\boldsymbol{p}')^{\mathrm{T}}\Omega \otimes I^d \boldsymbol{p}' > 0\,,$$

where Ω is the stress matrix associated to ω.

It is not too hard to see that the condition for being pre-stress stable is the same as the condition as in Theorem 5.7 except that a *single* proper self stress must satisfy the condition. There are examples in Connelly and Whiteley (1990) of bar frameworks that are second-order rigid but not pre-stress stable. Another such example is the complete bipartite graph $K_{3,3}$ on a line in the plane as in fig. 5.5. All six points of $K_{3,3}$ lie on a line with one partition of three points on the negative part of the line, and the other partition on the positive part. Each point of one partition is joined to each other point of the other partition. This bar framework is second-order rigid in the plane, and is a mechanism (i.e., a flexible framework) in three-space. In fig. 5.5 the bars are curved so as to be more visible.

In such examples it is as if there is a demon in the framework that somehow senses what first-order flex (or load) will be encountered and then provides an

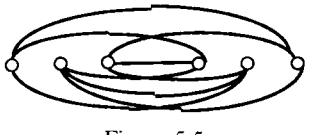


Figure 5.5.

appropriate self stress to "block" that flex. Different flexes require different self stresses. Because of this, it seems to me much more reasonable to regard a framework as truly "stable" if it is pre-stress stable, than if it is just second-order rigid.

An application of the second-order stress test (Theorem 5.7) is to show Roth's Statement 4.3. Suppose that $G(\boldsymbol{p})$ is a tensegrity polygonal framework in the plane with bars on the external edges and cables on the inside. If $G(\boldsymbol{p})$ is not infinitesimally rigid, let $\boldsymbol{p}'$ be a non-trivial first-order flex of $G(\boldsymbol{p})$. Let ω be any proper self stress for $G(\boldsymbol{p})$. The result in Theorem 5.6 says that if Ω is the associated stress matrix, then $-\Omega$ is positive semi-definite of rank $n-3$. So Ω is negative definite of rank $n-3$. By an argument similar to the one sketched in section 5.4, since there are no non-trivial infinitesimal flexes that are affine images of $\boldsymbol{p}$, the infinitesimal flex $\boldsymbol{p}'$ is not in the kernel of $\Omega \otimes I^2$. Thus

$$(\boldsymbol{p}')^{\mathrm{T}} \Omega \otimes I^2 \boldsymbol{p}' < 0 ,$$

and $\boldsymbol{p}'$ must extend to a second-order flex $(\boldsymbol{p}', \boldsymbol{p}'')$ of $G(\boldsymbol{p})$ by the second-order stress test (Theorem 5.7). An extension of the second-order stress test implies that since the inequality above is strict, any smooth flex $\boldsymbol{p}(t)$ of the points of $\boldsymbol{p}$, with the first and second derivatives at $t=0$ the same as $\boldsymbol{p}'$ and $\boldsymbol{p}''$, respectively, will have all of the cable distances decrease. It is possible to make sure that, in fact, $\boldsymbol{p}(t)$ does not change the length of the bars of $G(\boldsymbol{p})$ as well. Then $\boldsymbol{p}(t)$ is a non-trivial flex of all of $G(\boldsymbol{p})$. If $G(\boldsymbol{p})$ is rigid, this non-trivial flex $\boldsymbol{p}(t)$ cannot exist. So $\boldsymbol{p}'$ cannot exist, and $G(\boldsymbol{p})$ is infinitesimally rigid, showing Roth's Statement. This argument is explained in detail in Connelly and Whiteley (1991).

5.6. Polyhedral surfaces revisited

Another application of the second-order stress test is to triangulated polygons, thought of as "paper membranes". This in turn can be used to strengthen our result Theorem 5.2 that triangulated polyhedral surfaces are second-order rigid. First we need a different kind of stability result from Connelly (1991). A *bar simplex* in $\mathbb{R}^d$ is a framework where all the vertices of the configuration are affine independent (no $k+2$ vertices lie in an affine k-dimensional subspace), and every pair of vertices is joined by a bar. For example, bar triangles and bar tetrahedra are bar simplices.

Theorem 5.8. *A tensegrity framework $G(\boldsymbol{p})$ in $\mathbb{R}^d$ is second-order rigid in $\mathbb{R}^{d'}$ for all $\mathbb{R}^{d'} \supset \mathbb{R}^d$, $d' \geq d$, if and only if either $G(\boldsymbol{p})$ is a bar simplex in $\mathbb{R}^d$ or there is a proper self stress ω such that $G(\boldsymbol{p})$ is ω globally rigid.*

This result allows one to use results about the second-order rigidity of a framework to know that there is some particular self stress ω that works for ω global rigidity. Although the proof is non-constructive, the result can be useful. The following result follows easily from the techniques in Connelly (1980).

Theorem 5.9. *Suppose that a bar framework $G(\boldsymbol{p})$ in $\mathbb{R}^2$ consists of the vertices and edges of some triangulation of a convex polygon in the plane, with the boundary vertices fixed. Then $G(\boldsymbol{p})$ is second-order rigid in any $\mathbb{R}^d$ for $d \geq 2$. Thus there is a self stress ω such that $G(\boldsymbol{p})$ is ω globally rigid*

At first sight this result seems like the statement about the ω global rigidity of a spider web. But for a spider web all the coordinates of the stress are positive. For the triangulation of a triangle as in fig. 5.6 it turns out that no proper self stress can be positive on all the internal bars. See Connelly and Henderson (1980) for an easy proof. Nevertheless, Theorem 5.9 does say that there is a self stress ω that makes $G(\boldsymbol{p})$ ω globally rigid.

We generalize Theorem 5.9 to a more delicate property of ω. Suppose that $G_0(\boldsymbol{p})$ is the bar framework obtained from a triangulation of a convex planar polygon P_0. Attach the boundary vertices of $G_0(\boldsymbol{p})$ to a system of parallel bars in three-space which are in turn attached to fixed (pinned) vertices in three-space as in fig. 5.7. (The pinned vertices are black.) This gives us a bar framework $G(\boldsymbol{p})$.

Suppose that the triangulation of P_0 is the orthogonal projection of a face of a convex polyhedral surface. We assume also that each vertex in the interior of P_0 corresponds to a strictly convex vertex on the surface. (That is, a support plane intersects the surface just at the vertex.) From the discussion of the Maxwell–Cremona Theory in section 4.6 this implies that there is a self stress ω_0 on $G_0(\boldsymbol{p})$ that is positive on interior edges that correspond to the edges of the surface. (It may be negative on the boundary.) Each interior vertex is adjacent to at least two members of $G(\boldsymbol{p})$ with a strictly positive stress. Let ω be the self stress for $G(\boldsymbol{p})$ obtained by adjoining to ω_0 a large positive self stress on each vertical bar.

Suppose that $\boldsymbol{p}'$ is a first-order flex of $G(\boldsymbol{p})$ in any dimension $d \geq 3$. We calculate

$$(\boldsymbol{p}')^{\mathrm{T}}\Omega \otimes I^d \boldsymbol{p}' = \sum_{i,j} \omega_{ij} |p_i' - p_j'|^2 .$$

On the boundary of P_0 the vertical stresses dominate, and thus the first-order flexes at those vertices must evaluate to zero. For the interior vertices of P_0 all the stresses on members adjacent to the vertices are positive, so anything non-zero will be strictly positive. Thus all the first-order flexes must be zero, and by Theorem 5.8, $G(\boldsymbol{p})$ is ω globally rigid.

If we further subdivide this triangulation of P_0 by only adding new vertices along the existing edges of first triangulation, then the same argument shows that the corresponding $G(\boldsymbol{p})$ will be ω globally rigid.

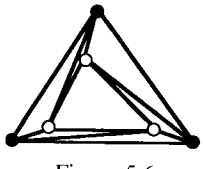


Figure 5.6.

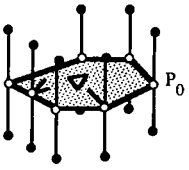


Figure 5.7.

Now suppose that $G_0(\boldsymbol{p})$ and $G(\boldsymbol{p})$ come from *any* triangulation of P_0. We claim that $G(\boldsymbol{p})$ is second-order rigid in any dimension $d \geq 3$. If not, then there must be a second-order flex $(\boldsymbol{p}', \boldsymbol{p}'')$ of $G(\boldsymbol{p})$, where $\boldsymbol{p}'$ is non-trivial. But it is easy to see that the second-order flex must extend to any subdivision of the triangulation of P_0. But we can choose a subdivision of this triangulation that is also a triangulation coming from a convex surface as mentioned above. This contradicts the argument above. Hence, this $G(\boldsymbol{p})$ must also be second-order rigid in all higher dimensions. Our stability result (Theorem 5.8) then says that there is a self stress ω such that $G(\boldsymbol{p})$ is ω globally rigid.

Using such a self stress for each face of an arbitrarily triangulated convex polytope and an argument similar to the one above gives the following basic result relating convexity and rigidity. This result and details of the argument above will appear in Connelly (1991).

Theorem 5.10. *Every triangulation of a convex polytope in three-space is pre-stress stable.*

For example, the frameworks of figs. 5.4 and 5.1 are seen to be pre-stress stable in three-space.

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CHAPTER 1.8

Convex Surfaces, Curvature and Surface Area Measures

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Contents

Introduction 275
1. First order boundary structure of convex bodies 275
2. Pointwise curvatures 279
3. Curvature measures and surface area measures 281
4. Special cases and relations to local shape 284
5. Minkowski's theorem and other existence problems 287
6. Uniqueness and stability results 292
References 296

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Introduction

The boundary of a convex body with interior points is a closed convex hypersurface. Due to the convexity assumption, many geometric properties of such a hypersurface can be proved without any additional analytic assumption. Even a kind of "differential geometry without differentiability assumptions" is possible for convex hypersurfaces. The present article is devoted to a description of the boundary structure of a general convex body. We distinguish between first order and second order properties. A convex body is called *strictly convex* if its boundary contains no segments, and *smooth* if through each boundary point there is only one supporting hyperplane. If a convex body is either not strictly convex or not smooth, its boundary points can be classified under different aspects, regarding the kind or degree of deviation from strict convexity or smoothness, respectively. Assertions connected to this deviation from strictly convex or smooth behaviour are considered as describing first order properties of convex hypersurfaces. Second order properties are those referring to the curvature behaviour. Since no differentiability assumptions are made, principal curvatures in the sense of differential geometry can in general not be defined. They exist, however, almost everywhere. More fruitful is the replacement of pointwise defined principal curvatures by the indefinite integrals of their elementary symmetric functions, since these exist, in the form of measures, for general convex bodies. Using these curvature measures and their counterparts, the surface area measures, one can develop a curvature theory of convex bodies and obtain existence and uniqueness results in the spirit of classical differential geometry of convex hypersurfaces, but for general convex bodies.

Part of the present article is covered by an earlier survey of Schneider (1979b), where additional references up to 1978 can be found.

1. First order boundary structure of convex bodies

By $\mathcal{K}^d$ we denote the space of convex bodies (nonempty, compact, convex subsets) in d-dimensional real Euclidean vector space $\mathbb{E}^d$ and by $\mathcal{K}_0^d$ the subset of convex bodies with interior points. As usual, $\mathbb{E}^d$ is equipped with its standard scalar product $\langle \cdot, \cdot \rangle$ and induced norm $\| \cdot \|$, and $\mathcal{K}^d$ carries the Hausdorff metric δ^{H}. $B^d := \{x \in \mathbb{E}^d : \| x \| \leqslant 1\}$ is the unit ball and $S^{d-1} := \{x \in \mathbb{E}^d : \| x \| = 1\}$ is the unit sphere of $\mathbb{E}^d$. By λ_k we denote the k-dimensional Hausdorff measure on $\mathbb{E}^d$.

The main topics of this section are faces, that is, certain convex subsets in the boundary of a convex body, and singularities, which are boundary points with more than one support plane passing through. Most of the results given without references can be found in standard books like Bonnesen and Fenchel (1934), Eggleston (1958), Valentine (1964), Rockafellar (1970), and Leichtweiß (1980). For convenience, we consider only convex bodies, although the boundedness is generally not needed. The reader will easily check which of the results hold for closed convex sets.

Let a convex body $K \in \mathcal{K}^d$ be given. A *face* of K is a convex subset F of K with the property that each segment in K whose relative interior meets F is entirely contained in F. Equivalently, $F \subset K$ is a face of K if it is convex and if $x, y \in K$ and $(x+y)/2 \in F$ implies $x, y \in F$. The empty set and K itself are faces of K; the other faces are called *proper*. Immediately from the definition it follows that faces are closed, proper faces are subsets of the relative boundary, intersections of faces are faces, and the face relation is transitive. Further, the system of the relative interiors of all faces of K is a disjoint decomposition of K, and each nonempty relatively open convex subset of K is contained in the relative interior of a unique face of K.

The intersection of K with a support plane to K is called an *exposed face* of K. Each exposed face of K is a face of K. The converse is not generally true, but each proper face is contained in an exposed face. The nonempty intersection of exposed faces of K is an exposed face of K. The relation of being an exposed face is in general not transitive.

One-pointed faces lead to the notions of extreme and exposed points. The point $x \in K$ is called an *extreme point* of K if it cannot be written in the form $x = (1-\lambda)y + \lambda z$ with $y, z \in K$ and $0 < \lambda < 1$, in other words, if it is not a nontrivial convex combination of points of K. Thus x is an extreme point of K if and only if $\{x\}$ is a face of K. Similarly, x is an *exposed point* of K if $\{x\}$ is an exposed face of K, that is, if $H \cap K = \{x\}$ for a suitable support plane H of K. These notions generalize as follows. Let $r \in \{0, \dots, d-1\}$. The point x is an *r-extreme* (*r-exposed*) point of K if it lies in some face (exposed face) of K of dimension at most r. The set of r-extreme (r-exposed) points of K is denoted by $\operatorname{ext}_r K$ ($\exp_r K$) and called its *r-skeleton (exposed r-skeleton)*. The importance of extreme points stems from the following fundamental result.

Theorem 1.1 (Minkowski). *Every convex body is the convex hull of its set of extreme points.*

The proof is easily done by induction with respect to the dimension. For zero-dimensional convex bodies the assertion is trivial. Let $K \in \mathcal{K}^d$ be of dimension d and assume that the assertion is true for convex bodies of smaller dimension. Clearly each point of K is a convex combination of boundary points. Each boundary point of K lies in a support plane H of K and hence, by the induction hypothesis, is a convex combination of extreme points of $H \cap K$, which are extreme points of K. The assertion follows.

A convex body is also the closed convex hull of its set of exposed points, since each extreme point is a limit of exposed points; more generally we have the following.

Theorem 1.2. *For $r = 0, \dots, d-1$, each r-extreme point of a convex body is a limit of r-exposed points of the body.*

For $r = 0$ this result is known as the theorem of Straszewicz (1935); the general case is due to Asplund (1963).

Each point $x \in K$ is contained in the relative interior of a unique face, say F_x, of K. This defines a map $x \mapsto F_x$ from K into $\mathcal{H}^d$, called the *face function* of K. Its semicontinuity properties were investigated by Klee and Martin (1971), and Larman (1971). The face function of $K \in \mathcal{H}_0^d$ is lower semicontinuous if and only if the midpoint map $(x,y) \mapsto (x+y)/2$ from $K \times K$ onto K is open; an equivalent condition is that all skeletons $\operatorname{ext}_r K$, $r = 0, \ldots, d-1$, are closed (Papadopoulou 1977, 1982). Convex bodies with these properties are called *stable*.

While a general convex body can be far from being strictly convex or smooth, it is an interesting task to find quantitative manifestations of the fact that these deviations cannot be too far. A first example is given by the following result, which restricts the directions of the line segments in the boundary of a convex body.

Theorem 1.3. *Let $K \in \mathcal{H}^d$ be a convex body and let $U(K) \subset S^{d-1}$ be the set of all unit vectors which are parallel to a segment in the boundary of K. Then $U(K)$ is of σ-finite $(d-2)$-dimensional Hausdorff measure.*

This theorem is due to Ewald, Larman and Rogers (1970). Their proof uses an interesting method of covering the boundary of a convex body by caps in a certain economical way. Even for the weaker assertion that $U(K)$ is a set of measure zero on the sphere S^{d-1}, no simple proof is known. The example of a convex body $K \in \mathcal{H}_0^d$ with a countable number of $(d-1)$-dimensional convex subsets in its boundary shows that the assertion of Theorem 1.3 is best possible.

Theorem 1.3 can be extended to higher-dimensional convex subsets in the boundary of a convex body. To formulate a general version, let $K \in \mathcal{H}^d$ and $1 \leqslant s \leqslant r \leqslant d-1$. Then the set of all r-dimensional linear subspaces of $\mathbb{E}^d$, parallel to some supporting r-flat of K which contains an s-dimensional convex subset of K, has σ-finite $(r(d-r)-s)$-dimensional Hausdorff measure. Here the latter refers to a rotation invariant Riemannian metric on the Grassmannian of all r-dimensional linear subspaces of $\mathbb{E}^d$. The result was proved by Ewald, Larman and Rogers (1970) for $s = r$ and by Zalgaller (1972) for $s = 1$, with a sketch for the general case; a detailed proof was given by Kaltenbach (1984).

We turn to singularities. First we introduce some additional notation. Let $K \in \mathcal{H}^d$ be a convex body. By

$$h(K,u) := \max\{\langle x,u\rangle : x \in K\}, \quad u \in \mathbb{E}^d,$$

the *support function* $h(K,\cdot)$ of K is defined. The hyperplane

$$H(K,u) := \{x \in \mathbb{E}^d : \langle x,u\rangle = h(K,u)\},$$

where $u \in \mathbb{E}^d \setminus \{0\}$, is the *support hyperplane* of K with outer normal vector u, and for $x \in K$, any vector $u \neq 0$ with $\langle x,u\rangle = h(K,u)$ is called an *outer normal vector* of K at x. The set $N(K,x)$ of all outer normal vectors of K at x, together with the zero vector, is a closed convex cone, called the *normal cone* of K at x. For $u \in \mathbb{E}^d \setminus \{0\}$, we write $F(K,u) := H(K,u) \cap K$ for the exposed face of K with outer normal vector u. If F is a nonempty convex subset of K, we define $N(K,F) := N(K,x)$, where x is a point in the relative interior of F; the cone $N(K,F)$ does not depend

on the choice of x. In particular, the cone $N(K, F(K,u))$ depends only on K and the direction of $u \neq 0$. The unique face of the cone $N(K, F(K,u))$ containing u in its relative interior is denoted by $T(K,u)$ and called the *touching cone* of K at u. This name is chosen because, for each vector v in the relative interior of $T(K,u)$, the support plane $H(K,v)$ touches K at the same set, namely $F(K,v) = K \cap D$, where D is the flat through relint $F(K,u)$ totally orthogonal to $T(K,u)$.

The point x of the convex body $K \in \mathcal{K}_0^d$ is called an *r-singular* point of K if $\dim N(K,x) \geqslant d-r$ $(r = 0, \ldots, d-2)$. Any r-singular point with $r \leqslant d-2$ is called *singular*, and a boundary point which is not singular is called *smooth* (or *regular*). The set of r-singular points of a convex body is restricted in size, as the following theorem shows.

Theorem 1.4. *For a convex body $K \in \mathcal{K}_0^d$ and for $r \in \{0, \ldots, d-2\}$, the set of r-singular points of K can be covered by countably many compact sets of finite r-dimensional Hausdorff measure.*

In particular, the set of singular points of a convex body is of $(d-1)$-dimensional Hausdorff measure zero (Theorem of Reidemeister 1921). The following proof of Theorem 1.4 is essentially due to Anderson and Klee (1952). Let $p_K : \mathbb{E}^d \to K$ be the nearest-point map. Let M_r denote the intersection of a closed ball containing K in its interior with the union of all r-dimensional flats which are spanned by points with rational coordinates. If x is an r-singular point of K, the translated cone $N(K,x) + x = p_K^{-1}(x)$ is of dimension at least $d-r$ and hence meets M_r. Thus the set of r-singular points of K is covered by the image of M_r under p_K. The assertion follows since p_K is a Lipschitz map.

Next, we classify the normal vectors of a nonsmooth convex body. This classification is dual to the classification of boundary points of a convex body which is not strictly convex. However, we prefer to give an independent definition.

Let $K \in \mathcal{K}^d$ and $u \in \mathbb{E}^d \backslash \{0\}$. The vector u is called an *r-extreme normal vector* of K if $\dim T(K,u) \leqslant r+1$, and u is an *r-exposed normal vector* of K if $\dim N(K, F(K,u)) \leqslant r+1$. A support plane $H(K,u)$ of K is called r-extreme (r-exposed) if its outer normal vector u is r-extreme (r-exposed). A reformulation of the definition shows that u is an r-extreme normal vector of K if and only if there do not exist $r+2$ linearly independent normal vectors $u_1, \ldots, u_{r+2}$ at the same boundary point of K such that $u = u_1 + \cdots + u_{r+2}$. An equivalent condition is that $u = u_1 + \cdots + u_{r+2}$ with linearly independent vectors $u_1, \ldots, u_{r+2}$ implies

$$h(K,u) < h(K,u_1) + \cdots + h(K,u_{r+2}).$$

The mentioned duality is established as follows. Let $K \in \mathcal{K}^d$ be a convex body with $0 \in \operatorname{int} K$. Its polar body is defined by

$$K^* := \{x \in \mathbb{E}^d : \langle x, y \rangle \leqslant 1 \quad \text{for all } y \in K\}.$$

Let $u \in \mathbb{E}^d \backslash \{0\}$. We may assume (after multiplying u by a uniquely determined positive factor) that the hyperplane $H_{u,1} := \{x \in \mathbb{E}^d : \langle x, u \rangle = 1\}$ supports K. Then u is an r-extreme (r-exposed) point of K^* if and only if u is an r-extreme (r-exposed)

normal vector of K. This fact can be used, in particular, to obtain a dual version of Theorem 1.2: if $K \in \mathcal{K}_0^d$ and $r \in \{0,\dots,d-2\}$, then each r-extreme support plane of K is a limit of r-exposed support planes of K. We may also dualize Theorem 1.3, but we mention only a special case. The vector $u \in \mathbb{E}^d \backslash \{0\}$ is called a *singular* normal vector of K if $\dim F(K,u) > 0$. Then the set of singular unit normal vectors of a convex body is a set on S^{d-1} of $(d-1)$-dimensional Hausdorff measure zero.

The r-extreme support planes play a role in the theory of mixed volumes and mixed area measures. For example, the following notion is useful there. Let $K, L \in \mathcal{K}^d$ be convex bodies such that $L \subset K$. For $p \in \{0,\dots,d-1\}$, the body K is called a *p-tangential body* of L if each $(n-p-1)$-extreme support plane of K is also a support plane of L. An equivalent description is contained in the following theorem, whose proof makes use of the dual of Theorem 1.2 (see Schneider 1978b).

Theorem 1.5. *Let $K, L \in \mathcal{K}^d$, $L \subset K$ and $p \in \{0,\dots,d-1\}$. Then K is a p-tangential body of L if and only if each support plane of K which is not a support plane of L contains only $(p-1)$-singular points of K.*

2. Pointwise curvatures

For a convex body that is strictly convex and smooth, the first order boundary behaviour is trivial. The second order behaviour, which we study from now on, is governed by the notion of curvature. The present section is devoted to a brief discussion of the type of curvatures that can be defined at a single point.

We assume that $K \in \mathcal{K}_0^d$ is a convex body and x is a smooth boundary point of K. Let u be the outer unit normal vector to K at x and let t be an arbitrary unit vector orthogonal to u. For small $\tau > 0$, let $f(\tau) \geqslant 0$ be the smaller number for which $y(\tau) := x + \tau t - f(\tau)u$ is in the boundary of K. Let $\rho(\tau) > 0$ be such that $x - \rho(\tau)u$ is the centre of the circle through x and $y(\tau)$ touching the support plane $H(K,u)$. If $\operatorname{bd} K$ is a smooth hypersurface of differentiability class C^2 in a neighbourhood of x, then $\lim_{\tau \to 0} \rho(\tau)$ exists in $\mathbb{R} \cup \{\infty\}$ and is equal to the reciprocal normal curvature (that is, the normal radius of curvature) of $\operatorname{bd} K$ at x in direction t, as defined in differential geometry. In the general case, one defines

$$\kappa_i(x,t) := 1/\limsup_{\tau \to +0} \rho(\tau),$$

$$\kappa_s(x,t) := 1/\liminf_{\tau \to +0} \rho(\tau)$$

(∞ admitted) and calls $\kappa_i(x,t)$ the *lower* and $\kappa_s(x,t)$ the *upper curvature* of K at x in direction t. If $\kappa_i(x,t) = \kappa_s(x,t) < \infty$, the common value is denoted by $\kappa(x,t)$ and called the *curvature* of K at x in direction t.

Again in the C^2 case, it is well known that the dependence of $\kappa(x,t)$ on the tangent vector t is quadratic: There are orthonormal tangent vectors $e_1,\dots,e_{d-1}$ and nonnegative numbers $k_1,\dots,k_{d-1}$ such that

$$\kappa(x,t) = k_1\langle t, e_1\rangle^2 + \cdots + k_{d-1}\langle t, e_{d-1}\rangle^2 \tag{2.1}$$

for each unit vector $t \perp u$. Equality (2.1) is called *Euler's theorem*. The vectors $e_1, \ldots, e_{d-1}$ determine the *principal directions*, and $k_1, \ldots, k_{d-1}$ are the *principal curvatures* of $\operatorname{bd} K$ at x.

For a general convex body K and a smooth boundary point x one says that x is a *normal point* or *Euler point* of K if the curvature $\kappa(x,t)$ exists for each tangent vector t at x and (2.1) holds for suitable orthonormal vectors $e_1, \ldots, e_{d-1}$ and numbers $k_1, \ldots, k_{d-1}$. These numbers are then called the *principal curvatures* of K at x. Their jth normalized elementary symmetric function is denoted by H_j, thus

$$H_j = \binom{d-1}{j}^{-1} \sum_{1 \leqslant i_1 < \cdots < i_j \leqslant d-1} k_{i_1} \cdots k_{i_j}$$

for $j = 1, \ldots, d-1$, and $H_0 = 1$. In particular, H_1 is the *mean curvature* and H_{d-1} is the *Gauss–Kronecker curvature*.

By Reidemeister's theorem, almost all boundary points of a convex body $K \in \mathcal{K}_0^d$, in the sense of $(d-1)$-dimensional Hausdorff measure, are smooth. Even more is true.

Theorem 2.1. *For each convex body, almost all boundary points (in the sense of $(d-1)$-dimensional Hausdorff measure) are normal.*

This was proved for $d \leqslant 3$ by Busemann and Feller (1936) and for arbitrary dimension d by Aleksandrov (1939c). For alternative proofs, see Rešetnjak (1968) and Bangert (1979).

The example of a polytope shows that it can happen that at all normal points all the curvatures are zero. In fact, this behaviour is generic: most convex bodies, in the sense of Baire category, have the property that they are smooth and strictly convex and their curvatures are zero wherever they exist. Moreover, most convex bodies have the property that at most of their boundary points (again in the sense of Baire category) and for every tangent direction, the lower curvature is zero and the upper curvature is infinite. For these results, which are due to Zamfirescu, and for more in the same spirit, see chapter 4.10.

The preceding observations show that, for general convex bodies, curvatures that can be defined pointwise do not carry much information on the local shape. It is for this reason that, in the subsequent sections, curvature functions will be replaced by certain measures.

Instead of curvatures at given boundary points of convex bodies, it is often more natural and more useful to consider radii of curvature as functions of the normal vector. One reason is the transparent behaviour under Minkowski addition.

Let $K \in \mathcal{K}_0^d$ be a convex body of class C_+^2, that is, whose boundary is a smooth hypersurface of class C^2 with everywhere positive principal curvatures. For a unit vector $u \in S^{d-1}$, let $x(u)$ be the unique boundary point of K at which u is attained as outer normal vector. The principal radii of curvature of K at u, say $R_1, \ldots, R_{d-1}$, can be defined as the reciprocal values of the principal curvatures of K at $x(u)$. The

jth normalized elementary symmetric function of the principal radii of curvature will be denoted by P_j, thus

$$P_j = \binom{d-1}{j}^{-1} \sum_{1 \leqslant i_1 < \cdots < i_j \leqslant d-1} R_{i_1} \cdots R_{i_j}.$$

Since K is of class C_+^2, its support function $h_K = h(K, \cdot)$ is of class C^2 on $\mathbb{E}^d \backslash \{0\}$. The Hessian matrix of h_K (with respect to some orthonormal basis of $\mathbb{E}^d$) at $u \in S^{d-1}$ has u as an eigenvector, with corresponding eigenvalue 0; the other eigenvectors determine the principal directions of K at $x(u)$, and the corresponding eigenvalues are the principal radii of curvature. In particular, $\binom{d-1}{j} P_j(u)$ is equal to the sum of the principal minors of order j of the Hessian of h_K at u.

If the support function of the convex body K is of class C^2 on $\mathbb{E}^d \backslash \{0\}$ (without K necessarily being of class C_+^2), principal directions and principal radii of curvature of K can be defined as above, via the Hessian of h_K at unit vectors. If all principal radii of curvature are everywhere positive, it can be shown that K is of class C_+^2.

There is a "dual" version of Euler's theorem (2.1). Let K be a convex body with support function of class C^2. Let $u \in S^{d-1}$ and a unit vector t orthogonal to u be given. The *tangential radius of curvature* at u in direction t, denoted by $R(u,t)$, is defined as the radius of curvature, at u, of the image of K under orthogonal projection onto the linear subspace spanned by u and t. Then

$$R(u,t) = R_1 \langle t, e_1 \rangle^2 + \cdots + R_{d-1} \langle t, e_{d-1} \rangle^2 \tag{2.2}$$

for each unit vector $t \perp u$, where $e_1, \ldots, e_{d-1}$ are the vectors determining the principal directions and R_i is the principal radius of curvature corresponding to e_i. Equation (2.2) is sometimes called *Blaschke's theorem* (Blaschke 1956, p. 117).

3. Curvature measures and surface area measures

Since pointwise defined curvatures or radii of curvature do not carry enough information on the geometric shape of a general convex body, they will now be replaced by locally defined measures. For a convex body K of class C_+^2 we may define, for $j = 0, \ldots, d-1$,

$$C_j(K, \beta) := \int_{\beta \cap \mathrm{bd}\, K} H_{d-1-j} \, \mathrm{d}\lambda_{d-1} \tag{3.1}$$

for $\beta \in \mathcal{B}(\mathbb{E}^d)$ and

$$S_j(K, \omega) := \int_{\omega} P_j \, \mathrm{d}\lambda_{d-1} \tag{3.2}$$

for $\omega \in \mathcal{B}(S^{d-1})$. Here we have denoted by $\mathcal{B}(X)$ the σ-algebra of Borel sets of a topological space X. (Recall that λ_{d-1} is $(d-1)$-dimensional Hausdorff measure.) Then $C_j(K, \cdot)$ is a finite measure on $\mathcal{B}(\mathbb{E}^d)$ and $S_j(K, \cdot)$ is a finite measure on

$\mathcal{B}(S^{d-1})$. Knowledge of $C_j(K,\cdot)$ is equivalent to knowledge of the curvature function H_{d-1-j} on $\operatorname{bd} K$, and a similar remark applies to $S_j(K,\cdot)$. The advantage of introducing the measures $C_j(K,\cdot)$, $S_j(K,\cdot)$ lies in the fact that they can be defined, in a natural way, for arbitrary convex bodies K. This is most conveniently done by considering the volume of a suitable local parallel set at a variable distance ε. It is intuitively clear that the growth of this volume in dependence upon the parameter ε reflects curvature properties.

It is appropriate to introduce a common generalization of the measures C_j and S_j, by considering sets of support elements. Let $K \in \mathcal{K}^d$ be a convex body. A *support element* of K is a pair (x,u) where x is a boundary point of K and u is an outer unit normal vector of K at x. The set $\operatorname{Nor} K$ of all support elements of K, also called the *generalized normal bundle* of K, is considered as a subspace of $\Sigma := \mathbb{E}^d \times S^{d-1}$ with the product topology. (Clearly, the map $(x,u) \mapsto x+u$ is a homeomorphism from $\operatorname{Nor} K$ onto $\operatorname{bd}(K+B^d)$.)

For $x \in \mathbb{E}^d$, we denote the unique point in K nearest to x by $p(K,x)$; thus $p(K,\cdot) = p_K$ is the nearest point map or metric projection onto K. Further, $r(K,x) := \| x - p(K,x) \|$ is the distance of x from K, and for $x \notin K$, the unit vector $u(K,x) := (x - p(K,x))/r(K,x)$ points from $p(K,x)$ to x. For a set $\eta \subset \Sigma$ and a number $\varepsilon > 0$, we define the *local parallel set* of K with respect to η at distance ε by

$$M_\varepsilon(K,\eta) := \{x \in \mathbb{E}^d : 0 < r(K,x) \leqslant \varepsilon,\ (p(K,x), u(K,x)) \in \eta\}.$$

If η is a Borel set, then $M_\varepsilon(K,\eta)$ is a Borel set, and we may consider its d-dimensional (Hausdorff or Lebesgue) measure,

$$\mu_\varepsilon(K,\eta) := \lambda_d(M_\varepsilon(K,\eta)).$$

For the function $\mu_\varepsilon : \mathcal{K}^d \times \mathcal{B}(\Sigma) \to \mathbb{R}$ thus defined one establishes the following properties. For fixed K, $\eta_\varepsilon(K,\cdot)$ is a finite measure. It depends weakly continuously on K, that is, if a sequence $(K_n)_{n\in\mathbb{N}}$ of convex bodies converges to K (in the Hausdorff metric), then the sequence $(\mu_\varepsilon(K_n,\cdot))_{n\in\mathbb{N}}$ of measures converges weakly to $\mu_\varepsilon(K,\cdot)$. For fixed $\eta \in \mathcal{B}(\Sigma)$, the function $\mu_\varepsilon(\cdot,\eta)$ on $\mathcal{K}^d$ is (Borel) measurable, and it is additive or a valuation (see chapter 3.6).

Most important is the way in which $\mu_\varepsilon(K,\eta)$ depends on the parameter ε: it is, in fact, a polynomial of degree at most $d-1$. Therefore, choosing a suitable normalization, one may write

$$\mu_\varepsilon(K,\eta) = \frac{1}{d} \sum_{j=0}^{d-1} \varepsilon^{d-j} \binom{d}{j} \Theta_j(K,\eta) \tag{3.3}$$

for $\varepsilon > 0$. That $\mu_\varepsilon(K,\eta)$ is a polynomial, is seen by direct computation in the case where K is a polytope. For a general convex body K, it is proved by approximating K by a sequence of polytopes and using the weak continuity of the measures μ_ε.

The identity (3.3) defines $\Theta_j(K,\eta)$ for $j = 0,\dots,d-1$. From the properties of μ, corresponding properties of Θ_j are deduced. For fixed K, each $\Theta_j(K,\cdot)$ is a

finite measure on $\mathcal{B}(\Sigma)$, called the jth *generalized curvature measure* of K. The map $K \mapsto \Theta_j(K, \cdot)$ from $\mathcal{K}^d$ into the space of Borel measures on Σ is weakly continuous and a valuation. For fixed $\eta \in \mathcal{B}(\Sigma)$, the real function $\Theta_j(\cdot, \eta)$ on $\mathcal{K}^d$ is measurable. For convenience, $\Theta_j(K, \cdot)$ was defined on Σ, but clearly $\Theta_j(K, \cdot)$ is concentrated on the generalized normal bundle $\operatorname{Nor} K$.

Now we specialize the generalized curvature measures, either to sets of boundary points or to sets of normal vectors, by writing

$$C_j(K, \beta) := \Theta_j(K, \beta \times S^{d-1}) \tag{3.4}$$

for Borel sets $\beta \in \mathcal{B}(\mathbb{E}^d)$ (of which only $\beta \cap \operatorname{bd} K$ is essential), and

$$S_j(K, \omega) := \Theta_j(K, \mathbb{E}^d \times \omega) \tag{3.5}$$

for Borel sets $\omega \in \mathcal{B}(S^{d-1})$ of unit normal vectors. The measures $C_0(K, \cdot)$, $\ldots, C_{d-1}(K, \cdot)$ are called the *curvature measures*, and $S_0(K, \cdot), \ldots, S_{d-1}(K, \cdot)$ are the *surface area measures* of K. Sometimes, variants with a different normalization are more convenient; one puts

$$d\kappa_{d-j} \Phi_j(K, \cdot) := \binom{d}{j} C_j(K, \cdot), \tag{3.6}$$

$$d\kappa_{d-j} \Psi_j(K, \cdot) := \binom{d}{j} S_j(K, \cdot), \tag{3.7}$$

where κ_m denotes the volume of the m-dimensional unit ball. The advantage of this normalization lies, for example, in the fact that $\Phi_j(K, \beta)$ does not change under embedding into a space of higher dimension.

The measures introduced so far can be considered as local versions of certain mixed volumes. In fact,

$$M_\varepsilon(K, \Sigma) = \{x \in \mathbb{E}^d : 0 < r(K, x) \leqslant \varepsilon\} = (K + \varepsilon B^d) \backslash K,$$

hence

$$\begin{aligned} \mu_\varepsilon(K, \Sigma) &= \lambda_d(K + \varepsilon B^d) - \lambda_d(K) \\ &= \sum_{j=0}^{d-1} \varepsilon^{d-j} \binom{d}{j} V(\underbrace{K, \ldots, K}_{j}, \underbrace{B^d, \ldots, B^d}_{d-j}), \end{aligned}$$

where V denotes the mixed volume (see chapter 1.2). Comparison with (3.3) shows that

$$\begin{aligned} \Theta_j(K, \Sigma) &= C_j(K, \mathbb{E}^d) = S_j(K, S^{d-1}) \\ &= dV(\underbrace{K, \ldots, K}_{j}, \underbrace{B^d, \ldots, B^d}_{d-j}) = dW_{d-j}(K), \end{aligned} \tag{3.8}$$

where W_{d-j} is a quermassintegral, or alternatively,

$$\Phi_j(K,\mathbb{E}^d) = \Psi_j(K,S^{d-1}) = V_j(K), \tag{3.9}$$

where V_j is an intrinsic volume.

The surface area measures also arise by specialization from the mixed surface area measure S (see chapter 1.2), namely by

$$S_j(K,\cdot) = S(\underbrace{K,\dots,K}_{j},\underbrace{B^d,\dots,B^d}_{d-1-j},\cdot). \tag{3.10}$$

Above we have listed a number of properties of the curvature measures and of the surface area measures. They are reminiscent of the properties appearing in Hadwiger's axiomatic characterization of the linear combinations of the quermassintegrals (see chapter 1.2). Similar characterizations can be obtained for the curvature and surface area measures, if their obvious invariance properties and local character are taken into account. We formulate such a result for the curvature measures.

Theorem 3.1. *Let φ be a map from $\mathcal{K}^d$ into the space of finite Borel measures on $\mathbb{E}^d$ which is additive, weakly continuous, rigid motion invariant (that is, $\varphi(gK)\circ g = \varphi(K)$ for $K \in \mathcal{K}^d$ and each rigid motion g of $\mathbb{E}^d$) and defined locally in the following sense: if $\beta \subset \mathbb{E}^d$ is open and $K \cap \beta = L \cap \beta$, then $\varphi(K) = \varphi(L)$ on $\mathcal{B}(\beta)$. Under these conditions,*

$$\varphi(K) = \sum_{i=0}^{d} a_i C_i(K,\cdot) \qquad \text{for } K \in \mathcal{K}^d,$$

where $a_0,\dots,a_d$ are nonnegative real constants and $C_d(K,\cdot) := \lambda_d(K \cap \cdot)$.

This result was proved by Schneider (1978a), and an analogous result for surface area measures by Schneider (1975a).

Historically, the surface area measures were first introduced in connection with the theory of mixed volumes; this was done independently by Aleksandrov (1937a) and Fenchel and Jessen (1938). The curvature measures are due to Federer (1959), who defined and treated them for sets of positive reach. A unified and simplified treatment of both types of measures for convex bodies can be found in Schneider (1978a). The generalized curvature measures on sets of support elements appear in Schneider (1979a).

4. Special cases and relations to local shape

Having introduced curvature measures and surface area measures as substitutes for pointwise defined curvature functions, we now look at their geometric meaning in special cases and at the information on local shape that is available from these measures.

First we remark that the definitions are consistent with the notation introduced at the beginning of section 3. If the convex body K is of class C^2, the curvature measure $C_j(K,\cdot)$ is the indefinite integral of the curvature function H_{d-1-j}, as expressed by (3.1). If the support function of K is of class C^2 on $\mathbb{E}^d\backslash\{0\}$, then $S_j(K,\cdot)$ can be represented in the form (3.2) as an indefinite integral on the sphere S^{d-1}. These assertions can be proved by computing the measure of the local parallel set $M_\varepsilon(K,\eta)$ by standard methods of differential geometry. For general convex bodies, in fact for sets of positive reach (see Federer (1959) for a definition), it is possible to define principal curvatures as functions on the generalized normal bundle and to express the curvature measures as integrals, with respect to $(d-1)$-dimensional Hausdorff measure, of elementary symmetric functions of such curvatures over the normal bundle. This was done by Zähle (1986) and further investigated in Zähle (1987).

Let $K \in \mathcal{K}_0^d$ be a general convex body. If x is a normal boundary point of K, then the $(d-1-j)$th normalized elementary symmetric function of the principal curvatures at x is defined, and one may ask whether it can be computed from the curvature measure $C_j(K,\cdot)$. For suitable sequences $(\beta_i)_{i\in\mathbb{N}}$ of compact subsets of $\operatorname{bd} K$ shrinking to x it is, in fact, true that

$$\lim_{i\to\infty} \frac{C_j(K,\beta_i)}{\lambda_{d-1}(\beta_i)} = H_{d-1-j}(x). \tag{4.1}$$

This was proved for $j=0$ by Aleksandrov (1939c) and for general j by Schneider (1979a); at both places one finds the conditions that the sequence $(\beta_i)_{i\in\mathbb{N}}$ has to satisfy.

If $P \in \mathcal{K}^d$ is a polytope, then the direct computation of the local parallel volume $\mu_\varepsilon(K,\eta)$ leads to simple explicit expressions for the curvature measures and the surface area measures. Let $\mathcal{F}_j(P)$ denote the set of j-dimensional faces of the polytope P. One finds that

$$C_j(P,\beta) = \frac{d\kappa_{d-j}}{\binom{d}{j}} \sum_{F\in\mathcal{F}_j(P)} \gamma(F,P)\lambda_j(F\cap\beta) \tag{4.2}$$

for $\beta \in \mathcal{B}(\mathbb{E}^d)$, where $\gamma(F,P)$ denotes the normalized external angle of P at its face F, and

$$S_j(P,\omega) = \binom{d-1}{j}^{-1} \sum_{F\in\mathcal{F}_j(P)} \lambda_j(F)\lambda_{d-1-j}(N(P,F)\cap\omega) \tag{4.3}$$

for $\omega \in \mathcal{B}(S^{d-1})$.

In the extreme cases $j=0$ and $j=d-1$, it is clear from (4.2) and (4.3) that C_j and S_j have a particularly simple geometric meaning, and this extends to general convex bodies. Let $K \in \mathcal{K}^d$. For $\beta \subset \mathbb{E}^d$, let $\sigma(K,\beta)$ be the set of all outer unit normal vectors of K at points of β. For $\omega \subset S^{d-1}$, let $\tau(K,\omega)$ be the set of all boundary points of K at which there exists an outer unit normal vector from

the set ω. We call $\sigma(K,\beta)$ the *spherical image* of K at β and $\tau(K,\omega)$ the *reverse spherical image* of K at ω. If K is smooth, its *spherical image map* $\sigma_K : \operatorname{bd} K \to S^{d-1}$ associates with $x \in \operatorname{bd} K$ the unique outer unit normal vector of K at x, and if K is strictly convex, then its *reverse spherical image map* $\tau_K : S^{d-1} \to \operatorname{bd} K$ is defined by $\tau_K(u) = x$, where x is the unique boundary point of K at which u is attained as outer normal vector. With these notations, we can state for $\beta \in \mathcal{B}(\mathbb{E}^d)$ and $\omega \in \mathcal{B}(S^{d-1})$ that

$$C_0(K,\beta) = \lambda_{d-1}(\sigma(K,\beta)), \tag{4.4}$$

thus C_0 is the measure of the spherical image or Gaussian curvature measure, and

$$S_0(K,\omega) = \lambda_{d-1}(\omega). \tag{4.5}$$

If K is not of dimension $d-1$, then

$$C_{d-1}(K,\beta) = \lambda_{d-1}(\beta \cap \operatorname{bd} K), \tag{4.6}$$

$$S_{d-1}(K,\omega) = \lambda_{d-1}(\tau(K,\omega)). \tag{4.7}$$

The latter relation says that $S_{d-1}(K,\omega)$ is the surface area of the reverse spherical image of K at ω; this is the reason for calling $S_{d-1}(K,\cdot)$ the surface area measure of K. From S_{d-1}, the measures S_j can be derived via the Steiner type formula

$$S_{d-1}(K + \varepsilon B^d, \cdot) = \sum_{j=0}^{d-1} \varepsilon^{d-1-j} \binom{d-1}{j} S_j(K,\cdot); \tag{4.8}$$

therefore, $S_j(K,\cdot)$ is called the surface area measure of order j. Equality (4.8) is a special case of the multilinearity properties of the mixed area measure, as a consequence of (3.10). We mention that a Steiner type formula holds also for the generalized curvature measures, if formulated in a suitable way. For $\varepsilon > 0$, define $t_\varepsilon(x,u) := (x + \varepsilon u, u)$ for $(x,u) \in \Sigma$. For $K \in \mathcal{K}^d$ and $\eta \in \mathcal{B}(\Sigma)$ we then have

$$\Theta_j(K + \varepsilon B^d, t_\varepsilon \eta) = \sum_{i=0}^{j} \varepsilon^i \binom{j}{i} \Theta_{j-i}(K,\eta). \tag{4.9}$$

Under special assumptions, the curvature measures and surface area measures can be computed from each other: If K is smooth, then $S_j(K,\cdot)$ is the image measure of $C_j(K,\cdot)$ under the spherical image map σ_K. If K is strictly convex, then $C_j(K,\cdot)$ is the image measure of $S_j(K,\cdot)$ under the reverse spherical image map. Finally, if K is smooth and strictly convex, then the spherical image map σ_K is a homeomorphism, which transforms $C_j(K,\cdot)$ into $S_j(K,\cdot)$.

Since the curvature measures, by their definition and as seen in special cases, describe local shape properties, an open set on which such a measure vanishes should be of particular geometric type. The following theorem describes the support of the jth curvature measure. By definition, the support of a Borel measure is the complement of the largest open set on which the measure vanishes.

Theorem 4.1. *For $K \in \mathcal{K}_0^d$ and $j \in \{0,\dots,d-1\}$, the support of $C_j(K,\cdot)$ is the closure of the j-skeleton* $\operatorname{ext}_j K$ *of K.*

This was proved by Schneider (1978a). The following counterpart for surface area measures was obtained by Schneider (1975b).

Theorem 4.2. *For $K \in \mathcal{K}_0^d$ and $j \in \{0,\dots,d-1\}$, the support of $S_j(K,\cdot)$ is the closure of the set of $(d-1-j)$-extreme unit normal vectors of K.*

Both proofs make use of integral-geometric mean value formulae (see chapter 5.1). Integral-geometric methods can also be used to describe the local behaviour of curvature and surface area measures under a different aspect, namely the fact that they cannot be too big on small sets. More precisely, if $K \in \mathcal{K}^d$ and $j \in \{1,\dots,d-1\}$, then

$$C_j(K,\beta) \leqslant a_1 \lambda_j(\beta) \tag{4.10}$$

for $\beta \in \mathcal{B}(\mathbb{E}^d)$, where a_1 is a constant depending only on d and j. Similarly,

$$S_j(K,\omega) \leqslant a_2 \lambda_{d-1-j}(\omega) \tag{4.11}$$

under suitable assumptions on ω, where now a_2 depends on d, j, and K. (See Schneider 1979b, §8, for references.) In particular, $C_j(K,\cdot)$ for $j > 0$, and $S_j(K,\cdot)$ for $j < d-1$, cannot have point masses.

5. Minkowski's theorem and other existence problems

Since curvature measures and surface area measures replace curvature functions which can be defined in the C^2 case, it is a natural question whether one can prove uniqueness and existence results for convex bodies in terms of these measures which are analogous to theorems of global differential geometry for convex hypersurfaces. For example, if some result involves a condition on elementary symmetric functions of principal curvatures and this condition can be formulated in terms of curvature measures, by means of (3.1), then it is a natural question whether the result extends to general convex bodies. Such generalizations are of an inherent interest, as expressed in the following quotation from Busemann (1958, p. 60), which refers to the general version of Minkowski's theorem (Theorem 5.1 below):

> "... we have here a first example of a deeper theorem of differential geometry in the large proved for a geometrically natural class of surfaces, i.e., without smoothness requirements necessitated by the methods rather than the problem."

The rest of this article will be devoted to existence, uniqueness and stability results for general convex bodies, involving surface area measures and curvature measures. For the former type of measures, their close connection to the theory of mixed volumes plays an essential role.

First we state Minkowski's existence theorem, a well-known and powerful result. It gives necessary and sufficient conditions for a measure on the sphere S^{d-1} to be the surface area measure $S_{d-1}(K,\cdot)$ of a convex body $K \in \mathcal{K}_0^d$.

Theorem 5.1. *A measure φ on $\mathcal{B}(S^{d-1})$ is the surface area measure $S_{d-1}(K,\cdot)$ of some convex body $K \in \mathcal{K}_0^d$ if and only if*

$$\int_{S^{d-1}} u \, \mathrm{d}\varphi(u) = 0 \tag{5.1}$$

and

$$\varphi(\Omega) < \varphi(S^{d-1}) \tag{5.2}$$

for each great subsphere Ω of S^{d-1}.

A d-dimensional convex body with given surface area measure S_{d-1} is unique up to a translation. Uniqueness theorems and corresponding stability results will be treated in the next section; here we concentrate on existence problems.

The necessity of the conditions of Theorem 5.1 is easy to see. To prove sufficiency, one can use a variational method which is motivated by the theory of mixed volumes. Suppose that there is a convex body $K \in \mathcal{K}^d$ with $S_{d-1}(K,\cdot) = \varphi$. By the integral representation of mixed volumes and by Minkowski's inequality, any convex body $L \in \mathcal{K}_0^d$ then satisfies

$$\frac{1}{d}\int_{S^{d-1}} h(L,u) \, \mathrm{d}\varphi(u) = V(L,K,\dots,K) \geqslant V(L)^{1/d} V(K)^{1-(1/d)},$$

where $V(\cdot,\dots,\cdot)$ is the mixed volume and $V(\cdot)$ is the volume. The equality sign in the inequality holds if and only if K and L are homothetic. It follows that up to a homothety the convex body K is characterized by the fact that at K the functional

$$L \mapsto \int_{S^{d-1}} h(L,u) \, \mathrm{d}\varphi(u)$$

attains a minimum under the condition $V(L) = 1$. It is not difficult to show that this minimum problem has a solution K, and it is then possible, but less immediate, to show by some variational argument that $S_{d-1}(\mu K,\cdot) = \varphi$ for a suitable number $\mu > 0$. The variational argument was given by Aleksandrov (1938a).

Minkowski first proved the special case of Theorem 5.1 referring to polytopes. We state it as a separate theorem.

Theorem 5.2. *Let $u_1,\dots,u_N \in S^{d-1}$ be pairwise distinct unit vectors linearly spanning $\mathbb{E}^d$, and let $f_1,\dots,f_N$ be positive real numbers satisfying*

$$\sum_{i=1}^{N} f_i u_i = 0. \tag{5.3}$$

Then there is a convex polytope $P \in \mathcal{K}_0^d$ for which $u_1,\dots,u_N$ are the outer normal vectors of the facets and f_i is the $(d-1)$-dimensional volume of the facet with normal vector u_i, for $i = 1,\dots,N$.

Minkowski (1897) (see also 1903, §9) proved this for $d = 3$ (the proof extends immediately to general $d \geqslant 2$), essentially by the method sketched above. For polytopes, the variational argument is considerably easier. The essential idea of Minkowski's proof can be interpreted as minimizing a linear functional over a convex region. Therefore, iterative methods for such optimization problems can be used to provide algorithmic versions for a constructive solution of Minkowski's problem for polytopes. Such an algorithm for $d = 3$ was described by Little (1983).

Slightly different versions of Minkowski's proof for polytopes were given by Aleksandrov (1950) and McMullen (1973). An alternative proof by a different idea, essentially a topological argument (and probably not leading to an algorithmic version), is due to Aleksandrov (1939a).

Minkowski himself (1903, §10) extended his existence theorem from polytopes to more general convex bodies. Since the notion of surface area measure was not yet available at that time, his result was restricted to convex bodies K for which $S_{d-1}(K,\cdot)$ has a continuous density with respect to spherical Lebesgue measure. The general version of Theorem 5.1 was obtained by Fenchel and Jessen (1938) in their paper where they introduced the surface area measures. For their extension, they used approximation and weak continuity, as did Aleksandrov (1939b) in a later paper. Independently, Aleksandrov (1938a) had given the proof mentioned above.

Similar existence results for nonclosed convex hypersurfaces are occasionally mentioned in the literature. Unbounded polyhedra and convex surfaces are treated in Aleksandrov (1958); a similar existence result for convex caps was obtained by Busemann (1959).

In differential geometry, Minkowski's existence problem takes a different form. Let K be a convex body of class C_+^2 with $S_{d-1}(K,\cdot) = \varphi$. By (3.2), φ has a density with respect to spherical Lebesgue measure, namely P_{d-1}, the product of the principal radii of curvature, which is the reciprocal of the Gauss-Kronecker curvature H_{d-1}. Hence, the differential-geometric version of Minkowski's problem requires to show the existence of a convex body of class C_+^2 for which a given positive function on the sphere S^{d-1} (satisfying the condition equivalent to (5.1)) is the Gauss-Kronecker curvature as a function of the outer unit normal vector. Minkowski's theorem 5.1 does not give such a result immediately, since it would remain to show that K is sufficiently regular to have a Gauss-Kronecker curvature in the differential-geometric sense. For treatments of the differential-geometric version of Minkowski's problem we refer to the classical papers of Lewy (1938), Miranda (1939), Nirenberg (1953) and to the more recent achievements of Gluck (1972), and Cheng and Yau (1976), further to the books of Pogorelov (1973, 1978), the survey of Gluck (1975), and the article of Oliker (1982).

For the lower order surface area measures $S_j, 1 \leqslant j \leqslant d-2$, similar existence problems are unsolved, except for $j = 1$. In order that a measure φ on S^{d-1} be the measure $S_j(K,\cdot)$ for some convex body $K \in \mathcal{K}_0^d$, the conditions (5.1) and (5.2) are again necessary. That they are not sufficient, can be seen in different ways. For example, the inequality (4.11) puts additional restrictions on $S_j(K,\cdot)$. From Theorem 4.2 and a result of Larman and Rogers (1970), applied to the polar

body, it follows that the support of $S_j(K,\cdot)$, for $j < d-1$, is arcwise connected. Aleksandrov (1938a) showed that even a measure φ satisfying (5.1) and with a positive analytic density need not be a jth surface area measure for $j < d-1$.

The existence problem for $j = 1$ is called *Christoffel's problem*, since Christoffel (1865) first treated it, for $d = 3$ and under regularity assumptions. If K is a convex body whose support function is of class C^2 on $\mathbb{E}^d \setminus \{0\}$, then the first order surface area measure $S_1(K,\cdot)$ is the indefinite integral of P_1, the arithmetic mean of the principal radii of curvature, as a function of the outer unit normal vector. This function is obtained from $h = h_K|S^{d-1}$, the restriction of the support function, by

$$h + \frac{1}{d-1}\Delta h = P_1, \tag{5.4}$$

where Δ denotes the Laplace-Beltrami operator on the sphere S^{d-1}. For given P_1, one may solve the elliptic linear differential equation (5.4) and then express the condition that h has to be the restriction of a positively homogeneous convex function, in terms of conditions on P_1; see Firey (1967). By extending this procedure to general convex bodies, Firey (1968) was able to find necessary and sufficient conditions for a measure φ on S^{d-1} to be the first order area measure of a convex body. Independently, Berg (1969) found such conditions within his theory of potentials and subharmonic functions on the sphere. For convex polytopes, an entirely different and elementary treatment of Christoffel's (generalized) existence problem is possible, see Schneider (1977).

Let $\mathcal{S}_j$ denote the set of all surface area measures of order j of d-dimensional convex bodies in $\mathbb{E}^d$. Clearly $\mathcal{S}_j$ is a cone, and for $j = 1$ and $j = d-1$ it is convex. For $j = 1$, this is trivial, since

$$S_1(K,\cdot) + S_1(L,\cdot) = S_1(K+L,\cdot).$$

If $K, L \in \mathcal{K}_0^d$ are convex bodies, the sum of their $(d-1)$st surface area measures satisfies the sufficient conditions of Theorem 5.1, hence there is a convex body, denoted by $K\sharp L$, for which

$$S_{d-1}(K,\cdot) + S_{d-1}(L,\cdot) = S_{d-1}(K\sharp L,\cdot).$$

The body $K\sharp L$, which is unique up to a translation, is usually called the *Blaschke sum* of K and L, because this kind of addition was mentioned by Blaschke (1956, p. 112); for polytopes, however, it was already introduced by Minkowski (1897, p. 117). References to applications of Blaschke sums in the literature are given in the survey by Schneider (1979b, p. 48); more recent applications are found, e.g., in McMullen (1980), and Lutwak (1988, 1991).

For $2 \leqslant j \leqslant d-2$, no complete characterization of the set $\mathcal{S}_j$ is known. It is definitely not convex, that is, there exist convex bodies $K, L \in \mathcal{K}_0^d$ such that $S_j(K,\cdot) + S_j(L,\cdot)$ is not the jth surface area measure of any convex body. This was proved independently by Fedotov (1979, 1982), and by Goodey and Schneider (1980). Further available information on $\mathcal{S}_j$ is only of a qualitative nature, for example, concerning the position of $\mathcal{S}_j$ within the set $\mathcal{M}$ of all finite measures φ

on $\mathcal{B}(S^{d-1})$ satisfying $\int_{S^{d-1}} u \, d\varphi(u) = 0$. Weil (1980) remarked that $\mathcal{S}_j$ is not dense in $\mathcal{M}$ (endowed with the weak topology) for $j < d-1$, and he proved that $\mathcal{S}_j - \mathcal{S}_j$ is dense in $\mathcal{M} - \mathcal{M}$. On the other hand, Goodey (1981) showed for $j < d-1$ that $\mathcal{S}_j - \mathcal{S}_j$ is a set of first category in the Banach space $\mathcal{M} - \mathcal{M}$ with the total variation norm. He also has results on the weak closure of $\mathcal{S}_j$.

Firey (1970) gave necessary and sufficient conditions on a function on S^{d-1} in order that it be the elementary symmetric function P_j of a sufficiently regular convex body of revolution.

We turn to curvature measures, where less is known. The following existence theorem for the curvature measure C_0, due to Aleksandrov (1939a), can be considered as a counterpart to Minkowski's theorem. For a convex body K with $0 \in \operatorname{int} K$, the *radial map*

$$\rho_K : S^{d-1} \to \operatorname{bd} K$$

is defined by demanding that $\rho_K(u) = \lambda u$ with $\lambda > 0$ and $\lambda u \in \operatorname{bd} K$.

Theorem 5.3. *Let κ be a measure on $\mathcal{B}(S^{d-1})$ with the following properties:*
(a) $\kappa(S^{d-1}) = \lambda_{d-1}(S^{d-1})$,
(b) $\kappa(S^{d-1} \backslash \omega) > \lambda_{d-1}(\omega^*)$, *if $\omega \subset S^{d-1}$ is a spherically convex, compact set and ω^* is its polar set.*
Then there exists a convex body $K \in \mathcal{K}^d$ with $0 \in \operatorname{int} K$ such that $C_0(K, \rho_K(\omega)) = \kappa(\omega)$ for $\omega \in \mathcal{B}(S^{d-1})$.

The conditions (a), (b) are also necessary for the existence of K. The body K is unique up to a dilatation with centre 0 (see the next section). Aleksandrov proved the theorem first for polytopes, where he used his "mapping lemma", essentially an application of the invariance of domain. The general case is then obtained by approximation.

Methods developed in the proof of this theorem and of its generalizations and analogues have led to a geometric theory of elliptic Monge-Ampère equations. We refer the reader to the books of Pogorelov (1964, 1973, ch. VIII), Bakel'man (1965, ch. IV), and also to the paper of Oliker (1986).

Treibergs (1990) proved a kind of a priori estimate for the solution in Aleksandrov's theorem. For a given number $\alpha \in (0, \pi/2)$ and for $\omega \subset S^{d-1}$, let ω_α be the set of all points of S^{d-1} at spherical distance at most α from ω. If in addition to the assumptions of Theorem 5.3 one assumes that

$$\kappa(\omega) \leqslant \lambda_{d-1}(\omega_\alpha) \quad \text{for } \omega \in \mathcal{B}(S^{d-1}),$$

then Treibergs showed the existence of a constant $c(\alpha, d)$ depending only on α and d such that the solution K in Theorem 5.3 satisfies

$$\frac{\sup\{\| \rho_K(u) \| : u \in S^{d-1}\}}{\inf\{\| \rho_K(u) \| : u \in S^{d-1}\}} \leqslant c(\alpha, d).$$

6. Uniqueness and stability results

The main classical uniqueness result involving surface area measures is the following *Theorem of Aleksandrov–Fenchel–Jessen.*

Theorem 6.1. *Let $j \in \{1, \dots, d-1\}$, and let $K, L \in \mathcal{K}^d$ be convex bodies of dimension at least $j+1$ for which*

$$S_j(K, \cdot) = S_j(L, \cdot). \tag{6.1}$$

Then K and L are translates of each other.

As a special case, this implies the following differential-geometric uniqueness theorem. Suppose that K and L are of class C_+^2 and that the jth elementary symmetric functions of the principal radii of curvature of K and L coincide at points with the same outer unit normal vector. Then K and L differ only by a translation. This special case can also be proved by a differential-geometric argument, see Chern (1959). The general Theorem 6.1 was proved independently by Aleksandrov (1937b), and Fenchel and Jessen (1938). Both proofs, which apply similar principles, rely on the Aleksandrov–Fenchel inequalities for mixed volumes and are, in particular, closely connected to the general Brunn–Minkowski theorem. Mixed volumes can also be used to obtain variants of the Aleksandrov–Fenchel–Jessen theorem for convex hypersurfaces with boundaries, see Busemann (1959) and Schneider (1989a).

Theorem 6.1 contains as cases $j = d-1$ and $j = 1$, respectively, the uniqueness assertions for the problems of Minkowski and Christoffel.

The assumption (6.1) can be generalized to one involving mixed area measures. Let a number $j \in \{1, \dots, d-1\}$ and $d-1-j$ convex bodies $C_{j+1}, \dots, C_{d-1}$ be given. If $K, L \in \mathcal{K}^d$ are convex bodies of dimension at least $j+1$ satisfying

$$S(\underbrace{K, \dots, K}_{j}, C_{j+1}, \dots, C_{d-1}, \cdot) = S(\underbrace{L, \dots, L}_{j}, C_{j+1}, \dots, C_{d-1}, \cdot), \tag{6.2}$$

does it follow that K and L are translates? This can certainly not be deduced without additional assumptions on $C_{j+1}, \dots, C_{d-1}$. Aleksandrov (1938b) proved that the answer is affirmative if $C_{j+1}, \dots, C_{d-1}$ are of class C_+^2. Combining the classical proof of Theorem 6.1 with a recent result of Schneider (1990a) on the equality case in the Aleksandrov–Fenchel inequalities, one can show that it suffices to assume that $C_{j+1}, \dots, C_{d-1}$ are smooth.

If the equality (6.1) is not satisfied exactly, but only to within an error of size ε, it is still possible to deduce that K and L are almost translates, with an explicit estimate for their translative deviation in terms of ε. Results of this kind are usually called *stability results*. Often such stability estimates involve constants that can be controlled only if degeneracies are avoided. For a convenient formulation, we denote by $\mathcal{K}^d(r, R)$ the set of convex bodies in $\mathcal{K}^d$ with inradius at least r and circumradius at most R, where $0 < r < R$ are given numbers. It is not a severe restriction to consider only bodies of a given class $\mathcal{K}^d(r, R)$.

Theorem 6.2. *Let $j \in \{1, \dots, d-1\}$ and $\varepsilon_0, r, R > 0$ be given. Let $K, L \in \mathcal{K}^d(r, R)$ be convex bodies for which*

$$| S_j(K, \omega) - S_j(L, \omega) | \leqslant \varepsilon \tag{6.3}$$

for all $\omega \in \mathcal{B}(S^{d-1})$ with some $\varepsilon \in [0, \varepsilon_0]$. Then

$$\delta^{\mathrm{H}}(K, L') \leqslant \gamma \varepsilon^{\alpha} \quad \text{with } \alpha = \frac{1}{(d+1)2^{j-1}} \tag{6.4}$$

for a suitable translate L' of L, with a constant γ that depends only on d, ε_0, r, R.

The proof, given by Schneider (1989b), uses some manipulations with the Aleksandrov–Fenchel inequalities to deduce from (6.3) the inequality

$$v(K, L)^2 - v(K, K)v(L, L) \leqslant c\varepsilon^{(1/2)^{j-1}},$$

where $v(K, L) := V(K, L, B^d, \dots, B^d)$ and c is a constant depending on d, ε_0, r, R. That this implies the estimate (6.4), follows from a stability result for the special Aleksandrov–Fenchel inequality $v(K, L)^2 \geqslant v(K, K)v(L, L)$ (see chapter 1.4, in particular its section 4.2).

A stability version of the Aleksandrov–Fenchel–Jessen theorem was first proved by Diskant (1985), but under a strong additional assumption and with a smaller exponent α in (6.4). The optimal (that is, biggest possible) exponent α in (6.4) is not known, but one has considerably sharper results for the special cases $j = d-1$ and $j = 1$.

Theorem 6.3. *Let $0 < r < R$ be given. Let $K, L \in \mathcal{K}^d(r, R)$ be convex bodies for which*

$$| S_{d-1}(K, \omega) - S_{d-1}(L, \omega) | \leqslant \varepsilon \tag{6.5}$$

for all $\omega \in \mathcal{B}(S^{d-1})$, with some $\varepsilon \in [0, \varepsilon_0]$. Then

$$\delta^{\mathrm{H}}(K, L') \leqslant \gamma \varepsilon^{1/d} \tag{6.6}$$

for some translate L' of L; here $\varepsilon_0 > 0$ and γ are suitable numbers depending on d, r, R.

This result is due to Diskant (1972). A slightly weaker version (with exponent $1/(d+2)$ instead of $1/d$) was proved earlier by Volkov (1963). Again the optimal exponent is unknown. Under regularity assumptions on K and L and with a stronger form of (6.5), Diskant (1979) achieved the exponent $1/(d-1)$. A stability result with optimal exponent is known for the case of the Christoffel problem (see Schneider 1979b for references):

Theorem 6.4. *Let $K, L \in \mathcal{K}^d$ be convex bodies with the same Steiner point. If*

$$| S_1(K, \omega) - S_1(L, \omega) | \leqslant \varepsilon \lambda_{d-1}(\omega) \tag{6.7}$$

for all $\omega \in \mathcal{B}(S^{d-1})$ with some $\varepsilon > 0$, then

$$\delta^{\mathrm{H}}(K, L) \leqslant \gamma \varepsilon,$$

where the constant γ depends only on d.

One should observe the factor $\lambda_{d-1}(\omega)$ on the right side of (6.7), which makes the assumption more restrictive for small sets ω.

Another special case where sharper stability results can be obtained is when one of the bodies is a ball. In this case, the powerful method of symmetrization can be applied. The following result of Diskant (1971) was obtained in this way.

Theorem 6.5. *Let $j \in \{1, \dots, d-1\}$; let $K \in \mathcal{K}^d$ be a convex body for which*

$$| S_j(K, \omega) - \lambda_{d-1}(\omega) | \leqslant \varepsilon \lambda_{d-1}(\omega) \tag{6.8}$$

for all $\omega \in \mathcal{B}(S^{d-1})$, with some $\varepsilon \in [0, \varepsilon_0]$. Then

$$\delta^{\mathrm{H}}(K, B) \leqslant \gamma \varepsilon^{1/(d-1)}$$

for some unit ball B; here $\varepsilon_0 > 0$ and γ are suitable numbers depending only on d.

The assumption (6.8) is satisfied, especially, if the convex body K is of class C_+^2 and its jth elementary symmetric function of the principal radii of curvature, P_j, satisfies $| P_j - 1 | \leqslant \varepsilon$. Thus Theorem 6.5 includes also a stability version of the classical differential-geometric result saying that $P_j =$ constant (on a closed convex hypersurface of class C_+^2) holds only for spheres. The characterization is easy to prove, e.g., by the method of integral formulas, but it appears that no differential-geometric proof of the stability result is known.

The differential-geometric characterization of spheres by the condition $P_j =$ constant has a counterpart in terms of principal curvatures: A closed convex hypersurface, smooth and of class C^2, on which H_j, the jth elementary symmetric function of the principal curvatures, is constant for some $j \in \{1, \dots, d-1\}$, is necessarily a sphere. For $d = 3$, the theorem is usually associated with the name of Liebmann. The condition $H_j = \alpha$, for a convex body K of class C_+^2, is equivalent to the condition

$$C_{d-1-j}(K, \cdot) = \alpha C_{d-1}(K, \cdot) \tag{6.9}$$

for curvature measures, and also to the condition

$$S_{d-1-j}(K, \cdot) = \alpha S_{d-1}(K, \cdot) \tag{6.10}$$

for surface area measures. This follows from (3.1) and (3.2), together with the fact that the principal curvatures at $x \in \operatorname{bd} K$ are the reciprocal values of the principal radii of curvature, taken at the outer unit normal vector at x. The conditions (6.9) and (6.10) make sense for arbitrary convex bodies K and may thus replace the condition $H_j = \alpha$ in the general case. In this way, one obtains a general form of Liebmann's theorem, without any differentiability assumptions.

Theorem 6.6. *Let $j \in \{1, \dots, d-1\}$; let $K \in \mathcal{K}_0^d$ be a convex body for which*

$$C_{d-1-j}(K, \cdot) = \alpha C_{d-1}(K, \cdot) \tag{6.11}$$

with some constant α. Then K is a ball.

On the other hand, condition (6.10) does not characterize balls.

Theorem 6.7. *Let $j \in \{1, \dots, d-1\}$. A convex body $K \in \mathcal{K}_0^d$ satisfies*

$$S_{d-1-j}(K, \cdot) = \alpha S_{d-1}(K, \cdot) \tag{6.12}$$

with some positive constant α if and only if K is a $(d-1-j)$-tangential body of a ball.

Theorem 6.7 was proved by Schneider (1978b) and Theorem 6.6 by Schneider (1979a). The proof of Theorem 6.7 exploits the connection with the theory of mixed volumes, in particular information on the equality case in special Aleksandrov–Fenchel inequalities. For the proof of Theorem 6.6 one first shows that (6.9) implies (6.10), then applies Theorem 6.7, and finally one has to show (using (4.1) and Theorem 2.1) that (6.9) cannot hold on tangential bodies of balls which are not balls.

The case $j = d-1$ of Theorem 6.6 admits also a stability version.

Theorem 6.8. *Let $K \in \mathcal{K}_0^d$ be a convex body for which*

$$| C_0(K, \beta) - C_{d-1}(K, \beta) | \leqslant \varepsilon C_{d-1}(K, \beta) \tag{6.13}$$

for all $\beta \in \mathcal{B}(\mathbb{E}^d)$, with some $\varepsilon \in [0, \frac{1}{2}]$. Then

$$\delta^{\mathrm{H}}(K, B) \leqslant \gamma \varepsilon$$

for some unit ball B, where γ depends only on d.

This was proved by Diskant (1968), also using symmetrization. The assumption (6.13) is satisfied, in particular, if K is a smooth body of class C^2 whose Gauss–Kronecker curvature satisfies $| H_{d-1} - 1 | \leqslant \varepsilon$. A stability result for convex bodies of class C^2 whose mean curvature satisfies $| H_1 - 1 | \leqslant \varepsilon$, is proved in Schneider (1990b), using the theory of mixed volumes; in this case, however, no extension to general convex bodies is known.

We conclude this article with a uniqueness result in terms of curvature measures that holds for pairs K, L of convex bodies. In this case, the correspondence between boundary points of K and L is not established by means of coinciding normal vectors as in the Aleksandrov–Fenchel–Jessen theorem, but by coinciding rays through the origin.

Theorem 6.9. *Let $j \in \{0, \dots, d-1\}$, let $K, L \in \mathcal{K}_0^d$ be convex bodies with the origin in the interior, and define the radial projection $f : \operatorname{bd} K \to \operatorname{bd} L$ so that $f(x) = \lambda(x)x$ with $\lambda(x) > 0$ and $f(x) \in \operatorname{bd} L$ for $x \in \operatorname{bd} K$. If*

$$C_j(K, \beta) = C_j(L, f(\beta \cap \operatorname{bd} K))$$

for all $\beta \in \mathcal{B}(\mathbb{E}^d)$, then $K = L$ if $j > 0$, while $K = \lambda L$ with some $\lambda > 0$ if $j = 0$.

The case $j = 0$, which is the uniqueness theorem corresponding to Aleksandrov's existence result, Theorem 5.3, was proved by Aleksandrov (1942). The case $j > 0$ was treated by Schneider (1978a).

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CHAPTER 1.9

The Space of Convex Bodies

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Contents

1. Introduction 303
2. Lattice structure 303
3. Semigroup structure and embedding 304
4. Metrics 307
5. Topology 310
6. Measure and category 311
7. Entropy and generalized dimension 311
References 312

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1. Introduction

A *convex body* in $\mathbb{E}^d$ is a compact convex subset of $\mathbb{E}^d$. It is called *proper* if it has non-empty interior. Let $\mathcal{K} = \mathcal{K}(\mathbb{E}^d)$ denote the space of all convex bodies in $\mathbb{E}^d$ (sometimes including $\emptyset$) and let $\mathcal{K}_p = \mathcal{K}_p(\mathbb{E}^d)$ be the subspace of all proper convex bodies.

The spaces $\mathcal{K}, \mathcal{K}_p$ and some of their subspaces have been investigated for many decades, an early result being Blaschke's selection theorem. The main features of these investigations will be presented below. We will proceed from algebraic to geometric and, finally, to more analytic topics.

2. Lattice structure

Let conv stand for convex hull. The definitions

$$C \wedge D = C \cap D, \qquad C \vee D = \operatorname{conv}(C \cup D)$$

for $C, D \in \mathcal{K}$, make $\mathcal{K}$ into an (algebraic) lattice $\langle \mathcal{K}, \wedge, \vee \rangle$ with the empty set $\emptyset$ as zero. This lattice is atomic and the atoms are precisely the convex bodies consisting of single points.

Lattices of convex bodies were studied mainly by Belgian mathematicians. In most of their results $\mathbb{E}^d$ is replaced by topological vector spaces of arbitrary dimensions.

A mapping $E: \mathcal{K} \to \mathcal{K}$ is an *endomorphism* of the lattice $\langle \mathcal{K}, \wedge, \vee \rangle$ if for all $C, D \in \mathcal{K}$

$$E(C \wedge D) = E(C) \wedge E(D), \qquad E(C \vee D) = E(C) \vee E(D).$$

Refining the finite-dimensional cases of earlier results of Fourneau (1975, 1979; see also 1976b), Gruber (1991) gave a characterization of endomorphisms:

Theorem 1. *A mapping $E: \mathcal{K} \to \mathcal{K}$ is an endomorphism of the lattice $\langle \mathcal{K}, \wedge, \vee \rangle$ precisely in the following cases:*

(i) *There is a fixed convex body $D \in \mathcal{K}$ such that $E(C) = D$ for each $C \in \mathcal{C}$.*

(ii) *$d = 1$ and there is a strictly monotone function $\varphi: \mathbb{R} (= \mathbb{E}^1) \to \mathbb{R}$ such that*
   (a) *$E(\emptyset) = \emptyset$,*
   (b) *$E(x)$ is a (compact) interval (possibly a point) contained in the interval $\varphi(x - 0) \vee \varphi(x + 0)$ for each $x \in \mathbb{R}$,*
   (c) *$E(x \vee y) = E(x) \vee E(y)$ for each interval $x \vee y \subset \mathbb{R}$.*

(iii) *$d > 1$ and there is a non-singular affine transformation $a: \mathbb{E}^d \to \mathbb{E}^d$ such that $E(C) = a(C)$ for each $C \in \mathcal{K}$.*

For a related result for norms see the author (1992b).

Fourneau (1984a) gave a characterization of those pairs $C, D \in \mathcal{K}$ for which C is *subperspective* to D, i.e.,

$$C \subset D \vee E, \quad C \wedge E = \emptyset \qquad \text{for suitable } E \in \mathcal{K}.$$

Filters were considered by Fourneau (1984b), and Fourneau (1976a) also investigates relations between lattice isomorphisms and semigroup isomorphisms from $\mathcal{K}(\mathbb{E}^c)$ onto $\mathcal{K}(\mathbb{E}^d)$. (For the semigroup structure see below.)

It seems that many further results on lattices of convex bodies still wait for their discovery. One problem is the following: characterize the lattice $\langle \mathcal{K}, \wedge, \vee \rangle$ or sublattices of it by algebraic and geometric properties.

3. Semigroup structure and embedding

For $C, D \in \mathcal{K}$ ($\mathcal{K}$ without $\emptyset$) and $\alpha \geqslant 0$ let

$$C + D = \{x + y : x \in C, y \in D\}, \qquad \alpha C = \{\alpha x : x \in C\}.$$

($C + D$ is called the *Minkowski sum* of C, D.) Then $\langle \mathcal{K}, + \rangle$ is an Abelian semigroup with zero $\{o\}$ (where o is the origin of $\mathbb{E}^d$) and with operators (multiplication with non-negative reals). It thus can be considered a convex cone.

In $\langle \mathcal{K}, + \rangle$ the *cancellation law* holds: for $C, D, E \in \mathcal{K}$

$$C + D = C + E \text{ yields } D = E$$

(cf. Rådström 1952, Pinsker 1966). Borwein and O'Brien (1978) proved that the cancellation law implies convexity: for any non-convex non-empty compact set D in $\mathbb{E}^d$, there are $C, E \in \mathcal{K}$ such that $C + D = C + E$ in spite of $D \neq E$.

A convex body C is *indecomposable* if for $D, E \in \mathcal{K}$

$$C = D + E \text{ implies } D, E \text{ are non-negative homothets of } C$$

(see Gale 1954). In geometric terms indecomposability of C means that the ray $\alpha C : \alpha \geqslant 0$ is an extreme ray of the convex cone $\mathcal{K}$. For results on indecomposability of convex bodies and, in particular, of polytopes see, among others, Gale (1954), Shephard (1963), Meyer (1974), Weil (1974b), Smilansky (1986, 1987), McMullen (1987), and Yost (1991). Schneider (1976) communicated that most convex bodies (in the sense of Baire categories; see section 6 below and chapter 4.10) are indecomposable.

Different extremal results for norms are due to Grzaślewicz (1984, 1988, 1990).

There is a series of results on homomorphic mappings of the semigroup $\langle \mathcal{K}, + \rangle$ into the semigroups $\langle \mathcal{K}, + \rangle$, $\langle \mathbb{E}^d, + \rangle$, $\langle \mathbb{R}, + \rangle$. We state a characteristic result for each of these cases and refer to McMullen and Schneider (1983, § 13) for a survey of further results. From now on we assume that $\mathcal{K}$ is endowed with the topology induced by the Hausdorff metric δ^{H} (see section 4).

Schneider (1974) proved the following result.

Theorem 2. *Let $d \geqslant 3$ and let $H: \mathcal{K} \to \mathcal{K}$ have the following properties:*
(i) $H(C+D) = H(C) + H(D)$ *for all* $C, D \in \mathcal{K}$,
(ii) H *is continuous,*
(iii) $H(a(C)) = a(H(C))$ *for any* $C \in \mathcal{K}$ *and any surjective affinity* $a: \mathbb{E}^d \to \mathbb{E}^d$.
Then there is a real constant $\lambda \geqslant 0$ *such that*

$$H(C) = C + \lambda(C - C) \quad \text{for all } C \in \mathcal{K}.$$

$C - C = \{x - y: x, y \in C\}$ is called the *difference body* of C. The *Steiner point* $s(C)$ of a convex body C is defined by:

$$s(C) = \frac{1}{\kappa_d} \int_{S^{d-1}} u h_C(u)\, \mathrm{d}\sigma(u),$$

where κ_d is the volume of the solid Euclidean unit ball $\mathbb{B}^d$ of $\mathbb{E}^d$, S^{d-1} denotes the $(d-1)$-dimensional unit sphere of $\mathbb{E}^d$, and σ is the ordinary surface area measure on S^{d-1}. Finally, $h_C: S^{d-1} \to \mathbb{R}$ is the *support function* of C, i.e.,

$$h_C(u) = \sup\{u \cdot x: x \in C\} \quad \text{for } u \in S^{d-1}.$$

Here the dot $\cdot$ stands for the inner product of $\mathbb{E}^d$. A characterization of the Steiner point is due to Schneider (1971) and Posicel'skii (1973):

Theorem 3. *Let $h: \mathcal{K} \to \mathbb{E}^d$ be a mapping with the following properties:*
(i) $h(C+D) = h(C) + h(D)$ *for all* $C, D \in \mathcal{K}$,
(ii) h *is continuous,*
(iii) $h(m(C)) = m(h(C))$ *for any* $C \in \mathcal{K}$ *and any rigid motion* m *of* $\mathbb{E}^d$.
Then $h = s$.

Next we state a result of Hadwiger (1957, p. 213).

Theorem 4. *Let $\eta: \mathcal{K} \to \mathbb{R}$ satisfy the following properties:*
(i) $\eta(C+D) = \eta(C) + \eta(D)$ *and* $\eta(\alpha C) = \alpha \eta(C)$ *for all* $C, D \in \mathcal{K}$ *and* $\alpha \geqslant 0$,
(ii) η *is continuous,*
(iii) $\eta(m(C)) = \eta(C)$ *for all* $C \in \mathcal{K}$ *and any rigid motion* m *of* $\mathbb{E}^d$.
Then $\eta = \alpha W$, *where* α *is a real constant and* W *denotes the mean width:*

$$W(C) = \frac{1}{d\kappa_d} \int_{S^{d-1}} [h_C(u) + h_C(-u)]\, \mathrm{d}\sigma(u) \quad \text{for } C \in \mathcal{K}.$$

The mapping $C \to h_C$ is injective and maps the convex cone $\mathcal{K}$ onto the set $\mathcal{H}$

of support functions on S^{d-1}. Note that $\mathcal{H}$ is a closed convex cone in the space $\mathcal{C}(S^{d-1})$ of real continuous functions on S^{d-1}. Clearly,

$$h_{C+D} = h_C + h_D, \quad h_{\alpha C} = \alpha h_C \qquad \text{for } C, D \in \mathcal{H}, \ \alpha \geqslant 0.$$

$\mathcal{L} = \mathcal{H} - \mathcal{H}$ is dense in $\mathcal{C}(S^{d-1})$, see Hörmander (1955). The geometric counterpart of $\mathcal{L}$ is the space of all pairs of convex bodies modulo the congruence relation

$$(C, D) \equiv (E, F) \text{ if } C + F = D + E,$$

see, e.g., Pinsker (1966). For general results in this context see Radström (1952), Hörmander (1955), Schmidt (1986a,b). A characterization of $\mathcal{H}$ in $\mathcal{L}$ is due to Weil (1974a).

Any (multi-)linear functional on $\mathcal{H}$ (on $\mathcal{H} \times \cdots \times \mathcal{H}$) can be extended (multi-) linearly to $\mathcal{L}$ (to $\mathcal{L} \times \cdots \times \mathcal{L}$); in particular this holds for mixed volumes, mixed surface area measures and McMullen's mixed valuations. Such extensions were applied by Aleksandrov (1937) to obtain results on mixed volumes for convex bodies. Mixed volumes and mixed surface area measures lack certain continuity properties and thus cannot be extended to $(\mathcal{C}(S^{d-1}))^d$, respectively to $(\mathcal{C}(S^{d-1}))^{d-1}$, but may be extended to distributions; see Weil (1981), McMullen and Schneider (1983), Goodey and Weil (1984), and Burago and Zalgaller (1988).

Ewald and Shephard (1966) and Ewald (1965/66) proposed a congruence relation on $\mathcal{H}$ which can also be used to define a linear structure:

$$C \equiv D \text{ if } C + S = D + T \quad \text{for suitable centrally symmetric } S, T \in \mathcal{H}.$$

The mapping $C \to h_C - h_{-C}$ maps the space of congruence classes $\mathcal{H}/\equiv$ isomorphically into a dense subspace of $\mathcal{C}(S^{d-1})$. See also Geivaerts (1976).

A second concept of linear combination of convex bodies is due to Blaschke: to each convex body $C \in \mathcal{H}_p$ there is associated a Borel measure $S(C, \cdot)$ on the unit sphere S^{d-1} in the following way. Let $\sigma : \operatorname{bd} C \to S^{d-1}$ be the (possibly multi-valued) mapping which maps each $x \in \operatorname{bd} C$ onto the set of exterior normal unit vectors of supporting hyperplanes of C at x. For a Borel set $\omega \subset S^{d-1}$ let $S(C, \omega)$ be the $(d-1)$-dimensional Hausdorff measure of $\sigma^{-1}(\omega)$. The measure $S(C, \cdot)$ is (sometimes) called the (*Gaussian*) *curvature measure* of C. It determines C up to translations. A theorem of Minkowski says the following: let μ be a finite positive Borel measure on S^{d-1} not concentrated on a great circle and such that componentwise

$$\int_{S^{d-1}} u \, \mathrm{d}\mu(u) = o.$$

Then there is a convex body $C \in \mathcal{H}_p$ with $S(C, \cdot) = \mu$, unique up to translations. For $C, D \in \mathcal{H}_p$ and $\lambda > 0$, the curvature measures $S(C, \cdot) + S(D, \cdot)$ and $\lambda S(C, \cdot)$

determine up to translations convex bodies $C \dot{+} D$ and $\lambda \cdot C$, respectively. These are called the *Blaschke sum* of C, D and the *Blaschke scalar product* of C with λ. ($\lambda \cdot C = \lambda^{1/(d-1)} C$ up to translations.) $\langle \mathcal{K}_p, \dot{+}, \mathbb{R}^+ \rangle$ is a convex cone. For references on Blaschke addition, up to 1979, see Firey (1967) and Schneider (1979, §10). Selected further references are Lutwak (1986, 1988, 1990, 1991), Oliker (1992) and Petty (1974).

An embedding different from the above embedding $C \to h_C$ of $\mathcal{K}$ into $\mathcal{C}(S^{d-1})$ is due to Nadler (1978): let C be a convex body of dimension $\geqslant 2$ and denote by $\mathcal{K}(C)$ the subspace of $\mathcal{K}$ consisting of all convex bodies contained in C. Then $\mathcal{K}(C)$ is homeomorphic to the Hilbert cube Q and $\mathcal{K}$ ($d \geqslant 2$) is homeomorphic to Q with one point removed. Montejano (1987) showed that for an open set U in $\mathbb{E}^d$ ($d \geqslant 2$) the space $\mathcal{K}(U)$ of all convex bodies in U is homeomorphic to $U \times Q \times [0,1)$.

4. Metrics

Lists of metrics or other notions of distance on $\mathcal{K}$ or subspaces or quotient spaces of it are contained in Grünbaum (1963), Shephard and Webster (1965), and Gruber (1983).

The (*Pompeiu–*) *Hausdorff* (*–Blaschke*) *metric* δ^{H} on $\mathcal{K}$ is defined as follows:

$$\delta^{\mathrm{H}}(C,D) = \max\{\sup_{x\in C} \inf_{y\in D} |x-y|, \sup_{y\in D} \inf_{x\in C} |x-y|\},$$

where $|\cdot|$ denotes the Euclidean norm on $\mathbb{E}^d$. Related to it is the (*Pompeiu–*) *Eggleston metric* δ^{E}:

$$\delta^{\mathrm{E}}(C,D) = \sup_{x\in C} \inf_{y\in D} |x-y| + \sup_{y\in D} \inf_{x\in C} |x-y|.$$

In terms of support functions, respectively using Minkowski addition,

$$\begin{aligned}
\delta^{\mathrm{H}}(C,D) &= \sup_{u\in S^{d-1}} |h_C(u) - h_D(u)| = \|h_C - h_D\|_\infty \\
&= \inf\{\lambda \geqslant 0 : C \subset D + \lambda \mathbb{B}^d, D \subset C + \lambda \mathbb{B}^d\}, \\
\delta^{\mathrm{E}}(C,D) &= \max\{0, \sup_{u\in S^{d-1}} h_C(u) - h_D(u)\} \\
&\quad + \max\{0, \sup_{u\in S^{d-1}} h_D(u) - h_C(u)\} \\
&= \inf\{\lambda \geqslant 0 : C \subset D + \lambda \mathbb{B}^d\} + \inf\{\lambda \geqslant 0 : D \subset C + \lambda \mathbb{B}^d\}.
\end{aligned}$$

Clearly, $\delta^{\mathrm{H}}, \delta^{\mathrm{E}}$ may be defined using any norm on $\mathbb{E}^d$ instead of $|\cdot|$. More generally, they can be defined for the space of bounded closed subsets of an arbitrary metric space.

Refining earlier work of Schneider (1975), Gruber and Lettl (1980) characterized the isometries of $\langle \mathcal{K}, \delta^{\mathrm{H}} \rangle$:

Theorem 5. *A mapping $I:\mathcal{K}\to\mathcal{K}$ is an isometry with respect to δ^{H} precisely when there is a rigid motion m of $\mathbb{E}^d$ and a convex body $D\in\mathcal{K}$ such that $I(C)=m(C)+D$ for all $C\in\mathcal{K}$.*

For related results, see Gruber and Lettl (1979), Lettl (1980), Gruber and Tichy (1982), and Bandt (1986).

A *segment* (in the sense of Menger) in a metric space $\langle M,\varrho\rangle$ is a curve in M with a continuous parameterization $p(\tau)$: $\tau\in[0,1]$ such that for any $\tau\in[0,1]$

$$\varrho(p(0),p(\tau))+\varrho(p(\tau),p(1))=\varrho(p(0),p(1)).$$

Answering a question of Jongmans, Schneider (1981) characterized segments in $\langle\mathcal{K},\delta^{\mathrm{H}}\rangle$ in the following way.

Theorem 6. *Two convex bodies $C,D\in\mathcal{K}$ are connected in the space $\langle\mathcal{K},\delta^{\mathrm{H}}\rangle$ by a unique segment with respect to δ^{H} if and only if either*

(i) *$C=D+\lambda\mathbb{B}^d$ for suitable $\lambda\geqslant 0$, or*

(ii) *$D=C+\mu\mathbb{B}^d$ for suitable $\mu\geqslant 0$, or*

(iii) *C and D lie in parallel hyperplanes and $C=D+x$, where the vector x is orthogonal to the hyperplanes.*

McClure and Vitale (1975) proposed (for $d=2$) the metric δ^p ($p\geqslant 1$): for $C,D\in\mathcal{K}$

$$\delta^p(C,D)=\left[\int_{S^{d-1}}|h_C(u)-h_D(u)|^p\,\mathrm{d}\sigma(u)\right]^{1/p}=\|h_C-h_D\|_p.$$

See also Gruber (1983).

Shephard (1960) used the following affine-invariant metric δ^{SH} on $\mathcal{K}_p$:

$$\delta^{\mathrm{SH}}(C,D)=\log\big(1+2\inf\{\lambda\geqslant 0\colon C\subset D+\lambda(D-D),\ D\subset C+\lambda(C-C)\}\big).$$

Let $V(\cdot),S(\cdot)$ denote volume and surface area. The *symmetric-difference* or *Nikodym metric* δ^{S} on $\mathcal{K}_p$ is defined as follows:

$$\delta^{\mathrm{S}}(C,D)=V(C\triangle D),$$

where $\triangle$ denotes the symmetric difference.

δ^{H} and δ^{S} induce the same topology on $\mathcal{K}_p$. The isometries with respect to δ^{S} were determined by Gruber (1978a):

Theorem 7. *A mapping $I:\mathcal{K}_p\to\mathcal{K}_p$ is an isometry with respect to δ^{S} precisely when there is a volume preserving affinity $a:\mathbb{E}^d\to\mathbb{E}^d$ such that $I(C)=a(C)$ for all $C\in\mathcal{K}_p$.*

Related results were obtained by Lettl (1980), Gruber (1982), and Tichy (1983). The problem of characterizing the isometries with respect to the symmetric difference metric for a general measure space was treated by Graf and Mägerl (1984), thereby verifying a conjecture of Gruber.

The following *measure of surface deviation* δ^{SD} on $\mathcal{K}_p$ is due to Eggleston (1957):

$$\delta^{SD}(C,D) = S(C \cup D) - S(C \cap D).$$

δ^{SD} fails to satisfy the triangle inequality and thus is not a metric.

Florian (1989) investigated the metric δ^1 (for general d), and remarked that it can be expressed in the following forms:

$$\delta^1(C,D) = \begin{cases} 2S(\operatorname{conv}(C \cup D)) - S(C) - S(D) & \text{for } d = 2, \\ d\kappa_d(2W(\operatorname{conv}(C \cup D)) - W(C) - W(D)) & \text{for } d \geqslant 2. \end{cases}$$

A version of δ^S is the *homogeneous symmetric difference metric* δ^{MS} on $\mathcal{K}_p$ introduced by Marczewski and Steinhaus (1958):

$$\delta^{MS}(C,D) = \delta^S(C,D)/V(C \cup D).$$

Note that δ^{MS} is bounded.

In the following, three bounded measures of distance are presented where the convex bodies are considered only up to linear transformations, respectively affinities.

The *Banach–Mazur metric* δ^{BM} on the space $\mathcal{K}_{po}/\sim$ of the equivalence classes of proper o-symmetric convex bodies with respect to linear transformations is defined as follows:

$$\delta^{BM}(C,D) = \log \inf\{\lambda \geqslant 1 : \exists l : \mathbb{E}^d \to \mathbb{E}^d \text{ linear}, C \subset l(D) \subset \lambda C\}.$$

It has attracted much interest in the theory of finite-dimensional normed spaces; see, e.g., Tomczak-Jaegermann (1989). John (1948) and Gluskin (1981) proved the following result.

Theorem 8. *The δ^{BM}-diameter of $\mathcal{K}_{po}/\sim$ is bounded between $\log d - \alpha$ and $\log d$, where $\alpha > 0$ is an absolute constant.*

Slightly more general than δ^{BM} is the metric δ^A on the space $\mathcal{K}_p/\approx$ of affine-equivalence classes in $\mathcal{K}_p$ due to Asplund (1960):

$$\delta^A(C,D) = \log \inf\{\lambda \geqslant 1 : \exists a : \mathbb{E}^d \to \mathbb{E}^d \text{ affine}, \\ t \in \mathbb{E}^d, C \subset a(D) \subset \lambda C + t\}.$$

Macbeath (1951) defined the following metric δ^M on $\mathcal{K}_p/\approx$:

$$\delta^M(C,D) = \log \inf\{|\det ab| : \exists a, b : \mathbb{E}^d \to \mathbb{E}^d \text{ affine}, \\ C \subset a(D), D \subset b(C)\}.$$

He gave the upper estimate $2d \log d$ for the δ^{M}-diameter of $\mathcal{K}_p/\approx$ and showed that $\langle \mathcal{K}_p/\approx, \delta^{\mathrm{M}} \rangle$ is compact.

There is a large body of various approximation results with respect to the metrics and measures of distance considered in this subsection. For a survey of results up to 1983, see the author (1983). Selected references to more recent work are Schneider (1987), Linhart (1989), Gruber (1992a) for δ^{H}; Bourgain and Lindenstrauss (1992), Bourgain, Lindenstrauss and Milman (1989), Gruber (1992a) for δ^{BM}; Florian (1986), Gruber (1992a) for δ^{S}; Florian (1992) for δ^1; Arnold (1989) for δ^2; and Arnold and Wellerding (1992) for Sobolev distance.

In recent years geometric stability problems of the following type were studied: assume that in a geometric inequality relating two convex bodies (such as the isoperimetric inequality or the Brunn–Minkowski inequality) the two sides differ only slightly. What can then be said about the deviation of the two bodies? For problems of this sort one typically uses measures of distance of the form

$$\begin{aligned} \vartheta(C,D) &= \inf\{\delta(C,D+t)\colon t \in \mathbb{E}^d\}, \text{ or} \\ &= \inf\{\delta(C,s(D))\colon s\colon \mathbb{E}^d \to \mathbb{E}^d \text{ similarity}\}, \text{ or} \\ &\ldots \end{aligned}$$

where δ is one of the metrics or measures of distance considered above. Some results of this type are due to Diskant (1972,1973), Groemer (1987,1988a), Bourgain and Lindenstrauss (1988), and Campi (1988). Compare also chapter 1.10.

Grünbaum (1963) listed bounds for certain measures of deviation. Shephard and Webster (1965) determined those metrics among $\delta^{\mathrm{H}}, \delta^{\mathrm{E}}, \delta^{\mathrm{S}}, \delta^{\mathrm{SH}}, \delta^{\mathrm{MS}}$ which are continuous with respect to a given one. Florian (1989) considered this problem for $\delta^1, \delta^{\mathrm{H}}$ and Vitale (1985) investigated the metrics δ^p $(p \geqslant 1)$.

For relations of the concepts of convergence determined by some of the above metrics and of other notions of convergence in $\mathcal{K}$ or $\mathcal{K}_p$, see Wijsman (1966), Salinetti and Wets (1979), Beer and Klee (1987), Beer (1989), Baronti and Papini (1986), and the references cited in these articles.

5. Topology

Let $\mathcal{K}$ be endowed with the topology induced by δ^{H} or some equivalent metric. The main topological result is the *selection* or *compactness theorem of Blaschke* (1916):

Theorem 9. *$\mathcal{K}$ is boundedly compact.*

Thus $\mathcal{K}$ is complete and $\mathcal{K}_p$ is locally compact. The proof published by Blaschke (1916) follows an idea of Carathéodory who essentially suggested to use ε-nets. Blaschke's original proof was based on a result of Hilbert which was a forerunner of the theorem of Arzelà–Ascoli. Bol (1965) and Heil (1967) proved that each of the results of Blaschke and Arzelà–Ascoli is a consequence of the other one.

Classical applications of the selection theorem show the existence of solutions of extremum problems (including, surprisingly, a proof of the isodiametric inequality). More recent is Groemer's (1971) proof of Mahler's selection theorem in the geometry of numbers.

Among the numerous generalizations we mention those of Urysohn (1921/22), Vietoris (1922), and Michael (1951). Particular classes of subsets of $\mathbb{E}^d$ were considered by Hirose (1965), Dreševič (1970), Spiegel (1975), Beer (1975), and Beer and Klee (1987).

Vitale (1985) considered the space $\langle \mathcal{K}, \delta^p \rangle$ and proved the corresponding compactness theorem, and Macbeath (1951) showed the compactness of $\langle \mathcal{K}_p / \approx, \delta^{\mathrm{M}} \rangle$.

6. Measure and category

A natural problem in convexity is to find tools by which one can distinguish between "large" and "small" sets of convex bodies. In this Handbook three such tools are discussed: measure, Baire categories and porosity. Let $\mathcal{K}$ be endowed with the topology induced by δ^{H}.

$\mathcal{K}$ being a locally compact space there are many Borel measures on it. Unfortunately they fail to satisfy a natural invariance property for Bandt and Baraki (1986) proved the following negative result.

Theorem 10. *Let $d > 1$. Then there is no non-trivial σ-finite Borel measure on $\mathcal{K}$ which is invariant with respect to all isometries of $\langle \mathcal{K}, \delta^{\mathrm{H}} \rangle$.*

This extends a result of Schneider (1978) on the non-existence of well-behaved Hausdorff measures on $\mathcal{K}$ – which in turn disproved a conjecture of the author (1978b).

Since $\mathcal{K}$ is complete and $\mathcal{K}_p$ locally compact, a version of Baire's category theorem shows that meager subsets of $\mathcal{K}$, respectively $\mathcal{K}_p$, may be considered "small"; a set being *meager* or of *first Baire category* if it is a countable union of nowhere dense sets. For a body of pertinent results, see chapter 4.10.

A concept between measure and category is that of σ-porous sets, see the same chapter.

7. Entropy and generalized dimension

Besides the tools mentioned in section 6, there exist several other means for measuring the size of a set. In the following, we treat ε-entropy and generalized Hausdorff dimension. While these seem to be too coarse to distinguish between interesting subsets of $\mathcal{K}$ or $\mathcal{K}_p$, they may be used to give an idea of the size of $\mathcal{K}$ and $\mathcal{K}_p$ or, rather, of the subspaces $\mathcal{K}(\mathbb{B}^d)$, respectively $\mathcal{K}_p(\mathbb{B}^d)$, of convex bodies contained in the solid unit ball.

The size of a totally bounded metric space is reflected in the growth rate of the

minimum number of elements of an ε-net as $\varepsilon \to +0$. (The 2-logarithm of this number is the *ε-entropy* of the space.)

Refining an estimate of Dudley (1974), Bronshtein (1976) proved the following result.

Theorem 11. *There are constants α, β, $0 < \alpha < \beta$, depending on d, such that for all sufficiently small $\varepsilon > 0$ any ε-net on the space $\langle \mathcal{H}(\mathbb{B}^d), \delta^H \rangle$ contains at least $\lceil 2^{\alpha/\varepsilon^{(d-1)/2}} \rceil$ elements and there are ε-nets consisting of at most $\lfloor 2^{\beta/\varepsilon^{(d-1)/2}} \rfloor$ elements.*

A similar (slightly weaker) result of Dudley (1974) applies to $\langle \mathcal{H}_p(\mathbb{B}^d), \delta^S \rangle$. We also refer to Bronshtein (1978).

Let $h: [0, +\infty[\to [0, +\infty[$ be a non-decreasing function which is continuous on the right and satisfies $h(\tau) = 0$ precisely for $\tau = 0$. Given a metric space, let μ_h denote its Hausdorff measure corresponding to h. If A is a subset of the metric space, call δ, $0 \leqslant \delta \leqslant +\infty$, its *Hausdorff dimension* if

$$\begin{aligned} &\mu_g(A) > 0 \quad \text{for } g(\tau) = \tau^\eta, \quad \text{where } \eta < \delta, \\ &\mu_h(A) < +\infty \quad \text{for } h(\tau) = \tau^\eta, \quad \text{where } \eta > \delta. \end{aligned}$$

In case the Hausdorff dimension of A is $+\infty$ (as holds for $\langle \mathcal{H}(\mathbb{B}^d), \delta^H \rangle$ and $\langle K_p(\mathbb{B}^d), \delta^S \rangle$) a pair of functions g, h $(g \geqslant h)$ satisfying

$$\mu_g(A) > 0, \qquad \mu_h(A) < +\infty,$$

still sheds some light on the "size" of A. Call such functions *g, h bounds for the dimension* of A. Theorem 11 together with a result of Bandt (1990), can be expressed as follows.

Theorem 12. *The functions g, h, defined by:*

$$g(\tau) = 2^{-\gamma/\tau^{(d-1)/2}(\log \tau)^2}, \quad h(\tau) = 2^{-\beta/(\tau/2)^{(d-1)/2}} \qquad \text{for } \tau > 0,$$

are bounds for the dimension of $\langle \mathcal{H}(\mathbb{B}^d), \delta^H \rangle$. Here $\beta > 0$ is as in Theorem 11 and $\gamma > 0$ is an arbitrary constant.

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CHAPTER 1.10

Aspects of Approximation of Convex Bodies

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Contents

1. Introduction 321
2. Some definitions 321
3. Properties of best approximating polytopes for $d = 2$ 323
4. Upper bounds 324
5. Asymptotic estimates 326
6. Algorithmic and asymptotic step-by-step approximation 332
7. Approximation of special (classes of) convex bodies 334
8. Symmetrization 336
9. Miscellanea 337
References 339

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1. Introduction

Approximation of convex bodies is frequently encountered in geometric convexity, discrete geometry, the theory of finite-dimensional normed spaces, in geometric algorithms and optimization, as well as in the realm of engineering. Approximation problems in optimization arise often from more practical problems of operations research and pattern recognition.

While we have tried to touch the more important topics in this area or say where they are discussed in detail, it is out of reach even to locate the major applications of approximation.

The emphasis of this report is on more recent developments, but many older results will also be described to make the survey more balanced. A comparison of this article with the book of L. Fejes Tóth (1953) and Gruber's (1983) survey reveals where progress took place in recent years.

For random approximation compare the reviews of Buchta (1985), Schneider (1988), Affentranger (1992) and chapter 5.2 by Weil and Wieacker. Results on the number of lattice points in "large" convex bodies may also be considered approximation results and are described by Gruber and Lekkerkerker (1987, pp. 140–147). For approximation results in the sense of Dvoretzky's sphericity theorem and related material on normed spaces we refer to Tomczak-Jaegermann (1989), Pisier (1989) and chapter 4.5 by Lindenstrauss and Milman. Approximation properties of "typical" convex bodies and a general principle underlying these are stated in chapter 4.10 by the author.

2. Some definitions

Denote by $\mathscr{C} = \mathscr{C}(\mathbb{E}^d)$ the space of all (compact, proper) convex bodies in $\mathbb{E}^d$. If $d = 2$ we also speak of convex discs. By $\mathscr{P}$ is meant the set of all convex polytopes in $\mathscr{C}$ and for $n = d+1, d+2, \ldots$, let $\mathscr{P}_n$ and $\mathscr{P}_{(n)}$ be the sets of convex polytopes in $\mathscr{C}$ having at most n vertices, respectively facets. Given $C \in \mathscr{C}$, the sets of those convex polytopes in $\mathscr{P}_n$ and $\mathscr{P}_{(n)}$ which are contained in C, respectively contain it are denoted by $\mathscr{P}_n^{\mathrm{i}}, \mathscr{P}_n^{\mathrm{c}}$ and $\mathscr{P}_{(n)}^{\mathrm{i}}, \mathscr{P}_{(n)}^{\mathrm{c}}$, respectively. Clearly, $\mathscr{P}_n = \mathscr{P}_{(n)}, \mathscr{P}_n^{\mathrm{i}} = \mathscr{P}_{(n)}^{\mathrm{i}}$ and $\mathscr{P}_n^{\mathrm{c}} = \mathscr{P}_{(n)}^{\mathrm{c}}$ if $d = 2$.

Let δ denote a metric or another measure of distance on $\mathscr{C}$ or on its subspace $\mathscr{C}_o$ consisting of all convex bodies which are centrally symmetric with respect to the origin o of $\mathbb{E}^d$. Call a polytope $P \in \mathscr{Q}_n$ where $\mathscr{Q}_n \in \{\mathscr{P}_n, \ldots, \mathscr{P}_{(n)}^{\mathrm{c}}\}$ a *best approximating polytope* in $\mathscr{Q}_n$ of a convex body C with respect to δ if

$$\delta(C,P) = \delta(C,\mathscr{Q}_n) = \inf\{\delta(C,Q)\colon Q \in \mathscr{Q}_n\}.$$

In all cases considered below, best approximating polytopes exist, at least for sufficiently large n. For the measures of distance δ treated in this article there are

always best approximating polytopes in $\mathscr{P}_n^{\mathrm{i}}$ and $\mathscr{P}_{(n)}^{\mathrm{c}}$ which are *inscribed*, respectively *circumscribed* to C, i.e., all vertices are on the boundary of C, respectively all facets touch C.

To make this section self-contained we list the notions of distance on $\mathscr{C}$, respectively $\mathscr{C}_0$ which have appeared more frequently in the context of approximation of convex bodies.

Hausdorff metric δ^{H}. Given $C, D \in \mathscr{C}$, define $\delta^{\mathrm{H}}(C, D)$ to be the maximum (Euclidean) distance which a point of one of the bodies C, respectively D can have from the other body. If h_C and h_D are the support functions of C, respectively D on the unit sphere S^{d-1}, then

$$\delta^{\mathrm{H}}(C, D) = \|h_C - h_D\|_\infty = \sup\{|h_C(u) - h_C(u)|\colon u \in S^{d-1}\}.$$

Eggleston metric δ^{E}. Let $C, D \in \mathscr{C}$. Then $\delta^{\mathrm{E}}(C, D)$ is the sum of the maximum distances of a point of C from D and of a point of D from C.

p-metric δ^p $(p \geqslant 1)$. If $C, D \in \mathscr{C}$, let

$$\delta^p(C, D) = \|h_C - h_D\|_p = \left(\int_{S^{d-1}} |h_C(u) - h_D(u)|^p \, \mathrm{d}\sigma(u) \right)^{1/p}$$

where σ is the ordinary surface area measure. For $A \subset \mathbb{E}^d$ we denote the surface area measure of the boundary $\mathrm{bd}\, A$ of A simply by $\sigma(A)$. A natural concept related to δ^2 is the *Sobolev distance* on $\mathscr{C}$ introduced by Arnold and Wellerding (1992).

Surface deviation (perimeter deviation in case $d = 2$*)* δ^{P}. For $C, D \in \mathscr{C}$ define $\delta^{\mathrm{P}}(C, D) = \sigma(C \cup D) - \sigma(C \cap D)$. δ^{P} is not a metric. In case $d = 2$ it is easy to see that $\delta^1(C, D) = 2\sigma(\mathrm{conv}(C \cup D)) - \sigma(C) - \sigma(D)$ where conv stands for convex hull. Thus for $d = 2$ the metric δ^1 is also a sort of perimeter deviation, compare Florian (1989a).

Symmetric difference metric δ^{S}. Let μ be the Lebesgue measure on $\mathbb{E}^d$. Then if $C, D \in \mathscr{C}$, let $\delta^{\mathrm{S}}(C, D) = \mu(C \cup D) - \mu(C \cap D)$.

Banach–Mazur distance δ^{BM}. For $C, D \in \mathscr{C}_o$,

$$\delta^{\mathrm{BM}}(C, D) = \inf\{\lambda > 1\colon C \subset l(D) \subset \lambda C, l : \mathbb{E}^d \to \mathbb{E}^d \text{ linear}\}.$$

δ^{BM} is not a metric. Applying it to unit balls of norms on $\mathbb{E}^d$ the common Banach–Mazur distance for norms results. It does not distinguish between isometric norms. Instead of δ^{BM} some authors prefer to use $\log \delta^{\mathrm{BM}}$, since in the latter case the triangle inequality holds.

For more detailed information on measures of distance see Grünbaum (1963), Shephard and Webster (1965), the stimulating article of Davis, Vitale and Ben-Sabar (1977), Vitale (1985) or chapter 1.9.

3. Properties of best approximating polytopes for $d = 2$

Aside from the bounds and the asymptotic estimates which will be described in the next two subsections, some interesting general properties of best approximation are known only in the case $d = 2$; their number is small and they are not of recent origin.

Equioscillation properties. Kenderov (1980, 1983) proved that for $d = 2$ each best approximating polygon $P \in \mathcal{P}_n$ in the sense of the metric δ^H of a convex disc $C \notin \mathcal{P}_n$ has the following properties: Each vertex of P is outside of C and its distance to C equals $\delta^H(C, P)$. Each edge of P meets the interior of C and the "depth of penetration" is $\delta^H(C, P)$. An extension of this result to Hausdorff metrics defined by norms other than Euclidean was given by Zhivkov (1982, 1987). We point out the resemblance of these results with well-known properties of Chebychev approximation of continuous functions by polynomials.

A remark of Eggleston (1957b) is the following. Let $P \in \mathcal{P}_n$ be a best approximating polygon of a strictly convex disc C with respect to the metric δ^S. Then any edge v_1v_2 of P intersects bd C in two points w_1, w_2 such that $\|v_1 - w_1\| = \|w_2 - v_2\| = \|v_1 - v_2\|/4$ where $\|\cdot\|$ denotes the Euclidean norm. A related higher-dimensional result for circumscribed polytopes is due to Klee (1986).

The properties of best approximating polygons just mentioned are necessary but not sufficient. In special cases they may be helpful in determining "well approximating" or even "asymptotically best approximating" polygons.

Dowker type theorems. Let $\delta = \delta^H, \delta^E, \delta^P, \delta^S$, further let $C \in \mathcal{C}$ and $\mathcal{Q}_n = \mathcal{P}_n, \mathcal{P}_n^i, \mathcal{P}_n^c$. In what cases are the inequalities

$$2\delta(C, Q_n) \leqslant \delta(C, \mathcal{Q}_{n-1}) + \delta(C, \mathcal{Q}_{n+1}) \quad \text{for } n = 4, 5, \ldots \tag{1}$$

fulfilled? If (1) is satisfied then the sequence $\delta(C, \mathcal{Q}_3), \delta(C, \mathcal{Q}_4), \ldots$ is "convex".

For $\delta = \delta^H$ and $\mathcal{Q}_n = \mathcal{P}_n, \mathcal{P}_n^i, \mathcal{P}_n^c$, Arias informs us that (1) does not hold generally. For $\delta = \delta^E$ and $\mathcal{Q}_n = \mathcal{P}_n, \mathcal{P}_n^i, \mathcal{P}_n^c$ the corresponding negative statement is due to Eggleston (1957a,b). For $\delta = \delta^P$ and $\mathcal{Q}_n = \mathcal{P}_n, \mathcal{P}_n^i, \mathcal{P}_n^c$ the inequalities (1) hold by results of Molnár (1955) and Eggleston (1957a,b). For $\delta = \delta^S$ and $\mathcal{Q}_n = \mathcal{P}_n^i, \mathcal{P}_n^c$ Dowker (1944) proved the validity of (1) and for $\mathcal{Q}_n = \mathcal{P}_n$ this was shown by Eggleston (1957a,b). A refinement of the latter's result is due to G. Fejes Tóth (1977).

Dowker type inequalities for convex sets on S^2 were given by L. Fejes Tóth (1958/59).

4. Upper bounds

For many concepts δ of distance on $\mathscr{C}$ or $\mathscr{C}_o$ and certain classes $\mathscr{Q}_n$ of polytopes the following inequalities hold for each convex body C in $\mathscr{C}$, respectively $\mathscr{C}_o$:

$$\delta(C,\mathscr{Q}_n) \leqslant \frac{\text{constant}}{n^{2/(d-1)}} \quad \text{for } n = d+1, d+2, \ldots . \tag{2}$$

(In this article constant always denotes a positive constant which may depend on a convex body as in the present case, on the dimension, or it may be an absolute constant as in the "high-dimensional" results of sections 7 and 8.) If C is a convex body (with $\operatorname{bd} C$) of class $\mathscr{C}^2$ for which the Gaussian curvature κ_C is positive then, in general, (2) is best possible, i.e., there is a constant > 0 such that

$$\delta(C,\mathscr{Q}_n) \geqslant \frac{\text{constant}}{n^{2/(d-1)}} \quad \text{for } n = d+1, d+2, \ldots .$$

Hausdorff metric δ^{H}. Dudley (1974) and Bronshtein and Ivanov (1975) (and independently Betke and Wills 1979) established (2) for $\delta = \delta^{\mathrm{H}}$ and $\mathscr{Q}_n = \mathscr{P}^{\mathrm{c}}_{(n)}, \mathscr{P}^{\mathrm{i}}_n$. From this one can deduce that (2) also holds in the cases $\mathscr{Q}_n = \mathscr{P}_n, \mathscr{P}^{\mathrm{c}}_n, \mathscr{P}_{(n)}, \mathscr{P}^{\mathrm{i}}_{(n)}$. More precise bounds for $d = 2$ were derived by Popov (1968, 1970).

The asymptotic results in the next section show that (2) is best possible for all $\mathscr{Q}_n$, assuming that $C \in \mathscr{C} \cap \mathscr{C}^2$ where $\kappa_C > 0$. (The case $\mathscr{Q}_n = \mathscr{P}^{\mathrm{i}}_n$ follows also from a result of a different type due to Schneider and Wieacker 1981.)

Following Dudley (1974) we sketch a proof of (2) for $\delta = \delta^{\mathrm{H}}$ and $\mathscr{Q}_n = \mathscr{P}^{\mathrm{c}}_{(n)}$. For convenience it is assumed that $C \in \mathscr{C} \cap \mathscr{C}^1$. For $x \in \operatorname{bd} C$ let $n_C(x)$ denote the exterior normal unit vector of C at x. Then $S = \{x + n_C(x) : x \in \operatorname{bd} C\}$ is the outer parallel surface of $\operatorname{bd} C$ at distance 1.

First, the following proposition will be established.

> Let $0 < \varepsilon < \frac{1}{2}\sqrt{3}$ and $y, z \in \operatorname{bd} C$ such that $\|y + n_C(y) - z - n_C(z)\| \leqslant \varepsilon$. Then the distance from y along the normal of C at y to the supporting hyperplane of C at z is at most $2\varepsilon^2$. (3)

By squaring the inequality $\|y + n_C(y) - z - n_C(z)\| \leqslant \varepsilon$ and taking into account that the inner products $\langle n_C(y), y - z\rangle$ and $\langle n_C(z), z - y\rangle$ are nonnegative, one shows that

$$\|y - z\|, \|n_C(y) - n_C(z)\| \leqslant (\|y + n_C(y) - z - n_C(z)\| \leqslant)\varepsilon.$$

Since $0 < \varepsilon < \frac{1}{2}\sqrt{3}$ the inequality $\|n_C(y) - n_C(z)\| \leqslant \varepsilon$ yields that

$$\tan(\operatorname{angle}(n_C(y), n_C(z))) \leqslant 2\varepsilon.$$

Let f denote the foot of y on the plane common to the supporting hyperplanes of C at y, respectively z. Clearly $\|f - y\| \leqslant \|y - z\|$. Noting that $\|y - z\| \leqslant \varepsilon$, we obtain,

$$\|f - y\| \leqslant \varepsilon.$$

Thus,

$$\|f-y\|\tan(\text{angle}(n_C(y),n_C(z)))\leqslant 2\varepsilon^2.$$

Since the left-hand side of this inequality is the desired distance, the proof of (3) is complete.

Second, since S is a compact surface of dimension $d-1$, there is a constant $\alpha>0$ such that for $n=d+1,d+2,\ldots$, one can cover S with n (Euclidean) balls of radius $\alpha/n^{1/(d-1)}$ and centers in S, say at $z_i+n_C(z_i), i=1,\ldots,n$. Let P_n be the intersection of the supporting halfspaces of C at $z_1,\ldots,z_n$. If n is sufficiently large, then $P_n\in\mathcal{P}^{\mathrm{c}}_{(n)}$.

Third, in order to prove that $\delta^{\mathrm{H}}(C,P_n)\leqslant\alpha^2/n^{2/(d-1)}$ for all sufficiently large n, choose n such that $P_n\in\mathcal{P}^{\mathrm{c}}_{(n)}$ and $\varepsilon=\alpha/\sqrt{2}n^{1/(d-1)}<\frac{1}{2}\sqrt{3}$. Let $x\in\operatorname{bd}P_n$ and let $y\in\operatorname{bd}C$ be the foot of x on $\operatorname{bd}C$. The point $y+n_C(y)\in S$ is contained in a ball of radius ε and center at $z_i+n_C(z_i)\in S$, say, such that a facet of P_n touches C at z_i. By (3) the distance from y along the normal of C and the hyperplane containing this facet and thus a fortiori the distance from x to $y\in\operatorname{bd}C$ is at most $2\varepsilon^2=\alpha^2/n^{2/(d-1)}$. Since this holds for each $x\in\operatorname{bd}P$ we see that $\delta^{\mathrm{H}}(C,P_n)\leqslant\alpha^2/n^{2/(d-1)}$, concluding the proof.

Gritzmann and Lassak (1989) established the following result. Let $C\in\mathcal{C}$ have minimum width $w(C)$ equal to 1. Then

$$w(C,\mathcal{P}^{\mathrm{i}}_n)=\sup\{w(P):P\in\mathcal{P}^{\mathrm{i}}_n\}\geqslant 1-\delta^{\mathrm{H}}(B^d,\mathcal{P}^{\mathrm{i}}_{\lfloor n/2\rfloor})$$

where B^d is the unit ball of $\mathbb{E}^d$. Since (2) holds for $\delta=\delta^{\mathrm{H}}$ and $\mathcal{Q}_n=\mathcal{P}^{\mathrm{i}}_{\lfloor n/2\rfloor}$ we thus see that

$$w(C,\mathcal{P}^{\mathrm{i}}_n)\geqslant 1-\frac{\text{constant}}{n^{2/(d-1)}}\quad\text{for } n=d+1,\ldots$$

where constant >0 depends only on d.

Symmetric difference metric δ^{S}. Let $C\in\mathcal{C}$ and $\mathcal{Q}_n=\mathcal{P}_n,\ldots,\mathcal{P}^{\mathrm{c}}_{(n)}$. Then

$$\delta^{\mathrm{S}}(C,\mathcal{Q}_n)\leqslant\text{constant}\,\delta^{\mathrm{H}}(C,\mathcal{Q}_n)\quad\text{for } n=d+1,\ldots$$

with a suitable constant >0 depending on C. Hence (2) holds for $\delta=\delta^{\mathrm{S}}$ and $\mathcal{Q}_n=\mathcal{P}_n,\ldots,\mathcal{P}^{\mathrm{c}}_{(n)}$. Given $C\in\mathcal{C}$ choose a ball $B\in\mathcal{C}$ with $\mu(B)=\mu(C)$. Then

$$\delta^{\mathrm{S}}(C,\mathcal{P}^{\mathrm{i}}_n)\leqslant\delta^{\mathrm{S}}(B,\mathcal{P}^{\mathrm{i}}_n)\quad\text{for } n=d+1,\ldots.$$

From this result of Blaschke (1917), Sas (1939) and Macbeath (1951a) one can obtain in the case $d=2$ a precise upper bound for $\delta^{\mathrm{S}}(C,\mathcal{P}^{\mathrm{i}}_n)$.

As a consequence of the asymptotic estimates for δ^S in section 5 below, (2) is best possible for $\delta = \delta^S$ and $\mathcal{Q}_n = \mathcal{P}_n^i, \mathcal{P}_{(n)}^c$ whenever $C \in \mathcal{C} \cap \mathcal{C}^2$ and $\kappa_C > 0$. Gruber and Kenderov (1982) proved this also for the case $\mathcal{Q}_n = \mathcal{P}_n$. For a class of convex bodies which properly contains the class $\{C \in \mathcal{C} \cap \mathcal{C}^2 : \kappa_C > 0\}$ Schütt (1990) gave a lower bound which is slightly weaker than constant$/n^{2/(d-1)}$.

Let C be a convex polygon. An algorithm for finding a triangle $P \in \mathcal{P}_3^i$ for which $\delta^S(C, \mathcal{P}_3^i)$ is attained was established by Dobkin and Snyder (1979).

The concepts of distance $\delta^E, \delta^p, \delta^P, \delta^{BM}$. The inequality (2) is valid for $\delta = \delta^E, \ldots, \delta^{BM} - 1$ and $\mathcal{Q}_n = \mathcal{P}_n, \ldots, \mathcal{P}_{(n)}^c$, respectively $\mathcal{Q}_n = \mathcal{P}_n, \mathcal{P}_{(n)}$ in the case of δ^{BM}. This is implied by the corresponding result for δ^H and the following relations:

(i) $\delta^H \leqslant \delta^E \leqslant 2\delta^H$,

(ii) $\delta^p \leqslant \sigma(S^{d-1})^{1/p}\delta^H$,

(iii) the Steiner formula for the surface area of parallel surfaces and the monotonicity of the surface area of convex bodies,

(iv) the implication: $C, D \in \mathcal{C}_o$ and $\alpha B^d \subset C \subset D$ implies $\delta^{BM}(C, D) \leqslant 1 + \delta^H(C, D)/\alpha$.

For $d = 2$ more precise inequalities for $\delta = \delta^P$ and $\mathcal{Q}_n = \mathcal{P}_n^i, \mathcal{P}_n^c$ were established by Schneider (1971).

In some cases the above relations together with the corresponding property of δ^H show that (2) is best possible for $C \in \mathcal{C} \cap \mathcal{C}^2$ where $\kappa_C > 0$.

5. Asymptotic estimates

Considering the upper bounds in the last subsection and the fact that in some cases they are best possible (which in part can be proved without using asymptotic estimates!) the following claim is highly plausible: if C is a sufficiently smooth convex body with $\kappa_C > 0$ then

$$\delta(C, \mathcal{Q}_n) \sim \frac{\text{constant}}{n^{2/(d-1)}} \quad \text{as } n \to \infty, \tag{4}$$

for certain δ and $\mathcal{Q}_n$ and suitable constants.

In his fundamental treatise "Lagerungen in der Ebene, auf der Kugel und im Raum" L. Fejes Tóth (1953) stated for $d = 2, 3$ many asymptotic formulae of the form (4) and in some cases indicated proofs. Through the efforts of L. Fejes Tóth, McClure and Vitale, Gleason, Schneider, Enomoto, Gruber and Ludwig most of these formulae have been proved, some new ones were added and several asymptotic formulae were extended to general dimensions. The following is a description of what is known.

Hausdorff metric δ^H. Let $C \in \mathcal{C} \cap \mathcal{C}^2$ where $\kappa_C > 0$. Then (4) holds for $\delta = \delta^H$

and $\mathfrak{Q}_n = \mathcal{P}_n^{\mathrm{i}}, \mathcal{P}_{(n)}^{\mathrm{c}}$ where

$$\text{constant} = \frac{1}{2}\left(\frac{\vartheta_{d-1}}{\kappa_{d-1}}\int_{\mathrm{bd}\,C}\kappa_C(x)^{1/2}\,\mathrm{d}\sigma(x)\right)^{2/(d-1)}. \tag{5}$$

Here ϑ_k is the minimum density of covering $\mathbb{E}^k$ with balls of fixed radius and κ_k is the k-dimensional volume of B^k. Note that $\vartheta_1 = 1, \kappa_1 = 2, \vartheta_2 = 2\pi/\sqrt{27}, \kappa_2 = \pi$; no other values of ϑ_k are known. For more precise information on ϑ_k see, e.g., Erdős, Gruber and Hammer (1989) or chapter 3.3 by G. Fejes Tóth and Kuperberg in this Handbook.

Since for $C \in \mathcal{C}^1$ it is possible to prove that

$$\delta^{\mathrm{H}}(C, \mathcal{P}_n^{\mathrm{i}}) \sim \delta^{\mathrm{H}}(C, \mathcal{P}_n^{\mathrm{c}}) \sim 2\delta^{\mathrm{H}}(C, \mathcal{P}_n),$$

$$\delta^{\mathrm{H}}(C, \mathcal{P}_{(n)}^{\mathrm{i}}) \sim \delta^{\mathrm{H}}(C, \mathcal{P}_{(n)}^{\mathrm{c}}) \sim 2\delta^{\mathrm{H}}(C, \mathcal{P}_{(n)})$$

as $n \to \infty$, (4) holds also for $\delta = \delta^{\mathrm{H}}$ and $\mathfrak{Q}_n = \mathcal{P}_n, \mathcal{P}_n^{\mathrm{c}}, \mathcal{P}_{(n)}, \mathcal{P}_{(n)}^{\mathrm{i}}$ for each $C \in \mathcal{C} \cap \mathcal{C}^2$ where $\kappa_C > 0$.

The asymptotic formulae for δ^{H} were proved by Schneider (1981, 1987) for $C \in \mathcal{C} \cap \mathcal{C}^3$ with $\kappa_C > 0$; Gruber (1992) established them for the present case. For $d = 2$ see L. Fejes Tóth (1948, 1953) and McClure and Vitale (1975).

Let C be a curve in $\mathbb{E}^d$ of class $\mathcal{C}^2$, finite length λ and (first) curvature κ_C and let $\mathfrak{Q}_n$ be the class of polygons inscribed into C, with the same endpoints as C and having at most n vertices. Extend δ^{H} to the space of compact sets in $\mathbb{E}^d$. Then a result of L. Fejes Tóth (1948) says that

$$\delta^{\mathrm{H}}(C, \mathfrak{Q}_n) \sim \frac{1}{8}\left(\int_0^{\lambda}\kappa_C(s)^{1/2}\,\mathrm{d}s\right)^2\frac{1}{n^2} \quad \text{as } n \to \infty$$

(s is the arc-length on C).

An application of the Cauchy–Schwarz inequality to the integral in (5) together with the property that $\kappa_C =$ constant precisely for balls shows that among all convex bodies $C \in \mathcal{C} \cap \mathcal{C}^2$ with $\kappa_C > 0$ of given surface area it is only the balls for which the integral is maximal.

Let $C \in \mathcal{C} \cap \mathcal{C}^2$ with $\kappa_C > 0$ be given. We now describe the underlying idea of the proofs of Schneider and Gruber of the asymptotic formula for $\delta^{\mathrm{H}}(C, \mathcal{P}_n^{\mathrm{i}})$, omitting many details. Let bd$C$ be endowed with the Riemannian metric determined by the second fundamental form.

First, the following proposition will be shown.

> Let P be a convex polytope inscribed into C. Then the geodesic discs with centers at the vertices of P and radius $(2\delta^{\mathrm{H}}(C, P))^{1/2}$ cover bd C. (6)

For the proof of this proposition, it is clearly sufficient to show the following assertion.

Let $x \in \operatorname{bd} C$, then $\varrho(x,v) \leqslant (2\delta^{\mathrm{H}}(C,P))^{1/2}$ for a suitable vertex v of P. (7)

Let S be the supporting hyperplane of C at x and let T be the supporting hyperplane of P with the same exterior normal as S. By a well-known property of δ^{H} the distance of S and T is at most $\delta^{\mathrm{H}}(C,P)$. The supporting hyperplane T of the polytope P contains a face and thus, in particular, a vertex of P, say v. Clearly $v \in \operatorname{bd} C$. The second fundamental form measures the distance from supporting hyperplanes. Hence $\varrho(x,v)^2/2$ is (approximately) equal to the distance of v from the supporting hyperplane of $\operatorname{bd} C$ at x, i.e., from v to S or, equivalently, of S and T. Since the latter is at most $\delta^{\mathrm{H}}(C,P)$, the proof of (7) is complete. (7) implies (6).

A similar reasoning establishes the next proposition.

Let $v_1,\ldots,v_n \in \operatorname{bd} C$ and $\varrho > 0$ be such that the geodesic discs with centers at $v_1,\ldots,v_n$ and radius ϱ cover $\operatorname{bd} C$. Then (8)

$$\delta^{\mathrm{H}}(C,\operatorname{conv}\{v_1,\ldots,v_n\}) \leqslant \varrho^2/2.$$

Comparing (6) and (8) it is plausible that best approximating polytopes $P \in \mathcal{P}_n^{\mathrm{i}}$ of C correspond to thinnest coverings of $\operatorname{bd} C$ with geodesic discs of equal radius. Note that $\operatorname{bd} C$ endowed with the Riemannian metric $\varrho(\cdot,\cdot)$ is locally almost a Euclidean space of dimension $d-1$. Thus the density of a thinnest covering of $\operatorname{bd} C$ with a large number n of geodesic discs of small radius ϱ is approximately ϑ_{d-1} or,

$$\frac{n\varrho^{d-1}\kappa_{d-1}}{\operatorname{area}(\operatorname{bd} C)} \sim \frac{\text{total area of the discs}}{\operatorname{area}(\operatorname{bd} C)} \sim \vartheta_{d-1}$$

or

$$\frac{\varrho^2}{2} \sim \frac{1}{2}\left(\frac{\vartheta_{d-1}}{\kappa_{d-1}}\operatorname{area}(\operatorname{bd} C)\right)^{2/(d-1)} \frac{1}{n^{2/(d-1)}}$$

as $n \to \infty$. Of course, here the area of $\operatorname{bd} C$ is the area determined by the Riemannian (locally Euclidean) metric $\varrho(\cdot,\cdot)$, i.e.,

$$\operatorname{area}(\operatorname{bd} C) = \int_{\operatorname{bd} C} \kappa_C(x)^{1/2}\,\mathrm{d}\sigma(x).$$

Combining these remarks one finally arrives at the asymptotic formula for $\delta^{\mathrm{H}}(C, \mathcal{P}_n^{\mathrm{i}})$.

As is clear from the foregoing, the problem of approximation of C by inscribed polytopes with respect to δ^{H} is intimately connected with thinnest covering of $\operatorname{bd} C$ with geodesic discs determined by a suitable Riemannian metric on $\operatorname{bd} C$. Thus it is related to the thinnest covering of $\mathbb{E}^{d-1}$ with balls of fixed radius. An

analogous relation turns up for δ^{H}-approximation by circumscribed polytopes and in δ^{BM}- and δ^{SCH}-approximation where δ^{SCH} is a notion of distance introduced by Schneider (1986), see below.

Very precise asymptotic formulae for $d = 2$ are due to Ludwig (1992).

Banach–Mazur distance δ^{BM}. Let $C \in \mathscr{C}_0 \cap \mathscr{C}^2$ where $\kappa_C > 0$. Then the asymptotic formula (4) holds for $\delta = \delta^{\mathrm{BM}} - 1$ and $\mathfrak{Q}_n = \mathscr{P}_n, \mathscr{P}_{(n)}$ by a result of Gruber (1992) where in both cases

$$\text{constant} = \frac{1}{2}\left(\frac{\vartheta_{d-1}}{2\kappa_{d-1}} \int_{\mathrm{bd}\, C} \left(\frac{\kappa_C(x)}{h_C(n_C(x))^{d-1}}\right)^{1/2} \mathrm{d}\sigma(x)\right)^{2/(d-1)}. \tag{9}$$

The appropriate Riemannian metric on $\mathrm{bd}\, C$ is that of central affine geometry. (For the notions and results of affine differential geometry used in this article see Blaschke 1923 or Salkowski 1934.) The integral in (9) is the corresponding surface area of $\mathrm{bd}\, C$. Like δ^{BM} it is invariant with respect to linear, i.e., central affine transformations of $\mathbb{E}^d$ and also with respect to polarity.

The Cauchy–Schwarz inequality for integrals and the Blaschke–Santaló inequality for the product of the volume of a convex body and the volume of its polar body show that the integral in (9) is maximal precisely for ellipsoids.

Schneider's notion of distance δ^{SCH}. If a convex polytope P is inscribed into a convex body C, let $\delta^{\mathrm{SCH}}(C, P)$ be the maximum volume of a cap of C determined by a hyperplane through a facet of P. Schneider (1986) established the asymptotic estimate

$$\delta^{\mathrm{SCH}}(C, \mathscr{P}^{\mathrm{i}}_{(n)}) \sim \frac{\kappa_{d-1}}{d+1}\left(\frac{\vartheta_{d-1}}{\kappa_{d-1}} \int_{\mathrm{bd}\, C} \kappa_C(x)^{1/(d+1)}\, \mathrm{d}\sigma(x)\right)^{(d+1)/(d-1)} \frac{1}{n^{(d+1)/(d-1)}} \tag{10}$$

as $n \to \infty$, assuming that $C \in \mathscr{C} \cap \mathscr{C}^3$ and $\kappa_C > 0$. In the proof of (10) $\mathrm{bd}\, C$ is endowed with the Riemannian metric defined by the fundamental form of equiaffine (i.e., volume-preserving) differential geometry. The related concept of area on $\mathrm{bd}C$, the so-called *affine surface area* determined by the integral in (10) is invariant with respect to volume preserving affinities, a property shared by δ^{SCH}.

Considering the asymptotic formulae for δ^{S} to be cited below the following formula suggests itself (assuming that $C \in \mathscr{C} \cap \mathscr{C}^2$ and $\kappa_C > 0$):

$$\delta^{\mathrm{S}}(C, \mathscr{P}^{\mathrm{i}}_{(n)}) \sim \frac{\text{constant}}{n^{2/(d-1)}} \quad \text{as } n \to \infty. \tag{11}$$

For a best approximating polytope $P \in \mathscr{P}^{\mathrm{i}}_{(n)}$ with respect to δ^{S} the individual caps of C determined by P most probably have roughly the same volume and do not overlap too much. Since their total number is about n, the exponent of n in the asymptotic formula (10) is in conformity with the exponent of n in (11).

The affine isoperimetric inequality of Blaschke shows that the integral in (10) is maximal among all convex bodies of class $\mathscr{C}^2$ with given volume precisely for ellipsoids (see chapter 1.5 by Lutwak).

Symmetric difference metric δ^S. For $C \in \mathscr{C} \cap \mathscr{C}^2$ where $\kappa_C > 0$ the asymptotic formula (4) holds for $\delta = \delta^S$ and $\mathfrak{Q}_n = \mathscr{P}_n^i, \mathscr{P}_{(n)}^c$. The corresponding constants are

$$\begin{aligned} &\tfrac{1}{2}\mathrm{del}_{d-1}\left(\int_{\mathrm{bd}\,C} \kappa_C(x)^{1/(d+1)}\,\mathrm{d}\sigma(x)\right)^{(d+1)/(d-1)}, \\ &\tfrac{1}{2}\mathrm{div}_{d-1}\left(\int_{\mathrm{bd}\,C} \kappa_C(x)^{1/(d+1)}\,\mathrm{d}\sigma(x)\right)^{(d+1)/(d-1)}. \end{aligned} \tag{12}$$

del_{d-1} and div_{d-1} are quantities related to Delone triangulations and Dirichlet–Voronoi tilings in $\mathbb{E}^{d-1}$, respectively. For $d = 2$ this result was indicated by L. Fejes Tóth (1953) and proved by McClure and Vitale (1975). Here $\mathrm{del}_1 = \frac{1}{6}, \mathrm{div}_1 = \frac{1}{12}$. More precise asymptotic formulae in the planar case are due to Ludwig (1992). Proofs for $d = 3$ are due to Gruber (1988, 1991) (in the case $\mathfrak{Q}_n = \mathscr{P}_n^i$ under the assumption $C \in \mathscr{C} \cap \mathscr{C}^3$ and $\kappa_C > 0$). His results imply that $\mathrm{del}_2 = 1/2\sqrt{3}$ and $\mathrm{div}_2 = 5/18\sqrt{3}$, thereby confirming a conjecture of L. Fejes Tóth (1953). The general case was obtained by Gruber (1992), however it seems extremely difficult to determine explicitly the values of $\mathrm{del}_k, \mathrm{div}_k$ for $k \geqslant 3$.

Again, in (12) the constants are maximal only for ellipsoids.

The proofs of the asymptotic formulae for $\delta^S(C, \mathscr{P}_n^i)$ and $\delta^S(C, \mathscr{P}_{(n)}^c)$ are complicated. The approximation problem is considered first for paraboloids. In this case one makes use of an argument of the following sort: Given a thin covering in $\mathbb{E}^{d-1}$ with balls of different sizes, one may delete balls in such a way that the remaining balls form a packing where the total volume of the balls of the packing is not too small. For paraboloids the case of inscribed polytopes corresponds to Delone triangulations in $\mathbb{E}^{d-1}$. The transition from paraboloids to the convex body C is achieved by a version of Blaschke's "Schüttelung" operation; best approximating polytopes of paraboloids are transformed into polytopes which well approximate C locally and vice versa.

Mean width deviation. It seems to be a natural problem to measure the deviation of a convex body and its inscribed or circumscribed polytopes in terms of their quermassintegrals, such as volume, surface area and mean width. An asymptotic result of the form (4) for mean width approximation is due to Gruber (1994, in preparation).

Remarks. Perhaps the reader will ask why in the asymptotic results for δ^H, δ^{BM}, δ^{SCH} and δ^S it was necessary to suppose that $\kappa_C > 0$? Most probably the results are true without this assumption, but in the existing proofs it is necessary to compare quadratic ("fundamental") forms corresponding to different points of $\mathrm{bd}\,C$. This is possible only if the quadratic forms are positive definite or, equivalently, if $\kappa_C > 0$.

A second natural question is whether one may replace the class $\mathscr{C}^2$ by $\mathscr{C}^\alpha$ where $\alpha < 2$ while still retaining asymptotic formulae of the form (4). Most probably

the answer is no. The results on the irregular behaviour of $\delta(C, \mathfrak{P}_n)$ as $n \to \infty$ for "typical" convex bodies C show that it is not possible to take $\alpha = 1$, see chapter 4.10.

p-metric δ^p. Assume that $d = 2$ and that $C \in \mathfrak{C}$ has support function of class $\mathfrak{C}^2$. Let $\operatorname{bd} C$ have length λ. McClure and Vitale (1975) confirmed (4) for $\delta = \delta^p$ and $\mathfrak{P}_n = \mathfrak{P}_n^{\mathrm{i}}, \mathfrak{P}_n^{\mathrm{c}}$ where the constants are

$$\frac{1}{8(2p+1)^{1/p}} \left(\int_0^\lambda \kappa_C(s)^{(p+1)/(2p+1)} \, \mathrm{d}s \right)^{(2p+1)/p},$$

$$\frac{1}{2} \left(\int_0^1 t^p (1-t)^p \, \mathrm{d}t \right)^{1/p} \left(\int_0^\lambda \kappa_C(x)^{(p+1)/(2p+1)} \, \mathrm{d}s \right)^{(2p+1)/p}.$$

(McClure and Vitale express the constants in terms of the support function of C.)

Perimeter deviation δ^{P}. For $d = 2$ and $C \in \mathfrak{C}$ with support function of class $\mathfrak{C}^2$ the asymptotic estimate (4) holds where $\delta = \delta^{\mathrm{P}}$ and $\mathfrak{P}_n = \mathfrak{P}_n^{\mathrm{i}}, \mathfrak{P}_n^{\mathrm{c}}$, the constants being

$$\frac{1}{12} \left(\int_0^\lambda \kappa_C(s)^{2/3} \, \mathrm{d}s \right)^3, \qquad \frac{1}{24} \left(\int_0^\lambda \kappa_C(x)^{2/3} \, \mathrm{d}s \right)^3.$$

This was stated by L. Fejes Tóth (1953) and proved by McClure and Vitale (1975).

Gleason (1979) showed that the result for $\delta^{\mathrm{P}}(C, \mathfrak{P}_n^{\mathrm{i}})$ continues to hold if C is a curve in $\mathbb{E}^d$ of class $\mathfrak{C}^2$ with (first) curvature κ_C, further $\mathfrak{P}_n^{\mathrm{i}}$ is the class of polygons inscribed into C which have the same endpoints as C and at most n vertices and where for $P \in \mathfrak{P}_n^{\mathrm{i}}$ the quantity $\delta^{\mathrm{P}}(C, P)$ is the difference in length of C and P. (The corresponding result for δ^{H} was proved by L. Fejes Tóth 1948, see above.) An extension of Gleason's theorem to approximation of curves on a Riemannian manifold by geodesic polygons was established by Enomoto (1986).

Explicit constructions of "well" approximating polytopes. Let $C \in \mathfrak{C} \cap \mathfrak{C}^2$, respectively $C \in \mathfrak{C}_o \cap \mathfrak{C}^2$ be given where $\kappa_C > 0$.

For $d = 2$ the proofs of the above results implicitly contain constructions of polygons $P_n \in \mathfrak{P}_n$ such that

$$\delta(C, P_n) \sim \delta(C, \mathfrak{P}_n) \quad \text{as } n \to \infty.$$

More sophisticated constructions of such "asymptotically best approximating" sequences of polygons for $\delta = \delta^{\mathrm{H}}, \delta^{\mathrm{S}}, \delta^{\mathrm{P}}$ and $\mathfrak{P}_n = \mathfrak{P}_n^{\mathrm{i}}, \mathfrak{P}_n^{\mathrm{c}}$ are due to McClure and Vitale (1975).

For $d = 3$ results of this type were not described explicitly, but, in principle, may be obtained from the work of Schneider (1981, 1986, 1987) and Gruber (1988, 1991, 1992) for $\delta = \delta^{\mathrm{H}}, \delta^{\mathrm{S}}$ and δ^{BM} where $\mathfrak{P}_n = \mathfrak{P}_n^{\mathrm{i}}, \mathfrak{P}_{(n)}^{\mathrm{c}}$ and $\mathfrak{P}_n, \mathfrak{P}_{(n)}$, respectively. The

construction of asymptotically best approximation requires the following two ingredients. First, one has to know for what configurations in $\mathbb{E}^2$ are $\vartheta_2, \mathrm{del}_2$ and div_2 attained. The well-known (but difficult to establish) answers are lattices based on regular triangles and tilings with congruent regular triangles, respectively hexagons. Second, for any $\varepsilon > 0$ one must be able to dissect $\mathrm{bd}\, C$ into finitely many pieces with piecewise smooth boundary, such that on every piece the values of each principal curvature differ by less than ε.

For general d no explicit construction of asymptotically best approximating sequences have been given up to now. This is due to the fact that to date the configurations for which $\vartheta_{d-1}, \mathrm{del}_{d-1}$ and div_{d-1} are attained are not known for $d \geqslant 4$. However, it is possible to construct sequences $P_n \in \mathfrak{D}_n$ such that

$$\delta(C, P_n) \leqslant \frac{\text{constant}}{n^{2/(d-1)}} \quad \text{for } n = d+1, \ldots.$$

Since in the next section a sharper result of this type will be discussed we abstain from a detailed description. A related slightly weaker result for more general convex bodies is due to Sonnevend (1983a).

6. Algorithmic and asymptotic step-by-step approximation

In this section we describe several effective approximation algorithms. They are formulated for convex functions or convex bodies. In the former case the approximation is considered with respect to the maximum norm, in the latter case with respect to the Hausdorff metric.

Sandwich algorithms for convex functions. Let $f: [a,b] \to \mathbb{R}$ be convex. Sandwich algorithms provide for f piecewise linear upper and lower approximations; see, e.g., Sonnevend (1983b), Fruhwirth, Burkard and Rote (1989), Rote (1990, 1992) and Burkard, Hamacher and Rote (1991): Assume that the right and left derivatives $f^+(a)$ and $f^-(b)$ of f at a, respectively b are finite.

Step 0: Define $u_o, l_o: [a,b] \to \mathbb{R}$ by

$$u_o(t) = f(a) + \frac{f(b) - f(a)}{b - a}(t - a) \quad \text{for } t \in [a,b],$$

$$l_o(t) = \max\{f(a) + f^+(a)(t-a), f(b) + f^-(b)(t-b)\} \quad \text{for } t \in [a,b].$$

Step 1: Dissect $[a,b]$ into intervals $[a,c], [c,b]$ where c is chosen according to a fixed one of the following rules:

(i) *Interval bisection rule:* $c = (a+b)/2$.

(ii) *Slope bisection rule:* $c \in (a,b)$ and $(c, f(c))$ is a point of the graph of f where it is supported by a line with slope $(f^+(a) + f^-(b))/2$.

(iii) *Maximum error rule:* $c \in (a,b)$ and $f(a) + f^+(a)(c-a) = f(b) + f^-(b)(c-b)$.

(iv) *Chord rule:* $c \in (a,b)$ and the graph of f is supported at $(c,f(c))$ by a line with slope $(f(b)-f(a))/(b-a)$.

Now approximate $f|[a,c]$ and $f|[c,b]$ by piecewise linear functions as was done for f in Step 0. By fitting together the upper approximations on $[a,c]$ and $[c,b]$ we obtain an upper approximation u_1 of f on $[a,b]$. In an analogous way one constructs a lower approximation l_1.

Step 2: Of the intervals into which $[a,b]$ has been dissected choose (the) one on which $u_1 - l_1$ attains its maximum. Dissect this interval into two intervals as was done for $[a,b]$ in Step 1. This leads (as in Step 1) to upper and lower approximations u_2 and l_2 of f which coincide with u_1 respectively l_1 outside the interval that was dissected just before.

Continuing in this way one obtains piecewise linear approximations u_n, l_n such that

$$l_n \leqslant f \leqslant u_n \quad \text{for } n = 0,1,\ldots .$$

If this algorithm is applied with the rules (i) or (ii), then

$$\|u_n - l_n\|_\infty \leqslant \frac{9(f^-(b) - f^+(a))(b-a)}{8n^2} \quad \text{for } n = 2,3,\ldots$$

if it is applied with the rules (iii) or (iv), then

$$\|u_n - l_n\|_\infty \leqslant \frac{(f^-(b) - f^+(a))(b-a)}{n^2} \quad \text{for } n = 2,3,\ldots .$$

For related material see Johnson and Vogt (1980) and Lew and Quarles (1989) and the references in Burkard, Hamacher and Rote (1991) and Rote (1992).

Asymptotic step-by-step approximation of convex bodies. Given $C \in \mathcal{C}$, how well can C be approximated by $\operatorname{conv}\{x_1,\ldots,x_n\}$ or $H_C^+(x_1) \cap \cdots \cap H_C^+(x_n)$ where $x_1, x_2, \ldots$ is a sequence in $\operatorname{bd} C$? (For $x \in \operatorname{bd} C$ let $H_C^+(x)$ denote a supporting halfspace of C at x.)

In order to state a pertinent result we need the concept of *dispersion* $d_n(x_i)$ of a sequence (x_i) in a totally bounded metric space $\langle X, \varrho(\cdot,\cdot)\rangle$:

$$d_n(x_i) = \inf\{\varrho > 0 \colon X \subset \{x \colon \varrho(x,x_1) \leqslant \varrho\} \cup \cdots \cup \{x \colon \varrho(x,x_n) \leqslant \varrho\}\},$$
$$n = 1,2,\ldots .$$

Note that the concept of dispersion is related to the concept of "empty balls" in the context of Delone triangulations.

The following result is due to Gruber (1992): let $C \in \mathcal{C} \cap \mathcal{C}^2$ where $\kappa_C > 0$ and endow $\operatorname{bd} C$ with the Riemannian metric determined by the second fundamental form. Then, if (x_i) is a sequence in $\operatorname{bd} C$ with $d_n(x_i) \to 0$ as $n \to \infty$, the following asymptotic relations hold as $n \to \infty$:

$$\delta^{\mathrm{H}}(C, \operatorname{conv}\{x_1,\ldots,x_n\}), \delta^{\mathrm{H}}(C, H_C^+(x_1) \cap \cdots \cap H_C^+(x_n)) \sim \tfrac{1}{2} d_n(x_i)^2.$$

Moreover, there is a sequence (y_i) in $\operatorname{bd} C$ such that

$$d_n(y_i) \leqslant \frac{\text{constant}}{n^{1/(d-1)}} \quad \text{for } n = 1, 2, \ldots.$$

The sequence (y_i) can be given in a more or less explicit way, using such a sequence in the unit cube $[0,1]^{d-1}$ specified by Niederreiter (1984). For related results concerning δ^{BM} and δ^{S} consult Gruber (1992) and for $d = 2$ see Müller (1992).

For other problems of stepwise approximation compare Button and Wilker (1978); see also Taylor (1976).

Step-by-step approximation of convex functions. From the last result, its proof and the fact that any two Riemannian metrics on the compact space $[0,1]^{d-1}$ are equivalent up to constants one may infer the following:

Let $y_1, y_2, \ldots \in [0,1]^{d-1}$ be a sequence such that its dispersion with respect to the Euclidean metric is $O(1/n^{1/(d-1)})$ as $n \to \infty$ (see Niederreiter 1984). We may suppose that $y_1, \ldots, y_{2^{d-1}}$ are the vertices of the cube $[0,1]^{d-1}$. Let $f: [0,1]^{d-1} \to \mathbb{R}$ be a convex function of class $\mathscr{C}^2$ with positive definite Hessian. For $n = 2^{d-1}, 2^{d-1}+1, \ldots$ define upper and lower piecewise linear convex approximations $u_n, l_n: [0,1]^{d-1} \to \mathbb{R}$ of f: the graph of u_n is the "lower" side of $\operatorname{conv}\{(y_1, f(y_1)), \ldots, (y_n, f(y_n))\}$ and $l_n(x) = \max\{h_1(x), \ldots, h_n(x)\}$ for $x \in [0,1]^{d-1}$ where $x^{(d)} = h_i(x)$ is the tangent hyperplane of the graph of f at $(y_i, f(y_i))$. Then we have the following sandwich approximation of f:

$$l_n \leqslant f \leqslant u_n \quad \text{and} \quad \|u_n - l_n\|_\infty \leqslant \frac{\text{constant}}{n^{2/(d-1)}} \qquad \text{for } n = 1, 2, \ldots.$$

7. Approximation of special (classes of) convex bodies

Many approximation problems in convexity are of the following kind. Let C be a convex body of a special type. Can C be well approximated by convex bodies of similar but simpler characteristics? This subsection contains a selection of such results.

Zonoids. Before discussing zonoids a remark is in order. The results on zonoids and the results on symmetrization in the next subsection belong to so-called "high-dimensional" problems. These are problems on the borderline of convexity and the theory of normed spaces where the limiting behaviour as $d \to \infty$ is investigated. This is reflected by the fact that the constants and the O- and o-symbols appearing in such results do not depend on d or, if so, in a definite way, e.g., polynomially.

A *zonotope* Z is a polytope admitting a representation of the form

$$Z = S_1 + \cdots + S_n \ (= \{x_1 + \cdots + x_n : x_i \in S_i\})$$

with o-symmetric line segments S_i. A convex body in $\mathscr{C}_o$ is a *zonoid* if it is the Hausdorff limit of a sequence of zonotopes. For $d = 2$ each convex body in $\mathscr{C}_o$ is

a zonoid. An example of a zonoid for $d \geqslant 3$ is B^d. For $d \geqslant 3$ the set of zonoids is closed and nowhere dense in $\mathscr{C}_o$. For general information on zonoids see Bolker (1969), Schneider and Weil (1983) and chapter 4.9 by Goodey and Weil. For $n = d, d+1, \ldots$, let $\mathscr{Z}_n$ denote the set of all zonoids in $\mathscr{C}_o$ which are sums of at most n line segments.

Improving upon results of Bourgain, Lindenstrauss and Milman (1989a), Bourgain and Lindenstrauss (1988) established the following estimates in an even more precise form. For any zonoid Z,

$$\delta^{\mathrm{BM}}(Z, \mathscr{Z}_n) \leqslant 1 + \frac{1}{n^{(5/4)-o(1)}} \quad \text{as } n \to \infty \quad (d = 3),$$

$$\delta^{\mathrm{BM}}(Z, \mathscr{Z}_n) \leqslant 1 + \frac{1}{n^{1-o(1)}} \quad \text{as } n \to \infty \quad (d = 4),$$

$$\delta^{\mathrm{BM}}(Z, \mathscr{Z}_n) \leqslant 1 + \frac{1}{n^{(d/2(d-1))-o(1)}} \quad \text{as } n \to \infty \quad (d \geqslant 5).$$

For $Z = B^d$ this may be improved, see Betke and McMullen (1983), Linhart (1989) and Bourgain and Lindenstrauss (1988). On the other hand a result of Bourgain, Lindenstrauss and Milman (1989a) says that

$$\delta^{\mathrm{BM}}(B^d, \mathscr{Z}_n) \geqslant 1 + \frac{\text{constant}}{n^{(d+1)/2(d-1)}} \quad \text{for } n = d, d+1, \ldots.$$

An older estimate for $\delta^{\mathrm{S}}(B^d, \mathscr{P}^{\mathrm{i}}_{(n)})$ as $n, d \to \infty$ is due to Rogers (1953).

Special classes of convex discs. Let $d = 2$. First, for $a, p > 0$ satisfying $4\pi a \leqslant p^2$ (isoperimetric inequality) let

$$\mathscr{C}(a,p) = \{C \in \mathscr{C} : \mu(C) \geqslant a, \sigma(C) \leqslant p\}.$$

Florian (1986, 1992) determined those pairs C, P in $\mathscr{C}(a,p) \times \mathscr{P}_n$ for which

$$\delta^{\mathrm{S}}(\mathscr{C}(a,p), \mathscr{P}_n) = \inf\{\delta^{\mathrm{S}}(C,P) : (C,P) \in \mathscr{C}(a,p) \times \mathscr{P}_n\}$$

is attained and similarly for δ^{P} and δ^{1}. Note that these results are of interest only if $p^2/a < 4n\tan(\pi/n)$ since otherwise $\mathscr{C}(a,p) \cap \mathscr{P}_n \neq \emptyset$ and thus the infima are 0. For related results see L. Fejes Tóth (1957), L. Fejes Tóth and Heppes (1960), G. Fejes Tóth and Florian (1984) and Florian (1989b).

Second, for $a, p > 0$ satisfying $p^3 \leqslant 8\pi^2 a$ (affine isoperimetric inequality) define

$$\mathscr{D}(a,p) = \{C \in \mathscr{C} \cap \mathscr{C}^2 : \mu(C) = a, \text{ affine perimeter of } C \text{ is } \geqslant p\}.$$

L. Fejes Tóth (1980) determined the pairs (C,P) for which $\delta^{\mathrm{S}}(\mathscr{D}(a,p), \mathscr{P}^{\mathrm{i}}_n)$ and $\delta^{\mathrm{S}}(\mathscr{D}(a,p), \mathscr{P}^{\mathrm{c}}_n)$ are attained.

Rotatory symmetry. Let $d = 2$ and assume that $C \in \mathscr{C}$ has k-fold rotatory symmetry.

Do the best approximating polygons of C in $\mathfrak{Q}_{nk}$ with respect to a metric δ also have k-fold rotatory symmetry? The answer is affirmative for $\delta = \delta^S$ and $\mathfrak{Q}_{kn} = \mathfrak{P}_{kn}, \mathfrak{P}^i_{kn}, \mathfrak{P}^c_{kn}$, as well as for $\delta = \delta^P$ and $\mathfrak{Q}_{kn} = \mathfrak{P}^i_{kn}, \mathfrak{P}^c_{kn}$. This has been established by G. Fejes Tóth and L. Fejes Tóth (1973) and G. Fejes Tóth (1977).

Constant width and symmetry. Schneider (1984) proved that any convex body C of constant width can be approximated arbitrarily closely by analytic convex bodies of constant width having the same symmetries as C. For $d = 2$ see Tanno (1976) and Wegner (1977).

Curvature. Given $C \in \mathcal{C}$, can C be approximated arbitrarily closely by convex bodies $C_1, C_2, \ldots$, such that the principal, Gaussian, or mean curvatures of $C_1, C_2, \ldots$ "converge" to that of C? For results of this type see Weil (1973) and, in a more general setting, Brehm and Kühnel (1982).

Geometric inequalities; stability. In many geometric inequalities equality holds only for special convex bodies such as balls, ellipsoids, simplices, Reuleaux triangles, etc. Do the convex bodies for which there is almost equality approximate those convex bodies well for which equality is attained? Are there quantitative results? A survey of the large body of pertinent results has been given by Groemer in chapter 1.4.

8. Symmetrization

For certain extremal problems in convexity there are transformations $s_1, s_2, \ldots : \mathcal{C} \to \mathcal{C}$ such that for $C \in \mathcal{C}$ the sequence $s_1 C, s_2 C, \ldots$ or a suitable subsequence of it converges to a solution. A problem arising in this context is that of the convergence rate.

Precise definitions of Steiner and Blaschke symmetrizations are contained in chapter 1.2 by Sangwine-Yager.

Refining an older "Kugelungstheorem" of Gross (1917), Lyusternik (1935) and Hadwiger (1952, 1957), Bourgain, Lindenstrauss and Milman (1988, 1989b) gave a precise answer to the following question: given $\varepsilon > 0$, how many Steiner or Blaschke symmetrizations are necessary to transform a convex body C into a convex body which differs from B^d by less than $1 + \varepsilon$ with respect to δ^{BM}?

Schmuckenschläger (1991a) proved the following result. There is an absolute constant such that for any $C \in \mathcal{C}$ and $n = 1, 2, \ldots$, there are rigid motions $m_1, \ldots, m_n$ with

$$\delta^{BM}(C, \frac{1}{n}(m_1 C + \cdots + m_n C)) \leqslant 1 + \frac{\text{constant}}{n^{1/2}}.$$

For related earlier results see Hadwiger (1957) and Bourgain, Lindenstrauss and Milman (1989b).

9. Miscellanea

In the last part of this article we present a selection of topics which have been investigated in recent years and which, possibly, will lead to further research.

Approximation by smooth bodies. An old result of Minkowski (1903) says that the convex bodies of the form

$$\{x \in \mathbb{E}^d : \varphi(x) \leqslant 1\}, \quad \varphi : \mathbb{E}^d \to \mathbb{R} \text{ analytic},$$

are dense in $\mathcal{C}$. Alternative proofs of this or of related results are due to Radon (1936) and Schmuckenschläger (1991b). Hammer (1963) showed that Minkowski's theorem continues to hold when the functions φ are polynomials. Firey (1974) proved that the convex bodies with algebraic support functions are dense in $\mathcal{C}$. For smooth approximation with additional properties see section 7 above and, in particular, Schneider's article (1984). See also Faro Rivas (1986).

Approximation by polytopes with bounded valences. Ewald raised the following question: Do there exist nonnegative integers p, q such that the set of all polytopes for which each vertex belongs to at most p edges and each facet contains at most q vertices, is dense in $\mathcal{C}$? Solutions of this problem are due to Bokowski (1977) ($d = 3$), Bokowski and Schulz (1978) ($d = 4$) and Bokowski and Mani (1987) ($d \geqslant 5$).

Optimization and optimal control. Several problems on best approximation of convex bodies have been solved by formulating them as optimization and optimal control problems and solving the latter; see, e.g., Englisch and Voigt (1988) or Juhnke (1990). Other geometric problems which are related to approximation and dealing with inradius, incircle or inball, minimal width and the like have also been treated in this way, cf. Klötzler (1983), Karkakis and Karagiorgis (1986), Hoàng (1987), Orlowski and Pachter (1987) and Juhnke (1988).

Approximation by images of a convex body. There are geometric and analytic problems in the context of which one has to compare two convex bodies, but due to the nature of the problems the convex bodies are of interest only up to rigid motions or other transformations. As an example consider norms on $\mathbb{E}^d$ where isometric norms are not considered as being essentially different. This led to the introduction of the Banach–Mazur metric. Other natural examples of such problems may be found in the context of chapter 1.4 by Groemer.

Let δ be a measure of distance on $\mathcal{C}$. Then for $C, D \in \mathcal{C}$ what can be said about

$$\begin{aligned} &\inf\{\delta(C, D + t) : t \in \mathbb{E}^d\}, \\ &\inf\{\delta(C, mD) : m \text{ rigid motion on } \mathbb{E}^d\}, \\ &\vdots \end{aligned} \tag{13}$$

Clearly this may be used to define new notions of distance on $\mathscr{C}$. It turns out that the latter are valuable tools for stability results, see the corresponding section by Groemer.

Arnold (1989) proved the following theorem which generalizes an earlier article of Inzinger (1950); see also an unfortunately unpublished result of Davis (1982). If for the metric δ^2 the infima in (13) are attained for t, respectively m then the Steiner points of C and $D+t$, respectively mD coincide. (The *Steiner point* of a convex body C is

$$\frac{1}{\kappa_d}\int_{S^{d-1}} uh_C(u)\,\mathrm{d}\sigma(u).)$$

For translations this solves the problem stated above. A consequence of this property of δ^2-approximation and its proof is the following: If $P \in \mathscr{P}_n$ is a best approximating polytope of a convex body C with respect to the metric δ^2, then the mean widths of C and P coincide and similarly for the Steiner points. In the case of rigid motions Arnold specifies for $d=2$ further necessary conditions in terms of Fourier expansions of h_C and h_{mD}. We also refer to Davis, Vitale and Ben-Sabar (1977).

Approximation by ellipsoids. Given $C \in \mathscr{C}$, a classical result of John (1948) with many applications says that there is an ellipsoid E, say with center c, such that

$$E \subset C \subset \lambda(E-c)+c \quad \text{for a suitable } \lambda \leqslant d.$$

Moreover, if $C \in \mathscr{C}_o$ then E may be chosen in $\mathscr{C}_o$ such that

$$E \subset C \subset \lambda E \quad \text{for a suitable } \lambda \leqslant \sqrt{d}.$$

It is the role which this and related ellipsoids play in the different versions of the "ellipsoid algorithm" of Shor and Khachiyan in optimization which in recent years spurred the work on approximation by ellipsoids. Selected references are Silverman and Titterington (1980), Tarasov, Khachiyan and Èrlikh (1988), Shor and Berezowskii (1989), Juhnke (1990) and Khachiyan (1990).

Approximation by parallelotopes. Improving upon results of Hadwiger (1955) and others, Kashin (1989) established the following estimate. For any $C \in \mathscr{C}_o$ there is a parallelotope $P \in \mathscr{C}_o$ with

$$C \subset P \quad \text{and} \quad \mu(P) \leqslant \text{constant}^d d^{d/2}\mu(C),$$

where the constant is absolute. See also Pelczynski and Szarek (1991).

Approximation by cylinders. A *generalized cylinder* Z is a convex body which can be represented in the form $Z = B+S$ where B is a convex body in a subspace of $\mathbb{E}^d$ of codimension 1 and S a line segment. We state a result of Gritzmann

(1985) which has implications for packing and covering. For any $C \in \mathcal{C}$ there is a generalized cylinder Z such that

$$Z \subset C \quad \text{and} \quad \mu(C) \leqslant (1 + 1/(d-1))^{d-1} \mu(Z).$$

General older results which are similar in spirit to the results cited in the last three paragraphs are due to Macbeath (1951b) and Thomas (1979). Recently Lassak (1989) considered the problem of δ^{BM}-approximation by o-symmetric discs.

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CHAPTER 1.11

Special Convex Bodies

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Contents

1. Introduction 349
   1.1. Regular polytopes 349
   1.2. Balls 352
   1.3. Ellipsoids 353
2. Simplices 355
   2.1. Characterizing simplices by intersections of homothets 355
   2.2. Measures of symmetry 356
   2.3. Further characterizations of simplices 357
3. Ellipsoids 358
   3.1. Characterizations by sections and projections 358
   3.2. Further characterizations 359
   3.3. Approximation of convex bodies by ellipsoids 360
   3.4. Further results on ellipsoids 360
4. Centrally symmetric convex bodies 361
   4.1. Characterizations by sections and projections 361
   4.2. Measures of symmetry 362
   4.3. Further characterizations of central symmetry 363
5. Convex bodies of constant width 363
   5.1. Introduction 363
   5.2. Basic properties and characterizations 364
   5.3. Special bodies and approximations 364
   5.4. Completion and completeness 364
   5.5. Intersection properties 365

HANDBOOK OF CONVEX GEOMETRY
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5.6. Curvature and mixed volumes 365
5.7. Inequalities 366
5.8. Inscribed and circumscribed bodies 367
5.9. Packing and covering 367
5.10. Rotors in polytopes 367
5.11. Concepts related to constant width 367
References 368

1. Introduction

Before we report on the literature about simplices, ellipsoids, centrally symmetric bodies, and bodies of constant width, we give a more elementary and detailed presentation of regular polytopes, balls and ellipsoids. Special bodies will of course also appear at other places of this book, e.g., in chapters 4.9 and 2.4. Some topics of the following chapter are treated in greater detail in Martini's survey (to be published).

1.1. Regular polytopes

A closed set in Euclidean space $\mathbb{E}^d$ bounded by a hyperplane is called *half space*. By a *polytope* we mean any bounded and non-empty intersection of finitely many closed half spaces. The 2-polytopes are the polygons, the 3-polytopes are often called *polyhedra*. We do not have to explain what the vertices, edges and faces of a 3-polytope are. In $\mathbb{E}^d$ we have the k-faces, $k = 0, \ldots, d$, where the 0-faces are the vertices, the 1-faces the edges, the $(d-1)$-faces the facets, and the d-face is the polytope itself.

1.1.1. The five Platonic solids

A plane polygon is called *regular* if all its edges and angles are congruent. A 3-polytope is called *regular* if all its faces are regular and congruent and if all its vertex figures are congruent. (A vertex figure consists of all faces adjacent to this vertex.)

If the faces are congruent equilateral triangles, then we can join together 3, 4, or 5 triangles to form a vertex. In the first case we can close the opening by a single triangle, and we obtain a *tetrahedron*. In the second case the opening is a square and can be closed by a congruent figure. We thus get an *octahedron*. In the third case we can add 15 further triangles and obtain the *icosahedron* which closes with altogether 20 triangles. It is not obvious that this really is true. We will soon return to this question of existence. There are no other regular polyhedra with triangular faces because 6 equilateral triangles laid together give a flat hexagon. If the faces are squares, then one may combine three at a vertex, and one gets a *cube*. Four squares already give a plane figure. The inner angles of a regular pentagon are 108°. One can join 3 of them (and not more) to form a vertex. Twelve pentagons form the *dodecahedron*. This is intimately connected with the existence of the icosahedron. Since already 3 regular hexagons form a plane figure, there cannot be further regular polyhedra.

Figures of the 5 regular polyhedra can, e.g., be found in Coxeter (1948), Hilbert and Cohn-Vossen (1932) or Fejes Tóth (1964). They are also called *Platonic solids* because Plato mentioned them in the "Timaeos". But they were already known before (Tropfke 1924, p. 47, Waterhouse 1972), even in prehistoric times (Critchlow 1979). The construction of the regular solids in Book XIII is one of the highlights of Euclid's Elements. The most elegant construction of the icosahedron was given by Luca Pacioli in 1509, see Stillwell (1989, p. 16).

A lot of literature on the Platonic solids and their history is collected by Toepell (1991).

The existence of the icosahedron and the dodecahedron may of course be exhibited by giving coordinates of their vertices. But the following simple argument will show their existence without long calculations or ingenious constructions. Imagine two horizontal pentagons one above the other. Rotate the upper one about an angle of 36° so that it is still horizontal. Then connect the vertices by 10 isosceles triangles. This gives a polytope called an *antiprism*. For a certain distance of the pentagons the triangles will be equilateral. Now replace the pentagons by two caps formed by 5 equilateral triangles. This completes the construction of an icosahedron. (Since this construction is not symmetric, the congruence of the vertex figures must still be shown.) If we connect the midpoints of adjacent triangular faces, then we obtain 12 regular pentagons which are the faces of a dodecahedron. Hereby the existence of the 5 Platonic solids is established.

In the same way we obtain an icosahedron from a dodecahedron: the midpoints of adjacent pentagons are the vertices of an icosahedron. In this sense icosahedron and dodecahedron and likewise cube and octahedron are *dual* to each other, whereas the tetrahedron is self-dual.

There are not more than 5 regular 3-polytopes, not only because of the metric relations in $\mathbb{E}^3$ but also because there are not more than 5 combinatorially regular 3-polytopes, i.e., 3-polytopes such that all faces have the same number of edges and such that at every vertex the same number of faces meet. This is a consequence of *Euler's Formula*

$$V - E + F = 2$$

where V is the number of vertices, E of edges and F of faces of a general 3-polytope (see, e.g., Coxeter 1948, p. 9). Indeed, if all faces have m edges and if at each vertex k faces meet, then

$$E = (m/2)F\,, \qquad V = (m/k)F\,.$$

Inserting this into Euler's Formula we see $(m/k) - (m/2) + 1 > 0$, i.e.,

$$(m-2)(k-2) < 4\,.$$

Since m, $k \geqslant 3$ there remain just 5 possibilities.

Various extremal properties of regular polydedra are discussed in chapter 1.6.

1.1.2. Regular polytopes in higher dimensions

A d-polytope is called regular if all its facets, the $(d-1)$-faces, are regular and congruent and if all its vertex figures are congruent. (This definition works by induction).

In $\mathbb{E}^d = \mathbb{R}^d$ a *cube* can be given by its 2^d vertices, e.g., $(x_1, \ldots, x_d)$ with $x_i = 0$

or 1, $i = 1, \ldots, d$. The convex hull of these points equals the intersection of the $2d$ half spaces $x_i \leqslant 1$, $x_i \geqslant 0$, $i = 1, \ldots, d$. A *cross polytope* is for instance given by the $2d$ vertices $(\pm 1, 0, \ldots, 0), \ldots, (0, \ldots, \pm 1)$. In $\mathbb{E}^3$ this is an octahedron. The *regular simplex* in $\mathbb{E}^d$ is the generalization of the tetrahedron in $\mathbb{E}^3$. It may be constructed in the following way. In $\mathbb{R}^{d+1}$ we choose the $d+1$ points $(1, 0, \ldots, 0), \ldots, (0, \ldots, 0, 1)$. They lie in the d-dimensional hyperplane $x_1 + \cdots + x_{d+1} = 1$ and are the vertices of the d-dimensional regular simplex. The regularity follows here as in the other two cases from the fact that there is no distinction among the vertices.

In $\mathbb{E}^d$, $d \geqslant 5$, there are no other regular polytopes, but in $\mathbb{E}^4$ there are 3 others (Rosenfeld and Jaglom 1971, p. 378, Berger 1978, vol. 3, p. 116). In table 1, e_0, e_1, e_2, e_3 mean the number of vertices, edges, 2-faces and facets, respectively. Projections of them are given by Coxeter (1948, plates V–VIII), Hilbert and Cohn-Vossen (1932, p. 133), and Banchoff (1990). The table reflects that cube and cross-polytope as well as 120-cell and 600-cell are dual to each other whereas simplex and 24-cell are self-dual. They all are centrally symmetric, except the simplex. One can also verify Euler's Formula. This was given for $\mathbb{E}^d$ by Schläfli and, in order to avoid distinction between even and odd dimensions, may best be written in the form

$$e_0 - e_1 + - \cdots \pm e_{d-1} \mp e_d = 1$$

(here $e_d = 1$, see, e.g., Coxeter 1948, p. 165).

A polytope in $\mathbb{E}^d$ which is combinatorially regular is already combinatorially isomorphic to a regular polytope, (McMullen 1967, see also Dress and Schulte 1987). But for $d > 3$ no reasoning is known which is similar to the one given above for $d = 3$ and which directly allows us to exclude the existence of other combinatorially regular polytopes.

Tables for volume, surface area, and other values for the regular polytopes can be found in Coxeter (1948); for $d = 3$, see also Hadwiger (1955, p. 36).

1.1.3. Polytopes which are less regular

A polytope is called semi-regular if all facets are regular (but not necessarily congruent) and all vertex figures are congruent (cf. section 1.1.1). In $\mathbb{E}^3$ the semi-regular polytopes are the 5 Platonic solids, certain prisms and antiprisms (cf.

Table 1
Regular polytopes in $\mathbb{E}^4$

Name	e_0	e_1	e_2	e_3 facets
regular simplex = 5-cell	5	10	10	5 tetrahedra
cube = 8-cell	16	32	24	8 cubes
cross-polytope = 16-cell	8	24	32	16 tetrahedra
24-cell	24	96	96	24 octahedra
120-cell	600	1200	720	120 dodecahedra
600-cell	120	720	1200	600 tetrahedra

section 1.1.1), the 13 Archimedean solids and one body closely related to one of them. For a complete enumeration and description of the Archimedean solids we refer to Aschkinuse (1969, p. 447), Cundy and Rollet (1961), Fejes Tóth (1964), p. 108), Lyusternik (1963, p. 147), Roman (1968) and others. For the history, see Tropfke (1924, p. 52), Hofmann (1963), and Waterhouse (1972). Aschkinuse, Cundy and Rollet, and Roman also treat the duals of the Archimedean solids. These duals are also called Catalan solids, because Catalan was the first mathematician who described all of them, cf. Brückner (1900, p. 160). Some of the Catalan solids are much older, e.g., the rhombic dodecahedron, which plays a role in crystallography (cf. chapter 3.7), and the rhombic triacontahedron, which is of some importance also in the theory of quasicrystals (cf. chapter 3.5).

In 1900, Gosset gave a list of semi-regular polytopes in $\mathbb{E}^d$, $d \geq 4$. That this list is complete was shown for $d = 4$ by Makarov in 1988 and for $d \geq 4$ by Blind and Blind (1991), see this paper for references. These proofs use the complete lists of regular-faced polytopes (i.e., of polytopes all of whose facets are regular), which were given by Zalgaller for $d = 3$, and Blind and Blind for $d \geq 4$, for references see Grünbaum (1967, p. 414), and Blind and Blind (1991).

Further classes arise if the bodies need not be convex. For instance, in $\mathbb{E}^3$ there are the 4 regular star polyhedra or Kepler–Poinsot polyhedra, see, e.g., Aschkinuse (1969, p. 450). Fejes Tóth (1964, no. 16), Roman (1968, p. 41), and Fischer (1986, chapter 5). Models of all such polytopes are described by Cundy and Rollet (1961), and Wenninger (1971, 1979, 1983). Related topics are also treated by Grünbaum (1977) and in chapter 2.4.

1.2. Balls

A ball in $\mathbb{E}^d = \mathbb{R}^d$ is given by $x_1^2 + \cdots + x_d^2 \leq r^2$. The unit ball ($r = 1$) is denoted by B^d. Its boundary is the sphere S^{d-1}. Since B^d is a body of revolution, its volume κ_d may be calculated recursively by

$$\kappa_d = \kappa_{d-1} \int_{-1}^{1} \sqrt{1 - x^2}^{\,d-1} \, dx = \kappa_{d-1} \int_0^{\pi} \sin^d \varphi \, d\varphi$$

$$= \begin{cases} \dfrac{\pi^{d/2}}{(d/2)!} & \text{if } d \text{ even}, \\ \dfrac{2^{(d+1)/2} \pi^{(d-1)/2}}{1 \cdot 3 \cdots d} & \text{if } d \text{ odd}. \end{cases}$$

This may be combined to

$$\kappa_d = \frac{\pi^{d/2}}{\Gamma(d/2 + 1)} .$$

For a more elegant derivation see Fejes Tóth (1964, chapter IV), and, also for the volume of balls in spaces of constant curvature, Gray (1990, p. 253).

Characterizations of balls or spheres are collected by Bonnesen and Fenchel (1934, no. 69), Giering (1971), and Bigalke (1986). Balls often furnish the equality case of a geometric inequality, and in this way they will appear in several chapters of this book, see also Burago and Zalgaller (1988, chapter 2). Hilbert and Cohn-Vossen (1932, pp. 190–205) describe properties of balls that do not characterize them, e.g., "constant width" which we treat in section 5. Here we also mention Blaschke's "Wiedersehensflächen", cf. Blaschke and Leichtweiss (1973, p. 310), and Berger and Gostiaux (1988, p. 422).

Sphere and ball also serve as underlying point sets for non-Euclidean geometries: Bigalke (1984) develops spherical geometry on S^2, Berger (1977, vol. 5, chapter 18) on S^d. Elliptical geometry is obtained from spherical geometry by identifying antipodal points, hyperbolic geometry takes place within the ball, see Klein (1928), Berger (1977, vol. 5, chapter 19), and Hilbert and Cohn-Vossen (1932, pp. 207–228). Spheres may also serve as basic elements of a geometry, playing the role that points play in more familiar geometries. Blaschke (1929) and Takasu (1938, 1939) are devoted to this type of geometry, see also Berger (1977, vol. 5, chapter 20), and Burau (1985).

Despite the vast literature on balls and spheres there are still open problems. Many of the characterizations quoted in Bonnesen and Fenchel (1934), Giering (1971) and Bigalke (1986) are given for 3 dimensions, and it is not always clear how much effort the extension to $\mathbb{E}^d$ will take (and the same is true for several characterizations mentioned in the following sections on ellipsoids and centrally symmetric bodies). Ulam's floating body problem is still unsolved: is a body of uniform density which will float in water in every position a ball? For more information see Mauldin (1981, p. 90), Croft, Falconer and Guy (1991, p. 19), and Gilbert (1991), who also gives interesting examples. Recently, we learned that Òdor (to be published) has a solution. We close with two equivalent questions of P. Gruber: is a surface, for instance of the topological type of the sphere, a sphere if there is no gravitational force on inner points for a constant mass distribution on the surface? Assuming a conducting surface, is this true if an electric charge has constant density? For the two-dimensional case see Martensen (1992).

1.3. Ellipsoids

An ellipsoid $\mathscr{E}^d$ in $\mathbb{E}^d$, $d \geq 2$, can be defined as an affine (linear) image of a ball: $\mathscr{E}^d = TB^d$. (Sometimes it is more convenient to call the boundary ellipsoid.) From this we have immediately for the volume

$$V(\mathscr{E}^d) = |\det T| \kappa_d ,$$

κ_d being the volume of B^d, see section 1.2. This is a special case ($i = 0$) of the following formula for the quermassintegrals (see chapter 1.2)

$$W_i(TB^d) = |\det T| W_{d-i}(T^{-1}B^d)\,,$$

which follows from the representation as mixed volumes. For the surface area S $(=dW_1)$ there is no simple device for calculation. For an ellipse with half axes a, b and length L $(=2W_1)$ of its circumference we have

$$L = 4aE(e, \tfrac{1}{2}\pi)\,,$$

where $e = \sqrt{a^2 - b^2}/a$ and $E(\cdot, \frac{1}{2}\pi)$ is the complete elliptic integral of the 2nd kind, see Lawden (1989, p. 95); for the surface area $S(\mathscr{E}^3)$, see Lawden (1989, p. 102). For ellipsoids of revolution, $S(\mathscr{E}^3)$ may be given explicitely, see also Hadwiger (1955, p. 36). We give further references in section 3.4.

Ellipses can be defined in various ways. For Appollonius they were *conic sections*. The configuration of Dandelin shows immediately that conic sections may be constructed by the *gardener's* or *string construction*. There also is a string construction for ellipsoids (Hilbert and Cohn-Vossen 1932, pp. 2, 7, 17, Klein 1926, p. 32, Berger and Gostiaux 1988, p. 367).

If the lengths a, b, c of the axes of an ellipsoid $\mathscr{E}$ in $\mathbb{E}^3$ are different, say $a > b > c$, then there are 2 families of parallel circles on $\mathscr{E}$. This is easily seen in the following way. Cut $\mathscr{E}$ by planes containing the axis of length b. The resulting ellipses all have one axis of length b. The other axis has length between a and c and equal to b for exactly 2 planes. We will now show that parallel planes cut $\mathscr{E}$ in circles, too. $\mathscr{E}$ is the image of a sphere S^2 under an invertible linear map A. The corresponding intersections on S are circles. As shown, in both of these families of circles there is one which is mapped onto a circle on $\mathscr{E}$. Then A will map all circles of these families onto circles on $\mathscr{E}$.

For historical references on these circles, see Staude (1904, p. 181). A description is also given by Hilbert and Cohn-Vossen (1932, p. 16). We now collect several other beautiful properties of ellipsoids.

The 2 families of parallel circles on an ellipsoid $\mathscr{E}$ have 4 points as limiting cases, the *circular points* or *umbilics* of $\mathscr{E}$. If we pick out 2 of them that are not diametrical, then we may perform a string construction on $\mathscr{E}$, the string tightened along $\mathscr{E}$. In this way we obtain 2 families of closed curves on $\mathscr{E}$, which in differential geometry are called *curvature lines*.

They occur also in the following way: the ellipsoid $\mathscr{E}$ may be embedded into a triply orthogonal system of surfaces such that through every point of $\mathbb{E}^3$ passes an ellipsoid, a one-sheeted and a two-sheeted hyperboloid. Then the 2 families of hyperboloids cut $\mathscr{E}$ in the lines of curvature (Hilbert and Cohn-Vossen 1932, p. 166, Berger and Gostiaux 1988, p. 398, Fischer 1986, chapter 3, §2).

The shortest lines (geodesics) have been extensively studied. All geodesics starting from one of the circular points pass through the opposite point. (For references, see Berger and Gostiaux 1988, p. 381.)

In section 3 we report extensively on the literature about characterizations of ellipsoids.

2. Simplices

A simplex in $\mathbb{E}^d$ is defined as the convex hull of $d+1$ affinely independent points. We list some books with chapters on simplices and papers containing a great number of references: Neuberg (1884), Zacharias (1914), Coolidge (1916), Roché and Comberousse (1922), Court (1932), Couderc and Ballicioni (1953), Thébault (1955), Mandan (1961/62), Satyanarayana (1962), Baston (1965), Benson (1966), Grünbaum (1967), Boltjanski and Jaglom (1971), Rosenfeld and Jaglom (1971), Gerber (1977), Kelly and Weiss (1979), Lay (1982), Brøndsted (1983), and Mitrinović, Pečarić and Volenec (1989). This long and not even complete list shows the vastness of this field. We will thus restrict ourselves to those properties of simplices that characterize them within the set K^d of convex bodies (or a sufficiently large subset) in $\mathbb{E}^d$ $(d \geqslant 2)$. For the characterization of convex polytopes among convex sets, see Klee (1959) and Gale, Klee and Rockafellar (1968). Formulas for the volume of simplices and its lower dimensional faces in terms of equations or vertices are provided by Klebaner, Sudbury and Watterson (1989).

2.1. Characterizing simplices by intersections of homothets

We start by joining some characterizations scattered in the literature. We need the following concepts and notations. A convex body $K \in K^d$ is called a *Choquet simplex* if, whenever the intersection of two positive homothets of K is at least one-dimensional, this intersection is positively homothetic to K. Let $V(K)$ denote the volume of K, $\overline{V}_{d-1}(K, u)$ the area of the orthogonal projection of K onto $\{x \mid \langle x, u\rangle = 0\}$, $\underline{V}_1(K, u)$ the maximal chord length of K in the direction $u \in S^{d-1}$, and h_K the support function of K, i.e., $h_K(u) = \max\{\langle x, u\rangle \mid x \in K\}$. The *projection body* ΠK is defined by its support function $h_{\Pi K}(u) = \overline{V}_{d-1}(K, u)$, the *polar body* K^* of K is $\{u \mid h_K(u) \leqslant 1\}$ and the *difference body* DK of K is $K + (-K)$. A d-dimensional vector sum of a $(d-1)$-dimensional convex body and a line segment is called a *cylinder*, whereas a compact d-dimensional set in $\mathbb{E}^d$ is called a *generalized cylinder* if all non-empty intersections with lines of a fixed direction u are segments of the same positive length. Martini (1990) showed the equivalence of the following properties of a convex body $K \in K^d$ $(d \geqslant 2)$ in an elementrary way:

(1) K is a simplex.
(2) K is a Choquet simplex.
(3) ΠK and DK are polars with respect to a sphere with radius $\sqrt{dV(K)}$.
(4) Each cylinder Z with $K \subset Z$ satisfies $V(Z)/V(K) \geqslant d$.
(5) Each generalized cylinder $\overline{Z}$ with $\overline{Z} \subset K$ satisfies

$$V(K)/V(\overline{Z}) \geqslant (1 + (d-1)^{-1})^{d-1} \quad (<e)\,.$$

The equivalence of (1) and (2) was first noticed by Choquet (1956) and used for the definition of simplices in infinite-dimensional spaces. Independently, Rogers

and Shephard (1957) proved the same equivalence and derived an estimate for $V(K)/V(DK)$ from it, cf. also Leichtweiss (1980). The implication (1) $\Rightarrow$ (4) goes back to Butler (1971), its converse was proved by Schneider (1973). For a fixed segment direction u of $\overline{Z}$, Gritzmann (1985) showed that (5) is equivalent to $\overline{V}_{d-1}(K, u)\underline{V}_1(K, u) = dV(K)$ and applied this to finite coverings. Martini and Weissbach (1992) showed (1) $\Rightarrow$ (3), Martini (1989a) its converse. Some of these equivalences allow interesting generalizations. The first possibility is to consider more general classes of point sets. Gruber (1969) found that (1) and (2) are still equivalent in the class of all compact sets in $\mathbb{E}^d$ with non-empty interior. Eggleston (1971) showed that a bounded measurable Choquet simplex of positive measure is precisely the interior of a simplex together with the relative interior of some of its r-faces, $0 \leq r \leq d-1$. For further generalizations in this sense we refer to Fourneau (1985) and the references given there. On the other hand, one can investigate (1) $\Leftrightarrow$ (2), but modify the intersection property in the definition of Choquet simplices. For example, if K and the d-dimensional intersections of K with translates of K have congruent normal cones at corresponding points, then K is a d-dimensional direct linear sum of simplices (Schneider 1967); the same is true if K and these intersections are polytopes with the same number of vertices (Schneider 1969) or if these intersections are affine images of K (Gruber 1970a). For related results and applications see Eggleston, Grünbaum and Klee (1964), Gruber (1970b, 1976), McMullen, Schneider and Shephard (1974), and Wilker (1975). Furthermore, the equivalences (1) $\Leftrightarrow$ (2) and (1) $\Leftrightarrow$ (3) were sharpened by Gruber (1971) and Martini (1989b), respectively. The question for which convex bodies K the bodies ΠK and DK are polars of each other [without the choice of radius in (3)] was raised by Petty (1972) and partially answered by Martini (1991): a convex d-polytope ($d \geq 3$) has this property if and only if it is a simplex. Thus one might ask whether for $d \geq 3$ a non-polytopal convex body with this property is necessarily an ellipsoid. Here we should also mention a recent result of Zhang (1991): if $\Pi^* K$ denotes the polar of ΠK, then the lower bound of $V(K)^{d-1} \cdot V(\Pi^* K)$ characterizes d-simplices within K^d. Finally, we mention some surveys on Choquet simplices in infinite-dimensional spaces, important for functional analysis, but also (e.g., in view of the methods) interesting for geometric convexity: Edwards (1971), Alfsen (1971), Rogalski (1971), Bair and Fourneau (1980), Ellis (1984), Rosenthal (1988).

2.2. *Measures of symmetry*

Measures of symmetry will be discussed more extensively in section 3. However, the following papers are mainly devoted to that side of a symmetry measure which characterizes simplices. For instance, the sharp lower estimate of $V(K)/V(DK)$ of Rogers and Shephard (1957) was proved again, with characterization of the equality case, or brought into a more general context by Rogers and Shephard (1958a), Eggleston (1958), Chakerian (1967), Schneider (1970a), and Leichtweiss (1980, p. 243). For the historical background, we also refer to Bonnesen and Fenchel (1934, p. 105), Godbersen (1938) and the references of Rogers and

Shephard (1957) and Grünbaum (1963). The sharp lower bound of $V(K)/V(DK)$ could also be derived from a stronger inequality independently conjectured by Godbersen (1938) and Makai (1974). A result related to measures of symmetry in the plane was given by Tsintsifas (1985).

Inspired by symmetry measures, Schneider (1975) introduced

$$c(M) = \inf\{\lambda \geqslant 0 \mid M + \lambda(\operatorname{conv} M) \text{ is convex}\}$$

as a measure of convexity for arbitrary sets M in $\mathbb{E}^d$. In particular, he showed $c(M) \leqslant d$ with equality just for the vertex sets of d-dimensional simplices.

2.3. Further characterizations of simplices

Leichtweiss (1959) used the infimum of the quotients of circumradius and inradius over classes of affinely equivalent convex bodies, Santaló (1957/58) and Eggleston (1961) the area and a certain width integral of a plane convex body in order to characterize simplices. Related results were obtained by Behrend (1937, 1938) and Sholander (1952).

A chord of a convex body K in $\mathbb{E}^d$ is called an affine diameter if both of its endpoints lie in different parallel supporting hyperplanes of K. A convex body K in $\mathbb{E}^2$ is a triangle if and only if each inner point belongs to exactly three (Eggleston 1957, chapter 3) or to at least two, but finitely many (Soltan and Nguen 1990) affine diameters. A convex d-polytope is a simplex precisely if every inner point lies on exactly $2^d - 1$ affine diameters (Martini, Nguen and Soltan 1990). Similar characterizations were obtained by Klee (1953), Zamfirescu (1966), and Soltan and Nguen (to be published). For two characterizations of simplices in combinatorial geometry we refer to Grünbaum (1961) and Boltyanskiĭ and Soltan (1978, chapter 8).

For which convex bodies K in $\mathbb{E}^d$ of fixed volume does the expected volume of the convex hull of n random points, chosen independently and uniformly from K, attain its maximum? In the plane this question was answered for $n = 3$ by Blaschke in 1917, see (1923, p. 55), for $n = 4$ by Buchta (1983), and for arbitrary n by Dalla and Larman (1991). For $d \geqslant 3$, Dalla and Larman settled the problem for all d-polytopes with at most $d + 2$ vertices and Bárány and Buchta (1990, to be published) for all d-dimensional convex bodies K and sufficiently large n. In all these cases the maximal volume is attained if K is a simplex. H. Groemer gives an interesting discussion in *Math. Reviews* 84g: 60 019; see also chapter 5.2 and, for historical remarks, Klee (1969). The corresponding problem for minimal volume leads to a characterization of ellipsoids, see Petty (1983), cf. also section 3.2. below.

Further simplex characterizations were given by Leichtweiss (1955), and Rogers and Shephard (1958b). We finish this section by some characterizations of *regular* simplices. The first one is connected with a reverse isoperimetric problem for volume and surface area. This makes sense if one looks at classes of affinely equivalent convex bodies. Namely, let K be a convex body and T a regular

simplex in $\mathbb{E}^d$. Then an affine image $\overline{K}$ of K exists with $V(\overline{K}) = V(T)$, $S(\overline{K}) \leqslant S(T)$, where V denotes the volume and S the surface area, cf. Ball (1991) and, for a related result in $\mathbb{E}^2$, Gustin (1953). A characterization of the edge system of the regular simplex is obtained by Heller (1957), similar (discrete) results by Witsenhausen (1974), Chung and Gilbert (1976) and Alexander (1977). An out-of-the-way characterization of special tetrahedra, namely by means of geodesics, is given by Post (1970). Several geometric inequalities characterizing (regular) simplices are given by Bonnesen and Fenchel (1934, §10), and Mitrinović, Pečarić and Volenec (1989).

3. Ellipsoids

For studying ellipsoids from the viewpoint of convexity one has a good starting point in the surveys of Bonnesen and Fenchel (1934, no. 70), Gruber and Höbinger (1976), and Petty (1983), which summarize results on characterizations of ellipsoid
- by means of sections and projections
- by extremal properties
- using the group of projectivities or subgroups thereof,

and also on ellipsoids in Minkowski geometry and Hilbert geometry. With a few exceptions we shall only present results which are not contained in these references. One of these exceptions is the famous Blaschke–Marchaud characterization of ellipsoids by plane shadow-boundaries. Blaschke (1916a) gave it for regular convex bodies with regular supporting planes, Marchaud (1959) for general convex bodies. For a convex body K and a straight line g through the origin o the (generalized) shadow boundary is defined by $\partial K \cap \partial(K + g)$. If for every g containing o there is a hyperplane H such that the shadow boundary contains $H \cap \partial K$, then K is an ellipsoid. We add some applications. A normed space is Euclidean if orthogonality is symmetric (Blaschke 1916a, Birkhoff 1935) or if for every k-dimensional ($k \geqslant 2$, closed in infinite dimensions) subspace there is a projection of norm 1 onto this subspace (Kakutani 1939 for finite, Bruck 1974 for infinite dimensions). More literature can be found in the surveys quoted above and in Gruber (1974). Krautwald (1980) and Martini (1989c) have related results. Here we also remark that in affine differential geometry shadow boundaries may be described as geodesics of suitable linear connections, see, e.g., Simon (1985, p. 53). We also mention the characterization of balls by Pamfilos (1988) which, as a special case, contains the characterization of ellipsoids by plane shadow boundaries.

3.1. Characterizations by sections and projections

Inspired by several characterizations of ellipsoids using intersecting hyperplanes through a fixed point, Bianchi and Gruber (1987) proved the following. Let K be a convex body in $\mathbb{E}^d$ ($d \geqslant 3$) and δ be a continuous real function on the unit sphere

S^{d-1} such that for any $u \in S^{d-1}$ the intersection of K with the hyperplane $\langle x, u \rangle = \delta(u)$ is non-degenerate and ellipsoidal. Then K is an ellipsoid. Moreover, they prove analogous statements for projections. Gruber (1990) asks if ellipsoids are the only (convex) billiard tables in $\mathbb{E}^d$ ($d \geq 3$) having caustics, or slightly more generally: is a convex body an ellipsoid if it appears centrally symmetric from any point of a surrounding closed surface? See also Klee and Wagon (1991, p. 76). Finally we mention the following remarkable result. Let K_1 and K_2 be convex bodies in $\mathbb{E}^d$ ($d \geq 3$). Then for all translates K_2' of K_2 (with $K_2' \neq K_1$) bd $K_2' \cap$ bd K_1 is contained in a hyperplane if and only if K_1 and K_2 are homothetic ellipsoids. This was shown by Shaĭdenko (1980), in fact for compact manifolds in $\mathbb{E}^d$ with tame boundary points, by Goodey and Woodcock (1981) for $d = 3$, and by Goodey (1983) for $d \geq 3$. Schneider (1980) characterizes balls by congruent hyperplane sections. This is related to characterizing ellipsoids by affinely equivalent sections. See Schneider's paper for further references. More generally, Montejano (1990b) characterizes balls by homothetic sections. Meyer and Reisner (1989) generalize the Brunn–Blaschke characterization of ellipsoids by straight lines of gravity centers ("gerade Schwerlinien"; cf. Blaschke 1923, pp. 18, 23, 213, Bonnesen and Fenchel 1934, p. 143, Kneser 1949, Gruber and Höbiger 1976); for a related result see Vincensini (1934). Characterizations of ellipsoids by means of inner and outer cross-sectional measures are given or conjectured by Busemann and Petty (1956), Rogers and Shephard (1958b), and Martini (1989c).

3.2. Further characterizations

There are a lot of geometric inequalities yielding ellipsoids as extremal bodies. For the general background, see Burago and Zalgaller (1988), and also Mitrinović, Pečarić and Volenec (1989). Affine isoperimetric inequalities, including the Blaschke–Santaló inequality, are treated in chapter 1.5. Various characterizations of ellipses are given in chapter 1.6.

Approximation of convex bodies by convex polytopes may lead to ellipsoids as extremal bodies. For volume approximation see, e.g., Groemer (1982), Petty (1983) and section 2.3 above. In particular, if P is a polytope with n vertices and maximal volume inscribed in a convex body $K \subset \mathbb{E}^d$ and if $g_{n,d}(K) = \text{vol}(P)/\text{vol}(K)$, then Macbeath (1951) shows $g_{n,d}(K) \geq g_{n,d}(B^d)$; inf $g_{4,3}$ can be found in Blaschke (1923, p. 191), and Bianchi (1986) determines $g_{5,3}$. In the last two cases the infimum is only attained for ellipsoids. For approximation in the sense of Hausdorff metric, Banach–Mazur distance, and symmetric difference metric see Gruber (1983) and Gruber (to be published) and the literature quoted there. In the second and third case the approximation is asymptotically (i.e., for $n \to \infty$) worst for ellipsoids. The ellipsoids here occur as the equality cases in the Blaschke–Santaló inequality and the affine isoperimetric inequality, respectively. In the first case (Hausdorff metric) balls are approximated worst.

Ellipsoids may also be characterized by the product of mean width and surface area or by the quotient of inradius and circumradius (Santaló 1957/58, Leicht-

weiss 1959, see also Behrend 1937, 1938). Several characterization theorems or conjectures from the theory of zonotopes and zonoids involve ellipsoids, see Schneider and Weil (1983). Characteristic properties of the ellipsoid are interesting for X-ray problems (Giering 1962). Characterizations of Euclidean spaces (see, e.g., Gołąb and Tamassy 1960) or, in infinite dimensions, of inner product spaces also give characterizations of ellipsoids (see Day 1958, Amir 1986). Berger (1936) characterizes ellipses by means of inscribed and circumscribed triangles in perspective position. Kubota (1937) extends this to higher dimensions. Characterizations based on the position of the vertices of all circumscribed boxes were given by Blaschke (1916b) and Chakerian (1965). Kiener (1986) characterizes ellipsoids by an extremal property of an integral over characteristic functions.

3.3. Approximation of convex bodies by ellipsoids

It is well known that for each convex body K in $\mathbb{E}^d$ ($d \geq 2$) there is a unique ellipsoid of minimal volume containing K (also called Loewner ellipsoid), likewise a maximal inscribed ellipsoid, see Behrend (1938), John (1948), Danzer, Laugwitz and Lenz (1957), Zaguskin (1958), Gruber (1988), Juhnke (1990), Berger (1990), and Ball (1992). There are a number of refinements and generalizations. For instance, one may restrict the class of approximated bodies. In the case of convex polyhedra one obtains characterizations of affine regular polyhedra in this way, cf. Fejes Tóth (1948a,b). For simplices the equations of both ellipsoids can be derived from the vertex coordinates in a convenient manner, see Nüesch (1983). For volume estimates of these ellipsoids we refer to Bourgain and Milman (1987), Milman (1987), Ball (1989) and, in connection with optimization theory, to Grötschel, Lovász and Schrijver (1988), Tarasov, Khachiyan and Èrlikh (1988) and Khachiyan (1990), Gruber (1988) showed that the boundaries of most (in the sense of Baire category) convex bodies have precisely $\frac{1}{2}d(d+3)$ points in common with the boundaries of both ellipsoids. There are analogous ellipsoids if volume is replaced by another quermassintegral (intrinsic volume). The question of uniqueness is then still open. Firey (1964) considers such ellipsoids if in addition their midpoints are prescribed. Further results on approximation involving ellipsoids are given by Guderley and Keller (1971) and Taylor (1966), see also the literature given by Gruber (1983). Bárány and Larman (1990) approximate finite point sets by ellipsoids.

3.4. Further results on ellipsoids

Lee and Shih (1990) give a formula for the volume of hyperplane sections of ellipsoids. Metrical properties of ellipsoids play an important part in stereology, see Weil (1983) for a survey. The stereological determination of ellipsoids by their linear sections is considered by Møller (1988) and Sallee (1990). One can find numerical calculations of the surface area and other quantities of ellipsoids in Lehmer (1950), Keller (1979), Gendzwill and Stauffer (1981), and Moran (1982); see also the hints given in section 1.3, Pfiefer (1988) and Lutwak (1991). A simple derivation of well-known estimates for the surface area is given by Klamkin

(1971). Investigations on the solid angle under which an ellipsoid is seen from boundary points of another ellipsoid might have interesting applications, see Nikolaev (1970). The convexity of generalized or transformed ellipsoids is studied by Wilker (1980) and Kapur (1986). The investigation of ellipsoids associated in a certain manner with given convex bodies also shows an interesting link with the theory of normed spaces (Milman and Schechtman 1986, Milman and Pajor 1989, and Pisier 1989).

4. Centrally symmetric convex bodies

As in the case of simplices (section 2) we restrict our attention to characterizations. We shall repeat only a few of the references given by Bonnesen and Fenchel (1934, §14). We also refer to section 2 of chapter 3.2 and to chapter 4.9.

4.1. Characterizations by sections and projections

The False Center Theorem says that if for some k, $2 \leq k < d$, every k-dimensional linear section of a convex body K in $\mathbb{E}^d$ $(d \geq 3)$ through $x \in \mathbb{E}^d$ is centrally symmetric, then K itself has a center of symmetry [cf. Petty (1983) for a brief discussion]. For $x \in \operatorname{int} K$ and $k = 2$ this was shown by Rogers (1965). Burton (1975) extended this to any $x \in \mathbb{E}^d$. For $k > 2$ and $x \in \mathbb{E}^d$ it is proved by Burton and Mani (1978). The corresponding characterization using orthogonal k-dimensional projections was given by Blaschke and Hessenberg in 1917, see Bonnesen and Fenchel (1934, p. 125). Bolker (1970) gave a sharpening of the Blaschke–Hessenberg Theorem for convex polytopes and further characterizations of central symmetry. Inspired by the question whether a plane convex body can have two distinct equichordal points (i.e., points such that all chords through them have the same length), Rogers (1981) showed: if x_1, x_2 are distinct inner points of the plane convex body K and if each chord of K through x_1 has the same length as the parallel chord through x_2, then the midpoint of x_1x_2 is a center of symmetry for K. Larman and Tamvakis (1981) generalized this to $\mathbb{E}^d$ using $(d-1)$-volumes of hyperplane sections of K through distinct points $x_1, x_2 \in \mathbb{E}^d$. Fejes Tóth and Kemnitz (1987) showed the analogous result in planes of constant curvature.

Let K be a convex body in $\mathbb{E}^d$, $d \geq 3$, and assume that for each direction u and every pair of parallel supporting hyperplanes containing u there is a longest chord of direction u lying midway between both hyperplanes. Then, as Khassa (1977) shows, K is centrally symmetric. For $d = 2$ he can show this only if K has constant width. Related results were derived by Dharmadhikhari and Jogdeo (1973) and Converse (1977). Harazišvili (1978) characterized centrally symmetric plane convex bodies by means of affine diameters. Furthermore, we refer to Kubota (1940) and Viet (1956).

In 1917, Blaschke [cf. Bonnesen and Fenchel (1934, p. 125)] showed that a convex body K in $\mathbb{E}^3$ is centrally symmetric if all planes through some interior point x halve the volume of K (or if their intersections with K have x as center of gravity). Related characterizations of central symmetry were given by Süss

(1950), Hammer and Smith (1964), Schneider (1970b), Falconer (1977, 1983), Lutwak (1980), and Goodey (1982). In particular, Schneider shows an analogue to Blaschke's result, replacing volume by surface area. He also generalizes a result of Kubota (1920): a convex body has central symmetry if each shadow-boundary bisects the surface area of K.

Zarankiewicz (1959) and Menon (1966) show that a plane convex body is symmetric with respect to x if and only if x is the only point lying on three area bisecting lines. Metrical properties of intersections of bd K with hyperplanes lead Meyer and Reisner (1991) to a characterization of central symmetry, which is also useful for the theory of flotation surfaces.

Klee (1963) and Bolker (1970) characterized centrally symmetric polytopes by intersection and projection properties.

4.2. Measures of symmetry

For results on measures of symmetry published before 1963 the reader should consult the extensive survey of Grünbaum (1963). Some more recent papers are also cited by Schneider (1989). According to Grünbaum a measure of symmetry is a continuous function $f: K^d \to [0, 1]$ for which $f(K) = 1$ if and only if K is centrally symmetric. K^d is the set of convex bodies in $\mathbb{E}^d$. Sometimes the superminimality condition,

$$f(K_1 + K_2) \geqslant \min\{f(K_1), f(K_2)\}$$

for all $K_1, K_2 \in K^d$, is added. As an example we give one of the best known symmetry measures, the affine-invariant volume ratio

$$f(K) = V(K)/V(\tfrac{1}{2}DK), \quad K \in K^d,$$

where $DK = K + (-K)$ is the difference body of K. Rogers and Shephard (1957) prove

$$2^d/\binom{2d}{d} \leqslant V(K)/V(\tfrac{1}{2}DK) \leqslant 1$$

with equality precisely for centrally symmetric convex bodies (right-hand side) and simplices. Firey (1965, 1967) replaced $K + (-K)$ by the Blaschke sum of K and $-K$ and gave partial results. Further related results were presented by Grünbaum (1964), Chakerian and Stein (1965), Tennison (1967), Lillington (1974b), Gardner, Kwapien and Laurie (1982), Cieślak and Zając (1985/86), and Lassak (1989). De Valcourt (1966a,b) assumes $f(K) = 1$ if and only if $K \subset \mathbb{E}^2$ is axially symmetric, and he studies f also for special convex bodies as those of constant width (see also section 5.8). Kind (1970) considers cyclic measures of symmetry for plane convex bodies, whereas Chui and Parnes (1970) generalize the notion of symmetry in order to get a more general measure of symmetry.

Kutateladze (1976) describes symmetry measures which satisfy a condition stronger than superminimality, namely superadditivity with respect to Blaschke addition.

4.3. Further characterizations of central symmetry

The following characterizations are largely independent of each other. Burton (1979) (for $d = 3$) and Montejano (1990b) show that $K \in K^d$ is centrally symmetric "if K can be completely turned in $\mathbb{E}^{d+1}$". Bol (1940) and Medjanik (1966) characterize central symmetry by curvature properties of $\operatorname{bd} K$ in $\mathbb{E}^2$ or $\mathbb{E}^3$, respectively. Zalgaller and Sudakov (1974) characterize special classes of (not necessarily compact) convex sets with centers of symmetry. Inequalities yielding centrally symmetric convex bodies as extremal cases were given, e.g., by Chakerian and Talley (1969) and Barthel and Pabel (1987). Schneider (1989) characterizes a class of planar convex bodies which consists of centrally symmetric bodies and some special polygons. We also mention results on degrees of symmetry where arbitrary centrally symmetric bodies stand on one side and zonoids on the other side. We refer to Schneider and Weil (1983), Goodey (1984a,b), Goodey and Weil (1991) and chapter 4.9.

We close with characterizations of centrally symmetric convex polytopes. In 1897, Minkowski (see Bonnesen and Fenchel 1934, p. 125) showed that a convex d-polytope P with pairwise parallel $(d-1)$-faces of equal $(d-1)$-volumes has a center of symmetry. He deduced that P is centrally symmetric if one can compose P by centrally symmetric (not necessarily convex) polytopes. If all k-dimensional faces (for some $k \in \{2, \ldots, d-1\}$) of a polytope P are centrally symmetric, then P belongs to a certain subclass of centrally symmetric polytopes. For $k = d-1$ this was shown by Aleksandrov (1933), see also Burckhardt (1940), Shephard (1967), and McMullen (1976). This subclass consists precisely of those polytopes which are equidecomposable to a d-cube with respect to the group of translations (Mürner 1977). On the other hand, for $k \in \{2, \ldots, d-2\}$ each such polytope is equal to a d-dimensional Minkowski sum of line segments, i.e., a zonotope (McMullen 1970). For zonotopes and their interesting properties, see Schneider and Weil (1983).

5. Convex bodies of constant width

5.1. Introduction

If the distance between parallel supporting hyperplanes of a convex body $K \subset \mathbb{E}^d$ is always the same, then K is said to be of constant width, and we denote the set of such bodies by W^d. There are a lot of books which devote special chapters to W^d, e.g., Jordan and Fiedler (1912), Yaglom and Boltyanskiĭ (1961), Lyusternik (1963), Eggleston (1957, 1958), Benson (1966), Boltyanskiĭ and Soltan (1978), Lay (1982), Fischer (1986), Böhm and Quaisser (1991); see also the comments of

E. Krätzel and B. Weissbach in the appendix of Minkowski (1989), and chapter 1.4 of this Handbook. A complete report on the literature is given by Bonnesen and Fenchel (1934, §15), and continued by Chakerian and Groemer (1983). Chakerian and Groemer discuss various equivalent definitions, also for constant width in finite dimensional normed spaces (=Minkowski spaces). We refer to this article for all those explanations which we have to omit here. We list some articles which give surveys, at least on certain aspects: Meissner (1911), Jordan and Fiedler (1913), Tiercy (1920, 1921), Bückner (1936, 1937), Beretta and Maxia (1940), Weissbach (1990), and (with regard to teaching aspects) Blatter (1981), Fisher (1987), and Gardner (1963). For applications in engineering, see, e.g., Wunderlich (1970, pp. 48, 246), the references given by Mayer (1932), Frank and Pfanzl (1985), and DIN 32711 (1979).

We give a few additions to Chakerian and Groemer (1983) and try to give a complete account up to now. In order to facilitate comparison we use their grouping of subjects though sometimes we could add only little.

5.2. Basic properties and characterizations

Lu and Ding (1983), Zamfirescu (1988), Makai and Martini (1990) (continuing Schmitz 1985 and posing an interesting problem for $d \geq 3$) characterize plane sets of constant width by different properties. Böröczky (1990) shows that a body C in W^2 is a circle if any 2 points can be joined within C by a non-degenerate rectangle. Wegner (1973) characterizes the ball in $\mathbb{E}^3$, Montejano (1990a, 1991) in $\mathbb{E}^d$, by the property that suitable hyperplanes cut the body in sets of constant width.

5.3. Special bodies and approximations

The simplest body in W^2 apart from a circle is the Reuleaux triangle. It is bounded by 3 circular arcs of radius b which are centered at the vertices of a triangle with side-length b. Reuleaux (1875, §155) ascribes its first mechanical usage to Hornblower, the inventor of the compound steam-engine. Reuleaux triangles or polygons are considered by Weissbach (1977), Giebler and Weissbach (1983, 1985), Sgheri (1984), and Groemer (1988). The only reference for special bodies in W^3 which we can give is Böhm (to be published). Approximating convex bodies by smooth ones, Schneider (1984) pays special attention to those of constant width. Aumann (1988) gives algorithms for determining width, inradius and circumradius of bodies in W^2 and approximates them by Bézier splines. Gritzmann and Lassak (1989) give estimates for the minimal width of polytopes inscribed in convex bodies, especially those which are centrally symmetric and those which have constant width.

5.4. Completion and completeness

A set is called complete if it cannot be enlarged without increasing the diameter (longest chord). In $\mathbb{E}^d$ completeness is equivalent to constant width, but this is not

true in Minkowski spaces (Chakerian and Groemer 1983, p. 62). Completion refers to the fact that any bounded set can be enlarged so that it becomes complete. In particular, part of the boundary can be preassigned fairly arbitrarily. Gajowski and Krawczyk (1981) deal with completion in $\mathbb{E}^2$, Schulte and Vrećica (1983) in $\mathbb{E}^d$, Sallee (1988) in d-dimensional Minkowski space. Groemer (1986) deals with uniqueness of completions in d-dimensional Euclidean and Minkowski spaces. Uniqueness of completion plays an essential role on Borsuk's partition problem in $\mathbb{E}^2$, see Boltyanskiĭ and Soltan (1977, 1978). Whereas in Euclidean spaces there are many complete sets (see section 5.3) it may happen in Minkowski spaces that the only complete sets are balls. In fact, this happens exactly if the unit ball is a parallelotope (Chakerian and Groemer 1983, p. 62). Dalla and Tamvakis (1985) ask for the same characterization in infinite-dimensional normed spaces. According to *Zentralblatt der Mathematik* 619.52001 this question was already answered by Franchetti (1977). Apparently it is not known in which Minkowski spaces sets of constant width are balls. Stefani (1987) (and earlier papers) is concerned with measure-theoretical questions which are related to completeness.

The following concept is dual to completeness: a convex body which does not properly contain a convex body with the same minimal width is called reduced. An equilateral triangle is reduced. A reduced body with C^1-smooth boundary has constant width [Groemer, see Chakerian and Groemer (1983, p. 61) and, for a simpler proof, Lassak (1990)]. The conjecture that strictly convex reduced bodies have constant width was proved by Dekster (1986a) for $\mathbb{E}^2$. Proofs for $\mathbb{E}^d$, $d \geqslant 3$, were announced by Böhm (1985, 1986). Dekster (1986b) shows that for $d \geqslant 3$ a reduced body K has constant width if at almost all boundary points K is supported by a sphere enclosing K. Bodies of given minimal width with minimal volume, surface area or other quermassintegrals are reduced. But nothing is known about these difficult problems. Concerning the volume there is a conjecture in Gruber and Schneider (1979, p. 261); see also Böhm and Quaisser (1991, p. 72), and Croft, Falconer and Guy (1991, p. 29). Compare the analogous problems for bodies of constant width (see section 5.7).

5.5. Intersection properties

Bodies of constant width b have concentric circumsphere and insphere and their radii have sum b. Sallee (1986) generalizes this to Minkowski spaces and shows that this does not depend on constant width, but rather on the more general spherical intersection property (which is explained by Chakerian and Groemer 1983, p. 62). The adjoint transform of Bavaud (to be published) is also related to the spherical intersection property.

5.6. Curvature and mixed volumes

Barbier's theorem says that the boundary of a body in W^2 of width b has length πb. Saban (1983) derives it from Fenchel's theorem on the total curvature of closed curves in $\mathbb{E}^3$ by applying this to pairs of closed Bertrand curves. Groemer

(1988) estimates the deviation of a plane convex body with diameter b from a body in W^2 if its perimeter is close to πb.

For bodies in W^d $(d \geq 2)$ of width b we have

$$2W_{d-k} = \sum_{i=0}^{k-1} (-1)^i \binom{k}{i} b^{k-i} W_{d-i}, \quad 1 \leq k \leq d, \ k \text{ odd}, \tag{*}$$

where W_i are the quermassintegrals (see chapter 1.2). Barbier's theorem is the case $d=2$, $k=1$. Schneider (1972) generalizes (*) in the following way. Let $p_0, \ldots, p_d$ be the curvature centers of a convex body $K \subset \mathbb{E}^d$ (p_0, p_1 are the centers of gravity of K, bd K, respectively, p_n is the Steiner point). For W^2 one has $p_1 = p_2$. Schneider (1972) extends this to $\mathbb{E}^d$ $(d \geq 2)$:

$$2W_{d-k} p_{d-k} = \sum_{i=0}^{k-1} (-1)^i \binom{k}{i} b^{k-i} W_{d-i} p_{d-i}, \quad 1 \leq k \leq d, \ k \text{ odd}.$$

These relations are translation invariant and therefore immediately imply (*). Arnold (1991) extends them to regions of constant width on the sphere S^2. Nádeník (1969) gives an analogue to (*) for the envelope of cylindrical surfaces of constant width.

Falconer (1983) gives a simple proof of Minkowski's Theorem which says that a convex body K in $\mathbb{E}^3$ has constant width b if all its projections have circumference πb. A result of Makai (1973) is similar: a smooth convex body is a rotational body in W^3 if $\frac{1}{2}\pi$ times its diameter is equal to the geodesic diameter of the boundary. Goodey and Groemer (1990) treat the stability problem associated with Minkowski's Theorem (does K have almost constant width if the perimeter of its projections is nearly constant?) and its appropriate extension to $\mathbb{E}^d$.

5.7. *Inequalities*

In W^2 Reuleaux triangles have minimal area. Groemer (1988) (see also chapter 1.4) investigates the associated stability problem: how much does a set in W^2 deviate from a Reuleaux triangle if its area is close to the minimal area? Groemer (1990, p. 388) gives a sharpened form of Bonnesen's inequality for W^2.

The smallest possible volume of a body in W^d, $d \geq 3$, of width b is not known. The state is still as described by Chakerian and Groemer (1983, p. 68), though Schramm (1988a) has improved the lower bounds for $d > 4$ (see also Croft, Falconer and Guy 1991, p. 34). Also almost nothing is known about minimal surface area etc., compare the corresponding problems for general convex bodies of given minimal width mentioned in section 5.4.

Lutwak (1975, 1983) gives inequalities in which equality is attained just for bodies in W^d or their affine images, respectively. De Castro Feitosa (1983) treats inequalities in Minkowski spaces. Ghandehari (to be published) gives inequalities for the quermassintegrals of polars of bodies in W^d and in other special classes.

5.8. Inscribed and circumscribed bodies

De Valcourt (1966a,b, 1967) defines measures of axial symmetry for ovals by means of inscribed figures of different types. He pays special attention to ovals in W^2 and poses several questions.

While all circumscribed rectangular parallelotopes of a body in W^d have the same volume, this property does not characterize W^d. Such bodies may even be centrally symmetric. Petty and McKinney (1987) characterize them for $d \geq 3$, but for $d = 2$ this problem is still unsolved.

Bezdek and Connelly (1989) show that any plane closed curve of length 1 can be covered by a translate of the plane body K if K has constant width $\frac{1}{2}$, and conversely, that K must have constant width if its perimeter is not larger than $\frac{1}{2}\pi$. Chakerian and Logothetti (1988) show that the smallest regular n-gon ($n \geq 4$) which can cover any plane set of diameter 1 is the smallest regular n-gon circumscribed about a Reuleaux triangle of width 1.

5.9. Packing and covering

Blind and Blind (1983, 1987) consider packings by Reuleaux triangles. Schramm (1988b) estimates the number of directions that are necessary to illuminate a body in W^d. Moreover, this gives an estimate for Borsuk's partition problem in $\mathbb{E}^d$, which we already mentioned in section 5.4. For Borsuk's problem, see Croft, Falconer and Guy (1991) p. 123, and p. 38 for another problem on sets of constant width also related to Borsuk's conjecture. One of the numerous corollaries of Dolnikov (1987) concerns the partition of sets of constant width and leads to a generalization of Borsuk's problem. For covering problems in connection with constant width, see also Weissbach (1990) and the references collected there, and Makeev (1989).

5.10. Rotors in polytopes

The bodies in W^d are rotors in cubes (and some other polytopes). But there are polytopes which have rotors not in W^d. A survey is given by Goldberg (1960). Rotors in regular triangles and hexagons appear as extreme cases in Lillington (1974a). Weissbach (1982) considers rotors in regular polygons and generalizations, see also Schaal (1962/63), Cieślak and Góźdź (1987). Croft (1980) considers convex curves in which triangles can rotate, the vertices always lying on the curves. It seems unknown whether there is any polyhedron P and closed non-spherical surface S such that P can revolve unlimited with its vertices always on S. Rotors are also discussed in chapter 4.8.

5.11. Concepts related to constant width

For curves of constant width in $\mathbb{E}^d$, $d \geq 3$, and the related concept of transnormal manifolds we mention Nádeník (1968), Hoschek (1974), Robertson (1984), Köse

(1986), Wegner (1988, 1990), and Cieślak (1988). Pottmann (1987) treats ruled surfaces of constant width in $\mathbb{E}^3$, Elkholy (1985) curves of constant width on the sphere S^2. Pinkall (1984) gives several equivalent definitions of curves of constant width in the hyperbolic plane. These curves correspond to Radon curves in the affine plane, and in this way an estimate for the isoperimetric deficit results from an affine inequality. Affine images of bodies in W^d occur in Alexander (1968) and Lutwak (1983). Maehara (1984) says that 2 convex bodies X, Y form a pair of constant width if $X + (-Y)$ is a ball. He shows that the insphere of X and the circumsphere of Y are concentric, as they are in W^d ($X = Y$). Sallee (1987) treats such pairs in Minkowski spaces. Obviously, there is a connection with the question whether some convex body is a summand of another. Cieślak and Mozgawa (1987) call a locally convex plane closed curve a rosette. They show that rosettes of constant width w with winding number j have length πwj. Related is Cieślak and Zając (1986). Dekster (1990, 1991) (and the papers quoted there) considers sets of constant width in Riemannian spaces of non-constant curvature. Adhikari and Pitman (1989) study shortest planar arcs of width 1 and pose a lot of questions. Rodríguez Palacios (1990) points out that results on summands of Banach spaces may be interpreted in terms of sets of constant width in Banach spaces.

There is an old problem which is still not satisfactorily solved: is a body in W^d a ball if it has constant brightness? Matsumura's proof, reproduced in Bonnesen and Fenchel (1934, p. 140), works with umbilical points and therefore needs a smooth boundary. A related problem is described by Chakerian (1970).

If a closed curve in $\mathbb{E}^2$ appears from all points of a circle under the same angle, it need not be a circle. Nitsche (1990) shows that it must be a circle if it appears from 2 concentric circles under constant angle. Similar questions may be posed in $\mathbb{E}^d$. We also mention the paper of Cieślak, Miernowski and Mozgawa (1991) on isoptic curves of a given convex curve in the plane.

Inequalities for convex sets in $\mathbb{E}^d$ with an equichordal point (and with similar points) are shown by Ghandehari (to be published); see also chapter 1.4 in this Handbook. Craveiro de Carvalho and Robertson (1988) consider equichordal curves in $\mathbb{E}^d$ and their relations to transnormal manifolds; Cieślak (1991) gives inequalities for such curves. We mention that the problem of whether there are plane curves with 2 equichordal points is still unsolved, see Chakerian and Groemer (1983, p. 83), Michelacci and Volčič (1990), and Klee and Wagon (1991, p. 12). Schäfke and Volkmer (1992) announce a (negative) solution, which, as may be seen from their published partial solution, will be very long. But there are such curves on the sphere, see Spaltenstein (1984).

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CHAPTER 2.1

Helly, Radon, and Carathéodory Type Theorems

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Contents

1. Introduction 391
Part I. Helly's Theorem, and the combinatorial geometry of families of convex sets
2. Helly's Theorem 392
3. Generalizations of Helly's Theorem 395
4. Spherical and topological Helly-type theorems 400
4.1. Spherical Helly-type theorems 400
4.2. Topological Helly-type theorems 402
5. Other Helly-type theorems 404
6. Piercing and coloring properties of convex sets 406
6.1. Gallai-type problems 406
6.2. (p, q)-problems 408
6.3. Coloring problems 409
7. Common transversals 410
7.1. Transversal problems in the plane 411
7.2. Transversal problems in $\mathbb{R}^d$ 413
8. Intersection patterns of convex sets 415
8.1. Nerves and intersection graphs 416
8.2. f-vectors 419
Part II. Radon's and Carathéodory's Theorem, and the convexity properties of configurations of points
9. Radon's Theorem and its relatives 421
9.1. Radon partitions and Radon types 422
9.2. The number of Radon partitions and Radon types 424
9.3. Tverberg's theorem and its relatives 427
10. The theorems of Carathéodory and Steinitz 430
10.1. Carathéodory's Theorem 430
10.2. Steinitz's theorem 432
11. The theorems of Kirchberger and Krasnosel'skiĭ 433
References 436

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1. Introduction

The classical theorems of Helly, Radon, and Carathéodory stand at the origin of what is known today as the combinatorial geometry of convex sets. They were discovered during the first quarter of this century and may be formulated as follows.

Theorem 1.1 (Helly's Theorem). *Let $\mathcal{K}$ be a family of convex sets in $\mathbb{R}^d$, and suppose $\mathcal{K}$ is finite or each member of $\mathcal{K}$ is compact. If every $d+1$ or fewer members of $\mathcal{K}$ have a common point, then there is a point common to all members of $\mathcal{K}$.*

Theorem 1.2 (Radon's Theorem). *Let X be a set of $d+2$ or more points in $\mathbb{R}^d$. Then X contains two disjoint subsets whose convex hulls have a common point.*

Theorem 1.3 (Carathéodory's Theorem). *Let X be a set in $\mathbb{R}^d$ and p a point in the convex hull of X. Then there is a subset Y of X consisting of $d+1$ or fewer points such that p lies in the convex hull of Y.*

The term "combinatorial geometry" seems to have been coined by H. Hopf (see Hadwiger 1955a) and was made popular by the book of Hadwiger, Debrunner and Klee (1964) and others. It is used here in a slightly restricted sense which emphasizes the purely combinatorial aspects of convexity and largely ignores its metrical or topological features.

This chapter is concerned with applications and generalizations of the three classical theorems as well as their ramifications in the context of combinatorial convexity theory. The chapter falls into two parts although in view of the many existing interrelationships between the two, the distinction is somewhat artificial. Part I reviews the combinatorial properties of families of convex sets in $\mathbb{R}^d$. In most cases, the families will be finite; the extension to infinite families of compact sets is usually standard. The prototypical result in Part I is, of course, Helly's Theorem. Topics to be discussed include Helly-type theorems, fixing and coloring properties, common transversals, and intersection patterns of convex sets. Part II deals with convexity properties of (finite or infinite) sets of points in $\mathbb{R}^d$, particularly with separation properties and the generation of convex hulls. Here the prototypical results are those of Radon and Carathéodory, and generalizations include the well-known theorems of Tverberg, Steinitz, Kirchberger, and Krasnosel'skiĭ.

Due to limitation of space we cannot hope to give a comprehensive account of all the different aspects of combinatorial geometry. Nevertheless we believe that the main areas of research have been covered or at least been touched. (Two important exceptions are the combinatorial theory of convex polytopes and the well-known Erdős–Szekeres problem. These will be treated in separate chapters.) The bibliography contains over 300 references which can be used for further study. (A full list of references would have to be at least twice as long.) For much

of the early work on the subject, we must refer to existing surveys such as the famous article "Helly's theorem and its relatives" by Danzer, Grünbaum and Klee (1963). This paper had a great impact on the development of the field and is still a very valuable source of information. Other more specialized surveys are those by Reay (1965a) on "Generalizations of Carathéodory's theorem", by Eckhoff (1979) on "Radon's theorem revisited" and, more recently, by Goodman, Pollack and Wenger (1993) on "Geometric transversal theory". The present article should be read in conjunction with these reports.

The natural setting of this chapter is the d-dimensional real linear space $\mathbb{R}^d$. Only in a few cases will it be necessary to consider other spaces (such as spherical space) as well, or to equip $\mathbb{R}^d$ with its usual metric in which case we are really dealing with Euclidean d-dimensional space $\mathbb{E}^d$. The standard abbreviations aff X, conv X, dim X, card X, int X, and bd X will be used for the affine hull, convex hull, dimension, cardinality, interior, and boundary of a set X in $\mathbb{R}^d$, respectively. A k-flat in $\mathbb{R}^d$ is a k-dimensional affine subspace, a k-simplex a k-dimensional simplex, and so forth. The unit ball in $\mathbb{E}^d$ is B^d, the unit sphere in $\mathbb{E}^{d+1}$ is S^d. If $\mathcal{K}$ is a family of sets, then $\bigcap \mathcal{K}$ denotes the intersection and $\bigcup \mathcal{K}$ the union of all the members of $\mathcal{K}$. A *disjoint* family is one whose members are pairwise disjoint. If P is a property of families of sets, we write $\mathcal{K} \in P$ to indicate that $\mathcal{K}$ has property P, and $\mathcal{K} \in P_k$ to mean that every subfamily of $\mathcal{K}$ consisting of k or fewer members has property P. Throughout, the property of having nonempty intersection will be denoted by Π. Thus $\mathcal{K} \in \Pi$ and $\bigcap \mathcal{K} \neq \emptyset$ can be used interchangeably.

PART I. HELLY'S THEOREM, AND THE COMBINATORIAL GEOMETRY OF FAMILIES OF CONVEX SETS

2. Helly's Theorem

Theorem 1.1, commonly known as "Helly's Theorem on the intersection of convex sets", was discovered by Helly in 1913 and communicated by him to Radon who published a first proof (Radon 1921). To judge from its appearance in textbooks, monographs, survey articles, and research papers, Helly's Theorem is perhaps the most widely known result in finite-dimensional convexity, rivaled only by Minkowski's lattice point theorem. It expresses a fundamental property of the space $\mathbb{R}^d$ and is intimately connected with the theorems of Radon and Carathéodory. Over the years, a vast body of applications, analogues and far-reaching generalizations of Helly's Theorem has been assembled in the literature. Among the many proofs of Theorem 1.1, Helly's (1923) own proof is still the most geometric and intuitive one. It proceeds by induction on the dimension d of the space and uses the separation theorem for convex sets. Essentially the same proof was given by König (1922). Notice that it suffices to establish the result for finite families. Indeed, the "finite intersection property" of compact sets assures that a

family of compact sets has nonempty intersection provided each of its finite subfamilies has. By contrast, Radon's proof is by induction on the cardinality of $\mathcal{K}$ and is probably the simplest possible. It is based on Radon's Theorem 1.2 which in turn is an easy consequence of the fact that a system of $d+1$ homogeneous equations in $d+2$ unknowns has a nontrivial solution.

Many other approaches to Helly's Theorem can be found in the literature; the more interesting ones are described in Danzer, Grünbaum and Klee (1963). Rademacher and Schoenberg (1950) deduce Helly's Theorem from Carathéodory's Theorem 1.3. Their proof is reproduced in Eggleston (1958) where it is shown that, conversely, Helly's Theorem implies Carathéodory's Theorem. The latter theorem is also used by Sandgren (1954) and Valentine (1963) (who in addition employ the duality theory of convex cones) and by Bohnenblust, Karlin and Shapley (1950). To the other proofs of Helly's Theorem mentioned in Danzer, Grünbaum and Klee (1963), we add those of Chakerian (1969), Sallee (1975), Mullin (1979), and Dol'nikov (1987).

Helly's Theorem is the best possible in the sense that its conclusion may fail for infinite families of closed convex sets, or if the "Helly number" $d+1$ is replaced by a smaller number. This is shown by the family of all translates of a closed halfspace in $\mathbb{R}^d$ and the family of all $(d-1)$-faces of a d-simplex. On the other hand, the hypothesis in Theorem 1.1 can be weakened. For example, it is enough to assume that the members of $\mathcal{K}$ are closed and have no common direction of recession (Fenchel 1953). (A *direction of recession* of X is a vector $u \neq 0$ such that $p + \lambda u \in X$ for all $p \in X$, $\lambda \geqslant 0$.) The condition is obviously satisfied if at least one member of $\mathcal{K}$ is bounded (Radon 1921). For stronger results and applications to systems of convex inequalities, see Rockafellar (1965, 1970). There remains the question of whether the convexity of the sets in Theorem 1.1 is really necessary. Certainly, it cannot be dropped altogether. On the other hand, the theorem can be extended to families of more general sets in many different ways. Helly (1930) himself proved a topological generalization which applies to well-behaved families of homology cells in $\mathbb{R}^d$ (see section 4). Nevertheless, the convexity of the members is, in a sense, the only condition which is compatible with the affine structure of $\mathbb{R}^d$. This is the content of the "converse Helly theorem" due to Dvoretzky (1955). A family of sets $\mathcal{K}$ is said to have the *Helly property* $H(k)$ if some k or fewer members of $\mathcal{K}$ have empty intersection, or else $\mathcal{K}$ has nonempty intersection. Thus a family of compact convex sets in $\mathbb{R}^d$ has property $H(d+1)$. The converse is false, but Dvoretzky proved the following.

Theorem 2.1. *Suppose $\mathcal{K}$ is a family of compact sets in $\mathbb{R}^d$, each having interior points, such that whenever the members of $\mathcal{K}$ are replaced (individually) by nondegenerate affine images, the resulting family has property $H(d+1)$. Then the members of $\mathcal{K}$ are convex.*

For a variant of this theorem and a simpler proof, see Guggenheimer (1971).

We turn now to a brief description of the ways in which Helly's Theorem can be (and has been) applied in convexity theory. Following Danzer, Grünbaum and

Klee (1963), we distinguish two main types of application. First, Helly's Theorem can be used to establish general Helly-type theorems (see below). Many of these are direct generalizations of Helly's Theorem and will be dealt with in the next section. Others, like Kirchberger's or Krasnosel'skiĭ's theorem, are not. Second, Helly's Theorem can be applied to simplifying proofs by making a reduction to an especially simple case of the assertion. This type of application was first illustrated by Helly (1921) himself in connection with systems of linear equations. An instructive example is the proof of a famous result of Jung according to which each set of diameter 1 in $\mathbb{R}^d$ lies in a ball of radius $\sqrt{2d/(d+1)}$. Using Theorem 1.1, the assertion can be reduced to sets of at most $d+1$ points; the rest is then linear algebra (see Danzer et al. 1963). Rado (1946) and Grünbaum (1960a) have applied Helly's Theorem to obtain versions of the following result.

Theorem 2.2. *Suppose μ is a σ-additive probability measure defined on the σ-algebra of Borel sets in $\mathbb{R}^d$. Then there exists a point $p \in \mathbb{R}^d$ such that for each halfspace $H \subset \mathbb{R}^d$ containing p, $\mu(H) \geq 1/(d+1)$.*

The result is sharp. See also Birch (1959) and, for a recent deep generalization which includes the "ham sandwich theorem", Živaljević and Vrećica (1990).

Again, lack of space prevents us from doing justice to all the applications of Helly's Theorem in the literature. To the references listed in Danzer, Grünbaum and Klee (1963), we add Hare and Kenelly (1968), Levin (1969), McMullen (1976), Uhrin (1977), Brøndsted (1978), and Amir (1985).

One of the remarkable and unique features of Helly's Theorem is its logical structure. There is now a rich variety of Helly-type problems and results which are distinguished by the same (or almost the same) structure. Loosely speaking, a *Helly-type theorem* is a statement of the following form: a family of sets (or a set of points) has a certain property provided each of its subfamilies (subsets) containing no more than a certain fixed number of sets (points) has this property. In the case of Theorem 1.1, the family is a family of compact convex sets in $\mathbb{R}^d$, the property is Π and the number can be taken to be $d+1$. Results of this sort are by no means confined to convexity. For example, numerous "Helly theorems" have been obtained in recent years in graph theory, combinatorics, and related fields; these will not be considered here.

For the purpose of the present article, the following abstract scheme seems appropriate. Suppose P is a property of families of sets, and assume that P is *hereditary*, i.e., $\mathcal{K} \in P$ and $\mathcal{F} \subset \mathcal{K}$ imply $\mathcal{F} \in P$. *The Helly number of $\mathcal{K}$ with respect to P* is the least positive integer h such that if $\mathcal{F} \subset \mathcal{K}$ is finite and $\mathcal{G} \in P$ for all $\mathcal{G} \subset \mathcal{F}$ with card $\mathcal{G} \leq h$, then $\mathcal{F} \in P$. Since the property of special interest in connection with Helly's Theorem is Π, we denote the Helly number of $\mathcal{K}$ w.r.t. Π by $h(\mathcal{K})$ and call it the *Helly number* of $\mathcal{K}$. Thus the family of all convex sets in $\mathbb{R}^d$ has Helly number $d+1$. Of course, it may happen that $h(\mathcal{K})$ does not exist. This is usually due to the fact that $\mathcal{K}$ is not intersectional, i.e., $\mathcal{F} \subset \mathcal{K}$ does not necessarily imply $\bigcap \mathcal{F} \in \mathcal{K}$. In such a case, a more suitable notion is that of the *Helly order* $h^\circ(\mathcal{K})$ of $\mathcal{K}$ which takes into account that the intersection of members

of $\mathcal{K}$ can be more complicated in structure than the individual members. By definition, $h^o(\mathcal{K})$ is the least integer h such that the following holds: if $\mathcal{F} \subset \mathcal{K}$ is finite and $\emptyset \neq \bigcap \mathcal{G} \in \mathcal{K}$ for all $\mathcal{G} \subset \mathcal{F}$ with card $\mathcal{G} \leq h$, then $\emptyset \neq \bigcap \mathcal{F} \in \mathcal{K}$. Two prominent examples are Theorems 3.10 and 4.6. For refinements of the above notions, see Danzer, Grünbaum and Klee (1963) and Grünbaum and Motzkin (1961).

Speaking of Helly numbers, we wish to draw the reader's attention to what is called (in a series of papers by Boltyanskiĭ and his students) the "Szökefalvi–Nagy problem". This consists in characterizing those compact convex sets $K \subset \mathbb{R}^d$ for which the family of all translates of K has some fixed Helly number $k+1$ $(1 \leq k \leq d)$. The case $k = 1$ was settled in the 1950s by Hanner and Szökefalvi and Nagy; in that case, K must be a parallelotope (see Danzer et al. 1963, section 7). While the general problem is still open, a solution has been obtained when $k \leq 4$ and K is centrally symmetric (see Kincses 1987).

3. Generalizations of Helly's Theorem

This section describes some of the numerous generalizations of Helly's Theorem which have been established over the years by various authors. Several of these appeared already in Danzer, Grünbaum and Klee (1963), while others are fairly recent. The list is by no means complete; more results will appear in later sections. For convenience, it is usually assumed that the family in question is finite.

We begin with a well-known result due to Vincensini (1939) and Klee (1953). This is a straightforward consequence, and at the same time an extension, of Theorem 1.1.

Theorem 3.1. *Let $\mathcal{K}$ be a finite family of convex sets in $\mathbb{R}^d$, and let $K \subset \mathbb{R}^d$ be convex. If for every $d+1$ or fewer members of $\mathcal{K}$ there exists a translate of K contained in them, then there is a translate of K contained in all members of $\mathcal{K}$.*

In the above, one may replace the words "contained in" by "containing" or by "intersecting". The result reduces to Helly's Theorem when K is a single point. It has proved useful in many applications. As noted by Buchman and Valentine (1982), it easily implies the following "metrical" generalization of Helly's Theorem. Recall that the *minimal width* of a convex set K is the smallest possible distance between two parallel supporting hyperplanes of K.

Theorem 3.2. *Suppose $\mathcal{K}$ is a finite family of convex sets in $\mathbb{R}^d$ such that every $d+1$ or fewer members of $\mathcal{K}$ simultaneously contain a set of minimal width ω. Then $\bigcap \mathcal{K}$ contains a set of minimal width ω.*

Helly's Theorem is obtained by letting $\omega = 0$. To prove Theorem 3.2, apply Theorem 3.1 and observe that a set has minimal width ω if and only if it contains

a translate of every line segment of length ω. A first (and more complicated) proof had been given by Sallee (1975). The paper of Buchman and Valentine establishes an analogue of Theorem 3.2 in which the words "contain a set of minimal width ω" are replaced by "are contained in a set of constant width ω". The proof uses Meissner's theorem which asserts that any set of diameter ω can be enclosed in a set of constant width ω. The result fails if "minimal width" is substituted for "constant width". It would be interesting to know what hypotheses, if any, should be added to Theorem 3.2 in order to assure that the members of $\mathcal{K}$ simultaneously *contain* a set of constant width ω.

We turn next to a different type of generalization of Helly's Theorem. What happens if the Helly number $d+1$ is replaced by a fixed smaller number? Of course, the conclusion of the theorem will no longer be true but other significant conclusions can still be drawn.

Theorem 3.3. *Suppose* $0 \leq k < d$, *and let* $\mathcal{K}$ *be a finite family of convex sets in* $\mathbb{R}^d$. *The following assertions are equivalent*:

(i) $\mathcal{K} \in \Pi_{d-k+1}$;

(ii) *each k-flat in* $\mathbb{R}^d$ *has a translate which intersects every member of* $\mathcal{K}$;

(iii) *each* $(k-1)$*-flat in* $\mathbb{R}^d$ *lies in a k-flat which intersects every member of* $\mathcal{K}$.

When $k = 0$, this becomes Helly's Theorem since in that case both (ii) and (iii) assert that $\bigcap \mathcal{K} \neq \emptyset$. The implication (i) $\Rightarrow$ (ii) was first proved by Horn (1949) and later by Karlin and Shapley (1950); the rest is due to Klee (1951). Karlin and Shapley employ Helly's Theorem and a version of Brouwer's fixed-point theorem. For alternate proofs using duality techniques, see Sandgren (1954) and Valentine (1963). As pointed out in Danzer, Grünbaum and Klee (1963), Theorem 3.3 may also be approached by means of spherical convexity, or in a projective setting. Recently, Dol'nikov (1988) has proved results which include the following variant of Theorem 3.3 as a special case: if $\mathcal{K}_1, \ldots, \mathcal{K}_{k+1}$ are families as above, each having property Π_{d-k+1}, then there is a k-flat intersecting every member of $\bigcup_{i=1}^{k+1} \mathcal{K}_i$.

The following theorem bears some resemblance to the preceding one. It was proved by De Santis (1957) who derived it from an analogue of Radon's Theorem. (See also Valentine 1963.)

Theorem 3.4. *Suppose* $0 \leq k < d$, *and let* $\mathcal{K}$ *be a finite family of convex sets in* $\mathbb{R}^d$. *If every* $d-k+1$ *or fewer members of* $\mathcal{K}$ *contain a common k-flat, then* $\bigcap \mathcal{K}$ *contains a k-flat.*

For each k, the number $d-k+1$ is the smallest which has the stated property. Helly's Theorem is the case $k = 0$. As De Santis remarks, his theorem can be put in a form in which the dimension of the space does not enter. The following refinement of Theorem 3.4 is due to Netrebin (1979): if the maximum dimension of a set in $\mathcal{K}$ is n, then $d-k+1$ above may be replaced by $n-k+2$ if $d = n+1$, and $n-k+1$ in all other cases; this is best possible.

In passing, we mention a Helly-type result of Katchalski (1978b) which should be compared, for $k = 1$, with Theorems 3.4 and 3.5: if a finite family of convex sets in $\mathbb{R}^d$ is such that the intersection of every $2d$ or fewer members contains a ray, then the same is true of all members. For a generalization to "k-half-flats" with $1 \leq k \leq d$ (and refinements), see Kołodziejczyk (1984b).

Helly's Theorem may be regarded as saying, for $k = 0$, that if every $d + 1$ members of a finite family of convex sets in $\mathbb{R}^d$ intersect in a set of dimension at least k, then the intersection of all the members is at least k-dimensional. The question of whether a similar result might hold for other values of k, with $d + 1$ replaced by a suitable function $g(d, k)$, was posed by Grünbaum (1962). Clearly, both $g(d, 0)$ and $g(d, d)$ exist and equal $d + 1$; the latter is seen by applying Helly's Theorem to the interiors of the sets considered. Already the case $k = 1$ is less obvious and has been investigated by several authors (see Grünbaum's paper for a historical account). It turns out that $g(d, 1) = 2d$; this is essentially Steinitz's theorem 10.3. For $1 < k < d$, Grünbaum showed that $g(d, k) \leq 2d - k$ and claimed that the opposite inequality is also true. Except for $k = 2$ and $k = d - 1$, however, his claim was wrong. [The incorrect values are quoted in Danzer, Grünbaum and Klee (1963).] The problem was taken up by Katchalski (1971) who obtained the following solution.

Theorem 3.5. $g(d, 0) = d + 1$; $g(d, k) = \max(d + 1, 2(d - k + 1))$ *if* $1 \leq k \leq d$.

Thus Grünbaum's bound and $g(d, k)$ are farthest apart when k is roughly half the dimension of the space. The assertion is first reduced to the case where the sets in question are polyhedral convex cones having apex at the origin and intersecting in a linear subspace of $\mathbb{R}^d$; the proof then proceeds by induction on d. Analogues and refinements of Theorem 3.5 were established by Falconer (1977) and Netrebin (1979). For an application in connection with Krasnosel'skiĭ's theorem, see Breen (1986b). In a sense, the critical case in the proof of Theorem of 3.5 occurs when the family under consideration has $g(d, k) + 1$ members. A detailed analysis of this case can be found in Katchalski (1977a).

The foregoing results lead to an interesting Helly-type problem of a different nature, namely, that of "reconstructing" the dimension of intersections of convex sets. It follows from Theorem 3.5 that for any finite family $\mathcal{K}$ of convex sets in $\mathbb{R}^d$,

$$\dim \bigcap \mathcal{K} = \min(\dim \bigcap \mathcal{L} : \mathcal{L} \subset \mathcal{K}, \operatorname{card} \mathcal{L} \leq 2d) .$$

Perles had asked whether the number $2d$ in this statement can be reduced. This is indeed the case, as was shown by Katchalski (1978a). In fact, $2d$ may be replaced by $d + 1$, which is clearly the best possible. The argument relies on properties of "nondegenerate families of convex cones" as developed in Katchalski (1977b, 1979). Such families will be discussed in section 4.

For the next two generalizations of Helly's Theorem it is assumed that the sets considered be closed. Without this restriction, the assertions fail even for finite families. The first result is due to Levi (1953).

Theorem 3.6. *For a finite family of closed convex sets in $\mathbb{R}^d$ with the property that the union of every $d+1$ or fewer of the sets has connected complement in $\mathbb{R}^d$, Π_d implies* II.

The proof of this theorem is based on the following lemma which is an interesting Helly-type result in its own right: if the union of $n+1$ closed convex sets is convex and every n of the sets intersect, then all the sets intersect. The lemma, in turn, follows from a well-known topological theorem (see Danzer, Grünbaum and Klee 1963), and it is for topological reasons that the closedness of the sets cannot be dispensed with. It appears that a natural setting for the lemma is the "Konvexring" in $\mathbb{R}^d$, i.e., the class of all finite unions of closed convex sets. Many results in combinatorial geometry stem from the fact that the Konvexring admits an Euler–Poincaré characteristic (Hadwiger 1955a, Klee 1963). For elementary proofs of the lemma, see Klee (1951) and Levi (1953); for certain extensions (some, but not all, of Helly-type), see Berge (1959), Ghouila–Houri (1961), and Fan (1968). A far-reaching "combinatorial" generalization of the lemma is established in Eckhoff (1987).

The second result is fairly recent. It was discovered by Breen (1990) and is similar in spirit to Theorem 3.6.

Theorem 3.7. *Let $\mathcal{K}$ be a finite family of closed convex sets in $\mathbb{R}^d$ such that every $d+1$ or fewer members of $\mathcal{K}$ have a starshaped union. Then $\bigcap \mathcal{K} \neq \emptyset$.*

This is a generalization of Helly's Theorem (for closed sets) since the stated condition is obviously implied by the condition that $\mathcal{K} \in \Pi_{d+1}$. Without the restriction that the members of $\mathcal{K}$ be convex, one can at least assert that $\bigcup \mathcal{K}$ is starshaped. This is the case $k=0$ of the following Helly-type theorem of Kołodziejczyk (1987): if $\dim(\ker \bigcup \mathcal{L}) \geqslant k$ for each family $\mathcal{L} \subset \mathcal{K}$ with $\operatorname{card} \mathcal{L} \leqslant g(d, k)$, then $\dim(\ker \bigcup \mathcal{K}) \geqslant k$. Here $g(d, k)$ is Katchalski's function defined in Theorem 3.5 and $\ker X$ is the kernel of X (see section 11). The result is best possible.

We now address a question of a rather different nature. Can Helly's Theorem be generalized to meaningful subclasses of the class of convex sets in $\mathbb{R}^d$ and, if so, what are the resulting Helly numbers? The answer, of course, depends on what is considered a "meaningful" class. Several such answers were given in the literature; we content ourselves with describing a particularly pleasing one. It is concerned with H-convex sets in $\mathbb{R}^d$, a notion introduced by Boltyanskiĭ and investigated by him and Soltan in a series of papers.

Let H be a nonempty subset of the unit sphere S^{d-1} in $\mathbb{R}^d$, fixed in what follows. A set $X \subset \mathbb{R}^d$ is called *H-convex* if X is the intersection of closed halfspaces whose (unit) outer normal vectors fall in H. The space $\mathbb{R}^d$ is also regarded as H-convex. Clearly, each H-convex set is closed and convex in the usual sense; the converse is false unless $H = S^{d-1}$. Now define $m(H)$ to be the largest integer m such that H contains $m+1$ minimally (positively) dependent vectors, i.e., m linearly independent vectors together with a sum of negative

multiples of these vectors. In particular, $m(S^{d-1}) = d$. The following was proved by Boltyanskiĭ (1976).

Theorem 3.8. *Suppose $H \subset S^{d-1}$ is closed and not contained in a hemisphere. Then the family of all H-convex sets in $\mathbb{R}^d$ has Helly number $m(H)+1$.*

This reduces to Helly's Theorem for closed sets when $H = S^{d-1}$. Examples show that the stated conditions are necessary for the result to hold. In Boltyanskiĭ (1977), Theorem 3.8 is applied to obtain H-convex analogues of Theorem 3.1. For a general review of H-convexity and the closely related concept of *d-convexity*, the reader should consult Boltyanskiĭ and Soltan (1978).

There are many ways of generalizing Helly's Theorem to families of not necessarily convex sets in $\mathbb{R}^d$. One instance is given by Helly's topological theorem discussed in section 4. Compare also Peterson (1967) and Dol'nikov (1988). Here we present yet another example. Let $\mathscr{C}_k^d$ denote the family of all sets in $\mathbb{R}^d$ which are expressible as the union of at most k pairwise disjoint closed convex sets. Clearly, $\mathscr{C}_k^d$ is nonintersectional when $k > 1$, and one expects the Helly number $h(\mathscr{C}_k^d)$ to be infinite. This is indeed the case (see Santaló 1951). Grünbaum and Motzkin (1961) made the natural conjecture that $h^0(\mathscr{C}_k^d) = k(d+1)$; they proved this for $k = 2$. The case $k = 1$ is Helly's Theorem. The conjecture was confirmed by Larman (1968) for $k = 3$ and finally by Morris (1973) for all k, in the following slightly stronger form. (See also Larman 1980.)

Theorem 3.9. *For a finite family $\mathscr{K} \subset \mathscr{C}_k^d$ which is such that every k or fewer members of $\mathscr{K}$ have their intersection in $\mathscr{C}_k^d$, $\Pi_{k(d+1)}$* implies Π.

Examples in Larman (1968) show that the hypotheses of Theorem 3.9 cannot be weakened. Also, the word "closed" in the definition of $\mathscr{C}_k^d$ cannot be deleted (though it may be replaced by "open"). As for the proof of Theorem 3.9, some comments seem in order. Grünbaum and Motzkin actually conjectured a more general result which applies to "abstract convexity spaces". Morris was able to reduce their claim to a combinatorial lemma, a sort of "generalized pigeonhole principle". In this manner, Theorem 3.9 turns out to be a purely combinatorial consequence of Helly's Theorem in $\mathbb{R}^d$. However, Morris' proof of the lemma is extremely involved, and the validity of some of his arguments is, at best, doubtful. So an independent proof would be desirable. For a step in this direction, see Hare and Thompson (1988).

We remark that Grünbaum and Motzkin (1961) contains other "abstract" Helly-type results, with applications to convexity. For example, if $\mathscr{K}$ is a family of sets in $\mathbb{R}^d$ and the intersection of any d members of $\mathscr{K}$ lies in $\mathscr{C}_d^d$, then so does $\bigcap \mathscr{K}$. Grünbaum (1960b) associates a "measure of nonconvexity" $\delta(K)$ with each compact set K in $\mathbb{R}^d$, where $\delta(K) = 0$ characterizes the convex sets and $\delta(K) < \infty$ if and only if $K \in \mathscr{C}_k^d$ for some k. For each finite $\varepsilon \geq 0$, the family of all sets K with $\delta(K) \leq \varepsilon$ is intersectional and has finite Helly number. The case $\varepsilon = 0$ corresponds to Helly's Theorem.

We conclude this section with still another generalization of Helly's Theorem, due to Bárány (1982), which may be called a "multiplied" version of Theorem 1.1.

Theorem 3.10. *Let $\mathcal{K}_1, \ldots, \mathcal{K}_{d+1}$ be (nonempty) finite families of convex sets in $\mathbb{R}^d$. Suppose for each choice of $K_i \in \mathcal{K}_i$, $i = 1, \ldots, d+1$, we have $\bigcap_{i=1}^{d+1} K_i \neq \emptyset$. Then, for some $i \in \{1, \ldots, d+1\}$, $\bigcap \mathcal{K}_i \neq \emptyset$.*

This result is sharp in the sense that the number of $\mathcal{K}_i$'s cannot be decreased. Helly's Theorem is recovered by letting $\mathcal{K}_1 = \cdots = \mathcal{K}_{d+1}$. It was observed by Lovász that Theorem 3.10 follows from a "multiplied" version of Carathéodory's Theorem (see Theorem 10.1) much as Theorem 1.1 follows from Theorem 1.3. In fact, the two results are essentially equivalent. Bárány's paper contains a "multiplied" version of the Levi–Klee lemma mentioned in connection with Theorem 3.6 as well as some applications to convex functions. Speaking of convex functions, we point out that Helly's Theorem can be deduced from the fact that certain systems of convex inequalities have a solution. This idea originated with Bohnenblust, Karlin and Shapley around 1950 (see Danzer, Grünbaum and Klee 1963). For far-reaching generalizations, compare Rockafellar (1970).

4. Spherical and topological Helly-type theorems

4.1. Spherical Helly-type theorems

A subset X of the d-sphere S^d is called *convex* if it contains, with every two nonantipodal points, the small arc of the great circle joining them. Equivalently, X is the intersection of S^d with a convex cone in $\mathbb{R}^{d+1}$ having apex at the origin. If, in addition, X does not contain antipodal points, then X is called *strongly convex*. Both the convex and the strongly convex subsets of S^d form intersectional families. [For other notions of convexity in S^d, see Danzer, Grünbaum and Klee (1963).]

The convex sets in S^d are sometimes referred to as *Robinson-convex*, in view of the following theorem due to Robinson (1942). This was the first significant Helly-type result on the sphere.

Theorem 4.1. *For a finite family of convex sets in S^d, Π_{2d+2} implies Π. The same is true with Π_{2d+2} replaced by Π_{2d+1} unless the family consists of $d+1$ independent pairs of opposite closed hemispheres of S^d. If some member contains no subspace S^k, $0 \leq k < d$, then Π_{d+k+2} implies Π.*

It should be clear what is meant by "independent pairs of opposite hemispheres". Theorem 4.1 remains valid for infinite families of closed convex sets. Examples show that none of the conditions stated above can be dropped. In particular, the Helly number of the family of all convex sets in S^d is $2d+2$ (while

for most of its proper subfamilies it is $2d + 1$). The first part of Theorem 4.1 is closely related to Steinitz's theorem 10.3.

The paper of Robinson contains other analogues of Helly's Theorem on the sphere dealing with convex sets of restricted (spherical) diameter. These results can be employed to study the *congruence indices* of spherical caps. More generally, the problem of determining the *congruence orders of metric spaces* is a Helly-type problem in the sense of section 2. For a discussion of these matters [which go back to Menger (1928)] and references to the literature, see Danzer, Grünbaum and Klee (1963).

Far-reaching generalizations of Theorem 4.1 were obtained by Hansen and Klee (1969), and Sharaburova and Shashkin (1975, 1978a). Rather than stating in full their results which depend on the characterization of strongly positively independent sets in $\mathbb{R}^{d+1}$, we present two of the more interesting corollaries. For a convex set $X \subset S^d$, let $\lambda(X)$ denote the least number k such that X contains no subspace S^k. This is the *lineality* of the cone in $\mathbb{R}^{d+1}$ spanned by X (Sandgren 1954). Also, let $\delta(X)$ denote the dimension of X w.r.t. S^d. The following results are sharp in the sense that the stated Helly numbers cannot be reduced.

Theorem 4.2. *For a finite family $\mathcal{K}$ of convex sets in S^d, Π_{j+2} implies Π, where $j = \min_{X \in \mathcal{K}}[\lambda(X) + \delta(X)]$.*

Theorem 4.3. *Suppose $\mathcal{K}$ is a finite family of convex sets in S^d, $k = \max_{X \in \mathcal{K}} \lambda(X)$, and assume that $0 < k < d$ or $\bigcup \mathcal{K} \neq S^d$. For such a $\mathcal{K}$, Π_{d+k+1} implies Π.*

Thus the Helly number $d + k + 2$ in Theorem 4.1 may be lowered by one provided the hypotheses there are suitably strengthened. The restriction on k in Theorem 4.3 is necessary. For a variant of Theorem 4.1, see Sharaburova and Shashkin (1978b).

Turning now to strongly convex sets, we remark that the family of all such sets in S^d has Helly number $d + 2$ (rather than $d + 1$). This follows at once from Helly's Theorem. In fact, since X is strongly convex if and only if X is convex and $\lambda(X) = 0$, a sharper result holds (see Theorem 4.1 for $k = 0$). Molnár (1957c) has shown that for a finite family of strongly convex sets in S^d, Π_{d+1} implies Π provided no $d + 2$ of the sets cover S^d. The latter condition is clearly necessary. Compare also Watson (1973).

Results of a different nature were obtained by Baker (1967a,b). Though not formally of Helly-type, they demonstrate that significant conclusions can be drawn even when the crucial number $d + 2$ is replaced by $d + 1$. Baker uses the separation theorem for strongly convex sets in S^d as well as Molnár's theorem quoted above. His results were considerably strengthened by Katchalski (1977c) who removed a technical restriction from the hypothesis and, more importantly, extended the assertion to (not necessarily strongly) convex sets in S^d.

Theorem 4.4. *Let $\mathcal{K}$ be a family of at least $d + n + 1$ convex sets in S^d, where $n > 0$, and suppose $\mathcal{K}$ has property Π_{d+1}. Then some $d + [\frac{1}{2}n] + 1$ members of $\mathcal{K}$ have nonempty intersection.*

The number $d + [\frac{1}{2}n] + 1$ cannot be reduced. A further improvement is possible when $n = 2$ and $\mathcal{K}$ consists of strongly convex sets.

Theorem 4.5. *If a family of $d + 3$ strongly convex sets in S^d has property Π_{d+1}, then – with at most two exceptions – each subfamily of size $d + 2$ has nonempty intersection.*

Katchalski's proof of Theorems 4.4 and 4.5 uses rather sophisticated techniques. His approach is based on properties of systems of convex cones called *nondegenerate families* (NDFs) and the existence of neighborly polytopes. A finite family $\mathcal{C}$ of cones in $\mathbb{R}^d$ is said to be an NDF if the intersection of any k members of $\mathcal{C}$ is at least $(d - k + 1)$-dimensional, for $k = 1, \ldots, d$, whereas $\bigcap \mathcal{C}$ is the origin. (A cone is convex and has apex at the origin.) A *face* of $\mathcal{C}$ is a subfamily $\mathcal{D} \subset \mathcal{C}$ such that $\bigcap (\mathcal{C} \backslash \mathcal{D})$ is a subspace of $\mathbb{R}^d$. The set of all faces of $\mathcal{C}$ defines the *face lattice* of $\mathcal{C}$. Katchalski (1979) proved that if $\mathcal{C}$ has $d + n + 1$ members, then there exists a convex polytope in $\mathbb{R}^n$ whose face lattice is isomorphic to that of $\mathcal{C}$. The converse is also true. The isomorphism enables one to transfer known properties of polytopes to properties of NDFs, and vice versa. For example, the fact that the bound in Theorem 4.4 is tight corresponds to the fact that there are neighborly polytopes in $\mathbb{R}^n$ with any given number of vertices. For an application of these ideas in a different context, see the remarks following Theorem 3.5 above.

4.2. Topological Helly-type theorems

It may be argued that Helly's Theorem on the intersection of convex sets has a purely topological background and depends on the higher connectedness (rather than the convexity) of the intersections. Helly (1930) himself was the first to establish such a topological version of his geometric theorem. It applies to families of homology cells in $\mathbb{R}^d$. A space X is a *homology cell* (or *acyclic*) if its singular homology groups are isomorphic to those of a point, i.e., X is nonempty and $\tilde{H}_q(X)$ vanishes for all q. Here $\tilde{H}_q(X)$ denotes the qth reduced (singular) homology group of X. Since nonempty convex sets are homology cells, Theorem 1.1 is clearly contained in the following.

Theorem 4.6. *Let $\mathcal{K}$ be a finite family of closed sets in $\mathbb{R}^d$ such that the intersection of every k members of $\mathcal{K}$ is a homology cell for $k \leq d$ and is nonempty for $k = d + 1$. Then $\bigcap \mathcal{K}$ is a homology cell.*

Thus the family of closed homology cells in $\mathbb{R}^d$ has Helly order $d + 1$. Helly's proof of Theorem 4.6 is rather lengthy, and a simpler proof of a slightly weaker combinatorial version was given by Aleksandrov and Hopf (1935). Debrunner (1970) remarks that these proofs do not stand at the basis of singular homology theory as presented in introductory courses. Debrunner's own proof works with

open sets and employs only standard tools from algebraic topology such as the exactness of the Mayer–Vietoris sequence. The dimension of the space enters via the following lemma: if $Y \subset \mathbb{R}^d$ is nonempty and open, then $\tilde{H}_q(Y) = 0$ for $q \geq d$. (It is sometimes erroneously stated that this holds for closed sets as well.)

In the plane, Theorem 4.6 deals with simply connected sets and can be proved in an elementary fashion (Helly 1930). The following improvement is due to Molnár (1957a).

Theorem 4.7. *For a finite family of simply connected sets in* $\mathbb{R}^2$, *each two of which have connected intersection,* Π_3 *implies* Π.

In fact, if $\mathcal{K}$ is such a family, then $\bigcap \mathcal{K}$ is also simply connected. A further strengthening was obtained by Haddock (1963).

For applications of Theorem 4.6 to common transversals and a Krasnosel'skiĭ-type result, see Grünbaum (1960c) and Breen (1991).

The method employed in Debrunner's paper can be exploited to yield an analogue of Helly's Theorem on manifolds. The following definitions are required. A space X is *p-acyclic* if $X \neq \emptyset$ and $\tilde{H}_q(X) = 0$ for $q \leq p$. Thus homology cells are p-acyclic for all p, 0-acyclic means path-connected, and (-1)-acyclic means nonempty. By a *d-manifold* we understand a d-dimensional connected manifold without boundary. A *homology d-sphere* is a manifold having the homology of S^d. Debrunner's result can be stated as follows.

Theorem 4.8. *Let* $\mathcal{K}$ *be a finite family of open subsets of a d-manifold* M^d. *Suppose each intersection of k members of* $\mathcal{K}$ *is* $(d-k)$*-acyclic, for* $k \leq d+1$. *If* M^d *is not a homology d-sphere, or no* $d+2$ *members of* $\mathcal{K}$ *cover* M^d, *then* $\bigcap \mathcal{K}$ *is a homology cell.*

This includes theorems of Molnár, Berstein, and others as special cases [see Danzer, Grünbaum and Klee (1963) for references]. Similar results were obtained by Dugundji (1966, 1967).

It follows from Theorem 4.8 that the family of (open) homology cells in M^d has Helly order at most $d+1$ unless M^d is a homology d-sphere. In that case, the Helly order is $d+2$. Thus homology d-spheres play an exceptional role in the context of Helly's topological theorem. Under certain circumstances, the exception can be narrowed down even further. Suppose M^d is a *Riemannian* manifold. Extending an earlier definition, we say that a set $X \subset M^d$ is *strongly convex* if for any two points of X there is exactly one shortest geodesic in M^d joining them, and this geodesic is contained in X. For a review of convexity in Riemannian manifolds, see Walter (1981). Using Theorem 4.8 and results from differential geometry, Kleinjohann (1981) proved that if the family of open, relatively compact and strongly convex sets in M^d has Helly number $d+2$, then M^d is *homeomorphic* to S^d.

5. Other Helly-type theorems

This section presents a selection of Helly-type problems and results which are not covered by other sections. The selection reflects the author's tastes and prejudices and cannot claim to be exhaustive. Two more well-known results, the theorems of Kirchberger and Krasnosel'skiĭ (and their relatives), will be treated in Part II.

Our first result deals with families of algebraic sets (in the sense of algebraic geometry). An *algebraic set* in $\mathbb{R}^d$ consists of all common zeros of finitely many polynomials in the d coordinate variables. Let $\mathscr{A}_k^d$ denote the family of all such sets whose defining polynomials have degree at most k. Clearly, $\mathscr{A}_k^d$ is intersectional; $\mathscr{A}_1^d$ is the family of all flats in $\mathbb{R}^d$.

Theorem 5.1. *The Helly number of* $\mathscr{A}_k^d$ *is* $\binom{d+k}{d}$.

This was proved by Motzkin (1955) in connection with Hilbert's Nullstellensatz. A direct (and shorter) proof was given by Deza and Frankl (1987). Both proofs apply to algebraic sets in projective d-space which in a sense is the appropriate setting. Note that $\binom{d+k}{d}$ is the dimension of the space of real polynomials of degree at most k in d variables. A related Helly-type theorem was obtained by Frankl (1989). When specialized to families of spheres in $\mathbb{R}^d$, it yields the first of the following two assertions proved in an elementary way by Maehara (1989).

Theorem 5.2. *For a family of spheres in* $\mathbb{R}^d$, Π_{d+2} *implies* Π. *When the spheres are distinct and there are at least* $d+3$ *of them,* Π_{d+1} *implies* Π.

The numbers $d+2$ and $d+1$ appearing above cannot be decreased. The second conclusion fails for families of $d+2$ spheres. Maehara's paper contains other Helly-type results and gives some evidence to support his conjecture that if $d>2$, Π_{d+1} implies Π for families of *congruent* spheres. This is false for $d=2$.

Following Dawson and Edelstein (1990), we say that a family $\mathscr{K}$ of compact convex sets in $\mathbb{R}^d$ has a *definite common support* if there exists a hyperplane H such that each member of $\mathscr{K}$ is supported by H and all the members lie on one side of H. Denote this property by Δ. In order to obtain Helly-type theorems with respect to Δ, some general position assumption must be imposed. We call $\mathscr{K}$ *separated* if no d members of $\mathscr{K}$ are met by a common $(d-2)$-flat. Dawson and Edelstein prove the following result which closely resembles Theorem 5.2.

Theorem 5.3. *For a separated family of compact convex sets in* $\mathbb{R}^d$ *with interior points,* Δ_{d+2} *implies* Δ. *For such a family with at least* $d+3$ *distinct members,* Δ_{d+1} *implies* Δ.

As in Theorem 5.2, the "threshold" of Theorem 5.3 cannot be relaxed. We remark that similar results in which the supporting hyperplanes are replaced by spheres were established by Kramer and Neméth (1973a). These authors conjecture (and prove for $d=2$) that if $\mathscr{K}$ consists of at least $d+3$ members, no $d+1$ of

which are met by a hyperplane, then the sets in $\mathcal{K}$ have a common supporting unit sphere if and only if this is true of any $d+1$ of them. For an extension to Minkowski planes, see Kramer and Neméth (1973b).

We turn now to a quantitative version of Helly's Theorem. By "quantitative" we mean that the "size" of the intersections considered is measured by a suitable numerical function. This can of course be done in many different ways. For example, the function could be the inradius, diameter, width, dimension, surface area, or volume of the sets. In this sense, Theorems 3.2 and 3.5 are quantitative results. The following "Helly theorem with volume" was established by Bárány, Katchalski and Pach (1982, 1984).

Theorem 5.4. *There exists a constant $v(d)>0$ such that if $\mathcal{K}$ is a finite family of convex sets in $\mathbb{R}^d$ and $\bigcap \mathcal{L}$ has volume at least* 1, *for every subfamily $\mathcal{L} \subset \mathcal{K}$ with* card $\mathcal{L} \leqslant 2d$, *then $\bigcap \mathcal{K}$ has volume at least $v(d)$.*

The proof of this theorem uses Helly's Theorem and a quantitative version of Steinitz's theorem 10.3. A similar result holds for diameters. It is easily seen that the number $2d$ above is the smallest which has the stated property. Let $v_0(d)$ denote the supremum of those constants $v(d)$ for which the assertion holds. The true order of magnitude of $v_0(d)$ is not known. The authors show that $v_0(d) \geqslant d^{-2d^2}$, but conjecture that $v_0(d) \approx d^{-cd}$ for some positive constant c. There is reason to believe that $v_0(2)$ is attained by a family of five halfplanes which intersect in a regular pentagon. A relative of Theorem 5.4 was proved by Polikanova and Perel'man (1986).

Many of the older Helly-type results are described in Danzer, Grünbaum and Klee (1963). We mention, in particular, the covering theorems due to Santálo, Robinson, Kelly, and others which appear in section 6 of that report. For a variety of later Helly-type theorems, the reader is referred to Fan (1966), Baker (1968), Breen (1972b, 1985c), Larman (1974), Buchman and Valentine (1982), Goodman and Pollack (1982), Tattersall (1985), Wieacker (1988), and Avis and Wenger (1988).

Our final example is concerned with k-pierceability. This property will be studied in detail in section 6 where we define a family of sets to be k-pierceable if there is a set of k points intersecting every member of the family. Thus Helly's Theorem deals with 1-pierceable families. The following yields one of the rare instances of a Helly-type result for k-pierceability with $k>1$. Let $\mathcal{Q}^d$ be the family of parallelotopes in $\mathbb{R}^d$ with edges parallel to the coordinate axes, and let $h(d, k)$ be the Helly number of $\mathcal{Q}^d$ with respect to the property "k-pierceable". The problem of determining these numbers was investigated by Danzer and Grünbaum (1982). Their somewhat unexpected solution can be stated as follows.

Theorem 5.5. $h(d, 1)=2$, $h(1, k)=k+1$, $h(d, 2)=3d$ *or* $3d-1$, *according as d is odd or even, and* $h(2, 3)=16$. *In all other cases, $h(d, k)$ does not exist.*

For other Helly-type results involving k-pierceability, see Grünbaum (1974) and Breen (1986a, 1988).

6. Piercing and coloring properties of convex sets

In the combinatorial geometry of convex sets, much interest has centered around the notions of "pierceability" and "colorability" (and related notions). Above all, the question of how efficiently a family of convex sets can be "pierced" by points has led to various problems (such as Gallai-type and (p, q)-problems) which may be regarded as extensions of the Helly-type problems encountered in the preceding sections.

A general set-up for piercing and coloring problems is as follows (see, e.g., Berge 1973). Let $\mathcal{H}$ be a family of sets, and suppose $\mathcal{H}$ can be split into k or fewer subfamilies, each of which has nonempty intersection. Then $\mathcal{H}$ is said to be *k-pierceable* (or *k-fixable*, or to have a *k-partition*), and the smallest number k with this property is called the *piercing number of $\mathcal{H}$*. If on the other hand, $\mathcal{H}$ can be split into k or fewer subfamilies each consisting of pairwise disjoint sets, then $\mathcal{H}$ is said to be *k-colorable*, and the smallest such k is the *coloring number* of $\mathcal{H}$. Finally, the *stability* (respectively *clique*) *number* of $\mathcal{H}$ is the maximum number of pairwise disjoint (pairwise intersecting) sets in $\mathcal{H}$. We denote the property of being k-pierceable by Π^k, and the piercing, coloring, stability, and clique number of $\mathcal{H}$ by $\tau(\mathcal{H})$, $\chi(\mathcal{H})$, $\nu(\mathcal{H})$, and $\omega(\mathcal{H})$, respectively.

The problem is to evaluate these numbers (which need not be finite) for meaningful families of sets and to study the possible relationships between them. Notice that for each $\mathcal{H}$, $\tau(\mathcal{H}) \geq \nu(\mathcal{H})$ and $\chi(\mathcal{H}) \geq \omega(\mathcal{H})$. Here we shall restrict ourselves to families of convex sets in $\mathbb{R}^d$. In general, Helly-type results for k-pierceable or k-colorable families with $k > 1$ cannot be expected; Theorem 5.5 is an exception. Thus one is led to the wider class of *Gallai-type* and (p, q)-*problems* and to the dual class of (p, q)-*coloring problems*.

6.1. Gallai-type problems

Some forty years ago, T. Gallai raised the following question: given a family $\mathcal{H}$ of pairwise intersecting circular disks in the plane, what is the smallest number of needles required to simultaneously pierce all members of $\mathcal{H}$? In our notation, this asks for the smallest number k such that $\tau(\mathcal{H}) \leq k$ for any such $\mathcal{H}$ (if it exists at all). The problem was first published in the book by Fejes Tóth (1953) and later by Hadwiger (1957b); it became the prototype of a variety of Gallai-type problems and results in combinatorial geometry [see Danzer, Grünbaum and Klee (1963) for an exposition]. Gallai's original problem was eventually solved by Danzer who proved that "four needles suffice", while in some cases three needles are not enough. This improved earlier results of Ungár and Szekeres, Heppes, and Stachó.

Theorem 6.1. *For a family of circular disks in* $\mathbb{R}^2$, Π_2 *implies* Π^4. *The piercing number* 4 *cannot be reduced to* 3.

That three needles will not suffice in general is demonstrated by examples of Grünbaum (1959) and Danzer (1986), consisting of 21 and 10 disks, respectively.

As noted by Danzer, and Stachó (1965), if five disks intersect pairwise, then some three of them intersect; hence a counterexample must have at least 9 members.

Danzer's proof of Theorem 6.1 is a remarkable achievement. It was presented by him at an Oberwolfach conference in 1956 but never published. (For a historical account, see Danzer 1986.) The first published proof is due to Stachó (1981–84). Finally, Danzer (1986) presented a new proof which is different from his original one. He actually established a stronger result, namely, that Π_2 implies Π^4 for families of spherical caps in S^2. The reasoning is difficult and rather involved, making use of cross ratios to distinguish between certain triples of disks and arriving at the desired piercing set by a construction which is invariant under Möbius transformations. Unfortunately, one of the two crucial inequalities to which the proof is reduced has been verified by numerical computations only.

We turn now to a discussion of general Gallai-type problems. The *kth Gallai number* of a family of sets $\mathcal{K}$, denoted by $\alpha_k(\mathcal{K})$, is the smallest number n such that for each finite subfamily $\mathcal{F}$ of $\mathcal{K}$, Π_k implies Π^n. Of course, $\alpha_k(\mathcal{K})$ need not exist. Most of the known results in this area deal with the case $k = 2$ and confine their attention to the families $\mathcal{H}(K)$, and $\mathcal{F}(K)$, of all positive homothets, and translates, respectively, of a given compact convex set $K \subset \mathbb{R}^d$. Following Grünbaum (1959), we write $H(K) = \alpha_2(\mathcal{H}(K))$, $h(K) = \alpha_2(\mathcal{F}(K))$ and $H(d) = \sup H(K)$, $h(d) = \sup h(K)$, where the supremum is over all $K \subset \mathbb{R}^d$. Note that Theorem 6.1 can be stated as $H(B^2) = 4$. In Hadwiger, Debrunner and Klee (1964) it is shown that $h(B^2) = 3$ (for a sharpening, see Schopp 1962). Grünbaum (1959) established the finiteness of $H(d)$, hence $h(d)$, and proved that $H(K) \leqslant 7$, $h(K) \leqslant 3$ when $K \subset \mathbb{R}^2$ is centrally symmetric. More generally, he conjectured that $h(2) = 3$. This is among the prettiest of the Gallai-type problems remaining unsolved.

Conjecture 6.2. For a family of translates of a compact convex set in $\mathbb{R}^2$, Π_2 implies Π^3.

The literature contains some partial solutions to Grünbaum's conjecture, e.g., $h(K) = H(K) = 3$ when K is a triangle (Chakerian and Stein 1967), and $h(K) = 3$ when K has constant width (Chakerian and Sallee 1969). Analogous results for sets of constant relative width were obtained by Loomis (1980). Chakerian and Stein also prove that $h(d) \leqslant d^d$. This bound can be substantially reduced by the Erdős–Rogers method (see Danzer, Grünbaum and Klee 1963) but it shows that $h(2)$ is either 3 or 4. A more general conjecture to the effect that $h(d) \leqslant d + 1$ was disproved by Chakerian and Stein (1965). Since $h(3) \geqslant 7$, it is false already for $d = 3$. Now the most important Gallai numbers are certainly those of the ball in $\mathbb{R}^d$, $h(B^d)$ and $H(B^d)$. A plausible conjecture (also attributed to Grünbaum) asserted that $h(B^d) = d + 1$. This is true for $d = 2$ and was confirmed for $d = 3$ by Katzarova-Karanova (1967). For large d, however, the conjecture is far off since Danzer (1967) gave an example showing that $h(B^d) > 1.003^d$. (This bound has since been improved, see Bourgain and Lindenstrauss 1991.) Thus the number of points required to pierce a family of pairwise intersecting unit balls in $\mathbb{R}^d$ grows exponentially in d.

Very little is known about the higher Gallai numbers $\alpha_k(\mathcal{H}(K))$ and $\alpha_k(\mathcal{T}(K))$. It seems that significant results exist only for the case $K = B^d$; they are mainly due to Danzer (1960, 1961). The Gallai problem for translates is equivalent (and that for homothets at least related) to a covering problem. A thorough and still useful account of these matters is given in section 7 of Danzer, Grünbaum and Klee (1963).

6.2. (p, q)-problems

Let p and q be integers with $p \geq q \geq 2$. We say that a family of sets $\mathcal{K}$ has the (p, q)-property and write $\mathcal{K} \in \Pi_{p,q}$ provided $\mathcal{K}$ has at least p members, and among every p members some q have a common point. How large can the piercing number $\tau(\mathcal{K})$ under these circumstances be (if it exists at all)? This problem was first treated for families of convex sets by Hadwiger and Debrunner (1957); it is referred to as the (p, q)-problem. Notice that $\Pi_{p,p} = \Pi_p$ and so the (p, q)-problem extends the Gallai problem.

Let $M(p, q; d)$ denote the smallest number k such that for any finite family of convex sets in $\mathbb{R}^d$, $\Pi_{p,q}$ implies Π^k. If no such k exists, let $M(p, q; d) = \infty$. By Helly's Theorem, $M(p, p; d) = 1$ when $p \geq d + 1$, and $M(p, q; d) = \infty$ when $q \leq d$. The question is whether meaningful results can be obtained in the range $p > q \geq d + 1$. The following answer was found by Hadwiger and Debrunner (1957); it generalizes Helly's Theorem.

Theorem 6.3. *$M(p, q; d) = p - q + 1$ if $p \geq q \geq d + 1$ and $p(d-1) < (q-1)d$.*

Thus if $d = 1$ or p and q do not differ too widely, exact results are available. Note that, trivially, $M(p, q; d) \geq p - q + 1$. This yields $M(4, 3; 2) \geq 2$; however, examples of Danzer and Grünbaum show that $M(4, 3; 2) \geq 3$, if indeed this number exists (see Hadwiger et al. 1964). Until recently, the existence problem was unsettled in all cases not covered by Theorem 6.3, and this was one of the intriguing open questions in the area. A positive answer has now been given by Alon and Kleitman (1993).

The (p, q)-problem may of course be studied for suitable restricted classes of convex sets. Of considerable interest is the family of parallelotopes with edges parallel to the coordinate axes in $\mathbb{R}^d$. Let $N(p, q; d)$ be defined for parallelotopes in the same way as $M(p, q; d)$ is defined for general convex sets. The following results appeared in Hadwiger, Debrunner and Klee (1964) (strictly speaking, for $d = 2$ only); they show that $N(p, q; d)$ is always finite.

Theorem 6.4. *$N(p, q; d) \leq \binom{p-q+d}{d}$ if $2 \leq q \leq p$.*

Theorem 6.5. *$N(p, q; d) = p - q + 1$ if $2 \leq q \leq p \leq 2q - 2$.*

A new approach to the (p, q)-problem in $\mathbb{R}^d$ was taken by Wegner (1965). Working in the setting of abstract simplicial complexes, he proved that Theorem

6.3 is a purely combinatorial consequence of Helly's Theorem provided p and q satisfy slightly stronger restrictions than those stated there. Unfortunately, there is a gap in his proof of "Satz 1" which makes it necessary to impose even stronger conditions when $d > 2$ (Wegner 1985). Some of the many important results obtained by Wegner were rediscovered and slightly strengthened by Dol'nikov (1972) who works in a graph-theoretic setting. For example, the validity of Theorem 6.5 is extended to $p < \binom{q+1}{2}$. Among others, Wegner proposes the following appealing conjecture.

Conjecture 6.6. If $\mathcal{K}$ is a family of parallel rectangles in the plane, no p of which are pairwise disjoint, then $\mathcal{K}$ can be pierced by $2p - 3$ points.

This is true for families of unit squares in $\mathbb{R}^2$ (see Dol'nikov 1972) and was confirmed by Wegner for $p \leq 4$. Note that the condition on $\mathcal{K}$ is just $\Pi_{p,2}$ so that Conjecture 6.6 can be formulated as $N(p, 2; 2) \leq 2p - 3$. If true in general, the conjecture would have a very strong consequence, namely, $N(p, q; 2) = p - q + 1$ for $p \geq q \geq 4$.

For additional results on parallelotopes, see Wegner (1967) and Grünbaum (1974). Families of homothets of a convex set were treated by Wegner (1965) and Dol'nikov (1972), families of unit disks in the plane by Wegner (1985). Related results appear in Fullbright (1974) and Netrebin (1980). See also Trotter (1971) for a topological (p, q)-theorem, and Perles and Rabinovich (1993) for a nice characterization of the (p, q)-property for families of intervals.

6.3. Coloring problems

If a family of sets $\mathcal{K}$ is k-colorable, then each member of $\mathcal{K}$ can be assigned one of k given colors in such a way that intersecting members get different colors. Thus $\chi(\mathcal{K})$ is the *chromatic number* of the intersection graph $G(\mathcal{K})$ of $\mathcal{K}$ (see section 8). Similarly, $\omega(\mathcal{K})$ is the *clique number* of $G(\mathcal{K})$. Recall that $\omega(\mathcal{K}) \leq \chi(\mathcal{K})$. Most of the known results on colorability of families of convex sets deal with the question of whether $\chi(\mathcal{K})$ can in turn be bounded by some function of $\omega(\mathcal{K})$.

If $\mathcal{K}$ is a family of intervals on the line, then $\chi(\mathcal{K}) = \omega(\mathcal{K})$. This fact is usually attributed to Gallai and was independently discovered by Bielecki (1948) and Rado (1949). It implies that the "interval graph" $G(\mathcal{K})$ is perfect (see section 8 for interval graphs). Now suppose $p \geq q \geq 2$, and let $C(p, q; d)$ denote the smallest number k such that the following statement is true: if $\mathcal{K}$ is a family of parallelotopes with edges parallel to the coordinate axes in $\mathbb{R}^d$ and each subfamily $\mathcal{F} \subset \mathcal{K}$ with card $\mathcal{F} \leq p$ satisfies $\chi(\mathcal{F}) \leq q$, then $\chi(\mathcal{K}) \leq k$. When no such k exists, let $C(p, q; d) = \infty$. The problem of determining these numbers (reminiscent of a Helly-type problem) is far from being solved. Of special interest is the case $p = q + 1$ where the assumption on $\mathcal{K}$ is simply $\omega(\mathcal{K}) \leq q$. Gallai's result above reads $C(q + 1, q; 1) = q$; more generally, we have $C(p, q; 1) = q$. Answering a question of Bielecki, Asplund and Grünbaum (1960) established the following "six-color theorem".

Theorem 6.7. $C(3, 2; 2) = 6$.

Remarkably, the smallest family of rectangles the authors could find – in order to demonstrate that five colors will not suffice – contains about 50 000 sets. Is there a family of a more reasonable size? One of the open problems raised by Asplund and Grünbaum concerned the value of $C(3, 2; 3)$. It turned out later that $C(3, 2; 3)$ does not exist. In fact, Burling (1965) used an ingenious construction (neatly described in Grünbaum 1974) to prove the following.

Theorem 6.8. $C(q + 1, q; d) = \infty$ *if* $d \geq 3$, $q \geq 2$.

Another result of Burling's is $C(q + 1, q; 2) \leq 2q^2 - q$; this improves the upper bound obtained by Asplund and Grünbaum.

Very little is known regarding the colorability of other families of convex sets. For example, there seems to be no analogue of Theorem 6.7 for arbitrary convex sets in $\mathbb{R}^2$. When $\mathcal{K}$ is a family of homothets in $\mathbb{R}^2$, $\omega(\mathcal{K}) \leq 2$ implies $\chi(\mathcal{K}) \leq 3$. This follows from a deep result of Grötzsch (1958/59) which asserts that every triangle-free planar graph is 3-colorable. See Grünbaum (1963) for a simplified proof and slight generalization of Grötzsch's theorem. [A gap in Grünbaum's reasoning was filled by Aksionov; compare Aksionov and Mel'nikov (1978).] For additional results on coloring families of homothets and translates in higher dimensions, we refer to Brudnyi and Kotlyar (1970) and Dol'nikov (1971, 1972).

7. Common transversals

Let $\mathcal{K}$ be a family of convex sets in $\mathbb{R}^d$. A *k-transversal* of $\mathcal{K}$ is a k-flat in $\mathbb{R}^d$ which intersects every member of $\mathcal{K}$. Thus Helly's Theorem deals with 0-transversals, and Theorem 3.3 establishes a connection between 0- and k-transversals of $\mathcal{K}$. The property of having a k-transversal will be denoted by T^k; according to our general convention, $\mathcal{K} \in T_r^k$ will then mean that each subfamily of $\mathcal{K}$ containing at most r members has a k-transversal.

This section is concerned with Helly-type (and related) results involving k-transversals in $\mathbb{R}^d$. Perhaps due to the fact that the manifold of all k-flats in $\mathbb{R}^d$ is not contractible when $k > 0$, there is a clear distinction in flavor between the two cases $k = 0$ and $k > 0$. Here we shall deal exclusively with the latter case. The history of the subject begins with the following question raised by Vincensini (1935): is there a finite number $r = r(d, k)$ such that for any family of compact convex sets in $\mathbb{R}^d$, T_r^k implies T^k? In other words, does the family of compact convex sets in $\mathbb{R}^d$ have finite Helly number w.r.t. property T^k? Vincensini claimed that the answer is positive for $d = 2$, with T_6^1 implying T^1, but his proof was in error. Easy examples in Hadwiger, Debrunner and Klee (1964) show that $r(2, 1)$ does not exist. It soon became clear that "Helly-type theorems with transversals" cannot be expected unless rather severe restrictions are placed on the shapes and the mutual position of the sets considered. A typical condition, for instance, is

that the sets be pairwise disjoint and translates of one another. After a period of relative inactivity, the theory of common transversals has flourished in recent years. In part, this is due to the advent of computational geometry. We cannot possibly cover all the different facets of the subject. For much of the early work on transversals, we refer to the broad discussion in section 5 of Danzer, Grünbaum and Klee (1963) and to the book by Hadwiger, Debrunner and Klee (1964). (See also Grünbaum 1974.) An excellent recent survey is that of Goodman, Pollack and Wenger (1993) which describes also the combinatorial complexity of the space of transversals as well as algorithmic aspects.

7.1. Transversal problems in the plane

In this subsection, we abbreviate T^1 as T and T_r^1 as T_r. To warm up, let us present the planar case of a well-known Helly-type theorem due to Santaló (1940).

Theorem 7.1. *For a family of parallel rectangles in* $\mathbb{R}^2$, T_6 *implies* T.

The Helly number 6 in this statement is the best possible, as examples of Frucht, Santaló, Grünbaum, and others show [see Danzer, Grünbaum and Klee (1963) for references]. Hadwiger, Debrunner and Klee (1964) deduce Theorem 7.1 from the following result: if every three members of the family are intersected by an ascending line (i.e., a line having nonnegative slope), then all the members are so intersected. If the assumption is added to Theorem 7.1 that the rectangles be pairwise disjoint, then T_6 can be replaced by T_5. This was proved in Grünbaum (1958) and was one of the incentives for a famous and long-standing conjecture proposed in that paper. After earlier attempts had met with little success, a full proof of Grünbaum's conjecture was given by Tverberg (1989). [Two similar conjectures of Grünbaum have been refuted by Lewis (1980).] Tverberg's theorem is as follows.

Theorem 7.2. *For a disjoint family of translates of a compact convex set in* $\mathbb{R}^2$, T_5 *implies* T.

A few historical comments seem justified. For families of unit disks in $\mathbb{R}^2$, Theorem 7.2 had been proved by Danzer (1957), answering a question of Hadwiger (1955b). Danzer's proof uses metrical properties of the (Euclidean) plane and cannot be extended to the general case. Even for disks, none of the special conditions above (disjoint, translates, T_5) may be omitted. On the other hand, Grünbaum observed that as soon as the family contains more than five disks, T_4 can take the place of T_5. It became apparent that new ideas would be needed to establish the general conjecture. That a finite "stabbing number" exists at all was shown by Katchalski (1986) who obtained T_{128} as a crude upper bound.

A brief outline of Tverberg's proof is given in Goodman, Pollack and Wenger (1993). Assuming the existence of a counterexample, he uses Hadwiger's (or

rather Klee's) "shrinking process" and other reductions to manufacture a counterexample with only six members. These six translates are centrally symmetric polygons, and their centers form a convex hexagon. Tverberg then studies the conditions under which the "partial" transversals of the counterexample (i.e., transversals of five of the six sets) can coexist. A delicate analysis of myriads of cases leads to the conclusion that the restrictions obtained are not compatible. Hence a counterexample cannot exist, and the proof is complete.

This is a good place to consider the notion of a *geometric permutation* introduced by Katchalski, Lewis and Liu (1986). A 1-transversal of a family of disjoint convex sets in $\mathbb{R}^2$ meets the sets in some definite order and therefore defines two permutations of the sets, one being the reverse of the other. The resulting pair is called a geometric permutation, or GP for short. How many different GPs can a family of n members have at most? The answer is $2n-2$, when $n \geq 4$. This was proved by Edelsbrunner and Sharir (1990) and is best possible, as an example of Katchalski, Lewis and Zaks (1985) shows. The corresponding number for segments is n, and that for translates of a convex set is 3 (see Katchalski, Lewis and Zaks 1985, Katchalski, Lewis and Liu 1987). Geometric permutations have already proved their usefulness in the study of transversals. Actually, what is needed is a more detailed picture of the collection of all GPs realized by a given family. For instance, 1234 and 2143 cannot be both GPs of the same four sets. Such incompatible pairs play a crucial role in Tverberg's proof of Theorem 7.2. (See also Tverberg 1991.) The whole area remains to be explored.

Now let us say that a family of convex sets in $\mathbb{R}^2$ has property $T-k$ if there is a common 1-transversal for all but at most k of the sets. The following result is due to Katchalski and Lewis (1980).

Theorem 7.3. *There exists a positive integer k_0 such that for any disjoint family of translates of a compact convex set in $\mathbb{R}^2$, T_3 implies $T-k_0$.*

This should be compared with Theorem 7.2. The authors remark that if the sets are merely assumed to be congruent, or if the disjointness hypothesis is dropped, then the result is no longer true. They provide an upper bound on k_0 which is certainly too large. In fact, they conjecture that $k_0 = 2$ will do. This is the best one can hope for (see Goodman, Pollack and Wenger 1993). A weaker form of the conjecture for parallelograms, with T_3 replaced by T_4, was established by Katchalski and Lewis (1982). For a promising approach to the Katchalski–Lewis conjecture, as well as partial results, see Tverberg (1991).

We turn now to an old theorem of Hadwiger (1957a) which has recently opened a new field of research in transversal theory.

Theorem 7.4. *Suppose $\mathcal{K}$ is a disjoint family of compact convex sets in $\mathbb{R}^2$. Then $\mathcal{K}$ admits a 1-transversal if, and only if, $\mathcal{K}$ can be linearly ordered in such a way that, for every three members of $\mathcal{K}$, there is a directed line meeting them in the specified order.*

Theorem 7.4 gives a necessary and sufficient condition for the existence of a 1-transversal in $\mathbb{R}^2$ and comes as close to a Helly-type theorem as it possibly can. The only restriction is that the members in question be pairwise disjoint. Several earlier results of Santaló, Vincensini, Grünbaum, and Klee are contained in Theorem 7.4 as special cases [see Danzer, Grünbaum and Klee (1963) for exact references]. For example, if $\mathcal{K}$ is a family of parallel segments in the plane, T_3 implies T. This counts among the oldest and most popular applications of Helly's Theorem. More generally, an order relation of the required sort is obtained when the plane is divided into parallel strips, each of which contains at most one member of $\mathcal{K}$. Most surprisingly, Wenger (1990) was able to remove the disjointness hypothesis from the statement of Theorem 7.4. He must make clear, of course, what it means for a directed line to meet three convex sets consistently with a given ordering when some of the sets intersect. For other striking features of Hadwiger's theorem and generalizations to higher dimensions, see Goodman, Pollack and Wenger (1993, section 2) and Theorem 7.8.

We end the subsection by describing a transversal problem of Gallai-type in the plane. Since for a general family of compact convex sets in $\mathbb{R}^2$, T_r does not imply T, no matter how large r may be, it is natural to consider the smallest number of subfamilies, each having property T, into which the family can be split. Denote this number by $g(r)$. The following is known.

Theorem 7.5. $3 \leq g(3) \leq 5$; $g(r) = 2$ *if* $r \geq 4$.

This improves and extends earlier results in Hadwiger, Debrunner and Klee (1964). The upper bound on $g(3)$ was established by Kramer (1974), the rest is due to Eckhoff (1969b, 1973). It is conjectured that $g(3) = 3$. For a variant, see Wegner (1973). We point out that the Gallai problem has also been studied for families of homothets, respectively translates, of a compact convex set. The corresponding "stabbing numbers" $h(3)$ and $t(3)$ satisfy $2 \leq h(3) \leq 4$ and $t(3) = 2$ (see Eckhoff 1973).

7.2. Transversal problems in $\mathbb{R}^d$

Much of the work to date on k-transversals in $\mathbb{R}^d$ has been concerned with the two extremes $k = 1$ and $k = d - 1$. A 1-transversal in $\mathbb{R}^d$ is also called a *line transversal*, or *stabbing line*; a $(d-1)$-transversal is called a *hyperplane transversal*.

Our first theorem collects some of the older Helly-type results on line transversals in $\mathbb{R}^d$. These are mainly due to Grünbaum (1960c).

Theorem 7.6. *Let $\mathcal{K}$ be a family of compact convex sets in $\mathbb{R}^d$ having property T_r^1. Then each of the following conditions ensures that $\mathcal{K}$ has property T^1:*

(i) $r = 2d - 1$, *and the members of $\mathcal{K}$ lie in distinct parallel hyperplanes*;

(ii) $r = d + 1$, *and the diameters of the members have a finite upper bound, while their union is unbounded*;

(iii) $r = 2d - 1$, *and the sets in* $\mathcal{K}$ *are balls such that the balls with the same centers and doubled radii are pairwise disjoint.*

Parts of Theorem 7.6 were obtained by various other authors at various times [see Danzer, Grünbaum and Klee (1963) for appropriate credits and remarks]. It can be shown that all the special conditions imposed on $\mathcal{K}$ are necessary for the result to hold. In particular, the "stabbing numbers" $2d - 1$ and $d + 1$ may not be replaced by smaller numbers.

Is there a finite Helly number r such that T_r^1 implies T^1 for any disjoint family of unit balls in $\mathbb{R}^d$? This is a question of Danzer (1957). Theorem 7.2 shows that the answer is positive for $d = 2$ but already the case $d = 3$ is not settled.

The following pair of Helly-type theorems was discovered some 50 years ago by Santaló (1940). It is regarded as the first significant result on common transversals. Both statements reduce to Theorem 7.1 when $d = 2$.

Theorem 7.7. *For a family of parallelotopes in* $\mathbb{R}^d$ *with edges parallel to the coordinate axes,* T_r^1 *implies* T^1 *and* T_s^{d-1} *implies* T^{d-1}, *where* $r = 2^{d-1}(2d - 1)$ *and* $s = 2^{d-1}(d + 1)$.

The proof of this theorem is patterned after Radon's proof of Helly's Theorem. It should be possible to establish an analogous theorem for k-transversals with $1 < k < d - 1$. This problem (raised by Santaló) is still open.

The statement about hyperplane transversals in Theorem 7.7 was extended by Grünbaum (1964) to families of polytopes *related* to a given convex polytope $P \subset \mathbb{R}^d$ [see Danzer, Grünbaum and Klee (1963) for definitions]. Santaló's result is recovered when P is a d-cube. Grünbaum proved that for a centrally symmetric polytope P with $2p$ vertices, the corresponding Helly number is at most $p(d + 1)$. That this is always the best possible (i.e., for each P) is a result of Kramer (1974). Until then, this was known for $d = 2$ only. By contrast, it seems to be unknown whether the line transversal number in Theorem 7.7 is also tight.

We point out that the restriction to convex polytopes in these results is not accidental. It was proved by Grünbaum (1964) for $d = 2$ and Eckhoff (1969b) in general that if $K \subset \mathbb{R}^d$ is compact and convex and the family of all translates of K has finite Helly number w.r.t. property T^{d-1}, then K must be a polytope. Are there analogues of Theorem 7.7 for other classes of polytopes? One peculiar example was found by Kramer (1974): for a family of homothets of a regular octahedron in $\mathbb{R}^3$, T_{35}^1 implies T^1.

We now describe the recent attempts to generalize Hadwiger's theorem 7.4 to hyperplane transversals in $\mathbb{R}^d$. These attempts have been successful, culminating in Theorem 7.8 below. Since the history of the problem is well documented (see Goodman, Pollack and Wenger 1993, section 2), we shall refrain from presenting the intermediate steps.

The problem is to find necessary and sufficient conditions on a family of compact convex sets in $\mathbb{R}^d$ which assure the existence of a hyperplane transversal of the family. A priori, it is not clear what sort of conditions (replacing

Hadwiger's) might be appropriate for this purpose. Such conditions were eventually found by Goodman and Pollack (1988). A few definitions are in order here. Suppose $\mathcal{K}$ is a family of compact convex sets in $\mathbb{R}^d$. A *k-ordering* of $\mathcal{K}$ is the *order type* [see Goodman, Pollack and Wenger (1993) for details] of a labelled set of points in $\mathbb{R}^k$, in one-to-one correspondence with the members of $\mathcal{K}$. A 1-ordering is the same as a linear ordering. The ordering is nontrivial if the points affinely span $\mathbb{R}^k$. Finally, an oriented k-flat $X \subset \mathbb{R}^d$ meets a subfamily $\mathcal{F} \subset \mathcal{K}$ *consistently with the k-ordering* if one can choose from each set $K \cap X$, $K \in \mathcal{F}$, a point such that the chosen points have the same order type as the corresponding points in $\mathbb{R}^k$.

Theorem 7.8. *A family $\mathcal{K}$ of compact convex sets in $\mathbb{R}^d$ has a hyperplane transversal if and only if, for some $k \in \{1, \ldots, d-1\}$, there is a nontrivial k-ordering of $\mathcal{K}$ such that every $k+2$ members are met by an oriented k-flat consistently with the k-ordering.*

Theorem 7.8 was proved by Pollack and Wenger (1990), using the Borsuk–Ulam theorem and Kirchberger's theorem 11.1. It subsumes the earlier Hadwiger-type results of Goodman and Pollack (1988) and Katchalski (1980a). However, these authors assumed (as did Hadwiger for $d = 2$ and $k = 1$) that no $k+1$ sets in $\mathcal{K}$ have a $(k-1)$-transversal. See Goodman, Pollack and Wenger (1993) for further ramifications.

Problem 7.9. For $1 < k < d-1$, find conditions in the spirit of Hadwiger's theorem for a family of compact convex sets in $\mathbb{R}^d$ to have a k-transversal.

To conclude, let us mention that Eckhoff (1969b) and Kramer (1974) have made a study of Gallai-type transversal problems for families of translates and homothets of a compact convex set in $\mathbb{R}^d$.

8. Intersection patterns of convex sets

Helly's Theorem may be regarded as saying something about the "intersection patterns" realized by families of convex sets in $\mathbb{R}^d$. In order to make this statement precise, we need to recall the definition (and some basic properties) of finite, abstract, simplicial complexes, or *complexes*, for short. A complex is a finite collection C of finite sets, called the *faces* of C, such that each subset of a face is again a face. A face of cardinality $k+1$ is said to be k-dimensional or a *k-face*; a 0-face is also called a *vertex*. Similarly, C is said to be k-dimensional, or a *k-complex*, if the maximum dimension of the faces of C is k. A *graph* is a k-complex with $k \leq 1$. Finally, the complex of all j-faces of C with $j \leq k$ is called the *k-skeleton* of C, denoted $\mathrm{skel}_k C$. For more information about complexes, see Wegner (1967), Eckhoff (1980), and Björner and Kalai (1989).

Now let $\mathcal{K}$ be a finite family of sets. Then the *nerve* of $\mathcal{K}$ is, by definition, the

following complex $N(\mathcal{K})$: its vertices represent the members of $\mathcal{K}$, and a set of vertices is a face if and only if the corresponding members of $\mathcal{K}$ intersect. In a combinatorial sense, $N(\mathcal{K})$ describes the intersection properties of $\mathcal{K}$. Two families $\mathcal{K}$ and $\mathcal{K}'$ are said to have the same *intersection pattern* if their nerves $N(\mathcal{K})$ and $N(\mathcal{K}')$ are isomorphic.

What are the intersection patterns of families of convex sets in $\mathbb{R}^d$? This asks for an intrinsic characterization of the class of d-representable complexes, where C is *d-representable* (Wegner 1975) if there exists a family $\mathcal{K}$ of convex sets in $\mathbb{R}^d$ such that $C = N(\mathcal{K})$. The problem is discussed in some detail in Danzer, Grünbaum and Klee (1963), section 4; a number of partial solutions will be described below. Helly's Theorem implies that a d-representable complex is determined by its d-skeleton. More precisely, a set of at least $d+2$ vertices is a face if and only if each of its $(d+1)$-subsets is a face. This is obviously a very weak restriction. The general characterization problem seems completely inaccessible. Only the case $d = 1$ has been settled.

This being so, it is natural to look for invariants of d-representable complexes which are more tractable. One such invariant is the *intersection graph* of the corresponding family $\mathcal{K}$, defined as $\mathrm{skel}_1 N(\mathcal{K})$ and denoted by $G(\mathcal{K})$. Another invariant is the *f-vector* of $N(\mathcal{K})$ which collects combinatorial data on the faces of $N(\mathcal{K})$. The f-vector of a complex C is the ultimately vanishing sequence $f(C) = (f_0, f_1, f_2, \ldots)$ in which $f_k = f_k(C)$ equals the number of k-faces of C. Hence if $C = N(\mathcal{K})$, then $f_k(C)$ counts the subfamilies of $\mathcal{K}$ of size $k+1$ with nonempty intersection. Clearly, $N(\mathcal{K})$ determines $f(N(\mathcal{K}))$ but not conversely. Helly's Theorem yields a small step towards the ultimate goal of characterizing the f-vectors of all d-representable complexes. This goal is reached in Theorem 8.5.

8.1. Nerves and intersection graphs

We denote the class of d-representable complexes by $\mathcal{K}^d$. The problem is whether or not a given complex belongs to $\mathcal{K}^d$. This problem is effectively decidable in the sense of Tarski (see Wegner 1967). In practical terms, however, it is rather hopeless. Note that the desired solution must depend on d since with unrestricted dimension, every intersection pattern is possible. The following theorem of Wegner (1967) shows, among others, that every k-complex with $k \leq [\frac{1}{2}(d-1)]$ is d-representable, and so is every complex with at most $d+2$ vertices except the d-skeleton of a $(d+1)$-simplex. [The first result was rediscovered by Perel'man (1985).]

Theorem 8.1. *Suppose the k-complex C is $(d+1)$-polyhedral, i.e., there is a convex polytope $P \subset \mathbb{R}^{d+1}$ such that C is isomorphic to the k-skeleton of the boundary complex of P. Then each subcomplex of C missing at least one vertex belongs to $\mathcal{K}^d$.*

In fact, any such complex is realizable by a family of convex polytopes having pairwise disjoint interiors. One of the two proofs of Theorem 8.1 given by

Wegner uses Rado's construction of neighborly families of polytopes (see Danzer, Grünbaum and Kleee 1963, section 8).

As a consequence of Theorem 8.1, it can be shown that *planar* complexes are 2-representable. More generally, Wegner conjectures that if C is geometrically embeddable in $\mathbb{R}^d$, then C is d-representable. In the opposite direction, it is well known that the k-skeleton of a $(2k+2)$-simplex cannot be embedded (not even topologically) in $\mathbb{R}^{2k+1}$, and it is very likely that no "strong subdivision" of that complex is d-representable. (See Wegner's dissertation for details.) This would prove that the bound $[\frac{1}{2}(d-1)]$ above is tight.

We turn now to necessary conditions for a complex to be d-representable. One such condition is provided by Helly's Theorem but a much stronger one is the *d-Leray condition* which states that all complexes in $\mathscr{K}^d$ are homologically trivial in dimensions $q \geq d$.

Theorem 8.2. *If $C \in \mathscr{K}^d$ and D is an induced subcomplex of C, then for each $q \geq d$, $\tilde{H}_q(D) = 0$.*

Here $\tilde{H}_q(D)$ is the qth reduced (simplicial) homology group of D. Theorem 8.2 follows from a special case of a famous theorem of Leray (1945): if a space X is covered by a finite family $\mathscr{K}$ of closed (or of open) sets, all of whose nonempty intersections are homology cells, then $\tilde{H}_q(X) \cong \tilde{H}_q(N(\mathscr{K}))$ for all q. (Homology cells and the groups on the left were defined in connection with Helly's topological theorem.) For modern proofs of various versions of Leray's theorem, see the references cited in Eckhoff (1987). The families $\mathscr{K}$ above are sometimes referred to as *good covers*. A complex C satisfying the conclusion of Theorem 8.2 is called a *d-Leray complex*, and the class of all such complexes is denoted by $\mathscr{L}^d$. Not surprisingly, $\mathscr{K}^d$ is a proper subclass of $\mathscr{L}^d$. A smallest example (in view of the number of vertices) of a 2-Leray but not 2-representable complex can be found in Wegner (1967). The question arises whether the d-Leray condition may be replaced by a stronger condition which is still necessary.

Such a condition was found by Wegner (1975). It asserts that every d-representable complex can be reduced to the void complex by a sequence of operations called *elementary d-collapsings*. This is now regarded as one of the fundamental properties shared by all finite families of convex sets in $\mathbb{R}^d$. Its importance in applications, notably in the study of f-vectors, cannot be overestimated. An elementary d-collapsing of C consists in deleting a face $S \in \mathrm{skel}_{d-1} C$ which lies in a unique maximal face of C, and all the faces containing S. A *d-collapsing* is a sequence of elementary d-collapsings, and the property referred to above is called *d-collapsibility*.

Theorem 8.3. *Every d-representable complex is d-collapsible.*

The proof of this theorem is by induction on d and uses a clever construction to find the sequence of faces S required in a d-collapsing. The case $d = 1$ can be rephrased as saying that every interval graph has a simplicial vertex (see the

discussion of intersection graphs below). Denoting the class of d-collapsible complexes by $\mathscr{C}^d$, we have $\mathscr{K}^d \subset \mathscr{C}^d \subset \mathscr{L}^d$; both inclusions are strict when $d > 1$. The right-hand one follows on noting that a d-collapsing does not affect the homology in dimensions $q \geq d$. Its strictness is demonstrated by any triangulation of a contractible yet not collapsible space such as (for $d = 2$) the dunce hat. For connections between d-collapsibility and *simplicial collapsibility* in the sense of Whitehead, see Wegner (1975), and Björner and Kalai (1989).

A strengthening of Wegner's theorem has been obtained by Eckhoff (1985). It states that for any vertex v of a complex $C \in \mathscr{K}^d$ there is a d-collapsing of C which reduces C *first* to the star of v in C (i.e., the subcomplex generated by all faces containing v) and *then* to the void complex. This property is called *strong d-collapsibility* and is, in fact, the strongest necessary condition to date for a complex to lie in $\mathscr{K}^d$. Wegner's complex mentioned above is an instance of a 2-collapsible complex which is not strongly 2-collapsible.

The rest of this subsection is devoted to a short discussion of intersection graphs. The question of whether a given graph may be regarded as the intersection graph of some (possibly restricted) family of convex sets in $\mathbb{R}^d$ has received considerable attention. A wealth of results is contained in Wegner's dissertation (1967). It was observed by Grünbaum, and follows from Theorem 8.1, that every graph is 3-representable and every planar graph is 2-representable. Wegner proved that if G is obtained from a nonplanar graph by inserting at least one vertex into every edge, then G is not 2-representable. The smallest such graphs have 15 vertices. It appears that the problem of determining all 2-representable graphs is very difficult.

The intersection graphs of finite families of convex sets (or intervals) in $\mathbb{R}^1$ are called *interval graphs* and have been extensively studied. The problem of characterizing them, posed by Hajós (1957), was encountered by Benzer (1959, 1962) in connection with a question in genetics. The following solution is due to Lekkerkerker and Boland (1962).

Theorem 8.4. *A graph G is an interval graph if and only if G is chordal (i.e., every cycle of length four or more has a chord) and among any three vertices of G, one is a vertex or adjacent to a vertex of any path in G joining the other two.*

A crucial step in the proof of this theorem is to show that chordal graphs have *simplicial vertices* (i.e., vertices with mutually adjacent neighbors). This fact was used in Wegner's proof of Theorem 8.3 and implies, incidentally, that $\mathscr{L}^1 = \mathscr{C}^1$. The authors provide a characterization of interval graphs in terms of "forbidden subgraphs" and discuss the complexity of deciding the 1-representability of a graph. For references to other proofs of Theorem 8.4 and a review of properties of interval graphs, see Fishburn (1985). The closely related problem of characterizing *circular-arc graphs* (i.e., intersection graphs of families of convex sets in S^1) is addressed in Klee (1969); a solution was provided by Tucker (1970).

Roberts (1969) defined the *boxicity* $b(G)$ of a graph G as the smallest dimension d such that G is the intersection graph of a family of parallelotopes in

$\mathbb{R}^d$ (with edges parallel to the coordinate axes). He found that $b(G) \leq [\frac{1}{2}n]$, when G has n vertices, and this is best possible. See Wegner (1967) and Witsenhausen (1980) for independent proofs and Trotter (1979) for a description of the graphs achieving the bound. The graphs with $b(G) \leq 1$ are of course the interval graphs. Recently, Quest and Wegner (1990) succeeded in characterizing the graphs with $b(G) \leq 2$. For other concepts of graph dimension such as the *cubicity* or the *sphericity* of G, the reader should consult West (1985). It was shown by Thomassen (1986) that every planar graph is the intersection graph of a family of nonoverlapping parallelotopes in $\mathbb{R}^3$. At the same time, every planar graph is the intersection graph of nonoverlapping disks in the plane. This was conjectured by Wegner (see Gruber and Schneider 1979) and independently by Jackson and Ringel (1984); it follows from the work of Andreev and Thurston on the existence of circle packings on Riemann surfaces (see, e.g., Rodin and Sullivan 1987) but had already been proved by Koebe (1936).

8.2. *f-vectors*

As mentioned before, the determination of all the possible f-vectors of d-representable complexes can be viewed as a far-reaching generalization of Helly's Theorem. In 1974, Eckhoff conjectured (and proved for $d \leq 2$) that these f-vectors are characterized by a certain system of linear and nonlinear inequalities (see Eckhoff 1980). A full proof of his conjecture was given about ten years later by Kalai (1984b, 1986). In order to formulate Kalai's theorem, we need some definitions. The *h-vector* of a complex $C \in \mathcal{K}^d$ is the sequence $h(C) = (h_0, h_1, h_2, \ldots)$, where the $h_k = h_k(C)$ are expressed as linear combinations of the $f_j = f_j(C)$ as follows: $h_k = f_k$ if $0 \leq k < d$, and

$$h_k = \sum_{j \geq k} (-1)^{j-k} \binom{j-d}{k-d} f_j \quad \text{if } k \geq d\,.$$

For fixed d, the correspondence between f- and h-vectors is one-to-one, so it is enough to consider the latter. Let n and k be positive integers. Then n can be written uniquely in the form

$$n = \binom{a_k}{k} + \binom{a_{k-1}}{k-1} + \cdots + \binom{a_i}{i}\,,$$

with $a_k > a_{k-1} > \cdots > a_i \geq i \geq 1$. Given this representation (sometimes termed *k-canonical*), define:

$$\partial_{k-1}(n) = \binom{a_k}{k-1} + \binom{a_{k-1}}{k-2} + \cdots + \binom{a_i}{i-1}\,.$$

Also, let $\partial_{k-1}(0) = 0$. Then Eckhoff's conjecture can be stated, in categorical formulation, as follows.

Theorem 8.5. *The integer sequence* $(h_0, h_1, h_2, \ldots)$ *is the h-vector of some d-representable complex if and only if the following conditions hold*:

(i) $h_k \geq 0$, $k \geq 0$;

(ii) $\partial_k(h_k) \leq h_{k-1}$, $1 \leq k \leq d-1$;

(iii) $\partial_{d-1}(h_k) \leq h_{k-1} - h_k$, $k \geq d$.

Kalai's proof of this theorem is a brilliant achievement. Unfortunately, we cannot present even a sketch of his reasoning which surpasses the realm of convexity. The necessity of the conditions in Theorem 8.5 is established in Kalai (1984b). The proof uses Wegner's theorem 8.3 and relies heavily on techniques from exterior algebra. It applies to all d-collapsible complexes. A crucial step consists in showing that certain "generalized homology groups" (which were invented for the purpose) vanish for d-collapsible complexes. The numbers h_k appear as dimensions of vector spaces. For a proof of the sufficiency of the above conditions, see Kalai (1986). The appropriate families of convex sets constructed there are related to the cyclic polytopes.

We point out that the "h-conjecture" in Eckhoff (1980) actually asserts that the conditions of Theorem 8.5 characterize not only the h-vectors of d-representable complexes, but those of all d-Leray complexes as well. This far-reaching generalization has also been proved by Kalai, using "algebraic shifting" methods. For references and a general survey on "f-vectors and homology", see Björner and Kalai (1989). It follows, e.g., that whenever $\mathcal{K}$ is a good cover in $\mathbb{R}^d$, there exists a family $\mathcal{L}$ of *convex* sets in $\mathbb{R}^d$ such that $N(\mathcal{K})$ and $N(\mathcal{L})$ have the same f-vector. Kalai's approach can be exploited to yield a characterization of the f-vectors of *acyclic* d-Leray complexes (see Eckhoff 1987). By Leray's theorem, the result applies to nerves of families of closed convex sets in $\mathbb{R}^d$ whose union is also convex.

An interesting consequence of Theorem 8.5 is that if $C \in \mathcal{K}^d$ has n vertices, then for $k \geq d$, $h_k(C) \leq \binom{n-k+d-1}{d}$. Translated back into the language of convex sets, this becomes the first part of the following assertion.

Theorem 8.6. *Suppose* $n \geq r \geq d$, *and let* $\mathcal{K}$ *be a family of n convex sets in* $\mathbb{R}^d$ *with* $f_r(N(\mathcal{K})) = 0$. *Then, for* $k = d, \ldots, r-1$,

$$f_k(N(\mathcal{K})) \leq \sum_{j=0}^{d} \binom{r-d}{k-j+1}\binom{n-r+d}{j}.$$

Moreover, if equality holds for some such k, then equality holds for each such k.

This was conjectured around 1980 by Katchalski and Perles and proved independently by Kalai (1984a) and Eckhoff (1985). (For the case $d = k = 1$, see Abbott and Katchalski 1979.) Both proofs employ Wegner's theorem 8.3. The result may be called an upper-bound theorem (UBT) for families of convex sets, by analogy with McMullen's famous UBT for simplicial convex polytopes. In fact, Eckhoff's proof [which uses the strong d-collapsibility of $N(\mathcal{K})$] closely resembles McMullen's proof, and Alon and Kalai (1985) derive both UBTs in a uniform fashion by reducing them to a combinatorial statement about collapsings in

complexes. Equality in Theorem 8.6 is attained when $\mathcal{K}$ consists of $r-d$ copies of $\mathbb{R}^d$ and $n-r+d$ hyperplanes in general position in $\mathbb{R}^d$, but not only in that case. A geometric characterization of such "extremal" families is not known.

In yet another generalization of Helly's Theorem, Katchalski and Liu (1979) have introduced the notion of a "fractional" Helly theorem. Roughly, this says that if $\mathcal{K}$ is a finite family of convex sets in $\mathbb{R}^d$, and if "most" subfamilies of $\mathcal{K}$ of size $d+1$ have nonempty intersection, then $\mathcal{K}$ has a subfamily with nonempty intersection containing "most" of the sets in $\mathcal{K}$. A sharp and detailed quantitative version of this result was obtained by Kalai (1984a) as an application of Theorem 8.6.

It would be interesting to find analogues of Theorems 8.5 and 8.6 for families of sets belonging to restricted classes of convex sets, such as flats, balls, translates, homothets, or parallelotopes with edges parallel to the coordinate axes in $\mathbb{R}^d$. Kalai (1984a) proposed a UBT for the latter class of sets; his conjecture was confirmed by Eckhoff (1988). Suppose $n \geq r \geq d$, and partition the number $n-r+d$ into d almost equal integer parts $n_1, \ldots, n_d$. Define the family $\mathcal{C}(n, d, r)$ to consist of n_i distinct hyperplanes perpendicular to the x_i axis, for $i = 1, \ldots, d$, and $r-d$ copies of $\mathbb{R}^d$. (These sets may be regarded as degenerate parallelotopes.) Then the following analogue of Theorem 8.6 holds.

Theorem 8.7. *Let $\mathcal{K}$ be a family of n parallelotopes in $\mathbb{R}^d$ with edges parallel to the coordinate axes, and suppose $f_r(N(\mathcal{K})) = 0$. Then, for $k = 1, \ldots, r-1$,*

$$f_k(N(\mathcal{K})) \leq f_k(N(\mathcal{C}(n, d, r))).$$

Moreover, if equality holds for some $k \geq d$, then equality holds for each $k \geq 1$.

Explicit expressions for the upper bounds appearing in this theorem can be found in Eckhoff's paper, which also contains a "fractional" Helly theorem for parallelotopes generalizing a result of Katchalski (1980b). Again, $\mathcal{C}(n, d, r)$ is only one of many families for which equality in Theorem 8.7 is achieved. The intersection patterns of all such "extremal" families have been classified (and counted) in Eckhoff (1991).

What are the f-vectors of finite families of parallelotopes in $\mathbb{R}^d$? This appears to be a difficult problem, and no conjecture as to its solution has, so far, been made. An answer to the above question would have important consequences in extremal graph theory.

PART II. RADON'S AND CARATHÉODORY'S THEOREM, AND THE CONVEXITY PROPERTIES OF CONFIGURATIONS OF POINTS

9. Radon's Theorem and its relatives

As already mentioned, Helly's Theorem 1.1 was first proved by Radon in 1921. The proof is both short and elegant. What is known today as Radon's Theorem,

i.e., Theorem 1.2, was employed in that proof as a lemma but not explicitly stated as such. Radon never went back to the subject, and it is safe to say that he did not mean the lemma to be a theorem in its own right. Nevertheless, "Radon's Theorem" turned out to be extremely useful in combinatorial convexity theory. It has seen numerous applications, frequently in proofs, and has led to a rich body of variants, refinements, and deep generalizations (such as Tverberg's theorem).

The development of the field is documented in Danzer, Grünbaum and Klee (1963) and in more detail in the later survey articles by Doignon and Valette (1975), Eckhoff (1979), and Reay (1982). Below we present an up-to-date review of most of the results, conjectures, and open problems which are pertinent to Radon's Theorem and its relatives.

9.1. Radon partitions and Radon types

The basic notion in connection with Radon's Theorem is that of a Radon partition. Let X be a set of points in $\mathbb{R}^d$. A *Radon partition* of X is a pair $\{X_1, X_2\}$ of subsets of X satisfying $X_1 \cup X_2 = X$, $X_1 \cap X_2 = \emptyset$, and $\operatorname{conv} X_1 \cap \operatorname{conv} X_2 \neq \emptyset$. Each of X_1 and X_2 is called *half a Radon partition* or a *Radon component* of X. A slight strengthening of Radon's Theorem 1.2 asserts that X has a Radon partition if and only if X is affinely dependent. This is the case $Y = \emptyset$ of the following theorem which was proved in various forms by Hare and Kenelly (1971), Petty (1975), and Doignon and Valette (1975).

Theorem 9.1. *A set $Y \subset X$ is contained in half a Radon partition of X if and only if either $X \setminus Y$ is affinely dependent or* $\operatorname{conv} Y \cap \operatorname{aff}(X \setminus Y) \neq \emptyset$.

In particular, when X consists of $d+2$ points in general position in $\mathbb{R}^d$, two points of X lie in the same Radon half of X precisely if they are separated by the hyperplane through the remaining points. Hence the Radon partition of X is unique. This fact was observed by many authors. [To the references listed in Eckhoff (1979), we add Rado (1952).] A similar description of the unique Radon partition in which hyperplanes are replaced by spheres is contained in Hare and Kenelly (1971) (for a more geometric proof, see Peterson 1972).

The paper by Hare and Kenelly introduced the notion of a *primitive Radon partition* (PRP) which has proved useful in various investigations. By a PRP in X we mean a Radon partition $\{Y_1, Y_2\}$ of some subset of X which does not extend any other Radon partition: if $Z_1 \subset Y_1$, $Z_2 \subset Y_2$ and $\operatorname{conv} Z_1 \cap \operatorname{conv} Z_2 \neq \emptyset$, then $Z_1 = Y_1$, $Z_2 = Y_2$. The basic properties of PRPs are listed in the following theorem [see Eckhoff (1979) for references].

Theorem 9.2. *If $\{Y_1, Y_2\}$ is a PRP in X, then $Y_1 \cup Y_2$ is in general position in* $\operatorname{aff}(Y_1 \cup Y_2)$, $\operatorname{card}(Y_1 \cup Y_2) = \dim(Y_1 \cup Y_2) + 2$, *and* $\operatorname{conv} Y_1 \cap \operatorname{conv} Y_2$ *is a single point. Every Radon partition of X extends a PRP in X.*

Thus X admits a unique Radon partition if and only if $\operatorname{card} X = \dim X + 2$ and X is in general position in $\operatorname{aff} X$. Properties of PRPs were used by Goodman and

Pollack (1988) and Pollack and Wenger (1990) in connection with common transversals and in a different context by Kallay (1984b). The first named paper contains an "exchange theorem" for PRPs.

The collection of all Radon partitions of a given set determines the *Radon type* of that set (Eckhoff 1974, 1975). We say that $X, Y \subset \mathbb{R}^d$ are of the same Radon type provided there is a one-to-one correspondence between the points of X and Y so that Radon partitions of X correspond to Radon partitions of Y, and vice versa. In a sense, this comes down to saying that X and Y have the same convexity properties. By Theorem 9.2, an equivalent definition is that PRPs in X correspond to PRPs in Y and vice versa. In fact, knowing the Radon partitions *of* X is equivalent to knowing the PRPs *in* X. It seems hopeless to try to characterize the possible Radon types of finite sets in $\mathbb{R}^d$, even of sets in general position. A solution to this problem has been obtained in a few rather special cases. These include vertex sets of cyclic polytopes in $\mathbb{R}^d$ (Breen 1973), sets of $d+3$ points in general position in $\mathbb{R}^d$ (Eckhoff 1975), sets having the property that for any Radon partition $\{X_1, X_2\}$, $\operatorname{conv} X_1 \cap \operatorname{conv} X_2$ is a single point (Eckhoff 1976), and "stable" sets in $\mathbb{R}^d$ (Kallay 1984b). A set is said to be *stable* if it includes, for each of its PRPs $\{Y_1, Y_2\}$, the singleton $\operatorname{conv} Y_1 \cap \operatorname{conv} Y_2$. For another example, see Eckhoff (1969a).

We point out that the Radon type of X coincides with the *semiequivalence type* of X introduced by Goodman and Pollack (1984b). The latter is defined in terms of the *semispaces* of X, i.e., the intersections of X with a halfspace. A set $Y \subset X$ is a semispace of X if and only if $\{Y, X \setminus Y\}$ is not a Radon partition of X (a *non-Radon partition*, for short). Theorem 1.2 amounts to saying that if $X \subset \mathbb{R}^d$ has at least $d+2$ points, then not all subsets of X are semispaces of X. (For a generalization, see Dudley 1979.)

There is an interesting relationship between the facial structure of a convex polytope P in $\mathbb{R}^d$ and the Radon partitions of its vertex set $\operatorname{vert} P$. It was proved by Breen (1972a) that the Radon type of $\operatorname{vert} P$ determines the combinatorial type of P. More precisely, Breen established the following result (for a shorter proof, see Doignon and Valette 1975).

Theorem 9.3. *For a convex polytope $P \subset \mathbb{R}^d$ and a set $Y \subset \operatorname{vert} P$, the following assertions are equivalent*:

(i) $\operatorname{conv} Y$ *is a face of* P;

(ii) *if* $\{Y_1, Y_2\}$ *is a PRP in* $\operatorname{vert} P$, *then* $Y_1 \subset Y$ *implies* $Y_2 \subset Y$.

Thus the faces of P can be read off from the list of PRPs in $\operatorname{vert} P$. The converse is also true when P has at most $d+3$ vertices in general position in $\mathbb{R}^d$, but not in general (Eckhoff 1975). Breen also showed that the combinatorial type of P can be "reconstructed" from suitable subcollections of PRPs in $\operatorname{vert} P$. When P is simplicial, assertion (ii) above takes the simpler form that no subset of Y is half a Radon partition of $\operatorname{vert} P$. For generalization to *matroid polytopes*, see Cordovil and Silva (1987).

We end this subsection by describing an intriguing problem of McMullen, which was partially solved by Larman (1972): what is the largest number $\nu(d)$ such that

any set of $\nu(d)$ points in general position in $\mathbb{R}^d$ can be mapped, by a permissible projective transformation, onto the vertices of a convex polytope? At first sight, this seems to have little to do with Radon's Theorem. However, the problem can be reformulated (via Gale diagrams) as a problem on Radon partitions, and Larman established the following.

Theorem 9.4. *A set X of $2d+3$ points in $\mathbb{R}^d$ admits a Radon partition $\{X_1, X_2\}$ such that, for each point $p \in X$, $\{X_1 \backslash p, X_2 \backslash p\}$ is a Radon partition of $X \backslash p$.*

This is surprisingly hard to prove. It is conjectured that $2d+3$ above may not be replaced by a smaller number but so far this has been verified for $d \leq 3$ only. If true in general, the answer to McMullen's question would be $\nu(d) = 2d+1$, $d > 1$. Cordovil and Silva (1985) gave a new proof of Larman's result, and Las Vergnas (1986) derived an upper bound on $\nu(d)$ which leads to a lower bound of (roughly) $d + \sqrt{2d}$ for the optimal number in Theorem 9.4. These authors work in the setting of oriented matroids; the application to sets $X \subset \mathbb{R}^d$ is made by considering the matroid of signed affine dependencies of X. For a discussion of (non-)Radon partitions in oriented matroids and references to other relevant papers, see Bienia and Cordovil (1987).

9.2. The number of Radon partitions and Radon types

In connection with Radon's Theorem, three main enumeration problems arise. The present subsection is devoted to a short review of these problems. (Compare also Eckhoff 1979.)

The first problem is concerned with the number of distinct Radon partitions of a finite set $X \subset \mathbb{R}^d$. Denote this number by $N(X)$, and suppose X has n points. A sharp lower bound on $N(X)$ in terms of d and n is given in the following theorem.

Theorem 9.5. *If X is a set of n points in $\mathbb{R}^d$, then*

$$N(X) \geq \sum_{j=0}^{n-d-2} \binom{n-1}{j},$$

with equality if and only if X is in general position in $\mathbb{R}^d$.

It seems that the equality assertion in this theorem was first proved by Harding (1967). Independent proofs of Theorem 9.5 are due to Eckhoff (1969a), Zaslavsky (1975), Brylawski (1976), and others. Sierksma (1976) extended the first part of Theorem 9.5 to arbitrary "convexity spaces" in which the analogue of Radon's Theorem holds.

A number of properties of $N(X)$ are listed in Eckhoff (1979, section 2). For example, the dualized version of a theorem of Winder (1966) states that $N(X)$ equals the number of subsets Y of X for which $\dim Y + \operatorname{card} Y$ is even. Most of these properties can be derived, in a uniform manner, from well-known results on

the Whitney numbers of geometric lattices (or matroids). A general reference on matroids is Welsh (1976). For a finite set $X \subset \mathbb{R}^d$, let $L(X)$ denote the lattice of all flats in $\mathbb{R}^d$ spanned by points of X and ordered by inclusion. This is the (geometric) lattice of closed sets in the "affine dependency matroid" of X. The connection with Radon partitions is given by the following formula due to Zaslavsky (1975) and, for general oriented matroids, Las Vergnas (1975):

$$N(X) = 2^{n-1} - \tfrac{1}{2} \sum_{j=0}^{d+1} |w_j| .$$

Here $n = \operatorname{card} X$, and w_j is the *jth Whitney number of the first kind* of $L(X)$. For a definition of these numbers, see Zaslavsky's memoir. Tight lower bounds on the $|w_j|$ were obtained by Dowling and Wilson (1974) for general geometric lattices and by Brylawski (1977) for irreducible geometric lattices. When substituted in the above formula, these estimates lead to sharp upper bounds on $N(X)$ in the following two cases: X spans $\mathbb{R}^d$; $X \backslash p$ spans $\mathbb{R}^d$ for each $p \in X$. See Eckhoff (1979, Theorems 2.5–2.7). It is natural to ask whether the individual Whitney numbers can also be interpreted for Radon partitions. This problem was raised in Eckhoff (1979), and a solution was obtained by Greene and Zaslavsky (1983). It reads as follows.

Theorem 9.6. *Suppose X spans $\mathbb{R}^d$, and let $A_0 \subset A_1 \subset \cdots \subset A_d$ be a chain of distinct flats in $\mathbb{R}^d$ in general position w.r.t. $L(X)$. Then, for $j = 0, \ldots, d+1$, $|w_j|$ equals the number of non-Radon partitions of X produced by hyperplanes containing A_{d-j-1}, but not A_{d-j+1}.*

We refer to the paper by Greene and Zaslavsky for any undefined terminology as well as additional results, open problems, and further references.

A related problem is that of counting the number of PRPs contained in a given set $X \subset \mathbb{R}^d$. Denote this number by $P(X)$. If X has n points, then $P(X) \leq \binom{n}{d+2}$, and the maximum is achieved precisely for sets in general position (see Welsh 1976). The problem of finding the corresponding minimum numbers was raised in Eckhoff (1979). Some partial answers were given by Doignon (1979, 1981). For odd dimensions d, Doignon conjectured that $P(X)$ attains its minimum when the points of X are distributed as evenly as possible among $\frac{1}{2}(d+1)$ skew lines. This was confirmed by Kallay (1984a). A similar conjecture for even dimensions is still open.

Our second enumeration problem deals with the number of Radon partitions of X of a fixed type. The *type* of $\{X_1, X_2\}$ is simply the pair $\{\operatorname{card} X_1, \operatorname{card} X_2\}$ (see Shephard 1969). In the equivalent setting of semispaces, the problem has increasingly attracted the interest of authors working in computational geometry. Let $f_k(X)$ denote the number of *k-sets* of X, i.e., semispaces of cardinality k, and let $g_k(X) = f_1(X) + \cdots + f_k(X)$. The problem of characterizing the possible sequences $(f_1(X), f_2(X), \ldots)$, where X runs through all sets of n points in general

position in $\mathbb{R}^d$, is discussed (in the dual language of "Radon vectors") in Eckhoff (1979). A solution is known for $n \leq d+3$ (Eckhoff 1975). For a wealth of upper and lower bounds on $f_k(X)$ and $g_k(X)$, see Edelsbrunner (1987, chapter 3). We content ourselves with a few examples.

Theorem 9.7. *For a set X of n points in $\mathbb{R}^2$, $g_k(X) \leq kn$ if $k < \frac{1}{2}n$, and $f_k(X) \leq (2k)^{3/2}$ if $k = \frac{1}{2}n$.*

The first assertion was proved by Alon and Györi (1986) and independently by Peck (1985) and Yang (1987). It replaces a weaker result of Goodman and Pollack (1984a) and is best possible for each k. The second assertion, due to Lovász (1971), concerns the maximum possible number of "halving lines" of X. The bound is almost certainly too large but it seems difficult to reduce it substantially. For the analogous problem in $\mathbb{R}^3$, see Bárány, Füredi and Lovász (1990) and Aronov et al. (1991).

Very little is known about the individual numbers $f_k(X)$ in higher dimensions. However, the asymptotic behavior of $g_k(X)$ in the worst case has recently been determined quite accurately. Using geometric algorithms based on random sampling, Clarkson (see Clarkson and Shor 1989) was able to show that

$$\max g_k(X) = O(n^{[d/2]}k^{[(d+1)/2]}) \quad \text{as } \frac{n}{k} \to \infty,$$

where d is fixed and the maximum is taken over all sets of n points in $\mathbb{R}^d$. This is an improvement over previous results by a number of authors. The new bound follows from the UBT for convex polytopes in $\mathbb{R}^d$; it is asymptotically sharp as the example of the vertex set of the cyclic polytope $C(n, d)$ shows. Guided by numerical evidence, we venture in fact the following:

Conjecture 9.8. For each set X of n points in $\mathbb{R}^d$, $g_k(X) \leq g_k(\text{vert } C(n, d))$, $k = 1, \ldots, [\frac{1}{2}n]$.

Finally, let us briefly mention the third enumeration problem connected with Radon's Theorem. This problem is concerned with the number of Radon types realized by sets of n points in $\mathbb{R}^d$. Denote this number by $t(d, n)$ and the corresponding number for sets in general position by $t_s(d, n)$. The problem of computing these numbers is hopelessly difficult, if not intractable. The values of $t_s(d, n)$ for $n \leq d+3$ are known (see Eckhoff 1979) but already $t_s(d, d+4)$ seems out of reach. Using Gale diagrams, it should at least be possible to determine $t(d, d+3)$. Quite recently, Goodman and Pollack (1986) and Alon (1986) have studied the closely related numbers of "order types" of (labelled) sets of n points in $\mathbb{R}^d$. Using results of Milnor and Warren on the number of connected components of certain algebraic sets, the authors were able to describe the asymptotic behavior of the numbers in question (as $n \to \infty$) almost precisely. Since the numbers $t(d, n)$ and $t_s(d, n)$ differ from the latter numbers by a factor of (at most) $n!$, their asymptotic behavior is now essentially known.

9.3. Tverberg's theorem and its relatives

The remainder of this section is centered on a famous generalization of Radon's Theorem proved by Tverberg (1966). Let $r \geq 2$ be a given integer.

Theorem 9.9. *Each set of* $(r-1)(d+1)+1$ *points in* $\mathbb{R}^d$ *can be divided into* r *pairwise disjoint subsets whose convex hulls have a common point. Some sets of* $(r-1)(d+1)$ *points cannot be divided in this manner.*

This becomes Radon's Theorem when $r = 2$. Unlike Radon's Theorem, however, Tverberg's theorem is by no means easy to prove. It improves earlier results of Rado (1952) (who introduced the problem) and Birch (1959); the latter conjectured the above solution and established it for $d = 2$.

We shall say that a set is *r-divisible* if it has the property stated in Theorem 9.9. Let X be a set of N points in $\mathbb{R}^d$, where $N = (r-1)(d+1)+1$. Tverberg's idea is to construct a sequence (X_i) of N-pointed sets converging to X. The initial set X_1 must be r-divisible and algebraically independent, i.e., the dN real coordinates of the points in X_1 must be algebraically independent over the rationals. Such a set is readily found. The sequence can be chosen in such a way that for each i, $X_i \cup X_{i+1}$ is algebraically independent and has $N+1$ points. A careful reasoning then shows that the r-divisibility carries over from X_i to X_{i+1} for each i and, in the limit, to X.

Tverberg's original proof can be somewhat simplified. In particular, the algebraic independence used by him may be replaced by the weaker and more geometric notion of *strong independence* introduced by Reay (1968), or the closely related notion of *full independence* due to Doignon and Valette (1975, 1977). A set $X \subset \mathbb{R}^d$ is called strongly independent provided the following is true: if $S_1, \ldots, S_t$ are flats spanned by pairwise disjoint subsets of X, then $\dim(S_1 \cap \cdots \cap S_t) = \max\{-1, \dim S_1 + \cdots + \dim S_t - (t-1)d\}$. This condition keeps the flats from forming pencils of lines, books of planes, etc.; it implies general position of X. The second statement in Theorem 9.9 can now be specified as follows: no strongly independent set of $N-1$ points in $\mathbb{R}^d$ is r-divisible. Incidentally, Tverberg (1981) found a new proof of Theorem 9.9 which proceeds by induction on r and is much simpler than the original one (though still not easy).

Theorem 9.9 has been extended in many different directions. The first such extension is due to Tverberg (1968) himself. It consists in replacing points with flats of a fixed dimension k, where $0 \leq k \leq d$.

Theorem 9.10. *Each set of* $(r-1)(d-k+1)+1$ *k-flats in* $\mathbb{R}^d$ *can be divided into* r *subsets whose convex hulls contain a common k-flat.*

By the convex hull of a set of flats we mean the convex hull of the union of the flats. Theorem 9.10 is best possible for all r and k and includes Theorem 9.9 as the special case $k = 0$. The case $r = 2$ was used by De Santis (1957) in his proof of Theorem 3.4. In a similar vein, one may seek conditions under which the convex

hulls in question admit a common k-transversal (see section 7). The following theorem of Stangeland (1978) [parts of which were proved earlier by Iversland (1969)] reduces to Tverberg's theorem when $k = 0$ and is also the best possible.

Theorem 9.11. *Each set of* $(r - k - 1)(d - k + 1) + k + 1$ *points in* $\mathbb{R}^d$ *can be divided into* r *pairwise disjoint subsets whose convex hulls admit a* k*-transversal.*

For a discussion of related matters, see Eckhoff (1979) and Reay (1982).

A very active line of research was opened by Reay (1968) when he asked for conditions to ensure that the convex hulls appearing in Theorem 9.9 intersect in a set of fixed dimension. Suppose $0 \leq k \leq d$. Then $X \subset \mathbb{R}^d$ is said to be (r, k)-*divisible* if there exists a partition of X into r subsets such that the intersection of the convex hulls of the subsets is at least k-dimensional. Thus $(r, 0)$-divisibility is the same as r-divisibility. The following conjecture of Reay (1979a), which he established for $d = 2$, is apparently still open.

Conjecture 9.12. Each set of $2d(r - 1) + 2$ points in $\mathbb{R}^d$ is $(r, 1)$-divisible.

For $r = 2$, this was proved by Eckhoff (1976). If true in general, the result would be sharp. Reay adduces some evidence to support his conjecture. For example, he notes that any set of the above cardinality contains two distinct "r-divisible points".

It is clear that for $k > 1$, other conditions besides a lower bound on the cardinality of X must be imposed if X is to be (r, k)-divisible. (For example, X must not lie in a $(k - 1)$-flat.) This motivated Reay (1968) to define the notion of strong independence (and other types of independence). He proved the following.

Theorem 9.13. *Each strongly independent set of* $(r - 1)(d + 1) + k + 1$ *points in* $\mathbb{R}^d$ *is* (r, k)*-divisible.*

In fact, if X is such a set, and X is partitioned into the vertex sets of r simplices, then the intersection of the simplices is either empty or k-dimensional. This implies Theorem 9.13 in view of Tverberg's theorem and Carathéodory's Theorem 1.3. Since no strongly independent set of $(r - 1)(d + 1) + k$ points in $\mathbb{R}^d$ can be (r, k)-divisible, the above bound is clearly best possible. However, the proof of Theorem 9.13 does not use the full power of strong independence, and Reay wondered whether a weaker condition might suffice to yield the same conclusion. The following conjecture is suggested by the examples in his paper and appeared explicitly in Reay (1979b, 1982) and Eckhoff (1979).

Conjecture 9.14. Theorem 9.13 remains true with strong independence replaced by general position.

The case $k = d$ of this conjecture is particularly appealing. It asserts that any set of $r(d + 1)$ points in general position in $\mathbb{R}^d$ splits into r sets of $d + 1$ points each so

that the simplices spanned by those sets have a common *interior* point. While a full proof of Conjecture 9.14 is still lacking, the following special cases have been confirmed: $d = 2$ and $r = 2$, both by Reay (1968), $d = r = 3$ by Reay (1979b), and $d = 3$ by Roudneff (1990). See also Petersen (1972) for the case $r = 2$, and Roudneff (1988) for a topological extension in the plane.

It appears that the general position hypothesis in Conjecture 9.14 is still too strong. But to what extent it may be relaxed is not known. When $k = 0$, no restriction at all is needed. Sufficient conditions for $(2, k)$-divisibility which are weaker than general position were obtained by Doignon (1980) and Gusak (1982).

Our next topic is a proposed topological generalization of Tverberg's theorem. Let Δ^k stand for a k-simplex in $\mathbb{R}^k$. Bajmóczy and Bárány (1979) proved that for every continuous map $\sigma : \Delta^{d+1} \to \mathbb{R}^d$, there exist two disjoint faces F_1, F_2 of Δ^{d+1} such that $\sigma(F_1) \cap \sigma(F_2) \neq \emptyset$. This can be viewed as a continuous analogue of Radon's Theorem which says (essentially) that any linear map has the above property. The proof uses the Borsuk–Ulam theorem. In the same spirit, the following assertion (conjectured by Bajmóczy and Bárány) would be a continuous analogue of Tverberg's theorem.

Conjecture 9.15. Suppose $\sigma : \Delta^{N-1} \to \mathbb{R}^d$ is a continuous map, where $N = (r - 1)(d + 1) + 1$. Then there exist r pairwise disjoint faces of Δ^{N-1} whose images under σ have a common point.

When r is an odd prime, the conjecture was established by Bárány, Shlosman and Szücs (1981). The idea of their proof is basically the same as that used for $r = 2$ except that the sphere in the Borsuk–Ulam theorem is replaced by a certain CW complex $X_{d,r}$ and instead of the antipodal map on S^d one has a cyclic group of order r acting freely on $X_{d,r}$. Unfortunately, the method does not work for arbitrary r. A simpler proof of the Bárány–Shlosman–Szücs result was found by Sarkaria (1991).

To conclude, let us briefly comment on an attractive new development which is concerned with a "colored" version of Tverberg's theorem. Interest in such a result arose in connection with the enumeration of k-sets discussed in the preceding subsection (see Bárány, Füredi and Lovász 1990). Suppose $n \geq r(d + 1)$, and let X be a set of n points in $\mathbb{R}^d$. Assume that each point of X is assigned one of $d + 1$ given colors and that each color is used at least r times. Call a subset of X *multicolored* if no two of its points are assigned the same color. Under these circumstances, Bárány and Larman (1992) conjecture that X can be divided into r multicolored subsets whose convex hulls have a common point. They prove this in the following cases: $d = 1$, $r = 2$, and $d = 2$. The planar case was also (essentially) settled by Jaromczyk and Świątek (1990). The approach adopted in these papers is similar to the one used by Tverberg in his original proof of Theorem 9.9. Despite some important progress made by Živaljević and Vrećica (1993), the higher-dimensional part of the conjecture has yet to be confirmed.

10. The theorems of Carathéodory and Steinitz

10.1. Carathéodory's Theorem

Carathéodory's Theorem 1.3 is the fundamental dimensionality result in convexity and one of the cornerstones of combinatorial geometry. It asserts that the convex hull of a set $X \subset \mathbb{R}^d$ is the union of all simplices (i.e., convex hulls of affinely independent points) having vertices in X. Clearly, the number $d+1$ in this statement cannot be replaced by a smaller number.

Theorem 1.3 originated, somewhat hidden for modern eyes, with Carathéodory (1907, 1911). A straightforward proof is based on Radon's Theorem 1.2; this is essentially the proof given by Aleksandrov and Hopf (1935) and reproduced in Danzer, Grünbaum and Klee (1963) and numerous books and articles. Another proof proceeds by induction on the dimension of the space and uses the existence of supporting hyperplanes at boundary points of convex sets. This is Carathéodory's original proof (see Bonnesen and Fenchel 1934). Finally, Theorem 1.3 can be deduced from Helly's Theorem 1.1 and, in turn, implies the latter (see Eggleston 1958). Thus the three classical theorems in the title of our chapter are not only closely related but, in fact, each of them can be derived from each of the others. (In abstract convexity theory, this has been the main incentive to study the interrelationships between the three results in an axiomatic setting.) For other proofs of Carathéodory's Theorem, see the references given in Danzer, Grünbaum and Klee (1963). To that list, we add Scherk (1966), and Cook and Webster (1972).

Carathéodory's Theorem has important applications in convexity theory and related fields. These applications are often not stated as such and are certainly too numerous to be included here. A useful and much-quoted consequence of Theorem 1.3 is the fact that the convex hull of a compact set is compact; see Danzer, Grünbaum and Klee (1963). In many applications, Carathéodory's Theorem can step in for Helly's Theorem in order to reduce and simplify a geometric situation. In view of what was said above, this should come as no surprise. A typical example is the proof of Jung's theorem described in Danzer, Grünbaum and Klee (1963). Among the applications of Theorem 1.3 in approximation theory, we mention the short proof of Haar's theorem on the uniqueness of best approximations due to Pták (1958). For additional references, see Danzer, Grünbaum and Klee (1963).

Carathéodory's Theorem plays a role in the study of hull-forming set operations and related areas. For $X \subset \mathbb{R}^d$ and $1 \leq k \leq d+1$, define $\operatorname{conv}_k X = \bigcup \{\operatorname{conv} Y: Y \subset X, \operatorname{card} Y \leq k\}$. When suitably iterated, such operations (with $k \leq d$) will generate the convex hull of X. See Bonnice and Klee (1963) for details. Carathéodory's Theorem implies that $\operatorname{conv}_{d+1} X = \operatorname{conv} X$. A similar concept is that of the *k-core* of X, denoted $\operatorname{core}_k X$ and defined as $\bigcap \{\operatorname{conv} Y: Y \subset X, \operatorname{card}(X \setminus Y) \leq k\}$. Here k is a nonnegative integer. The "Carathéodory number for the k-core" is the smallest integer $c(d, k)$ such that for every $X \subset \mathbb{R}^d$, $\operatorname{core}_k X = \bigcup \{\operatorname{core}_k Y: Y \subset X, \operatorname{card} Y \leq c(d, k)\}$. By Theorem 1.3, $c(d, 0) = d+1$. It seems

extremely difficult to determine $c(d, k)$ in general. For interesting partial results, see Bárány and Perles (1990).

We turn now to some of the many other ramifications of Carathéodory's Theorem. Several authors have observed that the set Y which appears in this theorem may be assumed to include an arbitrarily prescribed point of X (see, e.g., Watson 1973). Tverberg (1979) proved that if X is the union of convex sets in $\mathbb{R}^d$, all meeting a fixed hyperplane, then $p \in \operatorname{conv} X$ implies $p \in \operatorname{conv} Y$ for some $Y \subset X$ with card $Y \leq d$. This becomes Theorem 1.3 (in $\mathbb{R}^{d-1}$) when the convex sets are single points. Another area of research is concerned with the question of how "thickly" the convex hull of X is covered with simplices having vertices in X, or *X-simplices*, as we shall say. Assume, for convenience, that X is the vertex set of a convex d-polytope in $\mathbb{R}^d$. For each k-face F of P there is a collection $\mathcal{S}$ of X-simplices such that $P = \operatorname{conv}_{k+1} \bigcup \mathcal{S}$ and $\dim(F \cap (\bigcap \mathcal{S})) = k$. Moreover, $\dim \bigcap \mathcal{S} = d$ when $k = d - 1$. This is a result of Klee (1980) which includes Carathéodory's Theorem as the case $k = 0$. Similar results were obtained by Motzkin (1967). When P has n vertices, then each point of P lies in at least $n - d$ X-simplices (for an extension, see Baker 1979). Bárány (1982) employed Tverberg's theorem 9.9 and Theorem 10.1 to show that *some* point of P is contained in

$$\binom{n}{d+1}(d+1)^{-d} + O(n^d)$$

different X-simplices. Compare also Boros and Füredi (1984) and Bárány, Füredi and Lovász (1990).

The paper of Bárány (1982) just mentioned established the following pretty "multiplied" version of Carathéodory's Theorem (which is used in its proof).

Theorem 10.1. *Let $X_1, \ldots, X_{d+1}$ be sets in $\mathbb{R}^d$, and suppose $p \in \operatorname{conv} X_i$ for $i = 1, \ldots, d+1$. Then there exist points $x_i \in X_i$ such that $p \in \operatorname{conv}\{x_1, \ldots, x_{d+1}\}$.*

This reduces to Theorem 1.3 when $X_1 = \cdots = X_{d+1}$. The number of X_i's may not be decreased. Again, one of the points x_i can be prescribed in advance. As a consequence of Theorem 10.1, Bárány derives the "multiplied" version of Helly's Theorem which appeared in Theorem 3.10. He notes that the two results are essentially equivalent.

Various attempts have been made to reduce the "Carathéodory number" $d + 1$ in Theorem 1.3. Of course, this requires that certain restrictions be placed on the set X. For instance, Fenchel (1929) proved that $d + 1$ may be replaced by d provided X is connected. This was generalized in two slightly different directions by Stoelinga and Bunt in the 1930s [see Danzer, Grünbaum and Klee (1963) for references], and Hanner and Rådström (1951). The latter authors define a subset of $\mathbb{R}^d$ to be *convexly connected* if there is no hyperplane $H \subset \mathbb{R}^d$ such that the set meets both the open halfspaces bounded by H but not H itself. Connected sets are convexly connected, but not vice versa. Each set decomposes uniquely into convexly connected components.

Theorem 10.2. *Suppose* $p \in \operatorname{conv} X$, *where* $X \subset \mathbb{R}^d$ *has at most* d *components or* X *is compact and has at most* d *convexly connected components. Then* $p \in \operatorname{conv} Y$ *for some* $Y \subset X$ *with* $\operatorname{card} Y \leq d$.

Examples show that the conclusion of Theorem 10.2 may fail if X is convexly connected and bounded *or* closed, but not compact. Even under strong additional hypotheses on X, the cardinality of Y cannot be reduced any further (Reay 1965a). On the other hand, Fenchel (1955) was able to obtain another sharpening of Carathéodory's Theorem by imposing certain symmetry conditions on X. His approach is described and generalized in different directions in Reay (1965a, section 5).

10.2. Steinitz's theorem

Shortly after the publication of Carathéodory's second paper, Steinitz (1913, 1914, 1916) established the following analogue of Theorem 1.3.

Theorem 10.3. *If* X *is a set in* $\mathbb{R}^d$ *and* $p \in \operatorname{int} \operatorname{conv} X$, *then* $p \in \operatorname{int} \operatorname{conv} Y$ *for some set* Y *consisting of at most* $2d$ *points of* X.

Actually, Theorem 10.3 was originally formulated in the dual setting of systems of convex cones having apex at p. The number $2d$ which appears above is the smallest for which the stated property holds. This is shown by a set of $2d$ points in $\mathbb{R}^d$, collinear in pairs with p and spanning $\mathbb{R}^d$. The following sharper form of Theorem 10.3 was also proved in Steinitz's paper: either $2d$ may be replaced by $2d-1$, or else Y consists of exactly $2d$ points collinear in pairs with p. Another strengthening in the spirit of Theorem 10.2 is due to Reay (1965a). It asserts that if X is connected, $d \geq 3$, and $p \notin X$, then the conclusion of Steinitz's theorem holds with $2d$ replaced by $2(d-1)$; moreover, each of the stated conditions is essential to the result.

In various forms and degrees of generality, Theorem 10.3 has been proved by Dines and McCoy, Robinson, Gustin, Gale, and many others [see Danzer, Grünbaum and Klee (1963) for precise references]. The easiest proof is by induction on d, using Carathéodory's Theorem and the existence of supporting hyperplanes at boundary points of convex sets. This proof is described in Danzer, Grünbaum and Klee (1963); a similar proof in Peterson (1972) seems to be incomplete. For applications of Steinitz's theorem to Kirchberger- and Helly-type results, see Rademacher and Schoenberg (1950) and Bárány, Katchalski and Pach (1982).

In view of the obvious similarities between Theorems 1.3 and 10.3, it is natural to inquire whether they might possess a common generalization. This is indeed the case, as was shown by Bonnice and Klee (1963). For $X \subset \mathbb{R}^d$ and $0 \leq k \leq d$, define the *k-interior* of X to be the set $\operatorname{int}_k X$ of all points p such that p is contained in the relative interior of some X-simplex of dimension k. Thus $\operatorname{int}_0 X = X$ and $\operatorname{int}_d X = \operatorname{int} X$. The result of Bonnice and Klee can be stated as follows.

Theorem 10.4. *If* $X \subset \mathbb{R}^d$ *and* $p \in \operatorname{int}_k \operatorname{conv} X$, *then* $p \in \operatorname{int}_k \operatorname{conv} Y$ *for some set* $Y \subset X$ *consisting of at most* $\max(2k, d+1)$ *points.*

This becomes Carathéodory's Theorem when $k = 0$, and Steinitz's theorem when $k = d$. Easy examples show that the result cannot be improved in general. Bonnice and Klee raised the problem of characterizing the cases in which $\max(2k, d+1)$ points are actually needed in Theorem 10.4. Despite some progress made by Reay (1965a), this problem remains unsolved.

There is an intimate relationship between the foregoing results and the theory of *positive sets and bases* in $\mathbb{R}^d$. For general information about such sets, see Reay (1965a), Hansen and Klee (1969), and Shephard (1971), and the references quoted in these papers. For example, Steinitz's theorem translates into the fact that a positive basis in $\mathbb{R}^d$ has at most $2d$ elements; if it has precisely $2d$ elements, then it consists of a linear basis of $\mathbb{R}^d$ together with a negative multiple of each of the basis points. According to Reay (1965a), any positive basis can be partitioned in a certain simple fashion. (For a shorter proof by means of positive diagrams, see Shephard 1971.) This partitioning was used by Reay (1965b) to give a a simplified proof of the Bonnice–Klee theorem, and by Bonnice and Reay (1969) to obtain a significant strengthening. The latter takes into account the maximum dimension of an X-simplex having p in its relative interior. Say this dimension is l. Then the bound on the size of Y in Theorem 10.4 can be replaced by $l+1$ when $k \leq l$, and $\max(2k-l+1, \min(2l+2, d+2))$ when $k > l$. These bounds are tight. Compare also Bonnice and Reay (1966). For further extensive ramifications of Carathéodory's and Steinitz's theorem, the reader is referred to Reay's memoir (Reay 1965a).

11. The theorems of Kirchberger and Krasnosel'skiĭ

Apart from Helly's Theorem itself, the two most popular Helly-type results in convexity are probably the theorems of Kirchberger (1903) and Krasnosel'skiĭ (1946). They are treated as applications of Helly's Theorem in Danzer, Grünbaum and Klee (1963) and have appeared as such (but also in their own right) in many text books. Both results are concerned with convexity properties of sets of points (rather than families of convex sets). This is the reason why we decided to include them in Part II of the present survey.

We begin with Kirchberger's theorem.

Theorem 11.1. *Let* X *and* Y *be finite sets in* $\mathbb{R}^d$. *If for every subset* S *of* $X \cup Y$ *consisting of at most* $d+2$ *points, the sets* $S \cap X$ *and* $S \cap Y$ *are strictly separable by a hyperplane, then* X *and* Y *are so separable.*

It is easy to see that the number $d+2$ in this assertion cannot be reduced (see, e.g., Eggleston 1958). However, the result continues to hold when X and Y are any two closed sets in $\mathbb{R}^d$ provided one of them is bounded (Rado 1952). A short

proof of Theorem 11.1 by means of Helly's Theorem was given by Rademacher and Schoenberg (1950) and is reproduced, for example, in Eggleston's book and Danzer, Grünbaum and Klee (1963). Kirchberger's own proof (which could not possibly employ Helly's Theorem) is more than 20 pages long. Another simple proof can be based on Radon's Theorem, more precisely, on the fact that any Radon partition in $\mathbb{R}^d$ extends a primitive Radon partition of $d+2$ or fewer points. Notice, on the other hand, that Carathéodory's Theorem may be viewed as the special case of Kirchberger's theorem in which one of X or Y reduces to a single point. For other proofs of Theorem 11.1, see Shimrat (1955), Watson (1973), Baker (1979), and Webster (1983).

Several authors have remarked that in order to ensure that X and Y are separable, not all the $(d+2)$-element subsets of $X \cup Y$ need be tested for local separability. It suffices, e.g., to test only those subsets which contain an arbitrarily chosen fixed point of $X \cup Y$ (see Watson 1973, Houle 1991). As noted by Rademacher and Schoenberg (1950), Kirchberger's theorem remains valid if "strictly separable" in its statement is replaced by "separable" and the number $d+2$ is increased to $2d+2$. This follows at once from Steinitz's theorem 10.3. An interesting generalization of Theorem 11.1 was recently proved by Houle (1991) who considers, instead of two sets X and Y, two finite families of subsets of $\mathbb{R}^d$. For extensions in a different direction, see Dol'nikov (1988).

Kirchberger's theorem deals with separation of sets by hyperplanes. One may ask whether similar results hold if the hyperplanes (or rather the corresponding halfspaces) are replaced by other geometric figures. This question was raised by Valentine (1964), and several answers were obtained in the literature. For example, Rennie (1970) considered separation of three sets in the plane by "wedges" instead of two sets by lines; the corresponding Helly number is then 6 rather than 4. Lay (1971, 1980) investigates separation by spherical surfaces and parallelotopes (in the obvious sense), the Helly number in these cases being $d+3$ and $d+1$. See also Lay (1972, 1982) and Houle (1991).

We turn now to Krasnosel'skiĭ's theorem. The following terminology is convenient. For points p and q of a set $X \subset \mathbb{R}^d$, we say that *p sees q via X*, or *q is visible from p in X*, provided the closed segment $[p, q]$ lies in X. The *kernel* of X, $\ker X$, is the set of all points p such that p sees each point of X via X. Thus X is convex (respectively starshaped) if and only if $\ker X = X$ ($\ker X \neq \emptyset$). Moreover, $\ker X$ is convex (possibly empty). The following is a slightly weaker form of Krasnosel'skiĭ's original theorem.

Theorem 11.2. *Suppose $X \subset \mathbb{R}^d$ is compact. If every $d+1$ or fewer points of X see a common point of X via X, then there is a point in X which sees all points of X via X, i.e., X is starshaped.*

Hence if for any three paintings in a gallery there is a place from which all three are visible, then there must be a place in the gallery from which all the paintings are visible. This nice (slightly inaccurate) interpretation of the planar case of Theorem 11.2 is due to Yaglom and Boltyanskiĭ (1951).

The proof of Krasnosel'skiĭ's theorem illustrates the first type of application of Helly's Theorem mentioned in section 2. Define, for each $p \in X$, the "star" S_p of all points q visible from p in X. The sets S_p are compact and, by hypothesis, every $d+1$ of them intersect. By Helly's Theorem, there is a point common to all sets conv S_p. This point is then shown to lie in all sets S_p (compare Danzer, Grünbaum and Klee 1963). Incidentally, the theorems of Helly and Krasnosel'skiĭ are essentially equivalent (Borwein 1977).

The number $d+1$ appearing above is the best possible. Also, the compactness assumption cannot be dropped (see Breen 1979). On the other hand, it suffices that any $d+1$ boundary points of X see a common point via X or any $d+1$ "ordinary" points in the sense of Krasnosel'skiĭ (see Valentine 1964). Further refinements and generalizations of Theorem 11.2 can be found in Molnár (1957b), Valentine (1964, 1967), Bollobás (1966), Foland and Marr (1966), and Chan (1988). For a "multiplied" version of Theorem 11.2 in the spirit of Theorem 3.10, see Kołodziejczyk (1984a).

The literature abounds in examples which may be viewed as variations on the theme of "visibility and starshape". For a short survey, see Breen (1985b). Not all of them are of Helly-type but usually Helly's Theorem plays a decisive role. One line of research has concentrated on the problem (posed by Valentine 1964) of finding sufficient conditions for ker X to be at least k-dimensional (k fixed). The case $k = 0$ is Krasnosel'skiĭ's theorem. Most of the results are too technical to be quoted here. It seems that satisfactory Helly-type statements can only be expected if strong visibility assumptions are imposed. We content ourselves with one example.

A point q is said to be *clearly visible* from a point p in X if there is a neighborhood N of q such that p sees each point of $N \cap X$ via X. This concept is due to Falconer (1977). Let $g(d, k)$ be Katchalski's function defined in connection with Theorem 3.5. Improving earlier results, Breen (1986b) established the following.

Theorem 11.3. *Suppose $X \subset \mathbb{R}^d$ is compact, $0 \leq k \leq d$. If every $g(d, k)$ or fewer points of* bd X *are clearly visible from a common k-dimensional subset of X, then* $\dim(\ker X) \geq k$.

The number $g(d, k)$ in this statement is always best possible. For $k = d$, this was already proved by Falconer who observed, moreover, that the word "clearly" can be omitted from Theorem 11.3 when X is smooth. Stronger versions of Theorem 11.3 hold in the plane (Breen 1982, 1985a).

In contrast to Krasnosel'skiĭ's theorem whose subject is "internal" visibility, our final Helly-type result is concerned with "external" visibility, i.e., visibility in the complement $\bar{X}$ of a set X. For $p \notin X$ and $q \in X$, we say that p *sees* q *via* $\bar{X}$ provided $X \cap [p, q] = \{q\}$. The following theorem is due to Valentine (1970); for a closely related result, see Breen (1989).

Theorem 11.4. *Suppose $X \subset \mathbb{R}^d$ is compact, smooth, and strictly convex, and let S*

be a subset of $\bar{X}$. *If every* $d+1$ *or fewer points of* S *see a common point of* bd X *via* $\bar{X}$, *then all of* S *sees a common point of* bd X *via* $\bar{X}$.

Again, the number $d+1$ in this assertion cannot be reduced. It is very likely, however, that the assumptions on X may be relaxed but to what extent is not known. Buchman and Valentine (1976) have constructed polytopes in $\mathbb{R}^d$, $d \geq 3$, for which no finite Helly number corresponding to $d+1$ above exists. Much more can be said in the plane. For a compact convex set $X \subset \mathbb{R}^2$, the Helly number is 3 when X has at most one extreme point, 4 when X has 2, 3, or 4 extreme points, and 5 in all other cases (Valentine 1970).

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CHAPTER 2.2

Problems in Discrete and Combinatorial Geometry

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Contents

Introduction 451
1. Sets of points and related topics 451
  1.1. Convex hulls of subsets 453
    1.1.1. Related results and generalizations 455
  1.2. The set of distances 457
    1.2.1. Few-distance sets 457
    1.2.2. Regularly distributed frequencies 458
  1.3. The set of directions 459
    1.3.1. Related results and generalizations 460
2. Finite tilings of sets 461
  2.1. Tilings by similar tiles: Squares 461
    2.1.1. Related problems and generalizations 463
  2.2. Tilings by congruent tiles 465
    2.2.1. Self-similarity 465
    2.2.2. Fault-free tilings of rectangles 466
  2.3. Tilings by polygons: k-gons 466
  2.4. Tilings by polygons: Regular polygons 467
  2.5. Tilings by smaller tiles: Diameter 468
    2.5.1. Related problems and generalizations 470
  2.6. Tilings by smaller tiles: Reduced copies 470
    2.6.1. Related problems 472
  2.7. Size restrictions: Equal area 472
    2.7.1. Tilings by triangles 473
    2.7.2. Tilings by rectangles 473
    2.7.3. Related questions 473
References 474

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Introduction

For most mathematicians, and for many laymen as well, combinatorial and discrete problems in geometry have a special appeal. They can easily be stated without formalism and without specialized terminology, and quite often the answer seems to be obvious, at least intuitively. But first impressions may be completely wrong! Many seemingly simple questions present challenging opportunities for mathematical research. They may be extremely difficult to solve and sometimes they resist every attack. But even when a problem can be solved by elementary means the solution usually requires some original idea, and thus provides a welcome exercise, or even entertainment, for the mathematical mind. In this respect combinatorial geometry closely resembles another field which, however, has a much longer tradition, namely elementary number theory. In fact, methods of analytic number theory are often used to treat a geometric problem.

No mathematical discipline is independent from the rest of mathematics. But for combinatorial and discrete problems in the context of convexity such interdependencies are abundant, and almost every question considered in convexity has its discrete (or combinatorial) aspects. This fact explains the great diversity of questions that can be classified as combinatorial or discrete, and the equally large diversity of methods used to deal with them. Therefore, in this chapter, we cannot give a unified exposition but only an overview of the subject. Moreover, this overview is necessarily far from complete. Nevertheless we hope that it suffices to show the character of this fascinating part of modern mathematics and its interaction both with convexity and with geometry as a whole.

General references

A standard reference on combinatorial geometry is the survey paper by Danzer, Grünbaum and Klee (1963). The choice of topics presented in this influential paper is rather different from that found below though most of them could (or even should) have been included here, as well. It also provides an extensive (and then virtually complete) bibliography. A quite recent reference is Croft, Falconer and Guy (1991). Concentrating on unsolved problems in intuitive geometry this book also presents an overview of the subject. The topics include those treated in this chapter and, of course, many more that are not mentioned here, but would also have fitted into this chapter.

1. Sets of points and related topics

In this section we shall discuss combinatorial properties of finite sets of points. Two typical examples of this kind of problems are 2-distance sets and the famous problem of Erdős-Szekeres:

A 2-distance set is a set such that the distance between any two of its points may take only one of two values. A set with this property cannot be arbitrarily

large. Therefore it is of interest to find examples of maximal 2-distance sets or, if this is not possible, at least to give (lower and upper) bounds for the cardinality of such sets.

The problem of Erdős–Szekeres is concerned with subsets which form a convex polygon: What is the smallest integer n such that every set of n points necessarily contains k points which are the vertices of a convex k-gon? Though put differently this is again a problem of the same type, i.e., that of finding a maximal set. In this case sets are considered if they do *not* contain the vertices of a convex k-gon. On the other hand, instead of looking for maximal 2-distance sets one could equivalently ask for the smallest integer n such that among the pairs of points of any set of n points there necessarily occur at least three with mutually distinct distances.

Both these problems can be seen as special cases of a more general task, namely that of counting the number of occurrences of subsets (or configurations) with specific properties. How many distinct values are taken by the distances between the points of a set? How many convex k-gons are formed by the points of a set?

There exists an extensive literature on this field. Therefore only a few selected problems can be presented here, and they have to be representative for all of them. A good source for information on this subject are the papers by Erdős on combinatorial problems from which only a selection is cited in the bibliography. There can be found details on the origin and on the progress of all these – and many related – questions, though in a non-systematic way.

Terminology and notation. Up to now no universally accepted mathematical language for speaking of these questions has evolved (as it is also the case for many other parts of discrete geometry). We shall employ the following – partially non-standard – terminology and notational conventions:

$\mathcal{P}$ denotes an n-point subset of the d-dimensional space $\mathbb{E}^d$. If f denotes some function defined on some set $\mathcal{S}$ derived from $\mathcal{P}$, then $F(\mathcal{P})$ is used for the extreme value (usually the maximum) of f taken over $\mathcal{S}$. Moreover, $\underline{F}(n)$ and $\overline{F}(n)$ denote the minimum and maximum, respectively, of $F(\mathcal{P})$ taken over all sets $\mathcal{P}$ ($\#\mathcal{P} = n$). The "dual" function associated with $\underline{F}$ (see the discussion in the preceding paragraphs) is denoted by $N_{[F=k]}$. It is defined as the maximum size (= cardinality) that is possible for a set $\mathcal{P}$ for which F takes the value k, i.e., $N_{[F=k]} = \max\{n \mid \underline{F}(n) = k\}$. Related functions are distinguished by superscripts (e.g., C^*), and subscripts (e.g., F_d) are used to indicate dependence on the dimension d of the set.

When point-sets $\mathcal{P}$ (or, more generally, some other geometric configurations) with a certain property are studied usually one or more integer parameters are involved (e.g., the number of points, the value taken by some function F as used in the preceding paragraph, etc.). If p is such a parameter then its *spectrum* is the set of all the values taken by it (over the family of configurations having the property considered).

The *dimension* of a finite set is the dimension of the *flat* spanned by these points (= smallest affine (hyper)space containing the points). A set of points is (and the points of a set are) *convexly independent* (or *convex*, for short, if no confusion is

possible) if each of its points is a vertex of its convex hull, and it is *weakly convexly independent* if none of its points lies in the interior of its convex hull.

When investigating some property of point-sets it is often convenient to concentrate on sets whose points are in a "general position", i.e., sets where certain special configurations of subsets are avoided. In this way it is often possible to avoid unnecessary (or unwanted) complications in the proof or in the statement of a theorem. (And sometimes, of course, such a restriction may even be essential for the validity of the proof or of the theorem.) The most frequent among these assumptions are that the set is *linefree* (no line passes through three points of the set) or *circlefree* (no circle passes through four points of the set), or (similarly, for a set in space) that it is *planefree* or *spherefree*. A set of points is *general of order* r if every subset of $r+2$ points has dimension $r+1$ (so that, for instance, general of order 1 means linefree). A set in $\mathbb{E}^d$ is *general* if it is general of order $d-1$.

1.1. Convex hulls of subsets

The convex hull $\operatorname{conv} S$ of each subset S of a 2-dimensional set $\mathcal{P}$ is (since S is finite) a (convex) polygon with $c(S)$ vertices for some $c(S) \leqslant \#S$. In this section we are interested in the maximal value of this number, i.e., $C(\mathcal{P}) = \max c(S)$, the maximum taken over all subsets of S. Of course, $C(\mathcal{P})$ could equivalently be defined as the maximum cardinality among the convexly independent subsets and in order to avoid points lying on the edges of $\operatorname{conv} S$, i.e., weakly independent subsets, it is usual to assume that $\mathcal{P}$ is linefree.

The object is, as already described in the introduction, to find lower bounds for $C(\mathcal{P})$ (in terms of n), or equivalently, to find upper bounds for $N_{[C=k]}$.

This question has first been raised in 1935 in a now classical paper by Erdős and Szekeres (1935) and therefore is often called the *problem of Erdős–Szekeres*. It was suggested by the following observation of Esther Klein (later Mrs. Szekeres): *From five points of the plane of which no three lie on the same straight line it is always possible to select four points determining a convex quadrilateral.*

The (easy) proof is given as follows: If the least convex polygon which encloses the points is a quadrilateral or a pentagon the theorem is trivial. Let therefore the enclosing polygon be a triangle ABC. Then the two remaining points D and E are inside ABC. Two of the given points (say A and C) must lie on the same side of the connecting straight line $\overline{DE}$. Then it is clear that $AEDC$ [– or $ADEC$ –] is a convex quadrilateral.

In the sequel Erdős and Szekeres show that $N_{[C=k]}$ is finite for every k. In fact, they give two proofs of this assertion, both of which are still of interest.

The first proof depends on Ramsey's theorem (see Ramsey 1930) that had been published some years before, but was still little known. (By the way, Ramsey's theorem is also proved in the paper, and was, in fact, independently rediscovered by the authors, see Szekeres 1973.) The argument is the following:

The 4-point subsets of $\mathcal{P}$ can be divided into two (disjoint) classes: the convex ones, whose convex hull has four vertices, and the non-convex ones, whose convex hull is a triangle. By Ramsey's theorem, if n is sufficiently large (depending only

on k) there is a k-point subset S of $\mathcal{P}$ such that all its 4-point subsets belong to the same class, i.e., they are either all convex or all non-convex. The second alternative is not possible for $k \geqslant 5$ since (as we have seen already) every 5-point set contains a convex 4-point set. Therefore all the 4-point subsets of S are convex, and this is equivalent to S being convex. Due to the nature of Ramsey's theorem, the bound obtained by this method is very weak, but this proof has the advantage that it immediately generalizes to higher dimensions.

The second proof – assuming $d = 2$ – yields a much better bound. It is obtained by explicitly calculating a closely related number: For some fixed coordinate system in the plane, consider the minimum number $a(i,j)$ such that every set of $a(i,j)$ points contains a convex polygonal arc of i points or a concave arc of j points. The function a is determined by the recursion

$$a(3,i) = a(i,3) = i \quad \text{and} \quad a(i,j) = a(i-1,j) + a(i,j-1) - 1,$$

and from the value

$$a(k,k) = \binom{2k-4}{k-2} + 1$$

of this function bounds for the original problem can be derived:

Obviously, it is $N_2[C=(k-1)] < a(k,k)$, since any such arc of k points determines a convex k-gon, too. On the other hand, since every convex $(2k-1)$-gon contains a convex or a concave arc of k points, there must exist a set of $a(k,k) - 1$ points that does not contain a convex $(2k-1)$-gon, i.e., $a(k,k) - 1 \leqslant N_2[C=(2k-2)]$.

This paper was one of the first in two now flourishing fields: in Ramsey theory and in combinatoric geometry. But as far as the original problem is concerned almost no progress has been made. The following theorem summarizes the known facts:

Theorem. $N_2[C=2] = 2$, $N_2[C=3] = 4$, $N_2[C=4] = 8$, *and*

$$2^{k-1} \leqslant N_2[C=k] \leqslant \binom{2k-2}{k-1}.$$

Remarks. Alternative proofs for the existence of the upper bound are due to Johnson (1986), and Tarsy (see Graham, Rothschild and Spencer 1980, p. 25). They both use Ramsey's theorem. The proposition that among nine points five belong to a convex quintuple is attributed to Makai, but the first published proof is found in Kalbfleisch, Kalbfleisch and Stanton (1970); for another proof see Bonnice (1974). The lower bound is due to Erdős and Szekeres (1960/61) [but see Kalbfleisch, Kalbfleisch and Stanton (1970) and Erdős (1981b) for a remark on inaccuracies]. The conjecture, however, that this lower bound is indeed the true value of N_2 dates back to the 1935 paper. A detailed exposition of the complete result as stated above can be found in Lovász (1979, §14, Problems 30 and 31). Goodman and Pollack

(1981) show that the problem can be interpreted as a combinatorial problem on sequences of permutations, see also Edelsbrunner (1987, section 2.5.)

1.1.1. Related results and generalizations

Algorithmic aspects. Chvátal and Klincsek (1980) investigate the computational aspects of the problem and show that the (convex or concave) arc used in the second proof can be found by an algorithm that requires at most $O(n^3)$ of time and at most $O(n^2)$ of storage resources. They also present some related experimental data. Edelsbrunner (1987, sections 12.1, and 12.7, Problems 12.1–3), discusses this algorithm, some improvements, and variants (empty convex polygons, see below). Compare also Edelsbrunner (1987, section 2.6, Problems 2.11 and 2.12) for bounds on the number of convex – and of empty convex – polygons, and the literature cited in Edelsbrunner (1987, sections 2.7 and 12.8). A special case of the latter – lower bounds on the number of convex quadrilaterals – is treated by Watson (1969/70).

Restrictions on the set $\mathcal{P}$. The gap between the lower and the upper bound can be decreased if attention is restricted to point sets for which the distances occurring between the points do not differ too much. More precisely, let

$$q(\mathcal{P}) = \max\{d(p,p')/d(q,q') \mid p,p',q,q' \in \mathcal{P}\}$$

and consider $\underline{C}^{(\alpha)}(n) = \min C(\mathcal{P})$ where the minimum is taken over all n-point sets satisfying $q(\mathcal{P}) \leqslant \alpha\sqrt{n}$. The following simple example illustrates the effect of a restriction of this type. If $\mathcal{P}$ is a set of four points in the plane and if $q(\mathcal{P}) < \sqrt{3}$, then these four points are the vertices of a convex quadruple. The following bounds are known (Alon, Katchalski and Pulleyblank 1989):

$$c_1\sqrt[4]{n} \leqslant c(\mathcal{P}) \leqslant c_2\sqrt{n}$$

or, equivalently,

$$c_1' k^2 \leqslant N_{[C_2^{(\alpha)}=k]} \leqslant c_2' k^4$$

for some positive constants c_1, c_2, c_1', c_2' depending on α. (Note that $N_{[C^\alpha=k]} \leqslant N_{[C=k]}$ since the maximum is taken over fewer sets $\mathcal{P}$.) If $\alpha \leqslant 4$ then the lower bound on N can be improved:

$$k^{3-\varepsilon} \leqslant N_{2[C_2^{(4)}=k]}$$

holds for every $\varepsilon > 0$ and sufficiently large $k > k(\varepsilon)$.

Empty polygons. An interesting variant of the Erdős–Szekeres problem was introduced by Erdős (1979). He suggested to consider only those subsets S of $\mathcal{P}$ that determine *empty polygons*, i.e., polygons S for which the interior of conv S

does not contain any point of $\mathcal{P}$: What can be said about $C^*(\mathcal{P}) = \max c(S)$ (the maximum taken over all empty polygons determined by $\mathcal{P}$)?

It turns out that the corresponding number $\underline{C}^*(n)$ exhibits quite a different behaviour from $\underline{C}(n)$: $N_{2[C^*=2]} = 2$ and $N_{2[C^*=3]} = 4$ are the same, but $N_{2[C^*=4]} = 9$ [obtained independently by Harborth (1978) and Morris; a simple proof that this number is bounded by $\binom{12}{6}$ is due to Ehrenfeucht (see Erdős 1979)], whereas $N_{2[C^*=6]} = \infty$, i.e., there are arbitrarily large (even infinite) sets which do not contain any empty 7-gon (see Horton 1984). It is not known whether $N_{2[C^*=5]}$ is finite or not.

See Bárány and Füredi (1987), Katchalski and Meir (1988), and Bialostocki, Dierker and Voxman (1991) for some related results.

Sets of convex sets. Bisztriczky and G. Fejes Tóth (1989a,b) transfer the problem to a more general setting. They replace the point-set $\mathcal{P}$ by a family of *ovals* (i.e., compact convex set) in the plane. A (sub)family of ovals is called *convexly independent* if none of its members is contained in the convex hull of the union of the remaining ones. Moreover, a family of ovals is *k-independent* if every k of them form a convexly independent subfamily. Obviously, a 3-independent family can be considered as the equivalent to a linefree point-set (and accordingly, three pairwise disjoint ovals are called *collinear*, if they are not independent, i.e., if one of them is contained in the convex hull of the other two). In this framework they obtain the following result analogous to $N_{[C=k]} < \infty$ and $N_{[C=4]} = 8$:

Theorem. *Every sufficiently large (i.e., with at least $n(k)$ elements) 3-independent family of pairwise disjoint ovals in the plane contains a convexly independent subfamily of k ovals. In particular, the following values of $n(k)$ are best possible: $n(4) = 5$ and $n(5) = 9$.*

The proof of this theorem uses Ramsey's theorem, therefore only very weak bounds on $n(k)$ are obtained. But much better bounds can be obtained if $\mathcal{P}$ is restricted to k-independent families ($k \geqslant 4$) (Bisztriczky and Fejes Tóth 1990).

Abstract convexity. A *box* in $\mathbb{R}^d$ is a product of d intervals of the real line. Since every intersection of boxes is a box, boxes define a convexity structure. Consider three points x, y, and z in $\mathbb{R}^d$: x is said to lie *between* y and z if x is contained in every box containing y and z. In this setting Cochand and Duchet (1983) prove the following theorem:

Theorem. *Among $2^{2^{d-1}}+1$ points in $\mathbb{R}^d$, there are necessarily three such that one of them lies between the other two. This is best possible since a set of $2^{2^{d-1}}$ independent points can be effectively constructed.*

This result is closely related to the following theorem on monotonic sequences contained and used in Erdős–Szekeres (1935) which is also a quite famous theorem and of independent interest [see Seidenberg (1959) for a short proof]:

Theorem. *A sequence of $pq+1$ terms $(p, q \geqslant 0)$ in a completely ordered set contains an increasing subsequence of $p+1$ terms, or a decreasing subset of $q+1$ terms.*

Another generalization is investigated by Korte and Lovász (1984). They consider a setting of abstract convexity called anti-exchange or *shelling structures* and prove that in every sufficiently large shelling structure (with some additional properties) there exists a "free set of size n" that corresponds to the convex polygon of the original theorem.

1.2. The set of distances

Associated with any finite point-set $\mathcal{P}$ is a finite subset of $\mathbb{R}$, the set

$$\mathcal{D} = \{d(P, Q) \mid P, Q \in \mathbb{E}^2, P \neq Q\}$$

of values taken by the distances between distinct points of the set $\mathcal{P}$. If among these values no one occurs twice, then this set has $n(n-1)/2$ elements. This can be considered as the general case. The other extreme case is represented by the set of the vertices of a regular simplex — in this case $\mathcal{D}$ is a singleton. Naturally, the more interesting case is the case where the number $D(P) = \#\mathcal{D}(\mathcal{P})$ of distinct distances is small.

1.2.1. Few-distance sets

Obviously, $N_{[D=1]} = d+1$ where the regular $(d+1)$-simplices in d-space are the only extreme sets. Therefore, the first non-trivial case are the 2-distance sets. Examples of such sets are

for $d = 1$: two points and their midpoint;

for $d = 2$: the five vertices of a regular pentagon;

for $d = 3$: the six vertices of an octahedron, or a regular triangular prism, or any six vertices of an icosahedron, provided there is no antipodal pair among them.

An example for arbitrary dimension d can be constructed in $\mathbb{R}^{d+1}$: The points $(x_0, \ldots, x_d)$ with coordinates such that $x_i = 1$ for precisely two indices, and $x_i = 0$ else, lie in a d-dimensional subspace (and on a sphere) and clearly form a 2-distance set. Another example of the same size is formed by the midpoints of the edges of a regular d-simplex.

These examples yield the lower bound of the following estimate:

$$d(d+1)/2 \leqslant N_{d[D=2]} \leqslant (d+1)(d+2)/2.$$

The upper bound is due to Blokhuis (1984b) and was obtained by modifying earlier results by Larman, Rogers and Seidel (1977).

Exact values are known only for $N_2 = 5$ (Kelly 1947) and $N_3 = 6$ (Croft 1962) (see the examples above). If the points of $\mathcal{P}$ are required to lie on a sphere then the upper bound can be sharpened to $d(d+3)/2$. This is best possible for some dimensions ($d = 2, 6, 22$). (Delsarte, Goethals and Seidel 1977).

For k-distance sets an analogous upper bound holds: $N_{d[D=k]} \leqslant \binom{d+k}{d}$ (Bannai and Bannai 1981, and Bannai, Bannai and Stanton 1983). And for point-sets in the plane the asymptotic behaviour of this function has also been studied: the initial estimate of $c_1 n^{1/2} < \underline{D} < c_2 n/(\log n)^{1/2}$ (Erdős 1946), or equivalently $N_{2[D=k]} < ck^2$, was improved in several steps (by Moser 1952, by Chung 1984, by Beck, and by Trotter and Szemerédi, see Erdős 1986), and now the best known bound is $N_{2[D=k]} < ck^{5/4}$, which is, however, still quite far from the conjectured bound $ck^{1+\varepsilon}$. For the related function $\underline{D}^*(n)$ obtained when the minimum is taken over all *general* sets only (rather than over *all* n-point sets) the estimate is $\underline{D}^*(n) \leqslant 4n^{\log 3/\log 2} = 4n^{1.585\ldots}$ (Erdős, Hickerson and Pach 1989).

Few-distance sets are studied by Seidel and Blokhuis in more general settings ($\mathbb{E}^d$, $\mathbb{H}^d$, and Lorentz spaces $\mathbb{R}^{p,q}$) (see the survey Seidel 1981), as well as in connection with related questions, e.g., equiangular lines, 2-angle sets (Blokhuis and Seidel 1986, Blokhuis 1984a).

1.2.2. Regularly distributed frequencies

A very special, but interesting, question was raised by Erdős (1981a, 1983, 1985b). He noticed the following curious property exhibited by some point sets.

Consider $n+1$ points P_i, $i = 0, \ldots, n$, situated regularly on a line (with constant distances between neighbours, i.e., $d(P_i, P_{i+1}) = s$, $i = 0, \ldots, n-1$. Then just n distinct distances occur. One of them ($d(P_0, P_n) = ns$, the largest) occurs just once, another one ($d(P_0, P_{n-1}) = d(P_1, P_n) = (n-1)s$) occurs just twice, and so on. The shortest distance, s, occurs n times. In other words, the frequencies with which the elements of $\mathfrak{D}$ occur are distributed with best possible regularity: They are just the numbers from 1 to n.

Erdős noted that another infinite family of points has the same property. Take n points P_i, $i = 1, \ldots, n$, situated regularly on a circle ($d(P_i, P_{i+1}) = s, i = 1, \ldots, n-1$), together with the centre P_0 of the circle. Assume further, that the radius r of the circle does not occur as the distance of two of the points P_i, $i = 0, \ldots, n$, as it is the case if, for instance, $d(P_1, P_n) = s_1 \leqslant r$. Then, again, one distance (in this case s_1) occurs once, another one ($d(P_1, P_{n-1}) = d(P_2, P_n)$) occurs twice, and so on. The distance s occurs $(n-1)$ times, and the radius r occurs n times.

A third infinite family with this property is the following as discovered by Hills (see Kelly 1987). In this case the points are distributed on two parallel lines, as in the following typical set of $2k$ or $2k+1$ points in $\mathbb{R}^2$: $P_i = (2i, 0)$, $i = 1, \ldots, k$ or $i = 0, \ldots, k$, and $Q_i = (2i-1, 1)$, $i = 1, \ldots, k$. (Note that the distance between two points on different lines does not occur among the distances between points on the same line.)

In view of these examples it is natural to ask: How large can such a set be if its points are in general position (in the plane: linefree and circlefree; in space: planefree, spherefree)? Only a few examples of this type are known: A 4-point example are the vertices of an isosceles triangle together with the centre of the circumscribed circle. The following 5-point set is due to Pomerance (see Palásti 1987). The vertices A, B, and C, of an equilateral triangle, its centre D, and one of the points E determined by $d(A, E) = d(D, E)$ and $d(B, E) = d(A, B)$. A sixth

point can be added to this set by $d(B,E) = d(D,E)$ and $d(C,E) = d(A,B)$, and further examples, including 7-point sets and an 8-point set, have been constructed by Palásti (1987, 1989a,b). Palásti (1989c) also describes sets in space (with 4, 5, and 6 points).

1.3. The set of directions

Any two points of a finite point-set $\mathcal{P}$ determine a straight line, and therefore a direction (= parallel class that is represented, for instance, by a vector). Thus, to any $\mathcal{P}$ there is a finite set, the set

$$\mathcal{S} = \{s(P,Q) \mid P,Q \in \mathbb{E}^2, P \neq Q\}$$

of slopes of these lines.

Trivially, for the number $S(\mathcal{P}) = \#\mathcal{S}(\mathcal{P})$ of distinct slopes, we have

$$1 \leqslant S(\mathcal{P}) \leqslant n(n-1)/2.$$

Clearly, this minimum is attained (if and) only if $\mathcal{P}$ is a 1-dimensional set.

The study of the set $\mathcal{S}$ of slopes was initiated by Scott (1970) (with an initial lower bound of order $\sqrt{2n}$) who already then had conjectured the correct result: n points in the plane determine at least $n-1$ distinct directions. More precisely, the following is true: If $\dim \mathcal{P} = 2$, then $2[n/2] \leqslant S(\mathcal{P})$, i.e., $n-1 \leqslant S(\mathcal{P})$ when n is odd, and (stronger) $n = 2k \leqslant S(\mathcal{P})$ when n is even, and these bounds are best possible (see the critical configurations described below). Therefore, $\underline{S}_2(n) = 2[n/2]$ or $N_{2[S\leqslant k]} = 2[k/2] + 1$.

After intermediate results $[S(\mathcal{P}) \geqslant (n-1)/2]$ obtained by Burton and Purdy (1979), the lower bounds in this theorem were finally proved by Ungar (1982). He applied successfully a new general combinatorial approach proposed by Goodman and Pollack (1981) (the *allowable sequences* introduced in Goodman and Pollack 1980).

Naturally, in connection with slopes, the interest concentrates on the classification of slope-minimal configurations, i.e., point-sets which determine as few directions as possible. A set $\mathcal{P}$ is called *slope-critical* if $s(\mathcal{P}) = n-1$ (and *near-critical* if $s(\mathcal{P}) = n$).

The following four (possibly all) infinite families of slope critical sets are known:

(a) *Centred 2t-gons.*

A regular $2t$-gon together with the centre

$$(0,0), \qquad (\sin k\pi/2t, \cos k\pi/2t):$$

$2t+1$ points, $t \geqslant 2$, $k = 1,\ldots,2t$. (Remark: A $2t$-gon is near-critical.)

(b) *Bipencil.*

A centrally symmetric configuration consisting of $2k-1$ points on a line and two more points, namely the points

$$(1,1), \qquad (-1,-1), \qquad (k,0):$$

$2t+1$ points, $t \geqslant 3$, $k = -t-1, \ldots, t-1$.

(c) *Tricolumnar array.*
The points

$$(\pm 1, k), \qquad (0, (t \pm 2i)/2), \qquad (0, (t \pm (2j+1))/2):$$

$2(t+r+s)+1$ points, $k = 0, \ldots, t$, $i = 1, \ldots, r$, $j = 0, \ldots, s-1$.

(d) *Exponential cross.*
The points

$$(0,0), \qquad (\pm\lambda^i, 0), \qquad (0, \pm\lambda^j):$$

$2(s+t)+3$ points, $i = 0, \ldots, s$ and $j = 0, \ldots, t$.

In addition to these four families 102 sporadic critical configurations are known (Jamison and Hill 1983). The list is complete for $n \leqslant 9$. The largest known sporadic configuration has 49 points ((i,j), $i = 0, \ldots, 6$, $j = 0, \ldots, 6$) in a square lattice. All but four of them are centrally symmetric, and all but four are bilaterally symmetric.

1.3.1. Related results and generalizations

Restrictions on $\mathcal{P}$. Special cases of the slope problem are obtained by restricting the sets $\mathcal{P}$ considered.

If a plane set $\mathcal{P}$ is linefree then $s(\mathcal{P}) \geqslant n$, and $s(\mathcal{P}) = n$ if and only if $\mathcal{P}$ consists of the vertices of a regular n-gon (Jamison).

If a plane set $\mathcal{P}$ is weakly convexly independent then $s(\mathcal{P}) \geqslant n$ (Scott 1970), and the sets for which $s(\mathcal{P}) = n$ holds have been classified (Jamison 1984).

The spectrum. In contrast to the situation concerning related problems (e.g., the line problem), for the slope problem the spectrum can be completely determined: for any arbitrary numbers n and m, $2 \leqslant n \leqslant m \leqslant n(n-1)/2$ there is a set $\mathcal{P}$ such that $\#\mathcal{P} = n$ and $s(\mathcal{P}) = m$. In addition, one may require that the set $\mathcal{P}$ is either linefree or a set of lattice points, or – provided that $2n-3 \leqslant m$ – even both (Jamison 1985).

Generalizations. Various generalizations (to matroids: Cordovil 1983, to higher dimensions, to the complex plane) and variants (e.g., for ordinary lines) have been considered (see Jamison 1985 for an overview):

For higher dimensions the following result holds (Jamison 1988):

Theorem. *Every d-dimensional point-set $\mathcal{P}$ determines at least $n + d(d+1)/2 - 4$ distinct directions.*

Jamison also considers an abstract analogue, namely edge-coloured complete graphs (where colours replace directions).

Direction trees. Using the results on slopes, Jamison (1987) has proved a related graph-theoretical result: The points of any 2-dimensional set can be connected by

edges to obtain a tree such that no two of these $(n-1)$ edges are parallel (direction tree).

2. Finite tilings of sets

It is intuitively clear what is meant by "cutting a geometric object into a number of parts". Many different terms are used for the result of this process: division, subdivision, dissection, partition, tiling, ... They all describe the same underlying idea that essentially coincides with the concept of a tiling (see chapter 3.5). We shall therefore use the term (finite) tiling (of) a (bounded) geometric object (a convex body or a closed connected domain). Consider some bounded closed domain D in $\mathbb{E}^d$. A finite tiling of D is a covering by a family $\mathcal{T} = \{T_i \mid i = 1, \ldots, n\}$ of closed topological balls, the tiles T_i, such that the interiors of the tiles form a packing. (This means: (1) each tile is homeomorphic to the closed unit ball, (2) the tiles are subsets of D, (3) each point of D belongs to some tile, and (4) no point of D lies in the interior of more than one tile.)

It is, of course, practically impossible to deal with finite tilings in full generality. To obtain interesting (and tractable) problems, one has to pose certain restrictions on the objects and/or the tiles considered, for instance by prescribing the shape or the size or the number of the tiles.

Restricting the size of the tiles leads to the following question: How many tiles are needed if one wants to tile D by sets that are (in some sense) smaller than D? The answer depends, of course, on the interpretation of the notion of size. If "size" means the measure (area, volume) of the sets involved then, obviously, two tiles are always sufficient. On the other hand, if the "size" of a set is its diameter, then the problem is extremely difficult and still not completely solved. It is equivalent to a famous problem of geometry (Borsuk's Problem).

If the shape of the tiles is important then it is quite natural to assume that both the tiles and the set D are polyhedra. Furthermore, usually it is assumed that at least the tiles are convex. The most fundamental case of this type is that of a *triangulation*, i.e., the tiling of a polygon by triangles (in the plane) or (in general) the tiling of a polyhedron by simplices. This case is out of the scope of this chapter – it is, however, extensively studied in the theory of polyhedra, and in computational geometry.

In combinatorial geometry most investigations in this area are confined to the plane, and often it is assumed that the tiles are pairwise congruent or, at least, similar. A typical example is (non-trivial) tilings of a rectangle (or square) by squares.

2.1. Tilings by similar tiles: Squares

The most basic shape, next to the triangle, are quadrilaterals, and among the quadrilaterals squares are special since any two squares are similar. It is, therefore, not surprising that tilings by squares have been studied for a long time. The first relevant result is due to Dehn (1903). He showed the following theorem:

Theorem. *If a rectangle can be tiled by squares, then the ratio $a : b$ of its sides can be expressed using integers a, b.*

Furthermore, the sides of the squares are rationally dependent (*commensurable*), as well. (Therefore, when studying tilings by squares one may restrict attention to squares whose sides have integer length.) Clearly, this necessary condition is also sufficient: Any such rectangle can trivially be tiled by ab congruent squares.

In view of this result it is not the mere existence of a tiling that is of interest, but the existence of a non-trivial tiling (with some specific property): A tiling of a rectangle (by squares) is called *perfect* if the tiles are pairwise incongruent. It is called *simple* if no proper subfamily of $\mathcal{T}$ (of two or more tiles) forms the tiling of a rectangle – otherwise it is called *compound*. "Squared rectangles (squares)", i.e., rectangles together with a tiling, are accordingly called perfect, simple, or compound, rectangles, respectively. The order of a squared rectangle is the number of the tiles used.

The search for squared rectangles and squares and, in particular, for a squared square of lowest order has a long history. It seems that nearly perfect rectangles first appeared in several geometric puzzles by Sam Lloyd and H.E. Dudeney. The existence of perfect rectangles, however, was established by Moroń (1925) in answer to a problem posed by Ruziewicz. He found two examples, one of order 9 (33×32), and one of order 10 (65×47). In an attempt to construct a perfect square Abe (1931) produced over 600 simple perfect rectangles, but no square. It was only in 1939 when the first example of a squared square was published by Sprague (Jaremkewycz, Mahrenholz and Sprague 1937). It was a compound square of order 55 (side 4205) and contained Moroń's rectangles as subsets. (See also Sprague 1939). Short time afterwards two further examples were published by Stone (1940). They had been found independently and were the result of a joint research effort that had been started about three years earlier. The approach of this group (Brooks et al. 1940/1987, a now classical paper of combinatorial theory), allowed a systematic search for squared rectangles.

They observed that to every squared rectangle there corresponds a flow in a network of a particular type: Each (maximal) horizontal line segment (formed by some edges of the tiles) is represented by a vertex of the underlying directed graph. Since both the top and the bottom edge of each square tile belong to some horizontal line segment, each tile determines two vertices. Accordingly it is represented by an edge joining these two vertices. The size of the flow in this edge from, say, the "top" to the "bottom" vertex is the length of a side of the corresponding square. An additional edge represents the rectangle and joins the vertex representing its bottom side to that of its top side. Obviously, the sum of the flow *to* a vertex equals the sum of the flow *from* this vertex (Kirchhoff's Law in the theory of electricity), i.e., it is equal to the length of the corresponding line segment. In this way, to every squared rectangle of order n there corresponds a flow on a planar (3-connected) graph with $n+1$ edges. Conversely, to every such graph there belong only $n+1$ squared rectangles: Each edge may represent the rectangle, and for each of these cases the flow can be calculated in terms of an unknown

total flow, i.e., the size of the horizontal sides of the rectangle. Consequently, since the graphs with $n+1$ edges can be effectively enumerated (an enumeration that, by the way, is also an enumeration of convex polyhedra: Federico 1975), squared rectangles of order n can also be enumerated.

The systematic search – initially done up to order 12 – was continued by Bouwkamp (1946/47, 1947a,b) (order 13), and later, with the help of computers, by Bouwkamp, Duijvestijn and Medema (1960) and Bouwkamp, Duijvestijn and Haubich (1964) (154490 simple squared rectangles of orders 9 through 18). The list of (simple) perfect rectangles starts with two specimens of order 9, six of order 10, and 22 of order 11. Finally, in 1978, the search produced the long awaited result – the (single) perfect square of lowest order, a (simple) square of order 21 (Duijvestijn 1978a).

Theoretical progress concerning squared squares was slow. Several squares of low order were constructed using additional theoretical tools. But the square of order 24 that was to hold the record for 30 years, from 1948 to 1978, had been discovered by Willcocks, an amateur mathematician, in 1948 (Willcocks 1948). Wilson (1967), using a newly developed method, produced several squares of order 25 and more, and Kazarinoff and Weitzenkamp (1973a) were able to prove that there are no compound squares of order 21 or less. In spite of all these efforts it is still not possible to directly construct (simple) squared squares by any general method. There are, however, (compound) perfect squares of any order $n \geqslant 24$: Müller (1989) gives a list for squares of any order from 24 to 43, and from a square of order n another one of order $n+20$ can be constructed by replacing the smallest tile by the square of order 21. Since Duijvestijn's square has order 21, and squares of order 22 are known (Duijvestijn 1978b), this result leaves only a small gap: Does there exist a squared square of order 23?

Recently the first pair of squared squares was found by Skinner (1992), two squares of order 29 and reduced side 1429.

More details can be found in the popular account by Tutte (1958). Federico (1979) is a detailed historical exposition with an extensive bibliography, and a shorter survey can be found in Mauldin (1981). Some more general references are Tutte (1950, 1965), and Kazarinoff and Weitzenkamp (1973b).

2.1.1. Related problems and generalizations

Remarks. (a) *Simple rectangles:* It is not known if every rectangle that can be squared (by Dehn's theorem) can also be squared simply.

(b) *A perfect squared plane:* Herda (1980) uses Fibonacci numbers and perfect squares to construct a tiling of the plane by distinct squares. See also Daykin (1967).

(c) *Higher dimensions:* It is not possible to tile a box by distinct cubes – "perfect boxes (cubes)" do not exist (see Honsberger 1970). Furthermore, there is no tiling of 3-space by mutually incongruent cubes (Dawson 1984, see also Daykin 1964).

Similar rectangles. If squares are replaced by similar rectangles not much is known:

Trivially, using homothetic rectangles is equivalent to using squares. Therefore, only those tilings are of interest where not all the tiles are homothetic. Federico (1970) and Müller (1990) study the case where the ratio of the sides is 2 : 1. For arbitrary rectangles the minimum order of a tiling by 2 : 1 rectangles is 5 (two non-equivalent – simple and perfect – tilings). There is no simple perfect tiling of order 6, and there are 42 simple perfect tilings of order 7. For 2 : 1 rectangles the minimum order of a perfect tiling is 8, and every order $n \geqslant 13$ is possible. (Nothing comparable is known about *simple* perfect tilings.)

Equilateral (regular) triangles. Another generalization is obtained if squares are replaced by equilateral triangles: A triangulation is considered to be perfect if no two tiles are congruent by translation. Congruent triangles cannot be excluded since any tiling of a rhomb contains necessarily a pair of congruent triangles having distinct aspects (i.e., orientations). From every squared rectangle a triangulation of a rhomb can be obtained: the rectangle is slanted to obtain a rhomb with acute angle $\frac{1}{3}\pi$. By this transformation the square tiles are also mapped to rhombs each of which can be dissected into two congruent triangles. Since there are triangulated rhombs which cannot be derived in this way, triangulations are the more general concept. Their theory can be developed on similar lines (as already observed by Brooks et al. 1940) – where triangulations correspond to networks which obey a modified condition on the flow – but they have attracted less attention. Tutte (1981) is an introduction to the subject, further references are Tutte (1948, 1973), and Brooks et al. (1975).

Tilings of the plane. There is no tiling of the plane by mutually incongruent equilateral triangles such that one triangle has minimal size (Scherer 1983).

Similar triangles. The general problem of tiling polygons by similar triangles is treated by Laczkovich (1990). The bases of his results is the following generalization of Dehn's theorem:

Theorem. *If a polygon is tiled by finitely many triangles, then the coordinates of the vertices of the triangles belong to the field generated by the coordinates of the vertices of the polygon and the cotangents of the angles of the triangles.*

Furthermore, if the polygon is convex then it is possible to choose the coordinate system so that the coordinates of its vertices belong to the field generated by the cotangents of the angles of the triangles.

Since, in this section, we are interested in tilings by similar triangles we shall say, for short, that a triangle tiles a polygon if the polygon can be tiled by finitely many similar copies of this triangle. There are some results on the general case (e.g., an arbitrary polygon is tiled by at most countably many distinct – with respect to similarity – triangles) but the main results concern tilings of rectangles.

With only four exceptions, a rectangle can only be tiled by right triangles. Moreover, a necessary condition is that the acute angle α of the triangle is such that $\tan\alpha$ is a totally positive algebraic number (i.e., $\tan\alpha$ must be algebraic, and all

its real conjugates must be positive). The four exceptional triangles are: $(\frac{1}{8}\pi, \frac{1}{4}\pi, \frac{5}{8}\pi)$, $(\frac{1}{4}\pi, \frac{1}{3}\pi, \frac{5}{12}\pi)$, and $(\frac{1}{12}\pi, \frac{1}{4}\pi, \frac{2}{3}\pi)$, (which tile the square) and $(\frac{1}{6}\pi, \frac{1}{6}\pi, \frac{2}{3}\pi)$, (which tiles a rectangle where the ratio of its sides is $a : b\sqrt{3}$ with integers a, b).

A side result states that a square cannot be tiled by finitely many triangles which all have angles whose size – if measured in degrees – is an even integer.

The special case where the polygon to be tiled is a triangle similar to the tiles has also been investigated: every triangle may be tiled using any number n of tiles but $n = 2,3,5$. From these exceptional values, 2 or 3, (and possibly also 5) tiles can be used only in the case of a right triangle (Freese, Miller and Usiskin 1970). Furthermore, if the triangle is not equilateral (see the discussion above) then there exist perfect tilings, and if it is neither a right nor an isosceles triangle, then the minimum order of a perfect tiling is 6 (Kaiser 1988). Kaiser (1991) further investigates perfect tilings of a triangle by triangles similar to it, and Tuza (1991) shows that for convex polygons perfect tilings by equilateral triangles do not exist.

Affinely equivalent tiles. This generalization is mentioned mainly for the sake of completeness. Tilings of this type by triangles and quadrilaterals always exist. Indeed, any two triangles are affinely equivalent, and this fact can also be used to construct a tiling by affinely equivalent quadrangles. (A tiling of an equilateral triangle by three congruent deltoids is transferred to each triangle of a triangulation.) Simple tilings of rectangles by n rectangles – note that any two rectangles are affinely equivalent – exist for $n = 5$ and $n \geqslant 7$, but not for $n = 6$ (Chung, Gilbert and Graham 1982 and Hertel 1989).

2.2. *Tilings by congruent tiles*

It would probably be most natural to require that the tiles used are all congruent. But whereas this case is fundamental for infinite tilings (*monohedral* tilings) it is too restrictive when finite tilings are concerned. Therefore only some quite special cases have been considered.

2.2.1. *Self-similarity*

A polygon is called *self-similar (of order n)* if it can be tiled by n congruent tiles that are, in addition, similar to the polygon itself. (Such tiles are also called *reptiles*.)

Among the convex polygons only triangles and quadrilaterals can be self-similar, The order of a triangle is either k^2 (any triangle), $k_1^2 + k_2^2$ (right triangles with shorter sides k_1 and k_2), and $3k^2$ (right triangle with angles $\frac{1}{2}\pi$, $\frac{1}{3}\pi$, and $\frac{1}{6}\pi$) (Snover, Waiveris and Williams 1991, proving a statement of Golomb and Posner), and quadrilaterals of order $n \leqslant 5$ have also been classified (Valette and Zamfirescu 1974, Doyen and Landuyt 1983, see also Hertel 1989).

Remark. Self-similar (convex or non-convex) tiles – and sets of prototiles with an analogous property – are an important tool to construct non-periodic tilings, both in the plane and in 3-space (*inflation* and *deflation*: Danzer 1989, see also Grünbaum and Shephard 1987).

2.2.2. Fault-free tilings of rectangles

A tiling of a rectangle by (congruent) rectangular tiles (often called *bricks*) is *fault-free* if it cannot be split into two rectangles, each tiled by a subset of the set of tiles. (In other words: a *fault-line* is a line that meets the interior of the rectangle, but does not meet the interior of any tile.)

Fault-free tilings of an $a \times b$ rectangle by $p \times q$ rectangles (with integer numbers a, b, p, and $q > 1$, $(p,q) = 1$) exist (Graham 1981) – with one exception: there is no fault-free tiling of the 6×6 square by 1×2 tiles – if and only if

(1) each of p and q divides a or b, and

(2) each of a and b can be expressed as $xp + yq$, $x, y > 0$, in (at least) two ways.

There are five families of these rectangles, and that with dimensions $3pq \times (pq + p + q)$ has minimal area (Scherer 1980/81 for $p=1$, and Robinson 1982) (The related problem of packing as many $p \times q$ rectangular tiles into a rectangle is treated by Holton and Rickard 1978.)

Remark. The related question of tiling the n-dimensional torus by *rods*, i.e., by n-dimensional boxes with $n - 1$ sides of equal length, is treated by Clivio (1991).

2.3. Tilings by polygons: k-gons

In this section we impose only a rather weak restriction on the shape of the tiles, namely, that they all are k-gons (for some fixed integer k). Bleicher (1984) investigates under what conditions (in terms of K, k, and n) a polygon (K-gon) can be tiled using n k-gons. (An exposition of these results can be found in Hertel 1989.)

In the general case (where arbitrary k-gons are allowed for the tiles) the existence of tilings is easily established: Clearly, every polygon can be tiled by triangles. And since every triangle can be dissected into two k-gons (of which at most one is convex), every polygon can be tiled by k-gons, as well. Moreover, even the number of tiles used can be prescribed, provided only that it is large enough: If the polygon is a convex K-gon then it can be tiled by $n \geqslant 2$ k-gons if and only if $n \geqslant (K-2)/(k-2)$. (For an arbitrary polygon, of course, it depends on its shape which small numbers n are possible.)

If the tiles are required to be convex the situation becomes more complicated. The lower bound on the number of tiles, valid in the general case, is no longer sharp and, in general (at least for $k \geqslant 7$), the number of tiles may no longer be arbitrarily large. The following theorem summarizes the known results (Bleicher 1984). In the cases $K = 3, 4$ and $k = 5$ necessity is due to Hertel (1989). Note the gaps in these results which still leave some cases undecided.

Theorem.

($k = 3$): *A convex K-gon can be tiled by n (convex) triangles if and only if* $n \geqslant K - 2$.

($k = 4$): *A triangle can be tiled by convex quadrilaterals n if and only if* $n \geqslant 3$. *A convex K-gon, $K \geqslant 4$, can be tiled by n convex quadrilaterals if and only if* $n \geqslant (K-2)/2$.

($k = 5$): *A triangle can be tiled by n convex pentagons if and only if* $n \geqslant 9$. *A convex quadrilateral can be tiled by n convex pentagons if and only if* $n \geqslant 8$. *A convex pentagon can be tiled by n convex pentagons if and only if* $n \geqslant 6$. *A convex K-gon,* $K \geqslant 6$ *can be tiled by* $n \geqslant 2$ *convex pentagons if and only if* $n \geqslant (K-2)/3$.

($k \geqslant 6$): *If a convex K-gon can be tiled by n convex k-gons, then*

$$(K-9)/(k-6) \geqslant n \geqslant (K-2)/(k-2).$$

The upper bound makes no sense for $k = 6$: In this case $n = 2$ is possible only for $8 \leqslant K \leqslant 10$, and $n \geqslant 3$ implies $9 \leqslant K$. Conversely, if

$$(K-6)/(k-5) \geqslant n \geqslant (K-2)/(k-2)$$

then every convex K-gon can be tiled by n k-gons. In particular, if $K \leqslant 2k-5$ then there is no tiling of the polygon.

It is not known if the existence of a tiling by n k-gons may depend on the shape of the K-gon (instead of on K alone).

2.4. Tilings by polygons: Regular polygons

Regular polygons have always attracted special attention because of their aesthetic appeal. For instance, tilings of the plane by regular polygons have been studied for a long time. And certainly this appeal was (at least) partially the reason why squared squares gained so much popularity. In view of this tendency it is surprising how little is known about finite tilings by regular polygons in general.

Malkevitch (1969, 1985) investigates which convex polygons (K-gons) can be tiled using equilateral triangles and squares with fixed common sidelength under the additional assumption that the tiles meet side-to-side. In that case at the vertices of the polygon only four values for the exterior angle are possible: $\frac{2}{3}\pi$, $\frac{1}{2}\pi$, $\frac{1}{3}\pi$, and $\frac{1}{6}\pi$. If the number of vertices of these types is v_1, v_2, v_3, and v_4, respectively, then the following diophantine equation must hold:

$$4v_1 + 3v_2 + 2v_3 + v_4 = 12.$$

Remark. Essentially the same equation occurs in connection with two enumeration problems on polytopes in space, namely, the enumeration of convex 3-valent polytopes with no face having more than six sides (Malkevitch 1970), and of polytopes all of whose faces are triangles (Beck, Bleicher and Crowe 1967). Discussion of the (finitely many) solutions of this equation shows that just the values $3 \leqslant K \leqslant 12$ are possible for the polygon, and that for all these values except 11 there are also equilateral convex polygons that can be tiled, but no classification of these polygons, especially for $K = 12$, is known.

Wildgrube (1987) allows arbitrary regular polygons (k-gons), again with fixed common side length. He shows that

(a) only polygons with at most 12 sides can be tiled, and that
(b) only regular polygons with 3, 4, 5, 6, or 12 sides can occur as tiles.
His proof is based on an enumeration of all possible vertex configurations.

2.5. *Tilings by smaller tiles: Diameter*

Among the various functions that measure the size of a set the diameter certainly is one of the most natural. Therefore, it is not surprising that tilings (or, equivalently, coverings) of a set by sets with a smaller diameter appear quite early in the mathematical literature. The main question in this connection is, of course the following:

Given some (usually convex) d-dimensional body B of diameter s – how many sets of diameter less than s are necessary to tile (cover) this body?

We shall denote by $t(B) = t_d(B)$ the minimum number of tiles necessary for a tiling of the d-dimensional bounded set B, and by $t(\mathfrak{B})$ the tiling number of $\mathfrak{B}$, i.e., the maximum value of $t(B)$, taken over some family $\mathfrak{B}$ of sets. If $\mathfrak{B}$ is the family of all bounded sets we shall use t_d instead of $t_d(\mathfrak{B})$. The size $s(\mathcal{T})$ of a tiling $\mathcal{T}$ is the largest value occurring amongst the diameters of the tiles. $S^n(B) = \min\{s(\mathcal{T})/s(B) \mid \#\mathcal{T} = n\}$, i.e., the smallest size of a tiling by n tiles (relative to the diameter of the set), indicates how "good" a particular set can be tiled. Accordingly, $\overline{S}^n(\mathfrak{B}) = \max\{S^n(B) \mid B \in \mathfrak{B}\}$ indicates how good the sets of some family can be tiled. The value $\overline{S}_d = \overline{S}^{t_d}$ is of particular interest.

The story of this problem started in 1932, at the International Congress in Zürich, when Borsuk (1932) presented the following theorem:

Theorem. *If a d-dimensional ball is tiled by d tiles, then at least one of the tiles contains an antipodal pair, i.e., it has the same diameter as the ball. Clearly, this assertion implies* $t_d \geqslant d + 1$.

(Essentially the same result is contained in a paper by Lyusternik and Schnirelman 1930.) One would, of course, expect that $t(B) = d + 1$ is true for every d-dimensional domain, or at least for every convex body, but this innocently looking conjecture turned out to be an extremely difficult problem. A recent result disproves this conjecture in its general form, but nonetheless the problem is still not solved satisfactorily. It was first formulated by Borsuk (1933), and is therefore well known as *Borsuk's Problem*. Only one case of the problem, namely $d = 2$, can be considered to be completely understood.

$d = 2$
$t_2 = 3$, i.e., $t(B) \leqslant 3$ for every bounded closed domain in the plane.

(Of course, for some sets two tiles suffice, e.g., for parallelograms and for non-circular ellipses.) The proof of this statement uses several easy and intuitively clear steps: (Pál 1920):

(a) Every plane set of diameter s can be embedded into a set of constant width s.
(b) Every set of constant width s can be embedded into a regular hexagon with

side $s/2$.

(c) A regular hexagon with side $s/2$ – and therefore all its subsets – can be tiled by three sets with diameter less than s (more precisely, less than $\sqrt{3}/2$).

For the proof of step (a) a special construction is used: For any plane set B the *disk extension* B^* is the intersection of all circular disks that enclose B ($B^* = \bigcup\{D \mid B \supset D,\ D \text{ disk}\}$). Clearly $\operatorname{diam} B = \operatorname{diam} B^*$. Moreover, $t(B) = t^*(B)$, and $t(B) = 3$ if and only if B^* is a set of constant width. Since the extension of B to a set of constant width is unique if and only if B^* is a set of constant width this leads to the following characterization of those sets for which $t(B) = 3$: For a plane domain B, $t(B) = 3$ if and only if the domain has a unique extension to a set of constant width. [In particular, $t(B) = 3$ for all sets of constant width (Boltyanskii 1969).]

$d = 3$

The first proof of Borsuk's conjecture in space is due to Eggleston (1957) (see also Perkal 1947). Much simpler proofs, similar to that outlined above for the 2-dimensional case, were given independently by Grünbaum (1957a) and Heppes and Révész (1956). The proof is based on the following result:

Theorem (Gale 1953). *Every 3-dimensional body of diameter s can be embedded into a right octahedron whose opposite faces are at distance s.*

This octahedron, however, cannot be directly used to construct a tiling – before that can be done some corners have to be cut away. In this way, Grünbaum was able to obtain a slightly better bound on $\bar{S}_3$, namely, 0.9887, compared to 0.9977 found by Heppes (1957).

$d \geqslant 4$

For higher dimensions only partial results are known: Hadwiger (1945/46, 1946/47) proved the conjecture for convex bodies with smooth boundaries, i.e., $t_d(B) = d+1$ for all such bodies. A corresponding result by Boltyansky (1960) on Hadwiger's problem (see below) extends this to convex bodies with at most d corners. (A *corner* is a point on the boundary in which the support hyperplane is not uniquely determined.)

Lenz (1955) showed that $t_d(B) \leqslant n$ if B has a smooth boundary, but does not have constant width. Conversely, $d+1 \leqslant t_d(B)$ holds for all sets of constant width.

Other partial results involve symmetry conditions instead of these smoothness conditions: $t_d(B) = d + 1$ holds for centrally symmetric bodies (Riesling 1971, see Kolodziejczyk 1988 for generalizations) and for bodies which have the same symmetry as the d-dimensional simplex (Rogers 1971).

For the general case only some bounds on t_d have been derived: Lenz (1955), Danzer (1961), Lassak (1982) ($t_4 \leqslant 2^{d-1} + 1$) and Schramm (1988), but they all are far from the expected value: They yield, for instance, $t_4 \leqslant 81$, $t_4 \leqslant 55$, $t_4 \leqslant 9$, and $t_4 \leqslant 685$, respectively. However, a very recent result of Kahn and Kalai (1992)

shows that $t_d \geqslant (1.1)^{\sqrt{d}}$. This implies that Borsuk's conjecture is false for high dimensions, in particular for $d = 1325$ and $d \geqslant 2014$.

2.5.1. Related problems and generalizations

Normed spaces. So far, only Euclidean spaces have been considered. But the problem also makes sense in any normed space, and the definitions can be transferred to this setting.

The case of a *Minkowski plane* (where the norm is defined by a (convex) unit disk G) was discussed by Grünbaum (1957b). As to be expected, the value of $t_G(B)$ depends both on G and B.

Theorem. *If G is a parallelogram then $t_G(B) = 4$ if and only if the convex hull of B is a parallelogram homothetic to G. In all other cases the bounds are $2 \leqslant t_G(B) \leqslant 3$. (The case of a circular disk B corresponds, of course, to the Euclidean case.)*

For an exposition of this subject (known results and problems) see Boltyanskiĭ and Gohberg (1980) (plane only) and Boltyanskiĭ and Gohberg (1985) (general case). (See also the survey Grünbaum 1963.)

Optimal tilings. Only for a few special cases of B and n optimal tilings, i.e., tilings $\mathcal{T}$ by n tiles for which the largest occurring diameter is as small as possible ($s(\mathcal{T})/s(B) = S^n(B)$), have been studied: see Krotoszyński (1992) for the circular disk ($n \leqslant 11$), Graham (1967) for the equilateral triangle, and Guy and Selfridge (1975) for the square ($n \leqslant 10$).

Bounds on S_d, i.e., on the diameters in a minimal tiling, have been derived for smooth bodies which satisfy some additional smoothness conditions (Hadwiger 1949 and Dekster 1989).

2.6. Tilings by smaller tiles: Reduced copies

A *reduced* copy of a set S is a set obtained by a homothety with a factor smaller than one. That means that it is similar to S, smaller, and has not been rotated. In analogy to Borsuk's Problem one may investigate the number of reduced copies that are necessary to cover some set.

Remark. Usually, reduced copies will not allow tilings, (but cf. the tilings by similar tiles) but Hadwiger's question on coverings can be turned into a question on finite tilings by means of a suitably defined notion of size. The size $s_D(A)$ of a set A (relative to D) is the least factor λ for which there is a set homothetic to D (with factor λ) that covers A. Clearly, the size of a reduced copy equals its factor of reduction λ, and it is $s_D(S) \leqslant \lambda$ for every subset. In the spirit of this chapter these definitions lead to the following question: Given some (usually convex) d-dimensional set C – how many sets of reduced size are necessary to tile (cover) this set? (Note that this question also makes sense if C is not bounded.) Similarly to the previous section, we shall denote by $r(C) = r_d(C)$ the minimum number of

reduced sets necessary for a tiling of the (d-dimensional) set C, and by $r(\mathscr{C})$ the maximum value of $r(C)$, taken over some family $\mathscr{C}$ of sets.

If B is a bounded convex set, then the diameter of a reduced copy is smaller than that of B: In this case, a tiling by sets of reduced size is also a tiling by sets with a smaller diameter, i.e., $t(B) \leqslant r(B)$. The converse of this statement is not true, and $r(B)$ may be (strictly) larger than $t(B)$. The extreme case seems to be represented by parallelotopes. Obviously, a d-dimensional parallelotope can be covered by 2^d reduced copies (one for each vertex), and this is optimal since a reduced copy cannot contain more than one of the vertices. These observations naturally lead to the following problem: (*Problem of Hadwiger*):

Problem. Is the following true: $r(B) \leqslant 2^d$ for every d-dimensional (bounded) convex body B, and $r(B) = 2^d$ if and only if B is a parallelotope? (Hadwiger 1957, 1960).

The situation concerning this problem is as bad as that concerning Borsuk's Problem (or even worse). Only the plane case has been solved, first (implicitly) by Levi (1955) even before it was explicitly stated, then (independently) by Gohberg and Markus (1960). But for higher dimensions (even for $d = 3$) – in spite of all efforts – only partial solutions are known.

$d = 3$

In 3-dimensional space $r(B) \leqslant 8$ holds (a) for every centrally-symmetric convex body (Lassak 1984), and (b) for every polyhedron with at least one non-trivial affine symmetry (Bezdek 1989).

$d \geqslant 3$

(a) If a d-dimensional convex body B has at most d corners then $r_d(B)$ [and consequently $t_d(B)$] equals $d+1$ (Boltyansky 1960) This result is almost best possible: if $d = 3$ the body B may even have $4 = d+1$ corners, but for $d \geqslant 4$ there exist bodies B with $d+1$ corners and $r(B) = d+2$ (Charazishvili 1973 and K. Bezdek 1992). On the other hand, Weissbach (1981) and Dekster (1988) describe classes of non-smooth convex bodies $\mathscr{B}$ (satisfying a certain condition on the projection cone of boundary points or on the set of directions of diameters, respectively) for which $r_d(\mathscr{B}) = d+1$.

(b) For any d-dimensional zonoid other than a parallelotope $r_d \leqslant 3 \cdot 2^{d-2}$ (Martini 1987 for zonotopes, Boltjanskii and Soltan 1990).

Moreover, it is known that all the expected values of $r_d(B)$, i.e., $d+1 \leqslant r_d \leqslant 2^d$, do indeed occur for polytopes B.

[Various bounds on $r(B)$ have been obtained by Levin and Petunin (see the recommended introductory monograph by Boltjansky and Gohberg 1985, p. 45) ($r_d(B) \leqslant (d+1)^d$ for centrally convex bodies), Rogers (see the survey Grünbaum 1963, p. 284) ($r_d(B) \leqslant 2^d d(\log d + \log\log d + 5)$), Lassak (1988) and Schramm (1988).]

For unbounded sets $r(C)$ is finite if and only if C is almost conic (Soltan 1963).

Boltyansky (1960) did not obtain his result in a straightforward way. Instead he solved a closely related *illumination problem*:

Given some (usually convex) d-dimensional domain C – how many light-sources (outside of C) are needed to illuminate the boundary of C?

Two distinct illumination numbers are associated with every domain C:

$l(C) = l_d(C)$, the minimum number of light-sources (in d-space), and

$l^\infty(C) = l_d^\infty(C)$, the minimum number of directions, i.e., light-sources at infinity, that are necessary to illuminate C. (Note that, in essence, this illumination problem is also a tiling (covering) problem, but presented in a special context with an intuitive appeal.)

Boltyanski's approach consisted of two steps: He proved (a) $l^\infty = d + 1$ for convex bodies with at most d corners (and $l^\infty \geqslant d + 1$ for convex bodies), and (b) $l^\infty(B) = r(B)$ for convex bodies.

This research was extended to unbounded sets by Soltan (1963): He proved (c) $l^\infty(C) \leqslant l(C) \leqslant r(C)$ for all convex domains C (therefore $l^\infty(B) = l(B) = r(B)$ for convex bodies). Moreover (K. Bezdek 1989), (d) $l(C)$ is finite if and only if $l(C)$ is finite, i.e., if and only if C is almost conic.

K. Bezdek (1991) introduced yet another problem equivalent to the problem of Hadwiger for convex bodies B – a separation problem (to be applied not directly to B, but on a body associated with B, namely the polar body of B): How many hyperplanes are needed to separate an interior point of a convex body from its faces?

2.6.1. Related problems

Optimal tilings. In the plane, the reduction factor of the four reduced copies needed to cover a convex body can be $\frac{1}{2}\sqrt{2}$ (sharp bound, Lassak 1986), and if the reduction factor is less than $\frac{1}{2}$ then five reduced copies are never sufficient (Krotoszyński 1987).

Total volume. For every d-dimensional convex body B there is a least constant $f(B) = f_d(B)$ with the following property: If a finite family B_i, $i = 1, \ldots, k$, of reduced copies of B is such that its total volume is at least $f(B)$, then B can be covered by this family, i.e., there is a cover B_i', $i = 1, \ldots, k$, of B such that each B_i' is a translate of B_i. Some special values of $f(B)$ are known:

(a) $f_2(B) = 2$ for triangles B,

(b) $2 \leqslant f_2(K) \leqslant 12$ for plane convex bodies B, and

(c) $f_d(B) = 2^d - 1$ for d-dimensional parallelotopes B (K. Bezdek and Füredi 1992 (a), and A. Bezdek and K. Bezdek 1984 (b,c)).

Denote by $f^*(B)$ the corresponding constant if the centrally symmetric image of B has to be covered instead. Then $f_2^* = 4$ (Vásárhelyi 1984).

2.7. Size restrictions: Equal area

In this section we shall be (mainly) concerned with tilings where all the tiles are

assumed to have equal area. (Such a tiling is sometimes called *equidissection*.) Clearly, not much can be said if only the area is restricted in some way. Therefore, in order to obtain interesting results, some additional conditions – usually restricting the shape – have to be posed on the tiles.

2.7.1. Tilings by triangles

Tilings by triangles of equal area have been studied for regular polygons.

Obviously, every triangle can be tiled by n triangles (of equal area) for any n. A square (or rectangle) can be tiled if and only if n is even (Richman and Thomas 1967, Thomas 1968, and (partially) Monsky 1970). And a regular K-gon, $K \geqslant 5$ can be tiled if and only if n is a multiple of K (Kasimatis 1989). (Note that sufficiency is obvious in this theorem.) See Stein (1989) for some results on centrally symmetric octagons.

For arbitrary polygons ($K \geqslant 4$), equidissections can exist only if the coordinates of the vertices are algebraically dependent, and if they exist it need not be the case that the spectrum (the set of possible numbers of tiles) is generated by a single integer (Kasimatis and Stein 1990) (with some generalizations to higher dimensions).

Mead generalized the first result to higher dimensions: A d-dimensional cube can be tiled using n (d-dimensional) simplices (all of equal d-dimensional volumes) if and only if n is a multiple of $n!$.

A negative result of this type is due to Pokrovskiĭ (1989): For every dimension $d \geqslant 2$ there exist d-dimensional convex polytopes with $d+2$ vertices which cannot be tiled by simplices of equal volume.

2.7.2. Tilings by rectangles

For any n there are only finitely distinct many ways in which a rectangle can be tiled by n rectangles of equal area.

This was shown (in answer to a question asked by Ihringer 1984) independently by Boros (1984) and Szegedy (1984). Boros has, in fact, proved a more general theorem:

Theorem. *Consider a d-dimensional cube (or box) of (d-dimensional) volume a, and n positive numbers v_i, $i = 1, \ldots, n$, whose sum equals the volume of the box. Then there are only finitely many distinct tilings by n boxes B_i, $i = 1, \ldots, n$, where the volume of B_i equals v_i for each i.*

The number of distinct tilings, however, increases at least exponentially with n (Scheithauer and Terno 1988).

2.7.3. Related questions

Tilings by rectangles: The perimeters. Rosenfeld (1986) considers the perimeters (another measure of size) of the tiles:

If the unit square is tiled by n rectangles of equal area then the largest of its

perimeters is at least

$$2\max((1/r + r/n), (1/(r+1) + (r+1)/n)),$$

where $r = [\sqrt{n}]$, i.e.,

$$p(m) = 2(1/r + r/n) \qquad \text{for } r(r+1) \leqslant n < (r+1)^2,$$

and

$$p(m) = 2(1/(r+1) + (r+1)/n) \quad \text{for } r^2 \leqslant n \leqslant r(r+1).$$

This bound is best possible for $n = r^2$ and $n = r(r+1)$ (as can be seen from the trivial tilings by $r \times r$ and $r \times (r+1)$ congruent rectangles).

(It is interesting to note that this problem originates from a model in communication theory: Each rectangle represents a potential message, and its perimeter measures the uncertainty of success when communicating this message. It was first treated by Alon and Kleitman (1986) without a restriction on the area of the rectangles. In this more general case they obtained $4(2r+1)/(n + r(r+1))$ as a lower bound for the largest perimeter.)

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CHAPTER 2.3

Combinatorial Aspects of Convex Polytopes

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Contents

1. Definitions and fundamental results 487
   1.1. Introduction 487
   1.2. Faces 487
   1.3. Polarity and duality 488
   1.4. Overview 488
2. Shellings 488
   2.1. Introduction 488
   2.2. Euler's relation 489
   2.3. Line shellings 489
   2.4. Shellable simplicial complexes 490
   2.5. The Dehn–Sommerville equations 490
   2.6. Completely unimodal numberings and orientations 492
   2.7. The upper bound theorem 493
   2.8. The lower bound theorem 493
   2.9. Constructions using shellings 494
   2.10. Notes 495
3. Algebraic methods 496
   3.1. Introduction 496
   3.2. The Stanley–Reisner ring 496
   3.3. Toric varieties 498

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HANDBOOK OF CONVEX GEOMETRY
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3.4. The McMullen conditions ... 499
3.5. Polytope pairs ... 500
3.6. Centrally symmetric simplicial polytopes ... 502
3.7. Flag vectors ... 502
3.8. Dimension four ... 504
3.9. Intersection homology ... 505
3.10. Kalai's convolutions ... 507
3.11. Other parameters ... 508
3.12. Algebraic shifting ... 509
3.13. Rigidity and stress ... 509
4. Gale transforms and diagrams ... 511
4.1. Introduction ... 511
4.2. Polytopes with few vertices ... 512
4.3. Subdivisions and triangulations ... 513
4.4. Oriented matroids ... 517
4.5. Lawrence polytopes ... 519
5. Graphs of polytopes ... 520
5.1. Introduction ... 520
5.2. Steinitz's Theorem ... 520
5.3. Hamiltonian circuits ... 521
5.4. Diameter ... 522
6. Combinatorial structure ... 523
6.1. Introduction ... 523
6.2. Regular polytopes ... 524
6.3. Numbers of combinatorial types ... 524
6.4. Isotopy ... 525
6.5. Realization ... 525
6.6. Barycentric subdivisions ... 527
6.7. p-vectors of 3-polytopes ... 528
References ... 528

1. Definitions and fundamental results

1.1. Introduction

A *convex polyhedron* is a subset of $\mathbb{R}^d$ that is the intersection of a finite number of closed halfspaces. A bounded convex polyhedron is called a *convex polytope*. Since most polyhedra under consideration will be convex, this adjective will usually be omitted. Space limitations prevent a comprehensive survey of the entire theory of polytopes; therefore, this chapter will concentrate primarily upon some techniques that have been successful in analyzing their combinatorial properties. Other aspects of the role of polytopes in convexity are treated in some of the other chapters of this volume.

The following may be regarded as the fundamental theorem of convex polytopes.

Theorem 1.1. *$P \subset \mathbb{R}^d$ is a polytope if and only if it is the convex hull of a finite set of points in $\mathbb{R}^d$.*

The problem of developing algorithms to convert from one description of a polytope to the other arises in mathematical programming and computational geometry. The above theorem and related results are foundational to the theory of linear programming duality, and one of the central themes of combinatorial optimization is to make this conversion for special polytopes related to specific programming problems. See, for example, Edelsbrunner (1987), Preparata and Shamos (1985), and chapters 2.7 and 2.8 in this handbook.

1.2. Faces

The *dimension*, $\dim P$, of a polyhedron P is the dimension of its affine span, and a k-dimensional polyhedron is called a *k-polyhedron* for brevity. The *faces* of P are $\emptyset$, P, and the intersections of P with its supporting hyperplanes. The empty set and P itself are *improper faces*; the other faces are *proper*. Each face of P is itself a polyhedron, and a face of dimension j is called a *j-face*. If $\dim P = d$, faces of P of dimension 0, 1, $d-2$ and $d-1$ are called *vertices*, *edges*, *subfacets* (or *ridges*), and *facets*, respectively. A polytope equals the convex hull of its vertices. The *f-vector* of P is the vector $f = (f_0, f_1, \ldots, f_{d-1})$, where $f_j = f_j(P)$ denotes the number of j-faces of P.

Theorem 1.2. *The collection of all the faces of a polyhedron P, ordered by inclusion, is a lattice.*

This lattice is called the *face lattice* or *boundary complex* of P, and two polytopes are (*combinatorially*) *equivalent* if their face lattices are isomorphic. Suppose $F \subset G$ are two distinct faces of a polytope P. Then the interval $[F, G]$ is isomorphic to the face lattice of some polytope Q, called a *quotient polytope* of P. In the case that $G = P$, we write $Q = P/F$.

1.3. Polarity and duality

Suppose $P \subset \mathbb{R}^d$ is a d-polytope containing the origin o in its interior. Then $P^* = \{x \in \mathbb{R}^d : \langle x, y \rangle \leqslant 1 \text{ for all } y \in P\}$ is also a d-polytope, called the *polar* of P (with respect to o).

Theorem 1.3. *The face lattices of P and P^* are anti-isomorphic.*

Two polytopes with anti-isomorphic face lattices are said to be *dual*. Two important dual classes of d-polytopes are the class of *simplicial* d-polytopes, those for which every proper face is a simplex, and the class of *simple* d-polytopes, those for which every vertex is incident to exactly d edges.

1.4. Overview

Our survey begins with a discussion of shellability (section 2), an influential notion which links early results in polytopes to some of the most important recent achievements. In section 3 we discuss the powerful tools in commutative algebra and algebraic geometry which have so successfully and dramatically enriched the theory of polytopes; see also chapter 2.6. Gale transforms, another early tool, and their relationship to the blossoming theory of oriented matroids, are treated in section 4 (see also chapter 2.5). Section 5 considers problems centered around the graphs (1-skeletons) of polytopes, and we conclude with section 6 which discusses some issues of realizability and combinatorial types.

The standard reference for the foundation of the theory of polytopes and results through 1967 is the influential book by Grünbaum (1967). McMullen and Shephard (1971) and Brøndsted (1983) are briefer introductions that also contain more information on face numbers. For more on regular polytopes, see Coxeter (1963). Klee and Kleinschmidt (1991) give a comprehensive survey of results in the combinatorial structure of polytopes.

2. Shellings

2.1. Introduction

A *shelling* of the boundary complex of a polytope is an ordering $F_1, F_2, \ldots, F_n$ of its facets such that $F_j \cap \bigcup_{i=1}^{j-1} F_i$ is homeomorphic to a $(d-2)$-dimensional ball or sphere for all j, $2 \leqslant j \leqslant n$. Many early "proofs" of Euler's relation exploited the intuitively appealing and seemingly obvious property that every polytope is shellable (see Grünbaum 1967), but this was not established until 1971, and until then examples of nonshellable simplicial 3-balls had suggested that in fact it might be false. See also Danaraj and Klee (1974), McMullen and Shephard (1971), and Stillwell (1980).

Theorem 2.1 (Bruggesser and Mani 1971). *The facets of any d-polytope P can be*

ordered $F_1, F_2, \ldots, F_n$ such that for all j, $2 \leqslant j \leqslant n-1$, $F_j \cap \bigcup_{i=1}^{j-1} F_i$ is the union of the first k facets of F_j in some shelling of F_j, for some k, $0 < k < f_{d-2}(F_j)$.

2.2. Euler's relation

Euler's relation is the generalization of the familiar equation $f_0 - f_1 + f_2 = 2$ for 3-polytopes, and provides a necessary condition for the f-vector.

Theorem 2.2 (Euler's relation, Poincaré 1893, 1899). *If P is a d-polytope, then*

$$\sum_{j=0}^{d-1} (-1)^j f_j = 1 - (-1)^d.$$

Moreover, this is the only affine relation satisfied by all f-vectors of d-polytopes.

Refer to Grünbaum (1967) for the history of this result. Though there now exist elementary combinatorial proofs of Euler's relation, the fact that the first real proof, by Poincaré, involved algebraic techniques foreshadowed the recent fruitful interaction among polytopes, commutative algebra, and algebraic geometry.

In three dimensions, Euler's relation with some simple inequalities characterizes f-vectors of 3-polytopes.

Theorem 2.3 (Steinitz 1906). *A vector (f_0, f_1, f_2) of nonnegative integers is the f-vector of a 3-polytope if and only if the following three conditions hold.*

(i) $f_1 = f_0 + f_2 - 2$.
(ii) $4 \leqslant f_0 \leqslant 2f_2 - 4$.
(iii) $4 \leqslant f_2 \leqslant 2f_0 - 4$.

On the other hand, for no $d \geqslant 4$ has the set of all f-vectors of d-polytopes been completely characterized, though considerable progress has been made in the case $d = 4$, see section 3.8.

2.3. Line shellings

To sketch the proof that a polytope P is shellable, without loss of generality assume $P = \{x \in \mathbb{R}^d : \langle a_i, x \rangle \leqslant 1 \text{ for all } i,\ 1 \leqslant i \leqslant n\}$, with a_i normal to facet F_i. Choose vector $c \in \mathbb{R}^d$ such that the inner products $\langle c, a_i \rangle$ are all distinct. Relabel the facets, if necessary, so that $\langle c, a_1 \rangle > \langle c, a_2 \rangle > \cdots > \langle c, a_n \rangle$. Then $F_1, F_2, \ldots, F_n$ is a shelling order, and is called a *line shelling*. Geometrically, one begins at the origin, travels along a line L in the direction c, and lists the facets of P in the order in which they become visible, i.e., in the order in which the corresponding supporting hyperplanes are crossed. Then one returns from the opposite direction, listing the facets on the other side of P in the order in which they disappear from view. This idea can be generalized to *curve shellings* in which one travels along an appropriate curve instead of a straight line.

That not all shellings are curve shellings is perhaps believable. In fact Smilansky (1988) proves that there exist shellings of some d-polytope P such that for no

polytope Q combinatorially equivalent to P are the corresponding shellings of Q curve shellings.

By exploiting line shellings, Seidel (1986) obtains an algorithm to compute the convex hull of a finite set of points in affinely general position in logarithmic cost per face.

2.4. *Shellable simplicial complexes*

The definition of shellability can be extended to more general complexes. A *simplicial complex* Δ is a nonempty collection of subsets of a finite set V such that $G \in \Delta$ whenever $G \subseteq F$ for some $F \in \Delta$. Members of Δ are its *faces* and the *dimension* $\dim F$ of a face F is $|F| - 1$. The *dimension* of Δ, $\dim \Delta$, is $\max\{\dim F\colon F \in \Delta\}$. So the boundary complex of a simplicial d-polytope is a simplicial $(d-1)$-complex. As with polytopes, faces of a simplicial $(d-1)$-complex of dimension 0, 1, $d-2$ and $d-1$ are called *vertices*, *edges*, *subfacets* (or *ridges*), and *facets* of Δ, respectively.

To say Δ is *shellable* means that all of its maximal faces are facets, and the facets can be ordered $F_1, F_2, \ldots, F_n$ such that

$$F_m \cap \bigcup_{j=1}^{m-1} F_j = \bigcup_{j=1}^{k_m} G_{mj} \quad \text{for all } m,\ 2 \leqslant m \leqslant n,$$

where $G_{m1}, G_{m2}, \ldots, G_{mk_m}$ are $k_m > 0$ distinct subfacets of Δ (facets of F_m) for all m, $2 \leqslant m \leqslant n$. Taking $k_0 = 0$, the numbers k_m readily determine the f-vector of Δ. For, let $h_i = \text{card}\{m\colon k_m = i\}$. Then McMullen and Walkup (1971) show

$$f_j = \sum_{i=0}^{j+1} \binom{d-i}{d-j-1} h_i \quad \text{for all } j,\ -1 \leqslant j \leqslant d-1. \tag{2.1}$$

Because these equations are invertible,

$$h_i = \sum_{j=0}^{i} (-1)^{j-i} \binom{d-j}{d-i} f_{j-1} \quad \text{for all } i,\ 0 \leqslant i \leqslant d, \tag{2.2}$$

the quantities h_i are independent of the shelling order, and can in fact be *defined* via eq. (2.2), even for nonshellable simplicial complexes or for more general collections of subsets of a finite set. The vector $h = (h_0, h_1, \ldots, h_d)$ is the *h-vector* of Δ and contains the same information as the f-vector. The above discussion shows that h-vectors of shellable simplicial complexes are nonnegative.

2.5. *The Dehn–Sommerville equations*

Let P be a simplicial d-polytope containing o in its interior, and take a line shelling $F_1, F_2, \ldots, F_n$ of P induced by a direction c. Then the vector $-c$ induces the shelling $F_n, F_{n-1}, \ldots, F_1$, showing that the shelling is reversible. Since each subfacet is contained in exactly two facets, if facet F_m contributes to h_i in the first shelling,

then it contributes to h_{d-i} in the second. As a consequence of the invariance of the h-vector, it must be symmetric.

Theorem 2.4 (Dehn–Sommerville equations, Sommerville 1927). *For a simplicial d-polytope,*

$$h_i = h_{d-i} \quad \text{for all } i,\ 0 \leqslant i \leqslant d.$$

Equivalently,

$$f_i = \sum_{j=i}^{d-1} (-1)^{d-j-1} \binom{j+1}{i+1} f_j \quad \text{for all } i,\ -1 \leqslant i \leqslant d-2.$$

Moreover, any affine relation satisfied by all f-vectors of simplicial d-polytopes is an affine combination of the above equations.

Hence the affine span of the set of all f-vectors of simplicial polytopes has dimension $\lfloor \frac{1}{2}d \rfloor$. Note that $1 = h_0 = h_d$ is equivalent to Euler's relation. The transformation of the f-vector into the h-vector and the above formulation of the Dehn–Sommerville equations in terms of the h-vector was already known to Sommerville, although he was not aware of the algebraic interpretation of the h-vector (discussed in section 3).

It is an easy matter to verify that if Δ is a shellable simplicial $(d-1)$-complex such that every subfacet is contained in exactly two facets, then such a complex must be a pl-sphere and every shelling of Δ is reversible. So it is easy to see that the Dehn–Sommerville equations hold for shellable spheres as well. In fact, the first proofs of the Dehn–Sommerville equations did not depend upon shellability and show that these equations hold for homological $(d-1)$-spheres as well as for some more general simplicial complexes. See Grünbaum (1967).

Although f-vectors of simplicial polytopes are not always unimodal, some inequalities are a consequence of the Dehn–Sommerville equations.

Theorem 2.5 (Björner 1981). *The f-vector of a simplicial $(d-1)$-sphere satisfies*

$$f_0 < f_1 < \cdots < f_{\lfloor d/2 \rfloor - 1} \leqslant f_{\lfloor d/2 \rfloor}$$

and

$$f_{\lfloor 3(d-1)/4 \rfloor} > \cdots > f_{d-2} > f_{d-1}.$$

It turns out, however, that the *h-vectors* of simplicial polytopes are unimodal, see section 3.

The equations imply that the f-vector of the boundary of a triangulated ball is determined by the f-vector of the ball itself (section 4.3). Let Q be any unbounded simple d-polyhedron with at least one vertex. Then there exists a simplicial d-polytope P with a vertex v such that the lattice of the nonempty faces of Q is anti-isomorphic to the lattice of faces of P that do not contain v. The next result is a consequence of this duality, and is mentioned in Billera and Lee (1981a).

Theorem 2.6. *If Q is a simple d-polyhedron with at least one vertex, then the number of unbounded k-faces of Q equals*

$$f_k - \sum_{j=0}^{k} (-1)^j \binom{d-j}{d-k} f_j, \quad 1 \leqslant k \leqslant d-1.$$

2.6. Completely unimodal numberings and orientations

Returning to P and the line shelling induced by c as in section 2.3, let Q be the simple d-polytope that is the polar of P. Any given acyclic orientation of the edges of Q and any given numbering of the vertices of Q from 1 to n are said to be *consistent* provided the edges are oriented from lower-numbered vertices to higher-numbered vertices. Associated with any numbering is a unique consistent orientation, and associated with any acyclic orientation is at least one consistent numbering.

Our indexing of the facets of P implies that $\langle c, a_i \rangle > \langle c, a_j \rangle$ if and only if $i < j$. Label the vertex a_i of Q with the number i, $i = 1, \ldots, n$, and consider the associated consistent orientation. Then h_i equals the number of vertices of Q having in-degree i, as well as the number of vertices having out-degree i.

This numbering of vertices also has the property that for every k-face of Q, $2 \leqslant k \leqslant d$, the restriction of the associated consistent orientation to that face has a unique vertex of in-degree zero. Any numbering of the vertices of Q possessing the above property will be called *completely unimodal*, and an acyclic orientation is said to be *completely unimodal* if any (equivalently, all) consistent numberings are completely unimodal. Such numberings and orientations may be regarded as abstract objective functions. See also Brøndsted (1983).

Theorem 2.7 (Williamson Hoke 1988). *The following are equivalent for a given numbering of the vertices of Q:*

(i) *The numbering is completely unimodal.*

(ii) *h_i equals the number of vertices having in-degree (out-degree) i in the associated consistent orientation.*

(iii) *For all k, $1 \leqslant k \leqslant n$, and any face F, the edge-graph induced by the set of vertices on F numbered less than (greater than) k is connected.*

(iv) *The induced ordering of the facets of P is a (not necessarily curve) shelling.*

(v) *In every k-face of P, $2 \leqslant k \leqslant d$, there is a unique vertex of in-degree (out-degree) zero with respect to the induced consistent orientation of the edges of that face.*

Kalai characterizes completely unimodal orientations in the case that the edge-graph of a simple d-polytope Q is given, but otherwise its facial structure is unknown. For any acyclic orientation O of the graph, let h_k^O be the number of vertices with in-degree k. Define $f^O = h_0^O + 2h_1^O + 4h_2^O + \cdots + 2^d h_d^O$.

Theorem 2.8 (Kalai 1988c). *The following are equivalent for an acyclic orientation* O^*.

(i) O^* *is completely unimodal.*
(ii) f^{O^*} *minimizes* f^O *over all acyclic orientations.*
(iii) f^{O^*} *equals the total number of nonempty faces of* Q.

From this Kalai obtains a new proof of a result first established by Blind and Mani-Levitska.

Theorem 2.9 (Blind and Mani-Levitska 1987). *The edge-graph of a simple polytope completely determines its entire combinatorial structure.*

2.7. The upper bound theorem

Given integers $d \geqslant 2$ and $n \geqslant d+1$, take the convex hull of any n distinct points on the moment curve $(t, t^2, \ldots, t^d)$. The combinatorial structure of the resulting simplicial d-polytope $C(n,d)$ is independent of the actual choice of points, and this polytope is referred to as the *cyclic d-polytope with n vertices*. It turns out that every subset of $k \leqslant \frac{1}{2}d$ vertices of $C(n,d)$ forms a face ($C(n,d)$ is *neighborly*), so it was conjectured that this polytope has the largest number of faces of all dimensions of all convex d-polytopes with n vertices. Explicit formulas for $f_j(C(n,d))$ can be found in Brøndsted (1983), Grünbaum (1967), or McMullen and Shephard (1971); we mention that

$$f_{d-1}(C(n,d)) = \binom{n - \lfloor \frac{1}{2}(d+1) \rfloor}{n-d} + \binom{n - \lfloor \frac{1}{2}(d+2) \rfloor}{n-d}.$$

Theorem 2.10 (Upper Bound Theorem, McMullen 1971). *Let* P *be a* d*-polytope with* n *vertices. Then* $f_j(P) \leqslant f_j(C(n,d))$ *for all* j, $1 \leqslant j \leqslant d-1$.

A perturbation argument shows that it suffices to prove this result for simplicial d-polytopes. McMullen uses properties of line shellings to show that $h_i(P) \leqslant \binom{n-d+i-1}{i}$ for all i, $0 \leqslant i \leqslant d$. But the fact that $h_i(C(n,d)) = \binom{n-d+i-1}{i}$ for all i, $0 \leqslant i \leqslant \lfloor \frac{1}{2}d \rfloor$, together with the Dehn–Sommerville equations, imply that $h_i(P) \leqslant h_i(C(n,d))$ for all i, $0 \leqslant i \leqslant d$. The result now follows immediately from the observation that the f_j are nonnegative combinations of the h_i. See McMullen and Shephard (1971) for an account of the solution of the Upper Bound Conjecture. The proof can also be found in Brøndsted (1983).

2.8. The lower bound theorem

Starting with a d-simplex, one can add new vertices by building shallow pyramids over facets to obtain a simplicial convex d-polytope with n vertices, called a *stacked* polytope. If $P(n,d)$ is such a polytope, then

$$f_j = \begin{cases} \binom{d}{j}n - \binom{d+1}{j+1}j, & \text{if } 0 \leqslant j \leqslant d-2, \\ (d-1)n - (d+1)(d-2), & \text{if } j = d-1. \end{cases}$$

It was conjectured that no simplicial d-polytope with n vertices can have fewer faces than $P(n,d)$, and a certain reduction implied that it was sufficient to show this result for f_1. Barnette proved this conjecture about the same time that McMullen established the Upper Bound Theorem. Barnette's argument does not use the full strength of shellability, but relies upon a weaker ordering of the facets of the dual simple polytope. The proof also appears in Brøndsted (1983).

Theorem 2.11 (Lower Bound Theorem, Barnette 1971, 1973). *Let P be a simplicial d-polytope with n vertices and $P(n,d)$ be a stacked d-polytope with n vertices. Then*

$$f_j(P) \geqslant f_j(P(n,d)) \quad \text{for all } j,\ 1 \leqslant j \leqslant d-1. \tag{2.3}$$

Moreover, if $d \geqslant 4$ and equality occurs in (2.3) *for any one value of j, then P must itself be stacked.*

The case of equality for $j = d-1$ was proved by Barnette and for the remaining values of j by Billera and Lee (1981b). Connections between the Lower Bound Theorem and rigidity were discovered by Kalai and will be discussed in section 3.13.

Another type of lower bound theorem was proved by Blind and Blind. Write C^d for the d-cube.

Theorem 2.12 (Blind and Blind 1990). *Let P be a d-polytope with no triangular faces. Then*

$$f_j(P) \geqslant f_j(C^d) \quad \text{for all } j,\ 0 \leqslant j \leqslant d-1. \tag{2.4}$$

Moreover, if equality occurs in (2.4) *for any one value of j, then P must itself be a d-cube.*

2.9. Constructions using shellings

Given any two positive integers h and i, there is a unique sequence of integers $n_i > n_{i-1} > \cdots > n_j \geqslant j \geqslant 1$ such that

$$h = \binom{n_i}{i} + \binom{n_{i-1}}{i-1} + \cdots + \binom{n_j}{j}.$$

The *ith pseudopower of h* is then defined as

$$h^{<i>} = \binom{n_i+1}{i+1} + \binom{n_{i-1}+1}{i} + \cdots + \binom{n_j+1}{j+1}.$$

For convenience we define $0^{<i>}$ to be 0 for any positive integer i.

Stanley characterized the h-vectors (and hence the f-vectors) of shellable simplicial complexes.

Theorem 2.13 (Stanley 1977). *A vector $h = (h_0, h_1, \ldots, h_d)$ of nonnegative integers*

is the h-vector of some shellable simplicial $(d-1)$*-complex if and only if* $h_0 = 1$ *and* $h_{i+1} \leqslant h_i^{<i>}$ *for all* i, $1 \leqslant i \leqslant d-1$.

The algebraic methods that imply the necessity of these conditions are discussed in section 3.2. But the sufficiency is much more straightforward. Given h satisfying the pseudopower conditions, let $n = h_1 + d$ and $V = \{1, 2, \ldots, n\}$. Take $\mathscr{F}$ to be the collection of all subsets of V of cardinality d, and $\mathscr{F}_i$ to be those members F of $\mathscr{F}$ such that $d+1-i$ is the smallest element of V not in F. For two members F and G of $\mathscr{F}$, say $F < G$ if the largest element of V in their symmetric difference is in G. For each i, $0 \leqslant i \leqslant d$, choose the first (in the given ordering) h_i members of $\mathscr{F}_i$. The resulting collection $\mathscr{C}$ consists of the facets of the desired shellable complex, and the given ordering induces the shelling order. This result and the accompanying construction are reminiscent of the characterization of the f-vectors of arbitrary simplicial complexes by Kruskal (1963) and Katona (1968).

Now suppose we have a vector $h = (h_0, h_1, \ldots, h_d)$ of nonnegative integers such that $h_i = h_{d-i}$ for all i, $1 \leqslant i \leqslant d$, and $h_{i+1} - h_i \leqslant (h_i - h_{i-1})^{<i>}$ for all i, $1 \leqslant i \leqslant \lfloor \frac{1}{2}d - 1 \rfloor$. These are known as the McMullen conditions; see section 3.4. Billera and Lee (1981b) show how to extend the above construction to obtain a shellable d-ball whose boundary is a simplicial $(d-1)$-sphere having h-vector h. Set $n = h_1 + d$, $V = \{1, 2, \ldots, n\}$, and regard V as the set of vertices of a cyclic $(d+1)$-polytope $C(n, d+1)$ with n vertices. For all i, $1 \leqslant i \leqslant \lfloor \frac{1}{2}d \rfloor$, define $\mathscr{F}_i$ to be the collection of all subsets F of V of cardinality $d+1$ corresponding to facets of the cyclic polytope $C(n, d+1)$ such that $d+2-2i$ is the smallest element of V not in F. For two members F and G of $\mathscr{F}_i$, say $F < G$ if the largest element of V in their symmetric difference is in G. For each i, $0 \leqslant i \leqslant \lfloor \frac{1}{2}d \rfloor$, choose the first (in the given ordering) $h_i - h_{i-1}$ members of $\mathscr{F}_i$ (with the convention that $h_{-1} = 0$). The resulting collection $\mathscr{C}$ consists of the facets of a d-ball Δ which is shellable in the given ordering. Further, $h(\Delta) = (h_0, h_1 - h_0, h_2 - h_1, \ldots, h_{\lfloor d/2 \rfloor} - h_{\lfloor d/2 \rfloor - 1}, 0, 0, \ldots, 0)$, whence the h-vector of the boundary $\partial\Delta$ of Δ equals $(h_0, h_1, \ldots, h_d)$ (see Theorem 4.9). In fact, $\partial\Delta$ can be realized as the boundary of a simplicial convex d-polytope; see section 3.4.

Kalai generalizes this construction to prove that there are many simplicial spheres.

Theorem 2.14 (Kalai 1988a). *For fixed* d*, the number of combinatorial types of triangulated* $(d-1)$*-spheres with* n *vertices is between* $e^{bn^{\lfloor (d-1)/2 \rfloor}}$ *and* $n^{cn^{\lfloor d/2 \rfloor}}$*, where* b *and* c *are positive constants.*

A comparison with Theorem 6.3 shows that there are many more simplicial spheres than polytopes.

2.10. Notes

Geometric analogues of Euler's relation and the Dehn–Sommerville equations can be found in Grünbaum (1967). Lawrence (1991) uses relatives of these to show that the volume of a polytope described by rational inequalities is not polynomially expressible in terms of the describing data.

The class of boundary complexes of simplicial polytopes is contained in the class of shellable spheres, which is contained in the class of pl-spheres, which is contained in the class of topological spheres, which is contained in the class of homological spheres. All inclusions are proper. For example, there exist shellable spheres that are not polytopal (Danaraj and Klee 1978b), and the nondecidability result of Volodin, Kuznetsov and Fomenko (1974) for determining whether or not a given complex is a pl-sphere implies that nonshellable pl-spheres must exist (see Mandel 1982).

In two dimensions the situation is somewhat simpler; see Danaraj and Klee (1978a), Grünbaum (1967), and section 5.2.

Theorem 2.15. *The following conditions are equivalent for a 2-complex.*

(i) *The complex is polytopal.*
(ii) *The complex is a sphere.*
(iii) *The complex is a shellable closed pseudomanifold.*

In three dimensions, all simplicial 3-spheres with at most 9 vertices are shellable (Danaraj and Klee 1978b).

3. Algebraic methods

3.1. Introduction

In this section we explore the developing relationship between techniques in commutative algebra and algebraic geometry and results in the combinatorial structure of polytopes. This interaction was launched by Stanley's use of the Stanley–Reisner ring to extend the Upper Bound Theorem to simplicial spheres, and was further propelled by Stanley's short and dramatic proof of the McMullen conditions based upon connections between the Stanley–Reisner ring of a polytope and the cohomology of an associated toric variety.

3.2. The Stanley–Reisner ring

The Stanley–Reisner ring of a simplicial complex encodes the simplices of the complex as monomials. Reisner's Theorem allows a translation of topological properties of the complex into algebraic properties of the ring. In particular, the Stanley–Reisner ring of the boundary complex of a simplicial convex polytope is Cohen–Macaulay. This is what enabled Stanley to prove the Upper Bound Theorem for simplicial spheres. He also used it in his proof of the necessity of the McMullen conditions. In what follows k is a fixed infinite field.

Let Δ be a simplicial complex with vertices $v_1, v_2, \ldots, v_n$, each simplex (face) being identified with its set of vertices. In the polynomial ring $k[x_1, x_2, \ldots, x_n]$, let I_Δ be the ideal generated by all monomials $x_{i_1} x_{i_2} \cdots x_{i_s}$ such that $\{v_{i_1}, v_{i_2}, \ldots, v_{i_s}\} \notin \Delta$. The *Stanley–Reisner ring* (or *face ring*) of Δ is the quotient ring $R_\Delta = k[x_1, x_2, \ldots, x_n]/I_\Delta$. See Stanley (1975).

Let R_m be the vector subspace of R_Δ generated by the monomials of degree m in R_Δ. This gives a grading of the Stanley–Reisner ring, $R_\Delta = \bigoplus_{m \geqslant 0} R_m$. As a k-algebra R_Δ is generated by the monomials of R_1, that is, by the variables $x_1, x_2, \ldots, x_n$ themselves. The graded component R_m is spanned by the degree m monomials whose supports are in the complex. The number of monomials of degree m with given support depends only on the size of the support. Thus we can write the Hilbert function of the graded algebra in terms of the f-vector of the complex. The *Hilbert function* of the graded algebra R_Δ is the function $H : \mathbb{N} \to \mathbb{N}$ given by $H_m = H(R_\Delta, m) = \dim_k R_m$. If Δ is a simplicial complex of dimension $d-1$ (for example, the boundary complex of a simplicial d-polytope) then (Stanley 1975)

$$H(R_\Delta, m) = \begin{cases} 1, & \text{if } m = 0, \\ \displaystyle\sum_{i=0}^{d-1} \binom{m-1}{i} f_i, & \text{if } m > 0. \end{cases} \tag{3.1}$$

Macaulay essentially gives a numerical characterization of the Hilbert functions of graded algebras generated by their degree 1 elements. Recall the definition of pseudopower given in section 2.9.

Theorem 3.1 (Macaulay 1927). *$H_0, H_1, H_2, \ldots$ is the Hilbert function of a graded algebra generated by degree 1 elements if and only if $H_0 = 1$ and, for $m > 0$, $0 \leqslant H_{m+1} \leqslant H_m^{<m>}$.*

We could apply Macaulay's Theorem to the Hilbert function of the Stanley–Reisner ring to get inequalities on the f-vectors of simplicial complexes. We are interested primarily in polytopes (or spheres), and in this case the inequalities say little. Instead we wish to apply Macaulay's Theorem to a quotient of the Stanley–Reisner ring.

A graded ring is called *Cohen–Macaulay* if its Krull dimension equals its depth. We do not have the space to elaborate on this; for more information see Stanley (1975). We note only the following property of Cohen–Macaulay rings. If a graded k-algebra R of the above form is Cohen–Macaulay of dimension d, then R has linear (degree 1) elements $\theta_1, \theta_2, \ldots, \theta_d$ such that R is a finitely generated, free module over $k[\theta_1, \theta_2, \ldots, \theta_d]$. In this case the Hilbert function h of the quotient algebra $R/(\theta_1, \theta_2, \ldots, \theta_d)$ is related to the Hilbert function H of R by the following simple relationship:

$$\sum_{i=0}^{d} h_i t^i = (1-t)^d \sum_{i=0}^{\infty} H_i t^i.$$

We apply this to the Stanley–Reisner ring of a simplicial sphere. The theorem of Reisner (1976) gives a topological criterion for the face ring of a simplicial complex to be Cohen–Macaulay. In particular, the Stanley–Reisner ring of a sphere is Cohen–Macaulay. The same is true for a shellable complex; see Kind and Kleinschmidt (1979) for a more elementary proof. The Hilbert function of the quotient

algebra $R/(\theta_1, \theta_2, \ldots, \theta_d)$, like that of R itself, can be expressed in terms of the f-vector of the simplicial complex. The Hilbert function of the quotient turns out to be the h-vector, and in fact is given by eq. (2.2) of section 2.4. Recall that the Dehn–Sommerville equations for simplicial convex polytopes have the simple form $h_i = h_{d-i}$.

Now Macaulay's Theorem can be applied to the quotient algebra $R/(\theta_1, \theta_2, \ldots, \theta_d)$, whose Hilbert function is the h-vector. This gives the relations $h_0 = 1$ and, for $m > 0$, $0 \leqslant h_{m+1} \leqslant h_m^{<m>}$.

These inequalities imply the following: for a simplicial d-polytope and for $0 \leqslant i \leqslant \frac{1}{2}d$,

$$h_i \leqslant \binom{n-d+i-1}{i}.$$

These inequalities imply the Upper Bound Theorem for simplicial spheres. The Upper Bound Theorem was first proved by McMullen for simplicial polytopes (see section 2.7). It was proved for arbitrary simplicial spheres by Stanley (1975) using the method outlined here.

3.3. Toric varieties

After he introduced the face ring of a simplicial complex and proved the Upper Bound Theorem for simplicial spheres, Stanley learned of a connection with algebraic geometry.

A certain type of algebraic variety, a projective toric variety, comes equipped with a moment map into real Euclidean space. The image of this map is a rational convex polytope (rationality here refers to the coordinates of the vertices). When the toric variety has only relatively mild singularities, the corresponding polytope is simplicial. From the combinatorial viewpoint, rationality places no restriction on *simplicial* polytopes – every simplicial polytope is combinatorially equivalent to a rational polytope. This is not the case for nonsimplicial polytopes – there are combinatorial types of polytopes not realized by any rational polytope (see sections 4.2 and 6.5).

The toric variety can be described explicitly in terms of the convex polytope. See Fine (1985) and chapter 2.6. Let P be a convex d-polytope with vertex set $V = \{v_1, v_2, \ldots, v_n\} \subset \mathbb{Q}^d$. Any affine dependence on V with integer coefficients can be written in the form $\sum_{i=1}^n b_i v_i = \sum_{i=1}^n c_i v_i$, where for all i, $b_i, c_i \in \mathbb{N}$ and $\sum_{i=1}^n b_i = \sum_{i=1}^n c_i$. Let $\mathcal{A}_P$ be the set of pairs (b, c) of coefficient vectors arising in this manner. Using the notation $x^b = x_1^{b_1} x_2^{b_2} \cdots x_n^{b_n}$, with the convention $0^0 = 1$, define E_P to be

$$E_P = \{x \in \mathbb{C}^n : x^b = x^c \text{ for all } (b, c) \in \mathcal{A}_P\}.$$

Although $\mathcal{A}_P$ is an infinite set, for a rational polytope P it is a finitely generated semigroup. Thus E_P is a subset of $\mathbb{C}^n$ defined by a finite number of polynomial equations, hence is an algebraic variety. Finally E_P is "projectivized": we define

T_P to be $E_P \setminus \{0\}$ modulo the relation $x \sim y$ if and only if $x = \lambda y$ for some $\lambda \in \mathbb{C}^*$. This is the toric variety associated with the polytope P (though not precisely the variety discussed by Danilov 1978).

The cohomology ring of T_P for simplicial (rational) polytopes P is computed by Danilov (1978). It turns out to be isomorphic to the quotient $A_\Delta = R_\Delta/(\theta_1, \theta_2, \ldots, \theta_d) = \bigoplus_{m \geqslant 0} A_m$ of the Stanley–Reisner ring modulo a certain linear system of parameters (here Δ is the boundary complex of P). Later Saito (1985) proved that the Hard Lefschetz Theorem holds for the varieties T_P (P simplicial). This ensures the existence of an element $\omega \in A_1$, called a *hyperplane section,* such that the maps $A_i \xrightarrow{\times\omega} A_{i+1}$ are injective for all i, $0 \leqslant i \leqslant \frac{1}{2}(d-1)$. Thus A_Δ/ω is a graded algebra with Hilbert function $H(A_\Delta/\omega, m) = h_m - h_{m-1}$ for $0 \leqslant m \leqslant \frac{1}{2}d$.

3.4. The McMullen conditions

We are now able to state the major theorem, conjectured by McMullen (1971) and proved by Billera and Lee (1981b) and Stanley (1980b).

Theorem 3.2 (The McMullen conditions). *A vector $(h_0, h_1, \ldots, h_d) \in \mathbb{N}^{d+1}$ is the h-vector of a simplicial polytope if and only if*

(i) $h_i = h_{d-i}$ *for all* i, $0 \leqslant i \leqslant d$.

(ii) $h_0 = 1$, *and* $h_i \leqslant h_{i+1}$ *for all* i, $0 \leqslant i \leqslant \frac{1}{2}d - 1$.

(iii) $h_{i+1} - h_i \leqslant (h_i - h_{i-1})^{<i>}$ *for all* i, $1 \leqslant i \leqslant \frac{1}{2}d - 1$.

Since the h-vector and f-vector are linearly equivalent, this theorem characterizes the f-vectors of simplicial polytopes. Of course, it also characterizes the f-vectors of simple polytopes, which are obtained by reversing the f-vectors of simplicial polytopes.

The "sufficiency" was proved constructively by Billera and Lee. They use a monomial algebra with Hilbert function $h_i - h_{i-1}$ to select a certain subset $\mathscr{C}$ of facets of the cyclic polytope $C(n, d+1)$ as described in section 2.9. Then they show that by selecting points $(t_i, t_i^2, \ldots, t_i^{d+1})$ on the moment curve such that $t_1 \ll t_2 \ll \cdots \ll t_n$, one can place a new point v beyond precisely the facets in $\mathscr{C}$. The construction is completed by taking the convex hull Q of $C(n, d+1)$ and v and passing to the vertex figure of v (the intersection of Q with a hyperplane strictly separating v from the other vertices of Q).

The "necessity" of the McMullen conditions, i.e., that all h-vectors of simplicial polytopes satisfy (i–iii), was proved by Stanley using toric varieties, as outlined in section 3.3. Note that condition (i) (the Dehn–Sommerville equations) reflects Poincaré duality of the cohomology of the toric variety, but we have seen much simpler proofs (section 2.5).

Stanley's proof of the necessity of the McMullen conditions is unsatisfying for several reasons. One reason is that the Hard Lefschetz Theorem, which is crucial to the proof, lies well beyond combinatorics. The proof depends heavily on the specific geometry of the polytope's embedding, not just on its combinatorial structure. In particular, it does not prove the McMullen conditions for h-vectors of

simplicial spheres that are not polytopal (see section 6.5). McMullen himself did not know of the toric variety connection when he made his conjecture. His idea arose from consideration of the geometry of polytopes coded in Gale diagrams (see section 4).

3.5. *Polytope pairs*

The McMullen conditions provide a characterization of the f-vectors of simple polytopes, which arise naturally in optimization. In this context it is also natural to ask about unbounded polyhedra. Just as duality establishes a correspondence between simple polytopes and simplicial polytopes, so also is there a correspondence between unbounded simple polyhedra and simplicial polytope pairs.

A (*simple*) *polytope pair of type* (d,v,d',v') is a pair (P^*,F^*), where P^* is a simple convex d-polytope with v facets and F^* is a d'-face of P^* with v' facets.

Associated with a polytope pair (P^*,F^*) is an unbounded simple d-polyhedron Q^*, obtained by applying a projective transformation that sends a supporting hyperplane for F^* onto the hyperplane at infinity. The polyhedron Q^* has recession cone of dimension $d'+1$. Conversely, every simple, pointed convex d-polyhedron with $(d'+1)$-dimensional recession cone can be associated to some polytope pair of type (d,v,d',v'), for some v and v'.

Methods for facial enumeration in simplicial polytopes were used by Klee (1974), Billera and Lee (1981a), Lee (1984), and Barnette, Kleinschmidt and Lee (1986) to develop bounds on the numbers of faces of polytope pairs. The pairs are first dualized.

Let $k=d-d'$ and $r=d-d'+v'$. The dual of a simple polytope pair (P^*,F^*) of type (d,v,d',v') is a *simplicial polytope pair* (P,F) of type (d,v,k,r), where P is a simplicial convex d-polytope with v vertices, and F is a $(k-1)$-face of P contained in $r-k$ k-faces of P. Let $\Gamma=\partial P\setminus F$, the simplicial complex obtained by deleting the face F (and all faces containing F) from the simplicial complex ∂P, the boundary of P. The faces of the boundary $\partial\Gamma$ of Γ correspond to the unbounded faces of Q^*. Thus to estimate the numbers of bounded and unbounded faces of Q^* we use estimates on the numbers of faces (or h-vectors) of P and Γ.

Theorem 3.3 (Upper Bound Theorem for polytope pairs). *Let* $3\leqslant d<r\leqslant v$ *and* $1\leqslant k\leqslant d-2$. *Put* $n=\lfloor\frac{1}{2}d\rfloor$. *Let* (P,F) *range over all (simplicial) polytope pairs of type* (d,v,k,r), *taking* $\Gamma=\partial P\setminus F$.

(i) *If* $1\leqslant k\leqslant\frac{1}{2}(d-1)$ *then*

$$\max h_i(P)=\begin{cases}\binom{v-d+i-1}{i}, & \text{if } 0\leqslant i\leqslant k,\\ \binom{v-d+i-1}{i}-\binom{v-d+i-k-1}{i-k}+\binom{r-d+i-k-1}{i-k}, & \text{if } k+1\leqslant i\leqslant n,\\ \binom{v-i-1}{d-i}-\binom{v-i-k-1}{d-i-k}+\binom{r-i-k-1}{d-i-k}, & \text{if } n+1\leqslant i\leqslant d-k-1,\\ \binom{v-i-1}{d-i}, & \text{if } d-k\leqslant i\leqslant d.\end{cases}$$

$$\max h_i(\Gamma) = \begin{cases} \binom{v-d+i-1}{i}, & \text{if } 0 \leqslant i \leqslant k-1, \\ \binom{v-d+i-1}{i} - \binom{v-d+i-k-1}{i-k}, & \text{if } k \leqslant i \leqslant n, \\ \binom{v-i-1}{d-i} - \binom{v-i-k-1}{d-i-k}, & \text{if } n+1 \leqslant i \leqslant d-k-1, \\ \binom{v-i-1}{d-i} - r + d, & \text{if } d-k \leqslant i \leqslant d-1, \\ 0, & \text{if } i = d. \end{cases}$$

(ii) *If* $\frac{1}{2}(d-1) < k \leqslant d-2$ *then*

$$\max h_i(P) = \begin{cases} \binom{v-d+i-1}{i}, & \text{if } 0 \leqslant i \leqslant n, \\ \binom{v-i-1}{d-1}, & \text{if } n+1 \leqslant i \leqslant d. \end{cases}$$

$$\max h_i(\Gamma) = \begin{cases} \binom{v-d+i-1}{i}, & \text{if } 0 \leqslant i \leqslant n, \\ \binom{v-i-1}{d-i}, & \text{if } n+1 \leqslant i \leqslant k-1, \\ \binom{v-i-1}{d-i} - 1, & \text{if } i = k, \\ \binom{v-i-1}{d-i} - r + d, & \text{if } k+1 \leqslant i \leqslant d-1, \\ 0, & \text{if } i = d. \end{cases}$$

Moreover, for each of the parts of (i) *and* (ii), *the maxima are simultaneously achievable.*

Theorem 3.4 (Lower Bound Theorem for polytope pairs). *Let* $4 \leqslant d < r \leqslant v$ *and* $2 \leqslant k \leqslant d-2$. *Put* $n = \lfloor \frac{1}{2}d \rfloor$ *and* $m = \lfloor \frac{1}{2}(d-k) \rfloor$. *Let* (P, F) *range over all (simplicial) polytope pairs of type* (d, v, k, r), *taking* $\Gamma = \partial P \setminus F$.

$$\min h_i(P) = \begin{cases} 1, & \text{if } i = 0, \\ v-d, & \text{if } 1 \leqslant i \leqslant n. \end{cases} \tag{3.2}$$

$$\min h_i(\Gamma) = \begin{cases} 1, & \text{if } i = 0, \\ v-d, & \text{if } 1 \leqslant i \leqslant k-1, \\ v-d-1, & \text{if } i = k, \\ v-r, & \text{if } k+1 \leqslant i \leqslant d-1, \\ 0, & \text{if } i = d. \end{cases} \tag{3.3}$$

For $k = 1$ *and* $3 \leqslant d < r \leqslant v$, *eq.* (3.2) *stays the same, but* (3.3) *becomes*

$$\min h_i(\Gamma) = \begin{cases} 1, & \text{if } i = 0, \\ v-d-1, & \text{if } i = 1, \\ v-r, & \text{if } 2 \leqslant i \leqslant d-1, \\ 0, & \text{if } i = d. \end{cases}$$

All bounds can be achieved.

Theorems 3.3 and 3.4 are proved in Barnette, Kleinschmidt and Lee (1986) and Lee (1984), respectively. Theorem 3.3 applies as well to the case when P is an arbitrary simplicial sphere, but the lower bounds of Theorem 3.4 depend in an essential way upon the McMullen conditions.

3.6. *Centrally symmetric simplicial polytopes*

A d-polytope P in $\mathbb{R}^d$ is *centrally symmetric* if for all points v in P, $-v$ is also in P. Björner conjectured (unpublished) that the h-vector of any centrally symmetric simplicial polytope satisfies the inequality $h_i - h_{i-1} \geqslant \binom{d}{i} - \binom{d}{i-1}$ for all i, $1 \leqslant i \leqslant \frac{1}{2}d$. Stanley (1987b) proves this conjecture using the connection with toric varieties. This also proves lower bounds on the f-vectors of centrally symmetric simplicial polytopes, conjectured earlier by Bárány and Lovász (1982).

Any centrally symmetric simplicial d-polytope is combinatorially equivalent to a centrally symmetric simplicial d-polytope with rational vertices. The associated toric variety T_P and its cohomology ring inherit the action of the group of order 2 on the polytope P by virtue of central symmetry. Furthermore, this group action on the cohomology ring commutes with multiplication by the hyperplane section ω (see section 3.3). So the cohomology ring decomposes as a direct sum of two graded vector spaces (one of which is a ring, the other of which is a module over this ring), on each of which multiplication by ω is injective. This gives Stanley's theorem.

Theorem 3.5 (Stanley 1987b). *If P is a centrally symmetric simplicial d-polytope, then*

$$h_i(P) - h_{i-1}(P) \geqslant \binom{d}{i} - \binom{d}{i-1} \quad \text{for all } i\ , 1 \leqslant i \leqslant \tfrac{1}{2}d.$$

We summarize some consequences of this theorem.

Corollary 3.6. *Let P be a centrally symmetric simplicial d-polytope, and let $(h_0, h_1, \ldots, h_d)$ be its h-vector.*

(i) $h_i \geqslant \binom{d}{i}$, *for all* i, $0 \leqslant i \leqslant d$.

(ii) $f_i \geqslant 2^{i+1}\binom{d}{i+1} + 2(n-d)\binom{d}{i}$, *for all* i, $0 \leqslant i \leqslant d-2$.

(iii) $f_{d-1} \geqslant 2^d + 2(n-d)(d-1)$.

(iv) *If for some* i, $1 \leqslant i \leqslant d-1$, $h_i(P) = \binom{d}{i}$, *then* $h_j(P) = \binom{d}{j}$ *for all* j, *and* P *is affinely equivalent to a crosspolytope.*

Part (i) of the corollary was first conjectured by Björner; parts (ii) and (iii) are the conjecture of Bárány and Lovász.

3.7. *Flag vectors*

In the previous sections of this chapter, the results on f-vectors of simplicial polytopes stemmed from interpretations of the h-vectors of the polytopes. One could

define the h-vector of a nonsimplicial polytope by the same linear transformation of the f-vector, but none of the interpretations of h-vectors would continue to hold. In fact the vector so defined has negative components for some nonsimplicial polytopes. Furthermore, the f-vector captures much less of the combinatorial structure of a nonsimplicial than of a simplicial polytope. Thus in the study of arbitrary polytopes, attention has focused on other parameters.

The one that most directly generalizes the f-vector is the flag vector. Let P be a d-polytope. A chain of faces of P, $\emptyset \subset F_1 \subset F_2 \subset \cdots \subset F_k \subset P$, is called an *$S$-flag*, where $S = \{\dim F_i : 1 \leqslant i \leqslant k\}$. The number of S-flags of P is denoted $f_S(P)$, and together these *flag numbers* form the *flag vector*, $(f_S(P))_{S \subseteq \{0,1,\ldots,d-1\}} \subseteq \mathbb{N}^{2^d}$. When writing a specific flag number we will drop the set brackets from the subscript. The f-vector is the projection of the flag vector onto the components with singleton indices. A flag number f_S of a d-simplex is a product of binomial coefficients depending only on d and S. Thus a flag number $f_S(P)$ of a simplicial d-polytope P depends only on the number of faces whose dimension is the largest element of S (and on d and S).

The problem of characterizing the f-vectors of polytopes extends to the problem of characterizing the flag vectors of polytopes. The main result on this problem is the specification of the affine hull of the flag vectors of polytopes of fixed dimension.

Theorem 3.7 (Bayer and Billera 1985). *The affine dimension of the flag vectors of d-polytopes is $e_d - 1$, where (e_d) is the Fibonacci sequence, $e_d = e_{d-1} + e_{d-2}$, $e_0 = e_1 = 1$. The affine hull of the flag vectors is determined by the equations*

$$\sum_{j=i+1}^{k-1} (-1)^{j-i-1} f_{S \cup \{j\}}(P) = \left(1 - (-1)^{k-i-1}\right) f_S(P),$$

where $i \leqslant k-2$, $i, k \in S \cup \{-1, d\}$, and S contains no integer between i and k.

These equations are called the *generalized Dehn–Sommerville equations*; their proof is similar to Sommerville's original proof of the Dehn–Sommerville equations for simplicial polytopes. For polytopes of dimension three, the generalized Dehn–Sommerville equations imply that the flag vector depends linearly on the f-vector. Thus Steinitz's characterization of the f-vectors of 3-polytopes (Theorem 2.3) extends to a characterization of the flag vectors of 3-polytopes.

Some inequalities are known to hold for the flag vectors of all polytopes. The most important of these was proved by Kalai using stress (see section 3.13).

Theorem 3.8 (Kalai 1987). *For all d-polytopes P*

$$f_{02}(P) - 3f_2(P) + f_1(P) - df_0(P) + \binom{d+1}{2} \geqslant 0.$$

The flag vector of a d-polytope P is a refinement of the f-vector of a simplicial d-polytope $\Delta(P)$, called the barycentric subdivision of P (see section 6.6);

the relationship is $f_i(\Delta(P)) = \sum_{|S|=i+1} f_S(P)$. The McMullen conditions applied to the barycentric subdivision thus give inequalities on the flag vector of the original polytope, but these are not sharp. The barycentric subdivision of a polytope is an example of a *completely balanced sphere*, studied in Stanley (1979). There Stanley defined a refined or *extended h-vector* of a completely balanced sphere (see sections 3.11 and 6.6). This extended h-vector is the Hilbert function of the Stanley–Reisner ring with respect to a fine grading. Unfortunately, no analogue of the Macaulay theorem (Theorem 3.1) is known for the extended h-vector. The extended h-vector of a shellable completely balanced sphere can also be calculated from a shelling.

3.8. Dimension four

Four is the lowest dimension for which f-vectors of polytopes have not been characterized, and the same is true of flag vectors. In the late sixties and early seventies, f-vectors of 4-polytopes were studied intensively, resulting in the characterizations of the projections of f-vectors of 4-polytopes onto all pairs of components (Grünbaum 1967, Barnette and Reay 1973, Barnette 1974). By duality only four projections need be determined.

Theorem 3.9. (i) *There exists a 4-polytope P with $(f_0(P), f_1(P)) = (f_0, f_1)$ if and only if f_0 and f_1 are integers satisfying*

$$10 \leqslant 2f_0 \leqslant f_1 \leqslant \binom{f_0}{2}$$

and $(f_0, f_1) \notin \{(6,12),(7,14),(8,17),(10,20)\}$.

(ii) *There exists a 4-polytope P with $(f_0(P), f_2(P)) = (f_0, f_2)$ if and only if f_0 and f_2 are integers satisfying*

$$\tfrac{1}{2}\left(2f_0 + 3 + (8f_0+9)^{1/2}\right) \leqslant f_2 \leqslant f_0^2 - 3f_0$$

$f_2 \neq f_0^2 - 3f_0 - 1$, and $(f_0, f_2) \notin \{(6,12),(6,14),(7,13),(7,15),(8,15),(8,16),(9,16),(10,17),(11,20),(13,21)\}$.

(iii) *There exists a 4-polytope P with $(f_0(P), f_3(P)) = (f_0, f_3)$ if and only if f_0 and f_3 are integers satisfying*

$$5 \leqslant f_0 \leqslant f_3(f_3-3)/2 \quad \text{and} \quad 5 \leqslant f_3 \leqslant f_0(f_0-3)/2.$$

(iv) *There exists a 4-polytope P with $(f_1(P), f_2(P)) = (f_1, f_2)$ if and only if f_1 and f_2 are integers satisfying*

$$f_2 \geqslant \tfrac{1}{2}f_1 + \lceil \sqrt{f_1 + \tfrac{9}{4}} + \tfrac{1}{2} \rceil + \tfrac{1}{2},$$

the pair (f_1, f_2) does not equal $(i^2 - 3i - 1, \frac{1}{2}(i^2 - i))$ for any i, and $(f_1, f_2) \notin \{(12,12),(14,13),(14,14),(15,15),(16,15),(17,15),(17,16),(18,18),(20,17),(21,19),(23,20),(24,20),(26,21)\}$.

The original proofs that the inequalities are satisfied by the f-vectors of 4-polytopes used arguments about chains of faces. The introduction of flag vectors thus simplifies the exposition of the proofs.

Here are the inequalities known to hold for the flag vectors of all 4-polytopes (Bayer 1987).

Theorem 3.10. *Let f_0, f_1, f_2 and f_{02} be flag numbers of a 4-polytope. Then*

(i) $f_{02} - 3f_2 \geqslant 0$,
(ii) $f_{02} - 3f_1 \geqslant 0$,
(iii) $f_{02} - 3f_2 + f_1 - 4f_0 + 10 \geqslant 0$,
(iv) $6f_1 - 6f_0 - f_{02} \geqslant 0$,
(v) $f_0 - 5 \geqslant 0$,
(vi) $f_2 - f_1 + f_0 - 5 \geqslant 0$,
(vii) $2(f_{02} - 3f_2) + f_1 \leqslant \binom{f_0}{2}$,
(viii) $2(f_{02} - 3f_1) + f_2 \leqslant \binom{f_2 - f_1 + f_0}{2}$,
(ix) $f_{02} - 4f_2 + 3f_1 - 2f_0 \leqslant \binom{f_0}{2}$,
(x) $f_{02} + f_2 - 2f_1 - 2f_0 \leqslant \binom{f_2 - f_1 + f_0}{2}$.

The linear inequalities (i) and (v) are obvious; inequalities (ii) and (vi) are their duals. Inequality (iii) is Kalai's inequality (Theorem 3.8). The proofs of the other inequalities are in Bayer (1987).

The projections of the inequalities of Theorem 3.10 onto f-vectors give all the inequalities appearing in Theorem 3.9, and one other linear inequality. This inequality was conjectured by Barnette (1972).

Theorem 3.11. *If (f_0, f_1, f_2, f_3) is the f-vector of a 4-polytope, then*

$$-3f_2 + 7f_1 - 10f_0 + 10 \geqslant 0.$$

In Bayer (1987) the tightness of the inequalities is analyzed.

3.9. Intersection homology

In the early parts of this section we discussed how Stanley proved the McMullen conditions using the interpretation of the h-vector as homology ranks for a toric variety. The toric variety is defined for a rational polytope even if it is not simplicial, but the singularities can be worse in this case. Different geometric realizations of the same combinatorial type of simplicial polytope have toric varieties with the same homology ranks. This is no longer the case for nonsimplicial polytopes. McConnell (1989) showed that the toric varieties associated with two different (rational) geometric realizations of the rhombododecahedron have different regular homology ranks.

The middle perversity intersection homology Betti numbers of a toric variety are, however, combinatorial invariants. A formula for these Betti numbers in terms

of the face lattice of the associated polytope was given independently by Bernstein, Khovanskiĭ and MacPherson (see Stanley 1987a). Stanley generalized these Betti number formulas to Eulerian posets. He and several other authors have applied them to the study of convex polytopes (Bayer and Klapper 1991, Kalai 1988b, Stanley 1987a). Here the coefficients h_i are defined to agree with the original h-vector in the simplicial case; h_i represents the rank of the $(2d-2i)$th intersection homology group.

For any d-polytope P are defined a *generalized h-vector* $(h_0, h_1, \ldots, h_d) \in \mathbb{N}^{d+1}$, with generating function $h(P,t) = \sum_{i=0}^{d} h_i t^{d-i}$, and *g-vector* $(g_0, g_1, \ldots, g_{\lfloor d/2 \rfloor}) \in \mathbb{N}^{\lfloor d/2 \rfloor + 1}$, with generating function $g(P,t) = \sum_{i=0}^{\lfloor d/2 \rfloor} g_i t^i$, related by $g_0 = h_0$ and $g_i = h_i - h_{i-1}$ for $1 \leqslant i \leqslant \lfloor \frac{1}{2} d \rfloor$. The generalized h-vector and g-vector are defined by the recursion

(i) $g(\emptyset, t) = h(\emptyset, t) = 1$, and

(ii) $$h(P,t) = \sum_{\substack{G \text{ face of } P \\ G \neq P}} g(G,t)(t-1)^{d-1-\dim G}$$

We summarize the known results on generalized h-vectors in the following theorem.

Theorem 3.12. (i) *The generalized h-vector of a rational polytope is the sequence of middle perversity intersection homology Betti numbers of the associated toric variety.*

(ii) *The generalized h-vector of a simplicial polytope is the same as its h-vector; hence it satisfies the McMullen conditions.*

(iii) *For any d-polytope,* $h_0 = 1$ *and for all* i, $0 \leqslant i \leqslant d$, $h_i = h_{d-i}$.

(iv) *For any rational d-polytope and any* i, $0 \leqslant i \leqslant \frac{1}{2} d - 1$, $h_i \leqslant h_{i+1}$.

(v) *There is a linear function from* $\mathbb{Q}^{e_d}$ *to* $\mathbb{Q}^{d+1}$ *that takes the flag vector of any d-polytope to the generalized h-vector of the polytope.*

Comments on the theorem. (i) Note that Stanley's definition of the generalized h-vector makes sense for all polytopes (or, more generally, for Eulerian posets), but the toric variety is only defined for rational polytopes. (ii) For the toric variety associated with a simplicial polytope, the middle perversity intersection homology is isomorphic to the ordinary homology. (iii) Stanley gives a purely combinatorial proof of this duality result. Thus it applies even when the toric variety is not defined. (iv) The proof of the unimodality of the generalized h-vector depends on the existence of primitive homology groups for the toric variety, hence on the rationality hypothesis. The first inequality holds trivially for all polytopes; the second is Kalai's Theorem 3.8. It is not known whether the other inequalities hold for nonrational polytopes. (iii), (v) According to (v), (iii) gives approximately $\frac{1}{2} d$ linear equations satisfied by the flag vectors of all d-polytopes. Theorem 3.7, however, gives all such linear equations, of which there are an exponential (in d) number. This suggests that the generalized h-vector should be embedded in a larger set of parameters equivalent to the flag vector. Kalai found one such set.

3.10. Kalai's convolutions

Kalai (1988b) creates new parameters by applying the g-vector transformation simultaneously to different intervals of a polytope. First we give his extended definition of the g-vector as a length $d+1$ vector. Write F_d for the real vector space with basis $\{f_S: S \subseteq \{0,1,\ldots,d-1\}\}$. An element of F_d (a "linear form") defines a real-valued function on the set of d-polytopes.

For $d \geqslant 0$ and $0 \leqslant i \leqslant d$, define linear functions G_i^d on the set of d-polytopes by the following recursion: $G_0^d(P) = 1$ for all P and

$$G_i^d(P) = (-1)^i \binom{d+1}{i} + \sum_{j=0}^{i-1} \sum_{s=0}^{i-j-1} (-1)^j \binom{d-i+j-s+1}{j} \times \sum_{\substack{F \text{ a face of } P \\ \dim F = i-j+s-1}} G_s^{i-j+s-1}(F).$$

The functions G_i^d have a natural representation as elements of F_d.

Kalai defines the *convolution* operation on the set $F = \bigcup_{d \geqslant 0} F_d$ as follows: for $S \subseteq \{0,1,\ldots,d-1\}$ and $T \subseteq \{0,1,\ldots,e-1\}$, let $f_S * f_T = f_{S \cup \{d\} \cup T+(d+1)}$, where by $T+(d+1)$ is meant $\{t+d+1: t \in T\}$. Thus extending linearly, the convolution of any linear forms is a linear form. Note that if P is a $(d+e+1)$-polytope, $S \subseteq \{0,1,\ldots,d-1\}$, and $T \subseteq \{0,1,\ldots,e-1\}$, then the convolution of f_S and f_T is

$$f_S * f_T(P) = \sum_{\substack{F \text{ face of } P \\ \dim F = d}} f_S(F) f_T(P/F).$$

Now consider the subset M_d of F_d, $M_d = \{G_{i_1}^{d_1} * G_{i_2}^{d_2} * \cdots * G_{i_k}^{d_k} : k \geqslant 1, 0 < i_j \leqslant d_j \text{ for } 1 \leqslant j \leqslant k-1, 0 \leqslant i_k \leqslant d_k, \text{ and } k-1+\sum d_j = d\}$.

Theorem 3.13 (Kalai 1988b). (i) $|M_d| = 2^d$.

(ii) *M_d is a basis for F_d.*

(iii) *Every element of M_d defines a nonnegative function on the set of rational d-polytopes.*

(iv) *Exactly $2^d - e_d$ of these functions are the zero function.*

Thus Kalai's convolutions extend the g-vector of a polytope to a vector that completely encodes the flag vector and incorporates the generalized Dehn–Sommerville equations. Note that among these are the equations $G_i^d = 0$ for $i > \lfloor \frac{1}{2} d \rfloor$, which are exactly the equations $h_i = h_{d-i}$.

The nonnegativity of the convolutions provides linear inequalities on the flag numbers. For any d-polytope P and its dual P^*, define $\overline{G}_i^d(P) = G_i^d(P^*)$. It is easy to see that $\overline{G}_i^d$ can be represented by an element of F_d, and that it is a nonnegative function on rational d-polytopes. Kalai suggests the following conjecture.

Conjecture 3.14. Every linear inequality on the flag numbers of polytopes is equivalent to the nonnegativity of some nonnegative linear combination of convolutions of the G_i^d and the $\overline{G}_i^d$.

3.11. Other parameters

There are two other sets of parameters that extend the generalized h-vector of a d-polytope. The first is to be found in Stanley (1987a), where he first introduced the generalized h-vector. Let $T \subseteq \{0,1,\ldots,d-1\}$, $\overline{T} = \{0,1,\ldots,d-1\} \setminus T$. The functions ϕ_T are certain linear forms in the flag numbers.

Theorem 3.15 (Stanley 1987a). $x^{d+1}\phi_T(P,1/x) = \phi_{\overline{T}}(P,x)$.

When $T = \emptyset$ the theorem gives the equations $h_i = h_{d-i}$ on the generalized h-vector of a d-polytope. Presumably, as T ranges over all subsets of $\{0,1,\ldots,d-1\}$, the equations of the theorem are equivalent to the generalized Dehn–Sommerville equations.

Another set of parameters comes not so directly from the h-vector. This is the *cd index* of a polytope. It was introduced by Fine, and derives from the *extended h-vector* of a polytope, mentioned in section 3.7.

Suppose $(f_S(P))_{S\subseteq\{0,1,\ldots,n-1\}} \in \mathbb{N}^{2^n}$ is the flag vector of an n-polytope P. The *extended h-vector* of P is the vector $(h_S(P))_{S\subseteq\{0,1,\ldots,n-1\}} \in \mathbb{N}^{2^n}$ given by

$$h_S(P) = \sum_{T\subseteq S} (-1)^{|S\setminus T|} f_T(P).$$

This transformation is invertible:

$$f_S(P) = \sum_{T\subseteq S} h_T(P).$$

The extended h-vector can be given by a generating function in the algebra of polynomials in the noncommuting variables a and b. For $S \subseteq \{0,1,\ldots,n-1\}$, write $w_i = a$ if $i \notin S$ and $w_i = b$ if $i \in S$; let $w_S = w_0 w_1 \cdots w_{n-1}$. The generating function for the extended h-vector is then $h(P) = \sum h_S(P) w_S$, the sum being over all $S \subseteq \{0,1,\ldots,n-1\}$. Now it turns out that for every polytope P, $h(P)$ is in the subalgebra generated by $c = a+b$ and $d = ab+ba$. This fact is essentially equivalent to the generalized Dehn–Sommerville equations. The coefficients of the cd words in $h(P)$ form a vector of length e_n; this is called the *cd index* of P.

The cd index can be computed recursively via a shelling of the polytope. Like the flag vector and the extended h-vector, the cd indices of a polytope and its dual have a simple relationship: the cd index of P^* is obtained by reversing every cd word in the cd index of P. Fine conjectured that the coefficients of the cd index of any polytope are nonnegative. This was proved for quasisimplicial polytopes and their duals by Purtill (1991). In Bayer and Klapper (1991) equations relating the cd index with the generalized h-vector are computed. They are used to give another proof of a result on the g-vectors of dual polytopes. This result was originally proved by Kalai, directly from the definition of the generalized h-vector.

Theorem 3.16. *Suppose n is even and let P and P^* be a pair of dual n-polytopes. Then $g_{n/2}(P) = g_{n/2}(P^*)$.*

3.12. Algebraic shifting

In his characterization of f-vectors of nerves of convex sets, Kalai (1984) considered an algebraic structure analogous to the Stanley–Reisner ring of a simplicial complex and defined the notion of algebraic shifting, which we briefly describe.

Let Δ be a simplicial complex with vertices $v_1, v_2, \ldots, v_n$, and let V be an n-dimensional vector space over a field k. Choose a basis $\{e_1, e_2, \ldots, e_n\}$ of V. Form the exterior algebra $\bigwedge V$ over V. Construct the ideal I_Δ spanned by $\{e_{i_1} \wedge e_{i_2} \wedge \cdots \wedge e_{i_s}\colon \{v_{i_1}, v_{i_2}, \ldots, v_{i_s}\} \notin \Delta\}$. Form the quotient algebra $\bigwedge(\Delta) = \bigwedge V / I_\Delta$. From another basis of V that is "generic" with respect to the first, another simplicial complex Δ' can be obtained that has the same f-vector as Δ but is "shifted" with respect to an appropriate partial order on subsets of $\{v_1, v_2, \ldots, v_n\}$.

Studying $\bigwedge(\Delta)$ has led to an impressive array of other results, including a simpler proof of the Upper Bound Theorem (Alon and Kalai 1985) and a complete characterization of (f-vector, Betti sequence) pairs for simplicial complexes (Björner and Kalai 1988).

3.13. Rigidity and stress

Let $v_1, v_2, \ldots, v_n$ be the vertices of a d-polytope $P \subset \mathbb{R}^d$. An *infinitesimal motion* of the vertices is a set of vectors $u_1, u_2, \ldots, u_n$ such that $\mathrm{d}(\|(v_i + tu_i) - (v_j + tu_j)\|^2)/\mathrm{d}t = 0$ when $t = 0$ for all edges $v_i v_j$. Equivalently, $\langle (v_i - v_j), (u_i - u_j) \rangle = 0$ for all edges. Dehn (1916) proves that there are no infinitesimal motions for convex simplicial 3-polytopes apart from the rigid motions, and this extends to arbitrary $d \geqslant 3$ (see Whiteley 1984).

Theorem 3.17. *For $d \geqslant 3$, simplicial convex d-polytopes are infinitesimally rigid.*

Given d-polytope P with vertex set V and edge set E, a *stress* is an assignment of numbers λ_{uv} to its edges $uv \in E$ such that the following equilibrium conditions hold:

$$\sum_{\{u \in V:\ vu \in E\}} \lambda_{vu}(v - u) = o \quad \text{for all } v \in V.$$

The set of stresses forms a vector space, called the *stress space.*

Dehn's Theorem is equivalent to the following.

Theorem 3.18. *Let P be a simplicial convex d-polytope, $d \geqslant 3$. Then the dimension of the stress space equals $f_1 - df_0 + \binom{d+1}{2}$. In particular, if $d = 3$ there are no nontrivial stresses.*

Note that the above dimension equals $h_2 - h_1$. Kalai (1987) observes that the nonnegativity of this quantity immediately yields another proof of the Lower Bound Theorem for simplicial polytopes, and he also extends the theorem to nonsimplicial polytopes (Theorem 3.8) and larger classes of complexes. See chapter 1.7 in this Handbook for more background on stress and rigidity in convexity.

Kalai's observation provides a new proof of the nonnegativity of $h_2 - h_1$ for simplicial d-polytopes, $d \geqslant 3$. On the other hand, the connection between the Stanley–Reisner ring and toric varieties proves $h_i - h_{i-1} \geqslant 0$ for all i, $1 \leqslant i \leqslant \lfloor \frac{1}{2}d \rfloor$. This foreshadowed a stronger connection between the Stanley–Reisner ring and stresses (Lee 1990, 1992).

Let $R_\Delta = R_1 \oplus R_2 \oplus \cdots$ be the Stanley–Reisner ring of any simplicial $(d-1)$-complex Δ with n vertices. For $\theta_1, \theta_2, \ldots, \theta_d \in R_1$ define $B = R_\Delta/(\theta_1, \theta_2, \ldots, \theta_d)$ and give B the grading induced by R_Δ. Then R_Δ is Cohen–Macaulay if and only if there exist $\theta_1, \theta_2, \ldots, \theta_d \in R_1$ such that $B = B_0 \oplus B_1 \oplus \cdots \oplus B_d$, where $h_i = \dim B_i$ for all i, $0 \leqslant i \leqslant d$. Regarding multiplication by θ_i as a linear map in R_Δ and dualizing, this condition can be reformulated.

Given $\theta_i = \sum_{j=1}^n a_{ji} x_j$ for all i, $1 \leqslant i \leqslant d$, define $v_j \in \mathbb{R}^d$ for all j, $1 \leqslant j \leqslant n$ by $v_j = (a_{j1}, a_{j2}, \ldots, a_{jd})$. For monomial $m = x_1^{r_1} x_2^{r_2} \cdots x_n^{r_n}$, define $\operatorname{supp}(m) = \{x_i : r_i \neq 0\}$. For all i, $0 \leqslant i \leqslant d$, let M_i be the set of monomials of degree i. Then for all i, $1 \leqslant i \leqslant d$, a *linear i-stress* on Δ (with respect to $v_1, \ldots, v_n$) is a homogeneous polynomial $b = \sum_{m \in M_i} b_m m$ such that the following two conditions hold:

(i) $b_m = 0$ if $\operatorname{supp}(m) \notin \Delta$.

(ii) $\sum_{j=1}^n b_{mx_j} v_j = o$ for every $m \in M_{i-1}$.

An *affine i-stress* is defined in exactly the same way, with the additional condition that $\sum_{j=1}^n b_{mx_j} = 0$ for every $m \in M_{i-1}$. (This condition corresponds to the conjecture that $\omega = x_1 + x_2 + \cdots + x_n$ is a hyperplane section.) A linear or affine 0-stress is defined to be any real number.

Let L_i (A_i) be the vector space of all linear (affine) i-stresses. In particular, L_1 (A_1) is the set of all linear (affine) relations on the points v_j.

Theorem 3.19 (Lee 1990). (i) *For simplicial $(d-1)$-complex Δ, R_Δ is Cohen–Macaulay if and only if there exist $v_1, v_2, \ldots, v_n$ such that $\dim L_i = h_i$ for all i, $0 \leqslant i \leqslant d$.*

(ii) *Suppose Δ is a simplicial $(d-1)$-sphere. If $\dim A_i = h_i - h_{i-1}$ for all i, $1 \leqslant i \leqslant \lfloor \frac{1}{2}d \rfloor$, then $h = (h_0, h_1, \ldots, h_d)$ satisfies the McMullen conditions.*

Condition (i) can be used for another proof that shellable simplicial complexes are Cohen–Macaulay. Using the fact that all simplicial pl-spheres can be obtained from the boundary of a simplex by a sequence of bistellar operations (see section 4.3), (i) leads to a more elementary proof that simplicial pl-spheres are Cohen–Macaulay.

In the case that Δ is the boundary complex of a simplicial d-polytope $P \subset \mathbb{R}^d$ containing o in its interior, the vectors v_j can be taken to be the actual vertices of P. Then A_2 is easily seen to be isomorphic to the classical stress space, and it can be shown that (i) holds.

The polar of P sheds light on some very interesting relationships among stress, Dehn's Theorem, Minkowski's Theorem, the Brunn–Minkowski Theorem, the Hard Lefschetz Theorem, the McMullen conditions, mixed volumes, and the Alexandrov–Fenchel inequalities. See Filliman (1992) and Lee (1992).

4. Gale transforms and diagrams

4.1. Introduction

Given d-polytope $P \subset \mathbb{R}^d$ with vertices $v_1, v_2, \ldots, v_n$, list these vectors as columns of a matrix and append a row of ones to obtain the $(d+1) \times n$ matrix A. Consider the nullspace of A, the space of all affine relations $(\lambda_1, \lambda_2, \ldots, \lambda_n)$ on the set of vertices; i.e., $\sum_{i=1}^n \lambda_i v_i = o$ and $\sum_{i=1}^n \lambda_i = 0$. Let $\overline{A}$ be an $(n-d-1) \times n$ matrix whose rows form a basis for this space, and denote its columns by $\overline{v}_1, \overline{v}_2, \ldots, \overline{v}_n$. This collection $\overline{V}$ of points in $\mathbb{R}^{n-d-1}$ is a *Gale transform* of V. The natural correspondence between vertices v_i and transform points $\overline{v}_i$ extends to a correspondence between subsets of V and subsets of $\overline{V}$. The key property of Gale transforms is the following.

Theorem 4.1. *Let X be a proper subset of V. Then* $\operatorname{conv}(X)$ *is a face of P if and only if* $o \in \operatorname{relint}(\operatorname{conv}(\overline{V} \setminus \overline{X}))$.

A collection $\overline{W} = \{\overline{w}_1, \overline{w}_2, \ldots, \overline{w}_n\}$ is a *Gale diagram* of V if it satisfies the property given in the above theorem. For example, Gale diagrams can be obtained by scaling the points in a Gale transform independently by positive amounts.

Gale transforms and diagrams are recognized for their usefulness in establishing results when n is not much larger than d, but even in the general case they are helpful tools. Both Grünbaum (1967) and McMullen and Shephard (1971) contain good introductions. For a more extensive survey of results than is presented here, refer to McMullen (1979). Note that the toric variety discussed in section 3.3 is the result of an algebraic analogue of the Gale transform.

Suppose one is given $V = \{v_1, v_2, \ldots, v_n\} \subset \mathbb{R}^d$ such that $o \in \operatorname{int}(\operatorname{conv}(V))$, and a spherical simplicial $(d-1)$-complex Δ on these n points. The next result characterizes when Δ can be realized convexly by positively scaling the points of V

Theorem 4.2 (Shephard 1971). *There exist positive numbers $\lambda_1, \lambda_2, \ldots, \lambda_n$ such that Δ is isomorphic to the boundary complex of* $\operatorname{conv}(\{\lambda_1 v_1, \lambda_2 v_2, \ldots, \lambda_n v_n\})$ *if and only if*

$$\bigcap \operatorname{relint}(\operatorname{conv}(\overline{V} \setminus \overline{F})) \neq \emptyset,$$

where $\overline{V}$ is a Gale transform of V and the intersection is taken over all facets F of Δ.

A direct translation of a theorem of Bárány via Gale transforms yields the following theorem; see Bigdeli (1991).

Theorem 4.3. *Given $V = \{v_1, v_2, \ldots, v_n\} \subset \mathbb{R}^d$ in linearly general position such that $o \in \operatorname{int}(\operatorname{conv}(V))$. Let $m(V)$ be the maximum number of facets of a polytope obtained by scaling the points in V independently by positive amounts. Then $m(V) \geqslant (1/k^k)\binom{n}{k} + O(n^{k-1})$, where $k = n - d$.*

Another illustration of the usefulness of affine relations in the structure of polytopes is the indecomposability characterization of Smilansky (1987).

4.2. Polytopes with few vertices

A d-polytope P with n vertices has a Gale transform of dimension $n-d-1$, and Sturmfels (1988) uses *affine* transforms to further reduce the dimension by one. Gale transforms have gained the (perhaps undeserved reputation) of being useful only when P has *few* vertices, i.e., when $n \leqslant d+3$. In this case the Gale transform is at most two-dimensional and is easier to analyze. This feeling is supported by the fact that there are many results that are easier to prove for polytopes with few vertices, and that quite often these results fail when $n = d+4$. Here is a small sample.

Theorem 4.4. *Every $(d-1)$-dimensional pl-sphere with at most $d+3$ vertices is polytopal. However, there exists a simplicial 3-sphere with 8 vertices that is not polytopal.*

The first part of the theorem is due to Mani (1972) for simplicial spheres and Kleinschmidt (1976b) for nonsimplicial spheres. The nonpolytopal sphere is discussed in Grünbaum (1967) and is due to Brückner, (who, however, thought it was polytopal). Kleinschmidt (1977) proves an analogue of the above theorem for $(d-1)$-spheres with at most $2d$ vertices possessing combinatorial involutions with no fixed points.

The following theorem is stated in Grünbaum (1967).

Theorem 4.5 (Perles). *For every d-polytope with at most $d+3$ vertices there exists a combinatorially equivalent d-polytope P such that every automorphism of the boundary complex of P is induced by a geometric symmetry of P.*

Theorem 4.6. *For every d-polytope with at most $d+3$ vertices and $\varepsilon > 0$ there exists a combinatorially equivalent polytope with rational vertices such that each vertex is a distance at most ε from the corresponding vertex of P. However, there exists a 6-polytope that is combinatorially equivalent to no polytope with rational vertices.*

The first part of the above theorem is due to Perles and stated in Grünbaum (1967), in which also a nonrational 8-polytope with 12 vertices discovered by Perles is described. Sturmfels (1987a) constructs lower-dimensional examples (see section 6.5).

One says that a facet F of a polytope P can be *preassigned* if, given any polytope F' combinatorially equivalent to F, there is a polytope P' combinatorially equivalent to P having F' as a facet corresponding to F.

Theorem 4.7 (Kleinschmidt 1976a). *If a d-polytope has at most $d+3$ vertices, then the shape of each of its facets can be preassigned. However, there exists a 4-polytope with 8 vertices such that the shape of one of its facets cannot be preassigned.*

Gale transforms can be used to count the number of different combinatorial types of polytopes with few vertices.

Theorem 4.8. *There are* $\lfloor \frac{1}{4}d^2 \rfloor$ *different combinatorial types of d-polytopes with* $d+2$ *vertices. There are* $\lfloor \frac{1}{2}d \rfloor$ *different combinatorial types of simplicial d-polytopes with* $d+2$ *vertices.*

Perles (see Grünbaum 1967) and Lloyd (1970) count the number of simplicial and general d-polytopes with $d+3$ vertices, respectively.

4.3. Subdivisions and triangulations

Given a d-polytope P with vertex set V, a *subdivision* of P is a collection Δ of d-polytopes such that (1) for every $Q_1, Q_2 \in \Delta$, $Q_1 \cap Q_2$ is a common face (possibly empty) of both Q_1 and Q_2; (2) P is the union of the polytopes in Δ; and (3) for every $Q \in \Delta$, the vertex set of Q is a subset of V. A subdivision Δ is a *triangulation* provided every member of Δ is a d-simplex.

The Dehn–Sommerville equations force relations among the h-vectors of a triangulation Δ, of the collection of its boundary faces $\partial\Delta$, and of the collection of its interior faces Δ°, which also hold for general simplicial balls.

Theorem 4.9 (McMullen and Walkup 1971). *Suppose* Δ *is a simplicial d-ball. Then*

(i) $h_i(\Delta) - h_{d-i+1}(\Delta) = h_i(\partial\Delta) - h_{i-1}(\partial\Delta)$ *for all* i, $1 \leqslant i \leqslant d$.

(ii) $h_i(\Delta) = h_{d-i+1}(\Delta^\circ)$ *for all* i, $0 \leqslant i \leqslant d+1$.

A simple corollary is mentioned in Lee (1991a).

Corollary 4.10. *Suppose* Δ *is a simplicial d-ball. Then* $f_d(\Delta) \geqslant h_{\lfloor d/2 \rfloor}(\partial\Delta)$.

There is a wide range of results on subdivisions and triangulations; we mention only a few that relate to Gale transforms.

Theorem 4.11 (McMullen 1979). *Let* P *be a d-polytope with vertex set* $V = \{v_1, v_2, \ldots, v_n\}$, *let* $\overline{V} \subset \mathbb{R}^{n-d-1}$ *be a Gale transform of* V, *and let* $\overline{z} \in \mathbb{R}^{n-d-1}$. *Consider the collection* Δ *of d-polytopes* $\operatorname{conv}(S)$ *such that* $S \subseteq V$ *and* $o \in \operatorname{relint}(\operatorname{conv}((\overline{V} \cup \{\overline{z}\}) \setminus \overline{S}))$. *Then* Δ *is a subdivision of* P.

Subdivisions and triangulations of the above form are called *regular*. An equivalent way to generate regular subdivisions of a d-polytope $P \subset \mathbb{R}^d$ with vertex set $\{v_1, v_2, \ldots, v_n\}$ is to choose real numbers $\lambda_1, \lambda_2, \ldots, \lambda_n$ and form the convex hull Q of $\{(v_1, \lambda_1), (v_2, \lambda_2), \ldots, (v_n, \lambda_n)\}$. Projecting the facets in the upper hull of Q into $(\mathbb{R}^d, 0)$ yields a regular subdivision of P.

We need some definitions. The *link* of a face F in a simplicial complex Δ is the set of faces G of Δ such that $F \cap G = \emptyset$ and $F \cup G$ is a face of Δ. The *stellar subdivision* of F in Δ is obtained by removing F from Δ and adding a new vertex

v along with all simplices formed from v, a proper subface of F, and a face in the link of F. A *bistellar* operation on a simplicial sphere is a certain combination of a stellar subdivision and inverse stellar subdivision at the same site. The following is then a consequence of a line shelling of the upper hull of Q.

Theorem 4.12 (Ewald 1978). *The boundary complex of any simplicial d-polytope can be obtained from that of a d-simplex by a sequence of bistellar operations, such that at each intermediate stage the simplicial complex is polytopal.*

Pachner (1990) proves that a simplicial complex is a pl-sphere if and only if it is obtainable from the boundary complex of a simplex by a sequence of bistellar operations. In fact, Pachner shows that simplicial pl-spheres are precisely boundaries of shellable balls. However, the undecidability result mentioned in section 2.10 implies that for simplicial pl-spheres, unlike for polytopes, no upper bound on the number of such operations needed can be computed from the given simplicial complex. On the other hand, properties of the h-vector imply that if Σ is a $(d-1)$-dimensional simplicial pl-sphere, then at least $h_{\lfloor d/2 \rfloor}(\Sigma)$ bistellar operations are necessary, generalizing Corollary 4.10.

Let P be a d-polytope, F be a facet of P with supporting hyperplane H, and x be a point in $\mathbb{R}^d$. Then x is *beyond* F if x and the interior of P lie in opposite open halfspaces determined by H, and *beneath* F if x and the interior of P lie in the same open halfspace determined by H. Now suppose that x lies beyond precisely one facet F of P and beneath all the others. Let $\mathscr{C}$ denote the boundary complex of P excluding F. Projecting $\mathscr{C}$ onto H centrally through x results in a polyhedral $(d-1)$-complex isomorphic to $\mathscr{C}$, called a *Schlegel diagram* of P. See Grünbaum (1967). Gale transforms provide a characterization of Schlegel diagrams.

Theorem 4.13 (Sturmfels 1986). *Let Δ be a subdivision of a convex $(d-1)$-polytope P, this time allowing the vertex set V of Δ to be a strict superset of the vertex set W of P, but on the other hand requiring that proper faces of P be faces of Δ. Then Δ is the Schlegel diagram of some d-polytope if and only if*

$$\left(\bigcap \operatorname{relint}(\operatorname{pos}(\overline{V} \setminus \overline{F}))\right) \cap \left(-\operatorname{relint}(\operatorname{pos}(\overline{V} \setminus \overline{W}))\right) \neq \emptyset,$$

where $\overline{V}$ is a Gale transform of V and the first intersection is taken over all $(d-1)$-polytopes F of Δ.

Not all subdivisions Δ of a $(d-1)$-polytope P satisfying the conditions in the first sentence of the theorem are Schlegel diagrams, even when P is two-dimensional. However, if P is two-dimensional, then Δ is isomorphic to the Schlegel diagram of some 3-polytope (Grünbaum 1967).

For polytopes with few vertices all subdivisions are regular, but this is not true in general.

Theorem 4.14 (Lee 1991a). *If P is a d-polytope with at most $d+3$ vertices, then every subdivision of P is regular. However, there exist 3-polytopes with 7 vertices that possess nonregular triangulations.*

On the other hand, the collection of all regular subdivisions of a given polytope has a nice structure, discovered by Gelfand, Kapranov, and Zelevinskiĭ in connection with their work on generalized discriminants and determinants.

Theorem 4.15 (Gelfand, Kapranov and Zelevinskiĭ 1989). *The collection of all regular subdivisions of a given d-polytope P with n vertices, partially ordered by refinement, is combinatorially equivalent to the boundary complex of some* $(n-d-1)$*-polytope Q.*

This polytope Q is called the *secondary polytope* of P, and can be constructed as follows. Let $\{v_1, v_2, \ldots, v_n\}$ be the set of vertices of P, and for any triangulation Δ of P (whether regular or not) define $z(\Delta) = (z_1, z_2, \ldots, z_n) \in \mathbb{R}^n$ by $z_i = \sum \mathrm{vol}(F)$, where the sum is taken over all d-simplices F containing v_i. Then $Q = \mathrm{conv}(\{z(\Delta) : \Delta \text{ is a triangulation of } P\})$. In particular, the vertices of Q correspond to the regular triangulations of P. Alternate constructions are described in Billera, Filliman and Sturmfels (1990). For example, Q can be expressed as a discrete or continuous Minkowski sum of polars of polytopes corresponding to various translates of a Gale transform of P. Generalizations of the secondary polytope appear in Billera and Sturmfels (1992).

In section 3.4 we learned that the h-vector of a simplicial d-polytope P satisfies $h_k \geqslant h_{k-1}$ for all k, $1 \leqslant k \leqslant \lfloor \frac{1}{2}d \rfloor$. From the Lower Bound Theorem 2.11 we also know that $h_2 = h_1$ if and only if P is stacked. In general, P is called *k-stacked* if P has a triangulation such that there is no interior face of dimension less than $d-k$. The McMullen conditions and Theorem 4.9 imply that if P is k-stacked, then $h_k = h_{k-1}$. McMullen and Walkup conjectured the converse as part of their Generalized Lower Bound Conjecture.

Conjecture 4.16 (McMullen and Walkup 1971). Let P be a simplicial d-polytope. For all k, $1 \leqslant k \leqslant \lfloor \frac{1}{2}d \rfloor$, $h_k = h_{k-1}$ if and only if P is $(k-1)$-stacked.

The following is a consequence of the construction in Billera and Lee (1981b).

Theorem 4.17 (Kleinschmidt and Lee 1984). *Let P be a simplicial d-polytope such that* $h_k = h_{k-1}$ *for some k,* $1 \leqslant k \leqslant \lfloor \frac{1}{2}d \rfloor$. *Then there exists a k-stacked simplicial d-polytope Q with the same h-vector as P.*

A few cases of the conjecture have been resolved by interpreting the differences $h_k - h_{k-1}$ as winding numbers in Gale transforms, but as a whole the conjecture remains unresolved.

Theorem 4.18 (Lee 1991b). *Conjecture 4.16 holds if* $f_0 \leqslant d+3$ *or if* $k < f_0/(f_0-d)$.

Suppose P is a d-polytope. Some particular regular triangulations of P, called *pulling* triangulations, can be obtained by first ordering the vertex set of P, $V = \{v_1, v_2, \ldots, v_n\}$. For every j-face F of P let $v(F)$ denote the vertex of smallest index that is in F. A *full flag* of P is a chain of faces $F_0 \subset F_1 \subset F_2 \subset \cdots \subset F_d = P$ such

that $\dim F_j = j$ for all j, $0 \leqslant j \leqslant d$, and $v(F_j) \notin F_{j-1}$ for all j, $1 \leqslant j \leqslant d$. Associate with each full flag the simplex $\operatorname{conv}(\{v(F_0), v(F_1), \ldots, v(F_d)\})$. Then these simplices are all d-dimensional and together determine a triangulation of P. This idea appears in Hudson (1969) in a more general context and has been frequently rediscovered in various guises.

Write $J(P,t) = 1 + \sum_{n=1}^{\infty} i(P,n)t^n$, where $i(P,n)$ denotes the number of points $x \in P$ such that $nx \in \mathbb{Z}^d$. Stanley uses pulling triangulations to prove the following, which strengthens earlier work of Ehrhart and McMullen.

Theorem 4.19 (Stanley 1980a). *Suppose every vertex of P is integral. Then $J(P,t) = W(P,t)/(1-t)^{d+1}$ where $W(P,t)$ is a polynomial of degree at most d with nonnegative integer coefficients.*

For integral d-polytope P, call an ordering σ of its vertices *compressed* if every d-simplex in the associated pulling triangulation has volume $1/d!$. P itself is *compressed* if every ordering σ is compressed. For example, the unit d-cube is compressed.

Theorem 4.20 (Stanley 1980a). *If P is an integral d-polytope with compressed ordering σ, then*

$$i(P,n) = \sum_{i=0}^{d} \binom{n-1}{i} f_i(\Delta),$$

and $W(P,t) = h_0(\Delta) + h_1(\Delta)t + \cdots + h_d(\Delta)t^d$, where Δ is the pulling triangulation induced by σ.

(Compare the above with eq. (3.1) of section 3.2.)

Corollary 4.21 (Stanley 1980a). *If P is a compressed integral d-polytope and σ is an ordering, then the f-vector of the triangulation induced by σ depends only on P, not on σ.*

The Cartesian product $T^m \times T^n$ of two simplices of any dimension is compressed, but unlike the d-cube has the property that *all* of its triangulations have the same f-vector, whether induced by an ordering as above or not; see Billera, Cushman, and Sanders (1988). Any polytope with this property is called *equidecomposable*. A *weakly neighborly* polytope is one for which every set of $k+1$ vertices is contained in a face of dimension at most $2k$. $T^m \times T^n$ is also weakly neighborly. The following results are drawn from Bayer (1990) and Stanley (1991).

Theorem 4.22. (i) *If P is a rational d-polytope and Δ is any subdivision of P, then $h(\Delta) \geqslant h(P)$, where h is the generalized h-vector, and P is regarded as a d-complex.*

(ii) *If P is a weakly neighborly d-polytope, then P is equidecomposable, and $h(\Delta) = h(P)$ for any triangulation Δ of P.*

(iii) *If P is a rational weakly neighborly d-polytope and Δ is any subdivision of P, then $h(\Delta) = h(P)$.*

(iv) *If P is a rational d-polytope and $h(\Delta) = h(P)$ for all triangulations of P, then P is weakly neighborly.*

Bayer (1990) uses Gale transforms to characterize equidecomposable and weakly neighborly d-polytopes with at most $d + 3$ vertices. All 2-polytopes are weakly neighborly. A 3-polytope is weakly neighborly iff it is a prism over a triangle or a pyramid over a polygon. A simplicial polytope is weakly neighborly iff it is a simplex or an even-dimensional neighborly polytope. Other classes of weakly neighborly polytopes include pyramids over weakly neighborly polytopes, subpolytopes of weakly neighborly polytopes, and Lawrence polytopes (section 4.5).

4.4. Oriented matroids

Matroids and oriented matroids provide a setting for a combinatorial abstraction of convexity, including analogues of Carathéodory's Theorem, Radon's Theorem, Helly's Theorem, and the Hahn–Banach Theorem; generalizations of point and hyperplane arrangements, convex polytopes, and Gale transforms; as well as a combinatorial derivation of linear programming. See chapter 2.5 in this Handbook or Björner et al. (1991) for details. Sturmfels (1986) discusses the relationship between oriented matroids and Gale transforms.

A *matroid* M is a pair $(E, \mathscr{C})$ consisting of a finite set E and a collection of nonempty incomparable subsets $\mathscr{C}$ of E (called the *circuits* of M) satisfying the following property: $C_1, C_2 \in \mathscr{C}$, $e \in C_1 \cap C_2$, and $e' \in C_1 \setminus C_2$ implies the existence of $C_3 \in \mathscr{C}$ such that $e' \in C_3 \subseteq (C_1 \cup C_2) \setminus \{e\}$. For example, the collection of supports of elementary vectors in a subspace V of $\mathbb{R}^n$ forms the circuits of a matroid on $E = \{1, 2, \ldots, n\}$. Given matroid (E, C), let $\mathscr{C}^*$ be the collection of all minimal nonempty subsets C^* of E such that $|C^* \cap C| \neq 1$ for all $C \in \mathscr{C}$. Then $M^* = (E, \mathscr{C}^*)$ is also a matroid, called the *dual* of M, and members of $\mathscr{C}^*$ are called the *cocircuits* of M. In the preceding example, $\mathscr{C}^*$ is the collection of supports of elementary vectors in $V^\perp$. When a matroid can be derived from a subspace of $\mathbb{R}^n$, it is called *representable* (over $\mathbb{R}$). So matroids provide a generalization of unsigned patterns of dependences of finite collections of vectors.

Oriented matroids, on the other hand generalize *signed* patterns of dependences. Let E be a finite set. A *signed set* X is an ordered pair (X^+, X^-) of disjoint subsets of E. The set $\underline{X} = X^+ \cup X^-$ is called the *underlying set* of X, and by $-X$ is meant (X^-, X^+). Two signed sets X, Y are said to be *orthogonal* if either their underlying sets are disjoint or else both $(X^+ \cap Y^+) \cup (X^- \cap Y^-) \neq \emptyset$ and $(X^+ \cap Y^-) \cup (X^- \cap Y^+) \neq \emptyset$. Let $\mathcal{O}, \mathcal{O}^*$ be two collections of signed sets in E. Then $M = (E, \mathcal{O})$ and $M^* = (E, \mathcal{O}^*)$ is a *dual pair of oriented matroids* provided the following conditions hold:

(i) The underlying sets of the members of $\mathcal{O}$ (respectively, $\mathcal{O}^*$) form the circuits (respectively, cocircuits) of a matroid (called the *underlying matroid* $\underline{M}$).

(ii) $X \in \mathcal{O}$ (respectively, $\mathcal{O}^*$) implies $-X \in \mathcal{O}$ (respectively, $\mathcal{O}^*$).

(iii) If $X, Y \in \mathcal{O}$ (respectively, $\mathcal{O}^*$) and $\underline{X} = \underline{Y}$, then $Y = \pm X$.

(iv) If $X \in \mathcal{O}$ and $Y \in \mathcal{O}^*$, then X and Y are orthogonal.

As in the unoriented case, members of $\mathcal{C}$ are referred to as the *circuits* of M and members of $\mathcal{C}^*$ as the *cocircuits*.

So, for example, the signed supports of elementary vectors in a pair $V, V^\perp$ of dual subspaces of $\mathbb{R}^n$ form the circuits and cocircuits of an oriented matroid on $\{1,2,\ldots,n\}$. This is true in particular for the null spaces of A and $\overline{A}$ used in section 4.1 to define the Gale transform of a polytope, and suggests the following definition.

Let $\mathcal{C}^*$ be the set of cocircuits of an oriented matroid M on E. A cocircuit X is *positive* if $X^- = \emptyset$, and M is *acyclic* if it has positive cocircuits. The *facets* of M are the sets of the form $E \setminus \underline{C}$ where C is a positive cocircuit of E. The *faces* of M are the intersections of finite numbers of facets of M. The collection of faces of M, ordered by inclusion, forms a lattice, called the *(Las Vergnas) face lattice* of M. Vertices of the lattice are faces of M that have rank one in the underlying matroid $\underline{M}$, and M is a *matroid polytope* provided all one-element subsets of E are vertices. In the case that M is derived from a dual pair of subspaces of $\mathbb{R}^n$, M is called *representable* (over $\mathbb{R}$) and the face lattice of M is isomorphic to the face lattice of a convex polytope.

Oriented matroids can alternately be defined by assigning sign patterns to ordered bases of a matroid (maximal subsets of E containing no circuit) which would not be inconsistent with the Plücker–Graßman relations should the oriented matroid be representable. All bases have the same cardinality, called the *rank* of the matroid. Much of the usefulness of oriented matroids in the theory of convex polytopes is related to realizability results, some of which are also discussed in section 6.5. Bokowski and Sturmfels (1987) developed algorithms based upon oriented matroids to test polytopality of spheres, which, combined with other results, has led to a complete classification of simplicial neighborly 3-spheres with 10 vertices into polytopal and nonpolytopal spheres.

We mention a few other results that are obtainable by matroid techniques. All of them are quoted from Björner et al. (1991).

Theorem 4.23 (Las Vergnas 1986). *For $d \geqslant 2$ there exists a set of $(d+1)(d+2)/2$ points in general position in $\mathbb{R}^d$ which is not projectively equivalent to the set of vertices of any d-polytope.*

The next theorem can be found in Cordovil and Duchet (1987).

Theorem 4.24 (Duchet and Roudneff). *Let n, d be integers with $n \geqslant d+1 \geqslant 3$. There exists an integer $N = N(n,d)$ such that every set of N points in general position in affine d-space contains the n vertices of a cyclic d-polytope. Moreover, cyclic polytopes are the only combinatorial types of polytopes with this property.*

Theorem 4.25 (Sturmfels 1987b). *Suppose the convex hull of $\{v_1, v_2, \ldots, v_n\}$ in $\mathbb{R}^d$ is combinatorially equivalent to $C(n,d)$. Then there exists a curve C containing $v_1, v_2, \ldots, v_n$ such that every hyperplane in $\mathbb{R}^d$ meets C in at most d points.*

Theorem 4.26 (Grünbaum 1967). (i) *Let M be a neighborly rank $2k+1$ matroid polytope with $n \leqslant 2k+3$. Then M is isomorphic to $C(n, 2k+1)$.*

(ii) *For all $k \geqslant 2$ there is a representable neighborly rank $2k+1$ matroid polytope with $2k+4$ vertices which is not isomorphic to a cyclic polytope.*

A matroid polytope M is called *rigid* if M is determined by its face lattice.

Theorem 4.27 (Shemer 1982). *Every neighborly rank $2k+1$ matroid polytope is rigid.*

Theorem 4.28 (Bokowski and Sturmfels 1987). *There exists a triangulated 3-sphere (the Barnette sphere) with 8 vertices which is not the face lattice of any matroid polytope.*

Theorem 4.29. *For all $d \geqslant 4$ and $n \geqslant d+5$ there exists a rank $d+1$ matroid polytope whose face lattice is a nonpolytopal simplicial $(d-1)$-sphere with n vertices.*

4.5. Lawrence polytopes

Bayer and Sturmfels (1990) is a good reference for the Lawrence construction, which provides an effective method for lifting matroid representability results into polytope realizability results. See also Björner et al. (1991). A polytope is called a *Lawrence polytope* if it has a centrally symmetric Gale transform. Let $\{\bar{v}_1, \bar{v}_2, \ldots, \bar{v}_n\}$ be a Gale transform of convex d-polytope P with n vertices. Let $\Lambda(P)$ be a polytope whose Gale transform is $\{\bar{v}_1, \bar{v}_2, \ldots, \bar{v}_n, -\bar{v}_1, -\bar{v}_2, \ldots, -\bar{v}_n\}$. Then $\Lambda(P)$ is a Lawrence $(d+n)$-polytope with $2n$ vertices which contains P as a quotient polytope. Hence every polytope is the quotient of a Lawrence polytope.

This construction can be extended in a natural way to oriented matroids M, so that if M is a rank r oriented matroid on n elements, then $\Lambda(M)$ is a rank $n+r$ oriented matroid on $2n$ elements. It turns out that the combinatorial structure of the face lattice of $\Lambda(M)$ depends strongly upon the matroid structure of M.

Theorem 4.30 (Bokowski and Sturmfels 1987). *Every Lawrence matroid polytope $\Lambda(M)$ is rigid. That is, $\Lambda(M)$ is uniquely determined by its face lattice.*

Theorem 4.31 (Bayer and Sturmfels 1990). *For any oriented matroid M, the f-vector and the flag vector of the Lawrence polytope $\Lambda(M)$ are functions of the underlying matroid $\underline{M}$. In the generic case where M is a uniform oriented matroid of rank r on n points, the f-vector and the flag vector can be expressed as functions depending only on n and r.*

Theorem 4.32 (Billera and Spellman Munson 1984). (i) *The face lattice of $\Lambda(M)$ is polytopal if and only if M is representable.*

(ii) *There exists a rank 12 matroid polytope with 16 vertices whose face lattice is not polytopal.*

(iii) *There exists a rank 12 matroid polytope M with 16 vertices which does not have a polar; i.e., there is no matroid polytope having a face lattice anti-isomorphic to the face lattice of M.*

Although oriented matroids capture and generalize the combinatorial flavor of convex polytopes very well, it is curious that the existence of polars does not generalize. Other realization problems will be mentioned in section 6.5.

Theorem 4.33 (Bayer and Sturmfels 1990). (i) *The convex realization space of the face lattice of $\Lambda(M)$ is homotopy equivalent to the realization space of M.*

(ii) *There exist two combinatorially equivalent Lawrence 19-polytopes P and Q that are not isotopy equivalent.*

We will return to the isotopy problem in section 6.4.

5. Graphs of polytopes

5.1. Introduction

In this section we mention only briefly the major results on the graphs of polytopes. The general subject is covered in Grünbaum (1975), and Klee and Kleinschmidt (1987) is an extensive survey of the d-step conjecture. See also Klee and Kleinschmidt (1991).

5.2. Steinitz's Theorem

The *graph* of a polytope is the set of vertices and edges of the polytope. The earliest major result on the graphs of polytopes was Steinitz's Theorem on the graphs of 3-polytopes. Recall that a graph is *d-connected* if every pair of vertices is connected by d internally disjoint paths or, equivalently, the removal of any $d-1$ vertices leaves a connected graph with at least two vertices. A graph is *planar* if it can be represented in $\mathbb{R}^2$ by a set of distinct points (for the vertices) and a set of curves (for the edges) which intersect only at common vertices.

Theorem 5.1 (Steinitz 1922). *A graph G is the graph of a 3-polytope P if and only if it is planar and 3-connected.*

See also Steinitz and Rademacher (1934) and Grünbaum (1967).

Mani (1971) showed that P can be chosen so that the isometries of P correspond to the automorphisms of the graph. Steinitz's Theorem has many other consequences on the realizability of 2-dimensional complexes (see section 6.5).

The connectivity condition can be extended to polytopes of arbitrary dimension.

Theorem 5.2 (Balinski 1961). *The graph of every d-polytope is d-connected.*

Another connectivity criterion is due to Klee (see Grünbaum 1967). For a polytope P with at least $n+1$ vertices, define $s_n(P)$ to be the maximum number of

connected components that can remain when n vertices are removed from the graph of P, and $s(n,d) = \max\{s_n(P)\colon P \text{ is a } d\text{-polytope}\}$.

Theorem 5.3 (Klee 1964). *For all n and d,*

$$s(n,d) = \begin{cases} 1, & \text{if } n \leqslant d-1, \\ 2, & \text{if } n = d, \\ f_{d-1}(C(n,d)), & \text{if } n \geqslant d+1. \end{cases}$$

Klee then used the above result to show that for every d there is a graph of a d-polytope that cannot be the graph of an e-polytope for any $e \neq d$. Moreover, such *dimensionally unambiguous* graphs can have arbitrarily large numbers of vertices.

The *k-skeleton* of a d-polytope is the polyhedral complex generated by the k-faces of the polytope. Thus the 1-skeleton is the graph of the polytope. The following result, originally observed for graphs, was proved in general by Grünbaum.

Theorem 5.4 (Grünbaum 1965). *The k-skeleton of a d-polytope* $(1 \leqslant k \leqslant d-1)$ *contains a subdivision of the k-skeleton of the d-simplex.*

When does the graph of a polytope determine the entire combinatorial structure of the polytope? Steinitz's Theorem implies that it does when the polytope is 3-dimensional. In general this is not the case, however. For example all neighborly polytopes with the same number of vertices have the same graph, namely, the complete graph. (For neighborly polytopes, see Grünbaum 1967.) Recall that Theorem 2.9 of section 2.6 shows that the graph of a *simple* polytope uniquely determines its face lattice. The same is true for zonotopes (Minkowski sums of finite collections of line segments) (Björner, Edelman and Ziegler 1990). These cases are quite special. In general, one cannot reconstruct a d-polytope even from its $(d-3)$-skeleton. The $(d-2)$-skeleton does, however, determine the combinatorial type of any d-polytope, and the same is true for the $\lfloor \frac{1}{2}d \rfloor$-skeleton of a simplicial d-polytope. See Grünbaum (1967) and Perles (1970).

5.3. Hamiltonian circuits

An important issue in graph theory is the existence of Hamiltonian circuits (closed paths containing all vertices), which began with Hamilton's observations about circuits on the dodecahedron. It is natural, therefore, to ask whether graphs of polytopes have Hamiltonian circuits. Already in the last century Kirkman knew of polytopes without Hamiltonian circuits. Tutte (1946) found the first example of a simple polytope without a Hamiltonian circuit. His example is a 3-polytope; it is still open whether all simple polytopes of dimension higher than three have Hamiltonian circuits. The following classes of 3-polytopes are known, however, to have Hamiltonian circuits: those with 4-connected graphs; simple 3-polytopes with at most 36 vertices; simple 3-polytopes with at most two types of 2-faces: 3-gons,

4-gons or 6-gons; and simplicial 3-polytopes with maximum vertex degree six. See Grünbaum (1967) and Klee and Kleinschmidt (1991). A Hamiltonian path in a graph is a spanning tree with maximum degree two. Thus, the following theorem is related in a natural way.

Theorem 5.5 (Barnette 1966). *The graph of every 3-polytope has a spanning tree with maximum degree 3.*

Given integers n, d such that $n > d \geqslant 3$, is there any simple d-polytope with n vertices that has a Hamiltonian circuit? A more basic question is: Does there even exist a simple d-polytope with n vertices? Of course, the relation $f_1 = dn/2$ shows that n must be even if d is odd. The McMullen conditions provide other restrictions on the possible values of n. Perles and Prabhu (see Prabhu 1991) address both questions simultaneously.

Theorem 5.6 (Perles and Prabhu). (i) *There is a constant c such that for all $n > cd^{5/2}$ (where n is even if d is odd) there exists a simple d-polytope with n vertices that has a Hamiltonian circuit.*

(ii) *For all d there exists an integer $n(d) = O(d^{5/2})$ (where $n(d)$ is even if d is odd) such that there is no simple d-polytope with $n(d)$ vertices.*

5.4. Diameter

Probably the most intensively studied question on polytope graphs is that of the diameter (see Klee and Kleinschmidt 1987). The *diameter* $\delta(P)$ of a polytope P is the maximum length of a shortest edge-path between two vertices of the polytope. Write $\Delta(d,n)$ for the maximum diameter of d-polytopes with n facets.

Much of the interest in diameters of polytopes comes from the search for efficient linear programming algorithms. If the function $\Delta(d,n)$ is not bounded by a polynomial in d and n, then no edge-following linear programming algorithm with arbitrary starting vertex could have polynomial complexity. A proof of a polynomial bound for $\Delta(d,n)$ might, on the other hand, suggest an efficient linear programming algorithm.

In computing Δ we can restrict the class of polytopes.

Theorem 5.7. (i) *For $n > d \geqslant 2$, $\Delta(d,n)$ is the maximum diameter of a simple d-polytope with n facets.*

(ii) *For $n \geqslant 2d \geqslant 4$, $\Delta(d,n)$ is realized as the distance between two vertices not on a common facet, in a simple d-polytope with n facets.*

Following are equivalent conjectures concerning the diameter.

Conjecture 5.8. (i) Hirsch Conjecture (Dantzig 1963). For $n > d \geqslant 2$, $\Delta(d,n) \leqslant n - d$.

(ii) d-Step Conjecture (Dantzig 1963). For $d \geqslant 2$, $\Delta(d,2d) = d$.

(iii) Nonrevisiting Conjecture (Klee and Walkup 1967). Between any two vertices of a simple polytope, there is a path that does not revisit any facet.

Remarkably little is known about these conjectures. They can fail for spheres (Walkup 1978) and for unbounded pointed polyhedra (Klee and Walkup 1967), though no d-polyhedron with n facets is known with diameter greater than $2n-2d$. For many values of d and n there are d-polytopes with n facets with diameter equal $n-d$ (e.g., d-cubes). Barnette (1969) found an upper bound for Δ for general d and n that is exponential in d, and then Larman (1970) obtained a better (but still exponential) bound. This was recently improved by Kalai (1990b) and further by Kalai and Kleitman (1992). The lower bound below is due to Adler (1974).

Theorem 5.9. *For* $n > d \geqslant 2$,
(i) $\Delta(d,n) \leqslant \min\{n2^{d-3}, n^{2+\log d}\}$.
(ii) $\Delta(d,n) \geqslant \lfloor (n-d) - (n-d)/\lfloor 5d/4 \rfloor \rfloor - 1$.

Precise values of $\Delta(d,n)$ are known only for small d and n (see Klee and Kleinschmidt 1987).

Theorem 5.10. (i) $\Delta(d,n) = \lfloor (d-1)n/d \rfloor - d + 2$, *if* $d \leqslant 3$ *or* $n \leqslant d+4$.
(ii) $\Delta(4,9) = \Delta(4,10) = 5$, $\Delta(5,9) = 4$, *and* $\Delta(5,11) = 6$.

Note that when $n - d \leqslant d \leqslant 5$, $\Delta(d,n)$ attains the Hirsch bound, $\Delta(d,n) = n-d$.

The final result of this section is an easy consequence of a particular class of regular triangulations.

Theorem 5.11 (Lee 1991a). *Every simple d-polytope P with n facets can be realized as a facet of a simple $(d+1)$-polytope Q with $n+1$ facets such that the diameter of Q does not exceed $2n-2d$.*

The proof of the theorem implies that, given any linear program with bounded feasible region and arbitrary starting vertex, by augmenting the problem with one variable and one constraint an optimum vertex can be reached in a linear number of pivots.

6. Combinatorial structure

6.1. Introduction

Finally we come to the broadest problem, that of classifying the combinatorial types of all polytopes. This section deals with asymptotic formulas for the number of combinatorial types, isotopy, and the realization of types of spheres as polytopes, rational polytopes and spherical polytopes. We discuss equifacetted polytopes, barycentric subdivisions, and the numbers of n-gons in a 3-polytope. We cover only briefly some topics discussed in greater depth in Klee and Kleinschmidt (1991).

6.2. Regular polytopes

There has been some speculation that the regular or Platonic solids were the primary motivation for the *Elements* of Euclid. The symmetry group of P is *flag transitive* if, for any two flags of P, there exists a symmetry which maps one flag onto the other. The polytope P is *regular* if its symmetry group is flag transitive. It is *semiregular* if it is not regular, but each of its facets is regular and the symmetry group of P is vertex transitive. See Coxeter (1963).

In three dimensions, Euler's relation easily implies that the only regular polytopes are the five Platonic solids. The three-dimensional semiregular polytopes consist of the thirteen Archimedean solids, together with the two infinite classes of the prisms and the antiprisms.

Theorem 6.1. *Up to rigid motion and scaling, there are five regular 3-polytopes and six regular 4-polytopes. For all dimensions $d > 4$ there are only three regular d-polytopes: the d-cube, the d-crosspolytope, and the regular d-simplex.*

6.3. Numbers of combinatorial types

See Klee and Kleinschmidt (1991) for a good summary of this topic. We briefly mention what is found there.

The combinatorial types of 3-polytopes are well-understood. By Steinitz's Theorem, classifying 3-polytopes is equivalent to classifying 3-connected planar graphs. Exact numbers of combinatorial types of 3-polytopes with at most 22 edges are given in Duivestijn and Federico (1981). Asymptotic formulas in terms of number of edges, number of vertices or numbers of facets and vertices are summarized in Bender (1987); these are the product of several people's work over 25 years.

Theorem 6.2. *The number of combinatorial types of 3-polytopes with $i+1$ vertices and $j+1$ facets is asymptotically*

$$\frac{1}{972ij(i+j)}\binom{2i}{j+3}\binom{2j}{i+3}.$$

Gale diagrams have been used to count the combinatorial types of simplicial or arbitrary polytopes with $d+2$ or $d+3$ vertices; see section 4.2. A d-polytope is *neighborly* if every set of $\lfloor \frac{1}{2}d \rfloor$ vertices forms a face. For even dimension d, every neighborly d-polytope with $d+2$ or $d+3$ vertices is equivalent to a cyclic polytope. The face lattice of a cyclic polytope is specified by Gale's evenness criterion (see Grünbaum 1967). For odd d, the numbers of types of neighborly d-polytopes with at most $d+3$ vertices are given in McMullen (1974). The numbers of d-polytopes and neighborly d-polytopes with $d+4$ vertices are known only for $d \leqslant 4$.

Asymptotic upper and lower bounds for the number of combinatorial types of d-polytopes have been brought surprisingly close in the last few years. Let $c(n,d)$

be the number of types of d-polytopes with n vertices, $c_s(n,d)$ the number of these that are simplicial.

Theorem 6.3.

$$\left(\frac{n-d}{d}\right)^{nd/4} \leqslant c_s(n,d) \leqslant c(n,d) \leqslant (n/d)^{d^2 n\left(1+O\left(\frac{1}{\log(n/d)}+\frac{\log\log(n/d)}{d\log(n/d)}\right)\right)}$$

and

$$c(n,d) \leqslant 2^{n^3+O(n^2)}.$$

This is based on Shemer's (1982) estimate of the number of simplicial neighborly polytopes, and Goodman and Pollack's (1986) application to configurations of Betti number estimates by Milnor, with improvements by Alon (1986). Comparing this with Theorem 2.14, we see that there are many more spheres than polytopes.

6.4. Isotopy

Björner et al. (1991) is a good reference for the isotopy problem. Steinitz (Steinitz and Rademacher 1934) proved the isotopy property for 3-polytopes. Represent a polytope in $\mathbb{R}^3$ with n vertices in some order, $v_1, v_2, \ldots, v_n$ by a length $3n$ vector $(v_1, v_2, \ldots, v_n)$.

Theorem 6.4. *Suppose P and Q are two combinatorially equivalent polytopes in $\mathbb{R}^3$ with n vertices in corresponding order. Then there is a path in $\mathbb{R}^{3n}$ connecting P with either Q or the reflection of Q, such that each point of the path represents a polytope combinatorially equivalent to P.*

This isotopy property fails in dimensions higher than 3 (see section 4.5). Already in $\mathbb{R}^4$ there is a simplicial polytope with 10 vertices for which it fails. The polytope was first described in Bokowski, Ewald and Kleinschmidt (1984); that it fails the isotopy property is due to Mnëv (1988) and Bokowski and Guedes de Oliveira (1990); for a good account see Bokowski and Sturmfels (1989).

A group of mathematicians in Leningrad (Viro 1988) has worked on a more general study of realization spaces. The *realization space* of a combinatorial type of polytope is the set of vector representations of all realizations of the combinatorial type. A polytope satisfies the isotopy property if its realization space is connected. Mnëv showed that the general situation is very far from the isotopy property.

Theorem 6.5 (Mnëv 1988). *For any semi-algebraic variety V there exists a convex polytope whose realization space is homotopy equivalent to V.*

6.5. Realization

Steinitz's Theorem says that every polyhedral complex homeomorphic to the 2-dimensional sphere can be realized convexly, i.e., as the boundary of a convex

3-polytope. In section 4.2 we saw that the same was true for $(d-1)$-dimensional pl-spheres with at most $d+3$ vertices. There are two simplicial and forty nonsimplicial 3-spheres with 8 vertices that cannot be realized as boundaries of polytopes; see Altshuler and Steinberg (1985) for a complete list of these. Klee and Kleinschmidt (1991) summarize the numbers of polytopal and nonpolytopal spheres of various types.

Tarski's decision method (Grünbaum 1967) implies the following.

Theorem 6.6. *There is a decision procedure to determine whether a given complex is polytopal or not.*

However, this method is far from efficient. Though the techniques of oriented matroids (section 4.4) are much better, still they cannot handle spheres with large numbers of vertices relative to the dimension. Further, Sturmfels proved that the polytopality of a sphere (of dimension at least 5) cannot be determined locally.

Theorem 6.7 (Sturmfels 1987a)**.** *For infinitely many different nonpolytopal 5-spheres, every subcomplex on fewer vertices can be extended to the boundary of a polytope.*

In section 4.2 it was noted that all d-polytopes with at most $d+3$ vertices can be realized rationally, i.e., with vertices in $\mathbb{Q}^d$. This is also the case for all simplicial polytopes, and also for all 4-polytopes with at most 8 vertices (Altshuler and Steinberg 1985). Another consequence of Steinitz's Theorem is that the same holds for 3-polytopes.

Theorem 6.8. *All combinatorial types of 3-polytopes can be realized with rational vertices.*

We do not know if this continues to hold for 4- and 5-polytopes, but it fails in higher dimensions.

Theorem 6.9 (Sturmfels 1987c)**.** *The decidability of the existence of a rational realization of a lattice as the face lattice of a polytope is equivalent to the decidability of the existence of rational roots of polynomials with integer coefficients.*

Theorem 6.10. (i) *All d-polytopes can be realized in $\mathbb{A}^d$, where $\mathbb{A}$ is the field of real algebraic numbers.*

(ii) *For every proper subfield Φ of $\mathbb{A}$, there is a 6-polytope not realizable in Φ^6.*

Part (i) of the above theorem is due to Lindström (1971), and part (ii) to Sturmfels (1987a).

There are two main questions concerning the realization of facets of polytopes. One asks whether the shape of a facet F of a d-polytope P can be preassigned. In section 4.2 we saw this was always the case if P had no more than $d+3$ vertices. This holds also when $d=3$.

Theorem 6.11 (Barnette and Grünbaum 1970). *The shape of any facet of any 3-polytope can be preassigned.*

A polytope is *equifacetted* if all its facets are of the same combinatorial type. A d-polytope is *facet-forming* (or a *d-facet*) if it is the combinatorial type of the facets of some equifacetted $(d+1)$-polytope; otherwise it is a *nonfacet*. It is easy to classify 2-polytopes using the condition on p-vectors (see section 6.7). The triangle, quadrilateral and pentagon are facet-forming, while the n-gon is a nonfacet for every $n \geqslant 6$. For higher dimensions no classification is known. See Perles and Shephard (1967), Barnette (1980), and Schulte (1985). Any d-polytope with $d+2$ vertices is facet-forming. Facet-forming polytopes with large numbers of vertices are also known.

Among the equifacetted polytopes are the (combinatorially) regular polytopes, for which vertex-figures are also all of the same combinatorial type. The icosahedron is not yet classified as facet-forming or a nonfacet; all other regular 3-polytopes are known to be facet-forming. The simplex and cube are the only facet-forming regular 4-polytopes (Kalai 1990a). For general d, the d-simplex and d-cube are, of course, facet-forming; the d-crosspolytope is a nonfacet for $d \geqslant 4$.

6.6. *Barycentric subdivisions*

Let P be a convex d-polytope. Perform a stellar subdivision of P with respect to each of its proper faces in succession, going from high- to low-dimensional faces. The result is the *barycentric subdivision* of P, $\Delta(P)$, a simplicial d-polytope with vertices corresponding to proper faces of P and faces corresponding to chains of faces of P. As a simplicial complex this is also known as the order complex of the face lattice of P (with least and greatest elements omitted). If each vertex of $\Delta(P)$ is labeled with the dimension of the corresponding face of P, then each facet of $\Delta(P)$ has exactly one vertex with each of the labels $0, 1, \ldots, d-1$.

A simplicial $(d-1)$-complex Δ is *balanced* if, under some labeling of vertices, each facet has one vertex of each label. By connectedness, the labeling is essentially unique in the case that Δ is a balanced sphere. Not all balanced simplicial d-polytopes arise as barycentric subdivisions of polytopes. Those that are barycentric subdivisions of regular CW spheres have been characterized using flag vectors (Bayer 1988). The definition of flag vectors is extended to balanced simplicial complexes as follows. Let Δ be a balanced simplicial $(d-1)$-complex with vertices labeled by $0, 1, \ldots, d-1$. For each subset $S \subseteq \{0, 1, \ldots, d-1\}$ let $f_S(\Delta)$ be the number of simplices in Δ whose vertices have exactly the labels of S. Note that for a d-polytope P, and the labeling described above for the vertices of $\Delta(P)$, $f_S(\Delta(P))$ agrees with the flag number $f_S(P)$. Thus the numbers $f_S(\Delta(P))$ satisfy the generalized Dehn–Sommerville equations (see section 3.7). For an arbitrary balanced simplicial polytope Δ the numbers $f_S(\Delta)$ do not necessarily satisfy these equations.

For the characterization of barycentric subdivisions we must go beyond polytopes. Regular CW spheres share some of the combinatorial properties of poly-

topes. For the definition and motivation for regular CW spheres, see Björner (1984).

Theorem 6.12 (Bayer 1988). *For any simplicial polytope Δ, Δ is the barycentric subdivision of a regular CW sphere if and only if, for some vertex labeling of Δ, the numbers $f_S(\Delta)$ satisfy the generalized Dehn–Sommerville equations.*

It is an open problem to distinguish barycentric subdivisions of polytopes (or even of polyhedral spheres) among those of regular CW spheres. We know of no example of a pl regular CW sphere whose barycentric subdivision is not polytopal.

6.7. p-vectors of 3-polytopes

We conclude our survey with a simple question of the combinatorics of 3-polytopes that remains open. What are the possible distributions of n-gons as facets of 3-polytopes? A partial answer was given a hundred years ago by Eberhard (see Grünbaum 1967). For a 3-polytope P and an integer $n \geqslant 3$, let $p_n(P)$ be the number of P's facets that are n-gons. The sequence $(p_n)_{n \geqslant 3}$ is the *p-vector* of P. Call a sequence $p = (p_3, p_4, p_5, p_7, p_8, \ldots)$ a *reduced (simple) p-vector* if some value of p_6 can be inserted to get the p-vector of some (simple) 3-polytope.

Theorem 6.13 (Eberhard 1891). *A sequence $(p_3, p_4, p_5, p_7, p_8, \ldots)$ of natural numbers, only finitely many nonzero, is a reduced (simple) p-vector if and only if $\sum_{n \geqslant 3}(6-n)p_n$ is even and is at least (equal to) 12.*

The values of p_6 that complete a given reduced simple p-vector are now fairly well understood. The following result is from Jendrol' (1983); it incorporates contributions by various people. For $p = (p_3, p_4, p_5, p_7, p_8, \ldots)$ a reduced simple p-vector, write $\sigma = \sum_{j \neq 6} p_j$, $\rho = \sum_{j \not\equiv 0 \pmod 3} p_j$, and $\mathcal{P}(p) = \{p_6\colon p_6 \text{ completes } p \text{ to a simple } p\text{-vector}\}$.

Theorem 6.14. *Let p be a reduced simple p-vector.*

(i) *If $\rho \leqslant 2$, then for some integer m, $\mathcal{P}(p)$ contains every integer $k \geqslant m$ of the same parity as σ and no integer of the opposite parity.*

(ii) *If $\rho \geqslant 3$, then for some integer m, $\mathcal{P}(p)$ contains every integer $k \geqslant m$.*

In both cases m can be chosen to be at most $\sum_{j \neq 6} j p_j$.

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CHAPTER 2.4

Polyhedral Manifolds

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Contents

1. Preliminaries 537
1.1. Introduction and history 537
1.2. Definitions 538
2. Embeddability 540
2.1. The basic problems 540
2.2. Embeddability 540
3. Minimality properties of polyhedra 542
3.1. Minimal number of vertices 542
3.2. Minimal number of faces and edges 543
3.3. Other minimality properties 544
4. Combinatorially regular polyhedra and related topics 545
4.1. Combinatorially regular polyhedra 545
4.2. Equivelarity 546
4.3. Polyhedra with vertex transitive symmetry group 547
4.4. Other realizations of regular maps 547
References 548

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1. Preliminaries

1.1. Introduction and history

This articles deals with polyhedral manifolds or briefly polyhedra which are compact 2-manifolds built up of planar convex or nonconvex polygons as, e.g., the Möbius–Császár-torus; a precise definition will be given in the next section. So this topic is closely related to combinatorial topology, PL-topology, combinatorics and graph theory. But the underlying topology is mainly elementary, and the basic ideas and constructions are similar to those of the theory of convex polytopes. So it is appropriate to subsume this chapter to convexity in a wider sense. There are some relations and small overlaps with the chapters by Bokowski (2.5), Bayer and Lee (2.3), Heil and Martini (1.11), and Schulte (3.5) in this Handbook.

The history of polyhedral manifolds began with classical convex polyhedra as pyramids and some of the Platonic solids in the time of ancient Sumer and Egypt; for the history see, e.g., Coxeter (1973), Malkevitch (1988), and Saffaro (1976). The first nonconvex polyhedra were discovered by Renaissance artists who considered them as highly symmetric objects of art. Paolo Uccello discovered about 1420, almost 200 years before Kepler, the great stellated dodecahedron, one of the four Kepler–Poinsot-solids, a polyhedral manifold of genus 4 with self-intersections. (Polyhedra with self-intersections will only be considered in section 4.4.) The Kepler–Poinsot-solids are also called star-polyhedra and they are the most famous of this type. In 1500–1503 Leonardo da Vinci drew the edge-skeletons of the Platonic solids and some Archimedean solids as polyhedral manifolds; these are perhaps the first and obviously the most famous non-intersecting polyhedral manifolds of positive genus. Leonardo's polyhedra influenced various artists from Renaissance to our century, e.g., S. Dalí and M.C. Escher (see Senechal 1988), but until recently no mathematicians. In 1861 Möbius (cf. Möbius 1886) described the combinatorial structure of a vertex-minimal torus, whose polyhedral realization was given by Császár in 1949. At the beginning of this century, some particular polyhedral manifolds were investigated (cf., e.g., Brückner 1900); we mention in particular Petrie's and Coxeter's regular skew polyhedra (see Coxeter 1937). But a more systematic investigation of nonconvex polyhedra began only around 1970 with papers by Altshuler (1971, 1972), Banchoff (1965), Barnette (1973), and Grünbaum (1967). They were influenced by purely geometric and topological problems, but in particular also by Heawood's map coloring problem and symmetry and regularity problems arising from regular maps.

Today's research on polyhedral manifolds is roughly reflected by the headings of the sections and subsections of this article. We do not mention relations to computational geometry, flexible polyhedra or applications to tool designs, finite element methods and robotics and refer to Chazelle and Palios (1990) and the literature quoted there (cf., e.g., Bokowski and Sturmfels 1989). In this field definitions and notations are not canonical, so it needs some space to introduce them carefully which will be done in the next subsection.

1.2. Definitions

We give some abstract definitions in full generality, although we shall consider later mainly the case of polyhedral 2-manifolds in $\mathbb{R}^3$. So the reader may skip this section in a first reading and come back to it later.

A *polyhedral complex* (or *convex cell complex*) Γ is a finite set of non-empty convex polytopes in $\mathbb{R}^d$ such that

(i) if $P \in \Gamma$ and if F is a face of P, then $F \in \Gamma$, and

(ii) if $P_1, P_2 \in \Gamma$ and $P_1 \cap P_2 \neq \emptyset$, then $P_1 \cap P_2$ is a face of P_1 and of P_2.

The elements of Γ are called *cells*. The *underlying space* of Γ is $\|\Gamma\| = \bigcup \Gamma$, with the topology induced as a subset of $\mathbb{R}^d$. If each cell in Γ is a simplex, then Γ is called a (*geometric*) *simplicial complex*. The *dimension* of a cell equals the dimension of its affine hull.

The *dimension* of Γ is $\dim \Gamma = \max_{P \in G} (\dim P)$.

The 0-dimensional cells are called *vertices*, the 1-dimensional cells *edges*, the maximal cells (w.r. to inclusion) are the *facets*. Γ is called *pure* if all facets have the same dimension.

The *face poset* $P(\Gamma)$ of Γ is the poset of all cells of Γ ordered by inclusion. Two polyhedral complexes are called *combinatorially equivalent* or of the *same combinatorial type* if their face posets are isomorphic.

A polyhedral complex Γ_1 is a *subdivision* of another such complex Γ_2 if $\|\Gamma_1\| = \|\Gamma_2\|$ and each cell of Γ_1 is a subset of some cell of Γ_2. Every polyhedral complex can be simplicially subdivided without introducing new vertices.

A *polyhedral manifold* is a polyhedral complex Γ such that there exists a simplicial subdivision of Γ which is a combinatorial manifold in the sense of PL-topology (cf., e.g., Glaser 1970).

The *boundary complex* of any polytope (i.e., set of all proper faces) is a polyhedral manifold.

A *polyhedron* P is a subset of $\mathbb{R}^d$ such that there exists a polyhedral complex Γ with $\|\Gamma\| = P$. From a polyhedron $P \subseteq \mathbb{R}^d$ one cannot construct in general a polyhedral complex in a reasonable canonical way. We suggest the following definition suited for all kinds of "pathological" situations: a *k-face* F of a polyhedron P is a maximal non-empty connected open subset of $P \cap U$ in U, where $U \subseteq \mathbb{R}^d$ is a k-dimensional affine subspace such that there exists a cone $\hat{F} \subseteq U^\perp$ such that $\hat{F}$ is either not a linear subspace of $U^\perp$ or $\hat{F} = \{0\}$ and such that $P \cap O = (F \oplus \hat{F}) \cap O$ for some neighborhood O of U in $\mathbb{R}^d$. However, a k-face may be topologically nontrivial. In fact, any relatively open set of the form $P_1 \backslash P_2$, where P_1, P_2 are polyhedra in a k-dimensional affine subspace of $\mathbb{R}^d$ (where $d > k$) can occur as a k-face of some polyhedron $P \subset \mathbb{R}^d$. Thus polyhedral complexes have much nicer combinatorial properties.

A *diagram* $\Gamma \cup \{P\}$ is a polyhedral complex Γ together with a polytope P such that $P = \bigcup \Gamma$ and

(i) if F is a proper face of P, then $F \in \Gamma$;

(ii) if $G \in \Gamma$ and $G \cap \operatorname{bd} P \neq \emptyset$, then $F \cap \operatorname{bd} P \in \Gamma$.

P is called the *basis* of the diagram $\Gamma \cup \{P\}$.

If P is a d-polytope, the diagram is called a *d-diagram*.

As a generalization of polyhedral complexes Mandel (1982) defines: a *ball complex* is a finite family C of topological balls (homeomorphs of B^d, $d>0$) in a Hausdorff space X such that the (relative) interiors of the balls partition X and such that the boundary of each ball is a union of other balls. The underlying space X is denoted by $\|C\|$.

The interiors of the balls of a ball complex form what is called a regular cell complex (cf. Björner 1992).

Ball complexes are more general than polyhedral complexes in several ways. For instance, it is allowed that the intersection of two of the balls can have a nontrivial topological structure.

Let C be a ball complex and $P(C)$ the associated partially ordered set (i.e., C ordered by inclusion).

Then $\|C\| \cong \|\Delta(P(C))\|$ (cf. Lundell and Weingram (1969) for proofs for regular cell complexes), i.e., the order complex of $P(C)$ is homeomorphic to $\|C\|$, where the order complex of C is the simplicial complex on the vertex set of C, whose k-faces are the k-chains $x_0 < x_1 < \cdots < x_k$ in $P(C)$. Geometrically this means that ball complexes admit "barycentric subdivisions". Combinatorially this means that ball complexes can be regarded as a class of posets without any loss of topological information.

For the case of a 2-dimensional manifold we recall some definitions: a *map* on a compact 2-manifold M is a decomposition of M into f_2 non-overlapping simply connected regions (called *faces*) by f_1 arcs (called *edges*) joining pairs of f_0 points (called *vertices*) such that the intersection of any two edges is empty or one or both common endpoints (vertices) and such that each edge belongs to just two faces.

A map is called *polyhedral*, if the intersection of any two (closed) faces is empty, one common vertex or one common edge. The set of closed faces, edges and vertices of a polyhedral map is a ball complex C with the property that the intersection of any two elements of C is empty or a ball of C. We shall regard polyhedral maps as special ball complexes. In particular they can be regarded as purely combinatorial objects.

A polyhedral map is called *weakly neighborly* if each pair of vertices is contained in some common face.

Polyhedral maps and in particular weakly neighborly polyhedral maps have been investigated from a combinatorial point of view in several papers, cf. Altshuler and Brehm (1986a,b, 1987), Brehm and Altshuler (1985, 1986), and Barnette (1983, 1987).

Let C be a ball complex, e.g., a polyhedral map. A continuous mapping $\gamma : |C| \to \mathbb{R}^d$ is called *polyhedral* if each ball of C is mapped onto a polytope, it is called *nc-polyhedral* if each k-dimensional ball (k-ball) of C is mapped onto a polyhedron with a k-dimensional affine hull.

We recall two standard definitions from topology.

A mapping γ is called an *embedding* if it is a homeomorphism onto the image of γ.

A mapping γ is called an *immersion* if it is continuous and locally injective.

Note that the image of a polyhedral (or an nc-polyhedral) mapping γ is a

polyhedron, because the union of finitely many polyhedra is a polyhedron. A *polyhedral realization* of a ball complex is a polyhedral complex which is combinatorially equivalent to the given ball complex. A polyhedral realization Γ is called *strict* if any two k-dimensional facets of Γ with a common $(k-1)$-face do not have the same affine hull.

2. Embeddability

2.1. The basic problems

(a) *Embeddability problems.* When is a given finite poset isomorphic to the face poset of some polyhedral complex?

When can a given ball complex be polyhedrally embedded (polyhedrally immersed, nc-polyhedrally embedded, nc-polyhedrally immersed, respectively) in $\mathbb{R}^d$?

Note that the geometric embeddability problems as stated here differ from the embeddability problems extensively studied in PL-topology, as we do not allow simplicial subdivisions. See Brehm and Sarkaria (1992) for examples of d-dimensional simplicial complexes which are PL-embeddable in $\mathbb{R}^{2d}$ but which cannot be polyhedrally embedded in $\mathbb{R}^{2d}$.

(b) *Polytopality problem.* When is a given finite lattice isomorphic to the face lattice of a polytope? The problem is treated in some detail in chapter 2.5 in this Handbook, where the recent computational methods are stressed. A related question is to decide whether a diagram is combinatorially equivalent to the boundary complex of a polytope. Steinitz' theorem (see below) implies that each 2-diagram is combinatorially equivalent to the boundary complex of a 3-polytope. However, the higher-dimensional analogues are not true (even in the case of simplicial diagrams cf. Sturmfels 1987b). For partial results and necessary conditions see Grünbaum's book (1967), Ewald et al. (1979), Bokowski and Garms (1987), and Bokowski and Sturmfels (1987).

(c) *Isotopy problems.* Is the space of all d-polytopes of a given combinatorial type connected (regarded as a subspace of $\mathbb{R}^{dn}$, where n is the number of vertices)? This and similar problems are called isotopy problems and are treated in some detail in chapter 2.5 in this Handbook (cf. also Mnëv 1985, 1988, and Goodman and Pollack 1985). Here we do not consider isotopy problems.

2.2. Embeddability

The embeddability problem and the polytopality problem are solved in the case of the 2-sphere by the well-known theorem of Steinitz:

Theorem. *Each polyhedral map on S^2 is combinatorially isomorphic to the boundary complex of a 3-polytope.*

A graph-theoretic version of Steinitz' theorem is the following.

Theorem. *A finite graph is isomorphic to the graph of edges of a 3-polytope if and only if it is planar and 3-connected.*

A graph is called 3-*connected* if it has at least 4 vertices and if it remains connected after removing any 2 vertices. For higher dimensions and for 2-manifolds different from S^2 there exist hardly any general results (cf. in particular Sturmfels 1987b).

One easy result is that any d-dimensional simplicial complex is polyhedrally embeddable in $\mathbb{R}^{2d+1}$, merely because it is isomorphic to a subcomplex of the boundary complex of a $(2d+2)$-dimensional cyclic polytope.

Another simple result (cf. Altshuler 1971, and Barnette 1982a) is that the image of a polyhedral embedding of the dual complex of a simplicial combinatorial d-manifold in $\mathbb{R}^n$ $(n > d)$ is necessarily the boundary of a $(d+1)$-polytope, hence if the d-manifold is not homeomorphic to a d-sphere, then there exists no polyhedral embedding at all.

Even in the simplest case more general than S^2, namely for orientable simplicial 2-manifolds of genus $g > 0$, the embeddability problem in $\mathbb{R}^3$ is completely open (cf. Barnette 1973, and Grünbaum 1967). For no orientable simplicial 2-manifold of genus $g > 0$ has been shown that it is not polyhedrally embeddable in $\mathbb{R}^3$.

On the other hand it has not been shown for any $g > 0$ (not even for the torus) that each simplicial 2-manifold of genus g can be polyhedrally embedded in $\mathbb{R}^3$ (or in $\mathbb{R}^4$).

In the non-orientable case much more is known: Barnette (1983) showed that each simplicial projective plane can be polyhedrally embedded in $\mathbb{R}^4$.

A surprising result by Brehm (1983) states that the Möbius strip and hence every non-orientable 2-manifold (with or without boundary) can be triangulated such that it cannot be polyhedrally immersed (hence not embedded) in $\mathbb{R}^3$ with respect to that triangulation.

The fact that no general results are known for the embedding problem for polyhedral maps on 2-manifolds makes it necessary and interesting to construct explicitly polyhedra with specific properties like minimality or combinatorial regularity which will be treated in the next paragraphs, and to develop methods such as described in chapter 2.5 in this Handbook, which allow the effective use of a computer. However, the polyhedral embeddability in $\mathbb{R}^3$ of fairly small examples such as a 12-vertex triangulation of the orientable 2-manifold of genus 6 has not yet been decided using these computational methods, which shows that the embeddability problem can be hard even for particular examples.

A useful method for finding a polyhedral embedding in $\mathbb{R}^d$ of a given ball complex C is to show that C is combinatorially isomorphic to a (proper) subcomplex of a $(d+1)$-polytope P. In this case we say that M is *embeddable* in bd P. Then an explicit construction can be done via the *Schlegel-diagram* of P, i.e., the central projection of the boundary complex of P onto one of its facets which is not in the subcomplex.

If it works, this method gives an elegant explicit description. However, even in the 2-dimensional case this method does not always work, as there exist polyhedral manifolds in $\mathbb{R}^3$ which are not isomorphic to a subcomplex of any convex

polytope in any $\mathbb{R}^d$. The first such example was a polyhedral Möbius strip in $\mathbb{R}^3$ (hence giving examples for all non-orientable 2-manifolds in $\mathbb{R}^4$) given in Betke, Schulz and Wills (1975); the first orientable such example, a polyhedral torus, was given in Altshuler and Brehm (1984). The first example where the method works, is the Möbius-torus which is embeddable in the cyclic 4-polytope $C(7, 4)$. Also for the direct higher-dimensional analogue of the Möbius-torus this method works with an appropriate cyclic polytope (cf. Kühnel 1986a). Betke, Schulz and Wills (1975, 1976, 1978) obtained several results on orientable and non-orientable polyhedra with or without boundary in the boundary complex of 4- and 5-polytopes. In particular they investigated analogues of Hamiltonian circuits, i.e., polyhedra which contain the 1-skeleton of the boundary complex. They further investigated polyhedra in the boundary complexes with minimal numbers of vertices, faces or edges (cf. also sections 3.1 and 3.2).

For the case of the torus, Gritzmann (1980, 1982, 1983) (cf. also Barnette, Gritzmann and Höhne 1991) determined the exact range of f-vectors for polyhedral embeddings of the torus: for each triple (f_0, f_1, f_2) of nonnegative numbers there exists a polyhedral embedding of the torus in $\mathbb{R}^3$ with f_ν ν-faces if and only if

$$f_0 - f_1 + f_2 = 0\,, \qquad \tfrac{1}{2} f_2(11 - f_2) \leqslant f_0 \leqslant 2f_2\,,$$
$$\tfrac{1}{2} f_0(11 - f_0) \leqslant f_2 \leqslant 2f_0 \quad \text{and} \quad 2f_1 - 3f_0 \geqslant 6\,.$$

The sufficiency of these conditions has been shown by Simutis (1977). For polyhedral embeddings of 2-manifolds in $\mathbb{R}^d$ bounds for the *valence functional* $2f_1 - 3f_0$ have been given by Gritzmann (1982, 1984) and Barnette, Gritzmann and Höhne (1991).

3. Minimality properties of polyhedra

3.1. Minimal number of vertices

For any triangulation of an orientable 2-manifold of genus g holds the inequality for the number f_0 of vertices

$$f_0 \geqslant \lceil \tfrac{1}{2}(7 + \sqrt{48g + 1}) \rceil\,, \tag{3.1}$$

where $\lceil x \rceil$ denotes the smallest integer $\geqslant x$. The inequality is an easy consequence of Euler's formula. Jungerman and Ringel (1980) have shown that for each $g \neq 2$ there exists a triangulation such that equality holds in (3.1) and for $g = 2$ the minimal number of vertices is 10. This result is closely related to the famous map coloring theorem of Ringel and Youngs (1968) (cf. also Ringel 1974).

The question is, for which g does there exist a polyhedron attaining this minimal number. It is also natural to ask for a symmetry group of maximal order of such a polyhedron. For the trivial case $g = 0$ we have $f_0 = 4$ and the regular tetrahedron attains this minimal number and has maximal symmetry.

For $g = 1$ the problem is solved by the Császár-torus with 7 vertices (1949). For

$g = 2$ and 3 Brehm (1981, 1987b) found polyhedra with 10 vertices which is the minimal number, and for $g = 4$ Bokowski and Brehm (1989) found a minimal polyhedron with 11 vertices. In the cases $g = 1$ and $g = 4$ the maximal possible symmetry group is of order 2 (rotation). In the cases $g = 2$ and $g = 3$ the maximal possible symmetry group is of order 4; 3 axes of rotational symmetry are possible (cf. Brehm 1987b, for $g = 2$ Brehm, unpublished). In the cases $g = 1, 2, 3$ the methods were intuitive, whereas for $g = 4$ computational methods (see chapter 2.5 in this Handbook) were used. For no $g > 4$ it is known whether there is a polyhedron which attains the lower bound (3.1), although at least for $g = 6$ several attempts were made. For $g = 6$ the 1-skeleton of the minimal polyhedral map is the complete graph K_{12} with 12 vertices and 66 edges.

For larger g there are only few results, but McMullen, Schulz and Wills (1983, 1985) obtained with a general method a series of polyhedra M_g of genus g with $f_0(M_g) = O(g/\log g)$. So for sufficiently large g the manifold M_g has more handles than vertices. The smallest value obtained with this method is $g = 577$, namely an M_{577} with 576 vertices only. There is still a large gap between the surprising bound $O(g/\log g)$ and the combinatorial bound $O(\sqrt{g})$ of the Ringel–Jungerman theorem.

For triangulations of non-orientable 2-manifolds of Euler-characteristic χ holds the analogue of (3.1) [with g replaced by $(2 - 2\chi)$] and according to Ringel (1955) for each non-orientable 2-manifold except the cases $\chi = 0$ (Klein-bottle) and $\chi = -1$ there exists a triangulation for which equality holds in (3.1). However, for the real projective plane $\mathbb{R}P^2$ the number of vertices of a polyhedral map which can be polyhedrally immersed in $\mathbb{R}^3$ is at least 9, because according to a result of Banchoff (1974) an immersion of $\mathbb{R}P^2$ has generically a triple point. Brehm (1990) has shown that there exist indeed polyhedral immersions of $\mathbb{R}P^2$ with 9 vertices and with an axis of symmetry of order 3. These immersions are minimal polyhedral versions of the Boy surface, which can also be regarded as nc-polyhedral immersions with only 10 facets.

For the Klein-bottle Cervone (1993) showed that there exists a polyhedral immersion with 9 vertices and that this is the minimal vertex number.

For combinatorial manifolds of dimension $d \geq 3$ only some results are known concerning lower bounds for the number of vertices (cf. Brehm and Kühnel 1987, 1993). However, there exists a direct higher-dimensional analogue of the Császár-torus found by Kühnel (1986). There exists a simplicial combinatorial d-manifold with $2d + 3$ vertices, which can be embedded in the boundary of a cyclic polytope and thus in $\mathbb{R}^{d+1}$, if d is even and $\mathbb{R}^{d+2}$, if d is odd. The manifold is homeomorphic to $S^1 \times S^{d-1}$, if d is even and to the twisted S^1-bundle over S^{d-1}, if d is odd.

The number of vertices is even minimal along all d-manifolds ($d \geq 3$) which are not simply connected according to Brehm and Kühnel (1987).

3.2. Minimal number of faces and edges

For simple combinatorial reasons (duality), Heawood's bound (3.1) also holds for polyhedral maps of genus g with f_2 instead of f_0. Again for $g = 0$ the tetrahedron is the minimal realization with its 4 facets and 6 edges.

For $g = 1$ Szilassi (1986) already in 1978 constructed an nc-polyhedral embedding of the dual of the unique 7-vertex triangulation of the torus with 7 nonconvex hexagons in $\mathbb{R}^3$. However (cf., e.g., Barnette 1982a, or Grünbaum and Shephard 1988) there does not exist a polyhedral embedding of the dual of a triangulated 2-manifold for $g > 0$ in any $\mathbb{R}^d$, thus one has to distinguish between polyhedral embeddings and nc-polyhedral embeddings, concerning the minimal number of facets. For $g \geq 2$ no examples of nc-polyhedral embeddings of polyhedral maps attaining the lower bound (3.1) are known.

McMullen, Schulz and Wills (1983) constructed an infinite sequence M_g of polyhedra in $\mathbb{R}^3$ of genus g with $f_2(M_g) = O(g/\log g)$ convex facets, which are even combinatorially regular (see section 4.1). In particular $f_2(M_{4093}) = 4092$, i.e., the polyhedral manifold M_{4093} has more handles than convex facets.

The minimal number of edges of a polyhedral map on a given 2-manifold is known only in some cases (cf. Brehm 1990a, and Gritzmann 1980, 1982, where also general lower bounds are given). So the minimality problem is even combinatorially unsolved in general.

For $g = 0$ the tetrahedron with 6 edges again is the trivial solution.

For $g = 1$ the "triangular picture frame" with 18 edges is the (combinatorially) unique polyhedral torus attaining the minimal number. Ljubic (1987) gave another nc-polyhedron (the twisted triangular picture frame) of genus 1 with 18 edges.

For $g > 1$ the minimal number of edges for a polyhedron in $\mathbb{R}^3$ is unknown.

For 2-manifolds with boundary the problems are much easier. Moreover there is no essential difference in the embedding problem between oriented and non-oriented 2-manifolds with boundary: Both are topologically embeddable in E^3. A simple result is (cf. Wills 1979): Any 2-manifold (oriented or non-oriented) with at least one boundary component can be embedded in E^3 with 2 nc-polygons. Here the number of vertices and edges depends linearly on the genus g.

Embeddings with convex faces are somewhat more complicated. Betke, Schulz and Wills (1975) and Schulz and Wills (1979) determine the minimal embeddings of the Möbius-strip and the Klein-bottle (with boundary) with respect to vertex-, face-, and edge-numbers.

3.3. Other minimality properties

(a) *Colorability*. It is well known that the four-color-theorem for maps on the sphere is equivalent to the four-color-conjecture for convex polytopes in E^3. From Heawood's map coloring problem and the results by Ringel, Jungerman and Youngs (cf. section 3.1) it is known that, e.g., every map on the torus can be colored with 7 colors and the Szilassi-torus is an example which requires exactly 7 colors. By a coloring we mean that any two faces with the same color have no common edge.

One might expect that there is also a toroidal polyhedron (i.e., with $g = 1$) built up of convex polygons which requires also 7 colors. However, Barnette (1985) showed that every toroidal polyhedron with convex faces requires at most 6

colors. Observations for higher genera led Barnette to his "Four-Color-Conjecture": any polyhedral manifold with convex faces is 4-colorable. Although the known polyhedra seem to support Barnette's conjecture, no proof is known.

(b) *Nonconvex vertices.* A vertex v of a polyhedral 2-manifold M is called convex, if at least one of the two components into which M divides a sufficiently small ball with center v, is convex. First results on the minimal number of nonconvex vertices in 2-manifolds are due to Altshuler (1971), Simutis (1977), Gritzmann (1980) and Barnette (1982b).

Betke and Gritzmann (1984) solved the minimality problem and proved: every polyhedral 2-manifold in E^3 of genus $g \geq 1$ contains at least 5 nonconvex vertices and for every genus $g \geq 1$ this bound is attained.

(c) *Tight embeddings.* A polyhedral 2-manifold M is called tight, if every hyperplane which contains a point of M and no nearby points is a global supporting hyperplane of M. First results are due to Banchoff (1965, 1967), see also Kühnel (1986). In a tight polyhedral 2-manifold (or embedding) the number of nonconvex vertices cannot be too small and Gritzmann (1981) showed for the minimal number $H_m(g)$ of nonconvex vertices in a tight polyhedral 2-manifold of genus $g \geq 1$:

$$6 \leq H_m(g) \quad \text{and} \quad 1 + (1 + 4g)^{1/2} \leq H_m(g) \leq 3g + 3 .$$

4. Combinatorially regular polyhedra and related topics

4.1. Combinatorially regular polyhedra

A map is called regular if its combinatorial automorphism group acts transitively on its flags. There is an extraordinary large literature on regular maps and their groups and generators (cf., e.g., Coxeter and Moser 1980, Wilson 1976, and Edmonds, Ewings and Kulkarni 1982, and the references given there), but here we are only interested in polyhedral embeddings of regular maps and weaker modifications, which we briefly discuss in the following sections (cf. also McMullen 1989, McMullen and Schulte 1990, Buekenhout 1979, Weiss 1989, Schulte and Wills 1991).

A polyhedral embedding of a regular map is called a combinatorially regular polyhedron. Each combinatorially regular polyhedral map on the 2-sphere is combinatorially equivalent to one of the 5 Platonic solids and hence possesses even a geometrically regular realization, i.e., the symmetry group acts transitively on the flags. [For higher-dimensional geometrically regular polytopes see McMullen (1967, 1989), Coxeter (1973), Banchoff (1988) or chapter 2.3 in this Handbook.] In all other cases the geometric symmetry group is a proper subgroup of the automorphism group of the regular map. The index of the subgroup is called the index of polyhedral embedding and is a measure for its symmetry.

For the torus ($g = 1$) all regular maps are known (cf. Coxeter and Moser 1980, and Altshuler 1973), and each regular map which is a cell-complex is realizable as

a combinatorially regular polyhedron (cf. Schwörbel 1988, and Schwörbel and Wills 1988). Much less is known in the hyperbolic case $g \geq 2$. Only for $g \leq 7$ all regular maps are known, see Garbe (1969). But several infinite series are known, see Coxeter and Moser (1980). At least two of them can be realized (see McMullen, Schulz and Wills 1983; McMullen, Schulte and Wills 1988); they are the series in Theorem 2. Further, some famous regular maps could be embedded as polyhedra, e.g. (cf. Schulte and Wills 1985, 1986, Bokowski 1989, 1991, and Brehm 1987), Klein's $\{3,7\}_8$, Dyck's $\{3,8\}_6$, Coxeter's $\{4,6|3\}$, $\{6,4|3\}$, $\{4,8|3\}$ and $\{8,4|3\}$. With the exception of the torus and the two mentioned series only very few combinatorially regular polyhedra are known, and for each $g \geq 2$ only finitely many are possible.

Clearly all these polyhedral 2-manifolds are interesting objects for computer-graphics and for models (cf., e.g., Bokowski and Sturmfels 1989, Saffaro 1976, 1983, 1987, and Leytem 1989). For more details we refer to the surveys by Bokowski and Wills (1988), Schulte and Wills (1991), and the book by Bokowski and Sturmfels (1989). Because of their close relation to the Platonic solids regular maps and their various realizations have always deserved particular interest.

So also weaker polyhedral variants besides the combinatorially regular polyhedra are of some interest. In the following three subsections we consider some of them.

4.2. *Equivelarity*

A map is said to be of type $\{p, q\}$, if all its faces are topological p-gons, q meeting at each vertex. If we also wish to note the genus g of the map, we write $\{p, q; g\}$. We underline that this notation does not determine a map combinatorially unique. A polyhedral embedding with convex faces of a map of type $\{p, q\}$ in E^3 (hence oriented) is called equivelar, i.e., with equal flags. In the spherical case $g = 0$ this is equivalent to regularity. So the only equivelar polyhedra of $g = 0$ are the 5 Platonic solids. The $\{p, q\}$-property and hence equivelarity can be interpreted as local regularity whereas the combinatorial regularity in section 4.1 is a global property. So in this sense the case $g = 0$ differs from all other genera. Besides the Euler-equation for equivelar polyhedra hold the simple equations $qf_0 = 2f_1 = pf_2$.

In contrast to the simple combinatorial properties the problems of geometric realization are more complicated, in particular for small genera. The following two theorems give the list of the known equivelar polyhedra.

Theorem (McMullen, Schulz and Wills 1982). *The following equivelar polyhedra exist*:

(a) $\{3, 7; g\}$ *for* $g \geq 2$,
(b) $\{3, 8; g\}$ *for* $g \geq 3$ ($g = 3$: Grünbaum and Shephard 1984),
(c) $\{4, 5; g\}$ *for* $g \geq 4$ ($g = 4, 6, 8$: Wanka 1981),
(d) $\{5, 4; g\}$ *for* $g = 5, 7$ *and* $g \geq 9$,
(e) $\{3, 9; g\}$, $\{4, 6; g\}$ *and* $\{6, 4; g\}$ *for* $g = 6, 9, 10$ *and* $g \geq 12$.

Theorem (McMullen, Schulz and Wills 1983). *For each of the following classes there are infinitely many g such that a corresponding equivelar polyhedron exists*:
(a) $\{3, q; g\}$, $q \geq 7$,
(b) $\{4, q; g\}$, $q \geq 5$,
(c) $\{p, 4; g\}$, $p \geq 5$.

Equivelarity requires convex faces, so part (c) of the theorem is surprising. In fact, the asymmetry of these theorems in p and q is mainly due to the fact that the faces are convex, whereas there is no restriction on the vertices. Each 3-valent vertex is convex, hence no equivelar polyhedron $\{p, 3; g\}$ with $g > 0$ exists. (This asymmetry can be avoided by introducing nonconvex planar faces; cf., e.g., Szilassi 1986 and Schwörbel 1988.)

A remarkable open problem is, if equivelar polyhedra $\{p, q; g\}$ with $p > 4$ and $q > 4$ exist. Betke and Gritzmann (1982) proved the non-existence of some classes $\{p, q; g\}$. In the following sections some equivelar polyhedra with additional transitivity properties are given.

4.3. Polyhedra with vertex transitive symmetry group

With the exception of the 5 Platonic solids the symmetry group does not act transitively on the flags of a combinatorially regular polyhedron. In 1978, Brehm found a series of vertex transitive tori ("flat" tori) (cf. Brehm and Kühnel 1982). Schwörbel (1988) showed that there is even a combinatorially regular and vertex transitive torus. Wills (1983, 1984, 1986) found equivelar polyhedra of genus 3, 5, 7, 11, 19, where the vertices or faces form two orbits under the symmetry group. Grünbaum and Shephard (1984) showed that in the case of faces this is best possible, because there are no polyhedra whose symmetry group acts transitively on their faces or edges. Moreover, they found polyhedra of genus 3, 5, 7, 11, 19 with vertex-transitivity; none of them combinatorially regular, those of genus 7 and 19 are not even equivelar. These are the only known polyhedra with vertex transitive symmetry. Leytem (1989) gave several computer-aided constructions of these polyhedra.

4.4. Other realizations of regular maps

Only in this section we consider polyhedra with self-intersections and even more degenerate "polyhedral realizations" of regular maps. This is justified by the importance of Kepler–Poinsot-solids and other star-polyhedra, the regular compounds, etc. as regular geometric objects. We very briefly mention a few examples. Cauchy proved that the 4 Kepler–Poinsot-solids are the only regular polyhedra with self-intersections. Coxeter, Longuet-Higgins and Miller (1954) investigated the 53 uniform polyhedra with self-intersection and vertex transitivity. Wenninger (1983) investigated their duals. Coxeter (1939) found 4 combinatorially regular polyhedra with self-intersections of index 2, hence the closest neighbours of the Kepler–Poinsot-solids; Wills (1987) found a fifth one and

showed that no other ones exist. For further results see Shephard (1952), Isbell (1986), Schulte and Wills (1986, 1987), Wills (1988a,b), Buekenhout and Dony (1990). Grünbaum (1977) investigated regular polyhedra with nonplanar faces. Dress (1981, 1985) found a further one and proved that these are all of this type. Many of these polyhedra are infinite, and in fact infinite polyhedra with regularity properties have a long tradition, starting with Petrie's and Coxeter's regular skew polyhedra (cf. Coxeter 1937). Later, Gott III (1967) found some closely related infinite polyhedra with local regularity properties. For further results see the references in Grünbaum (1977), Dress (1985) and also Harker (1991).

We finally mention polyhedral realizations (of nonregular maps), where all faces are regular or congruent, and refer for this, e.g., to Stewart (1980), Szilassi (1986), Bilinski (1985), Fejes Tóth (1965), Pugh (1976), Wells (1977), and Zalgaller (1969).

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CHAPTER 2.5

Oriented Matroids

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Contents

1. Introduction 557
1.1. Generalizations leading to oriented matroids 558
1.1.1. Farkas' Lemma 558
1.1.2. Circuit properties of oriented graphs 558
1.1.3. Incidence properties of lines in $\mathbb{R}P^2$ 559
1.1.4. Incidence properties of hyperplanes in $\mathbb{R}P^d$ 560
1.1.5. Determinant signs 561
1.2. On getting accustomed with oriented matroids 564
2. Models in oriented matroid theory 568
2.1. Sphere systems 569
2.2. Signed vectors of a vectorspace 570
2.3. Grassmann–Plücker polynomials 572
3. Matroid polytopes 575
3.1. Basic concepts in oriented matroid theory 576
3.2. Duality for oriented matroids 583
3.2.1. Duality in general 583
3.2.2. d-Polytopes with $d+4$ vertices 584
4. Matroid theory in convexity 585
4.1. The missing axiom for polytopal spheres 585
4.2. On the isotopy problem 586
4.3. Convexity theorems with extensions 586
4.3.1. Non-properties of matroid polytopes 587
5. Polytopes versus matroid polytopes 588
5.1. How to prove realizability 588
5.2. The final polynomial method 590
References 592

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1. Introduction

Convexity of a finite point set E in Euclidean space can be expressed by relations of the form (p, H) where p is any of the points of E and H is a certain oriented (supporting) hyperplane also derived from E. For problems involving such or similar assertions, it has proven useful to think of the point configuration in an abstract manner and to study this within the framework of the theory of oriented matroids. As a first approximation matroid theory can be considered as coordinate-free linear algebra, and as such it has been applied to various fields in mathematics. The objective of this article is to motivate and to prepare the reader for further studies in this field where emphasis is placed on its relation to combinatorial convexity. Once an overview from the more general concept of oriented matroids has been accomplished, we can look back to the polytopes, we started with.

Convexity in oriented matroids, the title of an article by Las Vergnas (1980a), was an early concept of oriented matroid theory, two independent origins being Las Vergnas (1974) and Bland (1974), see Bland and Las Vergnas (1978). In *Convexity and Its Applications* (edited by Gruber and Wills), oriented matroid theory is mentioned in the very first sentence, Bachem (1983). Bland's approach to introducing oriented matroids was motivated from operations research with its close connection to convexity. In the conclusion of his survey article *Transforms, diagrams and representations,* McMullen dedicated two pages to oriented matroids, see McMullen (1978).

Since the time of these early contributions, many important combinatorial theorems and concepts about convex polytopes have been extended to the setting of oriented matroids. Radon partitions, supporting hyperplanes, face lattices, Euler's formula, Gale diagrams, Farkas' Lemma from operations research, duality, shellability, the upper bound theorem, and many other well-known concepts in combinatorial convexity have been carried over to and studied within the theory of oriented matroids. The combinatorial part of the theory of convex polytopes can be viewed as part of oriented matroid theory, and the theory of oriented matroids benefits from a well established theory of convex polytopes, see Grünbaum (1967) and the development thereafter described in this volume.

The theory of convex polytopes, on the other hand, considered in its own right, has gained much impetus through oriented matroid theory. We mention, e.g., the negative solution of Sturmfels (1987a) to what Ewald always put forward as the Steinitz problem.

Oriented matroid theory has grown up to be a powerful tool in combinatorial convexity. In more than one decade of studying oriented matroids from various points of view, this structure with its internal convexity property can be viewed as an appropriate framework for combinatorial convexity. The fact that all matroid polytopes can be generated in a purely combinatorial fashion shows a closure property which the set of ordinary convex polytopes seems to lack. Compare Grünbaum (1967, section 5.5): it follows that it may be necessary to use different representatives of a given combinatorial type in order to obtain all the polytopes having

one vertex more which are obtainable from polytopes of the given combinatorial type. In fact, it was the enumeration problem for ordinary convex polytopes with its contributions by Altshuler (1974) and others which led the author to considering oriented matroids. Matroid polytopes include all polytopes but they are not as general as abstract schemes in the sense of Grünbaum (1967, section 5.5) or many other concepts to generalize polytopes used, e.g., by Danzer, Schulte, Ivic Weiß, and McMullen.

Today we are facing a comprehensive theory of oriented matroids, see Björner et al. (1991) and the references given in section 1.2. But there is still a lack of introductory material which helps keeping alive the motivation whilst studying this structure. Not everybody doing research in convexity is willing to study the structure of oriented matroids with all its aspects in full detail. This article aims at helping the reader to understand some basic ideas while providing some selected highlights of results within the theory of oriented matroids with emphasis placed on their convexity properties. Chapter 9 of Björner et al. (1991) can be viewed as a similar contribution. It is assumed that the latter book will be available when this article appears. There is of course some overlap but the author hopes that different views of a survey might be of interest.

1.1. Generalizations leading to oriented matroids

Natural generalizations have been a common idea behind many sources in mathematics which led independently to oriented matroids. We list five properties $\mathcal{P}_1, \mathcal{P}_2, \ldots, \mathcal{P}_5$ such that looking for the most general structure with property $\mathcal{P}_i$ leads to the concept of oriented matroids. Within a very rough introduction, we cannot meet the desire of a historical report.

1.1.1. Farkas' Lemma

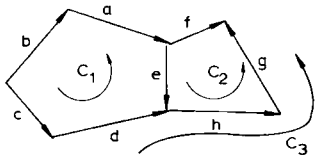
($\mathcal{P}_1$) Farkas' Lemma from operations research: Let a be an (m,n) matrix and $b \in \mathbb{R}^m$. Either there is a point $x \in \mathbb{R}^n$ with $ax \leqslant b$ or there is a point $u \in \mathbb{R}^m_+$ with $a^{\mathrm{T}}u = 0$ and $u^{\mathrm{T}}b < 0$.

Oriented matroids can thus be considered as generalizing duality of orthogonal linear subspaces. See Bachem (1983) and Bachem and Kern (1992) for a detailed exposition of oriented matroids when viewed as a tool in operations research.

1.1.2. Circuit properties of oriented graphs

($\mathcal{P}_2$) Given any two circuits C_1 and C_2 in an oriented graph with a common edge e, there exists a third circuit C_3 with edges from the union of C_1 and C_2 but without the common edge e. Figure 1 gives an example.

When looking at the circuit axioms of Las Vergnas, it is useful to have this concept in mind. Oriented matroids are used to generalize certain results for directed graphs and convex polytopes, see Las Vergnas (1974, p. 74), Bland and Las Vergnas (1978).



a	b	c	d	e	f	g	h	
−	−	+	+	−	0	0	0	C_1
0	0	0	0	+	−	+	+	C_2
−	−	+	+	0	−	+	+	C_3

Figure 1. Example

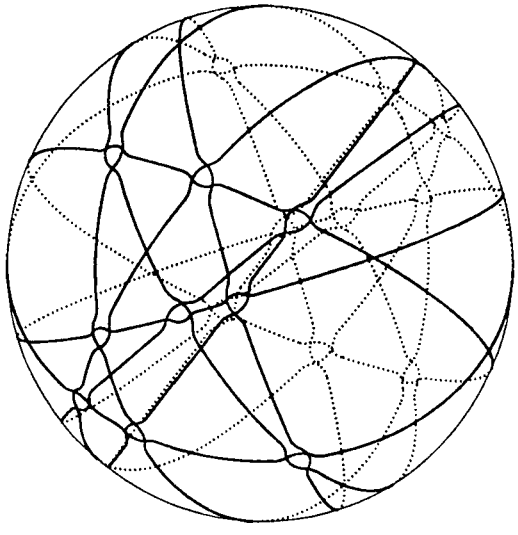


Figure 2. Uniform Pappos configuration.

1.1.3. Incidence properties of lines in $\mathbb{R}P^2$

($\mathcal{P}_3$) Two different lines in the projective plane cross each other exactly once where they meet.

When considering n closed curves in the projective plane with the same incidence properties as n lines, we have a model for a *reorientation class of an oriented matroid in rank* 3. The collection of n such closed curves is called a *pseudoline arrangement.* When we assume the lines to be oriented, we have an oriented matroid of rank 3. See fig. 2 for a first example. When half of the sphere is projected onto a plane we do not lose any combinatorial information of the reorientation class. The bad pentagon of Goodman and Pollack in fig. 3 is depicted in this way.

The research monograph, *Arrangements and Spreads*, Grünbaum (1972), provides a comprehensive description of what was known up to that time. Levi's article (1926), can be viewed as an origin. A more recent collection of additional results can be found in Roudneff (1987a). A recent connection of pseudolines with spreads was established by Goodman, Pollack, Wenger, and Zamfirescu showing that every finite arrangement of pseudolines can be enlarged to a spread of pseudolines. In an important article, Lawrence (1982) showed the one-to-one correspondence of arrangements of pseudoplanes and equivalence classes of oriented matroids, for pseudolines (rank 3), see also Cordovil (1982a).

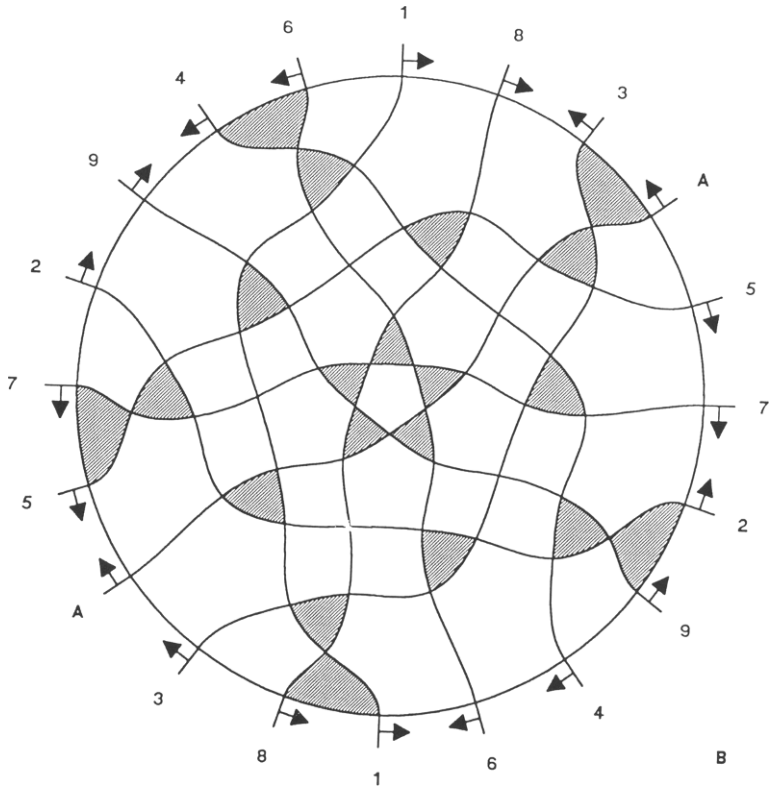


Figure 3. Bad pentagon of Goodman and Pollack.

1.1.4. Incidence properties of hyperplanes in $\mathbb{R}P^d$

($\mathcal{P}_4$) Any two different great $(d-2)$-spheres on an ordinary $(d-1)$-unit sphere in Euclidean $\mathbb{R}^d$ intersect in a $(d-3)$-sphere.

In replacing the spheres with topological spheres while requiring the intersection properties as in the metrical case, we arrive at the sphere system model for oriented matroids. The model in the colored fig. 4 shows a smallest non-realizable pseudo-plane arrangement in projective 3-space with 8 pseudo-planes which has the smallest number of tetrahedral regions, see Bokowski and Richter-Gebert (1990) for details.

Eight planes are defined by the affine hull of the 4 facets of the central inner tetrahedron together with the affine hull of the 4 facets of its homothetic outer tetrahedron. Towards the plane at infinity, these 8 planes are deformed to pseudo-planes as indicated in the picture in order to have a *simple arrangement of pseudo-planes,* i.e., only 3 pseudo-planes meet in a common point. When the tetrahedral symmetry is maintained, we see for each edge of the simplex a tetrahedral region having a 4-gon as intersection with the plane at infinity. Together with the central tetrahedron, we have 7 tetrahedral cells of maximal dimension. There is a long standing open question of Las Vergnas of whether or not there is always at least

one tetrahedral region within a pseudo-(hyper)plane arrangement in rank $r \geqslant 4$, see Roudneff (1987b). The above example occurred in Roudneff's studying pseudo-lines and in Bokowski's studying spheres and their polytopality, see Roudneff and Sturmfels (1988) and Bokowski and Sturmfels (1985).

1.1.5. Determinant signs
($\mathcal{P}_5$) The signs of $(d \times d)$-subdeterminants of an $(n \times d)$-matrix do not violate any Grassmann–Plücker relation.

Example 1.1. The 2-dimensional vectorspace $V \subset W = \mathbb{R}^4$ with basis (x, y),

$$x = (x_1, x_2, x_3, x_4)^{\mathrm{T}} = (1, 0, a, c)^{\mathrm{T}},$$
$$y = (y_1, y_2, y_3, y_4)^{\mathrm{T}} = (0, 1, b, d)^{\mathrm{T}}.$$

We use the bracket notation for determinants, see White (1975),

$$[i, j] := \det \begin{pmatrix} x_i & y_i \\ x_j & y_j \end{pmatrix}.$$

The realizable oriented matroid

$$\chi = \chi(V) = \operatorname{sign}(x \wedge y) \in \{-1, 0, 1\}^{\binom{\dim W}{\dim V}}$$

of 4 points in rank 2 is defined as

$$\chi(V) = (\operatorname{sign}[1,2], \operatorname{sign}[1,3], \operatorname{sign}[1,4], \operatorname{sign}[2,3], \operatorname{sign}[2,4], \operatorname{sign}[3,4]).$$

The vanishing relation for determinants (Grassmann–Plücker relation)

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} - \det \begin{pmatrix} 1 & 0 \\ a & b \end{pmatrix} \det \begin{pmatrix} 0 & 1 \\ c & d \end{pmatrix} + \det \begin{pmatrix} 1 & 0 \\ c & d \end{pmatrix} \det \begin{pmatrix} 0 & 1 \\ a & b \end{pmatrix} = 0$$

or

$$[1,2][3,4] - [1,3][2,4] + [1,4][2,3] = 0, \qquad (**)$$

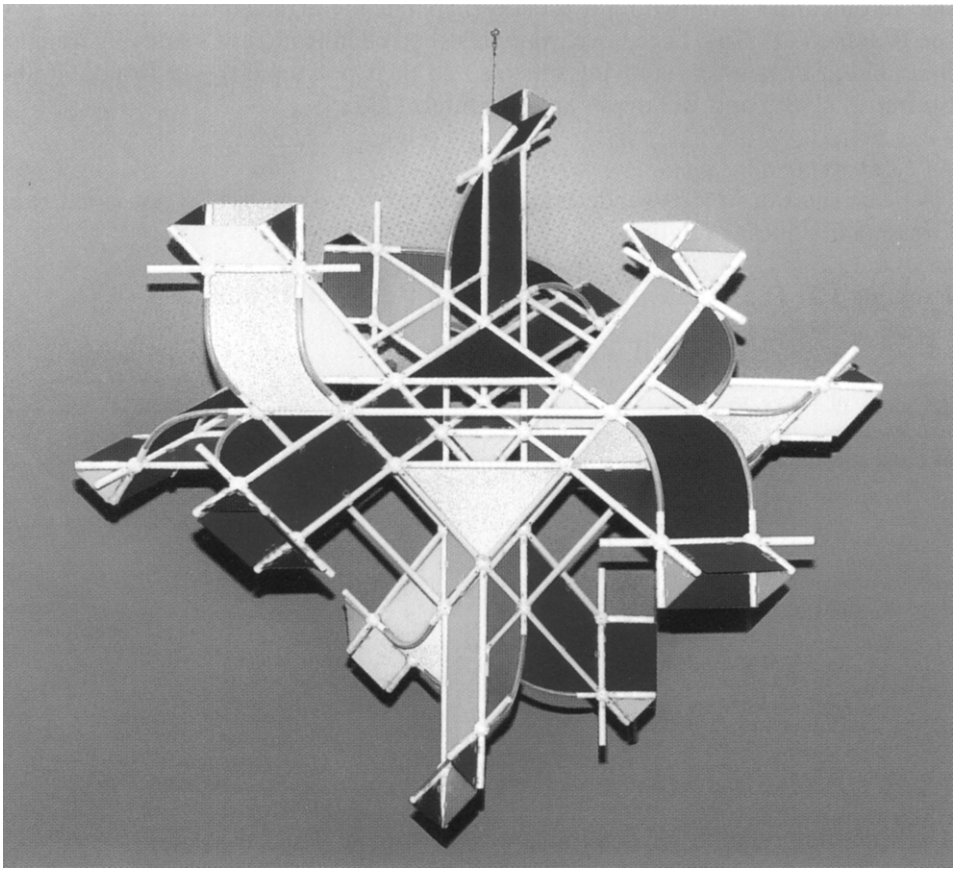
leads to a sign condition $(*)$, the necessary condition for the signs of the determinants:

$$A + B + C + ABC = 0, \qquad A, B, C \in \{-1, 0, 1\}, \qquad (*)$$

$A := \operatorname{sign}[1,2] \operatorname{sign}[3,4]$, $B := -\operatorname{sign}[1,3] \operatorname{sign}[2,4]$, $C := \operatorname{sign}[1,4] \operatorname{sign}[2,3]$. The determinant signs $(\operatorname{sign}[1,2], \operatorname{sign}[3,4], \operatorname{sign}[1,3], \operatorname{sign}[2,4], \operatorname{sign}[1,4], \operatorname{sign}[2,3])$ do not violate the Grassmann–Plücker relation $(**)$ if the sign condition $(*)$ holds.

A *signed vector*

$$\chi \in \{-1, 0, 1\}^{\binom{n}{d}},$$



i.e., a set of possible determinant signs, is an oriented matroid if and only if no Grassmann–Plücker relation is violated in the above sense. This definition leads in a straightforward way to a geometric model, the hyperline sequences which will be discussed later. They are closely related to the order types and the Λ-matrix concept of Goodman and Pollack (1983).

This concludes our five chosen properties which lead to the structure of oriented matroids. It was our aim to show some selected aspects for why mathematicians began investigating oriented matroids. As a consequence, this description does not reflect the history in this field. A detailed report about additional motivations, early contributions by other authors like Edmonds, Gelfand, Goodman, Mandel, Munson, Pollack, Rockafellar, Vershik, and others lies beyond the scope of this article. Even the same concept was often found independently by several authors with different motivations.

Whereas convexity in general attracts many mathematicians by its clear geometrical and intuitive ideas; at first sight, oriented matroids seem to lack this property. But there do exist many concepts and models to handle oriented matroids. They are contained in various articles some of which are not easy to read or difficult

Figure 4. *Model of a pseudo-plane arrangement in projective space.* (Center for Interdisciplinary Research, Bielefeld University) made by Jürgen Bokowski. We are concerned with an ordered set of n oriented planes in Euclidean 3-space embedded in projective 3-space, e.g., we consider the supporting planes of a convex polytope. For each of the planes E_j, $j = 1, \ldots, n$, it is known which of the two adjacent open halfspaces has to be regarded as positive (E_j^+), or negative (E_j^-), respectively.

We assume that the planes are in an arbitrary position. For each pair of these planes (E_i, E_j), $i < j$, there exists a line of intersection $g_{ij} := E_i \cap E_j$. This line of intersection itself has an orientation (direction vector v_{ij}) which we define in terms of the fixed oriented planes as follows: *together with the direction vector v_{ij} of g_{ij} in the sequence n_i, n_j, v_{ij}, the normal vectors n_i, n_j of the planes E_i, E_j pointing towards their positive halfspaces, should form a right handed coordinate system.*

If we proceed along the line of intersection g_{ij} in the direction v_{ij}, then we will meet, one after the other, each of the planes which are different from E_i and E_j . For each line g_{ij}, we note the sequence of appearance of the indices of the other planes. The indices will be positive, or negative, respectively, depending on whether the plane E_k is transversed from E_k^- to E_k^+ or vice versa. We complete the sequence of positive and negative indices of the planes to a periodic sequence with a length of $2(n-2)$ by repeating the same sequence but with opposite signs.

The ordered set of n oriented planes thus defines a set of cyclic sequences which is in fact the series of plane intersections on all oriented lines of intersection for two planes at a time. The direction of passage is also recorded. These cyclic sequences are the *oriented matroid of rank 4 with n elements associated with the ordered set of n oriented planes.*

In this sense, all ordered sets of n oriented planes having the same oriented matroid are classified as being equivalent and therefore can be studied without regard to specific position, so that important combinatorial characteristics emerge.

Even after topological distortions of the projective space, it is possible to form equivalence classes in the same manner. In accordance with Felix Klein's Erlanger program, it is necessary to consider the appropriate formulation of the above concept such that the formulation remains invariant in the face of topological transformations of the space. This leads to universal oriented matroids. In our particular case they are uniform because of our assumption regarding the general position.

It turns out that the former set of n oriented planes in the definition of an oriented matroid will then become an ordered set of *oriented pseudo-planes*, i.e., a set of n images of n topological maps $I_1(E_1)$, $I_2(E_2), \ldots, I_n(E_n)$ of certain oriented planes $E_1, E_2, \ldots, E_n$, with the property that each subset S of 6 such images of planes can be achieved also simultaneously by just one topological map I_S of 6 oriented planes $E_1^S, \ldots, E_6^S$.

Oriented matroids which give rise to one another only by reorientation of the planes belong to the same so called reorientation class of oriented matroids. They are represented by a system of pseudo-planes devoid of information about their orientations.

The model presents a smallest reorientation class of oriented matroids by its pseudo-planes which could not have been obtained by a set of planes. Compare also section 1.1.4.

to obtain. Besides giving an idea as to what has been established in oriented matroid theory, this article aims at showing several basic tools in oriented matroid theory and provides motivations for why the setting of oriented matroids is useful in convexity.

Above all, we emphasize again, the connection of combinatorial convexity with the theory of oriented matroids is twofold. Some methods, first used in convexity, have proven useful in the theory of oriented matroids in a wider class of applications, e.g., the *final polynomial method* of Bokowski, see Bokowski and Sturmfels (1989a). On the other hand, tools from oriented matroid theory have been successfully applied in combinatorial convexity, e.g., the negative solution to the Steinitz problem, Sturmfels (1987a).

1.2. On getting accustomed with oriented matroids

Whereas matroid theory plays a well established role in mathematics, oriented matroid theory seems to deserve a better recognition. Perhaps this article can serve as a bridge from those working in combinatorial convexity to those already familiar with the tools of oriented matroid theory.

For getting accustomed with oriented matroids, one needs to know the original concept before its generalization. Looking at the definition of oriented matroids in terms of axioms is one possible approach to understand the structure. In oriented matroid theory this should not be the only way. A set of concrete examples is needed to serve also as a source for conjectures and theorems; axioms do not reveal the full geometric significance of oriented matroid theory. Unfortunately, some useful examples or main ideas are not available for the reader when looking only at early research papers in the field. In addition, connections and proofs of equivalence which made the theory of oriented matroids so exciting during the last decade, were not known at early times, and the mathematical language in early papers reflects only the corresponding field in mathematics in which the structure was applied due to the knowledge of the author. Moreover, equivalent concepts have often had different notions.

Being aware of all these difficulties for someone not in touch with any school of oriented matroids, our attempt in this article is to clarify some underlying *geometrical* concepts. We recommend to spend some time analyzing the matroid polytopes in section 3. These matroid polytopes help in introducing major concepts.

For further studying, we recommend books on oriented matroid theory as well as Ph.D. dissertations. We list also several books on *ordinary* matroid theory which are of some help when the generalization of the matroid concept to oriented matroids is pursued.

Now, we give an overview of some literature on matroid theory.

- Books with Chapters on *Oriented* Matroid Theory.

  Bachem and Kern (1992), Björner et al. (1991), Bokowski and Sturmfels (1989a), Crippen and Havel (1988), Bachem (1983), Grünbaum (1972).

Very good dissertations and master theses in oriented matroid theory seem to be a characteristic in the field. In general they are self-contained monographs and thus perhaps easier to read.

- Some Dissertations about Topics in *Oriented* Matroid Theory.

  Bland (1974), Cordovil (1981), Fukuda (1982), Guedes de Oliveira (1990), Halsey (1971), Jensen (1985), Kern (1985), Lawrence (1975), Mandel (1981), Miller (1983), Morris (1986), Richter (1989), Richter-Gebert (1992), Roudneff (1987a), Spellman-Munson (1984), Sturmfels (1985, 1987b,c), Wanka (1986),

- Some Books on *Ordinary* Matroid Theory.

Aigner (1976), Crapo and Rota (1970), Kung (1986), Tutte (1971), Welsh (1976), White (1986, 1987).

Example 1.2 (*Convex 5-gon*). As a very easy example we consider a convex 5-gon with vertices $1,2,\ldots,5$ as depicted. We embed the affine case in the projective plane, and we replace the former five points with corresponding vectors pointing to them again as depicted. Now we find unique oriented great circles on the unit sphere orthogonal to the vectors and oriented always in the same manner with respect to the vectors. These five oriented great circles still contain the metrical information of the original 5-gon (here the affine plane is considered to be fixed). Now we forget the metrical information by considering the oriented circles to be oriented closed curves in the projective plane only, and we write down only combinatorial information, namely the 5 circular sequences of signed incidences of these great circles with the others when we pass along them, see table 1.

Table 1

Great circle	Order of oriented intersections							
1	+2	+3	+4	+5	−2	−3	−4	−5
2	+3	+4	+5	+1	−3	−4	−5	−1
3	+4	+5	+1	+2	−4	−5	−1	−2
4	+5	+1	+2	+3	−5	−1	−2	−3
5	+1	+2	+3	+4	−1	−2	−3	−4

Table 1 should be clear from figs. 5 and 6. The signs indicate the direction of the crossing of the other great circles. This scheme is one way of storing the information of the oriented matroid which corresponds to the given 5-gon. It still contains the convexity property of the original 5-gon as will be seen in Example 1.3.

Example 1.3 (*Non-realizable matroid polytope in rank 4*). We close this session with an example of a non-realizable matroid polytope in rank 4. We consider the following boundary structure of a 3-dimensional simplicial polytope presented as a list of 12 triangular facets of altogether 8 points. This combinatorial sphere can be described as a double pyramid over a 6-gon.

$$\begin{array}{llllll} 1,3,7 & 1,3,8 & 1,5,7 & 1,5,8 & 2,4,7 & 2,4,8 \\ 2,6,7 & 2,6,8 & 3,4,7 & 3,4,8 & 5,6,7 & 5,6,8 \end{array}$$

We can think of homogeneous coordinates for eight points forming such a boundary structure, and we think of them as an 8×4 matrix. Furthermore, we can imagine that we have calculated the signs of all 4×4 subdeterminants of this matrix leading to table 2.

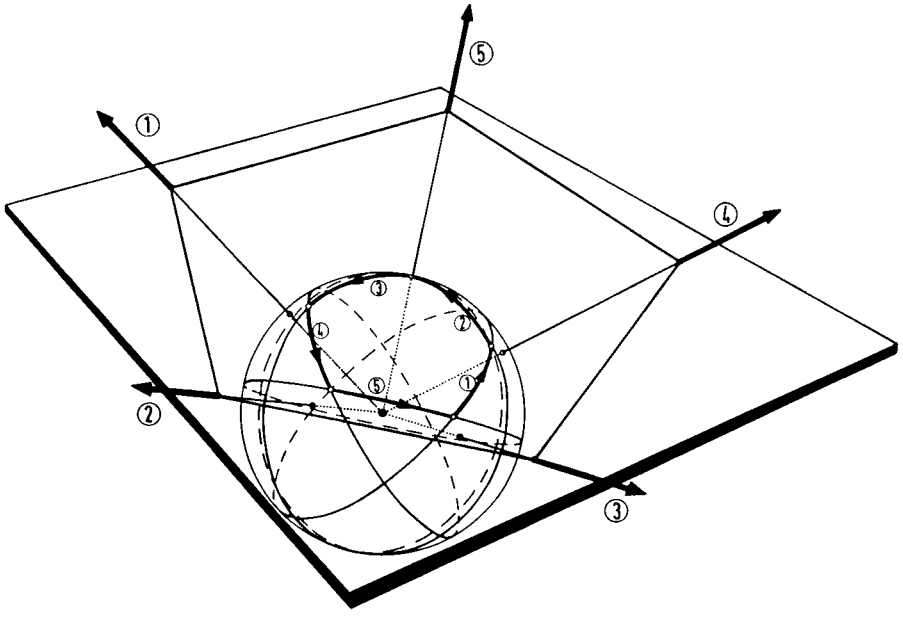


Figure 5. Convex 5-gon and its sphere system representation.

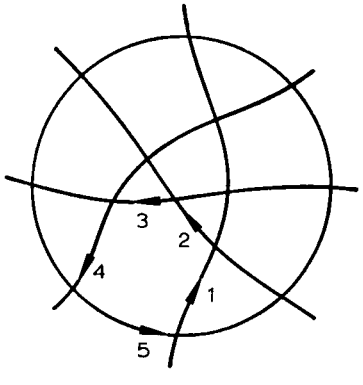


Figure 6. Essential information of the sphere system representation.

The sign of a determinant $[a,b,c,d]$ tells us the side on which the point d lies with respect to the oriented plane (a,b,c) defined as oriented affine hull of these points. Thus we list all these relations like in table 2. On the other hand this scheme gives also back the signs of all determinants. It turns out that the above signs obey the axioms for oriented matroids. The convexity property is encoded in the sign structure as well. Note in particular that the face structure coincides with the given one. Moreover, we can read off all hyperplanes (cocircuits) and all Radon partitions (circuits) from this combinatorial information without using the actual coordinates, see table 3.

Table 2
Cocircuit information about the matroid polyhedron

Negative	Plane	Positive	Negative	Plane	Positive
5,6,8	1,2,3	4,7	3,8	1,2,4	5,6,7
4,6,7	1,2,5	3,8	4,7	1,2,6	3,5,8
3,4,8	1,2,7	5,6	5,6	1,2,8	3,4,7
7	1,3,4	2,5,6,8	2,4,6,7	1,3,5	8
2,4,7	1,3,6	5,8		1,3,7	2,4,5,6,8
2,4,5,6,7	1,3,8		6,7	1,4,5	2,3,8
7	1,4,6	2,3,5,8	3	1,4,7	2,5,6,8
2,5,6,7	1,4,8	3	2,3,4,8	1,5,6	7
2,3,4,6,8	1,5,7			1,5,8	2,3,4,6,7
2,3,4,8	1,6,7	5	5	1,6,8	2,3,4,7
2,5,6	1,7,8	3,4	1,5,6,7	2,3,4	8
8	2,3,5	1,4,6,7	5,8	2,3,6	1,4,7
1,5,6,8	2,3,7	4	4	2,3,8	1,5,6,7
1,3,8	2,4,5	6,7	1,3,5,8	2,4,6	7
1,3,5,6,8	2,4,7			2,4,8	1,3,5,6,7
8	2,5,6	1,3,4,7	6	2,5,7	1,3,4,8
1,3,4,7	2,5,8	6		2,6,7	1,3,4,5,8
1,3,4,5,7	2,6,8		3,4	2,7,8	1,5,6
1,6,7	3,4,5	2,8	1,7	3,4,6	2,5,8
	3,4,7	1,2,5,6,8	1,2,5,6,7	3,4,8	
2,4,8	3,5,6	1,7	2,4,6,8	3,5,7	1
1	3,5,8	2,4,6,7	2,4	3,6,7	1,5,8
1,5,7	3,6,8	2,4	1,5	3,7,8	2,4,6
2,8	4,5,6	1,3,7	2,6,8	4,5,7	1,3
1,3	4,5,8	2,6,7	2	4,6,7	1,3,5,8
1,3,5,7	4,6,8	2	1,3,5	4,7,8	2,6
1,2,3,4,8	5,6,7			5,6,8	1,2,3,4,7
2,6	5,7,8	1,3,4	2,3,4	6,7,8	1,5

Once more the use of the table: for each oriented *affine plane* written in the center column, we write the points lying on the negative side on the left and the remaining on the right. This leads, e.g., to the sign of the determinant [1,3,8,4] as follows: we look for the row with the oriented plane (1,3,4), and we find, e.g., $\text{sign}[1,3,4,7] = -1$ and $\text{sign}[1,3,4,8] = +1$ which yields $\text{sign}[1,3,8,4] = -1$.

The Radon partitions are written in a similar fashion meaning that the points 4,5,6 and 7,8 form a Radon partition, etc.

This might be a useful matroid polytope to play with as an example. It is among the smallest non-realizable matroid polytopes, i.e., it cannot be realized with coordinates in real 3-dimensional vectorspace. But this example already shows the duality concept of hyperplanes (cocircuits) and Radon partitions (circuits) in a very clear way. The fact that there are no coordinates involved help to emphasize essential combinatorial properties of convex polytopes. The facial structure is kept, and perhaps the reader might study this example first on his own before reading further.

Table 3
Circuit information about the matroid polyhedron

Negative	Plane	Positive	Negative	Plane	Positive
4,5,6	1,2,3	7,8	7,8	1,2,4	3,5,6
4,7,8	1,2,5	3,6	4,5	1,2,6	3,7,8
3,6,8	1,2,7	4,5	3,6	1,2,8	4,5,7
2,5	1,3,4	6,7,8	2,7,8	1,3,5	4,6
2,4,5	1,3,6	7,8	4,6,8	1,3,7	2,5
4,6	1,3,8	2,5,7	3,6	1,4,5	2,7,8
7,8	1,4,6	2,3,5	2,5	1,4,7	3,6,8
2,5,7	1,4,8	3,6	4,7,8	1,5,6	2,3
2,3	1,5,7	4,6,8	2,3,7	1,5,8	4,6
4,5,8	1,6,7	2,3	4,5	1,6,8	2,3,7
2,3,6	1,7,8	4,5	5,7,8	2,3,4	1,6
7,8	2,3,5	1,4,6	4,5	2,3,6	1,7,8
1,6,8	2,3,7	4,5	1,6	2,3,8	4,5,7
1,3,6	2,4,5	7,8	1,7,8	2,4,6	3,5
3,5	2,4,7	1,6,8	3,5,7	2,4,8	1,6
1,4	2,5,6	3,7,8	3,6	2,5,7	1,4,8
3,6,7	2,5,8	1,4	1,4,8	2,6,7	3,5
1,4	2,6,8	3,5,7	3,6	2,7,8	1,4,5
6,7,8	3,4,5	1,2	1,2	3,4,6	5,7,8
1,6,8	3,4,7	2,5	1,6	3,4,8	2,5,7
1,2,4	3,5,6	7,8	4,6,8	3,5,7	1,2
4,6	3,5,8	1,2,7	1,2	3,6,7	4,5,8
1,2,7	3,6,8	4,5	4,5	3,7,8	1,2,6
7,8	4,5,6	1,2,3	1,2,8	4,5,7	3,6
1,2	4,5,8	3,6,7	3,5	4,6,7	1,2,8
3,5,7	4,6,8	1,2	1,2,5	4,7,8	3,6
2,3	5,6,7	1,4,8	2,3,7	5,6,8	1,4
1,2	5,7,8	3,4,6	3,4,5	6,7,8	1,2

2. Models in oriented matroid theory

The theory of convexity uses many tools encoding no metrical but only combinatorial information such as graphs and face lattices. Combinatorial properties of convex cones are also studied by means of Gale diagrams. But here the link between an ordered set X of points in a linear space and a linear transform $\tilde{X}$ uses the underlying vectorspaces. An oriented matroid and its dual can be viewed as the pure combinatorial abstraction of X together with a linear transform $\tilde{X}$, see McMullen (1978, 8a2 and 8a3). Just as such diagram techniques require a certain familiarity before becoming a powerful tool in the combinatorial theory of convexity, the same holds true for oriented matroids. In the following exposition of some basic concepts in oriented matroid theory, we aim at providing links between various ways of defining oriented matroids.

We start with a d-dimensional vectorspace $V \subset \mathbb{R}^n$ and its corresponding orthogonal vectorspace V^* of dimension $n - d$. Different ways of representing the

vectorspace $V \subset \mathbb{R}^n$, or $V^* \subset \mathbb{R}^n$, and successive abstractions lead to corresponding ways of defining oriented matroids. Duality of oriented matroids corresponds to switching from V to V^*.

2.1. Sphere systems

A basis $M = (c_1, c_2, \ldots, c_d)$ of V, written as an $(n \times d)$-matrix M with column vectors c_i can be interpreted as n ordered points $M = (v_1, v_2, \ldots, v_n)^{\mathrm{T}} \in \mathbb{R}^d$ in Euclidean d-space $\mathbb{R}^d$. If the positive hull $\mathrm{pos}\{v_1, v_2, \ldots, v_n\}$ lies in an open half space of $\mathbb{R}^d$, the basis of V can be interpreted as n ordered points in $\mathbb{R}^{d-1}$ with homogeneous coordinates. Furthermore, if these n points in $\mathbb{R}^{d-1}$ are the vertices of a convex polyhedron P in $\mathbb{R}^{d-1}$ (but not only in this case), we see that V contains essential combinatorial information of P: *the linear dependencies of row vectors of* M; and they do not depend on the particular choice of the basis $(c_1, c_2, \ldots, c_d)$ of V.

When assuming $v_1, v_2, \ldots, v_n$ as normal vectors of oriented hyperplanes in $\mathbb{R}^d$, $H_1, H_2, \ldots, H_n$, with H_i^+ denoting the open halfspace in direction v_i and H_i^- the one in direction $-v_i$, and when considering the spheres

$$S^{d-1} := \{x \in \mathbb{R}^d \mid \|x\| = 1\},$$

$$S_i^+ := S^{d-1} \cap H_i^+, \qquad S_i^- := S^{d-1} \cap H_i^-, \qquad S_i^0 := S^{d-1} \cap H_i,$$

we arrive at a *sphere system in the realizable case.* Its topological generalization, the *Edmonds–Folkman–Lawrence representation* (S, E, H) *of an oriented matroid* reads as follows:

Definition 2.1 (*Sphere system representation of an oriented matroid*)**.** A tripel (S, E, H) consisting of a $(d-1)$-dimensional sphere S, a finite set $E = \{1, 2, \ldots, n\}$, and a family H of subsets of S, $H := \{S_e^j \mid e \in E, j \in \{0, +, -\}\}$, is called an *oriented matroid with n points in rank d* if and only if

(1) S_e^0 is a topological hypersphere of S with sides S_e^+ and S_e^-.

(2) For each $a \subset E, a \neq \emptyset$, $F := \bigcap\{S_e^0 \mid e \in a\}$ is again a topological sphere.

(3) For any F of (2) and any $e \in E$, we have either $F \subset S_e^0$, or $S_e^0 \cap F$ is a hypersphere of F.

This representation of an oriented matroid is a very useful model to work with. Compare fig. 4 in section 1.1. It shows the essential part to understand a smallest non-linear sphere system for $n = 8, d = 4$.

Figure 7 is a sphere system representation of an oriented matroid with 6 points in rank 4. The 6 circles show homeomorphic images of halfspheres of $S_1, S_2, \ldots, S_6$ with oriented curves illustrating the intersection with the corresponding other 2-spheres. The 3-dimensional cell on S defined by

$$C := S_1^+ \cap S_2^+ \cap \cdots \cap S_6^+$$

is indicated by its 2-dimensional boundary cells.

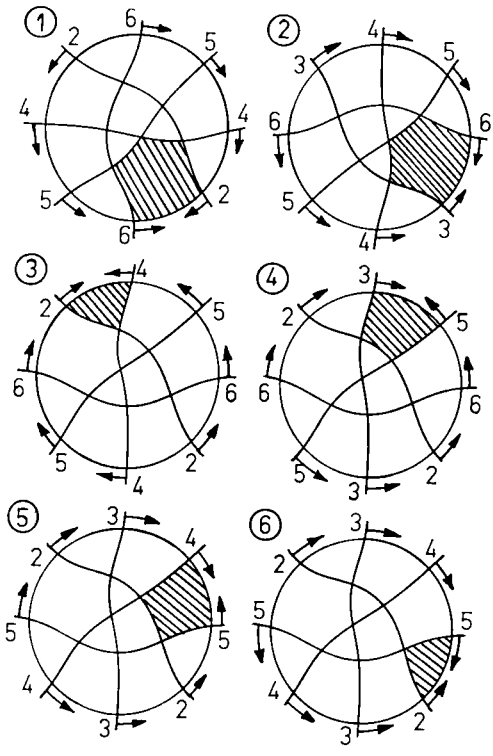


Figure 7.

Cells of maximal dimension on S like C are called *topes*. We observe that all spheres $S_1, S_2, \ldots, S_6$ occur as part of the boundary of this tope C. Therefore, the tope is called a *maximal cell*. Starting with a realizable affine convex polytope P in $\mathbb{R}^d$ and computing its oriented matroid as a sphere system representation would yield a maximal cell combinatorially equivalent to its polar P^*. Compare again Example 1.2.

We mention a survey with emphasis on topological methods where the generalization of the foregoing concept is discussed in a wider context, Björner (1993).

2.2. *Signed vectors of a vectorspace*

We start again with a d-dimensional vectorspace $V \subset \mathbb{R}^n$. The vectorspace V can be written in the form $V = \{\sum_{i=1}^{d} p_i c_i \mid p_i \in \mathbb{R}\}$ with $(c_1, c_2, \ldots, c_d)$ forming a basis of $V \subset \mathbb{R}^n$. The finite set $E = \{1, 2, \ldots, n\}$ together with the set $\mathcal{C}$ of *signed vectors*

$$\mathcal{C} := \left\{ \operatorname{sign}\left(\sum_{i=1}^{d} p_i c_i \right) \mid p \in \mathbb{R}^d \right\} = \left\{ \begin{pmatrix} \operatorname{sign}\langle p, v_1 \rangle \\ \operatorname{sign}\langle p, v_2 \rangle \\ \vdots \\ \operatorname{sign}\langle p, v_n \rangle \end{pmatrix} \mid p \in \mathbb{R}^d \right\},$$

where v_j is the vector of the jth components of the vectors c_i, $\langle \cdot, \cdot \rangle$ denotes the

scalar product, $v_j = (c_1^j, c_2^j, \ldots, c_d^j)$, is another way of defining a (realizable) oriented matroid. The generalization is given in Definition 2.2. We first define the composition $C_1 \cdot C_2$ of two signed vectors C_1 and C_2 to be the signed vector with components of C_1 except the zero components of C_1, which are replaced with the corresponding elements of C_2.

Definition 2.2 (*Oriented matroid, cocircuit span*). A finite set E together with a set $\mathcal{C}$ (*cocircuit span*) of signed vectors with the following properties is an *oriented matroid*. The corresponding properties as part of the vectorspace properties are given in brackets.

(1) $0 \in \mathcal{C}$ $(0 \in V)$,

(2) $X \in \mathcal{C}$ implies $-X \in \mathcal{C}$ ($x \in V$ implies $-x \in V$),

(3) $X, Y \in \mathcal{C}$ implies $X \cdot Y \in \mathcal{C}$ ($x, y \in V$ implies $x + \varepsilon y \in V$, $\varepsilon > 0$, small enough),

(4) $X, Y \in \mathcal{C}$, $X_f = -Y_f \neq 0$ (f separates X and Y) implies: ex. $Z \in \mathcal{C}$ such that $Z_f = 0$, and for all $e \in E$ not separating X and Y, we have $Z_e = (X \cdot Y)_e$ ($x, y \in V$ implies $(x_e y - y_e x)_e = 0$ for an $e \in E$).

Equal signed vectors correspond to the same cell in the cell decomposition of S by the oriented spheres S_e^j. In the following example, we can think of the vectors $v_j = (c_1^j, c_2^j, \ldots, c_d^j)$ as normal vectors pointing always to the left when passing along the great circles in the given direction. The signed vector

$$\begin{array}{rccccccccccc} & & 1 & & 2 & & 3 & & 4 & & 5 & \\ \mathcal{C}^{\mathrm{T}} = & (& +, & & 0, & & -, & & +, & & 0, & \\ & & 6 & & 7 & & 8 & & 9 & & & \\ & & -, & & +, & & 0, & & -, & &) & \end{array}$$

corresponds to the midpoint of the Pappos configuration, see fig. 8.

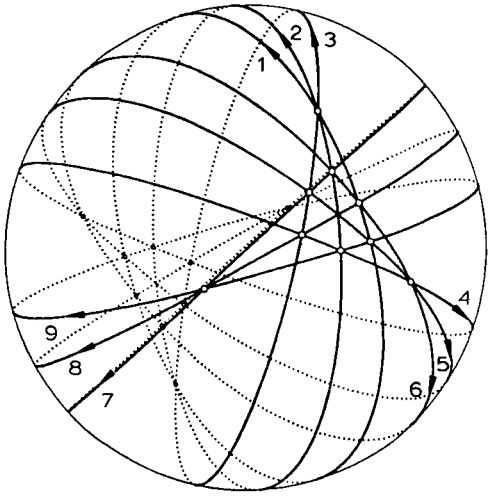


Figure 8.

The signed vectors of the zero-dimensional cells are called cocircuits and they correspond to hyperplanes in the realizable case.

2.3. Grassmann–Plücker polynomials

Representing our given vectorspace V as a point p on the Grassmann manifold leads to another approach of oriented matroids. We start with a straightforward definition of oriented matroids in the realizable (= representable, = linear) case. We use the exterior algebra of an n-dimensional vectorspace $\mathbb{R}^n$ and the notion of Grassmann coordinates, i.e., simple (or decomposable) d-vectors in the exterior product space $\wedge_d\mathbb{R}^n$, or $\mathbb{R}^{\binom{n}{d}}$. We consider the Grassmann manifold $\mathcal{G}_{n,d}^{\mathbb{R}}$ over $\mathbb{R}$ of all d-dimensional (unoriented) subspaces of $\mathbb{R}^n$. A point p on the Grassmann manifold, representing such a subspace, and represented by a simple d-vector in $\wedge_d\mathbb{R}^n$, can be used to define a realizable oriented matroid or realizable chirotope.

Definition 2.3 (*Realizable oriented matroid*). A *realizable oriented matroid* χ with n points of rank d on $E = \{1,2,\ldots,n\}$ is a map $\chi : E^d \to \{-1,0,+1\}$ with $\chi = \text{sign}(p)$ for some point p on the Grassmann manifold.

For the general case we have to fix some more notations. We write $\Lambda(n,d)$ for the set of all coordinates $[\lambda] = [\lambda_1,\lambda_2,\ldots,\lambda_d] \in E^d$, $1 \leqslant \lambda_1 < \lambda_2 < \cdots < \lambda_d \leqslant n$, (*bracket variables*) in $\mathbb{R}^{\binom{n}{d}}$. These coordinates define similar expressions $[\pi(\lambda)] = [\lambda_{\pi(1)},\lambda_{\pi(2)},\ldots,\lambda_{\pi(d)}]$ for all permutations π of the d numbers $\lambda_1,\lambda_2,\ldots,\lambda_d$, $[\pi(\lambda)] = \text{sign}(\pi)[\lambda]$. We define $[\lambda_1,\ldots,\lambda_d] := 0$ if for any $1 \leqslant i \neq j \leqslant n, \lambda_i = \lambda_j$. For a given field $\boldsymbol{K}$ let $\boldsymbol{K}[\Lambda(n,d)]$ be the polynomial ring freely generated by the elements of $\Lambda(n,d)$ with coefficients in $\boldsymbol{K}$.

Definition 2.4 (*k-term Grassmann–Plücker polynomial*). For an integer k with $3 \leqslant k \leqslant d+1$ and tuples $A := (a_1,\ldots,a_{k-2})$, $B := (b_1,\ldots,b_k)$, $C := (c_1,\ldots,c_{d-k+1})$ of altogether pairwise different elements of E, we define the *k-term Grassmann Plücker polynomial* $\{C|A|B\} \in \boldsymbol{K}[\Lambda(n,d)]$ on E, to be

$$\{C|A|B\} := \sum_{i=1}^{k}(-1)^{i+1}[c_1,\ldots,c_{d-k+1},a_1,\ldots,a_{k-2},b_i] \cdot [c_1,\ldots,c_{d-k+1},b_1,\ldots,\hat{b_i},\ldots,b_k],$$

$\hat{b_i}$ denotes that b_i has to be deleted. The set of all k-term Grassmann–Plücker polynomials in $\boldsymbol{K}[\Lambda(n,d)]$, $3 \leqslant k \leqslant d+1$, with $k \leqslant t$ will be denoted by $\text{GP}_{\boldsymbol{K}}^{n,d,t}$.

Definition 2.5 (*Grassmann variety*). The solution set in $\boldsymbol{K}^{\Lambda(n,d)}$ for all equations $\{C|A|B\} = 0$, i.e., the variety of the ideal generated by $\text{GP}_{\boldsymbol{K}}^{n,d,d+1}$ is called the *Grassmann variety*. The homogeneous coordinates $[\lambda]$ of the points $p \in \boldsymbol{K}^{\Lambda(n,d)}$ in this variety are precisely those for which $[\lambda] = \det(X_{\lambda_1},\ldots,X_{\lambda_d})$ for a certain $X := (X_1,\ldots,X_n) \in \boldsymbol{K}^{d\times n}$, $[\lambda] \in \Lambda(n,d)$.

For an alternating function $\chi : E^d \to \{-1,0,+1\}$, and a k-term Grassmann–Plücker polynomial $\{C|A|B\}$, we count the number of positive ($\#p$) and negative ($\#n$) monomials in the polynomial $\{C|A|B\}(\chi)$.

Definition 2.6 (*Oriented matroid, chirotope*). An *oriented matroid* (E,χ) *on* $E = \{1,\dots,n\}$ *of rank* d is an alternating function $\chi : E^d \to \mathrm{GF}_3 = \{-1,0,+1\}$, such that for each k-term Grassmann–Plücker-polynomial $\{C|A|B\}$ on E with $3 \leqslant k \leqslant d+1$, and $A := (a_1,\dots,a_{k-2})$, $B := (b_1,\dots,b_k)$, $C := (c_1,\dots,c_{d-k+1})$, we have either ($\#p = 0$, $\#n = 0$) or ($\#p \neq 0$, $\#n \neq 0$).

Before providing a non-trivial example of an oriented matroid in terms of bases, we compare the definition with that of a matroid in terms of bases. It formulates Steinitz' exchange lemma from linear algebra as an axiom (weak basis-exchange axiom). For that reason matroid theory is considered also as coordinate-free linear algebra.

Definition 2.7 (*Matroid*). A *matroid* $(E,\mathcal{B})$ *on* $E = \{1,\dots,n\}$ *of rank* d is a set $\mathcal{B}$ of d-element subsets of E such that we have: for any sets $A,B \in \mathcal{B}$ (the *bases*) and any element $a \in A \setminus B$, there is an element $b \in B$ such that $(A \setminus \{a\}) \cup \{b\} \in \mathcal{B}$.

There is a close connection between matroids and oriented matroids, see Bokowski, Guedes de Oliveira and Richter-Gebert (1991a): matroids and oriented matroids differ exactly by the underlying fields GF_2 and GF_3, respectively, i.e., a modified definition like Definition 2.6 with GF_2 instead of GF_3 is equivalent to Definition 2.7 of a matroid. For an oriented matroid (E,χ) of Definition 2.6 the absolute value of χ defines a matroid, the *underlying matroid of* (E,χ).

In the following we mainly consider uniform oriented matroids, i.e., $\chi(\lambda) \neq 0$ for all $\lambda \in \Lambda(n,d)$. Given a permutation σ on E, and $\lambda \in \Lambda(n,d)$, we define $\sigma\lambda \in E^d$ by $(\sigma\lambda)_i = \sigma(\lambda_i)$. For such a permutation σ and an oriented matroid χ in rank d with n vertices, we define a new oriented matroid $\sigma\chi : E \to \{-1,0,+1\}$ by $\lambda \mapsto \sigma\chi(\lambda) := \chi\big(\sigma^{-1}(\lambda)\big)$. Thus $\sigma\chi$ represents the oriented matroid χ with renumbered vertices according to σ. The permutation group S_n acts on the set of all oriented matroids of rank d with n vertices.

We call two oriented matroids χ, χ' of rank d with n vertices *A-equivalent* with respect to the subgroup A of S_n, if $\sigma\chi = +\chi'$ or $\sigma\chi = -\chi'$ for some $\sigma \in A$.

In particular we are interested in A_{FL}-equivalence, where A_{FL} denotes the group of automorphisms of the face lattice of the oriented matroid (of the matroid polytope). These concepts will be introduced in section 3. In this case we call an element $\sigma : E \to E$ of the automorphism group A_{FL} a *symmetry of the matroid polytope* χ if and only if $\sigma\chi = \chi$ or $\sigma\chi = -\chi$.

Example 2.8 (*Oriented matroid with* $n = 10$ *points in rank* $d = 5$). The map χ is given as signs of brackets ($10 = 0$, s = sign).

$$s[1,2,3,4,5] = +1, s[1,2,3,4,6] = +1, s[1,2,3,4,7] = +1, s[1,2,3,4,8] = -1,$$
$$s[1,2,3,4,9] = -1, s[1,2,3,4,0] = +1, s[1,2,3,5,6] = +1, s[1,2,3,5,7] = +1,$$
$$s[1,2,3,6,7] = -1, s[1,2,3,6,8] = -1, s[1,2,3,6,9] = -1, s[1,2,3,6,0] = +1,$$
$$s[1,2,3,7,8] = -1, s[1,2,3,7,9] = -1, s[1,2,3,7,0] = +1, s[1,2,3,8,9] = -1,$$
$$s[1,2,4,5,6] = -1, s[1,2,4,5,8] = -1, s[1,2,4,5,9] = -1, s[1,2,4,5,0] = -1,$$
$$s[1,2,4,6,7] = -1, s[1,2,4,6,8] = -1, s[1,2,4,6,9] = -1, s[1,2,4,6,0] = +1,$$
$$s[1,2,4,7,8] = -1, s[1,2,4,7,9] = -1, s[1,2,4,8,0] = +1, s[1,2,5,6,9] = -1,$$
$$s[1,2,5,7,9] = -1, s[1,2,6,7,0] = -1, s[1,3,5,7,9] = -1, s[2,4,6,8,0] = -1.$$

The remaining signs are defined by the symmetry group G generated by σ and τ,

$$\sigma := (1,3,5,7,9)(2,4,6,8,0), \qquad \tau := (0)(1)(2,8)(3,9)(4,6)(5,7),$$

$$\operatorname{sign}[b_1,b_2,b_3,b_4,b_5] := \operatorname{sign}(\pi)\operatorname{sign}[\pi(b_1),\ldots,\pi(b_5)] \quad \text{for all } \pi \in G.$$

In order to check the oriented matroid property, we consider a 4-term Grassmann–Plücker polynomial

$$\begin{aligned}\{C|A|B\} &= \{7,8|1,2|3,4,5,6\}\\ &= +[7,8,1,2,3][7,8,4,5,6] - [7,8,1,2,4][7,8,3,5,6]\\ &\quad + [7,8,1,2,5][7,8,3,4,6] - [7,8,1,2,6][7,8,3,4,5].\end{aligned}$$

The sign function χ for our bases $b \in \Lambda(10,5)$ leads to

$$\begin{array}{llll}
\chi[7,8,1,2,3] & \chi[7,8,4,5,6] & \chi[7,8,1,2,4] & \chi[7,8,3,5,6]\\
= -1, & \chi[5,6,7,8,4] & = -1, & \chi[5,6,7,8,3]\\
 & \chi[1,2,3,4,0] & & \chi[1,2,3,4,9]\\
 & = +1, & & = -1,
\end{array}$$

$$\begin{array}{llll}
\chi[7,8,1,2,5] & \chi[7,8,3,4,6] & \chi[7,8,1,2,6] & \chi[7,8,3,4,5]\\
\chi[1,2,5,6,9] & \chi[3,4,6,7,8] & -\chi[1,8,6,7,2] & \chi[3,4,5,7,8]\\
= -1, & \chi[1,2,4,5,6] & -\chi[1,2,4,5,8] & \chi[1,2,3,5,6]\\
 & = -1, & = +1, & = +1.
\end{array}$$

We have $\#p = 1 \neq 0, \#n = 3 \neq 0$. The condition of the last definition of an oriented matroid holds. The same is true for all Grassmann–Plücker polynomials; for the images under elements of the symmetry group G this is immediate.

In order to check all conditions of an oriented matroid, computational support seems to be essential. But even in this case, the following result due to Las Vergnas is of significance.

Theorem 2.9 (Las Vergnas). *In the uniform case, i.e., $\chi(\lambda) \neq 0$ for all $\lambda \in \Lambda(n,d)$, the 3-term Grassmann–Plücker relations are sufficient to check the oriented matroid property.*

See Bokowski, Guedes de Oliveira and Richter-Gebert (1991a) for an algebraic proof of Las Vergnas' result as well as for the introduction of an algebraic variety characterizing oriented matroids. The original proof of Las Vergnas uses non-algebraic methods, and it uses the non-trivial equivalence of oriented matroids defined in terms of circuits with oriented matroids defined by Grassmann–Plücker relations. When working with oriented matroids, the following problems occur naturally.

Problem 2.10. Find a fast algorithm for testing whether a given map $\chi : \Lambda(n,d) \to \mathrm{GF}_3$ is an oriented matroid.

Problem 2.11. Find a fast algorithm for testing whether two oriented matroids are isomorphic up to relabeling the vertices and up to sign reversal.

We have by now defined oriented matroids in three completely different looking but isomorphic and essential ways: via the sphere system representation, via the cocircuit span approach, and via the chirotope approach using Grassmann–Plücker relations. Like in ordinary matroid theory, there are many more ways to define oriented matroids, the appropriate one depends on the context and the purpose. Here, we cannot accomplish a comprehensive report about all possibilities to define oriented matroids. We do not aim at providing proofs of equivalence, either. Our goal is to introduce the structure by peeling off a very often technical looking machinery in order to show some underlying main ideas. We recommend now to work with the examples and to discover how the transition from one model to another has to be done, thus becoming more familiar with the abstract definitions. The concept of hyperline sequences introduced in the next section might serve as a bridge between the sphere systems and the chirotopes.

3. Matroid polytopes

This section deals with generalizations of ordinary polytopes to the oriented matroid setting. There are two ways to do so. This is done in both cases by introducing a face lattice for an acyclic oriented matroid which coincides in the realizable case with the face lattice of the polytope. The first one is a generalization of taking the convex hull of all its extreme points of a point set, and in the second one, a polar approach, one proceeds by taking the intersection of all supporting halfspaces. The first approach was introduced and studied first by Las Vergnas whereas the second version of generalizing ordinary polytopes was investigated by Edmonds, Lawrence, Mandel, Fukuda and others. A contribution of Billera and Spellman-Munson (1984b), has shown that both concepts do not coincide. When we use the Las Vergnas lattice in the following, it is because the example is easier to handle. The cells of maximal dimension on the sphere in the sphere system representation seem to be a very natural concept as well.

For working with matroid polytopes, hyperline sequences are a useful concept. This concept is of importance in its own right, and it is closely related to the last

definition of an oriented matroid; for technical reasons emphasis will be placed on the uniform case.

3.1. Basic concepts in oriented matroid theory

We write again for the finite set of n points $E = \{1, \ldots, n\}$, and we use $\Lambda(n,d)$ for the set of all tuples $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_d) \in E^d$, $1 \leqslant \lambda_1 < \lambda_2 < \cdots < \lambda_d \leqslant n$. To each oriented hyperline $l = (l_1, \ldots, l_{d-2}) \in \Lambda(n, d-2)$, we assign a *normalized hyperline sequence*

$$h_l = (l_1, \ldots, l_{d-2} | s^l_{d-1} l_{d-1}, \ldots, s^l_n l_n)$$

with $\{l_1, \ldots, l_n\} = E$, $l_{d-1} < l_d, \ldots, l_n$, and $s^l_i \in \{-1, 1\}, d-1 \leqslant i \leqslant n$ $(s^l_{d-1} = 1)$; and we assign corresponding *bracket signs,*

$$\operatorname{sign}[l_1, \ldots, l_{d-2}, s^l_j l_j, s^l_k l_k] := 1$$

for all j,k: $d-1 \leqslant j < k \leqslant n$, such that for all bases $b \in \Lambda(n,d)$ and for all permutations $\pi : \{1, \ldots, d\} \to \{1, \ldots, d\}$

$$\operatorname{sign}[b_1, \ldots, b_d] = \operatorname{sign}(\pi) \operatorname{sign}[b_{\pi(1)}, \ldots, b_{\pi(d)}],$$

$$-\operatorname{sign}[b_1, \ldots, b_j, \ldots, b_d] = \operatorname{sign}[b_1, \ldots, -b_j, \ldots, b_d].$$

The last conditions tell us that the $\binom{d}{2}$ definitions of signs for a fixed bracket $b = [b_1, \ldots, b_d] \in \Lambda(n,d)$ have to be compatible. An easy argument shows: the hyperline sequences are determined by the bracket signs and vice versa. Especially, Definition 2.6 of an oriented matroid using Grassmann–Plücker relations leads to a corresponding hyperline sequence.

Definition 3.1 (*Oriented matroid in terms of hyperline sequences*)**.** The set E together with a set of hyperline sequences with compatible bracket signs is a *uniform oriented matroid of n points in rank d.*

Compare again Example 1.2. Half of the circular sequence is redundant there. The following non-trivial example of a matroid polytope shows some more concepts in oriented matroid theory. The reader not familiar with oriented matroids is invited to study this example carefully.

Example 3.2 (*Matroid polytope*)**.** We define an oriented matroid as a set HL of sign compatible hyperline sequences for $E = \{1, 2, \ldots, 10 = 0\}$ in rank 5 as follows:

$$
\begin{array}{llllllll}
(1,2,3|4, & 5, & 7, & 6, & 0, & -9, & -8 &) \\
(1,2,4|3, & -7, & -6, & -0, & -5, & 8, & 9 &) \\
(1,2,6|3, & 5, & 4, & 7, & 8, & 9, & -0 &) \\
(1,2,7|3, & 5, & 4, & -6, & 8, & 9, & -0 &) \\
(1,2,9|3, & -6, & -7, & -0, & -5, & -4, & -8 &) \\
(1,2,5|3, & -7, & -6, & -0, & 4, & 8, & 9 &) \\
(1,2,0|3, & 5, & 4, & 8, & 9, & 7, & 6 &) \\
(1,2,8|3, & -6, & -7, & -0, & -5, & -4, & 9 &) \\
(1,3,5|2, & 7, & 6, & 0, & -4, & -8, & -9 &) \\
(1,3,6|2, & -5, & -4, & -7, & -8, & -9, & 0 &) \\
(2,4,8|1, & 3, & -5, & -7, & -6, & -9, & -0 &) \\
(1,3,7|2, & -5, & -4, & 6, & -8, & -9, & 0 &) \\
(1,4,5|2, & 3, & -0, & -6, & -7, & -9, & -8 &) \\
(1,4,6|2, & 3, & 5, & -7, & -9, & -8, & 0 &) \\
(1,4,0|2, & 3, & 5, & -8, & -9, & -7, & -6 &) \\
(2,4,6|1, & -5, & 9, & 8, & -3, & 7, & -0 &)
\end{array}
$$

The complete set of all hyperline sequences $\{\pi(h_l) \mid \pi \in G, h_l \in \mathrm{HL}\}$ are images of hyperline sequences of HL, i.e., of the above list, under elements π of the permutation group G which was defined in Example 2.8. In general, the *normalized* hyperline sequence condition is violated for the images of elements of G, $\pi(l) \notin \Lambda(n, d-2)$ but the bracket sign condition carries over as

$$\mathrm{sign}[\pi(l_1), \ldots, \pi(l_{d-2}), s_j^l \pi(l_j), \ldots, s_k^l \pi(l_k)] := 1.$$

We can assume a transition to the normalized form whilst retaining this sign property. The oriented matroid, we have thus defined, is equal to the foregoing Example 2.8 defined by using Grassmann–Plücker sign conditions. In the uniform case it turns out that a single 3-term Grassmann–Plücker sign condition requires for a corresponding hyperline together with 4 additional points a coherent ordering. In other words, a violation of a 3-term Grassmann–Plücker sign condition can be checked already locally, see Antonin (1982) for a first reference.

In order to work with the given set of hyperline sequences, it is useful to have the following picture in mind (fig. 9): we assume the point set of 10 points lying in $\mathbb{R}^4$, and we use an orthogonal projection of these 10 points along $L := \mathrm{aff}\{l_1, \ldots, l_{d-2}\}$ onto the 2-dimensional linear space L^* orthogonal to L. The circular occurrence of the points $E \setminus \{l_1, \ldots, l_{d-2}\}$ is depicted.

The set of these hyperline sequences are closely related to the sphere system model and to the allowable sequences of Goodman and Pollack and the λ-matrix concept, see Goodman and Pollack (1983, 1984b). In the sphere system model, when passing along the 1-sphere $S_1 \cap S_3 \cap S_7$ in the order given by the hyperline sequence, we come to the cocircuits

$$(1,3,7,2), (1,3,7,-5), (1,3,7,-4), (1,3,7,6), (1,3,7,-8), (1,3,7,-9), (1,3,7,0)$$

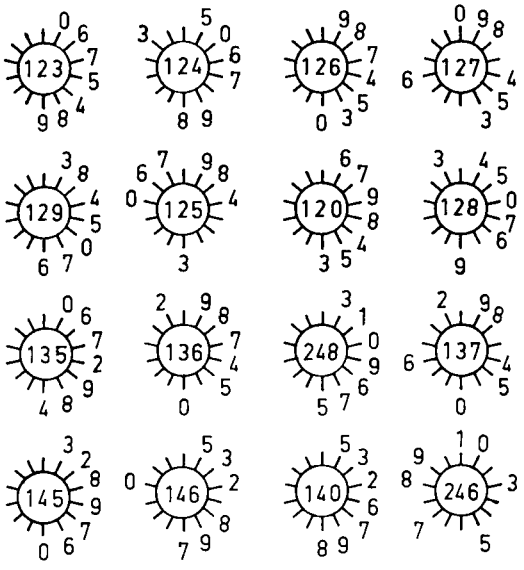


Figure 9. Hyperline sequences considered geometrically.

(intersections with the remaining oriented spheres). The way we pierce these spheres, from the positive side to the negative, or vice versa, is given by the signs in the hyperline sequences. $L = \text{aff}\{1,3,7\}$.

For an oriented matroid we define *circuits* and *cocircuits* as follows. For $\lambda \in \Lambda(n,d-1)$ and $\mu \in \Lambda(n,d+1)$, we define the ith component of $C^*_\lambda, C_\mu \in \{-1,0,+1\}^n$ to be

$$C^*_\lambda(i) := \chi(\lambda_1,\ldots,\lambda_{d-1},i) \quad \text{for } i = 1,\ldots,n,$$

$$C_\mu(i) := \begin{cases} (-1)^j \chi(\mu_1,\ldots,\widehat{\mu_j},\ldots,\mu_{d+1}) & \text{if } i = \mu_j, \\ 0 & \text{otherwise,} \end{cases}$$

or by using the δ-function, we write

$$C_\mu(i) := (-1)^j \chi(\mu_1,\ldots,\widehat{\mu_j},\ldots,\mu_{d+1})\delta(\widehat{\mu_j},i).$$

Then

$$\mathcal{C}(\chi) := \{\pm C_\mu \mid \mu \in \Lambda(n,d+1)\}$$

is the set of circuits of χ, and

$$\mathcal{C}^*(\chi) := \{\pm C^*_\lambda \mid \lambda \in \Lambda(n,d-1)\}$$

is the set of cocircuits of χ.

Table 4
Compare hyperline sequence for hyperline (1,2,3)

	1	2	3	4	5	6	7	8	9	0	
$s\cdot$	(0,	0,	0,	+,	+,	+,	+,	+,	0,	+)	$= C_a$
$s\cdot$	(0,	0,	0,	+,	+,	+,	+,	0,	−,	+)	
$s\cdot$	(0,	0,	0,	0,	+,	+,	+,	−,	−,	+)	
$s\cdot$	(0,	0,	0,	−,	0,	+,	+,	−,	−,	+)	
$s\cdot$	(0,	0,	0,	−,	−,	+,	0,	−,	−,	+)	
$s\cdot$	(0,	0,	0,	−,	−,	0,	−,	−,	−,	+)	
$s\cdot$	(0,	0,	0,	−,	−,	−,	−,	−,	−,	0)	$= C_b$

The signed vectors in table 4 are cocircuits ($s = +1$ or -1) in the above example.

C_a and C_b are supporting cocircuits, supporting hyperplanes in the realizable case. $E \setminus \operatorname{supp} C_a$ and $E \setminus \operatorname{supp} C_b$ are *facets*. χ is *acyclic* if and only if there is a positive cocircuit, χ is *cyclic* otherwise.

An *acyclic* oriented matroid is also called affine. This can be expressed in the following form: an oriented matroid χ in rank d with n vertices is called *affine* if for all $\lambda \in \Lambda(n, d+1)$ the set $\{(-1)^i \cdot \chi(\lambda_1, \ldots, \widehat{\lambda_i}, \ldots, \lambda_{d+1}) \mid i \in \{1, \ldots, d+1\}\}$ either contains $\{-1, +1\}$ or equals $\{0\}$.

$\{1,2,4\}$ is not a face which is easily seen by looking at the cocircuits in table 5.

Table 5
Compare hyperline sequence for hyperline (1,2,4)

	1	2	3	4	5	6	7	8	9	0
$s\cdot$	(0,	0,	0,	0,	−,	−,	−,	+,	+,	−)
$s\cdot$	(0,	0,	−,	0,	−,	−,	0,	+,	+,	−)
$s\cdot$	(0,	0,	−,	0,	−,	0,	+,	+,	+,	−)
$s\cdot$	(0,	0,	−,	0,	−,	+,	+,	+,	+,	0)
$s\cdot$	(0,	0,	−,	0,	0,	+,	+,	+,	+,	+)
$s\cdot$	(0,	0,	−,	0,	+,	+,	+,	0,	+,	+)
$s\cdot$	(0,	0,	−,	0,	+,	+,	+,	−,	0,	+)

$$
\begin{aligned}
E \setminus \operatorname{supp} C_c &= E \setminus \operatorname{supp}((0,0,0,+,+,+,+,+,0,+) \\
&\qquad \cdot (0,0,0,+,+,+,+,+,+,0)) \\
&= \{1,2,3\}
\end{aligned}
$$

is a face as an intersection of two facets. The faces form a lattice, the *convex hull* of the oriented matroid. The facets in our case are $F_1 = \{1,2,3,0\}, F_2 = \{1,2,6,9\}, F_3 = \{1,3,5,0\}, F_4 = \{1,4,5,0\}, F_5 = \{1,4,8,0\}$ and the images under the symmetry, $g(F_i)$ for $g \in G$.

In the face lattice $\mathrm{FL}(\chi)$ all points are involved, there are no *interior points*,

i.e., all points are *extreme*. Starting with the face lattice and reconstructing the oriented matroid from the given sphere is a good exercise. It turns out that the oriented matroid is uniquely defined by the sphere. In other words: the sphere is *rigid*. The sphere was investigated by Altshuler, Bokowski and Steinberg (1980), see also Bokowski and Garms (1987). The rigidity can be established beforehand due to the following result of Shemer presented also in Perles survey on Shemer's results, Oberwolfach (1984).

Theorem 3.3 (Shemer 1982). *The face structure of a neighborly m-polytope determines the face structure of all its subpolytopes.*

This result was carried over to matroid polytopes.

Theorem 3.4 (Sturmfels 1987c). *Every neighborly* $(2k+1)$ *matroid polytope is rigid.*

A minimal matroid polytope with respect to dimension and number of vertices which is no ordinary polytope was implicitly given in Altshuler, Bokowski and Steinberg (1980). The essential oriented matroid showing its non-realizability was mentioned before in the introduction. It was first published in Bokowski and Sturmfels (1985). Roudneff's example is the contraction of the matroid polytope by a single point, see Roudneff and Sturmfels (1988).

Definition 3.5 (*Hull operator*). Matroids occur in our context as underlying matroids of oriented matroids. We have the *hull operator* $\langle\cdot\rangle$ of a set of abstract points generalizing the affine hull in vectorspaces. It is the largest set of points with the same rank as the given set.

Definition 3.6 (*Outer simplex*). A basis $\lambda \in \Lambda(n,d)$ is called an *outer simplex* of an acyclic oriented matroid if for some $i \in \{1,\ldots,d\}$, $\langle \lambda \setminus \{\lambda_i\}\rangle$ is a facet of χ.

If $\mathcal{A}(\chi)$ denotes the set of outer simplices of an oriented matroid, we have

$$\mathrm{FL}(\chi_1) = \mathrm{FL}(\chi_2) \quad \text{implies} \quad \mathcal{A}(\chi_1) = \mathcal{A}(\chi_2), \chi_1|_{\mathcal{A}} = \pm\chi_2|_{\mathcal{A}}.$$

This observation is important when determining matroid polytopes from given combinatorial spheres like in the above example.

Definition 3.7 (*Mutation, mutant*). When changing a single sign of a basis of an oriented matroid such that the new set of signs of bases forms an oriented matroid again, we call according to a suggestion of Grünbaum, the basis a *mutation* and the pair of oriented matroids a pair of *mutants*.

In the above example of a matroid polytope, $g(1,2,3,8,9)$ and $g(1,2,3,6,0)$ are mutations for $g \in G$. This can be confirmed by checking corresponding hyperline

sequences. Any two cocircuits corresponding to facets of the simplex (the basis) must be adjacent. On the other hand, this is a sufficient condition for not violating any Grassmann–Plücker relation.

The mutant obtained when changing the signs of $g(1,2,3,8,9)$ has the following facets: $g(1260), g(1203), g(1269), g(1238), g \in G$. In changing the sign of $g([1,2,3,8,9])$ for all $g \in G$, we have again a matroid polytope with the same symmetry and 30 facets. The facets are

$$g(\{1,2,6,0\}), \qquad g(\{1,2,3,0\}), \qquad g(\{1,2,3,8\}), \qquad g(\{1,2,6,9\}).$$

The matroid polytope we started with has 35 facets.

Las Vergnas asked the following: is it possible to gain all uniform oriented matroids by just switching single signs of brackets in the above sense? This question is still open whereas the realizable case from geometric considerations is trivial. However, the author has shown recently that a counterexample to the Las Vergnas conjecture must have at least 13 elements.

The notion of a reorientation class for oriented matroids (Definition 3.9) makes no sense when considering polytopes only. The study of reorientation classes can be viewed as the study of the combinatorial abstraction of projectively equivalent properties. A question of McMullen deals with a corresponding problem in the realizable case which might be solved by methods from oriented matroids. McMullen's problem formulated in the oriented matroid language says the following.

Problem 3.8 (*McMullen*). Find a uniform arrangement (sphere system representation) of $2d+2$ (pseudo)-hyperplanes in projective d-space which has no cell of maximal dimension such that each (pseudo)-hyperplane has a boundary point in common with this cell. See Roudneff's (1987b) list of problems, No. 12.

Definition 3.9 (*Reorientation class*). The *reorientation class* of an oriented matroid is a class of oriented matroids containing the given one which is closed under taking negative signs of its points. In the sphere system representation, the reorientation class can be represented by non-fixing any orientation for the set of spheres. Reorienting one or several spheres corresponds to reorienting the oriented matroid at corresponding points. The oriented matroid χ' obtained from χ by a sign reversal of point k is defined as

$$\chi'([b_1,\ldots,b_d]) = \begin{cases} -\chi([b_1,\ldots,b_d]) & \text{if } b_j = k \text{ for one } j, \\ +\chi([b_1,\ldots,b_d]) & \text{otherwise.} \end{cases}$$

There are two more operations which are considered frequently in oriented matroid theory and which have easy meanings in the polytopal realizable case: deleting a point (or a set of points) corresponds to taking only the convex hull

of the remaining points into account (the opposite operation is called extension); contracting a point leads to the projection from this point onto a hyperplane (the opposite operation is called lifting).

The following definition of an *oriented matroid manifold* generalizes the notion of a *matroid polytope*. For technical reasons this is defined only in the rank 4 case.

We call the pair of an affine oriented matroid $\chi : \Lambda(n,4) \to \{-1,0,+1\}$ and a (triangulated) combinatorial manifold a (triangulated) *oriented matroid manifold* if for every pair of an edge $\{i,j\}$ and a triangle $\{k,l,m\}$ of the combinatorial manifold, there is no corresponding circuit X in χ with $X^+ := X^{-1}(\{+1\}) = \{i,j\}$ and $X^- := X^{-1}(\{-1\}) = \{k,l,m\}$.

The oriented matroid manifold property implies that in case of realizability of χ there are no self-intersections. See also chapter 2.4 in this Handbook.

In this context, it is a very natural question to ask whether there exist "non-embeddable Platonic solids" in this sense. Las Vergnas as well as Dress have found, independently, that this is not the case (private communication).

An early application of Grassmann–Plücker polynomials to convexity was used in the following example.

Example 3.10 (*Kleinschmidt's sphere*)**.** Can combinatorial symmetry groups of face lattices of polyhedra always be obtained from their geometric symmetry groups inherited from a suitable realization of the polytope and suitable affine transformations of the polytope? In other words, can we find a convex polyhedron with hidden symmetries in the sense of Wills, see, e.g., Bokowski and Wills (1988)? The negative answer in dimension 3 was given in 1971 by Mani (1971). The corresponding question in higher dimensions was open for a long time.

Kleinschmidt's sphere cannot be realized with its full combinatorial symmetry. For the combinatorial input, it was a challenging task to prove that it is the boundary structure of a convex polyhedron in 4-dimensional space.

Kleinschmidt's sphere is given by the following list of simplicial facets.

{1,2,3,4}	{1,2,3,7}	{1,2,4,8}	{1,2,6,7}	{1,2,6,8}	{1,3,4,7}	{1,4,5,6}
{1,4,5,8}	{1,4,6,7}	{1,5,6,8}	{2,3,4,8}	{2,3,7,0}	{2,3,8,9}	{2,3,9,0}
{2,6,7,9}	{2,6,8,9}	{2,7,9,0}	{3,4,5,7}	{3,4,5,8}	{3,5,7,0}	{3,5,8,0}
{3,8,9,0}	{4,5,6,7}	{5,6,7,9}	{5,6,8,0}	{5,6,9,0}	{5,7,9,0}	{6,8,9,0}

Three independent methods were successfully applied by Altshuler, Bokowski, and Ewald to confirm Kleinschmidt's hope that the sphere is polytopal. These methods can be found in two articles about this sphere Bokowski, Ewald and Kleinschmidt (1984) and Altshuler (1984). Sturmfels pointed out in 1989 (private communication) that there are faults (probably misprints) in the two sets of coordinates given in Bokowski, Ewald and Kleinschmidt (1984). We provide, in the following, correct coordinates.

The face lattice of the sphere defines a partial chirotope, i.e., there are several oriented matroids having the same face lattice.

One realization of Kleinschmidt's sphere has the following coordinates:

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ -0.2 & -0.0259 & -1 & +0.4 & +0.8 \\ 0 & 1 & 0 & 0 & 0 \\ +5.2636 & +0.5 & +1 & -0.402 & -0.6 \\ 0 & 0 & 1 & 0 & 0 \\ +1 & -1 & -1 & +1 & +1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ -10 & -0.99 & -1 & +3 & +38.645 \\ -2 & -0.19 & +1 & +0.25 & +7.695 \end{pmatrix}$$

For details how these coordinates were found, see Bokowski (1991). These coordinates might be helpful to find a corresponding non-simplicial sphere with similar properties, thus answering a question posed by Billera recently.

Finally, we come back to the matroid polytope of Example 3.2, and we pose the open question whether its polar is a matroid polytope as well. In other words, reverse its boundary lattice (upside down) and find a matroid polytope (necessarily not uniform) with precisely this face lattice.

3.2. Duality for oriented matroids

3.2.1. Duality in general

Duality for oriented matroids can be seen by looking at circuits and cocircuits, the Grassmann–Plücker relation, and its sign analogon. The Grassmann–Plücker relation reads

$$\sum_{i=1}^{n} [\lambda_1, \ldots, \lambda_{d-1}, i](-1)^j [\mu_1, \ldots, \widehat{\mu_j}, \ldots, \mu_{d+1}] \delta(\widehat{\mu_j}, i) = 0.$$

We define the product of $C_\lambda^*(i) \cdot C_\mu(i)$ in the usual way as if + were +1 and − were −1. The sum

$$\sum_{i=1}^{n} C_\lambda^*(i) \cdot C_\mu(i)$$

becomes zero by definition if and only if either all summands are zero or +1 and −1 occur both at least once as a summand. By using this *scalar product,* we can say, a circuit is a signed vector which is orthogonal to all its cocircuits. And the reverse is true, a cocircuit is orthogonal to all circuits. These properties can be used to characterize circuits from cocircuits and vice versa. Now when we interpret a set of circuits as a set of cocircuits of another oriented matroid, we switch to the *dual oriented matroid.* Duality for oriented matroids can be seen from this point

of view. Bland studied this abstract duality in a wider context, see Bland and Dietrich (1988) and the papers cited there. This was Bland's original approach of characterizing oriented matroids. By looking at linear programming techniques in a purely combinatorial manner, he found his new finite pivot rule, Bland (1977b). In this chapter we do not seek to accomplish a report about oriented matroid programming. The interested reader is referred to Fukada (1982), Björner et al. (1991), and Bachem and Kern (1992).

Circuits correspond to minimal Radon-partitions and cocircuits correspond to hyperplanes in the realizable case. This correspondence is in the realizable case well known when applying the Gale diagram technique, the corresponding pure combinatorial setting was worked out by Sturmfels (1987c) in his Darmstadt Ph.D. thesis. We refer the reader also to chapter 2.3 in this Handbook.

3.2.2. d-Polytopes with $d+4$ vertices

The affine Gale diagram technique, see Bokowski and Shemer (1987) for its first application in this context and later Sturmfels (1988b), can be used for investigating polytopes where the number of vertices does not exceed the dimension very much. The surprising fact that the planar realizability question applies to results for polytopes in all dimensions was always used by the author (like in the case of Sturmfels) to attract students to enter this field.

Theorem 3.11 (Bokowski and Sturmfels 1989b). *Given a combinatorial $(d-1)$-sphere S with $d+4$ vertices, there is a finite set $\{\chi_1, \ldots, \chi_k\}$ of affine oriented matroids of rank 3 with $d+4$ vertices such that S is polytopal if and only if at least one χ_i is realizable.*

In coordinates this construction can be described as follows. Assuming that a $(d-1)$-sphere is isomorphic to the boundary complex of a d-polytope P with $d+4$ vertices, we can write homogeneous coordinates for the vertices of P as rows of a $(d+4) \times (d+1)$-matrix M. Without loss of generality, and if necessary, after relabeling the vertices of P, we can further assume that $M = \binom{1}{A}$, where $\mathbf{1}$ denotes the $(d+1) \times (d+1)$-unit matrix, and A is a $3 \times (d+1)$-matrix. The isomorphism from S to the boundary complex of P imposes certain sign conditions on the subdeterminants of A. In fact, if P is evendimensional and neighborly, compare Altshuler (1974), then for *all* subdeterminants of A the sign is prescribed due to a result of Shemer (1982).

If χ is the chirotope associated with P, then the dual χ^* corresponds to a Gale transform of P, the coordinates of which are given by the rows of the $(d+4) \times 3$-matrix

$$\begin{pmatrix} -A^{\mathrm{T}} \\ \mathbf{1} \end{pmatrix}.$$

Here $\mathbf{1}$ denotes the 3×3-unit matrix. After replacing some of the row vectors of this matrix by their negative, we can interpret this matrix as homogeneous

coordinate matrix of a configuration of $d+4$ points in the affine plane. The sphere S is polytopal if and only if a configuration can be found which fulfills the above conditions on the minors of $-A^T$ which are (up to some sign reversals) the minors of A.

4. Matroid theory in convexity

Once the reader is familiar with at least one concept of defining oriented matroids and its corresponding language, he probably appreciates having results in combinatorial convexity expressed in terms of oriented matroids. We have seen this already in case of Shemer's result (Theorem 3.3). For this section we have selected results which might serve to support the importance of oriented matroid theory in convexity. Besides these results, the reader will find more in the book of Björner et al. (1991), especially in chapter 9. Before we deal with specific theorems, here are again some main advantages of oriented matroid theory:

(i) Oriented matroid theory clarifies the duality concept.

(ii) Oriented matroid theory serves as a sieve for theorems which are certain to be of a purely combinatorial nature like the upper bound theorem for matroid polytopes.

(iii) Oriented matroid theory can be considered as a coordinate-free linear algebra with an internal convex hull structure (e.g., the Las Vergnas lattice). Oriented matroid programming is possible (linear programming without coordinates).

(iv) Some conjectures in convexity have its generalizations to oriented matroids and might be solvable within a wider context, see e.g., Problem 3.8 and section 4.1.

(v) Oriented matroid theory in its computational applications (e.g., extensions in the enumeration problem for polytopes and in the realizability problem, see section 5) leads to automatic theorem proving methods.

4.1. The missing axiom for polytopal spheres

The realizability problem for matroids is a fundamental question in matroid theory in general (see also section 5). One is tempted to look for an additional (local) axiom to characterize those matroids which are realizable, say over the field $\mathbb{R}$. But as Vamos has shown, there is no such axiom.

Theorem 4.1 (Vamos 1978). *The missing axiom for matroids is lost forever.*

Now one can ask the same for oriented matroids. Here again the answer has been found in different ways.

Theorem 4.2 (Dress; Bokowski and Sturmfels 1989b; Mandel). *The missing axiom for oriented matroids is lost forever.*

These results made it very likely that a local characterization of spheres which are boundaries of realizable matroid polytopes (the Steinitz problem, see Ewald et

al. 1978) is not possible as well, compare Bokowski and Sturmfels (1989b) which was written years before it was published.

It was Sturmfels who found additional techniques in his Darmstadt Ph.D. thesis to tackle this problem. The Lawrence polytope construction, Lawrence (1975), the equivalence of planar point configurations with spheres in all dimensions, compare Theorem 3.11, and an additional concept, combining all these methods, lead to a negative answer of the Steinitz problem due to Sturmfels. This can be viewed as an essential contribution where the theory of oriented matroids caused a remarkable development in convexity.

Theorem 4.3 (Sturmfels 1987a,c). *The missing axiom for polytopal spheres is lost forever.*

This provides a solution to the Steinitz problem in the negative sense. The uniform case and lower-dimensional cases are still open according to the Lawrence polytope technique. A new detailed description can be found in Björner et al. (1991). It is remarkable that the same method yields also an improvement for polytopes without rational coordinates.

4.2. On the isotopy problem

The result about the hidden symmetry of Kleinschmidt's sphere, compare Example 3.10, stimulated Mnëv when he studied realization spaces of oriented matroids, see Mnëv (1988). In using Smith's theory from topology about finite group actions on topological spaces, he realized that the realization space for this sphere is not contractible. Mnëv gave a very general proof about realization spaces of oriented matroids. His result, Mnëv (1985), together with Bokowski, Ewald and Kleinschmidt (1984), or Altshuler (1984) tells us that there are two polytopes having Kleinschmidt's sphere as its boundary structure but which cannot be moved continuously into each other without changing its combinatorial boundary structure. An elementary proof using methods from oriented matroid theory for this fact as well as more references about the general isotopy problem can be found in Bokowski and Guedes de Oliveira (1990), and Jaggi et al. (1989).

It is well known that the above phenomenon cannot happen in 3-space. Let P and Q be two different realizations of the same combinatorially given boundary complex of a simplicial convex polytope in dimension 3. According to Steinitz' theorem, there is always a continuous path connecting corresponding vertices of P and Q, such that the face lattice does not change along this path (modulo reflections).

4.3. Convexity theorems with extensions

There are many classical theorems which have been carried over to the theory of oriented matroids. Within this article, we have tried to reduce the technical calculus of oriented matroid theory as much as possible. Many definitions have been

omitted in order to provide geometrical ideas and to formulate main ideas, and we do not aim at providing a complete overview of the results where a complete book can only sketch the results. Many results can be found in Sturmfels' Darmstadt Ph.D. thesis.

In chapter 9 of Björner et al. (1991) we find the following headlines and corresponding contributions.

(i) Lawrence construction.
(ii) Cyclic and neighborly matroid polytopes.
(iii) The Steinitz problem and its relatives, non-rational 6-polytope.
(iv) Polyhedral subdivisions and triangulations.

For Helly type theorems see, e.g., Goodman and Pollack (1982). The papers of Cordovil contain convexity theorems, especially we have Euler's formula in Cordovil, Las Vergnas and Mandel (1982).

A remarkable result of Mandel is contained in his Ph.D. thesis.

Theorem 4.4 (Mandel 1981). *The face lattice (in the sense of Mandel) of an acyclic oriented matroid is shellable.*

The same result can be found in Fukuda's thesis.

Theorem 4.5 (Fukuda 1982). *The face lattice $L(t)$ of every d-dimensional oriented matroid cell t is a shellable d-ball and its boundary $\delta L[t] = L[t] \setminus \{l_{L[t]}\}$ is a shellable $(d-1)$-sphere.*

This implies:

Theorem 4.6 (Fukuda 1982). *Every oriented matroid cell t of dimension d satisfies Euler's relation*

$$\sum_{i=0}^{d} (-1)^i f_i(t) = 1.$$

Here we have $f_k(M)$ = number of k-dimensional faces, $f_0(M)$ = number of cocircuits, $f_d(M)$ = number of topes. As an example of results characterizing possible f vectors, we mention a recent result of Fukuda, Saito, Tamura and Tokuyama:

$$f_k(M) < \frac{k+1}{d-k} f_{k+1}(M), \quad 0 \leqslant k < d.$$

4.3.1. Non-properties of matroid polytopes

In the last section, we have listed some properties of matroid polytopes. Now we list some concepts in convexity which do not carry over to the oriented matroid generalization.

Whereas the polar of the face lattice of a convex polytope is isomorphic to the face lattice of some convex polytope, this is no longer true for general matroid polytopes.

Theorem 4.7 (Billera and Spellman-Munson 1984b, Spellman-Munson 1984). *Polarity does not extend to oriented matroid polytopes.*

There are also separation theorems which fail to hold in the theory of oriented matroids, see Bachem and Wanka (1988).

In the oriented matroid programming case, Fukuda has shown that there exist non-degenerate cycles in the linear program, Fukuda (1982). This is equivalent to the existence of an extension of the oriented matroid in such a way that the formal point of intersection of a line with a hyperplane (cocircuit) occurs. The corresponding subclass of oriented matroids is called Euclidean.

Theorem 4.8 (Fukuda 1982). *Every non-Euclidean oriented matroid has a non-degenerate cycle in the linear program.*

Richter-Gebert has shown that the existence of a bi-quadratic final polynomial (see section 5 for this notion) is equivalent to the existence of a non-degenerate cycle in the linear program.

Theorem 4.9 (Richter-Gebert 1991). *Every non-Euclidean oriented matroid has a bi-quadratic final polynomial.*

5. Polytopes versus matroid polytopes

We have seen that matroid polytopes share basic properties with ordinary, i.e., realizable polytopes such as, e.g., face lattices. The natural question arises: up to what extent do properties of ordinary polytopes carry over to the more general setting of matroid polytopes? This question is part of the more general major question in oriented matroid theory which deals with the interaction of the definition in the realizable case and the definition in the generalized case.

5.1. How to prove realizability

Whereas many theorems in oriented matroid theory provide contributions to the realizability question, a general practical tool is not available. But any property of a matroid polytope which can be shown not to be valid for realizable convex polytopes provides a non-realizability proof. We think of the fact that Levi's enlargement lemma does not hold for general oriented matroids according to Goodman and Pollack (1981b). Or we think of the non-degenerate cycling shown by Fukuda (1982), or we think of the existence of a bi-quadratic final polynomial which was

mentioned in Theorem 4.9. There are also asymptotic results which show the existence of much more non-realizable oriented matroids as there are realizable ones, Alon (1986).

A general applicable method for proving realizability seems to be out of reach or at least very hard. In fact, the problem of deciding realizability is known to be NP-hard, see Shor (1990).

When assuming realizability for the (uniform) oriented matroid, we have a matrix with homogeneous coordinates of the points of the oriented matroid. By multiplying this matrix with the inverse of a $(d \times d)$-submatrix (w.l.o.g. the first d rows), we find the matrix

$$A = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \\ a & b & \dots & \dots \\ \vdots & \vdots & \vdots & \vdots \\ \dots & \dots & \dots & z \end{pmatrix}$$

where the unknown variables $a, b, \dots, z$ can be expressed as brackets of the matrix A, i.e., as $(d \times d)$-subdeterminants of A. What was just said for the variables, the (1×1)-subdeterminants of the so called *Tucker matrix* or *standard representative matrix*

$$\begin{pmatrix} a & b & \dots & \dots \\ \vdots & \vdots & \vdots & \vdots \\ \dots & \dots & \dots & z \end{pmatrix},$$

the same is true for *all* subdeterminants of this matrix. Knowing the signs of all subdeterminants of the Tucker matrix is equivalent to knowing the signs of all $(d \times d)$-subdeterminants of the original matrix and thus equivalent to knowing the oriented matroid. When looking for a solution for the entries of the Tucker matrix, such that all bracket signs of the oriented matroid are fulfilled, we have posed the realizability question in algebraic terms. By looking at the corresponding topological types of the solution sets of the algebraic (equations) and inequalities, Mnëv has shown that even for the simplest case, the rank 3 case, there occur all topological types, see Mnëv (1988). In other words, the special (in a sense linear) structure of the inequalities at the beginning does not lead to a reduction (in terms of a complexity result) in view of general solution techniques for solving non-linear equations.

Nevertheless, the successful attempt of the author to decide the polytopality question for spheres of Altshuler, Kleinschmidt, Perles, and Shemer depends very much on the linear property at the beginning. It led with computational support (implemented by Sturmfels) to a large class of such decisions, see Bokowski and

Sturmfels (1985, 1986a, 1987, 1989a). The easiest and straightforward decisions covering also the combinatorial method of Roudneff (covariant pair and contravariant pair extensions) were later summarized under the name *solvability sequence method.*

Again, there are many ways to prove realizability for oriented matroids for particular subclasses but a satisfactory general answer remains unknown. Combinatorial criteria extending the solvability sequence method are provided in Richter-Gebert (1989, 1992). The algebraic final polynomial method just mentioned above does have algorithmical variants but there are many more attempts to tackle the problem. Several authors gave infinite classes of non-realizable oriented matroids. Fukuda's method to reveal a non-degenerate cycle in the linear program, Fukuda (1982) comes down to the existence of a bi-quadratic final polynomial, a notion defined in Bokowski and Richter (1990), see Richter-Gebert (1991) for this assertion, we have also Lawrence polytopes (Bayer and Sturmfels 1990), and Goodman's and Pollack's class (1983).

For small rank and small number of points complete classifications are available. Table 6 shows that within this class the non-realizable examples are very rare. In addition one would like to have minor-minimal non-realizable examples like those given in Bokowski and Sturmfels (1989b).

Table 6
Reorientation classes of oriented matroids, total number : number of non-realizable ones

Rank	Points					
	5	6	7	8	9	10
7					1 : 0	
6				1 : 0	4382 : 1	
5			1 : 0	135 : 0	unknown	
4		1 : 0	11 : 0	2628 : 24 Bo/RG	unknown	
3	1 : 0	4 : 0	11 : 0 Gr	135 : 0 Go/Po, (Ha/Ca)	4382 : 1 RG, La/GS	312.356 : 242 Bo/La/RG

Authors in the table above: Bo = Bokowski, Ca = Canham, Go = Goodman, Gr = Grünbaum, GS = Gonzalez-Sprinberg, Ha = Halsey, La = Laffaille, Po = Pollack, RG = Richter-Gebert.

5.2. *The final polynomial method*

Major parts of the contribution *Computational synthetic geometry*, Bokowski and Sturmfels (1989a), are devoted to the realizability question of oriented matroids,

especially chapter IV with its algebraic criteria for geometric realizability. Dress and Sturmfels have proved that a so called *final polynomial* exists whenever the oriented matroid is non-realizable.

Many sections of this research monograph are devoted to the question of *how to find* final polynomials. This method grew out of Steinitz problem type investigations during the last decade, and additional attempts to further research in this direction have proven useful in the more general setting of oriented matroids.

The notion of final polynomials can be formally introduced as follows. Let $K[\Lambda(n,d)]$ be the polynomial ring over the field K. We consider the ideal I_χ^K in $K[\Lambda(n,d)]$ generated by the set of all non-bases $\{[\lambda] \in \Lambda(n,d), \chi(\lambda) = 0\}$. N_χ^K is the set of all products with factors of the form $\{[\lambda], \chi(\lambda) = +1\}$, $\{-[\lambda], \chi(\lambda) = -1\}$, or positive elements in the ordered field K. P_χ^K is the quadratic semiring in $K[\Lambda(n,d)]$ generated by N_χ^K and the set of all squares of elements in $K[\Lambda(n,d)]$.

Definition 5.1 (*Final polynomial*). A *final polynomial for* χ is an element in the ring $K[\Lambda(n,d)]$ with the property $f \in I_{n,d}^K \cup (I_\chi^K + N_\chi^K + P_\chi^K)$.

The existence of a final polynomial in every non-realizable case is a very nice theoretical result. However, the proof is not constructive, and we are looking for algorithmical methods. A report about many aspects of the realizability problem together with new combinatorial methods which grew out of the solvability sequence method can be found in Richter-Gebert (1992).

The crucial question about final polynomials is how to find them. We first mention a simple example to show the *method of proof* which can be applied to prove that a sphere is not polytopal.

Example 5.2 (*Final polynomial for Pappos' theorem*). We consider the uniform Pappos configuration depicted in section 1.1.3 in the projective plane. In the realizable case the closed curves can be stretched to great circles, and under this assumption, we can work with Grassmann–Plücker identities which tell us that polynomials of the type

$$\{a|b,c,d,e\} := [a,b,c][a,d,e] - [a,b,d][a,c,e] + [a,b,e][a,c,d]$$

vanish. Multiplying and canceling opposite summands in the following polynomial leads to the bracket identity written after the first equality sign.

$$\begin{aligned}
&+\{1|2479\}[148][157][347][467] \quad + \{1|3478\}[149][247][157][467]\\
&+\{1|4567\}[148][247][347][179] \quad + \{4|1267\}[148][157][347][179]\\
&+\{4|1357\}[148][247][167][179] \quad + \{4|1789\}[247][157][167][134]\\
&+\{7|1459\}[247][148][167][134] \quad + \{7|1468\}[247][157][149][134]\\
&+\{7|1234\}[149][148][157][467]
\end{aligned}$$

$$
\begin{aligned}
= &+ [147][129][148][157][347][467] \quad + [147][138][149][247][157][467] \\
&+ [147][156][148][247][347][179] \quad + [147][246][148][157][347][179] \\
&+ [147][345][148][247][167][179] \quad + [147][489][247][157][167][134] \\
&+ [147][579][247][148][167][134] \quad + [147][678][247][157][149][134] \\
&+ [147][237][149][148][157][467] = 0.
\end{aligned}
$$

From fig. 2, one can derive that all 9 summands have equal signs different from zero. This contradiction to the bracket identity shows that the uniform Pappos configuration is not realizable.

On the other hand, we know that this example is unique in the following sense.

Theorem 5.3 (Richter 1989 and Gonzalez-Sprinberg and Laffaille 1989). *Excluding the Pappos' theorem oriented matroid implies realizability for uniform oriented matroids in rank* 3 *with* 9 *points.*

By using duality, we formulate the corresponding result for matroid polytopes. A matroid polytope with 9 points in rank 6 with face lattice not isomorphic to the boundary complex of a convex polytope must have a final polynomial isomorphic to the above Pappos theorem one. Since all oriented matroids with 8 points in rank 3 are realizable, we have: all matroid polytopes with 8 points in rank 5 (=dimension 4) are realizable. This shows in particular that Altshuler's non-polytopal sphere M_9^{963} is a minimal non-realizable one, compare Bokowski and Garms (1987). A corresponding problem was posed in Bland and Las Vergnas (1978). The investigation of Altshuler's spheres was one origin of applying algebraic methods in oriented matroid theory to prove non-realizability. For another independent and earlier approach see the work of Whiteley in rigidity, scene analysis and geometry.

Finally, we mention the Ph.D. thesis of Richter-Gebert (1992) as a detailed report about the final polynomial method. The idea to detect bi-quadratic final polynomials by means of linear programming is implemented (in connection with MINOS), and so far it seems to be the best practical way to confirm oriented matroids to be non-realizable.

Problem 5.4. Find an oriented matroid which is non-realizable and which does not have a bi-quadratic final polynomial.

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CHAPTER 2.6

Algebraic Geometry and Convexity

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Contents

1. Introduction and historical notes 605
2. Definition of toric varieties 607
3. Blow ups and stellar subdivisions 612
4. Resolution of singularities 614
5. Invertible sheaves, Picard group, and projective toric varieties 616
6. Homology 619
7. Research problems 621
   7.1. Classification problems 621
   7.2. Fano varieties and Fano polytopes 621
   7.3. Existence problems 622
   7.4. Oda's conjectures 622
   7.5. Singularities 623
   7.6. Quotients of toric varieties 623
   7.7. Homology and intersection numbers 623
8. Dictionary 624
References 624

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1. Introduction and historical notes

During the last two decades a surprising development has taken place which interrelates the combinatorial theory of convex polytopes in many ways to algebraic geometry. It is the purpose of this article to outline this development in some of its basic issues. Let $\mathbb{R}$ be the field of real numbers, $\mathbb{C}$ that of complex numbers.

We begin by an example. If $K_1, \dots, K_r$ are polytopes (or, more generally, convex bodies) in $\mathbb{R}^n$, and if $\lambda_1, \dots, \lambda_r$ are nonnegative real numbers, a classical result by Minkowski states ($V(\cdot)$ denoting n-dimensional volume)

$$V(\lambda_1 K_1 + \cdots + \lambda_r K_r) = \sum V(K_{\varrho_1}, \dots, K_{\varrho_n}) \lambda_{\varrho_1} \cdots \lambda_{\varrho_n},$$

where the summation is taken independently for $\varrho_1, \dots, \varrho_n = 1, \dots, r$. The coefficients $V(K_{\varrho_1}, \dots, K_{\varrho_n})$ are called *mixed volumes*. They do not have, in general, a plausible geometric meaning, except $V(K, \dots, K) = V(K)$ which denotes volume or $V(K, \dots, K, B)$, B a unit ball, which characterizes the surface area of K times $1/n$. One of the most important applications of mixed volumes lies in the solution of the isoperimetric problem, that is, finding the body of maximal volume if its surface area is prescribed.

A totally different application has been found in the middle of the seventies by Bernstein, based on work of Kouchnirenko (1976) and Khovanskij (1988). Suppose $f_1 = \cdots = f_n = 0$ is a system of polynomial equations in n complex variables. To each polynomial $f = \sum c_j z^j$ where $j = (j_1, \dots, j_n) \in \mathbb{Z}_+^n$, $z^j := z_1^{j_1} \cdots z_n^{j_n}$ and $c_j \in \mathbb{C}$ we assign its *support* $\operatorname{supp} f := \{j \mid c_j \neq 0\}$. Then $\Delta(f) := \operatorname{conv} \operatorname{supp} f$ is a lattice polytope called the *Newton polytope* of f. Let $l(f_1, \dots, f_n)$ be the number of those solutions of $f_1 = \cdots = f_n = 0$, counted with multiplicity, which lie in $\mathbb{C}^{*n} := (\mathbb{C} \setminus \{0\})^n$. We assume $l(f_1, \dots, f_n)$ is finite. $\mathbb{C}^{*n}$ is called an *(algebraic) torus*. Bernstein (1976) proved:

$$l(f_1, \dots, f_n) = n! V(\Delta(f_1), \dots, \Delta(f_n)).$$

If $f_1 = 1 + z_1^2 + 4z_2^2$, $f_2 = -z_1 + z_2 - z_1^3$, then $2V(\Delta(f_1), \Delta(f_2)) = V(\Delta(f_1) + \Delta(f_2)) - V(\Delta(f_1)) - V(\Delta(f_2)) = 9.5 - 2 - 1.5 = 6$, the number of points in which the above curves intersect (see fig. 1a).

Bernstein's formula implies that $l(f_1, \dots, f_n)$ does not depend on the value of the coefficients of $f_1, \dots, f_n$ but only on their being "generic". In the above example this is a known fact: the intersection number even depends only on the degrees of f_1 and f_2, according to Bézout's Theorem if f_1, f_2 are made homogeneous.

However, if we choose

$$f_1 = z_1 z_2 - 1, \qquad f_2 = z_1 z_2 - 2,$$

$l(f_1, f_2) = 0$, and also the mixed volume of the line segments $\Delta(f_1)$, $\Delta(f_2)$ that join $(0,0)$ to $(1,1)$ is zero. Here Bézout's Theorem does not apply. In fact, Bézout's

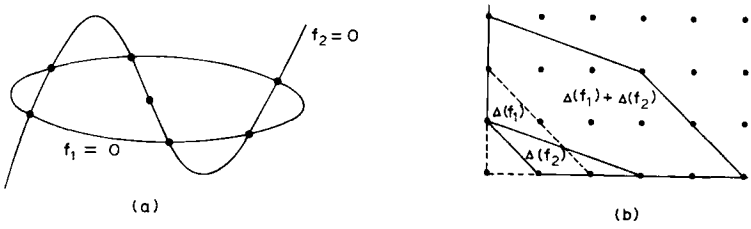

Figure 1.

Theorem can be considered a special case of Bernstein's Theorem: if we consider only homogeneous polynomials and apply an appropriate bijective linear transformation to the $(n+1)$-dimensional vector space that represents the projective extension of $\mathbb{R}^n$, then each f_j can be brought into a form which contains all monomials $z_1^n, \ldots, z_{n+1}^n$. If we now take the affine specialization $z_{n+1} = 1$, $\Delta(f_1) = r_1\Delta, \ldots, \Delta(f_n) = r_n\Delta$ are all simplexes homothetic to the convex hull Δ of 0 and the canonical basis points (see fig. 2).

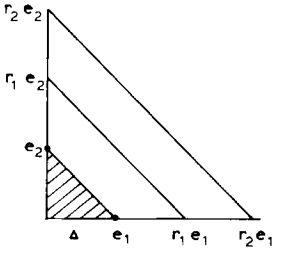

Figure 2.

Since $n!V(\Delta) = 1$ we obtain

$$l(f_1, \ldots, f_n) = n!V(r_1\Delta, \ldots, r_n\Delta) = n!V(\Delta)r_1 \cdots r_n$$
$$= r_n \cdots r_n = (\deg f_1) \cdots (\deg f_n).$$

When Newton worked on expressing implicit functions $f(z_1, z_2) = 0$ by $z_2 = \varphi(z_1)$ he introduced the polytope $\Delta(f)$. His ideas were made explicit for $n = 2$ by Minding in 1841 (compare Khovanskij 1988).

This interesting work has been forgotten until its rediscovery and generalization by Kouchnirenko and Bernstein.

Nevertheless, it was already in 1970 Demazure (1970) discovered a relationship between polyhedral and algebraic geometry under a different view point, followed in 1973 by generalizations due to Kempf et al. (1973) and Oda (1988). It may be contrasted to Bernstein's work by another interpretation of the above statement about properties that do not depend on the coefficients of a polynomial but only on their being generic. Let $c \neq 0$. Then c can be replaced by a $d \neq 0$ if we multiply each coordinate by a constant $\neq 0$: let $d = c\alpha_1^{-j_1} \cdots \alpha_n^{-j_n}$, then $cz^j = cz_1^{j_1} \cdots z_n^{j_n} = d(\alpha_1 z_1)^{j_1} \cdots (\alpha_n z_n)^{j_n}$.

In other words: We consider invariants under the group of transformations

$$A: \mathbb{C}^n \longrightarrow \mathbb{C}^n$$

$$(z_1, \ldots, z_n) \longmapsto (\alpha_1 z_1, \ldots, \alpha_n z_n)$$

$$= (z_1, \ldots, z_n) \begin{pmatrix} \alpha_1 & & \\ & \ddots & \\ & & \alpha_n \end{pmatrix} = (z_1, \ldots, z_n)A.$$

The group of all such (diagonal) matrices is isomorphic to $\mathbb{C}^{*n}$ (under coordinatewise multiplication). The group of all A is also called a *torus action*.

There may be different embeddings of tori into the same algebraic variety X such that each torus acts as an algebraic group on X. Under appropriate assumptions the respective varieties can be described by combinatorial concepts and do not preassume algebraic groups. They are called *toric varieties*.

We shall first define toric varieties (section 2) and then (sections 3–8) list a number of correspondences between geometric and algebraic-geometric concepts. In fact, a sort of a dictionary already has been set up which translates terms and theorems from one language into the other (section 8).

So far, combinatorial methods are usually used as a tool to solve algebraic-geometric problems. There are only a few examples of the opposite relationship. One is Stanley's proof of the necessity of McMullen's conjecture on "f-vectors". Teissier's theory (Teissier 1982) based on toric varieties which provides a new (and perhaps deeper than Bernstein's) approach to the relationship between mixed volumes and intersection numbers, has led to new proofs of geometric facts, but did not provide new results on convex polytopes.

In 1988 a book by Oda, *Convex Bodies and Algebraic Geometry*, has appeared (Oda 1988). It provides a compact survey of the material which we introduce here shortly and extend by some recent results. (For a general background in algebraic geometry see, for example, Brodmann 1989, Fulton 1984, Hartshorne 1977.)

A word on the use of "combinatorial" should be added. Two polytopes are said to be combinatorially equivalent if they have isomorphic boundary complexes (under inclusion). In the present article we shall often have to deal with lattice polytopes and lattice cones with the group of unimodular transformations as a "permissible" group. So we use the word "combinatorial" in a somewhat stronger sense than suggested by "combinatorial equivalence".

2. Definition of toric varieties

Let us first consider the ordinary ring $\mathbb{C}[\xi_1, \ldots, \xi_n]$ of polynomials in n variables $\xi_1, \ldots, \xi_n$. Any polynomial is a sum of monomials $\alpha \xi_1^{j_1} \cdots \xi_n^{j_n}$, where $\alpha \in \mathbb{C}$ and $(j_1, \ldots, j_n) \in \mathbb{Z}_+^n$ (n-tuples of nonnegative integers). As we have already mentioned in the introduction, the correspondence

$$j := (j_1, \ldots, j_n) \quad \longleftrightarrow \quad \xi_1^{j_1} \cdots \xi_n^{j_n} =: z^j \tag{1}$$

is fundamental for our considerations.

If $e_1 := (1,0,\ldots,0),\ldots,e_n = (0,\ldots,0,1)$ is the canonical basis of $\mathbb{R}^n$, we have, in particular,

$$\xi_1 = z^{e_1} = z_1,\ldots,\xi_n = z^{e_n} = z_n.$$

So (1) can be looked at as assigning to a linear combination

$$[j_1e_1 + \cdots + j_ne_n] \quad \text{the monomial} \quad [(z^{e_1})^{j_1}\cdots(z^{e_n})^{j_n}]$$

with nonnegative integers $j_1,\ldots,j_n$.

We may generalize this interpretation of monomials and choose instead of $e_1,\ldots,e_n$ some lattice vectors $m_1,\ldots,m_k$ as generators of a semi-group $\mathbb{Z}_+m_1 + \cdots + \mathbb{Z}_+m_k$, set $u_1 := z^{m_1},\ldots,u_k := z^{m_k}$, and assign to each nonnegative integral linear combination

$$\lfloor j_1m_1 + \cdots + j_km_k \rfloor \quad \text{a monomial} \quad \lfloor u_1^{j_1}\cdots u_k^{j_k} \rfloor.$$

In general, $m_1,\ldots,m_k$ will not be linearly independent as $e_1,\ldots,e_n$ are. Therefore, the polynomial ring $\mathbb{C}[u_1,\ldots,u_k]$ is obtained from $\mathbb{C}[\xi_1,\ldots,\xi_k]$ with free variables $\xi_1,\ldots,\xi_k$ as

$$\mathbb{C}[u_1,\ldots,u_k] \cong \mathbb{C}[\xi_1,\ldots,\xi_k]/I$$

for some ideal I. Hereby I can be obtained from the nonnegative integral linear combinations

$$i_1m_1 + \cdots + i_km_k = j_1m_1 + \cdots + j_km_k$$

by using

$$u_1^{i_1}\cdots u_k^{i_k} - u_1^{j_1}\cdots u_k^{j_k} \tag{2}$$

as generators of I.

A basic example is that of

$$m_1 = e_1,\ldots,m_n = e_n, \qquad m_{n+1} = -e_1,\ldots,m_{2n} = -e_n.$$

From

$$m_1 + m_{n+1} = 0,\ldots,m_n + m_{2n} = 0$$

we obtain that I is generated by $\xi_1\xi_{n+1},\ldots,\xi_n\xi_{2n}$. Hence

$$u_1u_{n+1} = 1,\ldots,,u_nu_{2n} = 1.$$

We set

$$u_{n+1} =: u_1^{-1},\ldots,u_{2n} =: u_n^{-1}.$$

So

$$\mathbb{C}[u_1,\ldots,u_{2n}] \cong \mathbb{C}[\xi_1,\ldots,\xi_{2n}]/I \cong \mathbb{C}[u_1,\ldots,u_n,u_1^{-1},\ldots,u_n^{-1}],$$

the ring of all *Laurent polynomials.*

Other examples are the following. Let $\sigma := \mathbb{R}_+m_1 + \cdots + \mathbb{R}_+m_r$ be a (closed) real cone in $\mathbb{R}^n$. Then $\sigma \cap \mathbb{Z}^n$ is a semi-group. By Gordan's lemma, $\sigma \cap \mathbb{Z}^n$ has a finite system of generators $a_1,\ldots,a_s$ which are simple lattice vectors and generate $\sigma \cap \mathbb{Z}^n$ nonnegatively.

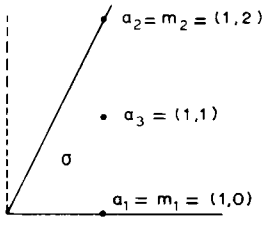

Figure 3.

We may suppose $m_1 = a_1, \ldots, m_r = a_r$ to be part of the generators. In general, however, $s > r$, as is seen in the example of fig. 3 where $\sigma = \mathbb{R}_+ a_1 + \mathbb{R}_+ a_2$, but a_1, a_2 do not generate $\sigma \cap \mathbb{Z}^n$ as a semi-group.

In order to free oneself from a specific system of generators, it is also possible to start with the ring $\mathbb{C}[\xi_1, \xi_2, \ldots]$ in infinitely many variables, assign to each element of $\sigma \cap \mathbb{Z}^n$ (provided $\dim \sigma > 0$) biuniquely a ξ_i, and then introduce the ideal I generated by all polynomials of type (2). Then $R_\sigma = \mathbb{C}[\xi_1, \xi_2, \ldots]/I$ is a coordinate-free introduction of the above ring. Hereafter *coordinates* $u_1, \ldots, u_k$ can possibly be introduced in several ways.

Now we turn to the sets of zeros of polynomials by which geometric objects are defined. Whereas to any polynomial there corresponds uniquely its set of zeros, there are many polynomials vanishing at the same point set. Their totality is an ideal in the ring under consideration. The larger the ideal, the smaller the set of common zeros of the polynomials of the ideal. To the maximal ideals (which are not the whole ring) we assign points, and call the set of all maximal ideals of R_σ the *spectrum* $\operatorname{spec} R_\sigma$ of R_σ. A prime ideal defines a variety, in case of R_σ this is called an *affine toric variety* X_σ. Its points are represented by all maximal ideals in which it is contained.

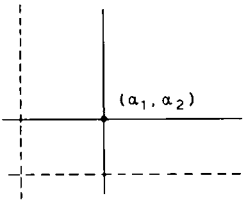

Figure 4.

In order to shift back from the algebraic language of describing a variety to the geometric one we apply *Hilbert's Nullstellensatz* to the ring $R := \mathbb{C}[\xi_1, \ldots, \xi_n]$ of all polynomials in $\xi_1, \ldots, \xi_n$ (as free variables): it says (for an elementary proof see Brodmann 1989) that any maximal ideal $I_{\max}$ has a presentation

$$I_{\max} = (\xi_1 - \alpha_1)R + \cdots + (\xi_n - \alpha_n)R$$

that is, the common zeros of all functions in $I_{\max}$ are characterized by the intersection of hyperplanes parallel to the coordinate hyperplanes in $\xi_1, \ldots, \xi_n$-space, thus

providing coordinates $(\alpha_1, \ldots, \alpha_n)$ (compare fig. 4). In other words: $\operatorname{spec} R = \mathbb{C}^n$ is an affine space. (We identify $\operatorname{spec} R$ and the points represented by its maximal ideals.)

If we restrict our attention to maximal ideals that contain a common ideal I_0 we obtain a subset of $\mathbb{C}^n$, described by finitely many equations (assigned to generating polynomials of I_0) in coordinates $u_1, \ldots, u_k$ (in general, $k > n$)

$$\begin{aligned} u_1^{i_{11}} \cdots u_k^{i_{k1}} &= u_1^{j_{11}} \cdots u_k^{j_{k1}} \\ &\vdots \\ u_1^{i_{1m}} \cdots u_k^{i_{km}} &= u_1^{j_{1m}} \cdots u_k^{j_{km}} \end{aligned} \tag{3}$$

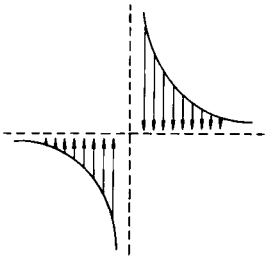

Figure 5.

Examples. (1) For $R = \mathbb{C}[u_1, \ldots, u_n, u_1^{-1}, \ldots, u_n^{-1}]$ we obtain as $\operatorname{spec} R$ in $\mathbb{C}^{2n}$ the higher-dimensional analogue of the hyperbola $u_1 \cdot u_{n+1} = 1$ in case $n = 1$. It is convenient to replace (up to a "rational transformation") this variety by the set of points of $\mathbb{C}^{*n}$ (as indicated in fig. 5). So $\operatorname{spec} R$ is usually identified with the torus $T := \mathbb{C}^{*n}$.

(2) In R_σ with σ given as in fig. 3 we have 3 coordinates u_1, u_2, u_3 which satisfy an equation

$$u_1 u_3 = u_2^2$$

and hence represent a quadratic cone in $\mathbb{C}^3$.

Now we turn to the general definition of toric variety obtained by "gluing together" affine toric varieties. In terms of the cones σ this is done as follows: to each σ we assign its dual cone

$$\check{\sigma} := \{x \mid \langle x, y \rangle \geqslant 0 \text{ for all } y \in \sigma\}.$$

Suppose $\check{\sigma}_1, \ldots, \check{\sigma}_r$ are cones which have all apex 0 and form a cell complex Σ, called a *fan*. If $\check{\sigma}_i$ is a face of $\check{\sigma}_j$, then $\sigma_j \subset \sigma_i$, and R_{σ_j} is a subring of R_{σ_i}. This, in turn, implies

$$X_{\sigma_i} = \operatorname{spec} R_{\sigma_i} \text{ is a subset of } X_{\sigma_j} = \operatorname{spec} R_{\sigma_j}.$$

In this way, the face relation in Σ induces a gluing together of the X_{σ_i}. The result is called a *toric variety* X_Σ. Since $\{0\}^\vee = \mathbb{R}^n$, the torus $T = \mathbb{C}^{*n}$ is an open subset of each X_{σ_i}; so all affine pieces X_Σ have an open and dense subset in common.

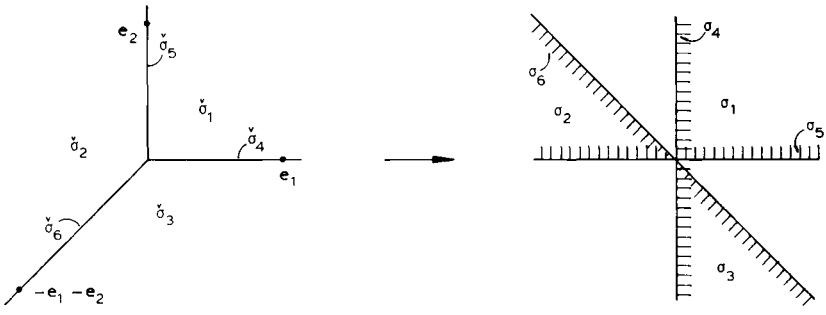

Figure 6.

Examples. (3) *Projective plane.* $X_{\sigma_1}, X_{\sigma_2}, X_{\sigma_3}$ (according to fig. 6) are affine planes which are charts of a projective plane. We remark that in a way toric varieties generalize the gluing together of an n-dimensional projective space by $n+1$ affine subspaces.

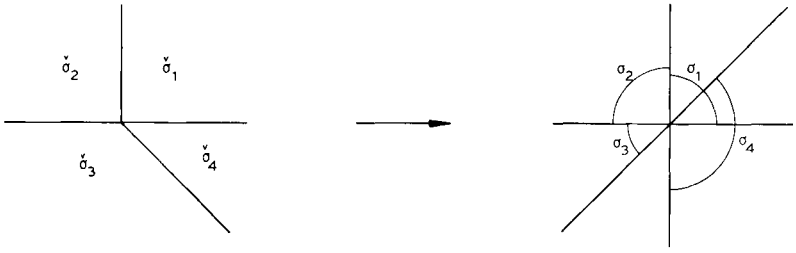

Figure 7.

(4) *Hirzebruch variety.* $X_{\sigma_1} = \operatorname{spec}\mathbb{C}[u_1, u_2]$, $X_{\sigma_2} = \operatorname{spec}\mathbb{C}[u_1^{-1}, u_2]$, $X_{\sigma_3} = \operatorname{spec}\mathbb{C}[u_1^{-1}u_2^{-k}, u_2^{-1}]$, $X_{\sigma_4} = \operatorname{spec}\mathbb{C}[u_2^{-1}, u_1 u_2^k]$ (according to fig. 7).

For $k = 0$ we obtain four affine planes covering $\mathbb{P}^1 \times \mathbb{P}^1$.

An *affine toric subvariety* of an affine toric variety X_σ is defined by a face τ of σ in the following way: if $a_1, \ldots, a_k$ are generators of the monoid $\tau \cap \mathbb{Z}^n$, they can be augmented to a system of generators $(a_1, \ldots, a_k, \ldots, a_m)$ of $\sigma \cap \mathbb{Z}^n$. Then for the corresponding coordinates $(u_1, \ldots, u_k)$, $(u_1, \ldots, u_k, \ldots, u_m)$ we can define a natural inclusion

$$\varphi : X_\tau \longrightarrow X_\sigma$$

by

$$\varphi : X_\tau \longrightarrow X_\sigma (u_1, \ldots, u_k) \longmapsto (u_1, \ldots, u_k, 0, \ldots, 0).$$

By gluing together we obtain *toric subvarieties*, correspondingly.

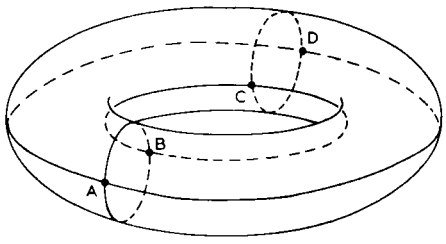

Figure 8.

Any k-dimensional toric subvariety X_τ of X_σ contains a k-dimensional torus as an open and dense subset. As indicated in the introduction, each torus can even be used to define automorphisms $X_\Sigma \longrightarrow X_\Sigma$ called *torus actions* whose orbits are tori. The following is true.

Theorem 2.1. *Any toric variety is a disjoint union of finitely many tori which are orbits of torus actions.*

Figure 8 illustrates the case of $\mathbb{P}^1 \times \mathbb{P}^1$ (see example (4) for $k = 0$). Two "meridians" intersect two "equators" in 0-dimensional tori A, B, C, D (the 2 meridians and 2 equators represent projective lines). The meridians and equators without A, B, C, D are 1-dimensional tori; the complement of the four projective lines is a 2-dimensional torus.

Finally we mention the following fact:

Theorem 2.2. *A toric variety X_Σ is compact if and only if Σ covers all of $\mathbb{R}^n$.*

3. Blow ups and stellar subdivisions

An n-dimensional affine toric variety X_σ is an affine space if and only if there is no linear relation between the generators of $\sigma \cap \mathbb{Z}^n$ and hence no eq. (3). We call, in this case, X_σ *regular*. It is characterized by $\sigma = \mathbb{R}_+ a_1 + \cdots + \mathbb{R}_+ a_n$ where $a_1, \ldots, a_n$ are primitive lattice vectors satisfying

$$\dim \sigma := \dim(a_1, \ldots, a_n) = \pm 1.$$

We say σ is *regular*. Then also $\check{\sigma}$ is regular.

If $k := \dim \sigma < n$, and if $\check{\sigma}$ is face of a regular cone, X_σ is isomorphic to

$$\mathbb{C}^k \times \mathbb{C}^{*n-k}$$

and also called *regular*. Finally, we say a toric variety X_Σ is *regular* if all X_σ, $\check{\sigma} \in \Sigma$, are regular.

Figure 9.

There is a fundamental operation for regular varieties, in particular, regular toric varieties, the idea of which can be illustrated as follows. Let in real geometry $\mathcal{S}$ be an ordinary sphere in $\mathbb{R}^3$ placed onto a Euclidean plane $\mathcal{E}$ with "south pole" S as point of tangency and "north pole" N opposite to S. We map $\mathcal{S} \setminus \{N\}$ onto $\mathcal{E}$ by stereographic projection. Then consider a semisphere $\mathcal{M}$ with boundary circle κ in a plane parallel to $\mathcal{E}$ such that $\mathcal{M}$ is again tangent to $\mathcal{E}$. By central projection from the center of the circle κ we map $\mathcal{E}$ onto $\mathcal{M} \setminus \kappa$. By combination of the maps, $\mathcal{S} \setminus \{N\}$ is mapped topologically onto $\mathcal{M} \setminus \kappa$.

By natural compactifications we add N to $\mathcal{S} \setminus \{N\}$ and κ to $\mathcal{M}$, and replace in this way a point N of $\mathcal{S}$ by a circle κ with diametrical points identified, that is, by a (real) projective line (fig. 9). The complex extension of this assignment is said to be a *blow up* of a point. Its inverse is called a *blow down* and represents a continuous map which is a homeomorphism in a neighborhood of κ with κ left out.

It is not essential that we have dealt with a sphere; blow ups can be defined more generally. In case of toric varieties the combinatorial analogue is the following. Let $\check{\sigma}$ first be an n-dimensional cone of a regular fan Σ, and let $\check{\sigma} = \mathbb{R}_+a_1 + \cdots + \mathbb{R}_+a_n$, a_i simple lattice vectors with determinant ± 1. We apply a *stellar subdivision* in direction $a_1 + \cdots + a_n =: p$, that is, we split $\check{\sigma}$ into the cones $\check{\sigma}_i := \mathbb{R}_+a_1 + \cdots + \mathbb{R}_+a_{i-1} + \mathbb{R}_+p + \mathbb{R}_+a_{i+1} + \cdots + \mathbb{R}_+a_n$, $i = 1, \ldots, n$ (compare fig. 10a). The $\check{\sigma}_i$ are again regular.

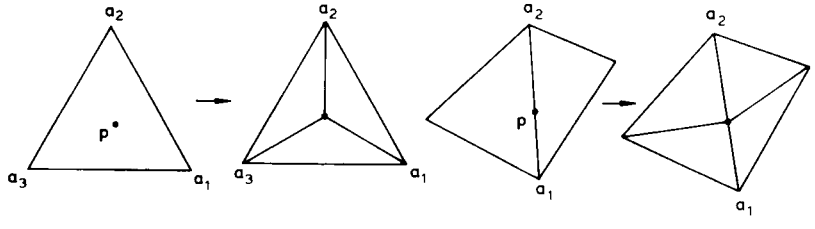

Figure 10.

If Σ' is obtained hereby from Σ then Σ' is again regular, and $X_{\Sigma'}$ is obtained from X_Σ in that a point (0-dimensional orbit) is replaced by a projective hyperplane, that is, by a *blow-up*.

In a similar way we proceed in case $p = a_1 + \cdots + a_k$ for the generators of a lower-dimensional face of the fan Σ. (Compare fig. 10b). We obtain generally a *regular stellar subdivision* of Σ. In X_Σ it induces a blow-up along a toric subvariety.

There is a number of interesting results about blow-ups and blow-downs of toric varieties shown by means of regular stellar subdivisions. We mention one by Danilov (1983) (with a correction in Ewald 1987): any two regular compact toric 3-varieties can be transformed one into the other by a finite sequence of operations consisting of blow-ups and blow-downs. For higher-dimensional toric varieties the analogous problem has also been solved, by Włodarczyk (1992).

4. Resolution of singularities

Any affine piece X_σ, $\dim \sigma = n$, of an n-dimensional toric variety is either an affine space and hence free of singularities, or it is described by monomial equations (3) (see section 2) which represent generalized algebraic cones with singularities. The simplest case is that of example (2) in section 2 where the apex 0 of the quadratic (conical) surface is a singularity.

Theorem 2.1 implies the following: if a singularity occurs there is a whole torus (orbit of torus action) full of singularities. In example (2) of section 2 this torus happens to be $\{0\}$; so we have an isolated singularity.

As we have mentioned at the beginning of the preceding section, regularity of an affine toric variety X_σ, $\dim \sigma = n$, is characterized by σ being a simplicial cone generated by simple lattice vectors with determinant ± 1. Consequently, singularities occur if σ is not such. Hereby two phenomena occur: σ is not a simplex cone or is a simplex cone with associated determinant $\neq \pm 1$.

Resolving singularities will, therefore, naturally have to do with splitting cones into regular ones. The general frame is given by the following theorem which we state without defining algebraic morphisms:

Theorem 4.1. *Let Σ, Σ' be fans such that each cone of Σ' is contained in a cone of Σ, and such that their point sets $|\Sigma|$, $|\Sigma'|$ are the same. Then there exists an algebraic morphism*

$$\Phi : X_{\Sigma'} \longrightarrow X_\Sigma$$

which maps toric subvarieties of $X_{\Sigma'}$ onto toric subvarieties of X_Σ, and commutes with torus actions.

In section 3 we have studied an example of such morphisms Φ, namely that of a blow-down. (A blow-up is not a morphism since it is not a map.)

By a *resolution* of the singularities of a toric variety we understand a morphism according to Theorem 4.1 such that $X_{\Sigma'}$ is regular.

The resolution can be achieved stepwise by using stellar subdivisions, hence by generalized blow-ups. The direction in which the stellar subdivision is applied need not be represented by a simple lattice vector p. Gordan's lemma assures that finitely many stellar subdivisions $s_1, \ldots, s_k$ of Σ exist such that

$$\Sigma \xrightarrow{s_1} \cdots \xrightarrow{s_k} \Sigma'$$

leads to a regular fan Σ'. Hereby stellar subdivisions are included in which p already spans a 1-dimensional cone: we obtain a nontrivial operation if the cones in the star of $\mathbb{R}_+p$ are not all simplicial.

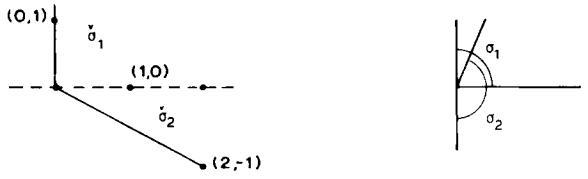

Figure 11.

As an example, we resolve the singularity in example (2) of section 2. From $\sigma = \mathbb{R}_+(1,0)+\mathbb{R}_+(1,2)$ we obtain $\check{\sigma} = \mathbb{R}_+(0,1)+\mathbb{R}_+(2,-1)$ (fig. 11). We split $\check{\sigma}$ in direction $p = (1,0)$ and obtain cones $\check{\sigma}_1 = \mathbb{R}_+(1,0) + \mathbb{R}_+(0,1)$, $\check{\sigma}_2 = \mathbb{R}_+(2,-1) + \mathbb{R}_+(1,0)$. Clearly, $\det\check{\sigma}_1 = \det\check{\sigma}_2 = 1$, hence also $\det\sigma_1 = \det\sigma_2 = 1$. $X_\sigma = \operatorname{spec} R_\sigma$ has coordinates (u_1,u_2,u_3) satisfying $u_1u_3 = u_2^2$, or, in parameter form (parameters z_1, z_2): $(u_1,u_2,u_3) = (z^{(1,0)}, z^{(1,1)}, z^{(1,2)}) = (z_1, z_1z_2, z_1z_2^2)$.

X_{σ_1} has coordinates

$$(v_1,v_2) = (z^{(1,0)}, z^{(0,1)}) = (z_1,z_2),$$

and X_{σ_2} is represented by

$$(w_1,w_2) = (z^{(0,-1)}, z^{(1,2)}) = (z_1^{-1}, z_1z_2^2).$$

We obtain an algebraic morphism

$$\Phi : X_{\sigma_1} \longrightarrow X_\sigma$$

by $(v_1,v_2) \mapsto (v_1, v_1v_2, v_1v_2^2)$.

Under Φ the line $\{(0,v_2)\}$ is mapped onto the singularity 0. Outside this line Φ has an inverse and is hence bijective.

Besides resolving singularities of a toric variety itself we can use blow-ups of regular toric varieties X_Σ in order to resolve singularities of algebraic varieties (not necessarily toric) immersed in X_Σ. We restrict ourselves to an example.

Let the complex projective plane $\mathbb{P}^2$ be given as a toric variety (see section 2, example (3)). If $(x_1 : x_2 : x_3)$ are the homogeneous coordinates of $\mathbb{P}^2$ the affine coordinates of the three charts X_{σ_1}, X_{σ_2}, X_{σ_3} can be given for $z_1 = (x_1/x_3)$, $z_2 = (x_2/x_3)$ by

$$(u_1,u_2) = (z_1,z_2),$$
$$(u_1,u_3) = (z_1, z_1^{-1}z_2),$$
$$(u_1,u_3) = (z_2, z_1z_2^{-1}).$$

We consider in $\mathbb{P}^2$ a curve given by the equation

$$x_1^2x_2^2 + x_1^2x_3^2 + x_2^2x_3^2 = 0,$$

(see fig. 12). It has three isolated singularities, namely the zero point of each of

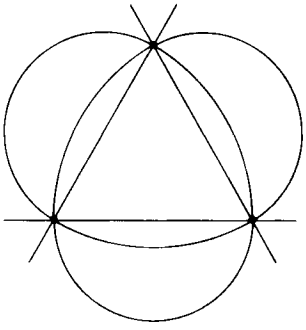

Figure 12.

the charts X_{σ_1}, X_{σ_2}, X_{σ_3}. We apply a blow-up in each singularity represented by stellar subdivisions s_1, s_2, s_3, respectively. We obtain a surface X_Σ (called *del Pezzo surface*). Then we add three blow-downs represented by inverse stellar subdivisions s_4^{-1}, s_5^{-1}, s_6^{-1} which do not create new singularities, so that, as a result, the above curve is turned into an ellipse (fig. 13).

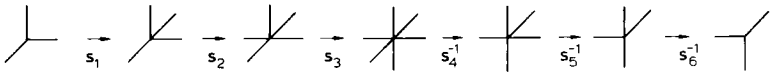

Figure 13.

One of the three new charts has coordinates

$$(v_1, v_2) = (z_1^{-1}, z_2^{-1}) = \left(\frac{x_3}{x_1}, \frac{x_3}{x_2}\right)$$

so that the transformed curve obtains the equation

$$1 + v_1^2 + v_2^2 = 0$$

which is an ellipse.

5. Invertible sheaves, Picard group, and projective toric varieties

Any affine piece X_σ of a toric variety X_Σ, $\check{\sigma} \in \Sigma$, has been defined by the spectrum of the ring R_σ of polynomials in monomials whose support lies in σ, hence of Laurent polynomials in $z_1, \ldots, z_n$. Each Laurent polynomial in R_σ is then a *regular function* on X_σ, that is, a quotient f/g of two polynomials, g nonvanishing on X_σ. (Clearly, any Laurent polynomial is the quotient of a polynomial f and a polynomial g.) It can even be shown that R_σ is exactly the set of all regular functions on X_σ. Any such function represents a rational function on X_Σ.

It is useful to assign to any open subset (in some given topology on X_Σ) of X_Σ the set of its regular functions. We denote the operator thus defined by $\mathbb{O}$:

$$U \longmapsto \mathbb{O}(U) \quad \longleftrightarrow \quad U \text{ open in } X_\Sigma.$$

So in particular

$$\mathbb{O}(X_\sigma) = R_\sigma \quad \text{for any } \check{\sigma} \in \Sigma.$$

$\mathbb{O}$ is an example of assigning to the open sets of a topological space algebraic structures such as rings, groups, or modules, satisfying the axioms of a so-called *sheaf*. In particular, the above operator is the *structure sheaf* of the toric variety. It expresses in a somewhat different language the relationship between the rings R_σ and the building stones X_σ of X_Σ. Although many more open subsets and corresponding rings are involved than the X_σ and R_σ, the structure of X_Σ is (by definition) uniquely determined by the R_σ. The extension of $\mathbb{O}$ to all open subsets of X_Σ is a matter of routine; in fact, practically all considerations about sheaves on toric varieties are sufficiently handled by restriction to the X_σ.

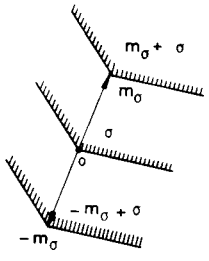

Figure 14.

We introduce now the most important examples of sheaves on toric varieties. To any $\check{\sigma} \in \Sigma$ we assign a lattice point m_σ and a translated cone (fig. 14)

$$m_\sigma + \sigma.$$

The Laurent polynomials f with support in $m_\sigma + \sigma$ can be written as $f = z^{m_\sigma} f_0$ where $f_0 \in R_\sigma$. The totality of such Laurent polynomials is denoted by

$$\mathcal{F}(X_\sigma) := z^{m_\sigma} \mathbb{O}(X_\sigma).$$

Each $\mathcal{F}(X_\sigma)$ can be considered a module over $R_\sigma = \mathbb{O}(X_\sigma)$. As in case of the structure sheaf $\mathbb{O}$ we can extend the assignment $\mathcal{F}$ to all open subsets of X_Σ. As a result we obtain a *sheaf*, provided, the m_σ satisfy the following compatibility condition:

(*) If $\check{\tau}$ is a face of $\check{\sigma}$, then $m_\sigma - m_\tau$ is contained in the maximal linear space on the boundary of τ (called cospan τ; compare fig. 15).

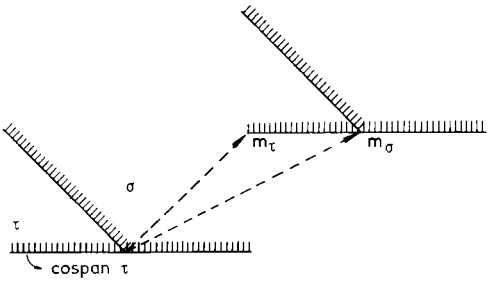

Figure 15.

The sheaf $\mathcal{F}$ is *invertible* as is illustrated in fig. 14:

$$(m_\sigma + \sigma) + (-m_\sigma + \sigma) = \sigma,$$

hence for $\mathcal{F}^{-1}(X_\sigma) := z^{-m_\sigma}\mathbb{C}(X_\sigma)$ we have

$$\mathcal{F}^{-1}(X_\sigma) \cdot \mathcal{F}(X_\sigma) = (\mathcal{F}^{-1} \otimes \mathcal{F})(X_\sigma) = \mathbb{C}(X_\sigma).$$

Clearly, if any two such invertible sheaves $\mathcal{F}$, $\mathcal{G}$ are given, their *tensor product* is defined if we set for any $\check{\sigma} \in \Sigma$:

$$(\mathcal{F} \otimes \mathcal{G})(X_\sigma) := \mathcal{F}(X_\sigma) \cdot \mathcal{G}(X_\sigma) = z^{m_\sigma + m'_\sigma}\mathbb{C}(X_\sigma).$$

The compatibility condition $(*)$ carries over from $\mathcal{F}$ and $\mathcal{G}$ to $\mathcal{F} \otimes \mathcal{G}$. We readily obtain:

Theorem 5.1. *The invertible sheaves as defined above form a commutative group under the tensor product "$\otimes$" as operation.*

One of the most striking facts in the relation between combinatorial and algebraic geometry is the following (which is not difficult to prove):

Theorem 5.2. *Any invertible sheaf on a toric variety, invariant under all torus actions on X_Σ, belongs to the group in Theorem* 5.1.

The properties of the group in Theorem 5.1 do in general not change if we add one and the same lattice vector to all m_σ, $\sigma \in \Sigma$. This corresponds to the fact that properties of convex sets are usually invariant under translations. So if Γ is the group of invertible sheafs according to Theorem 5.1, we set

$$\operatorname{Pic} X_\Sigma := \Gamma/\mathbb{Z}^n$$

and call it the *Picard group* of X_Σ.

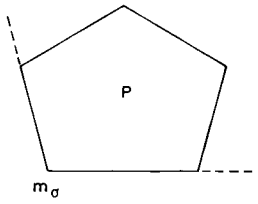

Figure 16.

Because of Theorems 5.1 and 5.2 the Picard group of a toric variety can completely be described by the associated systems of cones $m_\sigma + \sigma$ where the m_σ satisfy (*). The most important case is that in which for the n-dimensional cones $\check{\sigma}_1, \ldots, \check{\sigma}_r \in \Sigma$,

$$P := (m_{\sigma_1} + \sigma_1) \cap \cdots \cap (m_{\sigma_r} + \sigma_r)$$

is a polytope from which the fan Σ can be reconstructed in the following way. Let $m_{\sigma_1}, \ldots, m_{\sigma_r}$ be the vertices of P, and set $\sigma_i := \operatorname{pos}(P - m_\sigma)$, $i = 1, \ldots, r$ (fig. 16). Then $\Sigma' := \{\tau \mid \tau$ is a $\check{\sigma}_i$ or a face of a $\check{\sigma}_i$, $i = 1, \ldots, r\}$ is a fan. It need not be the original fan Σ. If $\Sigma' = \Sigma$, we call $-P$ an *associated polytope* of $\Sigma = \Sigma(P)$, and call Σ *strongly polytopal*. One can show:

Theorem 5.3. *X_Σ is projective, that is, can be embedded into a finite-dimensional projective space by an algebraic morphism if and only if Σ is strongly polytopal.*

If the above polytope P is not associated with Σ there may exist another polytope Q associated with Σ. In this case, up to multiplying (dilatate) Q by a positive integer, P is a Minkowski summand of Q. In fact, the rank of the Picard group of X_Σ can be calculated explicitly with the aid of decomposition Theorems by McMullen (1973) and Smilansky (1987). This has been worked out by Eikelberg (1990, 1992a,b).

Summarizing the ideas sketched in this section the following can be said: one of the keys for the relation between combinatorial and algebraic geometry is the correspondence between Minkowski addition of polytopes P, Q and tensor products of invertible sheaves $\mathcal{F}_P$, $\mathcal{F}_Q$ defined by the polytopes:

$$P + Q \quad \longleftrightarrow \quad \mathcal{F}_P \otimes \mathcal{F}_Q.$$

The next section will provide an example of how powerful this relation is (Theorem 6.2).

6. Homology

It is a natural question in how far the topological structure of a toric variety X_Σ is determined by the geometric structure of the fan Σ. In order to answer it we

consider X_Σ to be a complex-analytic space. A basic result is that of Danilov (1978) [with a correction in Eikelberg (1990) and partly going back to Ehlers (1975); see also Jurkiewicz (1985)] that the cohomology groups of a regular compact toric variety can be calculated as follows.

Theorem 6.1. *Let $H^{2k}(X_\Sigma, \mathbb{Z})$ denote the cohomology group of X_Σ in $2k$ real dimensions and integral coefficients (Čech-cohomology or ordinary cohomology groups). Then the odd Betti numbers are 0 and the even ones are*

$$h_{2k} = \operatorname{rank} H^{2k}(X_\Sigma, \mathbb{Z}) = \sum_{i=k}^{n} (-1)^{i-k} \binom{i}{k} a_{n-i}$$

where a_i is the number of i-dimensional cones of Σ.

Very little is known in the non-regular case, except that examples have been found for non-vanishing odd Betti numbers (McConnell 1989) and that Eikelberg (1990) has given a simple proof of the following.

Theorem 6.2. *$H^2(X_\Sigma, \mathbb{Z}) \cong \operatorname{Pic} X_\Sigma$ provided X_Σ is compact and projective (Σ complete and strongly polytopal). In particular, $H^2(X_\Sigma, \mathbb{Z})$ is torsion-free (which is trivial for $\mathbb{Q}$ instead of $\mathbb{Z}$).*

The proof makes use of another result of Danilov saying that for $k > 0$ all Čech-cohomology groups of a compact toric variety with coefficients in the structure sheaf $\mathcal{O} = \mathcal{O}_{X_\Sigma}$ vanish:

$$\check{H}^k(X_\Sigma, \mathcal{O}) = 0. \tag{1}$$

If $\mathcal{O}^*$ denotes the multiplicative version of the structure sheaf, and if $\mathcal{Z}$ denotes the sheaf of constant functions on X_Σ, then the sequence of sheaves (defined in the usual way)

$$0 \longrightarrow \mathcal{Z} \longrightarrow \mathcal{O} \longrightarrow \mathcal{O}^* \longrightarrow 0$$

is exact. One obtains from it a long exact homology sequence from which by using (1) and the standard theorem

$$\operatorname{Pic} X_\Sigma \cong \check{H}^1(X_\Sigma, \mathcal{O}^*)$$

one deduces

$$H^2(X_\Sigma, \mathbb{Z}) \cong \check{H}^2(X_\Sigma, \mathcal{Z}) \cong \check{H}^1(X_\Sigma, \mathcal{O}^*) \cong \operatorname{Pic} X_\Sigma.$$

So the combinatorial calculation of the structure of $\operatorname{Pic} X_\Sigma$ explained in the preceding section applies to the calculation of $H^2(X_\Sigma, \mathbb{Z})$.

7. Research problems

In our survey we did not discuss the applications of toric varieties in analytic number theory (see Ash et al. 1975, Namikawa 1980) or those in the theory of algebraic groups. Numerous papers have been written about these applications or using them; so it is a large field of research. Also we leave out the theory of Stanley–Reisner rings and their applications (see, for example, Stanley 1983), and relations to moment maps or Morse theory (compare Atiyah 1983, Audin 1991).

We restrict ourselves here mainly to explaining some research problems which are accessible to those coming from combinatorial convexity and not having a far-reaching background in algebraic geometry.

7.1. Classification problems

In the work of Oda and Ishida (for reference, see Oda 1988) an extensive study has been made about classification of regular compact toric varieties. Starting point is usually a "coarse" classification based on the cell complex structure ("combinatorial equivalence classes") of the fan or the polytope which spans the fan in the strongly polytopal case. Then a refinement is made for each combinatorial class of fans taking care of properties invariant under unimodular transformations of $\mathbb{Z}^n \subset \mathbb{R}^n$. A contribution also has been made by Kleinschmidt (1988). About the use of linear transforms see Oda and Park (1991). Many promising questions are open.

7.2. Fano varieties and Fano polytopes

A special type of regular compact projective toric varieties is that of a Fano variety, given by polytopes whose vertices are simple lattice vectors, called Fano polytopes. The algebraic-geometric condition defining them is the ampleness of the anticanonical sheaf of the variety.

For $n = 2$ there are, up to unimodular transformations, only 5 Fano polytopes (fig. 17).

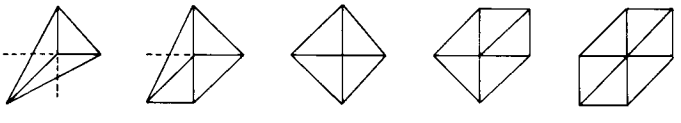

Figure 17.

For $n = 3$ there are 18 types (Batyrev 1982) and for $n = 4$ one has 127 Fano polytopes or Fano varieties, respectively (Batyrev in a forthcoming paper). A partial classification can be found in Evertz (1988). Contributions to classification in higher dimensions have been made in Ewald (1988), Batyrev (1986), and in Klyachko and Voskresenskij (1985). In Ewald (1988) some conjectures have been stated.

7.3. Existence problems

Kleinschmidt, Gretenkort and Sturmfels (1990) have shown that there does not exist a regular complete projective variety whose fan is spanned by a polytope combinatorially isomorphic to a cyclic polytope with at least $n+3$ vertices, $n \geqslant 4$. So a new "upper bound problem" exists for simplicial polytopes which represent regular toric varieties. For $n = 3$, it is not known if all simplicial polytopes occur, up to combinatorial isomorphism, among the spanning polytopes of regular, complete, projective toric varieties.

In a joint paper with Schulz (Ewald and Schulz 1992) we have shown that there exists a simplicial 3-sphere which cannot be embedded into $\mathbb{R}^4$ as boundary complex of a star-shaped polytope. So in the classification of nonprojective complete toric varieties not every combinatorial sphere shows up. Not much is known yet about the respective classification problems.

Another existence problem refers to Cartier divisors whose "partners" in combinatorial geometry are piecewise linear, positive homogeneous and continuous functions, especially support functions of lattice polytopes. A 3-dimensional toric variety without effective Cartier divisors is presented in Eikelberg (1992b). What are the conditions for their existence?

7.4. Oda's conjectures

Given any two regular, complete, 3-dimensional toric varieties X_Σ, $X'_{\Sigma'}$ Oda (1988) has stated two conjectures.

Conjecture I (*Strong version*). There exist regular stellar subdivisions $s_1, \ldots, s_p$, $s'_1, \ldots, s'_q$ such that

$$s_p \circ \cdots \circ s_1 \Sigma = s'_q \circ \cdots \circ s'_1 \Sigma',$$

hence, for the blow-downs $\varphi_1, \ldots, \varphi_p$, $\varphi'_1, \ldots, \varphi'_q$ induced by $s_1, \ldots, s_p$, $s'_1, \ldots, s'_q$, respectively,

$$\varphi_p \circ \cdots \circ \varphi_1 X_\Sigma = \varphi'_q \circ \cdots \circ \varphi'_1 X_{\Sigma'}.$$

Conjecture II (*Weak version*). There exist operations $t_1, \ldots, t_r$, either regular stellar subdivisions or inverse regular stellar subdivisions such that

$$t_r \circ \cdots \circ t_1 \Sigma = \Sigma'$$

or, for the induced blow-ups and blow-downs $\varphi_1, \ldots, \varphi_r$:

$$\varphi_r \circ \cdots \circ \varphi_1 X_\Sigma = X_{\Sigma'}.$$

The weak version has been proved in all dimensions (see end of section 3). The strong version has been reduced in Ewald (1987) to the projective case. The analogous problem in higher dimension is unsolved. (About a combinatorial theorem corresponding to Conjecture I see Ewald 1986.)

7.5. Singularities

In a paper by Ishida and Iwashita (1986) a combinatorial classification of a class of singularities of 3-dimensional toric varieties is presented. Their method cannot directly be extended to higher dimensions. Perhaps modifications will be successful to find at least partial results for dimension greater than 3.

7.6. Quotients of toric varieties

There is an interesting relationship between so-called fiber polytopes, Chow polytopes, and quotients of toric varieties (Billera and Sturmfels 1992, Kapranov, Sturmfels and Zelevinsky 1991a,b) which involves new problems.

7.7. Homology and intersection numbers

The transfer of homology questions into combinatorial problems as we outlined in section 6 is a promising field of further research. It is also related to intersection numbers. There is a good chance that large parts of homology and intersection theory for toric varieties can be developed in purely combinatorial terms. As one

Table 1
Concepts of a fan Σ and the toric variety X_Σ

Fan Σ	Toric variety X_Σ
unimodular transformation	isomorphism
complete	compact
strongly polytopal	projective
regular	regular
stellar subdivision which preserves regularity	blow-up along a toric subvariety
join in the sense of p.l. topology: $\Sigma = \Sigma' \cdot \Sigma''$	fibration with a toric subvariety as fiber
associated polytope Δ of Σ or summand of it	invertible sheaf $\mathcal{F}$ of rational functions
Minkowski sum $\Delta + \Delta'$	tensor product $\mathcal{F} \otimes \mathcal{F}'$
group generated by polytopes Δ up to common translations	Picard group $\mathrm{Pic}\, X_\Sigma \cong H^2(X_\Sigma, \mathcal{Z})$
support function, p.l. on Σ	effective Cartier divisor
mixed volume $V(\Delta_1, \ldots, \Delta_n)$	intersection number $I(\mathcal{F}_1, \ldots, \mathcal{F}_n)$

special problem we mention the following: a conjecture by Schneider on the equality case in the Alexandrov–Fenchel inequality has been partly established (see Schneider 1985, 1988, Ewald 1993, chapter 4). What are the consequences for equality in the Hodge inequality that corresponds to the Alexandrov–Fenchel inequality?

8. Dictionary

Some of the corresponding concepts of a fan Σ and the toric variety X_Σ have been listed in table 1.

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CHAPTER 2.7

Mathematical Programming and Convex Geometry

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Contents

Introduction 629
1. Terminology for convex optimization 630
2. Some qualitative aspects of convex minimization 632
3. Some qualitative aspects of convex maximization 633
4. General background for linear programming 635
5. Duality; primal-dual algorithm 636
6. Fourier–Motzkin elimination 639
7. The simplex method 640
8. Center of gravity cuts and the ellipsoid method 647
9. Karmarkar's projective algorithm 653
10. Analytic center, path-following methods 659
11. Strongly polynomial algorithms 663
References 664

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Introduction

As the term is used here, mathematical programming consists of methods for finding or approximating the global optimum of a real-valued function φ over a domain D in real n-space $\mathbb{R}^n$ or over $D \cap \mathbb{Q}^n$ or $D \cap \mathbb{Z}^n$ (the rational or integer points in D). Since the most successful methods involve convexity conditions on φ or D, it is certainly appropriate to discuss mathematical programming in a Handbook of Convex Geometry. However, because of the Handbook's focus, our emphasis is on the geometric essence of optimization methods and not on a fully detailed description of the resulting algorithms. Algorithms are of course described, but we assume the reader to be interested in convex geometry *per se* rather than in optimization *per se*, and we attempt to explain some of the interactions of convex geometry with an area to which it has been applied with great success.

We begin with three short sections on convex optimization. The main purpose of the first is to establish our terminology. The second and third survey some contrasting qualitative aspects of convex minimization and convex maximization, partly to point out the contrast and partly to exhibit, in a more general setting, some geometric facts that have influenced the design of linear programming algorithms. In all the later sections, we concentrate on the case of *linear* programming, which is generally understood to imply that the objective function φ is linear, that the domain D is a convex polyhedron given as the intersection of a finite number of closed halfspaces, and that the optimization is over D or $D \cap \mathbb{Q}^n$. It is there that the convex-geometric aspects are most strongly in evidence, and in any case, integer programming (where the optimization is over $D \cap \mathbb{Z}^n$) is treated in chapter 2.8 by Burkard, while Roberts treats certain aspects of convex optimization (where the objective function is convex but not necessarily linear) in chapter 4.2.

This chapter represents personal choices, and we do not pretend to provide complete coverage of the parts of convex geometry that are important for optimization. (That would require much more than an entire volume!) Additional information on relations between convex geometry and optimization can be found in the following articles from this or other handbooks published by North-Holland.

Author(s)	Title of Article	Subject of Handbook
M. Bayer and C.W. Lee	Combinatorial aspects of convex polytopes	Convex Geometry
U. Brechtken-Manderscheid and E. Heil	Convexity and calculus of variations	Convex Geometry
R.E. Burkard	Convexity and discrete optimization	Convex Geometry
H. Edelsbrunner	Geometric algorithms	Convex Geometry
P.E. Gill, W. Murray, M.A. Saunders and M.H. Wright	Constrained nonlinear programming	Operations Research and Management Science
D. Goldfarb and M.J. Todd	Linear programming	Operations Research and Management Science
P. Gritzmann and J.M. Wills	Lattice points	Convex Geometry

P.M. Gruber	Geometry of numbers	Convex Geometry
V. Klee and P. Kleinschmidt	Convex polytopes and related complexes	Combinatorics
A.W. Roberts	Convex functions	Convex Geometry
A. Schrijver	Polyhedral combinatorics	Combinatorics
F.F. Yao	Computational geometry	Computer Science

1. Terminology for convex optimization

Throughout this section, X denotes a vector space over an arbitrary ordered field $\mathbb{F}$. A function $\varphi: C \to \mathbb{F}$ is *convex* (resp. *concave*) if its domain C is a convex subset of X and

$$\varphi\big(\lambda x + (1-\lambda)y\big) \leqslant \lambda\varphi(x) + (1-\lambda)\varphi(y)$$

$$\big(\text{resp. } \varphi\big(\lambda x + (1-\lambda)y\big) \geqslant \lambda\varphi(x) + (1-\lambda)\varphi(y)\big)$$

for all $x, y \in C$ and $0 < \lambda < 1$. An *affine* function is one that is both convex and concave. The *strictly convex* and *strictly concave* functions are those for which, in the defining inequalities, strict inequality ($<$ or $>$) holds whenever $x \neq y$.

In addition to the basic notions of convexity or concavity, many relaxations of these notions have been studied. In particular, a function $\varphi: C \to \mathbb{F}$ is *quasiconvex* if C is convex and

$$\varphi(\lambda x + (1-\lambda)y) \leqslant \max\{\varphi(x), \varphi(y)\}$$

for all $x, y \in C$ and $0 < \lambda < 1$. Again, the strict notion is defined by requiring strict inequality when $x \neq y$. (Quasiconvexity is equivalent to the requirement that for each $\mu \in \mathbb{F}$ the set $\{x \in C: \varphi(x) \leqslant \mu\}$ is convex.) For other generalizations of convexity motivated by optimization problems, see the survey of Schaible and Ziemba (1981), the book by Avriel et al. (1988), and more recent notions of Hu, Klee and Larman (1989) and Phu (1991).

Many of the basic properties of convex functions are surveyed in books by Rockafellar (1970) and Roberts and Varberg (1973), and in chapter 4.2 in this handbook. Here we focus on the properties most closely related to optimization.

Since the negative of a linear function is also linear, the essential features of linear optimization are the same whether minimization or maximization is contemplated. However, since the essential features of convex minimization are very different from those of convex maximization, we deplore the common practice of using the terms "convex programming" and "concave programming" to refer respectively to convex minimization and to (the equivalent) concave maximization. That leaves no term of the form "... programming" to refer to the equally important and often more difficult problems of convex maximization or concave minimization. In any case, it seems preferable to avoid the vague and misleading word "programming," and to refer instead to minimization or maximization and to indicate whether a convex or concave function is involved. In the present article,

we use the term "programming" only for the linear case, and the term "convex optimization" only when both minimization and maximization of convex functions are under consideration.

For a function $\varphi : C \to \mathbb{F}$, a number $\mu \in \mathbb{F}$ is called the *(global) minimum* of φ if there is a point $p \in C$ such that

$$\varphi(p) = \mu \quad \text{and} \quad \mu \leqslant \varphi(x) \text{ for all } x \in C \setminus \{p\}.$$

Each such point p is called a *minimizer* of φ. *Strict minimum* and *strict minimizer* are defined by using $<$ in place of $\leqslant$, and of course the notions related to maximization are defined similarly.

With respect to a given topology for X or for the domain C, a number μ is called a *local minimum* for φ if there is a point $p \in C$ and a neighborhood U of p such that μ is the minimum of φ's restriction to the set $U \cap C$; each such p is a *local minimizer* of φ. *Local maxima* and *local maximizers* are defined similarly. In each case here, the global notions are intended except where the adjective "local" is attached.

If φ is an affine function, then φ is convex, so the problem of minimizing φ has some aspects in common with convex minimization. At the same time, φ is concave, so the problem of minimizing φ has aspects in common with concave minimization, or, equivalently, with convex maximization. The next two sections describe some important and contrasting aspects of convex minimization and convex maximization. The following aspects are included:

(i) relationship between local and global extrema;

(ii) extent to which optimization is an interior- or boundary-point phenomenon;

(iii) extent to which optimization reduces to one-dimensional considerations;

(iv) complexity of quadratic optimization over polytopes.

The reader will find several traces of these aspects in the linear programming methods described in later sections. In particular, the simplex method, with its emphasis on boundary behavior and on extreme points, is reminiscent of convex maximization, while the ellipsoid method and the various interior-point methods are more reminiscent of convex minimization.

For the continuity result that ends this section, and for each of the main results stated in the next two sections, the following is to be understood.

Standing hypotheses until the end of section 3.

(i) X is a finite-dimensional vector space over the ordered field $\mathbb{R}$;

(ii) the convex subset C of X has nonempty interior;

(iii) the function $\varphi : C \to \mathbb{R}$ is convex.

Most of the stated results are well known in this setting. Also, as shown by Gritzmann and Klee (1993), with appropriately extended definitions many of them remain valid when the real field $\mathbb{R}$ is replaced by an arbitrary ordered field.

No generality is lost in assuming that $\operatorname{int}(C) \neq \emptyset$, because a finite-dimensional convex set always has nonempty interior relative to its own affine hull. We do not

assume that the convex function φ is continuous, and without this assumption the boundary behavior of φ may be very wild. However, continuity on C's interior is a consequence of convexity.

Theorem (Continuity of convex functions). *The restriction of φ to* $\operatorname{int}(C)$ *is continuous, and for each polytope $P \subset C$ the restriction of φ to P is upper semi-continuous.*

2. Some qualitative aspects of convex minimization

With respect to global optimization, the following is the most important property of convex functions.

Theorem (Local minima are global). *Each local minimizer of φ is in fact a global minimizer.*

In the following sense, recognizing a local (and hence a global) minimizer of a convex function can be reduced to a collection of one-dimensional tests.

Theorem (One-dimensional nature of minimization). *If $p \in C$ and if, for each ray R issuing from p, p is a minimizer of the restriction of φ to R, then p is a minimizer of φ.*

Although minimizers are characterized by the totality of one-dimensional tests described in the preceding result, for a general convex function no finite set of such tests suffices. For suppose that $\dim X \geqslant 2$ and $R_0, R_1, \ldots, R_k$ are (pairwise different) rays issuing from the origin in X. Then there exists a piecewise-linear convex function $\varphi : X \to \mathbb{R}$ such that for $1 \leqslant i \leqslant k$ the origin is a strict global minimizer of the restriction of φ to R_i, but nevertheless the origin is a strict global maximizer of the restriction of φ to R_0.

A priori, any point p of the domain C may be a minimizer of φ (consider the function $\varphi(x) = \|x - p\|$). However, because of the following fact, convex minimization may be regarded as an interior-point phenomenon. (To treat the case in which $p \notin \operatorname{int}(C)$, use the semicontinuity result at the end of section 1.)

Theorem (Minimization depends on interior behavior). *A point $p \in C$ is a minimizer of φ if and only if p is a minimizer of the restriction of φ to the set* $\operatorname{int}(C) \cup \{p\}$.

The next result deals with the computational complexity of convex minimization over polytopes. Throughout this article (and as is common in mathematical programming) the binary (Turing machine) model is used. Hence, the size of the input of a problem (usually denoted by L) is the number of binary digits needed to encode the input data, and, when we speak of a polynomial-time algorithm we mean an algorithm whose worst-case running time is bounded above by a polynomial in

the size of the input. For the underlying concepts of the theory of computation, see Aho, Hopcroft and Ullman (1974), Garey and Johnson (1979) and Johnson (1990).

Theorem (Efficient minimization when the domain is a polytope). *There is a polynomial-time algorithm for convex quadratic minimization over a polytope given as an intersection of halfspaces.*

The first such polynomial-time algorithm was devised by Kozlov, Tarasov and Khachiyan (1979), using the ellipsoid method. Others, using interior-point methods, have been produced by Goldfarb and Liu (1991), Granot and Skorin-Kapov (1990), Kojima, Mizuno and Yoshise (1989b), Monteiro and Adler (1989b), and Ye and Tse (1989).

3. Some qualitative aspects of convex maximization

We turn now to the maximization of convex functions. As before, we assume that X is a finite-dimensional vector space over the ordered field $\mathbb{R}$, that the convex subsets C of X under consideration have nonempty interior, and that the functions $\varphi : C \to \mathbb{R}$ are convex.

The problem of maximization of convex functions leads naturally to the notion of a face of a convex set C. For each point $x \in C$, let $F(x)$ denote the union of all closed line-segments in C that cross x, with $F(x) = \{x\}$ when there are no such segments. In other words,

$$F(x) = \bigcup \left\{ [p,q] \colon p \in C, q \in C, x \in]p,q[\right\}.$$

The set $F(x)$ is the *face* of x in C, and the point x is an *extreme point* of C when $F(x) = \{x\}$. Faces are important in convex maximization.

Theorem (A property of faces). *If x is a nonextreme point of C, then the affine hull of $F(x)$ is the union of all lines L in X such that L contains a segment in C that crosses x. The face $F(x)$ is convex, and x belongs to the relative interior* relint $F(x)$. *If C is closed, then so is $F(x)$.*

Theorem (Faces and local maximization). *If x is a local maximizer of φ, then φ is constant on the set* relint $F(x)$, *and each point of this set is a local maximizer of φ.*

As the following result shows, local and global maximization may be extremely different for convex functions.

Theorem (Local and global maximization). *Suppose that C is closed, $F_1, \ldots, F_k$ are pairwise disjoint faces of C, and $\mu_1, \ldots, \mu_k$ are real numbers. Then there exists*

a continuous convex function ψ on C such that $\psi(x) = \mu_i$ for all $x \in F_i$ and $\bigcup_{i=1}^{k} F_i$ is the set of all local maximizers of ψ on C.

Theorem (Extreme points and global maximization). *Suppose that C is closed and contains no line, and φ is bounded above on C. Then for each point $x \in C$ there exists an extreme point p of C such that $\varphi(x) \leqslant \varphi(p)$. In particular, if φ attains a global maximum, then some extreme point of C is a global maximizer.*

The above result is due to Bauer (1958) and to Hirsch and Hoffman (1961), and it remains valid when φ is merely quasiconvex. To extend it to the case of an arbitrary ordered field, it is necessary to replace the assumption of closedness with the stronger notion of *segmental closedness* introduced by Klee (1964a) in dealing with extremal structure.

In the following sense, local maximization of convex functions may be regarded as a boundary-point phenomenon.

Theorem (Local maximization depends on boundary behavior). *Suppose that C is closed and p is an extreme point of C. Then p is a local maximizer for φ if and only if p is a local maximizer for the restriction of φ to C's boundary.*

For a general convex function, one-dimensional tests do not suffice for the recognition of global or local maximizers. In particular, when C is a circular disk in the plane $\mathbb{R}^2$, it is possible to construct a continuous convex function ψ on C and a boundary point p of C such that the following two conditions are satisfied: (a) for each ray R issuing from p, p is a strict local maximizer for the restriction of ψ to R; (b) p is the unique global minimizer for the restriction of ψ to C's boundary. This is in marked contrast to the situation with respect to convex minimization. However, the following result shows that in the case of a polyhedral domain, there is a finite number of one-dimensional tests which (taken together) suffice for the recognition of local maximizers. That, too, is in contrast with the situation for convex minimization.

Theorem (One-dimensional nature of maximization). *Suppose that C is a polyhedron, p is an extreme point of C, and $R_1, \ldots, R_k$ are the rays that issue from p and run along edges or extreme rays of C. Then p is a local maximizer of φ if and only if, for each i, p is a local maximizer of the restriction of φ to the intersection $C \cap R_i$.*

The following result of Bodlaender et al. (1990) (see also Gritzmann and Klee 1989) is in contrast to the final result in section 2.

Theorem (Difficulty of efficient maximization). *Maximization of φ is* NP*-hard, even when φ is the square of the Euclidean norm and the domain C is a full-dimensional parallelotope with the origin as its center or as a vertex.*

For the complexity of some related uniqueness problems see Gritzmann and Klee (1992).

4. General background for linear programming

The general linear programming problem is the task of minimizing or maximizing a linear functional over a subset of a finite-dimensional real (or rational) vector space that is defined by a finite number of linear restrictions, equality constraints or (nonstrict) inequality constraints. The basic ideas of linear programming were formulated by Fourier (1826, 1827), [see also Grattan-Guinness (1970) and Dorfman (1984)], but the modern development of the subject began with the work of Dantzig (1951, 1963), Kantorovich (1939), Koopmans (1951), and von Neumann (1947).

For the sake of simplicity, most of the discussion in this section is set in real n-space $\mathbb{R}^n$, and some of the geometric results seem to require that. At the same time, it should be recognized that for precise computation, the rational n-space $\mathbb{Q}^n$ is usually essential. (But see Adler and Beling (1992a,b) for linear programming over the algebraic numbers.) Many of the statements below are valid in $\mathbb{F}^n$ for an arbitrary ordered field $\mathbb{F}$, while others deal more specifically with $\mathbb{Q}^n$ as usually embedded in $\mathbb{R}^n$. We shall not further belabor this aspect of linear programming, but shall instead concentrate on its geometric aspects.

Each linear programming algorithm operates on some "standardized" form of the general linear programming problem, and the first computational step in dealing with a general problem is to transform it into the standardized form that is appropriate for the algorithm in question. For elaboration of the underlying geometry, the following form is most appropriate.

The linear programming problem. For given $m, n \in \mathbb{N}$, a real $m \times n$-matrix A, a vector $b \in \mathbb{R}^m$ and a vector $c \in \mathbb{R}^n$, maximize the inner product $\varphi(x) = \langle c, x \rangle$ over the polyhedron $P = \{x \in \mathbb{R}^n\colon Ax \leqslant b\}$.

For a general convex φ, the set of all (global) maximizers of φ on P would be a union of faces, but in the case of linearity it is a single face F. If the polyhedron P is nonempty and φ is bounded above on P, then F is nonempty, and if P is pointed (i.e., P has at least one extreme point, or, equivalently for nonempty closed convex sets, P contains no line), then F includes a vertex of P. The latter is the case precisely when $P \neq \emptyset$ and the matrix A has n linearly independent rows.

Each vertex v of P is determined by a set of hyperplanes of the form $ax = \beta$ where a is a row of A and β is the corresponding entry of the column-vector b. (Note that if P has interior points and if the system $Ax \leqslant b$ does not contain any redundant inequality, then all such hyperplanes are facet hyperplanes.) Hence v is the solution of a system $A'x = b'$ of linear equalities where A' is a nonsingular $n \times n$ submatrix of A and b' is the appropriate subvector of b. Thus an application of Cramer's rule yields an upper bound on the components of v in terms of both an upper bound on the entries of A and b and a nontrivial lower bound on the absolute value of the determinants of the nonsingular submatrices of A. This simple fact is particularly important in the case of rational data, for then we obtain, for the binary sizes of the coordinates of vertices of P, an upper bound that is polynomial in the

binary size of the input to the problem. We also obtain a nontrivial polynomial lower bound for the inradius of P if P has any interior points at all. These facts are basic in the derivation of exact algorithms for linear programming that employ approximation schemes such as those of the ellipsoid method or interior-point methods to be discussed in later sections.

5. Duality; primal-dual algorithm

The main duality result to be included here can be derived as a simple consequence of the Karush–Kuhn–Tucker theorem, see, e.g., Fletcher (1987) or Peressini, Sullivan and Uhl (1988). However, the present approach is especially elementary and intuitive.

Let us assume first that P is nonempty and that the linear objective function $\varphi(x) = \langle c, x\rangle$ is bounded above on P. Geometrically, the maximum value μ^* of φ on P is characterized by the fact that the set

$$H^{+*} = \{x\colon\ \langle c, x\rangle \leqslant \mu^*\}$$

is the supporting halfspace of P with outer normal c. Let

$$F = \{x \in P\colon\ \langle c, x\rangle = \mu^*\}$$

and let $a_{i_1}^{\mathrm{T}}, \ldots, a_{i_k}^{\mathrm{T}}$ be the row vectors of A for which the corresponding inequality constraints are satisfied with equality for all points of the flat $\operatorname{aff}(F)$. Then the cone C of outer normals of P at F (with apex at 0) is generated by $a_{i_1}, \ldots, a_{i_k}$; i.e., $C = \operatorname{pos}\{a_{i_1}, \ldots, a_{i_k}\}$. Since $c \in C$ there are nonnegative reals $\eta_{i_1}, \ldots, \eta_{i_k}$ such that $c = \sum_{j=1}^{k} \eta_{i_j} a_{i_j}$. This means that for any point $x^* \in F$ we have

$$\mu^* = \max_{Ax \leqslant b} \langle c, x\rangle = \max_{x \in F} \langle c, x\rangle = \langle c, x^*\rangle = \sum_{j=1}^{k} \eta_{i_j} \langle a_{i_j}, x^*\rangle = \sum_{j=1}^{k} \eta_{i_j} \beta_{i_j}$$

$$\geqslant \min_{A^{\mathrm{T}} y = c,\, y \geqslant 0} \langle y, b\rangle.$$

Now, consider any other positive combination $A^{\mathrm{T}} y, y \geqslant 0$ of the row vectors of A with $A^{\mathrm{T}} y = c$. Then for $x \in P$ we have

$$\langle c, x\rangle = \langle A^{\mathrm{T}} y, x\rangle = \langle y, Ax\rangle \leqslant \langle y, b\rangle,$$

with equality in the last inequality if and only if for each index i the ith coordinate of y is 0 or the ith inequality in $Ax \leqslant b$ is satisfied with equality. (This is the *complementary slackness* condition.) The above reasoning implies in particular that

$$\max_{Ax \leqslant b} \langle c, x\rangle = \min_{A^{\mathrm{T}} y = c,\, y \geqslant 0} \langle b, y\rangle.$$

These two linear programming problems are *dual* to each other.

Observe that $\dim(F)+\dim(C) = n$. Hence F is not a singleton if and only if there are minimizers of the dual program that have fewer than n nonzero coefficients (*dual degeneracy*). On the other hand, if there is an optimal vertex of the primal (i.e., the original) program that satisfies more than n constraints with equality (*primal degeneracy*) then the cone C is not simplicial and the optimal face of the dual program is not a singleton.

We note in passing that the transformation that associates with a given finite sequence of vectors (the vectors $a_{i_1}, \ldots, a_{i_k}, c$ in our case) the affine or positive relations among these vectors (the $\eta_{i_1}, \ldots, \eta_{i_k}$ of our example) is called a *Gale transformation*. For a survey of such transformations and some related "diagrams," see McMullen (1979).

Now, if the objective function $\varphi(x) = \langle c, x\rangle$ is not bounded above on P then there is no solution to the dual system $A^{\mathrm{T}}y = c, y \geqslant 0$. Further, if $\langle y, b\rangle$ is not bounded below on the set $\{y\colon A^{\mathrm{T}}y = c, y \geqslant 0\}$ then the system $Ax \leqslant b$ is infeasible. All the other combinations (of boundedness and feasibility for the primal and the dual programs) can actually occur.

The following theorem collects all the statements that have been derived in the preceding paragraphs. It is due to von Neumann (1947) and Gale, Kuhn and Tucker (1951).

Theorem (Linear programming duality). *For the two linear programs*

$$\text{(P)} \quad \max_{Ax \leqslant b} \langle c, x\rangle, \qquad \text{(D)} \quad \min_{A^{\mathrm{T}}y = c, y \geqslant 0} \langle y, b\rangle$$

there are the following possibilities:

– both problems are feasible and have finite solutions; in this case, the optimal values of the objective functions coincide, and $y^{*\mathrm{T}}(b - Ax^*) = 0$ *for each pair* x^*, y^* *of optimizers;*

– problem (P) *is infeasible and the objective function of* (D) *is not bounded below on its domain;*

– problem (D) *is infeasible and the objective function of* (P) *is not bounded above on its domain;*

– both problems are infeasible.

It is clear from the underlying geometry that duality is involutory.

The duality theorem implies many other results, in particular some "theorems of the alternative" (part of the duality theorem is itself such a theorem). One of the basic theorems of the alternative is the lemma of Farkas (1893/4, 1902) stated below (see also Minkowski 1896). (There are other formulations of Farkas's lemma and other theorems of the alternative; we state the following only as an example.)

Lemma (Farkas). *There exists a vector* y *with* $A^{\mathrm{T}}y = c, y \geqslant 0$ *if and only if* $\langle c, x\rangle \leqslant 0$ *whenever* $Ax \leqslant 0$.

By the duality theorem applied with $b = 0$ there exists such a vector y if and

only if

$$\infty > \min_{A^{\mathrm{T}}y=c,\, y\geqslant 0} \langle 0, y\rangle = \max_{Ax\leqslant 0} \langle c, x\rangle.$$

But this maximum is less than ∞ if and only if $\langle c, x\rangle \leqslant 0$ whenever $Ax \leqslant 0$.

From a geometric viewpoint, Farkas's lemma is just a *separation theorem*: a point outside a polyhedral cone can be separated from this cone by a hyperplane.

Another important consequence of the duality theorem is that the optimality of a putative solution of a linear programming problem can be established just by solving *feasibility problems*. If (P) and (D) are both feasible then every pair (x^*, y^*) consisting of a primal optimizer x^* and a dual optimizer y^* is feasible for the system

$$\langle c, x\rangle \geqslant \langle y, b\rangle, \quad Ax \leqslant b, \quad A^{\mathrm{T}}y = c, \quad y \geqslant 0$$

and vice versa. Thus, the linear programming problem belongs to the complexity class NP$\cap$co-NP. It is known, however, that if the complement of an NP-complete problem is in NP then NP = co-NP. Since the latter identity is widely believed to be false, one could say that, long before an actual polynomial-time LP algorithm was known, LP duality provided evidence that linear programming problems could be solved in polynomial time (or at least that LP is not NP-complete).

Further, the duality theorem shows that the linear programming problem is equivalent (in the sense of polynomial-time transformations) to the feasibility problem, which again can be transformed into various "standardized" forms.

Still another important aspect of duality is that it provides justification and framework for a variety of algorithms for linear programming, see Dantzig, Ford and Fulkerson (1957).

In principle one could proceed as follows. Suppose (with notation as before) that x_0 is feasible for $Ax \leqslant b$, i.e., $x_0 \in P = \{x\colon Ax \leqslant b\}$, and suppose (for simplicity) that P is nonempty and bounded. As we have seen before, x_0 is optimal if and only if c can be expressed in the form $c = \sum_{j=1}^{k} \eta_{i_j} a_{i_j}$ with $\eta_{i_1}, \ldots, \eta_{i_k} \geqslant 0$, where $a_{i_1}^{\mathrm{T}}, \ldots, a_{i_k}^{\mathrm{T}}$ denote the row vectors of A for which the corresponding inequality constraints are satisfied with equality for the point x_0. If this is not the case, then there is a vector v such that

$$\langle a_{i_j}, v\rangle \leqslant 0 \quad \text{for } j = 1, \ldots, k,$$
$$\langle c, v\rangle > 0,$$

i.e., the cone $x_0 + \mathrm{pos}(P - x_0)$ has a nonempty intersection with the open halfspace $\{x\colon \langle c, x\rangle > \langle c, x_0\rangle\}$. Clearly, we can use v to improve the objective function value: choose $\theta > 0$ (as large as possible) such that $x_1 = x_0 + \theta v \in P$. Of course, there is an abundance of possible specifications as how to choose a suitable v, and one must be careful in order to obtain reasonable algorithms. One possible specification leads to the simplex algorithm that is described in section 7. In what follows we will

describe a kind of "recursive" possiblity that involves the solution of a modified linear program.

Note first that, as stated, the auxiliary problem is not a linear programming problem, for it involves a strict inequality. However, it is easy to derive various equivalent linear programming formulations. A formulation that is particularly useful for applications to network flow problems (see Papadimitriou and Steiglitz 1982) is the program

$$\begin{aligned} &\max \langle c, v\rangle \\ &\langle a_{i_j}, v\rangle \leqslant 0 \quad \text{for } j = 1, \ldots, k, \\ &v \leqslant \mathbb{1}. \end{aligned}$$

In fact, the optimum of this linear program is 0 if and only if the above auxiliary problem is infeasible (and then x_0 is optimal in the original linear problem). Further, if the optimum is greater than 0 then each maximizer v is feasible for the auxiliary problem.

Hence, this primal–dual approach reduces the original linear program to a sequence of new LPs, and it does not seem that one has gained anything. However, the right-hand side of the new linear programs is now a 0–1-vector, and thus the problems have become "more combinatorial". In particular, it turns out that when applied to the shortest path problem in a weighted graph with nonnegative weights, the original problem (which is the dual of the standard LP-formulation of the shortest path problem) is replaced by a sequence of connectivity tests; we obtain essentially Dijkstra's (1959) algorithm.

Further, a primal-dual approach to the max-flow problem leads to the Ford and Fulkerson (1962) algorithm. For details and further applications see Papadimitriou and Steiglitz (1982).

6. Fourier–Motzkin elimination

The duality theorem shows that the linear programming problem is equivalent (in the sense of polynomial-time transformations) to the feasibility problem, which again can be transformed into the form $Ax \leqslant b$. Hence a natural approach to linear programming is to try to solve $Ax \leqslant b$ by an extension of Gaussian elimination, a method that works so well for a system of linear equations. This approach, historically a predecessor of linear programming, is now called *Fourier–Motzkin elimination*. It was suggested by Fourier (1827), Dines (1918/19) and Motzkin (1936).

Suppose that the feasibility problem is scaled so that the last column of A has components in $\{-1, 0, 1\}$. Let $a_1'^{\mathrm{T}}, \ldots, a_m'^{\mathrm{T}}$ denote the row vectors of the matrix A' that is obtained from A by deleting the last column, let $b = (\beta_1, \ldots, \beta_m)^{\mathrm{T}}$, and let us write $x = \begin{pmatrix} x' \\ \xi_n \end{pmatrix}$. Then, after possibly renumbering the inequalities, our problem reads explicitly

$$\begin{aligned}
&\langle a'_1, x'\rangle - \xi_n \leqslant \beta_1 \quad \dots \quad \langle a'_k, x'\rangle - \xi_n \leqslant \beta_k,\\
&\langle a'_{k+1}, x'\rangle \leqslant \beta_{k+1} \quad \dots \quad \langle a'_l, x'\rangle \leqslant \beta_l,\\
&\langle a'_{l+1}, x'\rangle + \xi_n \leqslant \beta_{l+1} \quad \dots \quad \langle a'_m, x'\rangle + \xi_n \leqslant \beta_m.
\end{aligned}$$

Hence, there is a point $x = \binom{x'}{\xi_n}$ that solves $Ax \leqslant b$ if and only if there is a vector x' such that

$$\begin{aligned}
&\max_{i=1,\dots,k} (\langle a'_i, x'\rangle - \beta_i) \leqslant \xi_n\\
&\langle a'_i, x'\rangle \leqslant \beta_i \quad \text{for } i = k+1, \dots, l,\\
&\xi_n \leqslant \min_{i=l+1,\dots,m} (-\langle a'_i, x'\rangle + \beta_i).
\end{aligned}$$

Thus, the original system is equivalent to the following system of inequalities in $n-1$ variables:

$$\begin{aligned}
&\langle a'_i, x'\rangle - \beta_i \leqslant -\langle a'_j, x'\rangle + \beta_j \quad \text{for } i = 1, \dots, k \text{ and } j = l+1, \dots, m,\\
&\langle a'_i, x'\rangle \leqslant \beta_i \qquad\qquad\qquad\qquad\quad \text{for } i = k+1, \dots, l.
\end{aligned}$$

This elimination technique can be iterated, and after $n-1$ steps we obtain a system of inequalities in one variable. This last system can be solved easily, and backtracking then yields a solution of the original system.

Note, however, that there is a price to pay for each step that decreases the dimension of the system by 1: the number of constraints increases to $k(m-l)+l-k$ which may be as large as $O(m^2)$. This increase is not just due to a generation of a large number of redundant inequalities, but it is an intrinsic feature of the method. This becomes more transparent when the elimination method is interpreted geometrically: each elimination step is a projection onto a hyperplane. In fact, in the step described above the polyhedron $P = \{x \in \mathbb{R}^n\colon Ax \leqslant b\}$ is projected onto the hyperplane $\{x \in \mathbb{R}^n\colon \xi_n = 0\}$. As is well known, the orthogonal projection of an n-dimensional cube onto a k-dimensional subspace in general position (with respect to the cube) is a zonotope which has $2\binom{n}{k-1}$ facets. Hence, in general it is not possible to avoid an exponential blow-up in the number of constraints. That is why the Fourier–Motzkin elimination method does not run in polynomial time.

7. The simplex method

We are now going to describe the first and still most widely used method for linear programming, the *simplex algorithm* developed by Dantzig (1951). The algorithm is usually applied to a linear programming problem over a polyhedron that is given in its *standard form* $\{x\colon Ax = b, x \geqslant 0\}$. The algorithm splits into two phases. The first phase decides whether the problem is feasible, and if this is the case it

finds a vertex of the feasible region. (An advantage of the standard form is that if the polyhedron is nonempty, it must have a vertex.) The actual optimization is accomplished in the second phase. The methods are the same for the two phases; in fact, in phase I the phase II algorithm is just applied to an auxiliary problem for which an initial vertex is known. Since we are here concerned only with the geometric aspects of the algorithm, we shall discuss only the second phase.

Let us consider the problem again in the form $\max_{Ax \leqslant b} \langle c, x \rangle$, assuming also that the polyhedron $P = \{x: Ax \leqslant b\}$ is nonempty and that we are given an initial vertex v_0. We shall first give a geometric description of the basic idea under the additional assumption that P is a full-dimensional simple polytope, that none of the constraints is redundant, and that no two vertices of P yield the same value for the objective function. After doing that, we shall give a complete, geometrically oriented, description of phase II, including Bland's noncycling pivoting rule, Bland (1977), pointing out why certain steps are different from those in the basic version.

Let us assume, then, that P is as indicated and a vertex v_0 of P is given. The simplex algorithm constructs an edge-following path from v_0 to an optimizing vertex v^*.

Let $e_1, \ldots, e_n \in \mathbb{R}^n$ such that the rays $(v_0 + [0, \infty[e_i)$ are precisely those that issue from v_0 and contain edges of P. Such vectors e_i can be computed by solving the $n \times n$ system $A'x = b'$ given by all n of the inequalities from the system of $Ax \leqslant b$ that are satisfied with equality for v_0. Assume (for simplicity of notation) that these are the first n inequalities and that the e_i are obtained by omitting the ith constraint of $A'x = b'$.

If $\langle c, e_i \rangle \leqslant 0$ for all $i = 1, \ldots, n$ then the vertex v_0 is already optimal. Hence, assume that there is an index i_0 with $\langle c, e_{i_0} \rangle > 0$, which means that the objective function increases along the ray $v_0 + [0, \infty[e_{i_0}$. A way of describing this situation without explicitly using e_{i_0} is as follows: the affine hull of $\{a_i: i = 1, \ldots, n; i \neq i_0\}$ strictly separates a_{i_0} from c. The ray $v_0 + [0, \infty[e_{i_0}$ contains precisely one additional vertex v_1 of P, and v_1 is characterized as the closest to v_0 of the points of intersection of $v_0 + [0, \infty[e_{i_0}$ with any of the other constraint hyperplanes $\{x: \langle a_i, x \rangle = \beta_i\}$ $(i = n+1, \ldots, m)$. Hence we compute

$$\min \left\{ \frac{\beta_i - \langle a_i, v_0 \rangle}{\langle a_i, e_{i_0} \rangle}: \; n+1 \leqslant i \leqslant m; \langle a_i, e_{i_0} \rangle > 0 \right\}.$$

Then the procedure is repeated with v_1 in place of v_0. Since the objective function increases at each step, and since there are only finitely many vertices, the procedure finds an optimal vertex of P in finitely many steps.

In general, there may be several "improving edges" issuing from the vertex v_0, and hence there may be several admissible choices for the index i_0. A procedure for making this choice at each step, and thus producing the edge-following path from the initial vertex to the optimizing vertex, is called a *pivot rule*. Many different pivot rules have been tested experimentally, and several have found their way into practical implementations of the simplex method.

Looking at the above computation from a geometric viewpoint, and recalling the results of sections 2 and 3, we may identify the following salient features:

(i) Since the (linear) objective function φ is convex and continuous, and since (in the special case under consideration) P is compact, φ's global maximum on P is attained at an extreme point of P.

(ii) Since φ is convex and P is a polyhedron, an extreme point v of P can be recognized as a local maximizer of φ by means of tests involving the edges of P incident to v.

(iii) Since the function $-\varphi$ is convex, its local minima (and hence the local maxima of φ) are in fact global extrema.

The *linearity* of φ (as opposed to mere convexity or concavity) is responsible for our ability to travel along the edges of P, from the starting vertex to an optimizing vertex, in such a way that the objective function steadily increases. This essential geometric aspect of the simplex algorithm has been extended by Larman and Rogers (1971) to an arbitrary convex body B and linear functional φ on B: in the 1-skeleton of B, there is a φ-increasing path that leads from a minimizer to a maximizer.

The above computational discussion was based on the simplifying assumption that each vertex of P satisfies exactly n constraints with equality, hence lies on exactly n facets and exactly n edges (or extreme rays) of P. However, this assumption fails in a truly geometric fashion if P is not simple and hence has a vertex incident to more than $\dim(P)$ facets. (For example, if P is a pyramid over an $(n-1)$-dimensional cube with apex $v_0 = (0,\ldots,0,1)^T$, there are $2(n-1)$ facets and 2^{n-1} edges of P incident to v_0.) Even when P is simple, the assumption fails in an algebraic fashion when there are redundant constraint hyperplanes that pass through a vertex. Either sort of failure implies that there is no efficient way to consider all edges at v_0 simultaneously. A remedy for this difficulty is to focus not on vertices per se, but rather on the n-tuples of constraints that determine vertices.

Let $M = \{1,\ldots,m\}$. For each nonempty subset I of M, let A_I denote the submatrix of A of rows with indices in I and let b_I denote the corresponding right-hand side. For each vertex v of P there is a subset $I \subset M$ of cardinality n such that A_I is nonsingular, $A_I v = b_I$ and $A_{M\setminus I} v \leqslant b_{M\setminus I}$. (The same vertex may correspond to many such index sets I and each such set I (or each such submatrix A_I) is called a *basis* for the vertex v.) We then proceed as before, trying to increase the objective function on one of the rays issuing from v that are determined by deleting one of the equality constraints from the system, $A_I x = b_I$. Such a ray may fail to contain any edge of P, for it may happen that v itself is the only feasible point on the ray. Nevertheless, we replace one constraint by another, hoping that this new system will lead to a ray along which the objective function can be improved while maintaining feasibility. It should be noted that dealing with this kind of degeneracy is in a sense as hard as the general linear programming problem, Megiddo (1986b). Also, the problem of detecting whether there is degeneracy at any vertex of the feasible region is NP-hard, Chandrasekaran, Kabadi and Murty (1982).

If the pivot steps at a degenerate vertex v are not chosen with care, they may lead to the phenomenon known as *cycling*. That is, the algorithm may simply move among various presentations of the same vertex v, never leaving v even though v is not optimal, Beale (1955), Hoffman (1953). However, cycling can be avoided

by the use of various pivot rules. The first of these was the *lexicographic rule* (see Dantzig 1963), which follows the pivots that would occur if the right-hand side b were perturbed by addition of the vector $(\varepsilon, \varepsilon^2, \ldots, \varepsilon^n)^T$ for a positive ε that is sufficiently small relative to the other data in the problem. An especially interesting noncycling pivot rule is due to Bland (1977). It will be described later.

Since we are now not concerned with all edges of P that contain v, but only with an $n \times n$ system $A_I x = b_I$ that determines v, we are faced with the problem of deciding whether v is optimum. In fact, we might not notice optimality until we find an index-set set I that actually has the property that $c \in \text{pos}\{a_i\colon i \in I\}$. And again, the problem of recognizing optimality is no easier than the general linear programming problem, Megiddo (1986b). Note that this optimality test is a test of dual feasibility. Since A_I is nonsingular there is a unique solution to the system $A_I^T y = c$. If $c \notin \text{pos}\{a_i\colon i \in I\}$ at least one component of y is negative – say the i_0th component. But this means that the hyperplane $\text{lin}\{a_i\colon i \in I \setminus \{i_0\}\}$ strictly separates c from a_{i_0}.

Let us now describe a complete phase II simplex algorithm.

Algorithm (*Simplex algorithm – with Bland's pivoting rule*).

Input: Positive integers m, n with $n \leqslant m$, a real $m \times n$ matrix A, a vector $b \in \mathbb{R}^m$, a vector $c \in \mathbb{R}^n$.

Output: The message that $\langle c, x\rangle$ is not bounded on $P = \{x \in \mathbb{R}^n\colon Ax \leqslant b\}$ from above or a vertex v^* of P such that $\langle c, v^*\rangle = \max_{x \in P}\langle c, x\rangle$.

Initialization: Let v_0 be a vertex of P, set $k = 0$.

Iteration: Choose a subset $I_k \subset M$ of cardinality n such that $A_{I_k} v_k = b_{I_k}$ and such that A_{I_k} is nonsingular.

Solve $A_{I_k}^T y_k' = c$ and let $y_k \in \mathbb{R}^m$ coincide with y_k' on all components that correspond to the indices in I_k while the other components are 0.

If $y_k \geqslant 0$ then v_k is an optimal vertex of P and the procedure terminates.

Otherwise choose the smallest index i_k for which the corresponding component η_{i_k} of y_k is negative.

Compute e_k such that $\langle a_i, e_k\rangle = 0$ for $i \in I_k \setminus \{i_k\}$ and $\langle a_{i_k}, e_k\rangle < 0$.

If $\langle a_i, e_k\rangle \leqslant 0$ for all $i \in M$ then the ray $v_k + [0, \infty[e_k$ is contained in P and we terminate with the message that the objective function is unbounded on P from above.

Otherwise let j_k be the smallest index for which $(\beta_j - \langle a_j, v_k\rangle) \times (\langle a_j, e_k\rangle)^{-1} = \mu_j$ is minimum for all $j \in M$ with $\langle a_j, e_k\rangle > 0$.

Set $I_{k+1} = (I_k \setminus \{i_k\}) \cup \{j_k\}$, $v_{k+1} = v_k + \mu_k e_k$.

Repeat the procedure with k replaced by $k + 1$.

As was first shown by Bland (1977) (see also Schrijver 1986), this variant of the general simplex algorithm terminates after finitely many steps.

Theorem (Finiteness of the simplex algorithm). *The simplex algorithm (with Bland's pivot rule, as described above) terminates.*

For an elegant use of Bland's pivot rule in an algorithm for finding all vertices of a polytope, see Avis and Fukuda (1992). Among the many other vertex-finding algorithms in the literature, those of Mattheiss (1973) and Seidel (1987, 1991) may be of special geometric interest. Mattheiss and Rubin (1980) survey and compare most of the methods up to 1980. It is useful to know, before setting out to find all vertices of an n-polytope presented as an intersection of m halfspaces, that the maximum possible number of such vertices is

$$\binom{m - \lfloor (n+1)/2 \rfloor}{m-n} + \binom{m - \lfloor (n+2)/2 \rfloor}{m-n}.$$

When the polytope is simple and the presentation is not redundant, the minimum possible number of vertices is $(m-n)(n-1)+2$. These sharp bounds are due respectively to McMullen (1970) and Barnette (1971). The literature also contains many results on the *expected* numbers of vertices of polytopes presented in various ways; see chapter 5.2. In attempting to maximize a convex function over a polytope, all vertices are a priori of interest. With clever search methods (see, e.g., Horst and Tuy 1990), it is usually possible to avoid looking explicitly at all vertices, but even in very special cases the computational situation is much worse than for linear optimization (see the last result stated in section 3).

In most implementations of the simplex algorithm, one does not actually stay in the same space and move along edges, but instead applies affine transformations that map the current vertex into the origin and the current "active" constraints $A_{I_k} x = b_{I_k}$ into nonnegativity constraints $x \geqslant 0$. In this way the optimality test is reduced to a sign test. Further, the vector e_k can then be chosen to be the negative of the i_kth column of $A_{I_k}^{-1}$. In each step, only one constraint is replaced by another, and thus the actual effort per step (in the real model of computation) is bounded by a quadratic function of the dimension n. Hence for the overall complexity of the method, the decisive factor is the number of pivot steps required. When the right-hand side is perturbed as outlined in order to obtain a simple polyhedron, this comes down to the question of the length of longest edge-paths that are increasing with respect to a given objective function and a given choice of pivot rule. A lower bound for this number (in the bounded case) is the number $\Delta(n,m)$, which is defined as the maximum of the edge-diameter $\delta(P)$ over all n-dimensional polytopes P with m facets. In other words, $\Delta(n,m)$ is the smallest integer k such that whenever v and w are vertices of an n-polytope P with m facets, then v and w are connected by a path formed from k or fewer edges of P. W.M. Hirsch conjectured in 1957 that $\Delta(n,m) \leqslant m-n$ whenever $m > n$ (the *Hirsch conjecture*), and in particular that $\Delta(n,2n) = n$ for all n (see Dantzig 1963). The first conjecture has been proved (for all m) only for $n \leqslant 3$, and the second only for $n \leqslant 5$, though it is known that in their full strength the two conjectures are equivalent, Klee and Walkup (1967). In certain directions, the best results currently known are as follows, due respectively to Klee (1964b), Larman (1970), and Kalai and Kleitman (1992).

Theorem (Results on the Hirsch conjecture).

$$\Delta(3,m) = \lfloor \frac{2m}{3} \rfloor - 1;$$
$$\Delta(n,m) \leqslant 2^{(n-3)}m;$$
$$\Delta(n,m) \leqslant m^{\log n+2}.$$

When P is an n-polytope with m facets and v is a point not in the affine hull of P, the pyramid $Q = \text{conv}(P \cup \{v\})$ is an $(n+1)$-polytope with $m+1$ facets. For each facet G of Q other than P itself, the intersection $F_G = G \cap P$ is a facet of P. Because Q is so "degenerate" at v, the fact that $\delta(P) \leqslant 2$ does not seem to have any significance for linear programming or for the Hirsch conjecture. However, Lee (1991) has shown that if P is *simple*, then by suitably "rotating" each facet G around F_G, one can obtain an approximation Q' of Q such that Q' is a *simple* $(n+1)$-polytope with $m+1$ facets and with $\delta(Q') \leqslant 2(m-n)$. It will be interesting to see whether this procedure will eventually have some computational significance.

For other results related to the Hirsch conjecture, see the survey of Klee and Kleinschmidt (1987) and the more recent papers of Kalai (1991, 1992a). The conjecture is now widely believed to be false in general, though no counterexample has been constructed and it has been proved for various special classes of polytopes. For 0–1 polytopes (those for which all vertex-coordinates are 0 or 1), the elegant observations of Naddef (1989) may be reformulated as follows, where an n-polytope with m facets is called an *N-polytope* if its diameter is at most n and an *H-polytope* if its diameter is at most $m-n$.

Theorem (N-polytopes and H-polytopes). *Suppose that a polytope P has two faces F and G such that each vertex of P belongs to $F \cup G$. If F and G are both N-polytopes, so is P. If, in addition, each facet of P is an H-polytope, then so is P.*

The above statements are easily proved, and induction shows that 0–1 polytopes are both N-polytopes and H-polytopes. Kleinschmidt and Onn (1992) have extended the first conclusion by showing that if $X \subset \mathbb{R}$, $V \subset X^n$ and $P = \text{conv}(V)$ then $\delta(P) \leqslant (|X|-1)\dim(P)$.

The most-studied 0–1 polytopes are those associated with *traveling salesman problems*. With $n = \binom{k}{2}$, let $\{e_1, \ldots, e_n\}$ denote the edge-set of the complete graph K_k on k nodes, and let a point $(\xi_1, \ldots, \xi_n)^T$ of $\{0,1\}^n$ belong to V if and only if the set $\{e_i\colon\ \xi_i = 1\}$ forms a Hamilton cycle in K_k. (Thus $|V| = (k-1)!/2$.) For the polytope $P = \text{conv}(V)$, it was conjectured by Grötschel and Padberg (1985) that $\delta(P) = 2$ for all $k \geqslant 5$. This was proved for $5 \leqslant k \leqslant 12$ by Sierksma and Tijssen (1992), who also showed that $\delta(P) \leqslant \lfloor \frac{1}{2}k \rfloor$ for all k. For the analogous polytope associated with a complete *directed* graph, the inequality $\delta(P) \leqslant 2$ was established by Padberg and Rao (1974). However, it should be emphasized that the results and conjectures about small diameters do not imply that linear programming over these polytopes is easy. That is emphasized by the result of Papadimitriou (1978),

according to which even deciding adjacency of two vertices on these polytopes is NP-hard. For optimization over these and many other 0–1 polytopes, the most effective methods are those of *Polyhedral Combinatorics*, which is surveyed in a handbook article by Schrijver (1993).

An upper bound for the number of pivot steps of the simplex method is given by the length of a longest edge-following path in P that is increasing with respect to some linear objective function. Since the general method (which need not employ Dantzig's or Bland's pivot rule) can proceed along any such increasing edge-path, any such length will be attained by some variant of the method. An example of Klee and Minty (1972) shows that a suitably distorted cube contains an edge-path, increasing with respect to a certain objective function, that involves all vertices and hence is of length $2^n - 1$. In their example, the feasible region is a polytope P that is projectively (and hence combinatorially) equivalent to an n-cube, and for a suitable presentation of P the edges of the path are those followed by the original pivot rule of Dantzig. This shows that Dantzig's pivot rule leads to exponential worst-case behavior of the simplex method. The idea behind the Klee–Minty example has been modified to show that the same sort of exponential behavior is exhibited by certain other pivot rules (see Jeroslow (1973) for running along the edge that gives the greatest improvement in the value of the objective function, and Avis and Chvátal (1978) for Bland's pivot rule).

Theorem (Exponential worst-case behavior)**.** *The number of steps of the simplex algorithm (with Dantzig's or Bland's pivot rule) is not bounded by any polynomial in n and m, and not even by any polynomial in the binary size of the input.*

In spite of this negative result, the simplex method works very well in practice. An explanation is that even though the worst-case behavior is exponential with the most common pivot rules, the *average-case* behavior (in a very natural probabalistic model) is favorable: the expected number of steps is $O(\min\{m^2, n^2\})$, Adler and Megiddo (1985), Todd (1986), Adler, Karp and Shamir (1986, 1987). The average-case behavior of the simplex method has been studied in many other papers, including Borgwardt (1977), Smale (1983), Haimovich (1983), Vershik and Sporyshev (1983), and Megiddo (1986a). For these studies, a particularly popular pivot procedure has been the one originally proposed by Gass and Saaty (1955). Haimovich has an especially clear description of the relevant geometry, which is discussed also by Klee and Kleinschmidt (1990). For a comprehensive discussion and analysis of average-case behavior, see Borgwardt (1987).

The Gass–Saaty pivot procedure is deterministic, and even though its average-case behavior is favorable, its worst-case behavior is exponentially bad, Murty (1980). That appears to be true of all the pivot rules used in the average-case analyses mentioned above. On the other hand, the randomized pivot rule of Kalai (1992b) is such that for every linear programming problem with n variables and m constraints, the expected number of arithmetic operations is subexponential. A "dual" subexponential randomized algorithm for linear programming (and related problems) was independently devised by Sharir and Welzl (1992); its analysis is

given in Matoušek, Sharir and Welzl (1992). It is still unknown whether there is a pivot rule (deterministic or randomized) for which the number of arithmetic operations is bounded by a fixed polynomial in n and m.

8. Center of gravity cuts and the ellipsoid method

It is still unknown whether there is a polynomial-time algorithm for linear programming in the real arithmetic model of computation, or a strongly polynomial-time algorithm (see section 11) in the binary model. It is known, however, that linear programming problems can be solved in polynomial time in the binary model. This was first proved by Khachiyan (1979), who made extensive use of the fact that in this model the input data are integers. He showed that an algorithm previously developed by Shor (1970a,b, 1977, see also 1985) and Nemirovski and Yudin (1976, see also 1983) in the context of convex minimization yields a polynomial-time procedure for linear programming.

All known polynomial-time algorithms for linear programming are iterative methods (based on techniques for nonlinear programming problems) that are approximative in nature and that lead to a finite (and in fact polynomial-time) algorithm only because the integrality assumption allows rounding good enough approximations to obtain an exact optimizer of the problem. Hence, we begin by describing conditions under which an iterative method leads to a polynomial-time algorithm.

If arbitrary real numbers are allowed as input the polyhedral feasible region will in general not have rational vertices and hence there is no way to guarantee that an approximative method will terminate. However, when the input is restricted to integers, the situation is as follows.

Lemma. *Let A be an integer $m \times n$-matrix, let $b \in \mathbb{Z}^m$, and let L denote the binary size of all input data. Then for each vertex v of the polyhedron $P = \{x: Ax \leqslant b\}$ there are integers $\tau_1, \ldots, \tau_n, \rho_1, \ldots, \rho_n$ such that*

$$v = \left(\frac{\tau_1}{\rho_1}, \ldots, \frac{\tau_n}{\rho_n}\right)^{\mathrm{T}} \quad \text{and} \quad \| v \|, | \tau_1 |, \ldots, | \tau_n |, | \rho_1 |, \ldots, | \rho_n | < 2^L.$$

This result implies that if a given linear programming problem has an optimal vertex v^*, then v^* is the unique rational vector with absolute values of denominators and numerators bounded by 2^L in a neighborhood $v^* + 2^{-2L}\mathbb{B}^n$ of v^* (where $\mathbb{B}^n$ denotes the Euclidean unit ball). Hence, once we reach this neighborhood, we can terminate whatever approximative method is used, and simply round to the (uniquely determined) rational vector of this form. There is a *continued fraction* rounding procedure that runs in polynomial time (in L), see, e.g., Schrijver (1986). The lemma implies also that in the following we may assume that P is bounded, since no vertex of P is cut off by intersection with $2^L[-1,1]^n$ and hence it is easy to decide whether the objective function is bounded above on P.

Now assume we have an approximative procedure that uses only polynomial time per step and that converges linearly; e.g., in the kth step it generates in polynomial time a vector x_k such that $\|x_k - v^*\| \leqslant \rho^k$, with $\rho \in]0,1[$. Then we need only perform

$$k_0 = \left\lceil \frac{2L}{\log(1/\rho)} \right\rceil$$

steps to be sufficiently close to v^*. Hence we obtain a polynomial-time algorithm whenever $\rho \leqslant 2^{-1/\pi(L)}$, where π is a fixed polynomial. In particular, this condition is satisfied when ρ is of the form $1 - 1/\alpha(m+n)^r$ with $r \in \mathbb{N}_0$ and $\alpha \geqslant 1$.

These observations enlarge the class of algorithms that one might consider as candidates for polynomial-time procedures for linear programming, in the binary model of computation; all known polynomial-time algorithms make use of them.

We turn now to some features that are more specifically related to the ellipsoid method. We begin by describing the "predecessor" method of *center-of-gravity sections*. It goes back to Levin (1965) and Newman (1965), and is based on the following result of Grünbaum (1960).

Theorem (Center-of-gravity sections decrease the volume quickly). *Let K be a convex body, let g denote K's center of gravity, let H be a hyperplane through g with corresponding halfspaces H^+ and H^-, and let $K^+ = K \cap H^+$ and $K^- = K \cap H^-$. Then*

$$V(K^{\pm}) \leqslant \left(1 - \left(1 - \frac{1}{n+1}\right)^n\right) V(K).$$

Observe that the factor $1 - (1 - 1/(n+1))^n$ is bounded above by $1 - 1/\mathrm{e} \sim 0.63$, which is of the form given above. In order to make use of this volume decrease, we further assume that a lower bound on the inradius $r_n(K)$ of K and an upper bound on the circumradius $R_n(K)$ of K are given. These bounds will enter our estimates for the number of steps of the algorithm. Actually, one of these bounds is enough if we are willing to accept the dependence on the volume of K. (This follows easily from classical inequalities (see Bonnesen and Fenchel 1934)). However, since the transition to the ellipsoid method for linear programming is more natural when both bounds are given, we make that assumption. The method of center-of-gravity sections might be regarded as a generalization of 1-dimensional *binary search*. It works as follows.

Algorithm (*The method of center-of-gravity sections*).

Input: A positive integer n, a convex body K in $\mathbb{R}^n$, positive rationals ρ and R such that $\rho \leqslant r_n(K)$ and $R_n(K) \leqslant R$, a vector c in the Euclidean unit sphere $\mathbb{S}^{n-1}$, a positive real number ε.

Output: A vector v' in K such that $\langle c, v' \rangle + \varepsilon \geqslant \max_{x \in K} \langle c, x \rangle$.

Initialization: Set $k = 0$, $K_0^+ = K$.

Iteration: Compute the center of gravity g_k of K_k^+.
Set $H_k^+ = \{x \in \mathbb{R}^n : \langle c, x\rangle \geqslant \langle c, g_k\rangle\}$, $K_{k+1}^+ = K_k^+ \cap H_k^+$.
If $V(K_k^+) \leqslant (1/n)\varepsilon^n (\rho/2R)^{n-1} V(\mathbb{B}^{n-1})$ terminate.

Note that the general concave maximization (or convex minimization) problem can be transformed into the form considered above.

Now let v^* be a maximizer of our problem and let $z + \rho\mathbb{B}^n \subset K$. By considering the volume of the set $\operatorname{conv}(\{v^*\} \cup (z + \rho\mathbb{B}^n)) \cap H_k^+$, we see that

$$V(K_{k+1}^+) \geqslant \frac{1}{n}\left(\frac{\rho}{2R-\rho}\right)^{n-1} (\langle c, v^* - g_k\rangle)^n V(\mathbb{B}^{n-1}).$$

Hence, for

$$V(K_{k+1}^+) \leqslant \frac{1}{n}\varepsilon^n \left(\frac{\rho}{2R}\right)^{n-1} V(\mathbb{B}^{n-1})$$

we have

$$\langle c, g_k\rangle + \varepsilon \geqslant \langle c, v^*\rangle,$$

which implies that the above algorithm solves the problem correctly. Observe further that after k steps the volume of K is reduced by at least the factor $(1-1/\mathrm{e})^k$, and since $V(K)$ is bounded above by $V(R\mathbb{B}^n)$ the algorithm terminates after at most $O(n(\log(1/\varepsilon) + \log(R^2/\rho)))$ steps. Hence this method needs only a number of steps that is bounded by a polynomial in n and the logarithms of ε, ρ and R. However, an important drawback is that there does not seem to be any easy way to compute the center of gravity, and as shown by Dyer and Frieze (1988) the problem of computing the volume deterministically even for an integer polytope is #P-hard. (See the book of Garey and Johnson (1979) for information about this notion of hardness, and see Dyer, Frieze and Kannan (1991), Lovász and Simonovits (1990) and the survey Dyer and Frieze (1991) for polynomial-time randomized methods of volume computation.)

Let us now turn to the variations of the central-cut method that led to the ellipsoid algorithm and subsequently to the first polynomial-time algorithm for linear programming. The basic underlying idea of the ellipsoid method for concave maximization is that in each step the set K_k^+ in the above analysis should be replaced by a suitable ellipsoid and the center-of-gravity g_k should be replaced by the center of this ellipsoid.

Algorithm (*The ellipsoid-method for concave maximization*).
Input: A positive integer n, a convex body K in $\mathbb{R}^n$, positive rationals ρ and R such that $\rho \leqslant r_n(K)$ and $K \subset R\mathbb{B}^n$, a vector $c \in \mathbb{R}^n$, a positive real number ε.
Output: A vector v' in K such that $\langle c, v'\rangle + \varepsilon \geqslant \max_{x \in K}\langle c, x\rangle$.
Initialization: Set $k = 0$, $e_0 = 0$, $E_0 = e_0 + R\mathbb{B}^n$.

Iteration: If $e_k \notin K$, let H_k be any hyperplane through e_k such that K is contained in the corresponding closed halfspace H_k^+.
If $e_k \in K$, let $H_k^+ = \{x \in \mathbb{R}^n : \langle c, x\rangle \geqslant \langle c, e_k\rangle\}$.
Set $E_{k+1}^+ = E_k \cap H_k^+$, let E_{k+1} be the smallest ellipsoid that contains E_{k+1}^+ and let e_{k+1} be its center.
If $V(E_{k+1}) \leqslant \min\{\rho^n V(\mathbb{B}^n), (1/n)\varepsilon^n (\rho/2R)^{n-1} V(\mathbb{B}^{n-1})\}$ terminate.
Otherwise repeat the procedure with k replaced by $k+1$.

Note that the condition $V(E_{k+1}) \leqslant \rho^n V(\mathbb{B}^n)$ guarantees that the algorithm finds at least one point of K, no matter how large ε may be.

It turns out that the volumes of the ellipsoids are still decreasing by a factor $\exp(-1/O(n))$ at each step, and this is enough to guarantee, as before, a polynomial bound on the number of steps. We have now reduced the problem to testing membership in K and, when $e_k \notin K$, of finding a separating hyperplane through e_k; this is a so-called *strong separation problem*. Various "oracles" for describing convex bodies have been studied by Grötschel, Lovász and Schrijver (1981, 1988) in their development of the *algorithmic theory of convex bodies*. A basic result is the polynomial-time equivalence of the (weak) optimization and the (weak) separation problem when an upper bound for the circumradius is known.

We have seen earlier that the polytope $P = \{x \in \mathbb{R}^n : Ax \leqslant b\}$ (with integer data in A and b) is contained in $2^L\mathbb{B}^n$, where L is the binary size of the input data. Furthermore, if the polytope P is full-dimensional then it contains a simplex given as the convex hull of $n+1$ affinely independent vertices of P. It is now easy to see that this simplex contains a ball of radius at least $2^{-O(L)}$. Hence, if approximation is required only up to an error of $\varepsilon = 2^{-\pi(L)}$ with a polynomial π (when applied to the polytope P), then the number of steps of the algorithm is bounded by a polynomial in the size L of P.

For a polytope $P = \{x \in \mathbb{R}^n : Ax \leqslant b\}$, membership of a point e_k can be tested easily by evaluating $b - Ae_k$. If all components are nonnegative, $e_k \in P$. Otherwise, at least one of the constraints is violated, say $\langle a, x\rangle \leqslant \beta$. Then the set $H_k = \{x : \langle a, x\rangle = \langle a, e_k\rangle\}$ is a separating hyperplane that can be used in the algorithm. (If $e_k \in P$ we use c instead of a. The following formulas change accordingly.)

The updating formulas for the ellipsoid and its center are as follows. Let B_k be a symmetric positive definite $n \times n$ matrix such that $E_k = \{x : (x - e_k)^{\mathrm{T}} B_k^{-1}(x - e_k) \leqslant 1\}$, then

$$e_{k+1} = e_k - \tau \frac{B_k a}{(a^{\mathrm{T}} B_k a)^{1/2}},$$

$$B_{k+1} = \delta\left(B_k - \sigma \frac{B_k a a^{\mathrm{T}} B_k}{a^{\mathrm{T}} B_k a}\right),$$

where

$$\tau = \frac{1}{n+1}, \qquad \delta = \frac{n^2}{n^2 - 1} \quad \text{and} \quad \sigma = \frac{2}{n+1}.$$

There are still two problems to be dealt with. Since the updating formula for e_{k+1} contains $\sqrt{a^T B_k a}$, we cannot actually compute the ellipsoids and their centers precisely, for we are only allowed to use finite precision. Hence we have to approximate e_k and B_k. It turns out that rounding to $O(L)$ bits works if we slightly enlarge the ellipsoid. The number of iterations increases slightly but not essentially. For details see, e.g., Khachiyan (1980) or Grötschel, Lovász and Schrijver (1981, 1988).

The second question is whether it is possible to assure that the centers e_k of the ellipsoids converge to an optimal vertex v^* of P. Let us assume for a moment that v^* is uniquely determined. Then, again by our lemma, $\langle c, v^* \rangle \geqslant \langle c, v \rangle + 2^{-2nL}$ for any other vertex v of P. Since, on the other hand, $P \subset 2^L \mathbb{B}^n$, the bound $\langle c, v^* - e_k \rangle \leqslant \varepsilon$ implies that $\|v^* - e_k\| \leqslant \varepsilon 2^{2nL} 2^L$. Hence, we can do the necessary rounding when $\varepsilon \leqslant 2^{-(2n+3)L}$. As shown by Grötschel, Lovász and Schrijver (1981), this situation can be obtained by perturbing the objective function by adding again a point $(\delta, \delta^2, \ldots, \delta^n)^T$ of the moment curve, where δ is of size that is bounded by a polynomial in L.

Theorem (The ellipsoid algorithm runs in polynomial time). *Let m, n be positive integers, let A be an integer $m \times n$ matrix, let $b \in \mathbb{Z}^m$, and let $c \in \mathbb{Z}^n$. Then the ellipsoid method can be adapted to solve the linear programming problem* $\max_{\{x: Ax \leqslant b\}} \langle c, x \rangle$ *in polynomial time.*

It turns out that the ellipsoid method has numerous applications to combinatorial optimization problems and it plays an important role in the algorithmic theory of convex bodies developed by Grötschel, Lovász and Schrijver (1988). Also, as was mentioned in section 2, the ellipsoid method yields a polynomial-time algorithm for minimizing a convex quadratic function subject to a finite number of linear constraints. However, for all practical purposes the ellipsoid method is inferior to the simplex method for solving linear programming problems. For surveys and further results on the ellipsoid algorithm see Bland, Goldfarb and Todd (1981), Gácz and Lovász (1981), Goldfarb and Todd (1982, 1989), Schrader (1982, 1983), Khachiyan (1982), Akgül (1984), Goffin (1984), Lovász (1986), Ye (1987), and Grötschel, Lovász and Schrijver (1988).

We have introduced the ellipsoid method as a relaxation of the method of center-of-gravity sections. There is, however, another "predecessor algorithm", the *relaxation method*, studied by Agmon (1954) and by Motzkin and Schoenberg (1954). This method was introduced for deciding whether a system $Ax \leqslant b$ of linear inequalities is feasible. The method starts with an arbitrary point e_0 and checks successively whether e_k is feasible. If not, a violated inequality $\langle a, x \rangle \leqslant \beta$ of the system is chosen and a step is taken in direction of the projection of e_k onto the hyperplane $\{x: \langle a, x \rangle = \beta\}$. More precisely, with a step-length parameter η we choose

$$e_{k+1} = e_k + \eta \frac{\beta - \langle a, e_k \rangle}{\|a\|^2} a.$$

With the choice $\eta = 2$, and provided the polytope $P = \{x: Ax \leqslant b\}$ has interior points, the relaxation method terminates, while for general nonempty P and $0 < \eta < 2$ the algorithm either terminates or generates a sequence that converges linearly, Agmon (1954), Motzkin and Schoenberg (1954), and also Goffin (1980). Todd (1979) and Goffin (1982) showed, however, that the factor that determines the linear convergence rate is too large; for any η there is a sequence of full-dimensional polytopes for which the relaxation method requires exponentially many steps to reach distance 1 from the feasible region. On the positive side, Maurras, Truemper and Akgül (1981) showed that a variant of the relaxation method solves any linear program with a totally unimodular constraint matrix in polynomial time.

To see how the relaxation method is related to the ellipsoid algorithm, let us consider the choice $\eta = 1$; i.e., we project directly onto a violated constraint hyperplane. Let us begin with an arbitrary point e_0 and assume that we have a radius R such that $P \subset e_0 + R\mathbb{B}^n$. (As we have seen before, this assumption can easily be satisfied if the input is integer.) Suppose that the constraint $\langle a, x\rangle \leqslant \beta$ is violated, let $H = \{x: \langle a, x\rangle = \beta\}$, and let $H^+ = \{x: \langle a, x\rangle \leqslant \beta\}$. Then the projection e_1 of e_0 onto the hyperplane H is the center of the smallest ball that contains $H^+ \cap (e_0 + R\mathbb{B}^n)$. This shows that the difference between the relaxation method and the ellipsoid method lies in the application of affine transformations used in the latter to speed up the former. These "space dilatations" were first applied by Shor (1970a,b). A generalization of this approach of minimizing distances was studied by Betke and Gritzmann (1992) and Betke and Henk (1991); see also the survey of Megiddo (1987).

Let us point out that the same principle that works with ellipsoids in the ellipsoid algorithm can also be applied with some other classes of bodies. In particular, Yamnitsky (1982) (see also Yamnitsky and Levin 1982) showed that a polynomial-time analogue of the ellipsoid method can be based on simplices in place of ellipsoids.

The (unrelaxed) general center-of-gravity section method is only of theoretical use, for the center of gravity of a convex body or even a polytope is hard to determine. This led to similar approaches with other notions of centers; the *analytic center* being particularly important. In general, we will use the term *method of central sections* for any method that is similar to the method of center-of-gravity sections but where the center of gravity is replaced by some other notion of center. Since the recent developments in this area are mainly motivated by Karmarkar's *logarithmic barrier function* approach, we will study Karmarkar's projective algorithm first and then come to some other central-section methods in section 9. (For a survey of various centers and their applications, see Kaiser, Morin and Trafalis 1991.) However, let us end this section by describing one more central-section method that uses a geometric center. This one is based on the result of Behrend (1938) (who treated the case $n = 2$), Danzer, Laugwitz and Lenz (1957), and Zaguskin (1958) asserting that for each convex body K in $\mathbb{R}^n$, there is a unique ellipsoid $E(K)$ of maximum volume inscribed in K. Let $z(K)$ denote the center of $E(K)$. Using $z(K)$ for central cuts, Tarasov, Khachiyan and Èrlikh (1988) obtain the following analogue of Grünbaum's theorem for (as we will call them) *Löwner–John sections*.

Theorem (Löwner–John sections decrease the volume quickly). *Let K be a convex body, let H be a hyperplane through $z(K)$ with corresponding halfspaces H^+ and H^-, and let $K^+ = K \cap H^+$ and $K^- = K \cap H^-$. Then*

$$V(K^{\pm}) \leqslant 0.843 \cdot V(K).$$

Tarasov, Khachiyan and Èrlikh (1988) show that for polytopes P, suitable approximations of $E(P)$ and $z(P)$ can be found in polynomial time, and that therefore the cutting plane method based on such approximate Löwner–John sections leads to a polynomial-time algorithm for linear programming.

9. Karmarkar's projective algorithm

In view of the ellipsoid method it is quite natural to ask whether there is a "dual" algorithm for linear programming, using the fact that one can easily maximize linear functionals over ellipsoids. The idea, which is analogous to that of *trust region methods* (see, e.g., Gill, Murray and Wright 1981), can be described as follows.

Algorithm (*A maximization-over-inscribed-ellipsoid approach*).

Input: Positive integers m, n, an integer $m \times n$-matrix A, $b \in \mathbb{Z}^m$, $c \in \mathbb{Z}^n$, such that $P = \{x: Ax \leqslant b\}$ is an n-dimensional polytope, a point $e_0 \in \text{int}(P)$, a positive rational ε.

Output: A possibly finite sequence (e_k) of points in P with strictly increasing objective function values $\langle c, e_k \rangle$.

Initialization: Set $k = 0$.

Iteration: Construct a maximal ellipsoid E_k with center e_k inscribed P;
Let e_{k+1} be such that $\langle c, e_{k+1} \rangle = \max_{x \in E_k} \langle c, x \rangle$;
If $e_{k+1} \in \text{bd}(P)$ terminate. Otherwise repeat the procedure with k replaced by $k+1$.

It is clear that if the above algorithm terminates with a point e_{k_0} then this point is a maximizer over P. Further, it is quite plausible (and it has been shown by Ye and Tse (1989) for a variant of this approach) that the method does converge. On the other hand, it is also quite plausible that this method will not converge very rapidly. One of the basic ideas underlying Karmarkar's polynomial-time algorithm is to apply *projective transformations* to "throw e_k back into the center of P", Karmarkar (1984).

Before we describe how this is actually done, let us try to give an idea of how much progress per step might be expected. The basis for our consideration is the following theorem of John (1948). (A strengthening of this result for symmetric bodies appeared in John (1942), and using classical inequalities this result already implies a slightly weaker form of the following theorem.)

Theorem (The Löwner–John ellipsoid). *For each convex body K in $\mathbb{R}^n$, there exist a point a and a linear transformation α such that*

$$a + \alpha(\mathbb{B}^n) \subset K \subset a + n\alpha(\mathbb{B}^n).$$

The ellipsoid $a+\alpha(\mathbb{B}^n)$ can be taken to be the ellipsoid of maximum volume inscribed in K.

Observe that the dilatation factor n in John's theorem is best possible for the simplex. The same factor n holds also if the ball in the above theorem is replaced by the regular simplex or the cube. Algorithmic approximations of the Löwner–John ellipsoid can be obtained by use of the ellipsoid method (see Grötschel, Lovász and Schrijver 1988), and also by the Löwner–John-section approach of Tarasov, Khachiyan and Èrlikh (1988). In fact, the ellipsoid used in the result called "Löwner–John sections decrease the volume quickly" is the same as the ellipsoid used here.

Let us mention that John also established a theorem of Helly type, asserting that for each convex body K in $\mathbb{R}^n$ there is a subset S of K, of cardinality at most $n(n+3)/2$, such that the ellipsoid of smallest volume containing S already contains K and hence is the smallest ellipsoid that contains K. For special bodies this number can be considerably smaller. Todd (1982) showed that for convex bodies K that are obtained from an ellipsoid by a hyperplane cut, the cardinality $n(n+3)/2$ can be replaced by $2n-1$, and if K is obtained from an ellipsoid by two parallel hyperplane cuts then this number is at most $4n-4$. Gruber (1988), on the other hand, proved that for most convex bodies (in the sense of Baire categories) the boundary of the smallest containing ellipsoid and the boundary of the largest contained ellipsoid each have precisely $n(n+3)/2$ points in common with the boundary of K.

Now, suppose that we had an efficient procedure for finding a and α. Then with

$$\gamma^- = \max_{a+\alpha(\mathbb{B}^n)} \langle c,x\rangle, \qquad \gamma^+ = \max_{a+n\alpha(\mathbb{B}^n)} \langle c,x\rangle \quad \text{and} \quad \mu^* = \max_K \langle c,x\rangle,$$

we have $n(\gamma^- - \langle c,a\rangle) = \gamma^+ - \langle c,a\rangle$, and this implies

$$\begin{aligned}\mu^* - \gamma^- &= (\mu^* - \langle c,a\rangle) - (\gamma^- - \langle c,a\rangle) = (\mu^* - \langle c,a\rangle) - \frac{1}{n}(\gamma^+ - \langle c,a\rangle)\\ &\leqslant \left(1 - \frac{1}{n}\right)(\mu^* - \langle c,a\rangle).\end{aligned}$$

Hence, moving from a to the maximizer a' of $\langle c,x\rangle$ over $a+\alpha(\mathbb{B}^n)$ results in progress by a factor $(1-1/n)$. As was explained in the previous section, progress of this kind (when followed by suitable rounding) yields a polynomial number of steps.

In order to be able to continue this procedure beyond the first step, we would have to apply a transformation τ such that $\tau(a')$ plays the role of a for $\tau(K)$. The class of affine transformations is not rich enough for this purpose, though it does

have the property that our concave maximization problem is invariant under this class. However, the collection of convex bodies is also preserved by the larger class of transformations known as *admissible projective transformations*. These have the disadvantage of not preserving parallelism (i.e., the family $\{\{x\colon \langle c,x\rangle = \gamma\}\colon \gamma \in \mathbb{R}\}$ of parallel hyperplanes is not transformed into some such family by a general projective transformation), but that difficulty can be overcome in a way that we shall describe after first telling how projective mappings can be used for the mapping task.

The key idea is based on the observation that, given a d-dimensional simplex $S = \operatorname{conv}\{s_0,\dots,s_d\}$ and two points $x_1, x_2 \in \operatorname{int}(S)$, there is an admissible projective transformation τ that has $\tau(x_1) = x_2$ and $\tau(s_i) = s_i$ for $i = 0,\dots,d$. Under such a transformation, the simplex is carried onto itself and hence the centers of its largest contained ellipsoid and smallest containing ellipsoid are invariant. (In this case, these ellipsoids are actually concentric balls.) So, if our polytope is a simplex we know how to construct the projective transformation.

Karmarkar transforms the given polytope into the intersection of a regular d-simplex in $\mathbb{R}^{d+1}$ with a linear subspace through the centroid of the simplex. This transformation is reminiscent of a result of Davis (1952, 1953) asserting that every n-polytope P is a section of an $(f_{n-1}(P)-1)$-dimensional simplex, where $f_{n-1}(P)$ denotes the number of facets of P; see also Naumann (1956). In our context the transformation works as follows. Suppose (without loss of generality) that our polyhedron P is given in the form $P = \{y \in \mathbb{R}^n\colon Ay \leqslant b, y \geqslant 0\}$, with an $m\times n$ matrix A. Add slack variables to transform P into the set $Q = \{x \in \mathbb{R}^d\colon Ax = b, x \geqslant 0\}$, where $d = n+m$. As we saw in the preceding section, we may assume that Q is bounded ($Q = Q \cap 2^L\mathbb{B}^d$), and hence after scaling and adding another slack variable ξ we reach the situation in which the set of constraints is of the form $Ax = b, \langle \mathbb{1}, x\rangle + \xi = 1, x, \xi \geqslant 0$. By subtracting suitable multiples of $\langle \mathbb{1}, x\rangle + \xi = 1$ from the constraints of $Ax = b$, we can achieve $b = 0$. Hence we can assume without loss of generality that our polytope is given in the form $S \cap M$ where S is the (lower-dimensional regular) simplex $\{x \in \mathbb{R}^{d+1}\colon \langle \mathbb{1}, x\rangle = 1, x \geqslant 0\}$ and M is the linear subspace $\{x \in \mathbb{R}^{d+1}\colon Ax = 0\}$. Setting $H = \{x \in \mathbb{R}^d\colon \langle \mathbb{1}, x\rangle = 1\}$ we have

$$\frac{1}{d+1}\,\mathbb{1} + \frac{1}{\sqrt{d(d+1)}}\,(\mathbb{B}^{d+1} \cap H) \subset S \subset \frac{1}{d+1}\,\mathbb{1} + \sqrt{\frac{d}{d+1}}\,(\mathbb{B}^{d+1} \cap H).$$

Observe that this transformation (though being convenient for our mapping task) leads to a weaker bound for the possible progress at each step; the ratio is now $1/d = 1/(n+m)$ rather than $1/n$, and hence depends on the number of constraints rather than only on the dimension of the original polyhedron P. Using this transformation and an analysis based on the ratio of the inradius and circumradius of the regular simplex, we lose the inherent potential of proving a result asserting that the number of steps taken by the algorithm (which, admittedly, has not been described yet in full detail) is not strongly dependent on the number of constraints. (If such a result could be proved, it would be very interesting for the use of Karmarkar's algorithm in the algorithmic theory of convex bodies.)

Using the fact that (as was pointed out in section 5) it suffices to solve the feasibility problem, we see that with respect to polynomial-time transformations, the linear programming problem is equivalent to the task of deciding whether there is a point with $\lambda \leqslant 0$ in the polytope

$$Ax - \lambda A\mathbb{1} = 0, \qquad \langle \mathbb{1}, x\rangle + \lambda = 1, \qquad x, \lambda \geqslant 0.$$

Since the centroid of the corresponding simplex (the point $x_0 = 1/(d+1)\mathbb{1}, \lambda_0 = 1/(d+1)$), is feasible, we obtain, after switching back to our earlier notational convention, the following form of the problem which is appropriate for describing Karmarkar's algorithm.

Karmarkar's canonical form. For given positive integers m, n, an integer $m \times n$-matrix A with $A\mathbb{1} = 0$ and a vector $c \in \mathbb{Z}^n$ such that $\langle c, x\rangle \geqslant 0$ for all points in $P = \{x: Ax = 0, \langle \mathbb{1}, x\rangle = 1, x \geqslant 0\}$, minimize $\langle c, x\rangle$ over P, and if the minimum is 0, find a vector $x^* \in P$ with $\langle c, v^*\rangle = 0$.

(For variants of Karmarkar's algorithm that avoid some of the assumptions made in Karmarkar's canonical form see, e.g., Anstreicher 1986, Todd and Burrell 1986, Gay 1987, Ye and Kojima 1987.)

Let us now return to the problem that linear functionals are not invariant under projective transformations. Let D be the diagonal matrix whose diagonal entries are n times the coordinates of a point e' which is relatively interior to the simplex $S = \{x: \langle \mathbb{1}, x\rangle = 1, x \geqslant 0\}$. With $s = (1/n)\mathbb{1}$ we have $Ds = e'$, i.e., D transforms the center of S into the current point e'. Hence the inverse D^{-1} transforms e' into s. The simplex S is mapped onto some other simplex that is the intersection of the positive orthant and a hyperplane with positive normal vector. In order to keep S fixed (as a set) we apply a radial projection onto $H = \{x: \langle \mathbb{1}, x\rangle = 1\}$. This is done by the mapping $1/\langle \mathbb{1}, D^{-1}x\rangle$. Hence the projective transformation that we use and its inverse are given as

$$x \mapsto \frac{D^{-1}x}{\langle \mathbb{1}, D^{-1}x\rangle} \quad \text{and} \quad y \mapsto \frac{Dy}{\langle \mathbb{1}, Dy\rangle}.$$

This transforms our original problem to the problem

$$\min \frac{\langle c, Dy\rangle}{\langle \mathbb{1}, Dy\rangle} \quad \text{over } \{y: ADy = 0, y \in S\}.$$

(Note that this problem is invariant under multiplication of D by a positive factor; and we will, indeed, use $(1/n)D$ in the description of Karmarkar's projective algorithm given below.) If our original problem has optimum 0 then so has this new problem, which means the numerator of the new fractional objective function has to have minimum 0. Therefore we take $\langle c, Dy\rangle$ as a linear approximation of the fractional function and consider the problem

$$\min\langle c, Dy\rangle \quad \text{over } \{y: ADy = 0, y \in S\}.$$

Since this linear programming problem is again in Karmarkar's canonical form, the procedure can now be repeated.

Let us point out that in order to have a close enough approximation of the fractional by the linear objective function, the insphere of S is actually replaced by a smaller sphere; a factor $\frac{1}{3}$ is enough.

In order to carry out the analysis that leads to the desired estimate for the progress at each step, Karmarkar introduced the "potential function"

$$f(x) = n \ln\langle c, x\rangle - \sum_{i=1}^{n} \ln \xi_i.$$

It is shown that this function is reduced by an additive constant at each step and that this implies that (upon feasibility) the objective function value is reduced at a linear rate with small enough constant to warrant polynomiality. [The algorithm does not have a superlinear convergence, Bayer and Lagerias (1991); but see also Kovacevic-Vujcic (1991). However, a quadratically convergent interior-point algorithm for linear programming has been produced by Ye et al. (1991).] Summarizing the above description, Karmarkar's algorithm works as follows.

Algorithm (*Karmarkar's projective algorithm*).

Input: Positive integers m, n, an integer $m \times n$-matrix A with $A\mathbb{1} = 0$ and a vector $c \in \mathbb{Z}^n$ such that $\langle c, x\rangle \geqslant 0$ for all points in $P = \{x: Ax = 0, \langle \mathbb{1}, x\rangle = 1, x \geqslant 0\}$.

Output: A point $x^* \in P$ such that $\langle c, v^*\rangle = 0$ if such a point exists; a message "$\min_{x \in P}\langle c, x\rangle > 0$" otherwise.

Initialization: $k = 0$, $s = e_k = \frac{1}{n}\mathbb{1}$, $D_k = I_n$, $\eta = (3\sqrt{(n-1)n})^{-1}$.

Iteration: If $\langle c, e_k\rangle \leqslant 2^{-2L}\langle c, s\rangle$ terminate; the minimum of the objective function is 0; x^* can be obtained from e_k by a suitable rounding procedure.
Compute the minimizer e_{k+1} of $\langle c, D_k x\rangle$ over $\{x \in s + \eta\mathbb{B}^n: \langle \mathbb{1}, x\rangle = 1, AD_k x = 0\}$.
Let D_{k+1} be the diagonal matrix with diagonal entries $D_k e_{k+1} \times (\langle \mathbb{1}, D_k e_{k+1}\rangle)^{-1}$.
If $f(e_{k+1}) > f(e_k) - \frac{1}{5}$ terminate; the minimum of the objective function is greater than 0.
Otherwise, repeat the procedure with k replaced by $k+1$.

Let us first clarify how the minimization of $\langle c, Dx\rangle$ over $\{x \in e + \eta\mathbb{B}^n: \langle \mathbb{1}, x\rangle = 1, ADx = 0\}$ is carried out. Clearly, $s - \eta Dc/\|Dc\|$ is the minimizer of $\langle c, Dx\rangle$ over $s + \eta\mathbb{B}^n$. So, all we have to do is to project Dc onto the nullspace of the matrix $B = \begin{pmatrix} AD \\ \mathbb{1}^{\mathrm{T}} \end{pmatrix}$. That produces the vector

$$\hat{x} = (I_n - B^{\mathrm{T}}(BB^{\mathrm{T}})^{-1}B)Dc$$

and hence we obtain the desired minimizer in the form

$$s - \eta \frac{\hat{x}}{\|\hat{x}\|}.$$

(Observe that $\hat{x} = 0$ only if the objective function is constant on the set of feasible points.) This minimization step is in fact the part of the algorithm that is most time-consuming. However, Karmarkar (1984) showed that this step can be modified in such a way that the complexity is reduced from $O(n^3L)$ to an average $O(n^{5/2}L)$.

As is already indicated in the statement of the algorithm, in case the minimum is 0 Karmarkar's method generates a sequence $(e_k)_{k\in\mathbb{N}_0}$ of points in the relative interior of the feasible region that satisfy

$$f(e_{k+1}) \leqslant f(e_k) - \tfrac{1}{5};$$

Karmarkar (1984), see also Goldfarb and Todd (1989). Since $\sum_{i=1}^{n} \ln(\xi_i)$ attains its maximum over S in e_0 this implies that

$$\langle c, e_k\rangle \leqslant \mathrm{e}^{-k/(5n)}\langle c, e_0\rangle.$$

Hence, after $O(nL)$ iterations we are close enough to obtain an optimizer by rounding the current iterate to the nearest rational vector with numerators and denominators bounded above by 2^L.

Summarizing the results stated above we obtain the following theorem.

Theorem (Karmarkar's projective algorithm runs in polynomial time). *Karmarkar's algorithm can be adapted to solve the general linear programming problem in* $O(nL)$ *steps, where the average step-complexity is* $O(n^{5/2}L)$. *The required precision is* $O(L)$.

Let us add some further remarks on Karmarkar's potential function

$$f(x) = n\ln\langle c, x\rangle - \sum_{i=1}^{n} \ln \xi_i.$$

The first term reflects the fact that a decrease of $\langle c, x\rangle$ should be related to a decrease in f. The second term, however, is a *barrier function*. In fact, if one of the coordinates approaches 0 the second term tends to infinity. A *logarithmic* barrier function of the kind that is used here is quite popular in the area of *constrained nonlinear optimization*. It is usually attributed to Frisch (1955). See Fiacco and McCormick (1968), Gill, Murray and Wright (1981) and the recent survey of constrained nonlinear optimization by Gill et al. (1989). As Gill et al. (1986) showed, Karmarkar's algorithm can indeed be interpreted as a projected Newton barrier method; see also Bayer and Lagarias (1991). In a different direction, Stone and Tovey (1991) have developed a geometric approach according to which both Karmarkar's projective scaling algorithm and Dantzig's simplex algorithm may be regarded as "iteratively reweighted least squares methods". Tseng (1989) has produced a variant of Karmarkar's method that admits an unusually simple complexity analysis. For a survey of different search directions that have been used by various authors in interior-point methods, see Den Hertog and Roos (1991).

We began this section with the question whether there is a "dual" to the ellipsoid method that is based on optimizing linear functionals over ellipsoids. It turns out, in fact, that Karmarkar's potential function reflects this point of view quite neatly. As Ye (1987) has shown (based on earlier work of Todd and Burrell 1986 and Todd 1988), the potential function is the logarithm of the volume of an ellipsoid in the space of slack variables that contains all optimal dual solutions.

Let us close this section with some remarks on an affine variant of Karmarkar's algorithm, the *affine scaling method*. The idea is very similar to Karmarkar's projective method; the main difference is that one does not apply any radial projection. The method is usually described for problems in the form $\max_{x \in P} \langle c, x \rangle$, where $P = \{x: Ax = b, x \geqslant 0\}$. Let $e' \in \text{relint}(P)$ and let D denote the diagonal matrix with diagonal entries e'. Then the affine transformation

$$x \mapsto D^{-1}x$$

maps e' into $\mathbb{1}$, the point that is equidistant from all nonnegativity constraints. This mapping transforms the given linear program into

$$\max\langle c, Dx \rangle \quad \text{over } \{x: ADx = b, x \geqslant 0\}.$$

Then we take again a step in the projected gradient direction $(I_n - DA^{\mathrm{T}}(AD^2 \times A^{\mathrm{T}})^{-1}AD)Dc$.

This affine scaling method was developed by Dikin (1967, 1974) long before Karmarkar's algorithm. However, Dikin's algorithm and his convergence analysis were not well known and the method was rediscovered (as a natural modification of Karmarkar's projective algorithm) by various people including Barnes (1986) and Vanderbei, Meketon and Freeman (1986). Some new convergence results can be found in Tsuchiya (1991), while Ye (1991b) gives a probabalistic analysis. No polynomial bound is known for this algorithm (and in view of the results of Megiddo and Shub (1989) the method is not very likely to run in polynomial time); Gonzaga (1990) and Ye (1991a) showed, however, that using the above affine transformations but the projected gradient of a potential function rather than of the linear objective functional, one obtains a polynomial-time algorithm. Ye's method runs in $O(\sqrt{n}L)$ steps.

10. Analytic center, path-following methods

As we have seen earlier, the main drawbacks of the method of center-of-gravity sections are the algorithmic difficulties that arise when trying to compute or approximate the center of gravity. The ellipsoid method, a relaxation of the center-of-gravity section method, provided one way of circumventing these difficulties and led to the first polynomial-time algorithm for linear programming. As we shall now describe, another way of circumventing the difficulties leads to a different class of polynomial-time algorithms.

Let us begin with the general notion of a *center* of a convex body K. Intuitively, a center z should be an interior point of K, uniquely determined, which maximizes the "distance" (in some sense of that notion) to K's boundary. Let Δ be a function defined on the set of convex bodies in $\mathbb{R}^n$ that maps a convex body K to a (possibly extended) real valued functional $\delta_K(x)$ defined on K. Thinking of this functional as the "distance" in question, we call Δ a *center function* if it has the following properties for arbitrary convex bodies K of $\mathbb{R}^n$:

$$\delta_K(x') = \min_{x \in K} \delta_K(x) \quad \text{for } x' \in \mathrm{bd}(K),$$

$$\delta_K(x') > \min_{x \in K} \delta_K(x) \quad \text{for } x' \in \mathrm{int}(K),$$

$$\delta_K(x) \geqslant \delta_{K'}(x) \quad \text{for convex bodies } K' \subset K \text{ and all } x \in K',$$

$$\max_{x \in K} \delta_K(x) \text{ is attained for a unique point } x^*.$$

Given a center function, the functional δ_K is called *boundary-distance function* of K; the point x^* is called *center* of K with respect to Δ.

General centers have been considered by Huard (1967) who used them for a central section method similar to the method of center-of-gravity sections.

Algorithm (*Huard's methods of centers with respect to* Δ).
Input: A positive integer n, a convex body K in $\mathbb{R}^n$, a vector $c \in \mathbb{R}^n$.
Output: A sequence $(g_k)_{k \in \mathbb{N}_0}$ such that $\langle c, g_{k+1} \rangle \geqslant \langle c, g_k \rangle$.
Initialization: Set $k = 0$, $K_0^+ = K$.
Iteration: Compute the maximizer g_k of $\delta_{K_k^+}$.
Set $H_k^+ = \{x \in \mathbb{R}^n \colon \langle c, x \rangle \geqslant \langle c, g_k \rangle\}$, $K_{k+1}^+ = K_k^+ \cap H_k^+$.
Repeat with k replaced by $k+1$.

Now, suppose we are given a polytope $P = \{x \colon Ax \leqslant b\}$ with nonempty interior. A reasonable boundary-distance function would be the functional

$$\delta_P(x) = \prod_{j=1}^{m} \frac{\beta_j - \langle a_j, x \rangle}{\|a_j\|^*}.$$

However, maximization of this functional over P is a constrained optimization problem. Hence, it might be preferable to use the logarithm of δ_P and consider the functional

$$f_P(x) = \ln \left(\prod_{j=1}^{m} \frac{\beta_j - \langle a_j, x \rangle}{\|a_j\|^*} \right) = \left(\sum_{j=1}^{m} \ln(\beta_j - \langle a_j, x \rangle) \right) - \left(\sum_{j=1}^{m} \ln(\|a_j\|^*) \right).$$

In this way we may use any of the standard algorithms for unconstrained maximization of f_P with an interior starting point since we will not leave P. It turns out, indeed, that this *logarithmic barrier function* is strictly concave on P, whence the

maximum exists and is attained by a unique maximizer called the *analytic center* of P. The functionals δ_P, f_P or their variants

$$\delta'_P(x) = \prod_{j=1}^{m} (\beta_j - \langle a_j, x\rangle), \qquad f'_P(x) = \sum_{j=1}^{m} \ln(\beta_j - \langle a_j, x\rangle)$$

are often referred to as *potential functions*. The notion of an analytic center was first introduced by Sonnevend (1986, 1987, 1988).

Observe that, unlike the center of gravity, the analytic center depends on the *presentation* of the polytope rather than on the polytope itself. In particular, discarding redundant constraints does not change the center of gravity but does change the analytic center. However, the analytic center is an analytic function of the input data; and it is affinely invariant in the following sense: let $\mathcal{I}$ denote the set $\langle a_i, x\rangle \leqslant \beta_i$ $(i = 1, \ldots, n)$ of constraints, and let $c_{\mathcal{I}}$ be the corresponding analytic center. If $\Gamma: \mathbb{R}^n \to \mathbb{R}^n$ is an invertible affine transformation, and if $\mathcal{I}_\Gamma$ denotes the transformed set of constraints $\langle a_i, \Gamma^{-1} y\rangle \leqslant \beta_i$ $(i = 1, \ldots, n)$, then

$$c_{\mathcal{I}_\Gamma} = \Gamma(c_{\mathcal{I}}).$$

Futhermore, a theorem of Löwner–John-type holds for ellipsoids centered at the analytic center with a dilatation factor $m - 1$ rather than n, Sonnevend (1986); see also Vaidya (1987), Sonnevend and Stoer (1989), Ye and Todd (1990) and Jarre (1991).

For algorithms that approximate the analytic center of a polytope see Renegar (1988), Vaidya (1987), and the survey of Gonzaga (1992).

Goldstein and Ye (1990) showed that with respect to the potential function, the analytic center has a property similar to that of the center of gravity with respect to the volume (as stated in Grünbaum's theorem).

Theorem (Analytic center sections decrease the potential function quickly). *Let $P = \{x: Ax \leqslant b\}$ be a polytope with nonempty interior, let z denote P's analytic center, let $a \in \mathbb{S}^{n-1}$, let $P' = P \cap \{x: \langle a, x\rangle \leqslant \langle a, z\rangle\}$ and let z' denote the analytic center of P'. Then*

$$f_{P'}(z') \leqslant f_P(z) + \ln(0.84 \cdot \omega),$$

where ω is the a-breadth of the ellipsoid $\{x: \|DAx\| \leqslant 1\}$ with D being the diagonal matrix with diagonal entries $1/(\beta_j - \langle a_j, z\rangle)$.

Note that P contains the translate of the above-defined ellipsoid that is centered at z. Thus the problem can easily be normalized so as to have $\omega \leqslant 1$, and hence the above estimate leads to a fixed decrease of the potential function after the cut.

This result can be used to derive polynomial-time variants of Huard's method of centers. Goldstein and Ye (1990) use it as a key to the analysis of an algorithm that they devise for solving the feasibility problem for linear programming. The

algorithm uses increasing subsets of the constraints with a relaxation of the right-hand side, computes the analytic center, and then uses cuts of the current polytope by a constraint hyperplane moved into the center.

The relevance of analytic centers is underlined by the study of continuous versions of Karmarkar's algorithm. The general idea of studying trajectories associated with discrete algorithms has been pursued for some time (see, e.g., Garcia and Zangwill 1981). In the context of interior-point methods for linear programming, it has been carried out by several researchers, including Megiddo and Shub (1989), Bayer and Lagarias (1989a,b), Megiddo (1989) and Lagarias (1990).

In Bayer and Lagarias (1989a,b) and Lagarias (1990) the "nonlinear geometry of linear programming" underlying Karmarkar's projective algorithm and the affine scaling algorithm is studied. With the projective algorithm there is associated a vector field on the feasible region that specifies the direction taken by the algorithm when it reaches the corresponding points – called by Bayer and Lagarias the *projective scaling vector field* $\vec{v}(x)$. A *projective scaling trajectory* or *P-trajectory* is the point set covered by a solution to the differential equation

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \vec{v}(x), \quad x(0) = x_0$$

extended as far as possible. Bayer and Lagarias show that these P-trajectories are parts of real algebraic curves and that there is a metric defined on the relative interior of the polytope such that the P-trajectories are geodesics for the geometry induced by this metric. The study of the affine scaling algorithm leads to similar results for *A-trajectories*. It is shown that the sets of P-trajectories and A-trajectories for a given linear program have one trajectory in common, the *central trajectory*, which passes through the *analytic center* of the polytope. The central trajectory can be characterized as the trajectory of a parametrized family of logarithmic barrier function problems of the form

$$\max\left(\langle c,x\rangle + \mu \sum_{i=1}^{n} \ln(\xi_i)\right) \text{ over } Ax = b, \quad x > 0,$$

as studied by Megiddo (1989). Bayer and Lagarias (1989b) show that for polytopes given in the form $\{x: Ax \leqslant b\}$, the central trajectory is the locus of the analytic centers of the family of polytopes obtained from the original polytope by adding the parametrized objective value constraint $\langle c,x\rangle \geqslant \mu$.

Any algorithm for the linear programming problem that generates a sequence of points that approximate the central trajectory is called a *path-following method.* Path-following algorithms for linear (and convex) programs have been studied by many authors, among whom are Sonnevend (1986), Renegar (1988), Vaidya (1987), Roos and Vial (1988), Jarre (1989, 1991), Nesterov and Nemirovsky (1989), Kojima, Mizuno and Yoshise (1989a,b), Monteiro and Adler (1989a), Gonzaga (1989, 1990), Jarre, Sonnevend and Stoer (1990), Mehrotra and Sun (1991a,b); see Gonzaga (1992) for a survey of such methods.

Renegar's procedure, which follows the central path approximately by using *Newton's method*, terminates after at most $O(\sqrt{m}L)$ steps. Vaidya (1990) gave a variant of this approach that terminates again after $O(\sqrt{m}L)$ steps and that requires a total of at most $O(m^3L)$ arithmetic operations, a complexity result that was also obtained by Gonzaga (1989) who uses a single projected Newton step to follow the central path. As was shown by Sonnevend, Stoer and Zhao (1990, 1991) and Zhao and Stoer (1992), there exists a global bound on the total curvature of the central path that allows the implementation of efficient *predictor–corrector* path-following methods. Sonnevend, Stoer and Zhao express the total number of steps of their first-order extrapolation algorithms as an integral of a local "weighted curvature" of the path. This integral representation can be used to give improved complexity bounds for large classes of problems.

For a unified treatment of central path algorithms that combines aspects of methods of centers, of penalty functions, of potential reduction and of primal–dual methods, see Gonzaga (1992).

11. Strongly polynomial algorithms

All known polynomial-time algorithms for linear programming share the drawback that the number of steps depends on the size of the input. This means that it is not known whether the linear programming problem can be solved in *strongly polynomial time*, i.e., in polynomial time with respect to the binary model but in addition with the number of *elementary arithmetic operations* (addition, subtraction, multiplication, division, comparison, performed in the real model of computations) being bounded by a polynomial in m and n.

Megiddo (1983a) proved that linear programs with n variables and m constraints where the objective function and each constraint contains at most two nonzero coefficients can be solved in $O(mn^3 \log(m))$ elementary operations. Cohen and Megiddo (1991) improved on this and showed that such a system of inequalities can be solved in a total number of $O(mn^2 \log m + n^3 \log n + mn^2 \log^2 n)$ arithmetic operations. Tardos (1985, 1986) showed that every polynomial algorithm for linear programming can be modified to give an algorithm that solves linear programming problems $\max_{\{x: Ax \leqslant b\}} \langle c, x \rangle$ in a number of elementary arithmetic operations that is bounded by a polynomial only in the size of the constraint matrix A. The arithmetic operations work on numbers of a size that is polynomially bounded in L, the size of A, b and c. This means in particular that there exists a strongly polynomial algorithm for rational linear programming problems with a constraint matrix whose entries belong to $\{-1, 0, 1\}$. This includes network-flow problems.

Megiddo (1991) shows that there is a strongly polynomial algorithm for linear programming if there is a strongly polynomial algorithm for the following problem: Given an optimal solution for the linear program, find a subset I of $\{1, \ldots, m\}$ of cardinality n such that the solution of $A_{I_v} x = b_I$ is feasible and such that the solution y' of $A_{I_v}^{\mathrm{T}} y' = c$ is nonnegative. On the other hand, such a *basis* I that is both primal-optimal and dual-optimal can be found in strongly polynomial time

if any pair consisting of a primal-optimal solution and a dual-optimal solution is given, Megiddo (1991).

Let us finally mention a result in fixed dimensions. Extending earlier work of Megiddo (1983b) and Dyer (1984), Megiddo (1984) showed that for each fixed n there is an algorithm for solving linear programming problems in n variables and with m constraints in a total of $O(m)$ arithmetic operations performed on numbers of size that is bounded by a polynomial in the size of the input.

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CHAPTER 2.8

Convexity and Discrete Optimization

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Contents

1. Introduction 677
2. Convex, mixed-integer programming 678
3. Feasibility and computational complexity of integer programs 681
4. Integral polyhedra 682
5. Polyhedral combinatorics 684
6. Combinatorial optimization problems arising from convex geometric configurations 692
References 694

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1. Introduction

In discrete optimization problems functions in real variables are maximized or minimized where some or all of the variables are supposed to take only integer values. We distinguish two main problems: mixed-integer programming problems and combinatorial optimization problems.

A mixed-integer programming problem may be defined as follows: let F and f_i $(1 \leqslant i \leqslant m)$ be continuous functions defined on $\mathbb{R}^d \times \mathbb{R}^{d'}$. We want to find a point (x^*, y^*) such that

$$\begin{aligned} &F(x^*, y^*) = \min F(x, y) \qquad \text{(MIP)}\\ &\text{subject to} \quad f_i(x, y) \leqslant 0 \quad \text{for } 1 \leqslant i \leqslant m\,,\\ &\qquad\qquad x \text{ integer}\,, \quad y \text{ real}\,, \end{aligned}$$

or we have to show that this optimization problem has no solution. The most important special case is (mixed-) integer linear programming, where all involved functions F and f_i are affine-linear and x, y are nonnegative. When we assume that F and f_i $(1 \leqslant i \leqslant m)$ are convex, or more general quasiconvex yielding convex level sets, (MIP) can still be solved quite similar to the linear case. In the next section we shall discuss branch and bound methods, cutting plane methods and Benders' decomposition as basic approaches for solving integer programming problems.

The feasibility of an integer linear program is equivalent with the existence of a solution of a linear diophantine inequality system. Thus these questions are strongly connected with questions of the geometry of numbers and lattice points, see chapters 3.1 and 3.2 in this Handbook. We discuss some of these questions from the viewpoint of integer programming in section 3.

In section 4 we deal with integral polyhedra which contain an integral point in each of its nonempty faces. The importance of integral polyhedra lies in the fact that an integer linear program whose set of feasible points forms an integral polyhedron can be solved via linear programming.

Combinatorial optimization problems are also special cases of (MIP). But often one can take advantage of their special structure which leads to another way of describing them. Let M be a nonempty finite set and let $\mathcal{S}$ be a nonempty family of subsets of M. Any $S \in \mathcal{S}$ is called a feasible solution. Given an objective function $F: 2^M \to \mathbb{R}$ the combinatorial optimization problem asks for a feasible solution with least cost. Again, the case of a linear objective function is most important. In this case we attach cost c_j to any element $j \in M$ and define:

$$F(S) := \sum \{c_j\colon j \in S\} \quad \text{for all } S \in \mathcal{S}\,.$$

Thus the combinatorial optimization problem can be written in the form

$$\min\{F(S)\colon S \in \mathcal{S}\}\,.$$

By using binary variables x_j with $x_j = 1$ for $j \in S$ and $x_j = 0$ for $j \notin S$, a combinatorial optimization problem can often be converted into an integer program with 0–1 variables.

There are two main issues where convexity plays a role in combinatorial optimization. The first is the convexification of the set of all feasible solutions in $\mathbb{R}^d$. This approach has widely been used and enables one to find good bounds on the optimal objective function value or even to generate an optimal solution. This area, known as *polyhedral combinatorics*, will be discussed in section 5.

Finally, the data of a combinatorial optimization problem may arise from convex geometric configurations. In such instances an optimal solution can often be found faster than in the general case by using methods from computational geometry. This is a rather new, but promising aspect which will be dealt with in section 6.

2. Convex, mixed-integer programming

When in the early days of mathematical programming algorithms were developed for convex programs and integer programs, several authors observed that these methods could be combined for solving mixed-integer problems with convex constraints.

A convex, mixed-integer problem can be stated as follows: let F and f_i $(1 \leq i \leq m)$ be convex functions from $\mathbb{R}^d \times \mathbb{R}^{d'}$ to $\mathbb{R}$. We want to minimize the *objective function* F subject to the *constraints*

$$\begin{aligned} & f_i(\boldsymbol{x}, \boldsymbol{y}) \leq 0\,, \quad 1 \leq i \leq m\,, \\ & \mathbf{0} \leq (\boldsymbol{x}, \boldsymbol{y}) \leq \boldsymbol{b}\,, \\ & \boldsymbol{x} \text{ integer}\,, \quad \boldsymbol{y} \text{ real}\,. \end{aligned} \tag{2.1}$$

W.l.o.g. we can deal with a linear objective function subject to convex constraints by introducing an additional variable z and replacing inf $F(\boldsymbol{x}, \boldsymbol{y})$ by inf z subject to $F(\boldsymbol{x}, \boldsymbol{y}) - z \leq 0$.

There are two main possibilities for attacking a convex, mixed-integer problem, namely *branch and bound methods* and *cutting plane methods*. A further possibility is to decompose the problem into an integer and in a real part. This approach is known as *Benders' decomposition*.

In branch and bound methods (see Dakin 1965, Burkard 1972, Burkard and Enge 1975) one proceeds as follows. We need a subroutine which solves a convex problem without integrality constraints. We first apply this subroutine to the given problem and get either the response that a minimum does not exist or we get a minimizer $(\boldsymbol{x}^0, \boldsymbol{y}^0)$. In the first case the set of feasible solutions is empty and therefore the given problem has no solution either. If in the second case $\boldsymbol{x}^0$ is already integral, the minimizer is also feasible for the given problem and we terminate. Otherwise we choose a nonintegral component of $\boldsymbol{x}^0$, say x_1^0 and create

two new problems: in one problem we add the constraint $x_1 \leq [x_1^0]$, in the other problem we add the constraint $x_1 \geq [x_1^0] + 1$. Thus we partition the set which contains all feasible solutions ("*branching*"), but exclude the current solution. With the subproblems we proceed in a similar way. To accelerate the procedure a *bound* can be computed for any subproblem, e.g., by solving the corresponding convex program. If this lower bound for all feasible solutions of a subproblem is at least as large as an already known value of a feasible solution, then the whole subproblem can be deleted from further investigations, since it cannot contain the optimal solution.

The procedure terminates after all subproblems have been fathomed. This means that for every subproblem either the best feasible solution has been determined or it has been shown that the subproblem cannot contain the global optimum.

Instead of creating the subproblems as above, we could *fix* the variable x_1 to $[x_1^0]$ in the first problem and to $[x_1^0] + 1$ in the second problem. Using this strategy one has to create in the next branching step three subproblems, namely two subproblems for fixing a new variable and one subproblem for fixing x_1 to a new (the next) value. That this approach works is due to the convexity of the level sets (in coordinate direction). In particular we can conclude that no feasible solution with $x_1 \leq [x_1^0]$ exists, if we do not get a feasible point for the problem with x_1 fixed to $[x_1^0]$. A similar remark applies to the second problem. Problems with fixed variables are easier to solve, since they are lower dimensional.

This approach makes essential use of the convexity of level sets, but not of the global convexity. Thus we could also deal with problems having a quasiconvex objective function provided there is a procedure for solving the occurring continuous subproblems.

The idea behind *cutting plane methods* (Kelley 1960, see also Burkard 1972, Fabian 1976, and Richter 1976) is the following. For the sake of simplicity let us assume that the involved convex functions F and f_i have bounded partial derivatives in all points of the rectangle $\mathbf{0} \leq (x, y) \leq b$. (This assumption could be weakened by using subgradients instead of gradients.) Moreover we can assume w.l.o.g. that we deal with a linear objective function. First, a (mixed-integer) linear program subject to the constraints $\mathbf{0} \leq (x, y) \leq b$ is solved. If this solution (x^0, y^0) is feasible for the original problem, then an optimal solution for the given problem has been found and we terminate. Otherwise, one of the constraints $f_i(x, y) \leq 0$ is violated, say f_1. We generate a *cutting plane*

$$\langle (x, y), \operatorname{grad} f_1(x^0, y^0) \rangle = \langle (x^0, y^0), \operatorname{grad} f_1(x^0, y^0) \rangle - f_1(x^0, y^0) . \quad (2.2)$$

This hyperplane cuts the point (x^0, y^0) from the feasible region, but all feasible points of (2.1) satisfy the *cut*

$$\langle (x, y), \operatorname{grad} f_1(x^0, y^0) \rangle \leq \langle (x^0, y^0), \operatorname{grad} f_1(x^0, y^0) \rangle - f_1(x^0, y^0) . \quad (2.3)$$

We add the inequality (2.3) to the constraints and resolve the linear program.

Thus a sequence of nested convex polyhedra is constructed. Kelley (1960) has shown that the optimal solution of the given problem is an accumulation point for the sequence of optimal solutions of the subproblems. In the pure integral case we get after a finite number of steps either an integer-valued optimal solution or have shown that the problem is infeasible.

In *Benders' decomposition approach* (Benders 1962) the problem is split into a pure integer and in a continuous part. To explain this method, let us consider a mixed-integer linear program of the form

$$\begin{aligned} &\max \quad \langle \boldsymbol{c}, \boldsymbol{x} \rangle + \langle \boldsymbol{d}, \boldsymbol{y} \rangle \\ &\text{subject to} \quad A_1 \boldsymbol{x} + A_2 \boldsymbol{y} \leqslant \boldsymbol{b}\,, \\ &\qquad\qquad \boldsymbol{x} \text{ integer}\,, \quad \boldsymbol{y} \geqslant \boldsymbol{0} \text{ real}\,. \end{aligned} \tag{2.4}$$

For fixed $\boldsymbol{x}$ we get

$$\begin{aligned} &\max \quad \langle \boldsymbol{d}, \boldsymbol{y} \rangle \\ &\text{subject to} \quad A_2 \boldsymbol{y} \leqslant \boldsymbol{b} - A_1 \boldsymbol{x}\,, \quad \boldsymbol{y} \geqslant \boldsymbol{0}\,, \end{aligned} \tag{2.5}$$

and as its linear programming dual

$$\begin{aligned} &\min \quad \langle \boldsymbol{u}, \boldsymbol{b} - A_1 \boldsymbol{x} \rangle \\ &\text{subject to} \quad A_2^{\mathrm{T}} \boldsymbol{u} \geqslant \boldsymbol{d}\,, \quad \boldsymbol{u} \geqslant \boldsymbol{0}\,. \end{aligned} \tag{2.6}$$

Now, the constraint set $M = \{\boldsymbol{u}\colon A_2^{\mathrm{T}} \boldsymbol{u} \geqslant \boldsymbol{d},\ \boldsymbol{u} \geqslant \boldsymbol{0}\}$ of the dual (2.6) does not depend on the primarily chosen $\boldsymbol{x}$. If it is empty, then the original problem (2.4) has no finite optimum. Otherwise, the minimum of $\langle \boldsymbol{u}, \boldsymbol{b} - A_1 \boldsymbol{x} \rangle$ is taken in an extreme point of the convex set M. Let $\boldsymbol{u}^k$, $k = 1, \ldots, K$ be the extreme points of M. Then the problem (2.4) is equivalent with

$$\begin{aligned} &\max \quad z \\ &\text{subject to} \quad z \leqslant \langle \boldsymbol{c}, \boldsymbol{x} \rangle + \min_{1 \leqslant k \leqslant K} \langle \boldsymbol{u}^k, \boldsymbol{b} - A_1 \boldsymbol{x} \rangle\,, \\ &\qquad\qquad \boldsymbol{x} \geqslant \boldsymbol{0} \text{ integer}\,. \end{aligned} \tag{2.7}$$

This equivalence is used in the following algorithmic scheme: starting from an arbitrary vertex $\boldsymbol{u}^1$ of M, the pure integer program

$$\begin{aligned} &\max \quad z \\ &\text{subject to} \quad z \leqslant \langle \boldsymbol{c}, \boldsymbol{x} \rangle + \langle \boldsymbol{u}^1, \boldsymbol{b} - A_1 \boldsymbol{x} \rangle\,, \\ &\qquad\qquad \boldsymbol{x} \geqslant \boldsymbol{0} \text{ integer}\,, \end{aligned} \tag{2.8}$$

is solved and first values $\boldsymbol{x}^1, z^1$ are obtained. Now the linear program (2.5) is

solved for $x = x^1$ which yields a new extreme point u^2. If and only if

$$z^1 = \langle c, x^1 \rangle + \langle u^2, b - A_1 x^1 \rangle ,$$

then x^1 is an optimal solution of (2.4) with optimum value z^1. Otherwise, the *cut*

$$z \leqslant \langle c, x \rangle + \langle u^2, b - A_1 x \rangle$$

is added to the constraint set of (2.8) and a new iteration is started with solving (2.8) again.

There are several further approaches which deal with (mixed-)integer programs involving convex functions. Benders' decomposition approach can be generalized to mixed-integer convex programs, see Burkard, Hamacher and Tind (1985). Künzi and Oettli (1963) developed a method for solving quadratic integer programs with convex objective function by approximating the level sets by polygons and solving mixed-integer linear subproblems. Quite a bit of work has been done on 0–1 *maximization* of quadratic forms. Gritzmann and Klee (1989) show that the problem of maximizing a positive definite quadratic function in 0–1 variables is polynomially equivalent to computing the width of a simplex which is known to be NP-complete. On the other hand, Hansen and Simeone (1986) described a class of quadratic 0–1 problems which can be converted into network flow problems and therefore be solved in polynomial time.

3. Feasibility and computational complexity of integer programs

The feasibility of an integer linear program

$$\max\{\langle c, x \rangle : Ax \leqslant b, x \geqslant 0\} \tag{IP}$$

is equivalent with the solvability of the system of linear diophantine equations $Ax \leqslant b$, $x \geqslant 0$. Thus questions of feasibility are related with the geometry of numbers. For an account of these questions see Lovász (1989). Of particular interest is the question which computational effort is needed to decide whether an integer program has a feasible solution or not. A first result in this direction is the following theorem, see, e.g., Nemhauser and Wolsey (1988, Theorem I.5.4.1).

Theorem 3.1. *Let A be an integer $(m \times n)$-matrix and let $b \in \mathbb{Z}^m$ be an integral vector. Further let $\theta := \max\{|a_{ij}|, |b_i|\}$ and $\tilde{n} := \max(m, n)$. If $Ax \leqslant b$, $x \geqslant 0$ admits a feasible integral solution, then there is also a solution $x = (x_j)$ of this linear diophantine system with*

$$0 \leqslant x_j \leqslant [2\,\tilde{n}^2\theta]^{\tilde{n}} .$$

This means that it can be decided, whether the integer program $\max\langle c, x \rangle$

subject to $Ax \leq b$, $x \geq 0$ is infeasible, has a finite solution or is unbounded by just investigating the points $x = (x_j)$ with $0 \leq x_j \leq [2\tilde{n}^2\theta]^{\tilde{n}}$. Therefore, (IP) lies in the complexity class NP and can be solved by enumeration.

Bank and Mandel (1988) and Bank et al. (1990) showed that this remains true, even if the constraint set is described by (quasi-)convex polynomials with integer-valued coefficients. Let L be an upper bound for the binary length of the coefficients and let d (≥ 2) be a bound for the degree of the polynomials F_1, $F_2, \ldots, F_s$. Bank et al. showed that if the system $F_1 \leq 0$, $F_2 \leq 0, \ldots, F_s \leq 0$ admits an integral solution, then there exists an integral solution of this diophantine system whose binary length is bounded by $(sd)^{n^{O(1)}}L$. This bound does not depend on the special form of the polynomials considered, but only on the parameters s, d, n and the input length. For related results, see also Tarasov and Khachiyan (1980).

A remarkable result for integer programs with fixed dimension has been proved by Lenstra (1983). Lenstra showed that any linear integer program with fixed number of variables can be solved in polynomial time. A related result in this direction is due to Grötschel, Lovász and Schrijver (1984). They proved the following.

Theorem 3.2 (Grötschel, Lovász and Schrijver 1984). *Given a system $Ax \leq b$ with rational coefficients, then either an integral solution x of this system can be found in polynomial time or an integral vector $c \neq 0$ can be determined in polynomial time such that for any two (not necessarily integral) solutions x and $\bar{x}$ of this inequality system*

$$\langle c, x - \bar{x} \rangle < n(n+1)2^{0.5\binom{n}{2}+1}$$

holds.

This means that independent of the number of variables either a solution of the integer program can be found or a direction will be determined in which the set of feasible points is "flat".

4. Integral polyhedra

The set of feasible solutions of a linear program

$$\max\{\langle c, x \rangle : Ax \leq b, x \geq 0\} \tag{4.1}$$

is a polyhedron P. A nonempty polyhedron P is said to be *integral*, if each of its (nonempty) faces contains an integral point. In particular the extreme points (vertices) of P are integral. This is of great importance for integer linear programming. If the linear program of the form (4.1) has a finite optimum, then there is a vertex of P where it is attained. This implies that an integer linear program can be solved via linear programming provided its feasible set is explicitly given by linear inequalities. This latter restriction will be discussed in

the next section where we deal with polyhedral combinatorics and describe the convex hull of integral point sets via linear inequalities.

The discussion above shows that it is an important question to investigate in which cases systems of linear inequalities lead to integral polyhedra. A first important result in this respect is due to Hoffman and Kruskal (1956). An $(m \times d)$-matrix A is called *totally unimodular*, if any regular square submatrix of A has determinant $+1$ or -1. Hoffman and Kruskal showed the following.

Theorem 4.1. *Let A be an integral matrix. Then the following two statements are equivalent*:

(1) *The polyhedron $P(A, \boldsymbol{b}) := \{A\boldsymbol{x} \leqslant \boldsymbol{b},\ \boldsymbol{x} \geqslant \boldsymbol{0}\}$ has integral vertices for any integral vector $\boldsymbol{b}$ for which $P(A, \boldsymbol{b})$ is nonempty.*

(2) *The matrix A is totally unimodular.*

This theorem raises the question to determine whether a given matrix is totally unimodular or not. A first sufficient condition is given by the following result of Heller and Tompkins (1956).

Theorem 4.2. *Let A be a matrix with entries from $\{0, 1, -1\}$. A is totally unimodular, if every column from A has at most two elements $\neq 0$ and if there is a partition of the rows of A in two sets with the following property: if the two elements $\neq 0$ in a column have a different sign, then the corresponding rows belong to the same set; if the two elements of a column have the same sign, then the corresponding rows are in different sets.*

The question to decide whether a general matrix is totally unimodular can be decided in polynomial time by using a decomposition due to Seymour (1980).

The theorem of Hoffman and Kruskal settles the question which systems of linear inequalities lead for *arbitrary* right-hand side $\boldsymbol{b}$ to an integral polyhedron. The analogous question is more difficult when we specify the right-hand side $\boldsymbol{b}$ or allow $\boldsymbol{b}$ to vary in a certain set. Typical results in this respect concern set-packing, set-partitioning and set-covering problems. Let A be an $(m \times d)$-matrix with only 0 and 1 entries. The set-packing problem asks for integral solutions of $A\boldsymbol{x} \leqslant \boldsymbol{1}$, $\boldsymbol{x} \geqslant \boldsymbol{0}$, where the m components of the vector $\boldsymbol{1}$ all equal 1. The set-partitioning problem asks for integer solutions of $A\boldsymbol{x} = \boldsymbol{1}$, $\boldsymbol{x} \geqslant \boldsymbol{0}$ and the set-covering problem asks for integral solutions of $A\boldsymbol{x} \geqslant \boldsymbol{1}$, $\boldsymbol{x} \geqslant \boldsymbol{0}$. A $(0, 1)$-matrix is called *balanced*, if it does not contain a square submatrix of odd order with row and column sums 2. Fulkerson, Hoffman and Oppenheimer (1974) showed the following.

Theorem 4.3. *If A is a balanced matrix, the set-partitioning polyhedron $\{\boldsymbol{x}: A\boldsymbol{x} = \boldsymbol{1},\ \boldsymbol{x} \geqslant \boldsymbol{0}\}$ is integral.*

A result more in the style of Theorem 4.1 is due to Berge (1972). Berge considers slightly modified packing and covering polyhedra by allowing superfluous rows in the inequality systems. Obviously the constraint $\langle \boldsymbol{a}_i, \boldsymbol{x} \rangle \leqslant \infty$ is superfluous in the packing case and $\langle \boldsymbol{a}_i, \boldsymbol{x} \rangle \geqslant 0$ is superfluous in the covering case.

Theorem 4.4. *Let A be a matrix with 0 and 1 entries and no zero row or column. Then the following statements are equivalent:*

(1) *The polyhedron $\{x: Ax \leq b, x \geq 0\}$ is integral for all b with $b_i \in \{1, \infty\}$.*
(2) *The polyhedron $\{x: Ax \geq b, x \geq 0\}$ is integral for all b with $b_i \in \{0, 1\}$.*
(3) *The matrix A is balanced.*

Recently, a polynomial algorithm for testing whether a given $(0, 1)$-matrix is balanced or not has been described by Conforti, Cornuéjols and Rao (1991). A related stronger concept, namely *totally balanced matrices*, has been studied by Hoffman, Kolen and Sakarovitch (1985) and polynomial recognition algorithms are based on Lubiw (1987) and Paige and Tarjan (1987).

Integral polyhedra are also related to *totally dual integral* (TDI) systems of linear inequalities as introduced by Edmonds and Giles (1977, 1984). A system of rational inequalities $Ax \leq b$, $x \geq 0$ is called TDI, if for all integral vectors c, for which the linear program (4.1) has a finite solution, its linear programming dual $\min\{\langle b, y\rangle: A^T y \geq c, y \geq 0\}$ has an integral optimal solution. Edmonds and Giles showed the following.

Theorem 4.5. *If $Ax \leq b$, $x \geq 0$ is totally dual integral and b is integral, then the corresponding polyhedron $P = \{x: Ax \leq b, x \geq 0\}$ is integral.*

In the next section we shall mention several combinatorial optimization problems which lead to TDI systems and therefore admit a solution via linear programming.

5. Polyhedral combinatorics

In this section we deal with a geometric approach for solving combinatorial optimization problems. Let M be a nonempty finite set with d elements and let $\mathcal{S}$ be a nonempty family of subsets S of M, called feasible solutions. By attaching cost c_j to any element $j \in M$, we can formulate the combinatorial optimization problem as

$$\min_{S \in \mathcal{S}} \sum \{c_j: j \in S\} . \tag{5.1}$$

We introduce binary variables x_j with $x_j = 1$ for $j \in S$ and $x_j = 0$ for $j \notin S$. Thus the feasible solutions of a combinatorial optimization problem correspond to points in $\mathbb{R}^d$. The objective function can be rewritten as

$$F(x) = \sum_{j=1}^{d} c_j x_j ,$$

and the combinatorial optimization problem becomes

$$\min_{S \in \mathcal{S}} \sum_{j=1}^{d} c_j x_j^s \quad \text{where } x^s = (x_1^s, x_2^s, \ldots, x_d^s) \text{ corresponds to } S \in \mathcal{S} . \tag{5.2}$$

Instead of finitely many feasible solutions we consider now the convex hull of all points corresponding to $S \in \mathcal{S}$:

$$\operatorname{conv} \mathcal{S} := \operatorname{conv}\{\boldsymbol{x}^S : S \in \mathcal{S}\} .$$

Since the main theorem of linear programming states that an affine-linear function on a compact polytope attains its optimum (also) in a vertex of this set, we can confine our search to the (finitely many) extreme points of conv $\mathcal{S}$. This enables us to restate our combinatorial optimization problem in the form

$$\begin{aligned} &\min \quad \sum_{j=1}^{d} c_j x_j \\ &\text{subject to} \quad \boldsymbol{x} = (x_1, x_2, \ldots, x_d) \in \operatorname{conv} \mathcal{S} . \end{aligned} \tag{5.3}$$

This is an ordinary linear program which has only one drawback: *the explicit description of the polytope* conv $\mathcal{S}$ *via linear inequalities is in most cases not known*!

In order to solve a combinatorial optimization problem a main task becomes to describe the convex hull of all feasible solutions by linear inequalities. For certain problems such a description is known. A famous early example is Birkhoff's Theorem.

Theorem 5.1 (Birkhoff 1946). *The vertices of the set of all doubly-stochastic* $(n \times n)$*-matrices, i.e., nonnegative matrices* $X = (x_{ij})$ *with constant row and column sums equal to* 1, *are the permutation matrices.*

Due to this theorem, *linear assignment problems*

"Find a permutation ϕ which minimizes $\sum c_{j\phi(j)}$"

can be solved by using methods from linear programming in a very efficient way, see, e.g., Carpaneto, Martello and Toth (1988). This can also be shown by the theorem of Heller and Tompkins (Theorem 4.2) which immediately yields that the coefficient matrix of assignment problems is totally unimodular. The *assignment polytope*, i.e., the convex hull of all $(n \times n)$-permutation matrices has been studied by Balinski and Russakoff (1972). They showed, among others, that the graph determined by the vertices and edges of the assignment polytope is Hamiltonian and has diameter 2.

Birkhoff's Theorem has further recent applications in telecommunication for finding optimal solutions for time division multiple access systems, see Burkard (1985).

Another important class of combinatorial optimization problems which admit a complete polyhedral characterization are flow problems in networks. Let G be a directed graph with node set N and arc set A and two special nodes, the *source* s and the *sink* t. Every arc (i, j) has a certain *capacity* d_{ij}. A *flow* in this graph is a function f, defined on the arcs with the following two properties:

• *flow conservation law*: in any node $x \neq s, t$ the sum of incoming flow equals the sum of outgoing flow,

• *arc capacity constraints*: on any arc (i, j) the flow lies between 0 and d_{ij}.

The *value* of a flow is the difference between the outgoing flow and the incoming flow at the source. The *maximum flow problem* asks for finding a flow from source s to sink t with maximum value. Flows are closely related to *cuts* in networks. A *cut* is a partition $(X, \bar{X})$ of the set of nodes such that the source s belongs to X and the sink t belongs to $\bar{X}$. The *capacity* $d(X, \bar{X})$ *of a cut* is defined as sum of capacities for all arcs (x, y) with $x \in X$ and $y \in \bar{X}$. The basic *max flow–min cut theorem* of Ford and Fulkerson (1956) states that the value of a maximum flow equals the minimum capacity of a cut. From the proof of this theorem it follows that the maximum flow is integral provided the arc capacities are integral.

If we attach costs c_{ij} to the arcs and want to find a flow of fixed value with minimum cost, the solution is again integral, since the coefficient matrix of network flow problems is again totally unimodular. This follows directly from Theorem 4.2.

The theorem of Birkhoff and the max flow–min cut theorem of Ford and Fulkerson are closely related results. A nice account on their equivalency and related equivalent results can be found in Reichmeider (1984).

Edmonds and Giles (1977) introduced *submodular network flow problems*. A real-valued function h defined on the subsets of a finite set M is called *submodular*, if

$$h(S) + h(T) \geq h(S \cap T) + h(S \cup T) \tag{5.4}$$

holds for all $S, T \subseteq M$.

A *submodular flow* in a directed graph is a function f, defined on the arcs of this digraph which satisfies:

• arc capacity constraints: $l(i, j) \leq f(i, j) \leq u(i, j)$ for a lower capacity function l and an upper capacity function u defined on the arcs (i, j),

• submodular flow constraints, i.e., for all subsets S of the nodes, the incoming flow minus the outgoing flow is bounded by a given submodular function $h(S)$.

Edmonds and Giles (1977) proved that for integral input data the constraints form a totally dual integral (TDI) system and, hence the corresponding polyhedron is integral. This allows to minimize linear cost functions over the submodular flow polytope in polynomial time. Even strongly polynomial algorithms are known for this problem. Several combinatorial flow algorithms have been generalized to submodular flows, see, e.g., Cunningham and Frank (1985) or Fujishige, Röck and Zimmermann (1989).

All examples mentioned so far led directly from the description of the problem to systems of linear inequalities which correspond to integral polyhedra. The first nontrivial example of forming the convex hull of integral valued points stems from Edmonds (1965) and is related with matching problems in nonbipartite graphs. Let $G = (V, E)$ be an arbitrary undirected finite graph with vertex set V and edge

set E. A *matching* M in G is a collection of edges from E such that no two edges have a vertex in common. Let $\boldsymbol{x}(M)$ be the incidence vector of a matching M. The *matching polytope* $\mathcal{M}(G)$ is defined as convex hull of all incidence vectors $\boldsymbol{x}(M)$ where M is a matching in the graph G. A straightforward condition (inequality) for $\boldsymbol{x}(M)$ is the following: let $\delta(v)$ be the set of edges incident with the vertex $v \in V$. Obviously,

$$\sum \{x_j\colon j \in \delta(v)\} \leqslant 1 \quad \forall v \in V \tag{5.5}$$

is a necessary condition for any incidence vector $\boldsymbol{x}$ of a matching. These conditions together with the sign constraints $\boldsymbol{x} \geqslant \boldsymbol{0}$ also suffice, if G does not contain a cycle of odd length. If G is just a cycle with odd length r, a maximum matching uses every second edge and has therefore at most $\frac{1}{2}(r-1)$ matching edges. On the other hand, if we assign a weight $\frac{1}{2}$ to every edge of the cycle, all inequalities (5.5) are satisfied. This shows that the convex polyhedron defined by (5.5) and the sign constraints $\boldsymbol{x} \geqslant \boldsymbol{0}$ has nonintegral vertices, if G contains odd cycles. Edmonds showed that it is sufficient to add the following inequality for any subset N of vertices with $|N|$ odd and $|N| \geqslant 3$:

$$\sum \{x_j\colon j \in \gamma(N)\} \leqslant \tfrac{1}{2}(|N|-1) \quad \forall N \text{ with } |N| \geqslant 3, \text{odd}\,, \tag{5.6}$$

where $\gamma(N)$ contains all edges with both endpoints in N.

Theorem 5.2. *The convex hull of incidence vectors corresponding to all matchings of an arbitrary graph $G = (V, E)$ is described by the sign constraints $\boldsymbol{x} \geqslant \boldsymbol{0}$, and the constraints* (5.5) *and* (5.6).

Pulleyblank and Edmonds (1974) gave a nonredundant description of the facets of the matching polytope $\mathcal{M}(G)$. Theorem 5.2 allows to solve the *maximum matching problem* which asks for a matching in G with maximum cardinality and the *weighted matching problem* as linear programs. For an extensive treatment of matching theory see the monograph of Lovász and Plummer (1986).

A generalization of matching problems are the so-called *b-matching problems*. In b-matching problems the constraints (5.5) are modified to

$$\sum \{x_j\colon j \in \delta(v_i)\} \leqslant b_i \quad \forall v_i \in V\,, \tag{5.7}$$

saying that for any vertex v_i the sum of all weights x_j of edges incident with this vertex should not exceed an integer b_i. Here we allow x_j to take also values greater than 1. Of particular importance (in connection with travelling salesperson problems) are 2-matching problems with $b_i = 2$ for all vertices. There are two versions of 2-matching problems. In one version x_j is allowed to take the values $\{0, 1, 2\}$, in the other version – important for travelling salesperson problems – $x_j \in \{0, 1\}$. Edmonds and Pulleyblank showed that the convex hull of all b-

matchings is described by (5.7) together with nonnegativity conditions and the constraints

$$\sum \{x_j: j \in \gamma(N)\} \leq \lfloor \tfrac{1}{2} \sum \{b_j: j \in N\} \rfloor \quad \forall N \text{ with } \sum \{b_j: j \in N\} \text{ odd},$$

where $\gamma(N)$ contains all edges with both endpoints in the subset N of V.

Another important class of combinatorial optimization problems are connected with *matroids*. Matroids were originally introduced by Whitney (1935) as combinatorial systems resembling abstractly independent vectors. They play nowadays a basic role in combinatorial optimization, since many discrete optimization problems can be stated in the framework of matroids and optimization questions in connection with matroids can often be solved in an efficient way. For a detailed discussion of combinatorial properties of matroids see the monograph of Welsh (1976). Here we focus on the integrality of polytopes related to matroids.

Let M be a nonempty finite ground set. A nonempty class $\mathcal{F}$ of subsets of M is called an *independence system* if $\emptyset \in \mathcal{F}$ and with $A \in \mathcal{F}$ also any subset B of A belongs to $\mathcal{F}$. Any set $F \in \mathcal{F}$ is called *independent set*. The system $(M, \mathcal{F})$ is called a *matroid*, if the independent sets fulfill an exchange property of the form:

$$\forall A, B \in \mathcal{F} \text{ with } |A| < |B| \quad \exists e \in B \backslash A: A \cup \{e\} \in \mathcal{F}. \tag{5.8}$$

This exchange property corresponds to Steinitz' Exchange Lemma. The function r, defined on the subsets of M by:

$$r(A) := \max\{|F|: F \subseteq A, F \in \mathcal{F}\}, \tag{5.9}$$

is called *rank function* of the matroid $(M, \mathcal{F})$. (A matroid could also be defined via its rank function instead of via independent sets.) Obviously, the rank function r is nondecreasing and submodular with $r(\emptyset) = 0$.

Edmonds (1970, 1971) studied the *matroid polytope*, i.e., the convex hull of the incidence vectors corresponding to independent sets in a matroid. He showed the following.

Theorem 5.3. *The convex hull of the incidence vectors corresponding to independent sets of a matroid on the ground set M with rank function r is described by*

$$P(r) = \left\{ \boldsymbol{x}: \boldsymbol{x} \geq \boldsymbol{0}, \sum_{j \in S} x_j \leq r(S) \text{ for all } S \subseteq M \right\}. \tag{5.10}$$

The polytope $P(r)$ is integral.

Edmonds gave also a nonredundant description of $P(r)$, i.e., he specified combinatorially those inequalities in (5.10) which are redundant.

Edmonds generalized these ideas in various ways. Among others he showed

that for an arbitrary nondecreasing submodular function f defined on M with $f(\emptyset) = 0$ the inequality system

$$P(f) = \left\{ x: x \geqq \mathbf{0}, \sum_{j \in S} x_j \leqq f(S) \text{ for all } S \subseteq M \right\}$$

is totally dual integral. He called the associated polytope $P(f)$ *polymatroid* and f a *polymatroid function*. Moreover, Edmonds showed the following.

Theorem 5.4. *Let f_1 and f_2 be two polymatroid functions on M. Then the system of linear inequalities*

$$\left\{ x: x \geqq \mathbf{0}, \sum_{j \in S} x_j \leqq f_1(S), \sum_{j \in S} x_j \leqq f_2(S) \text{ for all } S \subseteq M \right\} \tag{5.11}$$

is totally dual integral (TDI).

The convex hull of all those incidence vectors which correspond to independent sets with respect to *two* given matroids on the ground set M can be described by

$$x \geqq \mathbf{0}, \quad \sum_{j \in S} x_j \leqq r_1(S), \quad \sum_{j \in S} x_j \leqq r_2(S) \qquad \text{for all } S \subseteq M,$$

where r_1 and r_2 are the rank functions of the given matroids. Theorem 5.4 implies now that the corresponding *matroid intersection polytope*

$$P = \left\{ x: x \geqq \mathbf{0}, \sum_{j \in S} x_j \leqq r_1(S), \sum_{j \in S} x_j \leqq r_2(S) \text{ for all } S \subseteq M \right\}$$

is integral.

The combinatorial optimization problems discussed so far are all polynomially solvable. Let us now turn to the hard combinatorial optimization problems for which no polynomial solution method is known. An immediate remark is the following: if the convex hull of such a problem would admit a description via a polynomial number (depending on the size of the ground set) of linear inequalities, then the problem would be polynomially solvable by linear programming. Thus we have to expect that the convex hull of the integer solutions has very many facets.

A classical hard problem is the *travelling salesperson problem* (TSP). The symmetric TSP can be stated as follows. Let an undirected graph $G = (V, E)$ be given. Every edge has some length $l(e)$. We ask for a *Hamiltonian circuit* (or *tour*) which passes once and only once through every vertex of G, and has minimum length. The feasible solutions S of this problem are Hamiltonian circuits. What can be said about their convex hull?

The ground set in this problem is the edge set E. Every incidence vector of a tour obviously satisfies

$$0 \leq x(e) \leq 1 \quad \text{for all } e \in E \tag{5.12}$$

and the degree constraints

$$\sum \{x_j\colon j \in \delta(v)\} = 2 \quad \forall v \in V\,. \tag{5.13}$$

For any nonempty subset W of the vertex set V, $W \neq V$, let $\delta(W)$ denote the set of all edges with one endpoint in W and the other endpoint in $V \backslash W$. In order to exclude solutions with more than one subcycle we can add the *subtour elimination constraints*

$$\sum \{x_j\colon j \in \delta(W)\} \geq 2 \quad \forall W \subseteq V \text{ with } 2 \leq |W| \leq |V| - 2\,. \tag{5.14}$$

The *integral* points lying in the convex polyhedron defined by (5.12)–(5.14) correspond exactly to tours. But this polyhedron is not integral, it has fractional vertices! So we need other classes of facet inducing inequalities. Such classes are the *comb inequalities* (Grötschel and Padberg 1979) and the *clique tree inequalities* (Grötschel and Pulleyblank 1986). A complete description of all facets of the symmetric TSP polytope (= the convex hull of all tours) is, however, not known in general. A complete description of the travelling salesperson polytope on eight nodes has been given by Christof, Jünger and Reinelt (1990).

Since even the number of all known facets of a TSP is too large to be used in a linear programming formulation for this problem, the following *cutting plane approach* is applied with much success. Some of the facet defining inequalities are used in a linear programming relaxation for the problem. If the corresponding linear program yields a fractional (or an infinite) solution, a *cutting plane* is constructed which separates this solution from the convex hull of all feasible solutions of the given problem. The cutting plane is added as an additional constraint to the linear program and this is resolved.

This cutting plane approach can generally be applied to combinatorial optimization problems, as soon as something about the facial structure of the polytope conv $\mathcal{S}$ is known. Therefore, a polyhedral description of polytopes arising from combinatorial optimization problems finds considerable interest, see, e.g., the monographs by Grötschel, Lovász and Schrijver (1988) or Nemhauser and Wolsey (1988). Of particular practical importance is the solution of the so-called *separation problem*: given an infeasible point, generate automatically a cutting plane which separates this point from conv $\mathcal{S}$. Problem-specific techniques have been developed to tackle this problem.

This approach has successfully been applied to large TSPs, see, e.g., Padberg and Rinaldi (1987), to the *linear ordering problem*, see Grötschel, Jünger and Reinelt (1984), to *set packing* and *set partitioning*, see Balas and Padberg (1976), or recently to the *three-index assignment problem*, see Balas and Saltzman (1989). It requires the availability of a code for solving large linear programs and needs detailed investigations on how the separation problem can be handled. A nice example of separation algorithms for the above-mentioned three-index assignment problem is contained in the recent paper of Balas and Qi (1990). Integral

polyhedra arising from practical network design problems are recently studied in Grötschel and Monma (1990).

We conclude this section with a short discussion of two further polytopes arising from basic combinatorial problems, namely the *asymmetric travelling salesperson problem* and the *stable set problem*.

The *asymmetric travelling salesperson problem* (asymmetric TSP) can be described in the following way. Let $G = (N, A)$ be a complete directed graph without loops, i.e., between any pair of nodes i and j, $i \neq j$, there is an arc (i, j). Every arc has a length $l(i, j)$ and the lengths can be different for oppositely directed arcs. The asymmetric TSP asks for a shortest directed tour. The convex hull of all incidence vectors corresponding to directed tours in G is called *asymmetric travelling salesperson polytope*. This polytope has for $n \geqslant 3$ the dimension $(n-1)^2 - n$ as stated in an abstract by Heller (1953) and proved by Grötschel and Padberg (1977). Bartels and Bartels (1989) give a complete facial description for the asymmetric 5-city polytope, Euler and Le Verge (1992) give a complete description of the 6-city polytope. It is known that the nonnegativity constraints $x_{ij} \geqslant 0$ define facets for all arcs $(i, j) \in A$, if $n \geqslant 5$. Further facets stem from the directed versions of the subtour elimination constraints (5.14). It is not known, however, if the directed version of the comb inequalities defines facets, too. For a survey on the symmetric and asymmetric travelling salesperson polytope, see Grötschel and Padberg (1985).

The last polytope we are discussing in more detail is the *stable set polytope*. Let an undirected graph $G = (V, E)$ with vertex set V and edge set E be given. A subset $S \subseteq V$ is called *stable*, if no two vertices of this set are adjacent. An important problem asks for finding a maximum stable set in G, or, given a weight function on the vertices, a stable set with maximum weight. Again, let $\boldsymbol{x} = (x_j)$ be the incidence vector of a subset of vertices. The *stable set polytope* is the convex hull of all incidence vectors corresponding to stable sets in G. Trivial inequalities for such vectors are

$$x_j \geqslant 0 \quad \text{for all } j \in V\,, \tag{5.15}$$

$$x_i + x_j \leqslant 1 \quad \text{for all } (i, j) \in E\,. \tag{5.16}$$

The inequalities (5.15) and (5.16) describe the stable set polytope if and only if G is bipartite and has no isolated vertices. Difficulties arise for nonbipartite graphs, but unlike for matchings, the maximum stable set problem for arbitrary graphs is NP-hard (Garey and Johnson 1979). For example, let G be an odd circuit. The point $(\frac{1}{2}, \frac{1}{2}, \ldots, \frac{1}{2})$ fulfills (5.15) and (5.16), but does not belong to the stable set polytope. This leads to new inequalities, the so-called *odd circuit constraints*

$$\sum \{x_j\colon j \in V(C)\} \leqslant \tfrac{1}{2}(|V(C)| - 1) \quad \text{for each odd circuit } C\,, \tag{5.17}$$

where $V(C)$ is the set of vertices of an odd circuit C.

Another class of constraints for this problem are the *clique constraints*. A *clique*

Q in G is a subset of vertices which are all adjacent. Therefore, a stable set can contain at most one vertex of a clique:

$$\sum \{x_j: j \in V(Q)\} \leq 1 \quad \text{for each clique } Q\,.$$

Further constraints are *odd wheel* and *odd antihole inequalities*. We refer the interested reader to Grötschel, Lovász and Schrijver (1988), where also algorithmic aspects of these inequalities for the stable set problem are discussed.

6. Combinatorial optimization problems arising from convex geometric configurations

The *planar travelling salesperson problem* can be stated as follows. We are given n points in the plane. The distances between any two points are just the Euclidean distances. We ask for a tour of shortest length which passes through everyone of the given points just once. The planar TSP is NP-*hard* in the general case. No polynomially bounded algorithm is known for its solution. This situation changes, however, if the given points lie on the boundary of a convex polygon. In this case the natural order of the points on the boundary leads immediately to a shortest tour.

Given n points in the plane it can be tested in $O(n \log n)$ time whether these points lie on the boundary of a convex polygon. If this test is affirmative, then the cyclic order of the points on the boundary can be determined in $O(n \log n)$ steps (see, e.g., Preparata and Shamos 1988). Thus the planar TSP can be solved in $O(n \log n)$ time in this case. A similar result holds if a distance matrix $A = (a_{ij})$ is given. It can be tested in $O(n^2)$ time whether this matrix is a distance matrix of the vertices of a convex polygon. If this test is affirmative, then the optimal tour can be found in $O(n \log n)$ steps as in the case above.

These remarks show that combinatorial optimization problems may become much easier, if the data arise from vertices of a convex polygon in the plane. In the following we shall mention three other classes of problems where such considerations have been applied. The first class concerns *spanning tree problems*, secondly we discuss *Steiner tree problems* and finally we deal with the *Euclidean matching problem*. These problems have nowadays found increasing interest in connection with the development of *computational geometry* as well as in connection with planar lay-out problems as a tool for VLSI design.

A *spanning tree* for n given points in the plane is a connected and cycle-free graph with the given points as vertices. The length of an edge in this graph is the distance between the two endpoints of this edge. Frequently used distance measures include the maximum norm, the Euclidean distance and the rectilinear distance (L_1-norm). The length of a tree is defined as sum of all lengths of its edges. The *minimum spanning tree problem* asks for a spanning tree with minimum length.

A basic result of computational geometry (see, e.g., Edelsbrunner 1987, Preparata and Shamos 1988) states that the minimum spanning tree for n points in

general position in the plane is a subgraph of the *relative neighbourhood graph D* which itself is a subgraph of the (planar) Delaunay triangulation. The vertices of D are the given points and two points P_1 and P_2 are connected by an edge, if there is no other vertex whose distance to P_1 and P_2 is less than the distance between P_1 and P_2. The relative neighborhood graph is planar and can be constructed in $O(n \log n)$ time. Since in a planar graph a minimum spanning tree can be found in $O(n)$ time (cf. Mehlhorn 1984), the minimum spanning tree problem for n points in the plane can be solved in $O(n \log n)$ steps.

One can do even better, if the given points are vertices of a convex polygon given in their cyclic order on the boundary. Due to a recent result of Aggarwal et al. (1989) the *Delaunay triangulation* of these points can be computed in linear time. From the Delaunay triangulation the relative neighborhood graph can also be computed in linear time. Since this graph is planar, a minimum spanning tree can be computed in linear time. Thus we get the following.

Theorem 6.1. *Let n points be given in the plane. A minimum spanning tree for these points can be computed in $O(n \log n)$ time. If the points lie on the boundary of a convex polygon and are given in their cyclic order, then a minimum spanning tree for these points can be constructed in linear time.*

A similar result has recently been shown for computing *maximum spanning trees*, i.e., trees whose total length is maximum. Though the geometric setting is quite different, Monma et al. (1990) could recently show that a maximum spanning tree for n points in the plane can be computed in $O(n \log h)$ time, where h denotes the number of points on the convex hull of the given set. If the given points are vertices of a convex polygon (given in order along the boundary), then the algorithm requires only a linear amount of time and space.

Let us now turn to the *Steiner tree problem*. Let V be a set of k points in the plane, called *terminals*. Again we want to connect these points by lines of minimum length, but now we allow that additional vertices may be introduced where the lines can branch. For example, given three vertices of an equilateral triangle of side length a, a minimum spanning tree always contains two sides and has length $2a$. For a minimum Steiner tree (with respect to Euclidean distance) we may insert the geometric center of the triangle as additional *Steiner point* and connect this Steiner point with the given vertices. Thus we get a minimum Steiner tree with total length $a\sqrt{3}$.

Again there are several versions of the Steiner tree problem according to the distance measure used. In the *Euclidean Steiner tree problem* the distance between two points is assumed to be the Euclidean distance. In the *rectilinear Steiner tree problem* the L_1-norm is used. The difficulty of the Steiner tree problem stems from the question how many and which Steiner points should be introduced such that the length of the tree becomes minimal. This problem is in general NP-*hard*. If the given points, however, lie on the boundary of a convex polygon it can be solved more efficiently. Aho, Garey and Hwang (1977) showed that the rectilinear Steiner tree problem is efficiently solvable when the terminals lie on the boundary of a rectangle. Provan (1988) generalized this result and showed that if

the terminals lie on the boundary of a rectilinearly convex region, then the rectilinear Steiner tree problem can be solved in polynomial time. Here a region R is called *rectilinearly convex*, if each pair of points P and Q in R is connected by some minimum length path of rectilinear line segments lying entirely in R. Equivalently, R is rectilinearly convex if every horizontal or vertical line intersects R in an interval.

Moreover, Provan showed the following result for Euclidean Steiner tree problems.

Theorem 6.2 (Provan). *If the n terminals of a Euclidean Steiner tree problem lie on the boundary of a convex polygon in the plane, then there is a fully polynomial approximation scheme for this problem, i.e., there is an algorithm which for any* $\varepsilon > 0$ *constructs a Steiner tree T of length* $l(T)$ *such that* $l(T) \leq (1 + \varepsilon) \cdot \text{Opt}$, *where* Opt *is the optimal value of the problem. The running time of the algorithm is polynomial in n and* $1/\varepsilon$.

Now let us turn to *matching problems* that were introduced in section 5. In general the weighted matching problem is solvable in $O(n^3)$ time. Matching problems in the plane are solvable by algorithms with lower complexity, see Vaidya (1989). Recently, it has been shown by Marcotte and Suri (1989a,b) that even a better complexity can be obtained if the data come from a convex polygon. Let the vertices $v \in V$ be vertices of a convex polygon in the plane and allow all possible edges between distinct vertices. The weight of an edge is the Euclidean distance between its endpoints. We get the following.

Theorem 6.3 (Marcotte and Suri). *Given a set V of* $2n$ *points on the boundary of a convex polygon whose edges are assigned weights equal to the Euclidean distance of their endpoints, then a minimum weight matching can be found in* $O(n \log n)$ *steps. A maximum weight matching can be determined in linear time* $O(n)$.

As a by-product of their method the authors get a fast algorithm for deciding whether a given matching is a minimum weight matching or not. This *verification problem* can be solved in $O(n\alpha(n))$ time, where $\alpha(n)$ is the extremely slow growing inverse Ackermann function.

When we color the vertices red and blue and allow only edges between a red and a blue vertex, we get an *assignment problem*. Marcotte and Suri (1989b) showed that this assignment problem can be solved in $O(n \log n)$ steps provided the vertices lie again on the boundary of a convex polygon in the plane.

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CHAPTER 2.9

Geometric Algorithms

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Contents

1. Introduction 701
1.1. Geometry and algorithms 701
1.2. A brief history of computational geometry 702
1.3. The style and structure of this chapter 702
2. Topics in computational geometry 703
2.1. Convex hulls 703
2.2. Divide-and-conquer 705
2.3. Incremental construction 707
2.4. Geometric transforms 709
2.5. k-sets 710
2.6. Delaunay triangulations 711
2.7. Flipping 713
2.8. Randomized incremental construction 715
2.9. Minimum spanning trees 716
2.10. Alpha shapes 717
2.11. Voronoi diagrams 718
2.12. Plane-sweep 719
2.13. Generalized Voronoi diagrams 721
2.14. Arrangements 723
2.15. Zones of arrangements 724
2.16. Applications of arrangements 726
2.17. Topological sweep 727
3. Concluding remarks 729
References 730

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1. Introduction

1.1. Geometry and algorithms

"Computational geometry" and "geometric algorithms" are synonymous terms that denote an active discipline within computer science studying algorithms for, or more generally the computational complexity of geometric objects and problems. It is sometimes poetically described as "the place where geometry and algorithms meet". Just like geometry, the study of algorithms can be traced back more than two-thousand years. For example, the ancient Chinese collection "Nine Chapters on the Mathematical Art" and Book VII of Euclid's "Elements" both give a description of the Euclidean algorithm for integer division. The word "algorithm" itself is derived from the name of the Persian textbook author Abu Ja'far Mohammed ibn Mûsâ al-Khowârizmî, see Knuth (1968). Indeed, many proofs of mathematical theorems are constructive and can be considered as or imply algorithms.

Still, the formulation of a specific algorithm remained a rare incidence until rather recent times, and there was only little interest in the study of computational complexity, that is, the *number* of elementary steps necessary to carry out an algorithm or to constructively solve a problem. Nevertheless, there was a fair amount of work dedicated to deciding whether or not a given collection of tools (e.g., ruler and compass) was *sufficient* to complete certain constructions (e.g., a regular n-gon or the trisection of an angle).

The situation changed dramatically with the beginning of on-going computerization of essentially every part of human life. Since most computers are based on the same or similar principles, it became possible to define the *complexity* of an algorithm as the number of elementary steps if executed on a computer. The complexity is commonly expressed as a function in the *size* of the input problem solved by the algorithm, or as a class of such functions. For example, the statement that some algorithm takes $O(f(n))$ steps means that there is a positive constant $c \in \mathbb{R}$ so that for every problem of size n the number of steps does not exceed $c \cdot f(n)$. $O(f(n))$ can be viewed as the class of functions $g(n)$ so that $g(n) \leqslant c \cdot f(n)$. The notation $\Omega(f(n))$ is used to express that *at least* $c \cdot f(n)$ steps are executed, and $\Theta(f(n)) = O(f(n)) \cap \Omega(f(n))$.

In order to make the study of algorithms independent of the specific computer used to implement them, a fair amount of effort went into developing abstract models that strive for generality as well as for an accurate though simplified simulation of the more complicated reality. The most popular model, the *random access machine* or *RAM*, was formulated in the early seventies, see Aho, Hopcroft and Ullman (1974), and is still widely used. Very common is also the *uniform cost* assumption that considers comparisons, arithmetic operations, and data movements as elementary operations.

In its early stages, the study of algorithms concentrated on numerical problems, the manipulation of sets and multisets, and the manipulation of text [see, e.g., the three volumes of Knuth (1968, 1969, 1973)]. This was followed by the system-

atic study of algorithms for more complex structures, such as graphs, groups, and geometric objects.

1.2. A brief history of computational geometry

A systematic study of geometric algorithms started with the influential dissertation of Shamos (1978). Before his work only a small number of geometric problems were studied from an algorithmic point of view. However, because of the multitude of connections between computational geometry and application areas of computer science (data bases, image processing, pattern recognition, computer graphics, visualization, computer aided geometric modeling, computer vision, robotics, etc.), it is difficult to draw a line that separates geometric algorithms from others.

Many problems in computational geometry are motivated by applications. Indeed, the many related application areas together with the great diversity of geometry itself set the stage for the development of computational geometry into a broad and multifaceted discipline, see Lee and Preparata (1984) and Yao (1990) for surveys of the area. The first textbook, volume III of Mehlhorn's trilogy on algorithms (Mehlhorn 1984a,b,c) concentrates on multi-dimensional searching. This part of computational geometry has particularly strong ties to the algorithmic study of searching as described in Knuth (1973). The book by Preparata and Shamos (1985) reflects the structure of the second author's thesis and focuses on the study of mostly 2- and occasionally 3-dimensional problems. The book by O'Rourke (1987) focuses on combinatorial and algorithmic problems for polygons. The relationship between discrete geometry and computational geometry is the main theme of the textbook by Edelsbrunner (1987). It is based on and built around the notion of an arrangement of hyperplanes, and many problems are discussed in arbitrary but constant dimensions, $\mathbb{R}^d$.

1.3. The style and structure of this chapter

The two most natural ways to categorize computational geometry use geometric problems and algorithmic paradigms[1] as classification criteria. The technical part of this survey paper consists of a mixed sequence of sections, some focusing on a geometric concept or problem and some organized around a method or paradigm. The sections are loosely connected. It is not our intention to represent the entire area of computational geometry. Quite to the contrary, the selection of topics is biased towards traditional themes in convex geometry. The main geometric concepts and problems discussed include convex polytopes, k-sets, Delaunay and regular triangulations, Dirichlet–Voronoi diagrams, generalized Voronoi diagrams, minimum spanning trees, alpha shapes, hyperplane arrangements, zones

[1] An algorithmic *paradigm* is a more abstract notion than an algorithm, just as an algorithm is more abstract a notion than a computer program. It refers to a method or technique that underlies the design of more than one algorithm.

and zonotopes. The (mostly algorithmic) methods and paradigms include divide-and-conquer, incrementation, randomization, geometric transforms, flipping and shelling, plane-sweep, and topological sweep.

Even though the scope of this survey paper is rather narrow, it is difficult to give a fair account of the covered topics. The author apologizes for the personal bias that is reflected in the selection of the material and the description of the results.

2. Topics in computational geometry

There are many beautiful connections between convex polytopes, Delaunay triangulations, Voronoi diagrams, and hyperplane arrangements. If anything, then this is the theme that underlies the following seventeen technical sections.

2.1. *Convex hulls*

A convex polytope is usually constructed as the convex hull of a finite set of points or the intersection of a finite set of half-spaces. We briefly discuss the former problem, and we will see later that the latter problem is really the same under a polar transformation. The problem of constructing the convex hull of a finite point set in $\mathbb{R}^d$ is probably the one that received the most attention in computational geometry. It generated a large variety of interesting results and techniques.

Let S be a set of n points in $\mathbb{R}^d$. The *convex hull* of S, $\text{conv}(S)$, is the smallest convex superset of S. Alternatively, $\text{conv}(S)$ can be defined as the set of all non-negative affine combinations of points of S, that is,

$$\text{conv}(S) = \{x \in \mathbb{R}^d \mid x = \sum_{p_i \in S} \lambda_i p_i, \sum \lambda_i = 1, \lambda_i \geqslant 0 \text{ for all } i\}.$$

What does it mean to "construct" $\text{conv}(S)$? The answer to this question depends on the representation of $\text{conv}(S)$ that is desired. In general, the choice of representation is very important in the area of algorithm design. It most often depends on how the object is going to be used, and different representations of the same object typically differ in their computational complexity. For example, the most implicit representation of $\text{conv}(S)$ is the set S itself, and this does not need any further computation. While computationally extremely inexpensive, this representation does not help much in answering questions about $\text{conv}(S)$, such as how many vertices it has, or how many facets bound it, or what its d-dimensional measure is.

A slightly less implicit representation is the set $\text{ext}(S)$ of *extreme* points of S, that is, vertices of $\text{conv}(S)$. A relatively straightforward way to construct $\text{ext}(S)$ is to decide for each point $p \in S$ whether or not $p \in \text{ext}(S)$. This is equivalent to solving a $(d-1)$-dimensional linear program with $n-1$ constraints for each point. Assuming d is a constant, this takes time $O(n)$ per point, see, e.g., Megiddo (1984), and therefore time $O(n^2)$ in total. The question whether or not faster algorithms exist for all (constant) dimensions was answered recently in the affirmative, and

the currently best result is an algorithm due to Matoušek and Schwarzkopf (1991) that runs in time $O(n^{2-(2/(\lfloor d/2 \rfloor+1))+\delta})$, for any $\delta > 0$.

In $\mathbb{R}^2$, $\text{conv}(S)$ is a convex polygon and its boundary can be explicitly represented by a linear list[2] that stores its vertices in a counterclockwise (ccw) order. This representation can be constructed in time $O(n \log n)$ – Graham (1972) gives the first algorithm that achieves this time bound – which is optimal under the algebraic decision tree model[3], see Yao (1981). Running times faster than $n \log n$ can be obtained if the points are sorted (see section 2.2), if certain distributions are assumed, or if the size of $\text{ext}(S)$ is significantly less than n. In particular, Kirkpatrick and Seidel (1986) show that $O(n \log h)$ time is sufficient and necessary, where $h = |\text{ext}(S)|$. All time bounds mentioned for computing $\text{conv}(S)$ in $\mathbb{R}^2$, lower and upper, also hold for computing $\text{ext}(S)$.

In $\mathbb{R}^3$, $\text{conv}(S)$ is a convex polytope with at most n vertices, at most $3n - 6$ edges, and at most $2n - 4$ facets[4]. Using a planar graph representation for $\text{conv}(S)$, e.g., the quad-edge data structure of Guibas and Stolfi (1985), thus takes only $O(n)$ storage. Section 2.2 will say a little bit about how to construct this representation in time $O(n \log n)$, see also Preparata and Hong (1977). An equally fast randomized[5] algorithm is described by Clarkson and Shor (1989) who also show that $O(n \log h)$ expected time is sufficient even in $\mathbb{R}^3$. Notice that in $\mathbb{R}^3$ the explicit construction of the boundary of $\text{conv}(S)$ is thus asymptotically no more expensive than constructing $\text{ext}(S)$.

In $\mathbb{R}^d$, the upper bound theorem for convex polytopes (see, e.g., McMullen and Shephard 1971 or Brøndsted 1983) implies that $\text{conv}(S)$ has at most $O(n^{\lfloor d/2 \rfloor})$ faces, and this bound is tight as exhibited by cyclic polytopes. This implies that at least this much time is needed if the explicit construction of all faces is desired. Most naturally, these faces are represented by the facial graph that represents each face by a node and uses arcs connecting nodes to indicate incidences between faces whose dimensions differ by one.

The historically first algorithm that constructs the facial graph of $\text{conv}(S)$ is due to Chand and Kapur (1970). Their "gift-wrapping" algorithm takes $O(n)$ time per facet, which yields time $O(n^{\lfloor (d+2)/2 \rfloor})$ in the worst case. Seidel (1981) gives an $O(n^{\lfloor (d+1)/2 \rfloor} + n \log n)$ time algorithm based on the beneath-beyond idea mentioned in Grünbaum (1967), see Edelsbrunner (1987) for a more accessible publication of the method. This algorithm is optimal in even dimensions. Using randomization, Clarkson and Shor (1989) improve the running time to $O(n^{\lfloor d/2 \rfloor} + n \log n)$ in the

[2] Depending on the application, one can store the list in a linear array, a linked list, or an even more sophisticated data structure, such as a search tree. We refer to Cormen, Leiserson and Rivest (1990) for a recent comprehensive text on algorithms and data structures.

[3] This is a model of algorithms that assumes that each decision is based on computing the sign of an algebraic expression, see, e.g., Ben-Or (1983). It concentrates on the number of decisions necessary to solve some problem and purposely ignores other computations such as data movements and arithmetic operations.

[4] These bounds are easily derived from Euler's formula which relates the number of vertices, v, the number of edges, e, and the number of facets, f, of a convex polytope in $\mathbb{R}^3$: $v - e + f = 2$. In $\mathbb{R}^d$, a *facet* is a $(d-1)$-dimensional face.

[5] See section 2.8 for a brief description of what a randomized algorithm is and for an example of one.

expected case, see section 2.8. Chazelle (1991) converts their randomized algorithm to a deterministic one with the same (worst-case) running time. An algorithm whose running time depends on the output-size (the size of the facial graph) can be found in Seidel (1986). It constructs $\text{conv}(S)$ in time $O(n^2 + F \log n)$, where F is the number of faces. A slower algorithm that finds all facets of $\text{conv}(S)$ with only constant extra working storage has been given by Avis and Fukuda (1991).

2.2. *Divide-and-conquer*

Divide-and-conquer is a very popular algorithmic paradigm that can be traced back to the time of the Romans when Caesar defeated enemy armies by splitting them up into smaller units. Abstractly, it can be described by the following three steps.

(i) *Divide.* Decompose the original problem into subproblems of the same kind as the original one.

(ii) *Recurse.* Solve the subproblems in the same way.

(iii) *Merge.* Combine the solutions to the subproblems to form a solution of the original problem.

Of course, the recursion has to stop at some point (a one soldier army?) and a different strategy must be used (what could that be in Caesar's application of the paradigm?).

One might be tempted to think that divide-and-conquer is just a fancy name for mathematical induction. Indeed, it is induction, and so are the incrementation paradigm (section 2.3) and other algorithmic design methods. The differences between the various ways to do induction algorithmically *are* important, however, as they influence the running time of the algorithms. Different problems have different structural properties and are thus amenable to different design methods.

Divide-and-conquer leads to $O(n \log n)$ time algorithms for constructing the convex hull of a set of n points in $\mathbb{R}^2$ or $\mathbb{R}^3$. The remainder of this section presents the 2-dimensional algorithm and ends with a few remarks about the 3-dimensional case. Details can be found in Preparata and Hong (1977), and Edelsbrunner (1987).

Let $S = \{p_i = (\pi_{i1}, \pi_{i2}) \mid 1 \leqslant i \leqslant n\}$ be the point set in $\mathbb{R}^2$, and assume for simplicity that no three points are collinear and that $\pi_{i1} \neq \pi_{j1}$ if $i \neq j$. A general technique that simulates this assumption with nominal extra cost can be found in Edelsbrunner and Mücke (1990).

(i) *Divide.* Find an x_1-coordinate μ so that $|S_L|, |S_R| \leqslant (n+1)/2$, where $S_L = \{p_i \in S \mid \pi_{i1} < \mu\}$ and $S_R = S \setminus S_L$.

(ii) *Recurse.* Compute $\text{conv}(S_L)$ and $\text{conv}(S_R)$ recursively.

(iii) *Merge.* Construct $\text{conv}(S)$ by finding the two common outer tangents of $\text{conv}(S_L)$ and $\text{conv}(S_R)$ and copying the relevant pieces of the two convex polygons.

In the case S contains only two or three points, the convex hull is constructed directly. We show below that the merge step can be done in time linear in the sizes of the two convex polygons. The time $T(n)$ of the entire algorithm is therefore $T(n) = O(n) + 2T(n/2) = O(n \log n)$.

Let $(u_1, u_2, \ldots, u_l)$ be the ccw sequence of the vertices of $\text{ext}(S_L)$, and let $(v_1, v_2, \ldots, v_r)$ be the ccw sequence of $\text{ext}(S_R)$. Assume also that u_1 is the rightmost point among the u_i (the one with the largest x_1-coordinate), and that v_1 is the leftmost point among the v_j. The edge $uv = u_1v_1$ thus intersects $\text{conv}(S_L)$ and $\text{conv}(S_R)$ only in its endpoints. The upper common tangent of the two convex polygons can be found by repeating the following steps. The method for finding the lower common tangent is symmetric. Let u^+ be the ccw next vertex of u and let v^- be the ccw previous vertex of v.

loop while uvv^- is a left-turn **do** $v := v^-$ **endwhile**;
  if vuu^+ is a right-turn **then** $u := u^+$ **else exit** loop **endif**
forever.

It is interesting to note that the edges uv generated during the construction of the upper and lower common tangents triangulate the part of $\text{conv}(S)$ between $\text{conv}(S_L)$ and $\text{conv}(S_R)$. Consequently, the edges uv generated during run time of the entire algorithm triangulate $\text{conv}(S)$. By Euler's relation for planar graphs there are at most $3n-6$ such edges, which implies that the algorithm spends a total of $O(n)$ time in step (3), summed over all recursive calls. If the points are sorted from left to right before starting the algorithm then step (1) can be done in constant time, assuming the input points are stored in a linear array. This implies that the entire algorithm takes only $O(n)$ time. Therefore, after sorting the 2-dimensional convex hull problem has complexity $O(n)$ as opposed to $\Omega(n \log n)$ before sorting.

As demonstrated by Preparata and Hong (1977), the same divide-and-conquer approach leads to an $O(n \log n)$ time algorithm also for points in $\mathbb{R}^3$. However, the merge step is considerably more involved than in $\mathbb{R}^2$ as it needs to construct the cyclic list of triangles that bound $\text{conv}(S)$ but neither $\text{conv}(S_L)$ nor $\text{conv}(S_R)$. Each such triangle is defined by a plane that contains three points of $S_L \cup S_R$, at least one point of each, so that both convex polytopes lie on the same side of this plane. We omit further details.

From a purely computational complexity viewpoint it thus seems that computing the convex hull in $\mathbb{R}^3$ is just as easy as in $\mathbb{R}^2$: both problems take time $O(n \log n)$ in the worst case. In contrast to $\mathbb{R}^2$ it is, however, not possible to reduce the running time to $O(n)$ if the points in $\mathbb{R}^3$ are assumed to be sorted, e.g., by x_1-coordinates. Indeed, Seidel (1985) proves an $\Omega(n \log n)$ lower bound even if sorting permutations for a constant number of directions are part of the input.

So we have seen that divide-and-conquer can be used to get efficient algorithms for constructing convex hulls in $\mathbb{R}^2$ and $\mathbb{R}^3$. In $\mathbb{R}^d$, for $d \geq 4$, other methods seem to lead to better results (see section 2.3). There are too many problems in computational geometry for which divide-and-conquer is used to attempt even a partial enumeration. In many of these applications, divide-and-conquer is used in combination with other paradigms, such as incremental construction (section 2.3) and plane-sweep (section 2.12).

2.3. Incremental construction

The idea of the incrementation paradigm is trivial but often effective: solve the problem by adding one object at a time. It thus mirrors the structure of straightforward induction. This implies the use of an ordering of the objects as $p_1, p_2, \ldots, p_n$ so that when p_i is added the solution for $\{p_1, p_2, \ldots, p_{i-1}\}$ is available.

Although the structure of the incrementation paradigm is extremely straightforward, there are many variations that differ in their fine details. For example, the objects might be added in an arbitrary order (which works well, e.g., for hyperplane arrangements, section 2.15), or one might initially spend some time to find a favorable sequence (e.g., a random sequence, section 2.8). Also, the solution for the first i objects might embed some knowledge about the remaining $n-i$ objects (as in the convex hull algorithm of Clarkson and Shor 1989), or one may insist on a truly incremental method that ignores or does not have available any such information.

In this section we describe an algorithm that incrementally constructs the convex hull of a set S of n points in $\mathbb{R}^d$. This algorithm is due to Seidel (1981) and has also been described in full detail in Edelsbrunner (1987). It will be convenient to sort the points, e.g., in the order of increasing x_1-coordinates, before starting the incremental construction. With a hint to the simulated perturbation technique of Edelsbrunner and Mücke (1990) we assume that the points are in general position, which includes that $n > d$, no $d+1$ points lie in a common hyperplane, and no two points share a common x_1-coordinate. Note that this implies that each face of $\text{conv}(S)$ is a simplex. In particular, each facet is a $(d-1)$-simplex.

Let $p_1, p_2, \ldots, p_n$ be the sorted sequence of the points. The construction starts with building the d-simplex defined by the first $d+1$ points and then proceeds by adding one point at a time. Here is a more formal description of the process. Define $S_i = \{p_1, p_2, \ldots, p_i\}$.

```
Construct conv(S_{d+1});
for i := d+2 to n do
    construct conv(S_i) by adding point p_i to conv(S_{i-1})
endfor.
```

By $\text{conv}(S_i)$ we mean the facial graph of it, but for the sake of simplicity we will ignore details about manipulating nodes and arcs of this graph.

How can we add p_i, that is, how can we construct $\text{conv}(S_i)$ out of $\text{conv}(S_{i-1})$? The main geometric idea is what is called beneath-beyond in Grünbaum (1967). Each facet of $\text{conv}(S_{i-1})$ is either visible or invisible from p_i. To be precise, define for each facet f the hyperplane $h_f = \text{aff}(f)$ which contains f, and let h_f^- be the half-space bounded by h_f that does not contain $\text{conv}(S_{i-1})$. Then f is *visible* from p_i if $p_i \in h_f^-$ and f is *invisible*, otherwise. Each ridge[6] of $\text{conv}(S_{i-1})$ is incident to exactly two facets, so three kinds of ridges can be distinguished: *invisible* ridges

[6] A *ridge* is a $(d-2)$-face.

incident to two invisible facets, *silhouette* ridges incident to one visible and one invisible facet, and the remaining *visible* ridges.

The facets of $\text{conv}(S_i)$ are the invisible facets of $\text{conv}(S_{i-1})$ and the new facets spanned by p_i and a silhouette ridge each. The visible facets are no longer part of the boundary and can be deleted. Alternatively, we can keep the visible facets and add the d-simplices spanned by p_i and a visible facet each. This effectively means that we not only construct $\text{conv}(S)$ but also a triangulation[7] of S. Indeed, the running time of the incremental algorithm is the same whether or not it keeps track of the triangulating d-simplices.

A key issue in obtaining an efficient implementation is how to find all visible facets in time independent of the total number of facets, and thus in particular of the number of invisible facets. Here it helps that the points are initially sorted. As a consequence, the edge $p_{i-1}p_i$ is disjoint from the interior of $\text{conv}(S_{i-1})$. We thus know that p_{i-1} belongs to the boundary of a visible facet. One such facet, f, can be found by exhaustively checking all facets that have p_{i-1} as a vertex. Starting with f, all other visible facets can be collected by breadth- or depth-first search[8] of the graph whose nodes are the visible facets connected by arcs that correspond to visible ridges.

The total effort for finding all facets visible from p_i is therefore at most proportional to the number of facets sharing p_{i-1} plus the number of facets visible from p_i. The vertex-figure[9] of p_{i-1} is combinatorially equivalent to a convex hull of $i-2$ points in $\mathbb{R}^{d-1}$. The upper bound theorem thus implies that at most $O(i^{\lfloor (d-1)/2 \rfloor})$ facets share p_{i-1}. Similarly, the number of new facets defined by p_i and the silhouette ridges is at most $O(i^{\lfloor (d-1)/2 \rfloor})$. Finally, each facet is constructed once, and when it is found to be visible from the new point then it is deleted, also only once. It is thus sufficient to pay for the effort to delete a facet already at the time of its creation. This implies that the total amount of time spent by the algorithm is bounded from above by $O(n \log n)$ for sorting plus

$$\sum_{i=1}^{n} O(i^{\lfloor (d-1)/2 \rfloor}) = O(n^{\lfloor (d+1)/2 \rfloor})$$

for maintaining the convex hull. Incidentally, this is also an upper bound on the number of d-simplices of any triangulation of n points in $\mathbb{R}^d$.

Recall that $\text{conv}(S)$ is a convex polytope with up to n vertices and thus has at most $O(n^{\lfloor (d/2) \rfloor})$ faces, and this bound is tight in the worst case. Thus, the incremental algorithm is worst-case optimal in even dimensions.

Similar to divide-and-conquer, the incremental construction paradigm has many

[7] A *triangulation* of S is a simplicial cell complex with vertex set S that decomposes $\text{conv}(S)$, see also section 2.6.

[8] *Breadth-first search* and *depth-first search* are two methods for traversing all nodes of a graph in time proportional to its size, that is, the number of nodes and arcs, see Cormen, Leiserson and Rivest (1990).

[9] The *vertex-figure* of a vertex p consists of all faces incident to p. If p is a vertex of a d-dimensional convex polytope, then there is a hyperplane that separates p from all other vertices. This hyperplane intersects the vertex-figure in a $(d-1)$-dimensional convex polytope.

applications. A randomized version will be described in section 2.8 for constructing Delaunay triangulations. Another variant will be sketched in section 2.15 where the construction of a hyperplane arrangement is discussed.

2.4. Geometric transforms

Geometric transforms are tools that can often be exploited in the design of algorithms for geometric problems. Although a geometric transform per se is neither algorithmic nor the opposite, it helps reduce a problem to another which may already have an adequate algorithmic solution. Two prime examples of geometric transforms that have found many applications in computational geometry are polarity and inversion.

Let p be any point in $\mathbb{R}^d$. If p is different from the origin, o, we can define its *polar hyperplane*

$$p^* = \{x \in \mathbb{R}^d \mid \langle x,p\rangle = 1\},$$

where $\langle x,p\rangle$ is the scalar product of the vectors x and p. Intuitively, p^* is the hyperplane with normal-vector p ($p - o$ points to the side of p^* that does not contain o) whose distance from o is $1/|po|$. Inversely, for a hyperplane h, with $o \notin h$, we define its *polar point* h^* so that $(h^*)^* = h$.

A most useful property of this transform, henceforth referred to as *polarity*, is that it preserves/reverses sidedness with respect to o. For a hyperplane h, $o \notin h$, define h^- as the closed half-space bounded by h that does not contain o. Then we have

$$(p \in h \text{ iff } h^* \in p^*) \quad \text{and} \quad (p \in h^- \text{ iff } h^* \in (p^*)^-). \tag{2.1}$$

A problem where this is useful is the intersection of half-spaces. Let H be a set of n hyperplanes in $\mathbb{R}^d$, all avoiding the origin, and consider the problem of constructing the intersection of all half-spaces h^+, where h^+ is the closure of the complement of h^-. Assume for simplicity that $\mathcal{P} = \bigcap_{h\in H} h^+$ is a polytope, that is, $\mathcal{P}$ is bounded.

Because of (2.1), x is a point of $\mathcal{P}$ iff x^* is a hyperplane so that $h^* \in (x^*)^+$, for all $h \in H$. Since $o \in \text{int}(\mathcal{P})$ this is equivalent to saying that $x \in \mathcal{P}$ iff its polar hyperplane, x^*, avoids or supports the convex hull of $H^* = \{h^* \mid h \in H\}$. The points $x \in \text{bd}(\mathcal{P})$ correspond to the hyperplanes x^* supporting $\text{conv}(H^*)$. It follows that $\mathcal{P}$ and $\text{conv}(H^*)$ are dual polytopes[10] and thus the facial graphs of $\mathcal{P}$ and $\text{conv}(H^*)$ are the same. More precisely, the facial graph of $\mathcal{P}$ is obtained by turning the facial graph of $\text{conv}(H^*)$ upside down so that the subface relation is reversed. Therefore, constructing $\mathcal{P}$ is the same as constructing $\text{conv}(H^*)$ and we do not have to repeat the results mentioned in section 2.1 which now also apply to intersecting half-spaces[11].

[10] Two d-dimensional convex polytopes $\mathcal{A}$ and $\mathcal{B}$ are *dual* if there is an incidence preserving one-to-one correspondence between the k-faces of $\mathcal{A}$ and the $(d-k-1)$-faces of $\mathcal{B}$.

[11] To be fair we should mention that there are complications if the origin does not belong to $\text{int}(\mathcal{P})$ or if $\mathcal{P}$ is not bounded; they are ignored in the above paragraph.

Another useful geometric transform is the *inversion*. For any point $p \in \mathbb{R}^d$, $p \neq o$, define the *inverse point* $p^\circ = p/|po|^2$. Intuitively, p° is the point collinear with o and p that lies on the same side of o as p so that its distance from o is the inverse of the distance between p and o. It is convenient to add a point at infinity to $\mathbb{R}^d$ and to extend the inversion so that this point corresponds to o. For a point set S its inverse is defined as $S^\circ = \{p^\circ \mid p \in S\}$. For example, the inverse of a hyperplane h that contains o is h itself, and if $o \notin h$ then h° is a $(d-1)$-sphere that contains o. A $(d-1)$-sphere s maps to a hyperplane s°, if $o \in s$, and to another $(d-1)$-sphere, otherwise. Of course, $s = s^\circ$ iff s is the unit-sphere, that is, $s = \{x \in \mathbb{R}^d \mid |xo| = 1\}$.

Let now h be a hyperplane that avoids the origin and let $s = h^\circ$ be its inverse $(d-1)$-sphere. As mentioned before $o \in s$. Each point $x \in s$, $x \neq o$, corresponds to a unique point $x^\circ \in h$. This mapping from s to h is often referred to as the *stereographic projection* from the center o. It has many nice properties that have been exploited for various purposes in the past. For example, a $(d-2)$-sphere t on s maps to a $(d-2)$-sphere t° in h, unless $o \in t$. This will be important in section 2.6 where Delaunay triangulations are discussed. Another nice property is that the stereographic projection preserves angles, which has been used by Miller and Thurston (1990) to prove results about separators of d-dimensional triangulations.

2.5. *k-sets*

Counting k-sets of a finite set of points in $\mathbb{R}^d$ is a combinatorial geometry problem that is relevant to the analysis of various geometric algorithms. One example is the randomized incremental construction of Delaunay triangulations discussed in section 2.8. k-sets are also related to order-k Voronoi diagrams, see section 2.13, and certain collections of cells in hyperplane arrangements, see section 2.14. We need some definitions.

Let S be a set of n points in $\mathbb{R}^d$. For any closed half-space H we call $S' = S \cap H$ an $|S'|$-set of S. For example, S is its only n-set and $\{p\}$ is a 1-set of S iff $p \in \text{ext}(S)$. Let $f_k(S)$ be the number of k-sets of S, for $0 \leqslant k \leqslant n$, and define $f_k^{(d)}(n)$ equal to the maximum $f_k(S)$ over all sets S of n points in $\mathbb{R}^d$.

It is fairly straightforward to show that $\sum_{k=0}^{n} f_k(S) = O(n^d)$ (see section 2.14) which implies the trivial bound $f_k^{(d)}(n) = O(n^d)$. Proving better bounds for $f_k^{(d)}(n)$ is a notoriously difficult problem and no tight bounds are known except in trivial cases. In $\mathbb{R}^2$, the best lower bound is $\Omega(n \log k)$, see Erdős et al. (1973), and Edelsbrunner and Welzl (1985) for a simpler construction. Both references also show that $O(n\sqrt{k})$ is an upper bound for $f_k^{(2)}(n)$. This bound has been improved to $O(n\sqrt{k}/\log^* k)$[12] in Pach, Steiger and Szemerédi (1992). It is commonly conjectured that $f_k^{(2)}(n)$ is closer to its current lower bound, and an upper bound of the form $O(nk^\varepsilon)$, for every $\varepsilon > 0$, might not be impossible to prove.

[12] The function $\log^* k$, also known as the *iterated logarithm* of k, counts the number of times it is necessary to iterate the (binary) logarithm before the result is less than 2.

In general, the most difficult case seems to be when $k = n/2$, assuming n is even. For this value of k we define

$$h(S) = f_{n/2}(S) \quad \text{and} \quad h^{(d)}(n) = f_{n/2}^{(d)}(n).$$

The 2-dimensional lower bound can be extended to higher dimensions giving $h^{(d)}(n) = \Omega(n^{d-1} \log n)$, see Edelsbrunner (1987). The first non-trivial upper bound for $h^{(3)}(n)$ is due to Bárány, Füredi and Lovász (1989). They prove $h^{(3)}(n) = O(n^{3-\delta})$, for some small $\delta > 0$, using a reduction to a three-colored point problem in $\mathbb{R}^2$. Using different techniques, Aronov et al. (1991) improve this bound to $h^{(3)}(n) = O(n^{8/3} \log^{5/3} n)$. The question whether or not there are non-trivial upper bounds in all dimensions has recently been answered in the affirmative by Živaljević and Vrécica (1991). Using the reduction to a multi-colored point problem in one lower dimension, as suggested by Bárány, Füredi and Lovász (1989), they prove the existence of $\delta(d) > 0$ so that $h^{(d)}(n) = O(n^{d-\delta(d)})$.

A significantly easier problem than proving bounds for $f_k^{(d)}(n)$ is to bound sums of the form $g_k(S) = \sum_{i=0}^{k} f_i(S)$. Define $g_k^{(d)}(n)$ as the maximum $g_k(S)$ over all sets S of n points in $\mathbb{R}^d$. In two dimensions, Alon and Györi (1986) show that $g_k^{(2)}(n) = nk + 1$ and thus completely solve the problem. Using a probabilistic counting argument, Clarkson and Shor (1989) extend this bound to $g_k^{(d)}(n) = O(n^{\lfloor d/2 \rfloor} k^{\lceil d/2 \rceil})$. This bound is asymptotically tight as can be seen from the following example.

Let S be a set of n points on the moment curve $M_d = (\tau, \tau^2, \ldots, \tau^d)$ in $\mathbb{R}^d$. Any d points of M_d span a unique hyperplane h and decompose M_d into $d+1$ pieces. Every other piece lies on a chosen side of h. So we can count the i-sets of S by considering the number of ways $\lceil d/2 \rceil$ pairwise disjoint intervals, one unbounded if d is odd, can cover i points of a set of n points on a line. For a given i there are $\Omega(n^{\lfloor d/2 \rfloor} i^{\lceil (d-2)/2 \rceil})$ such ways which implies the lower bound $\Omega(n^{\lfloor d/2 \rfloor} k^{\lceil d/2 \rceil})$. It should be mentioned that the analysis in Clarkson and Shor (1989) considers k-sets in a much more general context that admits objects other than points and ranges other than half-spaces.

2.6. Delaunay triangulations

Let S be a finite point set in $\mathbb{R}^d$. A natural way to decompose the convex hull of S is to use simplices spanned by points of S. Formally, a *triangulation* of S is a simplicial cell complex[13] with vertex set S that decomposes conv(S). In some circumstances, but not in this section, the condition is relaxed to allow subsets of S as vertex sets.

[13] A *cell complex* $\mathcal{C}$ is a collection of closed sets so that the boundary of each $A \in \mathcal{C}$ is the the union of sets in $\mathcal{C}$, and if $A, B \in \mathcal{C}$ then $A \cap B \in \mathcal{C}$ and the relative interiors of A and B are disjoint. $\mathcal{C}$ is *simplicial* if each $A \in \mathcal{C}$ is a simplex of some dimension $k \leqslant d$.

A special triangulation of S is the so-called Delaunay triangulation which can be defined unambiguously if the points are in general position. In this context this means that $n = |S| > d$, no $d+1$ points lie in a common hyperplane, and no $d+2$ points lie on a common $(d-1)$-sphere. A d-simplex defined by a subset S' of $d+1$ points of S is *Delaunay* if all points of $S \setminus S'$ lie outside the unique sphere through the points of S'. The *Delaunay triangulation* of S, $\mathcal{D}(S)$, is defined by the Delaunay d-simplices and the lower-dimensional simplices in their boundaries.

$\mathcal{D}(S)$ is named after the Russian mathematician Boris Delone (also Delaunay). In a seminal paper, Delaunay (1934) proves the following important property of $\mathcal{D}(S)$. Let $\mathcal{T}$ be a triangulation of S, and let $f = \text{conv}(S'')$ be a facet (i.e., $(d-1)$-simplex) of $\mathcal{T}$; so $S'' \subseteq S$ and $|S''| = d$. Either f is a facet of $\text{conv}(S)$ or it is incident to two d-simplices $\text{conv}(S'' \cup \{a\})$ and $\text{conv}(S'' \cup \{b\})$ of $\mathcal{T}$. Then f is said to be *locally Delaunay* unless in the latter case the unique $(d-1)$-sphere through the points in $S'' \cup \{a\}$ encloses b. Delaunay proves that if all facets of $\mathcal{T}$ are locally Delaunay then $\mathcal{T} = \mathcal{D}(S)$.

It is instructive to observe the following relationship between Delaunay triangulations in $\mathbb{R}^d$ and convex hulls in $\mathbb{R}^{d+1}$. Identify $\mathbb{R}^d$ with some hyperplane h_0 in $\mathbb{R}^{d+1}$ so that h_0 does not contain the origin o of $\mathbb{R}^{d+1}$. So $S \subseteq h_0$ and $\mathcal{D}(S)$ is a d-dimensional triangulation in h_0. Define $s_0 = h_0^\circ$, the inverse d-sphere of h_0, and recall that $o \in s_0$. Define $S^\circ = \{p^\circ \mid p \in S\}$ and notice that $S^\circ \subseteq s_0$. By assumption, no $d+2$ points of S lie on a common $(d-1)$-sphere, which implies that no $d+2$ points of S° lie on a common (d-dimensional) hyperplane in $\mathbb{R}^{d+1}$. Furthermore, if $R \subseteq S$, $|R| = d+1$, spans a d-simplex of $\mathcal{D}(S)$ then $\text{conv}(R^\circ)$ is a (d-dimensional) facet of $\text{conv}(S^\circ)$. The reverse is not necessarily true. However, if $\text{conv}(Q^\circ)$ is a facet of $\text{conv}(S^\circ)$ so that o and S° lie on the same side of the hyperplane $\text{aff}(Q^\circ)$ then $\text{conv}(Q)$ is a d-simplex of $\mathcal{D}(S)$. A possible algorithm for constructing $\mathcal{D}(S)$ thus maps S to S°, computes $\text{conv}(S^\circ)$ in $\mathbb{R}^{d+1}$, and maps the relevant facets of $\text{conv}(S^\circ)$ back to $\mathbb{R}^d$. This algorithm has first been suggested by Brown (1979).

Delaunay triangulations have many nice properties, and we spend the remainder of this section to describe some of them. In two dimensions, $\mathcal{D}(S)$ is the one triangulation of S that maximizes smallest angles. Let us be more specific. Every triangulation of $S \subseteq \mathbb{R}^2$ has the same number of triangles and therefore the same number of angles (three per triangle). For two triangulations $\mathcal{A}$ and $\mathcal{B}$ of S, write $\mathcal{A} \prec \mathcal{B}$ if the non-decreasing angle vector of $\mathcal{A}$ is lexicographically less than that of $\mathcal{B}$. Sibson (1978) proves that $\mathcal{T} \prec \mathcal{D}(S)$ for all triangulations $\mathcal{T} \neq \mathcal{D}(S)$. Interestingly, the symmetric property of minimizing largest angles is not optimized by $\mathcal{D}(S)$. An algorithm that constructs the triangulation with lexicographically minimum non-increasing angle vector in time $O(n^2 \log n)$ has been described in Edelsbrunner, Tan and Waupotitsch (1992). Both properties are important in the application of triangulations to finite element analysis.

Somewhat surprisingly, no generalization to three dimensions of the maxmin angle property of 2-dimensional Delaunay triangulations has been found. Indeed, there are simple counterexamples for dihedral angles, solid angles, and 2-dimensional face angles. Nevertheless, Rajan (1991) proves the following optimality result for Delaunay triangulations in arbitrary dimensions. The *miniball*

of a d-simplex is the smallest ball that contains it. Among all triangulations of a finite point set $S \subseteq \mathbb{R}^d$, $\mathcal{D}(S)$ minimizes the size of the largest miniball of any of its d-simplices. Interestingly, this result does not generalize to the lexicographical ordering of non-increasing miniball vectors, not even in $\mathbb{R}^2$.

2.7. *Flipping*

Possibly the first reported algorithm for 2-dimensional Delaunay triangulations goes back to Lawson (1972), see also Lawson (1977). It is based on Delaunay's lemma (see section 2.6) and "flips" edges of the current triangulation that are not locally Delaunay. We give a more formal description of this method and a brief analysis below. The algorithm starts by constructing an arbitrary initial triangulation $\mathcal{T}$ of the set S in $\mathbb{R}^2$. All edges of $\mathcal{T}$ are tested and the ones that are not locally Delaunay are pushed on a stack $\mathcal{S}$[14]. The main part of the algorithm consists of the following loop.

while $\mathcal{S}$ is non-empty **do**
  ab := *pop*;
  let c and d be the other vertices of the triangles incident to ab,
  if they exist;
  if ab is not locally Delaunay **then**
    remove ab from $\mathcal{T}$, add cd, and *push* edges ac, cb, bd, da
  endif
endwhile.

An argument is needed to prove that this algorithm is correct. First note that if ab is not locally Delaunay then the quadrilateral $acbd$ is convex. Replacing ab by cd thus maintains that $\mathcal{T}$ is a triangulation. However, after removing one edge from the stack four more are pushed on the stack if the removed edge is flipped. It is thus not obvious that the while-loop ever terminates. Using the inversion transform in $\mathbb{R}^3$ we now show that the while-loop iterates at most $O(n^2)$ times, $n = |S|$.

Embed $\mathbb{R}^2$ as a plane h_0 in $\mathbb{R}^3$ that avoids the origin. Let $s_0 = h_0^\circ$ be the inverse sphere of h_0 and recall that $S^\circ = \{p^\circ \mid p \in S\}$ is a set of n points on s_0. For every triangulation $\mathcal{T}$ of S in h_0 we can construct a (not necessarily convex) polytope $\mathcal{T}^\circ$ as follows. For each triangle abc of $\mathcal{T}$, $a^\circ b^\circ c^\circ = \text{conv}(\{a^\circ, b^\circ, c^\circ\})$ is a triangle of $\mathcal{T}^\circ$, and for every edge ab of conv(S), $a^\circ b^\circ o$ is a triangle of $\mathcal{T}^\circ$. An edge ab of $\mathcal{T}$ is not locally Delaunay iff $a^\circ b^\circ$ is a reflex edge of $\mathcal{T}^\circ$. Because $\mathcal{T}^\circ$ is star-shaped with o in its kernel, the tetrahedron $a^\circ b^\circ c^\circ d^\circ$ lies completely outside $\mathcal{T}^\circ$,

[14] A *stack* is a simple data structure for storing a collection of objects – in this case edges – that remembers the order in which the objects are added to the collection. It supports operations *push* (add an object as the new topmost object to the collection) and *pop* (return the topmost object on the stack and remove it). Each operation takes constant time.

except for the two triangles $a^\circ b^\circ c^\circ$ and $a^\circ b^\circ d^\circ$ which it shares with $\mathcal{T}^\circ$. Flipping ab is therefore the same as expanding $\mathcal{T}^\circ$ so it includes this tetrahedron. The while-loop thus generates a 3-dimensional triangulation of $\text{conv}(S^\circ \cup \{o\})$ and it terminates when $\mathcal{T}^\circ = \text{conv}(S^\circ \cup \{o\})$. This view of the algorithm illustrates the close relationship between the 2-dimensional flipping operation and the notion of shelling the 3-dimensional triangulation, see Bruggesser and Mani (1971). Since every triangulation of a set of n points in $\mathbb{R}^3$ has at most $O(n^2)$ tetrahedra it must be that Lawson's flip algorithm halts after at most $O(n^2)$ steps.

Attempts to generalize this algorithm to $\mathbb{R}^d$, $d \geqslant 3$, have failed, and Joe (1989) indeed found triangulations $\mathcal{T} \neq \mathcal{D}(S)$ of finite point sets $S \subseteq \mathbb{R}^3$ that cannot be improved by flipping a locally non-Delaunay triangle. To understand how this can happen let us briefly examine what it means to do a flip in a 3-dimensional triangulation. Let abc be a triangle and let d and e be the other vertices of the two incident tetrahedra. There are three cases.

Case (1). The union of the two tetrahedra, $abcd \cup abce$, is convex. Then abc can be removed and the edge de together with the incident triangles dea, deb, dec can be added.

Case (2). $abcd \cup abce$ is non-convex, ab is the only reflex edge of this polytope, and $abde$ is also a tetrahedron of the triangulation. Then abc, abd, abe can be replaced by the triangle cde. (This is the inverse of the operation in Case 1.)

Case (3). $abcd \cup abce$ is non-convex and has two reflex edges, or its only reflex edge is incident to at least four triangles. Then no flip is possible.

Joe (1989) shows that there are triangulations $\mathcal{T} \neq \mathcal{D}(S)$ so that all locally non-Delaunay triangles belong to Case 3. In another paper, Joe (1991) proves that such an example is impossible if $\mathcal{D}(S)$ is constructed incrementally, that is, the Delaunay triangulation of the first $i-1$ points is completed before the ith point is added. The order in which the points are added is arbitrary. Edelsbrunner and Shah (1992) generalize this result to so-called regular triangulations (see section 2.13) and to arbitrary dimensions d. Their result will be assumed in the next section which describes a randomized incremental algorithm for constructing Delaunay triangulations in $\mathbb{R}^d$.

The above results on flipping raise an interesting topological question about triangulations. Let S be a set of n points in $\mathbb{R}^d$ and consider the collection N of all triangulations of S. Let A be the set of pairs $\{\mathcal{A}, \mathcal{B}\}$, $\mathcal{A}, \mathcal{B} \in N$, so that $\mathcal{A}$ can be obtained from $\mathcal{B}$ by a single flip[15]. The question is whether or not the graph (N, A) is connected. In $\mathbb{R}^2$, it is connected because every triangulation can be reduced to $\mathcal{D}(S)$ as shown above. The diameter of the graph is $O(n^2)$ because Lawson's flip algorithm takes only $O(n^2)$ steps before it reaches $\mathcal{D}(S)$. There are examples that show that this bound is tight. It is not known whether or not the graph (N, A) is connected for every finite set of points in general position in $\mathbb{R}^3$.

[15] A *flip* in a d-dimensional triangulation involves a set S' of $d+2$ points. If the d-simplices of the triangulation defined by subsets of size $d+1$ of S' decompose $\text{conv}(S')$, then the flip replaces these d-simplices by the other d-simplices defined by S'. See also Lawson (1986) for a discussion of flipping in $\mathbb{R}^d$.

2.8. Randomized incremental construction

A *randomized* algorithm is one that occasionally uses chance to determine how to proceed. There are various different types of randomized algorithms that people use. The algorithm – or more generally, the paradigm – described in this section produces a predictable output, only the running time and the amount of storage it uses depends on chance. The paradigm is called *randomized incremental*, which indicates that the solution is constructed incrementally by adding one object at a time. Chance is used to decide in which *sequence* the objects are added.

Why should one choose to add the objects in a random order and give up on the idea of arranging them in a favorable order? The answer is that there are many problems where most sequences are good, but it is still difficult to deterministically construct a sequence that is guaranteed to be good. Notice that "random" is not the same as "arbitrary". A *random sequence* can be any one of the $n!$ orderings of the n objects, and the probability of any particular sequence to occur is $1/n!$. The *expected time* of a randomized incremental algorithm is $1/n!$ times the total amount of time it would take if it solved the problem for every possible permutation.

Before we discuss the construction of Delaunay triangulations as a specific example, we should stress that the terms "randomized" and "expected time" average over all *permutations* of the input and in no way assume any probability distribution. Indeed, randomized algorithms are frequently rather insensitive to different distributions of the input objects.

Let S be a set of n points in $\mathbb{R}^d$. The first step of the algorithm chooses a random permutation of the points; let this permutation be $p_1, p_2, \ldots, p_n$. In order to reduce all different cases to one, the algorithm uses $d+1$ dummy points, $p_{-d}, p_{-d+1}, \ldots, p_0$, at infinity in directions so that all finite points lie in the d-simplex they span. Define $S_i = \{p_{-d}, p_{-d+1}, \ldots, p_i\}$, for $0 \leqslant i \leqslant n$. The main part of the algorithm can now be described as follows.

Construct $\mathcal{D}(S_0) = \text{conv}(S_0)$;
for $i := 1$ **to** n **do**
  (1) find the d-simplex $s = \text{conv}(S')$ in $\mathcal{D}(S_{i-1})$ that contains p_i;
  (2) decompose s using the $d+1$ d-simplices spanned by p_i and the facets of s;
  (3) use flipping until the triangulation obtained is $\mathcal{D}(S_i)$
endfor.

We will ignore details of how steps (2) and (3) are performed and elaborate later on step (1), the point location step. The algorithm as given above is a fairly straightforward generalization of the 2-dimensional algorithm due to Guibas, Knuth and Sharir (1990). The most difficult step in obtaining this generalization is the proof that flipping actually works, which is briefly discussed in section 2.7. We indicate the important steps in the analysis of the algorithm which, again, is an extension of the analysis given by Guibas, Knuth and Sharir (1990).

Let us begin by estimating the expected number of different d-simplices of the sequence of Delaunay triangulations, $\mathcal{D}(S_0), \mathcal{D}(S_1), \ldots, \mathcal{D}(S_n)$, constructed by the

algorithm. Consider a particular subset S' of $d+1$ points of S, and let b be the open ball bounded by the unique $(d-1)$-sphere through the points of S'. We call $|S \cap b|$ the *width* w of S'. The crucial observation is that $\text{conv}(S')$ is a d-simplex of one of the constructed Delaunay triangulations iff the $d+1$ points of S' precede the w points of $S \cap b$ in the input sequence. The probability for this to happen is one over $\binom{d+1+w}{d+1}$. The set $S \cap b$ corresponds to a w-set of S°. Recall that S° is the set of n points in $\mathbb{R}^{d+1}$ obtained from S by use of the inversion transform. As mentioned in section 2.5, S° has at most $O(n^{\lfloor (d+1)/2 \rfloor} k^{\lceil (d+1)/2 \rceil})$ w-sets, for $w \leqslant k$. The last two observations can be combined to show that the expected number of d-simplices of the $\mathcal{D}(S_i)$, for $0 \leqslant i \leqslant n$, is $O(n^{\lfloor (d+1)/2 \rfloor})$.

But what about the *transient* d-simplices that are constructed while adding p_i to $\mathcal{D}(S_{i-1})$ and are deleted before $\mathcal{D}(S_i)$ is reached? There cannot be substantially more transient d-simplices than non-transient ones. The reason is that the former are created by flips, and each flip destroys a non-transient d-simplex while creating at most d transient d-simplices. Assuming d is a constant, the expected number of d-simplices ever constructed by the algorithm is therefore at most $O(n^{\lfloor (d+1)/2 \rfloor})$.

Finally, we address the question of how to implement step (1) of the algorithm. This is indeed the step where this algorithm differs most strikingly from earlier Delaunay triangulation algorithms of Watson (1981) and Bowyer (1981). The idea is to build a hierarchy of d-simplices on top of the current triangulation. More specifically, this hierarchy is a directed acyclic graph (a dag) whose nodes are d-simplices. The dag has a single root which is the initial infinite d-simplex $\text{conv}(S_0)$. The leaves of the dag are the d-simplices of the current triangulation. Whenever a flip is performed, in steps (2) or (3), the involved d-simplices are not really deleted but rather assigned some children nodes, the new d-simplices. This effectively means that these d-simplices no longer belong to the current triangulation. In order to add a new point p_i, a path is traced from the root down to the leaf level visiting only d-simplices that contain p_i. By an argument similar to the one above, the expected number of such d-simplices cannot be very large. The analysis indeed shows that the time spent in step (1), summed over all p_i, is $O(n^{\lfloor (d+1)/2 \rfloor} + n \log n)$ in the expected case. This is therefore also an upper bound on the expected running time of the entire algorithm.

The randomized incrementation paradigm has found a number of other applications in computational geometry. These include linear programming, Clarkson (1988), Seidel (1991), finding the smallest enclosing sphere of a point set, Welzl (1991), and constructing generalized Voronoi diagrams, Mehlhorn, Meiser and Ó'Dúnlaing (1991).

2.9. Minimum spanning trees

Computing the *minimum spanning tree*, or MST, of a connected graph is a classic problem in graph algorithms, and many different algorithms are known, see, e.g., Cormen, Leiserson and Rivest (1990) or Tarjan (1983). One of the earliest methods is known as Prim's algorithm. It constructs the MST by breadth-first search, which is an incremental method. Let (N, A) be the graph. At any stage of the algorithm,

the set of nodes, N, is partitioned into the set T of nodes of the current tree, and the set $\bar{T} = N \setminus T$ of nodes that have not yet been added to the tree. Initially, $T := \{a\}$, for an arbitrary node $a \in N$. The main part of the algorithm can be described as follows.

while $\bar{T} \neq \emptyset$ **do**
(1) find the shortest edge $ab \in A$ with $a \in T$ and $b \in \bar{T}$;
(2) add ab to the tree, and set $T := T \cup \{b\}$ and $\bar{T} := \bar{T} \setminus \{b\}$
endwhile.

Using a priority queue[16], this algorithm can be implemented so it runs in time $O(n \log n + m)$, where $n = |N|$ and $m = |A|$.

Consider now the problem of computing a minimum spanning tree of a set S of n points in $\mathbb{R}^d$. This tree is the MST of (S, A), where $A = \binom{S}{2}$ and the length of an arc $ab \in A$ is the Euclidean distance between a and b. Using Prim's algorithm, it can be proven that every MST of S is in fact a subgraph of the Delaunay triangulation of S. For consider an edge ab added to the tree in the general step of the algorithm. At that moment, b is a point in $\bar{T}$ closest to a, and a is a point in T closest to b. It follows that the smallest d-sphere through a and b does not contain or enclose any other points of S. This implies that ab is an edge of $\mathcal{D}(S)$.

This subgraph property has been observed by Shamos and Hoey (1975) who also notice that it implies an $O(n \log n)$ time algorithm for constructing the MST of a set of n points in the plane: first construct $\mathcal{D}(S)$ and then use any efficient MST algorithm. In $d \geqslant 3$ dimensions, the Delaunay triangulation can have up to $\binom{n}{2}$ edges and is thus not of much use in the MST construction. Other sparse supergraphs of the MST are defined in Yao (1982), along with algorithms that take subquadratic time for any constant d. The complexity was later improved to expected time $O(n^{2-(2/(\lceil d/2 \rceil + 1))+\varepsilon})$, for any positive ε, by Agarwal et al. (1991).

2.10. Alpha shapes

What is the shape of a point cloud, that is, a finite point set S in $\mathbb{R}^d$? Certainly, the convex hull of S gives some indication of what one might intuitively feel the shape of S should be. Still, this answer can hardly be satisfying as $\text{conv}(S)$ might include large regions void of any points of S. Edelsbrunner, Kirkpatrick and Seidel (1983) answer this question for the 2-dimensional case by defining a one-parametric family of shapes of S. Edelsbrunner and Mücke (1992) generalize their result to $\mathbb{R}^3$. They also report experiences with an implementation of this concept and mention applications to chemistry, astronomy, biology and other sciences. The reason for

[16] A *priority queue* stores a collection of objects with priorities. In Prim's algorithm, the objects are the nodes in $\bar{T}$ and the priority of a node is one over the length of its shortest arc connecting it to a node in T, if it exists. A particular data structure that implements a priority queue is the Fibonacci heap, see Cormen, Leiserson and Rivest (1990). It admits the insertion of a new object and the improvement of the priority of an object in constant amortized time. The object with highest priority can be deleted in time logarithmic in the size of the collection.

the diverse set of applications is the important role the notion of shape plays in our intuitive understanding of reality and space.

Let S be a set of n points in $\mathbb{R}^d$. Call an open ball with radius $\alpha > 0$ an *α-ball*. For simplicity assume general position of the points, which now includes that no $d+1$ points lie on the boundary of a common α-ball, for any fixed α considered. Let S' be a subset of at most d points of S. Then $\text{conv}(S')$ is a face of the *α-shape* of S if there is an α-ball b with $S \cap b = \emptyset$ and $S \cap \text{bd}(b) = S'$. It should be obvious that a face of the α-shape is thus also a face of $\mathcal{D}(S)$. More specifically, a face belongs to some α-shape of S, for $\alpha > 0$, iff it belongs to $\mathcal{D}(S)$.

In Edelsbrunner, Kirkpatrick and Seidel (1983) the notion of an α-shape is also defined for negative values of α. In these cases, the α-shapes are based on the so-called furthest-point Delaunay triangulation. This triangulation is a natural counterpart of the ordinary, or closest-point Delaunay triangulation. For example, the facets of $\text{conv}(S^\circ)$, with S° the inverse image of S embedded in $\mathbb{R}^{d+1}$ (see sections 2.4 and 2.6), that do not correspond to d-simplices of $\mathcal{D}(S)$ are exactly the ones that correspond to d-simplices of the furthest-point Delaunay triangulation of S.

We now return to the relationship between α-shapes, for $\alpha > 0$, and $\mathcal{D}(S)$. Since every face of every α-shape of S is also a face of $\mathcal{D}(S)$, it is natural to choose the latter as an implicit representation of the entire collection of α-shapes. Furthermore, one can define that a d-simplex of $\mathcal{D}(S)$ belongs to the α-shape if its circumsphere has radius less than α. These d-simplices thus automatically triangulate the interior of the α-shape. Such a triangulation is handy in many applications. Each face of $\mathcal{D}(S)$ can be associated with a non-empty interval so that it belongs to the α-shape iff α lies in its interval. The Delaunay triangulation of S, each face annotated with its interval, is an appropriate data structure that admits relatively quick extraction of the α-shape, once a value of α is given. As explained in section 2.8, $\mathcal{D}(S)$ can be constructed in expected time $O(n^{\lfloor (d+1)/2 \rfloor} + n \log n)$, and the intervals can be computed in constant amortized time per face.

2.11. Voronoi diagrams

As usual, let S be a set of n points in $\mathbb{R}^d$. For each $p \in S$ define its *Voronoi cell* as the set of points $x \in \mathbb{R}^d$ so that the distance of x to every point of S is at least $|xp|$. The n Voronoi cells define a cell complex called the *Voronoi diagram* of S, $\mathcal{V}(S)$. Voronoi diagrams have a long history and are also known as Dirichlet or Voronoi tessellations, honoring the pioneering work of Dirichlet (1850) and Voronoi (1907, 1908). Generalizations of these diagrams will be discussed in section 2.13.

Let us assume for simplicity that no $d+1$ points of S lie on a common hyperplane and that no $d+2$ points lie on a common $(d-1)$-sphere. Under these assumptions, $\mathcal{V}(S)$ is a *simple* cell complex, that is, each vertex is incident to exactly $d+1$ edges. Every vertex thus belongs to exactly $d+1$ Voronoi cells, and by construction, the points of S generating these $d+1$ cells lie on a common $(d-1)$-sphere centered at the vertex. This implies that $\mathcal{V}(S)$ and $\mathcal{D}(S)$ are in fact dual cell complexes: $\text{conv}(S')$, with $S' \subseteq S$ and $|S'| = k+1$, is a k-face of $\mathcal{D}(S)$ iff the intersection of

the Voronoi cells generated by the $k+1$ points of S' is a $(d-k)$-face of $\mathcal{V}(S)$. So $\mathcal{V}(S)$ can be computed using any one algorithm that constructs the facial graph of $\mathcal{D}(S)$. Then turn this graph upside down, which effectively reverses the subface relationship.

A Voronoi diagram in $\mathbb{R}^d$ can also be obtained by projecting the boundary complex of a convex polyhedron in $\mathbb{R}^{d+1}$ onto a hyperplane in this space. Recall the inversion transform that embeds $\mathbb{R}^d$ as a hyperplane h_0 avoiding the origin o of $\mathbb{R}^{d+1}$. It maps h_0 to $s_0 = h_0^\circ$, which is a d-sphere through o. As explained in section 2.6, some of the faces of $\text{conv}(S^\circ)$ corresponds to faces of $\mathcal{D}(S)$. Now apply the polarity transform of section 2.4. This transform maps every point p° on s_0 to a hyperplane $h = (p^\circ)^*$ that is tangent to a paraboloid of revolution whose axis is orthogonal to h_0. For each such hyperplane h define h^+ as the half-space that contains o and define $\mathcal{H} = \bigcap h^+$. If we now project the boundary complex of $\mathcal{H}$ orthogonally back onto h_0, then we obtain the Voronoi diagram of $T = \{2p \mid p \in S\}$. After scaling down this diagram by a factor of 2 we get $\mathcal{V}(S)$.

The remainder of this section describes another geometric transform that relates $\mathcal{V}(S)$ with the lower envelope of a collection of vertical cones. Using this correspondence, one can give a sweep algorithm for computing $\mathcal{V}(S)$. This has been observed by Fortune (1987) who also shows that this algorithm has an efficient implementation in $\mathbb{R}^2$. The next section uses this algorithm as an illustration for the popular plane-sweep paradigm.

Represent $\mathbb{R}^d$ as the hyperplane h_0: $x_{d+1} = 0$ in $\mathbb{R}^{d+1}$. For each point $p \in S$, $S \subseteq \mathbb{R}^d$ define a real-valued function $f_p(x) = |px|$ measuring the Euclidean distance between p and $x \in \mathbb{R}^d$. The graph f_p of this function is the boundary of a cone with apex p whose axis is orthogonal to h_0. All points of f_p, except its apex, lie above h_0. The *lower envelope* of the collection f_p, $p \in S$, is defined as the graph f of

$$f(x) = \min\{f_p(x) \mid p \in S, x \in \mathbb{R}^d\}.$$

Think of f as a complex of k-faces, for $0 \leqslant k \leqslant d$, where each k-face is a subset of the intersection of $d+1-k$ cone surfaces. It should be obvious that the orthogonal projection of f onto h_0 is indeed $\mathcal{V}(S)$.

2.12. *Plane-sweep*

The idea of the plane-sweep paradigm is to move a vertical line continuously from left to right over the geometric input data. Computations are (almost) exclusively performed for objects in the near vicinity of the sweep-line. This method has found many applications in planar computational geometry. Typical examples of problems amenable to the plane-sweep paradigm are various types of 2-dimensional intersection computations.

The first algorithmic application of plane-sweep to geometric problems goes back to Shamos and Hoey (1976). They show that plane-sweep can be used to decide in $O(n \log n)$ time whether or not any two of a set of n line segments intersect. This has been extended to an $O(n \log n + k \log n)$ time algorithm for reporting all

k intersecting pairs by Bentley and Ottmann (1979). Chazelle and Edelsbrunner (1992) solve the same problem in time $O(n \log n + k)$, which is optimal. Nievergelt and Preparata (1982) describe several problems that can be solved using plane-sweep, including the overlay of two subdivisions in the plane. Using a "topological" variant of the plane-sweep paradigm (see also section 2.17), Guibas and Seidel (1987) show that $O(n+k)$ time is sufficient to overlay two convex subdivisions (all regions are convex), where n is the total number of edges and k edge pairs intersect. Under the polarity transform, a plane-sweep appears as a continuous rotation of or around the input data. This has been used to construct the visibility graph of a set of n pairwise disjoint line segments in time $O(n^2)$; see Welzl (1985) for the first such algorithm and Edelsbrunner and Guibas (1989) for an improvement to $O(n)$ working storage. A rather untypical but very interesting application of plane-sweep is the construction of 2-dimensional Voronoi diagrams, Fortune (1987), which is discussed in some detail below.

The idea of sweeping some higher-dimensional space can also be found in the discrete geometry literature. For example, Hadwiger (1955) uses it to prove Euler's relation for the faces of a convex polytope, and Alexanderson and Wetzel (1978) apply the sweep paradigm to count the number of faces in an arrangement of hyperplanes (see also section 2.14). The gist of the idea is to reduce a d-dimensional problem to a sequence of $(d-1)$-dimensional problems. Because of the abundance of efficient 1-dimensional data structures, this is particularly effective for 2-dimensional problems. Very few efficient algorithmic extensions of the sweep idea to dimensions beyond two are known.

The difficulty with the plane-sweep approach for computing $\mathcal{V}(S)$, $S \subseteq \mathbb{R}^2$, is that a point $p \in S$ that lies far to the right of the sweep-line, and thus has not yet been processed, can already influence the part of the diagram intersected by the sweep-line. This is where the geometric transform that relates each point $p \in S$ with the graph of the function $f_p(x) = |px|$ comes in. Let ℓ_t: $x_1 = t$ be the sweep-line at some moment in time t. Define $h_t(x)$ as the function that maps a point $x = (\xi_1, \xi_2) \in \mathbb{R}^2$ to $t - \xi_1$. Each f_p is the surface of a cone and h_t is a plane in $\mathbb{R}^3$. The intersection $f_p \cap h_t$ is empty if p lies to the right of ℓ_t, equal to a half-line if $p \in \ell_t$, and equal to a parabola if p lies to the left of ℓ_t. Define $u_{p,t}$ as the orthogonal projection of $f_p \cap h_t$ onto the plane h_0: $x_3 = 0$. Unless $u_{p,t}$ is empty it is a (possibly degenerate) parabola in $\mathbb{R}^2$, with focus point p and directrix ℓ_t.

At any point in time, t, the sweep algorithm maintains the set of parabolas $u_{p,t}$. Most important about these parabolas is that the *front*, defined as their right envelope[17], is the orthogonal projection of $f \cap h_t$ onto h_0. The front is a linear sequence of parabola pieces that meet at intersection points between parabolas. The intersection points lie on edges of $\mathcal{V}(S)$. As t increases continuously, the parabolas change continuously and the intersection points move on straight lines, the edges of $\mathcal{V}(S)$. A single parabola can contribute more than one connected piece to the front, but the total number of pieces is never more than $2n-2$, $n = |S|$. This

[17] Recall that each parabola has a horizontal axis and is thus a function from x_2 to x_1. The *right envelope* is the pointwise maximum of these functions.

is because there cannot exist two parabolas u and v so that a piece of u precedes a piece of v, this piece in turn precedes a piece of u, which precedes another piece of v. In short, the parabola pieces form a Davenport–Schinzel sequence of order 2. We refer to Sharir (1988) for a general discussion of such sequences and some of their applications.

The sweep algorithm must support two types of operations. When the sweep-line ℓ_t encounters a new point $p \in S$, then the parabola $u_{p,t}$ is inserted into the collection defining the front. Initially, $u_{p,t}$ is infinitely narrow so that inserting it is like splitting an old node and adding a new node in a linear list. By storing the front in a search tree, this operation can be carried out in time $O(\log n)$. The other type of operation is necessary when two intersection points of the front collide. This corresponds to the event that the front encounters a vertex of $\mathcal{V}(S)$. A parabola piece must be deleted, which again takes time $O(\log n)$. The total number of operations is $O(n)$ since there is only one insertion for each point in S and one deletion for each vertex of $\mathcal{V}(S)$. This implies a running time of $O(n \log n)$.

2.13. *Generalized Voronoi diagrams*

Generalizations of Voronoi diagrams are abundant in the computational geometry literature. This section makes an attempt to briefly introduce the more important of these generalizations and to mention some of their properties. A general reference in this context is Aurenhammer (1991). All generalizations are obtained either by replacing the Euclidean distance function with some other way of measuring distance, or by defining the diagram for more general objects than points, or both. In each case, we describe the diagram as the lower envelope of functions $f_p(x)$.

Non-Euclidean distance functions. This natural generalization of Voronoi diagrams replaces the Euclidean metric by some other metric or maybe even by a distance function that is not a metric. Define $f_p(x)$ equal to the distance from $p \in S$ to $x \in \mathbb{R}^d$. This view of generalized Voronoi diagrams is described in Edelsbrunner and Seidel (1986). An efficient randomized algorithm for the 2-dimensional case can be found in Mehlhorn et al. (1991).

Additive weight. Suppose the distance between a point $p \in S$ and a point $x \in \mathbb{R}^d$ is the Euclidean distance minus a weight $w_p \in \mathbb{R}$ of p. The resulting Voronoi diagram can be obtained from the graphs of the functions $f_p(x) = |px| - w_p$. The graphs are boundaries of cones with opening angle of 45° and with axes orthogonal to the hyperplane h_0: $x_{d+1} = 0$, just as in the standard case. However, their apices are no longer points in h_0. In $\mathbb{R}^2$, the plane-sweep algorithm of Fortune (1987), see section 2.12, can easily be adapted to construct the diagram for n weighted points in time $O(n \log n)$. Rosenberger (1991) generalizes this algorithm to order-k diagrams for weighted points, see below, and uses it to prove bounds on their number of faces. A geometric transform that reduces the d-dimensional problem to the intersection

of the cells of a power diagram for n points in $\mathbb{R}^{d+1}$ with cones in the same space can be found in Aurenhammer and Imai (1988).

Multiplicative weight. Let each point $p \in S$ have a positive real weight w_p, and define $f_p(x) = |px|/w_p$. Intuitively, the weight represents an attractive force of the point. The graphs f_p define boundaries of cones whose opening angles vary with the point weights. It can be shown that the region of a single point p, that is, the set of points x where p minimizes $f_p(x)$, is not necessarily connected. Indeed, even in $\mathbb{R}^2$ the diagram for n points can have $\Omega(n^2)$ cells.

The diagram defined by the graphs f_p is referred to as the *multiplicative weight Voronoi diagram* of S. It models entities with different degrees of attraction or different growth rates, and has been used in urban studies, see, e.g., Boots (1979). The first $O(n^2)$ time algorithm is due to Aurenhammer and Edelsbrunner (1984). Their approach uses a geometric transform that reduces the problem to constructing a convex cell complex in $\mathbb{R}^3$; the 2-dimensional diagram can be obtained by stereographic projection of the intersection of the cells with a 2-sphere. The cell complex is, in fact, the projection onto $\mathbb{R}^3$ of the boundary complex of a convex polyhedron in $\mathbb{R}^4$. More generally, the multiplicative weight Voronoi diagram for n points in $\mathbb{R}^d$ can be derived from the convex hull of n points in $\mathbb{R}^{d+2}$.

Power diagrams. As explained in section 2.11, a standard Voronoi diagram in $\mathbb{R}^d$ can be obtained by projecting onto $\mathbb{R}^d$ the boundary complex of a convex polyhedron in $\mathbb{R}^{d+1}$. All facets of this polyhedron are tangent to a common paraboloid of revolution. More general cell complexes, called *power diagrams*, can be obtained if the restriction to tangent facets is dropped, see, e.g., Aurenhammer (1991). Equivalently, one can define functions $f_p(x) = |px|^2 - w_p$, where $w_p \in \mathbb{R}$ is the weight of point $p \in S$. The obtained diagram is also known as the Dirichlet tessellation or the Voronoi diagram under the Laguerre metric. Computing the power diagram of n weighted points in $\mathbb{R}^d$ is equivalent to constructing the convex hull of n (unweighted) points in $\mathbb{R}^{d+1}$, see section 2.1. In analogy to Voronoi diagrams and Delaunay triangulations, each power diagram for points and weights in general position has a dual *regular* triangulation, see, e.g., Lee (1991).

Objects other than points. The Voronoi diagram for points with additive weights can also be viewed as a diagram for $(d-1)$-spheres in $\mathbb{R}^d$. For each sphere s, the distance of $x \in \mathbb{R}^d$ from s is defined as $f_s(x) = \min\{|yx| \mid y \in s\}$ if x lies outside s, and the negative of this minimum if x lies inside s. One can consider such diagrams for collections of more general classes of objects, such as lines, line segments, etc. A divide-and-conquer algorithm that constructs the Voronoi diagram for n pairwise disjoint simple curve segments in $\mathbb{R}^2$ can be found in Yap (1987). Already in $\mathbb{R}^3$ the problem seems to be much more difficult. For instance, no tight bounds for the maximum number of faces of such diagrams even just for n lines in $\mathbb{R}^3$ are currently known.

Order-k diagrams. Let S be a set of n points in $\mathbb{R}^d$. For any integer k, $1 \leqslant k \leqslant n-1$, consider the collection of subsets $T_k = \{S' \subseteq S \mid |S'| = k\}$, and for each $S' \in T_k$

define $f_{S'}(x) = \max\{|px| \mid p \in S'\}$. The implied Voronoi diagram for subsets of size k is called the *order-k Voronoi diagram* of S, $\mathcal{V}_k(S)$. For any $S' \subset S$ with $|S'| = k$, a point x in the interior of the Voronoi cell of S' is closer to every point in S' than to any point in $S \setminus S'$.

In general, not every subset S' can have a non-empty cell. Indeed, finding bounds for the maximum number of cells in the order-k Voronoi diagram of n points in $\mathbb{R}^d$ is an interesting problem. This number is closely related to the maximum number of k-sets (section 2.5) of n points on a d-sphere in $\mathbb{R}^{d+1}$. In $\mathbb{R}^2$, the problem has been solved by Fejes Tóth (1976) and Lee (1982) who independently derive the asymptotically tight upper bound $O(k(n-k))$. The latter reference also contains an algorithm that constructs $\mathcal{V}_k(S)$ in time $O(k^2 n \log n)$. No good upper bounds on the size of order-k diagrams are known in $\mathbb{R}^d$, $d \geqslant 3$. There is a relationship between order-k diagrams in $\mathbb{R}^d$ and certain subsets of cells in hyperplane arrangements in $\mathbb{R}^{d+1}$. This is briefly described in the next section.

2.14. *Arrangements*

A set H of n hyperplanes in $\mathbb{R}^d$ defines a cell complex known as the *arrangement* $\mathcal{A}(H)$ of H. $\mathcal{A}(H)$ is said to be *simple* if the hyperplanes of H are in general position. This means that $n \geqslant d$, every d hyperplanes meet in a common point, and no $d+1$ hyperplanes have a non-empty common intersection. For convenience, we will use this simplifying assumption and mostly consider simple arrangements. Under this assumption, every subset of d hyperplanes defines a unique vertex, which implies that $\mathcal{A}(H)$ has exactly $\binom{n}{d}$ vertices.

There are many ways to derive extensions of this formula that count the number of k-faces, for $0 \leqslant k \leqslant d$. Different proofs of these face-counting formulas can be found in the following non-exhaustive list of general references about arrangements. Grünbaum (1972) gives an extensive survey of results on line arrangements in $\mathbb{R}^2$, and Grünbaum (1971) discusses hyperplane arrangements. Zaslavsky (1975) studies arrangements in general dimensions and offers an analysis of face-counting formulas covering also the non-simple case. Edelsbrunner (1987) describes combinatorial as well as algorithmic properties of hyperplane arrangements. One of the earliest references on face-counting formulas for arrangements is Steiner (1826).

We now present a sweep argument to derive a formula for $\Phi_k^{(d)}(n)$, that is, the number of k-faces in a simple arrangement of n hyperplanes in $\mathbb{R}^d$. The number of k-faces indeed depends only on the number of hyperplanes in H and not on their positions, as long as they are in *general* position. So $\Phi_k^{(d)}(n)$ is well-defined. If the hyperplanes are in non-general position then we obtain an upper bound on the number of faces. Choose a vector u so that every hyperplane normal to u contains at most one vertex of $\mathcal{A}(H)$, and if it contains no vertex then it is in general position together with the n hyperplanes in H. Define $g_t = \{x \in \mathbb{R}^d \mid \langle x, u \rangle = t\}$, for all $t \in \mathbb{R}$, and consider the continuous movement of g_t as t increases from $-\infty$ to $+\infty$. Initially, g_t intersects $\Phi_{k-1}^{(d-1)}(n)$ k-faces of $\mathcal{A}(H)$ because the n hyperplanes intersect g_t in a simple $(d-1)$-dimensional arrangement. When $g(t)$ passes through a vertex of $\mathcal{A}(H)$ then it encounters $\binom{d}{d-k} = \binom{d}{k}$ new k-faces. This is because d hyperplanes

intersect in $\binom{d}{d-k}$ k-flats each containing exactly one new k-face. Thus,

$$\Phi_k^{(d)}(n) = \Phi_{k-1}^{(d-1)}(n) + \binom{d}{k}\binom{n}{d} = \sum_{i=0}^{k} \binom{d-i}{k-i}\binom{n}{d-i}.$$

The general interest in hyperplane arrangements stems from their many relations to other geometric concepts. For example, a polyhedron, which is the intersection of finitely many half-spaces, is naturally embedded as a cell (that is, a d-face) in the arrangement defined by the hyperplanes bounding the half-spaces.

The polarity transform associates H with the point set $H^* = \{h^* \mid h \in H\}$. By property (2.1) of the transform, a point $x \neq o$ in a cell of $\mathcal{A}(H)$ corresponds to a hyperplane x^* that avoids all points of H^* and thus splits H^* into two subsets, one on each side of x^*. More specifically, a point $h^* \in H^*$ lies on the same side of x^* as o iff points x and o lie on the same side of h. Similarly, h^* and o lie on opposite sides of x^* iff h separates x and o. The number of ways a set of n points can be partitioned by a hyperplane should therefore be the same as the number of cells in the polar hyperplane arrangement. This is not quite so since we define arrangements in Euclidean rather than projective space. Two unbounded cells that are opposite each other in $\mathcal{A}(H)$ correspond to the same partition of H^*, which implies that only half of the unbounded cells should be counted. In any case, $\Phi_d^{(d)}(n) = O(n^d)$ is an upper bound on the number of ways n points in $\mathbb{R}^d$ can be partitioned with a hyperplane, which thus proves one of the claims in section 2.5.

The remainder of this section elaborates a bit further on the relationship between k-sets of a set S of n points in $\mathbb{R}^d$ and cells in the polar arrangement $\mathcal{A}(H)$, $H = S^*$. No hyperplane in H contains o, so there is a unique cell in $\mathcal{A}(H)$ that contains o. Assign to this cell the separation number 0. For a general cell c of $\mathcal{A}(H)$ define the *separation number*, $\sigma(c)$, equal to the number of hyperplanes that separate c from o. What we said above implies that a cell c with $\sigma(c) = k$ corresponds to a partition of S into a k-set and an $(n-k)$-set. Thus, $f_k^{(d)}(n)$ is an upper bound on the number of cells in $\mathcal{A}(H)$ with separation number k (see section 2.5 for a definition of $f_k^{(d)}(n)$).

How does all this relate to order-k Voronoi diagrams? Let S be a set of n points in $\mathbb{R}^d$ identified with the hyperplane h_0 in $\mathbb{R}^{d+1}$ that avoids the origin o. Define $H = (S^\circ)^*$. That is, take the inverse image p° of each point $p \in S$ and replace it by its polar hyperplane. With these definitions, there is a one-to-one correspondence between the d-faces f of the order-k Voronoi diagram, $\mathcal{V}_k(S)$, and the cells c of $\mathcal{A}(H)$ with $\sigma(c) = k$ so that f can be obtained by orthogonally projecting c onto h_0 and scaling it down by a factor 2.

2.15. Zones of arrangements

Let H be a set of n hyperplanes in $\mathbb{R}^d$. A subset Z of the $O(n^d)$ cells of $\mathcal{A}(H)$ is a *zone* if there exists a hyperplane b so that the interior of a cell intersects b iff it belongs to Z. Since the hyperplanes in H cut b in a $(d-1)$-dimensional arrangement,

there are $\Phi_{d-1}^{(d-1)}(n) = \Theta(n^{d-1})$ cells in Z if we assume general position. Zones are interesting because an incremental algorithm that adds b to the arrangement needs to visit and update exactly all cells in the zone of b. Of particular interest is the combinatorial complexity of a zone, that is, the total number of faces bounding cells in the zone.

An important result on arrangements is the so-called zone theorem which states that the number of faces bounding the cells of any zone in any arrangement defined by n hyperplanes in $\mathbb{R}^d$ is $O(n^{d-1})$. The 2-dimensional version of this theorem has been proved independently by Chazelle, Guibas and Lee (1985) and Edelsbrunner, O'Rourke and Seidel (1986). The latter reference also gives a proof for arbitrary dimensions, and this proof is repeated in Edelsbrunner (1987), but it turned out that it works only for facets bounding zone cells. A complete proof of the zone theorem has recently been given by Edelsbrunner, Seidel and Sharir (1992). A generalization of the zone theorem to curves in the plane can be found in Edelsbrunner et al. (1988). The combinatorial complexity of the zone of a $(d-1)$-dimensional manifold in a hyperplane arrangement has been studied in Aronov and Sharir (1991).

We present below a proof for the zone theorem in $\mathbb{R}^2$ and subsequently sketch the general proof for dimensions $d \geqslant 3$. Let H be a set of n lines in $\mathbb{R}^2$, let b be another line, and assume general position, that is, $\mathcal{A}(H \cup \{b\})$ is simple. Index the lines of H as $h_1, h_2, \ldots, h_n$ so that their intersections with b are sorted. Pretending that b is horizontal, we can classify each edge e of any cell c as *left* or *right* depending on whether c lies to the right or the left of the (non-horizontal) line that contains e. The left and the right edges are counted separately. That is, we only count the left edges assuming the intersections of the h_i with b are sorted from left to right.

Consider the zone Z_i of b in the arrangement $\mathcal{A}(H_i)$, where $H_i = \{h_1, h_2, \ldots, h_i\}$. The next line, h_{i+1}, may intersect several cells in Z_i, but it can contribute only one new left edge, namely the one that bounds the new rightmost cell. It also may intersect two old left edges that become two new left edges each. Thus, Z_{i+1} has at most three more left edges than Z_i, which implies that the $n+1$ cells of the zone in $\mathcal{A}(H)$ have at most $6n$ edges altogether.

The proof of the zone theorem in $\mathbb{R}^d$, for $d \geqslant 3$, is entirely different from the one in $\mathbb{R}^2$. Moreover, this different proof gives only very loose bounds if applied to $\mathbb{R}^2$. The just presented $O(n)$ bound is thus needed to establish the base case of the d-dimensional induction. The general proof also gives only poor bounds for vertices and edges so that another argument using Euler's relation for convex polytopes is needed to cover these cases. So what is the main idea of the general proof?

Consider the zone of a hyperplane b in $\mathcal{A}(H)$. Remove one of the hyperplanes $h \in H$ and count the number of k-faces of the new zone that contain k-faces of the old zone. Each k-face of the new zone contains only one k-face of the old zone, except if it intersects h in a $(k-1)$-face that belongs to the zone of $b \cap h$ in the arrangement defined by the $(d-2)$-flats $g \cap h$, $g \in H$ and $g \neq h$. Do this

experiment n times, once for each hyperplane $h \in H$. Since a k-face of the original zone belongs to $d-k$ hyperplanes, it is counted $n-(d-k)$ times. Furthermore, in order to compensate for the k-faces that contain two k-faces of the original zone, we add the number of $(k-1)$-faces in a zone of $n-1$ hyperplanes in $\mathbb{R}^{d-1}$. Putting all this together, one can derive recurrence relations that imply the zone theorem.

2.16. Applications of arrangements

As mentioned earlier, an important implication of the zone theorem is that the straightforward incremental construction of an arrangement of n hyperplanes in $\mathbb{R}^d$ takes time $O(n^d+n\log n)$. This result in turn has many interesting consequences, a few of which are mentioned below.

Lambda matrices. The *λ-matrix* of a set S of n points in $\mathbb{R}^d$ has an entry for every sequence of d points, see Goodman and Pollack (1983). If the d points span a unique hyperplane h then this entry is the number of points of S in the positive open half-space of h. Otherwise, the entry is undefined. Constructing this matrix can be thought of as a generalization of sorting n numbers, that is, points in $\mathbb{R}^1$. The matrix can be derived in time $O(n^d)$ from the arrangement $\mathcal{A}(S^*)$, which thus gives an optimal $O(n^d)$ time algorithm.

Order-k Voronoi diagrams. As explained in section 2.14, the order-k Voronoi diagram of a set S of n points in $\mathbb{R}^d$ corresponds to the collection of cells with separation number k in $\mathcal{A}((S^\circ)^*)$ in $\mathbb{R}^{d+1}$. The separation number of each cell can be computed in constant amortized time[18] by searching the graph of cells connected by arcs reflecting the adjacency relation. This leads to an optimal $O(n^{d+1})$ time algorithm for constructing all order-k Voronoi diagrams of S, for $1 \leqslant k \leqslant n-1$.

Sums of products. Consider $\mathcal{A}(H)$ in $\mathbb{R}^d$, and for a hyperplane $h \in H$ count the number of k-faces bounding cells that have a facet in h. The zone theorem implies that this quantity is $O(n^{d-1})$. The sum of these quantities over the n hyperplanes of H is therefore $O(n^d)$. The same sum can be obtained in a different way. For each cell of $\mathcal{A}(H)$ consider the number of facets times the number of k-faces. The above sum is the same as the sum of these products taken over all cells of $\mathcal{A}(H)$. It thus follows that $O(n^d)$ is also an upper bound for the latter sum.

Minimum simplices. Consider $S' \subseteq S$, $|S'| = d+1$, that defines the d-simplex $\mathrm{conv}(S')$ with smallest d-dimensional measure. This subset S' is unique if we assume general position of the points. Take a subset $S'' \subseteq S'$, $|S''| = d$, and let v be the intersection of the d hyperplanes in $(S'')^*$. This point v is a vertex of $\mathcal{A}(S^*)$. The point $p \in S' \setminus S''$ has the property that no other hyperplane in S^* intersects the line $\ell = \mathrm{aff}(\{v, o\})$ between v and $\ell \cap p^*$. All subsets $S' \subseteq S$ with this property can be

[18] Constant *amortized time* means that the total time, summed over all cells, is proportional to the number of cells. This does not exclude the possibility that the effort for one particular cell is more than constant time.

enumerated by testing all facet-vertex pairs of all cells of $\mathcal{A}(S^*)$. As argued above, the total number of such pairs is at most $O(n^d)$. Thus, the minimum measure d-simplex defined by $d+1$ points of S can be found in time $O(n^d)$.

Zonotopes. A *zonotope* is a particular kind of convex polytope that can be defined as the Minkowsky sum[19] of a finite collection of line segments in $\mathbb{R}^d$. Alternatively, let S be a finite point set in $\mathbb{R}^d$ and define

$$\mathcal{Z}(S) = \{x \in \mathbb{R}^d \mid x = \sum_{p_i \in S} \alpha_i p_i, \text{ with } \alpha_i \in \mathbb{R} \text{ and } -1 \leqslant \alpha_i \leqslant 1\}.$$

All translates of $\mathcal{Z}(S)$ are zonotopes. As demonstrated by Coxeter (1962), $\mathcal{Z}(S)$ is structurally related to the $(d-1)$-dimensional hyperplane arrangement $\mathcal{A}(H)$ obtained as follows. Let h_0 be a hyperplane in $\mathbb{R}^d$ that avoids the origin, and define h_p: $\langle x, p\rangle = 0$. Then $H = \{h_p \cap h_0 \mid p \in S\}$. This correspondence between zonotopes and arrangements implies an $O(n^{d-1} + n \log n)$ time algorithm for constructing $\mathcal{Z}(S)$.

Counting oriented matroids. We finally mention the relationship between hyperplane arrangements and oriented matroids, see Björner et al. (1992). An *oriented matroid* is a combinatorial description of a pseudo-hyperplane arrangement. In a nutshell, the latter is the same as a hyperplane arrangement, except that the pseudo-hyperplanes are not necessarily straight. In particular, the intersection patterns of pseudo-hyperplanes are locally the same as for hyperplanes. The study of pseudoline arrangements in $\mathbb{R}^2$ is fairly old and goes back at least as far as Levy (1926). The zone theorem also holds for pseudo-hyperplane arrangements, and the incremental algorithm can just as well be used for pseudo-hyperplanes, at least abstractly. This implies that there are at most $2^{O(n^d)}$ different arrangements of n pseudo-hyperplanes in $\mathbb{R}^d$, $d \geqslant 2$. There are examples that show that the asymptotic order of the exponent is tight. This bound should be contrasted with the result of Goodman and Pollack (1986) that there are only $2^{O(n \log n)}$ arrangements of n straight hyperplanes.

2.17. *Topological sweep*

As indicated in section 2.16, there is quite a number of computational geometry problems that can be solved by constructing a hyperplane arrangement. In many cases, however, it is not necessary to have the entire arrangement represented in storage. Often it suffices to examine the cells, one by one, without storing them after they have been inspected. A typical such example is the minimum measure d-simplex defined by a set of n points in $\mathbb{R}^d$, see section 2.15. Unfortunately, the incremental construction of an arrangement requires the entire cell complex be explicitly represented when the next hyperplane is added. An alternative method is based on the idea of sweeping the arrangement with another hyperplane. This idea has been used in section 2.14 to count the number of faces in a simple arrangement.

[19] The *Minkowsky sum* of $A, B \subseteq \mathbb{R}^d$ is the set $A + B = \{x = a + b \mid a \in A, b \in B\}$.

This section addresses the question how this idea can be made algorithmic. Most importantly, a topological variant of the sweep idea is discussed that leads to an algorithm that sweeps an arrangement in constant amortized time per face.

As usual, let H be a set of n hyperplanes in $\mathbb{R}^d$, and let g_t be the sweep hyperplane. For simplicity, assume that at any point in time, t, g_t contains at most one vertex of $\mathcal{A}(H)$, and that at all other times $\mathcal{A}(H \cup \{g_t\})$ is simple. Denote by $\mathcal{A}_t$ the $(d-1)$-dimensional arrangement defined by the $(d-2)$-flats $h \cap g_t$, $h \in H$. Intuitively, $\mathcal{A}_t$ is the intersection of g_t with $\mathcal{A}(H)$. Notice that although $\mathcal{A}_t$ changes continuously with t, its combinatorial structure changes only when g_t passes a vertex of $\mathcal{A}(H)$. At that time, a $(d-1)$-simplex of $\mathcal{A}_t$ collapses and its symmetric image appears. It is easy to change the facial graph that represents $\mathcal{A}_t$ accordingly. Call this an *elementary step*.

It is important that the elementary steps are performed in the correct sequence corresponding to the order the vertices of $\mathcal{A}(H)$ are encountered by g_t. This sequence can be found by maintaining a priority queue whose entries are the $(d-1)$-simplices is $\mathcal{A}_t$. There are at most $O(n^{d-1})$ such simplices. Each one is defined by d hyperplanes of H, and these hyperplanes intersect in a unique vertex. The $(d-1)$-simplex collapses exactly at the time g_t encounters this vertex. So the priority of a $(d-1)$-simplex should be higher for $(d-1)$-simplices that collapse earlier; simplices whose vertices have already been passed do not have to be stored in the priority queue. The total amount of storage is thus $O(n^{d-1})$, for $\mathcal{A}_t$ and for the priority queue, and the total amount of time is $O(n^d \log n)$, namely $O(\log n)$ time per elementary step.

An obvious question is whether or not the sweep can be improved to optimal time $O(n^d)$. It turns out this is possible, and to achieve this goal, it is necessary to replace the priority queue by some other means of deciding which elementary step to perform next. There is a danger of a premature elementary step because a $(d-1)$-simplex in $\mathcal{A}_t$ can get additional facets before it becomes a simplex again and collapses. The crucial dimension for the improvement is two. This is because a sweep of a d-dimensional arrangement, for $d \geqslant 3$, can be carried out by coordinating sweeps of the $\binom{n}{d-2}$ 2-dimensional subarrangements, see Anagnostou, Guibas and Polimenis (1990) for details of this idea in $\mathbb{R}^3$. The remainder of this section sketches the improved sweep method in $\mathbb{R}^2$, as described in Edelsbrunner and Guibas (1989).

Before proceeding to the details of the 2-dimensional sweep, let us point out that our approach does not attempt to find the elementary steps in their properly ordered sequence. For if we could find this order in time $O(n^2)$ – and this is the time bound that we are aiming for – we could also sort $X + Y$ [20] in time $O(n^2)$, see, e.g., Fredman (1976). It is still open whether or not this is possible.

Let $H = \{h_1, h_2, \ldots, h_n\}$ be the set of lines defining the arrangement. Assume

[20] Let $X = (x_1, x_2, \ldots, x_n)$ and $Y = (y_1, y_2, \ldots, y_n)$ be two sorted vectors of real numbers. $X + Y$ is the matrix $(a_{ij})_{1 \leqslant i,j \leqslant n}$ with $a_{ij} = x_i + y_j$. The problem is whether or not there is an algorithm that sorts the a_{ij} in time $O(n^2)$. Obviously, $O(n^2 \log n)$ time suffices, and Fredman (1976) shows that $O(n^2)$ comparisons are also sufficient.

that no line is vertical and that the slope of h_i is less than that of h_j if $i < j$. The main idea of the improved sweep algorithm is to replace the straight sweep line by a topological sweep line g_t. This topological line is a simple connected curve whose unbounded ends always reside in the topmost and the bottommost cells of $\mathcal{A}(H)$. Furthermore, g_t intersects every line in exactly one point. It follows that in general its intersection with $\mathcal{A}(H)$ is a sequence of n points, just as it is for a straight sweep line.

The second important idea is the data structure that can be used to decide which elementary step to take next. It consists of two trees, the so-called upper and lower horizon trees which identify possible elementary steps, and a stack, which stores all possible elementary steps. The trees are symmetric to each other, so it suffices to describe one of them. Extend each line h_i to the right, starting at its intersection point with g_t, until it hits a line h_j with $i < j$. Call h_j the *parent* of h_i. The parent relation defines a forest[21], and a tree can be obtained by artificially adding a line with infinitely large slope to the right of all vertices of $\mathcal{A}(H)$. This is the *upper horizon tree.*

Now orient g_t from top to bottom and let h_i and h_j be two lines so that h_i intersects g_t immediately before h_j does. Then h_i and h_j can be swapped in an elementary step iff $i < j$, h_j is parent of h_i in the upper horizon tree, and h_i is parent of h_j in the lower horizon tree. When such a configuration is detected it is pushed on the stack awaiting further processing.

The crucial question is how to maintain the two trees during the sweep. When h_i and h_j are swapped, then h_i loses its parent in the upper horizon tree, and a new parent needs to be found. This is done by traversing the tree upward, starting at line h_k that intersects g_t immediately after h_j, until the new parent of h_i is found. The traversed path can have length $\Omega(n)$, but it can be shown that the total length of all $O(n^2)$ search paths is $O(n^2)$, see Edelsbrunner and Guibas (1989, 1991). This implies the desired running time of $O(n^2)$.

Edelsbrunner and Guibas (1989) provide a fairly extensive list of applications of this topological sweep method to geometric problems. These include finding a longest convex chain and a largest convex subset of a finite point set in $\mathbb{R}^2$, two problems with prominent place in the history of combinatorial geometry, see Erdős and Szekeres (1935). Another application, described in Edelsbrunner and Souvaine (1990), is finding the regression line that minimizes the median distance to any point of a given finite set in $\mathbb{R}^2$. This application requires that the choice of the next elementary step is slightly more selective than in the original topological sweep method described above.

3. Concluding remarks

The above seventeen technical sections are barely enough to give the reader an impression of topics studied in the area of geometric algorithms. The author of

[21] A *forest* is a collection of disjoint trees.

this chapter still hopes that these sections convey some of the principles and ways of thinking and working that currently predominate. The goal is usually to find the fastest and most efficient method for carrying out geometric operations and constructions. The tools are mostly borrowed from algorithm design and analysis and the structural and combinatorial study of geometric concepts.

It should be added, however, that much more is necessary to come up with the final product, a computer program. Indeed, numerical or symbolic methods are needed to implement the geometric primitives neglected in this article. In order to produce a useful piece of software one also needs to worry about graphical user-interfaces, software design and maintenance, and other issues dealt with in applied areas of computer science. These final words are added in the hope they will convince researchers in mathematics and computer science that there is a continuum between our fields of interests. Once we cross the borders we discover that there are far fewer walls than imagined.

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Živaljević, R.T., and S.T. Vrécica
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Author Index

Abbott, H. 420
Abe, M. 462
Aberth, O. 207, 208
Abramowitz, M. 1142
Adams, R.A. 1139
Adhikari, A. 368
Adler, I. 523, 635, 646
Adler, I., *see* Monteiro, R.D.C. 633, 662
Aeppli, A. 1074
Affentranger, F. 154, 321, 1384, 1395, 1398, 1399, 1402
Agarwal, P.K. 717
Aggarwal, A. 693
Agmon, S. 651, 652
Aho, A.V. 633, 693, 701, 769, 872
Aigner, A. 565
Akgül, M. 651
Akgül, M., *see* Maurras, J.F. 652
Akhmerov, R.R. 1229, 1233
Aksionov, V.A. 410
Aksoy, A.G. 1226, 1232, 1233
Alagar, V.S. 1395
Aleksandrov, A.D. 8, 10, 12, 14, 23, 30, 52, 53, 57, 58, 61–63, 76, 153, 167, 229, 231, 232, 243, 245, 247, 250, 280, 284, 285, 288–292, 295, 306, 363, 916, 942, 1057, 1059, 1062, 1065, 1071, 1072, 1280, 1299, 1300, 1331–1333
Aleksandrov, P.S. 402, 430
Alekseev, V.M. 1120
Alexander, E. 1030
Alexander, J.M., *see* Saaty, T.L. 865, 885, 886
Alexander, R. 358, 368, 1320
Alexander, S., *see* Olamy, Z. 887
Alexanderson, G.L. 720
Alexandrova, D.A., *see* Kotzev, J.N. 1030
Alfsen, E.M. 356, 1307
Allen, C.S. 1231
Almgren, F.J. 6
Alon, N. 408, 420, 426, 455, 474, 509, 525, 589, 711, 1172
Altshuler, A. 526, 537, 539, 541, 542, 545, 558, 580, 582, 584, 586
Altshuler, A., *see* Brehm, U. 539
Ambartzumian, R.V. 12, 1320, 1351
Amiouny, S.V. 885
Amir, D. 360, 394, 1160, 1172, 1189
Ammann, R. 922
Ampère, A.M. 7
Anagnostou, E.G. 728
Andalefte, E.Z., *see* Freese, R.W. 30
Anderson, R.D. 278, 1331
Andreev, E.M. 889
Andriyasyan, A.K. 757
Anikonov, Yu.E. 148, 1284
Anstreicher, K.M. 656
Antonin, C. 577
Araujo, A. 1405
Archimedes 4
Arnold, R. 130, 139, 310, 322, 338, 366, 1277
Arnold, V.I. 781
Aronov, B. 426, 711, 725
Aronsson, G. 1141
Arrow, K.J. 1406
Artstein, Z. 1405, 1406
Arzelà, C. 11, 12
Ascher, E. 1024
Aschkinuse, W.G. 352
Ascoli, G. 11, 12
Ash, A. 621
Ash, P. 253
Ashbaugh, M.S. 1143
Asimow, L. 232, 239, 243, 245–247
Asplund, E. 276, 309, 409, 1096, 1331, 1338
Assouad, P. 1306
Atiyah, M.F. 621
Attouch, H. 1087
Audier, M., *see* Guyot, P. 1033
Audin, M. 621
Aumann, G. 26, 364
Aumann, R.J. 1404
Aurenhammer, F. 721, 722
Avis, D. 405, 644, 646, 705
Avriel, M. 630

Babai, L. 751
Babenko, I.K. 188
Babenko, V.F. 756
Bachem, A. 557, 558, 564, 584, 588
Bachem, A., *see* Kannan, R. 742
Backus, G. 1279
Baddeley, A.J. 975, 1428
Baddeley, A.J., *see* Jensen, E.B. 1428
Bagley, B.G. 887, 1033
Bair, J. 356
Baire, R. 12, 1329
Bajmóczy, E.G. 429
Bakel'man, I.Ja. 291
Baker, B.S. 883
Baker, B.S., *see* Brown, D.J. 884
Baker, M.J.C. 401, 405, 431, 434
Balas, E. 690, 890
Balinski, M.L. 520, 685
Ball, J.M. 1128, 1129
Ball, K. 15, 157, 163, 164, 166, 167, 358, 360, 1170, 1176, 1180, 1183, 1209, 1210, 1313
Ballicioni, A., *see* Couderc, P. 355
Bambah, R.P. 9, 14, 745, 750, 757, 771, 814, 839, 840, 848, 871, 887
Banach, S. 13, 15
Banaszczyk, W. 745, 751
Banchoff, T. 537, 543, 545, 942
Banchoff, T.F. 351
Bandle, C. 7, 56, 76, 1133
Bandt, Chr. 308, 311, 312
Bang, T. 1210
Bangert, V. 22, 29, 280, 1058
Bank, B. 682
Bannai, E. 458, 849
Bannai, E., *see* Bannai, E. 458
Bantegnie, R. 30
Baraki, G., *see* Bandt, Chr. 311
Baranovskiĭ, E.P. 752, 810, 917, 1008
Baranovskiĭ, E.P., *see* Ryškov, S.S. 752, 753, 756, 814, 844, 917
Bárány, I. 158, 357, 360, 400, 405, 426, 429, 431, 432, 456, 502, 711, 781, 784, 850, 1335, 1336, 1398–1402
Bárány, I., *see* Bajmóczy, E.G. 429
Bárány, I., *see* Böröczky, K. 818
Barnes, E.R. 659
Barnes, E.S. 756
Barnette, D.W. 13, 253, 494, 500, 502, 504, 505, 522, 523, 527, 537, 539, 541, 542, 544, 545, 644, 921
Baronti, M. 310
Barry, J.E., *see* Petty, C.M. 153
Bartels, H.G. 691
Bartels, S.G., *see* Bartels, H.G. 691
Barthel, W. 363
Bartholdi, J.J. 884
Bartholdi, J.J., *see* Amiouny, S.V. 885
Baston, V.J.D. 355
Batyrev, V.V. 621
Bauer, H. 15, 634
Baumgartner, L. 919
Bavaud, F. 365
Bayer, D. 657, 658, 662
Bayer, M.M. 503, 505, 506, 508, 516, 517, 519, 520, 527, 528, 590
Beale, E.M.L. 642
Beck, A. 467, 1161
Beck, J. 1205
Beckenbach, E.F. 1084
Beer, G.A. 33, 310, 311, 1087
Behrend, F. 15, 163, 164, 357, 360, 652
Belickii, G.R. 1247
Beling, P.A., *see* Adler, I. 635
Bell, D.E. 783
Ben-Or, M. 704
Ben-Sabar, E., *see* Davis, P.J. 323, 338
Bender, E.A. 524, 773
Benders, J.F. 680
Benguria, R.D., *see* Ashbaugh, M.S. 1143
Bennett, G. 1174
Benson, D.G. 64
Benson, R.V. 355, 363
Bentley, J.L. 720, 1401
Benyamini, Y. 1195, 1196
Benzer, S. 398, 418
Berard, P. 76
Berend, D. 926
Beretta, L. 364
Berezovskii, O.A., *see* Shor, N.Z. 338
Berg, C. 290, 971, 972, 1264, 1265, 1308–1310
Berge, C. 398, 406, 683
Berger, K.H. 360
Berger, M. 76, 84, 353, 354, 360, 1334
Berger, R. 922
Berman, J.D. 200
Bernshtein, D.N. 605, 781, 944
Bernstein, F. 131, 183
Berwald, L. 12, 1280, 1358, 1378
Besicovitch, A.S. 12, 183, 184, 193, 205, 208
Best, M.R. 818

Betke, U. 10, 66, 324, 335, 542, 544, 545, 547, 652, 768, 771, 773, 774, 778, 780–782, 870, 879, 943, 977, 979, 980, 1205, 1321, 1419
Bezdek, A. 472, 813, 821, 834, 841, 844, 846–848, 918
Bezdek, K. 183, 367, 471, 472, 803, 833, 847
Bezdek, K., *see* Bezdek, A. 472, 847, 848
Bialostocki, A. 456
Bianchi, G. 187, 358, 359
Bieberbach, L. 13, 65, 906, 995, 1013
Bielawski, R. 32
Bielecki, A. 409
Bienia, W. 424
Bieri, H. 60
Bigalke, H.-G. 353
Bigdeli, F. 511
Bilinski, S. 548, 913
Billera, L.J. 13, 34, 255, 491, 494, 495, 499, 500, 515, 516, 519, 575, 588, 623
Billera, L.J., *see* Bayer, M.M. 503
Billiet, Y. 1026
Birch, B.J. 394, 427
Birkhoff, G. 358, 685
Birnbaum, Z. 1089
Bishop, E. 21
Bisztriczky, T. 456
Björck, G. 29
Björner, A. 35, 415, 418, 420, 491, 509, 517–519, 521, 525, 528, 539, 558, 564, 570, 584–587, 727
Blanc, E. 31
Bland, R.G. 557, 558, 564, 584, 592, 641, 643, 651
Blaschke, W. 6, 7, 10–12, 15, 46, 48, 50, 55–57, 60, 64, 66, 76, 133, 154, 164, 166, 182, 187, 196, 247, 281, 290, 310, 325, 329, 353, 357–360, 941, 975, 1055, 1078, 1179, 1270, 1272, 1276, 1277, 1279, 1280, 1283, 1287–1289, 1299, 1301, 1308, 1310, 1329, 1351, 1355, 1358, 1359, 1375, 1381, 1383, 1396
Blatter, C. 364
Blech, I., *see* Shechtman, D. 925, 1031
Bleicher, M.N. 466, 820
Bleicher, M.N., *see* Beck, A. 467
Blichfeldt, H.F. 5, 9, 747, 770, 772, 810, 814
Blind, G. 352, 367, 494
Blind, R. 31, 493
Blind, R., *see* Blind, G. 352, 367, 494
Bloh, E.L. 825
Blokhuis, A. 457, 458
Bodlaender, H.L. 634, 789, 873
Boerdijk, A.H. 887
Bohm, J. 1030
Böhm, J. 214, 363–365, 939, 974
Bohnenblust, H.F. 393, 400
Bokowski, J. 9, 34, 51, 65, 337, 518, 519, 525, 537, 540, 543, 546, 560, 561, 563, 564, 573, 575, 580, 582–586, 590, 592, 773, 774, 1365
Bokowski, J., *see* Altshuler, A. 580
Bol, G. 11, 54, 59, 62, 63, 183, 209–211, 310, 363, 1270, 1286
Boland, J.C., *see* Lekkerkerker, C.G. 418
Bolker, E.D. 335, 362, 1204, 1299, 1300, 1305, 1310
Bolker, E.D., *see* Ash, P. 253
Boll, G. 131, 133
Bolle, U. 822
Bollobás, B. 435
Boltianskii, V.G. 46
Boltjanski, W.G. 355
Boltjanskii, V.G. 30, 471
Boltjansky, V.G. 470, 471
Boltyanskii, V.G. 470
Boltyanskiĭ, V.G. 357, 363, 365, 939
Boltyanskiĭ, V.G. 399
Boltyanskiĭ, V.G., *see* Yaglom, I.M. 363, 434
Boltyansky, V.G. 469, 471, 472
Bolza, O. 1108, 1120
Bonnesen, T. 6, 14, 45, 47, 50, 51, 55, 56, 58, 59, 64, 65, 76, 129, 131, 156, 181–184, 209, 275, 353, 356, 358, 359, 361, 363, 364, 368, 430, 648, 772, 1263, 1264, 1279, 1299, 1308
Bonnet, P.O. 14
Bonnice, W.E. 430, 432, 433, 454
Boots, B.N. 722
Borell, C. 1158, 1180
Borgwardt, K.H. 646, 1402
Borisovich, Yu.G. 1230
Böröczky, K. 14, 202, 364, 818, 831–834, 844, 848, 849
Böröczky Jr, K. 870, 871
Böröczky, K., *see* Bezdek, A. 847
Boros, E. 431, 473
Borsuk, K. 468
Borwein, J.M. 28, 304, 435, 1229
Bos, A. 9
Bourgain, J. 11, 15, 55, 138, 148, 154, 156, 167, 168, 310, 335, 336, 360, 407, 745, 751, 778, 1159, 1160, 1167, 1170, 1173, 1174,

Bourgain, J. (*cont'd*) 1178–1180, 1182, 1184, 1186–1188, 1190, 1192, 1196, 1197, 1199, 1200, 1202, 1205–1208, 1281, 1282, 1314, 1321
Bourgin, D.G. 25
Bourgin, R.D. 1097
Bouwkamp, C.J. 463
Bowyer, A. 716
Brands, J.J.A.M. 1143
Brascamp, H.J. 1136
Bravais, A. 1015, 1022, 1023
Brechtken-Manderscheid, U. 1112
Breen, M. 397, 398, 403, 405, 421, 423, 435
Brehm, U. 336, 539–541, 543, 544, 546, 547
Brehm, U., *see* Altshuler, A. 539, 542
Brehm, U., *see* Bokowski, J. 543
Bretagnolle, J. 1306
Brianchon, C.J. 961
Bricard, M. 11
Bricard, R. 967
Britton, S.C. 878
Brodmann, H. 607, 609
Brøndsted, A. 22, 355, 394, 488, 492–494, 704
Bronshtein, E.M. 29, 312, 324
Brooks, F. 29
Brooks, J.N. 1057
Brooks, R.L. 462, 464
Brothers, J.E. 1136
Brouwer, L.E.J. 15
Browder, F.E. 1230, 1233
Brown, A.L. 22
Brown, D.J. 884
Brown, D.J., *see* Baker, B.S. 883
Brown, H. 906, 1020, 1022, 1025
Brown, K.Q. 712
Bruck, R.E. 358, 1231
Brückner, M. 13, 352, 537
Brudnyi, Yu.A. 410
Bruggesser, H. 5, 7, 488, 714
Brunn, H. 3, 6, 7, 21, 58
Bryant, V.W. 32
Brylawski, T.H. 424, 425
Buchman, E.O. 395, 405, 436
Buchta, C. 15, 321, 357, 1395, 1396, 1399, 1402
Buchta, C., *see* Bárány, I. 357, 1399
Bückner, H. 364
Budach, L. 942
Buekenhout, F. 545, 548
Buerger, M.J. 1019
Bülow, J., *see* Brown, H. 1020, 1022, 1025
Bülow, R., *see* Brown, H. 906
Burago, Yu.D. 6, 33, 51, 55–58, 64, 76, 77, 130, 153, 180, 306, 353, 359
Burau, W. 353
Burckhardt, J.J. 363, 905, 991, 1018, 1021, 1025
Burckhardt, J.J., *see* van der Waerden, B.L. 1029
Burger, T. 148, 1289
Burkard, R.E. 332, 333, 678, 679, 681, 685
Burkard, R.E., *see* Fruhwirth, B. 332
Burke, J.G. 991
Burling, J.P. 410
Burrell, B.P., *see* Todd, M.J. 656, 659
Burton, G.R. 361, 363, 459, 1371
Burzlaff, H. 1007, 1013, 1026
Burzlaff, H., *see* Billiet, Y. 1026
Burzlaff, H., *see* Fischer, W. 1027
Busemann, H. 8, 12, 25, 30, 57, 153, 156, 160, 161, 167, 228, 280, 287, 289, 292, 359, 942, 1059, 1062, 1065, 1128, 1316
Buser, P. 906, 995, 1016
Butler, G.J. 188, 356
Button, L. 334

Caccioppoli, R. 90
Cahn, J.W., *see* Shechtman, D. 925, 1031
Cairns, S.S. 34
Calabi, E. 166
Calabi, L. 28, 29
Calladine, C.R. 259
Campi, S. 147, 148, 310, 1279, 1281–1283, 1314
Cano, J. 1229
Carathéodory, C. 11–13, 76, 430, 1126
Carl, B. 1183
Carleman, T. 1383
Carlsson, S. 1404
Carnal, H. 1402
Carpaneto, G. 685
Cassels, J.W.S. 9, 33, 741, 767, 770, 771
Catalan, E.C. 4
Cauchy, A.L. 5, 6, 15, 51, 225, 226
Cervone, D.P. 543
Cesari, L. 1108, 1118–1120
Chakerian, G.D. 64, 153, 182, 188, 190, 356, 360, 362–368, 393, 407, 1271, 1273, 1278, 1316, 1381
Chalk, J.H.H. 9, 747, 751, 777

Chan, J.B. 435
Chand, D.R. 704
Chandler, E. 1230
Chandrasekaran, R. 642
Charazishvili, A.B. 471
Charve, H.F. 752
Chaudhary, P. 887
Chavel, I. 84
Chazelle, B. 537, 705, 720, 725
Chazelle, B., *see* Aronov, B. 426, 711
Cheeger, J. 942
Chen, W., *see* Beck, J. 1205
Cheng, S.Y. 8, 289
Cherkasskii, V.L., *see* Zamanskii, L.Y. 788
Chern, S.S. 8, 12, 227, 292, 1051, 1071
Chernoff, P.R. 1271
Chevet, S. 1195
Chisini, O. 76
Chiti, G. 1143
Choquet, G. 15, 355, 1301, 1305, 1306
Christof, T. 690
Christoffel, E.B. 8, 290, 1072
Chui, C.K. 362
Chung, F.R.K. 358, 458, 465, 884
Chvátal, V. 455, 848
Chvátal, V., *see* Avis, D. 646
Cieślak, W. 362, 367, 368, 1271, 1274
Clack, R. 163
Clark, J.M.C., *see* Mallows, C.L. 1381
Clarke, F.H. 1108
Clarkson, K.L. 426, 704, 707, 711, 716
Clivio, A. 466
Cochand, M. 456
Cochran, W. 1031
Coffman, C.V. 1144, 1145
Coffman, E.G. 882–885
Coffman, E.G., *see* Baker, B.S. 883
Cohen, E. 663
Cohn, M.J. 821
Cohn-Vossen, S., *see* Hilbert, D. 349, 351, 353, 354, 991
Cohn-Vossen, S.E. 229
Coleman, R. 1321, 1382
Collier, J.B. 29
Conforti, M. 684
Connelly, R. 226, 227, 230, 232, 234–236, 239, 241, 242, 246, 251, 252, 256–266, 890
Connelly, R., *see* Bezdek, K. 367, 847
Converse, G. 361
Conway, J. 9
Conway, J.H. 741, 750, 756, 815, 818, 849, 867, 868, 876, 886, 888, 916, 1008, 1029, 1032
Cook, W.D. 430
Cook, W.J. 784, 788, 789
Coolidge, J.L. 355
Coppersmith, D. 884
Cordovil, R. 423, 424, 460, 518, 559, 564, 587
Cordovil, R., *see* Bienia, W. 424
Cormen, T.H. 704, 708, 716, 717
Corson, H. 29
Couderc, P. 355
Courant, R. 81, 1112
Court, N.A. 355
Cowan, R. 1428
Coxeter, H.S.M. 4, 187, 207, 208, 349–351, 488, 524, 537, 545–548, 727, 811, 849, 865, 869, 887, 903, 905, 917, 925, 1019, 1299
Cramer, G. 6
Crapo, H. 226, 232, 236–238, 252, 253
Crapo, H., *see* Ash, P. 253
Crapo, H.H. 565
Craveiro de Carvalho, F.J. 368
Cremona, L. 252
Cressie, N. 1405, 1406
Crick, F.H.C., *see* Cochran, W. 1031
Crippen, G.M. 226
Crippen, T. 564
Critchlow, K. 349
Croft, H.T. 208, 353, 365–367, 451, 457
Crofton, M.W. 12
Crowe, D.W., *see* Beck, A. 467
Császár, A. 537, 542
Cudia, D.F. 22
Cundy, H.M. 352
Cunningham, W.H. 686
Curie, P. 1024
Cushman, R., *see* Billera, L.J. 516

Dacorogna, B. 1108, 1119, 1126
Dacunha Castelle, D., *see* Bretagnolle, J. 1306
Dade, E.C. 1020
Dakin, R.J. 678
Daley, D.J. 1413
Dalla, L. 357, 365, 1396
Danaraj, G. 488, 496
Danicic, V.I. 750
Danilov, V.I. 499, 614, 620
Danilova, N.I. 6
Dantzig, G.B. 5, 522, 635, 638, 640, 643, 644

Dantzig, G.B., *see* Veinott, A.F. 783
Danzer, L. 13, 29, 31, 227, 360, 392–398, 400, 401, 403, 405–408, 411, 413, 414, 416, 417, 422, 430–435, 451, 465, 469, 652, 831, 834, 886, 908, 919–921, 925, 1033
Darboux, G. 238
Das Gupta, S. 8
Dauter, Z. 1001
Davenport, H. 9, 741, 743, 750, 773, 824
Davis, C. 655
Davis, P.J. 323, 338
Davis, W.J. 1189, 1195, 1196
Davy, P.J. 1382, 1383, 1418
Davy, P.J., *see* Miles, R.E. 1383
Dawson, R. 404
Dawson, R.J.M. 463
Day, M.M. 360
Daykin, D.E. 463
de Blasi, F.S. 1231, 1338, 1341
de Bruijn, N.G. 924, 925, 1032
de Buffon, G.L.L. 12
de Castro Feitosa, E. 366
de Comberousse, C., *see* Roché, E. 355
de Finetti, B. 26
de Giorgi, E. 77, 86, 90, 100
de Saint-Venant, B. 7, 1133
de Santis, R. 396, 427
de Valcourt, B.A. 362, 367
de Vries, H.L. 1143, 1276
de Wolff, P.M. 1012
Dean, D.W., *see* Davis, W.J. 1189
Debrunner, H.E. 402, 919, 921
Debrunner, H.E., *see* Hadwiger, H. 391, 407, 408, 410, 411, 413
Dehn, M. 10, 11, 13, 243, 461, 509, 967
Deicke, A. 12, 164, 1078
Dekster, B.V. 365, 368, 470, 471
Delone (Delaunay), B.N. 10, 13, 14, 712, 752, 755, 756, 814, 844, 901, 915, 917, 920, 992, 995, 1001, 1005, 1007, 1013
Delsarte, P. 457, 825
Deltheil, R. 1351
Demazure, M. 606
Democrit 4
den Hertog, D. 658
Descartes, R. 3, 5
Deschaseaux, J.P. 1194, 1195
Dette, H. 1402
Devroye, L.P. 1401
Deza, M. 404
Dharmadhikari, S.W. 361
Diaconis, P. 1208
Dierker, P., *see* Bialostocki, A. 456
Diestel, J. 1097
Dietrich, B.L., *see* Bland, R.G. 584
Dieudonne, J. 1099
Diewert, W.E., *see* Avriel, M. 630
Dijkstra, E.W. 639
Dikin, I.I. 659
Dines, L.L. 639
Ding, R. 779
Ding, Y.S., *see* Lu, Y.Y. 364
Dinghas, A. 6, 13, 56, 58, 64, 211, 1264, 1270, 1275, 1277, 1286–1288
Dirichlet, G.L. 6, 10
Dirichlet, P.G.L. 718, 752, 996, 1011
Diskant, V.I. 8, 65, 129, 131, 134, 137, 138, 141, 147, 293–295, 310
Dobkin, D.P. 326
Doignon, J.-P. 422, 423, 425, 427, 429, 783
Dolbilin, N.P., *see* Delone (Delaunay), B.N. 915, 992, 995
Dol'nikov, V.L. 31, 367, 393, 396, 399, 409, 410, 434
Dolženko, E. 1331
Donnay, J.D.H., *see* Fischer, W. 1027
Dony, E., *see* Buekenhout, F. 548
Dor, L.E. 1190, 1204
Dor, L.E., *see* Bennett, G. 1174
Dorfman, R. 635
Dostál, Z. 1245, 1254
Dowker, C.H. 323, 837
Dowling, A. 425
Doyen, J. 465
Drandell, M. 31
Dresevic, M. 311
Dress, A.W.M. 351, 548, 901, 904, 914, 915
Drobot, V. 1402
Du Val, P. 1019
Duchet, P. 32
Duchet, P., *see* Cochand, M. 456
Duchet, P., *see* Cordovil, R. 518
Dudley, R.M. 312, 324, 423
Duffin, R.J., *see* Coffman, C.V. 1144, 1145
Dugundji, J. 403
Duijvestijn, A.J.W. 463, 524
Duijvestijn, A.J.W., *see* Bouwkamp, C.J. 463
Dumir, V.C. 823, 848
Dumir, V.C., *see* Bambah, R.P. 757
Duneau, M. 1032

Duneau, M., *see* Katz, A. 925, 1031
Duneau, M., *see* Oguey, Ch. 925
Dunford, N. 89, 92
Dupont, J.L. 956, 965, 968
Dvoretzky, A. 15, 27, 188, 393, 1152, 1166, 1176
Dwyer, R.A. 1395, 1399, 1401, 1402
Dyck, W. 1019
Dyer, M.E. 649, 664, 788, 789, 874

Eberhard, V. 528
Eberlein, P. 833
Eckhoff, J. 13, 392, 398, 413–415, 417–426, 428
Eddy, W.F. 1395
Edelman, P.H. 32
Edelman, P.H., *see* Björner, A. 521
Edelsbrunner, H. 10, 412, 426, 455, 487, 692, 702, 704, 705, 707, 710–712, 714, 717, 718, 720, 721, 723–725, 728, 729
Edelsbrunner, H., *see* Agarwal, P.K. 717
Edelsbrunner, H., *see* Aronov, B. 426, 711
Edelsbrunner, H., *see* Aurenhammer, F. 722
Edelsbrunner, H., *see* Chazelle, B. 720
Edelstein, M., *see* Dawson, R. 404
Edler, F. 6, 76
Edmonds, A.L. 545
Edmonds, J. 684, 686, 688, 784, 890
Edmonds, J., *see* Pulleyblank, W.R. 687
Edwards, D.A. 356
Efimov, N.V. 226, 258
Efron, B. 1397, 1399
Eggleston, H.G. 45, 55, 85, 181, 188, 191, 275, 309, 323, 356, 357, 363, 393, 430, 433, 469, 848, 1263
Eggleston, H.G., *see* Besicovitch, A.S. 205
Egorychev, G.P. 57
Ehlers, F. 620
Ehlers, P.F. 1382
Ehlers, P.F., *see* Enns, E.G. 1382
Ehrhart, E. 10, 772, 774, 779, 782, 943
Eikelberg, M. 619, 620, 622
Eisenack, G. 1233
Eisenstein, G. 1011
Ekeland, I. 1084, 1089, 1119
Elkholy, E.M. 368
Elkies, N.D. 818
Ellis, A.J. 356
Elser, V. 925, 1032
Elsgolc, L.E. 1108, 1110
Enflo, P. 1161, 1193
Enge, H., *see* Burkard, R.E. 678
Engel, P. 13, 901, 905, 906, 917, 919, 992, 996–999, 1001, 1002, 1004, 1006, 1008, 1011–1013, 1015, 1019, 1021, 1025, 1027
Englisch, H. 337
Ennola, V. 836
Enns, E.G. 1382
Enns, E.G., *see* Ehlers, P.F. 1382
Enomoto, K. 331
Epstein, C.L. 227
Epstein, P. 9
Erdélyi, A. 1265
Erdős, P. 153, 327, 453–456, 458, 710, 729, 741, 744, 754, 767, 770, 771, 777, 821, 881, 916, 1339
Èrlikh, I.I., *see* Tarasov, S.P. 338, 360, 652–654
Eschenburg, J.H. 23
Euclid 4, 5, 11
Eudorus 4
Euler, L. 5, 7, 11, 13, 15
Euler, R. 691
Evertz, S. 621
Ewald, G. 34, 64, 277, 306, 514, 540, 585, 586, 614, 621–624, 1331
Ewald, G., *see* Bokowski, J. 525, 582, 586
Ewald, G., *see* Busemann, H. 25, 1316
Ewing, G.M. 1112
Ewings, J.H., *see* Edmonds, A.L. 545

Faber, G. 7, 76, 1133, 1136
Fabian, Cs. 679
Fabian, M. 1331
Falconer, K.J. 26, 362, 366, 397, 435, 1279, 1282, 1283, 1301, 1329
Falconer, K.J., *see* Croft, H.T. 353, 365–367, 451
Falikman, D.I. 57
Fan, K. 5, 398, 405
Farkas, G. 637
Farkas, J. 5
Faro Rivas, R. 337
Fáry, I. 188, 822
Faulkner, G., *see* Chandler, E. 1230
Favard, J. 54, 60–62, 65
Federer, H. 6, 22, 77, 85, 86, 91, 100, 102, 284, 285, 942, 1351, 1352, 1356, 1375
Federico, P.J. 3, 463, 464
Federico, P.J., *see* Duijvestijn, A.J.W. 524

Fedorov, E.S. 10, 13, 756, 901, 906, 917, 1000, 1007, 1025
Fedotov, V.P. 57, 290
Fedotov, V.P., *see* Okuneva, V.A. 207
Fejes Tóth, G. 14, 185, 187, 192, 193, 323, 335, 336, 361, 723, 803, 812, 823, 834, 840–843, 845, 846, 848, 865, 868, 870–872, 879, 885, 886, 888, 918, 1339
Fejes Tóth, G., *see* Bisztriczky, T. 456
Fejes Tóth, L. 14, 153, 179–190, 193, 196, 197, 199, 200, 202, 204–207, 212, 213, 321, 323, 326, 327, 330, 331, 335, 349, 352, 360, 406, 548, 741, 818, 820, 831, 834, 837, 838, 840–844, 846–850, 865, 868, 869, 871, 885–887, 905, 1275, 1335
Fejes Tóth, L., *see* Coxeter, H.S.M. 187, 207
Fejes Tóth, L., *see* Fejes Tóth, G. 336, 841, 842, 845
Feller, W., *see* Busemann, H. 280, 1059
Fenchel, W. 3, 8, 12, 14, 27, 33, 53, 57, 58, 61, 284, 289, 292, 393, 431, 432, 942, 1051, 1071, 1084, 1086, 1088, 1089
Fenchel, W., *see* Bonnesen, T. 45, 47, 50, 51, 55, 56, 58, 65, 129, 131, 156, 181, 182, 209, 275, 353, 356, 358, 359, 361, 363, 364, 368, 430, 648, 772, 1263, 1264, 1279, 1299, 1308
Fenske, C., *see* Eisenack, G. 1233
Fernandez de la Vega, W. 882
Festiopoulos, K., *see* Krätschmer, W. 886
Few, L. 822, 823
Few, L., *see* Coxeter, H.S.M. 811, 869
Fiacco, A.V. 658
Fiedler, R., *see* Jordan, C. 363, 364
Figiel, T. 15, 1152, 1167, 1174, 1177, 1187, 1190, 1194
Filliman, P. 202, 250, 255, 510
Filliman, P., *see* Billera, L.J. 515
Fillmore, J.R. 1289
Fine, J. 498
Firey, W.J. 8, 60, 66, 153, 206, 290, 291, 307, 337, 360, 362, 978, 1276, 1309, 1316, 1358, 1371
Fischer, G. 352, 354, 363
Fischer, K.G. 952
Fischer, W. 919, 1027, 1029
Fischer, W., *see* Koch, E. 919, 1026, 1027
Fishburn, P.C. 418
Fisher, J.C. 364, 1269, 1271, 1274, 1276
Flanders, H. 64, 133, 1245
Fleck, G., *see* Senechal, M. 906
Fleming, W.H. 6, 90, 102
Fleming, W.H., *see* Federer, H. 77
Fletcher, R. 636
Florian, A. 14, 185, 187, 189, 191, 194–197, 199–201, 204, 214, 309, 310, 322, 335, 822, 843, 846, 1275
Florian, A., *see* Fejes Tóth, G. 192, 193, 335, 843
Florian, A., *see* Fejes Tóth, L. 197, 843
Florian, H. 199
Focke, J. 1271, 1273, 1274
Fogelsanger, A. 251
Foland, N.E. 435
Folkman, J.H. 775, 840, 886
Fomenko, A.T., *see* Volodin, I.A. 496
Fomin, S.V., *see* Alekseev, V.M. 1120
Föppl, L. 919
Ford, L.R. 639, 686
Ford, L.R., *see* Dantzig, G.B. 638
Fortune, S.J. 719–721
Fourier, J.B. 5, 9, 635, 639
Fourneau, R. 303, 304, 356
Fourneau, R., *see* Bair, J. 356
Fowler, R.J. 889
Franchetti, C. 365
Frank, A. 364, 890
Frank, A., *see* Cunningham, W.H. 686
Frank, F.C. 887, 1033
Frankenheim, M.L. 992, 1015, 1022
Frankl, P. 404
Frankl, P., *see* Deza, M. 404
Franz, R. 915
Franz, R., *see* Dress, A.W.M. 915
Fredman, M.L. 728
Freedman, B., *see* Vanderbei, R. 659
Freese, R.W. 30, 465
Frehse, J. 1138
Freund, R.M. 787, 873
Fricker, F. 768
Friedman, A. 1136
Friedrich, W. 991
Friesen, D.K. 883
Frieze, A.M., *see* Dyer, M.E. 649, 788, 789, 874
Frisch, K.R. 658
Frobenius, F.G. 906, 995
Frobenius, G. 1249
Frobenius, G.F. 13
Fruhwirth, B. 332
Frumkin, M.A. 742

Fuglede, B. 8, 131, 142, 143, 1287
Fujishige, S. 686
Fujiwara, M. 1273, 1274
Fukuda, K. 564, 584, 587, 588, 590
Fukuda, K., *see* Avis, D. 644, 705
Fulkerson, D.R. 683
Fulkerson, D.R., *see* Dantzig, G.B. 638
Fulkerson, D.R., *see* Ford, L.R. 639, 686
Fullbright, B.E. 409
Fuller, R.B. 260
Fulton, W. 607
Funk, P. 1125, 1279, 1283, 1308
Füredi, Z., *see* Bárány, I. 426, 429, 431, 456, 711, 850, 1402
Füredi, Z., *see* Bezdek, K. 472
Füredi, Z., *see* Boros, E. 431
Furstenberg, H. 162

Gaćz, P. 651
Gadolin, A.V. 1015
Gage, M.E. 56, 131
Gähler, F. 1032
Gähler, F., *see* Korepin, V.E. 925
Gajowski, J. 365
Gale, D. 5, 304, 355, 469, 637
Gale, J.D., *see* Eddy, W.F. 1395
Galiulin, R.V. 1023, 1026, 1032
Galiulin, R.V., *see* Delone (Delaunay), B.N. 915, 992, 995
Gallot, S. 76
Gandini, P.M. 871, 1331
Garbe, D. 546
Garcia, C.B. 662
Gardner, M. 364, 922
Gardner, R.J. 167, 362, 1339
Garey, M.R. 633, 649, 683, 691, 769, 872, 882, 883
Garey, M.R., *see* Aho, A.V. 693
Garey, M.R., *see* Chung, F.R.K. 884
Garey, M.R., *see* Coffman, E.G. 882–885
Garling, D.J.H. 1164, 1176, 1195
Garms, K., *see* Bokowski, J. 540, 580, 592
Garsia, A.M. 1136
Gáspár, Zs., *see* Tarnai, T. 831
Gass, S. 646
Gates, J. 1381, 1384
Gauduchon, P., *see* Berger, M. 76
Gauss, C.F. 5, 7, 8, 14, 752, 802, 814
Gay, D. 656
Gečiauskas, E. 1381, 1382
Geiss, S. 1177
Geivaerts, M. 306
Gelfand, I.M. 515, 1316
Gel'man, B.D., *see* Borisovich, Yu.G. 1230
Gendzwill, D.J. 360
Geppert, H. 1270, 1276, 1286, 1287
Gerber, L. 355
Gericke, H. 76, 182, 1270, 1271, 1275, 1276
German, L.F. 30
Gerwien, P. 966
Ghandehari, M. 366, 368
Ghouila-Houri, A. 398
Giannopoulos, A.A. 167
Giaquinta, M. 1108
Giebler, P. 364
Giering, O. 353, 360, 1339
Giessen, B.C., *see* Chaudhary, P. 887
Giesy, D.P. 1160, 1171
Gilbert, E.N. 353, 1282, 1424
Gilbert, E.N., *see* Chung, F.R.K. 358, 458, 465
Gilbert, J.E. 1156
Giles, J.R. 1084, 1097, 1098
Giles, R., *see* Edmonds, J. 684, 686
Gill, P.E. 653, 658
Giné, E. 1405, 1406
Giné, E., *see* Araujo, A. 1405
Giusti, E. 91
Gizaslewic, R. 26
Glaser, L.C. 538
Gleason, A.M. 331
Gluck, H. 24, 230, 232, 243, 245, 246, 252, 289
Glur, P., *see* Hadwiger, H. 966
Gluskin, E.D. 15, 309, 1152, 1174, 1191–1193, 1197, 1198
Göbel, F. 881
Godbersen, C. 356, 357
Goebel, K. 1223–1228, 1230–1233
Goethals, J.M., *see* Delsarte, P. 457, 825
Goffin, J.L. 651, 652
Gohberg, I.T. 471
Gohberg, I.T., *see* Boltjansky, V.G. 470, 471
Gohberg, I.T., *see* Boltyanskii, V.G. 470
Gołąb, S. 29, 360
Goldberg, M. 199, 212, 213, 367, 831, 919, 1245
Goldfarb, D. 633, 651, 658
Goldfarb, D., *see* Bland, R.G. 651
Goldman, W.M. 35
Goldstein, A.A. 661

Golubyatnikov, V.P. 147
Gomory, R.E. 5
Gonzaga, C. 659, 661–663
Gonzalez-Sprinberg, G. 592
Goodey, P.R. 138, 140, 145, 147, 148, 182, 290, 291, 306, 359, 362, 363, 366, 977, 978, 1277, 1279, 1281, 1282, 1286, 1287, 1289, 1308–1310, 1312, 1314, 1315, 1317–1319, 1358, 1359, 1378, 1423
Goodey, P.R., *see* Betke, U. 977
Goodey, P.R., *see* Chakerian, G.D. 182
Goodman, J.E. 392, 405, 411–415, 422, 423, 426, 454, 455, 459, 525, 540, 562, 577, 587, 588, 590, 726
Goodman, V., *see* Bennett, G. 1174
Goodner, D.A. 1188
Gordan, P. 5
Gordon, Y. 168, 202, 1166, 1189, 1190, 1193
Gordon, Y., *see* Benyamini, Y. 1195, 1196
Gordon, Y., *see* Garling, D.J.H. 1176, 1195
Gornuéjols, G., *see* Conforti, M. 684
Görtler, H. 1270, 1275, 1276
Gostiaux, B., *see* Berger, M. 84, 354
Gott III, J.R. 548
Goursat, E. 1019
Gowers, W.T. 1172, 1200
Góźdź, S. 1274
Góźdź, S., *see* Cieślak, W. 367, 1271, 1274
Grace, D.W. 200
Graev, M.I., *see* Gelfand, I.M. 1316
Graf, S. 309
Graham, R.L. 454, 466, 470, 704, 840, 886, 888
Graham, R.L., *see* Chung, F.R.K. 458, 465
Graham, R.L., *see* Erdős, P. 881
Graham, R.L., *see* Folkman, J.H. 775, 840, 886
Gram, J.P. 961
Granot, F. 633
Gratias, D., *see* Shechtman, D. 925, 1031
Grattan-Guinness, I. 635
Gray, A. 352
Green, J.W. 163
Greene, C. 425
Grenander, U. 1395
Grenander, U., *see* Carlsson, S. 1404
Gretenkort, J., *see* Kleinschmidt, P. 622
Grinberg, E.L. 162, 169, 1316
Gritzmann, P. 60, 66, 325, 338, 339, 356, 364, 542, 544, 545, 631, 634, 681, 773, 775, 776, 787, 789, 810, 868, 870, 872–874, 876, 878, 879, 884, 887–890
Gritzmann, P., *see* Barnette, D.W. 542
Gritzmann, P., *see* Betke, U. 545, 547, 652, 774, 870, 879
Gritzmann, P., *see* Bodlaender, H.L. 634, 789, 873
Gritzmann, P., *see* Dyer, M.E. 788
Gritzmann, P., *see* Fejes Tóth, G. 865, 870, 871, 879, 888
Groemer, H. 8, 10, 15, 48, 60, 130, 131, 133–137, 140, 141, 143–148, 154, 310, 311, 359, 364–366, 743, 775, 803, 805, 813, 820, 842, 850, 868, 881, 885, 917, 938, 941, 953, 975, 1263, 1271, 1276, 1283, 1287, 1357, 1358, 1366, 1372, 1383, 1396
Groemer, H., *see* Chakerian, G.D. 364–366, 368, 1273, 1278
Groemer, H., *see* Goodey, P.R. 138, 140, 145, 147, 148, 366, 1277, 1279, 1282, 1286, 1287, 1314
Groeneboom, P. 1401
Gromov, M. 31, 251, 255, 1181, 1187
Gross, H., *see* van der Waerden, B.L. 752, 1014
Gross, W. 6, 55, 187, 188, 336
Grothendieck, A. 1164, 1166
Grötschel, M. 10, 360, 645, 650, 651, 654, 682, 690–692, 741, 746, 754, 785–787, 875, 876
Grötzsch, H. 410
Gruber, B., *see* de Wolff, P.M. 1012
Gruber, B., *see* Křivý, S. 1012
Gruber, P.M. 3, 12, 153, 158, 170, 188, 191, 303, 307–311, 321, 326, 327, 329–331, 333, 334, 356, 358–360, 365, 419, 654, 741, 751, 754, 756, 757, 767, 770, 771, 786, 807, 867, 916, 918, 943, 1014, 1231, 1290, 1331–1337, 1339, 1340, 1401
Gruber, P.M., *see* Bianchi, G. 358
Gruber, P.M., *see* Erdős, P. 153, 327, 741, 744, 754, 767, 770, 771, 777, 916, 1339
Grünbaum, B. 7, 13, 14, 26, 34, 153, 181, 211, 233, 240, 242, 246, 260, 307, 310, 323, 352, 355, 357, 362, 394, 395, 397, 399, 403, 405–407, 409–411, 413, 414, 465, 469–471, 488, 489, 491, 493, 495, 496, 504, 511–514, 518, 520–522, 524, 526, 528, 537, 540, 541, 544, 546–548, 557, 559, 564, 648, 704, 723, 768, 778, 779, 850, 873, 901, 902, 904–908,

Grünbaum, B. (*cont'd*)
910–915, 918, 919, 921, 922, 924, 925, 936, 971, 972, 992, 1000, 1029, 1030, 1189
Grünbaum, B., *see* Ammann, R. 922
Grünbaum, B., *see* Asplund, E. 409
Grünbaum, B., *see* Barnette, D.W. 527
Grünbaum, B., *see* Danzer, L. 29, 31, 392–398, 400, 401, 403, 405–408, 411, 413, 414, 416, 417, 422, 430–435, 451, 908, 920, 921
Grünbaum, B., *see* Eggleston, H.G. 356
Grzaslewicz, R. 304
Gubler, M. 1026
Guderley, K.G. 360
Guedes de Oliveira, A. 564
Guedes de Oliveira, A., *see* Bokowski, J. 525, 573, 575, 586
Guggenheimer, H. 27, 153, 167, 393
Guibas, L.J. 704, 715, 720
Guibas, L.J., *see* Aggarwal, A. 693
Guibas, L.J., *see* Anagnostou, E.G. 728
Guibas, L.J., *see* Aronov, B. 426, 711
Guibas, L.J., *see* Chazelle, B. 725
Guibas, L.J., *see* Edelsbrunner, H. 720, 725, 728, 729
Güler, O., *see* Ye, Y. 657
Gundersen, H.J.G., *see* Jensen, E.B. 1428
Gurari, V.E. 1186
Gusak, I.Ya. 429
Gustin, W. 163, 358
Guy, R.K. 470
Guy, R.K., *see* Croft, H.T. 353, 365–367, 451
Guyot, P. 1033
Györi, E., *see* Alon, N. 426, 711

Haase, R.W. 1033
Hackbusch, W. 1145
Hadamard, J. 6, 7, 225, 1048
Haddock, A.G. 403
Hadley, G. 1112
Hadwiger, H. 7–12, 15, 26, 35, 45, 46, 48, 50, 52, 55–58, 60, 64, 76, 180–182, 204, 209–212, 305, 336, 338, 351, 354, 391, 398, 406–408, 410–413, 469–471, 720, 768, 773, 774, 779, 780, 806, 850, 886, 936, 937, 939, 941, 944–946, 948, 966, 969–972, 974–977, 1208, 1209, 1351, 1356, 1364–1366, 1374, 1375, 1381, 1384
Hadwiger, H., *see* Bokowski, J. 773, 1365
Hadžić 1230
Haffman, D.R., *see* Krätschmer, W. 886
Hahn, F.H., *see* Arrow, K.J. 1406
Hahn, H. 13
Hahn, M.C., *see* Giné, E. 1405, 1406
Hahn, T. 1022, 1023, 1025, 1030
Haimovich, M. 646
Hajós, G. 9, 186, 418, 918
Hajós, G., *see* Davenport, H. 824
Hall, G.R. 1383
Hall, P. 1402
Hall, R.R. 1276
Halpern, B. 1334
Halsey, E. 564
Hamacher, H.W., *see* Burkard, R.E. 332, 333, 681
Hammer, J. 773
Hammer, J., *see* Erdős, P. 153, 327, 741, 744, 754, 767, 770, 771, 777, 916, 1339
Hammer, P.C. 337, 362, 1335
Hammersley, J.M. 205, 1382
Hanes, K., *see* Berman, J.D. 200
Hanner, O. 431
Hanrath, W., *see* Plesken, W. 1022
Hans-Gill, R.J. 9
Hans-Gill, R.J., *see* Bambah, R.P. 757
Hans-Gill, R.J., *see* Dumir, V.C. 823
Hansen, J.C., *see* Artstein, Z. 1406
Hansen, P. 681
Hansen, W. 401, 433
Harazišvili, A.B. 361
Harborth, H. 456
Harding, E.F. 424, 1428
Hardy, G.H. 1073, 1134, 1136
Hare, W.R. 394, 399, 422
Harker, D. 548, 1030
Hárs, L. 831
Hart, S., *see* Artstein, Z. 1406
Hartman, P. 1004
Hartmann, M., *see* Cook, W.J. 784, 788, 789
Hartnett, W.E., *see* Calabi, L. 28, 29
Hartshorne, R. 607
Haubich, J., *see* Bouwkamp, C.J. 463
Hausdorff, F. 11, 12
Haüy, R.-J. 1015
Havel, C.M., *see* Crippen, T. 564
Havel, T.F., *see* Crippen, G.M. 226
Hayashi, E. 1247
Hayashi, T. 1273, 1275, 1276
Hayes, A.C. 784
Heath, T.L. 225
Heesch, H. 14, 901, 919

Heil, E. 11, 51, 56, 153, 165, 166, 168, 310, 1271, 1334
Heil, E., *see* Bokowski, J. 51, 65
Heintz, J., *see* Bank, B. 682
Helgason, S. 1308, 1312
Heller, I. 358, 683, 685, 691
Hellner, E., *see* Fischer, W. 1027
Hellner, E., *see* Löckenhoff, H.D. 919, 1001
Helly, E. 13, 392–394, 402, 403
Henderson, D.W., *see* Connelly, R. 265
Henk, H. 750
Henk, M. 153, 771, 774
Henk, M., *see* Betke, U. 652, 771, 774, 778, 870
Henk, M., *see* Böröczky Jr, K. 870
Hensley, D. 782, 1180, 1209
Henstock, R. 8, 58
Henze, N. 1395
Henze, N., *see* Dette, H. 1402
Heppes, A. 203, 469, 821, 834, 841, 843
Heppes, A., *see* Fejes Tóth, L. 193, 335, 848, 850
Herda, H. 463
Herglotz, G. 15, 229, 1351
Hermann, C. 1016, 1017, 1030
Hermite, C. 9, 10
Hermite, Ch. 752
Herrmann, K., *see* Alexander, E. 1030
Hersch, J. 7, 76, 1133
Hertel, E. 465, 466
Hertel, E., *see* Böhm, J. 939, 974
Herz, C.S. 1306
Hess, C. 1405, 1406
Hessel, J.F.C. 992, 1015
Hiai, F. 1405
Hibi, T. 780, 781
Hickerson, D., *see* Erdős, P. 458
Hicks, N.J. 226, 229
Hilbert, D. 10, 11, 13, 46, 349, 351, 353, 354, 901, 918, 967, 991, 995, 1066, 1067
Hilbert, D., *see* Courant, R. 81, 1112
Hildebrandt, S. 1119
Hilden, K. 1136
Hile, G.N. 1143
Hill, D., *see* Jamison, R.E. 460
Hilliard, J.E. 1321
Hilliard, J.E., *see* Philofsky, E.M. 1321
Hingston, N. 1333
Hirabayashi, M., *see* Hiraga, K. 1033
Hiraga, K. 1033
Hirose, T. 311
Hirsch, W.M. 634
Hirshfeld, F.L. 1026
Hlawka, E. 9, 741, 747, 748, 757, 767, 807
Hoàng, X.P. 337
Höbinger, J., *see* Gruber, P.M. 153, 358, 359
Hochbaum, D.S. 889
Hochstadt, H. 1265
Hodge, W.V.D. 8
Hoey, D., *see* Shamos, M.I. 717, 719
Hoffman, A.J. 642, 683, 684, 783
Hoffman, A.J., *see* Fulkerson, D.R. 683
Hoffman, A.J., *see* Hirsch, W.M. 634
Hoffmann-Jørgensen, J. 1156
Hofman, G., *see* Hackbusch, W. 1145
Hofmann, J.E. 352
Hofreiter, N. 752
Höhne, R. 774
Höhne, R., *see* Barnette, D.W. 542
Holladay, J.C. 1250
Holmes, R.B. 1089, 1329, 1330
Holmes, R.D. 153, 160
Holton, D.A. 466
Hong, S.J., *see* Preparata, F.P. 704–706
Honsberger, R. 463
Hopcroft, J.E. 253
Hopcroft, J.E., *see* Aho, A.V. 633, 701, 769, 872
Hopf, H. 1075
Hopf, H., *see* Aleksandrov, P.S. 402, 430
Hoppe, R. 849
Hörmander, L. 306
Horn, A. 396
Horst, R. 644
Hortobágyi, I. 849
Horton, J.D. 456
Horváth, J. 844
Horváth, J., *see* Temesvári, A.H. 823
Hoschek, J. 367
Hostinsky, B. 1395
Houle, M.E. 434
Hovanski, A.G. 8
Howard, R., *see* Goodey, P.R. 1281, 1315, 1318, 1423
Howe, R. 1332
Howe, R., *see* Bárány, I. 784
Hsiang, W.Y. 24
Hsuing, C.-C. 251
Hu, T.C. 630
Huard, P. 660

Huck, H. 1075
Hudson, J.F.P. 516
Hufnagel, A., *see* Dyer, M.E. 788
Hurwitz, A. 76, 79, 1261, 1268, 1269, 1274, 1275, 1277, 1285, 1286
Husemoller, D., *see* Milnor, J. 745
Hüsler, J., *see* Carnal, H. 1402
Huson, D. 915
Huson, D., *see* Dress, A.W.M. 904, 915
Hwang, H.W., *see* Aho, A.V. 693

Iglesias, J.E. 1029
Ihringer, T. 473
Ilin, I.V., *see* Andriyasyan, A.K. 757
Imai, H., *see* Aurenhammer, F. 722
Imre, M. 187
Imrich, W. 27
Inoue, A., *see* Hiraga, K. 1033
Inzinger, R. 338, 972, 1276
Ioffe, A.D. 1090
Isbell, J. 548
Ishida, N.N. 623
Istrăţescu, V.I. 1233
Ivanov, L.D., *see* Bronshtein, E.M. 324
Ivanova-Karatopraklieva, I. 228
Iversland, L.M. 428
Iwashita, N., *see* Ishida, N.N. 623

Jackson, B. 419
Jacobi, C.G.J. 1010
Jaggi, B. 586
Jaglom, I.M., *see* Boltjanski, W.G. 355
Jaglom, I.M., *see* Rosenfeld, B.A. 351, 355
James, R.C. 1152, 1161, 1162
Jamison, R.E. 460
Jamison, R.E., *see* Edelman, P.H. 32
Jamnitzer, W. 4
Janner, A. 1031
Janner, A., *see* Ascher, E. 1024
Janson, S. 1320
Janssen, T., *see* Janner, A. 1031
Jaremkewycz 462
Jarnik, V. 750, 751
Jaromczyk, J.W. 429
Jarrat, J.D. 1030
Jarre, F. 661, 662
Jendrol', S. 528
Jensen, D.L. 564
Jensen, E.B. 1428
Jensen, J.L.W.V. 25, 1083
Jeroslow, R.G. 646
Jessen, B. 8, 11, 935, 938, 944, 949, 950, 955, 956, 959, 967, 968
Jessen, B., *see* Fenchel, W. 53, 61, 284, 289, 292, 942, 1071
Joe, B. 714
Jogdeo, K., *see* Dharmadhikari, S.W. 361
John, F. 15, 309, 338, 360, 653, 1153, 1175, 1179
Johnson, D.S. 633, 883, 889
Johnson, D.S., *see* Chung, F.R.K. 884
Johnson, D.S., *see* Coffman, E.G. 882–885
Johnson, D.S., *see* Garey, M.R. 633, 649, 683, 691, 769, 872, 882, 883
Johnson, H.H. 333
Johnson, K. 160
Johnson, S. 454
Johnson, W.B. 1164, 1173, 1174, 1200
Johnson, W.B., *see* Bennett, G. 1174
Johnson, W.B., *see* Figiel, T. 1174, 1194
Jordan, C. 5, 363, 364, 992, 1020
Juhnke, F. 337, 338, 360
Junge, M. 1159
Jünger, M., *see* Christof, T. 690
Jünger, M., *see* Grötschel, M. 690
Jungerman, M. 542
Jurkiewicz, J. 620
Juul, K. 27

Kabadi, S.N., *see* Chandrasekaran, R. 642
Kabatjanski, G.A. 9
Kabatjanskii, G.A. 745, 747, 749, 869
Kabatjanskiĭ, G.A. 811, 825
Kadec, M.I. 1188, 1190
Kadec, M.I., *see* Gurari, V.E. 1186
Kadec, V.M. 1162
Kahane, J.P. 1157
Kahn, J. 469
Kahn, P.J., *see* Hopcroft, J.E. 253
Kaiser, H. 465
Kaiser, M.J. 652
Kakeya, S., *see* Fujiwara, M. 1274
Kakutani, S. 11, 358, 1189
Kalai, G. 254, 255, 419–421, 423, 493, 495, 503, 506, 507, 509, 523, 527, 644–646
Kalai, G., *see* Alon, N. 420, 509
Kalai, G., *see* Björner, A. 415, 418, 420, 509
Kalai, G., *see* Kahn, J. 469
Kalbfleisch, J.D. 454

Kalbfleisch, J.G., *see* Kalbfleisch, J.D. 454
Kallay, M. 420–422, 425
Kallenberg, O. 1412
Kallenberg, O., *see* Janson, S. 1320
Kaltenbach, F.J. 277, 1403, 1404, 1427
Kalton, N.J., *see* Bourgain, J. 1159, 1200
Kamenezki, M. 1273
Kamenskii, M.I., *see* Akhmerov, R.R. 1229, 1233
Kanatani, K. 1321
Kanemitsu, S. 189
Kann, E. 247
Kannan, R. 10, 153, 741, 742, 745, 751, 753, 754, 769, 777, 782, 786, 787, 789
Kannan, R., *see* Cook, W.J. 784, 788, 789
Kannan, R., *see* Dyer, M.E. 649
Kantorovič, L.W. 5
Kantorovich, L.V. 635
Kapranov, M.M. 623
Kapranov, M.M., *see* Gelfand, I.M. 515
Kapur, J.N. 361
Kapur, S.S., *see* Chand, D.R. 704
Karagiorgis, P., *see* Karkakis, J. 337
Karkakis, J. 337
Karlin, S. 396
Karlin, S., *see* Bohnenblust, H.F. 393, 400
Karmarkar, N. 5, 653, 658, 783, 882
Karp, R.M. 787, 884
Karp, R.M., *see* Adler, I. 646
Karp, R.M., *see* Karmarkar, N. 882
Karr, A.F. 1413
Kashin, B.S. 338, 1169, 1203
Kasimatis, E.A. 473
Kasper, J.S., *see* Frank, F.C. 887, 1033
Kassay, G. 1228
Katchalski, M. 397, 401, 402, 411, 412, 415, 421, 456
Katchalski, M., *see* Abbott, H. 420
Katchalski, M., *see* Alon, N. 455
Katchalski, M., *see* Bárány, I. 405, 432
Kato, Y. 1247
Katok, A.B., *see* Zemlyakov, A.N. 1334
Katona, G.O.H. 495
Katseff, H.P., *see* Baker, B.S. 883
Katseff, H.P., *see* Brown, D.J. 884
Katsurada, Y., *see* Hopf, H. 1075
Katz, A. 925, 1031
Katz, A., *see* Duneau, M. 1032
Katz, A., *see* Oguey, Ch. 925
Katzarova-Karanova, P. 407
Kawohl, B. 1134
Kay, D.C. 32
Kazarinoff, N.D. 76, 463
Keller, C.L., *see* Guderley, K.G. 360
Keller, D.-H. 741
Keller, O.H. 767, 918
Keller, S.R. 360
Kellerer, A.M. 1418
Kellerer, H.G. 1418
Kelley, J.E. 679, 680
Kelly, D.G. 1402
Kelly, L.M. 457, 458
Kelly, P.J. 355
Kemnitz, A., *see* Fejes Tóth, G. 361
Kemp, M.C., *see* Hadley, G. 1112
Kempf, G. 606
Kendall, D.G. 12, 1394, 1407
Kendall, D.G., *see* Harding, E.F. 1428
Kendall, W.S., *see* Stoyan, D. 1320, 1394, 1406, 1413, 1419, 1423, 1428
Kenderov, P. 323, 1337
Kenderov, P., *see* Gruber, P.M. 326, 1336, 1337
Kenelly, J.W., *see* Hare, W.R. 394, 422
Kepler, J. 4, 907
Kern, W. 564
Kern, W., *see* Bachem, A. 558, 564, 584
Kershner, R. 811, 814, 838
Kershner, R.B. 14, 919
Kertész, G. 846
Kertész, G., *see* Bezdek, A. 841
Khachiyan, L.G. 5, 338, 360, 647, 651, 783, 789, 873, 874
Khachiyan, L.G., *see* Kozlov, M.K. 633
Khachiyan, L.G., *see* Tarasov, S.P. 338, 360, 652–654, 682
Khamsi, M.A. 1232
Khamsi, M.A., *see* Aksoy, A.G. 1226, 1232, 1233
Khan, M.A. 33
Khassa, D.S. 361
Khassa, D.S., *see* Dumir, V.C. 848
Khovanskij, A.G. 57, 605, 606
Kiener, K. 360
Kincses, J. 395, 1228
Kind, B. 202, 362, 497
Kinderlehrer, D., *see* Hildebrandt, S. 1119
Kingman, J.F.C. 154, 1381–1383, 1395
Kinnersley, N.G. 883
Kirchberger, P. 13, 433
Kirk, W.A. 1230, 1231

Kirk, W.A., *see* Goebel, K. 1223–1228, 1230–1233
Kirkpatrick, D.G. 704
Kirkpatrick, D.G., *see* Edelsbrunner, H. 717, 718
Kiržner, V. 1244, 1253
Klamkin, M.S. 202, 360, 361
Klapper, A., *see* Bayer, M.M. 506, 508
Klebaner, C.F. 355
Klee, V. 5, 12, 13, 15, 21, 22, 28, 29, 154, 277, 323, 355, 357, 359, 362, 368, 395, 396, 398, 418, 431, 488, 500, 520–524, 526, 634, 644–646, 902, 941, 1331, 1332
Klee, V., *see* Anderson, R.D. 278, 1331
Klee, V., *see* Beer, G.A. 310, 311
Klee, V., *see* Bodlaender, H.L. 634, 789, 873
Klee, V., *see* Bonnice, W.E. 430, 432
Klee, V., *see* Corson, H. 29
Klee, V., *see* Danaraj, G. 488, 496
Klee, V., *see* Danzer, L. 29, 31, 392–398, 400, 401, 403, 405–408, 411, 413, 414, 416, 417, 422, 430–435, 451
Klee, V., *see* Eggleston, H.G. 356
Klee, V., *see* Gale, D. 355
Klee, V., *see* Gritzmann, P. 631, 634, 681, 787, 789, 873, 874, 884, 889
Klee, V., *see* Hadwiger, H. 391, 407, 408, 410, 411, 413
Klee, V., *see* Hansen, W. 401, 433
Klee, V., *see* Hu, T.C. 630
Klein, F. 14, 353, 354, 1019
Kleiner, B., *see* Hsiang, W.Y. 24
Kleinjohann, N. 22, 403
Kleinschmidt, P. 512, 515, 621, 622, 645
Kleinschmidt, P., *see* Barnette, D.W. 500, 502
Kleinschmidt, P., *see* Bokowski, J. 525, 582, 586
Kleinschmidt, P., *see* Ewald, G. 34, 540, 585, 586
Kleinschmidt, P., *see* Kind, B. 202, 497
Kleinschmidt, P., *see* Klee, V. 488, 520, 522–524, 526, 645, 646
Kleitman, D.J. 881
Kleitman, D.J., *see* Alon, N. 408, 474
Kleitman, D.J., *see* Kalai, G. 523, 644
Kléman, M., *see* Olamy, Z. 887
Klima, V. 1331, 1332
Klincsek, G., *see* Chvátal, V. 455
Klingenberg, W. 1333
Klötzler, R. 337, 1108, 1125, 1126, 1274
Klyachko, A.A. 621
Kneser, A. 3
Kneser, M. 9, 10, 359
Kneser, M., *see* Betke, U. 780, 979, 980
Knipping, P., *see* Friedrich, W. 991
Knop, O., *see* Melnyk, T.W. 831
Knothe, H. 58, 62, 131, 209, 1182, 1276, 1384
Knothe, H., *see* Bol, G. 210
Knowles, K.M., *see* Conway, J.H. 1032
Knudson, F., *see* Kempf, G. 606
Knuth, D.E. 701, 702
Knuth, D.E., *see* Guibas, L.J. 715
Kobayashi, S. 1352
Koch, E. 919, 1001, 1026, 1027
Koebe, P. 419, 889
Kojima, M. 633, 662
Kojima, M., *see* Ye, Y. 656
Kolen, A., *see* Hoffman, A.J. 684
Kolmogoroff, A.N. 1183
Kolodziejczyk, D. 469
Kołodziejczyk, K. 397, 398, 435
Kolodziejczyk, K., *see* Ding, R. 779
Kömhoff, M. 207, 208
König, D. 392
König, H. 15, 1159, 1184, 1188
Koopmans, T.C. 635
Kopský, V. 1030, 1031
Koptsik, V.A. 1030
Korepin, V.E. 925
Korkin, A. 752, 753, 814
Korkin, A.N. 5, 10
Korte, B. 457
Köse, Ö. 367, 368
Kosinski, A. 26, 1335
Koszul, J.L. 35
Kotlyar, B.D., *see* Brudnyi, Yu.A. 410
Kötter, E. 259
Kotzev, J.N. 1030
Kouchnirenko, A.G. 605
Kovacevic-Vujcic, V.V. 657
Kozlov, M.K. 633
Krahn, E. 76, 1133, 1136
Kramer, D. 414
Kramer, H. 28, 404, 405
Kramer, L., *see* Haase, R.W. 1033
Kramer, P. 925, 1033
Kramer, P., *see* Haase, R.W. 1033
Krammer, G. 203
Krasnoselskii, M.A. 13
Krasnosel'skiĭ, M.A. 433

Krätschmer, W. 886
Krautwald, W. 358
Krawczyk, J., *see* Gajowski, J. 365
Krein, M.G. 15
Krick, T., *see* Bank, B. 682
Krickeberg, K. 90
Krieger, M.K., *see* Kleitman, D.J. 881
Krishna, P., *see* Verma, A.R. 996
Krivine, J.L. 1171, 1172
Krivine, J.L., *see* Bretagnolle, J. 1306
Křivý, S. 1012
Krotoszyński, S. 470, 472
Kruskal, J.B. 495
Kruskal, J.B., *see* Hoffman, A.J. 683, 783
Kubota, T. 6, 51, 55, 60, 63, 65, 136, 138, 360–362, 1267, 1274, 1276, 1277, 1279, 1280, 1282, 1284, 1286, 1287
Kuczumow, T. 1232
Kuczumow, T., *see* Goebel, K. 1230
Kuhn, H.W. 5, 25
Kuhn, H.W., *see* Gale, D. 637
Kühnel, W. 542, 545
Kühnel, W., *see* Brehm, U. 336, 543, 547
Kuiper, N.H. 230
Kulkarni, R.S., *see* Edmonds, A.L. 545
Kung, H.T., *see* Bentley, J.L. 1401
Kung, J.P.S. 565
Künzi, H.P. 681
Kuperberg, G. 833, 836
Kuperberg, K. 850
Kuperberg, W. 188, 835
Kuperberg, W., *see* Bezdek, A. 821, 844, 848, 918
Kuperberg, W., *see* Kuperberg, G. 836
Kuperberg, W., *see* Kuperberg, K. 850
Kürschák, J. 181
Kutateladze, S.S. 363
Kuttler, J.R. 1145
Kuznetsov, E.N. 259
Kuznetsov, V.E., *see* Volodin, I.A. 496
Kwapien, S. 1157, 1158
Kwapien, S., *see* Figiel, T. 1194
Kwapien, S., *see* Gardner, R.J. 362

Labourie, F. 23
Laczkovich, M. 464
Laffaille, G., *see* Gonzalez-Sprinberg, G. 592
Lafferriere, B. 1231
Lagarias, J.C. 10, 57, 662, 746, 751, 753, 767, 782, 918
Lagarias, J.C., *see* Bayer, D. 657, 658, 662
Lagarias, J.C., *see* Coffman, E.G. 884
Lagrange, J.L. 4, 752, 802, 814, 1010
Lalvani, H., *see* Haase, R.W. 1033
Lamb, L.D., *see* Krätschmer, W. 886
Landau, H.J. 187
Landuyt, M., *see* Doyen, J. 465
Lang, S. 767
Lange, L.H., *see* Chakerian, G.D. 188
Langston, M.A., *see* Friesen, D.K. 883
Langston, M.A., *see* Kinnersley, N.G. 883
Larman, D.G. 167, 277, 289, 361, 399, 405, 423, 457, 523, 642, 644, 1177, 1313, 1338
Larman, D.G., *see* Bárány, I. 158, 360, 429, 1336, 1399–1401
Larman, D.G., *see* Dalla, L. 357, 1396
Larman, D.G., *see* Ewald, G. 277, 1331
Larman, D.G., *see* Hayes, A.C. 784
Larman, D.G., *see* Hu, T.C. 630
Las Vergnas, M. 423, 425, 518, 557, 558
Las Vergnas, M., *see* Björner, A. 517–519, 525, 558, 564, 584–587, 727
Las Vergnas, M., *see* Bland, R.G. 557, 558, 592
Las Vergnas, M., *see* Cordovil, R. 587
Lashof, R., *see* Chern, S.S. 1051
Lassak, M. 30, 188, 339, 362, 365, 469, 471, 472, 881
Lassak, M., *see* Gritzmann, P. 325, 364
Lau, A.T.-M. 1232
Laugwitz, D., *see* Danzer, L. 360, 652
Laurie, D.P., *see* Gardner, R.J. 362
Laves, F. 913, 1000
Lawden, D.F. 354
Lawler, E.L. 784
Lawrence, J. 559, 564, 586, 874, 964
Lawson, C.L. 713, 714
Lay, S.R. 355, 363, 434
Lázár, D. 188
Lazutkin, V.F. 1334
Le Verge, H., *see* Euler, R. 691
Lebesgue, H. 3, 76, 79, 225, 1269
Ledoux, M. 1152
Lee, C.W. 13, 500, 502, 510, 513–515, 523, 645, 722
Lee, C.W., *see* Barnette, D.W. 500, 502
Lee, C.W., *see* Billera, L.J. 34, 255, 491, 494, 495, 499, 500, 515
Lee, C.W., *see* Kleinschmidt, P. 515

Lee, D.T. 702, 723
Lee, D.T., *see* Chazelle, B. 725
Lee, S.-N. 360
Leech, J. 9, 750, 815–817, 849
Legendre, A.M. 5, 225
Lehmer, D.H. 360
Leichtweiß, K. 8, 12, 15, 25, 27, 45, 47, 50, 55–58, 61, 129, 130, 153, 156, 158, 163, 166, 167, 180–182, 209, 275, 356, 357, 360, 975, 1053, 1071, 1078, 1263, 1329, 1330
Leichtweiß, K., *see* Blaschke, W. 353
Leiserson, C.E., *see* Cormen, T.H. 704, 708, 716, 717
Lekkerkerker, C.G. 76, 418, 743
Lekkerkerker, C.G., *see* Gruber, P.M. 153, 188, 321, 741, 751, 754, 757, 767, 770, 771, 786, 807, 867, 916, 918, 1014, 1290
Lemberg, H. 1171
Lenstra, A.K. 746, 752–754
Lenstra, H.W. 10, 776, 788, 789
Lenstra Jr, H.W. 682, 746, 754
Lenstra Jr, H.W., *see* Lagarias, J.C. 746, 751, 753
Lenstra Jr, H.W., *see* Lenstra, A.K. 746, 752–754
Lenstra, H.W., *see* Lenstra, J.K. 786, 789
Lenstra, J.K. 10, 786, 789
Lenstra, J.K., *see* Lawler, E.L. 784
Lenz, H. 469, 941
Lenz, H., *see* Danzer, L. 360, 652
Lequette, R., *see* Guyot, P. 1033
Leray, J. 417
Letac, G. 1271
Lettl, G. 308, 309
Lettl, G., *see* Gruber, P.M. 307, 308
Levenštein, V.I. 9, 810, 825, 830
Levenštein, V.I., *see* Kabatjanskii, G.A. 745, 747, 749, 869
Levenštein, V.I., *see* Kabatjanskiĭ, G.A. 811, 825
Levi, F. 559
Levi, F.W. 397, 398, 471, 727
Levin, L.A. 648
Levin, L.A., *see* Yamnitzky, B. 652
Levin, V.L. 394
Levine, D. 925
Lévy, P. 1167, 1300, 1305, 1306
Lew, J.S. 333
Lewis, D.R. 1165, 1186, 1188
Lewis, D.R., *see* Gordon, Y. 1190, 1193
Lewis, D.R., *see* König, H. 1188
Lewis, T. 411
Lewis, T., *see* Katchalski, M. 412
Lewy, H. 15, 289
Leytem, Ch. 546, 547
L'Huillier, S.A.J. 6
Liang, F.M. 884
Liberman, T. 25
Lieb, E.H. 1136, 1143
Lieb, E.H., *see* Brascamp, H.J. 1136
Liebmann, H. 228, 1073
Lillington, J.N. 206, 207, 362, 367
Lin, P.-K. 1224, 1231, 1232
Lindelöf, L. 3, 210
Lindenstrauss, J. 15, 153, 1152, 1158, 1164, 1166, 1188–1190, 1193, 1198–1200
Lindenstrauss, J., *see* Bourgain, J. 55, 138, 148, 156, 310, 335, 336, 407, 1167, 1173, 1174, 1202, 1205–1208, 1281, 1282, 1314, 1321
Lindenstrauss, J., *see* Figiel, T. 1167, 1187, 1190
Lindenstrauss, J., *see* James, R.C. 1162
Linderholm, C. 208
Lindquist, N.F. 1300, 1304
Lindsey, J.H. 811
Lindström, B. 526
Linhart, J. 187, 202–206, 310, 335, 823, 843–845, 849, 1205, 1321
Litsyn, S.N. 9
Little, J.J. 289
Littlewood, J.E., *see* Hardy, G.H. 1073, 1134, 1136
Liu, A., *see* Katchalski, M. 412, 421
Liu, J.D., *see* Hsuing, C.-C. 251
Liu, S., *see* Goldfarb, D. 633
Ljubic, D. 544
Ljubič, U.I. 1244
Ljubič, U.I., *see* Belickii, G.R. 1247
Lloyd, E.K. 513
Lloyd, S.P. 825
Löckenhoff, H.D. 919, 1001
Logothetti, D., *see* Chakerian, G.D. 367
Longinetti, M. 184
Longuet-Higgins, M.S., *see* Coxeter, H.S.M. 547
Loomis, P. 407
Loritz, A. 62
Lovász, L. 10, 426, 454, 649, 651, 681, 687, 741, 752–754, 786, 875

Lovász, L., *see* Bárány, I. 426, 429, 431, 502, 711, 784
Lovász, L., *see* Erdős, P. 710
Lovász, L., *see* Gaćz, P. 651
Lovász, L., *see* Grötschel, M. 360, 650, 651, 654, 682, 690, 692, 741, 746, 754, 786, 787, 875, 876
Lovász, L., *see* Kannan, R. 153, 751, 769, 777, 787
Lovász, L., *see* Korte, B. 457
Lovász, L., *see* Lenstra, A.K. 746, 752–754
Lovász, L., *see* Lenstra, J.K. 786, 789
Love, W.E., *see* Patterson, A.L. 1013
Lowen, R. 32
Löwner, K. 15
Lozanovskii, G. 1155
Lu, Y.Y. 364
Lubiw, A. 684
Luby, M., *see* Karp, R.M. 884
Ludwig, M. 329, 330
Lueker, G.S., *see* Coffman, E.G. 884
Lueker, G.S., *see* Fernandez de la Vega, W. 882
Lundell, A.T. 539
Lusternik, L.A. 55, 58
Luttinger, J.M., *see* Brascamp, H.J. 1136
Lutwak, E. 12, 15, 155, 156, 158–161, 163, 166, 169, 182, 190, 290, 307, 360, 362, 366, 368, 1271, 1313
Lyapunov, A.M. 15
Lyashenko, N.N. 1406
Lyusternik, L.A. 8, 227, 228, 250, 336, 352, 363, 468, 1333

Maak, W. 974
Maass, W., *see* Hochbaum, D.S. 889
Mačaev, V.E., *see* Gurari, V.E. 1186
Macaulay, F.S. 497
Macbeath, A.M. 8, 9, 12, 187, 188, 309, 311, 325, 339, 359, 757, 809, 876
Macbeath, A.M., *see* Henstock, R. 58
Macdonald, I.G. 10, 779, 944
Mackay, A.L. 887, 1033
Maehara, H. 368, 404
Mägerl, G., *see* Graf, S. 309
Magnus, W., *see* Erdélyi, A. 1265
Mahler, K. 9, 11, 76, 167, 188, 743, 745, 750, 772, 836
Mahrenholz, *see* Jaremkewycz 462
Makai Jr, E. 357, 364, 366, 820, 847, 881
Makai Jr, E., *see* Bezdek, A. 848
Makai Jr, E., *see* Böröczky, K. 818
Makeev, V.V. 367
Malkevitch, J. 3, 467, 537
Mallows, C.L. 1381
Malyshev, A.V. 9, 757
Malyshev, A.V., *see* Andriyasyan, A.K. 757
Mandan, S.R. 355
Mandel, A. 496, 539, 564, 587
Mandel, A., *see* Cordovil, R. 587
Mandel, R., *see* Bank, B. 682
Mani, P. 5, 7, 512, 520, 582, 999, 1208
Mani, P., *see* Bruggesser, H. 488, 714
Mani, P., *see* Burton, G.R. 361
Mani, P., *see* Hadwiger, H. 941
Mani, P., *see* Larman, D.G. 1177, 1338
Mani-Levitska, P. 55, 823, 824, 1356
Mani-Levitska, P., *see* Blind, R. 493
Mani-Levitska, P., *see* Bokowski, J. 337
Mani-Levitska, P., *see* Grünbaum, B. 921
Mani-Levitska, P., *see* Jaggi, B. 586
Mankiewicz, P. 1187, 1192, 1195
Marcellini, P. 1143
Marchaud, A. 358
Marchetti-Spaccamela, A., *see* Karp, R.M. 884
Marcotte, O. 694
Marcus, M. 1195
Marczewski, E. 309
Margulis, G.A. 755
Mařík, J. 1243, 1244, 1248
Marino, G. 1231
Markov, A. 754
Markus, A.S., *see* Gohberg, I.T. 471
Marr, J.M., *see* Foland, N.E. 435
Martello, S., *see* Carpaneto, G. 685
Martensen, E. 353
Martin, K., *see* Klee, V. 277
Martin, M., *see* Klee, V. 1331
Martini, H. 170, 171, 188, 202, 349, 355–359, 471, 875
Martini, H., *see* Makai Jr, E. 364
Martini, M. 356
Mase, S. 1406
Massa, S. 1228, 1229
Massera, J.L. 27
Matérn, B. 187
Mathai, A.M. 1402
Matheron, G. 944, 949, 958, 975, 1300, 1304, 1315, 1320, 1351, 1353, 1364, 1375, 1377,

Matheron, G. (*cont'd*)
1394, 1406, 1407, 1410, 1411, 1413, 1418, 1422, 1423, 1425, 1427, 1428
Matoušek, J. 647, 704
Matsumoto, M. 1121
Matsumoto, T. 1027
Matsumoto, T., *see* Engel, P. 992, 1027
Matsumoto, T., *see* Hiraga, K. 1033
Mattheiss, T.H. 644
Mattheiss, T.H., *see* Schmidt, B.K. 1402
Mauldin, R.D. 353, 463
Maurey, B. 1152, 1156, 1158, 1160, 1172, 1190
Maurey, B., *see* Johnson, W.B. 1200
Maurras, J.F. 652
Maxia, A., *see* Beretta, L. 364
Maxwell, J.C. 252
Mayer, A.E. 364
Mazet, F., *see* Berger, M. 76
Maz'ja, V.G. 77, 91, 100
Mazur, S. 15, 1095, 1153, 1330
McClure, D.E. 191, 308, 327, 330, 331
McConnell, M. 505, 620
McCormick, G.P., *see* Fiacco, A.V. 658
McDiarmid, C., *see* Cook, W.J. 784, 788, 789
McEliece, R.J. 825
McKinney, J.R. 188
McKinney, J.R., *see* Petty, C.M. 367
McLarnan, T.J. 1029
McLeod, B., *see* Friedman, A. 1136
McMullen, C., *see* Kuperberg, G. 833
McMullen, P. 10, 11, 13, 34, 48–50, 52, 202, 254, 255, 289, 290, 304, 306, 351, 356, 363, 394, 488, 490, 493, 499, 511, 513, 515, 524, 543–547, 557, 568, 619, 637, 644, 704, 768, 769, 774, 776, 777, 780, 781, 886, 916, 917, 925, 935, 936, 938, 941, 944, 949–951, 953, 954, 958, 960, 961, 963, 969, 970, 974, 976–978, 980, 1006, 1209, 1317, 1318, 1371
McMullen, P., *see* Betke, U. 335, 781, 782, 1205, 1321
McNulty, G. 32
Mecke, J. 1320, 1385, 1394, 1423, 1424, 1428
Mecke, J., *see* Pohlmann, S. 1320, 1420
Mecke, J., *see* Stoyan, D. 1320, 1394, 1406, 1413, 1419, 1423, 1428
Medema, P., *see* Bouwkamp, C.J. 463
Medjanik, A.I. 363
Megiddo, N. 642, 643, 646, 652, 659, 662–664, 703, 889
Megiddo, N., *see* Adler, I. 646
Megiddo, N., *see* Cohen, E. 663
Mehlhorn, K. 693, 702, 716, 721
Mehrotra, S. 662
Meier, C. 953
Meijering, J.L. 1424
Meilijson, I. 1398, 1402
Meir, A., *see* Katchalski, M. 456
Meiser, St., *see* Mehlhorn, K. 716, 721
Meissner, E. 364, 1271, 1275, 1276, 1284
Meketon, M., *see* Vanderbei, R. 659
Mel'nikov, L.S., *see* Aksionov, V.A. 410
Melnyk, T.W. 831
Melzak, Z.A. 208
Menger, K. 29, 401
Menon, V.V. 362
Mertens, F.C.J. 10
Meyer, D., *see* Berard, P. 76
Meyer, M. 166, 168, 359, 362, 1183, 1209
Meyer, M., *see* Bourgain, J. 154
Meyer, M., *see* Gordon, Y. 168, 202
Meyer, R.R. 783
Meyer, W. 304
Meyer, W.J. 971
Micallef, M.J. 24
Michael, E. 11, 311
Michel, L. 1022, 1025
Michel, L., *see* Engel, P. 1008, 1015
Michelacci, G. 368
Miernowski, A., *see* Cieślak, W. 368
Mikhlin, S.G. 1119
Mikshkis, A.D., *see* Borisovich, Yu.G. 1230
Miles, R.E. 12, 1357–1359, 1383–1385, 1393, 1401, 1402, 1418, 1424, 1428
Miles, R.E., *see* Davy, P.J. 1383
Miller, A.K., *see* Freese, R.W. 465
Miller, D.A. 564
Miller, G.L. 710
Miller, J.C.P., *see* Coxeter, H.S.M. 547
Milman, D. 15
Milman, V.D. 11, 15, 155, 161, 360, 361, 1152, 1166, 1167, 1170–1172, 1179, 1181, 1182, 1184–1186, 1201, 1203, 1210
Milman, V.D., *see* Alon, N. 1172
Milman, V.D., *see* Amir, D. 1160, 1172
Milman, V.D., *see* Bourgain, J. 55, 154, 168, 310, 335, 336, 360, 745, 751, 778, 1170, 1173, 1174, 1178, 1182, 1186, 1196, 1202, 1205–1208, 1282, 1321
Milman, V.D., *see* Davis, W.J. 1195, 1196

Milman, V.D., *see* Figiel, T. 1167, 1187, 1190
Milman, V.D., *see* Gromov, M. 1181
Milman, V.D., *see* König, H. 1184
Milman, V.D., *see* Lindenstrauss, J. 153
Milnor, J. 745, 906, 918
Minagawa, T. 247
Minasian 1334
Minkowski, H. 3, 6–10, 14, 15, 21, 45, 46, 50, 51, 58–60, 63, 76, 128, 132, 147, 209, 210, 289, 290, 337, 364, 637, 741, 746, 750, 752, 769–771, 807, 901, 916, 918, 943, 1010, 1013, 1020, 1072, 1278, 1308
Minty, G.J. 5
Minty, G.J., *see* Klee, V. 646
Miranda, C. 289
Miranda, M. 90, 91
Miranda, M., *see* Hildebrandt, S. 1119
Mitrinović, D.S. 76, 130, 180, 355, 358, 359
Mizuno, S., *see* Kojima, M. 633, 662
Mnëv, N.E. 34, 525, 540, 586, 589
Möbius, A.F. 537
Møller, J. 360, 1424, 1428
Molnar, E. 915
Molnar, E., *see* Dress, A.W.M. 915
Molnár, J. 323, 401, 403, 435, 831
Molter, U.M. 1371
Monge, G. 7
Monma, C.L. 693
Monma, C.L., *see* Grötschel, M. 691
Monsky, P. 473
Monteiro, R.D.C. 633, 662
Montejano, L. 307, 359, 363, 364
Montesinos Amilibia, A. 184
Moore, J.D., *see* Micallef, M.J. 24
Moran, P.A.P. 12, 360
Mordell, L.J. 9
Moreau, J.-J. 1084
Morin, T.L., *see* Kaiser, M.J. 652
Moroń, Z. 462
Morrey Jr, Ch.B. 1108, 1126, 1138
Morris, H.C. 399
Morris, W.D. 564
Mosco, U. 1087
Moser, L. 458
Moser, W.O.J., *see* Coxeter, H.S.M. 545, 546, 905
Mosseri, R., *see* Sadoc, J.F. 1033
Mossino, J. 76, 1133
Motzkin, T.S. 5, 13, 22, 404, 431, 639, 651, 652
Motzkin, T.S., *see* Grünbaum, B. 395, 399
Mount, D.M. 838, 876
Mozes, S. 926
Mozgawa, W., *see* Cieślak, W. 368
Mozrzymas, J., *see* Michel, L. 1022, 1025
Mücke, E.P., *see* Edelsbrunner, H. 705, 707, 717
Muder, D.J. 811
Müller, C. 463, 464, 1069, 1265, 1267
Müller, G. 1232
Müller, J.S. 334, 1402
Müller, J.S., *see* Buchta, C. 1395, 1402
Müller, W., *see* Cheeger, J. 942
Mullin, A.A. 393
Mumford, D., *see* Ash, A. 621
Mumford, D., *see* Kempf, G. 606
Murgolo, F.D. 883
Mürner, P. 363
Murray, W., *see* Gill, P.E. 653, 658
Murty, K.G. 646
Murty, K.G., *see* Chandrasekaran, R. 642
Myjak, J., *see* de Blasi, F.S. 1231, 1338, 1341

Nachbin, L. 1188
Naddef, D. 645
Nádeník, Z. 366, 367
Nadler Jr, S.B. 307, 1232
Nagel, W., *see* Mecke, J. 1320
Naidu, S.V.R., *see* Sastry, K.P.R. 1230
Naimpally, S.A. 1230
Nakajima, S. 1274, 1279, 1283
Namikawa, Y. 621
Namioka, I. 1097
Namioka, I., *see* Preiss, D. 1097
Nash, J. 230
Naumann, H. 655
Naumovich, N., *see* Warren, R. 1382
Nelson, D.R. 887, 925
Nelson, J.L. 1228, 1229, 1232, 1233
Németh, A.B., *see* Kramer, H. 404, 405
Nemhauser, G.L. 681, 690
Nemirovski, A.S. 647
Nemirovsky, A.S., *see* Nesterov, J.E. 662
Neri, R., *see* Kramer, P. 925
Nesterov, J.E. 662
Netrebin, A.G. 396, 397, 409
Netuka, I., *see* Klima, V. 1331, 1332
Neuberg, J. 355
Neubüser, J. 1022
Neubüser, J., *see* Brown, H. 906

Neubüser, R., *see* Brown, H. 1020, 1022, 1025
Newman, C.M., *see* Bennett, G. 1174
Newman, D.J. 648
Newton, I. 10
Nguen, M.H., *see* Martini, H. 357
Nguen, M.H., *see* Soltan, V.P. 357
Nguyen, X.X. 1393, 1415
Niederreiter, H. 334, 777, 877
Nievergelt, J. 720
Niggli, P. 992, 996, 1011, 1012, 1025, 1026, 1028, 1030
Nikišin, E.M. 1156
Nikolaev, E.G. 361
Nirenberg, L. 8, 15, 226, 289
Nishiura, T. 27
Nissen, H.-U. 1033
Nitsche, J.C.C. 368
Nomizu, K., *see* Kobayashi, S. 1352
Nordlander, G. 1155
Nosarzewska, M. 773, 774
Nowacki, W. 919, 1028, 1029
Nüesch, P. 360

Oberhettinger, F., *see* Erdélyi, A. 1265
Oberlin, D.M. 162
O'Brien, R.C., *see* Borwein, J.M. 28, 304
Obukhovskii, V.V., *see* Borisovich, Yu.G. 1230
Oda, T. 57, 65, 606, 607, 621, 622
Odlyzko, A.M. 10, 754, 825
Odlyzko, A.M., *see* Bokowski, J. 773
Odlyzko, A.M., *see* Elkies, N.D. 818
Òdor, T. 353
Odor, T., *see* Gritzmann, P. 890
Ó'Dúnlaing, C., *see* Mehlhorn, K. 716, 721
Oettli, W., *see* Künzi, H.P. 681
Oguey, Ch. 925
Ohmann, D. 8, 35, 58
Ohmann, D., *see* Hadwiger, H. 58
Ôishi, K. 1282
Okuneva, V.A. 207
Olamy, Z. 887
Oler, N. 773, 775, 840, 886, 888
Oliker, V. 8, 307
Oliker, V.I. 24, 289, 291, 1072, 1077
Olovianishnikoff, S. 232
Onn, S., *see* Kleinschmidt, P. 645
Oppenheim, A. 754, 755
Oppenheimer, R., *see* Fulkerson, D.R. 683
Orlicz, W., *see* Birnbaum, Z. 1089
Orlin, J.B., *see* Freund, R.M. 787, 873
Orlowski, M. 337
O'Rourke, J. 702
O'Rourke, J., *see* Edelsbrunner, H. 725
Osgood, W.F. 1329
Osserman, R. 6, 7, 65, 76, 85, 129, 131, 141, 182, 184, 209
Ostaszewski, A.J. 1340
Otsu, Y. 24
Ottmann, Th.A., *see* Bentley, J.L. 720
Overhagen, T. 774
Oxtoby, J.C. 1329

Pabel, H., *see* Barthel, W. 363
Pach, J. 710, 823
Pach, J., *see* Bárány, I. 405, 432, 781, 850
Pach, J., *see* Böröczky, K. 818
Pach, J., *see* Edelsbrunner, H. 725
Pach, J., *see* Erdős, P. 458
Pach, J., *see* Makai Jr, E. 881
Pach, J., *see* Mani-Levitska, P. 823, 824
Pachner, U. 514
Pachner, U., *see* Ewald, G. 34, 540, 585, 586
Pachter, M., *see* Orlowski, M. 337
Padberg, M.W. 645, 690, 785
Padberg, M.W., *see* Balas, E. 690, 890
Padberg, M.W., *see* Grötschel, M. 645, 690, 691, 785, 786
Paige, R. 684
Pajor, A. 15, 1202
Pajor, A., *see* Bourgain, J. 154, 1184
Pajor, A., *see* Meyer, M. 166, 1183, 1209
Pajor, A., *see* Milman, V.D. 155, 161, 361, 1179, 1181, 1210
Pál, J. 468
Palásti, I. 458, 459
Palios, L., *see* Chazelle, B. 537
Palmon, O. 1186
Pamfilos, P. 358
Panina, Y. 1310, 1320
Papaderou-Vogiatzaki, I. 1364
Papadimitrakis, M. 167
Papadimitriou, C.H. 639, 645, 787, 788
Papadopoulou, S. 15, 277, 278
Papini, P.L., *see* Baronti, M. 310
Pappus 4
Park, H.S., *see* Oda, T. 621
Parnes, M.N., *see* Chui, C.K. 362
Paterson, M., *see* Monma, C.L. 693
Paterson, M.S., *see* Fowler, R.J. 889
Patterson, A.L. 1013

Pauc, C.Y. 90
Payne, L.E. 7, 76, 1133, 1143
Pečarić, J.E., *see* Mitrinović, D.S. 76, 130, 180, 355, 358, 359
Peck, G.W. 426
Peck, N.T. 21
Pei, L.W. 32
Pelczynski, A. 15, 188, 338, 1152, 1164, 1165, 1177
Pelczynski, A., *see* Figiel, T. 1194
Pelczynski, A., *see* Kadec, M.I. 1190
Pelczynski, A., *see* Lindenstrauss, J. 1164, 1188
Pellegrino, S., *see* Calladine, C.R. 259
Penrose, R. 14, 922, 1033
Perel'man, G.Ya. 416
Perel'man, G.Ya., *see* Polikanova, I.V. 405
Peressini, A.L. 636
Peri, C. 1337
Perkal, J. 32, 469
Perles, M.A. 409, 521, 527, 782, 921, 938, 1244, 1251–1254, 1339
Perles, M.A., *see* Bárány, I. 431
Perron, O. 3, 918
Persson, A. 1164
Persson, O., *see* Matérn, B. 187
Peterson, B.B. 29, 399, 422, 429, 432
Petryshyn, W.V. 1229
Petryshyn, W.V., *see* Lafferriere, B. 1231
Petty, C.M. 12, 153, 155–158, 160, 161, 163, 164, 166–171, 188, 307, 356–359, 361, 367, 422, 1181, 1279, 1281, 1282, 1300, 1308, 1313
Petty, C.M., *see* Busemann, H. 153, 167, 359
Pfiefer, R.E. 360, 1383
Pflanzl, M., *see* Frank, A. 364
Phadke, B.B. 25
Phadke, B.B., *see* Busemann, H. 30, 153
Pham, Dinh-Tao 1252
Phelps, R.R. 15, 21, 1084, 1095, 1097, 1098
Phelps, R.R., *see* Bishop, E. 21
Phelps, R.R., *see* Namioka, I. 1097
Phelps, R.R., *see* Preiss, D. 1097
Phillips, R.S. 1189
Philofsky, E.M. 1321
Phu, H.X. 630
Pick, G. 12, 775, 778
Piefke, F. 1381
Pietramala, P., *see* Marino, G. 1231
Pietsch, A. 1163, 1164
Pietsch, A., *see* Persson, A. 1164
Pinkall, U. 31, 368
Pinsker, A.G. 304, 306
Pisier, G. 15, 153, 168, 321, 361, 1152, 1155, 1157, 1160–1162, 1164, 1171, 1173, 1177, 1182, 1189
Pisier, G., *see* Marcus, M. 1195
Pisier, G., *see* Maurey, B. 1156, 1158, 1160, 1172
Pisier, G., *see* Milman, V.D. 1170
Pitman, J., *see* Adhikari, A. 368
Plesken, W. 1020, 1022
Plesken, W., *see* Neubüser, J. 1022
Plummer, M.D., *see* Lovász, L. 687
Pogorelov, A.V. 7, 8, 12, 14, 24, 228, 231, 232, 255, 289, 291
Pohlmann, S. 1320, 1420
Pohst, M. 741, 752
Pohst, M., *see* Plesken, W. 1020
Poincaré, H. 7, 12, 489, 1133
Poinsot, L. 4
Pokrovskiĭ, V.G. 473
Polikanova, I.V. 24, 405
Polimenis, V.G., *see* Anagnostou, E.G. 728
Pollack, R. 415, 423
Pollack, R., *see* Edelsbrunner, H. 725
Pollack, R., *see* Goodman, J.E. 392, 405, 411–415, 422, 423, 426, 454, 455, 459, 525, 540, 562, 577, 587, 588, 590, 726
Pólya, G. 7, 76, 182, 1025, 1030, 1133, 1136
Pólya, G., *see* Hardy, G.H. 1073, 1134, 1136
Pólya, G., *see* Payne, L.E. 1143
Pompeiu, D. 11
Popov, V.A. 324
Porter, T.A. 76
Porter, T.I. 3
Posicel'skii, E.D. 305
Positselśkiĭ, E.D. 971
Post, K.A. 358
Potapov, S.I., *see* Akhmerov, R.R. 1229, 1233
Pottmann, H. 368
Prabhu, N. 522
Prachar, K., *see* Florian, A. 189, 1275
Preiss, D. 1097, 1331
Prékopa, A. 1402
Preparata, F.P. 487, 692, 702, 704–706
Preparata, F.P., *see* Lee, D.T. 702
Preparata, F.P., *see* Nievergelt, J. 720
Protter, M.H. 7, 1143
Protter, M.H., *see* Hile, G.N. 1143

Provan, J.S. 693
Przemieniecki, J.S. 259
Pták, V. 15, 430, 1242–1249, 1254, 1339
Pták, V., *see* Mařík, J. 1243, 1244, 1248
Pucci, C. 187
Pugh, A. 260, 263, 548
Pulleyblank, W.R. 687
Pulleyblank, W.R., *see* Alon, N. 455
Pulleyblank, W.R., *see* Grötschel, M. 690
Purdy, G.B., *see* Burton, G.R. 459
Puri, M.L. 1405, 1406
Purtill, M. 508

Qi, L., *see* Balas, E. 690
Quaisser, E., *see* Böhm, J. 363, 365
Quarles Jr, D.A., *see* Lew, J.S. 333
Quebbemann, H.-G. 750
Quest, M. 419
Quinto, E.T. 1312

Rabinovich, Yu., *see* Perles, M.A. 409
Rabinowitz, S. 782
Rademacher, H. 13, 393, 432, 434
Rademacher, H., *see* Steinitz, E. 33, 225, 227, 520, 525
Radin, C. 923, 926
Radin, C., *see* Berend, D. 926
Rado, R. 394, 409, 422, 427, 433
Radò, T. 77
Rado, T., *see* Minagawa, T. 247
Radon, J. 11–13, 337, 392, 393, 421
Rådström, H. 304, 306
Rådström, H., *see* Hanner, O. 431
Raghavan, P., *see* Coppersmith, D. 884
Rajan, V.T. 712
Ralescu, D.A., *see* Puri, M.L. 1405, 1406
Ralescu, S.S., *see* Puri, M.L. 1406
Ramharter, G. 757
Ramharter, G., *see* Gruber, P.M. 757
Ramsey, F.P. 453
Rankin, R.A. 9, 824
Rao, I.H.N., *see* Sastry, K.P.R. 1230
Rao, K.P.R., *see* Sastry, K.P.R. 1230
Rao, M.R., *see* Conforti, M. 684
Rao, M.R., *see* Padberg, M.W. 645, 785
Rapoport, M., *see* Ash, A. 621
Rataj, J. 1321
Rausenberger, O. 11
Rayleigh, Lord 7, 1133
Raynaud, H. 1398
Reay, J.R. 392, 422, 427–429, 432, 433
Reay, J.R., *see* Barnette, D.W. 504
Reay, J.R., *see* Bonnice, W.E. 433
Reay, J.R., *see* Ding, R. 779
Recski, A. 226
Rédei, L., *see* Fáry, I. 188
Reed, W.J. 1395
Reeve, J.A. 10
Reeve, J.E. 779, 782
Reich, S. 1233
Reich, S., *see* Goebel, K. 1233
Reichmeider, Ph.F. 686
Reidemeister, K. 12, 278, 1331
Reidemeister, K., *see* Blaschke, W. 1310
Reilly, R.C. 84
Reinelt, G., *see* Christof, T. 690
Reinelt, G., *see* Grötschel, M. 690
Reinermann, J., *see* Müller, G. 1232
Reinhardt, K. 13, 188, 836, 901, 909, 918, 919
Reisner, G. 497
Reisner, S. 167, 168, 1404
Reisner, S., *see* Gordon, Y. 168, 202
Reisner, S., *see* Meyer, M. 359, 362
Renegar, J. 661, 662
Rennie, B.C. 434
Rényi, A. 1396, 1399, 1402, 1403
Rešetnjak, Ju.G. 280
Retherford, J.R., *see* Gordon, Y. 1190
Reuleaux, F. 364
Révész, P., *see* Heppes, A. 469
Reznick, B. 782
Reznikov, A.G. 212
Rhee, W.T. 883
Rhoades, B.E. 1228
Rhyner, J., *see* Gähler, F. 1032
Rhyner, J., *see* Korepin, V.E. 925
Richman, F. 473
Richter, D. 679
Richter, J. 564, 592
Richter, J., *see* Bokowski, J. 590
Richter-Gebert, J. 564, 588, 590, 591
Richter-Gebert, J., *see* Bokowski, J. 560, 573, 575
Rickard, J.A., *see* Holton, D.A. 466
Rickert, N.W. 1300
Rieger, G. 66
Riemann, G.F.B. 14
Riesling, A.S. 469
Riesz, F. 1136
Rinaldi, G., *see* Padberg, M.W. 690
Ringel, G. 542, 543

Ringel, G., *see* Jackson, B. 419
Ringel, G., *see* Jungerman, M. 542
Rinnooy Kan, A.H.G., *see* Coffman, E.G. 884
Rinnooy Kan, A.H.G., *see* Lawler, E.L. 784
Ripley, B.D. 1353, 1410
Rishel, R., *see* Fleming, W.H. 102
Rivest, R.L., *see* Baker, B.S. 883
Rivest, R.L., *see* Cormen, T.H. 704, 708, 716, 717
Roberts, A.W. 23, 25, 630, 1083–1085, 1096
Roberts, F.S. 418
Roberts, J.W. 29
Robertson, S.A. 367
Robertson, S.A., *see* Craveiro de Carvalho, F.J. 368
Robinson, C.V. 400
Robinson, P.J. 466
Robinson, R.M. 922
Roché, E. 355
Röck, H., *see* Fujishige, S. 686
Rockafellar, R.T. 53, 275, 393, 400, 630, 1084, 1087, 1089, 1098, 1099
Rockafellar, R.T., *see* Gale, D. 355
Rodemich, E., *see* Garsia, A.M. 1136
Rodemich, E.R., *see* McEliece, R.J. 825
Rodin, B. 419
Rodríguez Palacios, A. 368
Rogalski, M. 356
Rogers, C.A. 9, 14, 86, 153, 335, 355–357, 359, 361, 362, 469, 741, 748, 775, 802, 806–810, 812, 817, 838, 840, 841, 869–871, 876, 1203
Rogers, C.A., *see* Bambah, R.P. 839, 840, 871
Rogers, C.A., *see* Coxeter, H.S.M. 811, 869
Rogers, C.A., *see* Dvoretzky, A. 188, 1176
Rogers, C.A., *see* Erdős, P. 821
Rogers, C.A., *see* Ewald, G. 277, 1331
Rogers, C.A., *see* Larman, D.G. 167, 289, 457, 642, 1313
Rohn, K. 995
Rolewicz, S. 1200
Rollet, A.P., *see* Cundy, H.M. 352
Roman, T. 352
Ronse, C. 1395
Roos, C. 662
Roos, C., *see* den Hertog, D. 658
Rosenberg, I.G. 256
Rosenberger, H. 721
Rosenfeld, B.A. 351, 355
Rosenfeld, M. 473
Rosenholtz, I. 1224
Rosenthal, H. 356
Rosu, R., *see* Gelfand, I.M. 1316
Rota, G.-C. 941, 954, 958, 963
Rota, G.-C., *see* Crapo, H.H. 565
Rote, G. 332, 333
Rote, G., *see* Burkard, R.E. 332, 333
Rote, G., *see* Fruhwirth, B. 332
Roth, B. 226, 232, 235–237, 242, 246
Roth, B., *see* Asimow, L. 232, 239, 243, 245–247
Roth, K.F. 881
Rother, W. 1356
Rothschild, B.L., *see* Graham, R.L. 454
Roudneff, J.-P. 429, 559, 561, 564, 580, 581
Roux, D. 1225, 1229, 1231
Roux, D., *see* Massa, S. 1228, 1229
Ruben, H. 1402
Rubin, D.S. 784
Rubin, D.S., *see* Mattheiss, T.H. 644
Rublev, V.S. 27
Rudelson, M. 1199
Rudin, W. 93, 116
Rumsey Jr, H., *see* McEliece, R.J. 825
Rund, H. 1121
Ruoff, D., *see* Fisher, J.C. 1269, 1276
Rush, J. 9
Rush, J.A. 750, 818
Rush, J.A., *see* Elkies, N.D. 818
Russakoff, A., *see* Balinski, M.L. 685
Rutovitz, D. 1189
Ryškov, S.S. 10, 14, 752, 753, 755, 756, 814, 844, 917, 1020
Ryškov, S.S., *see* Baranovskiĭ, E.P. 917, 1008
Ryškov, S.S., *see* Delone (Delaunay), B.N. 752, 755, 814, 844
Ryškov, S.S., *see* Gruber, P.M. 756

Saaty, T.L. 865, 885, 886
Saaty, T.L., *see* Gass, S. 646
Saban, G. 365
Sabitov, I.Kh. 228
Sabitov, I.Kh., *see* Ivanova-Karatopraklieva, I. 228
Sachs, H. 848
Sacksteder, R. 22, 23, 1050
Sadoc, J.F. 1033
Sadovskii, B.N., *see* Akhmerov, R.R. 1229, 1233
Saffaro, L. 537, 546

Šafranovskiĭ, I.I. 991
Sah, C.-H. 11, 944, 955, 956, 959, 960, 962, 964, 965
Sah, C.-H., *see* Dupont, J.L. 965, 968
Saint-Donat, B., *see* Kempf, G. 606
Saint-Raymond, J. 166, 167, 1404
Saito, M. 499
Sakaria, K.S., *see* Brehm, U. 540
Sakarovitch, M., *see* Hoffman, A.J. 684
Salinetti, G. 310
Salkowski, E. 329
Sallee, G.T. 360, 365, 368, 393, 396, 937, 938, 971
Sallee, G.T., *see* Chakerian, G.D. 407
Sallee, G.T., *see* Perles, M.A. 938
Saltzman, M.J., *see* Balas, E. 690
Sandakova, N.N., *see* Delone (Delaunay), B.N. 915, 1001
Sanders, J.A., *see* Billera, L.J. 516
Sandgren, L. 393, 396, 401
Sangwine-Yager, J.R. 60, 65, 144, 190
Sansone, G. 208
Santaló, L.A. 31, 52, 76, 153, 154, 164, 166, 167, 182, 183, 357, 359, 399, 411, 414, 745, 774, 778, 886, 941, 1351, 1355, 1381–1384, 1393, 1398, 1402
Santaló, L.J. 11, 12, 15
Sard, A. 1050
Sarkaria, K.S. 429
Sas, E. 187, 325, 837
Sastry, K.P.R. 1230
Satyanarayana, K. 355
Saunders, M.A., *see* Gill, P.E. 658
Sawyer, D.B. 772
Saxe, J., *see* Aggarwal, A. 693
Saxl, I., *see* Rataj, J. 1321
Scarf, H.E. 782, 783
Scarf, H.E., *see* Kannan, R. 787
Scarf, H.E., *see* Lovász, L. 786
Schaal, H. 367, 1273
Schaeffer, J.J. 15
Schäffer, J.J., *see* Massera, J.L. 27
Schäfke, R. 368
Schaible, S. 630
Schaible, S., *see* Avriel, M. 630
Scharlau, R., *see* Dress, A.W.M. 915
Schattschneider, D. 919
Schechtman, G. 1166, 1173, 1206, 1207
Schechtman, G., *see* Johnson, W.B. 1164, 1173, 1174, 1200
Schechtman, G., *see* Milman, V.D. 361, 1152, 1181
Scheithauer, G. 473
Scherer, K. 464, 466
Scherk, P. 430
Scherrer, W. 747, 750
Schkolnick, M., *see* Bentley, J.L. 1401
Schläfli, J. 1016
Schläfli, L. 7
Schmidt, B.K. 1402
Schmidt, E. 6, 8, 77, 183
Schmidt, K.D. 306
Schmidt, W.M. 9, 749, 773, 805, 807, 822, 1403
Schmitt, K.A. 971
Schmitt, P. 806
Schmitz, M. 364
Schmuckenschläger, M. 336, 337, 1207
Schneider, R. 7, 8, 11, 12, 24, 32, 33, 53, 54, 61–63, 130, 134, 136, 138, 139, 153–155, 158, 162, 165, 166, 168, 170, 188–190, 206, 275, 279, 284, 285, 287, 290, 292, 293, 295, 304, 305, 307, 308, 310, 311, 321, 324, 326, 327, 329, 331, 335–337, 356, 357, 359, 360, 362–364, 366, 624, 917, 942, 943, 954, 971–974, 978, 1057, 1059, 1077, 1078, 1262, 1265, 1267, 1271, 1273, 1275, 1279–1289, 1299–1302, 1305, 1308, 1309, 1311–1314, 1318, 1320, 1332, 1335, 1336, 1356–1358, 1364, 1366, 1368, 1371, 1374, 1377, 1379, 1382–1385, 1395, 1399, 1401–1404, 1422, 1423, 1427, 1428
Schneider, R., *see* Firey, W.J. 206
Schneider, R., *see* Goodey, P.R. 290
Schneider, R., *see* Groemer, H. 130, 140, 143, 1263, 1287
Schneider, R., *see* Gruber, P.M. 365, 419
Schneider, R., *see* McMullen, P. 48, 50, 52, 304, 306, 356, 768, 886, 935, 938, 953, 970, 976
Schneider, R., *see* Mecke, J. 1320, 1394, 1428
Schneider, R., *see* Papaderou-Vogiatzaki, I. 1364
Schnell, U. 769, 773
Schnirel'man, L.G., *see* Lyusternik, L.A. 468, 1333
Schnitzer, F., *see* Imrich, W. 27
Schnitzer, F., *see* Nishiura, T. 27
Schnitzer, F.J. 15
Schnorr, C.P. 10, 742, 746, 753

Schnorr, C.P., *see* Lagarias, J.C. 746, 751, 753
Schoenberg, I.J. 227, 1276, 1305
Schoenberg, I.J., *see* Motzkin, T.S. 651, 652
Schoenberg, I.J., *see* Rademacher, H. 393, 432, 434
Schoenflies, A. 901, 906, 995, 1025
Schöneberg, R., *see* Goebel, K. 1230
Schönhage, A. 753
Schöpf, P. 1396
Schöpf, P., *see* Leichtweiß, K. 27
Schopp, J. 407
Schrader, R. 651
Schrader, R., *see* Cheeger, J. 942
Schramm, O. 211, 249, 366, 367, 469, 471, 890
Schrijver, A. 10, 643, 646, 647, 767, 783, 784, 786–788, 890
Schrijver, A., *see* Grötschel, M. 360, 650, 651, 654, 682, 690, 692, 741, 746, 754, 786, 787, 875, 876
Schulte, E. 14, 211, 365, 527, 545, 546, 548, 920, 921
Schulte, E., *see* Dress, A.W.M. 351
Schulte, E., *see* Gritzmann, P. 890
Schulte, E., *see* McMullen, P. 545, 546
Schulz, Ch. 544
Schulz, Ch., *see* Betke, U. 542, 544
Schulz, Ch., *see* Bokowski, J. 337
Schulz, Ch., *see* Ewald, G. 34, 540, 585, 586, 622
Schulz, Ch., *see* McMullen, P. 543, 544, 546, 547
Schur, A. 227
Schur, I., *see* Bieberbach, L. 1013
Schürger, K. 1406
Schütt, C. 12, 158, 326, 1194, 1199
Schwartz, J.S., *see* Baker, B.S. 883
Schwartz, J.T., *see* Dunford, N. 89, 92
Schwarz, H.A. 6, 7, 56, 209
Schwarz, T. 1340
Schwarzenberger, R.L.E. 1024–1026, 1030
Schwarzenberger, R.L.E., *see* Jarrat, J.D. 1030
Schwarzkopf, O., *see* Agarwal, P.K. 717
Schwarzkopf, O., *see* Matoušek, J. 704
Schwermer, J. 767
Schwörbel, J. 546, 547
Scott, P.R. 459, 460, 776, 777, 782
Sedláček, J., *see* Pták, V. 1244, 1248
Seeber, L.A. 5, 752, 995, 996, 1010, 1011
Seeley, R.T. 1263
Segura Gomis, S., *see* Montesinos Amilibia, A. 184
Seidel, J.J. 458, 825, 826
Seidel, J.J., *see* Blokhuis, A. 458
Seidel, J.J., *see* Delsarte, P. 457, 825
Seidel, J.J., *see* Larman, D.G. 457
Seidel, R. 490, 644, 704–707, 716
Seidel, R., *see* Edelsbrunner, H. 717, 718, 721, 725
Seidel, R., *see* Guibas, L.J. 720
Seidel, R., *see* Kirkpatrick, D.G. 704
Seidenberg, A. 456
Selfridge, J.L., *see* Guy, R.K. 470
Selling, E. 10, 752, 1012
Semyanistyi, V.I. 1308
Senechal, M. 537, 906, 925, 991, 1030
Senechal, M., *see* Engel, P. 1008, 1015
Sen'kin, E.P. 228, 232
Sen'kin, E.P., *see* Aleksandrov, A.D. 229, 247
Serra, J.P. 1421
Serra, J.P., *see* Miles, R.E. 1393
Seymour, P.D. 683, 784
Sforza, G. 11
Sgheri, L. 364
Shaffer, D.H., *see* Coffman, C.V. 1144, 1145
Shafrir, I. 1231
Shah, N., *see* Edelsbrunner, H. 714
Shahshahani, M., *see* Diaconis, P. 1208
Shaĭdenko, A.V. 359
Shallcross, D.F., *see* Scarf, H.E. 782
Shamir, A. 786
Shamir, R., *see* Adler, I. 646
Shamos, M.I. 702, 717, 719
Shamos, M.I., *see* Preparata, F.P. 487, 692, 702
Shannon, C.E. 825
Shapiro, J., *see* Fischer, K.G. 952
Shapiro, M. 31
Shapley, L.S., *see* Bohnenblust, H.F. 393, 400
Shapley, L.S., *see* Karlin, S. 396
Sharaburova, L.G. 401
Sharir, M. 646, 721
Sharir, M., *see* Aronov, B. 426, 711, 725
Sharir, M., *see* Edelsbrunner, H. 412, 725
Sharir, M., *see* Guibas, L.J. 715
Sharir, M., *see* Matoušek, J. 647
Sharir, M., *see* Milman, V.D. 1172
Sharpe, R.W. 33
Shashkin, Yu.A., *see* Sharaburova, L.G. 401
Shechtman, D. 925, 1031
Shemer, I. 519, 525, 580, 584

Shemer, I., *see* Bokowski, J. 584
Sheng, T.K. 1382
Shephard, G.C. 14, 34, 304, 307, 308, 310, 323, 363, 425, 433, 511, 548, 917, 961, 971, 976, 1289, 1302, 1313, 1316
Shephard, G.C., *see* Ammann, R. 922
Shephard, G.C., *see* Busemann, H. 25, 1128, 1316
Shephard, G.C., *see* Danzer, L. 908, 920, 921
Shephard, G.C., *see* Ewald, G. 306
Shephard, G.C., *see* Grünbaum, B. 211, 233, 240, 242, 246, 260, 465, 544, 546, 547, 779, 901, 902, 904–908, 910–915, 918, 919, 921, 922, 924, 925, 992, 1000, 1029, 1030
Shephard, G.C., *see* McMullen, P. 356, 488, 493, 511, 704, 936
Shephard, G.C., *see* Perles, M.A. 527, 921
Shephard, G.C., *see* Rogers, C.A. 153, 355–357, 359, 362, 808, 1203
Shevchenko, V.N. 784
Shih, M.-H., *see* Lee, S.-N. 360
Shilleto, J., *see* Fisher, J.C. 1269, 1276
Shimrat, M. 434
Shiohama, K., *see* Otsu, Y. 24
Shirasaka, S., *see* Kanemitsu, S. 189
Shlosman, S.B., *see* Bárány, I. 429
Shmoys, D.B., *see* Lawler, E.L. 784
Sholander, M. 357
Shor, N.Z. 5, 338, 647, 652
Shor, P.W. 589
Shor, P.W., *see* Aggarwal, A. 693
Shor, P.W., *see* Clarkson, K.L. 426, 704, 707, 711
Shor, P.W., *see* Lagarias, J.C. 918
Shub, M., *see* Megiddo, N. 659, 662
Shultz, F.W. 29
Sibson, R. 712
Sidelnikov, V.M. 9, 810, 825
Siegel, C.L. 9, 743, 747, 749, 752, 770, 1333
Sierksma, G. 424, 645
Sierpiński, W. 3
Sigillito, V.G., *see* Kuttler, J.R. 1145
Silin, D.B. 1339
Silva, I.P., *see* Cordovil, R. 423, 424
Silverman, B.W. 338
Silverman, E. 25
Silverman, R., *see* Mount, D.M. 838, 876
Simeone, B., *see* Hansen, P. 681
Simmons, A., *see* Erdős, P. 710
Simon, U. 153, 358, 1078
Simon, U., *see* Oliker, V.I. 1077
Simonovits, M., *see* Lovász, L. 649
Simons, S. 1098
Simutis, J. 542, 545
Sine, R. 1225, 1231
Singer, I., *see* Davis, W.J. 1189
Singh, K.L. 1233
Singh, K.L., *see* Naimpally, S.A. 1230
Singh, K.L., *see* Nelson, J.L. 1228, 1229, 1232, 1233
Singmaster, D. 184
Sinogowitz, U. 1029
Skinner II, J.D. 463
Skorin-Kapov, J., *see* Granot, F. 633
Skubenko, B.F. 9, 757
Slepian, D., *see* Landau, H.J. 187
Sloane, N.J.A. 9, 1290
Sloane, N.J.A., *see* Bannai, E. 849
Sloane, N.J.A., *see* Barnes, E.S. 756
Sloane, N.J.A., *see* Conway, J.H. 741, 750, 756, 815, 818, 849, 867, 868, 876, 886, 888, 916, 1008, 1029
Sloane, N.J.A., *see* Elser, V. 925
Sloane, N.J.A., *see* Graham, R.L. 888
Sloane, N.J.A., *see* Leech, J. 750, 816
Sloane, N.J.A., *see* Odlyzko, A.M. 825
Sloane, N.J.A., *see* Rush, J.A. 818
Smale, S. 646
Smilansky, Z. 304, 489, 512, 619
Smith, C.A.B., *see* Brooks, R.L. 462, 464
Smith, F.W. 919
Smith, M.J. 850
Smith, T.J., *see* Hammer, P.C. 362
Smith, W.R., *see* Melnyk, T.W. 831
Snelson, K. 260
Snobar, M.G., *see* Kadec, M.I. 1188
Snover, S.L. 465
Snyder, L., *see* Dobkin, D.P. 326
Sobczyk, A., *see* Hammer, P.C. 1335
Sobolev, S.L. 9, 755
Sohncke, L. 992, 1023
Solernó, P., *see* Bank, B. 682
Soltan, P.S. 471, 472
Soltan, P.S., *see* Boltjanskii, V.G. 30, 471
Soltan, P.S., *see* Boltyanskiĭ, V.G. 357, 363, 365
Soltan, P.S., *see* Boltyanskiĭ, V.G. 399
Soltan, P.S., *see* German, L.F. 30
Soltan, V.P. 32, 357
Soltan, V.P., *see* Boltyanskiĭ, V.G. 365

Soltan, V.P., *see* Martini, H. 357
Somigliana, C. 1134
Sommerville, D.M.Y. 13, 491, 949, 961
Sonnevend, G. 332, 661–663
Sonnevend, G., *see* Jarre, F. 662
Sorger, H. 1336
Sorger, H., *see* Gruber, P.M. 1337
Souppouris, D.J., *see* Singmaster, D. 184
Souvaine, D.L., *see* Edelsbrunner, H. 729
Spaltenstein, N. 368
Speiser, A. 1029, 1030
Spellman-Munson, B. 564, 588
Spellman-Munson, B., *see* Billera, L.J. 519, 575, 588
Spencer, J.H., *see* Graham, R.L. 454
Sperner Jr, E. 1136
Spiegel, W. 311, 1136
Spivak, M. 226
Spivak, M.A. 23
Sporyshev, P.V., *see* Vershik, A.M. 646
Sprague, R. 462
Sprague, R., *see* Jaremkewycz 462
Stachó, L. 407
Stangeland, J. 428
Stanley, R.P. 13, 34, 35, 57, 494, 496–499, 502, 504, 506, 508, 516, 621, 780–782
Stanton, D., *see* Bannai, E. 458
Stanton, R.G., *see* Kalbfleisch, J.D. 454
Staude, O. 354
Stauffer, M.R., *see* Gendzwill, D.J. 360
Steenaertz, P. 1337
Stefani, O. 365
Stegun, I., *see* Abramowitz, M. 1142
Stehling, T. 911
Steiger, W., *see* Pach, J. 710
Steiglitz, K., *see* Papadimitriou, C.H. 639
Stein, E.M. 827, 1156, 1265
Stein, E.M., *see* Oberlin, D.M. 162
Stein, R. 919, 979
Stein, S.K. 14, 26, 208, 473, 918
Stein, S.K., *see* Chakerian, G.D. 362, 407
Stein, S.K., *see* Kasimatis, E.A. 473
Steinberg, L., *see* Altshuler, A. 526, 580
Steiner, J. 6, 7, 46, 49, 50, 54, 107, 204, 209, 723
Steinhardt, P.J., *see* Levine, D. 925
Steinhaus, H., *see* Marczewski, E. 309
Steinitz, E. 6, 13, 33, 211, 225, 227, 432, 489, 520, 525
Steinmann, G., *see* Engel, P. 992, 1027
Stepanov, V.N., *see* Anikonov, Yu.E. 148, 1284
Stephani, I., *see* Carl, B. 1183
Sternfeld, Y., *see* Lin, P.-K. 1232
Steurer, W. 925, 1031
Stewart, B.M. 548
Stiemke, E. 5
Stillwell, J. 349, 488
Stoer, J., *see* Jarre, F. 662
Stoer, J., *see* Sonnevend, G. 661, 663
Stoer, J., *see* Zhao, G. 663
Štogrin, M.I. 756, 917, 919, 920, 1001, 1007, 1008
Štogrin, M.I., *see* Delone (Delaunay), B.N. 915, 992, 995
Stoka, M. 12
Stoka, M.I. 1351
Stoker, J.J. 226, 228, 250, 1050
Stolfi, J., *see* Guibas, L.J. 704
Stone, A.H. 462
Stone, A.H., *see* Brooks, R.L. 462, 464
Stone, R.E. 658
Stoyan, D. 1320, 1394, 1405, 1406, 1413, 1419, 1423, 1424, 1428
Stoyan, D., *see* Mecke, J. 1320, 1394, 1428
Stoyan, D., *see* Pohlmann, S. 1320, 1420
Strantzen, J.B., *see* Brooks, J.N. 1057
Straszewicz, S. 276
Straus, E.G. 28
Straus, E.G., *see* Sacksteder, R. 22
Strauss, E.G., *see* Erdős, P. 710
Streit, F. 1359, 1384
Streit, F., *see* Hadwiger, H. 1384
Strichartz, R. 1308, 1309
Study, E., *see* Carathéodory, C. 76
Sturmfels, B. 13, 512, 514, 517, 518, 526, 540, 541, 557, 563, 564, 580, 582, 584, 586
Sturmfels, B., *see* Bayer, M.M. 519, 520, 590
Sturmfels, B., *see* Billera, L.J. 515, 623
Sturmfels, B., *see* Björner, A. 517–519, 525, 558, 564, 584–587, 727
Sturmfels, B., *see* Bokowski, J. 34, 518, 519, 525, 537, 540, 546, 561, 563, 564, 580, 584–586, 590
Sturmfels, B., *see* Gritzmann, P. 789, 873
Sturmfels, B., *see* Jaggi, B. 586
Sturmfels, B., *see* Kapranov, M.M. 623
Sturmfels, B., *see* Kleinschmidt, P. 622
Sturmfels, B., *see* Roudneff, J.-P. 561, 580
Stute, W. 1402
Su, B. 1274

Šubnikov, A.V. 913
Sucksdorff, C.-G. 211
Sudakov, V.N. 1202
Sudakov, V.N., *see* Zalgaller, V.A. 363
Sudbury, A., *see* Klebaner, C.F. 355
Sugihara, K. 226, 253
Sulanke, R. 1381, 1384, 1403
Sulanke, R., *see* Rényi, A. 1396, 1399, 1402, 1403
Sullivan, D., *see* Rodin, B. 419
Sullivan, F.E., *see* Peressini, A.L. 636
Sun, J., *see* Mehrotra, S. 662
Sundberg, R., *see* Jensen, E.B. 1428
Supowit, K.J., *see* Megiddo, N. 889
Suri, S., *see* Marcotte, O. 694
Suri, S., *see* Monma, C.L. 693
Süss, W. 59, 361, 362, 1072, 1076
Šverák, V. 1129
Świątek, G., *see* Jaromczyk, J.W. 429
Swinnerton-Dyer, H.P.F. 850
Sydler, J.-P. 11, 967
Sylvester, J.J. 12
Sz.-Nagy, B. 1246, 1264, 1269
Szabó, S. 14, 918
Szankowski, A., *see* Lindenstrauss, J. 1198–1200
Szarek, S.J. 1159, 1168, 1170, 1178, 1192, 1193, 1197
Szarek, S.J., *see* Bourgain, J. 1178, 1184, 1197
Szarek, S.J., *see* Pelczynski, A. 188, 338, 1177
Szegedy, M. 473
Szegö, G. 7, 1133, 1144
Szegö, G., *see* Pólya, G. 76, 182, 1136
Szekeres, Gy. 453
Szekeres, Gy., *see* Erdős, P. 453, 454, 456, 729
Szemerédi, E., *see* Pach, J. 710
Szenthe, J. 30
Szilassi, L. 544, 547, 548
Szücs, A., *see* Bárány, I. 429

Tabačnikov, M.I., *see* Kiržner, V. 1244, 1253
Tabačnikov, M.I., *see* Ljubič, U.I. 1244
Tai, Y., *see* Ash, A. 621
Takasu, T. 353
Talagrand, M. 1173
Talagrand, M., *see* Ledoux, M. 1152
Talagrand, M., *see* Szarek, S.J. 1178
Talenti, G. 7, 1134, 1136, 1140, 1145
Talenti, G., *see* Aronsson, G. 1141
Talley, W.K., *see* Chakerian, G.D. 363
Tamassy, L., *see* Gołąb, S. 360
Tammela, P. 10, 836
Tammes, P.M.L. 831
Tammes, R.M.L. 886
Tamvakis, N.K., *see* Dalla, L. 365
Tamvakis, N.K., *see* Larman, D.G. 361
Tan, T.S., *see* Edelsbrunner, H. 712
Tanimoto, S.L., *see* Fowler, R.J. 889
Tanner, R.M. 208
Tanno, S. 336
Tapia, R.A., *see* Ye, Y. 657
Tarasov, S.P. 338, 360, 652–654, 682
Tarasov, S.P., *see* Kozlov, M.K. 633
Tardos, E. 663
Tarjan, R.E. 716
Tarjan, R.E., *see* Coffman, E.G. 883
Tarjan, R.E., *see* Paige, R. 684
Tarnai, T. 831, 886, 890
Tattersall, J.J. 405
Tay, T.S. 251
Taylor, D.H. 360
Taylor, H. 334
Taylor, J., *see* Senechal, M. 925
Taylor, W., *see* McNulty, G. 32
te Riele, H. 10
te Riele, H., *see* Odlyzko, A.M. 754
Teissier, B. 8, 57, 65, 607
Temam, R., *see* Ekeland, I. 1084, 1089, 1119
Temesvári, A.H. 823
Tennison, R.L. 362, 1274
Terno, J., *see* Scheithauer, G. 473
Terrell, M., *see* Connelly, R. 260
Theaetet 4
Thébault, V. 355
Thomas, C. 1320, 1422
Thomas, C., *see* Mecke, J. 1423
Thomas, J. 473
Thomas, J., *see* Richman, F. 473
Thomas, R.H.K. 339
Thomassen, C. 419
Thomeier, S. 1232
Thompson, A.C. 153
Thompson, A.C., *see* Holmes, R.D. 153, 160
Thompson, A.C., *see* Johnson, K. 160
Thompson, C.D., *see* Bentley, J.L. 1401
Thompson, G.G., *see* Hare, W.R. 399
Thorup, A. 11
Thorup, A., *see* Jessen, B. 935, 938, 944, 949, 950, 955, 956, 959
Thue, A. 14, 814, 838

Thurston, W. 889, 890
Thurston, W., *see* Miller, G.L. 710
Tichomirov, V.M., *see* Kolmogoroff, A.N. 1183
Tichy, R.F. 309
Tichy, R.F., *see* Buchta, C. 1402
Tichy, R.F., *see* Gruber, P.M. 308
Tiercy, G. 364
Tietze, H. 21
Tijssen, G.A., *see* Sierksma, G. 645
Tikhomirov, V.M., *see* Alekseev, V.M. 1120
Tikhomirov, V.M., *see* Ioffe, A.D. 1090
Tind, J., *see* Burkard, R.E. 681
Titterington, D.M., *see* Silverman, B.W. 338
Todd, M.J. 646, 652, 654, 656, 659
Todd, M.J., *see* Bland, R.G. 651
Todd, M.J., *see* Goldfarb, D. 651, 658
Todd, M.J., *see* Ye, Y. 661
Toepell, M. 350
Tolle, J.W., *see* Kelly, D.G. 1402
Tomczak-Jaegermann, N. 15, 309, 321, 1152, 1159, 1186, 1187, 1195, 1197, 1199
Tomczak-Jaegermann, N., *see* Bourgain, J. 1184
Tomczak-Jaegermann, N., *see* Davis, W.J. 1195, 1196
Tomczak-Jaegermann, N., *see* Figiel, T. 1177
Tomczak-Jaegermann, N., *see* Gluskin, E.D. 1174
Tomczak-Jaegermann, N., *see* König, H. 1188
Tomczak-Jaegermann, N., *see* Mankiewicz, P. 1187
Tomczak-Jaegermann, N., *see* Pajor, A. 1202
Tomczak-Jaegermann, N., *see* Szarek, S.J. 1170
Tomlin, J.A., *see* Gill, P.E. 658
Tomor, B. 187, 203
Tompkins, C.B., *see* Heller, I. 683, 685
Tonelli, L. 77, 1118
Toth, P., *see* Carpaneto, G. 685
Totik, V. 1225
Totik, V., *see* Kincses, J. 1228
Tovey, C.A., *see* Stone, R.E. 658
Trafalis, T.B., *see* Kaiser, M.J. 652
Treibergs, A. 291
Tricomi, F., *see* Erdélyi, A. 1265
Tropfke, J. 349, 352
Trotter Jr, W.T. 409, 419
Troutman, J.L. 1112
Truemper, K., *see* Maurras, J.F. 652
Truskina, V.I. 249
Tse, E., *see* Ye, Y. 633, 653
Tseng, P. 658
Tsfasman, M.A. 9
Tsintsifas, G. 357
Tsintsifas, G.A., *see* Klamkin, M.S. 202
Tsuchiya, T. 659
Tucker, A.C. 418
Tucker, A.W. 5
Tucker, A.W., *see* Gale, D. 637
Turnbull, B., *see* Chaudhary, P. 887
Turner, Ph.H. 1334
Tutte, W.T. 463, 464, 521, 565
Tutte, W.T., *see* Brooks, R.L. 462, 464
Tuy, H., *see* Horst, R. 644
Tuza, Z. 465
Tverberg, H. 13, 411, 412, 427, 431, 946
Tzafriri, L. 15, 1191
Tzafriri, L., *see* Bourgain, J. 1159, 1174, 1199, 1200
Tzafriri, L., *see* Gluskin, E.D. 1174
Tzafriri, L., *see* Johnson, W.B. 1200
Tzafriri, L., *see* Lindenstrauss, J. 1158, 1189, 1193
Tzkoni, I., *see* Furstenberg, H. 162

Uccello, P. 4
Uhl, J.J., *see* Diestel, J. 1097
Uhl, J.J., *see* Peressini, A.L. 636
Uhrin, B. 394, 747
Ulam, S., *see* Mazur, S. 1153
Ullman, J.D., *see* Aho, A.V. 633, 701, 769, 872
Ullrich, D.C. 1181
Ungar, P. 459, 1283
Urysohn, P. 11, 65
Urysohn, P.S. 311
Usiskin, Z., *see* Freese, R.W. 465

Vaaler, J.D. 1209
Vaidya, P.M. 661–663, 694
Valentine, F.A. 21, 28, 45, 275, 393, 396, 434–436
Valentine, F.A., *see* Buchman, E.O. 395, 405, 436
Valentine, F.A., *see* Sacksteder, R. 22
Valentine, F.A., *see* Straus, E.G. 28
Valette, G. 32, 465, 973
Valette, G., *see* Blind, R. 31
Valette, G., *see* Doignon, J.-P. 422, 423, 427
Valiant, L.G. 788
Vamos, P. 585

van de Vate, J.H., *see* Amiouny, S.V. 885
van de Vate, J.H., *see* Bartholdi, J.J. 884
van der Corput, J.G. 747, 770
van der Waerden, B.L. 3, 752, 831, 1014, 1029
van Heijenoort, J. 1049
van Leeuwen, J., *see* Bodlaender, H.L. 634, 789, 873
van Tiel, J. 1084
van Wel, B.F. 1399
Vand, V., *see* Cochran, W. 1031
Vanderbei, R. 659
Varberg, D.E. 779
Varberg, D.E., *see* Roberts, A.W. 23, 25, 630, 1083–1085, 1096
Varga, O., *see* Berwald, L. 1358, 1378
Varga, R.S., *see* Holladay, J.C. 1250
Vásárhelyi, É. 472
Vaughan, R.C., *see* Roth, K.F. 881
Veinott, A.F. 783
Vekua, I.N. 228
Venkov, B.A. 10, 752, 916, 1006
Vere-Jones, D., *see* Daley, D.J. 1413
Verma, A.R. 996
Verner, A.I. 26
Vershik, A.M. 646
Vershik, A.M., *see* Bárány, I. 781
Vial, J.P., *see* Roos, C. 662
Viet, U. 361
Vietoris, L. 11, 311
Vincensini, P. 359, 395, 410
Viro, O.Y. 525
Vitale, R.A. 130, 153, 191, 310, 311, 323, 1263, 1320, 1405, 1406
Vitale, R.A., *see* Artstein, Z. 1405, 1406
Vitale, R.A., *see* Bárány, I. 1402
Vitale, R.A., *see* Davis, P.J. 323, 338
Vitale, R.A., *see* McClure, D.E. 191, 308, 327, 330, 331
Vlasov, L.P. 22
Voderberg, H. 919
Vogt, A., *see* Johnson, H.H. 333
Voigt, H., *see* Englisch, H. 337
Volčič, A. 1339
Volčič, A., *see* Michelacci, G. 368
Volenec, V., *see* Mitrinović, D.S. 76, 130, 180, 355, 358, 359
Volkmer, H., *see* Schäfke, R. 368
Volkov, Yu.A. 138, 141, 227, 232, 293
Volland, W. 938
Volodin, I.A. 496
von Laue, M., *see* Friedrich, W. 991
von Neumann, J. 635, 637, 1183
von Staudt, Ch. 6
Voronoi, G.F. 6, 10, 13, 718, 741, 752, 756, 901, 917
Voronoï, G.M. 996, 1005, 1006
Voskresenskij, V.E., *see* Klyachko, A.A. 621
Voss, K. 8, 1051, 1073, 1381
Voxman, B., *see* Bialostocki, A. 456
Vrećica, S., *see* Schulte, E. 365
Vrećica, S.T., *see* Živaljević, R.T. 394, 429
Vrécica, S.T., *see* Živaljević, R.T. 711

Wagner, C.H. 1233
Wagner, G. 1207
Wagon, S., *see* Klee, V. 359, 368
Waiveris, C., *see* Snover, S.L. 465
Waksman, P. 1381
Walfisz, A. 768
Walkup, D.W. 253, 523
Walkup, D.W., *see* Klee, V. 523, 644
Walkup, D.W., *see* McMullen, P. 254, 490, 513, 515
Wallen, L.J. 131, 133, 1286, 1288
Walter, R. 403, 1073
Wang, H. 922
Wáng, Y. 743
Wanka, A. 546, 564
Wanka, A., *see* Bachem, A. 588
Warren, R. 1382
Waterhouse, W.C. 349, 352
Watson, D. 401, 431, 434
Watson, D.F. 716
Watson, F.R. 455
Watterson, G.A., *see* Klebaner, C.F. 355
Waupotitsch, R., *see* Edelsbrunner, H. 712
Weber, L. 1030
Weber, O. 1014
Webster, R.J. 434
Webster, R.J., *see* Bryant, V.W. 32
Webster, R.J., *see* Cook, W.D. 430
Webster, R.J., *see* Shephard, G.C. 307, 310, 323
Wegmann, R. 33, 1276
Wegner, B. 237, 336, 364, 368
Wegner, G. 408, 409, 413, 415–419, 849, 868
Wegner, G., *see* Quest, M. 419
Weierstrass, K. 6
Weil, A. 9

Weil, W. 12, 48, 66, 170, 291, 304, 306, 336, 360, 972, 1301, 1302, 1304–1306, 1309, 1310, 1313, 1314, 1317–1320, 1359, 1364, 1366, 1369, 1371, 1406, 1410, 1416, 1418, 1419, 1428
Weil, W., *see* Betke, U. 66, 1419
Weil, W., *see* Goodey, P.R. 306, 363, 977, 978, 1289, 1308–1310, 1312, 1317–1319, 1358, 1359
Weil, W., *see* Mecke, J. 1320, 1394, 1428
Weil, W., *see* Schneider, R. 155, 335, 360, 363, 917, 1280, 1282, 1299, 1305, 1311, 1312, 1314, 1318, 1356, 1357
Weinberger, H.F. 1142
Weinberger, H.F., *see* Payne, L.E. 1143
Weingram, S.A., *see* Lundell, A.T. 539
Weinstein, A., *see* Osserman, R. 76
Weir, A.J. 1118
Weiss, A.I. 545
Weiss, C.S. 1015, 1022
Weiss, G. 208
Weiss, G., *see* Stein, E.M. 827, 1265
Weiss, M.L., *see* Kelly, P.J. 355
Weiss, V. 1424
Weissbach, B. 212, 364, 367, 471
Weissbach, B., *see* Giebler, P. 364
Weissbach, B., *see* Martini, M. 356
Weitzenkamp, R., *see* Kazarinoff, N.D. 463
Welch, L.R., *see* McEliece, R.J. 825
Wellerding, A., *see* Arnold, R. 130, 310, 322
Wells, A.F. 548
Welsh, D.J A. 565
Welsh, D.J.A. 425, 688
Welzl, E. 716, 720
Welzl, E., *see* Agarwal, P.K. 717
Welzl, E., *see* Edelsbrunner, H. 710
Welzl, E., *see* Matoušek, J. 647
Welzl, E., *see* Sharir, M. 646
Wenger, R. 413
Wenger, R., *see* Aronov, B. 426, 711
Wenger, R., *see* Avis, D. 405
Wenger, R., *see* Goodman, J.E. 392, 411–415
Wenger, R., *see* Pollack, R. 415, 423
Wenninger, M.J. 352, 547
Werner, E., *see* Schütt, C. 158
West, D.B. 419
Wets, R.J.-B., *see* Rockafellar, R.T. 1087
Wets, R.J.-B., *see* Salinetti, G. 310
Wetzel, J.E., *see* Alexanderson, G.L. 720
Weyl, H. 8, 15, 243
White, G.K. 782
White, N.L. 238, 561, 565
White, N.L., *see* Björner, A. 517–519, 525, 558, 564, 584–587, 727
White, N.L., *see* Jaggi, B. 586
Whiteley, W. 237, 245–247, 252–254, 262, 509
Whiteley, W., *see* Ash, P. 253
Whiteley, W., *see* Connelly, R. 242, 256, 257, 259, 260, 263, 264
Whiteley, W., *see* Crapo, H. 226, 232, 236–238, 252, 253
Whiteley, W., *see* Roth, B. 232, 235–237, 242
Whiteley, W., *see* Tay, T.S. 251
Whiteley, W., *see* White, N.L. 238
Whitfield, J.H.M., *see* Naimpally, S.A. 1230
Whitfield, J.H.M., *see* Nelson, J.L. 1228, 1229, 1232, 1233
Whitney, H. 688
Whittaker, E.J.W. 925, 1032
Whittaker, R.M., *see* Whittaker, E.J.W. 925, 1032
Wieacker, J.A. 405, 942, 1320, 1338, 1341, 1375, 1376, 1378, 1379, 1398, 1420–1423, 1426
Wieacker, J.A., *see* Affentranger, F. 1399
Wieacker, J.A., *see* Schneider, R. 324, 1332, 1336, 1371, 1399
Wieacker, J.A., *see* Weil, W. 1410, 1418
Wielandt, H. 1249
Wiener, Chr. 992
Wieting, T.W. 1030
Wijsman, R.A. 310
Wildgrube, E. 467
Wilker, J.B. 356, 361
Wilker, J.B., *see* Button, L. 334
Willcocks, T.H. 463
Williams, J.K., *see* Snover, S.L. 465
Williamson Hoke, K. 492
Wills, J.M. 9, 10, 15, 56, 64, 65, 153, 544, 547, 548, 747, 773, 774, 776, 781, 868, 869, 871, 887, 926, 1033
Wills, J.M., *see* Betke, U. 324, 542, 544, 768, 771, 778, 870, 879, 943
Wills, J.M., *see* Bokowski, J. 546, 582, 773, 1365
Wills, J.M., *see* Fejes Tóth, G. 865, 870, 871, 879, 888
Wills, J.M., *see* Gandini, P.M. 871
Wills, J.M., *see* Gritzmann, P. 66, 775, 776, 868, 870, 879, 887

Wills, J.M., *see* Hadwiger, H. 774, 779
Wills, J.M., *see* Henk, M. 153
Wills, J.M., *see* McMullen, P. 543, 544, 546, 547, 776, 777
Wills, J.M., *see* Niederreiter, H. 777, 877
Wills, J.M., *see* Perles, M.A. 782
Wills, J.M., *see* Schnell, U. 773
Wills, J.M., *see* Schulte, E. 545, 546, 548
Wills, J.M., *see* Schulz, Ch. 544
Wills, J.M., *see* Schwörbel, J. 546
Wilson, J.C. 463
Wilson, R.M., *see* Dowling, A. 425
Wilson, S.E. 545
Wimmer, H. 1245
Winder, R.O. 424
Winternitz, A. 12
Wintgen, P., *see* Sulanke, R. 1403
Witsenhausen, H.S. 358, 419, 1305, 1306, 1317
Witsenhausen, H.S., *see* Graham, R.L. 840, 886
Włodarczyk, J. 614
Wode, D. 208
Wolff, M., *see* Radin, C. 923
Wolfson, H., *see* Milman, V.D. 1186
Wolsey, L.A., *see* Nemhauser, G.L. 681, 690
Womble, E.W., *see* Kay, D.C. 32
Wondratschek, H., *see* Brown, H. 906, 1020, 1022, 1025
Wondratschek, H., *see* Engel, P. 992, 1027
Wondratschek, H., *see* Matsumoto, T. 1027
Wondratschek, H., *see* Neubüser, J. 1022
Woodcock, M.M., *see* Goodey, P.R. 359, 1378
Woods, A.C. 9, 750, 771
Woods, A.C., *see* Bambah, R.P. 750, 771, 848, 871, 887
Wrase, D., *see* Gritzmann, P. 66
Wright, M.H., *see* Gill, P.E. 653, 658
Wschebor, M. 1423
Wulff, G. 13, 1023
Wulff, L. 1024
Wunderlich, W. 237, 364
Wyckoff, R. 1026
Wyner, J.M. 825

Xu, Q., *see* Pisier, G. 1162
Xu, Y.J. 777

Yadin, M. 1421
Yaglom, I.M. 363, 434
Yakovlev, N.N., *see* Temesvári, A.H. 823
Yamaguchi, T., *see* Otsu, Y. 24
Yamnitzky, B. 652
Yang, Y.L. 426
Yannakakis, M., *see* Papadimitriou, C.H. 787, 788
Yao, A.C. 704, 717
Yao, F.F. 702
Yao, F.F., *see* Monma, C.L. 693
Yap, C.K. 722
Yates, D.L. 9, 747
Yau, S.S.T., *see* Xu, Y.J. 777
Yau, S.T. 8, 23
Yau, S.T., *see* Cheng, S.Y. 289
Ye, Y. 633, 651, 653, 656, 657, 659, 661
Ye, Y., *see* Goldstein, A.A. 661
Yoshishe, A., *see* Kojima, M. 633, 662
Yost, D. 304
Young, L.C. 1114
Youngs, J.W.T., *see* Ringel, G. 542
Yudin, D.B., *see* Nemirovski, A.S. 647

Zaanen, A.C. 1331
Zacharias, M. 355
Zacks, S., *see* Yadin, M. 1421
Zaguskin, V.L. 360, 652
Zähle, M. 285, 1410, 1416, 1418, 1423, 1428
Zähle, M., *see* Rother, W. 1356
Zähle, M., *see* Weiss, V. 1424
Zając, J., *see* Cieślak, W. 362, 368
Zaji̇ček, L. 1338
Zaji̇ček, L., *see* Fabian, M. 1331
Zaji̇ček, L., *see* Preiss, D. 1331
Zaks, J., *see* Katchalski, M. 412
Zaks, J., *see* Perles, M.A. 782
Zalgaller, V.A. 6, 227, 231, 277, 363, 548
Zalgaller, V.A., *see* Aleksandrov, A.D. 231
Zalgaller, V.A., *see* Burago, Yu.D. 33, 51, 55–58, 64, 76, 77, 130, 153, 180, 306, 353, 359
Zamanskii, L.Y. 788
Zamfirescu, T. 12, 28, 31, 32, 357, 364, 1332–1335, 1337–1340
Zamfirescu, T., *see* Bárány, I. 1335
Zamfirescu, T., *see* Blind, R. 31
Zamfirescu, T., *see* Fejes Tóth, G. 1339
Zamfirescu, T., *see* Gruber, P.M. 1340
Zamfirescu, T., *see* Schwarz, T. 1340
Zamfirescu, T., *see* Valette, G. 465
Zamfirescu, T., *see* Volčič, A. 1339

Zamorzaev, A.M. 1030
Zang, I., *see* Avriel, M. 630
Zangwill, W.I., *see* Garcia, C.B. 662
Zarankiewicz, K. 362
Zaremba, S.K., *see* Schoenberg, I.J. 227
Zaslavsky, T. 423, 425, 723
Zaslavsky, T., *see* Greene, C. 425
Zassenhaus, H.J. 9, 774, 1020, 1025
Zassenhaus, H.J., *see* Bambah, R.P. 750, 771, 839, 840, 871
Zassenhaus, H.J., *see* Brown, H. 906, 1020, 1022, 1025
Zassenhaus, H.J., *see* Graham, R.L. 840, 886
Zassenhaus, H.J., *see* Pohst, M. 741, 752
Zelevinskiĭ, A.V., *see* Gelfand, I.M. 515
Zelevinsky, A.V., *see* Kapranov, M.M. 623
Zelinskii, J.B. 27
Zemlyakov, A.N. 1334
Zenodorus 4
Zessin, H., *see* Nguyen, X.X. 1393, 1415
Zhang, G. 157, 356
Zhang, J., *see* Amiouny, S.V. 885
Zhang, J., *see* Bartholdi, J.J. 884
Zhang, J., *see* Lassak, M. 881
Zhang, Y., *see* Ye, Y. 657
Zhao, G. 663
Zhao, G., *see* Sonnevend, G. 663
Zhivkov, N.V. 323, 1337
Ziegler, G.M., *see* Björner, A. 517–519, 521, 525, 558, 564, 584–587, 727
Ziegler, G.M., *see* Lagarias, J.C. 782
Ziemba, W.T., *see* Schaible, S. 630
Ziemer, W.P. 91, 100
Ziemer, W.P., *see* Brothers, J.E. 1136
Ziezold, H. 1399, 1403
Zilberberg, A.A. 30
Zimmermann, H., *see* Billiet, Y. 1026
Zimmermann, H., *see* Burzlaff, H. 1007, 1013, 1026
Zimmermann, U., *see* Fujishige, S. 686
Zinn, J., *see* Giné, E. 1405, 1406
Zippin, M. 1188
Žitomirskiĭ, O.K. 10, 756, 917
Živaljević, R.T. 394, 429, 711
Zizler, V., *see* Fabian, M. 1331
Zolotareff, G. 5, 10
Zolotarev, G., *see* Korkin, A. 752, 753, 814
Zucco, A. 1337
Zucco, A., *see* Gandini, P.M. 1331
Zucco, A., *see* Peri, C. 1337
Zwas, G., *see* Goldberg, M. 1245
Zygmund, A. 1264

Subject Index

The index is based on the proposals of the contributors to the Handbook. Because of their heterogeneous standpoints it was necessary to homogenize the proposals. Names occurring in entries are always placed first, e.g., we cite "Aleksandrov–Fenchel inequality" instead of "inequality of Aleksandrov–Fenchel". Well-known geometric notions are cited in general in the form in which they are used, e.g., we take "affine surface area" and not "surface area, affine".

absolutely convex 1049
abstract convexity 32, 456
active constraints 644
acyclic oriented matroid 579
adapted convex body 63
additive function 936, 1372
additive setvalued function 1248
admissible lattice 743
admissible projective transformations 655
affine dependency matroid 425
affine exponent 1243, 1251
affine extremal problems 167
affine function 630
affine hull 631, 1051
affine i-stress 510
affine isoperimetric inequality 164, 165
affine length 196
affine mean curvature 1078
affine normal vector 1077
affine perimeter 196
affine quermassintegral 168, 169
affine scaling algorithm 662
affine scaling method 659
affine surface area 157, 329, 1078, 1284, 1329
affine transforms 512
Aleksandrov–Fenchel inequality 57, 139, 510, 1287
Aleksandrov–Fenchel–Jessen inequalities 292, 1062
algebraic decision tree 704
algebraic manifold 1244
algebraic methods 496
algebraic set 404
algebraic shifting 509
algorithm 701
algorithmic theory of convex bodies 650
algorithmic theory of packing and covering 872
allowable sequences 459
almost fixed point property 1227
almost periodic element 1250
alpha shapes 717
Ammann rhombohedra 925
amorphous glass 1031
amortized time 718
analytic center 652, 659, 661, 662
angle 957, 1056
angle cone 957
anisohedral tile 918
annular symmetrization 56
antipodal map 964
antiprism 350, 524
aperiodic set of tiles 921
apolarity conditions 1078
approximate algorithms 882
approximate fixed point sequence 1227
approximation 129, 1289
approximation schemes 636
approximative 647
arbitrary convex hypersurface 1057
Archimedean solids 352, 524
Archimedean tilings 907
area deviation 190
area function 53
arithmetic crystal class 1021
arrangement 723, 803
Arzelà–Ascoli's theorem 310
Asplund space 1096
assignment problems 685
associated lattice polytope 783
associated polytope 619
associated zonoid 1320
asymptotic center 1227
asymptotic estimate 326
asymptotic radius 1227
Attouch–Wets convergence 1087

Auerbach system 1154
automorphism group of a tiling 902
average-case behavior 646

b-matching problem 687
Baire category 1329, 1401
– first 311
Baire space 1329
balanced 527
balanced matrix 683
balanced packings 885
balanced simplicial polytope
balanced sphere 527
balanced tiling 910
Ball–Behrend inequality 163
balls 352
Banach spaces 1239, 1241, 1405
– local theory 1151
Banach–Mazur compactum 1191
Banach–Mazur distance 322, 1152, 1335
Banach–Mazur metric 309
bar 232
bar framework 232
bar simplex 264
Barbier's theorem 145, 366
Barnette sphere 519
barrier function 658
barycentric subdivision 523, 527
basis 576, 642, 741
– 1-unconditional 1154
– λ-symmetric 1155
– λ-unconditional 1155
basis constant 1192
belt 1006
Benders's decomposition 680
beneath-beyond 514, 704, 707
Bertrand's paradox 1393
best approximating polytope 321
Betti sequence 509
Bieberbach's inequality 65, 143
bilinear extension 1065
bilinear function 1065
billiard 1334
bin packing 863, 879
binary code 815
binary representation 816
binary search 648
binary sizes 635
binary Turing machine 632
Binet ellipsoid 1178
Birkhoff's theorem 685
bistellar operation 510, 514
Bland's pivoting rule 643, 644
Blaschke diagram 60
Blaschke product 1246
Blaschke scalar product 307
Blaschke selection theorem 46, 310, 1329
Blaschke sum 290, 307
Blaschke symmetrization 55, 1207
Blaschke–Groemer inequality 154
Blaschke–Petkantschin identity 1398
Blaschke–Santaló inequality 165–169, 329, 745, 1182, 1404
– complementary 167, 168
Blaschke's theorem on curvatures 281
Blichfeldt's enlargement method 811
Blichfeldt's theorem 747
blow-up 613
Bonnesen-type inequality 141, 142, 1288
Bonnesen's inequality 64, 130, 132
Bonnice–Klee theorem 433
Boolean model 1416, 1417, 1419, 1431
Borsuk's problem 367, 461, 468
boundary complex 487, 538
boundary-distance function 660
box 456
bracket signs 576
branch and bound methods 678
Bravais classes
– arithmetic 1022
– geometric 1020
Bravais point group 1020
breadth-first search 708, 716
Brillouin zone 997
Brunn–Minkowski function 1071
Brunn–Minkowski inequality, reverse 1184
Brunn–Minkowski theorem 57, 133, 134, 1180, 1287, 1288, 1406
brush set 53
Buckminster-Fullerites 878, 886
Busemann intersection inequality 161, 162
Busemann random simplex inequality 154, 155
Busemann–Petty centroid inequality 155, 157
Busemann–Petty list 167
$BV(R^n)$ 101

C_2-boundaries 1059
cable 232
Caccioppoli set 85, 91

cage 208
cancellation 28, 304
cap bodies 54, 144
capacity 143, 1137
capacity functional 1407, 1412, 1416, 1420
Carathéodory number 430, 431
Carathéodory's theorem 391
– multiplied 431
Cartier divisors 622
case 193, 194
Catalan solids 352
Cauchy's surface area formula 51, 148, 1282, 1363
Cavalieri's principle 102
Cayley–Hamilton theorem 1246
cd index 508
cell 538, 1424
cell complex 538
center 660, 1413
center function 660
center of gravity cuts 647
center of gravity sections 648
center polytope 869
centered bodies 1299
central affine geometry 329
central body 1261
central limit theorem 1405
central n-gon 194
central path 663
central symmetrization 55, 135
central trajectory 662
centro-affine surface area 165
centroid 134, 165, 974, 1276, 1283
centroid body 155, 171, 1181, 1282
characteristic equation 1048
characteristic function 88, 938
characteristic triangle 184
characterization of the f-vectors 35
Chebyshev set 22
chemical whiskers 875, 878
chirality character 993
chirotope 573
Choquet simplex 355
chord power integrals 1380
Christoffel problem 290, 1308, 1309
chromatic number 409
circle packings on Riemann surfaces 419
circlefree 453
circuit 517, 578
circular-arc graphs 418
circular arcs 1274
circular disk 129
circumradius 46, 648, 655, 776
circumscribed box 1271
circumscribed cylinder 1271, 1276
circumscribed polygon 211, 1271
circumscribed polytope 181
circumscribed rectangle 1271
circumsphere 888, 1337
class 944
classification of lattices 744
clearly visible 435
clique 691
clique constraints 691
clique number 409
clique tree inequality 690
closed system of eigenfunctions 1068
closure of a convex function 1087
co-NP 638
co-unit (augmentation) 965
coarea formula 101
cocircuit 517, 578
cocircuit span 571
codeword 815
cofactor 1067
Cohen–Macaulay 496, 497, 510
cohomology 496, 499
cohomology ring 499, 502
collinear (convex sets) 456
collision probability 1371
color group 1029
colorability 544
coloring number of $\mathcal{K}$ 406
comb inequality 690
combinatorial equivalence of tilings 902
combinatorial involutions 512
combinatorial optimization 651, 677, 782
combinatorial prototile 920
combinatorial types 488, 495, 513, 521, 523–526, 998
combinatorially equivalent 487, 512, 538
combinatorially equivalent polytopes 525
combinatorially regular polyhedra 545
communication 474
compact packing 846
complementary geominimal surface area inequality 168
complementary slackness 636
complete directed graph 645
complete graph 645

completely balanced sphere 504
completely unimodal 492
completeness 364
complex 415, 496
complexity class 638
composition of tilings 922
compressed integral polytope 516
computational complexity 632, 701, 706
computational geometry 692, 701
computing the volume 649
comultiplication 965
concave family 46
concave function 630
concavity 1071
condensed harmonic expansion 1266, 1277
conditionally positive definite 1305
cone class 950
cone group 950
configuration 232
configuration space 34
congruent tiling 902
conjugate convex function 1089
connected 492
connected arrangement 820
3-connected planar graphs 524
connected set C 1050
conormal vector 1075
constant brightness 368, 1280
constant width 144, 363, 1273, 1278, 1280, 1285
contact measure 1365, 1369
contact point 1175
container problem 882
continued fraction rounding procedure 647
continuity argument 1064
continuity method 1069
continuity of convex functions 632
continuous flex 233
contraction 1223, 1239
convergence in distribution 1406
convergence of sequences of convex bodies 310
convex 45, 1047
convex body 45, 179, 768, 936, 1049, 1133
convex cap 180, 189
convex disk 179, 834
convex domains 129, 1262
convex function 630, 1047
convex hull 487, 490, 703, 707, 1395, 1405
– of the oriented matroid 579
convex integral 1111, 1123
convex maximization 629
convex minimization 629
convex optimization 629
convex path 454
convex polyhedral metric 231
convex polyhedron 487, 629
convex polytopes 487, 703, 768
– enumeration 467
convex programming 1089
convex quadratic minimization 633
convex ring 1372
convex subset 454
convex surgery 1203
convexity structure 1230
convexly connected 431
convexly independent 452, 453, 456
– weakly 453
convolution 92, 507
coordinate array 816
corona 1001
corona vector 1005
cosine transform 1302, 1307, 1311
cotype Gaussian 1155, 1158
cotype q constant 1156
countably m-rectifiable 1375
counting measure 1411
covariance 1405
covering 146, 470, 801, 863
– k-fold 821
covering density 804, 864
covering minimum 751, 777
covering numbers 1183
covering property 998
covering radius 751
covering space 35
critical exponent 1237, 1239, 1339
critical lattices 754
Crofton formula 1275, 1355, 1358, 1409, 1416, 1429
cross-sectional measure 359
crossing disk 838
crosspolytope 351, 502, 524, 1154
crystal form 1023
crystallographic group 906, 994
crystallographic orbit 1026
crystallographic point group 1018
crystallographic symmetry operation 1015
cube 349, 494, 527
cubic fundamental form 1077

cubical ratio 1176
curvature 279, 1269
curvature center 366
curvature function 30, 157
curvature image 160, 164, 169, 1077
curvature measure 53, 282, 1363, 1394, 1408
– generalized 282
curve shelling 489
cut 679
cutting plane 679
cutting-stock 884
cyclic polytope 493, 518, 524, 704
cycling 642
cylinders 338, 1422

d-collapsible complexes 418
d-connected 520
d-Leray complex 417
d-nonfacet 920
d-nontile 920
d-representable complexes 416
d-skeleton 521
d-step conjecture 520, 522
Darboux transformation 238
Davenport–Schinzel sequence 721
De Giorgi's perimeter 85
decidability 526
decidability of the tiling problem 922
decomposition of tilings 922
decreasing rearrangement 1134
deflation process 465, 923
degeneracy 642
degree sequences 780
Dehn invariant
– total classical 965
– total spherical 964
Dehn–Sommerville equations 490, 491, 493, 495, 498, 499, 503, 513
– generalized 503, 507, 527
del Pezzo surface 616
Delaunay ..., *see* Delone ...
Delone cell 813
Delone mosaic 1428
Delone reduced 1012
Delone symbol 915
Delone triangulation 330, 693, 711, 717, 814
dendrites 878
density function 863
density of a lattice packing 743
density of a packing 804, 864, 1339
density of a set 100
density relative to a domain 803
depth-first search 708
designed-based 1429
determinant of a lattice 742, 769
deterministic algorithm 705
diagonal forms 1063
diagram 538
diameter 46, 522, 523, 776, 1262, 1335
dictator problem 865
Dido's problem 77
difference body 169, 305, 355, 362
difference set 1032
differentiable submanifold 1047
differential form valued 1053
dimension, bounds 312
dimensionally unambiguous graphs 521
directed acyclic graph 716
directed graph 462, 1248
directional distribution 1426
Dirichlet cell 769, 812, 916
Dirichlet domain 997
Dirichlet parallelotope 1005
Dirichlet region 916
Dirichlet tessellation 253, 722
Dirichlet–Voronoi cell 743, 769, 916
Dirichlet–Voronoi tiling 330, 756, 916
discontinuum 992
discrete group 905
discrete optimization 677
discrete set 742
disk 834
dispersion 333, 877
dissection 461, 939
distance ellipsoid 1177
distance function 1088
2-distance set 451
distribution 1395, 1402, 1404, 1407, 1411, 1412, 1419, 1424
distribution function 1134
divide-and-conquer 705, 722
dodecahedron 349, 521
domain of influence 997
dominate 233
dot pattern 992, 1027
Dowker type theorems 323
dual basis 1004
dual degeneracy 637
dual ideal norm 1164
dual indices 1159

dual lattice 743
dual mixed volume 158, 170
dual mixed volume inequality 158, 159, 161, 164
dual of a tiling 902
dual space 1153
duality 488, 636, 709, 718, 902
duality for oriented matroids 517, 583
Dupin's indicatrix 1059
d_X 1153

edge 487, 490, 538, 543
edge-convex 248
edge curvature 204
edge diameter 644
edge effects 1430
edge graph 492
edge paths 644
edge-to-edge plane tiling 902
edges of a plane tiling 902
effective domain 1086
Efron's identity 1396
Eggleston metric 307, 322
Ehrhart polynomial 780
eigenspace 1061
eigenvalue 1054, 1141
eigenvector 1054
elementary arithmetic operations 663
ellipsoid 338, 353, 358, 1058, 1078, 1396, 1404, 1422, 1427
ellipsoid algorithm 651
ellipsoid $\mathcal{E}_K$ 1153
ellipsoid method 633, 647
ellipsoid method for concave maximization 649
ellipsoid of maximum volume 654
elliptic cylinder 1058
elliptic type 1067
embeddability 540
embedding 539, 1049
empirical distribution method 1173
empty balls 333
endomorphism 972, 1289
endomorphism of lattice of convex bodies 303
enough symmetries 1194
enumeration of convex polyhedra 463
envelope 719, 720, 1077
epigraph 1083
ε-convex 32
ε-entropy 312

equiaffine differential geometry 1077
equiaffine map 1078
equiaffinities 955
equiangular polygon 1272
equichordal point 361, 368
equichordal sets 146
equidecomposable 516
equidissection 473
equifacetted polytopes 523
equilibrium force 236
equilibrium load 236
equioscillation properties 323
equitransitive tiling 908
equivalent 86, 118, 233, 1407, 1410
equivariant plane tiling 915
equivelarity 546, 547
equiwide 1279
Erdős–Szekeres's problem 452, 453
ergodic 1415, 1428
ergodic theorems 1406
essential variation 103
Euclidean Dehn invariant 967
Euclidean group 993
Euclidean Steiner tree problem 693
Euler characteristic 26, 778, 779, 941, 1419
Euler map 949
Euler point 280
Euler totient function 1016
Euler–Lagrange equation 1110, 1123
Eulerian posets 506
Euler's homogeneity relation 1069
Euler's relation 350, 488, 489, 491, 495, 524, 706, 720, 725
Euler's theorem, curvature 280
Euler's theorem for tilings 910, 912
evolute 1275
expectation 1404
expected numbers of vertices 644
expected time 715
exponential 947
exponential behavior 646
exposed face 276
exposed point 276
exposed skeleton 276
extended h-vector 504, 508
extension theorem for tilings 912
external visibility 435
extremal values of curvature 1057
extremals 1107, 1123
extreme direction 53

extreme point 29, 276, 633, 634, 642, 703
extreme rays 1246

f-vector 416, 487, 489–491, 494, 497–500, 502, 503, 505, 509, 516, 519
face 276, 349, 487, 490, 518, 633, 1252
face function 277
face lattice 487, 488, 518, 526, 998, 1250
– of a tiling 903
face numbers 488
face poset 538
face ring 497, 498
face-to-face tiling 756, 902, 1000
facet 349, 487, 490, 518, 538
facet-forming 527
facet hyperplanes 635
facet-to-facet tiling 756
facet vector 1005
facial graph 704, 707, 709, 728
Fano polytope 621
Fano variety 621
Farkas' lemma 637
farthest point 1332
Favard's inequality 144
feasibility 637, 681
feasibility problems 638
Federer coarea formula 102
Fenchel–Jessen area measure 942
Fenchel's inequality 33
few-distance set 457
Fibonacci number 784
Fibonacci sequence 503
fibre process 1411
final polynomial 591
finite covering 863, 864
finite element analysis 712
finite intersection property 392
finite packing 863
Finsler space 167, 1121
first fundamental form 1048
first-order flex 256
fixed point free map 1224
fixed point property (for a set) 1224
flag numbers 503
flag transitive 524
flag vector 502, 503, 519, 527, 999
flat 452
flat manifolds 35
flat points 1050
flex 230
flipping 713
floating body 353
flow in a network 462
foot condition 203
force 235
form polytope 210
four vertex theorem 1276
Fourier coefficients 1264
Fourier series 131, 1264
– finite 1276
Fourier–Motzkin elimination 639
frame functional 950
Frechet derivative 1094, 1096
frieze groups 905
full cone group 950
full flag 515
fully polynomial approximation scheme 694
functional 1065
fundamental formulae of stereology 1430
fundamental parallelepiped 742
fundamental parallelogram 912
fundamental region 769
Funk–Hecke formula 1279, 1281
Funk–Hecke theorem 1267, 1283
Furstenberg–Tzkoni formula 162
furthest-point Delaunay triangulation 718

G-equicomplementable 939
G-equidissectable 939
G-invariant 937
g-vector 506–508
Gale diagrams 500, 511, 524
Gale transformation 637
Gale transforms 488, 511, 513–515, 517–519
– centrally symmetric 519
Gallai number 407
Gallai's problem 406
gap size 843
Gass–Saaty pivot procedure 646
Gateaux differentiability spaces 1097
Gateaux differentiable 1093
Gateaux differential 1095
gauge for density 804
gauge function 1088
gauge hypersurface 1075
Gauss map, generalized 1059
Gauss–Bonnet map 962
Gauss–Green formula 101
Gauss–Kronecker curvature 24, 280

Gaussian curvature 1048, 1289
– generalized 1059
Gaussian curvature measure 306
Gaussian elimination 639
Gaussian measure 1406
Gaussian type p 1158
Gaussian variable 1406
general position 453
generalized contractive map 1223
generalized h-vector 506, 508, 516
generalized lower bound conjecture 515
generalized normal bundle 282
generalized spherical map 1051
generalized zonoid 1301–1304, 1309
generating distributions 1302, 1303, 1308
generating measures 1304
geodesic 1333
geodesic segment 1333
geodesic space 30
geometric algorithm 701
geometric class 1018
geometric permutation 412
geometric transform 709
geometrical probability 1393
geominimal surface area 163
gift-wrapping 704
glidebending condition 248
global maximization 633, 634
global maximum 642
global minimum 631
global optimum 629
globally rigid 233, 262
Gluskin spaces 1192
Golay code 817
Gordon and Lewis constant 1193
graded algebra 497, 499
graded volume 962
Gram matrix 992
graph 488, 520, 1248
graph algorithm 716
graph-theoretic diameter 785
Grassmann cones 24
Grassmann–Plücker 518
Grassmann–Plücker polynomial 572
Grassmann–Plücker relation 518, 561
Grassmannian 1168, 1359
Grothendieck's constant 1164
Grothendieck's inequality 1164
Grötzsch's theorem 410
Grünbaum's conjecture 407

H^∞ 1247
h-conjecture 420
H-convex sets 398
$\mathcal{H}$-polyhedron 768
$\mathcal{H}$-polytope 768, 873
$\mathcal{H}$-presentation 768, 873
h-vector 419, 490, 494, 498, 502, 508, 513
Haar measure 1271, 1369
Hadamard matrix 816, 817
Hadamard's determinant theorem 1103
Hadwiger functional 956
Hadwiger number 849, 850
Hadwiger-type results 415
Hadwiger's problem 471
Hadwiger's theorem 412
half-volumes 1283
Hamiltonian circuit 521, 689
Hamiltonian cycles 785
Hamming distance 815
hard Lefschetz theorem 499, 510
harmonic expansion 1266, 1277
harmonic homogeneous polynomials 1069
harmonic linear combination 158
harmonic polynomial 1265
Hausdorff dimension 312
Hausdorff distance 936, 1262, 1400
– translative 130
Hausdorff measure 86, 1351, 1394
Hausdorff metric 130, 307, 322, 1329
Hausdorff rectifiable sets 1423
Helly number 394
Helly order 394
Helly property 393
Helly theorem 27, 391
– converse 393
– fractional 421
– multiplied 400
– quantitative 405
– topological 402
Helly theorem on manifolds 403
Helly theorem with volume 405
Helly type 654
Helly-type theorem 394, 410, 783
hemispheres 1283
Hermite matrix 992
Hermite's constant 745
hierarchy 716
Hilbert cube 307
Hilbert function 497, 499, 504
Hilbert–Schmidt norm 1164

Hilbert's parametrix method 1066
Hilbert's 3rd problem 966
Hilbert's 18th problem 918
Hirsch conjecture 522, 644
Hirzebruch variety 611
Hlawka's inequality 1306
homeogonal tiling 913
homeohedral tiling 912
homeomeric type 999, 1012
homeomerism 914
homeomorphic mapping 1049
homeotoxal tiling 913
homogeneity 1054
homogeneity of an arrangement 846
homogeneous discontinuum 992
homogeneous equivariant 1077
homogeneous extension of degree 1065
homogeneous minimum 745
homogeneous of degree r 941
homogeneous tiling 912
homological spheres 491
homology 505
homology cell 26, 402
homothetic copies 45, 470
honeycomb 997
horizon tree 729
Huard's methods of centers 660
hull-forming set operations 430
hyperconvexity 31
hypercylinder 1050
hyperellipsoid 1064
hypermetric 1305
hyperplane arrangement 723
hyperplane network 1426
hyperplane process 1411
hyperplane section 499, 510
hyperplane transversal 413
hypersurface 1047

i-faces of a tiling 902
iceflowers 878
icosahedron 349, 527
ideal norm 1162
illuminated portion 1284
illumination 148, 472
image analysis 1428
immersion 23, 231, 539, 1047
improper faces 487
improving edges 641
incidence algebra 958
inclusion–exclusion principle 938, 1372
increasing edge-path 646
incremental construction 706, 707, 726
indecomposability 304, 512, 1248
indecomposable convex body 304
indecomposable graph 1248
independence system 688
independent set 688
index 1373
index of imprimitivity 1248
indicator function 1088
induced linear map 1050
induction 705, 707, 725
inertia ellipsoid 1179
infeasibility 637
infinitely divisible 1406
infinitesimal behavior 1070
infinitesimal flex 234
infinitesimal motion 509
infinitesimally rigid 234, 509
inflation process 465, 922, 923
inhomogeneous minimum 751
inner angle 957
inner density 803
inner parallel bodies 62
inner quermass 874
inradius 46, 636, 648, 655, 776, 1143
inscribed ellipsoid 653
inscribed polytope 181, 211
insphere 888, 1337
integer hull 783
integer linear program 677
integer polytope 780
integer programming 767
integral geometry 1393
integral polytope 516, 682
integrality assumption 647
integration constants 1056
intensity 1414, 1415, 1426, 1431
intensity measure 1412, 1413, 1425, 1426
interior-point methods 633
intersection body 160
intersection density 1420
intersection graph 416
intersection homology 505
intersection pattern 416
intersectional family 936
interval graph 418
intrinsic class 951
intrinsic inner angle 976

intrinsic inner cone 961
intrinsic outer angle 976
intrinsic outer cone 963
intrinsic volume 49, 768, 771, 941, 1354, 1394, 1408, 1422, 1427
inverse Ackermann function 694
inverse angle 957
inverse projection body 1425
inversion 709, 716, 719
isepiphanic inequality 59
isogonal tiling 905
isohedral tiling 905, 1000
isometry 520, 992, 1153
isomorphic symmetrization 1201
isomorphism 1153
– of crystallographic groups 906
– of tilings 902
isoparametric hypersurfaces 1073
isoperimetric deficit 142
isoperimetric inequality 75, 76, 84, 127, 130, 137, 140, 182, 208, 210, 1268, 1270, 1287, 1420, 1426
isoperimetric problem with constraints 184
isoperimetric theorem, standard 1133
isoperimetric theorems of mathematical physics 1133
isoptic curve 368
isotopy 520, 523, 525, 540, 586
isotoxal tiling 905
isotropic 1402, 1404, 1408, 1413, 1415–1417, 1421, 1422, 1424
isotropic position 1179
isotropic uniform random flat 1422, 1429
isotropy 1410, 1412
iterative methods 647
iterative procedure 1239

Jacobi condition 1115
Jensen's inequality 1084, 1100
job scheduling 885
join 961
join-endomorphisms 1250
joining hypersurfaces 1074

k-algebra 497
k-ball contraction 1229
k-canonical 419
K-convex space 1161
k-core 430
k-flat 1050
k-impassable arrangement 820
k-interior 432
k-pierceability 405
k-set 425, 710
k-set contraction 1229
k-skeleton 521
k-stacked 515
k-transversal 410
Kahane's inequality 1159
Kalai's convolutions 507
Karmarkar's canonical form 656
Karmarkar's projective algorithm 652, 653, 657, 658
Karush–Kuhn–Tucker theorem 636
Katchalski–Lewis conjecture 412
Kepler–Poinsot polyhedron 352
kernel 192–194, 196, 434
– of a star set 1340
kinematic formulae 1354
Kirchberger's theorem 433
kissing number 886
Klee–Minty example 646
knapsack polytope 784
Korkin–Zolotarev reduction 752
Krasnosel'skii's theorem 434
kth order neighbors 850
Kubota formula 65
Kubota's integral recursion 1363
Kugelungstheorem 336

L_2-distance 1262, 1270
L_p-distance 130
ℓ norm 1165
L-simple 937
l^∞ space 1243
ℓ-structure 1177
Lagrange's multiplicator method 1048
λ embedded 1165
Λ-matrix 562, 726
laminated packing 848
Laplace operator 1137, 1263
Laplace–Beltrami operator 1263, 1267
largest contained ellipsoid 654
largest convex subset 729
Las Vergnas circuit axioms 558
Las Vergnas face lattice 518
lattice 492, 741, 768, 1339
lattice basis 1003
lattice complex 1027
lattice constant 745, 754

lattice covering 774, 863
lattice covering density 805
lattice invariant 979
lattice of convex bodies 303
lattice packing 146, 743, 774, 863, 1339
lattice packing density 805
lattice point enumerator 767, 877, 943
lattice-point-free 776
lattice point problems 767
lattice polytope 767, 778, 781, 943
lattice tiling 756, 904
lattice translation 943
lattice width 777
Laurent polynomial 608
Laves tilings 908
law of the iterated logarithm 1406
Lawrence polytope 517, 519
Lebesgue–Blaschke inequality 127
Leech lattice 817, 886
Legendre condition 1124
Legendre ellipsoid 1179
Legendre polynomial 1278
Leray's theorem 417
level algorithm 883
lexicographic rule 643
Liebmann's theorem 294
limited semiconvex disk 841
Lindelöf's theorem 210
line process 1411
line shelling 489, 490, 492, 493, 514
line transversal 413
linear code 815, 816
linear differential operator 1066
linear diophantine equations 681
linear exponent 1243
linear family 46
linear i-stress 510
linear list 704
linear ordering problem 690
linear program 523, 703, 716
linear programming 522, 629, 635
linear programming bound 825
linear programming problem 635, 783
linefree 453
Lipschitz α-contraction 1229
Lipschitz map 1223
Lipschitz–Killing curvature 1052
LLL reduction 753
load 235
local isomorphism 923
local maximization 633
local maximizer 631, 642
local maximum 631
local minimizer 631
local minimum 631, 642
local parallel set 53, 282
local uniform continuity 970
locally convex 21, 1049
locally finite tiling 902
logarithm 947
logarithmic barrier function 652, 658, 660
Lovász reduction 753, 786
lower bound theorem 493, 509, 515
– for polytope pairs 501
lower density 804
lower orthogonal projection 1073
lower semicontinuity 1109, 1126
Löwner–John ellipsoid 360, 654, 1337
Löwner–John sections 652, 653
LP duality 638

M-ellipsoid 1182
m-rectifiable 1375
Mahler's conjecture 745
Mahler's selection theorem 311
map 539
marked prototile 922
marked tiling 904
matching 687
matching polytope 687
matching problem 686
mathematical programming 629
matroid 517–519, 573, 688
matroid intersection polytope 689
matroid polytope 518, 575, 576, 688
matroid representability 519
max-flow problem 639, 686
max flow–min cut theorem 686
maximum principle 1072
McMullen conditions 495, 499, 504–506, 510, 522
McMullen's problem 423
meager set 311, 1329
mean curvature 139, 280, 1048
mean projection measures 129, 136, 140, 1262, 1285
mean radii of curvature 1055
mean visible volume 1422
mean width 970, 1168, 1262, 1277, 1286
measure of convexity 32

measure of symmetry 356, 362
member directions 262
members 232
membership oracle 746
metallic glasses 878, 887
method of central sections 652
metric projection 282, 1338
metric tensor 992
midconvex 25, 1084
middle perversity intersection homology Betti numbers 505
Miller indices 1004
minimal invariant set 1227
minimal shell 1337
minimal width 46, 325
minimizer 631
minimum covering density 327
minimum ellipsoid 1337
minimum energy packing 887
minimum spanning tree 692, 716
Minkowski additive 937
Minkowski area 1420, 1426
Minkowski functional 1152
Minkowski inequality 58, 137, 138, 159, 163, 168
Minkowski linear combination 156
Minkowski plane 470
Minkowski space 364
Minkowski sum 45, 304, 515, 936, 1204, 1275, 1360
Minkowski symmetrization 1207
Minkowski–Fenchel–Aleksandrov inequalities 1422
Minkowski–Hlawka theorem 748, 807
Minkowskian geometry 163, 167, 1271
Minkowski's existence theorem 288
Minkowski's fundamental theorem 744, 769, 771
Minkowski's integral formulas 1053
– generalized 1072
Minkowski's linear form theorem 746
Minkowski's second theorem 750, 771
Minkowski's surface area 160
Minkowski's theorem on constant width 366
mirror image 1335
mixed area 52, 131, 1262, 1269, 1288
mixed area functions 942
mixed area inequalities 131, 1269, 1270
mixed discriminants 1060
mixed-integer programming 677
mixed moment vector 943
mixed polytope 953
mixed principal minors 1060
mixed projection body 1421
mixed volume 47, 137, 138, 156, 170, 469, 510, 605, 773, 940, 1059, 1285, 1286, 1419
model-based 1429
modulus of continuity 877
mollifier 97
moment curve 493, 499, 651, 711
moment map 498
moment of D 186
moment of inertia 1275
moment problem 755
moment vector 943
momentum lemma 186, 201, 207, 213
monohedral tiling 904, 918, 1000
monotone functional 969
monotone operator 1098
monotone valuation 969
monotypic tiling 920
Mordell's converse problem 757
most elements 1329
multi-dimensional Fourier 1290
multi-dimensional searching 702
multiple variational integrals 1122
multiplication 945
multiplicity 1069
multiprocessor scheduling 884
mutation 580
mutually orthogonal 1068

n-dimensional Hilbert space 1242
n-dimensional l^p 1254
n-gon associated with C 195
n-hedral tiling 904
n-neighbor packing 847
near-critical 459
nearest point map 21, 282, 1338
nearly all elements 1331
needle problem 1393
negative type 1305
neighbor 846
neighbor in a packing 1339
neighborly polytopes 493, 517, 521
nerve 415, 889
nerves of convex sets 509
network design problems 691
network flow problems 639, 685
Newton number 849, 850

Newton polytope 605
Newton's formulas 1073
Newton's method 663
Niggli reduced 1011
Nikodym metric 308
nondegenerate affinity 1077
nondegenerate families (NDFs) 402
nonexpansive map 1223
– generalized 1229
– multivalued 1230
nonfacet 527
nonperiodic tiling 904
nonpolytopal spheres 512, 518, 526
nonrational polytopes 506, 512
nonregular triangulations 514
nonrevisiting conjecture 523
nonseparable arrangement 820
nonshellable pl-spheres 496
nonshellable simplicial balls 488
nonsimplicial polytopes 503, 505, 509
nonsimplicial spheres 512
norm 1152, 1245
normal cone 277, 951
normal point 280, 1059
normal structure 1227
normal tiling 902
normal vector
– r-exposed 278
– r-extreme 278
normality lemma 909
normalization vector 1075
normalized 1068
normalized elementary symmetric functions 1048
normalized hyperline sequence 576
normalized moment 1395
normalizer 1026
NP-complete 638
NP-hard 634, 642
nuclear norm 1162
nullhomotopic differences 27
number density 805
number of Radon types 426
νth curvature measure in the sense of Federer 1053
νth surface area measure in the sense of Aleksandrov 1055

octahedron 349, 1285
Oda's conjectures 622
odd antihole inequality 692
odd circuit constraints 691
odd wheel inequality 692
on-line packings 882
one-sided curved 1049
Oppenheim's conjecture 754
optimal basis 1014
optimal control 337
optimal vertex 637
optimization 650
optimizing vertex 641
oracle 875
orbit type 1027
order complex 527
order types 562
ordered field 630, 634
ordinary point processes 1411
oriented hyperline 576
oriented matroid 424, 488, 517, 526, 559, 727
oriented matroid manifold 582
orthogonal functions 1265
orthogonal group 993
orthogonal projection 146, 1057, 1278
orthogonal transformations 1060
orthogonality defect 742
orthoscheme 198, 200, 201
osculating body 1284
osculating circle 1273
osculating domain 1271
outer angle 957
outer density 803
outer normal 100, 277, 636
outer parallel body 49
outer parallel hypersurface 1053
outer quermass 874
output sensitive 705
oval 456
ovaloid 1055
overlap size 843

(p,q)-problems 408
p-absolutely summing norm 1163
#P-hard 649
p-metric 322
p-tangential body 279
p-vectors 527, 528
packing 146, 743, 801, 863, 1027
– k-fold 821
packing constant 146

packing density 743, 804, 864, 1339
packing lattice 743
Palm measures 1424
paradigm 702, 706
parallel map 1074
parallelohedron 744, 756, 916
parallelopiped 1154
parallelotope 338, 634, 1005, 1285, 1403, 1422, 1427
1-parameter transformation groups 1075
Parseval's equation 1264, 1266
– generalized 1265, 1267
particle process 1411, 1424
patch of a tiling 908
path-following method 659, 662
pedal curve 1275
penny packing 888
Penrose tiling 922
pentagrid method 924
perfect matching 788
perfect square 462
perimeter 50, 94, 1269
perimeter deviation 191, 322
periodic arrangement 806
periodic element 1250
periodic group 905
periodic tiling 904
Peterson map 1075
Petty body 161, 170
Petty projection inequality 156, 157, 169
Petty's affine projection inequality 158, 160
Petty's geominimal surface area inequality 163, 166
phase space 1334
Picard group 618
Pick's identity 778
piercing number of $\mathcal{K}$ 406
pigeon hole principle 747
pivot rule 641, 646
pl-sphere 491, 512, 526
planar graph 520, 706
plane-sweep 706, 719
planefree 453
plank 1210
Platonic solids 349, 524, 545
plesiohedron 997
Poincaré duality 499
Poincaré-type formulae 1377
point density 805
point group 994
point process 1394, 1410, 1411, 1415, 1416
point trapping arrangement 818
points of twice differentiability 1058
Poisson hyperplane network 1427
Poisson hyperplane processes 1404
Poisson process 1413, 1415, 1417, 1421
polar body 156, 355, 1154, 1403, 1420
polar cone 964
polar coordinates 1058
polar intersection problem 170
polar lattice 743, 769
polar polyhedron 204
polar projection problem 169
polarity 488, 709, 719, 724, 964
polyconvexity 1126, 1128
polygon, self-similar 465
polygonal domain 1273
polyhedral combinatorics 646, 684, 784
polyhedral complex 538
polyhedral manifold 538
polyhedral map 539
polyhedral realization 540
polyhedral set 939
polyhedron 538, 936
polymatroid 689
polymatroid function 689
polynomial time 783
polynomial-time algorithm 632
polytopal set 1284
polytopal spheres 526
polytopality 496, 518, 540
polytope 181, 349, 471, 498, 517, 633, 996, 1248
3-polytope 489, 503, 514, 517, 520
polytope algebra 944
polytope graphs 522
polytope group 955
polytope pairs 500
polytope realizability 519
porosity 1331
positive sets and bases 433
potential function 657, 658, 661
power diagram 722
pre-stress stable 260
preassigned 512, 526
primal degeneracy 637
primal–dual algorithm 636, 639
primitive graph 1248
primitive parallelohedron 756, 916
Prim's algorithm 716

principal curvatures 280, 1048
– generalized 1059
principal directions 280
principal kinematic formula 1354, 1407, 1409, 1416, 1417, 1419
principal minors 1055
principal radii of curvature 1054
priority queue 717, 728
prism 524
probabilistic counting 711
processes of flats 1411
product of non-homogeneous linear forms 756
projection 146, 1280, 1360
projection body 48, 147, 156, 160, 170, 355, 1280, 1281, 1299, 1300, 1314, 1316, 1320, 1376
projection constant 1187
projection function 1315, 1316
projection generating measure 1315, 1317
projection method 924, 925, 1032
projective coordinates 1064
projective exponent 1243
projective scaling trajectory 662
projective space 724
projective toric variety 498, 616, 619
projective transformations 653, 656
proper affine sphere 1078
proper convex body 303, 1329
proper convex function 1086
proper face 487
proper stress 236
prototile 756, 904, 1000
pseudo-hyperplane arrangement 727
pseudoline arrangement 559
pseudopower 494, 497
pth order area function 942
pulling triangulations 515, 516
pyramid 642, 645
Pythagorean inequality 66

QS theorem 1171
quad-edge data structure 704
quadratic form 1003, 1009, 1047, 1062
quadric 1058
quasi-nonexpansive map 1228, 1229
quasiconvex 630, 634, 1126
quasicrystal 878, 925, 1031
quasiperiodic 925
quasiregular system 1032
quasiregular tiling 1033
quasisimplicial polytopes 508
quermass densities 1408, 1415, 1417, 1430
quermassintegral 49, 168, 366, 767, 773, 941, 1363, 1417
quotient polytope 487

(r,k)-divisible 428
(r,R)-system 992
r-convex disk 842
R-disk 841
r-divisible 427
r-exposed point 276
r-extreme point 276
r-singular point 278
r-skeleton 276
Rademacher functions 1156
Rademacher projection 1159
radial function 155, 160
radial map 291, 1074
radii 888
radius of curvature 1275
Radon curve 368
Radon measure 1408
Radon partition 422, 566
– primitive 422
Radon transform 1309, 1310, 1315
– spherical 1279, 1307
Radon type 423
Radon's theorem 391
Ramsey theory 454
Ramsey's theorem 453
random access machine 701
random closed set 1394, 1402, 1407
random convex body 1404, 1408
random curve 1419
random flat 1380, 1402, 1429
random hyperplane 1402
random hypersurface 1419, 1421
random mosaic 1423, 1428
random points 1395
random sequence 715
random set 1394, 1404, 1408, 1413
random surface 1419, 1423
randomized algorithm 704
randomized incremental 715
randomized methods of volume computation 649
randomized pivot rule 646
rank 518
rank function 688

rank one convex 1125, 1128
rational n-space 635
rational polytope 498, 505, 523, 769
realizability 488, 518, 540, 565, 588
realization space 525
rearrangement of functions 1134
Reay conjecture 428
recession cone 961
reciprocal basis 1004
reciprocity law 780, 943
recognizing optimality 643
rectilinear Steiner tree problem 693
rectilinearly convex 694
recursion 705
reduced basis 786
reduced body 365
reduced boundary 100
reduced copies 470
reduced covering 806
reduced νth mean curvature 1074
reflection theorem 1073
regression line 729
regular arc-sided n-gon 192–194
regular case 185, 186
regular conic n-gon 196
regular convex body 1059
regular convex reflected curvature image 1078
regular CW spheres 527
regular map 545
regular point 278
regular polygon 1274
regular polytopes 349, 488, 527
– 3-polytopes 349, 524
– 4-polytopes 524, 527
– d-polytopes 524
regular simplex 357, 655
regular solids 524
regular stellar subdivision 613
regular system of points 992
regular tiling 905, 1000
regular toric variety 612
regular triangulation 515, 523, 714, 722
regularity condition 995, 1001
relative differential geometry 163, 1075
relative geometric Minkowski's integral formulas 1076
relative geometric support function 1076
relative neighborhood graph 693
relative principal curvatures 1076
relative principal radii of curvature 1076
relative sphere 1076
relative surface area 1076
relaxation 1119
relaxation method 651
reorientation class 559, 581
representable matroid 517, 519
representation 419
reptile 465
residual sets 1329
resolution of the singularities of a toric variety 614
resolve 235
Reuleaux triangle 127, 145, 364, 366
reverse spherical image 286
rhombic domain 1274
rhombododecahedron 505
ridge 487, 490
Riemannian curvature tensor 1050
Riemannian metric 327, 1077
Riemannian spaces 1121
Riemannian target space R^d 1075
rigid 230, 234, 519
rigid motion 509
rigid motion translation equivariant 971
rigidifying pre-stress 260
rigidity 225, 494, 509
rigidity map 239
Robinson-convex 400
Robinson tiles 922
Rogerian disk 841
rolling freely 1056
root class 1019
rotation averaging 56
rotational symmetry 1274
rotatory symmetry 335, 336
rotor 367, 1271, 1273, 1284
rounding 647
rth weight space 948

sandwich algorithms 332
Santaló point 165, 166
saturated arrangement 848
saturated packing 806
sausage 848, 868
sausage catastrophe 871
sausage conjectures 869
sausage covering 863
sausage packing 863
sausage-skin 888
scalar multiple 45, 936

Schauder basis 1193
Schlegel diagram 514, 541
Schneider's notion of distance 329
Schwarz rounding 56
Schwarz symmetrization 1135
search directions 658
search tree 721
second fundamental form 327, 1048, 1076
second intersection body 160
second order 1067
second order flex 256
second order rigid 256
second projection body 160, 170
secondary polytope 515
sectional curvature 23, 1050
sections of a convex body 1289
segment in a metric space 308
segmental closedness 634
selection 1404
self-dual lattice 1004
self-stress 236
semialgebraic variety 525
semiconvex disk 841
semiequivalence type 423
semigroup of convex bodies 304
semiregular polytopes 351, 524
semispaces 423
separating hyperplane 650
separation problem 690
separation theorem 638
set-covering 683, 890
set of directions 459
set of distances 457, 458
set of first category 1329
set of second category 1329
set-packing 683, 890
set-partitioning 683
sets of positive reach 22, 1423
shadow boundary 358, 1074, 1337
shape distribution 1414, 1415, 1431
sheaf 617
shelf algorithms 883
shell 1053
shellability 488, 490, 504
shellable ball 495, 514
shellable complex 497
shellable pseudomanifold 496
shellable simplicial complex 490, 494, 510
shellable spheres 491, 496
shelling 488, 489, 492, 494, 508, 714
shelling structures 457
shift operator 1244
shortest path problem 639
Siegel's formula 747
Siegel's mean value theorem 749
σ-porous set 1331
signed area 1275
signed set 517
signed vector 561
simple perfect square 462
simple polytope 488, 493, 499, 521, 641
simplex 355, 527, 650, 652, 656, 1285
simplex algorithm 640
simplicial ball 513
simplicial complex 244, 490, 496, 497, 500, 509, 510, 513, 527, 538, 711
simplicial convexity 32
simplicial neighborly polytope 525
simplicial pl-sphere 510, 514
simplicial polytope 488, 491, 493–496, 498, 499, 503, 505, 509, 510
– centrally symmetric 502
simplicial sphere 491, 495, 500, 502
simulated perturbation 707
singular normal vector 279
singular point 278
six-color theorem 409
size 632
skeletal points 28
skeleton 488, 642
slope 459
slope-critical 459
smallest containing ellipsoid 654
smallest enclosing sphere 716
smooth convex body 46, 275, 337, 1330
smooth point 278, 1330
smooth regular n-gon 193, 194
smoothing 34
solid covering 834
solid packing 834
space dilatation 652
space fillers 919
space group 906, 994, 1024
space-group type 994, 1025
space ℓ_p^n 1151
space of lattices 741
spanning tree 522, 692
spatial statistics 1428
specific convexity number 1419
spectral radius 1239, 1339

spectrum 452, 460
sphere 352, 496, 518, 519, 525, 1055
sphere covering 867
sphere packing 867, 1290
sphere system 569
spherefree 453
spherical harmonics 138, 826, 1069, 1265
spherical image 286, 1054
spherical map 1049
spherical polytope 523
spherical simplicial complex 511
spherically symmetric rearrangement 1135
spider web 251
spin 249
spiral tiling 919
squared rectangle 462
stability estimates 1281
stability function 128
stability inequality 128
stability problem 128, 310
stabilizer 994
stable contraction 1242
stable convex body 277
stable random bodies 1406
stable set polytope 691
stable set problem 691
stable type p 1173
stack 713, 729
stacked polytopes 493, 515
standard form 640
standard sequence 1265
Stanley–Reisner ring 496, 497, 499, 504, 509, 510
star body 754, 1340
star set 1340
starshape 435
statically rigid 236
stationarity 1410, 1412
stationary 1408, 1413, 1416, 1419, 1421
Steiner ball 140, 1262, 1285
Steiner disk 1262, 1270
Steiner point 305, 338, 693, 971, 1269, 1274, 1286, 1413
– abstract 971
Steiner symmetral 54, 1339
Steiner symmetrization 26, 106, 154, 1208
Steiner tree problem 693
Steiner's continuous symmetrization 1074
Steiner's formula 49, 1053
– generalized 1406
Steinitz's exchange lemma 573
Steinitz's theorem 432, 520, 524–526
stellar subdivision 513, 527, 613
step-by-step approximation 332
stereographic projection 710, 722
stereohedron 920
stereology 1428
stereotope 1000
stiffening 56
stiffness matrix 259
stochastic geometry 1393
Stokes' theorem 1054
stress 235, 503, 509
stress matrix 259
stress space 509, 510
stressed direction 262
strict contraction 1223
strict minimizer 631
strictly concave function 630
strictly convex 46, 245, 275
strictly convex function 630
string construction 354
string of spheres 848
strip group 905
strong d-collapsibility 418
strong independence 427
strong law of large numbers 1405
strong separation problem 650
strongly balanced tiling 910
strongly convex 400
strongly polynomial algorithms 663
strongly polynomial-time algorithm 647
strongly polytopal 619
strut 232
subdifferentiable 1098
subdifferential 1094
subdivision 513, 516
subexponential randomized algorithm 646
subfacet 487, 490
sublattice 742
submodular network flow problems 686
subperiodic group 1030
subperspective pair 304
subtour elimination constraints 690
successive minima 750, 770
support element 282, 1361
support function 45, 277, 305, 1054, 1088, 1262, 1269, 1277, 1405, 1419
support functional 937
support hyperplane 21, 46, 277, 1049
support line 1091
support plane 1093

supporting halfspace 636, 1047
surface area 50, 768, 941, 1205, 1262, 1283
surface area measures 282, 1281
surface deviation 309, 322
Sylvester's law of inertia 1061
Sylvester's problem 154, 1396
symbol of a plane tiling 908
symmetric convex body 1152
symmetric difference metric 85, 190, 308, 322, 1335
– homogeneous 309
symmetry constant 1194
symmetry domain 1028
symmetry group 524
symmetry of a tiling 904, 905
symmetry operation 994
symmetry scaffolding 1019
symmorphic space group 1024
Szökefalvi–Nagy problem 395

tangential body 54
tangential polygon 1271
tangential radius of curvature 281
Tarski's decision method 526
Taylor's formula 1058
TDI 684
Teichmüller theory 35
tensegrity framework 232
tensegrity graph 232
tetrahedron 349, 1285
theorems of the alternative 637
third fundamental form 1072, 1075
three-index assignment problem 690
tight embedding 545
tile 805, 902
tiling 756, 801, 902
– finite 461
– non-periodic 465
– polygon 466
tilings by rectangles
– faultfree 466
– perimeter restriction 473
tilings by squares 461
tilings by tiles of equal area 472, 473
tilings by triangles 464
tomography 1339
topes 570
topological equivalence of tilings 904
topological manifold 1049
topological spheres 496
topological sweep 727
topological type of packing 1027
topology on the space of lattices 743
toric subvariety 611
– affine 611
toric variety 496, 498, 505, 511, 607, 611
– affine 609
torsional rigidity 1139
torus action 607
total absolute curvature 1052
total curvature 663
total edge-length 207, 208
total variation 101
totally balanced matrix 684
totally convex 1128
totally dual integral 684
totally separable packing 845
totally unimodular 783
totally unimodular matrix 683
touching cone 53, 278
touching probabilities 1371
tour 689
trajectories 662
transference theorems 745
translate 936, 1071
translation covariant 951
translation covering density 805
translation invariance 936, 1066
translation packing density 805
transnormal manifold 367
transversal problem 413
traveling salesman polytope 784, 785
– asymmetric 691
traveling salesman problem 645, 689
– asymmetric 691
– planar 692
triangular domains 1273
triangulated packing 847
triangulated sphere 491, 495
triangulation 464, 513, 516, 711
Tschebycheff-vector 1077
Tverberg's theorem 427
– colored 429
– topological 429
type of a tiling 906, 1000
type of parallelohedra 756
type of regular packing 1028
type p constant 806
types of crystal forms 1023
typical cell 1424
typical element 1329

umbilics 354
unbiased estimators 1430
unbounded pointed polyhedra 523
unbounded simple polyhedron 491
unconditional constant 1154, 1155
undecidability 514
underlying matroid 517
underlying space 244
uniform random flat 1429
uniform random point 1380, 1395
uniform tiling 907
uniformly convex 1161
uniformly k-Lipschitz map 1229
unimodality 491, 506
unimodular mapping 979
unimodular transformation 780, 1063
union set 1412, 1416
uniqueness problems 634
uniqueness theorems 1070
unit ball 1052, 1152, 1245
unit circle 1049
unit normal bundle 1051
unit normal vector 1047
universal convex figures 26
upper bound theorem 493, 496, 509, 704, 708
– for families of convex sets 420
– for polytope pairs 500
upper curvature 279
upper density 804
upper orthogonal projection 1073
Urysohn's inequality 65, 137, 143, 1210

$\mathcal{V}$-polytope 873
$\mathcal{V}$-presentation 873
– of a polytope 768
valence 337
valuation 48, 936
variances 1431
variation of constants 1056
vector sum 936
verification problem 694
vertex 487, 490, 538, 542
vertex figure 499
vertex-finding algorithms 644
vertex of a plane tiling 902
visibility 435, 1421
visibility graph 720
visible distance 1420
volume 46, 495, 768, 948, 1262
volume element 1052, 1055
volume product 1403, 1427
volume ratio 164, 1169
volume-weighted 1430
Voronoi diagram 718
– generalized 716, 721
– order-k 723, 724, 726
Voronoi mosaic 1424, 1428
Voronoi region 812, 916, 997
Voronoi's problem 756

wallpaper groups 905
Wang tiles 922
weak Asplund space 1096
weak distance 1199
weak Hilbert spaces 1171
weak membership oracle 875
weak separation problem 650
weak valuation 937
weakly continuous 969
weakly convex 1128
weakly neighborly polytope 516
wedge 1273, 1274
Weierstrass condition 1124
Weierstrass excess-function 1114, 1125
Weierstrass principle 86
weight of a word 815
weighted matching problem 890
Weingarten equations 1049
well approximating polytopes 331
Weyl problem 23
Whitney number 425
width 776, 874, 970
Wigner–Seitz cell 997
winding numbers 515
Wirkungsbereich 916
Wirtinger's inequality 81, 133, 138, 1287, 1288
Wyckoff position 1026
Wyckoff set 1026

Young's inequality 1088

Z-reduced 1010
zero set 1050
zeta-function 755
zonal characterisation 1310
zone 724, 1006
zonoid 156, 334, 1180, 1204, 1280, 1282, 1300, 1301, 1307, 1309, 1320, 1403, 1419
zonoidal functions 1307
zonotope 334, 363, 521, 640, 727, 774, 1007, 1204, 1299, 1301, 1310